Eclipses by Appointment - Supplement

Volume 4 of 4

R540-R675 and remaining reviewed evidence

Original continuous pages 2451-3269 of 3269

Navigation: each volume adds this one-page guide.

A page in this volume = original page minus 2450 plus 1.

The public supplement-review-pages.csv maps every R key to volume

and exact page. The scientific-review-register.csv maps status to report.

Volume 1: Original A catalogue and R001-R216 (original pages 1-1186).

Volume 2: Fixed magnitude comparison and R217-R409 (original pages 1187-1947).

Volume 3: R410-R539 (original pages 1948-2450).

Volume 4: R540-R675 and remaining reviewed evidence (original pages 2451-3269).

Original illustrated A entries and all R full reports remain in the

continuous source HTML and the public reports/evidence archive.

Cross-volume references use the page map above.

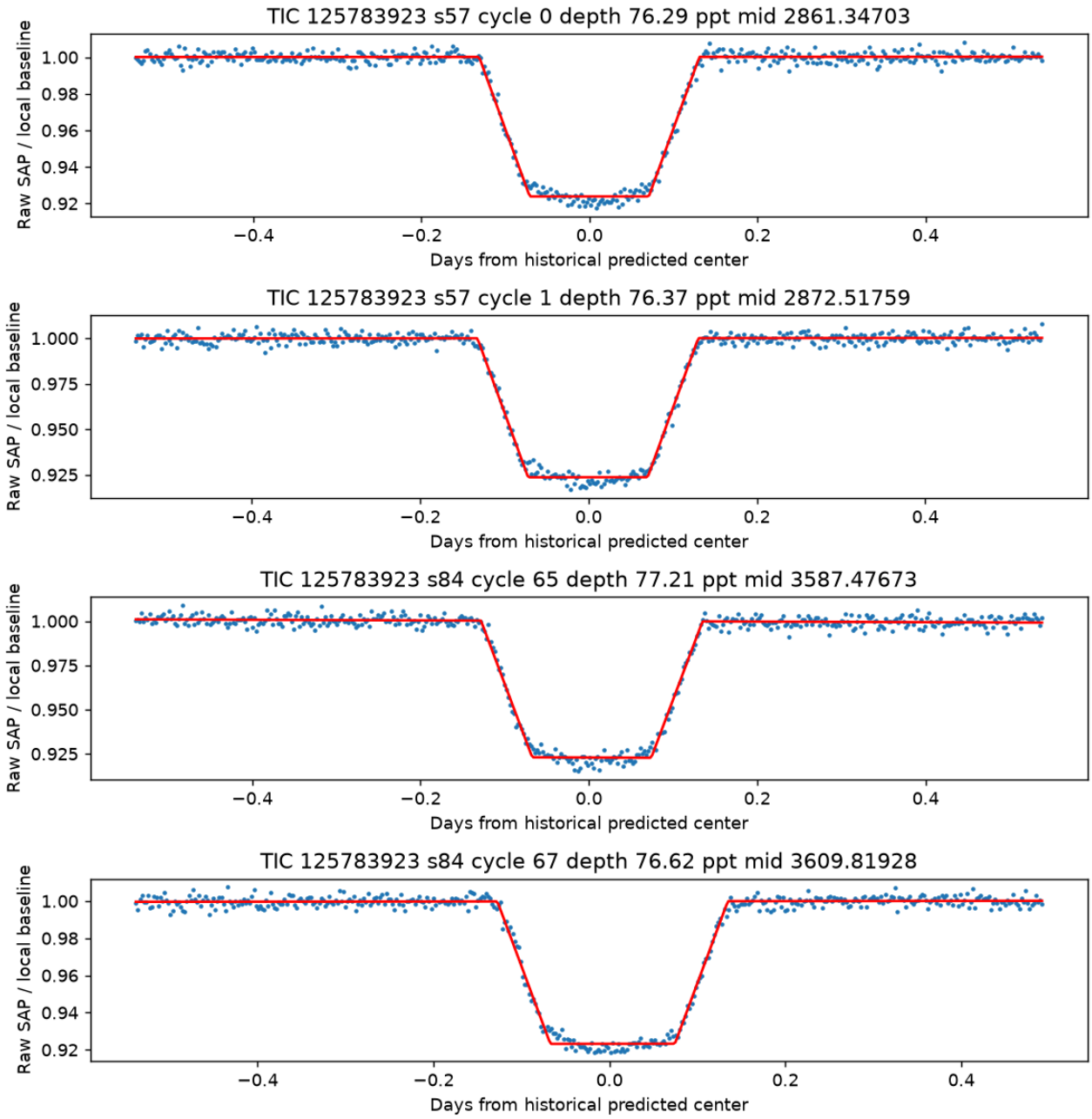
R540. TIC 125783923

physical_verdict: CONFIRMED

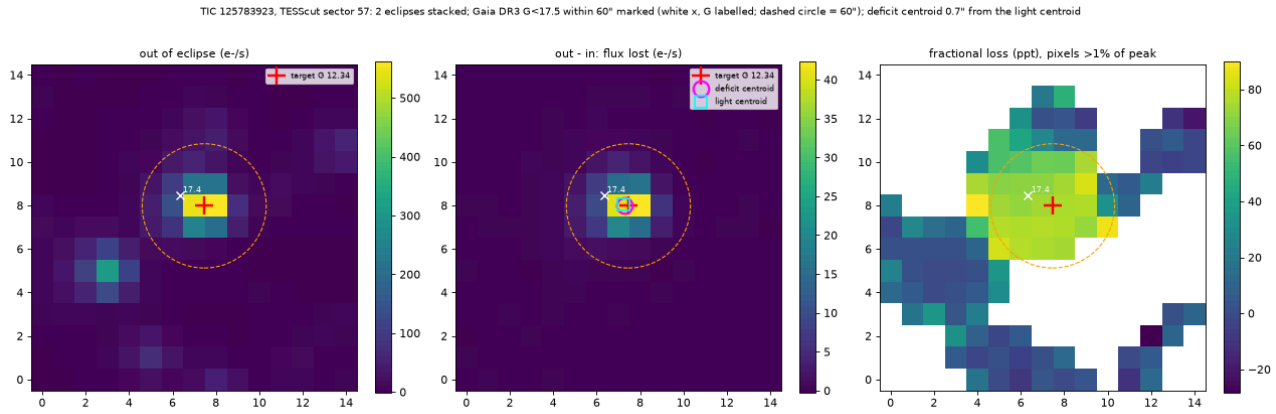
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision physical parameters.



R540; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R540; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 125783923: CONFIRMED

Full bounded physical review: **COMPLETE**. Physical classification is separate from catalogue recovery/novelty; historical screening was not an admission verdict.

1. **Raw events.** 4 free-center trapezoid measurements from quality-zero QLP SAP_FLUX in sectors [57, 84]; 4 enter the reported ephemeris. Representative raw light curves and numerical windows show repeated eclipse morphology. Predictions guided the windows and are not measurements. Flagged or uncovered windows are not clean nulls.
2. **Timing and aliases.** Measured recurring-eclipse timing supported. P=11.1712318772 +/- 0.0000176648 d; reference BTJD=2861.3466987. Epoch kind: Selected repeating eclipse; depth rank to be checked against other event kind. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The formal error column is the local free-center model covariance; the input error adds a half-cadence floor. Singular/unidentifiable centers are excluded. Cycle/parity counts are in ../alias-audit.json. Selected eclipse means the measured recurrence, with no assumption that it is the deeper primary. The bounded half/double check does not prove all possible aliases absent.
3. **Attribution.** Fresh difference stack in sector 57 has 2 events; target coordinate [7.461893783837478, 7.980070498167082], deficit [7.261013222514187, 8.018100919870943]; offset 0.204 pixel. The image-model fractional residual 0.380 is not a localization uncertainty. Every capable Gaia neighbor in the retained 60-arcsec table was inspected: None under the approximate G-band brightness screen within 60 arcsec. The intrinsic loss proxy uses a pairwise G-band flux budget and cannot be directly compared with diluted pixel fractional loss. Close blends and passband/aperture response limit exclusion claims. Four deep raw eclipses; compact target-centered deficit and no flux-capable Gaia neighbor within 60 arcsec.
4. **Alternative reduction.** Fresh TESScut background-subtracted target-centered 1x1/3x3/5x5 depths are 74.10/74.32/73.83 ppt; selected QLP fitted median 76.50 ppt. This reruns the historical extraction primitive using the fresh ephemeris and measured duration, with identity and source hashes checked. The alternative extraction and the QLP data for this sector share the same TESS photons; they test processing robustness and are not independent observations. Distinct sectors are

distinct observing occasions on one instrument. Depth differences can reflect dilution, aperture and background response; intrinsic binary depth is not inferred.

5. **Gaia geometry/physical plausibility.** Exact DR3 source 2851838871293063424, orbital P=11.1702778456 d, e=0.07352. Closest edge-on conjunction offset=-0.04497 d (-0.00403 phase), full source-specific covariance linear timing allowance=0.42043 d. Full 6x6 orbit correlation matrix is positive definite (helper receipt). Mass function=0.023717 solar masses; adopted primary mass=1.08, minimum secondary mass=0.368. Rough central duration=7.41 h versus observed median=6.30 h. Radius and main-sequence companion assumptions make this a plausibility calculation, not a fitted geometry or calibrated significance veto. Shorter observed duration can reflect impact parameter and model assumptions. Orbit agreement does not establish ownership among unresolved sources.

Deciding limit: Four deep raw eclipses; compact target-centered deficit and no flux-capable Gaia neighbor within 60 arcsec.

Next check: Optional resolved photometry and spectroscopy for precision physical parameters.

Catalogue recovery is recorded centrally in ../../shared-catalogues/coverage-matrix.csv. A successful cone with no exact eclipsing target classification is distinct from a failed query; known recoveries must not count as new additions. No exhaustive literature priority claim is made.

Sector	Search cycle	Measured midpoint BTJD	Formal center error d	Timing input error d	Depth ppt	Timing status
57	0	2861.3470318	0.000279202	0.00115738	76.29	selected
57	1	2872.5175937	0.000280544	0.00115738	76.37	selected
84	65	3587.4767277	0.000289568	0.00115739	77.21	selected
84	67	3609.8192812	0.000272936	0.00115739	76.62	selected

Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png, historical original receipts, ../measurement-source-receipts.json, ../identity-check.json, ../alias-audit.json, ../gaia-geometry.json and ../helper-audit.md. Original inputs and fresh outputs are covered by ../SHA256SUMS.json and ../source-hashes.json.

Stable review key: R540.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s57.json - historical/diff_s57.png - historical/gaia.json - measurements.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

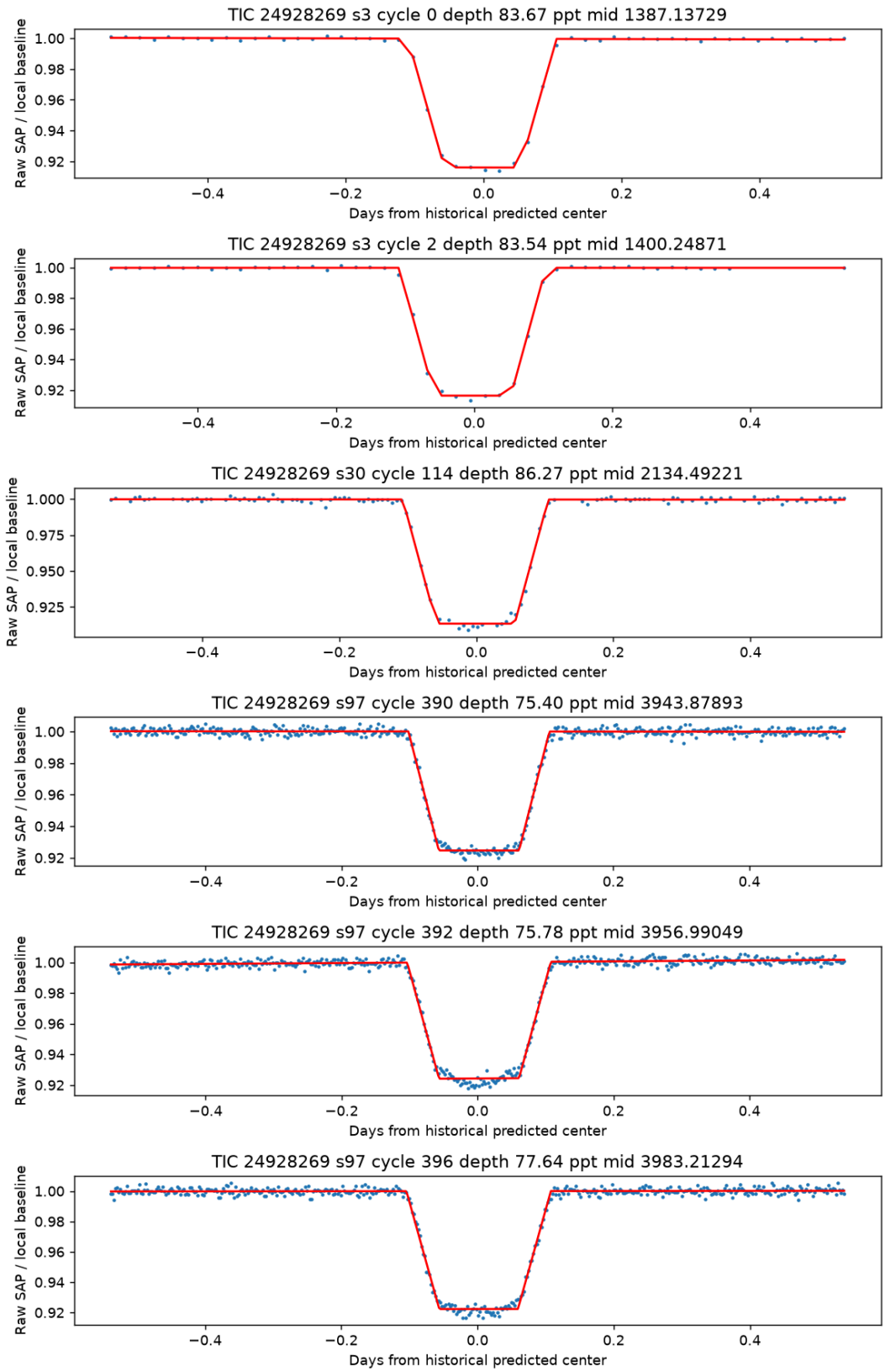
R541. TIC 24928269

physical_verdict: CONFIRMED

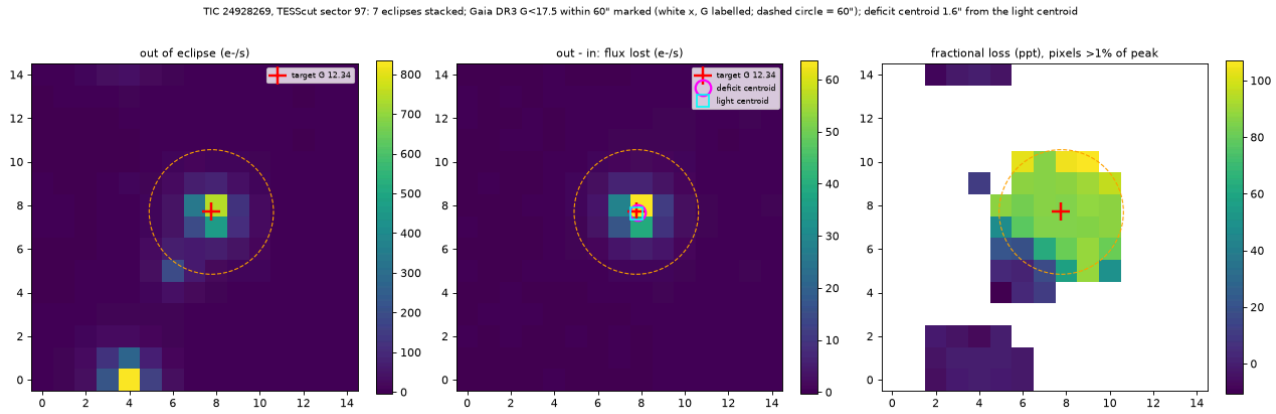
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision physical parameters.



R541; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R541; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 24928269: CONFIRMED

Full bounded physical review: **COMPLETE**. Physical classification is separate from catalogue recovery/novelty; historical screening was not an admission verdict.

1. **Raw events.** 12 free-center trapezoid measurements from quality-zero QLP SAP_FLUX in sectors [3, 30, 97]; 12 enter the reported ephemeris. Representative raw light curves and numerical windows show repeated eclipse morphology. Predictions guided the windows and are not measurements. Flagged or uncovered windows are not clean nulls.
2. **Timing and aliases.** Measured recurring-eclipse timing supported. P=6.5557481629 +/- 0.0000077663 d; reference BTJD=1387.1372325. Epoch kind: Selected repeating eclipse; depth rank to be checked against other event kind. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The formal error column is the local free-center model covariance; the input error adds a half-cadence floor. Singular/unidentifiable centers are excluded. Cycle/parity counts are in ../alias-audit.json. Selected eclipse means the measured recurrence, with no assumption that it is the deeper primary. The bounded half/double check does not prove all possible aliases absent.
3. **Attribution.** Fresh difference stack in sector 97 has 7 events; target coordinate [7.76964353045026, 7.698866310414914], deficit [7.846047728639347, 7.684373126419072]; offset 0.078 pixel. The image-model fractional residual 0.209 is not a localization uncertainty. Every capable Gaia neighbor in the retained 60-arcsec table was inspected: None under the approximate G-band brightness screen within 60 arcsec. The intrinsic loss proxy uses a pairwise G-band flux budget and cannot be directly compared with diluted pixel fractional loss. Close blends and passband/aperture response limit exclusion claims. Twelve deep raw eclipses over S3/30/97. Sector-dependent depth is consistent with changing dilution; isolated target-centered deficit.
4. **Alternative reduction.** Fresh TESScut background-subtracted target-centered 1x1/3x3/5x5 depths are 85.35/85.78/83.76 ppt; selected QLP fitted median 77.24 ppt. This reruns the historical extraction primitive using the fresh ephemeris and measured duration, with identity and source hashes checked. The alternative extraction and the QLP data for this sector share the same TESS photons; they test processing robustness and are not independent observations. Distinct sectors are

distinct observing occasions on one instrument. Depth differences can reflect dilution, aperture and background response; intrinsic binary depth is not inferred.

5. **Gaia geometry/physical plausibility.** Exact DR3 source 2460872435038706944, orbital P=6.5555008903 d, e=0.03484. Closest edge-on conjunction offset=-0.00266 d (-0.00041 phase), full source-specific covariance linear timing allowance=0.17924 d. Full 6x6 orbit correlation matrix is positive definite (helper receipt). Mass function=0.02874 solar masses; adopted primary mass=1.23, minimum secondary mass=0.429. Rough central duration=6.00 h versus observed median=5.07 h. Radius and main-sequence companion assumptions make this a plausibility calculation, not a fitted geometry or calibrated significance veto. Shorter observed duration can reflect impact parameter and model assumptions. Orbit agreement does not establish ownership among unresolved sources.

Deciding limit: Twelve deep raw eclipses over S3/30/97. Sector-dependent depth is consistent with changing dilution; isolated target-centered deficit.

Next check: Optional resolved photometry and spectroscopy for precision physical parameters.

Catalogue recovery is recorded centrally in ../../shared-catalogues/coverage-matrix.csv. A successful cone with no exact eclipsing target classification is distinct from a failed query; known recoveries must not count as new additions. No exhaustive literature priority claim is made.

Sector	Search cycle	Measured midpoint BTJD	Formal center error d	Timing input error d	Depth ppt	Timing status
3	0	1387.1372902	0.000277692	0.0104169	83.67	selected
3	1	1393.6936237	0.000339641	0.0104169	83.56	selected
3	2	1400.2487076	0.000163537	0.0104169	83.54	selected
30	112	2121.3811531	0.000281748	0.00347229	87.16	selected
30	114	2134.4922055	0.000213255	0.00347229	86.27	selected
97	389	3937.3234800	0.000208905	0.00115741	75.93	selected
97	390	3943.8789339	0.000189922	0.00115741	75.40	selected
97	391	3950.4348892	0.000246981	0.00115741	76.20	selected
97	392	3956.9904867	0.000210853	0.00115741	75.78	selected
97	393	3963.5466919	0.000365138	0.00115741	76.83	selected
97	394	3970.1019276	0.00018623	0.00115741	76.72	selected
97	396	3983.2129381	0.000171054	0.00115741	77.64	selected

Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png, historical original receipts, ../measurement-source-receipts.json, ../identity-check.json, ../alias-audit.json, ../gaia-geometry.json and ../helper-audit.md. Original inputs and fresh outputs are covered by ../SHA256SUMS.json and ../source-hashes.json.

Stable review key: R541.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s97.json - historical/diff_s97.png - historical/gaia.json - measurements.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

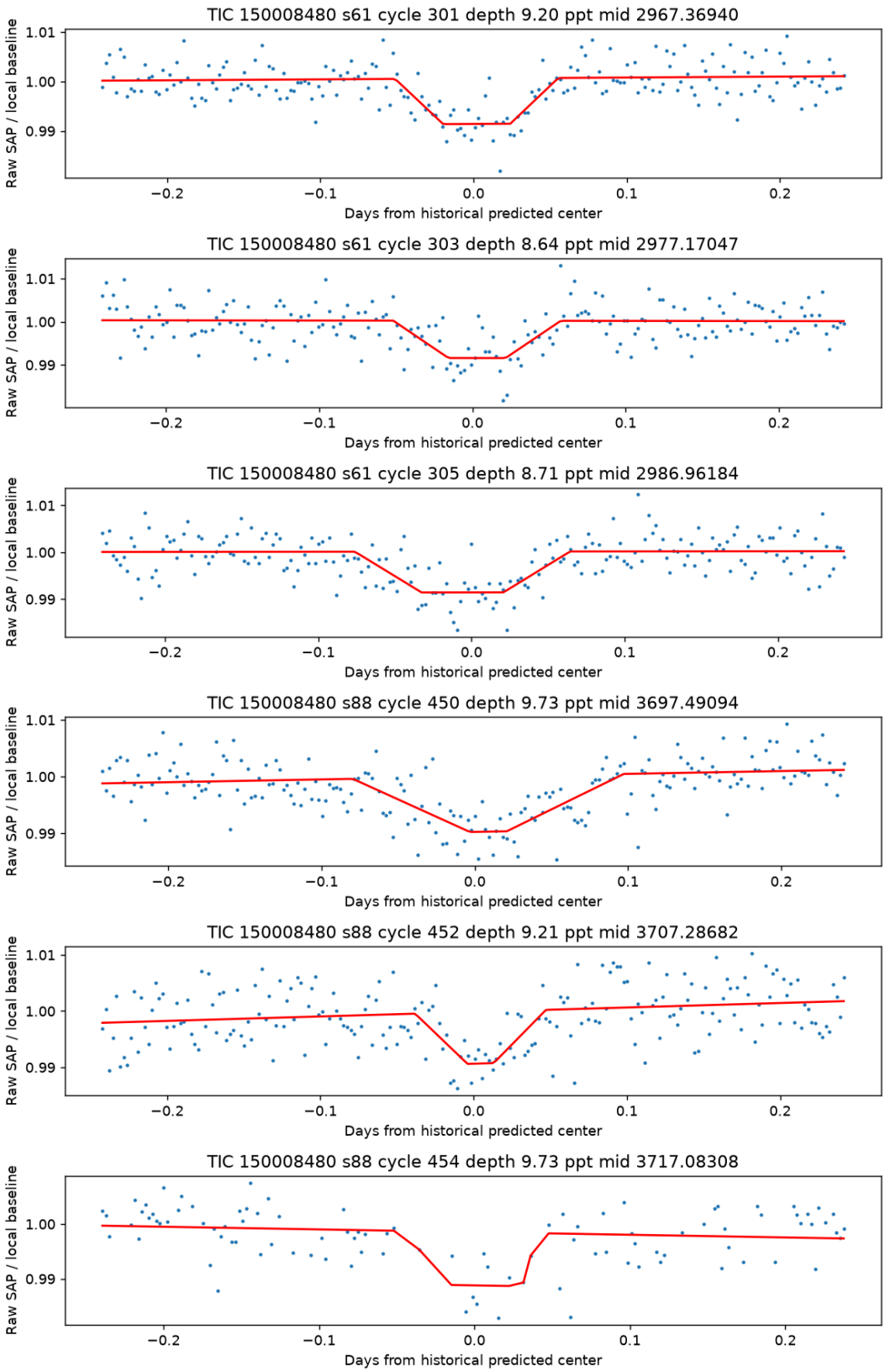
R542. TIC 150008480

physical_verdict: PROBABLE

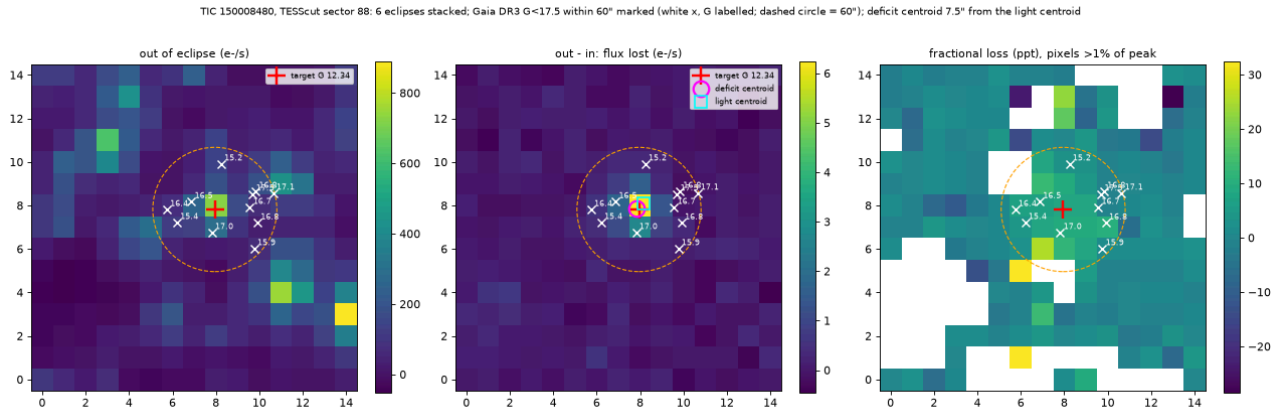
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or multi-source PRF localization; distinguish alternating event kinds.



R542; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R542; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 150008480: PROBABLE

Full bounded physical review: **COMPLETE**. Physical classification is separate from catalogue recovery/novelty; historical screening was not an admission verdict.

1. **Raw events.** 11 free-center trapezoid measurements from quality-zero QLP SAP_FLUX in sectors [61, 88]; 11 enter the reported ephemeris. Representative raw light curves and numerical windows show repeated eclipse morphology. Predictions guided the windows and are not measurements. Flagged or uncovered windows are not clean nulls.
2. **Timing and aliases.** Measured recurring-eclipse timing supported. P=4.9001176563 +/- 0.0000153100 d; reference BTJD=2967.3669874. Epoch kind: Selected repeating eclipse; depth rank to be checked against other event kind. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The formal error column is the local free-center model covariance; the input error adds a half-cadence floor. Singular/unidentifiable centers are excluded. Cycle/parity counts are in ../alias-audit.json. Selected eclipse means the measured recurrence, with no assumption that it is the deeper primary. The bounded half/double check does not prove all possible aliases absent.
3. **Attribution.** Fresh difference stack in sector 88 has 6 events; target coordinate [7.949368002849406, 7.814154538161146], deficit [7.840202825047673, 7.884281988246106]; offset 0.130 pixel. The image-model fractional residual 0.852 is not a localization uncertainty. Every capable Gaia neighbor in the retained 60-arcsec table was inspected: Gaia 5598671598997580160: 21.97 arcsec, G=16.992, intrinsic required loss proxy=0.640; Gaia 5598672049978646528: 23.92 arcsec, G=16.517, intrinsic required loss proxy=0.416; Gaia 5598671976954710528: 33.79 arcsec, G=16.674, intrinsic required loss proxy=0.480; Gaia 5598671672021524736: 38.52 arcsec, G=15.384, intrinsic required loss proxy=0.152; Gaia 5598671976956157696: 39.45 arcsec, G=17.455, intrinsic required loss proxy=0.976; Gaia 5598672015618461696: 42.89 arcsec, G=16.796, intrinsic required loss proxy=0.536; Gaia 5598671981258727424: 43.01 arcsec, G=16.760, intrinsic required loss proxy=0.518; Gaia 5598672084337939328: 43.79 arcsec, G=15.165, intrinsic required loss proxy=0.126; Gaia 5598671672021097856: 46.32 arcsec, G=16.372, intrinsic required loss proxy=0.365; Gaia 5598660230228212224: 53.18 arcsec, G=15.908, intrinsic required loss proxy=0.241; Gaia 5598672015618457600: 58.32 arcsec, G=17.149, intrinsic required loss

proxy=0.738. The intrinsic loss proxy uses a pairwise G-band flux budget and cannot be directly compared with diluted pixel fractional loss. Close blends and passband/aperture response limit exclusion claims. Eleven shallow repeated raw eclipses; eleven capable neighbors within 60 arcsec and a noisy deficit leave source attribution unresolved. The Gaia conjunction discrepancy is -0.043 phase, within the descriptive orbit timing uncertainty, and is not evidence of unique target ownership.

4. **Alternative reduction.** Fresh TESScut background-subtracted target-centered 1x1/3x3/5x5 depths are 8.61/8.46/6.69 ppt; selected QLP fitted median 8.71 ppt. This reruns the historical extraction primitive using the fresh ephemeris and measured duration, with identity and source hashes checked. The alternative extraction and the QLP data for this sector share the same TESS photons; they test processing robustness and are not independent observations. Distinct sectors are distinct observing occasions on one instrument. Depth differences can reflect dilution, aperture and background response; intrinsic binary depth is not inferred.
5. **Gaia geometry/physical plausibility.** Exact DR3 source 5598671976949355264, orbital P=4.9006621570 d, e=0.05643. Closest edge-on conjunction offset=-0.21183 d (-0.04322 phase), full source-specific covariance linear timing allowance=0.34169 d. Full 6x6 orbit correlation matrix is positive definite (helper receipt). Mass function=0.042527 solar masses; adopted primary mass=1.25, minimum secondary mass=0.509. Rough central duration=7.10 h versus observed median=2.85 h. Radius and main-sequence companion assumptions make this a plausibility calculation, not a fitted geometry or calibrated significance veto. Shorter observed duration can reflect impact parameter and model assumptions. Orbit agreement does not establish ownership among unresolved sources.

Deciding limit: Eleven shallow repeated raw eclipses; eleven capable neighbors within 60 arcsec and a noisy deficit leave source attribution unresolved. The Gaia conjunction discrepancy is -0.043 phase, within the descriptive orbit timing uncertainty, and is not evidence of unique target ownership.

Next check: Resolved eclipse photometry or multi-source PRF localization; distinguish alternating event kinds.

Catalogue recovery is recorded centrally in ../../shared-catalogues/coverage-matrix.csv. A successful cone with no exact eclipsing target classification is distinct from a failed query; known recoveries must not count as new additions. No exhaustive literature priority claim is made.

Sector	Search cycle	Measured midpoint BTJD	Formal center error d	Timing input error d	Depth ppt	Timing status
61	301	2967.3693978	0.0019375	0.0019375	9.20	selected
61	302	2972.2629428	0.00412332	0.00412332	8.41	selected
61	303	2977.1704736	0.00306238	0.00306238	8.64	selected
61	304	2982.0643101	0.00372941	0.00372941	8.12	selected
61	305	2986.9618408	0.00319197	0.00319197	8.71	selected
88	449	3692.5818366	0.00353184	0.00353184	12.12	selected
88	450	3697.4909404	0.00311603	0.00311603	9.73	selected
88	451	3702.3836062	0.00381891	0.00381891	7.77	selected
88	452	3707.2868200	0.00354984	0.00354984	9.21	selected
88	453	3712.1823973	0.00298151	0.00298151	7.65	selected
88	454	3717.0830767	0.00278977	0.00278977	9.73	selected

Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png, historical original receipts, ../measurement-source-receipts.json, ../identity-check.json, ../alias-audit.json, ../gaia-geometry.json and ../helper-audit.md. Original inputs and fresh outputs are covered by ../SHA256SUMS.json and ../source-hashes.json.

Stable review key: R542.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s88.json - historical/diff_s88.png - historical/gaia.json - measurements.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

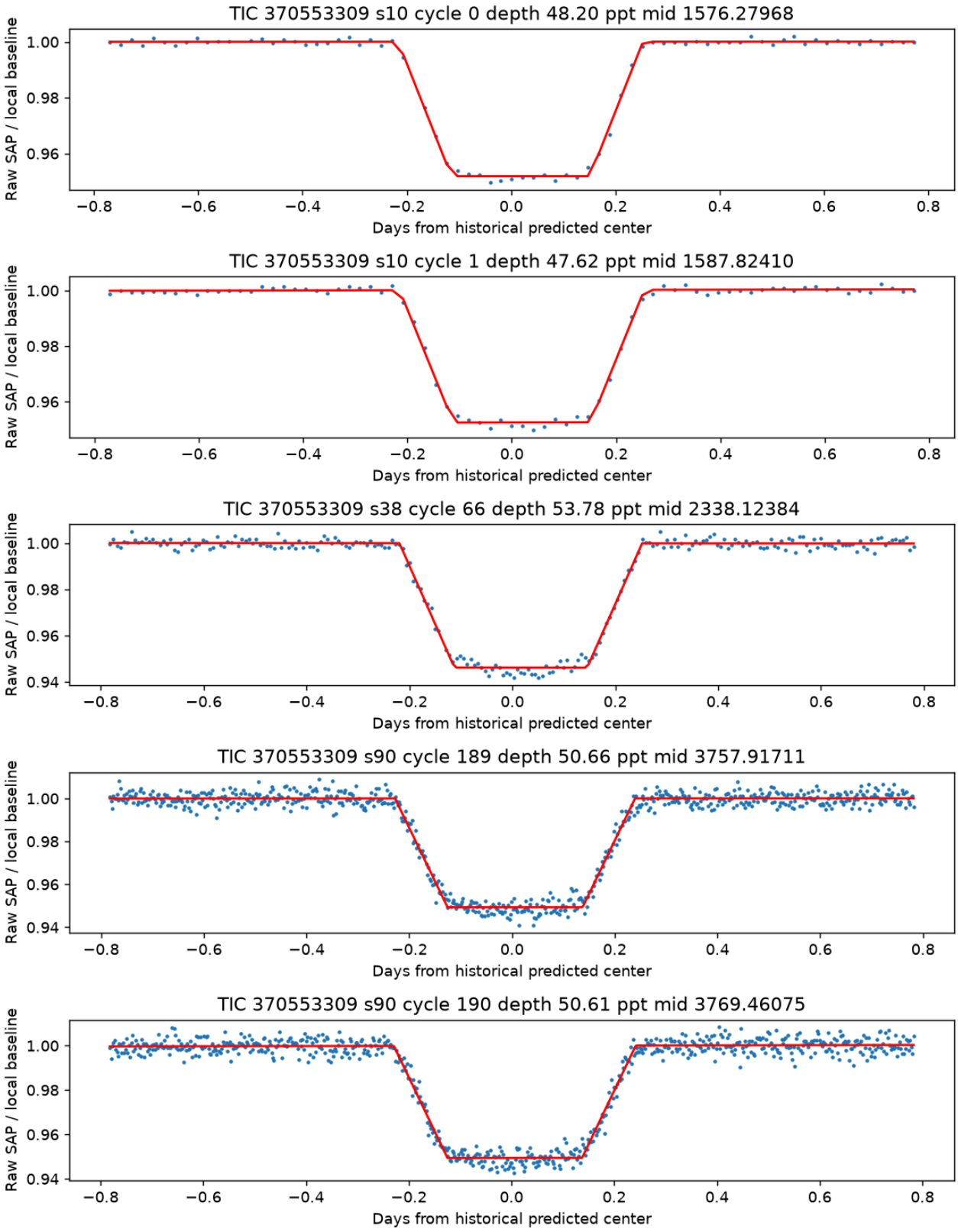
R543. TIC 370553309

physical_verdict: PROBABLE

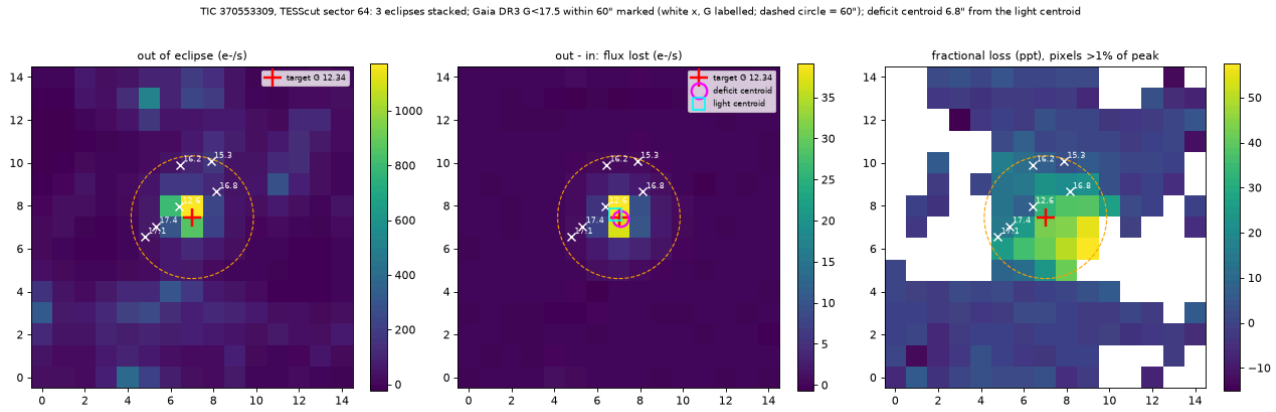
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or multi-source PRF localization; distinguish alternating event kinds.



R543; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R543; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 370553309: PROBABLE

Full bounded physical review: **COMPLETE**. Physical classification is separate from catalogue recovery/novelty; historical screening was not an admission verdict.

1. **Raw events.** 5 free-center trapezoid measurements from quality-zero QLP SAP_FLUX in sectors [10, 38, 90]; 5 enter the reported ephemeris. Representative raw light curves and numerical windows show repeated eclipse morphology. Predictions guided the windows and are not measurements. Flagged or uncovered windows are not clean nulls.
2. **Timing and aliases.** Measured recurring-eclipse timing supported. P=11.5430428545 +/- 0.0000235331 d; reference BTJD=1576.2823229. Epoch kind: Selected repeating eclipse; depth rank to be checked against other event kind. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The formal error column is the local free-center model covariance; the input error adds a half-cadence floor. Singular/unidentifiable centers are excluded. Cycle/parity counts are in ../alias-audit.json. Selected eclipse means the measured recurrence, with no assumption that it is the deeper primary. The bounded half/double check does not prove all possible aliases absent.
3. **Attribution.** Fresh difference stack in sector 64 has 3 events; target coordinate [7.012775182780933, 7.4639535261664705], deficit [7.105188643147638, 7.412381390520829]; offset 0.106 pixel. The image-model fractional residual 0.536 is not a localization uncertainty. Every capable Gaia neighbor in the retained 60-arcsec table was inspected: Gaia 5243406099139937664: 16.62 arcsec, G=12.646, intrinsic required loss proxy=0.117; Gaia 5243406064780208896: 57.94 arcsec, G=15.345, intrinsic required loss proxy=0.854. The intrinsic loss proxy uses a pairwise G-band flux budget and cannot be directly compared with diluted pixel fractional loss. Close blends and passband/aperture response limit exclusion claims. Five long eclipses repeat. A near-equal G=12.65 source at 16.62 arcsec overlaps the deficit core; target ownership is unresolved at this image resolution.
4. **Alternative reduction.** Fresh TESScut background-subtracted target-centered 1x1/3x3/5x5 depths are 42.79/30.88/29.06 ppt; selected QLP fitted median 50.61 ppt. This reruns the historical extraction primitive using the fresh ephemeris and measured duration, with identity and source

hashes checked. The retained QLP sectors do not include this cutout sector. This cutout is a distinct observing occasion on the same instrument; it is an alternative extraction of that sector FFI data, without a direct same-sector QLP comparison here. Distinct sectors are distinct observing occasions on one instrument. Depth differences can reflect dilution, aperture and background response; intrinsic binary depth is not inferred.

5. **Gaia geometry/physical plausibility.** Exact DR3 source 5243406094839663232, orbital P=11.5392602408 d, e=0.23457. Closest edge-on conjunction offset=0.31924 d (0.02767 phase), full source-specific covariance linear timing allowance=0.24094 d. Full 6x6 orbit correlation matrix is positive definite (helper receipt). Mass function=0.061528 solar masses; adopted primary mass=1.21, minimum secondary mass=0.583. Rough central duration=13.52 h versus observed median=11.27 h. Radius and main-sequence companion assumptions make this a plausibility calculation, not a fitted geometry or calibrated significance veto. Shorter observed duration can reflect impact parameter and model assumptions. Orbit agreement does not establish ownership among unresolved sources.

Deciding limit: Five long eclipses repeat. A near-equal G=12.65 source at 16.62 arcsec overlaps the deficit core; target ownership is unresolved at this image resolution.

Next check: Resolved eclipse photometry or multi-source PRF localization; distinguish alternating event kinds.

Catalogue recovery is recorded centrally in ../../shared-catalogues/coverage-matrix.csv. A successful cone with no exact eclipsing target classification is distinct from a failed query; known recoveries must not count as new additions. No exhaustive literature priority claim is made.

Sector	Search cycle	Measured midpoint BTJD	Formal center error d	Timing input error d	Depth ppt	Timing status
10	0	1576.2796808	0.00067316	0.0104168	48.20	selected
10	1	1587.8240974	0.000716633	0.0104168	47.62	selected
38	66	2338.1238386	0.000728331	0.00347219	53.78	selected
90	189	3757.9171089	0.00063774	0.00115743	50.66	selected
90	190	3769.4607506	0.0007356	0.00115743	50.61	selected

Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png, historical original receipts, ../measurement-source-receipts.json, ../identity-check.json, ../alias-audit.json, ../gaia-geometry.json and ../helper-audit.md. Original inputs and fresh outputs are covered by ../SHA256SUMS.json and ../source-hashes.json.

Stable review key: R543.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s64.json - historical/diff_s64.png - historical/gaia.json - measurements.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

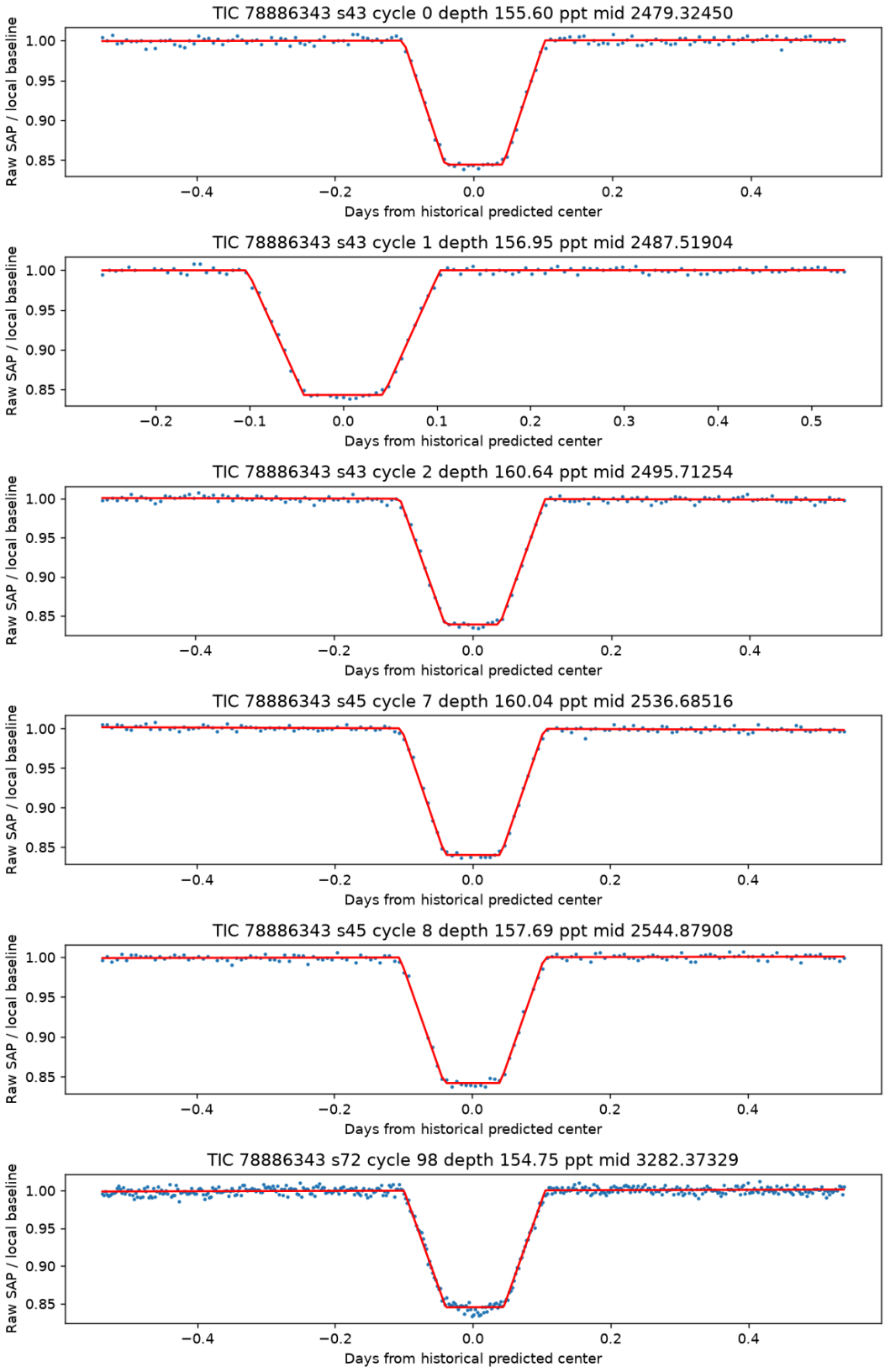
R544. TIC 78886343

physical_verdict: CONFIRMED

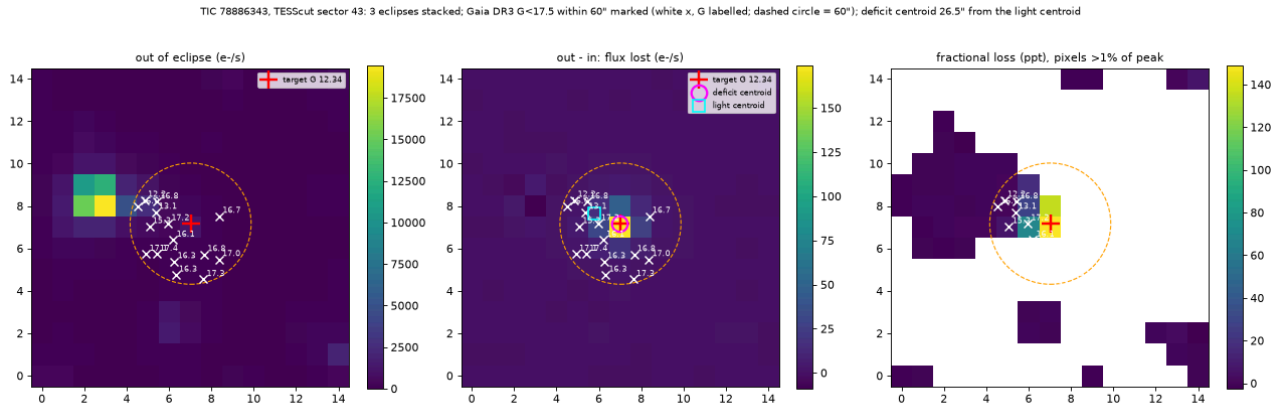
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision physical parameters.



R544; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R544; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 78886343: CONFIRMED

Full bounded physical review: **COMPLETE**. Physical classification is separate from catalogue recovery/novelty; historical screening was not an admission verdict.

1. **Raw events.** 8 free-center trapezoid measurements from quality-zero QLP SAP_FLUX in sectors [43, 45, 72]; 8 enter the reported ephemeris. Representative raw light curves and numerical windows show repeated eclipse morphology. Predictions guided the windows and are not measurements. Flagged or uncovered windows are not clean nulls.
2. **Timing and aliases.** Measured recurring-eclipse timing supported. P=8.1943751036 +/- 0.0000175843 d; reference BTJD=2479.3243437. Epoch kind: Selected repeating eclipse; depth rank to be checked against other event kind. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The formal error column is the local free-center model covariance; the input error adds a half-cadence floor. Singular/unidentifiable centers are excluded. Cycle/parity counts are in ../alias-audit.json. Selected eclipse means the measured recurrence, with no assumption that it is the deeper primary. The bounded half/double check does not prove all possible aliases absent.
3. **Attribution.** Fresh difference stack in sector 43 has 3 events; target coordinate [7.020139253913671, 7.16586789211822], deficit [6.954075516953401, 7.164599708152924]; offset 0.066 pixel. The image-model fractional residual 0.999 is not a localization uncertainty. Every capable Gaia neighbor in the retained 60-arcsec table was inspected: Gaia 3428059934359773440: 36.94 arcsec, G=13.122, intrinsic required loss proxy=0.479; Gaia 3428059930060420608: 51.67 arcsec, G=12.226, intrinsic required loss proxy=0.298. The intrinsic loss proxy uses a pairwise G-band flux budget and cannot be directly compared with diluted pixel fractional loss. Close blends and passband/aperture response limit exclusion claims. Eight deep raw eclipses; compact deficit is near the target, distinct from both capable neighbors (approximately 1.7 and 2.4 pixels away). Bright off-target light displaces the out-of-eclipse centroid; it does not move the deficit peak. Target attribution is favored at TESS resolution.
4. **Alternative reduction.** Fresh TESScut background-subtracted target-centered 1x1/3x3/5x5 depths are 148.98/108.88/46.83 ppt; selected QLP fitted median 157.32 ppt. This reruns the

historical extraction primitive using the fresh ephemeris and measured duration, with identity and source hashes checked. The alternative extraction and the QLP data for this sector share the same TESS photons; they test processing robustness and are not independent observations. Distinct sectors are distinct observing occasions on one instrument. Depth differences can reflect dilution, aperture and background response; intrinsic binary depth is not inferred.

5. **Gaia geometry/physical plausibility.** Exact DR3 source 3428060033139635072, orbital P=8.1931724592 d, e=0.07441. Closest edge-on conjunction offset=0.22692 d (0.02770 phase), full source-specific covariance linear timing allowance=0.43137 d. Full 6x6 orbit correlation matrix is positive definite (helper receipt). Mass function=0.042875 solar masses; adopted primary mass=1.31, minimum secondary mass=0.524. Rough central duration=5.65 h versus observed median=4.96 h. Radius and main-sequence companion assumptions make this a plausibility calculation, not a fitted geometry or calibrated significance veto. Shorter observed duration can reflect impact parameter and model assumptions. Orbit agreement does not establish ownership among unresolved sources.

Deciding limit: Eight deep raw eclipses; compact deficit is near the target, distinct from both capable neighbors (approximately 1.7 and 2.4 pixels away). Bright off-target light displaces the out-of-eclipse centroid; it does not move the deficit peak. Target attribution is favored at TESS resolution.

Next check: Optional resolved photometry and spectroscopy for precision physical parameters.

Catalogue recovery is recorded centrally in ../../shared-catalogues/coverage-matrix.csv. A successful cone with no exact eclipsing target classification is distinct from a failed query; known recoveries must not count as new additions. No exhaustive literature priority claim is made.

Sector	Search cycle	Measured midpoint BTJD	Formal center error d	Timing input error d	Depth ppt	Timing status
43	0	2479.3245025	0.000393849	0.00347256	155.60	selected
43	1	2487.5190390	0.000329102	0.00347256	156.95	selected
43	2	2495.7125372	0.0002809	0.00347256	160.64	selected
45	6	2528.4907727	0.000332102	0.0034724	160.15	selected
45	7	2536.6851645	0.000290897	0.0034724	160.04	selected
45	8	2544.8790784	0.000349266	0.0034724	157.69	selected
72	96	3265.9841647	0.000222825	0.00115746	155.35	selected
72	98	3282.3732894	0.000243797	0.00115746	154.75	selected

Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png, historical original receipts, ../measurement-source-receipts.json, ../identity-check.json, ../alias-audit.json, ../gaia-geometry.json and ../helper-audit.md. Original inputs and fresh outputs are covered by ../SHA256SUMS.json and ../source-hashes.json.

Stable review key: R544.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s43.json - historical/diff_s43.png - historical/gaia.json - measurements.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

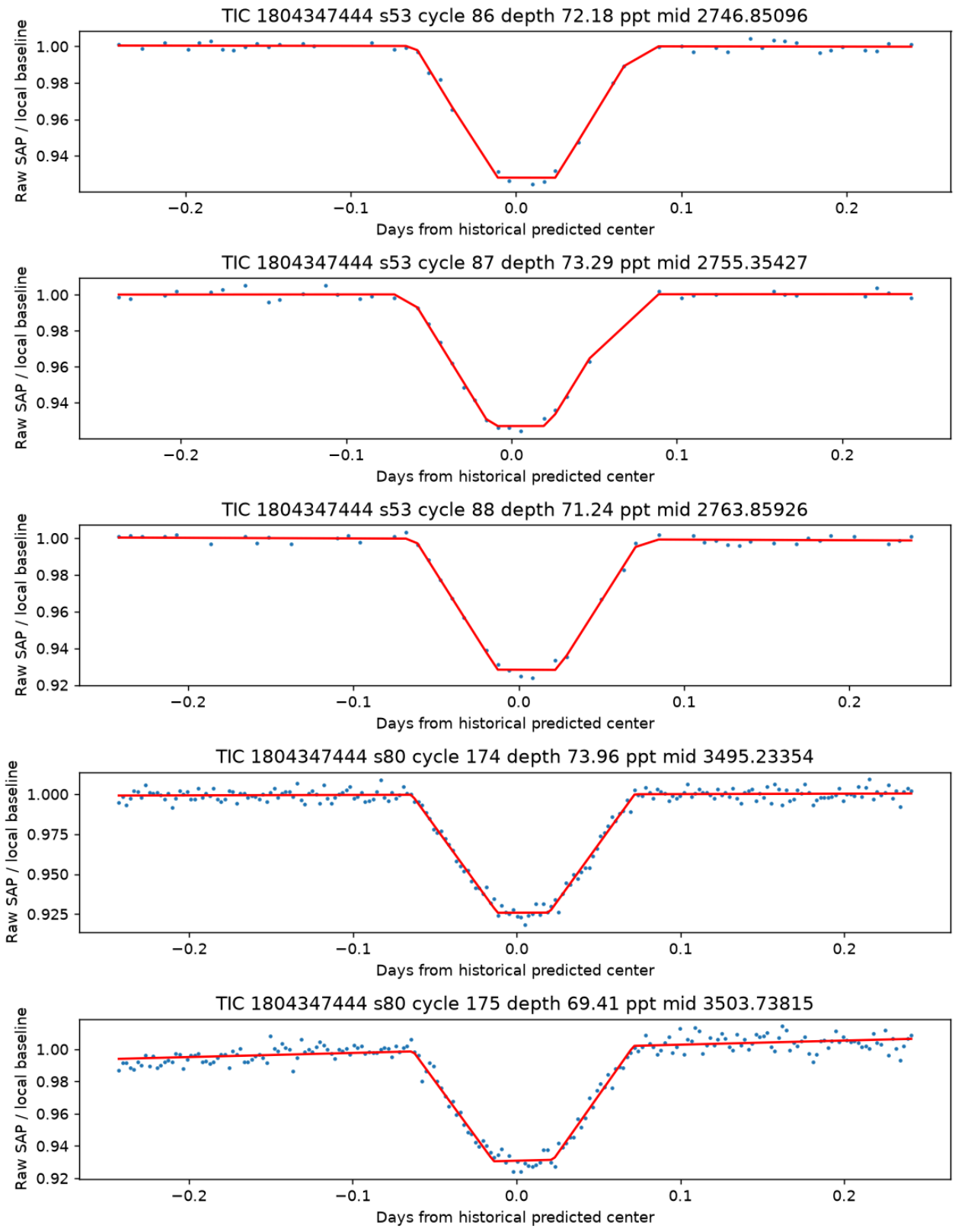
R545. TIC 1804347444

physical_verdict: PROBABLE

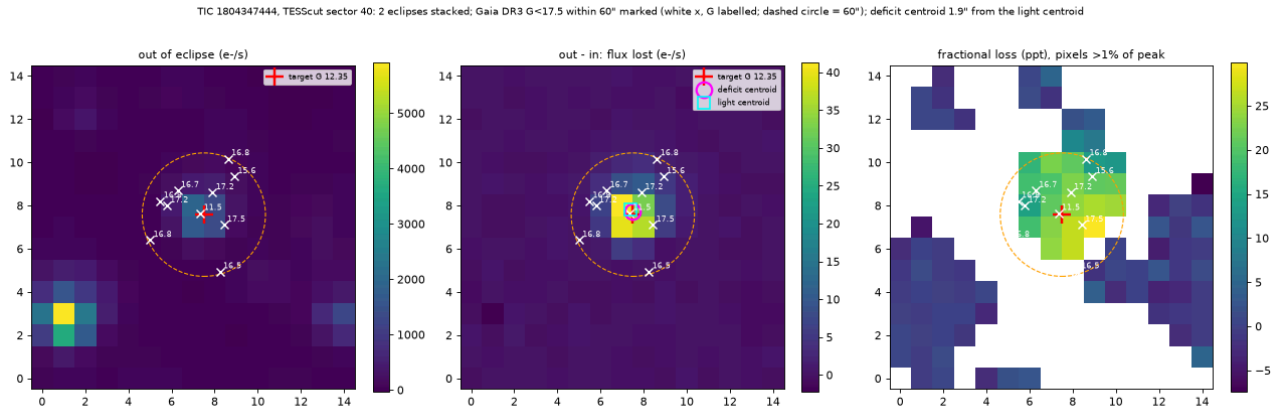
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or multi-source PRF localization; distinguish alternating event kinds.



R545; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R545; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 1804347444: PROBABLE

Full bounded physical review: **COMPLETE**. Physical classification is separate from catalogue recovery/novelty; historical screening was not an admission verdict.

1. **Raw events.** 5 free-center trapezoid measurements from quality-zero QLP SAP_FLUX in sectors [53, 80]; 5 enter the reported ephemeris. Representative raw light curves and numerical windows show repeated eclipse morphology. Predictions guided the windows and are not measurements. Flagged or uncovered windows are not clean nulls.
2. **Timing and aliases.** Measured recurring-eclipse timing supported. P=8.5043545697 +/- 0.0000247428 d; reference BTJD=2746.8504687. Epoch kind: Selected repeating eclipse; depth rank to be checked against other event kind. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The formal error column is the local free-center model covariance; the input error adds a half-cadence floor. Singular/unidentifiable centers are excluded. Cycle/parity counts are in ../alias-audit.json. Selected eclipse means the measured recurrence, with no assumption that it is the deeper primary. The bounded half/double check does not prove all possible aliases absent.
3. **Attribution.** Fresh difference stack in sector 40 has 2 events; target coordinate [7.4978386291745345, 7.580646367474408], deficit [7.47234432170148, 7.7392649746852875]; offset 0.161 pixel. The image-model fractional residual 0.211 is not a localization uncertainty. Every capable Gaia neighbor in the retained 60-arcsec table was inspected: Gaia 4510645061668258816: 3.45 arcsec, G=11.537, intrinsic required loss proxy=0.106. The intrinsic loss proxy uses a pairwise G-band flux budget and cannot be directly compared with diluted pixel fractional loss. Close blends and passband/aperture response limit exclusion claims. Five repeated eclipses; a brighter G=11.54 source only 3.45 arcsec from target is unresolved. Pixel-aperture depth differs from QLP because of strong dilution; this is not a failed eclipse detection.
4. **Alternative reduction.** Fresh TESScut background-subtracted target-centered 1x1/3x3/5x5 depths are 20.94/22.49/22.38 ppt; selected QLP fitted median 72.18 ppt. This reruns the historical extraction primitive using the fresh ephemeris and measured duration, with identity and source hashes checked. The retained QLP sectors do not include this cutout sector. This cutout is a distinct

observing occasion on the same instrument; it is an alternative extraction of that sector FFI data, without a direct same-sector QLP comparison here. Distinct sectors are distinct observing occasions on one instrument. Depth differences can reflect dilution, aperture and background response; intrinsic binary depth is not inferred.

5. **Gaia geometry/physical plausibility.** Exact DR3 source 4510645057369288960, orbital P=8.5041747949 d, e=0.11020. Closest edge-on conjunction offset=0.01159 d (0.00136 phase), full source-specific covariance linear timing allowance=0.38059 d. Full 6x6 orbit correlation matrix is positive definite (helper receipt). Mass function=0.12508 solar masses; adopted primary mass=1, minimum secondary mass=0.717. Rough central duration=4.89 h versus observed median=3.23 h. Radius and main-sequence companion assumptions make this a plausibility calculation, not a fitted geometry or calibrated significance veto. Shorter observed duration can reflect impact parameter and model assumptions. Orbit agreement does not establish ownership among unresolved sources.

Deciding limit: Five repeated eclipses; a brighter G=11.54 source only 3.45 arcsec from target is unresolved. Pixel-aperture depth differs from QLP because of strong dilution; this is not a failed eclipse detection.

Next check: Resolved eclipse photometry or multi-source PRF localization; distinguish alternating event kinds.

Catalogue recovery is recorded centrally in ../../shared-catalogues/coverage-matrix.csv. A successful cone with no exact eclipsing target classification is distinct from a failed query; known recoveries must not count as new additions. No exhaustive literature priority claim is made.

Sector	Search cycle	Measured midpoint BTJD	Formal center error d	Timing input error d	Depth ppt	Timing status
53	86	2746.8509567	0.000730067	0.00347229	72.18	selected
53	87	2755.3542658	0.000588564	0.00347229	73.29	selected
53	88	2763.8592557	0.000512613	0.00347229	71.24	selected
80	174	3495.2335438	0.000358219	0.0011574	73.96	selected
80	175	3503.7381514	0.000692737	0.0011574	69.41	selected

Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png, historical original receipts, ../measurement-source-receipts.json, ../identity-check.json, ../alias-audit.json, ../gaia-geometry.json and ../helper-audit.md. Original inputs and fresh outputs are covered by ../SHA256SUMS.json and ../source-hashes.json.

Stable review key: R545.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s40.json - historical/diff_s40.png - historical/gaia.json - measurements.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

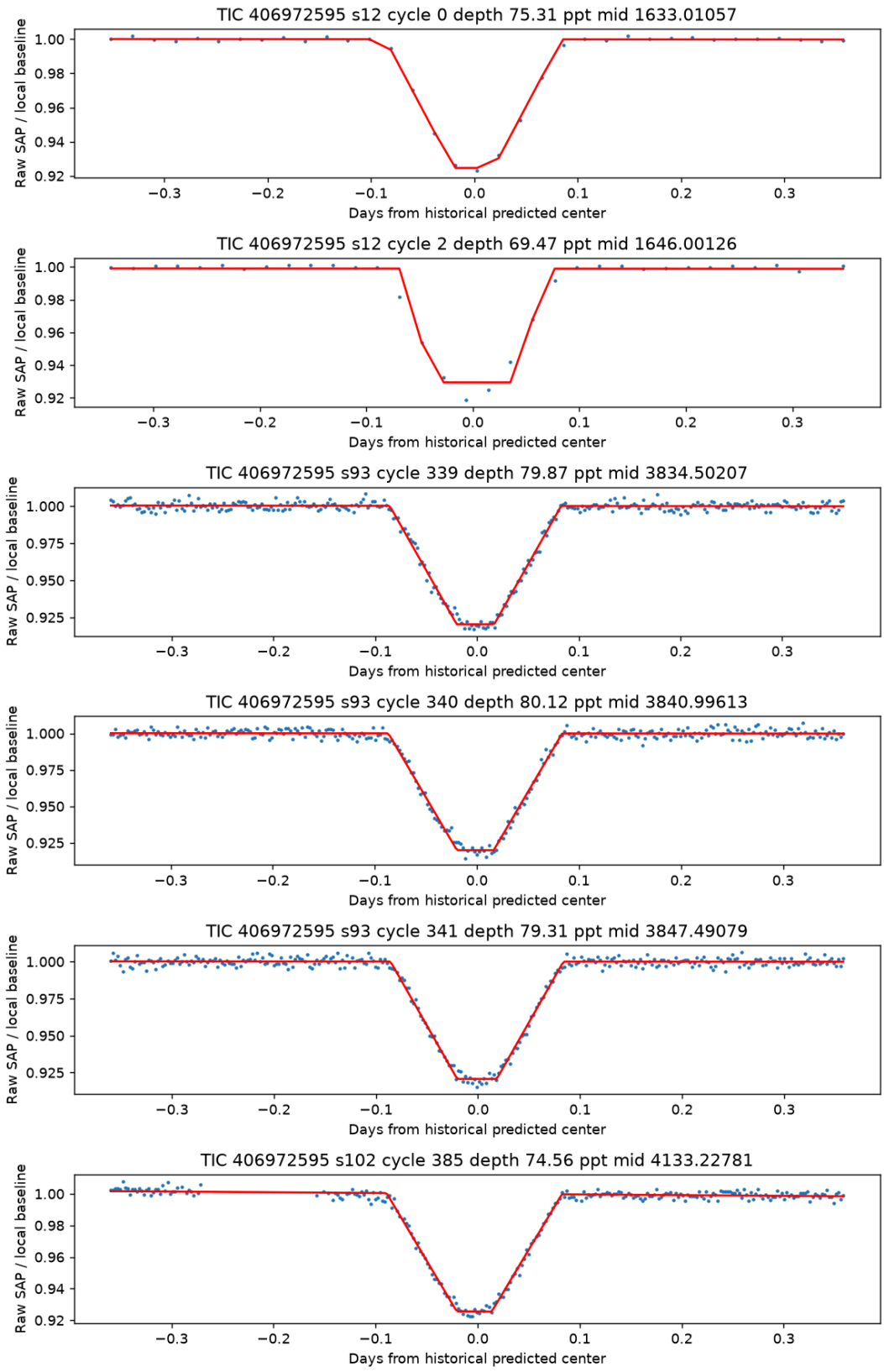
R546. TIC 406972595

physical_verdict: CONFIRMED

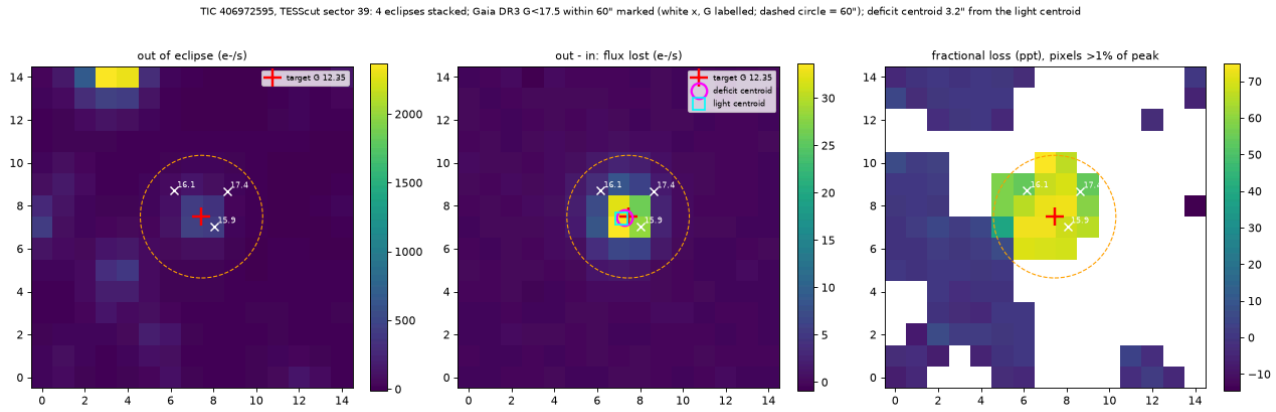
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision physical parameters.



R546; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R546; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 406972595: CONFIRMED

Full bounded physical review: **COMPLETE**. Physical classification is separate from catalogue recovery/novelty; historical screening was not an admission verdict.

1. **Raw events.** 7 free-center trapezoid measurements from quality-zero QLP SAP_FLUX in sectors [12, 93, 102]; 7 enter the reported ephemeris. Representative raw light curves and numerical windows show repeated eclipse morphology. Predictions guided the windows and are not measurements. Flagged or uncovered windows are not clean nulls.
2. **Timing and aliases.** Measured recurring-eclipse timing supported. P=6.4940582389 +/- 0.0000171180 d; reference BTJD=1633.0162055. Epoch kind: Selected repeating eclipse; depth rank to be checked against other event kind. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The formal error column is the local free-center model covariance; the input error adds a half-cadence floor. Singular/unidentifiable centers are excluded. Cycle/parity counts are in ../alias-audit.json. Selected eclipse means the measured recurrence, with no assumption that it is the deeper primary. The bounded half/double check does not prove all possible aliases absent.
3. **Attribution.** Fresh difference stack in sector 39 has 4 events; target coordinate [7.442694351830347, 7.490629109323532], deficit [7.259587537596887, 7.462001674428883]; offset 0.185 pixel. The image-model fractional residual 0.601 is not a localization uncertainty. Every capable Gaia neighbor in the retained 60-arcsec table was inspected: None under the approximate G-band brightness screen within 60 arcsec. The intrinsic loss proxy uses a pairwise G-band flux budget and cannot be directly compared with diluted pixel fractional loss. Close blends and passband/aperture response limit exclusion claims. Seven repeated raw eclipses with within-sector recurrence in S93; compact target deficit and no capable Gaia neighbor within 60 arcsec.
4. **Alternative reduction.** Fresh TESScut background-subtracted target-centered 1x1/3x3/5x5 depths are 73.25/71.22/66.21 ppt; selected QLP fitted median 79.31 ppt. This reruns the historical extraction primitive using the fresh ephemeris and measured duration, with identity and source hashes checked. The retained QLP sectors do not include this cutout sector. This cutout is a distinct observing occasion on the same instrument; it is an alternative extraction of that sector FFI data,

without a direct same-sector QLP comparison here. Distinct sectors are distinct observing occasions on one instrument. Depth differences can reflect dilution, aperture and background response; intrinsic binary depth is not inferred.

5. **Gaia geometry/physical plausibility.** Exact DR3 source 5779383275684367360, orbital P=6.4933125698 d, e=0.16654. Closest edge-on conjunction offset=0.17883 d (0.02754 phase), full source-specific covariance linear timing allowance=0.12773 d. Full 6x6 orbit correlation matrix is positive definite (helper receipt). Mass function=0.032901 solar masses; adopted primary mass=1, minimum secondary mass=0.401. Rough central duration=6.01 h versus observed median=4.07 h. Radius and main-sequence companion assumptions make this a plausibility calculation, not a fitted geometry or calibrated significance veto. Shorter observed duration can reflect impact parameter and model assumptions. Orbit agreement does not establish ownership among unresolved sources.

Deciding limit: Seven repeated raw eclipses with within-sector recurrence in S93; compact target deficit and no capable Gaia neighbor within 60 arcsec.

Next check: Optional resolved photometry and spectroscopy for precision physical parameters.

Catalogue recovery is recorded centrally in ../../shared-catalogues/coverage-matrix.csv. A successful cone with no exact eclipsing target classification is distinct from a failed query; known recoveries must not count as new additions. No exhaustive literature priority claim is made.

Sector	Search cycle	Measured midpoint BTJD	Formal center error d	Timing input error d	Depth ppt	Timing status
12	0	1633.0105672	0.000420498	0.0104167	75.31	selected
12	2	1646.0012616	0.000271617	0.0104167	69.47	selected
93	339	3834.5020662	0.000280772	0.0011574	79.87	selected
93	340	3840.9961318	0.000324632	0.0011574	80.12	selected
93	341	3847.4907862	0.000289572	0.0011574	79.31	selected
93	342	3853.9840881	0.000298791	0.0011574	80.41	selected
102	385	4133.2278059	0.000272035	0.00115741	74.56	selected

Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png, historical original receipts, ../measurement-source-receipts.json, ../identity-check.json, ../alias-audit.json, ../gaia-geometry.json and ../helper-audit.md. Original inputs and fresh outputs are covered by ../SHA256SUMS.json and ../source-hashes.json.

Stable review key: R546.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s39.json - historical/diff_s39.png - historical/gaia.json - measurements.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

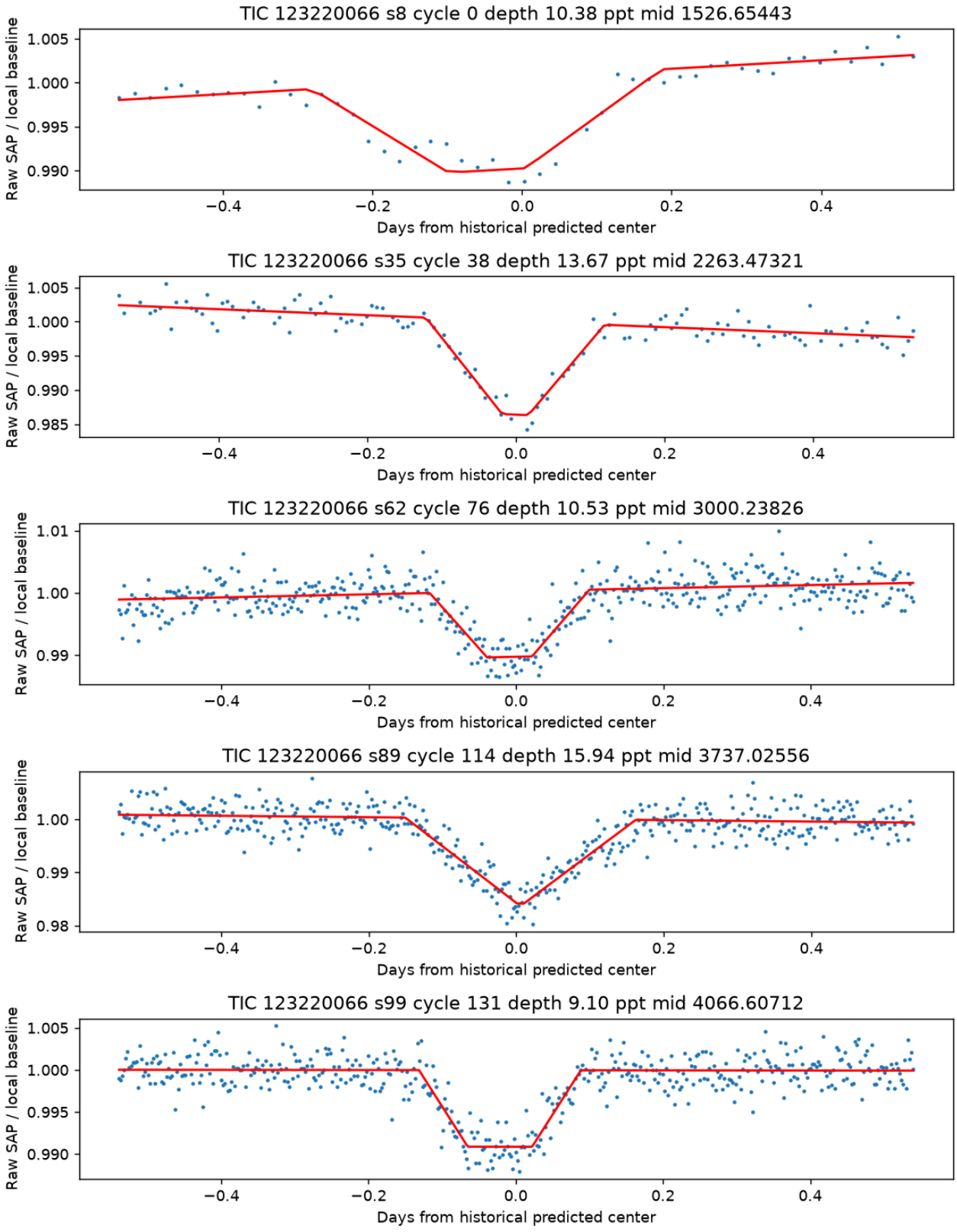
R547. TIC 123220066

physical_verdict: CONFIRMED

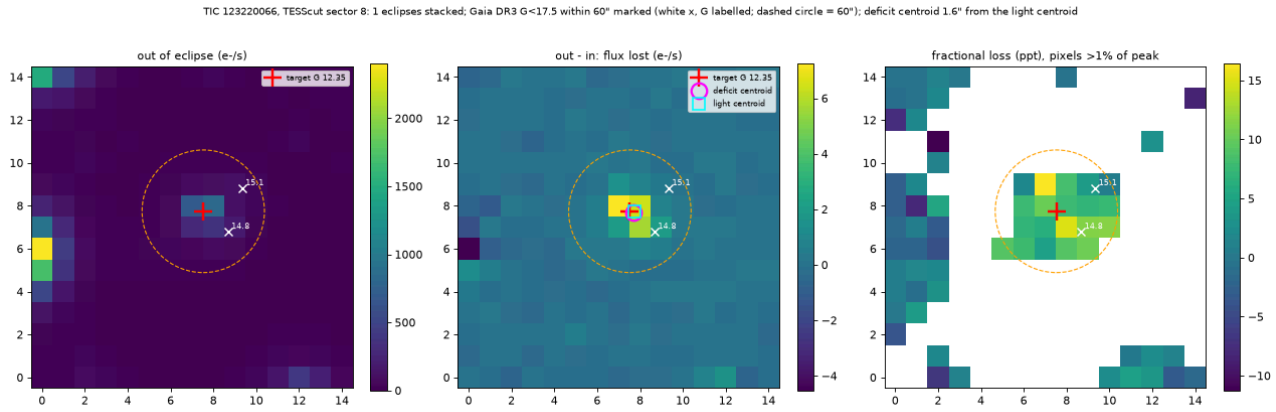
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision physical parameters.



R547; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R547; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 123220066: CONFIRMED

Full bounded physical review: **COMPLETE**. Physical classification is separate from catalogue recovery/novelty; historical screening was not an admission verdict.

1. **Raw events.** 5 free-center trapezoid measurements from quality-zero QLP SAP_FLUX in sectors [8, 35, 62, 89, 99]; 5 enter the reported ephemeris. Representative raw light curves and numerical windows show repeated eclipse morphology. Predictions guided the windows and are not measurements. Flagged or uncovered windows are not clean nulls.
2. **Timing and aliases.** Measured recurring-eclipse timing supported. P=19.3886399262 +/- 0.0002064827 d; reference BTJD=1526.7034798. Epoch kind: Selected repeating eclipse; depth rank to be checked against other event kind. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The formal error column is the local free-center model covariance; the input error adds a half-cadence floor. Singular/unidentifiable centers are excluded. Cycle/parity counts are in ../alias-audit.json. Selected eclipse means the measured recurrence, with no assumption that it is the deeper primary. The bounded half/double check does not prove all possible aliases absent. For this sparse case, ../sparse-alias-coverage.json and ../terra/123-half-null.png retain additional current-coverage and alias evidence; the null thresholds are heuristic, not confidence limits. Unequal shallower secondary eclipses are not excluded.
3. **Attribution.** Fresh difference stack in sector 8 has 1 events; target coordinate [7.524867716712931, 7.741832028869618], deficit [7.713426640385074, 7.662301616416583]; offset 0.205 pixel. The image-model fractional residual 0.280 is not a localization uncertainty. Every capable Gaia neighbor in the retained 60-arcsec table was inspected: Gaia 3824060626331605760: 32.06 arcsec, G=14.787, intrinsic required loss proxy=0.110; Gaia 3824060729410819456: 44.60 arcsec, G=15.069, intrinsic required loss proxy=0.139. The intrinsic loss proxy uses a pairwise G-band flux budget and cannot be directly compared with diluted pixel fractional loss. Close blends and passband/aperture response limit exclusion claims. Five actual raw midpoints after inspecting all five currently available QLP sectors (8/35/62/89/99). A bounded 0.8-40 day integer-alias test leaves only the 19.389-day recurrence without clean sampled nulls; the half-period predicts four nulls. Two capable sources at 32 and 45 arcsec are separated from the target-centered deficit. One-event image noise and non-calibrated localization remain limitations.

4. **Alternative reduction.** Fresh TESScut background-subtracted target-centered 1x1/3x3/5x5 depths are 7.84/9.61/9.09 ppt; selected QLP fitted median 10.53 ppt. This reruns the historical extraction primitive using the fresh ephemeris and measured duration, with identity and source hashes checked. The alternative extraction and the QLP data for this sector share the same TESS photons; they test processing robustness and are not independent observations. Distinct sectors are distinct observing occasions on one instrument. Depth differences can reflect dilution, aperture and background response; intrinsic binary depth is not inferred.
5. **Gaia geometry/physical plausibility.** Exact DR3 source 3824060695051082368, orbital P=19.3881864909 d, e=0.25411. Closest edge-on conjunction offset=0.18990 d (0.00979 phase), full source-specific covariance linear timing allowance=0.41755 d. Full 6x6 orbit correlation matrix is positive definite (helper receipt). Mass function=0.022158 solar masses; adopted primary mass=0.929, minimum secondary mass=0.327. Rough central duration=13.01 h versus observed median=5.78 h. Radius and main-sequence companion assumptions make this a plausibility calculation, not a fitted geometry or calibrated significance veto. Shorter observed duration can reflect impact parameter and model assumptions. Orbit agreement does not establish ownership among unresolved sources.

Deciding limit: Five actual raw midpoints after inspecting all five currently available QLP sectors (8/35/62/89/99). A bounded 0.8-40 day integer-alias test leaves only the 19.389-day recurrence without clean sampled nulls; the half-period predicts four nulls. Two capable sources at 32 and 45 arcsec are separated from the target-centered deficit. One-event image noise and non-calibrated localization remain limitations.

Next check: Optional resolved photometry and spectroscopy for precision physical parameters.

Catalogue recovery is recorded centrally in ../../shared-catalogues/coverage-matrix.csv. A successful cone with no exact eclipsing target classification is distinct from a failed query; known recoveries must not count as new additions. No exhaustive literature priority claim is made.

Sector	Search cycle	Measured midpoint BTJD	Formal center error d	Timing input error d	Depth ppt	Timing status
8	0	1526.6544279	0.0115875	0.0115875	10.38	selected
35	38	2263.4732149	0.00389949	0.00389949	13.67	selected
62	76	3000.2382630	0.0022037	0.0022037	10.53	selected
89	114	3737.0255629	0.00220041	0.00220041	15.94	selected
99	131	4066.6071194	0.00168496	0.00168496	9.10	selected

Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png, historical original receipts, ../measurement-source-receipts.json, ../identity-check.json, ../alias-audit.json, ../gaia-geometry.json and ../helper-audit.md. Original inputs and fresh outputs are covered by ../SHA256SUMS.json and ../source-hashes.json.

Stable review key: R547.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s08.json - historical/diff_s08.png - historical/gaia.json - measurements.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

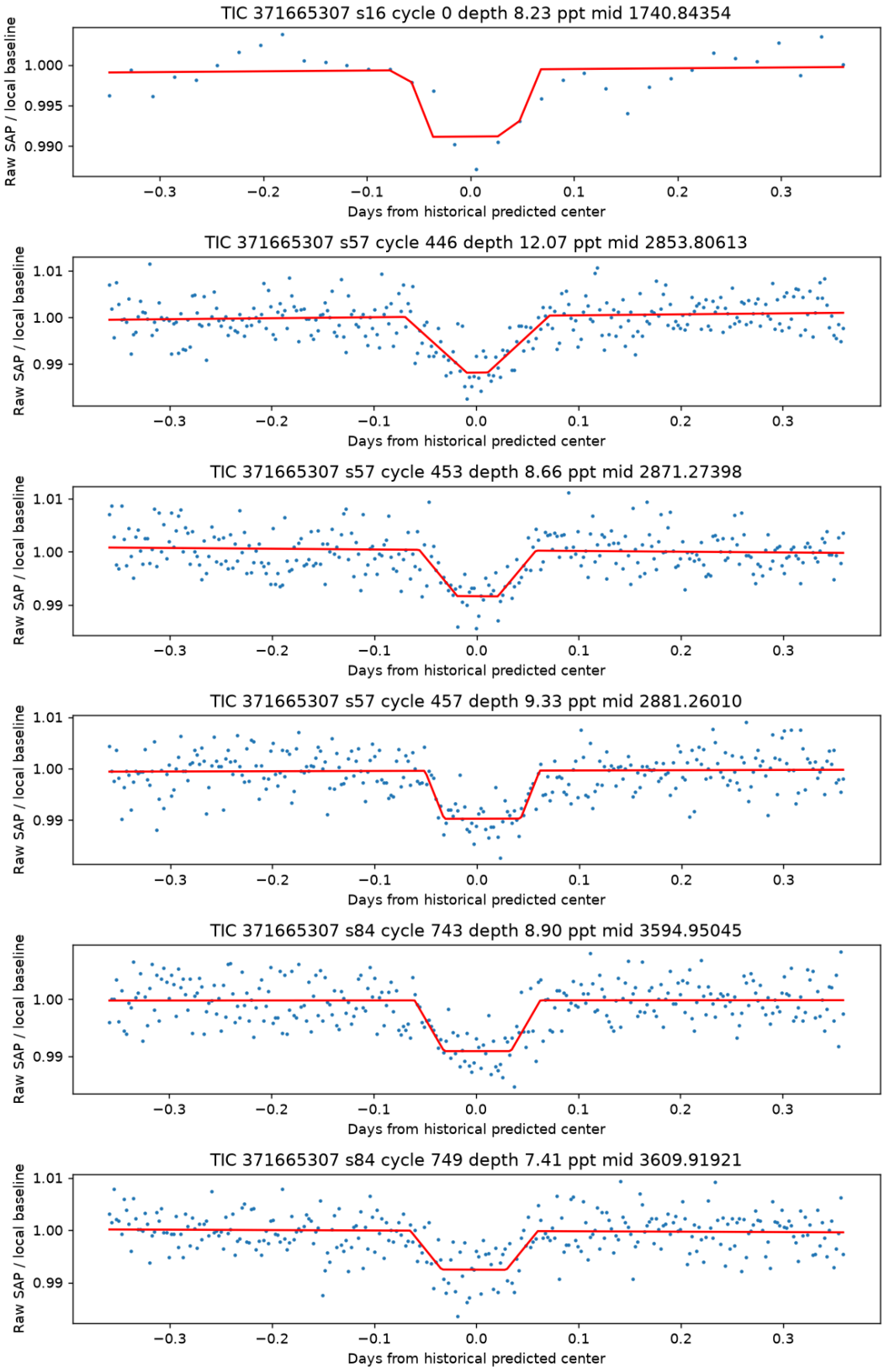
R548. TIC 371665307

physical_verdict: PROBABLE

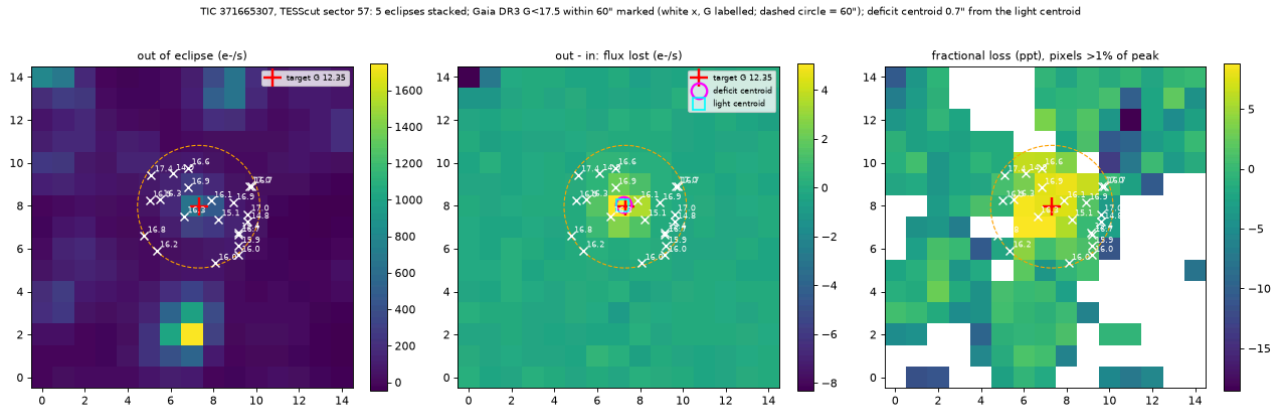
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or multi-source PRF localization; distinguish alternating event kinds.



R548; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R548; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 371665307: PROBABLE

Full bounded physical review: **COMPLETE**. Physical classification is separate from catalogue recovery/novelty; historical screening was not an admission verdict.

1. **Raw events.** 24 free-center trapezoid measurements from quality-zero QLP SAP_FLUX in sectors [16, 57, 84]; 10 enter the reported ephemeris. Representative raw light curves and numerical windows show repeated eclipse morphology. Predictions guided the windows and are not measurements. Flagged or uncovered windows are not clean nulls.
2. **Timing and aliases.** Candidate orbital timing; factor-two ambiguity retained. P=4.9908584340 +/- 0.0000131023 d; reference BTJD=1740.8463664. Epoch kind: Even-parity eclipse in candidate 4.99087-d orbit; deeper rank unestablished; half-period ambiguity retained. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The formal error column is the local free-center model covariance; the input error adds a half-cadence floor. Singular/unidentifiable centers are excluded. Cycle/parity counts are in ../alias-audit.json. Selected eclipse means the measured recurrence, with no assumption that it is the deeper primary. The bounded half/double check does not prove all possible aliases absent.
3. **Attribution.** Fresh difference stack in sector 57 has 5 events; target coordinate [7.311130466648379, 7.956803856146049], deficit [7.206752205666073, 8.051402306721076]; offset 0.141 pixel. The image-model fractional residual 0.630 is not a localization uncertainty. Every capable Gaia neighbor in the retained 60-arcsec table was inspected: Gaia 2007172155994105856: 12.99 arcsec, G=16.117, intrinsic required loss proxy=0.312; Gaia 2007172155994111872: 16.56 arcsec, G=16.343, intrinsic required loss proxy=0.382; Gaia 2007172911908346368: 20.49 arcsec, G=16.891, intrinsic required loss proxy=0.627; Gaia 2007172155994111232: 22.14 arcsec, G=15.053, intrinsic required loss proxy=0.123; Gaia 2007172155994105344: 33.25 arcsec, G=16.851, intrinsic required loss proxy=0.605; Gaia 2007172911908339328: 37.35 arcsec, G=16.597, intrinsic required loss proxy=0.480; Gaia 2007172052914892544: 37.89 arcsec, G=16.319, intrinsic required loss proxy=0.374; Gaia 2007172808829127552: 39.57 arcsec, G=14.885, intrinsic required loss proxy=0.107; Gaia 2007172190353853824: 44.61 arcsec, G=15.399, intrinsic required loss proxy=0.166; Gaia 2007172190353853952: 46.45 arcsec, G=16.691, intrinsic required loss proxy=0.523; Gaia 2007172190353846784: 47.65 arcsec, G=17.019, intrinsic required loss

proxy=0.704; Gaia 2007172048609299072: 47.73 arcsec, G=16.853, intrinsic required loss proxy=0.606; Gaia 2007172190353849088: 49.31 arcsec, G=14.828, intrinsic required loss proxy=0.102; Gaia 2007172946262005248: 52.00 arcsec, G=17.011, intrinsic required loss proxy=0.699; Gaia 2007172121634381952: 53.03 arcsec, G=15.878, intrinsic required loss proxy=0.252; Gaia 2007172941962552320: 53.93 arcsec, G=16.705, intrinsic required loss proxy=0.530; Gaia 2007172121634391552: 54.66 arcsec, G=15.969, intrinsic required loss proxy=0.273; Gaia 2007172087274650752: 57.52 arcsec, G=16.163, intrinsic required loss proxy=0.325; Gaia 2007172121634386048: 58.96 arcsec, G=15.966, intrinsic required loss proxy=0.273; Gaia 2007171984195429760: 59.03 arcsec, G=16.782, intrinsic required loss proxy=0.568. The intrinsic loss proxy uses a pairwise G-band flux budget and cannot be directly compared with diluted pixel fractional loss. Close blends and passband/aperture response limit exclusion claims. Repeated 2.49543-day dips are real. Depth parity is not decisive; the quoted 4.99086-day candidate orbital ephemeris uses even half-period parity only and retains a factor-two ambiguity. The shallow crowded deficit has twenty currently flux-capable neighbors and does not uniquely establish target ownership.

4. **Alternative reduction.** Fresh TESScut background-subtracted target-centered 1x1/3x3/5x5 depths are 7.82/7.46/5.81 ppt; selected QLP fitted median 9.42 ppt. This reruns the historical extraction primitive using the fresh ephemeris and measured duration, with identity and source hashes checked. The alternative extraction and the QLP data for this sector share the same TESS photons; they test processing robustness and are not independent observations. Distinct sectors are distinct observing occasions on one instrument. Depth differences can reflect dilution, aperture and background response; intrinsic binary depth is not inferred.
5. **Gaia geometry/physical plausibility.** Exact DR3 source 2007172151688083072, orbital P=4.9910192319 d, e=0.05318. Closest edge-on conjunction offset=0.02306 d (0.00462 phase), full source-specific covariance linear timing allowance=0.10982 d. Full 6x6 orbit correlation matrix is positive definite (helper receipt). Mass function=0.23406 solar masses; adopted primary mass=1.39, minimum secondary mass=1.15. Rough central duration=10.24 h versus observed median=3.09 h. Radius and main-sequence companion assumptions make this a plausibility calculation, not a fitted geometry or calibrated significance veto. Shorter observed duration can reflect impact parameter and model assumptions. Orbit agreement does not establish ownership among unresolved sources.

Deciding limit: Repeated 2.49543-day dips are real. Depth parity is not decisive; the quoted 4.99086-day candidate orbital ephemeris uses even half-period parity only and retains a factor-two ambiguity. The shallow crowded deficit has twenty currently flux-capable neighbors and does not uniquely establish target ownership.

Next check: Resolved eclipse photometry or multi-source PRF localization; distinguish alternating event kinds.

Catalogue recovery is recorded centrally in ../../shared-catalogues/coverage-matrix.csv. A successful cone with no exact eclipsing target classification is distinct from a failed query; known recoveries must not count as new additions. No exhaustive literature priority claim is made.

Sector	Search cycle	Measured midpoint BTJD	Formal center error d	Timing input error d	Depth ppt	Timing status
16	0	1740.8435444	0.00493599	0.0104168	8.23	selected
16	1	1743.3340499	0.00267524	0.0104168	8.73	other parity
16	5	1753.3187725	0.00176773	0.0104168	8.28	other parity
16	6	1755.8188187	0.00183474	0.0104168	8.38	selected
57	446	2853.8061349	0.0023894	0.0023894	12.07	selected
57	447	2856.2837163	0.0042196	0.0042196	9.39	other parity

Sector	Search cycle	Measured midpoint BTJD	Formal center error d	Timing input error d	Depth ppt	Timing status
57	448	2858.7885673	315465	315465	9.79	excluded center covariance
57	449	2861.2908682	0.00326148	0.00326148	7.04	other parity
57	452	2868.7849608	0.00283299	0.00283299	9.56	selected
57	453	2871.2739818	0.0027109	0.0027109	8.66	other parity
57	454	2873.7704600	0.00216753	0.00216753	11.45	selected
57	455	2876.2691219	0.00370213	0.00370213	8.73	other parity
57	456	2878.7618079	0.00254927	0.00254927	8.81	selected
57	457	2881.2600958	0.00158823	0.00158823	9.33	other parity
84	739	3584.9736429	0.00328012	0.00328012	10.43	other parity
84	740	3587.4662218	0.00254042	0.00254042	12.18	selected
84	741	3589.9605175	0.00298326	0.00298326	11.35	other parity
84	742	3592.4536804	0.00223542	0.00223542	10.06	selected
84	743	3594.9504504	0.00253229	0.00253229	8.90	other parity
84	745	3599.9338603	0.00323209	0.00323209	11.03	other parity
84	746	3602.4290268	0.00319896	0.00319896	9.27	selected
84	747	3604.9287199	0.00173011	0.00173011	9.60	other parity
84	748	3607.4305489	0.00232786	0.00232786	8.93	selected
84	749	3609.9192061	0.00259463	0.00259463	7.41	other parity

Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png, historical original receipts, ../measurement-source-receipts.json, ../identity-check.json, ../alias-audit.json, ../gaia-geometry.json and ../helper-audit.md. Original inputs and fresh outputs are covered by ../SHA256SUMS.json and ../source-hashes.json.

Stable review key: R548.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s57.json - historical/diff_s57.png - historical/gaia.json - measurements.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

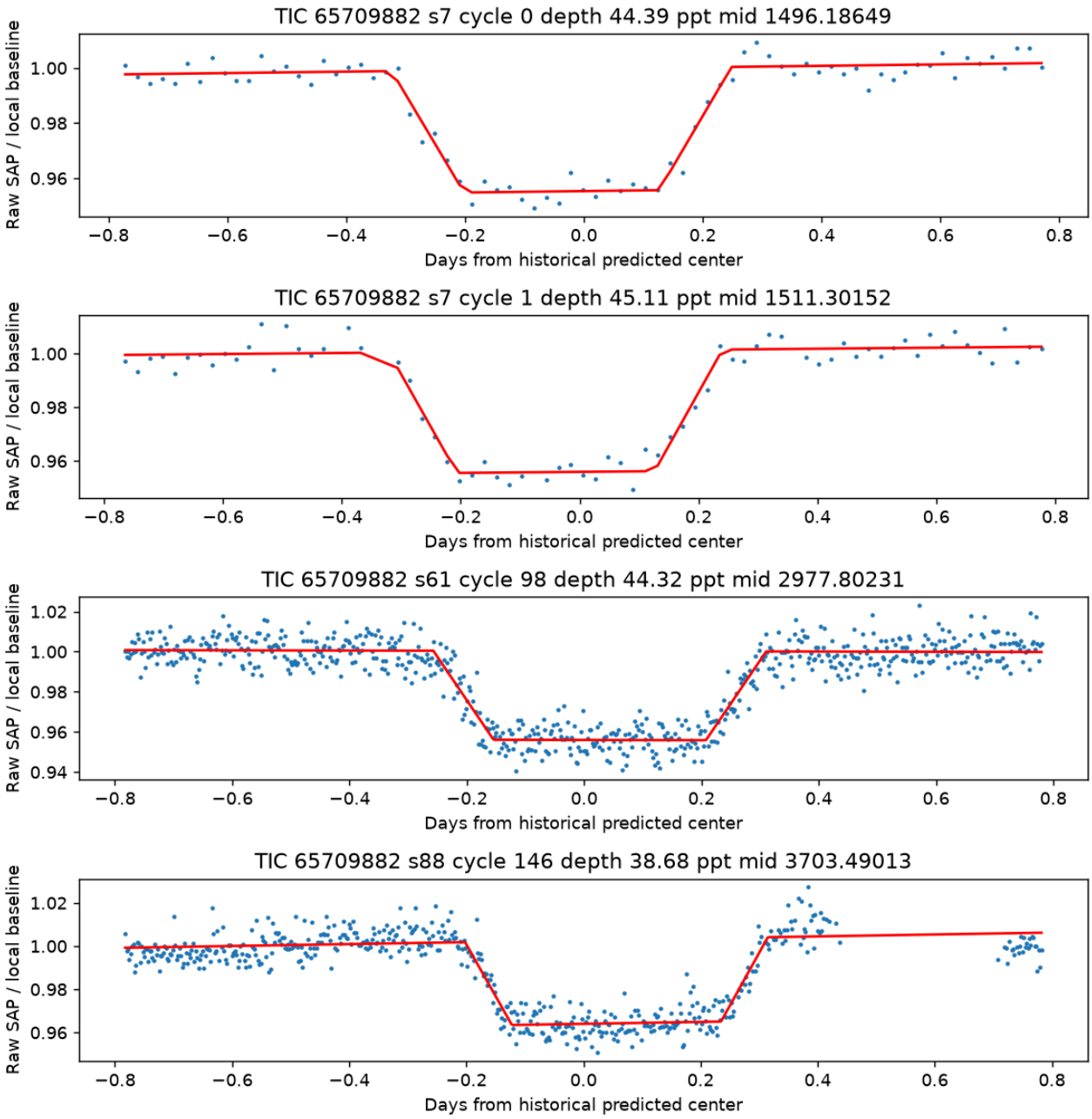
R549. TIC 65709882

physical_verdict: CONFIRMED

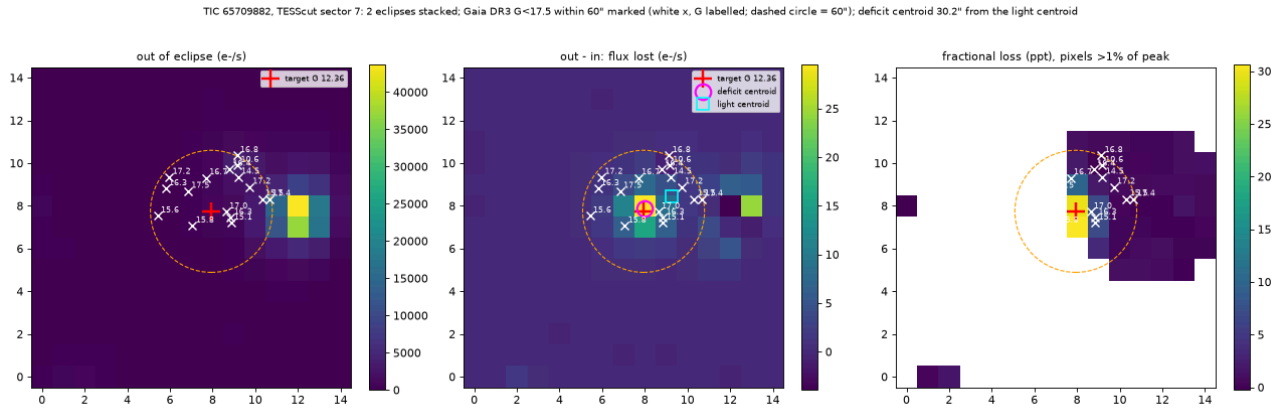
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision physical parameters.



R549; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R549; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 65709882: CONFIRMED

Full bounded physical review: **COMPLETE**. Physical classification is separate from catalogue recovery/novelty; historical screening was not an admission verdict.

1. **Raw events.** 4 free-center trapezoid measurements from quality-zero QLP SAP_FLUX in sectors [7, 61, 88]; 4 enter the reported ephemeris. Representative raw light curves and numerical windows show repeated eclipse morphology. Predictions guided the windows and are not measurements. Flagged or uncovered windows are not clean nulls.
2. **Timing and aliases.** Measured recurring-eclipse timing supported. P=15.1185128633 +/- 0.0000390171 d; reference BTJD=1496.1875304. Epoch kind: Selected repeating eclipse; depth rank to be checked against other event kind. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The formal error column is the local free-center model covariance; the input error adds a half-cadence floor. Singular/unidentifiable centers are excluded. Cycle/parity counts are in ../alias-audit.json. Selected eclipse means the measured recurrence, with no assumption that it is the deeper primary. The bounded half/double check does not prove all possible aliases absent.
3. **Attribution.** Fresh difference stack in sector 7 has 2 events; target coordinate [7.94178071976971, 7.742445486437363], deficit [7.965526147928077, 7.85884305040405]; offset 0.119 pixel. The image-model fractional residual 0.999 is not a localization uncertainty. Every capable Gaia neighbor in the retained 60-arcsec table was inspected: Gaia 5609864391149987072: 22.11 arcsec, G=15.147, intrinsic required loss proxy=0.624; Gaia 5609864425509719680: 41.71 arcsec, G=14.509, intrinsic required loss proxy=0.366; Gaia 5609864421205498240: 50.71 arcsec, G=10.613, intrinsic required loss proxy=0.053; Gaia 5609864356790241792: 50.73 arcsec, G=15.532, intrinsic required loss proxy=0.870; Gaia 5609911326554371712: 52.71 arcsec, G=15.649, intrinsic required loss proxy=0.964. The intrinsic loss proxy uses a pairwise G-band flux budget and cannot be directly compared with diluted pixel fractional loss. Close blends and passband/aperture response limit exclusion claims. Four deep raw eclipses including consecutive S7 events support recurrence. Fresh deficit lies 0.12 pixel from target while all five capable alternatives are at least about one pixel away. The very bright off-target light dominates the full image but not its compact deficit. Attribution favors target at TESS resolution; nearest capable neighbor limits subpixel claims.

4. **Alternative reduction.** Fresh TESScut background-subtracted target-centered 1x1/3x3/5x5 depths are 29.76/12.73/4.12 ppt; selected QLP fitted median 44.35 ppt. This reruns the historical extraction primitive using the fresh ephemeris and measured duration, with identity and source hashes checked. The alternative extraction and the QLP data for this sector share the same TESS photons; they test processing robustness and are not independent observations. Distinct sectors are distinct observing occasions on one instrument. Depth differences can reflect dilution, aperture and background response; intrinsic binary depth is not inferred.
5. **Gaia geometry/physical plausibility.** Exact DR3 source 5609864386845761152, orbital P=15.1280611937 d, e=0.25942. Closest edge-on conjunction offset=-0.40112 d (-0.02651 phase), full source-specific covariance linear timing allowance=0.35535 d. Full 6x6 orbit correlation matrix is positive definite (helper receipt). Mass function=0.097082 solar masses; adopted primary mass=1.19, minimum secondary mass=0.703. Rough central duration=15.17 h versus observed median=13.55 h. Radius and main-sequence companion assumptions make this a plausibility calculation, not a fitted geometry or calibrated significance veto. Shorter observed duration can reflect impact parameter and model assumptions. Orbit agreement does not establish ownership among unresolved sources.

Deciding limit: Four deep raw eclipses including consecutive S7 events support recurrence. Fresh deficit lies 0.12 pixel from target while all five capable alternatives are at least about one pixel away. The very bright off-target light dominates the full image but not its compact deficit. Attribution favors target at TESS resolution; nearest capable neighbor limits subpixel claims.

Next check: Optional resolved photometry and spectroscopy for precision physical parameters.

Catalogue recovery is recorded centrally in ../../shared-catalogues/coverage-matrix.csv. A successful cone with no exact eclipsing target classification is distinct from a failed query; known recoveries must not count as new additions. No exhaustive literature priority claim is made.

Sector	Search cycle	Measured midpoint BTJD	Formal center error d	Timing input error d	Depth ppt	Timing status
7	0	1496.1864876	0.00352353	0.0104166	44.39	selected
7	1	1511.3015204	0.00315238	0.0104166	45.11	selected
61	98	2977.8023105	0.00182987	0.00182987	44.32	selected
88	146	3703.4901278	0.00164383	0.00164383	38.68	selected

Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png, historical original receipts, ../measurement-source-receipts.json, ../identity-check.json, ../alias-audit.json, ../gaia-geometry.json and ../helper-audit.md. Original inputs and fresh outputs are covered by ../SHA256SUMS.json and ../source-hashes.json.

Stable review key: R549.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s07.json - historical/diff_s07.png - historical/gaia.json - measurements.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

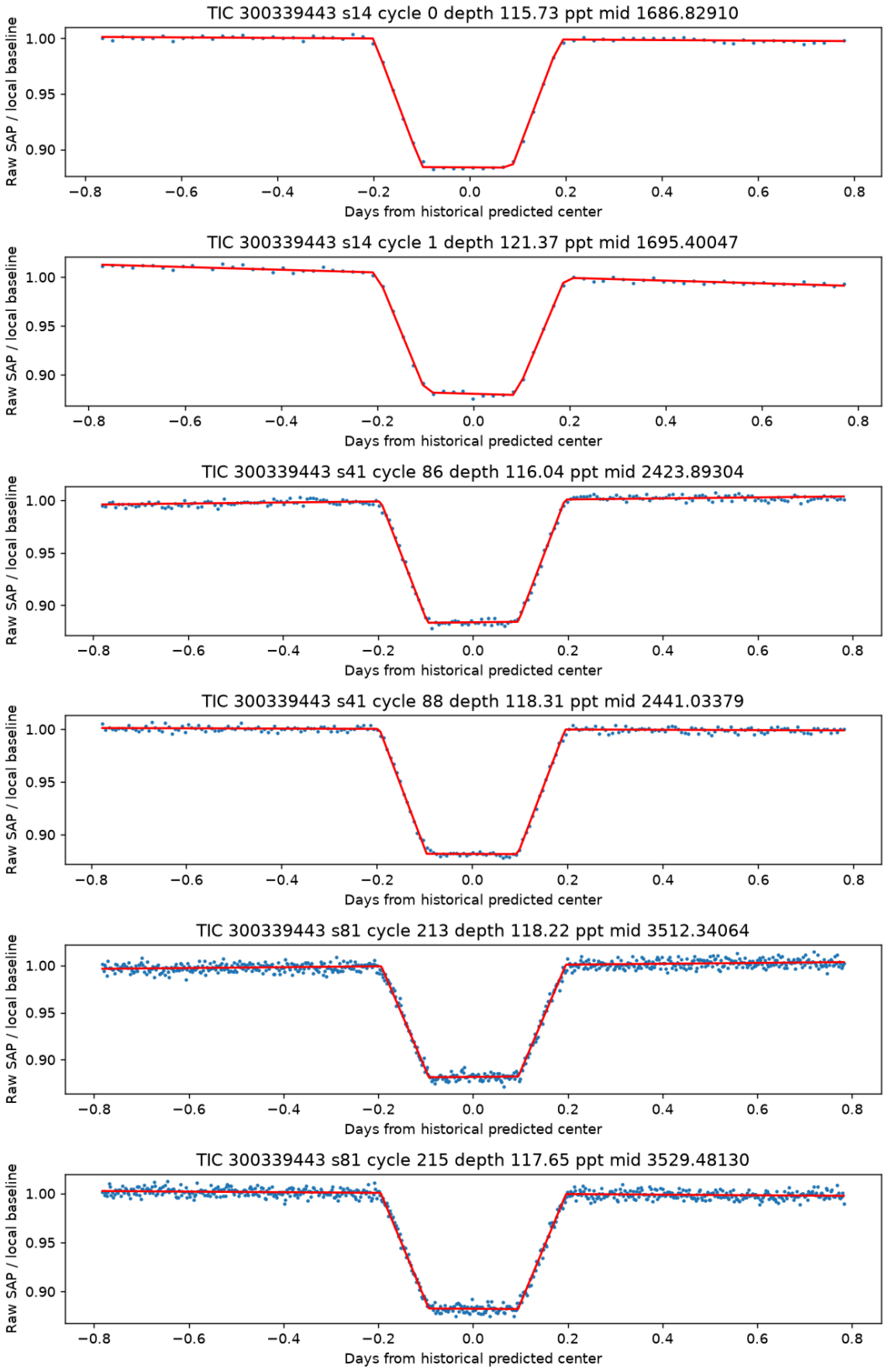
R550. TIC 300339443

physical_verdict: CONFIRMED

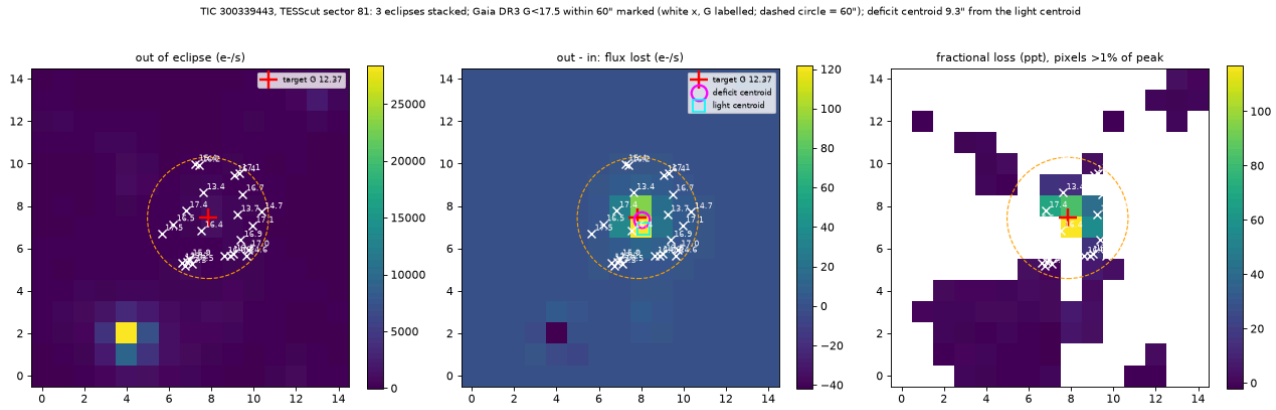
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and precision binary characterization.



R550; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R550; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 300339443: CONFIRMED

Full bounded physical review: COMPLETE. Catalogue novelty is separate from this verdict.

Raw QLP SAP_FLUX contains 7 complete free-center fitted eclipses in representative sectors[14, 41, 81]. Predicted historical centers guided extraction, not timing points. Fits use local linear baselines and trapezoid shapes; quality-zero cadences only. Flagged, partial and uncovered windows are not clean nulls. raw-events.png, raw-sector-fold.png and raw-windows.csv show the actual observations.

Measured recurrence P=8.5704599586 +/- 0.0000175154d. Named reference epoch BTJD1686.8328010: selected eclipse kind at phase zero, not an assumed deeper primary. The legacy primary_epoch_btjd column carries this epoch. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when positive-degree-of-freedom reduced chi square exceeds one. With only two centers, no reduced chi square or measured residual scatter exists; covariance propagates input timing allowances. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Consecutive same-kind measured pairs: 3. Same-sector parity and actual coverage are in alias-coverage.json. Other-kind free fits: 6 midpoints; measured phase range -0.43672 to -0.43624. See secondary-timing.json. The exact-half-phase baseline estimates are not interpreted as clean depths where they overlap the other eclipse. This is a bounded alias check, not a blind exhaustive search. Two separated sectors are distinct observing occasions on the same instrument.

Fresh raw TESScut pixel identity agrees with Gaia DR3 1827899598271026688 to 0.00000arcsec by saved WCS; all source bytes are hashed. In sector81, 3 separately recomputed event deficits average0.171px from the target. Event-scatter vector is[0.0014919499920669182, 0.018055773229612843]; this is descriptive sampling scatter, not calibrated positional confidence or pixel-response systematics. Lead visually inspected the historical full-field and freshly recomputed core images. Every capable neighbor in the saved60arcsec Gaia list: Gaia 1827899671327619072: separation25.56arcsec, G13.416, intrinsic required fraction0.428, deficit distance1.322px; Gaia 1827899602608139776: separation29.83arcsec, G13.748, intrinsic required fraction0.539, deficit distance1.274px; Gaia 1827898846705015680: separation43.38arcsec, G13.999, intrinsic required fraction0.649, deficit distance1.905px; Gaia 1827524291192242560: separation53.19arcsec, G13.346,

intrinsic required fraction0.409, deficit distance2.508px Intrinsic source loss requirements and diluted fractional pixel losses are different quantities and are never compared as a veto. The G-band flux screen is approximate and does not exclude unknown unresolved companions.

Alternative extraction: background-subtracted raw TESScut1x1/3x3/5x5 median event depths are115.55/74.46/45.53ppt; selected QLP raw median fit depth117.65ppt. Details, windows, individual centroids, count and depths are in pixel-evidence.json. This separately extracted reduction shares the same TESS photons as QLP and is not an independent instrument or independent observing occasion. Aperture dilution, cadence and event-core averaging explain differing apparent depths; no intrinsic binary depth is inferred.

Gaia orbit and physical plausibility: source-specific NSS P=8.5709502081d, e=0.13410, omega=214.3992deg, periastron BTJD391.66674. Closest edge-on conjunction offset at the measured epoch=-0.08621d (-0.01006phase); descriptive linear full-SB1-covariance timing error=0.10829d. The covariance uses source-specific P/e/omega/Tp correlations, not only diagonal errors. Mass function=0.19808solar; assumed primary mass=1.000solar gives minimum companion mass=0.892solar. Approximate central duration=12.32h versus observed9.35h. TIC radius/main-sequence companion assumptions and edge-on conjunction are approximations, not a fitted stellar model or arbitrary sigma admission rule. Orbit/geometry are consistent at this descriptive precision and do not establish the identity of an unresolved eclipse source.

Deciding interpretation: Two near-equal-depth event kinds recur at unequal phase separation. Nearby known EB Gaia DR3 1827899632630791296 is 48.55 arcsec away, G=18.463, has catalogue P=19.923 d, and would require an impossible approximate 32.52 fractional intrinsic loss to produce this depth. Its pixel (9.694,8.783) is away from the target core. This is a contamination check, not an exact target catalogue recovery. Other capable neighbors are spatially separated from the deficit core.

Next check: Optional resolved photometry and precision binary characterization.

Seven scoped catalogue groups were checked centrally for this identity; see ../../shared-catalogues/coverage-matrix.csv and verification receipts. A nearby catalogue source is not an exact target match, and the scoped result does not assert exhaustive publication priority.

Sector	Photometric cycle	Measured midpoint BTJD	Input timing allowance d	Fitted depth ppt
14	0	1686.8290999	0.010417	115.73
14	1	1695.4004679	0.010417	121.37
41	86	2423.8930350	0.003472	116.04
41	88	2441.0337929	0.003472	118.31
81	213	3512.3406415	0.001157	118.22
81	214	3520.9117055	0.001157	116.42
81	215	3529.4812969	0.001157	117.65

Evidence and provenance: measurements.json, raw-windows.csv, raw-events.png, raw-sector-fold.png, alias-coverage.json, pixel-evidence.json, pixel-recomputed.png, historical/diff_s*.json/png, historical/gaia.json; batch ../gaia-geometry.json, ../alias-audit.json, ../secondary-audit.json, ../measurement-source-receipts.json, ../download-receipts.json, ../source-hashes.json, ../LEAD-AUDIT.md and ../helper-audit.md. SHA256SUMS.json covers reports, scripts, raw FITS and original cutouts. Current availability receipt for246857844 is in shared-catalogues and copied locally with its hash. No canonical/public files were edited by this batch.

Stable review key: R550.

Deciding evidence and numeric artifacts: - alias-coverage.json - historical/diff_s81.json - historical/diff_s81.png - historical/gaia.json - measurements.json - pixel-evidence.json - pixel-recomputed.png - raw-events.png - raw-sector-fold.png - raw-windows.csv - secondary-timing.json

Batch audits and scripts. Depth correction provenance.

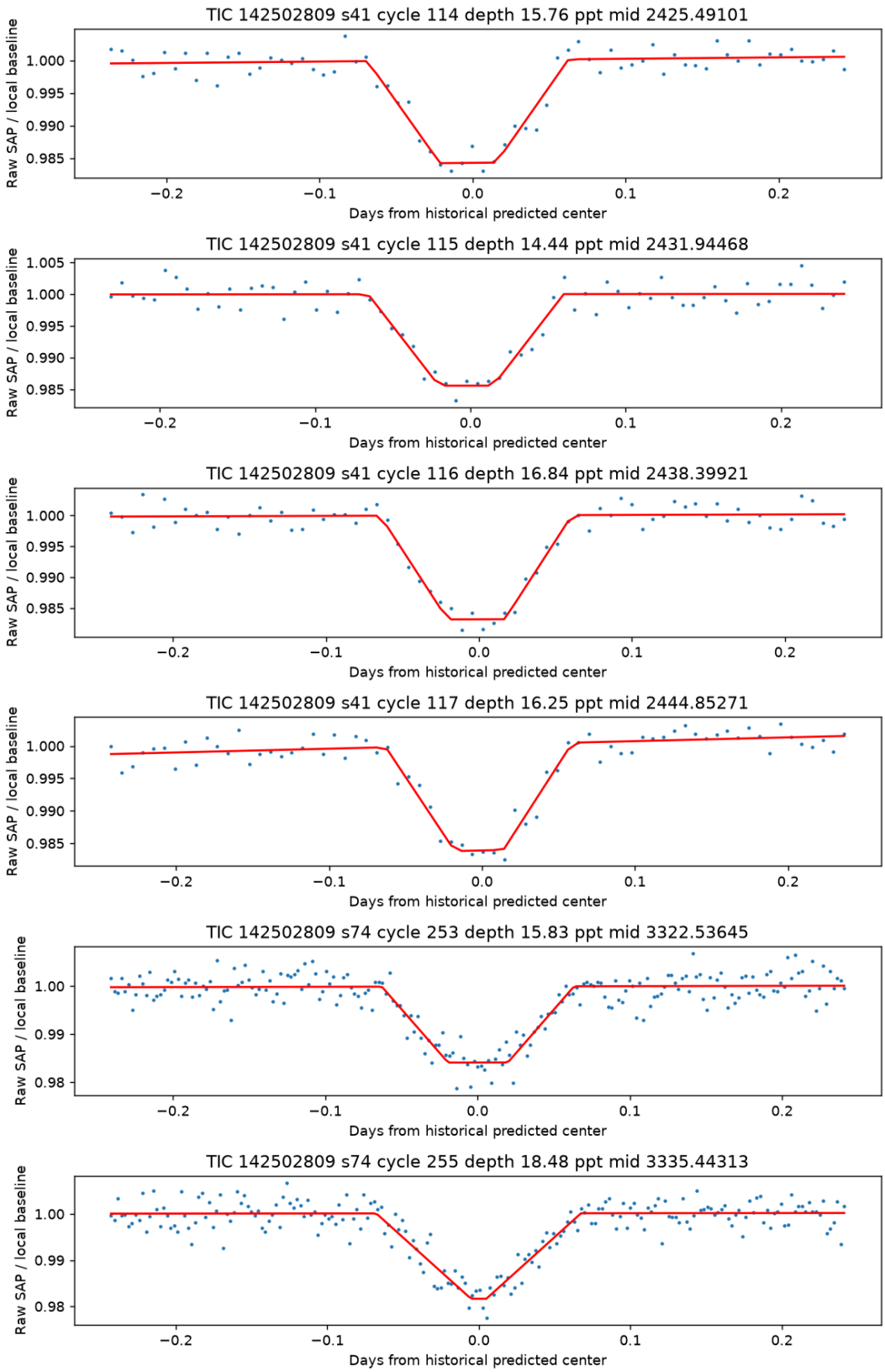
R551. TIC 142502809

physical_verdict: CONFIRMED

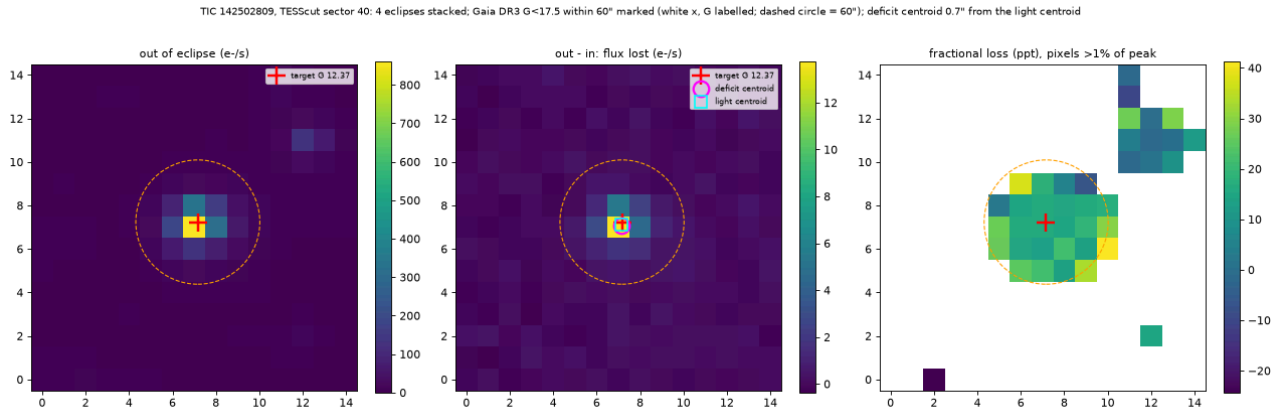
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and precision binary characterization.



R551; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R551; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 142502809: CONFIRMED

Full bounded physical review: COMPLETE. Catalogue novelty is separate from this verdict.

Raw QLP SAP_FLUX contains 6 complete free-center fitted eclipses in representative sectors[41, 74]. Predicted historical centers guided extraction, not timing points. Fits use local linear baselines and trapezoid shapes; quality-zero cadences only. Flagged, partial and uncovered windows are not clean nulls. raw-events.png, raw-sector-fold.png and raw-windows.csv show the actual observations.

Measured recurrence P=6.4535587013 +/- 0.0000140965d. Named reference epoch BTJD2425.4915658: selected eclipse kind at phase zero, not an assumed deeper primary. The legacy primary_epoch_btjd column carries this epoch. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when positive-degree-of-freedom reduced chi square exceeds one. With only two centers, no reduced chi square or measured residual scatter exists; covariance propagates input timing allowances. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Consecutive same-kind measured pairs: 3. Same-sector parity and actual coverage are in alias-coverage.json. No persistent deep alternating-depth pattern in the same-sector parity arrays. Covered half-phase windows lack an eclipse comparable to the selected kind. A perfectly identical two-eclipse2P harmonic is not mathematically excluded by single-kind morphology alone; Gaia orbital association and the lack of a second comparable sequence support the reported recurrence. This is a bounded alias check, not a blind exhaustive search. Two separated sectors are distinct observing occasions on the same instrument.

Fresh raw TESScut pixel identity agrees with Gaia DR3 1128177721274662400 to 0.00000arcsec by saved WCS; all source bytes are hashed. In sector40, 4 separately recomputed event deficits average0.149px from the target. Event-scatter vector is[0.03751658804580031, 0.05294847378631903]; this is descriptive sampling scatter, not calibrated positional confidence or pixel-response systematics. Lead visually inspected the historical full-field and freshly recomputed core images. Every capable neighbor in the saved60arcsec Gaia list: None within60arcsec under approximate G-band flux budget. Intrinsic source loss requirements and diluted fractional pixel losses are different quantities and are never compared as a veto. The G-band flux screen is approximate and does not exclude unknown unresolved companions.

Alternative extraction: background-subtracted raw TESScut1x1/3x3/5x5 median event depths are14.89/14.65/15.16ppt; selected QLP raw median fit depth16.04ppt. Details, windows, individual centroids, count and depths are in pixel-evidence.json. This separately extracted reduction shares the same TESS photons as QLP and is not an independent instrument or independent observing occasion. Aperture dilution, cadence and event-core averaging explain differing apparent depths; no intrinsic binary depth is inferred.

Gaia orbit and physical plausibility: source-specific NSS P=6.4537461386d, e=0.22283, omega=15.3736deg, periastron BTJD391.66061. Closest edge-on conjunction offset at the measured epoch=-0.01756d (-0.00272phase); descriptive linear full-SB1-covariance timing error=0.26484d. The covariance uses source-specific P/e/omega/Tp correlations, not only diagonal errors. Mass function=0.042903solar; assumed primary mass=1.210solar gives minimum companion mass=0.501solar. Approximate central duration=6.62h versus observed3.03h. TIC radius/main-sequence companion assumptions and edge-on conjunction are approximations, not a fitted stellar model or arbitrary sigma admission rule. Orbit/geometry are consistent at this descriptive precision and do not establish the identity of an unresolved eclipse source.

Deciding interpretation: Repeated sharp shallow eclipses across representative sectors. Raw full-sector plot includes additional under-sampled or incompletely bracketed events excluded from free timing. No capable Gaia neighbor within the historical 60-arcsec flux screen.

Next check: Optional resolved photometry and precision binary characterization.

Seven scoped catalogue groups were checked centrally for this identity; see ../../shared-catalogues/coverage-matrix.csv and verification receipts. A nearby catalogue source is not an exact target match, and the scoped result does not assert exhaustive publication priority.

Sector	Photometric cycle	Measured midpoint BTJD	Input timing allowance d	Fitted depth ppt
41	114	2425.4910086	0.003472	15.76
41	115	2431.9446790	0.003472	14.44
41	116	2438.3992124	0.003472	16.84
41	117	2444.8527064	0.003472	16.25
74	253	3322.5364466	0.001280	15.83
74	255	3335.4431326	0.001250	18.48

Evidence and provenance: measurements.json, raw-windows.csv, raw-events.png, raw-sector-fold.png, alias-coverage.json, pixel-evidence.json, pixel-recomputed.png, historical/diff_s*.json/png, historical/gaia.json; batch ../gaia-geometry.json, ../alias-audit.json, ../secondary-audit.json, ../measurement-source-receipts.json, ../download-receipts.json, ../source-hashes.json, ../LEAD-AUDIT.md and ../helper-audit.md. SHA256SUMS.json covers reports, scripts, raw FITS and original cutouts. Current availability receipt for246857844 is in shared-catalogues and copied locally with its hash. No canonical/public files were edited by this batch.

Stable review key: R551.

Deciding evidence and numeric artifacts: - alias-coverage.json - historical/diff_s40.json - historical/diff_s40.png - historical/gaia.json - measurements.json - pixel-evidence.json - pixel-recomputed.png - raw-events.png - raw-sector-fold.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

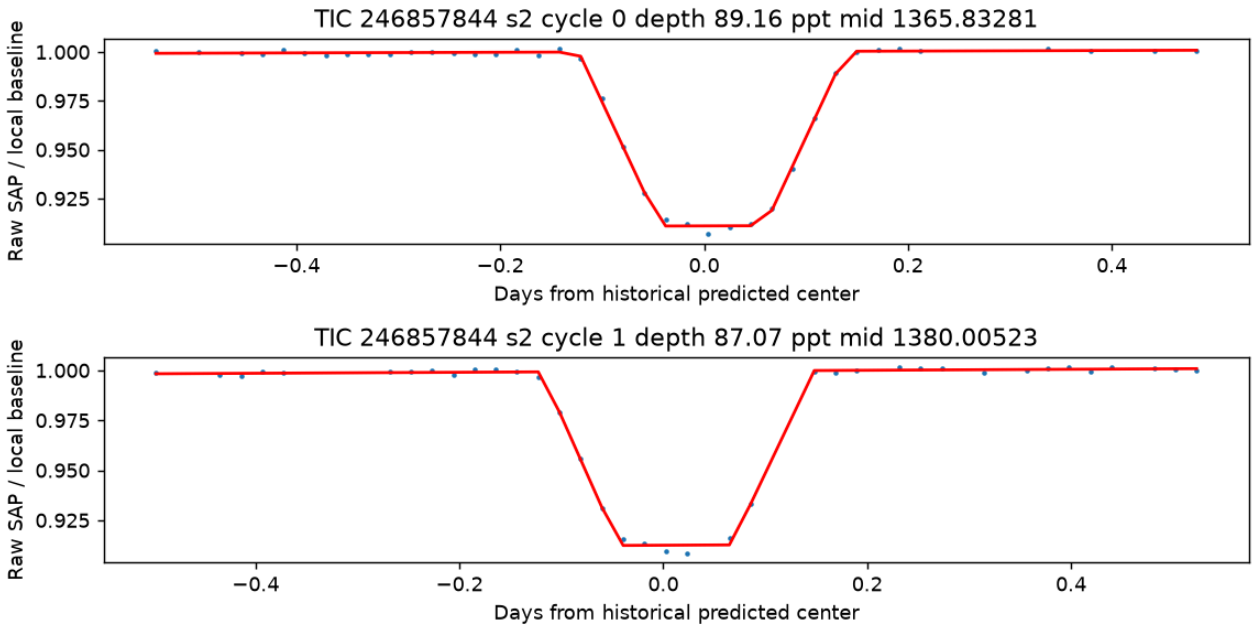
R552. TIC 246857844

physical_verdict: CONFIRMED

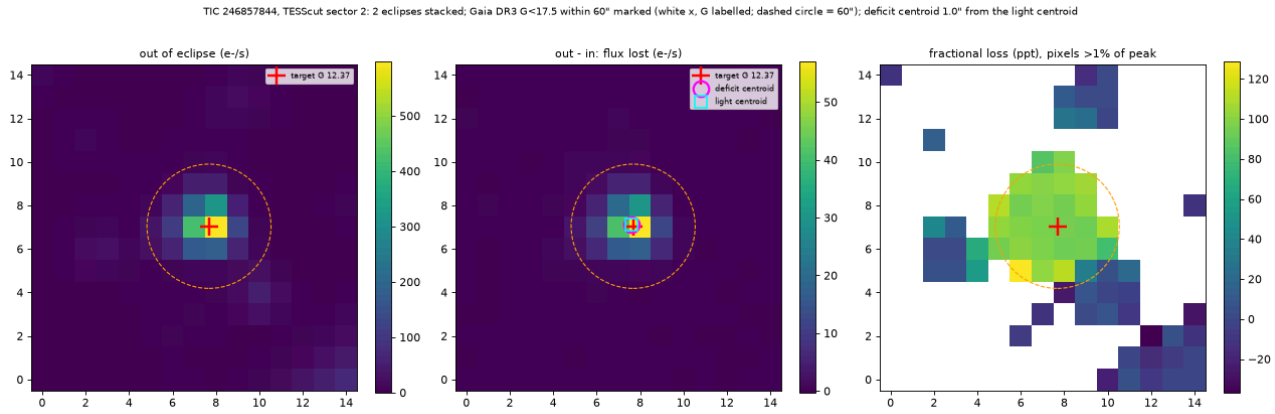
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: A complete new eclipse for a longer-baseline ephemeris.



R552; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R552; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 246857844: CONFIRMED

Full bounded physical review: COMPLETE. Catalogue novelty is separate from this verdict.

Raw QLP SAP_FLUX contains 2 complete free-center fitted eclipses in representative sectors[2]. Predicted historical centers guided extraction, not timing points. Fits use local linear baselines and trapezoid shapes; quality-zero cadences only. Flagged, partial and uncovered windows are not clean nulls. raw-events.png, raw-sector-fold.png and raw-windows.csv show the actual observations.

Measured recurrence P=14.1724200490 +/- 0.0147317170d. Named reference epoch BTJD1365.8328131: selected eclipse kind at phase zero, not an assumed deeper primary. The legacy primary_epoch_btjd column carries this epoch. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when positive-degree-of-freedom reduced chi square exceeds one. With only two centers, no reduced chi square or measured residual scatter exists; covariance propagates input timing allowances. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Consecutive same-kind measured pairs: 1. Same-sector parity and actual coverage are in alias-coverage.json. No persistent deep alternating-depth pattern in the same-sector parity arrays. Covered half-phase windows lack an eclipse comparable to the selected kind. A perfectly identical two-eclipse2P harmonic is not mathematically excluded by single-kind morphology alone; Gaia orbital association and the lack of a second comparable sequence support the reported recurrence. This is a bounded alias check, not a blind exhaustive search. Two separated sectors are distinct observing occasions on the same instrument.

Fresh raw TESScut pixel identity agrees with Gaia DR3 2347638783375829632 to 0.00000arcsec by saved WCS; all source bytes are hashed. In sector2, 2 separately recomputed event deficits average0.126px from the target. Event-scatter vector is[0.016565559748460818, 0.050466038760845267]; this is descriptive sampling scatter, not calibrated positional confidence or pixel-response systematics. Lead visually inspected the historical full-field and freshly recomputed core images. Every capable neighbor in the saved60arcsec Gaia list: None within60arcsec under approximate G-band flux budget. Intrinsic source loss requirements and diluted fractional pixel losses are different quantities and are never compared as a veto. The G-band flux screen is approximate and does not exclude unknown unresolved companions.

Alternative extraction: background-subtracted raw TESScut1x1/3x3/5x5 median event depths are93.40/94.04/93.69ppt; selected QLP raw median fit depth88.11ppt. Details, windows, individual centroids, count and depths are in pixel-evidence.json. This separately extracted reduction shares the same TESS photons as QLP and is not an independent instrument or independent observing occasion. Aperture dilution, cadence and event-core averaging explain differing apparent depths; no intrinsic binary depth is inferred.

Gaia orbit and physical plausibility: source-specific NSS P=14.1629385943d, e=0.27949, omega=235.6661deg, periastron BTJD393.16694. Closest edge-on conjunction offset at the measured epoch=0.30047d (0.02122phase); descriptive linear full-SB1-covariance timing error=0.85000d. The covariance uses source-specific P/e/omega/Tp correlations, not only diagonal errors. Mass function=0.02807solar; assumed primary mass=1.000solar gives minimum companion mass=0.376solar. Approximate central duration=8.38h versus observed6.30h. TIC radius/main-sequence companion assumptions and edge-on conjunction are approximations, not a fitted stellar model or arbitrary sigma admission rule. Orbit/geometry are consistent at this descriptive precision and do not establish the identity of an unresolved eclipse source.

Deciding interpretation: Only two complete timed eclipses, both in S2 and adjacent with spacing14.17242d, rather than two long-separated events with unresolved integer count. Current MAST query returns QLP sectors2,29,69, all inspected. S29 starts during a partial real eclipse; it is not used as a midpoint. S69 selected-eclipse windows lack clean coverage. Two-center period error propagates

timing allowances with zero fit degrees of freedom; it does not measure timing scatter. Half-cycle covered baselines show no comparable eclipse. This short-baseline photometric period remains imprecise; no all-sector long-baseline precision is claimed.

Next check: A complete new eclipse for a longer-baseline ephemeris.

Seven scoped catalogue groups were checked centrally for this identity; see ../../shared-catalogues/coverage-matrix.csv and verification receipts. A nearby catalogue source is not an exact target match, and the scoped result does not assert exhaustive publication priority.

Sector	Photometric cycle	Measured midpoint BTJD	Input timing allowance d	Fitted depth ppt
2	0	1365.8328131	0.010417	89.16
2	1	1380.0052332	0.010417	87.07

Evidence and provenance: measurements.json, raw-windows.csv, raw-events.png, raw-sector-fold.png, alias-coverage.json, pixel-evidence.json, pixel-recomputed.png, historical/diff_s*.json/png, historical/gaia.json; batch ../gaia-geometry.json, ../alias-audit.json, ../secondary-audit.json, ../measurement-source-receipts.json, ../download-receipts.json, ../source-hashes.json, ../LEAD-AUDIT.md and ../helper-audit.md. SHA256SUMS.json covers reports, scripts, raw FITS and original cutouts. Current availability receipt for246857844 is in shared-catalogues and copied locally with its hash. No canonical/public files were edited by this batch.

Stable review key: R552.

Deciding evidence and numeric artifacts: - alias-coverage.json - current-mast-availability.json - historical/diff_s02.json - historical/diff_s02.png - historical/gaia.json - measurements.json - pixel-evidence.json - pixel-recomputed.png - raw-events.png - raw-sector-fold.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

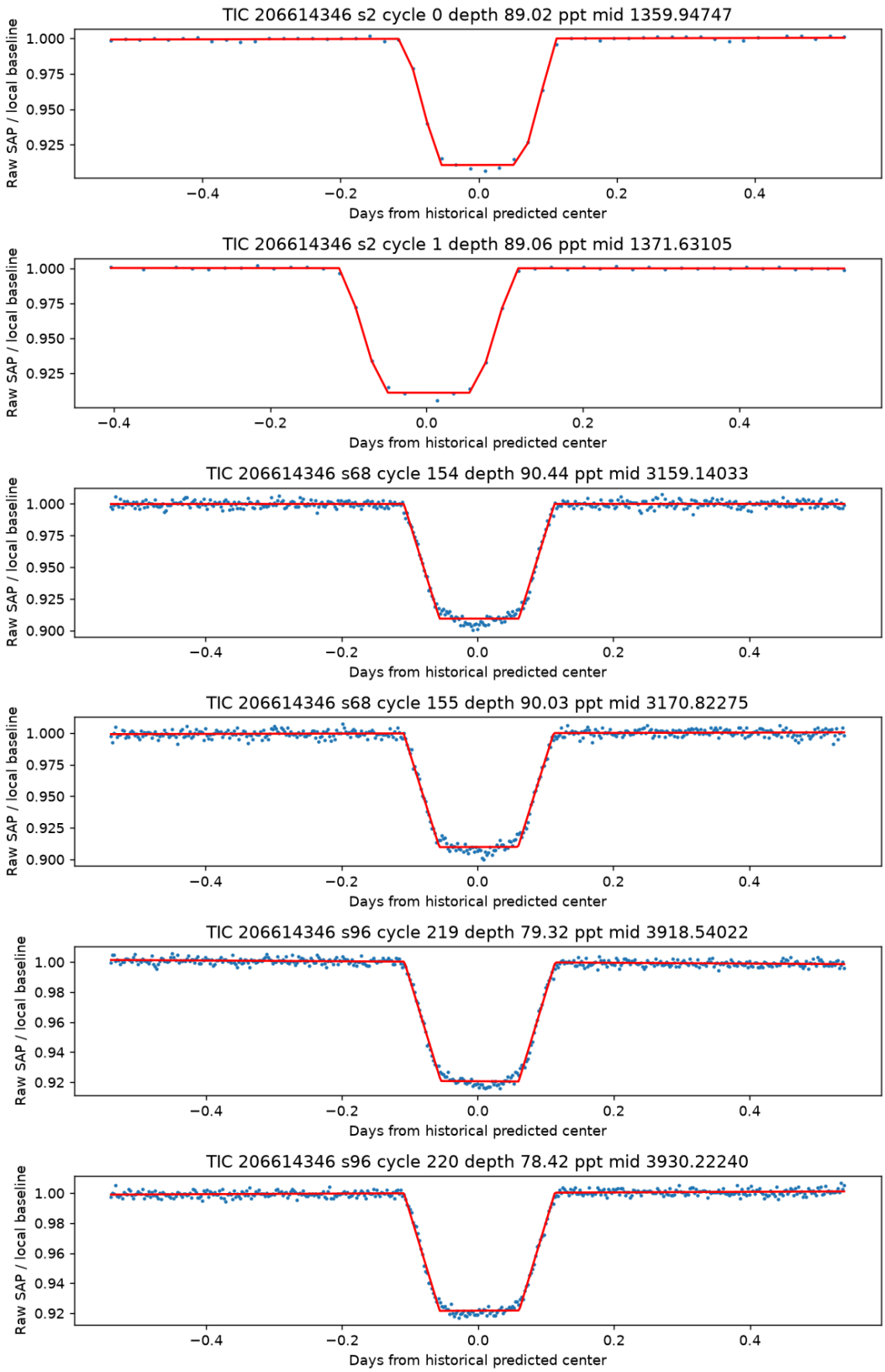
R553. TIC 206614346

physical_verdict: CONFIRMED

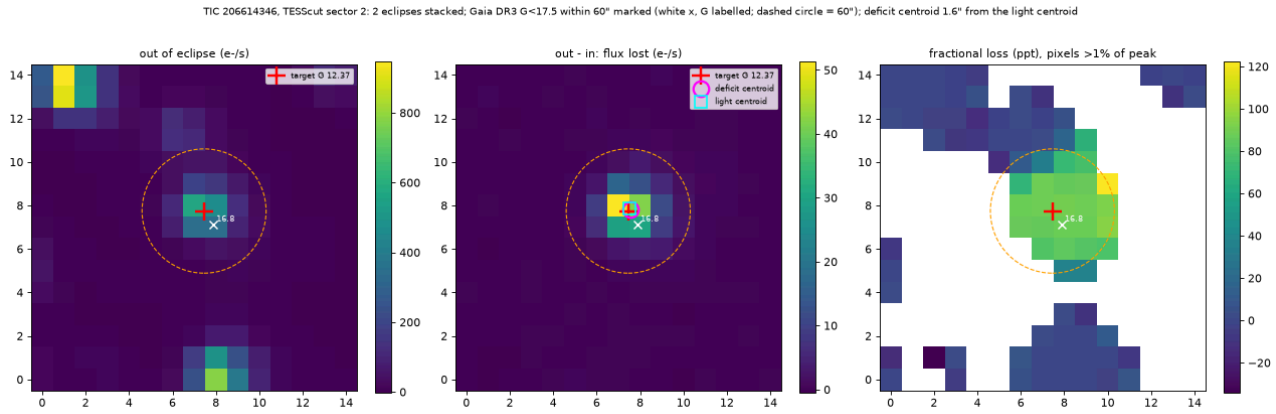
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and precision binary characterization.



R553; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R553; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 206614346: CONFIRMED

Full bounded physical review: COMPLETE. Catalogue novelty is separate from this verdict.

Raw QLP SAP_FLUX contains 6 complete free-center fitted eclipses in representative sectors[2, 68, 96]. Predicted historical centers guided extraction, not timing points. Fits use local linear baselines and trapezoid shapes; quality-zero cadences only. Flagged, partial and uncovered windows are not clean nulls. raw-events.png, raw-sector-fold.png and raw-windows.csv show the actual observations.

Measured recurrence P=11.6830723205 +/- 0.0000162395d. Named reference epoch BTJD1359.9469056: selected eclipse kind at phase zero, not an assumed deeper primary. The legacy primary_epoch_btjd column carries this epoch. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when positive-degree-of-freedom reduced chi square exceeds one. With only two centers, no reduced chi square or measured residual scatter exists; covariance propagates input timing allowances. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Consecutive same-kind measured pairs: 3. Same-sector parity and actual coverage are in alias-coverage.json. No persistent deep alternating-depth pattern in the same-sector parity arrays. Covered half-phase windows lack an eclipse comparable to the selected kind. A perfectly identical two-eclipse2P harmonic is not mathematically excluded by single-kind morphology alone; Gaia orbital association and the lack of a second comparable sequence support the reported recurrence. This is a bounded alias check, not a blind exhaustive search. Two separated sectors are distinct observing occasions on the same instrument.

Fresh raw TESScut pixel identity agrees with Gaia DR3 6607402809430700800 to 0.00000arcsec by saved WCS; all source bytes are hashed. In sector2, 2 separately recomputed event deficits average0.108px from the target. Event-scatter vector is[0.014919760833283377, 0.07233357074091407]; this is descriptive sampling scatter, not calibrated positional confidence or pixel-response systematics. Lead visually inspected the historical full-field and freshly recomputed core images. Every capable neighbor in the saved60arcsec Gaia list: None within60arcsec under approximate G-band flux budget. Intrinsic source loss requirements and diluted fractional pixel losses are different quantities and are never compared as a veto. The G-band flux screen is approximate and does not exclude unknown unresolved companions.

Alternative extraction: background-subtracted raw TESScut1x1/3x3/5x5 median event depths are88.53/87.44/84.64ppt; selected QLP raw median fit depth89.04ppt. Details, windows, individual centroids, count and depths are in pixel-evidence.json. This separately extracted reduction shares the same TESS photons as QLP and is not an independent instrument or independent observing occasion. Aperture dilution, cadence and event-core averaging explain differing apparent depths; no intrinsic binary depth is inferred.

Gaia orbit and physical plausibility: source-specific NSS P=11.6877344473d, e=0.05379, omega=49.7158deg, periastron BTJD388.97347. Closest edge-on conjunction offset at the measured epoch=-0.29090d (-0.02489phase); descriptive linear full-SB1-covariance timing error=0.31142d. The covariance uses source-specific P/e/omega/Tp correlations, not only diagonal errors. Mass function=0.017288solar; assumed primary mass=0.980solar gives minimum companion mass=0.306solar. Approximate central duration=5.91h versus observed5.31h. TIC radius/main-sequence companion assumptions and edge-on conjunction are approximations, not a fitted stellar model or arbitrary sigma admission rule. Orbit/geometry are consistent at this descriptive precision and do not establish the identity of an unresolved eclipse source.

Deciding interpretation: Six complete selected-kind raw eclipses span S2/68/96. Depth varies with aperture and sector. Narrow sharp eclipses are distinct from broad baseline variability; no capable Gaia neighbor in the saved 60-arcsec screen.

Next check: Optional resolved photometry and precision binary characterization.

Seven scoped catalogue groups were checked centrally for this identity; see ../../shared-catalogues/coverage-matrix.csv and verification receipts. A nearby catalogue source is not an exact target match, and the scoped result does not assert exhaustive publication priority.

Sector	Photometric cycle	Measured midpoint BTJD	Input timing allowance d	Fitted depth ppt
2	0	1359.9474701	0.010417	89.02
2	1	1371.6310539	0.010417	89.06
68	154	3159.1403325	0.001157	90.44
68	155	3170.8227458	0.001157	90.03
96	219	3918.5402168	0.001157	79.32
96	220	3930.2224031	0.001157	78.42

Evidence and provenance: measurements.json, raw-windows.csv, raw-events.png, raw-sector-fold.png, alias-coverage.json, pixel-evidence.json, pixel-recomputed.png, historical/diff_s*.json/png, historical/gaia.json; batch ../gaia-geometry.json, ../alias-audit.json, ../secondary-audit.json, ../measurement-source-receipts.json, ../download-receipts.json, ../source-hashes.json, ../LEAD-AUDIT.md and ../helper-audit.md. SHA256SUMS.json covers reports, scripts, raw FITS and original cutouts. Current availability receipt for246857844 is in shared-catalogues and copied locally with its hash. No canonical/public files were edited by this batch.

Stable review key: R553.

Deciding evidence and numeric artifacts: - alias-coverage.json - historical/diff_s02.json - historical/diff_s02.png - historical/gaia.json - measurements.json - pixel-evidence.json - pixel-recomputed.png - raw-events.png - raw-sector-fold.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

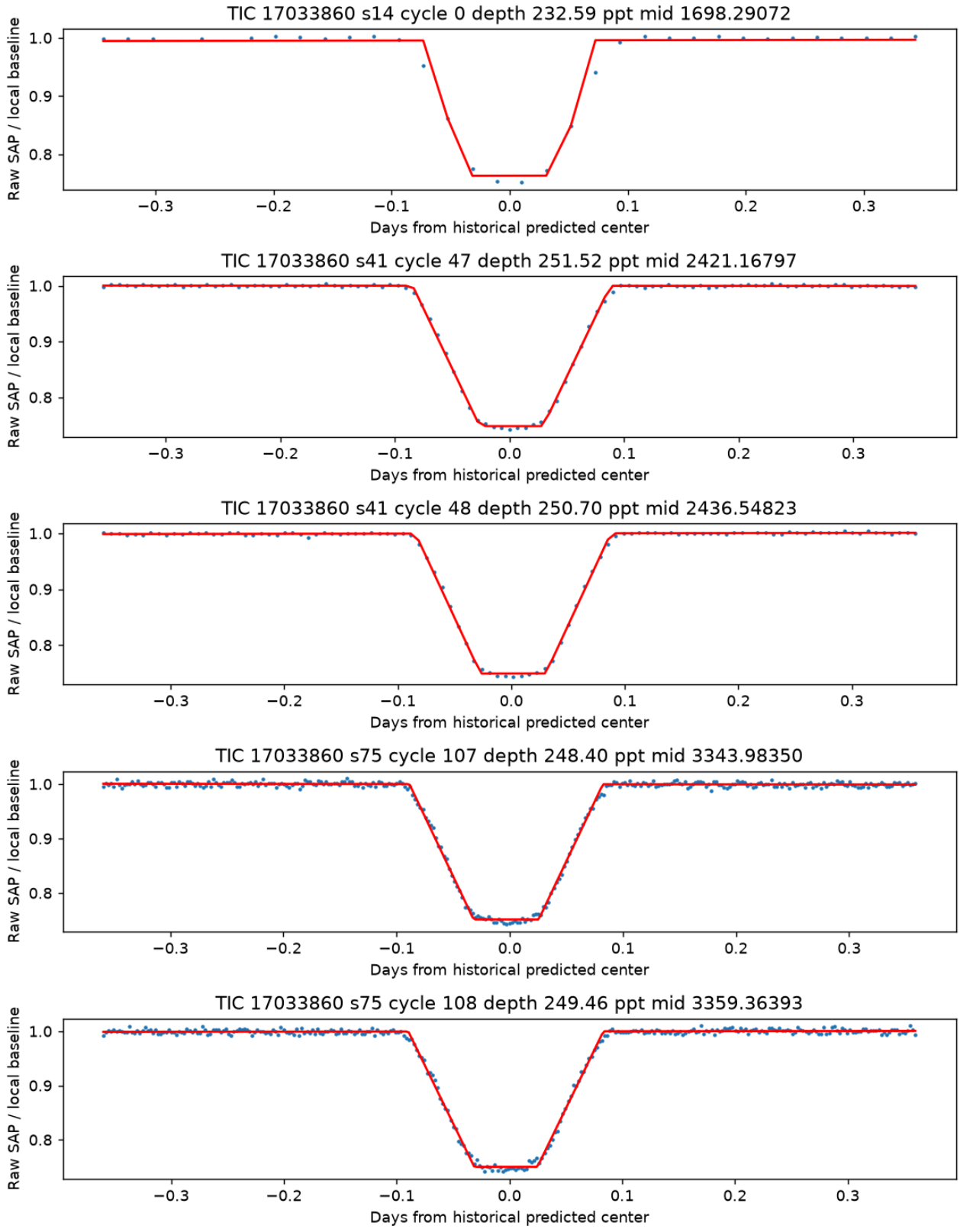
R554. TIC 17033860

physical_verdict: CONFIRMED

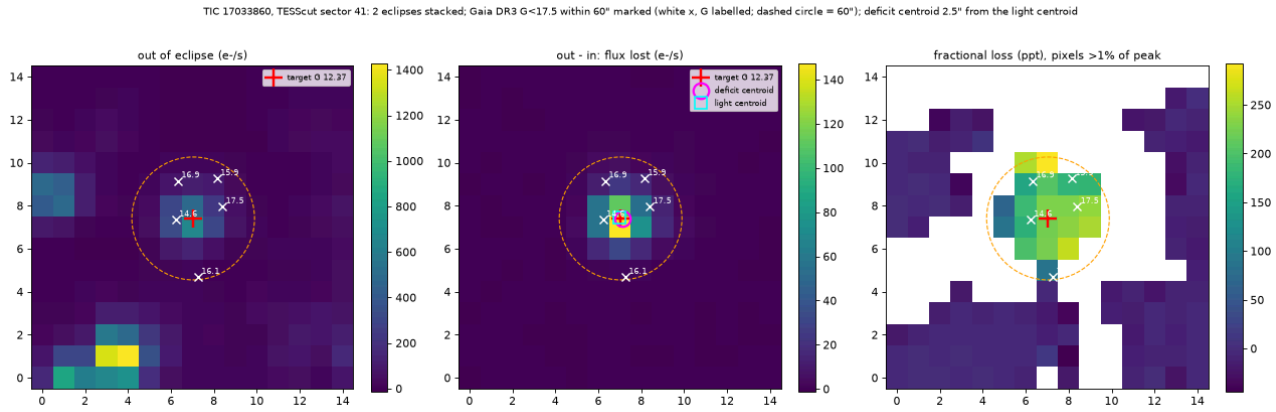
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and precision binary characterization.



R554; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R554; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 17033860: CONFIRMED

Full bounded physical review: COMPLETE. Catalogue novelty is separate from this verdict.

Raw QLP SAP_FLUX contains 5 complete free-center fitted eclipses in representative sectors[14, 41, 75]. Predicted historical centers guided extraction, not timing points. Fits use local linear baselines and trapezoid shapes; quality-zero cadences only. Flagged, partial and uncovered windows are not clean nulls. raw-events.png, raw-sector-fold.png and raw-windows.csv show the actual observations.

Measured recurrence P=15.3802677227 +/- 0.0000397687d. Named reference epoch BTJD1698.2949564: selected eclipse kind at phase zero, not an assumed deeper primary. The legacy primary_epoch_btjd column carries this epoch. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when positive-degree-of-freedom reduced chi square exceeds one. With only two centers, no reduced chi square or measured residual scatter exists; covariance propagates input timing allowances. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Consecutive same-kind measured pairs: 2. Same-sector parity and actual coverage are in alias-coverage.json. Other-kind free fits: 6 midpoints; measured phase range 0.27346 to 0.27389. See secondary-timing.json. The exact-half-phase baseline estimates are not interpreted as clean depths where they overlap the other eclipse. This is a bounded alias check, not a blind exhaustive search. Two separated sectors are distinct observing occasions on the same instrument.

Fresh raw TESScut pixel identity agrees with Gaia DR3 2067824378464107520 to 0.00000arcsec by saved WCS; all source bytes are hashed. In sector41, 2 separately recomputed event deficits average0.080px from the target. Event-scatter vector is[0.01026359917147296, 0.024579567158189125]; this is descriptive sampling scatter, not calibrated positional confidence or pixel-response systematics. Lead visually inspected the historical full-field and freshly recomputed core images. Every capable neighbor in the saved60arcsec Gaia list: None within60arcsec under approximate G-band flux budget. Intrinsic source loss requirements and diluted fractional pixel losses are different quantities and are never compared as a veto. The G-band flux screen is approximate and does not exclude unknown unresolved companions.

Alternative extraction: background-subtracted raw TESScut1x1/3x3/5x5 median event depths are220.62/214.36/208.21ppt; selected QLP raw median fit depth249.46ppt. Details, windows, individual centroids, count and depths are in pixel-evidence.json. This separately extracted reduction

shares the same TESS photons as QLP and is not an independent instrument or independent observing occasion. Aperture dilution, cadence and event-core averaging explain differing apparent depths; no intrinsic binary depth is inferred.

Gaia orbit and physical plausibility: source-specific NSS P=15.3825766034d, e=0.30673, omega=164.8905deg, periastron BTJD392.67385. Closest edge-on conjunction offset at the measured epoch=-0.04548d (-0.00296phase); descriptive linear full-SB1-covariance timing error=0.26778d. The covariance uses source-specific P/e/omega/Tp correlations, not only diagonal errors. Mass function=0.0419solar; assumed primary mass=0.979solar gives minimum companion mass=0.438solar. Approximate central duration=5.70h versus observed4.13h. TIC radius/main-sequence companion assumptions and edge-on conjunction are approximations, not a fitted stellar model or arbitrary sigma admission rule. Orbit/geometry are consistent at this descriptive precision and do not establish the identity of an unresolved eclipse source.

Deciding interpretation: Deep selected eclipse and a much shallower second kind at phase+0.2737 establish unequal event separation. No capable Gaia neighbor under the depth budget. Eclipse shapes remain sharp and recurring across S14/41/75.

Next check: Optional resolved photometry and precision binary characterization.

Seven scoped catalogue groups were checked centrally for this identity; see ../../shared-catalogues/coverage-matrix.csv and verification receipts. A nearby catalogue source is not an exact target match, and the scoped result does not assert exhaustive publication priority.

Sector	Photometric cycle	Measured midpoint BTJD	Input timing allowance d	Fitted depth ppt
14	0	1698.2907217	0.010417	232.59
41	47	2421.1679716	0.003472	251.52
41	48	2436.5482295	0.003472	250.70
75	107	3343.9835028	0.001157	248.40
75	108	3359.3639277	0.001157	249.46

Evidence and provenance: measurements.json, raw-windows.csv, raw-events.png, raw-sector-fold.png, alias-coverage.json, pixel-evidence.json, pixel-recomputed.png, historical/diff_s*.json/png, historical/gaia.json; batch ../gaia-geometry.json, ../alias-audit.json, ../secondary-audit.json, ../measurement-source-receipts.json, ../download-receipts.json, ../source-hashes.json, ../LEAD-AUDIT.md and ../helper-audit.md. SHA256SUMS.json covers reports, scripts, raw FITS and original cutouts. Current availability receipt for246857844 is in shared-catalogues and copied locally with its hash. No canonical/public files were edited by this batch.

Stable review key: R554.

Deciding evidence and numeric artifacts: - alias-coverage.json - historical/diff_s41.json - historical/diff_s41.png - historical/gaia.json - measurements.json - pixel-evidence.json - pixel-recomputed.png - raw-events.png - raw-sector-fold.png - raw-windows.csv - secondary-timing.json

Batch audits and scripts. Depth correction provenance.

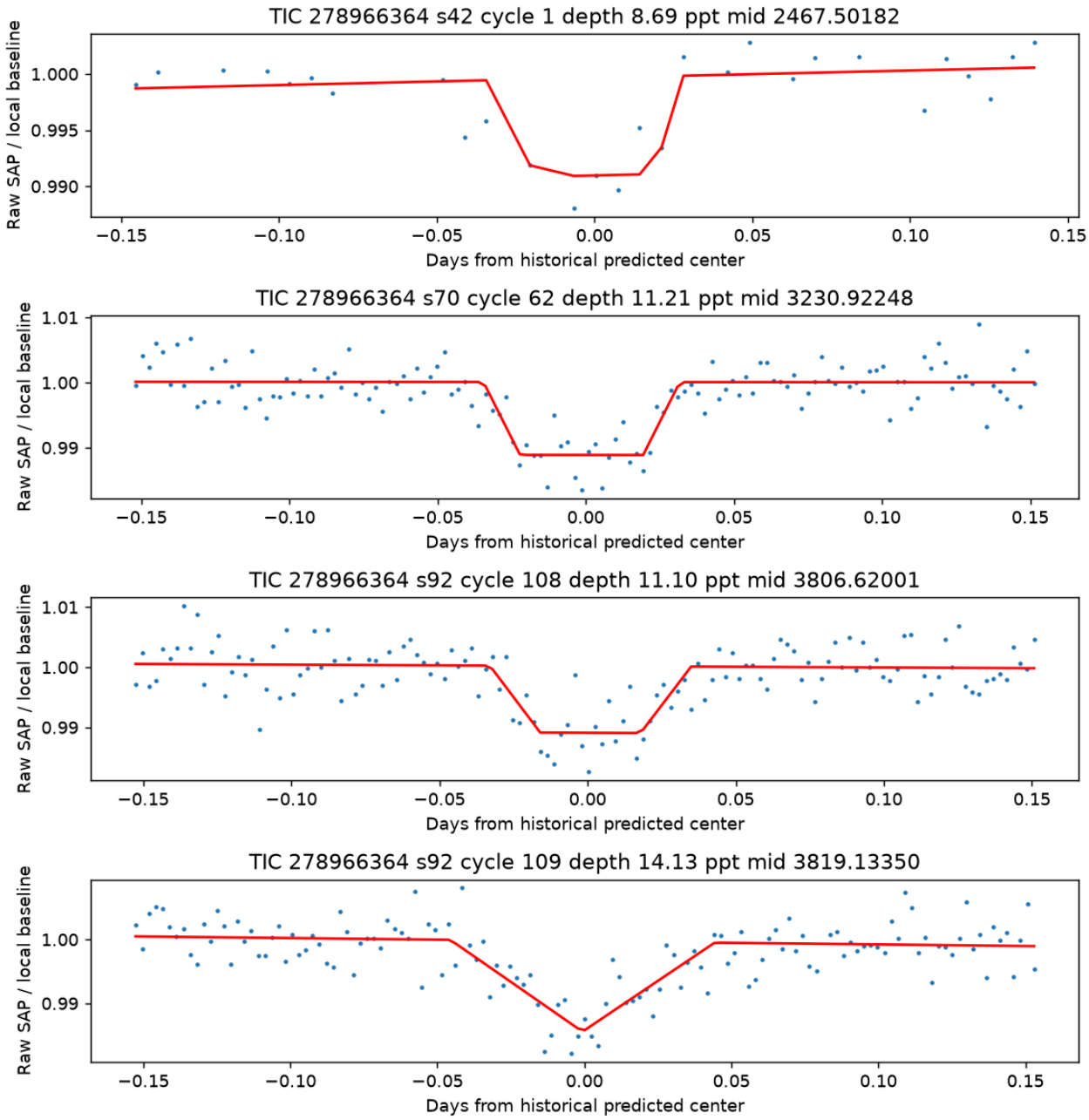
R555. TIC 278966364

physical_verdict: CONFIRMED

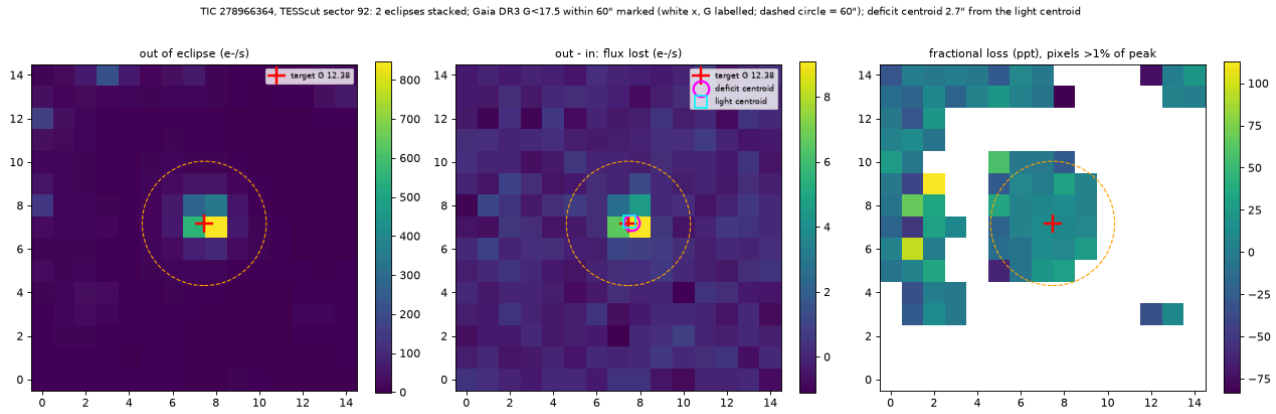
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and precision binary characterization.



R555; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R555; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 278966364: CONFIRMED

Full bounded physical review: COMPLETE. Catalogue novelty is separate from this verdict.

Raw QLP SAP_FLUX contains 4 complete free-center fitted eclipses in representative sectors[42, 70, 92]. Predicted historical centers guided extraction, not timing points. Fits use local linear baselines and trapezoid shapes; quality-zero cadences only. Flagged, partial and uncovered windows are not clean nulls. raw-events.png, raw-sector-fold.png and raw-windows.csv show the actual observations.

Measured recurrence P=12.5151308867 +/- 0.0000268573d. Named reference epoch BTJD2467.4999809: selected eclipse kind at phase zero, not an assumed deeper primary. The legacy primary_epoch_btjd column carries this epoch. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when positive-degree-of-freedom reduced chi square exceeds one. With only two centers, no reduced chi square or measured residual scatter exists; covariance propagates input timing allowances. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Consecutive same-kind measured pairs: 1. Same-sector parity and actual coverage are in alias-coverage.json. No persistent deep alternating-depth pattern in the same-sector parity arrays. Covered half-phase windows lack an eclipse comparable to the selected kind. A perfectly identical two-eclipse2P harmonic is not mathematically excluded by single-kind morphology alone; Gaia orbital association and the lack of a second comparable sequence support the reported recurrence. This is a bounded alias check, not a blind exhaustive search. Two separated sectors are distinct observing occasions on the same instrument.

Fresh raw TESScut pixel identity agrees with Gaia DR3 2638040248315742080 to 0.00000arcsec by saved WCS; all source bytes are hashed. In sector92, 2 separately recomputed event deficits average0.099px from the target. Event-scatter vector is[0.14133162900382695, 0.06136783897343869]; this is descriptive sampling scatter, not calibrated positional confidence or pixel-response systematics. Lead visually inspected the historical full-field and freshly recomputed core images. Every capable neighbor in the saved60arcsec Gaia list: None within60arcsec under approximate G-band flux budget. Intrinsic source loss requirements and diluted fractional pixel losses are different quantities and are never compared as a veto. The G-band flux screen is approximate and does not exclude unknown unresolved companions.

Alternative extraction: background-subtracted raw TESScut1x1/3x3/5x5 median event depths are10.93/11.09/10.45ppt; selected QLP raw median fit depth11.16ppt. Details, windows, individual centroids, count and depths are in pixel-evidence.json. This separately extracted reduction shares the same TESS photons as QLP and is not an independent instrument or independent observing occasion. Aperture dilution, cadence and event-core averaging explain differing apparent depths; no intrinsic binary depth is inferred.

Gaia orbit and physical plausibility: source-specific NSS P=12.5169206043d, e=0.09850, omega=89.9261deg, periastron BTJD389.89151. Closest edge-on conjunction offset at the measured epoch=-0.20245d (-0.01617phase); descriptive linear full-SB1-covariance timing error=0.80213d. The covariance uses source-specific P/e/omega/Tp correlations, not only diagonal errors. Mass function=0.077427solar; assumed primary mass=1.020solar gives minimum companion mass=0.584solar. Approximate central duration=5.68h versus observed1.61h. TIC radius/main-sequence companion assumptions and edge-on conjunction are approximations, not a fitted stellar model or arbitrary sigma admission rule. Orbit/geometry are consistent at this descriptive precision and do not establish the identity of an unresolved eclipse source.

Deciding interpretation: Small sharp eclipses repeat in S42/70/92, including adjacent S92 events. Large positive spikes and gaps are visible in the raw sectors; the selected eclipse cores are distinct from those artifacts. The weaker sparse S92 event increases model and timing uncertainty; timing covariance is scatter-inflated.

Next check: Optional resolved photometry and precision binary characterization.

Seven scoped catalogue groups were checked centrally for this identity; see ../../shared-catalogues/coverage-matrix.csv and verification receipts. A nearby catalogue source is not an exact target match, and the scoped result does not assert exhaustive publication priority.

Sector	Photometric cycle	Measured midpoint BTJD	Input timing allowance d	Fitted depth ppt
42	1	2467.5018153	0.003472	8.69
70	62	3230.9224845	0.001157	11.21
92	108	3806.6200069	0.001545	11.10
92	109	3819.1335000	0.001670	14.13

Evidence and provenance: measurements.json, raw-windows.csv, raw-events.png, raw-sector-fold.png, alias-coverage.json, pixel-evidence.json, pixel-recomputed.png, historical/diff_s*.json/png, historical/gaia.json; batch ../gaia-geometry.json, ../alias-audit.json, ../secondary-audit.json, ../measurement-source-receipts.json, ../download-receipts.json, ../source-hashes.json, ../LEAD-AUDIT.md and ../helper-audit.md. SHA256SUMS.json covers reports, scripts, raw FITS and original cutouts. Current availability receipt for246857844 is in shared-catalogues and copied locally with its hash. No canonical/public files were edited by this batch.

Stable review key: R555.

Deciding evidence and numeric artifacts: - alias-coverage.json - historical/diff_s92.json - historical/diff_s92.png - historical/gaia.json - measurements.json - pixel-evidence.json - pixel-recomputed.png - raw-events.png - raw-sector-fold.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

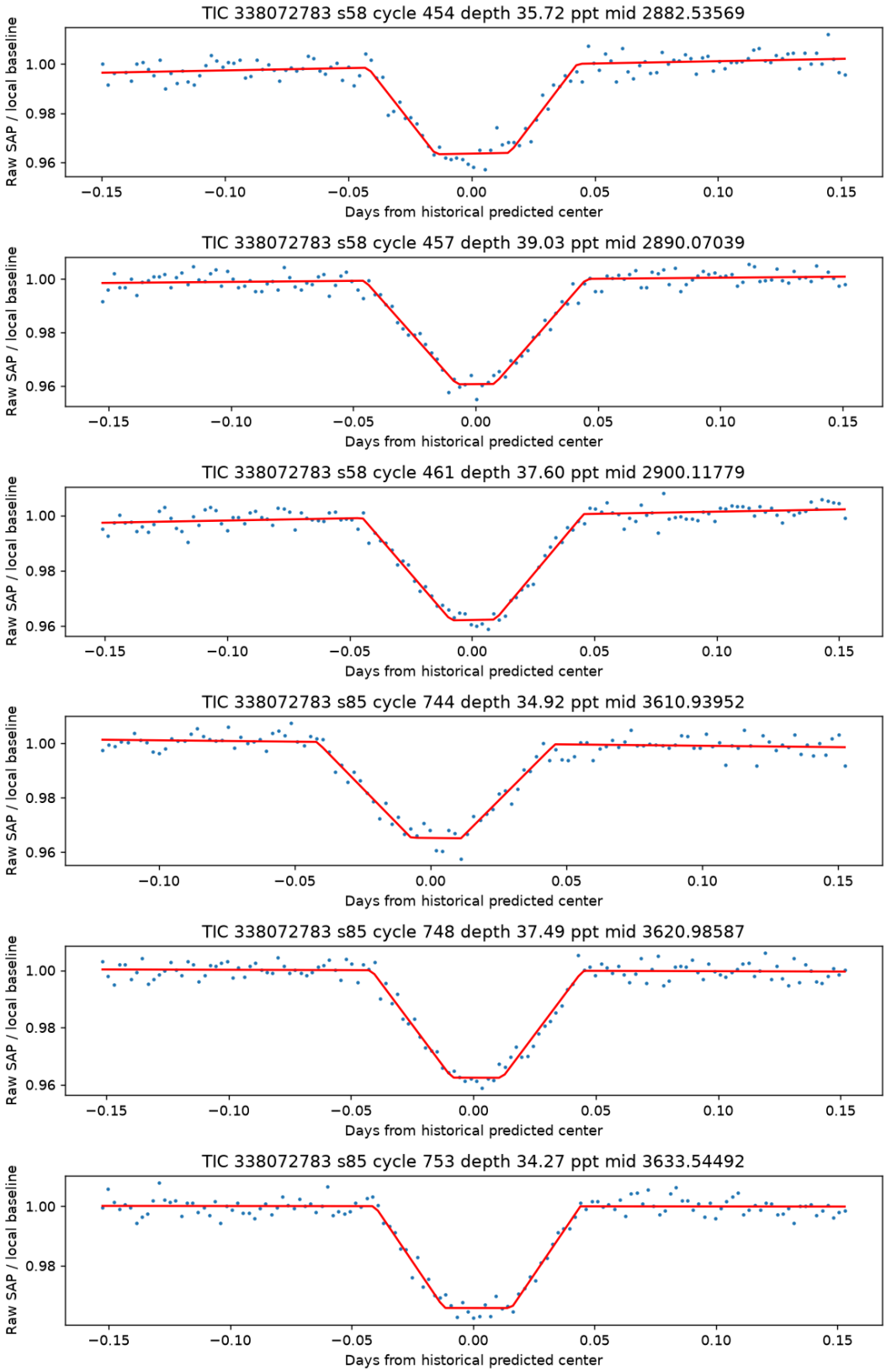
R556. TIC 338072783

physical_verdict: CONFIRMED

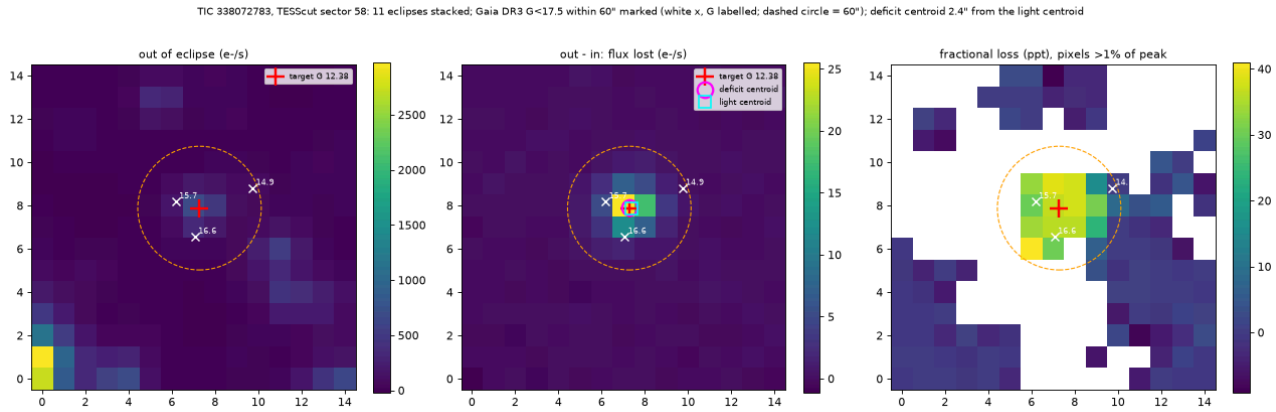
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and precision binary characterization.



R556; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R556; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 338072783: CONFIRMED

Full bounded physical review: COMPLETE. Catalogue novelty is separate from this verdict.

Raw QLP SAP_FLUX contains 20 complete free-center fitted eclipses in representative sectors[58, 85]. Predicted historical centers guided extraction, not timing points. Fits use local linear baselines and trapezoid shapes; quality-zero cadences only. Flagged, partial and uncovered windows are not clean nulls. raw-events.png, raw-sector-fold.png and raw-windows.csv show the actual observations.

Measured recurrence P=2.5117384941 +/- 0.0000017968d. Named reference epoch BTJD2882.5350707: selected eclipse kind at phase zero, not an assumed deeper primary. The legacy primary_epoch_btjd column carries this epoch. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when positive-degree-of-freedom reduced chi square exceeds one. With only two centers, no reduced chi square or measured residual scatter exists; covariance propagates input timing allowances. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Consecutive same-kind measured pairs: 17. Same-sector parity and actual coverage are in alias-coverage.json. No persistent deep alternating-depth pattern in the same-sector parity arrays. Covered half-phase windows lack an eclipse comparable to the selected kind. A perfectly identical two-eclipse2P harmonic is not mathematically excluded by single-kind morphology alone; Gaia orbital association and the lack of a second comparable sequence support the reported recurrence. This is a bounded alias check, not a blind exhaustive search. Two separated sectors are distinct observing occasions on the same instrument.

Fresh raw TESScut pixel identity agrees with Gaia DR3 2205518861011274240 to 0.00000arcsec by saved WCS; all source bytes are hashed. In sector58, 11 separately recomputed event deficits average0.012px from the target. Event-scatter vector is[0.03857414338079675, 0.0435990507643629]; this is descriptive sampling scatter, not calibrated positional confidence or pixel-response systematics. Lead visually inspected the historical full-field and freshly recomputed core images. Every capable neighbor in the saved60arcsec Gaia list: Gaia 2205518796593429248: separation22.99arcsec, G15.654, intrinsic required fraction0.769, deficit distance1.133px; Gaia 2205524946986588288: separation52.70arcsec, G14.917, intrinsic required fraction0.408, deficit distance2.618px Intrinsic

source loss requirements and diluted fractional pixel losses are different quantities and are never compared as a veto. The G-band flux screen is approximate and does not exclude unknown unresolved companions.

Alternative extraction: background-subtracted raw TESScut1x1/3x3/5x5 median event depths are34.84/34.89/33.24ppt; selected QLP raw median fit depth36.89ppt. Details, windows, individual centroids, count and depths are in pixel-evidence.json. This separately extracted reduction shares the same TESS photons as QLP and is not an independent instrument or independent observing occasion. Aperture dilution, cadence and event-core averaging explain differing apparent depths; no intrinsic binary depth is inferred.

Gaia orbit and physical plausibility: source-specific NSS P=2.5118018526d, e=0.13911, omega=37.9371deg, periastron BTJD388.00992. Closest edge-on conjunction offset at the measured epoch=0.02489d (0.00991phase); descriptive linear full-SB1-covariance timing error=0.16199d. The covariance uses source-specific P/e/omega/Tp correlations, not only diagonal errors. Mass function=0.021811solar; assumed primary mass=1.120solar gives minimum companion mass=0.363solar. Approximate central duration=3.57h versus observed2.08h. TIC radius/main-sequence companion assumptions and edge-on conjunction are approximations, not a fitted stellar model or arbitrary sigma admission rule. Orbit/geometry are consistent at this descriptive precision and do not establish the identity of an unresolved eclipse source.

Deciding interpretation: Twenty complete timing-usable eclipses and17 within-sector consecutive pairs establish recurrence. Two capable neighbors at23.0 and52.7arcsec are separated from the deficit core. Strong out-of-eclipse variability remains visible; local raw eclipse fits do not use detrended QLP flux.

Next check: Optional resolved photometry and precision binary characterization.

Seven scoped catalogue groups were checked centrally for this identity; see ../../shared-catalogues/coverage-matrix.csv and verification receipts. A nearby catalogue source is not an exact target match, and the scoped result does not assert exhaustive publication priority.

Sector	Photometric cycle	Measured midpoint BTJD	Input timing allowance d	Fitted depth ppt
58	454	2882.5356893	0.001157	35.72
58	455	2885.0471278	0.001157	35.85
58	456	2887.5584785	0.001157	38.68
58	457	2890.0703938	0.001157	39.03
58	458	2892.5816290	0.001157	35.16
58	459	2895.0932130	0.001157	36.16
58	460	2897.6056992	0.001157	37.99
58	461	2900.1177892	0.001157	37.60
58	462	2902.6287876	0.001157	36.38
58	463	2905.1405869	0.001157	39.31
58	464	2907.6519600	0.001157	39.36
85	744	3610.9395215	0.001157	34.92
85	745	3613.4520127	0.001157	37.47
85	746	3615.9624097	0.001157	36.16
85	747	3618.4742793	0.001157	37.29
85	748	3620.9858691	0.001157	37.49
85	750	3626.0102019	0.001157	34.23
85	751	3628.5204741	0.001157	36.48
85	752	3631.0329991	0.001157	37.44
85	753	3633.5449196	0.001157	34.27

Evidence and provenance: measurements.json, raw-windows.csv, raw-events.png, raw-sector-fold.png, alias-coverage.json, pixel-evidence.json, pixel-recomputed.png, historical/diff_s*.json/png, historical/gaia.json; batch ../gaia-geometry.json, ../alias-audit.json, ../secondary-audit.json, ../measurement-source-receipts.json, ../download-receipts.json, ../source-hashes.json, ../LEAD-AUDIT.md and ../helper-audit.md. SHA256SUMS.json covers reports, scripts, raw FITS and original cutouts. Current availability receipt for246857844 is in shared-catalogues and copied locally with its hash. No canonical/public files were edited by this batch.

Stable review key: R556.

Deciding evidence and numeric artifacts: - alias-coverage.json - historical/diff_s58.json - historical/diff_s58.png - historical/gaia.json - measurements.json - pixel-evidence.json - pixel-recomputed.png - raw-events.png - raw-sector-fold.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

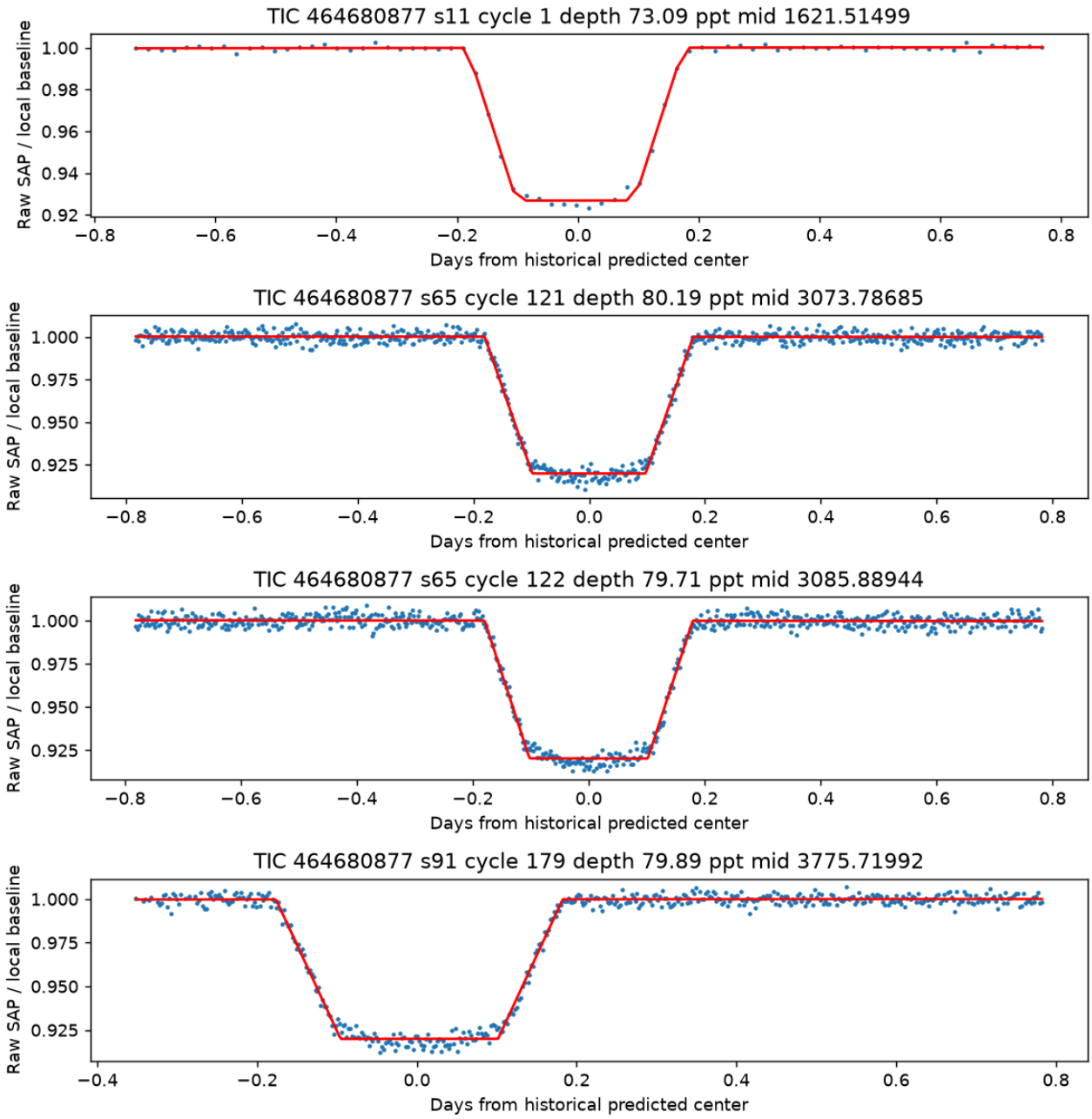
R557. TIC 464680877

physical_verdict: CONFIRMED

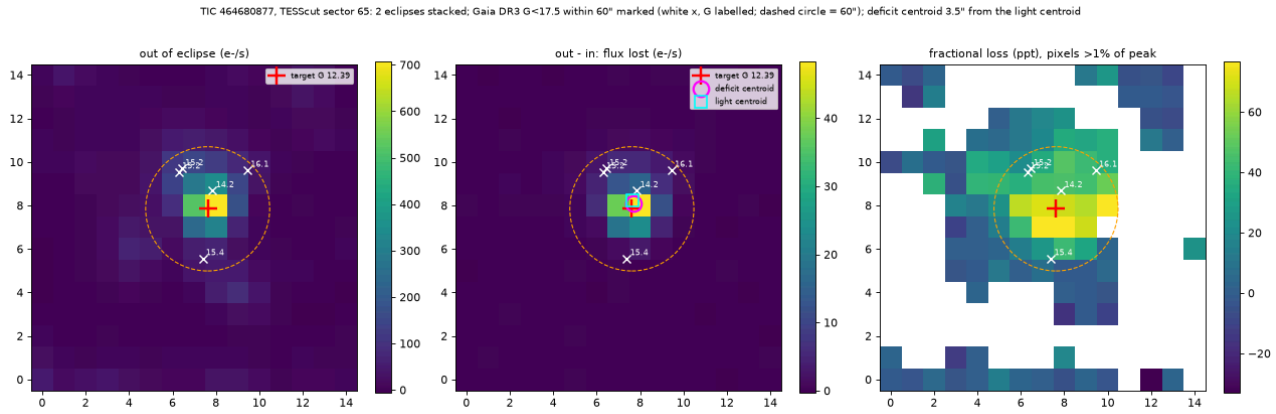
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and precision binary characterization.



R557; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R557; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 464680877: CONFIRMED

Full bounded physical review: COMPLETE. Catalogue novelty is separate from this verdict.

Raw QLP SAP_FLUX contains 4 complete free-center fitted eclipses in representative sectors[11, 65, 91]. Predicted historical centers guided extraction, not timing points. Fits use local linear baselines and trapezoid shapes; quality-zero cadences only. Flagged, partial and uncovered windows are not clean nulls. raw-events.png, raw-sector-fold.png and raw-windows.csv show the actual observations.

Measured recurrence P=12.1022896701 +/- 0.0000234101d. Named reference epoch BTJD1621.5122896: selected eclipse kind at phase zero, not an assumed deeper primary. The legacy primary_epoch_btjd column carries this epoch. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when positive-degree-of-freedom reduced chi square exceeds one. With only two centers, no reduced chi square or measured residual scatter exists; covariance propagates input timing allowances. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Consecutive same-kind measured pairs: 1. Same-sector parity and actual coverage are in alias-coverage.json. Other-kind free fits: 6 midpoints; measured phase range 0.48322 to 0.48384. See secondary-timing.json. The exact-half-phase baseline estimates are not interpreted as clean depths where they overlap the other eclipse. This is a bounded alias check, not a blind exhaustive search. Two separated sectors are distinct observing occasions on the same instrument.

Fresh raw TESScut pixel identity agrees with Gaia DR3 6213615984122331008 to 0.00000arcsec by saved WCS; all source bytes are hashed. In sector65, 2 separately recomputed event deficits average0.254px from the target. Event-scatter vector is[0.00914305465200767, 0.06164496078772161]; this is descriptive sampling scatter, not calibrated positional confidence or pixel-response systematics. Lead visually inspected the historical full-field and freshly recomputed core images. Every capable neighbor in the saved60arcsec Gaia list: Gaia 6213615984122329728: separation17.29arcsec, G14.180, intrinsic required fraction0.500, deficit distance0.617px Intrinsic source loss requirements and diluted fractional pixel losses are different quantities and are never compared as a veto. The G-band flux screen is approximate and does not exclude unknown unresolved companions.

Alternative extraction: background-subtracted raw TESScut1x1/3x3/5x5 median event depths are67.81/63.13/56.91ppt; selected QLP raw median fit depth79.80ppt. Details, windows, individual centroids, count and depths are in pixel-evidence.json. This separately extracted reduction shares the same TESS photons as QLP and is not an independent instrument or independent observing occasion. Aperture dilution, cadence and event-core averaging explain differing apparent depths; no intrinsic binary depth is inferred.

Gaia orbit and physical plausibility: source-specific NSS P=12.1023168179d, e=0.12316, omega=246.7988deg, periastron BTJD392.22420. Closest edge-on conjunction offset at the measured epoch=-0.08106d (-0.00670phase); descriptive linear full-SB1-covariance timing error=0.50725d. The covariance uses source-specific P/e/omega/Tp correlations, not only diagonal errors. Mass function=0.052033solar; assumed primary mass=1.160solar gives minimum companion mass=0.530solar. Approximate central duration=10.09h versus observed8.62h. TIC radius/main-sequence companion assumptions and edge-on conjunction are approximations, not a fitted stellar model or arbitrary sigma admission rule. Orbit/geometry are consistent at this descriptive precision and do not establish the identity of an unresolved eclipse source.

Deciding interpretation: Two recomputed S65 deficits locate the mean0.254px from target versus0.617px from a capable G14.18 neighbor17.29arcsec away (target-neighbor separation0.82px). The maximum core and aperture pattern favor the target. Only two deficit occasions and uncalibrated pixel-response systematics limit subpixel claims, but the neighbor lies above the target-centered core. The repeated shallower kind atphase+0.4834 distinguishes the measured orbital recurrence.

Next check: Optional resolved photometry and precision binary characterization.

Seven scoped catalogue groups were checked centrally for this identity; see ../../shared-catalogues/coverage-matrix.csv and verification receipts. A nearby catalogue source is not an exact target match, and the scoped result does not assert exhaustive publication priority.

Sector	Photometric cycle	Measured midpoint BTJD	Input timing allowance d	Fitted depth ppt
11	1	1621.5149901	0.010417	73.09
65	121	3073.7868492	0.001157	80.19
65	122	3085.8894398	0.001157	79.71
91	179	3775.7199181	0.001157	79.89

Evidence and provenance: measurements.json, raw-windows.csv, raw-events.png, raw-sector-fold.png, alias-coverage.json, pixel-evidence.json, pixel-recomputed.png, historical/diff_s*.json/png, historical/gaia.json; batch ../gaia-geometry.json, ../alias-audit.json, ../secondary-audit.json, ../measurement-source-receipts.json, ../download-receipts.json, ../source-hashes.json, ../LEAD-AUDIT.md and ../helper-audit.md. SHA256SUMS.json covers reports, scripts, raw FITS and original cutouts. Current availability receipt for246857844 is in shared-catalogues and copied locally with its hash. No canonical/public files were edited by this batch.

Stable review key: R557.

Deciding evidence and numeric artifacts: - alias-coverage.json - historical/diff_s65.json - historical/diff_s65.png - historical/gaia.json - measurements.json - pixel-evidence.json - pixel-recomputed.png - raw-events.png - raw-sector-fold.png - raw-windows.csv - secondary-timing.json

Batch audits and scripts. Depth correction provenance.

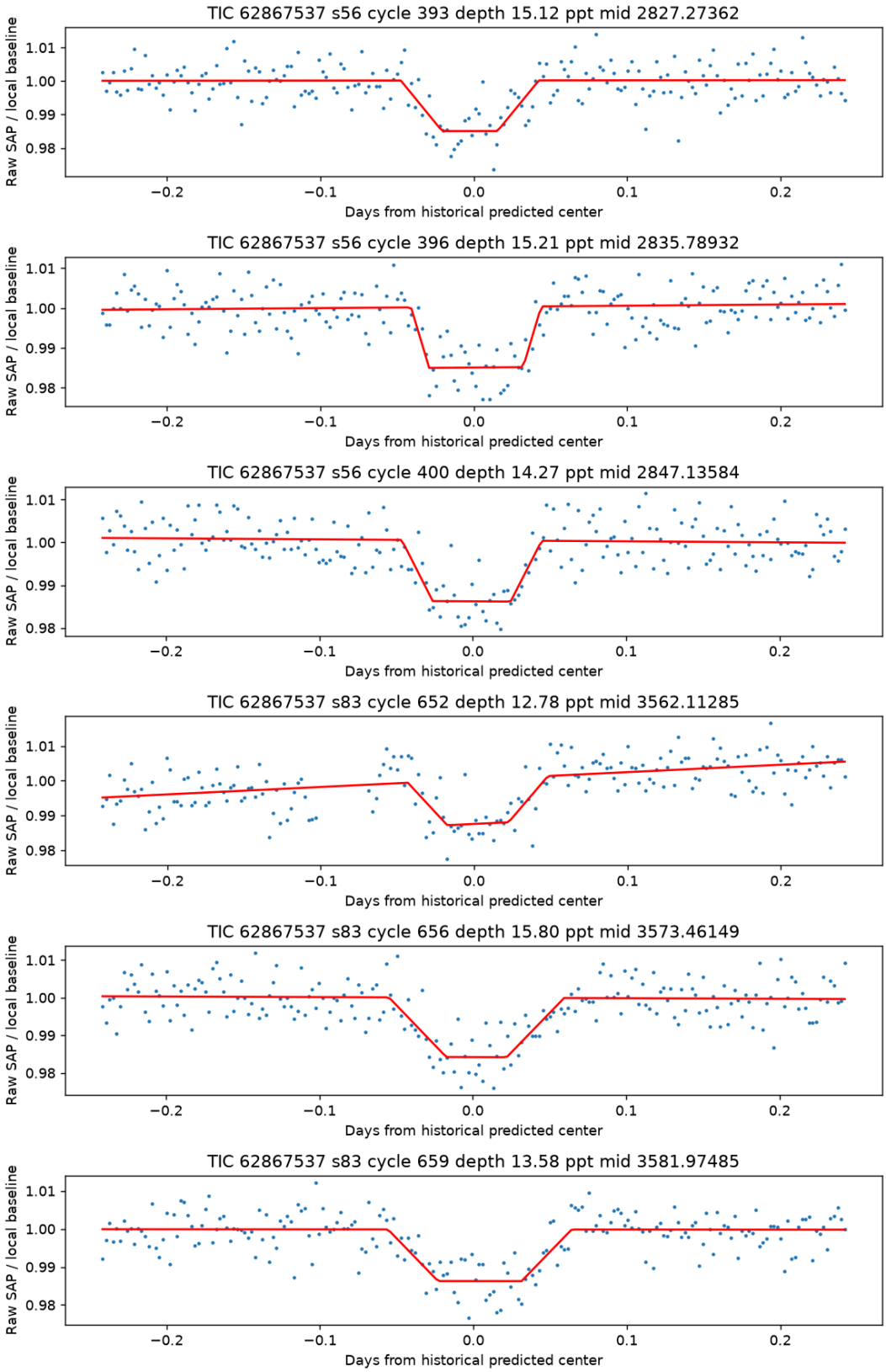
R558. TIC 62867537

physical_verdict: PROBABLE

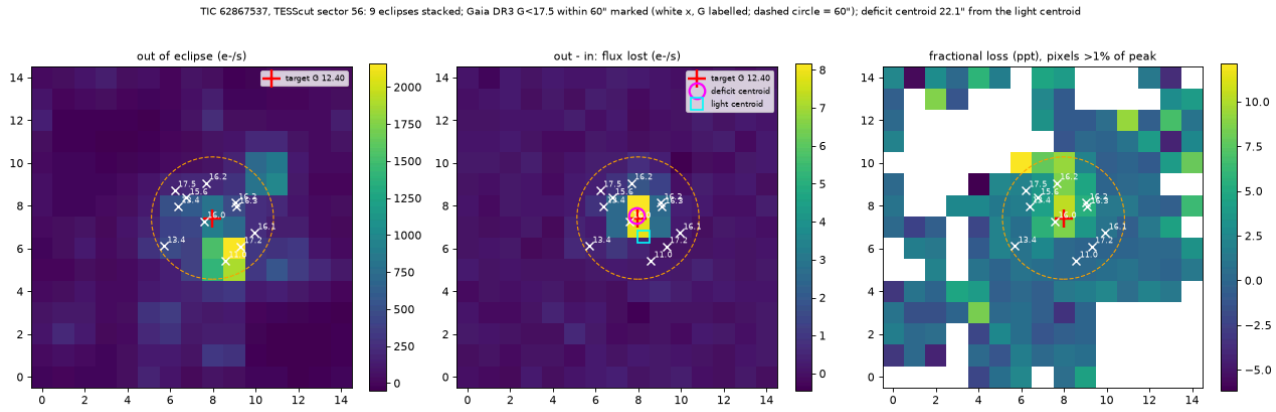
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or calibrated two-source PRF localization of8.8arcsec pair.



R558; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R558; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 62867537: PROBABLE

Full bounded physical review: COMPLETE. Catalogue novelty is separate from this verdict.

Raw QLP SAP_FLUX contains 16 complete free-center fitted eclipses in representative sectors[56, 83]. Predicted historical centers guided extraction, not timing points. Fits use local linear baselines and trapezoid shapes; quality-zero cadences only. Flagged, partial and uncovered windows are not clean nulls. raw-events.png, raw-sector-fold.png and raw-windows.csv show the actual observations.

Measured recurrence P=2.8372050009 +/- 0.0000035438d. Named reference epoch BTJD2827.2764461: selected eclipse kind at phase zero, not an assumed deeper primary. The legacy primary_epoch_btjd column carries this epoch. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when positive-degree-of-freedom reduced chi square exceeds one. With only two centers, no reduced chi square or measured residual scatter exists; covariance propagates input timing allowances. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Consecutive same-kind measured pairs: 12. Same-sector parity and actual coverage are in alias-coverage.json. No persistent deep alternating-depth pattern in the same-sector parity arrays. Covered half-phase windows lack an eclipse comparable to the selected kind. A perfectly identical two-eclipse2P harmonic is not mathematically excluded by single-kind morphology alone; Gaia orbital association and the lack of a second comparable sequence support the reported recurrence. This is a bounded alias check, not a blind exhaustive search. Two separated sectors are distinct observing occasions on the same instrument.

Fresh raw TESScut pixel identity agrees with Gaia DR3 2164885691563277824 to 0.00000arcsec by saved WCS; all source bytes are hashed. In sector56, 9 separately recomputed event deficits average0.029px from the target. Event-scatter vector is[0.0951169563249573, 0.10623889582676811]; this is descriptive sampling scatter, not calibrated positional confidence or pixel-response systematics. Lead visually inspected the historical full-field and freshly recomputed core images. Every capable neighbor in the saved60arcsec Gaia list: Gaia 2164885687254026880: separation8.79arcsec, G16.033, intrinsic required fraction0.407, deficit distance0.402px; Gaia 2164885691563276928: separation25.37arcsec, G16.305, intrinsic required fraction0.520, deficit distance1.252px; Gaia 2164885691563276160: separation26.40arcsec, G16.235, intrinsic required fraction0.488, deficit distance1.308px; Gaia 2164879816048014848: separation31.98arcsec, G15.562, intrinsic required

fraction0.269, deficit distance1.546px; Gaia 2164885897721704576: separation34.20arcsec, G16.191, intrinsic required fraction0.469, deficit distance1.696px; Gaia 2164879816048016512: separation35.38arcsec, G13.400, intrinsic required fraction0.049, deficit distance1.693px; Gaia 2164885622832174976: separation42.60arcsec, G11.017, intrinsic required fraction0.018, deficit distance2.065px; Gaia 2164885657192541824: separation43.26arcsec, G16.146, intrinsic required fraction0.451, deficit distance2.085px; Gaia 2164879816048019328: separation54.05arcsec, G13.404, intrinsic required fraction0.049, deficit distance2.594px Intrinsic source loss requirements and diluted fractional pixel losses are different quantities and are never compared as a veto. The G-band flux screen is approximate and does not exclude unknown unresolved companions.

Alternative extraction: background-subtracted raw TESScut1x1/3x3/5x5 median event depths are8.66/3.47/2.17ppt; selected QLP raw median fit depth14.00ppt. Details, windows, individual centroids, count and depths are in pixel-evidence.json. This separately extracted reduction shares the same TESS photons as QLP and is not an independent instrument or independent observing occasion. Aperture dilution, cadence and event-core averaging explain differing apparent depths; no intrinsic binary depth is inferred.

Gaia orbit and physical plausibility: source-specific NSS P=2.8372544030d, e=0.03049, omega=250.7928deg, periastron BTJD388.57332. Closest edge-on conjunction offset at the measured epoch=-0.07767d (-0.02738phase); descriptive linear full-SB1-covariance timing error=0.20597d. The covariance uses source-specific P/e/omega/Tp correlations, not only diagonal errors. Mass function=0.0027229solar; assumed primary mass=1.922solar gives minimum companion mass=0.233solar. Approximate central duration=3.79h versus observed2.32h. TIC radius/main-sequence companion assumptions and edge-on conjunction are approximations, not a fitted stellar model or arbitrary sigma admission rule. Orbit/geometry are consistent at this descriptive precision and do not establish the identity of an unresolved eclipse source.

Deciding interpretation: Real raw eclipses repeat in S15/56/83, with16 usable midpoints and12 consecutive pairs. Nine fresh S56 deficit centers average0.029px from target, so target is favored. A capable G16.03 neighbor8.79arcsec away shares the central TESS resolution element (separation0.42px); a calibrated two-source pixel response/localization uncertainty is unavailable. Deficit event scatter~0.10px and severe aperture dilution do not uniquely exclude that source. Attribution remains unresolved after completed check, so probable rather than confirmed.

Next check: Resolved eclipse photometry or calibrated two-source PRF localization of8.8arcsec pair.

Seven scoped catalogue groups were checked centrally for this identity; see ../../shared-catalogues/coverage-matrix.csv and verification receipts. A nearby catalogue source is not an exact target match, and the scoped result does not assert exhaustive publication priority.

Sector	Photometric cycle	Measured midpoint BTJD	Input timing allowance d	Fitted depth ppt
56	393	2827.2736150	0.001838	15.12
56	394	2830.1129900	0.002239	19.70
56	395	2832.9516742	0.001189	14.86
56	396	2835.7893179	0.001204	15.21
56	398	2841.4617981	0.002294	13.41
56	399	2844.2980121	0.002187	12.93
56	400	2847.1358365	0.001650	14.27
56	401	2849.9746955	0.002059	11.88
56	402	2852.8129364	0.002478	12.49
83	652	3562.1128513	0.002827	12.78
83	654	3567.7842873	0.001702	13.73
83	655	3570.6222924	0.001810	15.32

Sector	Photometric cycle	Measured midpoint BTJD	Input timing allowance d	Fitted depth ppt
83	656	3573.4614912	0.002189	15.80
83	657	3576.2991845	0.001643	13.63
83	658	3579.1380268	0.002092	15.50
83	659	3581.9748491	0.001696	13.58

Evidence and provenance: measurements.json, raw-windows.csv, raw-events.png, raw-sector-fold.png, alias-coverage.json, pixel-evidence.json, pixel-recomputed.png, historical/diff_s*.json/png, historical/gaia.json; batch ../gaia-geometry.json, ../alias-audit.json, ../secondary-audit.json, ../measurement-source-receipts.json, ../download-receipts.json, ../source-hashes.json, ../LEAD-AUDIT.md and ../helper-audit.md. SHA256SUMS.json covers reports, scripts, raw FITS and original cutouts. Current availability receipt for246857844 is in shared-catalogues and copied locally with its hash. No canonical/public files were edited by this batch.

Depth correction audit (2026-11-02): realized trapezoid decrement replaces its unconstrained amplitude parameter. Original input windows, fitted centers/widths, timing selections and physical verdict are retained; capable-neighbor membership is unchanged. See ../../depth-audit/README.md.

Stable review key: R558.

Deciding evidence and numeric artifacts: - alias-coverage.json - historical/diff_s56.json - historical/diff_s56.png - historical/gaia.json - measurements.json - pixel-evidence.json - pixel-recomputed.png - raw-events.png - raw-sector-fold.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

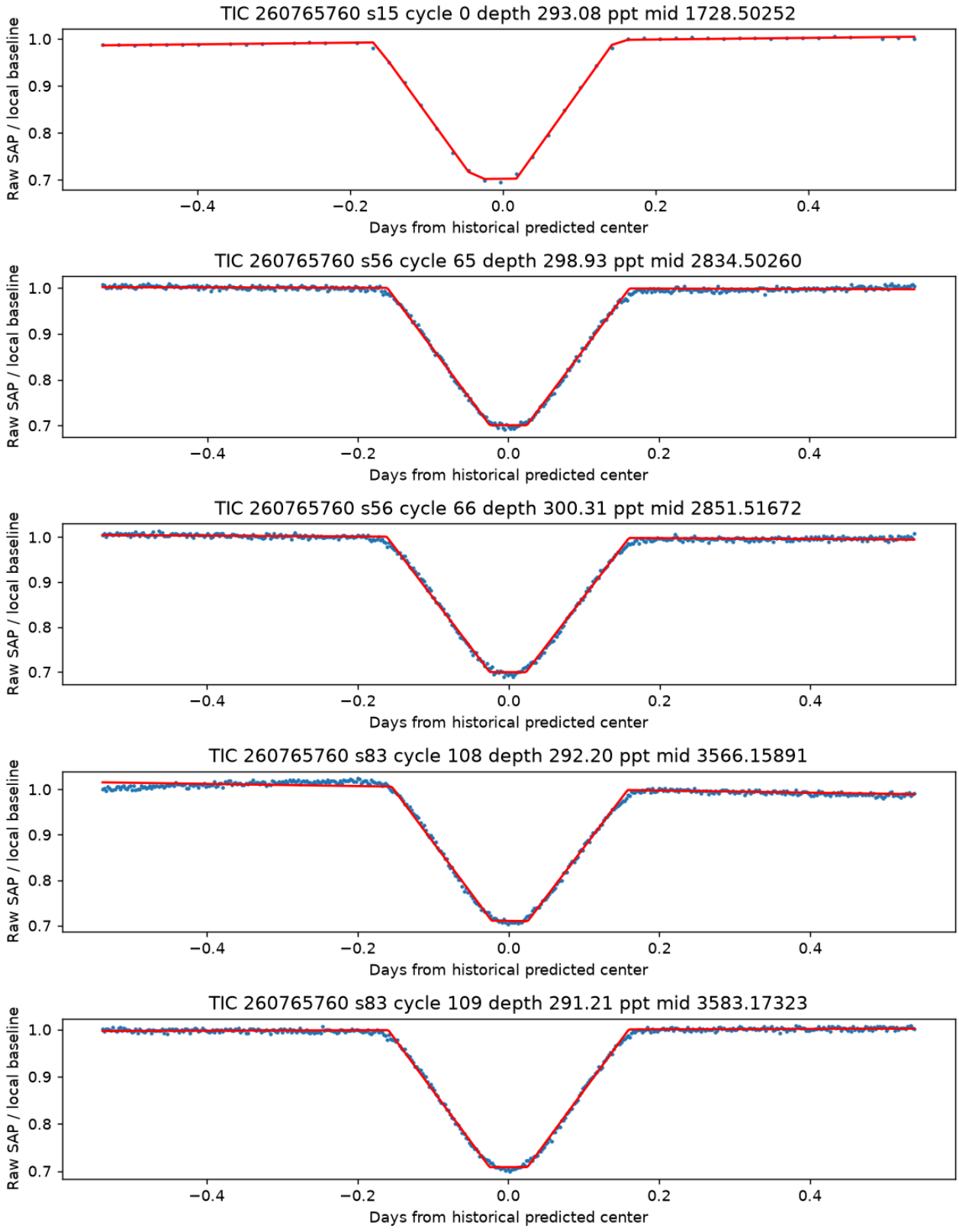
R559. TIC 260765760

physical_verdict: CONFIRMED

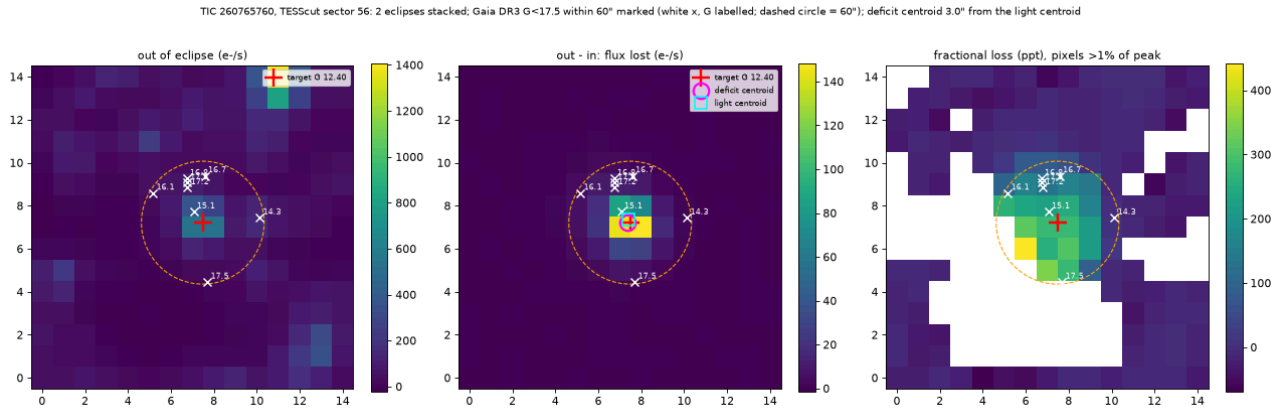
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and precision binary characterization.



R559; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R559; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 260765760: CONFIRMED

Full bounded physical review: COMPLETE. Catalogue novelty is separate from this verdict.

Raw QLP SAP_FLUX contains 5 complete free-center fitted eclipses in representative sectors[15, 56, 83]. Predicted historical centers guided extraction, not timing points. Fits use local linear baselines and trapezoid shapes; quality-zero cadences only. Flagged, partial and uncovered windows are not clean nulls. raw-events.png, raw-sector-fold.png and raw-windows.csv show the actual observations.

Measured recurrence P=17.0152689354 +/- 0.0000262557d. Named reference epoch BTJD1728.5094459: selected eclipse kind at phase zero, not an assumed deeper primary. The legacy primary_epoch_btjd column carries this epoch. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when positive-degree-of-freedom reduced chi square exceeds one. With only two centers, no reduced chi square or measured residual scatter exists; covariance propagates input timing allowances. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Consecutive same-kind measured pairs: 2. Same-sector parity and actual coverage are in alias-coverage.json. Other-kind free fits: 3 midpoints; measured phase range -0.46780 to -0.46769. See secondary-timing.json. The exact-half-phase baseline estimates are not interpreted as clean depths where they overlap the other eclipse. This is a bounded alias check, not a blind exhaustive search. Two separated sectors are distinct observing occasions on the same instrument.

Fresh raw TESScut pixel identity agrees with Gaia DR3 1952546902186887808 to 0.00000arcsec by saved WCS; all source bytes are hashed. In sector56, 2 separately recomputed event deficits average0.122px from the target. Event-scatter vector is[0.02533583839128054, 0.0035793617927881165]; this is descriptive sampling scatter, not calibrated positional confidence or pixel-response systematics. Lead visually inspected the historical full-field and freshly recomputed core images. Every capable neighbor in the saved60arcsec Gaia list: None within60arcsec under approximate G-band flux budget. Intrinsic source loss requirements and diluted fractional pixel losses are different quantities and are never compared as a veto. The G-band flux screen is approximate and does not exclude unknown unresolved companions.

Alternative extraction: background-subtracted raw TESScut1x1/3x3/5x5 median event depths are257.75/250.05/237.27ppt; selected QLP raw median fit depth293.08ppt. Details, windows, individual centroids, count and depths are in pixel-evidence.json. This separately extracted reduction

shares the same TESS photons as QLP and is not an independent instrument or independent observing occasion. Aperture dilution, cadence and event-core averaging explain differing apparent depths; no intrinsic binary depth is inferred.

Gaia orbit and physical plausibility: source-specific NSS P=17.0211399438d, e=0.23654, omega=43.7504deg, periastron BTJD382.53148. Closest edge-on conjunction offset at the measured epoch=-0.05966d (-0.00351phase); descriptive linear full-SB1-covariance timing error=0.45671d. The covariance uses source-specific P/e/omega/Tp correlations, not only diagonal errors. Mass function=0.19571solar; assumed primary mass=1.380solar gives minimum companion mass=1.049solar. Approximate central duration=10.48h versus observed7.65h. TIC radius/main-sequence companion assumptions and edge-on conjunction are approximations, not a fitted stellar model or arbitrary sigma admission rule. Orbit/geometry are consistent at this descriptive precision and do not establish the identity of an unresolved eclipse source.

Deciding interpretation: Deep selected eclipse and shallower second kind atphase-0.4677 recur with unequal depths. Generic exact-half-phase baseline windows partly overlap the secondary; their negative depth estimates are not clean eclipse depths. Fresh pixel aperture deficits remain deep and target-centered; no capable Gaia neighbor under the flux budget.

Next check: Optional resolved photometry and precision binary characterization.

Seven scoped catalogue groups were checked centrally for this identity; see ../../shared-catalogues/coverage-matrix.csv and verification receipts. A nearby catalogue source is not an exact target match, and the scoped result does not assert exhaustive publication priority.

Sector	Photometric cycle	Measured midpoint BTJD	Input timing allowance d	Fitted depth ppt
15	0	1728.5025163	0.010417	293.08
56	65	2834.5025963	0.001157	298.93
56	66	2851.5167175	0.001157	300.31
83	108	3566.1589146	0.001157	292.20
83	109	3583.1732303	0.001157	291.21

Evidence and provenance: measurements.json, raw-windows.csv, raw-events.png, raw-sector-fold.png, alias-coverage.json, pixel-evidence.json, pixel-recomputed.png, historical/diff_s*.json/png, historical/gaia.json; batch ../gaia-geometry.json, ../alias-audit.json, ../secondary-audit.json, ../measurement-source-receipts.json, ../download-receipts.json, ../source-hashes.json, ../LEAD-AUDIT.md and ../helper-audit.md. SHA256SUMS.json covers reports, scripts, raw FITS and original cutouts. Current availability receipt for246857844 is in shared-catalogues and copied locally with its hash. No canonical/public files were edited by this batch.

Stable review key: R559.

Deciding evidence and numeric artifacts: - alias-coverage.json - historical/diff_s56.json - historical/diff_s56.png - historical/gaia.json - measurements.json - pixel-evidence.json - pixel-recomputed.png - raw-events.png - raw-sector-fold.png - raw-windows.csv - secondary-timing.json

Batch audits and scripts. Depth correction provenance.

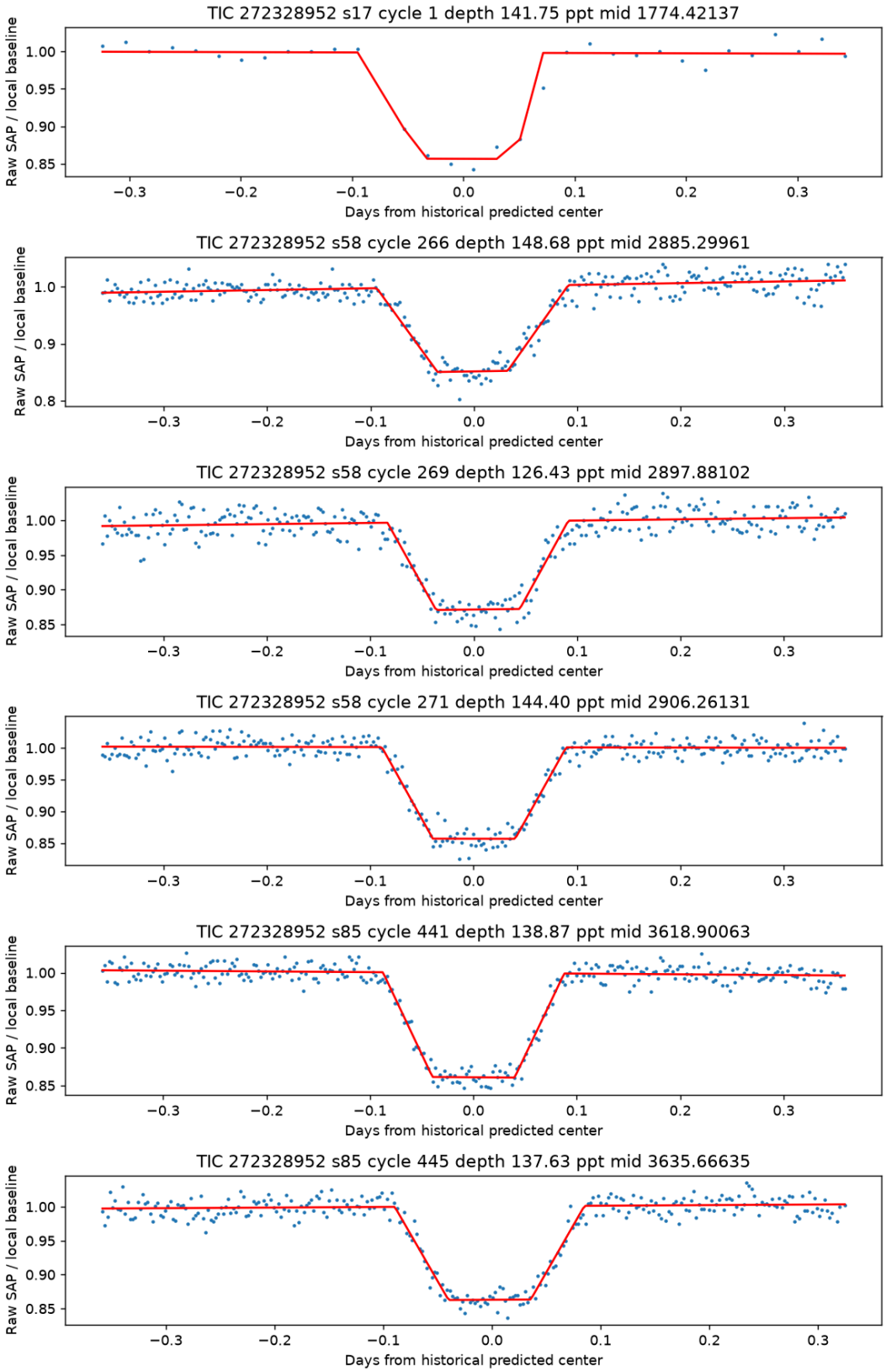
R560. TIC 272328952

physical_verdict: PROBABLE

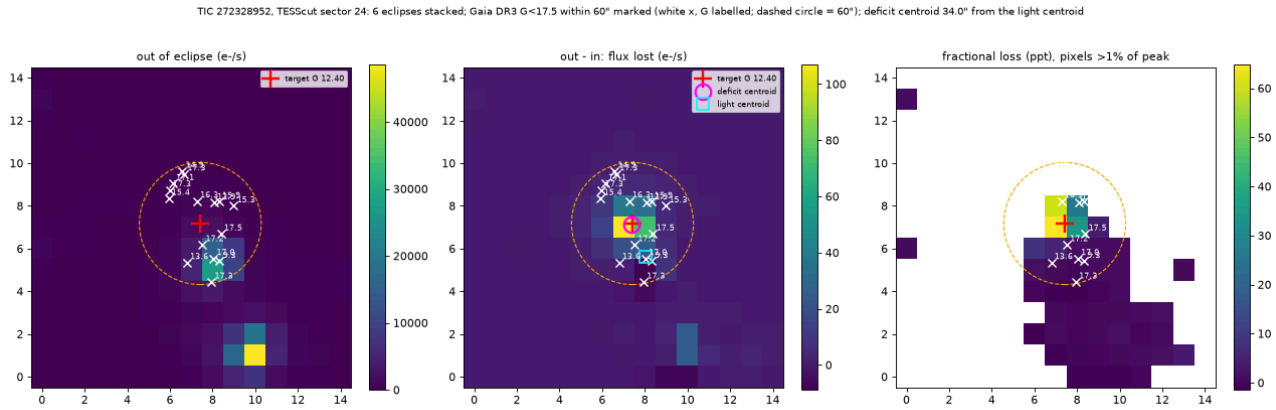
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved photometry through an eclipse to establish ownership; additional timing for remaining period/shape limits.



R560; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R560; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 272328952: PROBABLE

Full bounded physical review: COMPLETE. Physical verdict is separate from catalogue novelty. The epoch is the selected recurring eclipse; no deeper-primary rank is assumed.

Deciding evidence and limitation: Deep repeated raw events. Compact difference core lies on target. Capable G12.87 neighbor at24.5arcsec lies1.17pixels away and overlaps the broad core; bright G9.29 neighbor at41.8arcsec dominates baseline but not deficit. Attribution favors target but does not uniquely exclude the closest capable source. Broad off-eclipse variability changes across sectors; S24 custom1x1/3x3/5x5 depths64.9/12.5/4.65ppt differ greatly from QLP144ppt, a major aperture/sector dilution limitation rather than depth agreement.

Raw QLP SAP_FLUX representative windows contain 12 accepted free-center measurements in sectors [17, 58, 85]. Predicted windows are search guides, not timings. Quality-zero points enter fits; flagged or uncovered windows are not clean nulls. Rejected fits remain visible in numerical artifacts. Raw event plots and full recurrence phase plots were inspected; fitted dips alone do not establish eclipse morphology (TIC290749976 remains plausible).

Measured recurrence P=4.1919898515 +/- 0.0000054571 days; selected-eclipse reference BTJD=1774.4240109; fitted median depth=144.340ppt. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Singular/unidentifiable timing covariance is excluded. Within-sector cycle counts/parity are in ../alias-audit.json. Full-phase inspection tests secondary structure at its observed phase, without imposing phase0.5; unequal secondaries do not halve same-kind periods. Half/double ambiguity is described explicitly above where material. No exhaustive alias search is claimed. Sparse349452680 additionally has current availability and bounded sampled-null receipts.

Fresh TESScut sector24 stacks 6 actual covered events. Deficit-target offset=0.094pixel, target pixel=[7.421443498225445, 7.171143902239606], deficit centroid=[7.356824175782659, 7.102789688791751]. The image-model residual=0.988 is not a calibrated location error. Actual target pixel deficit=107.07 electron/s and baseline=1649.9 electron/s. All flux-capable Gaia neighbors within60arcsec are below; their actual signed deficit and baseline are evaluated in common electron/s

units at each source's rounded pixel. Overlapping PSFs prevent deblended ownership. Intrinsic required eclipse fraction uses pairwise Gaia G brightness; it is not directly comparable with diluted fractional pixel loss. Color, aperture response and unlisted faint/unresolved sources limit this approximate screen.

Gaia source	separation arcsec	G	intrinsic required loss proxy	actual signed pixel deficit e/s	pixel baseline e/s
2016009377616964736	24.49	12.870	0.3660	33.593	1268.9
2016009102739075712	40.52	13.608	0.5818	2.3201	5070.3
2016009102739426816	41.84	9.294	0.1526	-8.9918	26869

Alternative background-subtracted target-centered TESScut1x1/3x3/5x5 depths=64.896/12.534/4.653ppt. These custom extractions test reduction robustness. All QLP and cutouts use TESS photons; same-sector extractions are not independent observations. Distinct sectors are distinct observing occasions on one instrument. Depth changes can be dilution/aperture/background effects; these numbers are not intrinsic eclipse depths. Pixel-arrays.npz preserves actual stacked images.

Gaia DR3 source=2016009308897494272, P=4.191921516d, e=0.04489. Closest edge-on conjunction offset=0.09819d (0.02342phase); source-specific full covariance linear timing allowance=0.11238d. This descriptive linear allowance is not a calibrated significance veto. Geometry code retains both eccentric conjunctions. Mass function=0.042593 solar masses; TIC/adopted primary mass=1.396, minimum secondary mass=0.543, TIC radius=1.410 solar radii. Approximate central duration=4.862h versus measured median=4.186h. TIC radius and main-sequence minimum-mass companion assumptions make this a plausibility screen, not a fitted binary model. Orbit agreement alone does not establish target ownership.

Sector	Search cycle	midpoint BTJD	formal center error d	timing input error d	depth ppt	status
17	1	1774.4213748	0.00098113	0.010417	141.748	used
17	4	1786.9969705	0.0025041	0.010417	144.950	used
58	266	2885.2996104	0.0010226	0.0011574	148.681	used
58	268	2893.6848678	0.00069836	0.0011574	144.338	used
58	269	2897.8810207	0.0011191	0.0011574	126.427	used
58	270	2902.0678373	0.0006255	0.0011574	144.342	used
58	271	2906.2613082	0.00071153	0.0011574	144.403	used
85	440	3614.7076126	0.00056813	0.0011574	146.753	used
85	441	3618.9006331	0.00058542	0.0011574	138.867	used
85	443	3627.2825888	0.00054816	0.0011574	147.370	used
85	444	3631.4763546	0.00048087	0.0011574	142.839	used
85	445	3635.6663487	0.00074852	0.0011574	137.632	used

Next check: Resolved photometry through an eclipse to establish ownership; additional timing for remaining period/shape limits.

Seven-group catalogue receipts are in ../../shared-catalogues/coverage-matrix.csv; known recovery and physical outcome are separate. No exhaustive priority claim. Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png and pixel-arrays.npz; batch identity/hash receipts, geometry, phase and alias artifacts, helper audit. Source hashes and SHA256SUMS.json cover retained sources and outputs. Methods are preserved in a read snapshot rather than mutable manuscript hashes.

Stable review key: R560.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/pixel-arrays.npz - fresh-pixels/run.log - historical/diff_s24.json - historical/diff_s24.png - historical/gaia.json - measurements.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

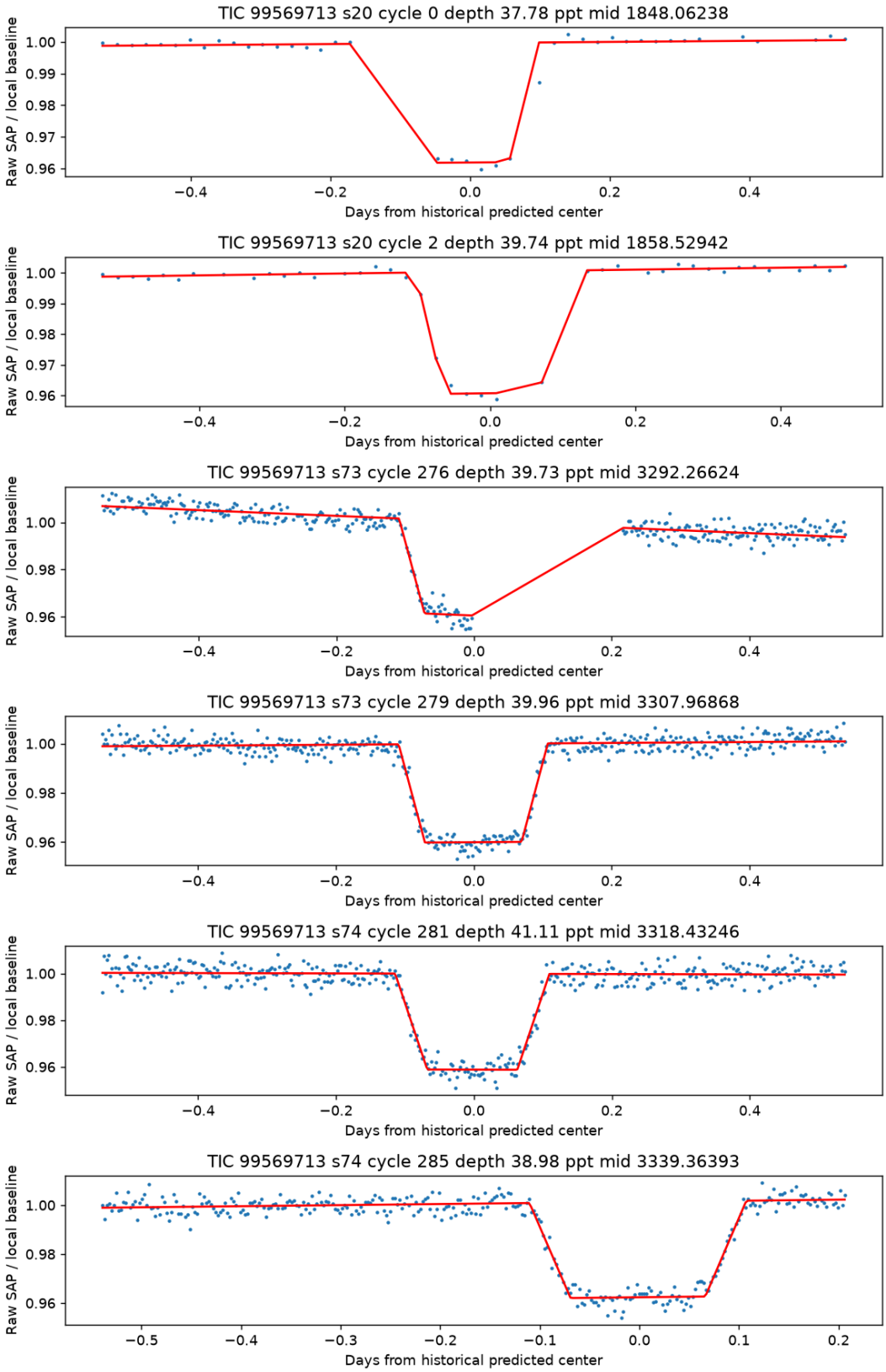
R561. TIC 99569713

physical_verdict: PROBABLE

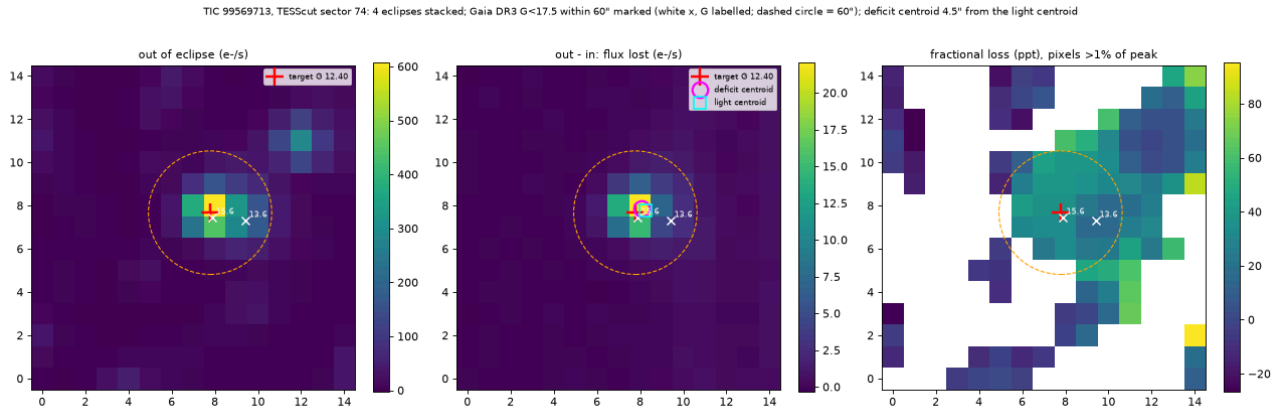
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved photometry through an eclipse to establish ownership; additional timing for remaining period/shape limits.



R561; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R561; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 99569713: PROBABLE

Full bounded physical review: COMPLETE. Physical verdict is separate from catalogue novelty. The epoch is the selected recurring eclipse; no deeper-primary rank is assumed.

Deciding evidence and limitation: Ten repeated raw eclipses. Capable G15.57 source at5arcsec is unresolved within the target deficit; ownership remains unresolved. A farther source33arcsec away overlaps the wings.

Raw QLP SAP_FLUX representative windows contain 10 accepted free-center measurements in sectors [20, 73, 74]. Predicted windows are search guides, not timings. Quality-zero points enter fits; flagged or uncovered windows are not clean nulls. Rejected fits remain visible in numerical artifacts. Raw event plots and full recurrence phase plots were inspected; fitted dips alone do not establish eclipse morphology (TIC290749976 remains plausible).

Measured recurrence P=5.2326290195 +/- 0.0000213523 days; selected-eclipse reference BTJD=1848.0644657; fitted median depth=39.098ppt. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Singular/unidentifiable timing covariance is excluded. Within-sector cycle counts/parity are in ../alias-audit.json. Full-phase inspection tests secondary structure at its observed phase, without imposing phase0.5; unequal secondaries do not halve same-kind periods. Half/double ambiguity is described explicitly above where material. No exhaustive alias search is claimed. Sparse349452680 additionally has current availability and bounded sampled-null receipts.

Fresh TESScut sector74 stacks 4 actual covered events. Deficit-target offset=0.344pixel, target pixel=[7.781927206513501, 7.671035884768308], deficit centroid=[8.07134657900216, 7.856092487187115]. The image-model residual=0.386 is not a calibrated location error. Actual target pixel deficit=22.078 electron/s and baseline=607.88 electron/s. All flux-capable Gaia neighbors within60arcsec are below; their actual signed deficit and baseline are evaluated in common electron/s units at each source's rounded pixel. Overlapping PSFs prevent deblended ownership. Intrinsic required eclipse fraction uses pairwise Gaia G brightness; it is not directly comparable with diluted fractional pixel loss. Color, aperture response and unlisted faint/unresolved sources limit this approximate screen.

Gaia source	separation arcsec	G	intrinsic required loss proxy	actual signed pixel deficit e/s	pixel baseline e/s
1102067481211251328	5.00	15.573	0.7623	14.868	442.2
1102064629352866304	33.29	13.634	0.1604	4.5959	286.98

Alternative background-subtracted target-centered TESScut1x1/3x3/5x5 depths=36.320/31.997/30.300ppt. These custom extractions test reduction robustness. All QLP and cutouts use TESS photons; same-sector extractions are not independent observations. Distinct sectors are distinct observing occasions on one instrument. Depth changes can be dilution/aperture/background effects; these numbers are not intrinsic eclipse depths. Pixel-arrays.npz preserves actual stacked images.

Gaia DR3 source=1102067476915290624, P=5.232723124d, e=0.12363. Closest edge-on conjunction offset=0.04468d (0.00854phase); source-specific full covariance linear timing allowance=0.23268d. This descriptive linear allowance is not a calibrated significance veto. Geometry code retains both eccentric conjunctions. Mass function=0.0098116 solar masses; TIC/adopted primary mass=1.070, minimum secondary mass=0.259, TIC radius=1.681 solar radii. Approximate central duration=5.507h versus measured median=5.210h. TIC radius and main-sequence minimum-mass companion assumptions make this a plausibility screen, not a fitted binary model. Orbit agreement alone does not establish target ownership.

Sector	Search cycle	midpoint BTJD	formal center error d	timing input error d	depth ppt	status
20	0	1848.0623800	0.00052327	0.010417	37.784	used
20	1	1853.2988967	0.0013999	0.010417	38.625	used
20	2	1858.5294235	0.0011585	0.010417	39.737	used
20	3	1863.7547087	1.442e+06	1.442e+06	38.525	rejected: Singular or unidentifiable center covariance: formal timing uncertainty exceeds search duration; excluded from ephemeris.
73	276	3292.2662357	3.6016e+05	3.6016e+05	39.725	rejected: Singular or unidentifiable center covariance: formal timing uncertainty exceeds search duration; excluded from ephemeris.
73	277	3297.5024285	0.00075714	0.0011574	39.579	used
73	279	3307.9686793	0.00043745	0.0011574	39.956	used
74	280	3313.2024396	0.0018421	0.0018421	39.027	used
74	281	3318.4324634	0.0006715	0.0011574	41.106	used
74	282	3323.6658271	0.00056952	0.0011574	38.985	used
74	284	3334.1305243	0.00077908	0.0011574	39.168	used
74	285	3339.3639341	0.00050783	0.0011574	38.978	used

Next check: Resolved photometry through an eclipse to establish ownership; additional timing for remaining period/shape limits.

Seven-group catalogue receipts are in ../../shared-catalogues/coverage-matrix.csv; known recovery and physical outcome are separate. No exhaustive priority claim. Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png and pixel-arrays.npz; batch identity/hash receipts, geometry, phase and alias artifacts, helper audit. Source hashes and SHA256SUMS.json cover retained sources and outputs. Methods are preserved in a read snapshot rather than mutable manuscript hashes.

Stable review key: R561.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/pixel-arrays.npz - fresh-pixels/run.log - historical/diff_s74.json - historical/diff_s74.png - historical/gaia.json - measurements.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

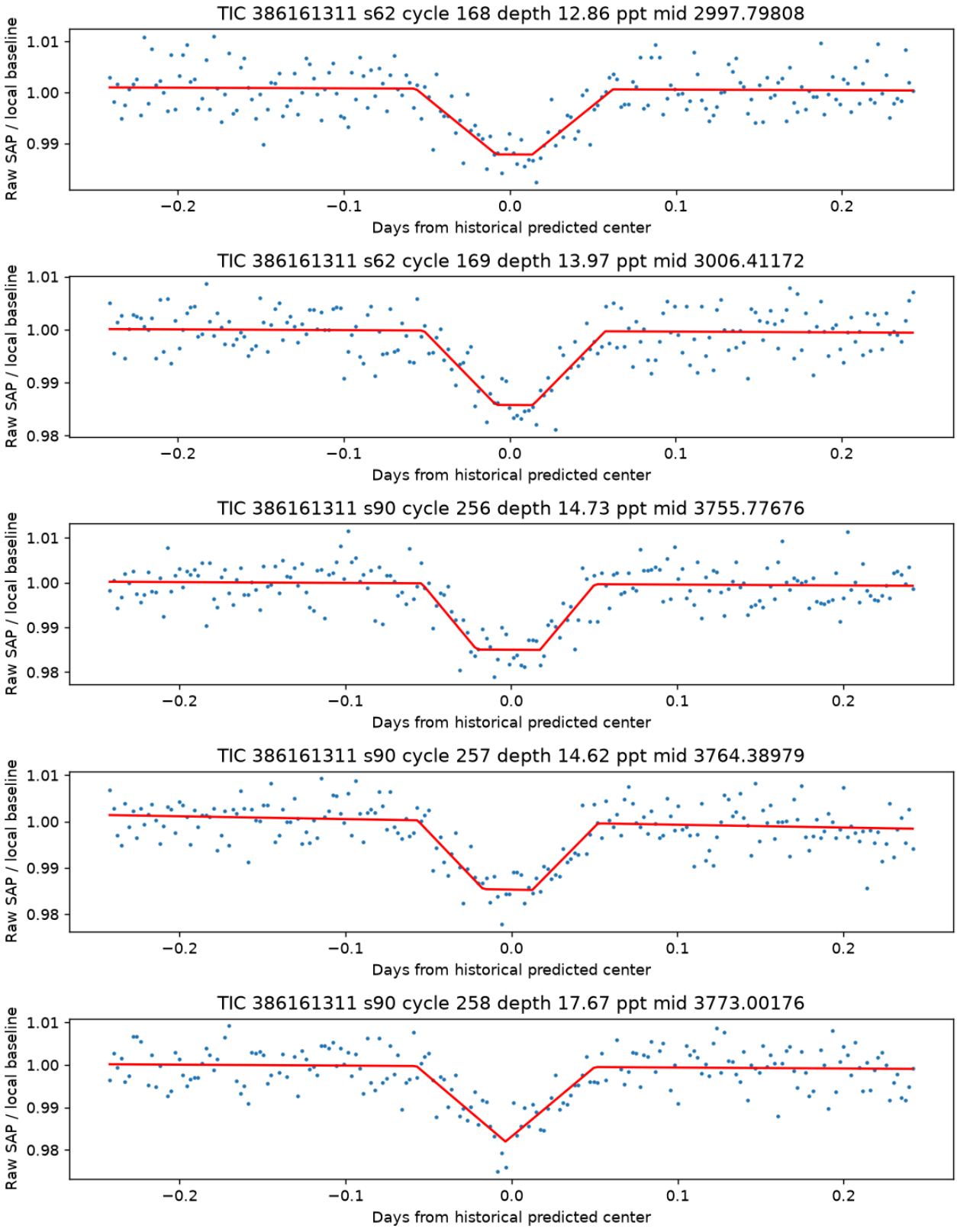
R562. TIC 386161311

physical_verdict: PROBABLE

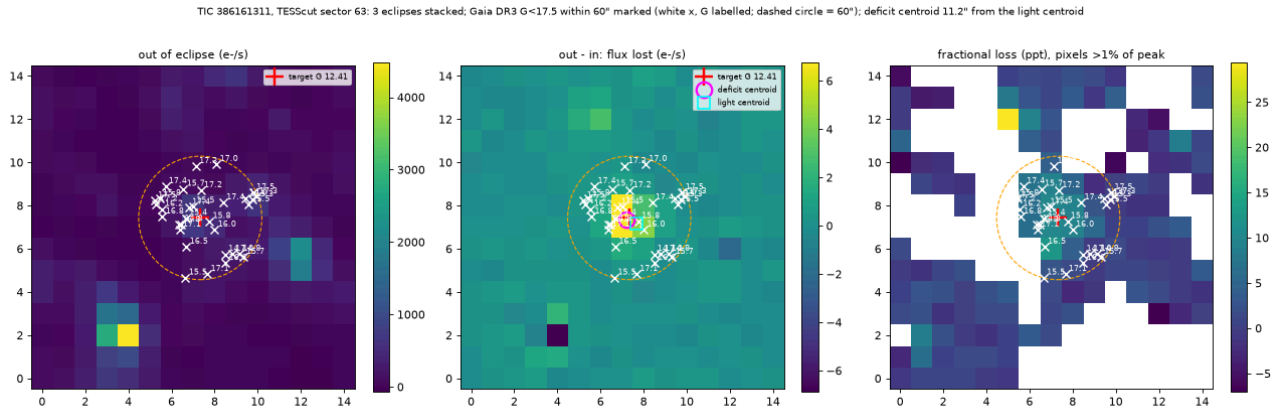
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved photometry through an eclipse to establish ownership; additional timing for remaining period/shape limits.



R562; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R562; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 386161311: PROBABLE

Full bounded physical review: COMPLETE. Physical verdict is separate from catalogue novelty. The epoch is the selected recurring eclipse; no deeper-primary rank is assumed.

Deciding evidence and limitation: Repeated eclipse-shaped raw dips with depth variation. Nineteen capable sources include sources9.4/13.4/14arcsec away, unresolved around the weak target-centered deficit. Target ownership remains unresolved; conjunction offset0.0505phase is comparable to propagated timing allowance, rather than a clean veto.

Raw QLP SAP_FLUX representative windows contain 5 accepted free-center measurements in sectors [62, 90]. Predicted windows are search guides, not timings. Quality-zero points enter fits; flagged or uncovered windows are not clean nulls. Rejected fits remain visible in numerical artifacts. Raw event plots and full recurrence phase plots were inspected; fitted dips alone do not establish eclipse morphology (TIC290749976 remains plausible).

Measured recurrence P=8.6133849971 +/- 0.0000236465 days; selected-eclipse reference BTJD=2997.7982325; fitted median depth=14.620ppt. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Singular/unidentifiable timing covariance is excluded. Within-sector cycle counts/parity are in ../alias-audit.json. Full-phase inspection tests secondary structure at its observed phase, without imposing phase0.5; unequal secondaries do not halve same-kind periods. Half/double ambiguity is described explicitly above where material. No exhaustive alias search is claimed. Sparse349452680 additionally has current availability and bounded sampled-null receipts.

Fresh TESScut sector63 stacks 3 actual covered events. Deficit-target offset=0.151pixel, target pixel= [7.331123068439474, 7.426947168527311], deficit centroid=[7.192118466223236, 7.3673959097231405]. The image-model residual=0.890 is not a calibrated location error. Actual target pixel deficit=6.7684 electron/s and baseline=678.42 electron/s. All flux-capable Gaia neighbors within60arcsec are below; their actual signed deficit and baseline are evaluated in common electron/s units at each source's rounded pixel. Overlapping PSFs prevent deblended ownership. Intrinsic required eclipse fraction uses pairwise Gaia G brightness; it is not directly comparable with diluted fractional pixel loss. Color, aperture response and unlisted faint/unresolved sources limit this approximate screen.

Gaia source	separation arcsec	G	intrinsic required loss proxy	actual signed pixel deficit e/s	pixel baseline e/s
5310692637375695232	9.42	15.832	0.3565	4.3713	454.01
5310692637375692288	13.42	15.508	0.2684	6.6649	663.52
5310692637375694464	13.98	15.400	0.2443	6.7684	678.42
5310692671735434752	18.03	16.049	0.4320	4.3713	454.01
5310692671735433472	19.50	14.754	0.1412	1.1349	174.25
5310692671735433728	21.88	16.827	0.8693	1.1349	174.25
5310692671735436800	31.24	16.504	0.6497	1.5526	125.03
5310692637375689472	32.29	15.742	0.3292	0.9311	102.35
5310692568656217344	36.80	16.800	0.8488	1.1349	174.25
5310692564333733120	38.40	16.236	0.5104	1.2721	196.79
5310692671735438336	42.56	14.127	0.0857	1.0559	252.9
5310692568656213504	43.73	15.798	0.3461	0.54704	41.072
5310693393289936000	48.46	13.982	0.0768	0.93649	367.35
5310693393289937536	48.69	15.538	0.2754	-0.12804	633.47
5310692671735440640	50.04	16.624	0.7238	0.27965	166.8
5310693427649682816	51.28	14.793	0.1460	-0.061834	345.61
5310693393289936768	53.42	13.277	0.0471	-0.12804	633.47
5310693427649684096	56.02	15.733	0.3267	-0.061834	345.61
5310692774814657408	59.98	15.479	0.2615	0.31248	119.71

Alternative background-subtracted target-centered TESScut1x1/3x3/5x5 depths=9.977/9.092/6.611ppt. These custom extractions test reduction robustness. All QLP and cutouts use TESS photons; same-sector extractions are not independent observations. Distinct sectors are distinct observing occasions on one instrument. Depth changes can be dilution/aperture/background effects; these numbers are not intrinsic eclipse depths. Pixel-arrays.npz preserves actual stacked images.

Gaia DR3 source=5310692633056834432, P=8.611193849d, e=0.08219. Closest edge-on conjunction offset=0.43524d (0.05054phase); source-specific full covariance linear timing allowance=0.34672d. This descriptive linear allowance is not a calibrated significance veto. Geometry code retains both eccentric conjunctions. Mass function=0.1171 solar masses; TIC/adopted primary mass=1.380, minimum secondary mass=0.830, TIC radius=1.599 solar radii. Approximate central duration=6.577h versus measured median=2.610h. TIC radius and main-sequence minimum-mass companion assumptions make this a plausibility screen, not a fitted binary model. Orbit agreement alone does not establish target ownership.

Sector	Search cycle	midpoint BTJD	formal center error d	timing input error d	depth ppt	status
62	168	2997.7980776	0.0026323	0.0026323	12.855	used
62	169	3006.4117155	0.0023901	0.0023901	13.974	used
90	256	3755.7767649	0.0017833	0.0017833	14.728	used
90	257	3764.3897947	0.0019994	0.0019994	14.620	used
90	258	3773.0017579	0.002024	0.002024	17.673	used

Next check: Resolved photometry through an eclipse to establish ownership; additional timing for remaining period/shape limits.

Seven-group catalogue receipts are in ../../shared-catalogues/coverage-matrix.csv; known recovery and physical outcome are separate. No exhaustive priority claim. Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png and pixel-arrays.npz; batch identity/hash receipts, geometry, phase and alias artifacts, helper audit. Source hashes and SHA256SUMS.json cover retained sources and outputs. Methods are preserved in a read snapshot rather than mutable manuscript hashes.

Depth correction audit (2026-11-02): realized trapezoid decrement replaces its unconstrained amplitude parameter. Original input windows, fitted centers/widths, timing selections and physical verdict are retained; capable-neighbor membership is unchanged. See ../../depth-audit/README.md.

Stable review key: R562.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/pixel-arrays.npz - fresh-pixels/run.log - historical/diff_s63.json - historical/diff_s63.png - historical/gaia.json - measurements.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

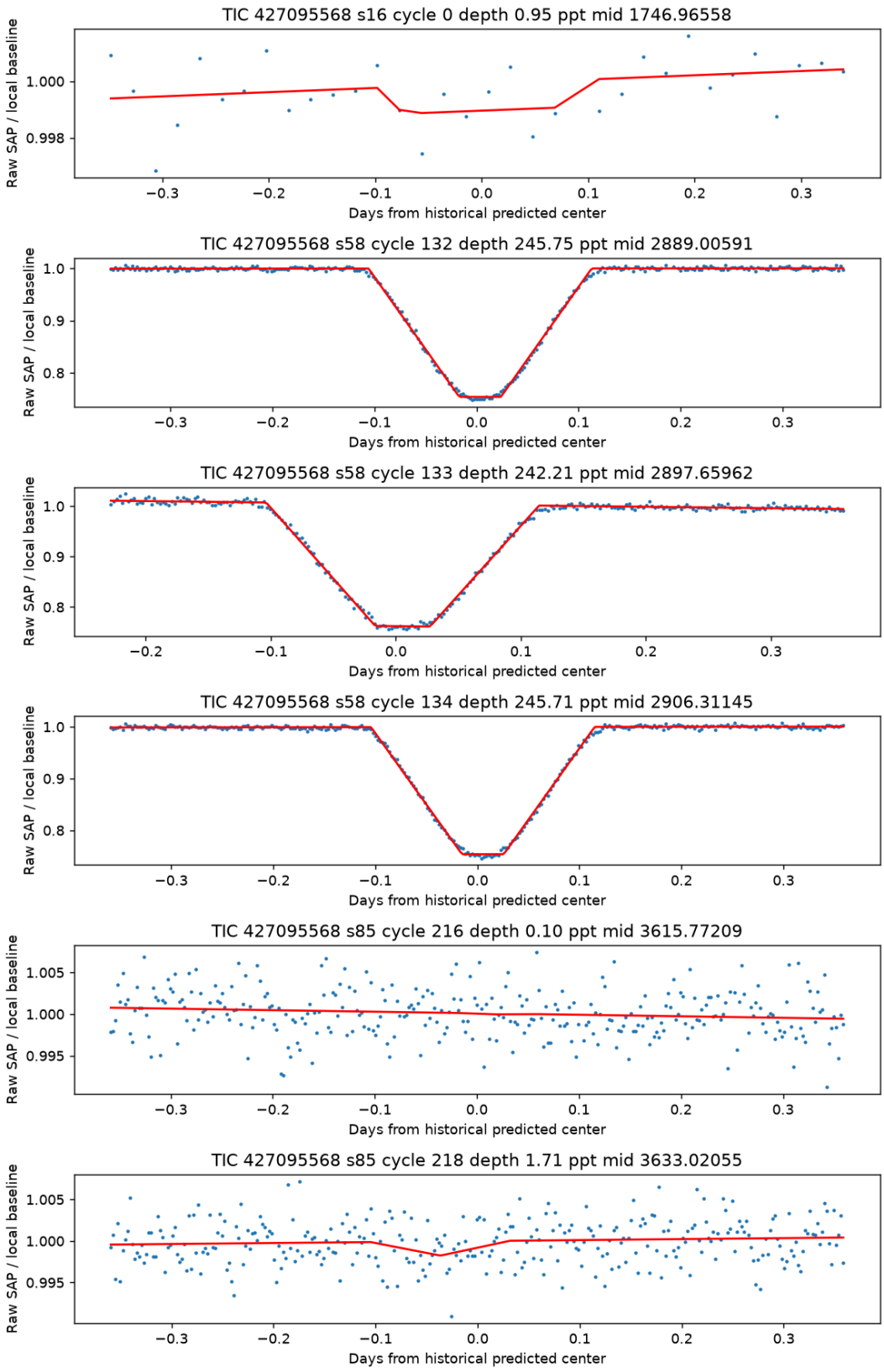
R563. TIC 427095568

physical_verdict: PROBABLE

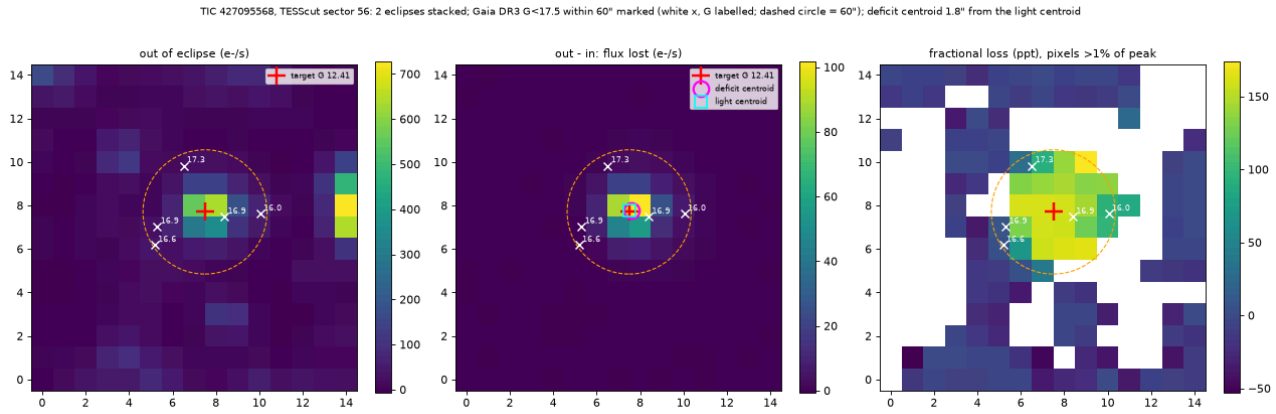
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved photometry through an eclipse to establish ownership; additional timing for remaining period/shape limits.



R563; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R563; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 427095568: PROBABLE

Full bounded physical review: COMPLETE. Physical verdict is separate from catalogue novelty. The epoch is the selected recurring eclipse; no deeper-primary rank is assumed.

Deciding evidence and limitation: Three deep S58 eclipses establish local recurrence; S16/85 small noise fits are explicitly rejected. Full quality-zero light curves lack any comparable deep eclipse anywhere (minima0.994/0.988 of median versus S58 0.746), independently of uncertain extrapolated phase. Including flagged finite cadences gives minima0.886/0.902 in S16/85, still no comparable25percent event; flagged windows are not clean nulls. Fresh S56 target-centered deficit also recovers eclipses. No capable Gaia source in60arcsec. Major cross-sector absence remains unexplained; local S58 timing is not a precise cross-year ephemeris.

Raw QLP SAP_FLUX representative windows contain 3 accepted free-center measurements in sectors [58]. Predicted windows are search guides, not timings. Quality-zero points enter fits; flagged or uncovered windows are not clean nulls. Rejected fits remain visible in numerical artifacts. Raw event plots and full recurrence phase plots were inspected; fitted dips alone do not establish eclipse morphology (TIC290749976 remains plausible).

Measured recurrence P=8.6527733498 +/- 0.0008183923 days; selected-eclipse reference BTJD=2889.0062191; fitted median depth=245.709ppt. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Singular/unidentifiable timing covariance is excluded. Within-sector cycle counts/parity are in ../alias-audit.json. Full-phase inspection tests secondary structure at its observed phase, without imposing phase0.5; unequal secondaries do not halve same-kind periods. Half/double ambiguity is described explicitly above where material. No exhaustive alias search is claimed. Sparse349452680 additionally has current availability and bounded sampled-null receipts.

Fresh TESScut sector56 stacks 2 actual covered events. Deficit-target offset=0.140pixel, target pixel= [7.497710466285785, 7.702597690902223], deficit centroid=[7.616144300055931, 7.776723749736388]. The image-model residual=0.198 is not a calibrated location error. Actual target pixel deficit=88.826 electron/s and baseline=552.83 electron/s. All flux-capable Gaia neighbors within60arcsec are below; their actual signed deficit and baseline are evaluated in common electron/s

units at each source's rounded pixel. Overlapping PSFs prevent deblended ownership. Intrinsic required eclipse fraction uses pairwise Gaia G brightness; it is not directly comparable with diluted fractional pixel loss. Color, aperture response and unlisted faint/unresolved sources limit this approximate screen.

Gaia source	separation arcsec	G	intrinsic required loss proxy	actual signed pixel deficit e/s	pixel baseline e/s
None capable within60arcsec					

Alternative background-subtracted target-centered TESScut1x1/3x3/5x5 depths=160.673/156.147/147.241ppt. These custom extractions test reduction robustness. All QLP and cutouts use TESS photons; same-sector extractions are not independent observations. Distinct sectors are distinct observing occasions on one instrument. Depth changes can be dilution/aperture/background effects; these numbers are not intrinsic eclipse depths. Pixel-arrays.npz preserves actual stacked images.

Gaia DR3 source=2192944949117566976, P=8.651007742d, e=0.04762. Closest edge-on conjunction offset=0.11688d (0.01351phase); source-specific full covariance linear timing allowance=0.18671d. This descriptive linear allowance is not a calibrated significance veto. Geometry code retains both eccentric conjunctions. Mass function=0.11101 solar masses; TIC/adopted primary mass=0.935, minimum secondary mass=0.655, TIC radius=1.819 solar radii. Approximate central duration=7.885h versus measured median=5.248h. TIC radius and main-sequence minimum-mass companion assumptions make this a plausibility screen, not a fitted binary model. Orbit agreement alone does not establish target ownership.

Sector	Search cycle	midpoint BTJD	formal center error d	timing input error d	depth ppt	status
16	0	1746.9655800	0.01132	0.01132	0.952	rejected: Visual raw-window rejection: no deep eclipse in S16/85; noise fit is not a measured event.
58	132	2889.0059057	0.0001121	0.0011574	245.750	used
58	133	2897.6596191	0.00028591	0.0011574	242.209	used
58	134	2906.3114524	0.00011079	0.0011574	245.709	used
85	216	3615.7720850	0.19036	0.19036	0.101	rejected: Visual raw-window rejection: no deep eclipse in S16/85; noise fit is not a measured event.
85	218	3633.0205453	0.013082	0.013082	1.710	rejected: Visual raw-window rejection: no deep eclipse in S16/85; noise fit is not a measured event.

Next check: Resolved photometry through an eclipse to establish ownership; additional timing for remaining period/shape limits.

Seven-group catalogue receipts are in ../../shared-catalogues/coverage-matrix.csv; known recovery and physical outcome are separate. No exhaustive priority claim. Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png and pixel-arrays.npz; batch identity/hash receipts, geometry, phase and alias artifacts, helper audit. Source hashes and SHA256SUMS.json cover retained sources and outputs. Methods are preserved in a read snapshot rather than mutable manuscript hashes.

Depth correction audit (2026-11-02): realized trapezoid decrement replaces its unconstrained amplitude parameter. Original input windows, fitted centers/widths, timing selections and physical verdict are retained; capable-neighbor membership is unchanged. See ../../depth-audit/README.md.

Stable review key: R563.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/pixel-arrays.npz - fresh-pixels/run.log - full-sectors.png - historical/diff_s56.json - historical/diff_s56.png - historical/gaia.json - measurements.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

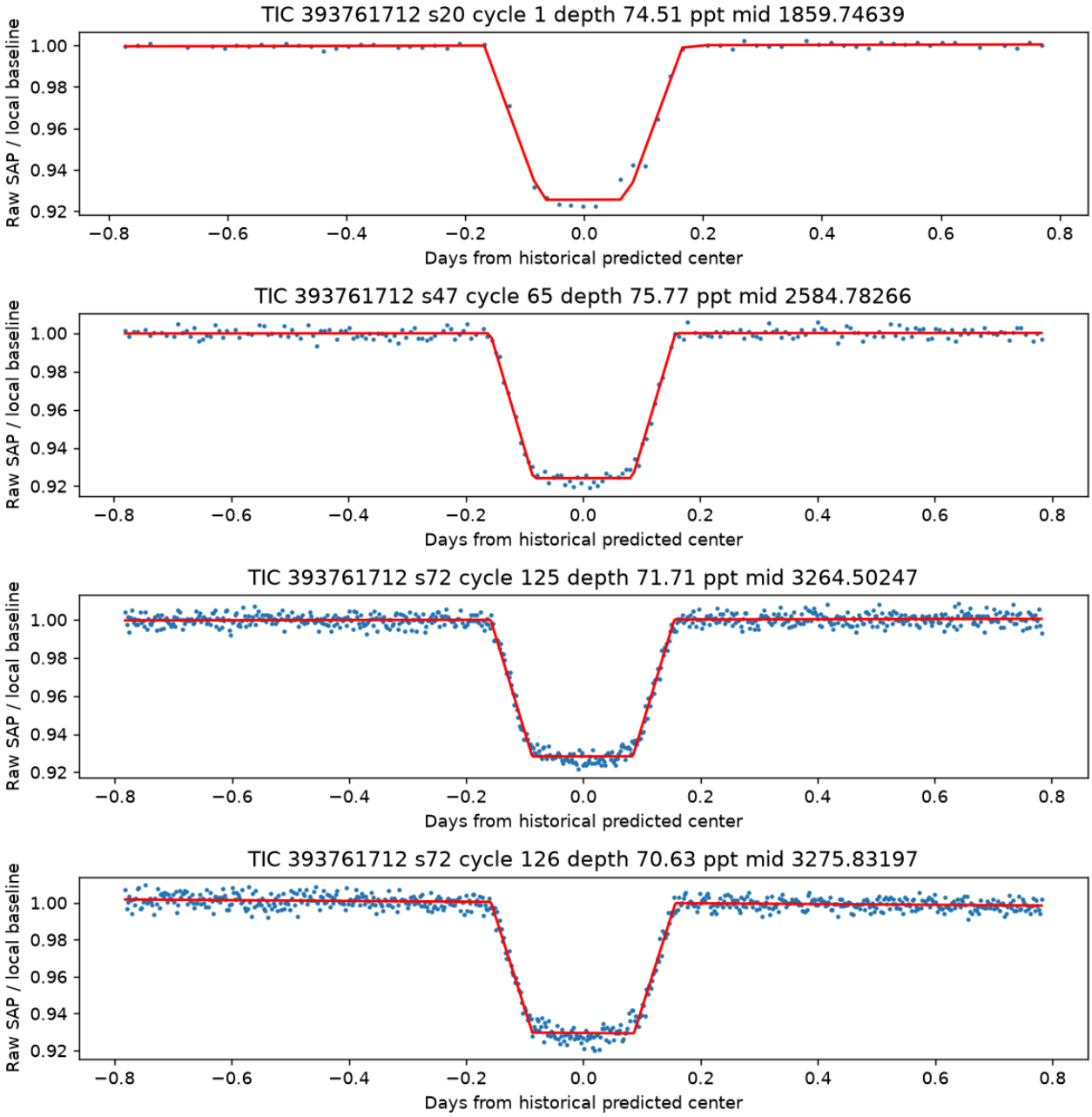
R564. TIC 393761712

physical_verdict: CONFIRMED

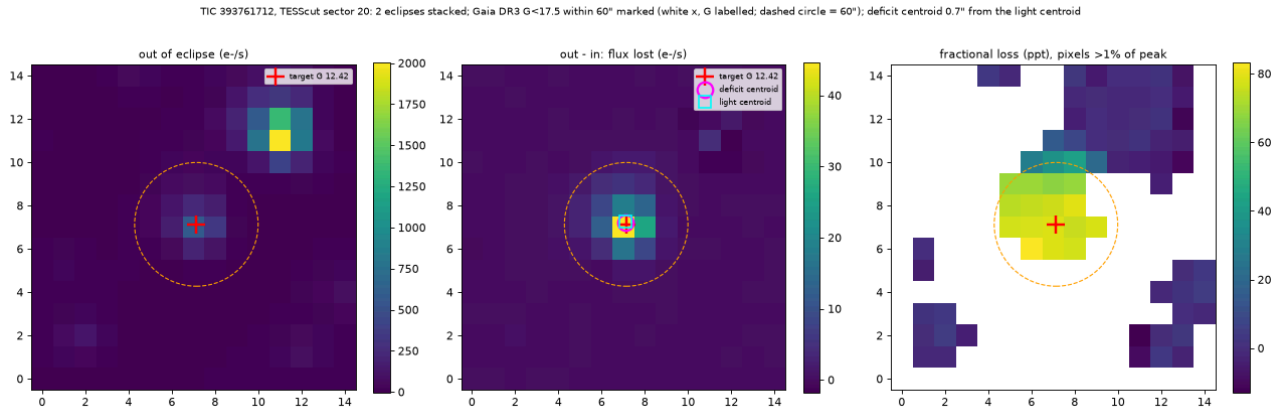
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved follow-up for physical parameters and orbital eclipse labeling.



R564; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R564; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 393761712: CONFIRMED

Full bounded physical review: COMPLETE. Physical verdict is separate from catalogue novelty. The epoch is the selected recurring eclipse; no deeper-primary rank is assumed.

Deciding evidence and limitation: Four deep eclipses repeat across three QLP sectors, with consecutive same-kind S72 events. Compact target deficit and no capable Gaia source in60arcsec support target attribution.

Raw QLP SAP_FLUX representative windows contain 4 accepted free-center measurements in sectors [20, 47, 72]. Predicted windows are search guides, not timings. Quality-zero points enter fits; flagged or uncovered windows are not clean nulls. Rejected fits remain visible in numerical artifacts. Raw event plots and full recurrence phase plots were inspected; fitted dips alone do not establish eclipse morphology (TIC290749976 remains plausible).

Measured recurrence P=11.3286747389 +/- 0.0000486314 days; selected-eclipse reference BTJD=1859.7472252; fitted median depth=73.110ppt. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Singular/unidentifiable timing covariance is excluded. Within-sector cycle counts/parity are in ../alias-audit.json. Full-phase inspection tests secondary structure at its observed phase, without imposing phase0.5; unequal secondaries do not halve same-kind periods. Half/double ambiguity is described explicitly above where material. No exhaustive alias search is claimed. Sparse349452680 additionally has current availability and bounded sampled-null receipts.

Fresh TESScut sector20 stacks 2 actual covered events. Deficit-target offset=0.105pixel, target pixel=[7.133785170811102, 7.130107863557841], deficit centroid=[7.077080695832156, 7.218069767439698]. The image-model residual=0.917 is not a calibrated location error. Actual target pixel deficit=44.682 electron/s and baseline=568.63 electron/s. All flux-capable Gaia neighbors within60arcsec are below; their actual signed deficit and baseline are evaluated in common electron/s units at each source's rounded pixel. Overlapping PSFs prevent deblended ownership. Intrinsic required eclipse fraction uses pairwise Gaia G brightness; it is not directly comparable with diluted fractional pixel loss. Color, aperture response and unlisted faint/unresolved sources limit this approximate screen.

Gaia source	separation arcsec	G	intrinsic required loss proxy	actual signed pixel deficit e/s	pixel baseline e/s
None capable within60arcsec					

Alternative background-subtracted target-centered TESScut1x1/3x3/5x5 depths=78.578/77.575/76.493ppt. These custom extractions test reduction robustness. All QLP and cutouts use TESS photons; same-sector extractions are not independent observations. Distinct sectors are distinct observing occasions on one instrument. Depth changes can be dilution/aperture/background effects; these numbers are not intrinsic eclipse depths. Pixel-arrays.npz preserves actual stacked images.

Gaia DR3 source=878893455728757760, P=11.329658980d, e=0.04789. Closest edge-on conjunction offset=-0.25755d (-0.02273phase); source-specific full covariance linear timing allowance=0.24832d. This descriptive linear allowance is not a calibrated significance veto. Geometry code retains both eccentric conjunctions. Mass function=0.040276 solar masses; TIC/adopted primary mass=1.080, minimum secondary mass=0.456, TIC radius=1.898 solar radii. Approximate central duration=8.191h versus measured median=7.565h. TIC radius and main-sequence minimum-mass companion assumptions make this a plausibility screen, not a fitted binary model. Orbit agreement alone does not establish target ownership.

Sector	Search cycle	midpoint BTJD	formal center error d	timing input error d	depth ppt	status
20	1	1859.7463853	0.00042639	0.010417	74.512	used
47	65	2584.7826617	0.00054338	0.0034722	75.771	used
72	125	3264.5024722	0.00037681	0.0011575	71.709	used
72	126	3275.8319704	0.00041103	0.0011575	70.634	used

Next check: Optional resolved follow-up for physical parameters and orbital eclipse labeling.

Seven-group catalogue receipts are in ../../shared-catalogues/coverage-matrix.csv; known recovery and physical outcome are separate. No exhaustive priority claim. Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png and pixel-arrays.npz; batch identity/hash receipts, geometry, phase and alias artifacts, helper audit. Source hashes and SHA256SUMS.json cover retained sources and outputs. Methods are preserved in a read snapshot rather than mutable manuscript hashes.

Stable review key: R564.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/pixel-arrays.npz - fresh-pixels/run.log - historical/diff_s20.json - historical/diff_s20.png - historical/gaia.json - measurements.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

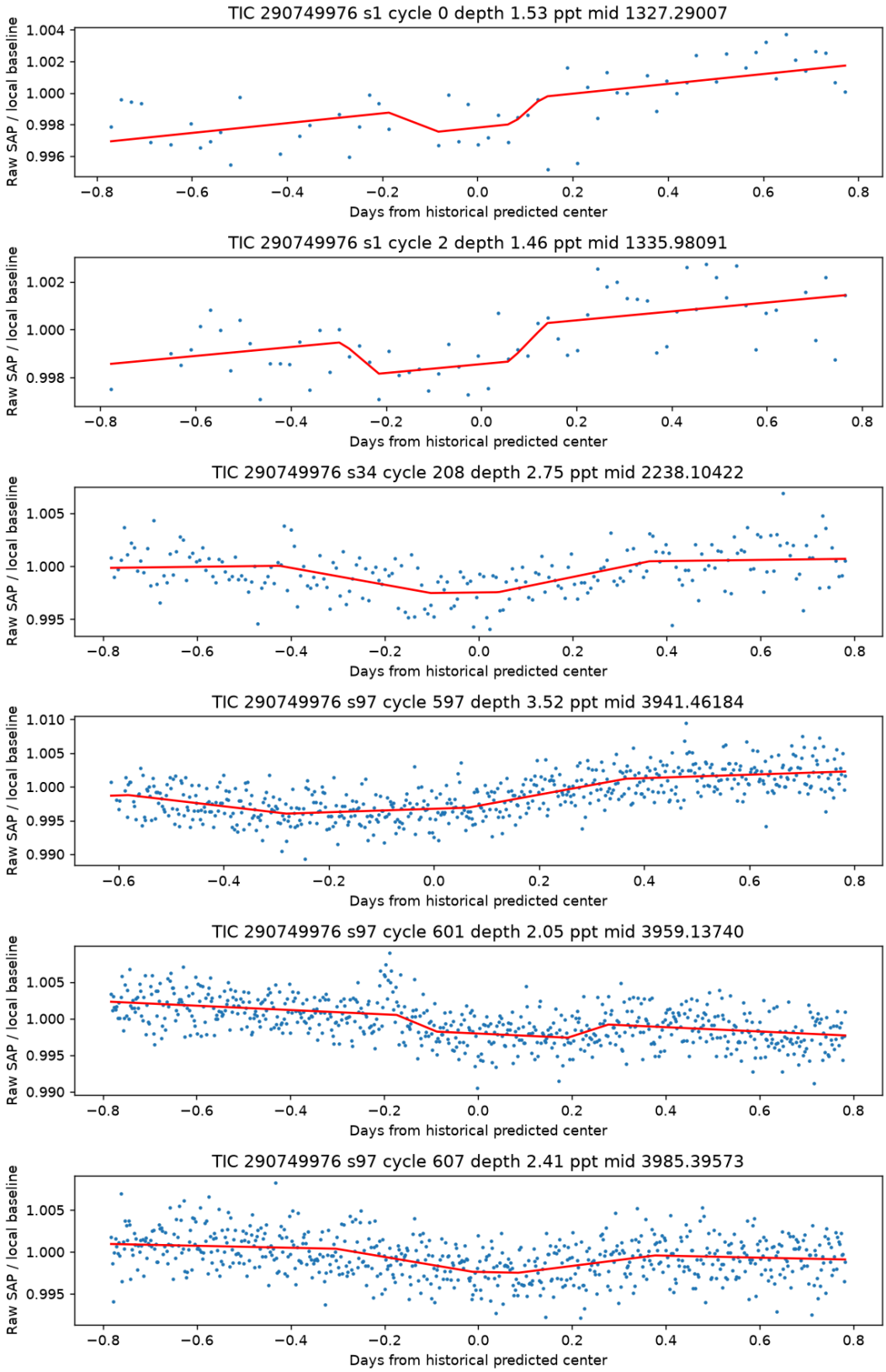
R565. TIC 290749976

physical_verdict: PLAUSIBLE

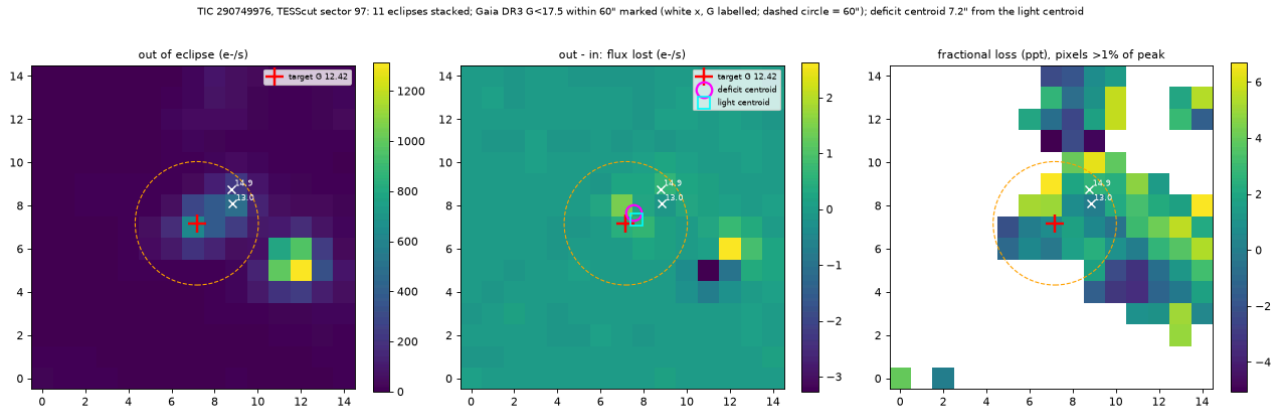
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Establish sharp eclipse morphology against rotational/tidal/baseline modulation, then resolve source ownership with eclipse photometry.



R565; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R565; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 290749976: PLAUSIBLE

Full bounded physical review: COMPLETE. Physical verdict is separate from catalogue novelty. The epoch is a fitted variability-dip reference, not a validated eclipse ephemeris.

Deciding evidence and limitation: Shallow recurring broad dips have variable baselines and a noisy offset deficit (0.62pixel). Three capable neighbors lie36-49arcsec away. Smooth morphology and variable fitted duration versus approximate central4.85h do not establish eclipse geometry; rotational, tidal or baseline modulation remains possible. No convincing sharp eclipse component is established, so this is PLAUSIBLE, not a demonstrated repeating eclipser. TIC radius2.113 solar radii is already used; broad shape/baseline and attribution remain unresolved. Period describes dip recurrence; trapezoid centers are model-dependent.

Raw QLP SAP_FLUX representative windows contain 13 accepted free-center measurements in sectors [1, 34, 97]. Predicted windows are search guides, not timings. Quality-zero points enter fits; flagged or uncovered windows are not clean nulls. Rejected fits remain visible in numerical artifacts. Raw event plots and full recurrence phase plots were inspected; fitted dips alone do not establish eclipse morphology (TIC290749976 remains plausible).

Measured variability-dip recurrence P=4.3790689451 +/- 0.0000392056 days; variability-dip reference BTJD=1331.6582646; fitted median depth=2.383ppt. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Singular/unidentifiable timing covariance is excluded. Within-sector cycle counts/parity are in ../alias-audit.json. Full-phase inspection tests secondary structure at its observed phase, without imposing phase0.5; unequal secondaries do not halve same-kind periods. Half/double ambiguity is described explicitly above where material. No exhaustive alias search is claimed. Sparse349452680 additionally has current availability and bounded sampled-null receipts.

Fresh TESScut sector97 stacks 11 actual covered events. Deficit-target offset=0.627pixel, target pixel=[7.169999815834171, 7.179607561524339], deficit centroid=[7.533701991063821, 7.690829547989072]. The image-model residual=0.958 is not a calibrated location error. Actual target pixel deficit=0.50129 electron/s and baseline=621.52 electron/s. All flux-capable Gaia neighbors within60arcsec are below; their actual signed deficit and baseline are evaluated in common electron/s

units at each source's rounded pixel. Overlapping PSFs prevent deblended ownership. Intrinsic required eclipse fraction uses pairwise Gaia G brightness; it is not directly comparable with diluted fractional pixel loss. Color, aperture response and unlisted faint/unresolved sources limit this approximate screen.

Gaia source	separation arcsec	G	intrinsic required loss proxy	actual signed pixel deficit e/s	pixel baseline e/s
4769034211197334912	36.22	13.012	0.0065	-0.047586	477.99
4769034211197335040	43.00	14.900	0.0258	0.63077	226.3
4769034206900744064	48.76	18.008	0.4120	0.63077	226.3

Alternative background-subtracted target-centered TESScut1x1/3x3/5x5 depths=0.807/1.477/1.291ppt. These custom extractions test reduction robustness. All QLP and cutouts use TESS photons; same-sector extractions are not independent observations. Distinct sectors are distinct observing occasions on one instrument. Depth changes can be dilution/aperture/background effects; these numbers are not intrinsic eclipse depths. Pixel-arrays.npz preserves actual stacked images.

Gaia DR3 source=4769034314276549632, P=4.378737543d, e=0.26791. Closest edge-on conjunction offset=0.01461d (0.00334phase); source-specific full covariance linear timing allowance=0.12464d. This descriptive linear allowance is not a calibrated significance veto. Geometry code retains both eccentric conjunctions. Mass function=0.022273 solar masses; TIC/adopted primary mass=1.490, minimum secondary mass=0.435, TIC radius=2.113 solar radii. Approximate central duration=4.850h versus measured median=10.845h. TIC radius and main-sequence minimum-mass companion assumptions make this a plausibility screen, not a fitted binary model. Orbit agreement alone does not establish target ownership.

Sector	Search cycle	midpoint BTJD	formal center error d	timing input error d	depth ppt	status
1	0	1327.2900745	9.483e+05	9.483e+05	1.531	rejected: Singular or unidentifiable center covariance: formal timing uncertainty exceeds search duration; excluded from ephemeris.
1	1	1331.6545971	0.0073051	0.010417	3.312	used
1	2	1335.9809144	0.041468	0.041468	1.457	used
1	4	1344.8274905	4.5622e+05	4.5622e+05	1.077	rejected: Singular or unidentifiable center covariance: formal timing uncertainty exceeds search duration; excluded from ephemeris.
34	207	2233.7636333	0.01971	0.01971	2.007	used
34	208	2238.1042212	0.025499	0.025499	2.746	used
34	210	2246.8722219	0.02951	0.02951	2.383	used
34	211	2251.2772694	0.011554	0.011554	1.742	used
97	597	3941.4618385	0.027857	0.027857	3.520	used
97	598	3945.9380403	0.0092929	0.0092929	1.899	used
97	600	3954.7704792	0.023275	0.023275	2.430	used
97	601	3959.1374038	0.014618	0.014618	2.046	used

Sector	Search cycle	midpoint BTJD	formal center error d	timing input error d	depth ppt	status
97	603	3967.9584083	0.035959	0.035959	1.141	used
97	604	3972.2320568	0.017947	0.017947	2.999	used
97	607	3985.3957278	0.018042	0.018042	2.407	used

Next check: Establish sharp eclipse morphology against rotational/tidal/baseline modulation, then resolve source ownership with eclipse photometry.

Seven-group catalogue receipts are in ../../shared-catalogues/coverage-matrix.csv; known recovery and physical outcome are separate. No exhaustive priority claim. Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png and pixel-arrays.npz; batch identity/hash receipts, geometry, phase and alias artifacts, helper audit. Source hashes and SHA256SUMS.json cover retained sources and outputs. Methods are preserved in a read snapshot rather than mutable manuscript hashes.

Depth correction audit (2026-11-02): realized trapezoid decrement replaces its unconstrained amplitude parameter. Original input windows, fitted centers/widths, timing selections and physical verdict are retained; capable-neighbor membership is unchanged. See ../../depth-audit/README.md.

Stable review key: R565.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/pixel-arrays.npz - fresh-pixels/run.log - historical/diff_s97.json - historical/diff_s97.png - historical/gaia.json - measurements.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

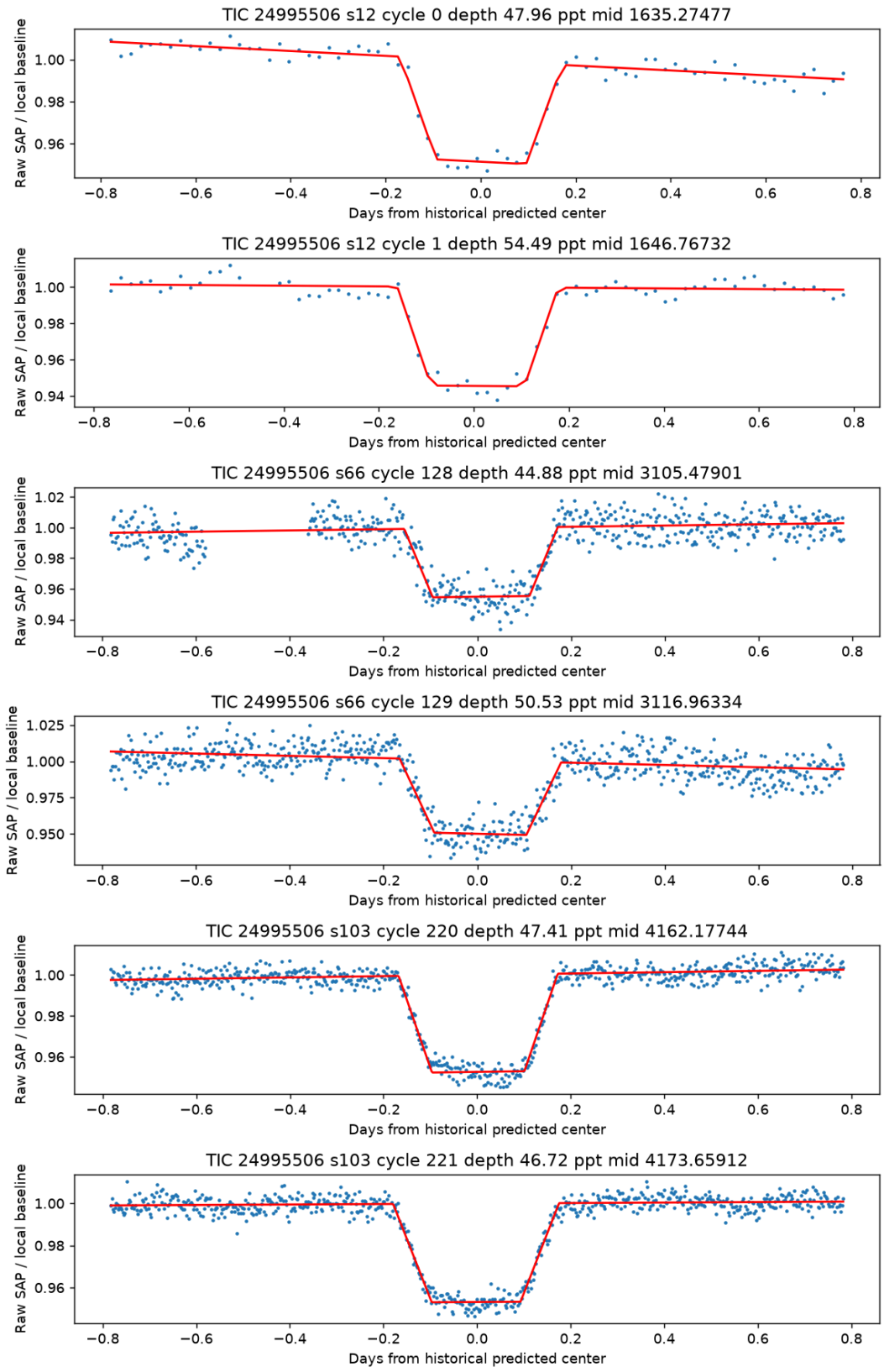
R566. TIC 24995506

physical_verdict: PROBABLE

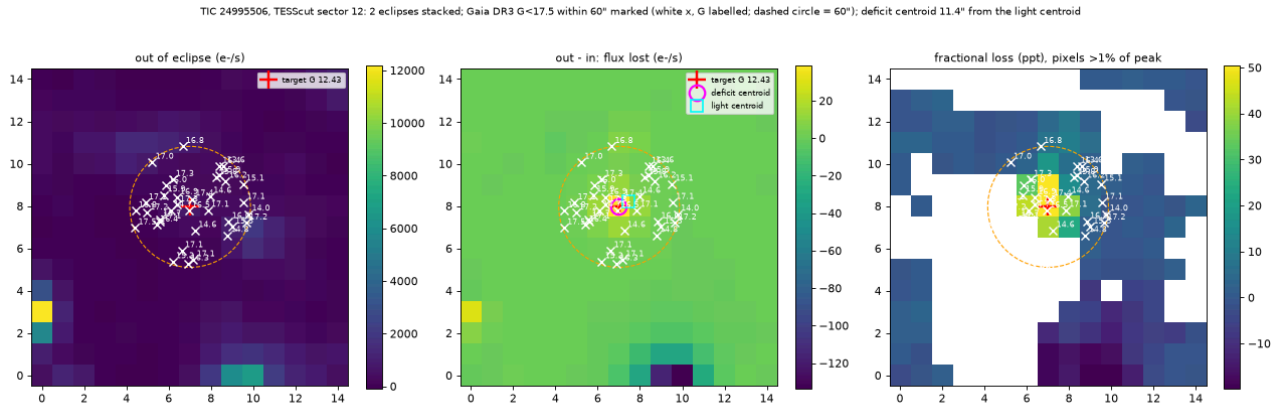
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved photometry through an eclipse to establish ownership; additional timing for remaining period/shape limits.



R566; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R566; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 24995506: PROBABLE

Full bounded physical review: COMPLETE. Physical verdict is separate from catalogue novelty. The epoch is the selected recurring eclipse; no deeper-primary rank is assumed.

Deciding evidence and limitation: Six repeated raw eclipses at11.48585d. Compact target-centered deficit overlaps three capable sources19-23arcsec away, so ownership remains unresolved. Nearby known EB Gaia5940721527468980608 at42.94arcsec has G17.062 and period0.72596d: G-band pairwise maximum diluted loss is1.38percent, below observed4.8percent; required intrinsic loss exceeds unity. Its period differs from observed recurrence and its location is distinct from deficit core. This excludes that catalogued EB under approximate passband budget, not every close neighbor.

Raw QLP SAP_FLUX representative windows contain 6 accepted free-center measurements in sectors [12, 66, 103]. Predicted windows are search guides, not timings. Quality-zero points enter fits; flagged or uncovered windows are not clean nulls. Rejected fits remain visible in numerical artifacts. Raw event plots and full recurrence phase plots were inspected; fitted dips alone do not establish eclipse morphology (TIC290749976 remains plausible).

Measured recurrence P=11.4858471797 +/- 0.0000230159 days; selected-eclipse reference BTJD=1635.2891562; fitted median depth=47.685ppt. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Singular/unidentifiable timing covariance is excluded. Within-sector cycle counts/parity are in ../alias-audit.json. Full-phase inspection tests secondary structure at its observed phase, without imposing phase0.5; unequal secondaries do not halve same-kind periods. Half/double ambiguity is described explicitly above where material. No exhaustive alias search is claimed. Sparse349452680 additionally has current availability and bounded sampled-null receipts.

Fresh TESScut sector12 stacks 2 actual covered events. Deficit-target offset=0.025pixel, target pixel=[7.013067868740087, 7.960893625032808], deficit centroid=[6.995820825831303, 7.978532655702426]. The image-model residual=0.969 is not a calibrated location error. Actual target pixel deficit=38.695 electron/s and baseline=767.54 electron/s. All flux-capable Gaia neighbors within60arcsec are below; their actual signed deficit and baseline are evaluated in common electron/s

units at each source's rounded pixel. Overlapping PSFs prevent deblended ownership. Intrinsic required eclipse fraction uses pairwise Gaia G brightness; it is not directly comparable with diluted fractional pixel loss. Color, aperture response and unlisted faint/unresolved sources limit this approximate screen.

Gaia source	separation arcsec	G	intrinsic required loss proxy	actual signed pixel deficit e/s	pixel baseline e/s
5940721561828733440	19.53	14.743	0.4506	14.243	325.22
5940721458749510272	22.29	14.624	0.4086	12.912	332.13
5940721355670290304	23.19	14.580	0.3945	13.502	325.26
5940721355670282240	45.47	15.524	0.8745	2.1286	927.05
5940721355670281088	45.66	14.784	0.4662	2.1286	927.05
5940721454416781440	51.50	13.560	0.1832	0.49117	273.01
5940721527468979840	55.49	15.338	0.7444	0.57377	-24.815
5940721458749504640	56.52	15.095	0.6050	0.24692	145.04
5940721458749618688	57.98	13.996	0.2502	-0.69164	507.25

Alternative background-subtracted target-centered TESScut1x1/3x3/5x5 depths=50.414/40.588/19.720ppt. These custom extractions test reduction robustness. All QLP and cutouts use TESS photons; same-sector extractions are not independent observations. Distinct sectors are distinct observing occasions on one instrument. Depth changes can be dilution/aperture/background effects; these numbers are not intrinsic eclipse depths. Pixel-arrays.npz preserves actual stacked images.

Gaia DR3 source=5940721561828728192, P=11.482085295d, e=0.11995. Closest edge-on conjunction offset=0.21671d (0.01887phase); source-specific full covariance linear timing allowance=0.30931d. This descriptive linear allowance is not a calibrated significance veto. Geometry code retains both eccentric conjunctions. Mass function=0.039523 solar masses; TIC/adopted primary mass=1.840, minimum secondary mass=0.621, TIC radius=2.263 solar radii. Approximate central duration=8.128h versus measured median=8.179h. TIC radius and main-sequence minimum-mass companion assumptions make this a plausibility screen, not a fitted binary model. Orbit agreement alone does not establish target ownership.

Sector	Search cycle	midpoint BTJD	formal center error d	timing input error d	depth ppt	status
12	0	1635.2747733	0.0056318	0.010417	47.961	used
12	1	1646.7673167	0.0016914	0.010417	54.494	used
66	128	3105.4790093	0.0016827	0.0016827	44.875	used
66	129	3116.9633384	0.001789	0.001789	50.533	used
103	220	4162.1774423	0.00072832	0.0011574	47.408	used
103	221	4173.6591234	0.00072991	0.0011574	46.721	used

Next check: Resolved photometry through an eclipse to establish ownership; additional timing for remaining period/shape limits.

Seven-group catalogue receipts are in ../../shared-catalogues/coverage-matrix.csv; known recovery and physical outcome are separate. No exhaustive priority claim. Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png and pixel-arrays.npz; batch

identity/hash receipts, geometry, phase and alias artifacts, helper audit. Source hashes and SHA256SUMS.json cover retained sources and outputs. Methods are preserved in a read snapshot rather than mutable manuscript hashes.

Stable review key: R566.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/pixel-arrays.npz - fresh-pixels/run.log - historical/diff_s12.json - historical/diff_s12.png - historical/gaia.json - measurements.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

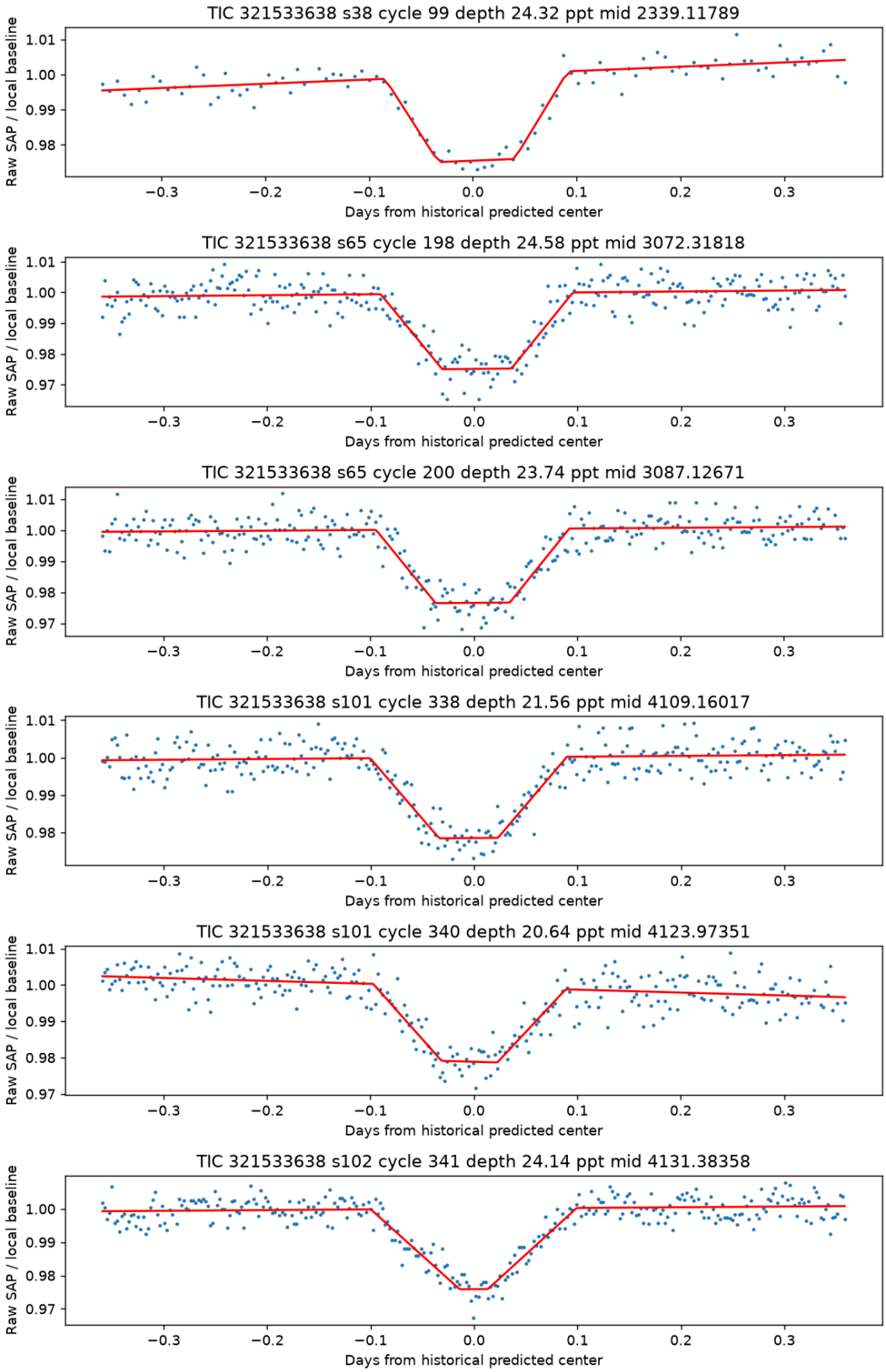
R567. TIC 321533638

physical_verdict: PROBABLE

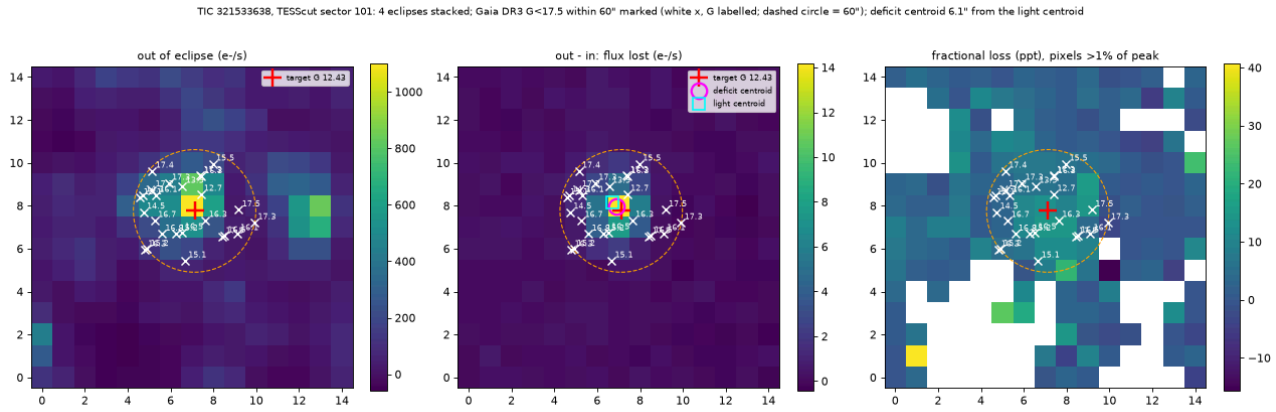
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved photometry through an eclipse to establish ownership; additional timing for remaining period/shape limits.



R567; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R567; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 321533638: PROBABLE

Full bounded physical review: COMPLETE. Physical verdict is separate from catalogue novelty. The epoch is the selected recurring eclipse; no deeper-primary rank is assumed.

Deciding evidence and limitation: Fresh availability query and actual additional S38/65/101 data increase timing sample to12 across all6 current QLP sectors. Consecutive within-sector recurrences establish7.40605d. Fourteen capable sources include one13.5arcsec away and near-equal source16.5arcsec away; crowded deficit does not uniquely establish target ownership.

Raw QLP SAP_FLUX representative windows contain 12 accepted free-center measurements in sectors [11, 38, 65, 100, 101, 102]. Predicted windows are search guides, not timings. Quality-zero points enter fits; flagged or uncovered windows are not clean nulls. Rejected fits remain visible in numerical artifacts. Raw event plots and full recurrence phase plots were inspected; fitted dips alone do not establish eclipse morphology (TIC290749976 remains plausible).

Measured recurrence P=7.4060531071 +/- 0.0000088853 days; selected-eclipse reference BTJD=1605.9177389; fitted median depth=22.319ppt. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Singular/unidentifiable timing covariance is excluded. Within-sector cycle counts/parity are in ../alias-audit.json. Full-phase inspection tests secondary structure at its observed phase, without imposing phase0.5; unequal secondaries do not halve same-kind periods. Half/double ambiguity is described explicitly above where material. No exhaustive alias search is claimed. Sparse349452680 additionally has current availability and bounded sampled-null receipts.

Fresh TESScut sector101 stacks 4 actual covered events. Deficit-target offset=0.289pixel, target pixel=[7.120895737749853, 7.759791546210337], deficit centroid=[6.900565557764185, 7.947501473652374]. The image-model residual=0.481 is not a calibrated location error. Actual target pixel deficit=14.181 electron/s and baseline=1100.3 electron/s. All flux-capable Gaia neighbors within60arcsec are below; their actual signed deficit and baseline are evaluated in common electron/s units at each source's rounded pixel. Overlapping PSFs prevent deblended ownership. Intrinsic required eclipse fraction uses pairwise Gaia G brightness; it is not directly comparable with diluted fractional pixel loss. Color, aperture response and unlisted faint/unresolved sources limit this approximate screen.

Gaia source	separation arcsec	G	intrinsic required loss proxy	actual signed pixel deficit e/s	pixel baseline e/s
5864551004026002688	13.49	16.270	0.7900	3.7188	251.71
5864550935306524032	16.51	12.716	0.0514	4.8774	815.8
5864551038385744000	23.61	16.524	0.9929	7.8261	527.19
5864550965337084160	25.32	13.538	0.0843	4.8774	815.8
5864551038385745280	27.62	15.166	0.3000	4.178	374.96
5864550930977340032	32.21	16.316	0.8234	4.8774	815.8
5864551004026003328	36.22	16.517	0.9860	0.16816	38.178
5864550969666265600	39.46	16.120	0.6909	1.2216	494
5864551004026002048	45.34	16.131	0.6977	0.16816	38.178
5864550935306519680	46.81	15.529	0.4103	1.2214	205.39
5864551210184439040	46.89	15.068	0.2761	0.52206	60.984
5864550965337097984	48.91	14.549	0.1796	1.2216	494
5864550969666266112	51.82	14.340	0.1521	1.2216	494
5864550969666266496	53.21	15.100	0.2837	1.2216	494
5864551038385748480	57.91	15.248	0.3220	0.58497	117.15
5864551038351575680	60.00	16.267	0.7879	0.58497	117.15

Alternative background-subtracted target-centered TESScut1x1/3x3/5x5 depths=12.889/10.038/8.588ppt. These custom extractions test reduction robustness. All QLP and cutouts use TESS photons; same-sector extractions are not independent observations. Distinct sectors are distinct observing occasions on one instrument. Depth changes can be dilution/aperture/background effects; these numbers are not intrinsic eclipse depths. Pixel-arrays.npz preserves actual stacked images.

Gaia DR3 source=5864551004026002304, P=7.403844033d, e=0.06346. Closest edge-on conjunction offset=0.09758d (0.01318phase); source-specific full covariance linear timing allowance=0.27983d. This descriptive linear allowance is not a calibrated significance veto. Geometry code retains both eccentric conjunctions. Mass function=0.0095563 solar masses; TIC/adopted primary mass=1.470, minimum secondary mass=0.312, TIC radius=1.759 solar radii. Approximate central duration=6.349h versus measured median=4.469h. TIC radius and main-sequence minimum-mass companion assumptions make this a plausibility screen, not a fitted binary model. Orbit agreement alone does not establish target ownership.

Sector	Search cycle	midpoint BTJD	formal center error d	timing input error d	depth ppt	status
38	99	2339.1178887	0.0025662	0.0034722	24.316	used
38	101	2353.9295947	0.0013049	0.0034722	20.800	used
65	198	3072.3181842	0.0014732	0.0014732	24.584	used
65	199	3079.7207283	0.002062	0.002062	21.798	used
65	200	3087.1267134	0.001446	0.001446	23.737	used

Sector	Search cycle	midpoint BTJD	formal center error d	timing input error d	depth ppt	status
101	337	4101.7562184	0.001581	0.001581	24.236	used
101	338	4109.1601703	0.0017417	0.0017417	21.558	used
101	339	4116.5699013	0.0013435	0.0013435	21.301	used
101	340	4123.9735134	0.0019689	0.0019689	20.645	used
11	0	1605.9189415	0.0034322	0.010417	22.839	used
100	334	4079.5432222	0.0015239	0.0015239	21.350	used
102	341	4131.3835802	0.001511	0.001511	24.143	used

Next check: Resolved photometry through an eclipse to establish ownership; additional timing for remaining period/shape limits.

Seven-group catalogue receipts are in ../../shared-catalogues/coverage-matrix.csv; known recovery and physical outcome are separate. No exhaustive priority claim. Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png and pixel-arrays.npz; batch identity/hash receipts, geometry, phase and alias artifacts, helper audit. Source hashes and SHA256SUMS.json cover retained sources and outputs. Methods are preserved in a read snapshot rather than mutable manuscript hashes.

Stable review key: R567.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/pixel-arrays.npz - fresh-pixels/run.log - historical/diff_s101.json - historical/diff_s101.png - historical/gaia.json - measurements.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

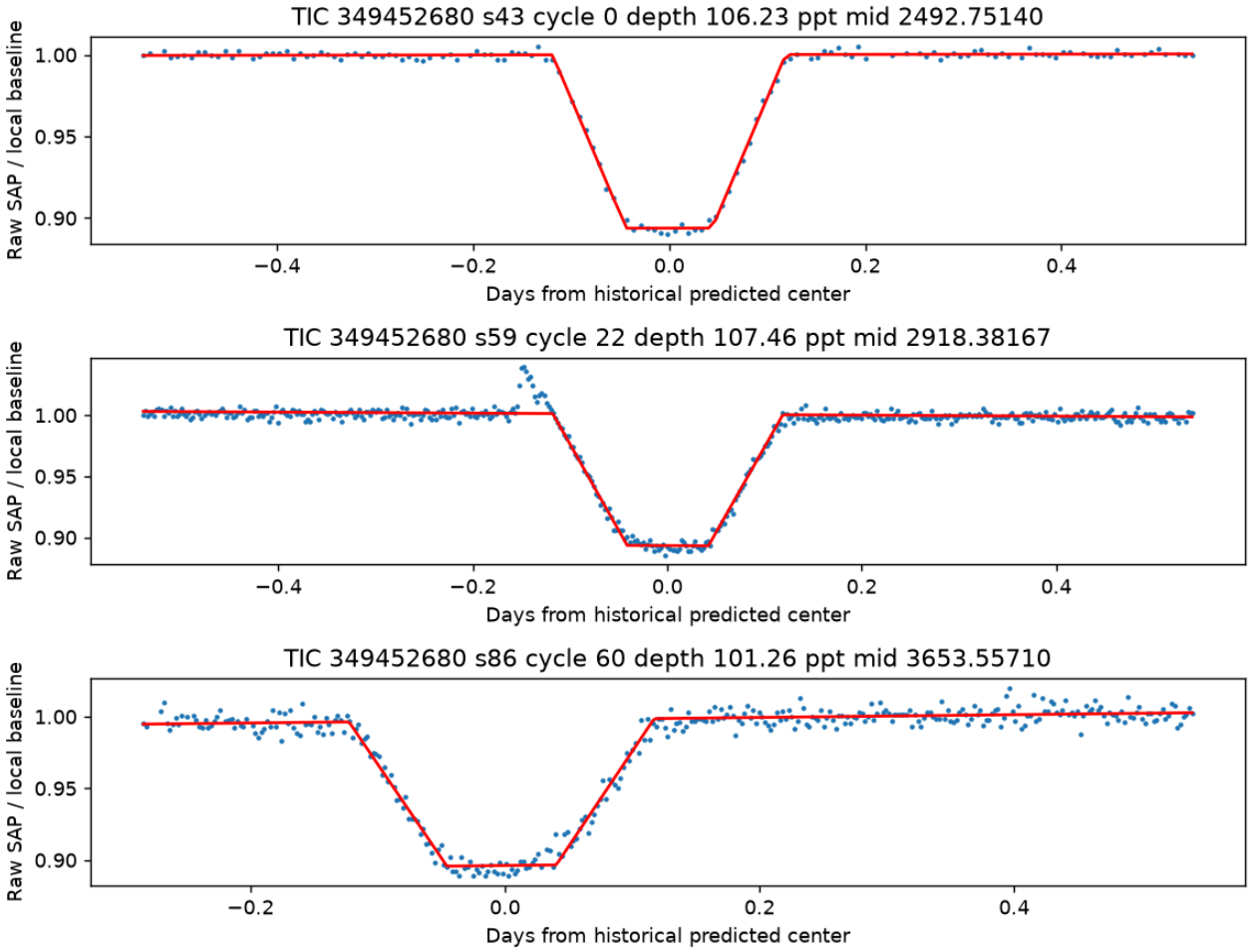
R568. TIC 349452680

physical_verdict: CONFIRMED

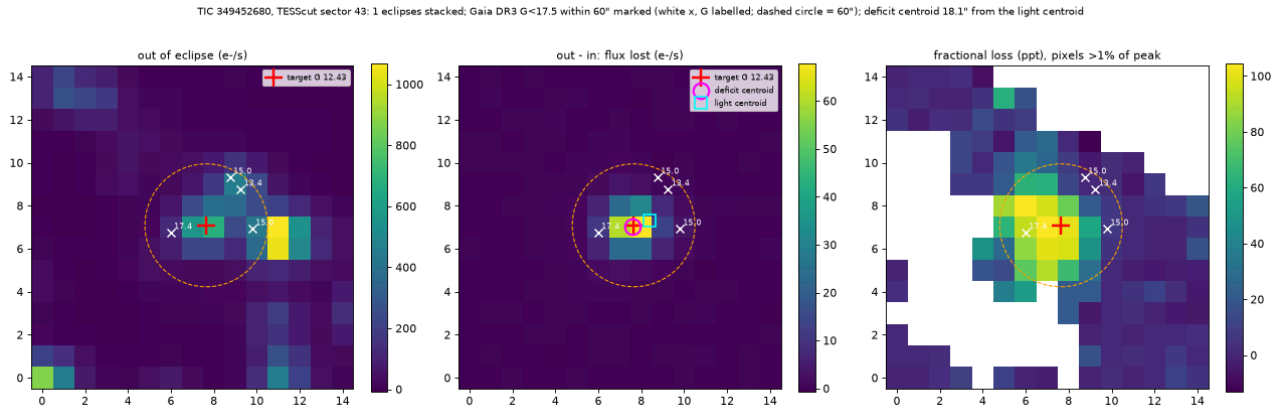
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved follow-up for physical parameters and orbital eclipse labeling.



R568; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R568; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 349452680: CONFIRMED

Full bounded physical review: COMPLETE. Physical verdict is separate from catalogue novelty. The epoch is the selected recurring eclipse; no deeper-primary rank is assumed.

Deciding evidence and limitation: All four currently available QLP sectors43/44/59/86 inspected; three clear actual events. Bounded integer aliases0.8-40d reject shorter alternatives through sampled clean nulls;19.34676d and38.69352d survive. Reported19.34673d is observed eclipse recurrence; double orbital-period equal-eclipse labeling remains unresolved. Sole capable neighbor47.1arcsec away is spatially distinct from compact target deficit.

Raw QLP SAP_FLUX representative windows contain 3 accepted free-center measurements in sectors [43, 59, 86]. Predicted windows are search guides, not timings. Quality-zero points enter fits; flagged or uncovered windows are not clean nulls. Rejected fits remain visible in numerical artifacts. Raw event plots and full recurrence phase plots were inspected; fitted dips alone do not establish eclipse morphology (TIC290749976 remains plausible).

Measured recurrence P=19.3467332486 +/- 0.0000386029 days; selected-eclipse reference BTJD=2492.7532205; fitted median depth=106.233ppt. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Singular/unidentifiable timing covariance is excluded. Within-sector cycle counts/parity are in ../alias-audit.json. Full-phase inspection tests secondary structure at its observed phase, without imposing phase0.5; unequal secondaries do not halve same-kind periods. Half/double ambiguity is described explicitly above where material. No exhaustive alias search is claimed. Sparse349452680 additionally has current availability and bounded sampled-null receipts.

Fresh TESScut sector43 stacks 1 actual covered events. Deficit-target offset=0.073pixel, target pixel=[7.647719502907066, 7.086909362988742], deficit centroid=[7.590605817088184, 7.042151821485786]. The image-model residual=0.854 is not a calibrated location error. Actual target pixel deficit=67.895 electron/s and baseline=675.3 electron/s. All flux-capable Gaia neighbors within60arcsec are below; their actual signed deficit and baseline are evaluated in common electron/s units at each source's rounded pixel. Overlapping PSFs prevent deblended ownership. Intrinsic required eclipse fraction uses pairwise Gaia G brightness; it is not directly comparable with diluted fractional pixel loss. Color, aperture response and unlisted faint/unresolved sources limit this approximate screen.

Gaia source	separation arcsec	G	intrinsic required loss proxy	actual signed pixel deficit e/s	pixel baseline e/s
1706465797534695 68	47.13	13.36 3	0.3571	0.89086	460.4

Alternative background-subtracted target-centered TESScut1x1/3x3/5x5 depths=100.540/83.084/54.365ppt. These custom extractions test reduction robustness. All QLP and cutouts use TESS photons; same-sector extractions are not independent observations. Distinct sectors are distinct observing occasions on one instrument. Depth changes can be dilution/aperture/background effects; these numbers are not intrinsic eclipse depths. Pixel-arrays.npz preserves actual stacked images.

Gaia DR3 source=170599644350858752, P=19.344353625d, e=0.35529. Closest edge-on conjunction offset=0.25468d (0.01317phase); source-specific full covariance linear timing allowance=0.75726d. This descriptive linear allowance is not a calibrated significance veto. Geometry code retains both eccentric conjunctions. Mass function=0.083882 solar masses; TIC/adopted primary mass=1.040, minimum secondary mass=0.612, TIC radius=1.922 solar radii. Approximate central duration=7.424h versus measured median=5.706h. TIC radius and main-sequence minimum-mass companion assumptions make this a plausibility screen, not a fitted binary model. Orbit agreement alone does not establish target ownership.

Sector	Search cycle	midpoint BTJD	formal center error d	timing input error d	depth ppt	status
43	0	2492.7513967	0.00021859	0.0034725	106.233	used
59	22	2918.3816718	0.00027642	0.0011574	107.458	used
86	60	3653.5570981	0.00049198	0.0011574	101.260	used

Next check: Optional resolved follow-up for physical parameters and orbital eclipse labeling.

Seven-group catalogue receipts are in ../../shared-catalogues/coverage-matrix.csv; known recovery and physical outcome are separate. No exhaustive priority claim. Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png and pixel-arrays.npz; batch identity/hash receipts, geometry, phase and alias artifacts, helper audit. Source hashes and SHA256SUMS.json cover retained sources and outputs. Methods are preserved in a read snapshot rather than mutable manuscript hashes.

Stable review key: R568.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/pixel-arrays.npz - fresh-pixels/run.log - historical/diff_s43.json - historical/diff_s43.png - historical/gaia.json - measurements.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

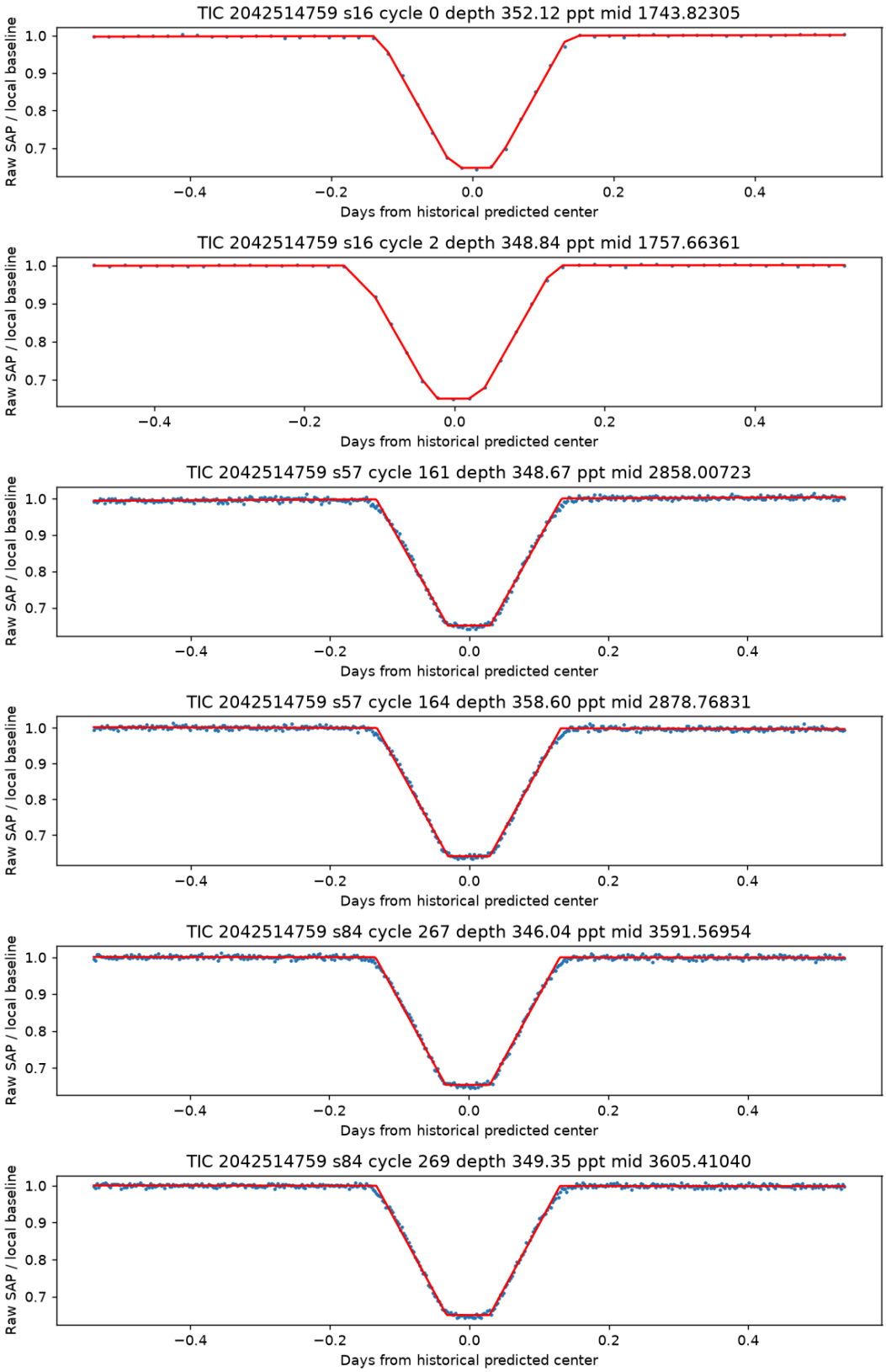
R569. TIC 2042514759

physical_verdict: PROBABLE

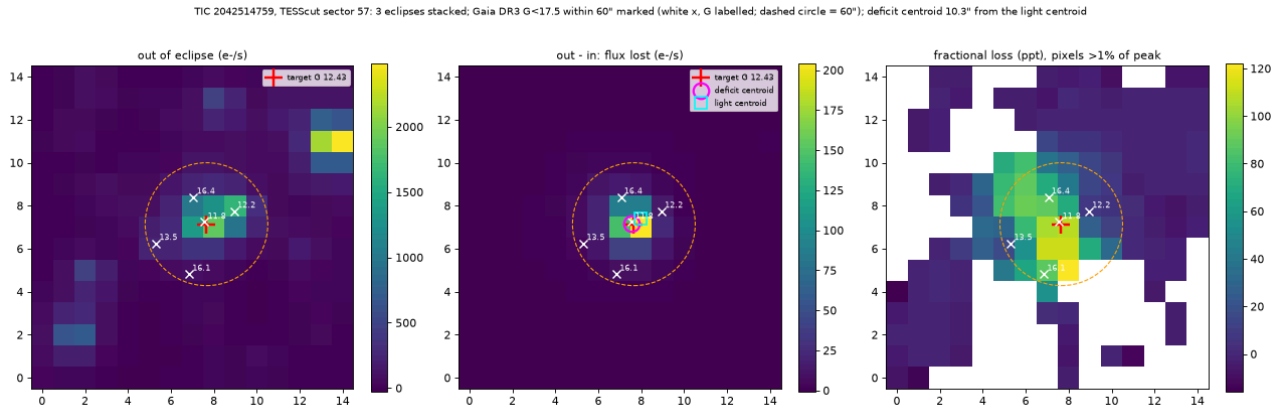
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved photometry through an eclipse to establish ownership; additional timing for remaining period/shape limits.



R569; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R569; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 2042514759: PROBABLE

Full bounded physical review: COMPLETE. Physical verdict is separate from catalogue novelty. The epoch is the selected recurring eclipse; no deeper-primary rank is assumed.

Deciding evidence and limitation: Eight deep repeated eclipses. Brighter G11.76 source only2.9arcsec away is unresolved within deficit core; target attribution remains unresolved. Depth in custom aperture is diluted and is not intrinsic binary depth.

Raw QLP SAP_FLUX representative windows contain 8 accepted free-center measurements in sectors [16, 57, 84]. Predicted windows are search guides, not timings. Quality-zero points enter fits; flagged or uncovered windows are not clean nulls. Rejected fits remain visible in numerical artifacts. Raw event plots and full recurrence phase plots were inspected; fitted dips alone do not establish eclipse morphology (TIC290749976 remains plausible).

Measured recurrence P=6.9203990000 +/- 0.0000086802 days; selected-eclipse reference BTJD=1743.8229790; fitted median depth=348.756ppt. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Singular/unidentifiable timing covariance is excluded. Within-sector cycle counts/parity are in ../alias-audit.json. Full-phase inspection tests secondary structure at its observed phase, without imposing phase0.5; unequal secondaries do not halve same-kind periods. Half/double ambiguity is described explicitly above where material. No exhaustive alias search is claimed. Sparse349452680 additionally has current availability and bounded sampled-null receipts.

Fresh TESScut sector57 stacks 3 actual covered events. Deficit-target offset=0.125pixel, target pixel=[7.6572319752954545, 7.139831340053767], deficit centroid=[7.540108476465893, 7.182339898383201]. The image-model residual=0.523 is not a calibrated location error. Actual target pixel deficit=204.39 electron/s and baseline=1883.1 electron/s. All flux-capable Gaia neighbors within60arcsec are below; their actual signed deficit and baseline are evaluated in common electron/s units at each source's rounded pixel. Overlapping PSFs prevent deblended ownership. Intrinsic required eclipse fraction uses pairwise Gaia G brightness; it is not directly comparable with diluted fractional pixel loss. Color, aperture response and unlisted faint/unresolved sources limit this approximate screen.

Gaia source	separation arcsec	G	intrinsic required loss proxy	actual signed pixel deficit e/s	pixel baseline e/s
1942541621251629952	2.89	11.755	0.5359	204.39	1883.1
1942541552532153472	29.88	12.242	0.6417	13.837	1651.2

Alternative background-subtracted target-centered TESScut1x1/3x3/5x5 depths=108.539/72.724/67.490ppt. These custom extractions test reduction robustness. All QLP and cutouts use TESS photons; same-sector extractions are not independent observations. Distinct sectors are distinct observing occasions on one instrument. Depth changes can be dilution/aperture/background effects; these numbers are not intrinsic eclipse depths. Pixel-arrays.npz preserves actual stacked images.

Gaia DR3 source=1942541625545683584, P=6.922654712d, e=0.41926. Closest edge-on conjunction offset=-0.15280d (-0.02207phase); source-specific full covariance linear timing allowance=0.12051d. This descriptive linear allowance is not a calibrated significance veto. Geometry code retains both eccentric conjunctions. Mass function=0.16023 solar masses; TIC/adopted primary mass=1.170, minimum secondary mass=0.875, TIC radius=1.685 solar radii. Approximate central duration=8.444h versus measured median=6.355h. TIC radius and main-sequence minimum-mass companion assumptions make this a plausibility screen, not a fitted binary model. Orbit agreement alone does not establish target ownership.

Sector	Search cycle	midpoint BTJD	formal center error d	timing input error d	depth ppt	status
16	0	1743.8230540	0.00024676	0.010417	352.116	used
16	2	1757.6636090	0.00012395	0.010417	348.843	used
57	161	2858.0072272	0.00018463	0.0011574	348.668	used
57	163	2871.8481102	0.00021824	0.0011574	337.638	used
57	164	2878.7683140	0.0001411	0.0011574	358.605	used
84	267	3591.5695354	0.00013508	0.0011574	346.037	used
84	268	3598.4898008	0.00014319	0.0011574	343.979	used
84	269	3605.4103957	0.00013209	0.0011574	349.354	used

Next check: Resolved photometry through an eclipse to establish ownership; additional timing for remaining period/shape limits.

Seven-group catalogue receipts are in ../../shared-catalogues/coverage-matrix.csv; known recovery and physical outcome are separate. No exhaustive priority claim. Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png and pixel-arrays.npz; batch identity/hash receipts, geometry, phase and alias artifacts, helper audit. Source hashes and SHA256SUMS.json cover retained sources and outputs. Methods are preserved in a read snapshot rather than mutable manuscript hashes.

Stable review key: R569.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/pixel-arrays.npz - fresh-pixels/run.log - historical/diff_s57.json - historical/diff_s57.png - historical/gaia.json - measurements.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

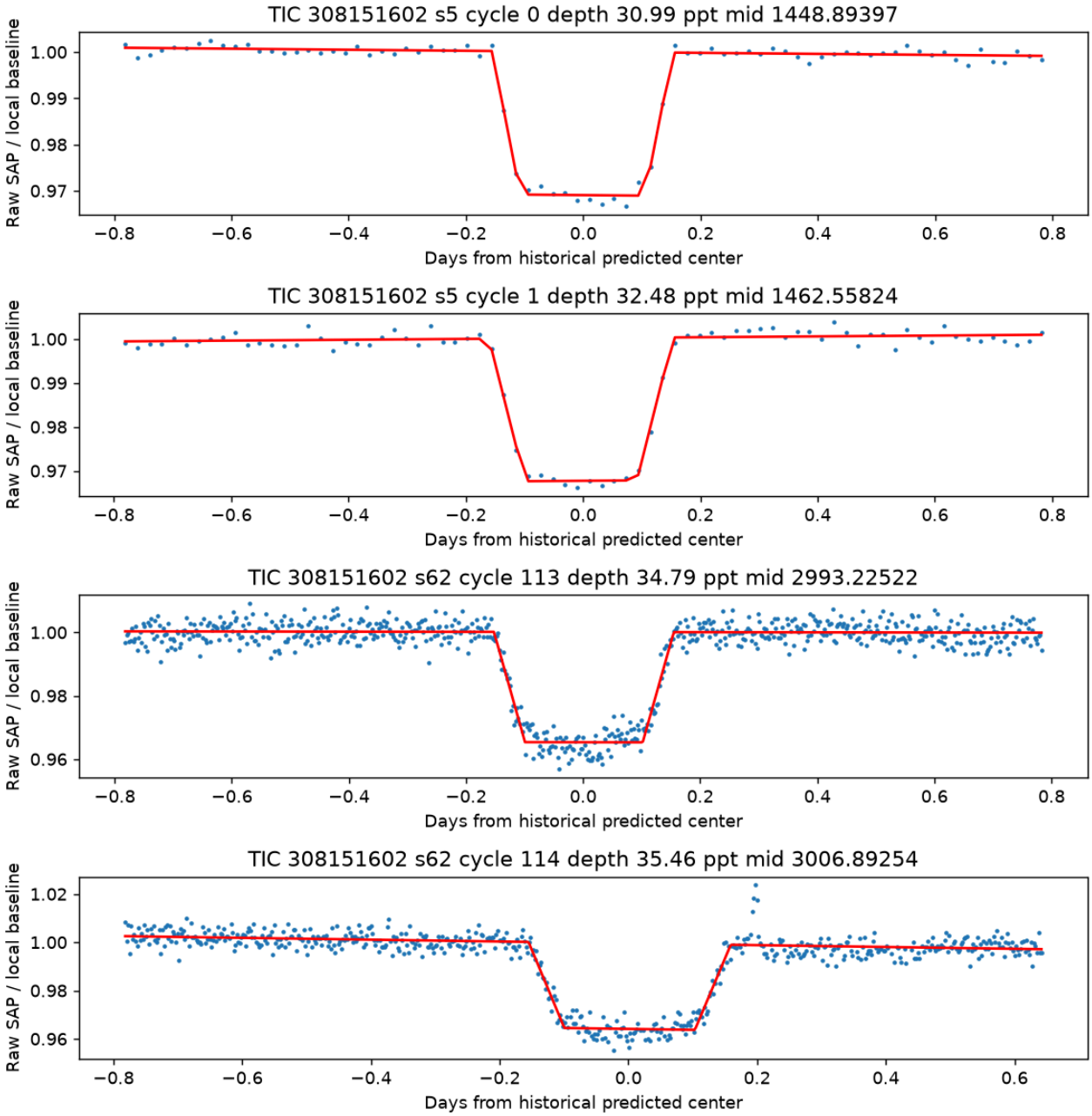
R570. TIC 308151602

physical_verdict: CONFIRMED

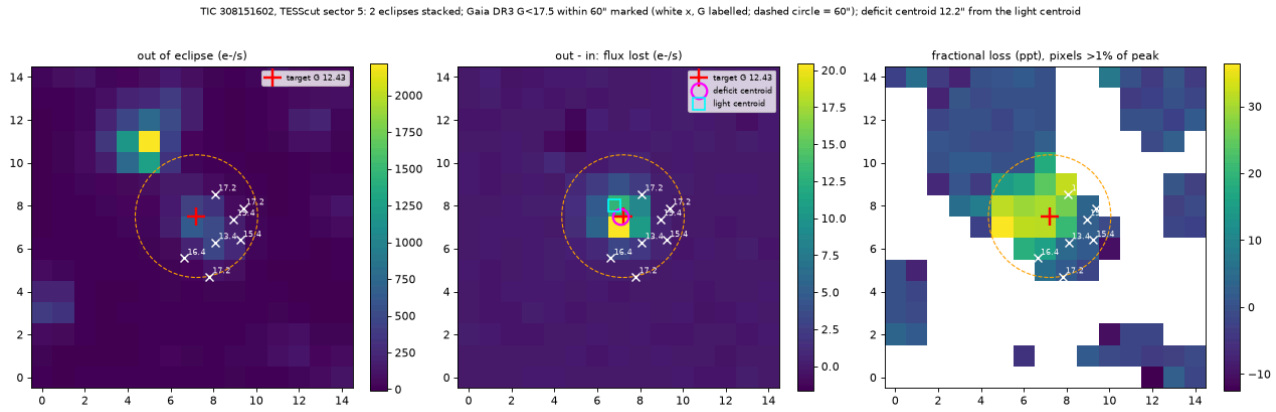
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for improved localization and physical parameters.



R570; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R570; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 308151602: CONFIRMED

Full bounded physical review: **COMPLETE**. Physical outcome and catalogue novelty are separate; a known recovery is not a new addition.

1. **Raw repeated events.** 4 free-center local trapezoid measurements in quality-zero QLP SAP_FLUX; 4 centers enter timing, in sectors [5, 62]. Predictions locate windows and are not measurements. Flagged or uncovered windows are not clean nulls. Four deep events include consecutive pairs. Target-centered deficit is distinct from three capable neighbors at 27.55-42.48 arcsec.
2. **Timing and aliases.** Measured recurring-eclipse period P=13.6666627352 +/- 0.0000655324 days, epoch BTJD=1448.8926609. Epoch kind is the selected recurring eclipse; deeper-primary rank is not assumed. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. There are 2 adjacent same-kind pairs within sectors. Selected depths occur comparably in both cycle parities; doubling does not reveal alternating selected-event kinds. Halving interleaves a shallow secondary or sampled baseline, not the selected eclipse. No significant repeated secondary center is claimed; this is not proof of absence. Actual phase scans and candidate fits are retained; secondary usability is a heuristic amplitude/local-scatter check, not calibrated significance. No exhaustive all-alias search or calibrated secondary-null confidence is asserted.
3. **Pixel attribution.** Fresh S5 difference stack contains 2 selected eclipses. Target WCS pixel= [7.20636019807713, 7.514049487813111], deficit centroid=[7.0611811359367, 7.516917427963772], descriptive offset=0.145 pixel. Neither offset nor image residual is a calibrated localization uncertainty or universal veto. All capable Gaia neighbors: Gaia 5272385342996188544: 27.55 arcsec, G=13.364, intrinsic-loss proxy=0.113; Gaia 5272385411716153728: 32.90 arcsec, G=15.378, intrinsic-loss proxy=0.541; Gaia 5272385342996678656: 42.48 arcsec, G=15.421, intrinsic-loss proxy=0.561. Capability is a G-band pairwise flux-budget proxy; color and aperture response limit precision. Intrinsic neighbor losses are never equated to diluted pixel fractional loss. Same-unit actual deficits and field morphology are independently recorded in ../terra-pixel-audit.json; close blends retain uncertainty.

4. **Alternative reduction.** Reused, read and rerun TESScut primitive subtracts faint-pixel background, with symmetric local side windows and selected eclipse core; fresh target-centered 1x1/3x3/5x5 depths=30.44/27.20/14.81 ppt. QLP fitted median=33.63 ppt. Aperture, dilution and background response explain why these are not intrinsic-depth measurements. This reduction and the retained QLP sector share TESS photons, testing processing robustness rather than independent observations. Distinct sectors remain occasions on one instrument.
5. **Gaia geometry.** Source 5272385411715661184; P=13.6657409443 +/- 0.007320072 days, e=0.05254. Closest edge-on conjunction offset=-0.21566 days (-0.01578 phase), source-specific full covariance linear allowance=0.54505 days. Other conjunction relative phase=0.48207, linear phase allowance=0.04094. Full correlation matrix minimum eigenvalue=0.00116081. These linear allowances are descriptive, not calibrated significance. Near Gaia reference the projected epoch offset=-0.28756 days, period-projection allowance alone=0.00511 days (ignores fit covariance/model systematics). Mass function=0.011536 solar masses; adopted primary mass=1.04, minimum secondary mass=0.271. Approximate central duration=7.58 hours versus observed median=7.46 hours. TIC radius, edge-on geometry and main-sequence secondary assumptions make this a plausibility test, not a fitted physical characterization. Impact parameter can shorten duration. Agreement does not by itself establish ownership.

Deciding limitation: Four deep events include consecutive pairs. Target-centered deficit is distinct from three capable neighbors at 27.55-42.48 arcsec.

Next check: Optional resolved photometry and spectroscopy for improved localization and physical parameters.

Shared catalogue results are in ../../shared-catalogues/coverage-matrix.csv. Successful no-matches are distinct from failed queries; no exhaustive priority claim is made. TIC 295771108 is a known exact ASAS-SN EA recovery.

Sector	Search cycle	Measured BTJD	Formal error d	Input allowance d	Depth ppt	Status
5	0	1448.8939705	0.000655853	0.0104169	30.99	used
5	1	1462.5582418	0.00152712	0.0104169	32.48	used
62	113	2993.2252159	0.000745839	0.00115743	34.79	used
62	114	3006.8925439	0.000782027	0.00115743	35.46	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png, historical originals, ../identity-check.json, ../alias-audit.json, ../secondary-measurements.json, ../secondary-phase-scan.json/png, ../gaia-geometry.json, ../terra-pixel-audit.json and ../helper-audit.md. Exact sources and outputs are covered by ../SHA256SUMS.json. The methods snapshot freezes the read methods without hashing a mutable canonical page.

Stable review key: R570.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s05.json - historical/diff_s05.png - historical/gaia.json - measurements.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

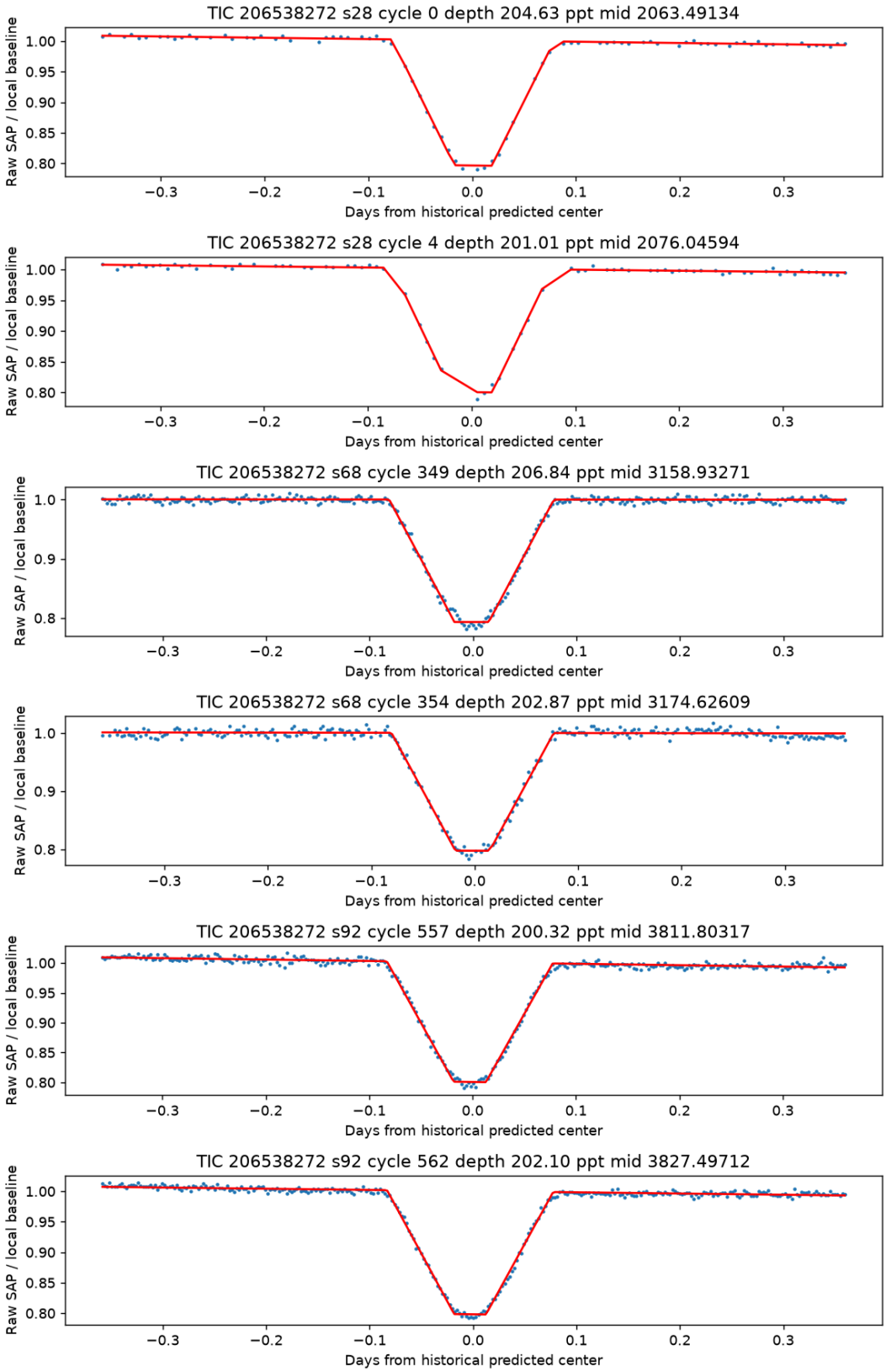
R571. TIC 206538272

physical_verdict: CONFIRMED

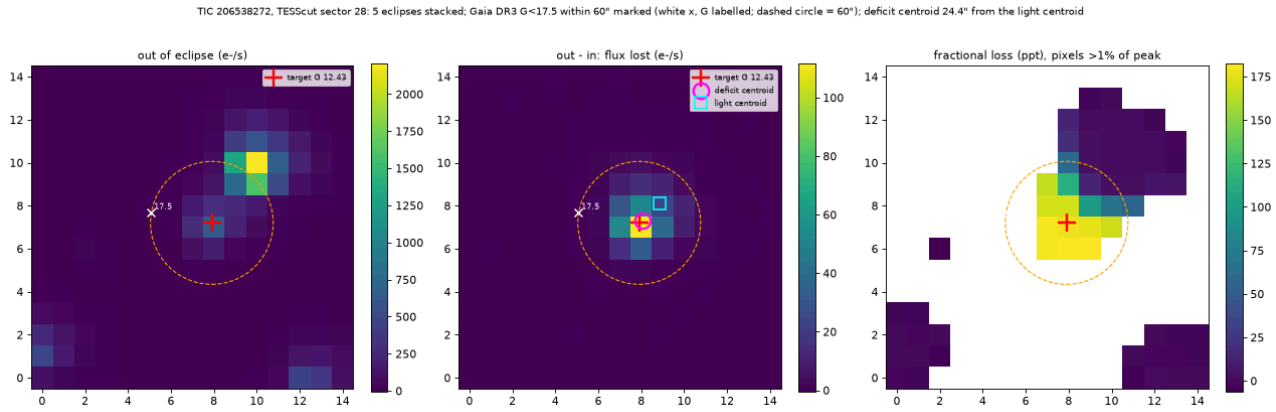
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for improved localization and physical parameters.



R571; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R571; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 206538272: CONFIRMED

Full bounded physical review: **COMPLETE**. Physical outcome and catalogue novelty are separate; a known recovery is not a new addition.

1. **Raw repeated events.** 21 free-center local trapezoid measurements in quality-zero QLP SAP_FLUX; 21 centers enter timing, in sectors [28, 68, 92]. Predictions locate windows and are not measurements. Flagged or uncovered windows are not clean nulls. Twenty-one deep events recur; nineteen shallower secondary fits at phase 0.50006 distinguish eclipse kinds. No capable Gaia neighbor within 60 arcsec.
2. **Timing and aliases.** Measured recurring-eclipse period P=3.1388019249 +/- 0.0000021104 days, epoch BTJD=2063.4906776. Epoch kind is the selected recurring eclipse; deeper-primary rank is not assumed. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. There are 16 adjacent same-kind pairs within sectors. Selected depths occur comparably in both cycle parities; doubling does not reveal alternating selected-event kinds. Halving interleaves a shallow secondary or sampled baseline, not the selected eclipse. Actual measured secondary-candidate phase=0.500060, depth=45.96 ppt, 19 usable fits. Actual phase scans and candidate fits are retained; secondary usability is a heuristic amplitude/local-scatter check, not calibrated significance. No exhaustive all-alias search or calibrated secondary-null confidence is asserted.
3. **Pixel attribution.** Fresh S28 difference stack contains 5 selected eclipses. Target WCS pixel= [7.906055696455212, 7.20140362062328], deficit centroid=[8.041218553144338, 7.290939336940613], descriptive offset=0.162 pixel. Neither offset nor image residual is a calibrated localization uncertainty or universal veto. All capable Gaia neighbors: None under the approximate pairwise G-band flux screen within 60 arcsec.. Capability is a G-band pairwise flux-budget proxy; color and aperture response limit precision. Intrinsic neighbor losses are never equated to diluted pixel fractional loss. Same-unit actual deficits and field morphology are independently recorded in ../terra-pixel-audit.json; close blends retain uncertainty.
4. **Alternative reduction.** Reused, read and rerun TESScut primitive subtracts faint-pixel background, with symmetric local side windows and selected eclipse core; fresh target-centered 1x1/3x3/5x5 depths=179.54/168.57/86.99 ppt. QLP fitted median=204.10 ppt. Aperture, dilution

and background response explain why these are not intrinsic-depth measurements. This reduction and the retained QLP sector share TESS photons, testing processing robustness rather than independent observations. Distinct sectors remain occasions on one instrument.

5. **Gaia geometry.** Source 6822324759628771840; P=3.1385994235 +/- 0.0001972352 days, e=0.05956. Closest edge-on conjunction offset=0.05815 days (0.01853 phase), source-specific full covariance linear allowance=0.09340 days. Other conjunction relative phase=0.47011, linear phase allowance=0.02561. Full correlation matrix minimum eigenvalue=0.0006272. These linear allowances are descriptive, not calibrated significance. Near Gaia reference the projected epoch offset=-0.04978 days, period-projection allowance alone=0.00112 days (ignores fit covariance/model systematics). Mass function=0.089806 solar masses; adopted primary mass=1.06, minimum secondary mass=0.637. Approximate central duration=4.87 hours versus observed median=3.82 hours. TIC radius, edge-on geometry and main-sequence secondary assumptions make this a plausibility test, not a fitted physical characterization. Impact parameter can shorten duration. Agreement does not by itself establish ownership.

Deciding limitation: Twenty-one deep events recur; nineteen shallower secondary fits at phase 0.50006 distinguish eclipse kinds. No capable Gaia neighbor within 60 arcsec.

Next check: Optional resolved photometry and spectroscopy for improved localization and physical parameters.

Shared catalogue results are in ../../shared-catalogues/coverage-matrix.csv. Successful no-matches are distinct from failed queries; no exhaustive priority claim is made. TIC 295771108 is a known exact ASAS-SN EA recovery.

Sector	Search cycle	Measured BTJD	Formal error d	Input allowance d	Depth ppt	Status
28	0	2063.4913360	0.000683353	0.00347229	204.63	used
28	1	2066.6299949	0.000479729	0.00347229	208.19	used
28	2	2069.7683520	0.000408709	0.00347229	203.07	used
28	3	2072.9013610	0.00326028	0.00347229	208.68	used
28	4	2076.0459376	0.000609328	0.00347229	201.01	used
28	5	2079.1848453	0.000345642	0.00347229	206.91	used
28	6	2082.3238183	0.00039525	0.00347229	203.69	used
68	348	3155.7945700	0.000287064	0.00115743	211.08	used
68	349	3158.9327084	0.000179889	0.00115743	206.84	used
68	350	3162.0717640	0.000175358	0.00115743	206.85	used
68	351	3165.2102760	0.000242832	0.00115743	205.94	used
68	353	3171.4878038	0.000182	0.00115743	208.00	used
68	354	3174.6260857	0.000334604	0.00115743	202.87	used
68	355	3177.7654193	0.000194385	0.00115743	210.57	used
92	555	3805.5257811	0.000255913	0.00115752	202.71	used
92	556	3808.6643363	0.000259275	0.00115752	204.10	used
92	557	3811.8031678	0.000368917	0.00115752	200.32	used
92	559	3818.0808595	0.000322586	0.00115752	203.57	used
92	560	3821.2197113	0.000309955	0.00115752	201.29	used
92	561	3824.3585864	0.000328223	0.00115752	200.65	used
92	562	3827.4971159	0.000284624	0.00115752	202.10	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png, historical originals, ../identity-check.json, ../alias-audit.json, ../secondary-measurements.json, ../secondary-phase-scan.json/png, ../gaia-geometry.json, ../terra-pixel-audit.json and ../helper-

audit.md. Exact sources and outputs are covered by ../SHA256SUMS.json. The methods snapshot freezes the read methods without hashing a mutable canonical page.

Stable review key: R571.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s28.json - historical/diff_s28.png - historical/gaia.json - measurements.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

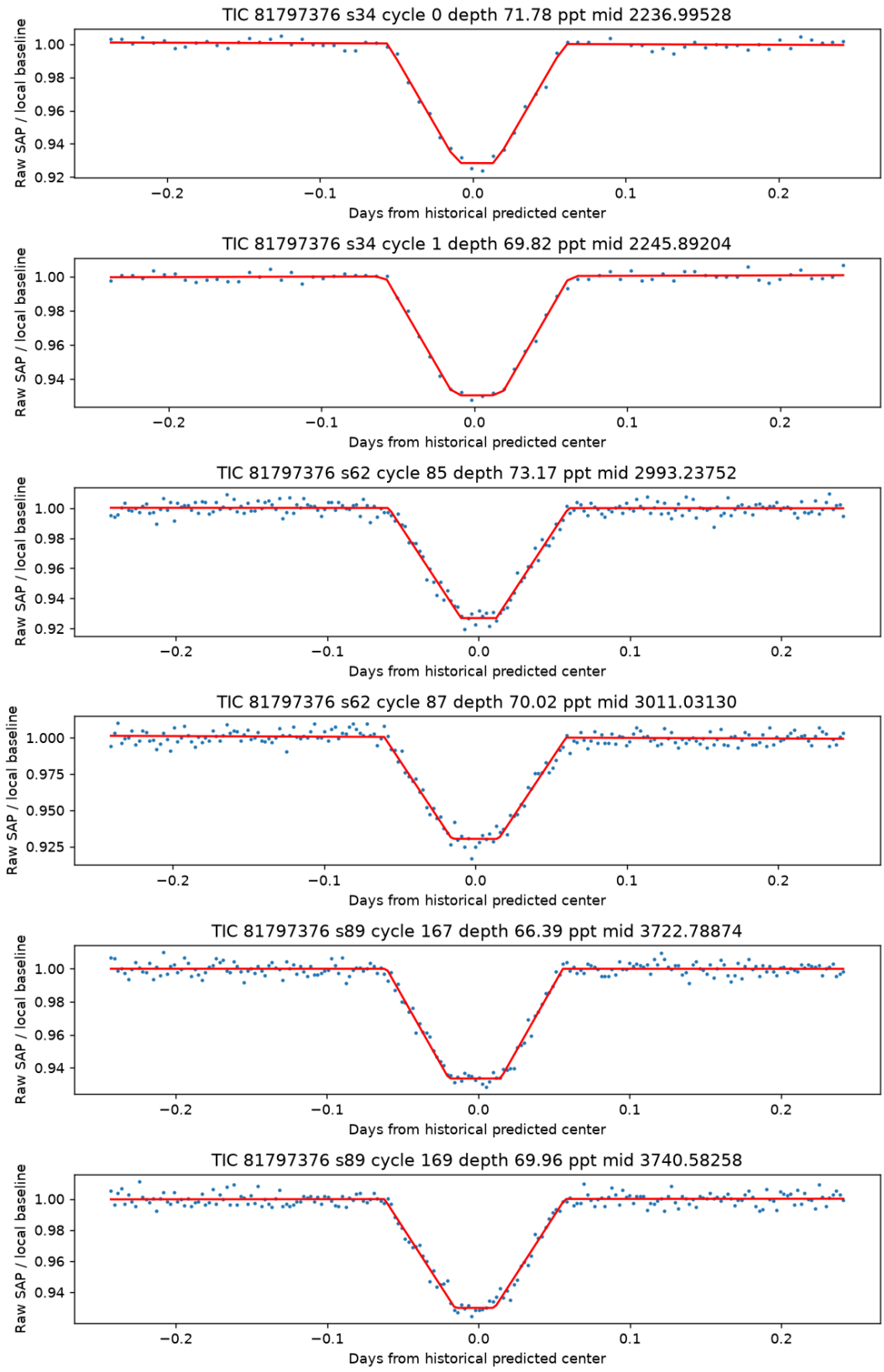
R572. TIC 81797376

physical_verdict: CONFIRMED

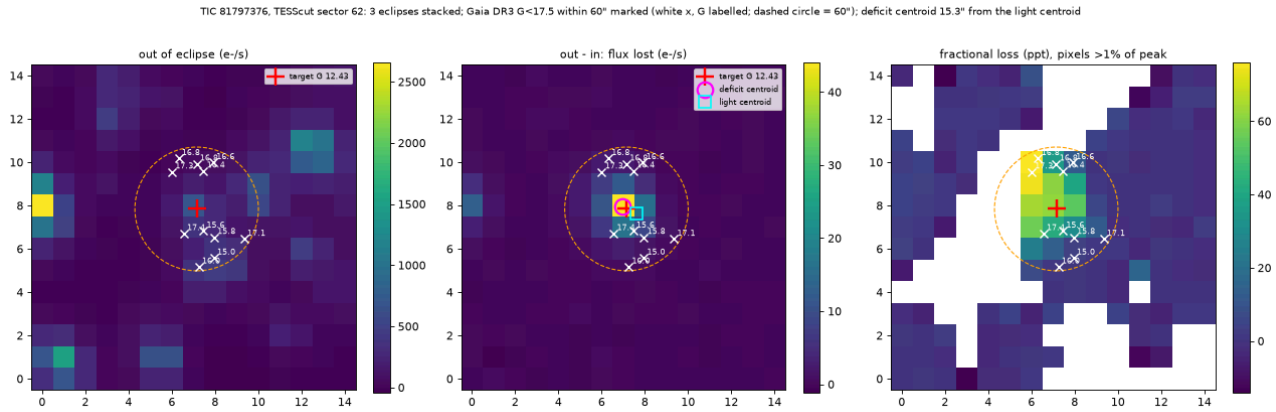
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for improved localization and physical parameters.



R572; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R572; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 81797376: CONFIRMED

Full bounded physical review: **COMPLETE**. Physical outcome and catalogue novelty are separate; a known recovery is not a new addition.

1. **Raw repeated events.** 8 free-center local trapezoid measurements in quality-zero QLP SAP_FLUX; 8 centers enter timing, in sectors [34, 62, 89]. Predictions locate windows and are not measurements. Flagged or uncovered windows are not clean nulls. Eight deep events repeat across three sectors. The one capable neighbor is 49.17 arcsec away, spatially distinct from the target deficit.
2. **Timing and aliases.** Measured recurring-eclipse period P=8.8969660081 +/- 0.0000099543 days, epoch BTJD=2236.9952672. Epoch kind is the selected recurring eclipse; deeper-primary rank is not assumed. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. There are 5 adjacent same-kind pairs within sectors. Selected depths occur comparably in both cycle parities; doubling does not reveal alternating selected-event kinds. Halving interleaves a shallow secondary or sampled baseline, not the selected eclipse. No significant repeated secondary center is claimed; this is not proof of absence. Actual phase scans and candidate fits are retained; secondary usability is a heuristic amplitude/local-scatter check, not calibrated significance. No exhaustive all-alias search or calibrated secondary-null confidence is asserted.
3. **Pixel attribution.** Fresh S62 difference stack contains 3 selected eclipses. Target WCS pixel= [7.1471920408251055, 7.843246791580194], deficit centroid=[6.966701022832995, 7.948381168614263], descriptive offset=0.209 pixel. Neither offset nor image residual is a calibrated localization uncertainty or universal veto. All capable Gaia neighbors: Gaia 5520899641172693504: 49.17 arcsec, G=15.021, intrinsic-loss proxy=0.830. Capability is a G-band pairwise flux-budget proxy; color and aperture response limit precision. Intrinsic neighbor losses are never equated to diluted pixel fractional loss. Same-unit actual deficits and field morphology are independently recorded in ../terra-pixel-audit.json; close blends retain uncertainty.
4. **Alternative reduction.** Reused, read and rerun TESScut primitive subtracts faint-pixel background, with symmetric local side windows and selected eclipse core; fresh target-centered 1x1/3x3/5x5 depths=61.94/53.21/35.07 ppt. QLP fitted median=70.14 ppt. Aperture, dilution and background response explain why these are not intrinsic-depth measurements. This reduction and

the retained QLP sector share TESS photons, testing processing robustness rather than independent observations. Distinct sectors remain occasions on one instrument.

5. **Gaia geometry.** Source 5520899602506878464; P=8.8972814589 +/- 0.001569795 days, e=0.20489. Closest edge-on conjunction offset=0.00129 days (0.00014 phase), source-specific full covariance linear allowance=0.31357 days. Other conjunction relative phase=0.57968, linear phase allowance=0.02107. Full correlation matrix minimum eigenvalue=0.0168949. These linear allowances are descriptive, not calibrated significance. Near Gaia reference the projected epoch offset=0.06690 days, period-projection allowance alone=0.00207 days (ignores fit covariance/model systematics). Mass function=0.046417 solar masses; adopted primary mass=1.57, minimum secondary mass=0.603. Approximate central duration=4.33 hours versus observed median=2.86 hours. TIC radius, edge-on geometry and main-sequence secondary assumptions make this a plausibility test, not a fitted physical characterization. Impact parameter can shorten duration. Agreement does not by itself establish ownership.

Deciding limitation: Eight deep events repeat across three sectors. The one capable neighbor is 49.17 arcsec away, spatially distinct from the target deficit.

Next check: Optional resolved photometry and spectroscopy for improved localization and physical parameters.

Shared catalogue results are in ../../shared-catalogues/coverage-matrix.csv. Successful no-matches are distinct from failed queries; no exhaustive priority claim is made. TIC 295771108 is a known exact ASAS-SN EA recovery.

Sector	Search cycle	Measured BTJD	Formal error d	Input allowance d	Depth ppt	Status
34	0	2236.9952778	0.000511889	0.00347228	71.78	used
34	1	2245.8920355	0.00047896	0.00347228	69.82	used
62	85	2993.2375189	0.000395094	0.0011574	73.17	used
62	86	3002.1342509	0.00041046	0.0011574	70.26	used
62	87	3011.0313011	0.000456843	0.0011574	70.02	used
89	167	3722.7887432	0.000373017	0.0011574	66.39	used
89	168	3731.6853257	0.000413564	0.0011574	70.37	used
89	169	3740.5825821	0.000429503	0.0011574	69.96	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png, historical originals, ../identity-check.json, ../alias-audit.json, ../secondary-measurements.json, ../secondary-phase-scan.json/png, ../gaia-geometry.json, ../terra-pixel-audit.json and ../helper-audit.md. Exact sources and outputs are covered by ../SHA256SUMS.json. The methods snapshot freezes the read methods without hashing a mutable canonical page.

Stable review key: R572.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s62.json - historical/diff_s62.png - historical/gaia.json - measurements.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

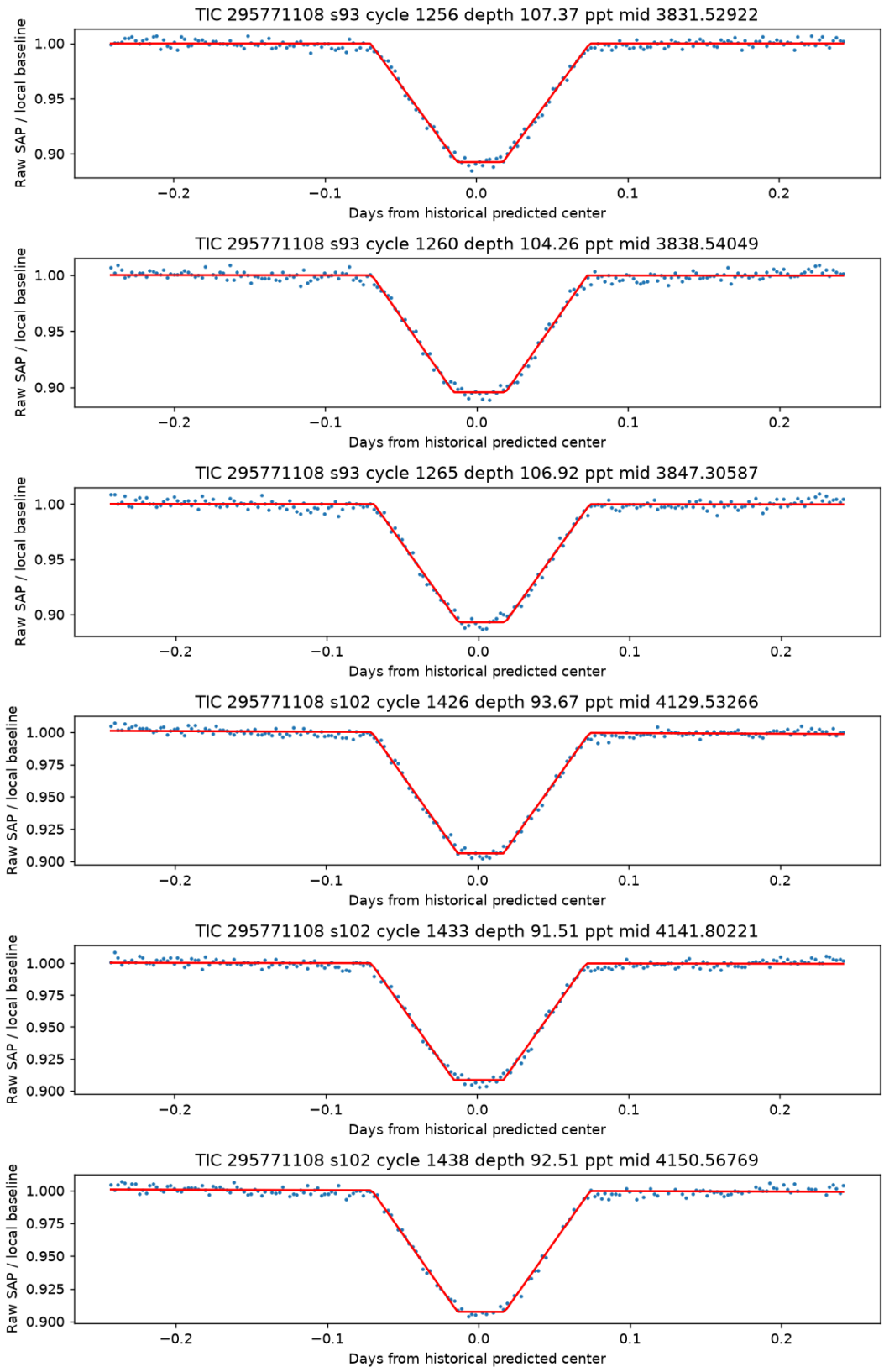
R573. TIC 295771108

physical_verdict: PROBABLE

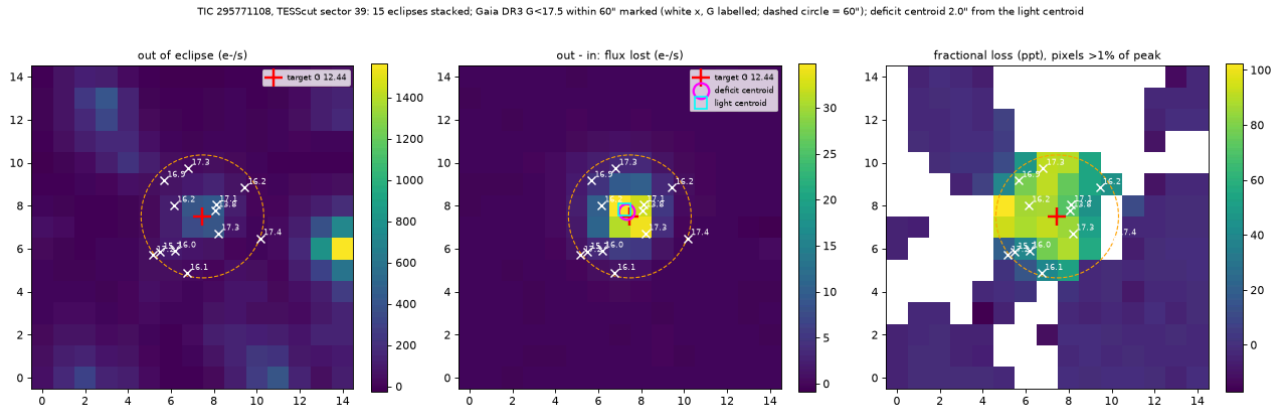
novelty_status: known prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or multi-source PRF analysis of the 13.45-arcsec blend, and a joint orbit/eclipse fit.



R573; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R573; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 295771108: PROBABLE

Full bounded physical review: **COMPLETE**. Physical outcome and catalogue novelty are separate; a known recovery is not a new addition.

1. **Raw repeated events.** 23 free-center local trapezoid measurements in quality-zero QLP SAP_FLUX; 23 centers enter timing, in sectors [93, 102]. Predictions locate windows and are not measurements. Flagged or uncovered windows are not clean nulls. Known ASAS-SN EA recovery, not a new addition. Deep eclipses and shallower phase-0.50052 events repeat, but the capable G=13.79 neighbor at 13.45 arcsec is unresolved: target and neighbor pixels lose 34.85 and 32.18 e-/s. Gaia predicts the other conjunction at phase 0.60848 +/-0.03919 (linear covariance allowance) versus measured 0.50052. This descriptive tension and blend leave geometry/ownership less secure.
2. **Timing and aliases.** Measured recurring-eclipse period P=1.7529583273 +/- 0.0000028609 days, epoch BTJD=3831.5290385. Epoch kind is the selected recurring eclipse; deeper-primary rank is not assumed. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. There are 18 adjacent same-kind pairs within sectors. Selected depths occur comparably in both cycle parities; doubling does not reveal alternating selected-event kinds. Halving interleaves a shallow secondary or sampled baseline, not the selected eclipse. Actual measured secondary-candidate phase=0.500519, depth=17.62 ppt, 24 usable fits. Actual phase scans and candidate fits are retained; secondary usability is a heuristic amplitude/local-scatter check, not calibrated significance. No exhaustive all-alias search or calibrated secondary-null confidence is asserted.
3. **Pixel attribution.** Fresh S39 difference stack contains 15 selected eclipses. Target WCS pixel= [7.468692683084559, 7.500484603313893], deficit centroid=[7.293330989219066, 7.725727339394298], descriptive offset=0.285 pixel. Neither offset nor image residual is a calibrated localization uncertainty or universal veto. All capable Gaia neighbors: Gaia 5782390478649218304: 13.45 arcsec, G=13.792, intrinsic-loss proxy=0.471. Capability is a G-band pairwise flux-budget proxy; color and aperture response limit precision. Intrinsic neighbor losses are never equated to diluted pixel fractional loss. Same-unit actual deficits and field morphology are independently recorded in ../terra-pixel-audit.json; close blends retain uncertainty.

4. **Alternative reduction.** Reused, read and rerun TESScut primitive subtracts faint-pixel background, with symmetric local side windows and selected eclipse core; fresh target-centered 1x1/3x3/5x5 depths=87.57/83.34/73.83 ppt. QLP fitted median=105.13 ppt. Aperture, dilution and background response explain why these are not intrinsic-depth measurements. The cutout sector is absent from the retained QLP files. It is a distinct observing occasion on TESS and an alternative extraction of that sector; no direct same-sector QLP comparison is claimed. Distinct sectors remain occasions on one instrument.
5. **Gaia geometry.** Source 5782390478649219456; P=1.7530448129 +/- 7.2839124e-05 days, e=0.17145. Closest edge-on conjunction offset=-0.06051 days (-0.03452 phase), source-specific full covariance linear allowance=0.12954 days. Other conjunction relative phase=0.60848, linear phase allowance=0.03919. Full correlation matrix minimum eigenvalue=0.00700217. These linear allowances are descriptive, not calibrated significance. Near Gaia reference the projected epoch offset=0.10935 days, period-projection allowance alone=0.00562 days (ignores fit covariance/model systematics). Mass function=0.0423 solar masses; adopted primary mass=1.3, minimum secondary mass=0.519. Approximate central duration=4.32 hours versus observed median=3.45 hours. TIC radius, edge-on geometry and main-sequence secondary assumptions make this a plausibility test, not a fitted physical characterization. Impact parameter can shorten duration. Agreement does not by itself establish ownership.

Deciding limitation: Known ASAS-SN EA recovery, not a new addition. Deep eclipses and shallower phase-0.50052 events repeat, but the capable G=13.79 neighbor at 13.45 arcsec is unresolved: target and neighbor pixels lose 34.85 and 32.18 e-/s. Gaia predicts the other conjunction at phase 0.60848 +/-0.03919 (linear covariance allowance) versus measured 0.50052. This descriptive tension and blend leave geometry/ownership less secure.

Next check: Resolved eclipse photometry or multi-source PRF analysis of the 13.45-arcsec blend, and a joint orbit/eclipse fit.

Shared catalogue results are in ../../shared-catalogues/coverage-matrix.csv. Successful no-matches are distinct from failed queries; no exhaustive priority claim is made. TIC 295771108 is a known exact ASAS-SN EA recovery.

Sector	Search cycle	Measured BTJD	Formal error d	Input allowance d	Depth ppt	Status
93	1256	3831.5292223	0.000250746	0.0011574	107.37	used
93	1257	3833.2816702	0.000231606	0.0011574	108.27	used
93	1258	3835.0348137	0.000263092	0.0011574	106.27	used
93	1259	3836.7879689	0.000281906	0.0011574	106.06	used
93	1260	3838.5404906	0.000256496	0.0011574	104.26	used
93	1261	3840.2938731	0.000275153	0.0011574	105.22	used
93	1263	3843.7997741	0.000327624	0.0011574	107.53	used
93	1264	3845.5529533	0.000243372	0.0011574	107.97	used
93	1265	3847.3058684	0.00028626	0.0011574	106.92	used
93	1266	3849.0588839	0.000270049	0.0011574	105.38	used
93	1267	3850.8113452	0.000260904	0.0011574	105.73	used
93	1268	3852.5644880	0.000291096	0.0011574	105.30	used
93	1269	3854.3175469	0.000327214	0.0011574	105.13	used
102	1426	4129.5326569	0.000243083	0.00115741	93.67	used
102	1427	4131.2852698	0.000214072	0.00115741	92.11	used
102	1429	4134.7911385	0.000219308	0.00115741	92.15	used
102	1430	4136.5442859	0.000224528	0.00115741	90.42	used
102	1433	4141.8022068	0.000252225	0.00115741	91.51	used
102	1434	4143.5554169	0.000261007	0.00115741	92.46	used

Sector	Search cycle	Measured BTJD	Formal error d	Input allowance d	Depth ppt	Status
102	1435	4145.3080866	0.000413906	0.00115741	89.59	used
102	1436	4147.0611214	0.000233417	0.00115741	91.88	used
102	1437	4148.8140190	0.000247007	0.00115741	91.84	used
102	1438	4150.5676908	0.00032802	0.00115741	92.51	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png, historical originals, ../identity-check.json, ../alias-audit.json, ../secondary-measurements.json, ../secondary-phase-scan.json/png, ../gaia-geometry.json, ../terra-pixel-audit.json and ../helper-audit.md. Exact sources and outputs are covered by ../SHA256SUMS.json. The methods snapshot freezes the read methods without hashing a mutable canonical page.

Stable review key: R573.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s39.json - historical/diff_s39.png - historical/gaia.json - measurements.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

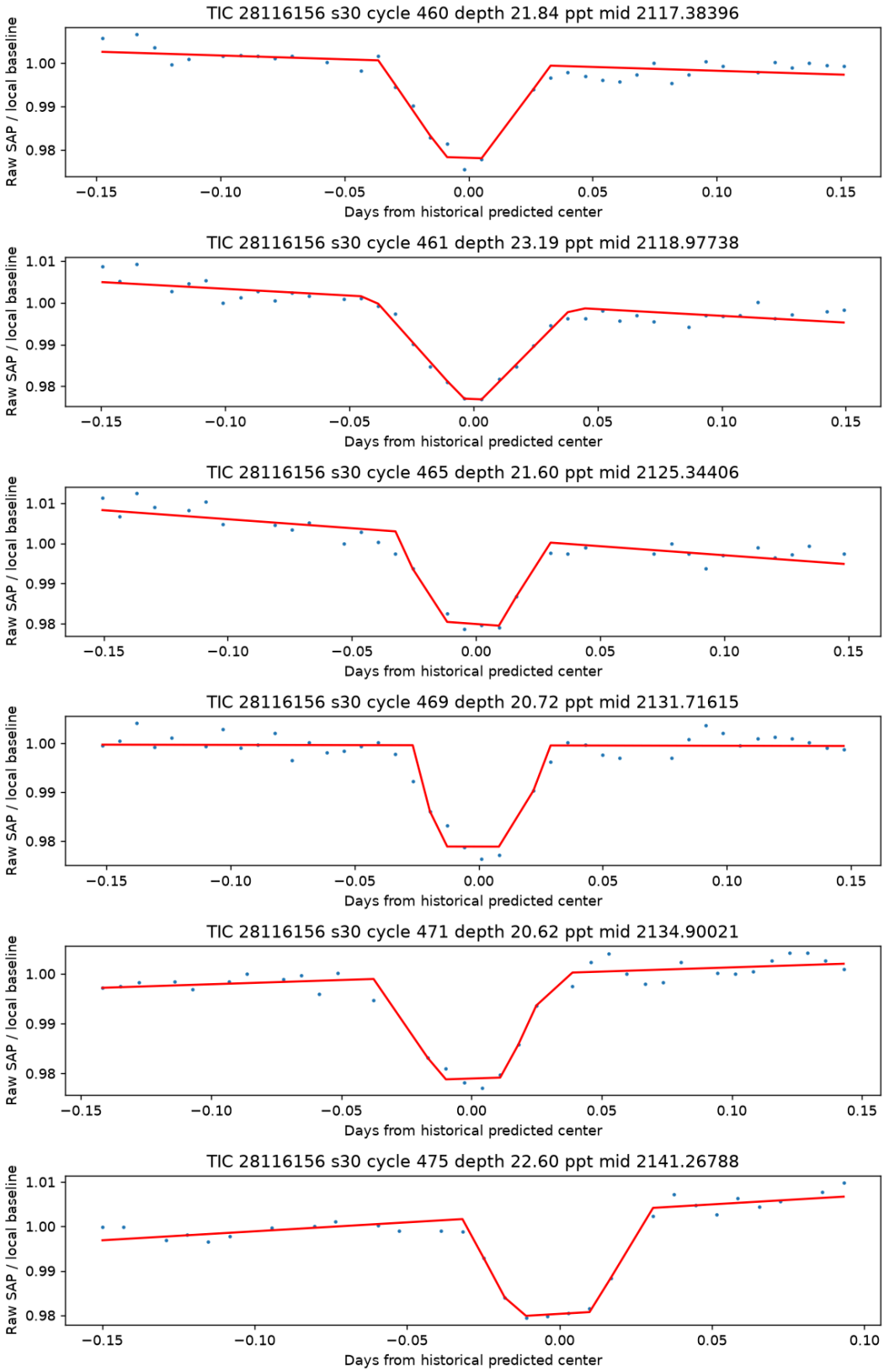
R574. TIC 28116156

physical_verdict: CONFIRMED

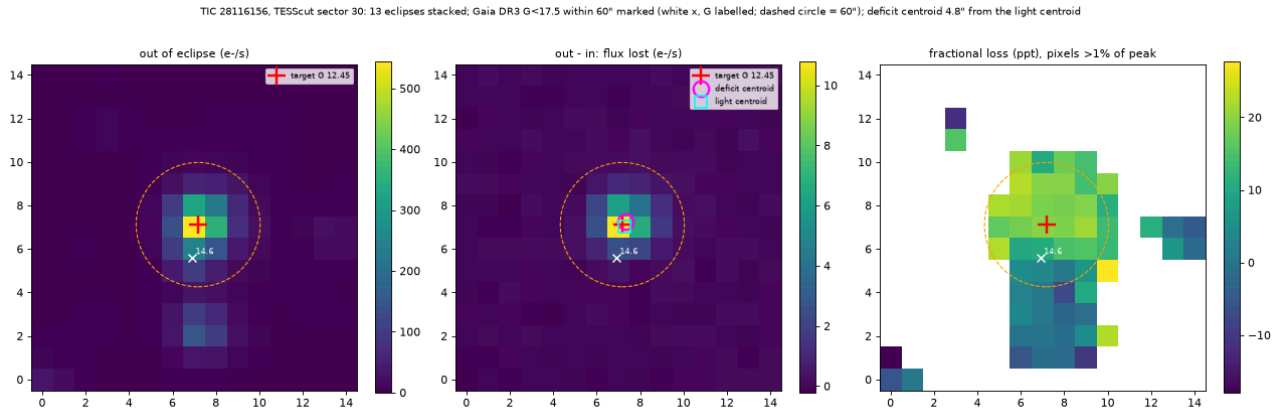
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for improved localization and physical parameters.



R574; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R574; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 28116156: CONFIRMED

Full bounded physical review: **COMPLETE**. Physical outcome and catalogue novelty are separate; a known recovery is not a new addition.

1. **Raw repeated events.** 9 free-center local trapezoid measurements in quality-zero QLP SAP_FLUX; 9 centers enter timing, in sectors [30]. Predictions locate windows and are not measurements. Flagged or uncovered windows are not clean nulls. Nine short eclipses in S30 establish same-kind recurrence; the period/error use this actual within-sector baseline. Coarse S3 does not contribute precise timings. The capable neighbor at 30.94 arcsec is spatially distinct from the compact deficit.
2. **Timing and aliases.** Measured recurring-eclipse period P=1.5922960846 +/- 0.0002318805 days, epoch BTJD=2117.3845444. Epoch kind is the selected recurring eclipse; deeper-primary rank is not assumed. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. There are 4 adjacent same-kind pairs within sectors. Selected depths occur comparably in both cycle parities; doubling does not reveal alternating selected-event kinds. Halving interleaves a shallow secondary or sampled baseline, not the selected eclipse. No significant repeated secondary center is claimed; this is not proof of absence. Actual phase scans and candidate fits are retained; secondary usability is a heuristic amplitude/local-scatter check, not calibrated significance. No exhaustive all-alias search or calibrated secondary-null confidence is asserted.
3. **Pixel attribution.** Fresh S30 difference stack contains 13 selected eclipses. Target WCS pixel= [7.182217395273183, 7.11712281687406], deficit centroid=[7.319444970737896, 7.254837847209099], descriptive offset=0.194 pixel. Neither offset nor image residual is a calibrated localization uncertainty or universal veto. All capable Gaia neighbors: Gaia 2355817466259768448: 30.94 arcsec, G=14.559, intrinsic-loss proxy=0.173. Capability is a G-band pairwise flux-budget proxy; color and aperture response limit precision. Intrinsic neighbor losses are never equated to diluted pixel fractional loss. Same-unit actual deficits and field morphology are independently recorded in ../terra-pixel-audit.json; close blends retain uncertainty.
4. **Alternative reduction.** Reused, read and rerun TESScut primitive subtracts faint-pixel background, with symmetric local side windows and selected eclipse core; fresh target-centered

1x1/3x3/5x5 depths=19.84/17.47/16.58 ppt. QLP fitted median=21.60 ppt. Aperture, dilution and background response explain why these are not intrinsic-depth measurements. This reduction and the retained QLP sector share TESS photons, testing processing robustness rather than independent observations. Distinct sectors remain occasions on one instrument.

5. **Gaia geometry.** Source 2355817565042831232; P=1.5923420791 +/- 8.154e-05 days, e=0.03598. Closest edge-on conjunction offset=-0.08013 days (-0.05032 phase), source-specific full covariance linear allowance=0.09512 days. Other conjunction relative phase=0.51027, linear phase allowance=0.02247. Full correlation matrix minimum eigenvalue=0.000493799. These linear allowances are descriptive, not calibrated significance. Near Gaia reference the projected epoch offset=-0.03023 days, period-projection allowance alone=0.25159 days (ignores fit covariance/model systematics). Mass function=0.059513 solar masses; adopted primary mass=1.17, minimum secondary mass=0.563. Approximate central duration=3.49 hours versus observed median=1.40 hours. TIC radius, edge-on geometry and main-sequence secondary assumptions make this a plausibility test, not a fitted physical characterization. Impact parameter can shorten duration. Agreement does not by itself establish ownership.

Deciding limitation: Nine short eclipses in S30 establish same-kind recurrence; the period/error use this actual within-sector baseline. Coarse S3 does not contribute precise timings. The capable neighbor at 30.94 arcsec is spatially distinct from the compact deficit.

Next check: Optional resolved photometry and spectroscopy for improved localization and physical parameters.

Shared catalogue results are in ../../shared-catalogues/coverage-matrix.csv. Successful no-matches are distinct from failed queries; no exhaustive priority claim is made. TIC 295771108 is a known exact ASAS-SN EA recovery.

Sector	Search cycle	Measured BTJD	Formal error d	Input allowance d	Depth ppt	Status
30	460	2117.3839584	0.00193591	0.00347219	21.84	used
30	461	2118.9773778	0.00210344	0.00347219	23.19	used
30	464	2123.7548855	0.00212481	0.00347219	21.65	used
30	465	2125.3440626	0.00265762	0.00347219	21.60	used
30	469	2131.7161471	0.000554862	0.00347219	20.72	used
30	470	2133.3073925	0.00053522	0.00347219	20.42	used
30	471	2134.9002135	0.00191663	0.00347219	20.62	used
30	473	2138.0851194	0.000953467	0.00347219	19.93	used
30	475	2141.2678764	0.00152062	0.00347219	22.60	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png, historical originals, ../identity-check.json, ../alias-audit.json, ../secondary-measurements.json, ../secondary-phase-scan.json/png, ../gaia-geometry.json, ../terra-pixel-audit.json and ../helper-audit.md. Exact sources and outputs are covered by ../SHA256SUMS.json. The methods snapshot freezes the read methods without hashing a mutable canonical page.

Stable review key: R574.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s30.json - historical/diff_s30.png - historical/gaia.json - measurements.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

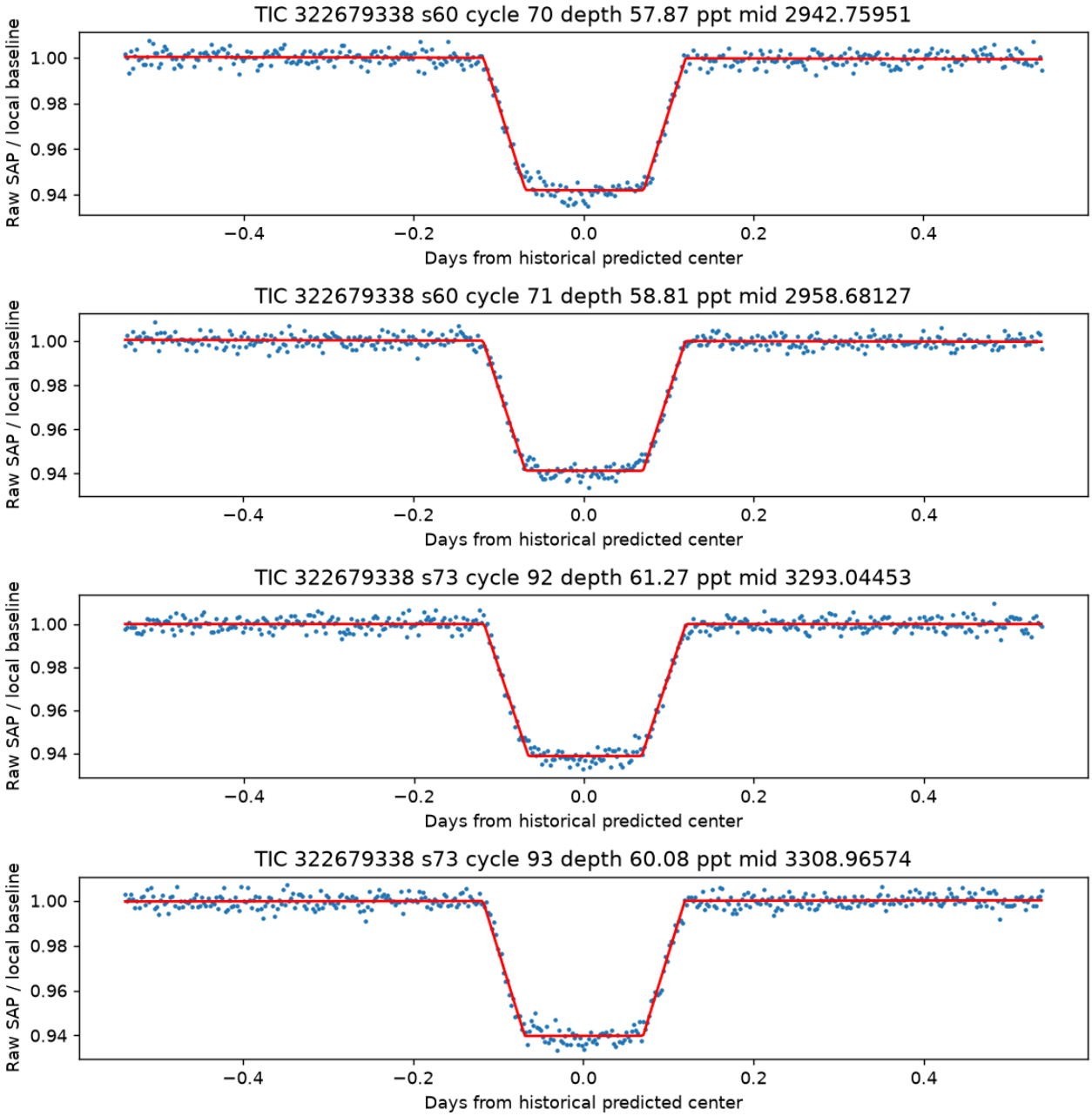
R575. TIC 322679338

physical_verdict: CONFIRMED

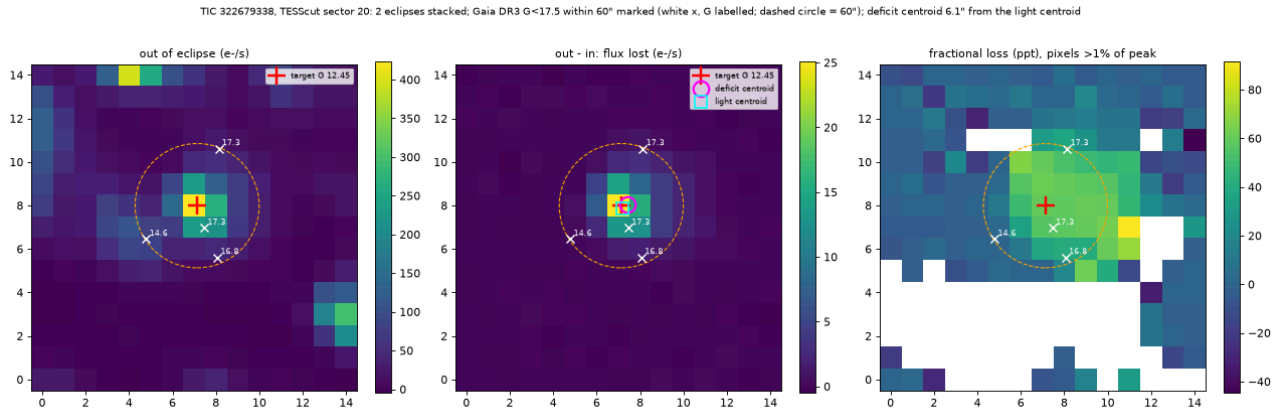
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for improved localization and physical parameters.



R575; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R575; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 322679338: CONFIRMED

Full bounded physical review: **COMPLETE**. Physical outcome and catalogue novelty are separate; a known recovery is not a new addition.

1. **Raw repeated events.** 4 free-center local trapezoid measurements in quality-zero QLP SAP_FLUX; 4 centers enter timing, in sectors [60, 73]. Predictions locate windows and are not measurements. Flagged or uncovered windows are not clean nulls. Four deep events form consecutive pairs in S60/S73. Two shallow secondary candidates in S19 lie at actual phase 0.54415, with other-sector confirmation limited. The capable neighbor at 54.93 arcsec is separate from the target deficit.
2. **Timing and aliases.** Measured recurring-eclipse period P=15.9220326620 +/- 0.0000525550 days, epoch BTJD=2942.7593854. Epoch kind is the selected recurring eclipse; deeper-primary rank is not assumed. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. There are 2 adjacent same-kind pairs within sectors. Selected depths occur comparably in both cycle parities; doubling does not reveal alternating selected-event kinds. Halving interleaves a shallow secondary or sampled baseline, not the selected eclipse. Actual measured secondary-candidate phase=0.544147, depth=5.39 ppt, 2 usable fits. Actual phase scans and candidate fits are retained; secondary usability is a heuristic amplitude/local-scatter check, not calibrated significance. No exhaustive all-alias search or calibrated secondary-null confidence is asserted.
3. **Pixel attribution.** Fresh S20 difference stack contains 2 selected eclipses. Target WCS pixel= [7.134597425825657, 7.989513142108965], deficit centroid=[7.426056128055689, 8.0361007363305], descriptive offset=0.295 pixel. Neither offset nor image residual is a calibrated localization uncertainty or universal veto. All capable Gaia neighbors: Gaia 1004675112045220352: 54.93 arcsec, G=14.588, intrinsic-loss proxy=0.485. Capability is a G-band pairwise flux-budget proxy; color and aperture response limit precision. Intrinsic neighbor losses are never equated to diluted pixel fractional loss. Same-unit actual deficits and field morphology are independently recorded in ../terra-pixel-audit.json; close blends retain uncertainty.

4. **Alternative reduction.** Reused, read and rerun TESScut primitive subtracts faint-pixel background, with symmetric local side windows and selected eclipse core; fresh target-centered 1x1/3x3/5x5 depths=59.23/57.02/50.67 ppt. QLP fitted median=59.45 ppt. Aperture, dilution and background response explain why these are not intrinsic-depth measurements. The cutout sector is absent from the retained QLP files. It is a distinct observing occasion on TESS and an alternative extraction of that sector; no direct same-sector QLP comparison is claimed. Distinct sectors remain occasions on one instrument.
5. **Gaia geometry.** Source 1004675146404805888; P=15.9191458295 +/- 0.007424117 days, e=0.15158. Closest edge-on conjunction offset=0.21015 days (0.01320 phase), source-specific full covariance linear allowance=1.11199 days. Other conjunction relative phase=0.51939, linear phase allowance=0.05055. Full correlation matrix minimum eigenvalue=0.00377364. These linear allowances are descriptive, not calibrated significance. Near Gaia reference the projected epoch offset=-0.25174 days, period-projection allowance alone=0.00841 days (ignores fit covariance/model systematics). Mass function=0.015095 solar masses; adopted primary mass=1.24, minimum secondary mass=0.334. Approximate central duration=5.87 hours versus observed median=5.70 hours. TIC radius, edge-on geometry and main-sequence secondary assumptions make this a plausibility test, not a fitted physical characterization. Impact parameter can shorten duration. Agreement does not by itself establish ownership.

Deciding limitation: Four deep events form consecutive pairs in S60/S73. Two shallow secondary candidates in S19 lie at actual phase 0.54415, with other-sector confirmation limited. The capable neighbor at 54.93 arcsec is separate from the target deficit.

Next check: Optional resolved photometry and spectroscopy for improved localization and physical parameters.

Shared catalogue results are in ../../shared-catalogues/coverage-matrix.csv. Successful no-matches are distinct from failed queries; no exhaustive priority claim is made. TIC 295771108 is a known exact ASAS-SN EA recovery.

Sector	Search cycle	Measured BTJD	Formal error d	Input allowance d	Depth ppt	Status
60	70	2942.7595051	0.000362647	0.00115739	57.87	used
60	71	2958.6812734	0.000314788	0.00115739	58.81	used
73	92	3293.0445300	0.000333904	0.00115742	61.27	used
73	93	3308.9657353	0.000331667	0.00115742	60.08	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png, historical originals, ../identity-check.json, ../alias-audit.json, ../secondary-measurements.json, ../secondary-phase-scan.json/png, ../gaia-geometry.json, ../terra-pixel-audit.json and ../helper-audit.md. Exact sources and outputs are covered by ../SHA256SUMS.json. The methods snapshot freezes the read methods without hashing a mutable canonical page.

Stable review key: R575.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s20.json - historical/diff_s20.png - historical/gaia.json - measurements.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

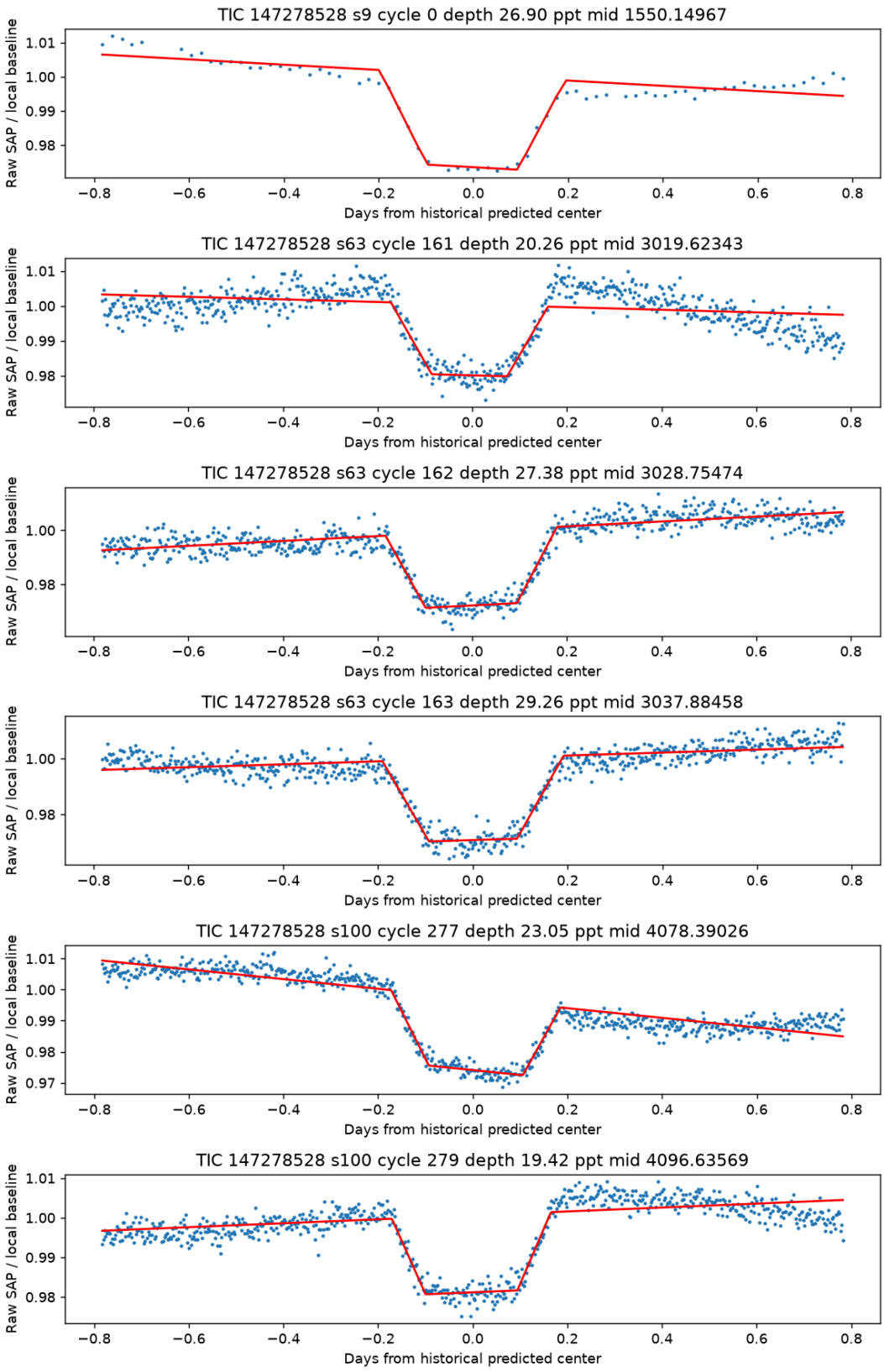
R576. TIC 147278528

physical_verdict: CONFIRMED

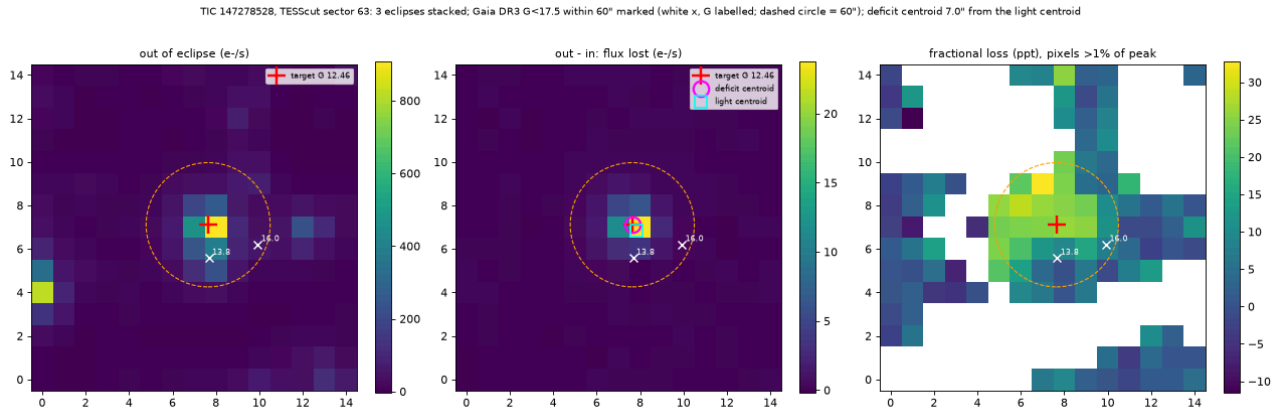
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for improved localization and physical parameters.



R576; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R576; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 147278528: CONFIRMED

Full bounded physical review: **COMPLETE**. Physical outcome and catalogue novelty are separate; a known recovery is not a new addition.

1. **Raw repeated events.** 6 free-center local trapezoid measurements in quality-zero QLP SAP_FLUX; 6 centers enter timing, in sectors [9, 63, 100]. Predictions locate windows and are not measurements. Flagged or uncovered windows are not clean nulls. Six selected eclipses repeat on a changing variable baseline. The target deficit is compact and separate from capable neighbors at 32.19 and 51.96 arcsec. Variability limits baseline, timing and depth precision.
2. **Timing and aliases.** Measured recurring-eclipse period P=9.1271967802 +/- 0.0000265159 days, epoch BTJD=1550.1490604. Epoch kind is the selected recurring eclipse; deeper-primary rank is not assumed. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. There are 2 adjacent same-kind pairs within sectors. Selected depths occur comparably in both cycle parities; doubling does not reveal alternating selected-event kinds. Halving interleaves a shallow secondary or sampled baseline, not the selected eclipse. No significant repeated secondary center is claimed; this is not proof of absence. Actual phase scans and candidate fits are retained; secondary usability is a heuristic amplitude/local-scatter check, not calibrated significance. No exhaustive all-alias search or calibrated secondary-null confidence is asserted.
3. **Pixel attribution.** Fresh S63 difference stack contains 3 selected eclipses. Target WCS pixel= [7.64172824112652, 7.1112813115194395], deficit centroid=[7.671350909433753, 7.133537740119036], descriptive offset=0.037 pixel. Neither offset nor image residual is a calibrated localization uncertainty or universal veto. All capable Gaia neighbors: Gaia 5387954219917479424: 32.19 arcsec, G=13.830, intrinsic-loss proxy=0.113; Gaia 5387954185557741696: 51.96 arcsec, G=15.980, intrinsic-loss proxy=0.666. Capability is a G-band pairwise flux-budget proxy; color and aperture response limit precision. Intrinsic neighbor losses are never equated to diluted pixel fractional loss. Same-unit actual deficits and field morphology are independently recorded in ../terra-pixel-audit.json; close blends retain uncertainty.
4. **Alternative reduction.** Reused, read and rerun TESScut primitive subtracts faint-pixel background, with symmetric local side windows and selected eclipse core; fresh target-centered

1x1/3x3/5x5 depths=26.18/21.68/18.54 ppt. QLP fitted median=24.98 ppt. Aperture, dilution and background response explain why these are not intrinsic-depth measurements. This reduction and the retained QLP sector share TESS photons, testing processing robustness rather than independent observations. Distinct sectors remain occasions on one instrument.

5. **Gaia geometry.** Source 5387954254277219968; P=9.1270980023 +/- 0.0010151658 days, e=0.01093. Closest edge-on conjunction offset=0.05706 days (0.00625 phase), source-specific full covariance linear allowance=0.13259 days. Other conjunction relative phase=0.50668, linear phase allowance=0.01718. Full correlation matrix minimum eigenvalue=4.38995e-05. These linear allowances are descriptive, not calibrated significance. Near Gaia reference the projected epoch offset=0.04451 days, period-projection allowance alone=0.00337 days (ignores fit covariance/model systematics). Mass function=0.038834 solar masses; adopted primary mass=1.08, minimum secondary mass=0.45. Approximate central duration=13.26 hours versus observed median=8.64 hours. TIC radius, edge-on geometry and main-sequence secondary assumptions make this a plausibility test, not a fitted physical characterization. Impact parameter can shorten duration. Agreement does not by itself establish ownership.

Deciding limitation: Six selected eclipses repeat on a changing variable baseline. The target deficit is compact and separate from capable neighbors at 32.19 and 51.96 arcsec. Variability limits baseline, timing and depth precision.

Next check: Optional resolved photometry and spectroscopy for improved localization and physical parameters.

Shared catalogue results are in ../../shared-catalogues/coverage-matrix.csv. Successful no-matches are distinct from failed queries; no exhaustive priority claim is made. TIC 295771108 is a known exact ASAS-SN EA recovery.

Sector	Search cycle	Measured BTJD	Formal error d	Input allowance d	Depth ppt	Status
9	0	1550.1496702	0.00725588	0.0104168	26.90	used
63	161	3019.6234282	0.00230331	0.00230331	20.26	used
63	162	3028.7547428	0.00264499	0.00264499	27.38	used
63	163	3037.8845799	0.00170881	0.00170881	29.26	used
100	277	4078.3902556	0.0052678	0.0052678	23.05	used
100	279	4096.6356940	0.00213345	0.00213345	19.42	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png, historical originals, ../identity-check.json, ../alias-audit.json, ../secondary-measurements.json, ../secondary-phase-scan.json/png, ../gaia-geometry.json, ../terra-pixel-audit.json and ../helper-audit.md. Exact sources and outputs are covered by ../SHA256SUMS.json. The methods snapshot freezes the read methods without hashing a mutable canonical page.

Stable review key: R576.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s63.json - historical/diff_s63.png - historical/gaia.json - measurements.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

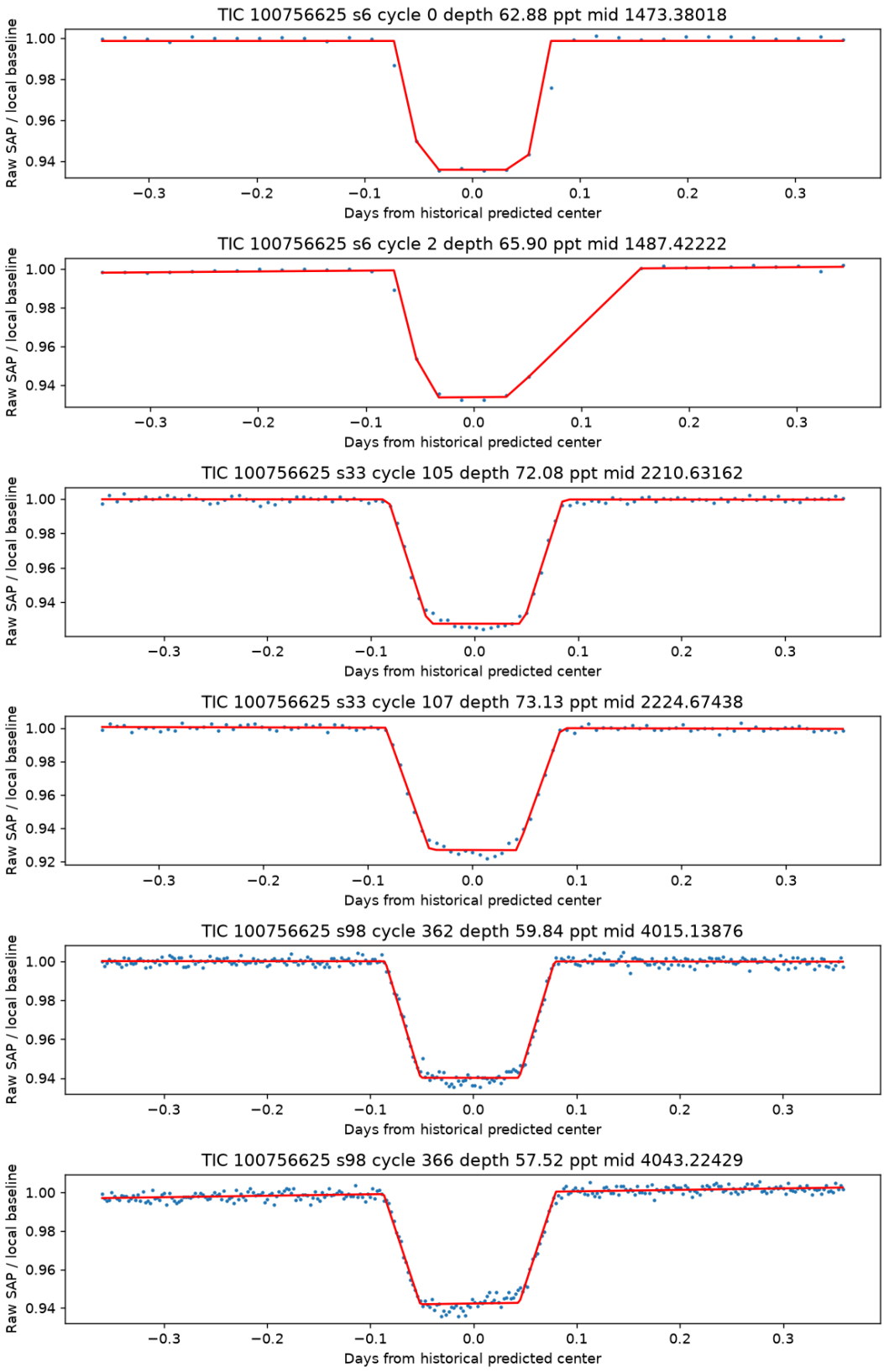
R577. TIC 100756625

physical_verdict: CONFIRMED

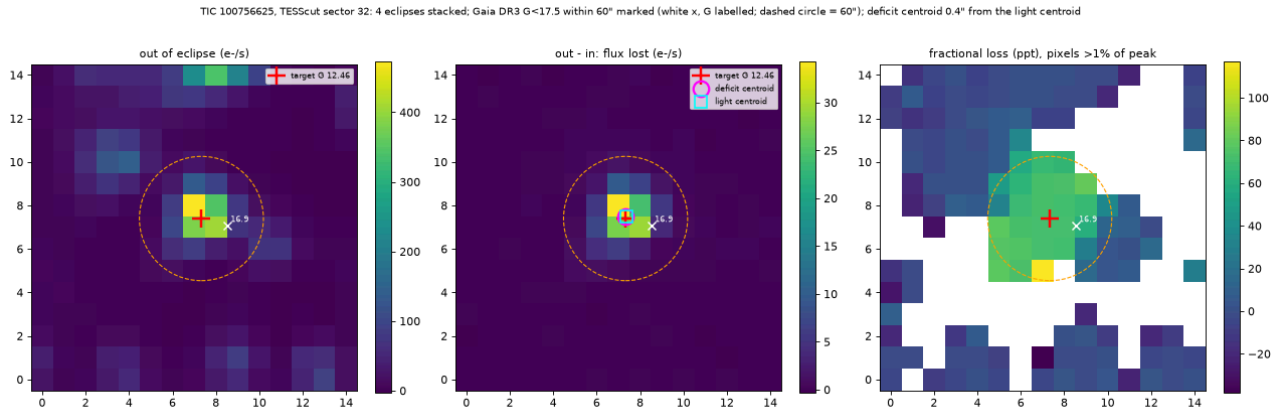
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for improved localization and physical parameters.



R577; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R577; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 100756625: CONFIRMED

Full bounded physical review: **COMPLETE**. Physical outcome and catalogue novelty are separate; a known recovery is not a new addition.

1. **Raw repeated events.** 11 free-center local trapezoid measurements in quality-zero QLP SAP_FLUX; 11 centers enter timing, in sectors [6, 33, 98]. Predictions locate windows and are not measurements. Flagged or uncovered windows are not clean nulls. Eleven deep selected events recur; four shallow secondary candidates lie near phase 0.49956. Compact target deficit and no capable neighbor within 60 arcsec.
2. **Timing and aliases.** Measured recurring-eclipse period P=7.0214288325 +/- 0.0000066060 days, epoch BTJD=1473.3814673. Epoch kind is the selected recurring eclipse; deeper-primary rank is not assumed. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. There are 5 adjacent same-kind pairs within sectors. Selected depths occur comparably in both cycle parities; doubling does not reveal alternating selected-event kinds. Halving interleaves a shallow secondary or sampled baseline, not the selected eclipse. Actual measured secondary-candidate phase=0.499562, depth=4.57 ppt, 4 usable fits. Actual phase scans and candidate fits are retained; secondary usability is a heuristic amplitude/local-scatter check, not calibrated significance. No exhaustive all-alias search or calibrated secondary-null confidence is asserted.
3. **Pixel attribution.** Fresh S32 difference stack contains 4 selected eclipses. Target WCS pixel= [7.331913353204918, 7.398452520054139], deficit centroid=[7.337889737414169, 7.510039391148359], descriptive offset=0.112 pixel. Neither offset nor image residual is a calibrated localization uncertainty or universal veto. All capable Gaia neighbors: None under the approximate pairwise G-band flux screen within 60 arcsec.. Capability is a G-band pairwise flux-budget proxy; color and aperture response limit precision. Intrinsic neighbor losses are never equated to diluted pixel fractional loss. Same-unit actual deficits and field morphology are independently recorded in ../terra-pixel-audit.json; close blends retain uncertainty.
4. **Alternative reduction.** Reused, read and rerun TESScut primitive subtracts faint-pixel background, with symmetric local side windows and selected eclipse core; fresh target-centered 1x1/3x3/5x5 depths=71.60/71.64/70.78 ppt. QLP fitted median=62.88 ppt. Aperture, dilution and

background response explain why these are not intrinsic-depth measurements. The cutout sector is absent from the retained QLP files. It is a distinct observing occasion on TESS and an alternative extraction of that sector; no direct same-sector QLP comparison is claimed. Distinct sectors remain occasions on one instrument.

5. **Gaia geometry.** Source 2903428510920674816; P=7.0205749150 +/- 0.0011569319 days, e=0.04190. Closest edge-on conjunction offset=0.05130 days (0.00731 phase), source-specific full covariance linear allowance=0.18513 days. Other conjunction relative phase=0.47572, linear phase allowance=0.02903. Full correlation matrix minimum eigenvalue=0.000540187. These linear allowances are descriptive, not calibrated significance. Near Gaia reference the projected epoch offset=-0.08021 days, period-projection allowance alone=0.00102 days (ignores fit covariance/model systematics). Mass function=0.010412 solar masses; adopted primary mass=0.994, minimum secondary mass=0.253. Approximate central duration=4.37 hours versus observed median=4.00 hours. TIC radius, edge-on geometry and main-sequence secondary assumptions make this a plausibility test, not a fitted physical characterization. Impact parameter can shorten duration. Agreement does not by itself establish ownership.

Deciding limitation: Eleven deep selected events recur; four shallow secondary candidates lie near phase 0.49956. Compact target deficit and no capable neighbor within 60 arcsec.

Next check: Optional resolved photometry and spectroscopy for improved localization and physical parameters.

Shared catalogue results are in ../../shared-catalogues/coverage-matrix.csv. Successful no-matches are distinct from failed queries; no exhaustive priority claim is made. TIC 295771108 is a known exact ASAS-SN EA recovery.

Sector	Search cycle	Measured BTJD	Formal error d	Input allowance d	Depth ppt	Status
6	0	1473.3801831	0.00018754	0.0104166	62.88	used
6	1	1480.4012630	0.000207135	0.0104166	62.32	used
6	2	1487.4222243	0.00039962	0.0104166	65.90	used
33	104	2203.6093144	0.000479887	0.00347218	74.57	used
33	105	2210.6316166	0.000226114	0.00347218	72.08	used
33	106	2217.6542908	0.000557113	0.00347218	70.63	used
33	107	2224.6743798	0.000277817	0.00347218	73.13	used
98	360	4001.0959270	0.000228484	0.00115742	59.39	used
98	362	4015.1387627	0.00018432	0.00115742	59.84	used
98	364	4029.1815326	0.000219175	0.00115742	57.56	used
98	366	4043.2242886	0.000318969	0.00115742	57.52	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png, historical originals, ../identity-check.json, ../alias-audit.json, ../secondary-measurements.json, ../secondary-phase-scan.json/png, ../gaia-geometry.json, ../terra-pixel-audit.json and ../helper-audit.md. Exact sources and outputs are covered by ../SHA256SUMS.json. The methods snapshot freezes the read methods without hashing a mutable canonical page.

Stable review key: R577.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s32.json - historical/diff_s32.png - historical/gaia.json - measurements.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

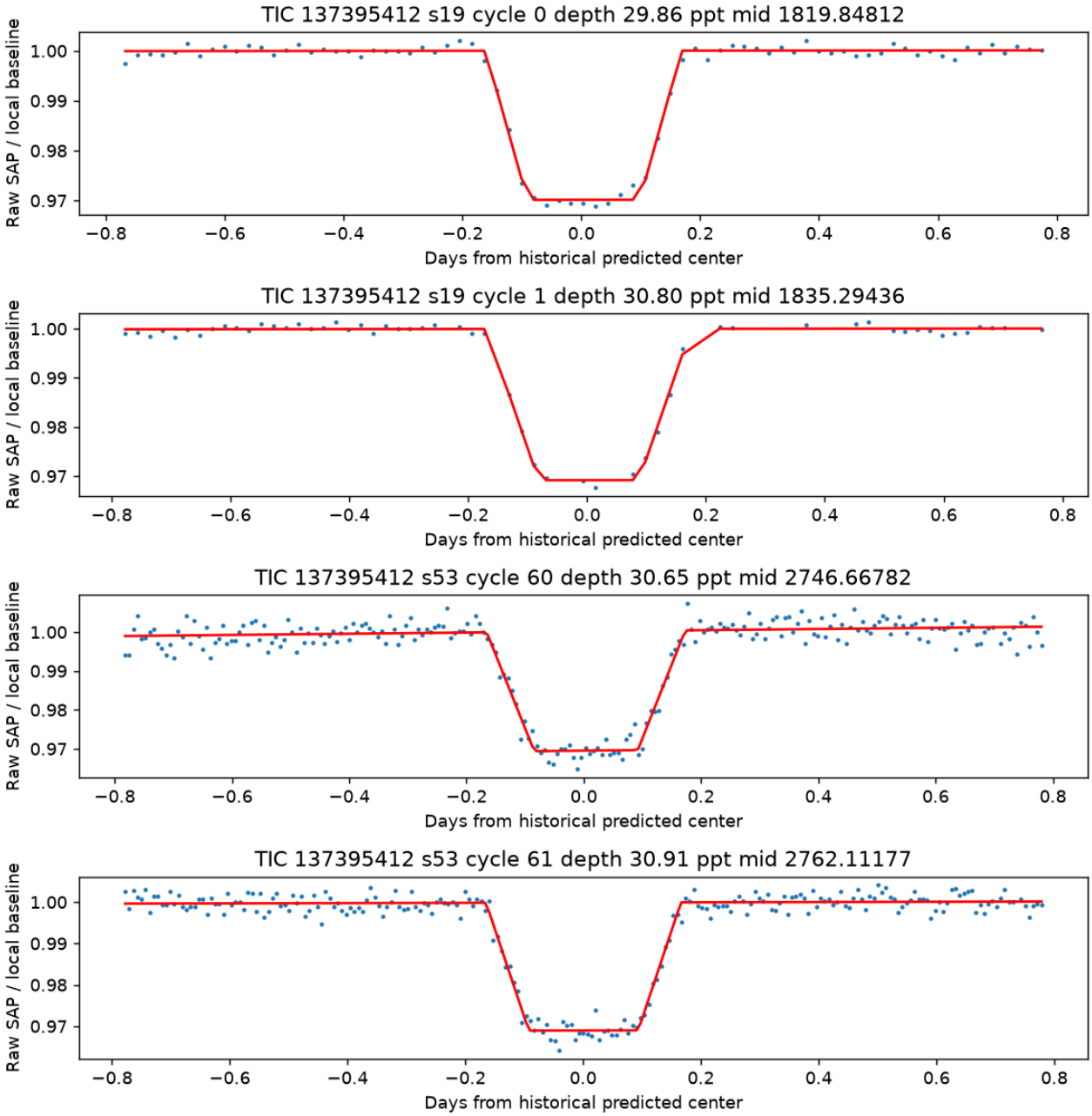
R578. TIC 137395412

physical_verdict: CONFIRMED

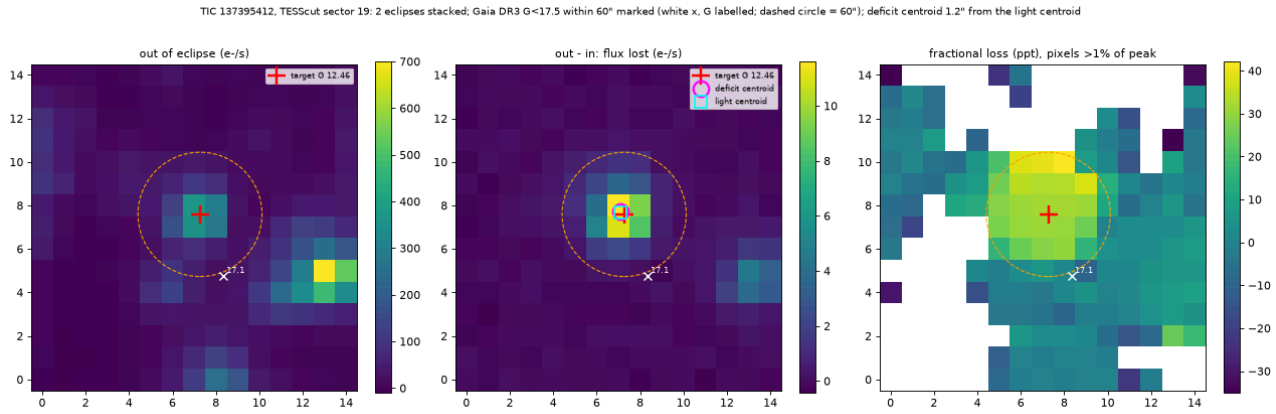
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for improved localization and physical parameters.



R578; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R578; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 137395412: CONFIRMED

Full bounded physical review: **COMPLETE**. Physical outcome and catalogue novelty are separate; a known recovery is not a new addition.

1. **Raw repeated events.** 4 free-center local trapezoid measurements in quality-zero QLP SAP_FLUX; 4 centers enter timing, in sectors [19, 53]. Predictions locate windows and are not measurements. Flagged or uncovered windows are not clean nulls. Four selected centers form consecutive pairs in S19/S53. No capable neighbor within 60 arcsec. No significant secondary center is claimed; Gaia predicts the other conjunction at actual phase 0.72348, so a half-phase null alone is not an eccentric-secondary test.
2. **Timing and aliases.** Measured recurring-eclipse period P=15.4469737423 +/- 0.0001293520 days, epoch BTJD=1819.8478710. Epoch kind is the selected recurring eclipse; deeper-primary rank is not assumed. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. There are 2 adjacent same-kind pairs within sectors. Selected depths occur comparably in both cycle parities; doubling does not reveal alternating selected-event kinds. Halving interleaves a shallow secondary or sampled baseline, not the selected eclipse. No significant repeated secondary center is claimed; this is not proof of absence. Actual phase scans and candidate fits are retained; secondary usability is a heuristic amplitude/local-scatter check, not calibrated significance. No exhaustive all-alias search or calibrated secondary-null confidence is asserted.
3. **Pixel attribution.** Fresh S19 difference stack contains 2 selected eclipses. Target WCS pixel= [7.266344681731502, 7.588371946702171], deficit centroid=[7.097606651246714, 7.7439308877700626], descriptive offset=0.230 pixel. Neither offset nor image residual is a calibrated localization uncertainty or universal veto. All capable Gaia neighbors: None under the approximate pairwise G-band flux screen within 60 arcsec.. Capability is a G-band pairwise flux-budget proxy; color and aperture response limit precision. Intrinsic neighbor losses are never equated to diluted pixel fractional loss. Same-unit actual deficits and field morphology are independently recorded in ../terra-pixel-audit.json; close blends retain uncertainty.
4. **Alternative reduction.** Reused, read and rerun TESScut primitive subtracts faint-pixel background, with symmetric local side windows and selected eclipse core; fresh target-centered

1x1/3x3/5x5 depths=31.90/31.57/30.79 ppt. QLP fitted median=30.73 ppt. Aperture, dilution and background response explain why these are not intrinsic-depth measurements. This reduction and the retained QLP sector share TESS photons, testing processing robustness rather than independent observations. Distinct sectors remain occasions on one instrument.

5. **Gaia geometry.** Source 544879277958019456; P=15.4431614446 +/- 0.014090715 days, e=0.42867. Closest edge-on conjunction offset=0.31577 days (0.02045 phase), source-specific full covariance linear allowance=1.20350 days. Other conjunction relative phase=0.72348, linear phase allowance=0.03700. Full correlation matrix minimum eigenvalue=0.0576975. These linear allowances are descriptive, not calibrated significance. Near Gaia reference the projected epoch offset=-0.03878 days, period-projection allowance alone=0.01203 days (ignores fit covariance/model systematics). Mass function=0.019994 solar masses; adopted primary mass=1.29, minimum secondary mass=0.382. Approximate central duration=11.78 hours versus observed median=8.03 hours. TIC radius, edge-on geometry and main-sequence secondary assumptions make this a plausibility test, not a fitted physical characterization. Impact parameter can shorten duration. Agreement does not by itself establish ownership.

Deciding limitation: Four selected centers form consecutive pairs in S19/S53. No capable neighbor within 60 arcsec. No significant secondary center is claimed; Gaia predicts the other conjunction at actual phase 0.72348, so a half-phase null alone is not an eccentric-secondary test.

Next check: Optional resolved photometry and spectroscopy for improved localization and physical parameters.

Shared catalogue results are in ../../shared-catalogues/coverage-matrix.csv. Successful no-matches are distinct from failed queries; no exhaustive priority claim is made. TIC 295771108 is a known exact ASAS-SN EA recovery.

Sector	Search cycle	Measured BTJD	Formal error d	Input allowance d	Depth ppt	Status
19	0	1819.8481178	0.000812293	0.0104166	29.86	used
19	1	1835.2943650	0.000838023	0.0104166	30.80	used
53	60	2746.6678217	0.00155186	0.00347228	30.65	used
53	61	2762.1117691	0.0010532	0.00347228	30.91	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png, historical originals, ../identity-check.json, ../alias-audit.json, ../secondary-measurements.json, ../secondary-phase-scan.json/png, ../gaia-geometry.json, ../terra-pixel-audit.json and ../helper-audit.md. Exact sources and outputs are covered by ../SHA256SUMS.json. The methods snapshot freezes the read methods without hashing a mutable canonical page.

Stable review key: R578.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s19.json - historical/diff_s19.png - historical/gaia.json - measurements.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

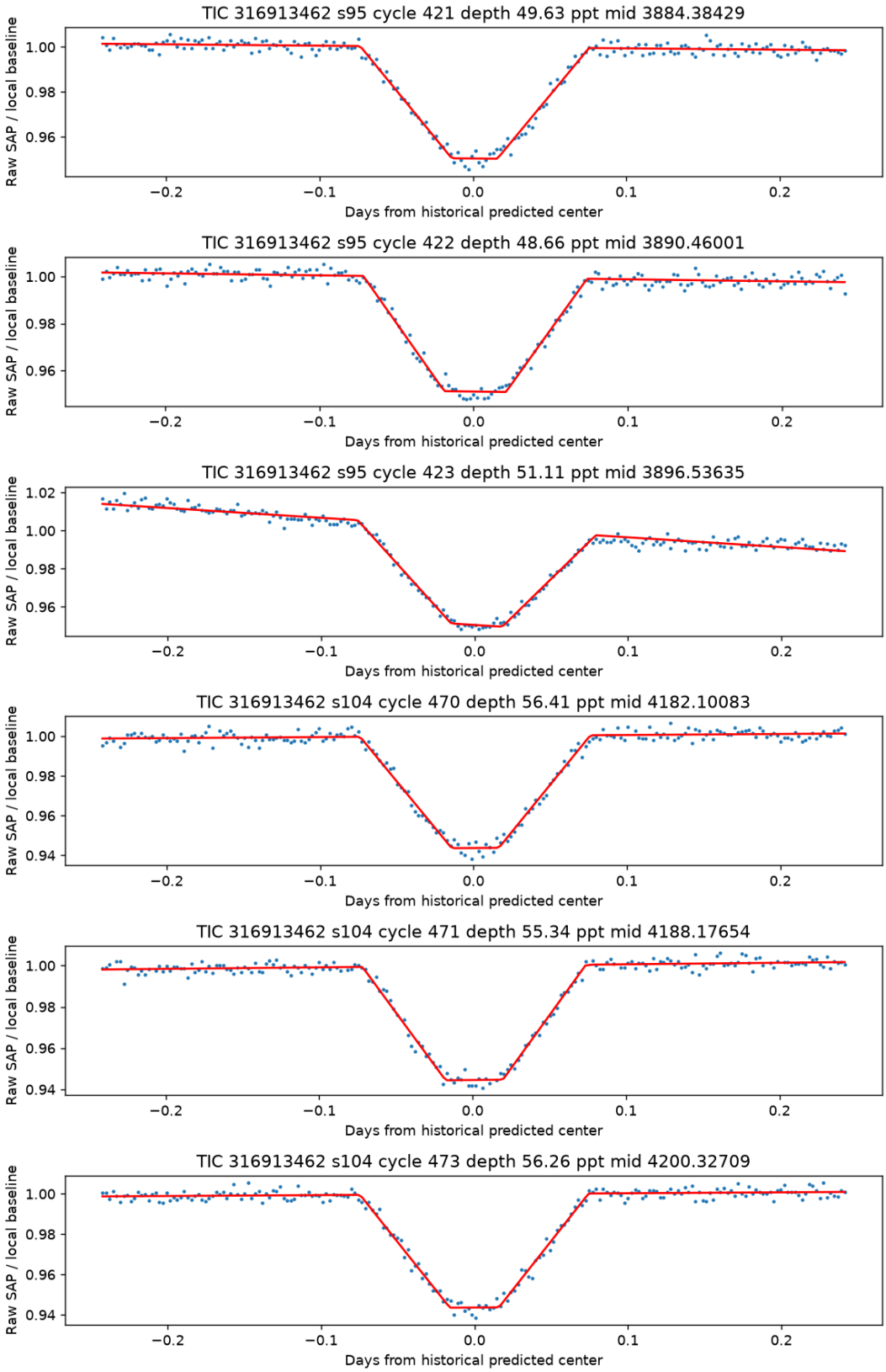
R579. TIC 316913462

physical_verdict: CONFIRMED

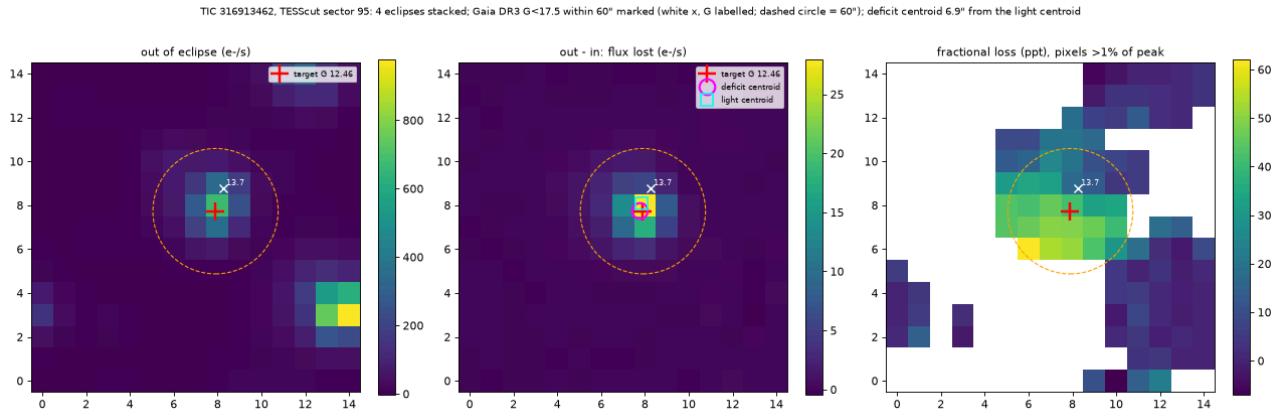
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for improved localization and physical parameters.



R579; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R579; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 316913462: CONFIRMED

Full bounded physical review: **COMPLETE**. Physical outcome and catalogue novelty are separate; a known recovery is not a new addition.

1. **Raw repeated events.** 8 free-center local trapezoid measurements in quality-zero QLP SAP_FLUX; 8 centers enter timing, in sectors [95, 104]. Predictions locate windows and are not measurements. Flagged or uncovered windows are not clean nulls. Eight selected events recur in S95/S104, plus five shallow secondary candidates at phase 0.49943, near Gaia prediction 0.49584. Target pixel loss 27.92 e-/s versus 5.85 e-/s at the capable 22.51-arcsec neighbor favors the target at TESS resolution despite overlapping wings. Modern Gaia conjunction offset 0.692 days reflects accumulated period difference: projection near Gaia reference agrees within 0.00132 days; P difference 0.001206 days versus Gaia error 0.000506 days. These descriptive tensions are not a definitive contradiction.
2. **Timing and aliases.** Measured recurring-eclipse period P=6.0758306405 +/- 0.0000174399 days, epoch BTJD=3884.3846087. Epoch kind is the selected recurring eclipse; deeper-primary rank is not assumed. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. There are 6 adjacent same-kind pairs within sectors. Selected depths occur comparably in both cycle parities; doubling does not reveal alternating selected-event kinds. Halving interleaves a shallow secondary or sampled baseline, not the selected eclipse. Actual measured secondary-candidate phase=0.499431, depth=6.60 ppt, 5 usable fits. Actual phase scans and candidate fits are retained; secondary usability is a heuristic amplitude/local-scatter check, not calibrated significance. No exhaustive all-alias search or calibrated secondary-null confidence is asserted.
3. **Pixel attribution.** Fresh S95 difference stack contains 4 selected eclipses. Target WCS pixel= [7.906782929874101, 7.729202811714394], deficit centroid=[7.740687137057815, 7.777263314860381], descriptive offset=0.173 pixel. Neither offset nor image residual is a calibrated localization uncertainty or universal veto. All capable Gaia neighbors: Gaia 6495928548171911168: 22.51 arcsec, G=13.663, intrinsic-loss proxy=0.214. Capability is a G-band pairwise flux-budget proxy; color and aperture response limit precision. Intrinsic neighbor losses are never equated to diluted pixel fractional loss. Same-unit actual deficits and field morphology are independently recorded in ../terra-pixel-audit.json; close blends retain uncertainty.

4. **Alternative reduction.** Reused, read and rerun TESScut primitive subtracts faint-pixel background, with symmetric local side windows and selected eclipse core; fresh target-centered 1x1/3x3/5x5 depths=41.36/37.43/36.21 ppt. QLP fitted median=53.23 ppt. Aperture, dilution and background response explain why these are not intrinsic-depth measurements. This reduction and the retained QLP sector share TESS photons, testing processing robustness rather than independent observations. Distinct sectors remain occasions on one instrument.
5. **Gaia geometry.** Source 6495928548171911424; P=6.0746245768 +/- 0.00050562446 days, e=0.08962. Closest edge-on conjunction offset=0.69217 days (0.11395 phase), source-specific full covariance linear allowance=0.29493 days. Other conjunction relative phase=0.49584, linear phase allowance=0.01389. Full correlation matrix minimum eigenvalue=0.00335541. These linear allowances are descriptive, not calibrated significance. Near Gaia reference the projected epoch offset=-0.00131 days, period-projection allowance alone=0.01003 days (ignores fit covariance/model systematics). Mass function=0.059768 solar masses; adopted primary mass=1.24, minimum secondary mass=0.584. Approximate central duration=5.73 hours versus observed median=3.55 hours. TIC radius, edge-on geometry and main-sequence secondary assumptions make this a plausibility test, not a fitted physical characterization. Impact parameter can shorten duration. Agreement does not by itself establish ownership.

Deciding limitation: Eight selected events recur in S95/S104, plus five shallow secondary candidates at phase 0.49943, near Gaia prediction 0.49584. Target pixel loss 27.92 e-/s versus 5.85 e-/s at the capable 22.51-arcsec neighbor favors the target at TESS resolution despite overlapping wings. Modern Gaia conjunction offset 0.692 days reflects accumulated period difference: projection near Gaia reference agrees within 0.00132 days; P difference 0.001206 days versus Gaia error 0.000506 days. These descriptive tensions are not a definitive contradiction.

Next check: Optional resolved photometry and spectroscopy for improved localization and physical parameters.

Shared catalogue results are in ../../shared-catalogues/coverage-matrix.csv. Successful no-matches are distinct from failed queries; no exhaustive priority claim is made. TIC 295771108 is a known exact ASAS-SN EA recovery.

Sector	Search cycle	Measured BTJD	Formal error d	Input allowance d	Depth ppt	Status
95	421	3884.3842949	0.000421103	0.00115742	49.63	used
95	422	3890.4600146	0.000477335	0.00115742	48.66	used
95	423	3896.5363538	0.00195652	0.00195652	51.11	used
95	424	3902.6128015	0.000414073	0.00115742	48.84	used
104	470	4182.1008316	0.000373483	0.00115741	56.41	used
104	471	4188.1765393	0.000333266	0.00115741	55.34	used
104	472	4194.2517742	0.000391464	0.00115741	56.76	used
104	473	4200.3270872	0.000319127	0.00115741	56.26	used

Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png, historical originals, ../identity-check.json, ../alias-audit.json, ../secondary-measurements.json, ../secondary-phase-scan.json/png, ../gaia-geometry.json, ../terra-pixel-audit.json and ../helper-audit.md. Exact sources and outputs are covered by ../SHA256SUMS.json. The methods snapshot freezes the read methods without hashing a mutable canonical page.

Stable review key: R579.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s95.json - historical/diff_s95.png - historical/gaia.json - measurements.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

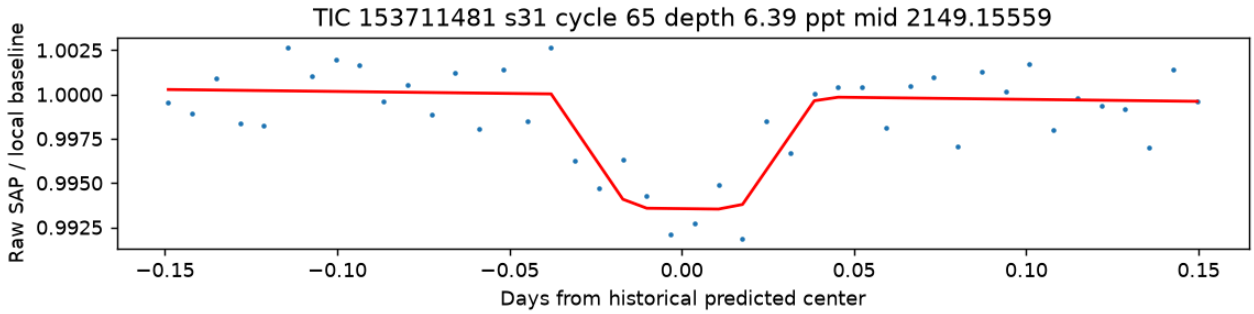
R580. TIC 153711481

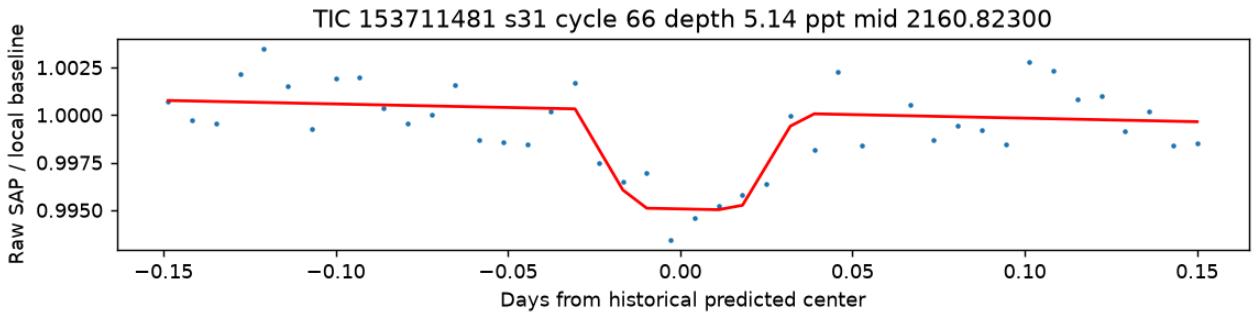
physical_verdict: CONFIRMED

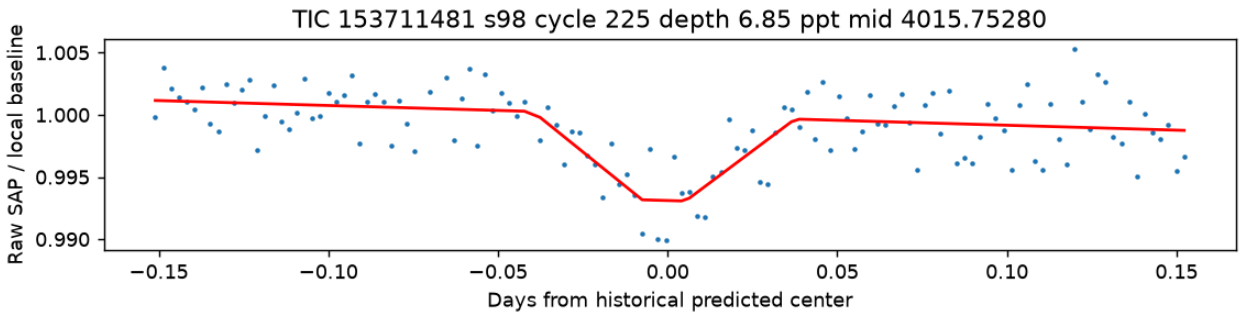
novelty_status: no identified prior EB

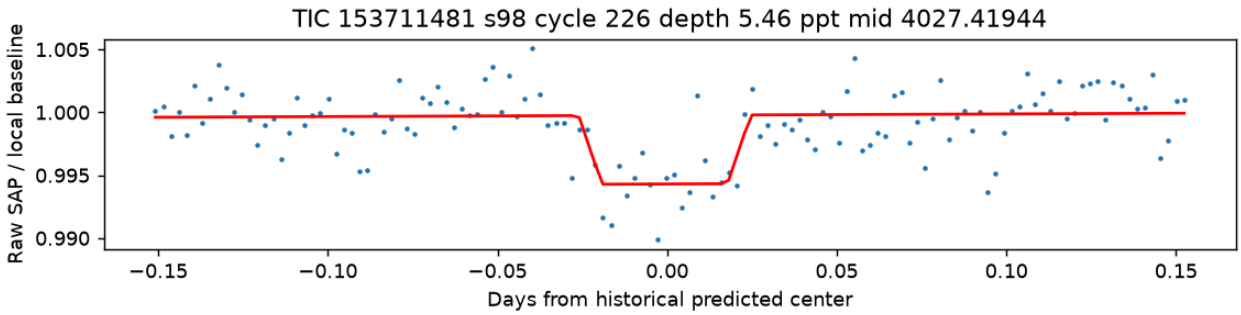
completion: COMPLETE_BOUNDED_REVIEW

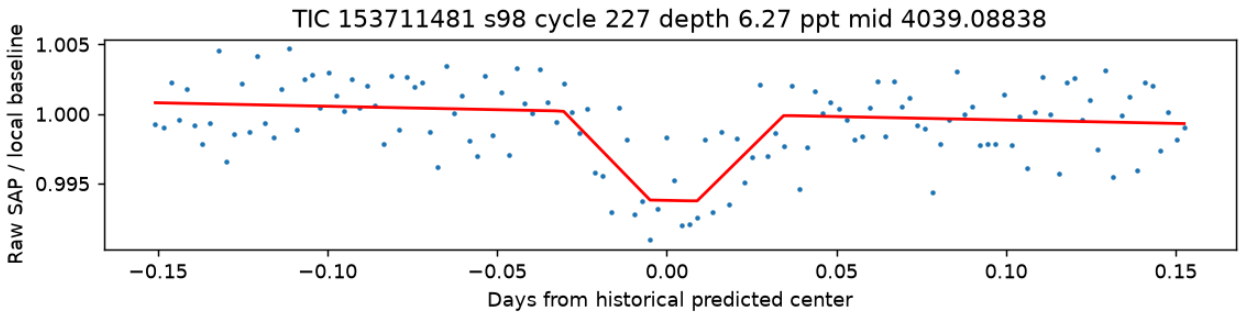
next_check: Optional resolved follow-up and physical binary modeling.



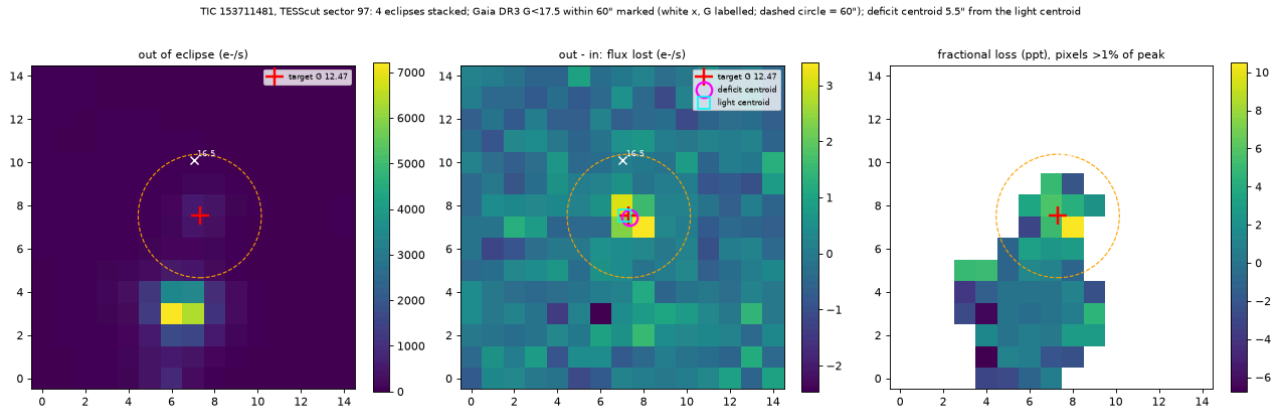








R580; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R580; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 153711481: CONFIRMED

Full bounded physical review: COMPLETE. Physical verdict and scoped catalogue novelty are separate. The epoch column names the selected repeating eclipse kind; deeper-primary rank is not assumed.

Deciding evidence: Five shallow but sharp recurring eclipses, including consecutive cycles in S31 and S98. Fresh S97 four-event deficit is centered on target (0.068pixel). Both capable neighbors, at33.4/54.0arcsec, are spatially distinct from the compact positive core; their pixels show weak target-wing/noise losses rather than distinct matching deficit peaks. A bright remote field source distorts the total light image, explaining high global PSF residual; it does not move the eclipse core. The shallow stack is noisy (target pixel deficit3.13e/s versus0.14e/s at closer capable source); this supports target attribution but no calibrated localization confidence is claimed. Raw per-point scatter is about2ppt, so coherent recurrence and multiple in-event samples, rather than any individual point, establish eclipse morphology.

5 free-center quality-zero raw QLP SAP_FLUX timings in sectors[31, 98] give P=11.6662342053 +/-0.0000164527d and selected-kind reference BTJD=2149.1561521. Median fitted diluted QLP depth=6.271ppt. Predicted windows guide the fit; they are not measured timings. Flagged/uncovered windows are not clean nulls. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Zero-degree-of-freedom reduced chi square is null, and singular center covariance is rejected. Representative raw plots and full raw phase plots were visually inspected. Within-sector consecutive same-kind counts and parity are recorded in ../alias-audit.json; half/double alternatives were checked against actual recurrence/secondary morphology. A double orbit with identical unseen labels cannot be exhaustively excluded; no exhaustive blind alias search is claimed.

Observed secondary check: No secure secondary timing is claimed; absence of a claim is not a clean null. Eccentric Gaia other conjunction is phase0.474648 +/-0.028672 descriptive linear allowance. This is evaluated at its actual phase, never assumed0.5. Weak/noisy box maxima in phase-review.json are search candidates, not detections.

Fresh TESScut S97 difference stack contains4 actually covered events. Deficit minus target= [0.03275525987827077, -0.06002281259825537]pixel (radial0.068); target pixel coordinate= [7.3102808916018756, 7.519448094808194]; deficit=[7.343036151480146, 7.4594252822099385].

Global PSF residual=0.981 is not a localization uncertainty. Actual target rounded-pixel signed deficit/baseline=3.1311/542.87 electron/s. All2 capable Gaia neighbors within60arcsec are assessed below from real pixels. Required intrinsic loss is a pairwise Gaia G passband proxy; it must not be compared directly with diluted fractional pixel loss. Actual deficit and baseline are reported in common electron/s units. Colors, aperture response, PSF overlap, and unlisted faint/unresolved sources limit this screen; no calibrated centroid confidence or full PSF deblend is claimed.

Gaia source	sep arcsec	G	required intrinsic loss proxy	signed pixel deficit e/s	pixel baseline e/s	distance to deficit px	attribution assessment
4788880945834740352	33.44	17.622	0.7301	0.1427	66.088	1.711	spatially distinct from compact target deficit; wings/noise may contribute
4788892696865262080	53.96	16.503	0.2644	0.056773	50.341	2.683	spatially distinct from compact target deficit; wings/noise may contribute

Alternative background-subtracted fixed target-centered TESScut1x1/3x3/5x5 depths=5.768/4.837/3.405ppt. Faintest40percent per-cadence background, symmetric local pre/post windows and actual covered cores provide an alternative extraction. Retained pixel-arrays.npz contains actual deficit/baseline. This is a reused numerical primitive with fresh ephemeris/duration, not independent-code replication. QLP and custom extractions share TESS photons in overlapping sectors; differing sectors are separate observing occasions on one instrument. Aperture/sector depth differences are dilution and reduction limitations, not intrinsic depth agreement.

Gaia DR3 source4788880941537833984 has P=11.666078802d, e=0.393535, omega=275.3435deg, Tp=387.767812BTJD. Both edge-on eccentric conjunctions and source-specific full NSS covariance are retained in ../gaia-geometry.json. Closest selected-kind offset=-0.120223d (-0.010305phase), linear covariance allowance=0.330198d. The SB1 lower triangle uses P,gamma,K,e,omega,Tp, and selected subindices0,3,4,5; minimum full correlation eigenvalue=0.028969. This descriptive linear approximation is not a calibrated rejection threshold. Mass function=0.0623192solar masses; assumed TIC/adopted M1=1.030, minimum M2=0.534; TIC/adopted radius=1.077solar radii. Approximate central duration=3.943h versus measured1.559h. Main-sequence minimum-mass companion and TIC mass/radius make this a plausibility screen, not a fitted geometry. Orbital agreement alone cannot establish source ownership.

Sector	same-kind search cycle	measured midpoint BTJD	formal center error d	timing input allowance d	depth ppt
31	65	2149.1555851	0.00235452	0.00347223	6.394
31	66	2160.8230012	0.00288377	0.00347223	5.144
98	225	4015.7527976	0.00232604	0.00232604	6.849
98	226	4027.4194406	0.00121109	0.00121109	5.463
98	227	4039.0883756	0.00229222	0.00229222	6.271

Limitations and next check: Optional resolved follow-up and physical binary modeling. Representative events suffice for this bounded review; no all-sector/all-event precision campaign is claimed. Sector-dependent dilution, finite cadence, baseline choice, color proxy, and linear Gaia propagation remain material limitations.

Catalogue checks use ../../shared-catalogues/coverage-matrix.csv and the shared receipt manifest; known recovery is separated from physical classification and no exhaustive priority claim is made. Evidence: measurements.json/raw-windows.csv/raw-events.png; fresh-pixels/diff_fresh.json/png/pixel-arrays.npz; neighbor-assessment.json; batch phase, secondary, alias, identity/source receipts, methods snapshot and audits. SHA256SUMS.json freezes exact sources and outputs.

Stable review key: R580.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/pixel-arrays.npz - fresh-pixels/run.log - historical/diff_s97.json - historical/diff_s97.png - historical/gaia.json - measurements.json - neighbor-assessment.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

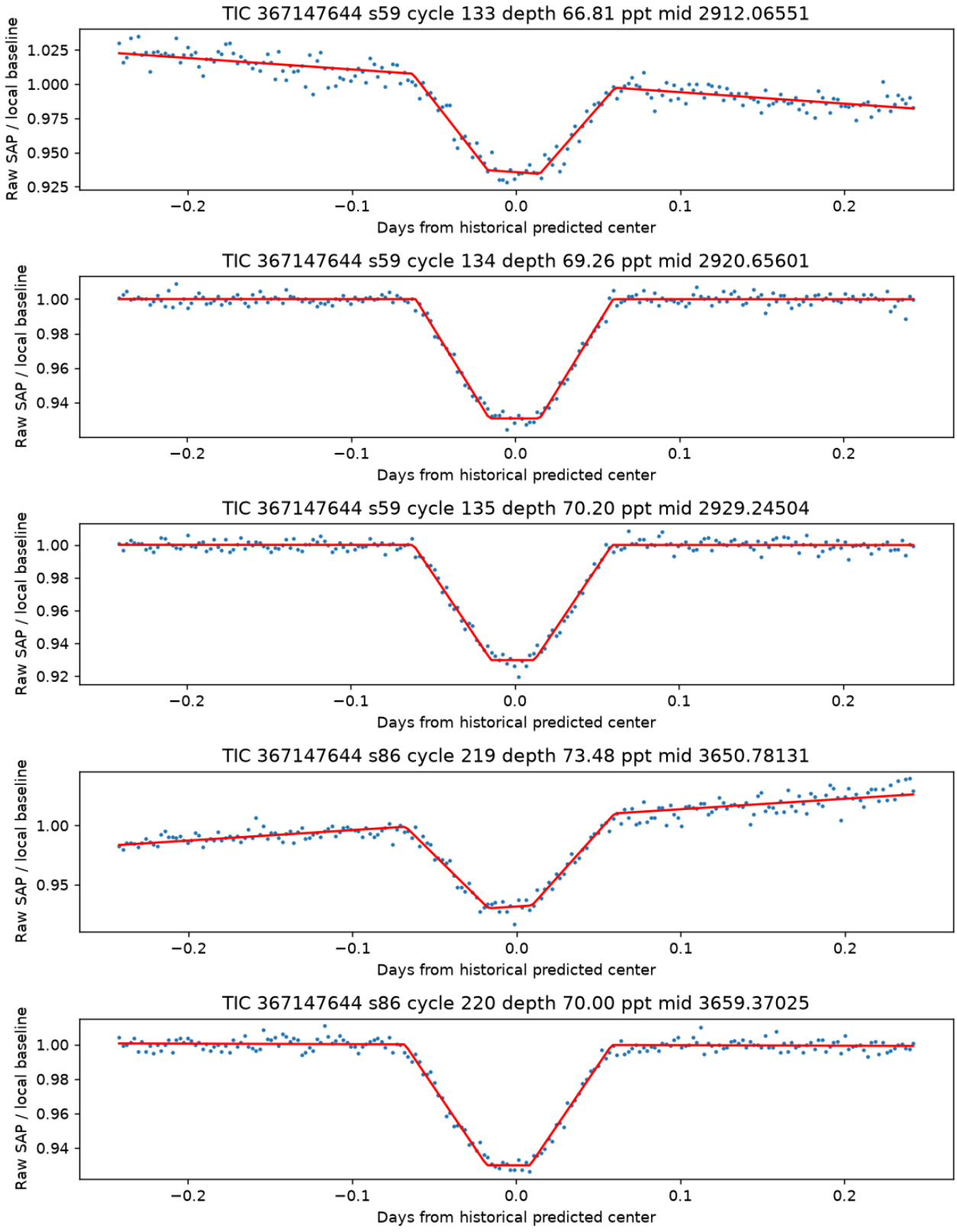
R581. TIC 367147644

physical_verdict: CONFIRMED

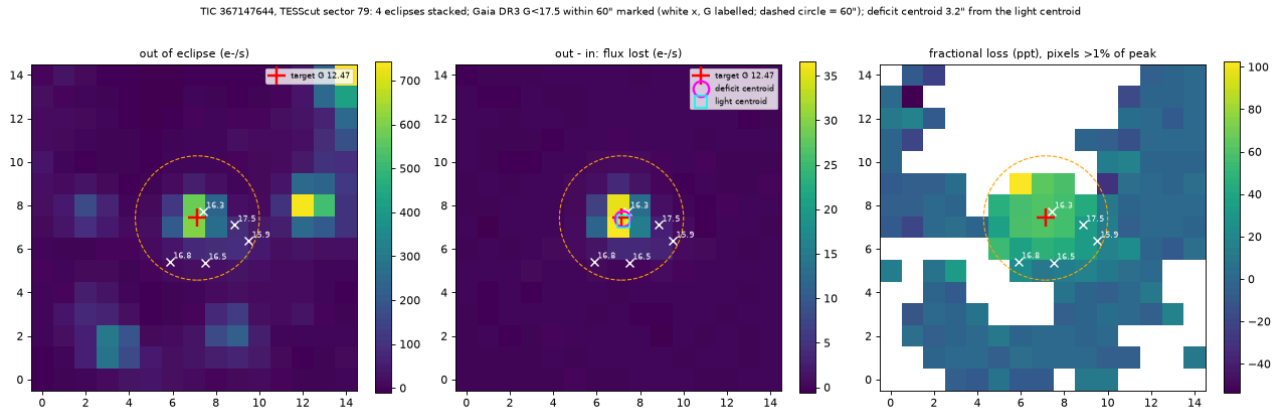
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved follow-up and physical binary modeling.



R581; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R581; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 367147644: CONFIRMED

Full bounded physical review: COMPLETE. Physical verdict and scoped catalogue novelty are separate. The epoch column names the selected repeating eclipse kind; deeper-primary rank is not assumed.

Deciding evidence: Five sharp repeated eclipses and consecutive S59 cycles. Fresh compact target deficit has no flux-capable Gaia neighbor within60arcsec. Strong raw baseline curvature in S17/S86 is not mistaken for a secondary. Gaia conjunction offset0.04136phase is comparable to its0.313d propagated timing allowance, not a hard0.03phase veto.

5 free-center quality-zero raw QLP SAP_FLUX timings in sectors[59, 86] give P=8.5897073212 +/-0.0000147088d and selected-kind reference BTJD=2912.0659088. Median fitted diluted QLP depth=70.005ppt. Predicted windows guide the fit; they are not measured timings. Flagged/uncovered windows are not clean nulls. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Zero-degree-of-freedom reduced chi square is null, and singular center covariance is rejected. Representative raw plots and full raw phase plots were visually inspected. Within-sector consecutive same-kind counts and parity are recorded in ../alias-audit.json; half/double alternatives were checked against actual recurrence/secondary morphology. A double orbit with identical unseen labels cannot be exhaustively excluded; no exhaustive blind alias search is claimed.

Observed secondary check: No secure secondary timing is claimed; absence of a claim is not a clean null. Eccentric Gaia other conjunction is phase0.436679 +/-0.016570 descriptive linear allowance. This is evaluated at its actual phase, never assumed0.5. Weak/noisy box maxima in phase-review.json are search candidates, not detections.

Fresh TESScut S79 difference stack contains4 actually covered events. Deficit minus target= [0.02081443348413803, -0.046120956791201984]pixel (radial0.051); target pixel coordinate= [7.1432496771200675, 7.423237145015856]; deficit=[7.164064110604206, 7.377116188224654]. Global PSF residual=0.259 is not a localization uncertainty. Actual target rounded-pixel signed deficit/baseline=36.643/607.69 electron/s. All0 capable Gaia neighbors within60arcsec are assessed below from real pixels. Required intrinsic loss is a pairwise Gaia G passband proxy; it must not be

compared directly with diluted fractional pixel loss. Actual deficit and baseline are reported in common electron/s units. Colors, aperture response, PSF overlap, and unlisted faint/unresolved sources limit this screen; no calibrated centroid confidence or full PSF deblend is claimed.

Gaia source	sep arcsec	G	required intrinsic loss proxy	signed pixel deficit e/s	pixel baseline e/s	distance to deficit px	attribution assessment
None capable within60arcsec							

Alternative background-subtracted fixed target-centered TESScut1x1/3x3/5x5 depths=60.299/57.637/54.604ppt. Faintest40percent per-cadence background, symmetric local pre/post windows and actual covered cores provide an alternative extraction. Retained pixel-arrays.npz contains actual deficit/baseline. This is a reused numerical primitive with fresh ephemeris/duration, not independent-code replication. QLP and custom extractions share TESS photons in overlapping sectors; differing sectors are separate observing occasions on one instrument. Aperture/sector depth differences are dilution and reduction limitations, not intrinsic depth agreement.

Gaia DR3 source2229977776348260864 has P=8.588454541d, e=0.233375, omega=114.6577deg, Tp=387.064765BTJD. Both edge-on eccentric conjunctions and source-specific full NSS covariance are retained in ../gaia-geometry.json. Closest selected-kind offset=0.355232d (0.041362phase), linear covariance allowance=0.313125d. The SB1 lower triangle uses P,gamma,K,e,omega,Tp, and selected subindices0,3,4,5; minimum full correlation eigenvalue=0.023597. This descriptive linear approximation is not a calibrated rejection threshold. Mass function=0.0570551solar masses; assumed TIC/adopted M1=1.177, minimum M2=0.555; TIC/adopted radius=1.371solar radii. Approximate central duration=4.957h versus measured2.962h. Main-sequence minimum-mass companion and TIC mass/radius make this a plausibility screen, not a fitted geometry. Orbital agreement alone cannot establish source ownership.

Sector	same-kind search cycle	measured midpoint BTJD	formal center error d	timing input allowance d	depth ppt
59	133	2912.0655062	0.00217901	0.00217901	66.815
59	134	2920.6560094	0.000253957	0.00115738	69.261
59	135	2929.2450394	0.000294189	0.00115738	70.203
86	219	3650.7813097	0.00195611	0.00195611	73.476
86	220	3659.3702500	0.000340905	0.00115739	70.005

Limitations and next check: Optional resolved follow-up and physical binary modeling. Representative events suffice for this bounded review; no all-sector/all-event precision campaign is claimed. Sector-dependent dilution, finite cadence, baseline choice, color proxy, and linear Gaia propagation remain material limitations.

Catalogue checks use ../../shared-catalogues/coverage-matrix.csv and the shared receipt manifest; known recovery is separated from physical classification and no exhaustive priority claim is made. Evidence: measurements.json/raw-windows.csv/raw-events.png; fresh-pixels/diff_fresh.json/png/pixel-arrays.npz; neighbor-assessment.json; batch phase, secondary, alias, identity/source receipts, methods snapshot and audits. SHA256SUMS.json freezes exact sources and outputs.

Stable review key: R581.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/pixel-arrays.npz - fresh-pixels/run.log - historical/diff_s79.json - historical/diff_s79.png - historical/gaia.json - measurements.json - neighbor-assessment.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

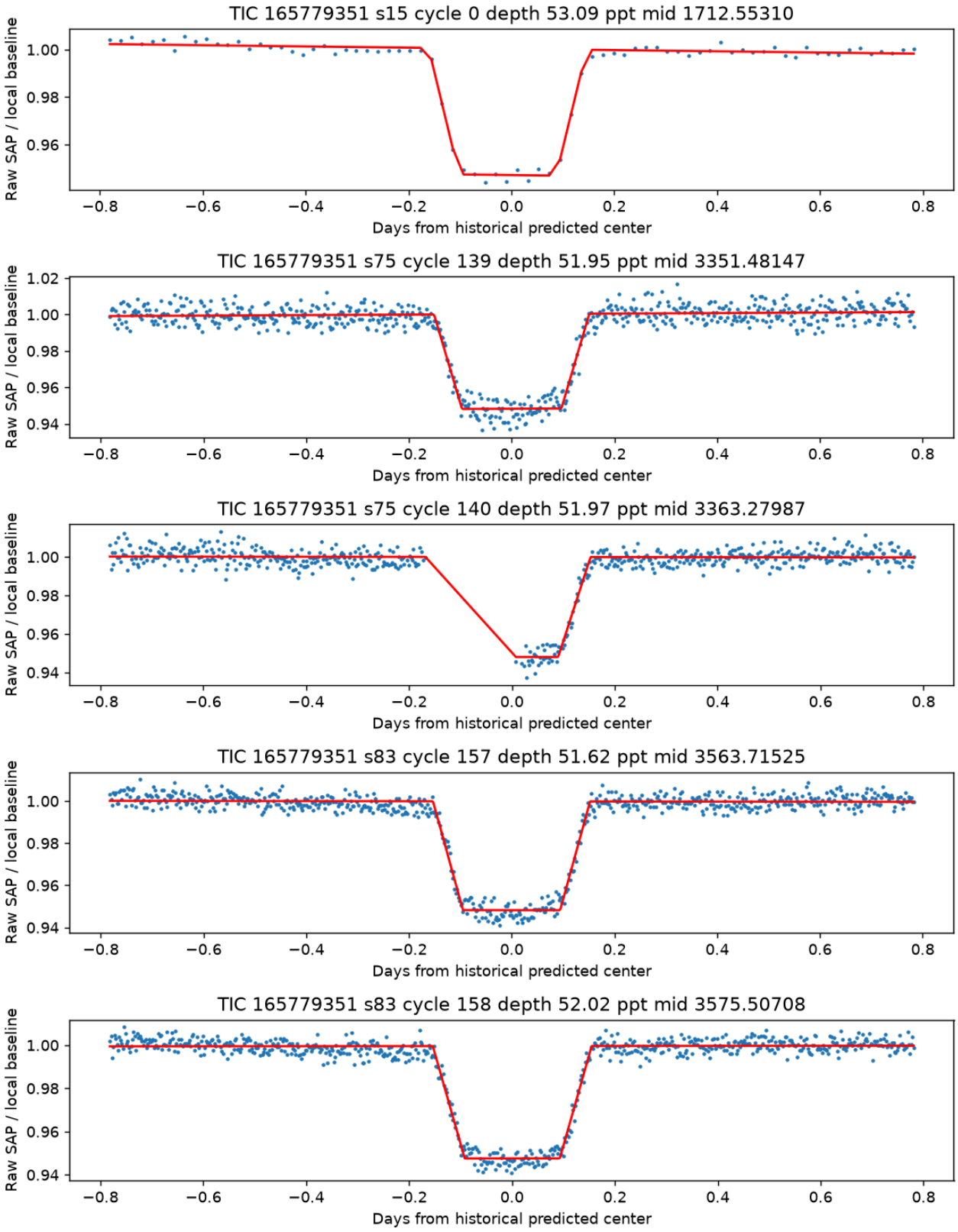
R582. TIC 165779351

physical_verdict: CONFIRMED

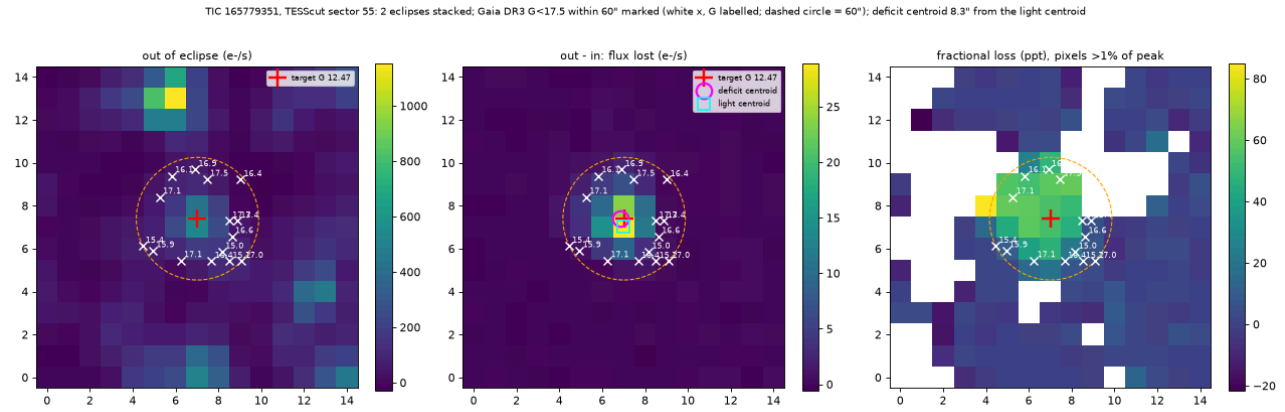
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved follow-up and physical binary modeling.



R582; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R582; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 165779351: CONFIRMED

Full bounded physical review: COMPLETE. Physical verdict and scoped catalogue novelty are separate. The epoch column names the selected repeating eclipse kind; deeper-primary rank is not assumed.

Deciding evidence: Four measured deep flat-bottomed eclipses establish same-kind recurrence. Fresh compact target deficit is distinct from all three capable neighbors at36.8-58.4arcsec. A weak recurring secondary candidate lies near0.72phase, consistent with eccentric Gaia0.7297 +/-0.0483; free-center secondary fits are not secure, so no secondary precision timing is claimed.

4 free-center quality-zero raw QLP SAP_FLUX timings in sectors[15, 75, 83] give P=11.7908244002 +/-0.0000512392d and selected-kind reference BTJD=1712.5564938. Median fitted diluted QLP depth=51.982ppt. Predicted windows guide the fit; they are not measured timings. Flagged/uncovered windows are not clean nulls. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Zero-degree-of-freedom reduced chi square is null, and singular center covariance is rejected. Representative raw plots and full raw phase plots were visually inspected. Within-sector consecutive same-kind counts and parity are recorded in ../alias-audit.json; half/double alternatives were checked against actual recurrence/secondary morphology. A double orbit with identical unseen labels cannot be exhaustively excluded; no exhaustive blind alias search is claimed.

Observed secondary check: No secure secondary timing is claimed; absence of a claim is not a clean null. Eccentric Gaia other conjunction is phase0.729723 +/-0.048314 descriptive linear allowance. This is evaluated at its actual phase, never assumed0.5. Weak/noisy box maxima in phase-review.json are search candidates, not detections.

Fresh TESScut S55 difference stack contains2 actually covered events. Deficit minus target= [-0.18050831937875778, 0.011143796944930173]pixel (radial0.181); target pixel coordinate= [7.024316492655553, 7.3960086776671545]; deficit=[6.843808173276795, 7.407152474612085]. Global PSF residual=0.413 is not a localization uncertainty. Actual target rounded-pixel signed deficit/baseline=28.893/531.88 electron/s. All3 capable Gaia neighbors within60arcsec are assessed below from real pixels. Required intrinsic loss is a pairwise Gaia G passband proxy; it must not be

compared directly with diluted fractional pixel loss. Actual deficit and baseline are reported in common electron/s units. Colors, aperture response, PSF overlap, and unlisted faint/unresolved sources limit this screen; no calibrated centroid confidence or full PSF deblend is claimed.

Gaia source	sep arcsec	G	required intrinsic loss proxy	signed pixel deficit e/s	pixel baseline e/s	distance to deficit px	attribution assessment
1969426952599000192	36.83	14.987	0.5786	3.6473	175	2.044	spatially distinct from compact target deficit; wings/noise may contribute
1969426746440569472	47.49	15.216	0.7021	0.25984	97.189	2.599	spatially distinct from compact target deficit; wings/noise may contribute
1969426918239256832	58.37	15.420	0.8368	0.08416	70.138	2.722	spatially distinct from compact target deficit; wings/noise may contribute

Alternative background-subtracted fixed target-centered TESScut1x1/3x3/5x5 depths=54.323/50.265/44.087ppt. Faintest40percent per-cadence background, symmetric local pre/post windows and actual covered cores provide an alternative extraction. Retained pixel-arrays.npz contains actual deficit/baseline. This is a reused numerical primitive with fresh ephemeris/duration, not independent-code replication. QLP and custom extractions share TESS photons in overlapping sectors; differing sectors are separate observing occasions on one instrument. Aperture/sector depth differences are dilution and reduction limitations, not intrinsic depth agreement.

Gaia DR3 source1969426948296573312 has P=11.789661906d, e=0.483596, omega=43.9817deg, Tp=391.507140BTJD. Both edge-on eccentric conjunctions and source-specific full NSS covariance are retained in ../gaia-geometry.json. Closest selected-kind offset=0.113792d (0.009652phase), linear covariance allowance=0.436573d. The SB1 lower triangle uses P,gamma,K,e,omega,Tp, and selected subindices0,3,4,5; minimum full correlation eigenvalue=0.015695. This descriptive linear approximation is not a calibrated rejection threshold. Mass function=0.0592116solar masses; assumed TIC/adopted M1=1.220, minimum M2=0.576; TIC/adopted radius=3.238solar radii. Approximate central duration=8.644h versus measured7.339h. Main-sequence minimum-mass companion and TIC mass/radius make this a plausibility screen, not a fitted geometry. Orbital agreement alone cannot establish source ownership.

Sector	same-kind search cycle	measured midpoint BTJD	formal center error d	timing input allowance d	depth ppt
15	0	1712.5530976	0.000854059	0.0104171	53.087
75	139	3351.4814696	0.000741592	0.00115739	51.949
83	157	3563.7152488	0.000490021	0.00115736	51.622
83	158	3575.5070826	0.000491709	0.00115736	52.016

Limitations and next check: Optional resolved follow-up and physical binary modeling. Representative events suffice for this bounded review; no all-sector/all-event precision campaign is claimed. Sector-dependent dilution, finite cadence, baseline choice, color proxy, and linear Gaia propagation remain material limitations.

Catalogue checks use ../../shared-catalogues/coverage-matrix.csv and the shared receipt manifest; known recovery is separated from physical classification and no exhaustive priority claim is made. Evidence: measurements.json/raw-windows.csv/raw-events.png; fresh-pixels/diff_fresh.json/png/pixel-arrays.npz; neighbor-assessment.json; batch phase, secondary, alias, identity/source receipts, methods snapshot and audits. SHA256SUMS.json freezes exact sources and outputs.

Stable review key: R582.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/pixel-arrays.npz - fresh-pixels/run.log - historical/diff_s55.json - historical/diff_s55.png - historical/gaia.json - measurements.json - neighbor-assessment.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

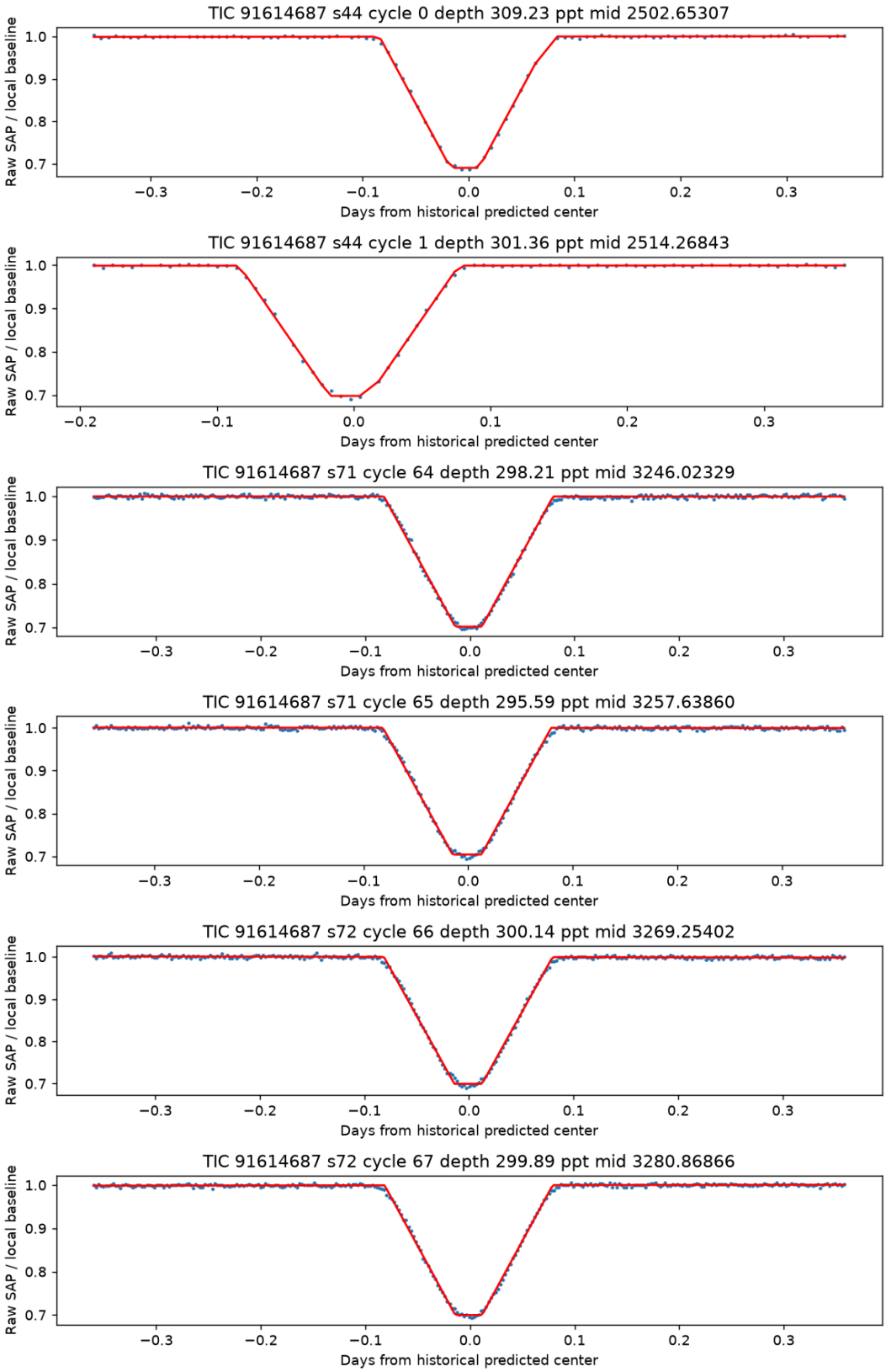
R583. TIC 91614687

physical_verdict: CONFIRMED

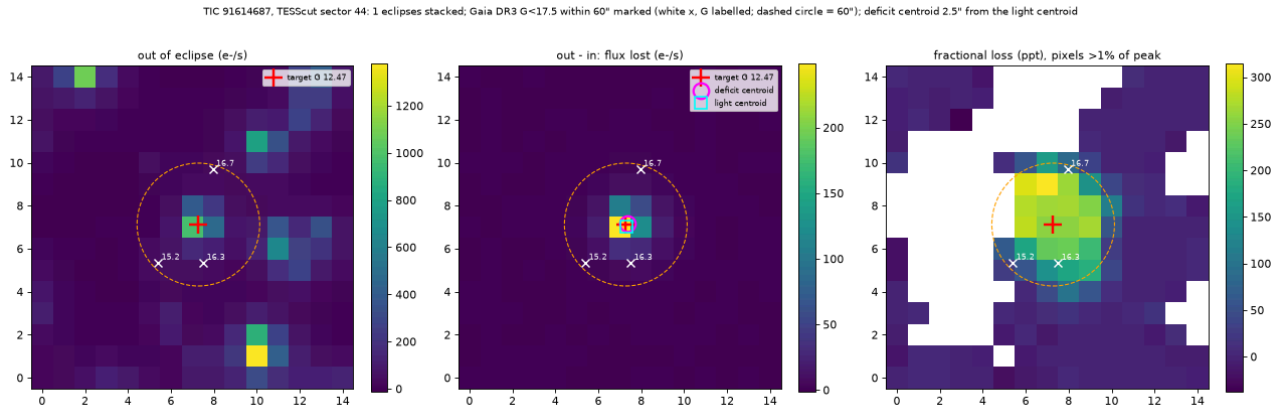
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved follow-up and physical binary modeling.



R583; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R583; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 91614687: CONFIRMED

Full bounded physical review: COMPLETE. Physical verdict and scoped catalogue novelty are separate. The epoch column names the selected repeating eclipse kind; deeper-primary rank is not assumed.

Deciding evidence: Six deep sharp same-kind eclipses and five measured unequal secondaries near0.50712phase distinguish the11.61516d orbit from its half. Compact target deficit has no capable Gaia neighbor within60arcsec.

6 free-center quality-zero raw QLP SAP_FLUX timings in sectors[44, 71, 72] give P=11.6151598290 +/-0.0000387017d and selected-kind reference BTJD=2502.6531731. Median fitted diluted QLP depth=300.014ppt. Predicted windows guide the fit; they are not measured timings. Flagged/uncovered windows are not clean nulls. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Zero-degree-of-freedom reduced chi square is null, and singular center covariance is rejected. Representative raw plots and full raw phase plots were visually inspected. Within-sector consecutive same-kind counts and parity are recorded in ../alias-audit.json; half/double alternatives were checked against actual recurrence/secondary morphology. A double orbit with identical unseen labels cannot be exhaustively excluded; no exhaustive blind alias search is claimed.

Observed secondary check: 5 accepted secondary-candidate free-center fits, median actual phase0.5071232. Eccentric Gaia other conjunction is phase0.505183 +/-0.017604 descriptive linear allowance. This is evaluated at its actual phase, never assumed0.5. Weak/noisy box maxima in phase-review.json are search candidates, not detections.

Fresh TESScut S44 difference stack contains1 actually covered events. Deficit minus target= [0.07055897604301808, 0.038209109051299706]pixel (radial0.080); target pixel coordinate= [7.279906285232881, 7.122872121388422]; deficit=[7.350465261275899, 7.161081230439722]. Global PSF residual=0.712 is not a localization uncertainty. Actual target rounded-pixel signed deficit/baseline=249.15/974.24 electron/s. All0 capable Gaia neighbors within60arcsec are assessed below from real pixels. Required intrinsic loss is a pairwise Gaia G passband proxy; it must not be compared directly with diluted fractional pixel loss. Actual deficit and baseline are reported in common electron/s units. Colors, aperture response, PSF overlap, and unlisted faint/unresolved sources limit this screen; no calibrated centroid confidence or full PSF deblend is claimed.

Gaia source	sep arcsec	G	required intrinsic loss proxy	signed pixel deficit e/s	pixel baseline e/s	distance to deficit px	attribution assessment
None capable within60arcsec							

Alternative background-subtracted fixed target-centered TESScut1x1/3x3/5x5 depths=255.738/255.186/244.213ppt. Faintest40percent per-cadence background, symmetric local pre/post windows and actual covered cores provide an alternative extraction. Retained pixel-arrays.npz contains actual deficit/baseline. This is a reused numerical primitive with fresh ephemeris/duration, not independent-code replication. QLP and custom extractions share TESS photons in overlapping sectors; differing sectors are separate observing occasions on one instrument. Aperture/sector depth differences are dilution and reduction limitations, not intrinsic depth agreement.

Gaia DR3 source3380745024139440512 has P=11.614338978d, e=0.024705, omega=70.7631deg, Tp=388.073164BTJD. Both edge-on eccentric conjunctions and source-specific full NSS covariance are retained in ../gaia-geometry.json. Closest selected-kind offset=0.179266d (0.015435phase), linear covariance allowance=0.437437d. The SB1 lower triangle uses P,gamma,K,e,omega,Tp, and selected subindices0,3,4,5; minimum full correlation eigenvalue=0.00025273. This descriptive linear approximation is not a calibrated rejection threshold. Mass function=0.121409solar masses; assumed TIC/adopted M1=0.724, minimum M2=0.596; TIC/adopted radius=0.930solar radii. Approximate central duration=5.824h versus measured3.863h. Main-sequence minimum-mass companion and TIC mass/radius make this a plausibility screen, not a fitted geometry. Orbital agreement alone cannot establish source ownership.

Sector	same-kind search cycle	measured midpoint BTJD	formal center error d	timing input allowance d	depth ppt
44	0	2502.6530699	8.91015e-05	0.00347255	309.231
44	1	2514.2684342	0.000101448	0.00347255	301.357
71	64	3246.0232888	0.000104519	0.00115751	298.212
71	65	3257.6385966	8.12183e-05	0.00115751	295.591
72	66	3269.2540183	8.30376e-05	0.00115748	300.139
72	67	3280.8686640	6.9207e-05	0.00115748	299.890

Limitations and next check: Optional resolved follow-up and physical binary modeling. Representative events suffice for this bounded review; no all-sector/all-event precision campaign is claimed. Sector-dependent dilution, finite cadence, baseline choice, color proxy, and linear Gaia propagation remain material limitations.

Catalogue checks use ../../shared-catalogues/coverage-matrix.csv and the shared receipt manifest; known recovery is separated from physical classification and no exhaustive priority claim is made. Evidence: measurements.json/raw-windows.csv/raw-events.png; fresh-pixels/diff_fresh.json/png/pixel-arrays.npz; neighbor-assessment.json; batch phase, secondary, alias, identity/source receipts, methods snapshot and audits. SHA256SUMS.json freezes exact sources and outputs.

Stable review key: R583.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/pixel-arrays.npz - fresh-pixels/run.log - historical/diff_s44.json - historical/diff_s44.png - historical/gaia.json - measurements.json - neighbor-assessment.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

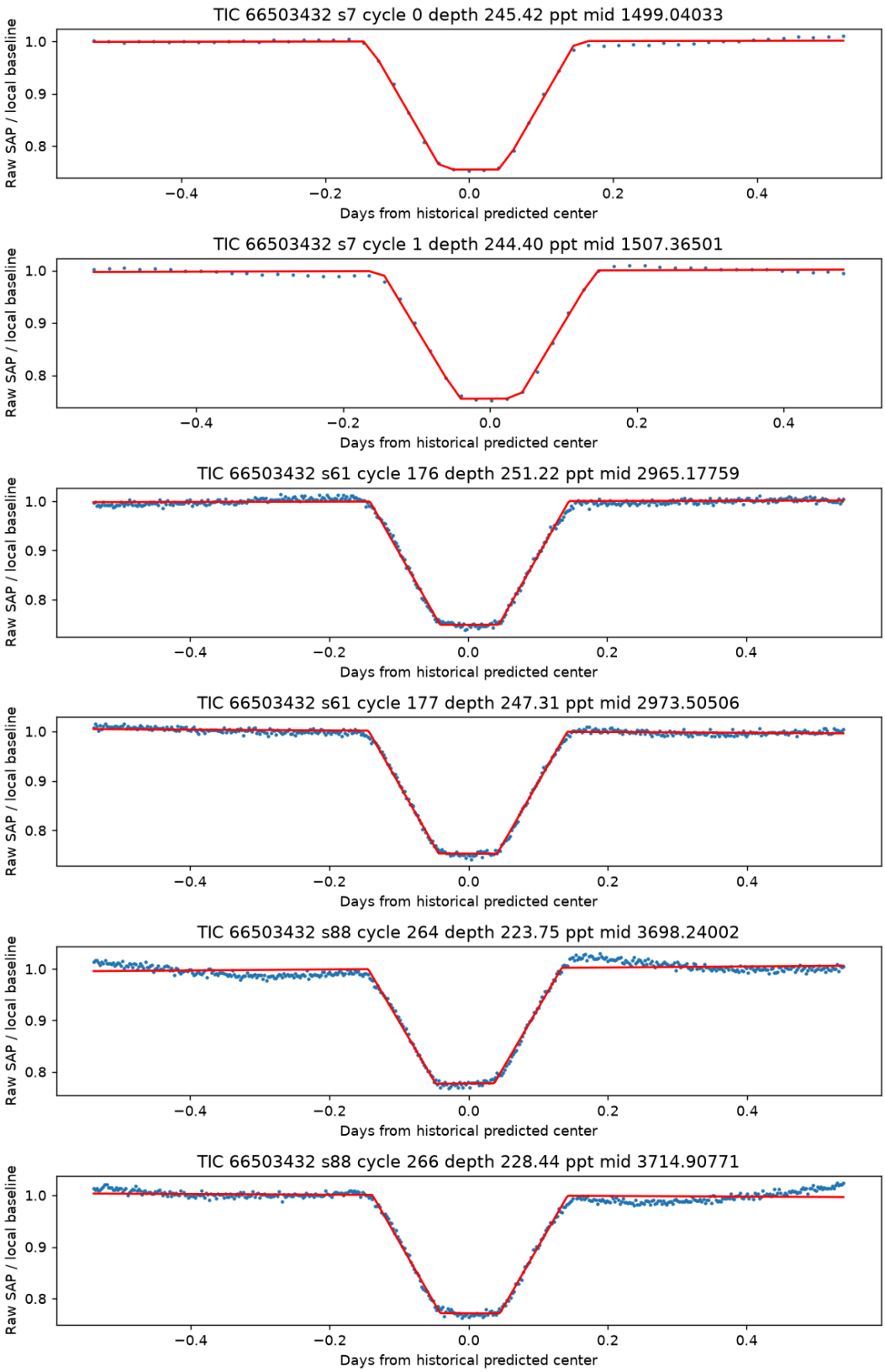
R584. TIC 66503432

physical_verdict: CONFIRMED

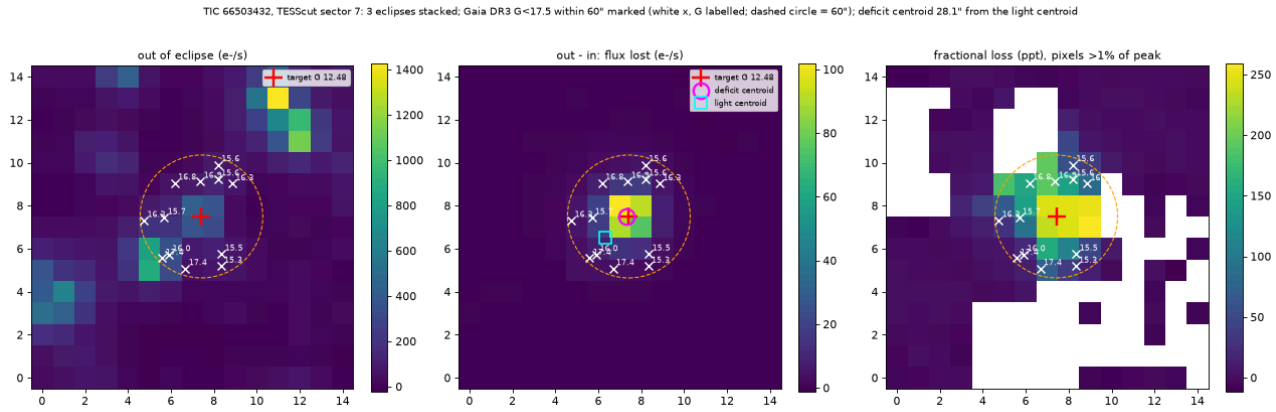
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Refine spectroscopic eccentric geometry and compare measured secondary phases; resolved physical modeling.



R584; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R584; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 66503432: CONFIRMED

Full bounded physical review: COMPLETE. Physical verdict and scoped catalogue novelty are separate. The epoch column names the selected repeating eclipse kind; deeper-primary rank is not assumed.

Deciding evidence: Nine deep selected eclipses and eight measured secondaries establish unequal spacings: secondary phase0.47448, then0.52552 back to selected kind. Half period cannot represent this event sequence. Compact target deficit has no capable Gaia neighbor within60arcsec. Source-specific Gaia predicts other phase0.51951 +/-0.01607 (linear allowance), a0.0450phase tension (~2.8 allowances). Argument-of-periastron and A/B conventions were checked against methods; observed centers, not Gaia predictions, define secondary phase. Primary conjunction agrees within0.0374d versus0.0965d allowance. This modest secondary geometry tension is retained as a limitation, not a calibrated rejection. Approximate minimum-mass central-duration model also remains a plausibility screen.

9 free-center quality-zero raw QLP SAP_FLUX timings in sectors[7, 61, 88] give P=8.3303238499 +/-0.0000193548d and selected-kind reference BTJD=1499.0390880. Median fitted diluted QLP depth=245.421ppt. Predicted windows guide the fit; they are not measured timings. Flagged/uncovered windows are not clean nulls. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Zero-degree-of-freedom reduced chi square is null, and singular center covariance is rejected. Representative raw plots and full raw phase plots were visually inspected. Within-sector consecutive same-kind counts and parity are recorded in ../alias-audit.json; half/double alternatives were checked against actual recurrence/secondary morphology. A double orbit with identical unseen labels cannot be exhaustively excluded; no exhaustive blind alias search is claimed.

Observed secondary check: 8 accepted secondary-candidate free-center fits, median actual phase0.4744845. Eccentric Gaia other conjunction is phase0.519507 +/-0.016070 descriptive linear allowance. This is evaluated at its actual phase, never assumed0.5. Weak/noisy box maxima in phase-review.json are search candidates, not detections.

Fresh TESScut S7 difference stack contains3 actually covered events. Deficit minus target= [-0.1339344344676583, -0.01917256934034217]pixel (radial0.135); target pixel coordinate= [7.4242021642934155, 7.498721557408501]; deficit=[7.290267729825757, 7.479548988068159].

Global PSF residual=0.855 is not a localization uncertainty. Actual target rounded-pixel signed deficit/baseline=90.161/382.86 electron/s. All0 capable Gaia neighbors within60arcsec are assessed below from real pixels. Required intrinsic loss is a pairwise Gaia G passband proxy; it must not be compared directly with diluted fractional pixel loss. Actual deficit and baseline are reported in common electron/s units. Colors, aperture response, PSF overlap, and unlisted faint/unresolved sources limit this screen; no calibrated centroid confidence or full PSF deblend is claimed.

Gaia source	sep arcsec	G	required intrinsic loss proxy	signed pixel deficit e/s	pixel baseline e/s	distance to deficit px	attribution assessment
None capable within60arcsec							

Alternative background-subtracted fixed target-centered TESScut1x1/3x3/5x5 depths=235.493/182.499/100.071ppt. Faintest40percent per-cadence background, symmetric local pre/post windows and actual covered cores provide an alternative extraction. Retained pixel-arrays.npz contains actual deficit/baseline. This is a reused numerical primitive with fresh ephemeris/duration, not independent-code replication. QLP and custom extractions share TESS photons in overlapping sectors; differing sectors are separate observing occasions on one instrument. Aperture/sector depth differences are dilution and reduction limitations, not intrinsic depth agreement.

Gaia DR3 source5616340171040985088 has P=8.331321017d, e=0.255453, omega=83.3363deg, Tp=390.922279BTJD. Both edge-on eccentric conjunctions and source-specific full NSS covariance are retained in ../gaia-geometry.json. Closest selected-kind offset=-0.037390d (-0.004488phase), linear covariance allowance=0.096489d. The SB1 lower triangle uses P,gamma,K,e,omega,Tp, and selected subindices0,3,4,5; minimum full correlation eigenvalue=0.011791. This descriptive linear approximation is not a calibrated rejection threshold. Mass function=0.17756solar masses; assumed TIC/adopted M1=1.250, minimum M2=0.951; TIC/adopted radius=3.074solar radii. Approximate central duration=8.802h versus measured6.875h. Main-sequence minimum-mass companion and TIC mass/radius make this a plausibility screen, not a fitted geometry. Orbital agreement alone cannot establish source ownership.

Sector	same-kind search cycle	measured midpoint BTJD	formal center error d	timing input allowance d	depth ppt
7	0	1499.0403290	0.00052515	0.0104166	245.421
7	1	1507.3650109	0.000934794	0.0104166	244.399
7	2	1515.7001850	0.000967621	0.0104166	242.915
61	176	2965.1775934	0.000232527	0.00115739	251.223
61	177	2973.5050634	0.000238581	0.00115739	247.312
61	178	2981.8367346	0.000337039	0.00115739	259.418
88	264	3698.2400203	0.000524037	0.00115739	223.748
88	265	3706.5768666	0.00042712	0.00115739	251.049
88	266	3714.9077080	0.000443278	0.00115739	228.439

Limitations and next check: Refine spectroscopic eccentric geometry and compare measured secondary phases; resolved physical modeling. Representative events suffice for this bounded review; no all-sector/all-event precision campaign is claimed. Sector-dependent dilution, finite cadence, baseline choice, color proxy, and linear Gaia propagation remain material limitations.

Catalogue checks use ../../shared-catalogues/coverage-matrix.csv and the shared receipt manifest; known recovery is separated from physical classification and no exhaustive priority claim is made. Evidence: measurements.json/raw-windows.csv/raw-events.png; fresh-

pixels/diff_fresh.json/png/pixel-arrays.npz; neighbor-assessment.json; batch phase, secondary, alias, identity/source receipts, methods snapshot and audits. SHA256SUMS.json freezes exact sources and outputs.

Stable review key: R584.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/pixel-arrays.npz - fresh-pixels/run.log - historical/diff_s07.json - historical/diff_s07.png - historical/gaia.json - measurements.json - neighbor-assessment.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

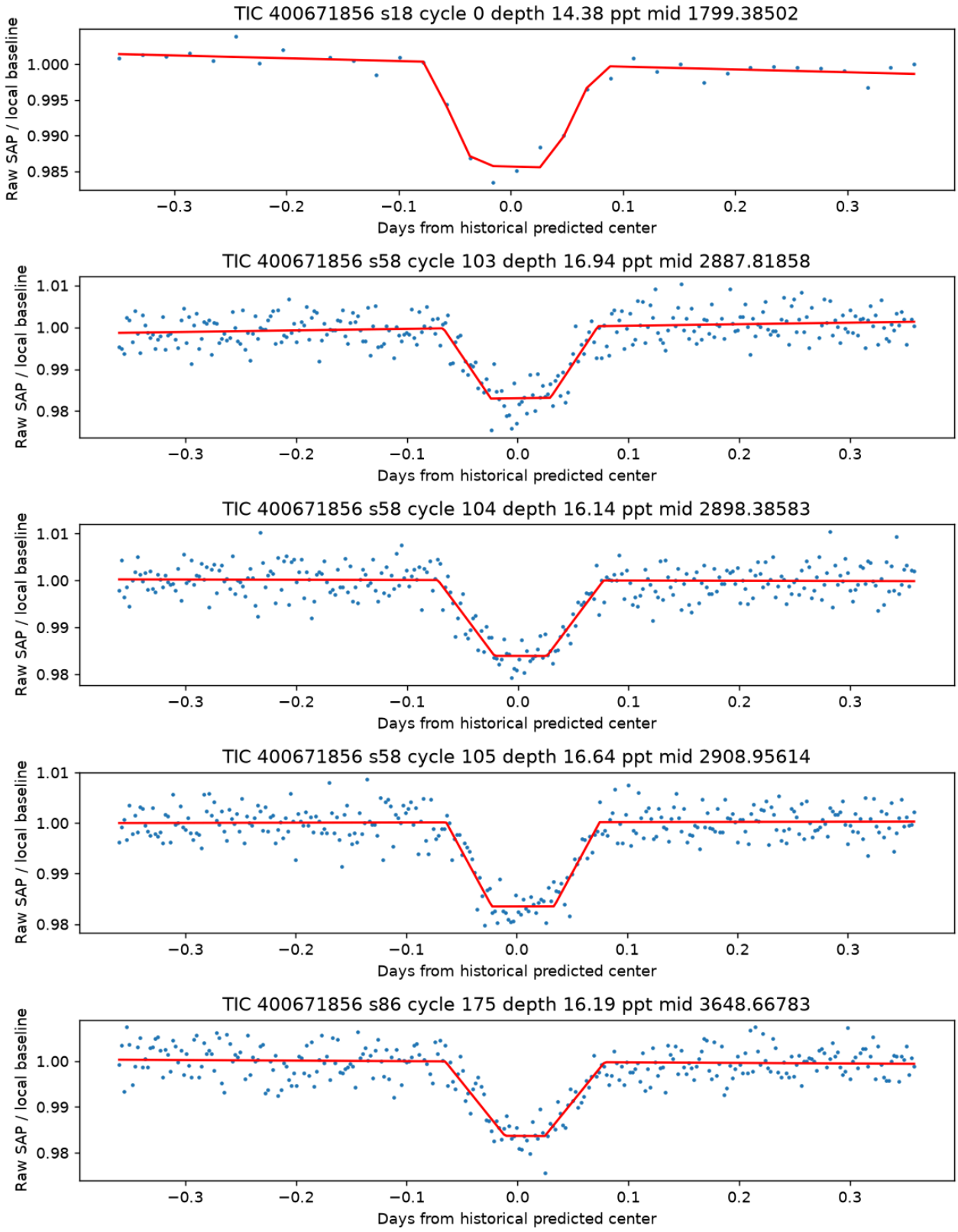
R585. TIC 400671856

physical_verdict: PROBABLE

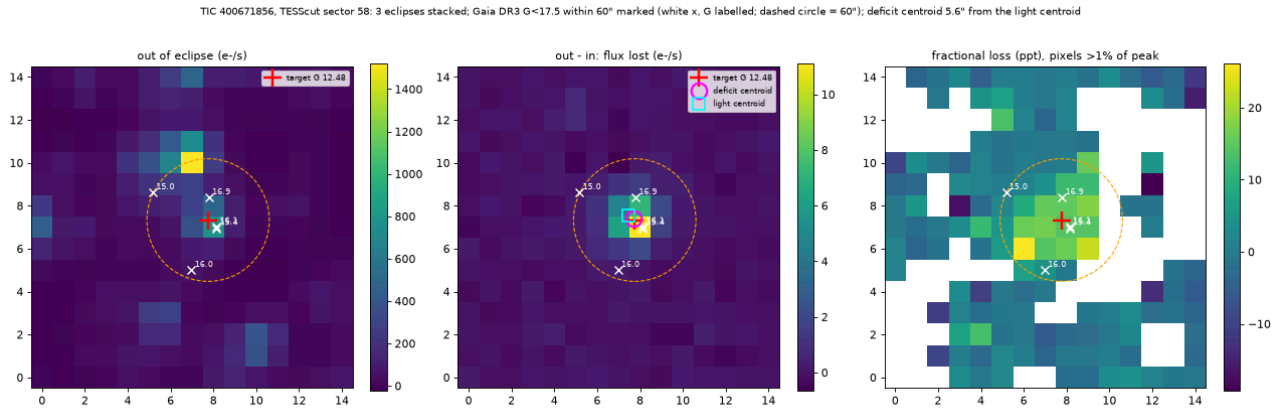
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry to establish ownership.



R585; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R585; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 400671856: PROBABLE

Full bounded physical review: COMPLETE. Physical verdict and scoped catalogue novelty are separate. The epoch column names the selected repeating eclipse kind; deeper-primary rank is not assumed.

Deciding evidence: Five sharp repeated selected eclipses establish10.56733d. Target-centered deficit favors target, but two capable sources at10.24/11.31arcsec overlap its PSF within0.54pixel, so ownership cannot be uniquely established. Two additional capable sources at50.7/60.0arcsec are distinct from the compact core. No secure secondary is claimed.

5 free-center quality-zero raw QLP SAP_FLUX timings in sectors[18, 58, 86] give P=10.5673332477 +/-0.0000275715d and selected-kind reference BTJD=1799.3845652. Median fitted diluted QLP depth=16.189ppt. Predicted windows guide the fit; they are not measured timings. Flagged/uncovered windows are not clean nulls. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Zero-degree-of-freedom reduced chi square is null, and singular center covariance is rejected. Representative raw plots and full raw phase plots were visually inspected. Within-sector consecutive same-kind counts and parity are recorded in ../alias-audit.json; half/double alternatives were checked against actual recurrence/secondary morphology. A double orbit with identical unseen labels cannot be exhaustively excluded; no exhaustive blind alias search is claimed.

Observed secondary check: No secure secondary timing is claimed; absence of a claim is not a clean null. Eccentric Gaia other conjunction is phase0.499782 +/-0.009856 descriptive linear allowance. This is evaluated at its actual phase, never assumed0.5. Weak/noisy box maxima in phase-review.json are search candidates, not detections.

Fresh TESScut S58 difference stack contains3 actually covered events. Deficit minus target= [-0.07521918222497614, 0.056124943154161 5]pixel (radial0.094); target pixel coordinate= [7.751384817213834, 7.333134434599639]; deficit=[7.676165634988858, 7.389259377753801]. Global PSF residual=0.906 is not a localization uncertainty. Actual target rounded-pixel signed deficit/baseline=11.112/769.02 electron/s. All4 capable Gaia neighbors within60arcsec are assessed below from real pixels. Required intrinsic loss is a pairwise Gaia G passband proxy; it must not be

compared directly with diluted fractional pixel loss. Actual deficit and baseline are reported in common electron/s units. Colors, aperture response, PSF overlap, and unlisted faint/unresolved sources limit this screen; no calibrated centroid confidence or full PSF deblend is claimed.

Gaia source	sep arcsec	G	required intrinsic loss proxy	signed pixel deficit e/s	pixel baseline e/s	distance to deficit px	attribution assessment
5185550454 41696512	10.24	16.1 02	0.4723	11.112	769.02	0.584	unresolved overlapping core
5185550454 48010112	11.31	15.4 45	0.2654	11.112	769.02	0.634	unresolved overlapping core
5185550798 07739392	50.70	15.9 76	0.4225	0.2832	66.347	2.494	spatially distinct from compact target deficit; wings/noise may contribute
5185542551 74025600	59.96	15.0 15	0.1838	0.064543	216.67	2.784	spatially distinct from compact target deficit; wings/noise may contribute

Alternative background-subtracted fixed target-centered TESScut1x1/3x3/5x5 depths=14.450/15.326/12.839ppt. Faintest40percent per-cadence background, symmetric local pre/post windows and actual covered cores provide an alternative extraction. Retained pixel-arrays.npz contains actual deficit/baseline. This is a reused numerical primitive with fresh ephemeris/duration, not independent-code replication. QLP and custom extractions share TESS photons in overlapping sectors; differing sectors are separate observing occasions on one instrument. Aperture/sector depth differences are dilution and reduction limitations, not intrinsic depth agreement.

Gaia DR3 source518554976728533760 has P=10.569061492d, e=0.158698, omega=270.1220deg, Tp=393.850304BTJD. Both edge-on eccentric conjunctions and source-specific full NSS covariance are retained in ../gaia-geometry.json. Closest selected-kind offset=-0.148350d (-0.014036phase), linear covariance allowance=0.126004d. The SB1 lower triangle uses P,gamma,K,e,omega,Tp, and selected subindices0,3,4,5; minimum full correlation eigenvalue=0.010093. This descriptive linear approximation is not a calibrated rejection threshold. Mass function=0.12066solar masses; assumed TIC/adopted M1=1.000, minimum M2=0.705; TIC/adopted radius=2.122solar radii. Approximate central duration=8.177h versus measured3.453h. Main-sequence minimum-mass companion and TIC mass/radius make this a plausibility screen, not a fitted geometry. Orbital agreement alone cannot establish source ownership.

Sector	same-kind search cycle	measured midpoint BTJD	formal center error d	timing input allowance d	depth ppt
18	0	1799.3850205	0.00206159	0.0104167	14.376
58	103	2887.8185848	0.0015294	0.0015294	16.937
58	104	2898.3858255	0.00154563	0.00154563	16.140
58	105	2908.9561447	0.00117193	0.00117193	16.644
86	175	3648.6678285	0.00174748	0.00174748	16.189

Limitations and next check: Resolved eclipse photometry to establish ownership. Representative events suffice for this bounded review; no all-sector/all-event precision campaign is claimed. Sector-dependent dilution, finite cadence, baseline choice, color proxy, and linear Gaia propagation remain material limitations.

Catalogue checks use ../../shared-catalogues/coverage-matrix.csv and the shared receipt manifest; known recovery is separated from physical classification and no exhaustive priority claim is made. Evidence: measurements.json/raw-windows.csv/raw-events.png; fresh-pixels/diff_fresh.json/png/pixel-arrays.npz; neighbor-assessment.json; batch phase, secondary, alias, identity/source receipts, methods snapshot and audits. SHA256SUMS.json freezes exact sources and outputs.

Stable review key: R585.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/pixel-arrays.npz - fresh-pixels/run.log - historical/diff_s58.json - historical/diff_s58.png - historical/gaia.json - measurements.json - neighbor-assessment.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

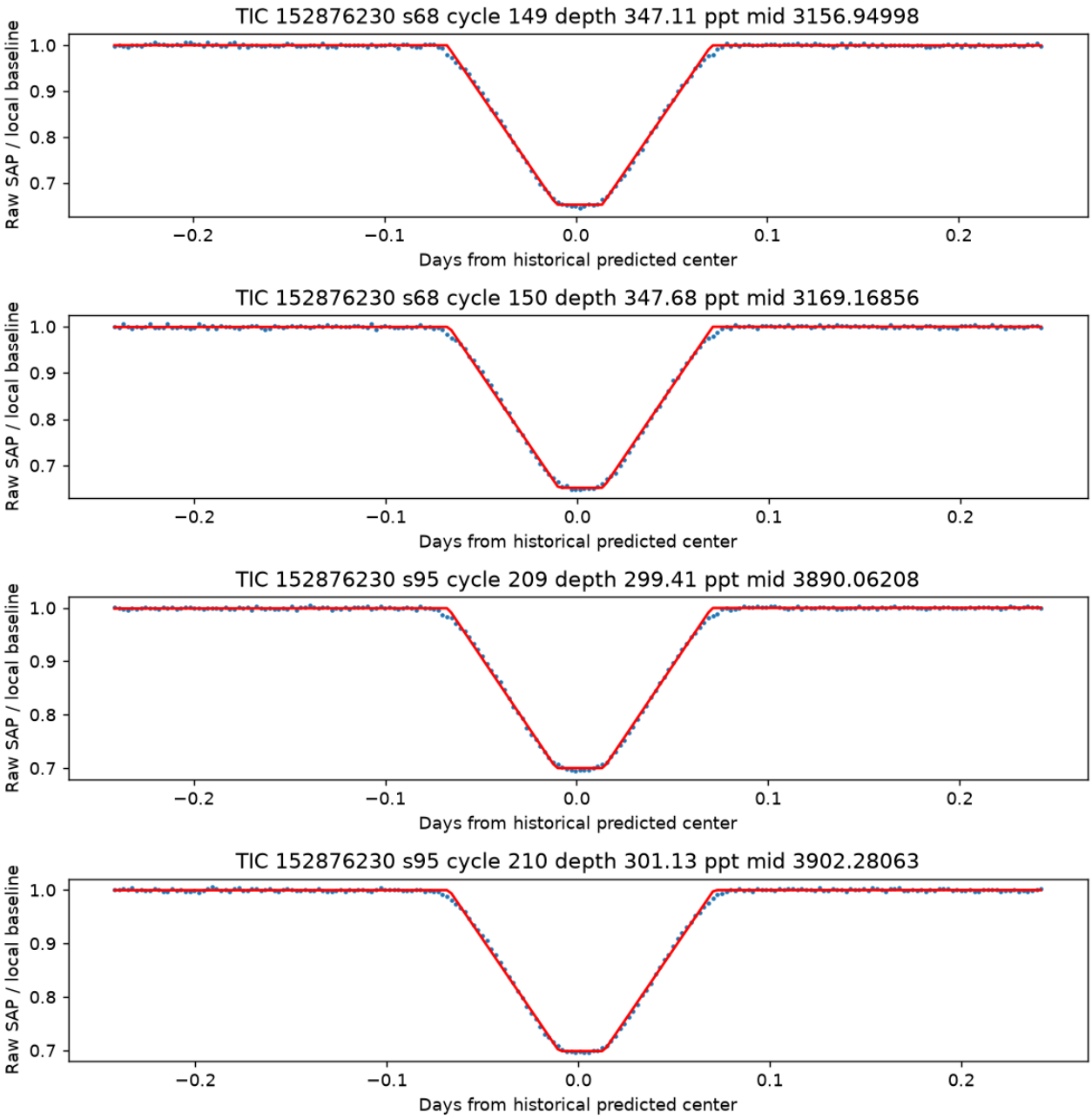
R586. TIC 152876230

physical_verdict: CONFIRMED

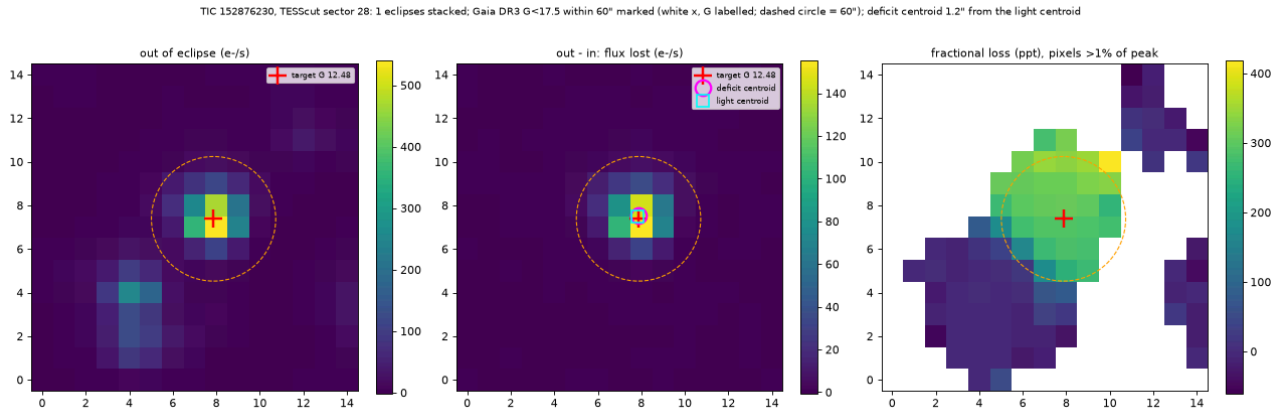
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved follow-up and physical binary modeling.



R586; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R586; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 152876230: CONFIRMED

Full bounded physical review: COMPLETE. Physical verdict and scoped catalogue novelty are separate. The epoch column names the selected repeating eclipse kind; deeper-primary rank is not assumed.

Deciding evidence: Four deep selected-kind events in S68/S95 and three measured secondaries near0.71238phase support eccentric12.21853d orbit and reject a half-period interpretation. Fresh target-centered deficit has no capable Gaia neighbor within60arcsec. S1 lacks an accepted selected-center fit but has secondary coverage, not a clean null. Selected depths347ppt in S68 versus300ppt S95 show sector dilution changes; reported median is a pipeline-dependent quantity.

4 free-center quality-zero raw QLP SAP_FLUX timings in sectors[68, 95] give P=12.2185346745 +/-0.0000192876d and selected-kind reference BTJD=3156.9500050. Median fitted diluted QLP depth=324.118ppt. Predicted windows guide the fit; they are not measured timings. Flagged/uncovered windows are not clean nulls. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Zero-degree-of-freedom reduced chi square is null, and singular center covariance is rejected. Representative raw plots and full raw phase plots were visually inspected. Within-sector consecutive same-kind counts and parity are recorded in ../alias-audit.json; half/double alternatives were checked against actual recurrence/secondary morphology. A double orbit with identical unseen labels cannot be exhaustively excluded; no exhaustive blind alias search is claimed.

Observed secondary check: 3 accepted secondary-candidate free-center fits, median actual phase0.7123817. Eccentric Gaia other conjunction is phase0.683747 +/-0.017816 descriptive linear allowance. This is evaluated at its actual phase, never assumed0.5. Weak/noisy box maxima in phase-review.json are search candidates, not detections.

Fresh TESScut S28 difference stack contains1 actually covered events. Deficit minus target= [-0.021333781357664172, 0.15011624243996202]pixel (radial0.152); target pixel coordinate= [7.872770821080806, 7.3841402672423655]; deficit=[7.851437039723142, 7.5342565096823275]. Global PSF residual=0.367 is not a localization uncertainty. Actual target rounded-pixel signed deficit/baseline=155.17/539.52 electron/s. All0 capable Gaia neighbors within60arcsec are assessed below from real pixels. Required intrinsic loss is a pairwise Gaia G passband proxy; it must not be

compared directly with diluted fractional pixel loss. Actual deficit and baseline are reported in common electron/s units. Colors, aperture response, PSF overlap, and unlisted faint/unresolved sources limit this screen; no calibrated centroid confidence or full PSF deblend is claimed.

Gaia source	sep arcsec	G	required intrinsic loss proxy	signed pixel deficit e/s	pixel baseline e/s	distance to deficit px	attribution assessment
None capable within60arcsec							

Alternative background-subtracted fixed target-centered TESScut1x1/3x3/5x5 depths=287.599/294.633/293.055ppt. Faintest40percent per-cadence background, symmetric local pre/post windows and actual covered cores provide an alternative extraction. Retained pixel-arrays.npz contains actual deficit/baseline. This is a reused numerical primitive with fresh ephemeris/duration, not independent-code replication. QLP and custom extractions share TESS photons in overlapping sectors; differing sectors are separate observing occasions on one instrument. Aperture/sector depth differences are dilution and reduction limitations, not intrinsic depth agreement.

Gaia DR3 source6593293356532115712 has P=12.217595220d, e=0.305264, omega=17.1561deg, Tp=394.254310BTJD. Both edge-on eccentric conjunctions and source-specific full NSS covariance are retained in ../gaia-geometry.json. Closest selected-kind offset=0.094190d (0.007709phase), linear covariance allowance=0.445736d. The SB1 lower triangle uses P,gamma,K,e,omega,Tp, and selected subindices0,3,4,5; minimum full correlation eigenvalue=0.023911. This descriptive linear approximation is not a calibrated rejection threshold. Mass function=0.0497709solar masses; assumed TIC/adopted M1=0.750, minimum M2=0.405; TIC/adopted radius=0.908solar radii. Approximate central duration=4.849h versus measured3.308h. Main-sequence minimum-mass companion and TIC mass/radius make this a plausibility screen, not a fitted geometry. Orbital agreement alone cannot establish source ownership.

Sector	same-kind search cycle	measured midpoint BTJD	formal center error d	timing input allowance d	depth ppt
68	149	3156.9499806	5.55939e-05	0.00115742	347.109
68	150	3169.1685645	6.10773e-05	0.00115742	347.685
95	209	3890.0620792	4.70031e-05	0.00115742	299.412
95	210	3902.2806258	5.79582e-05	0.00115742	301.127

Limitations and next check: Optional resolved follow-up and physical binary modeling. Representative events suffice for this bounded review; no all-sector/all-event precision campaign is claimed. Sector-dependent dilution, finite cadence, baseline choice, color proxy, and linear Gaia propagation remain material limitations.

Catalogue checks use ../../shared-catalogues/coverage-matrix.csv and the shared receipt manifest; known recovery is separated from physical classification and no exhaustive priority claim is made. Evidence: measurements.json/raw-windows.csv/raw-events.png; fresh-pixels/diff_fresh.json/png/pixel-arrays.npz; neighbor-assessment.json; batch phase, secondary, alias, identity/source receipts, methods snapshot and audits. SHA256SUMS.json freezes exact sources and outputs.

Stable review key: R586.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/pixel-arrays.npz - fresh-pixels/run.log - historical/diff_s28.json - historical/diff_s28.png - historical/gaia.json - measurements.json - neighbor-assessment.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

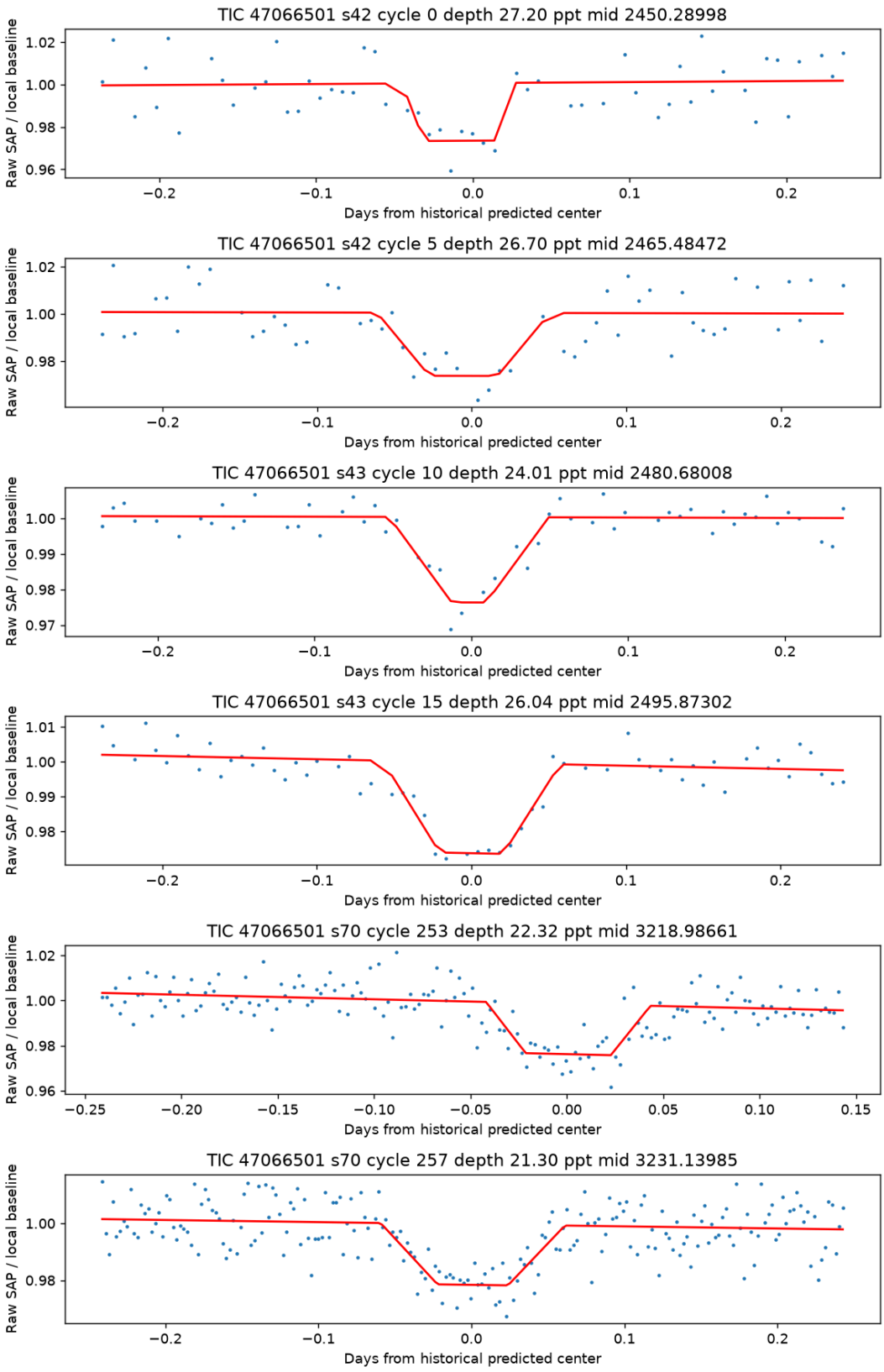
R587. TIC 47066501

physical_verdict: CONFIRMED

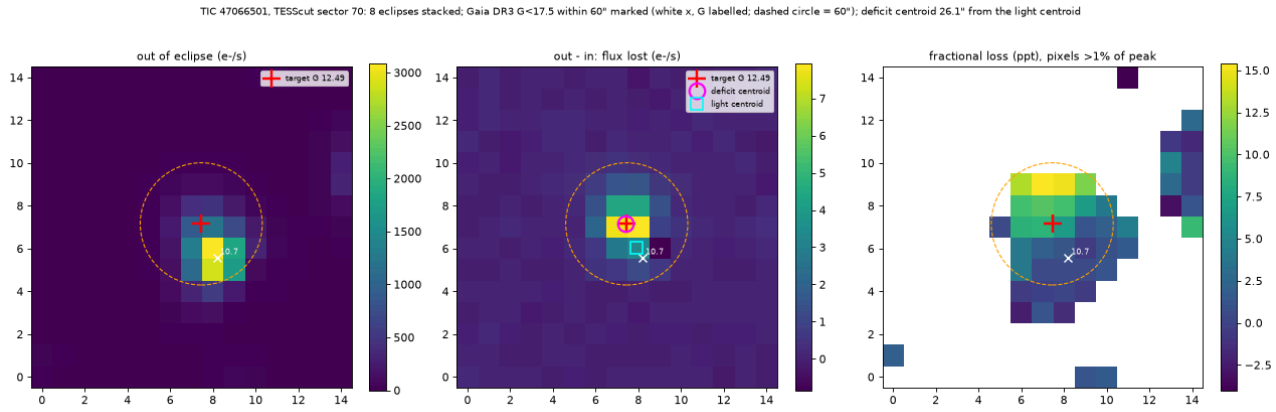
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved follow-up and physical binary modeling.



R587; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R587; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 47066501: CONFIRMED

Full bounded physical review: COMPLETE. Physical verdict and scoped catalogue novelty are separate. The epoch column names the selected repeating eclipse kind; deeper-primary rank is not assumed.

Deciding evidence: Eighteen sharp selected eclipses across three sectors establish3.03829d. Brighter G10.68 neighbor34.7arcsec away dominates the baseline image but the compact deficit is on target, spatially displaced from that neighbor. The1x1/3x3/5x5 depth dilution progression and the26.1arcsec light-to-deficit displacement are expected for that blend; source ownership follows actual deficit morphology. No secure secondary is claimed.

18 free-center quality-zero raw QLP SAP_FLUX timings in sectors[42, 43, 70] give P=3.0382901257 +/-0.0000064772d and selected-kind reference BTJD=2450.2999227. Median fitted diluted QLP depth=22.508ppt. Predicted windows guide the fit; they are not measured timings. Flagged/uncovered windows are not clean nulls. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Zero-degree-of-freedom reduced chi square is null, and singular center covariance is rejected. Representative raw plots and full raw phase plots were visually inspected. Within-sector consecutive same-kind counts and parity are recorded in ../alias-audit.json; half/double alternatives were checked against actual recurrence/secondary morphology. A double orbit with identical unseen labels cannot be exhaustively excluded; no exhaustive blind alias search is claimed.

Observed secondary check: No secure secondary timing is claimed; absence of a claim is not a clean null. Eccentric Gaia other conjunction is phase0.449798 +/-0.056911 descriptive linear allowance. This is evaluated at its actual phase, never assumed0.5. Weak/noisy box maxima in phase-review.json are search candidates, not detections.

Fresh TESScut S70 difference stack contains8 actually covered events. Deficit minus target= [-0.044238690131201075, 0.020629504697470047]pixel (radial0.049); target pixel coordinate= [7.454940439580826, 7.154530424464195]; deficit=[7.410701749449625, 7.175159929161665]. Global PSF residual=0.858 is not a localization uncertainty. Actual target rounded-pixel signed deficit/baseline=7.7313/893.81 electron/s. All1 capable Gaia neighbors within60arcsec are assessed below from real pixels. Required intrinsic loss is a pairwise Gaia G passband proxy; it must not be

compared directly with diluted fractional pixel loss. Actual deficit and baseline are reported in common electron/s units. Colors, aperture response, PSF overlap, and unlisted faint/unresolved sources limit this screen; no calibrated centroid confidence or full PSF deblend is claimed.

Gaia source	sep arcsec	G	required intrinsic loss proxy	signed pixel deficit e/s	pixel baseline e/s	distance to deficit px	attribution assessment
2586915362805716608	34.72	10.682	0.0268	3.2814	3081.1	1.777	spatially distinct from compact target deficit; wings/noise may contribute

Alternative background-subtracted fixed target-centered TESScut1x1/3x3/5x5 depths=8.650/4.378/2.628ppt. Faintest40percent per-cadence background, symmetric local pre/post windows and actual covered cores provide an alternative extraction. Retained pixel-arrays.npz contains actual deficit/baseline. This is a reused numerical primitive with fresh ephemeris/duration, not independent-code replication. QLP and custom extractions share TESS photons in overlapping sectors; differing sectors are separate observing occasions on one instrument. Aperture/sector depth differences are dilution and reduction limitations, not intrinsic depth agreement.

Gaia DR3 source2586915392870164224 has P=3.038142919d, e=0.081953, omega=195.6368deg, Tp=388.144755BTJD. Both edge-on eccentric conjunctions and source-specific full NSS covariance are retained in ../gaia-geometry.json. Closest selected-kind offset=0.070101d (0.023074phase), linear covariance allowance=0.293388d. The SB1 lower triangle uses P,gamma,K,e,omega,Tp, and selected subindices0,3,4,5; minimum full correlation eigenvalue=0.0031323. This descriptive linear approximation is not a calibrated rejection threshold. Mass function=0.00422369solar masses; assumed TIC/adopted M1=1.320, minimum M2=0.215; TIC/adopted radius=1.624solar radii. Approximate central duration=4.452h versus measured2.443h. Main-sequence minimum-mass companion and TIC mass/radius make this a plausibility screen, not a fitted geometry. Orbital agreement alone cannot establish source ownership.

Sector	same-kind search cycle	measured midpoint BTJD	formal center error d	timing input allowance d	depth ppt
42	0	2450.2899789	0.0232971	0.0232971	27.200
42	1	2453.3368357	0.00638327	0.00638327	31.715
42	4	2462.4512290	0.00642456	0.00642456	18.476
42	5	2465.4847177	0.00662085	0.00662085	26.696
43	8	2474.6079732	0.00319346	0.00347235	23.105
43	9	2477.6473411	0.00405423	0.00405423	21.306
43	10	2480.6800797	0.00184336	0.00347235	24.006
43	13	2489.7951346	0.00249847	0.00347235	21.636
43	14	2492.8435885	0.00408437	0.00408437	24.336
43	15	2495.8730218	0.00176639	0.00347235	26.037
70	250	3209.8757498	0.0014423	0.0014423	20.043
70	251	3212.9100979	0.00226656	0.00226656	22.406
70	252	3215.9486560	0.00215097	0.00215097	22.609
70	253	3218.9866121	0.00163928	0.00163928	22.320
70	254	3222.0237686	0.00238726	0.00238726	22.230
70	255	3225.0648092	0.00310524	0.00310524	23.172
70	256	3228.0981332	0.00229693	0.00229693	20.547
70	257	3231.1398479	0.00267587	0.00267587	21.301

Limitations and next check: Optional resolved follow-up and physical binary modeling. Representative events suffice for this bounded review; no all-sector/all-event precision campaign is claimed. Sector-dependent dilution, finite cadence, baseline choice, color proxy, and linear Gaia

propagation remain material limitations.

Catalogue checks use ../../shared-catalogues/coverage-matrix.csv and the shared receipt manifest; known recovery is separated from physical classification and no exhaustive priority claim is made. Evidence: measurements.json/raw-windows.csv/raw-events.png; fresh-pixels/diff_fresh.json/png/pixel-arrays.npz; neighbor-assessment.json; batch phase, secondary, alias, identity/source receipts, methods snapshot and audits. SHA256SUMS.json freezes exact sources and outputs.

Depth correction audit (2026-11-02): realized trapezoid decrement replaces its unconstrained amplitude parameter. Original input windows, fitted centers/widths, timing selections and physical verdict are retained; capable-neighbor membership is unchanged. See ../../depth-audit/README.md.

Stable review key: R587.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/pixel-arrays.npz - fresh-pixels/run.log - historical/diff_s70.json - historical/diff_s70.png - historical/gaia.json - measurements.json - neighbor-assessment.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

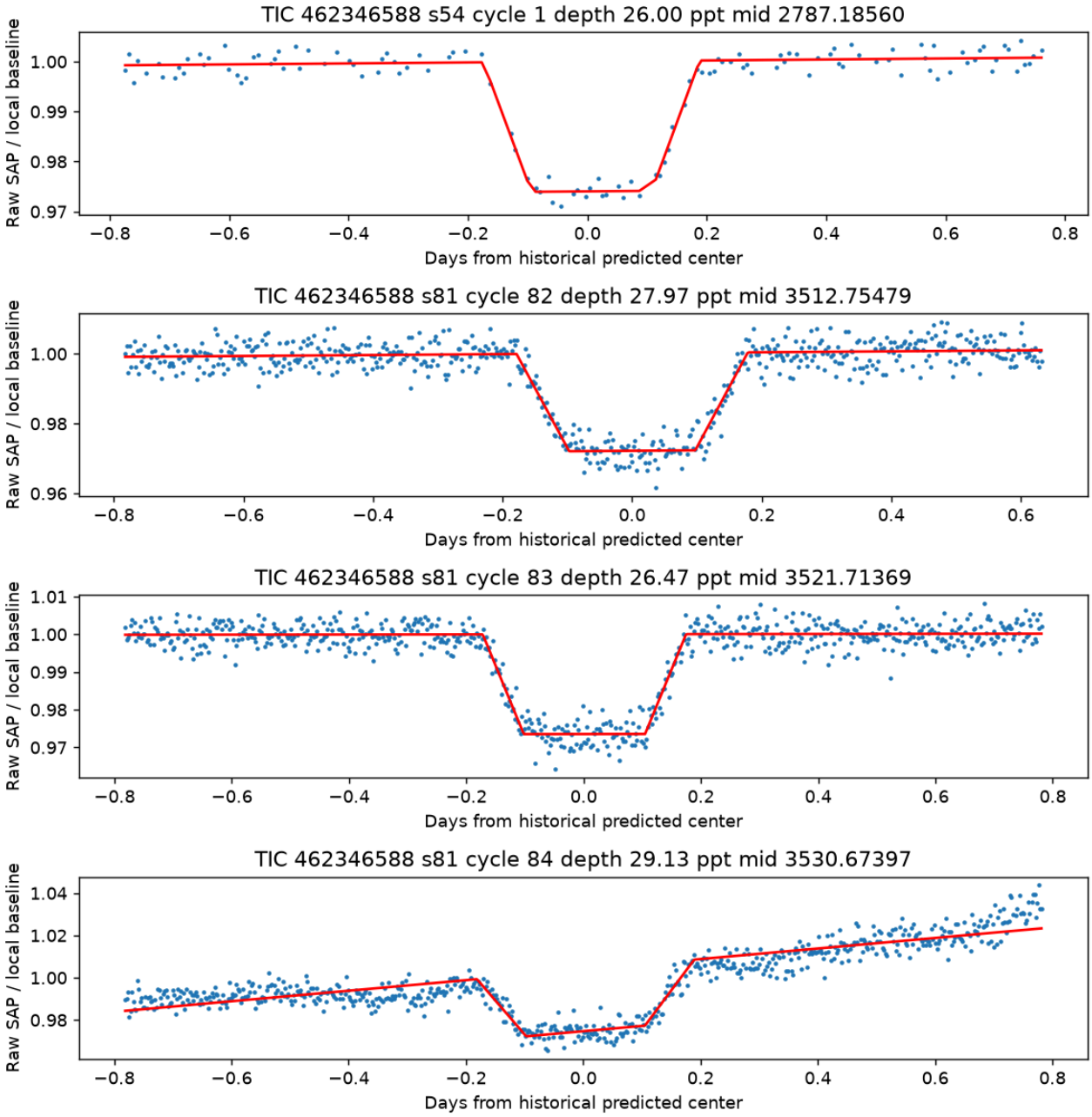
R588. TIC 462346588

physical_verdict: CONFIRMED

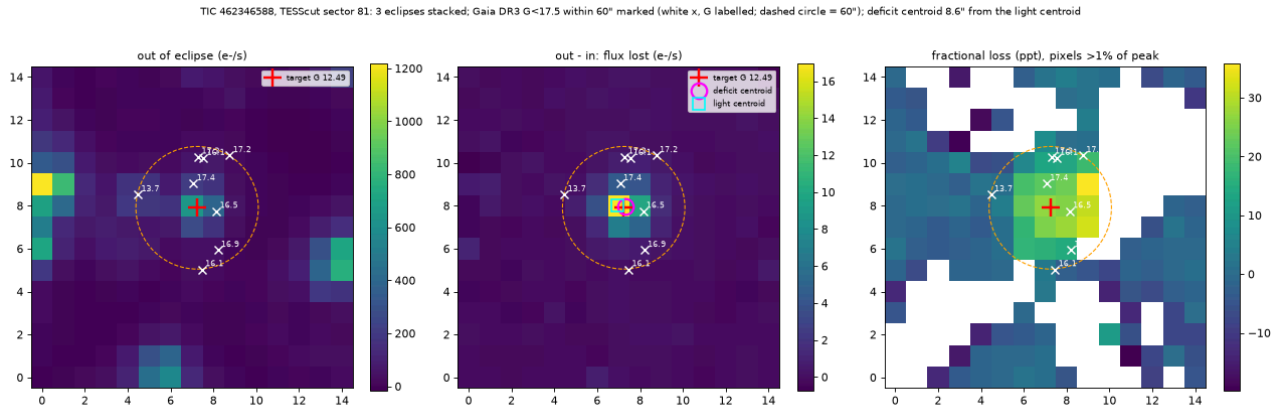
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved follow-up and physical binary modeling.



R588; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R588; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 462346588: CONFIRMED

Full bounded physical review: COMPLETE. Physical verdict and scoped catalogue novelty are separate. The epoch column names the selected repeating eclipse kind; deeper-primary rank is not assumed.

Deciding evidence: Four flat-bottomed selected eclipses include three consecutive S81 cycles. Compact target deficit is distinct from all three capable sources at47.3-59.0arcsec. Long selected duration8.63h is compatible with approximate central12.73h geometry. No secure secondary is claimed.

4 free-center quality-zero raw QLP SAP_FLUX timings in sectors[54, 81] give P=8.9576575673 +/-0.0000856199d and selected-kind reference BTJD=2787.1852462. Median fitted diluted QLP depth=27.222ppt. Predicted windows guide the fit; they are not measured timings. Flagged/uncovered windows are not clean nulls. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Zero-degree-of-freedom reduced chi square is null, and singular center covariance is rejected. Representative raw plots and full raw phase plots were visually inspected. Within-sector consecutive same-kind counts and parity are recorded in ../alias-audit.json; half/double alternatives were checked against actual recurrence/secondary morphology. A double orbit with identical unseen labels cannot be exhaustively excluded; no exhaustive blind alias search is claimed.

Observed secondary check: No secure secondary timing is claimed; absence of a claim is not a clean null. Eccentric Gaia other conjunction is phase0.559530 +/-0.030947 descriptive linear allowance. This is evaluated at its actual phase, never assumed0.5. Weak/noisy box maxima in phase-review.json are search candidates, not detections.

Fresh TESScut S81 difference stack contains3 actually covered events. Deficit minus target= [0.09023693059382865, 0.06710189477644501]pixel (radial0.112); target pixel coordinate= [7.23766146742701, 7.905209866766553]; deficit=[7.327898398020839, 7.972311761542998]. Global PSF residual=0.446 is not a localization uncertainty. Actual target rounded-pixel signed deficit/baseline=16.96/632.62 electron/s. All3 capable Gaia neighbors within60arcsec are assessed below from real pixels. Required intrinsic loss is a pairwise Gaia G passband proxy; it must not be compared directly with diluted fractional pixel loss. Actual deficit and baseline are reported in common electron/s units. Colors, aperture response, PSF overlap, and unlisted faint/unresolved sources limit this screen; no calibrated centroid confidence or full PSF deblend is claimed.

Gaia source	sep arcsec	G	required intrinsic loss proxy	signed pixel deficit e/s	pixel baseline e/s	distance to deficit px	attribution assessment
4251093071200793984	47.34	16.090	0.7767	0.96993	69.904	2.250	spatially distinct from compact target deficit; wings/noise may contribute
4251093109859804416	58.78	13.747	0.1138	0.26829	171.81	2.905	spatially distinct from compact target deficit; wings/noise may contribute
4251093384737714176	58.99	16.141	0.8125	0.13569	38.015	2.948	spatially distinct from compact target deficit; wings/noise may contribute

Alternative background-subtracted fixed target-centered TESScut1x1/3x3/5x5 depths=26.810/24.868/20.199ppt. Faintest40percent per-cadence background, symmetric local pre/post windows and actual covered cores provide an alternative extraction. Retained pixel-arrays.npz contains actual deficit/baseline. This is a reused numerical primitive with fresh ephemeris/duration, not independent-code replication. QLP and custom extractions share TESS photons in overlapping sectors; differing sectors are separate observing occasions on one instrument. Aperture/sector depth differences are dilution and reduction limitations, not intrinsic depth agreement.

Gaia DR3 source4251093144219543936 has P=8.956840784d, e=0.357786, omega=284.2110deg, Tp=391.724841BTJD. Both edge-on eccentric conjunctions and source-specific full NSS covariance are retained in ../gaia-geometry.json. Closest selected-kind offset=0.195713d (0.021851phase), linear covariance allowance=0.584442d. The SB1 lower triangle uses P,gamma,K,e,omega,Tp, and selected subindices0,3,4,5; minimum full correlation eigenvalue=0.029353. This descriptive linear approximation is not a calibrated rejection threshold. Mass function=0.0125728solar masses; assumed TIC/adopted M1=1.170, minimum M2=0.301; TIC/adopted radius=2.304solar radii. Approximate central duration=12.734h versus measured8.633h. Main-sequence minimum-mass companion and TIC mass/radius make this a plausibility screen, not a fitted geometry. Orbital agreement alone cannot establish source ownership.

Sector	same-kind search cycle	measured midpoint BTJD	formal center error d	timing input allowance d	depth ppt
54	1	2787.1855977	0.00227246	0.00694447	25.996
81	82	3512.7547875	0.0012022	0.0012022	27.971
81	83	3521.7136938	0.000976404	0.00115741	26.473
81	84	3530.6739685	0.00565037	0.00565037	29.126

Limitations and next check: Optional resolved follow-up and physical binary modeling. Representative events suffice for this bounded review; no all-sector/all-event precision campaign is claimed. Sector-dependent dilution, finite cadence, baseline choice, color proxy, and linear Gaia propagation remain material limitations.

Catalogue checks use ../../shared-catalogues/coverage-matrix.csv and the shared receipt manifest; known recovery is separated from physical classification and no exhaustive priority claim is made. Evidence: measurements.json/raw-windows.csv/raw-events.png; fresh-pixels/diff_fresh.json/png/pixel-arrays.npz; neighbor-assessment.json; batch phase, secondary, alias, identity/source receipts, methods snapshot and audits. SHA256SUMS.json freezes exact sources and outputs.

Stable review key: R588.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/pixel-arrays.npz - fresh-pixels/run.log - historical/diff_s81.json - historical/diff_s81.png - historical/gaia.json - measurements.json - neighbor-assessment.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

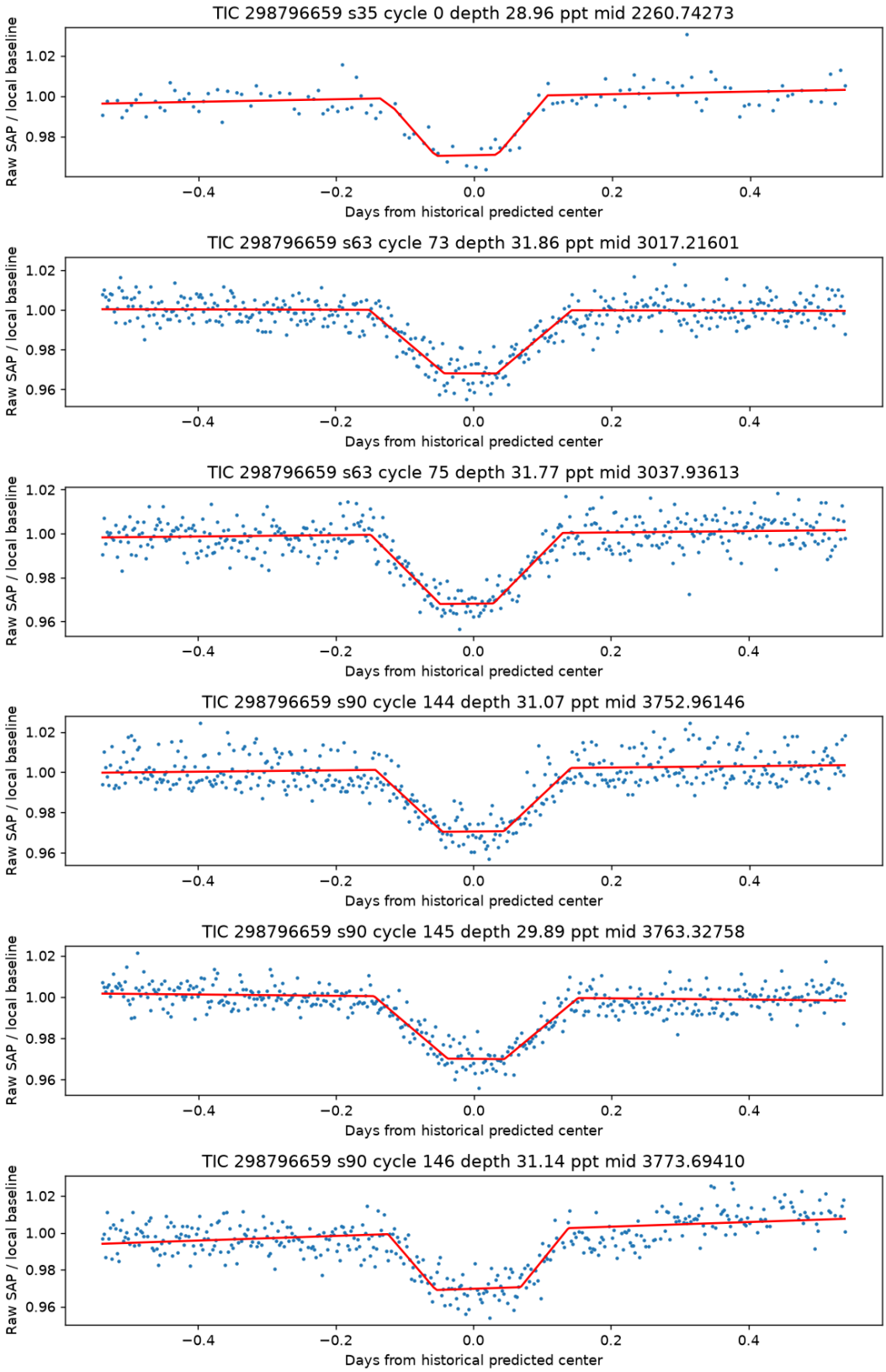
R589. TIC 298796659

physical_verdict: PROBABLE

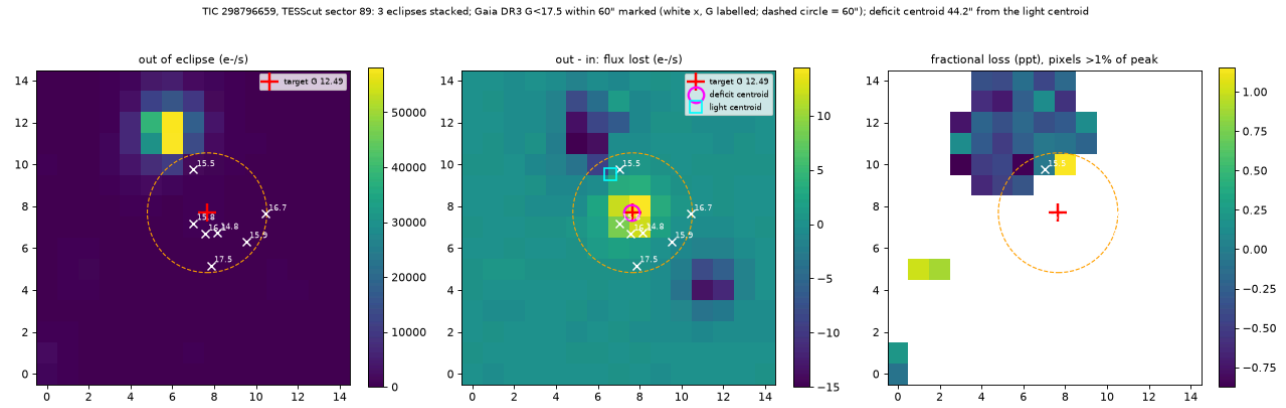
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry to establish ownership.



R589; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R589; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 298796659: PROBABLE

Full bounded physical review: COMPLETE. Physical verdict and scoped catalogue novelty are separate. The epoch column names the selected repeating eclipse kind; deeper-primary rank is not assumed.

Deciding evidence: Six sharp repeated selected eclipses establish10.36267d. Target-centered deficit favors target, but three capable sources at16.2-21.6arcsec overlap the core within roughly onepixel. Two farther capable sources43.7/47.5arcsec are spatially distinct. Unresolved close-source ownership prevents confirmation; large variable raw baselines are kept separate from selected eclipses.

6 free-center quality-zero raw QLP SAP_FLUX timings in sectors[35, 63, 90] give P=10.3626720256 +/-0.0000312949d and selected-kind reference BTJD=2260.7394561. Median fitted diluted QLP depth=31.102ppt. Predicted windows guide the fit; they are not measured timings. Flagged/uncovered windows are not clean nulls. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Zero-degree-of-freedom reduced chi square is null, and singular center covariance is rejected. Representative raw plots and full raw phase plots were visually inspected. Within-sector consecutive same-kind counts and parity are recorded in ../alias-audit.json; half/double alternatives were checked against actual recurrence/secondary morphology. A double orbit with identical unseen labels cannot be exhaustively excluded; no exhaustive blind alias search is claimed.

Observed secondary check: No secure secondary timing is claimed; absence of a claim is not a clean null. Eccentric Gaia other conjunction is phase0.484133 +/-0.034816 descriptive linear allowance. This is evaluated at its actual phase, never assumed0.5. Weak/noisy box maxima in phase-review.json are search candidates, not detections.

Fresh TESScut S89 difference stack contains3 actually covered events. Deficit minus target= [-0.04559348445274036, -0.004124933636912686 6]pixel (radial0.046); target pixel coordinate= [7.661210992027399, 7.689106870375202]; deficit=[7.615617507574659, 7.684981936738289]. Global PSF residual=0.812 is not a localization uncertainty. Actual target rounded-pixel signed deficit/baseline=14.463/525.57 electron/s. All5 capable Gaia neighbors within60arcsec are assessed below from real pixels. Required intrinsic loss is a pairwise Gaia G passband proxy; it must not be

compared directly with diluted fractional pixel loss. Actual deficit and baseline are reported in common electron/s units. Colors, aperture response, PSF overlap, and unlisted faint/unresolved sources limit this screen; no calibrated centroid confidence or full PSF deblend is claimed.

Gaia source	sep arcsec	G	required intrinsic loss proxy	signed pixel deficit e/s	pixel baseline e/s	distance to deficit px	attribution assessment
5410277295264120320	16.19	15.769	0.6667	8.5665	346.83	0.782	unresolved overlapping core
5410277295264120832	20.11	16.076	0.8745	12.21	502.28	1.001	unresolved overlapping core
5410277295264120448	21.62	14.817	0.2955	12.21	502.28	1.078	unresolved overlapping core
5410089588017992832	43.73	15.502	0.5280	-0.066141	1763.7	2.147	spatially distinct from compact target deficit; wings/noise may contribute
5410277363983597312	47.46	15.919	0.7608	-0.74326	9.0855	2.344	spatially distinct from compact target deficit; wings/noise may contribute

Alternative background-subtracted fixed target-centered TESScut1x1/3x3/5x5 depths=27.519/21.555/3.541ppt. Faintest40percent per-cadence background, symmetric local pre/post windows and actual covered cores provide an alternative extraction. Retained pixel-arrays.npz contains actual deficit/baseline. This is a reused numerical primitive with fresh ephemeris/duration, not independent-code replication. QLP and custom extractions share TESS photons in overlapping sectors; differing sectors are separate observing occasions on one instrument. Aperture/sector depth differences are dilution and reduction limitations, not intrinsic depth agreement.

Gaia DR3 source5410277295264120064 has P=10.361888573d, e=0.413866, omega=273.1440deg, Tp=385.199903BTJD. Both edge-on eccentric conjunctions and source-specific full NSS covariance are retained in ../gaia-geometry.json. Closest selected-kind offset=0.071883d (0.006937phase), linear covariance allowance=0.939139d. The SB1 lower triangle uses P,gamma,K,e,omega,Tp, and selected subindices0,3,4,5; minimum full correlation eigenvalue=0.096082. This descriptive linear approximation is not a calibrated rejection threshold. Mass function=0.0941635solar masses; assumed TIC/adopted M1=1.000, minimum M2=0.630; TIC/adopted radius=2.107solar radii. Approximate central duration=6.062h versus measured6.750h. Main-sequence minimum-mass companion and TIC mass/radius make this a plausibility screen, not a fitted geometry. Orbital agreement alone cannot establish source ownership.

Sector	same-kind search cycle	measured midpoint BTJD	formal center error d	timing input allowance d	depth ppt
35	0	2260.7427276	0.00391782	0.00391782	28.958
63	73	3017.2160093	0.00215781	0.00215781	31.861
63	75	3037.9361340	0.00221555	0.00221555	31.765
90	144	3752.9614589	0.00253884	0.00253884	31.066
90	145	3763.3275831	0.0020492	0.0020492	29.890
90	146	3773.6940954	0.0030345	0.0030345	31.138

Limitations and next check: Resolved eclipse photometry to establish ownership. Representative events suffice for this bounded review; no all-sector/all-event precision campaign is claimed. Sector-dependent dilution, finite cadence, baseline choice, color proxy, and linear Gaia propagation remain material limitations.

Catalogue checks use ../../shared-catalogues/coverage-matrix.csv and the shared receipt manifest; known recovery is separated from physical classification and no exhaustive priority claim is made. Evidence: measurements.json/raw-windows.csv/raw-events.png; fresh-

pixels/diff_fresh.json/png/pixel-arrays.npz; neighbor-assessment.json; batch phase, secondary, alias, identity/source receipts, methods snapshot and audits. SHA256SUMS.json freezes exact sources and outputs.

Stable review key: R589.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/pixel-arrays.npz - fresh-pixels/run.log - historical/diff_s89.json - historical/diff_s89.png - historical/gaia.json - measurements.json - neighbor-assessment.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

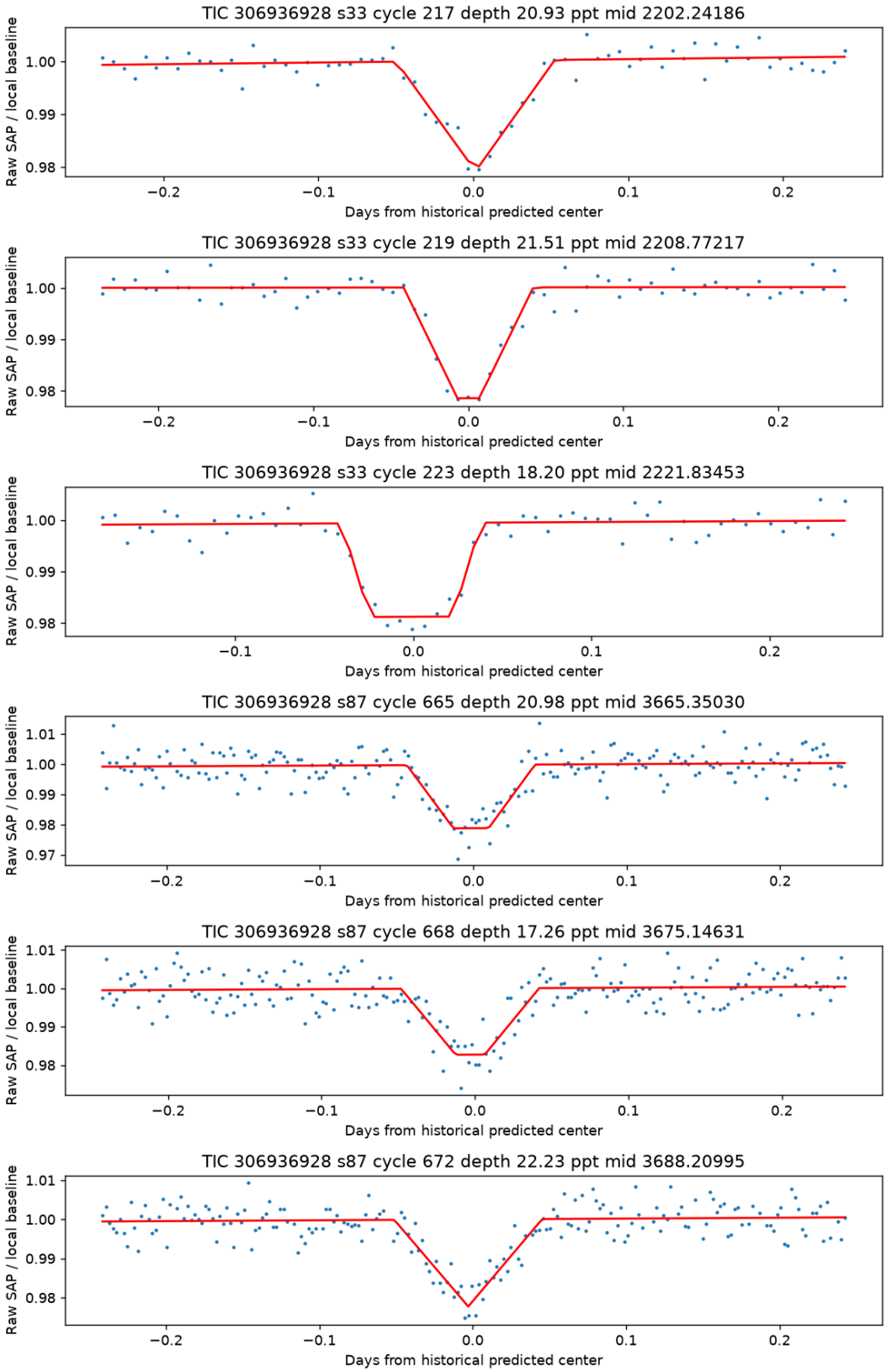
R590. TIC 306936928

physical_verdict: PROBABLE

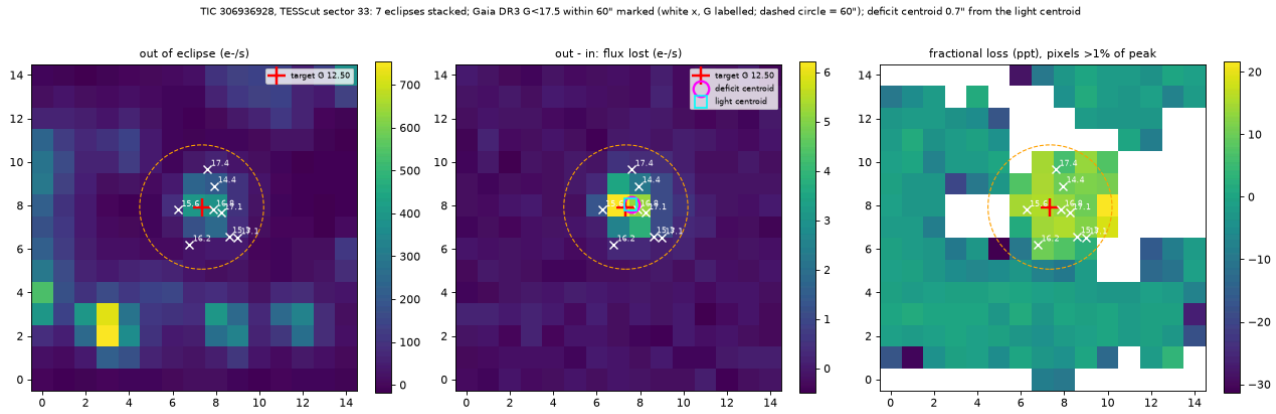
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or multi-source PRF localization; distinguish alternating event kinds.



R590; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R590; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 306936928: PROBABLE

Full bounded physical review: **COMPLETE**. Physical classification is separate from catalogue recovery/novelty; historical screening was not an admission verdict.

1. **Raw events.** 14 free-center trapezoid measurements from quality-zero QLP SAP_FLUX in sectors [33, 87]; 14 enter the reported ephemeris. Representative raw light curves and numerical windows show repeated eclipse morphology. Predictions guided the windows and are not measurements. Flagged or uncovered windows are not clean nulls.
2. **Timing and aliases.** Measured recurring-eclipse timing supported. Selected primary and unequal secondary morphology support the quoted orbital period; half/double hypotheses were checked. P=3.2658677781 +/- 0.0000031329 d; reference BTJD=2202.2402559. Epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The formal error column is the local free-center model covariance; the input error adds a half-cadence floor. Singular/unidentifiable centers are excluded. Cycle/parity counts are in ../alias-audit.json. The epoch kind above states whether the fitted events are identified as deeper primary or selected eclipse by convention. The bounded half/double check does not prove all possible aliases absent.
3. **Attribution.** Fresh difference stack in sector 33 has 7 events; target coordinate [7.343879930269948, 7.9267470273072576], deficit [7.582796384292077, 8.026741837058745]; offset 0.259 pixel. The image-model fractional residual 0.584 is not a localization uncertainty. Every capable Gaia neighbor in the retained 60-arcsec table was inspected: Gaia 3032561616123801728: 10.59 arcsec, G=16.765, intrinsic required loss proxy=0.982; Gaia 3032561616123796352: 21.85 arcsec, G=15.627, intrinsic required loss proxy=0.356; Gaia 3032561581764068224: 22.78 arcsec, G=14.392, intrinsic required loss proxy=0.127; Gaia 3032567697797482112: 36.58 arcsec, G=16.204, intrinsic required loss proxy=0.593; Gaia 3032561822283763840: 37.84 arcsec, G=15.291, intrinsic required loss proxy=0.267. The intrinsic loss proxy uses a pairwise G-band flux budget and cannot be directly compared with diluted pixel fractional loss. Close blends and passband/aperture response limit exclusion claims. Repeated narrow primary and secondary dips are secure, but five flux-capable

Gaia neighbors include a 10.59 arcsec G16.77 source requiring an almost total intrinsic eclipse. The compact blended deficit does not uniquely resolve target ownership; PROBABLE.

4. **Alternative reduction.** Fresh TESScut background-subtracted target-centered 1x1/3x3/5x5 depths are 15.54/13.83/12.35 ppt; selected QLP fitted median 18.87 ppt. This reruns the historical extraction primitive using the fresh ephemeris and measured duration, with identity and source hashes checked. The alternative extraction and the QLP data for this sector share the same TESS photons; they test processing robustness and are not independent observations. Distinct sectors are distinct observing occasions on one instrument. Depth differences can reflect dilution, aperture and background response; intrinsic binary depth is not inferred.
5. **Gaia geometry/physical plausibility.** Exact DR3 source 3032561611821997952, orbital P=3.2656610989 d, e=0.04672. Closest edge-on conjunction offset=0.05747 d (0.01760 phase), full source-specific covariance linear timing allowance=0.13828 d. Mass function=0.07895 solar masses; adopted primary mass=1.46, minimum secondary mass=0.722. Rough central duration=4.87 h versus observed median=2.25 h. Radius and main-sequence companion assumptions make this a plausibility calculation, not a fitted geometry or calibrated significance veto. Shorter observed duration can reflect impact parameter and model assumptions. Orbit agreement does not establish ownership among unresolved sources.

Secondary phase check: Pooled raw-window free-center phase=0.4995127844887043, depth=3.4525177588269473ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Weak secondary fits (especially314713331) describe inspected narrow dips and are not independent primary-eclipse evidence or calibrated detection significances. See ../secondary-phases.json and ../secondary-contact.png.

Deciding limit: Repeated narrow primary and secondary dips are secure, but five flux-capable Gaia neighbors include a 10.59 arcsec G16.77 source requiring an almost total intrinsic eclipse. The compact blended deficit does not uniquely resolve target ownership; PROBABLE.

Next check: Resolved eclipse photometry or multi-source PRF localization; distinguish alternating event kinds.

Catalogue recovery is recorded centrally in ../../shared-catalogues/coverage-matrix.csv. A successful cone with no exact eclipsing target classification is distinct from a failed query; known recoveries must not count as new additions. No exhaustive literature priority claim is made.

Sector	Search cycle	Measured midpoint BTJD	Formal center error d	Timing input error d	Depth ppt	Timing status
33	217	2202.2418554	0.00106251	0.00347228	20.93	selected
33	218	2205.5070455	0.00135498	0.00347228	19.67	selected
33	219	2208.7721713	0.000827799	0.00347228	21.51	selected
33	220	2212.0372461	0.00131487	0.00347228	19.04	selected
33	222	2218.5702969	0.0013536	0.00347228	14.80	selected
33	223	2221.8345328	0.00085435	0.00347228	18.20	selected
33	224	2225.0993307	0.00142478	0.00347228	19.35	selected
87	665	3665.3503026	0.00117801	0.00117801	20.98	selected
87	666	3668.6145142	0.0013157	0.0013157	18.70	selected
87	667	3671.8803126	0.00146109	0.00146109	16.39	selected
87	668	3675.1463076	0.00162138	0.00162138	17.26	selected
87	670	3681.6791059	0.00119067	0.00119067	16.52	selected
87	671	3684.9431140	0.00114514	0.00115742	18.33	selected
87	672	3688.2099483	0.00151347	0.00151347	22.23	selected

Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png, historical original receipts, ../download-receipts.json, ../identity-check.json, ../alias-audit.json, ../gaia-geometry.json and ../helper-audit.md. Original inputs and fresh outputs are covered by ../SHA256SUMS.json and ../source-hashes.json.

Depth correction audit (2026-11-02): realized trapezoid decrement replaces its unconstrained amplitude parameter. Original input windows, fitted centers/widths, timing selections and physical verdict are retained; capable-neighbor membership is unchanged. See ../../depth-audit/README.md.

Stable review key: R590.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s33.json - historical/diff_s33.png - historical/gaia.json - measurements.json - phase-check.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

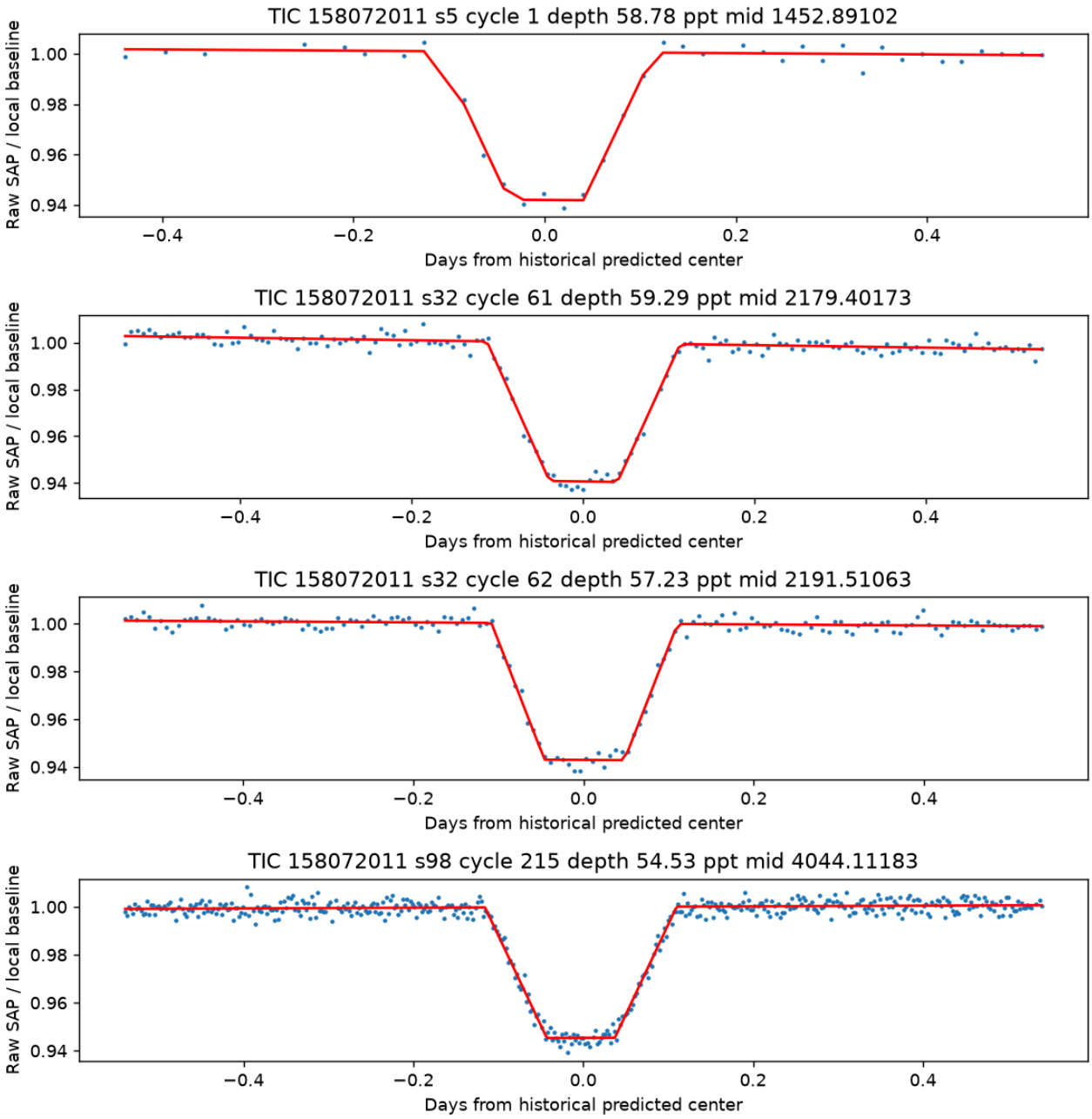
R591. TIC 158072011

physical_verdict: CONFIRMED

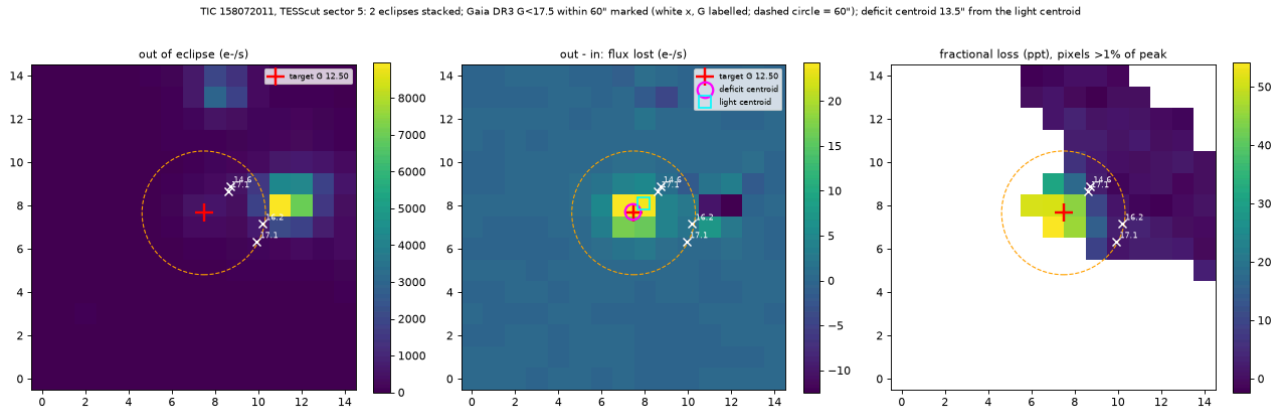
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision physical parameters.



R591; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R591; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 158072011: CONFIRMED

Full bounded physical review: **COMPLETE**. Physical classification is separate from catalogue recovery/novelty; historical screening was not an admission verdict.

1. **Raw events.** 4 free-center trapezoid measurements from quality-zero QLP SAP_FLUX in sectors [5, 32, 98]; 4 enter the reported ephemeris. Representative raw light curves and numerical windows show repeated eclipse morphology. Predictions guided the windows and are not measurements. Flagged or uncovered windows are not clean nulls.
2. **Timing and aliases.** Measured recurring-eclipse timing supported. Selected primary and unequal secondary morphology support the quoted orbital period; half/double hypotheses were checked. P=12.1085061732 +/- 0.0000168695 d; reference BTJD=1452.8915159. Epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The formal error column is the local free-center model covariance; the input error adds a half-cadence floor. Singular/unidentifiable centers are excluded. Cycle/parity counts are in ../alias-audit.json. The epoch kind above states whether the fitted events are identified as deeper primary or selected eclipse by convention. The bounded half/double check does not prove all possible aliases absent.
3. **Attribution.** Fresh difference stack in sector 5 has 2 events; target coordinate [7.476034565434107, 7.661629294345968], deficit [7.415096194745247, 7.727690072744489]; offset 0.090 pixel. The image-model fractional residual 0.999 is not a localization uncertainty. Every capable Gaia neighbor in the retained 60-arcsec table was inspected: Gaia 5571245106071082112: 36.05 arcsec, G=14.559, intrinsic required loss proxy=0.446. The intrinsic loss proxy uses a pairwise G-band flux budget and cannot be directly compared with diluted pixel fractional loss. Close blends and passband/aperture response limit exclusion claims. Repeated box eclipses and a secondary near phase 0.397. The deficit favors the target over the capable 36.05 arcsec neighbor; the bright blended out-of-eclipse image displaces the light centroid. Saturation/blend morphology limits precision localization.

4. **Alternative reduction.** Fresh TESScut background-subtracted target-centered 1x1/3x3/5x5 depths are 51.05/43.55/28.82 ppt; selected QLP fitted median 58.01 ppt. This reruns the historical extraction primitive using the fresh ephemeris and measured duration, with identity and source hashes checked. The alternative extraction and the QLP data for this sector share the same TESS photons; they test processing robustness and are not independent observations. Distinct sectors are distinct observing occasions on one instrument. Depth differences can reflect dilution, aperture and background response; intrinsic binary depth is not inferred.
5. **Gaia geometry/physical plausibility.** Exact DR3 source 5571244899912652416, orbital P=12.1129446814 d, e=0.10656. Closest edge-on conjunction offset=0.12594 d (0.01040 phase), full source-specific covariance linear timing allowance=0.62075 d. Mass function=0.023311 solar masses; adopted primary mass=1.38, minimum secondary mass=0.423. Rough central duration=6.84 h versus observed median=5.40 h. Radius and main-sequence companion assumptions make this a plausibility calculation, not a fitted geometry or calibrated significance veto. Shorter observed duration can reflect impact parameter and model assumptions. Orbit agreement does not establish ownership among unresolved sources.

Secondary phase check: Pooled raw-window free-center phase=0.3965392945105339, depth=4.975725065587859ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Weak secondary fits (especially314713331) describe inspected narrow dips and are not independent primary-eclipse evidence or calibrated detection significances. See ../secondary-phases.json and ../secondary-contact.png.

Deciding limit: Repeated box eclipses and a secondary near phase 0.397. The deficit favors the target over the capable 36.05 arcsec neighbor; the bright blended out-of-eclipse image displaces the light centroid. Saturation/blend morphology limits precision localization.

Next check: Optional resolved photometry and spectroscopy for precision physical parameters.

Catalogue recovery is recorded centrally in ../../shared-catalogues/coverage-matrix.csv. A successful cone with no exact eclipsing target classification is distinct from a failed query; known recoveries must not count as new additions. No exhaustive literature priority claim is made.

Sector	Search cycle	Measured midpoint BTJD	Formal center error d	Timing input error d	Depth ppt	Timing status
5	1	1452.8910175	0.00199866	0.0104169	58.78	selected
32	61	2179.4017264	0.000948661	0.00347227	59.29	selected
32	62	2191.5106308	0.00059826	0.00347227	57.23	selected
98	215	4044.1118344	0.000456651	0.00115742	54.53	selected

Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png, historical original receipts, ../download-receipts.json, ../identity-check.json, ../alias-audit.json, ../gaia-geometry.json and ../helper-audit.md. Original inputs and fresh outputs are covered by ../SHA256SUMS.json and ../source-hashes.json.

Stable review key: R591.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s05.json - historical/diff_s05.png - historical/gaia.json - measurements.json - phase-check.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

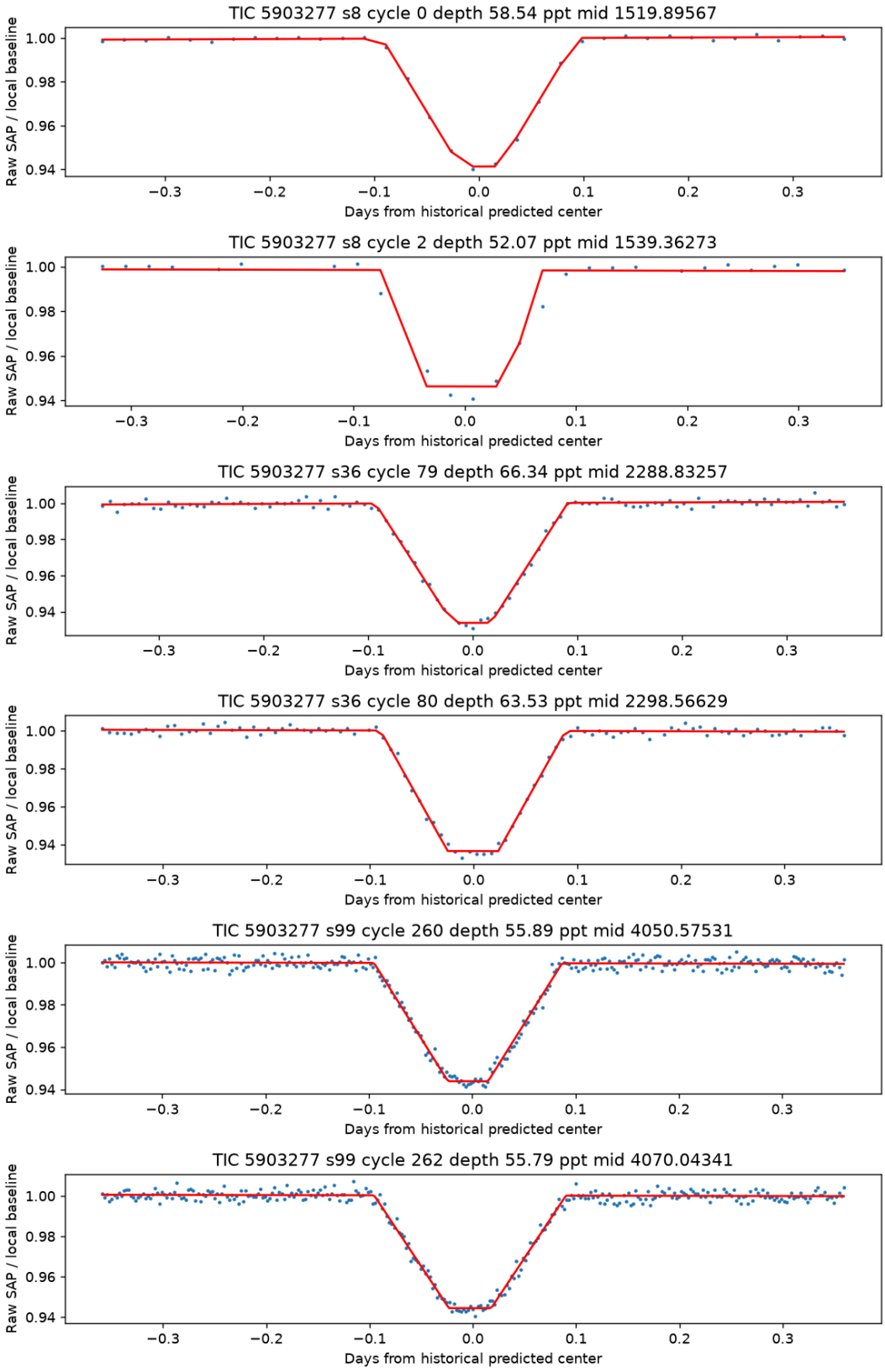
R592. TIC 5903277

physical_verdict: CONFIRMED

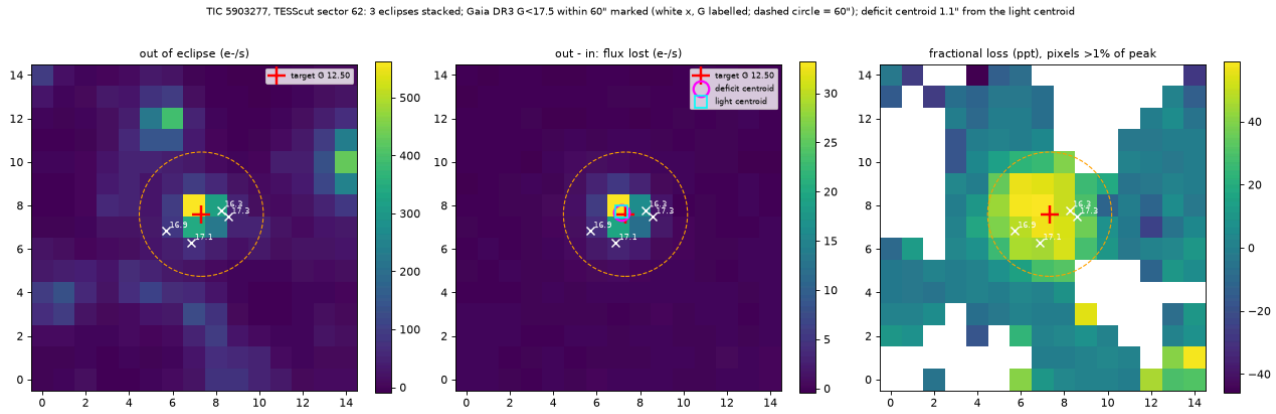
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision physical parameters.



R592; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R592; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 5903277: CONFIRMED

Full bounded physical review: **COMPLETE**. Physical classification is separate from catalogue recovery/novelty; historical screening was not an admission verdict.

1. **Raw events.** 6 free-center trapezoid measurements from quality-zero QLP SAP_FLUX in sectors [8, 36, 99]; 6 enter the reported ephemeris. Representative raw light curves and numerical windows show repeated eclipse morphology. Predictions guided the windows and are not measurements. Flagged or uncovered windows are not clean nulls.
2. **Timing and aliases.** Measured recurring-eclipse timing supported. Selected primary and unequal secondary morphology support the quoted orbital period; half/double hypotheses were checked. P=9.7333876663 +/- 0.0000129249 d; reference BTJD=1519.8951780. Epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The formal error column is the local free-center model covariance; the input error adds a half-cadence floor. Singular/unidentifiable centers are excluded. Cycle/parity counts are in ../alias-audit.json. The epoch kind above states whether the fitted events are identified as deeper primary or selected eclipse by convention. The bounded half/double check does not prove all possible aliases absent.
3. **Attribution.** Fresh difference stack in sector 62 has 3 events; target coordinate [7.326853365408729, 7.5975558525549065], deficit [7.153078836664321, 7.691826662532669]; offset 0.198 pixel. The image-model fractional residual 0.577 is not a localization uncertainty. Every capable Gaia neighbor in the retained 60-arcsec table was inspected: None under the approximate G-band brightness screen within 60 arcsec. The intrinsic loss proxy uses a pairwise G-band flux budget and cannot be directly compared with diluted pixel fractional loss. Close blends and passband/aperture response limit exclusion claims. Repeated eclipses with unequal secondary near phase 0.495. Compact target-centered deficit; no flux-capable Gaia neighbor within 60 arcsec under the approximate G-band screen.
4. **Alternative reduction.** Fresh TESScut background-subtracted target-centered 1x1/3x3/5x5 depths are 59.09/55.30/50.73 ppt; selected QLP fitted median 57.22 ppt. This reruns the historical

extraction primitive using the fresh ephemeris and measured duration, with identity and source hashes checked. The retained QLP sectors do not include this cutout sector. This cutout is a distinct observing occasion on the same instrument; it is an alternative extraction of that sector FFI data, without a direct same-sector QLP comparison here. Distinct sectors are distinct observing occasions on one instrument. Depth differences can reflect dilution, aperture and background response; intrinsic binary depth is not inferred.

5. **Gaia geometry/physical plausibility.** Exact DR3 source 5436115058297930624, orbital P=9.7313526090 d, e=0.01829. Closest edge-on conjunction offset=0.28129 d (0.02891 phase), full source-specific covariance linear timing allowance=0.18692 d. Mass function=0.12041 solar masses; adopted primary mass=1.44, minimum secondary mass=0.861. Rough central duration=7.92 h versus observed median=4.39 h. Radius and main-sequence companion assumptions make this a plausibility calculation, not a fitted geometry or calibrated significance veto. Shorter observed duration can reflect impact parameter and model assumptions. Orbit agreement does not establish ownership among unresolved sources.

Secondary phase check: Pooled raw-window free-center phase=0.4952360274681802, depth=20.356940954848675ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Weak secondary fits (especially314713331) describe inspected narrow dips and are not independent primary-eclipse evidence or calibrated detection significances. See ../secondary-phases.json and ../secondary-contact.png.

Deciding limit: Repeated eclipses with unequal secondary near phase 0.495. Compact target-centered deficit; no flux-capable Gaia neighbor within 60 arcsec under the approximate G-band screen.

Next check: Optional resolved photometry and spectroscopy for precision physical parameters.

Catalogue recovery is recorded centrally in ../../shared-catalogues/coverage-matrix.csv. A successful cone with no exact eclipsing target classification is distinct from a failed query; known recoveries must not count as new additions. No exhaustive literature priority claim is made.

Sector	Search cycle	Measured midpoint BTJD	Formal center error d	Timing input error d	Depth ppt	Timing status
8	0	1519.8956711	0.000459918	0.0104169	58.54	selected
8	2	1539.3627294	0.000185141	0.0104169	52.07	selected
36	79	2288.8325716	0.000523046	0.00347216	66.34	selected
36	80	2298.5662883	0.000425782	0.00347216	63.53	selected
99	260	4050.5753059	0.00041353	0.00115747	55.89	selected
99	262	4070.0434111	0.00035204	0.00115747	55.79	selected

Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png, historical original receipts, ../download-receipts.json, ../identity-check.json, ../alias-audit.json, ../gaia-geometry.json and ../helper-audit.md. Original inputs and fresh outputs are covered by ../SHA256SUMS.json and ../source-hashes.json.

Stable review key: R592.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s62.json - historical/diff_s62.png - historical/gaia.json - measurements.json - phase-check.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

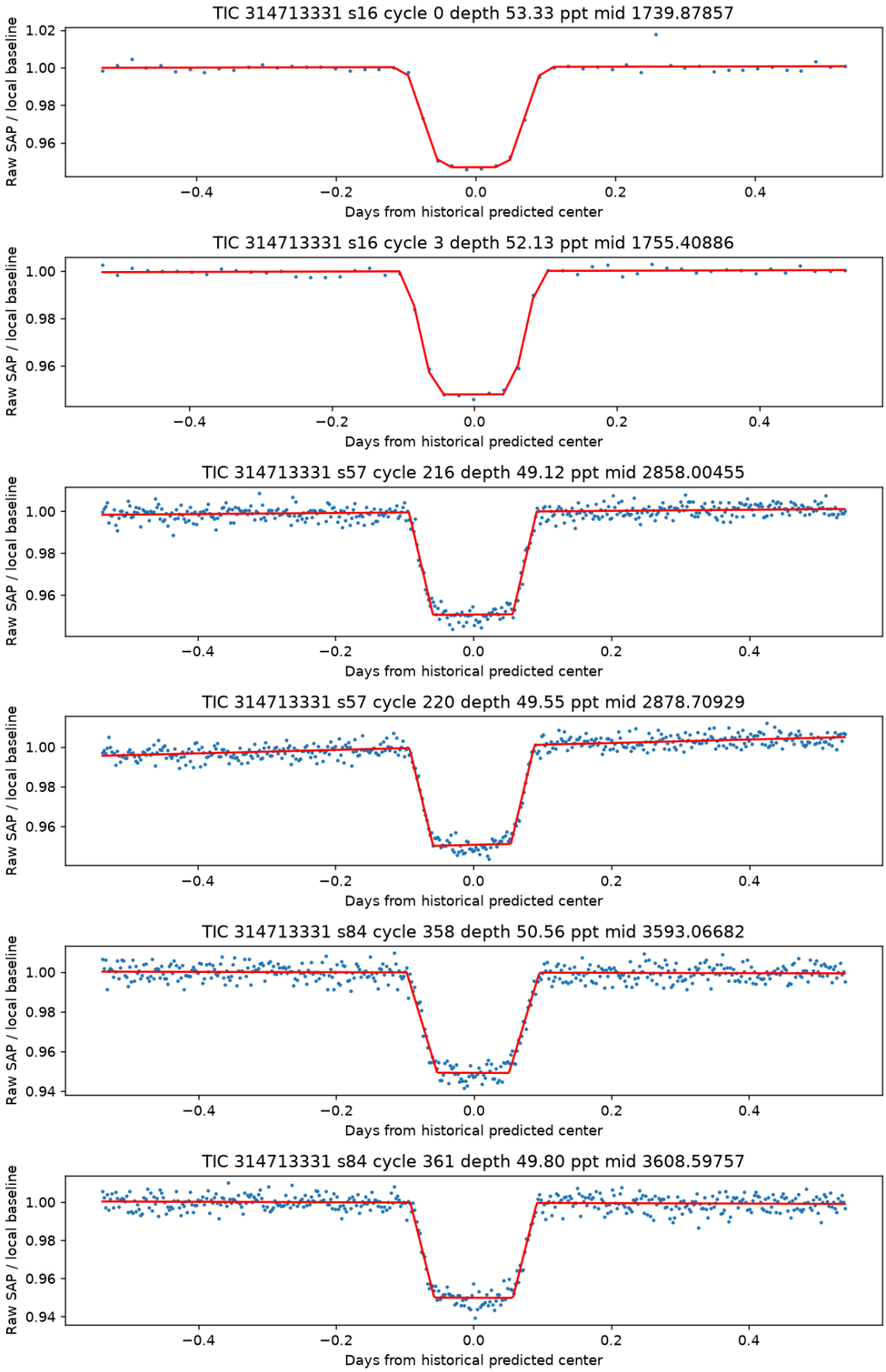
R593. TIC 314713331

physical_verdict: CONFIRMED

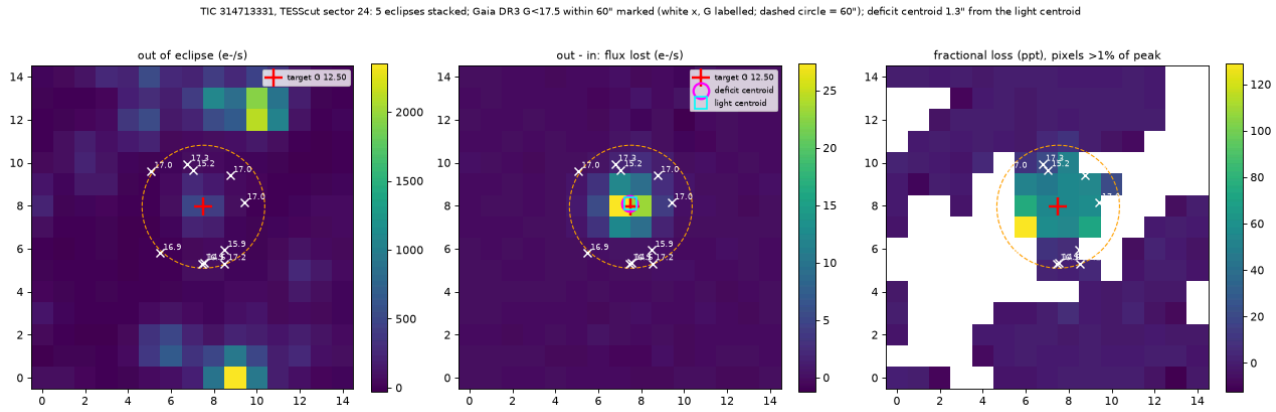
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision physical parameters.



R593; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R593; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 314713331: CONFIRMED

Full bounded physical review: **COMPLETE**. Physical classification is separate from catalogue recovery/novelty; historical screening was not an admission verdict.

1. **Raw events.** 13 free-center trapezoid measurements from quality-zero QLP SAP_FLUX in sectors [16, 57, 84]; 13 enter the reported ephemeris. Representative raw light curves and numerical windows show repeated eclipse morphology. Predictions guided the windows and are not measurements. Flagged or uncovered windows are not clean nulls.
2. **Timing and aliases.** Measured recurring-eclipse timing supported. Selected primary and unequal secondary morphology support the quoted orbital period; half/double hypotheses were checked. P=5.1765061266 +/- 0.0000052664 d; reference BTJD=1739.8785468. Epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The formal error column is the local free-center model covariance; the input error adds a half-cadence floor. Singular/unidentifiable centers are excluded. Cycle/parity counts are in ../alias-audit.json. The epoch kind above states whether the fitted events are identified as deeper primary or selected eclipse by convention. The bounded half/double check does not prove all possible aliases absent.
3. **Attribution.** Fresh difference stack in sector 24 has 5 events; target coordinate [7.513877286219605, 7.953249188522971], deficit [7.446741433788142, 8.09249674220127]; offset 0.155 pixel. The image-model fractional residual 0.967 is not a localization uncertainty. Every capable Gaia neighbor in the retained 60-arcsec table was inspected: Gaia 2008766035527664640: 36.57 arcsec, G=15.176, intrinsic required loss proxy=0.637; Gaia 2008754250137422336: 53.57 arcsec, G=14.639, intrinsic required loss proxy=0.408. The intrinsic loss proxy uses a pairwise G-band flux budget and cannot be directly compared with diluted pixel fractional loss. Close blends and passband/aperture response limit exclusion claims. Repeated box eclipses and weak secondary near phase 0.500; target-centered deficit distinct from two capable neighbors 36.57 and 53.57 arcsec away.

4. **Alternative reduction.** Fresh TESScut background-subtracted target-centered 1x1/3x3/5x5 depths are 55.25/54.64/51.47 ppt; selected QLP fitted median 49.80 ppt. This reruns the historical extraction primitive using the fresh ephemeris and measured duration, with identity and source hashes checked. The retained QLP sectors do not include this cutout sector. This cutout is a distinct observing occasion on the same instrument; it is an alternative extraction of that sector FFI data, without a direct same-sector QLP comparison here. Distinct sectors are distinct observing occasions on one instrument. Depth differences can reflect dilution, aperture and background response; intrinsic binary depth is not inferred.
5. **Gaia geometry/physical plausibility.** Exact DR3 source 2008765962505842304, orbital P=5.1766131923 d, e=0.03225. Closest edge-on conjunction offset=-0.01184 d (-0.00229 phase), full source-specific covariance linear timing allowance=0.16870 d. Mass function=0.017688 solar masses; adopted primary mass=1.18, minimum secondary mass=0.345. Rough central duration=4.94 h versus observed median=4.47 h. Radius and main-sequence companion assumptions make this a plausibility calculation, not a fitted geometry or calibrated significance veto. Shorter observed duration can reflect impact parameter and model assumptions. Orbit agreement does not establish ownership among unresolved sources.

Secondary phase check: Pooled raw-window free-center phase=0.4996902283812347, depth=1.7904809648658992ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Weak secondary fits (especially314713331) describe inspected narrow dips and are not independent primary-eclipse evidence or calibrated detection significances. See ../secondary-phases.json and ../secondary-contact.png.

Deciding limit: Repeated box eclipses and weak secondary near phase 0.500; target-centered deficit distinct from two capable neighbors 36.57 and 53.57 arcsec away.

Next check: Optional resolved photometry and spectroscopy for precision physical parameters.

Catalogue recovery is recorded centrally in ../../shared-catalogues/coverage-matrix.csv. A successful cone with no exact eclipsing target classification is distinct from a failed query; known recoveries must not count as new additions. No exhaustive literature priority claim is made.

Sector	Search cycle	Measured midpoint BTJD	Formal center error d	Timing input error d	Depth ppt	Timing status
16	0	1739.8785717	0.000689522	0.0104168	53.33	selected
16	1	1745.0559513	0.000627102	0.0104168	51.91	selected
16	3	1755.4088620	0.000534279	0.0104168	52.13	selected
16	4	1760.5843301	0.000867886	0.0104168	51.64	selected
57	216	2858.0045545	0.00035742	0.0011574	49.12	selected
57	218	2868.3568502	0.000414582	0.0011574	49.57	selected
57	219	2873.5332905	0.000422196	0.0011574	49.25	selected
57	220	2878.7092876	0.000580049	0.0011574	49.55	selected
84	357	3587.8909194	0.000532365	0.0011574	50.02	selected
84	358	3593.0668218	0.00057135	0.0011574	50.56	selected
84	359	3598.2456129	0.000813559	0.0011574	47.39	selected
84	360	3603.4203422	0.000469937	0.0011574	48.74	selected
84	361	3608.5975693	0.00050462	0.0011574	49.80	selected

Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png, historical original receipts, ../download-receipts.json, ../identity-check.json, ../alias-audit.json, ../gaia-geometry.json and ../helper-audit.md. Original inputs and fresh outputs are covered by ../SHA256SUMS.json and ../source-hashes.json.

Stable review key: R593.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s24.json - historical/diff_s24.png - historical/gaia.json - measurements.json - phase-check.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

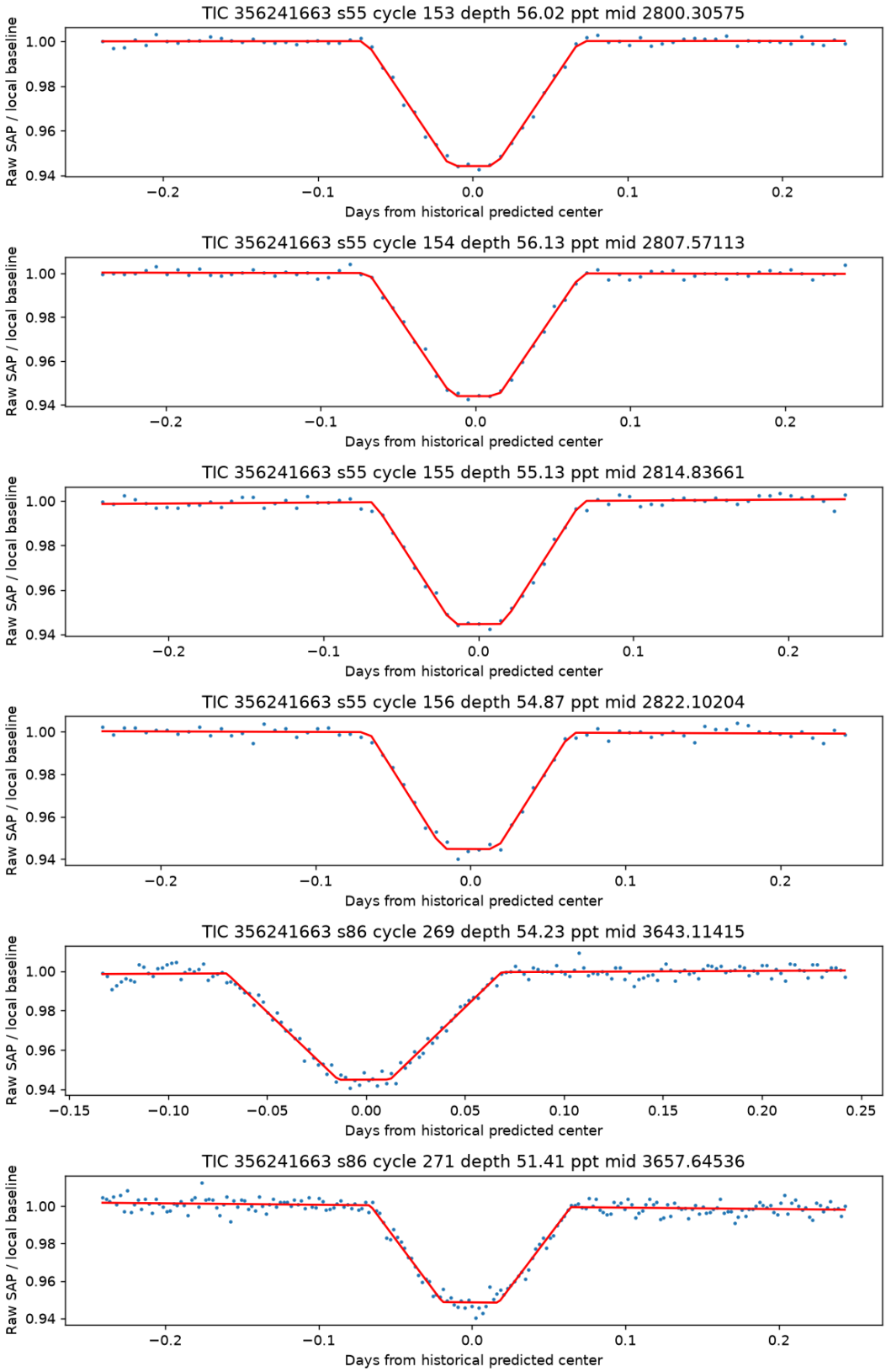
R594. TIC 356241663

physical_verdict: CONFIRMED

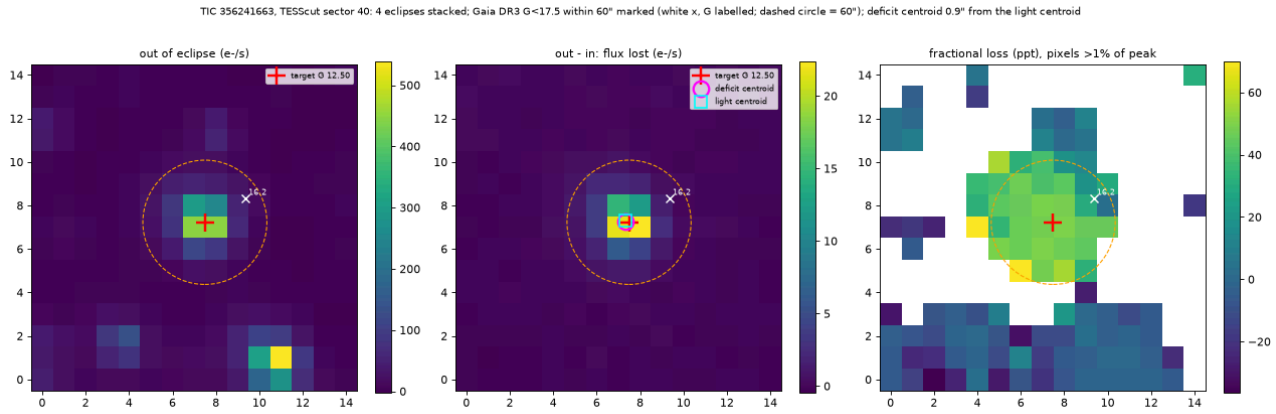
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision physical parameters.



R594; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R594; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 356241663: CONFIRMED

Full bounded physical review: **COMPLETE**. Physical classification is separate from catalogue recovery/novelty; historical screening was not an admission verdict.

1. **Raw events.** 7 free-center trapezoid measurements from quality-zero QLP SAP_FLUX in sectors [55, 86]; 7 enter the reported ephemeris. Representative raw light curves and numerical windows show repeated eclipse morphology. Predictions guided the windows and are not measurements. Flagged or uncovered windows are not clean nulls.
2. **Timing and aliases.** Measured recurring-eclipse timing supported. Selected primary and unequal secondary morphology support the quoted orbital period; half/double hypotheses were checked. P=7.2655940116 +/- 0.0000161020 d; reference BTJD=2800.3054925. Epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The formal error column is the local free-center model covariance; the input error adds a half-cadence floor. Singular/unidentifiable centers are excluded. Cycle/parity counts are in ../alias-audit.json. The epoch kind above states whether the fitted events are identified as deeper primary or selected eclipse by convention. The bounded half/double check does not prove all possible aliases absent.
3. **Attribution.** Fresh difference stack in sector 40 has 4 events; target coordinate [7.4906668002166406, 7.225294930459409], deficit [7.317329600098449, 7.250914606608196]; offset 0.175 pixel. The image-model fractional residual 0.116 is not a localization uncertainty. Every capable Gaia neighbor in the retained 60-arcsec table was inspected: None under the approximate G-band brightness screen within 60 arcsec. The intrinsic loss proxy uses a pairwise G-band flux budget and cannot be directly compared with diluted pixel fractional loss. Close blends and passband/aperture response limit exclusion claims. Repeated narrow eclipses plus weak secondary. Target-centered deficit without a capable neighbor within 60 arcsec. Gaia conjunction offset-0.0457 phase is descriptive and lies within the propagated orbit timing allowance; it is not a fixed phase-cut rejection.

4. **Alternative reduction.** Fresh TESScut background-subtracted target-centered 1x1/3x3/5x5 depths are 48.32/48.63/47.28 ppt; selected QLP fitted median 54.87 ppt. This reruns the historical extraction primitive using the fresh ephemeris and measured duration, with identity and source hashes checked. The retained QLP sectors do not include this cutout sector. This cutout is a distinct observing occasion on the same instrument; it is an alternative extraction of that sector FFI data, without a direct same-sector QLP comparison here. Distinct sectors are distinct observing occasions on one instrument. Depth differences can reflect dilution, aperture and background response; intrinsic binary depth is not inferred.
5. **Gaia geometry/physical plausibility.** Exact DR3 source 1416984104080742656, orbital P=7.2666480591 d, e=0.14183. Closest edge-on conjunction offset=-0.33181 d (-0.04566 phase), full source-specific covariance linear timing allowance=0.37945 d. Mass function=0.032544 solar masses; adopted primary mass=1.07, minimum secondary mass=0.416. Rough central duration=4.62 h versus observed median=3.30 h. Radius and main-sequence companion assumptions make this a plausibility calculation, not a fitted geometry or calibrated significance veto. Shorter observed duration can reflect impact parameter and model assumptions. Orbit agreement does not establish ownership among unresolved sources.

Secondary phase check: Pooled raw-window free-center phase=0.498700627984285, depth=4.505768399874709ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Weak secondary fits (especially314713331) describe inspected narrow dips and are not independent primary-eclipse evidence or calibrated detection significances. See ../secondary-phases.json and ../secondary-contact.png.

Deciding limit: Repeated narrow eclipses plus weak secondary. Target-centered deficit without a capable neighbor within 60 arcsec. Gaia conjunction offset-0.0457 phase is descriptive and lies within the propagated orbit timing allowance; it is not a fixed phase-cut rejection.

Next check: Optional resolved photometry and spectroscopy for precision physical parameters.

Catalogue recovery is recorded centrally in ../../shared-catalogues/coverage-matrix.csv. A successful cone with no exact eclipsing target classification is distinct from a failed query; known recoveries must not count as new additions. No exhaustive literature priority claim is made.

Sector	Search cycle	Measured midpoint BTJD	Formal center error d	Timing input error d	Depth ppt	Timing status
55	153	2800.3057467	0.000322308	0.00347215	56.02	selected
55	154	2807.5711287	0.000327691	0.00347215	56.13	selected
55	155	2814.8366134	0.000570428	0.00347215	55.13	selected
55	156	2822.1020401	0.000517738	0.00347215	54.87	selected
86	269	3643.1141473	0.000449884	0.0011574	54.23	selected
86	270	3650.3804733	0.000606962	0.0011574	53.47	selected
86	271	3657.6453556	0.000507138	0.0011574	51.41	selected

Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png, historical original receipts, ../download-receipts.json, ../identity-check.json, ../alias-audit.json, ../gaia-geometry.json and ../helper-audit.md. Original inputs and fresh outputs are covered by ../SHA256SUMS.json and ../source-hashes.json.

Stable review key: R594.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s40.json - historical/diff_s40.png - historical/gaia.json - measurements.json - phase-check.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

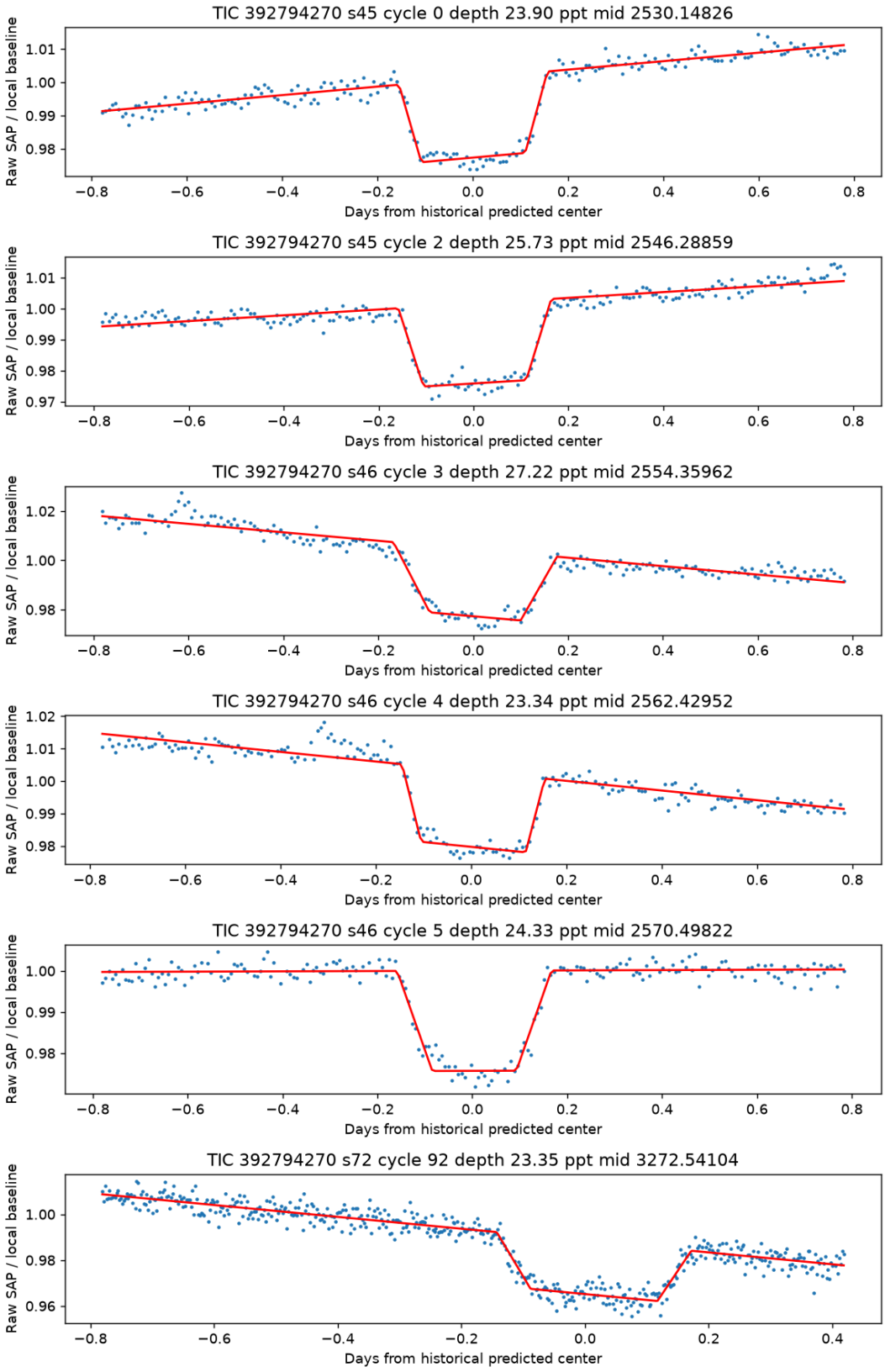
R595. TIC 392794270

physical_verdict: CONFIRMED

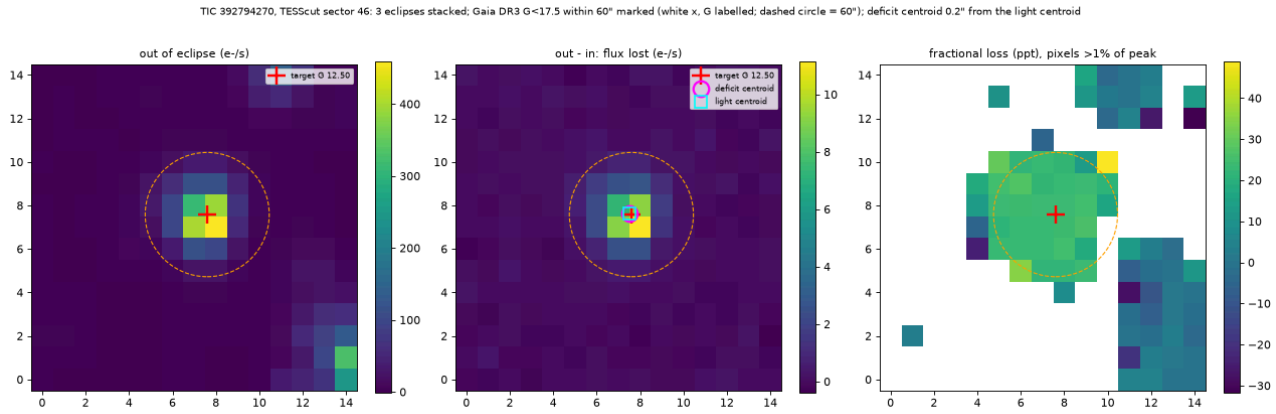
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision physical parameters.



R595; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R595; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 392794270: CONFIRMED

Full bounded physical review: **COMPLETE**. Physical classification is separate from catalogue recovery/novelty; historical screening was not an admission verdict.

1. **Raw events.** 7 free-center trapezoid measurements from quality-zero QLP SAP_FLUX in sectors [45, 46, 72]; 7 enter the reported ephemeris. Representative raw light curves and numerical windows show repeated eclipse morphology. Predictions guided the windows and are not measurements. Flagged or uncovered windows are not clean nulls.
2. **Timing and aliases.** Measured recurring-eclipse timing supported. Selected primary and unequal secondary morphology support the quoted orbital period; half/double hypotheses were checked. P=8.0693890282 +/- 0.0000305111 d; reference BTJD=2530.1503849. Epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The formal error column is the local free-center model covariance; the input error adds a half-cadence floor. Singular/unidentifiable centers are excluded. Cycle/parity counts are in ../alias-audit.json. The epoch kind above states whether the fitted events are identified as deeper primary or selected eclipse by convention. The bounded half/double check does not prove all possible aliases absent.
3. **Attribution.** Fresh difference stack in sector 46 has 3 events; target coordinate [7.599025882750558, 7.57969604350226], deficit [7.501264762064753, 7.610329271099777]; offset 0.102 pixel. The image-model fractional residual 0.075 is not a localization uncertainty. Every capable Gaia neighbor in the retained 60-arcsec table was inspected: None under the approximate G-band brightness screen within 60 arcsec. The intrinsic loss proxy uses a pairwise G-band flux budget and cannot be directly compared with diluted pixel fractional loss. Close blends and passband/aperture response limit exclusion claims. Repeated sharp box eclipses are distinct from strong changing baseline variability. No secure secondary phase is assigned. The target-centered difference image and no capable Gaia neighbor support ownership.
4. **Alternative reduction.** Fresh TESScut background-subtracted target-centered 1x1/3x3/5x5 depths are 23.48/23.56/23.59 ppt; selected QLP fitted median 24.33 ppt. This reruns the historical

extraction primitive using the fresh ephemeris and measured duration, with identity and source hashes checked. The alternative extraction and the QLP data for this sector share the same TESS photons; they test processing robustness and are not independent observations. Distinct sectors are distinct observing occasions on one instrument. Depth differences can reflect dilution, aperture and background response; intrinsic binary depth is not inferred.

5. **Gaia geometry/physical plausibility.** Exact DR3 source 3863614109736230656, orbital P=8.0698655652 d, e=0.09715. Closest edge-on conjunction offset=0.02801 d (0.00347 phase), full source-specific covariance linear timing allowance=0.97097 d. Mass function=0.011284 solar masses; adopted primary mass=0.9, minimum secondary mass=0.246. Rough central duration=7.27 h versus observed median=7.57 h. Radius and main-sequence companion assumptions make this a plausibility calculation, not a fitted geometry or calibrated significance veto. Shorter observed duration can reflect impact parameter and model assumptions. Orbit agreement does not establish ownership among unresolved sources.

Secondary phase check: No secure secondary identified amid strong non-eclipse variability; phase 0.5 tested visually, not a clean null. Weak secondary fits (especially314713331) describe inspected narrow dips and are not independent primary-eclipse evidence or calibrated detection significances. See ../secondary-phases.json and ../secondary-contact.png.

Deciding limit: Repeated sharp box eclipses are distinct from strong changing baseline variability. No secure secondary phase is assigned. The target-centered difference image and no capable Gaia neighbor support ownership.

Next check: Optional resolved photometry and spectroscopy for precision physical parameters.

Catalogue recovery is recorded centrally in ../../shared-catalogues/coverage-matrix.csv. A successful cone with no exact eclipsing target classification is distinct from a failed query; known recoveries must not count as new additions. No exhaustive literature priority claim is made.

Sector	Search cycle	Measured midpoint BTJD	Formal center error d	Timing input error d	Depth ppt	Timing status
45	0	2530.1482583	0.005159	0.005159	23.90	selected
45	2	2546.2885931	0.00316943	0.00347257	25.73	selected
46	3	2554.3596239	0.00667629	0.00667629	27.22	selected
46	4	2562.4295151	0.00595019	0.00595019	23.34	selected
46	5	2570.4982179	0.00123878	0.00347255	24.33	selected
72	91	3264.4636348	0.00116102	0.00116102	26.70	selected
72	92	3272.5410419	0.00286001	0.00286001	23.35	selected

Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png, historical original receipts, ../download-receipts.json, ../identity-check.json, ../alias-audit.json, ../gaia-geometry.json and ../helper-audit.md. Original inputs and fresh outputs are covered by ../SHA256SUMS.json and ../source-hashes.json.

Stable review key: R595.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s46.json - historical/diff_s46.png - historical/gaia.json - measurements.json - phase-check.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

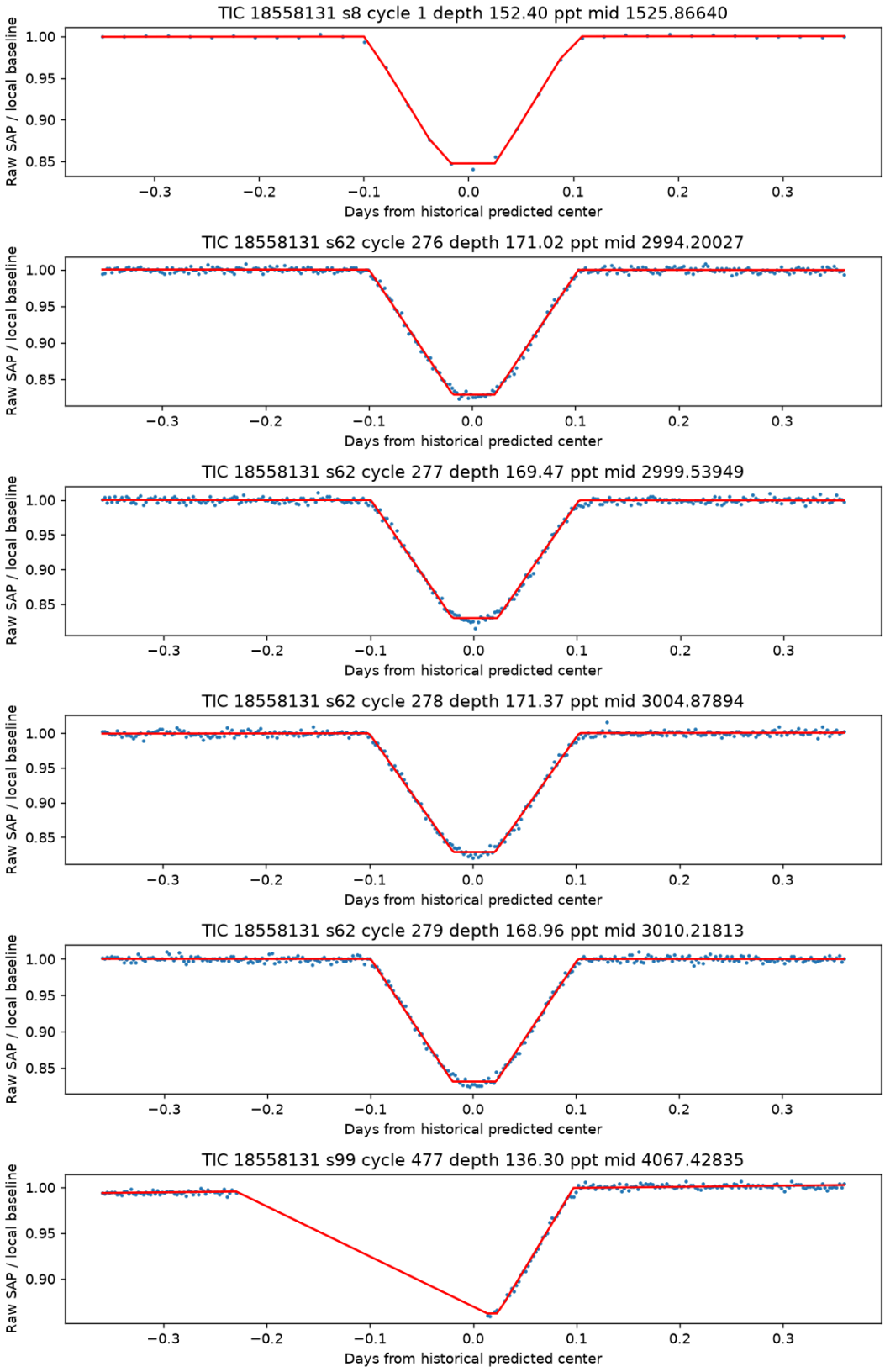
R596. TIC 18558131

physical_verdict: CONFIRMED

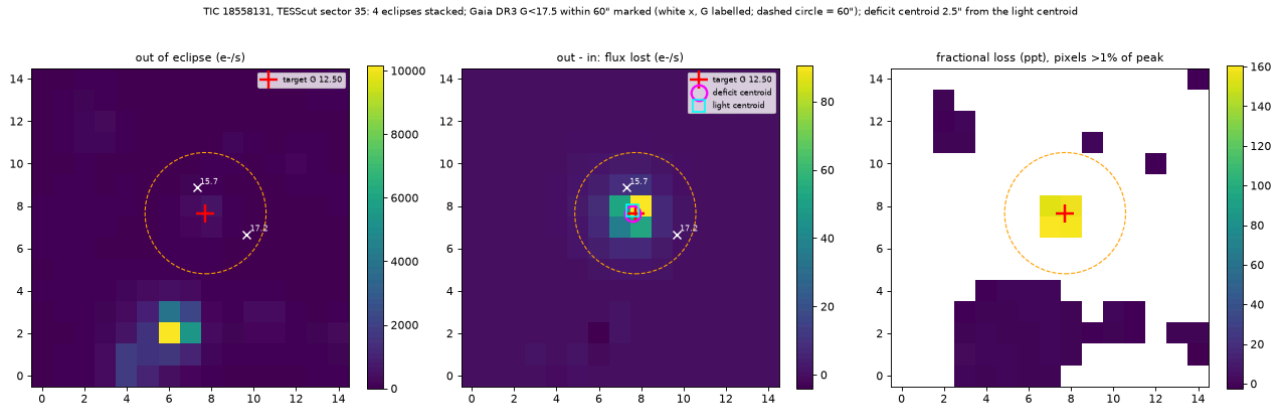
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision physical parameters.



R596; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R596; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 18558131: CONFIRMED

Full bounded physical review: **COMPLETE**. Physical classification is separate from catalogue recovery/novelty; historical screening was not an admission verdict.

1. **Raw events.** 6 free-center trapezoid measurements from quality-zero QLP SAP_FLUX in sectors [8, 62, 99]; 5 enter the reported ephemeris. Representative raw light curves and numerical windows show repeated eclipse morphology. Predictions guided the windows and are not measurements. Flagged or uncovered windows are not clean nulls.
2. **Timing and aliases.** Measured recurring-eclipse timing supported. Selected primary and unequal secondary morphology support the quoted orbital period; half/double hypotheses were checked. P=5.3393948228 +/- 0.0000376320 d; reference BTJD=1525.8665377. Epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The formal error column is the local free-center model covariance; the input error adds a half-cadence floor. Singular/unidentifiable centers are excluded. Cycle/parity counts are in ../alias-audit.json. The epoch kind above states whether the fitted events are identified as deeper primary or selected eclipse by convention. The bounded half/double check does not prove all possible aliases absent.
3. **Attribution.** Fresh difference stack in sector 35 has 4 events; target coordinate [7.723498086785394, 7.66106304495762], deficit [7.596665512534231, 7.668929771258732]; offset 0.127 pixel. The image-model fractional residual 0.697 is not a localization uncertainty. Every capable Gaia neighbor in the retained 60-arcsec table was inspected: None under the approximate G-band brightness screen within 60 arcsec. The intrinsic loss proxy uses a pairwise G-band flux budget and cannot be directly compared with diluted pixel fractional loss. Close blends and passband/aperture response limit exclusion claims. Repeated deep eclipses and shallower secondary near phase 0.500; compact target deficit. The nearby known EB Gaia 5634548560328478848 at 48.69 arcsec has period 0.64499 d, unlike this 5.33939 d signal, and is too faint under the G-band loss budget. It is a contamination alert, not a target catalogue recovery.

4. **Alternative reduction.** Fresh TESScut background-subtracted target-centered 1x1/3x3/5x5 depths are 157.64/155.14/159.17 ppt; selected QLP fitted median 169.47 ppt. This reruns the historical extraction primitive using the fresh ephemeris and measured duration, with identity and source hashes checked. The retained QLP sectors do not include this cutout sector. This cutout is a distinct observing occasion on the same instrument; it is an alternative extraction of that sector FFI data, without a direct same-sector QLP comparison here. Distinct sectors are distinct observing occasions on one instrument. Depth differences can reflect dilution, aperture and background response; intrinsic binary depth is not inferred.
5. **Gaia geometry/physical plausibility.** Exact DR3 source 5634548598985801088, orbital P=5.3397027022 d, e=0.04065. Closest edge-on conjunction offset=0.01833 d (0.00343 phase), full source-specific covariance linear timing allowance=0.06402 d. Mass function=0.11753 solar masses; adopted primary mass=1.39, minimum secondary mass=0.835. Rough central duration=6.49 h versus observed median=4.83 h. Radius and main-sequence companion assumptions make this a plausibility calculation, not a fitted geometry or calibrated significance veto. Shorter observed duration can reflect impact parameter and model assumptions. Orbit agreement does not establish ownership among unresolved sources.

Secondary phase check: Pooled raw-window free-center phase=0.4996555857506564, depth=54.45643492248451ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Weak secondary fits (especially314713331) describe inspected narrow dips and are not independent primary-eclipse evidence or calibrated detection significances. See ../secondary-phases.json and ../secondary-contact.png.

Deciding limit: Repeated deep eclipses and shallower secondary near phase 0.500; compact target deficit. The nearby known EB Gaia 5634548560328478848 at 48.69 arcsec has period 0.64499 d, unlike this 5.33939 d signal, and is too faint under the G-band loss budget. It is a contamination alert, not a target catalogue recovery.

Next check: Optional resolved photometry and spectroscopy for precision physical parameters.

Catalogue recovery is recorded centrally in ../../shared-catalogues/coverage-matrix.csv. A successful cone with no exact eclipsing target classification is distinct from a failed query; known recoveries must not count as new additions. No exhaustive literature priority claim is made.

Sector	Search cycle	Measured midpoint BTJD	Formal center error d	Timing input error d	Depth ppt	Timing status
8	1	1525.8664011	0.000237771	0.0104168	152.40	selected
62	276	2994.2002719	0.000192538	0.0011574	171.02	selected
62	277	2999.5394894	0.000194959	0.0011574	169.47	selected
62	278	3004.8789373	0.000190938	0.0011574	171.37	selected
62	279	3010.2181278	0.000182451	0.0011574	168.96	selected
99	477	4067.4283471	116392	116392	136.30	excluded center covariance

Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png, historical original receipts, ../download-receipts.json, ../identity-check.json, ../alias-audit.json, ../gaia-geometry.json and ../helper-audit.md. Original inputs and fresh outputs are covered by ../SHA256SUMS.json and ../source-hashes.json.

Stable review key: R596.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s35.json - historical/diff_s35.png - historical/gaia.json - measurements.json - phase-check.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

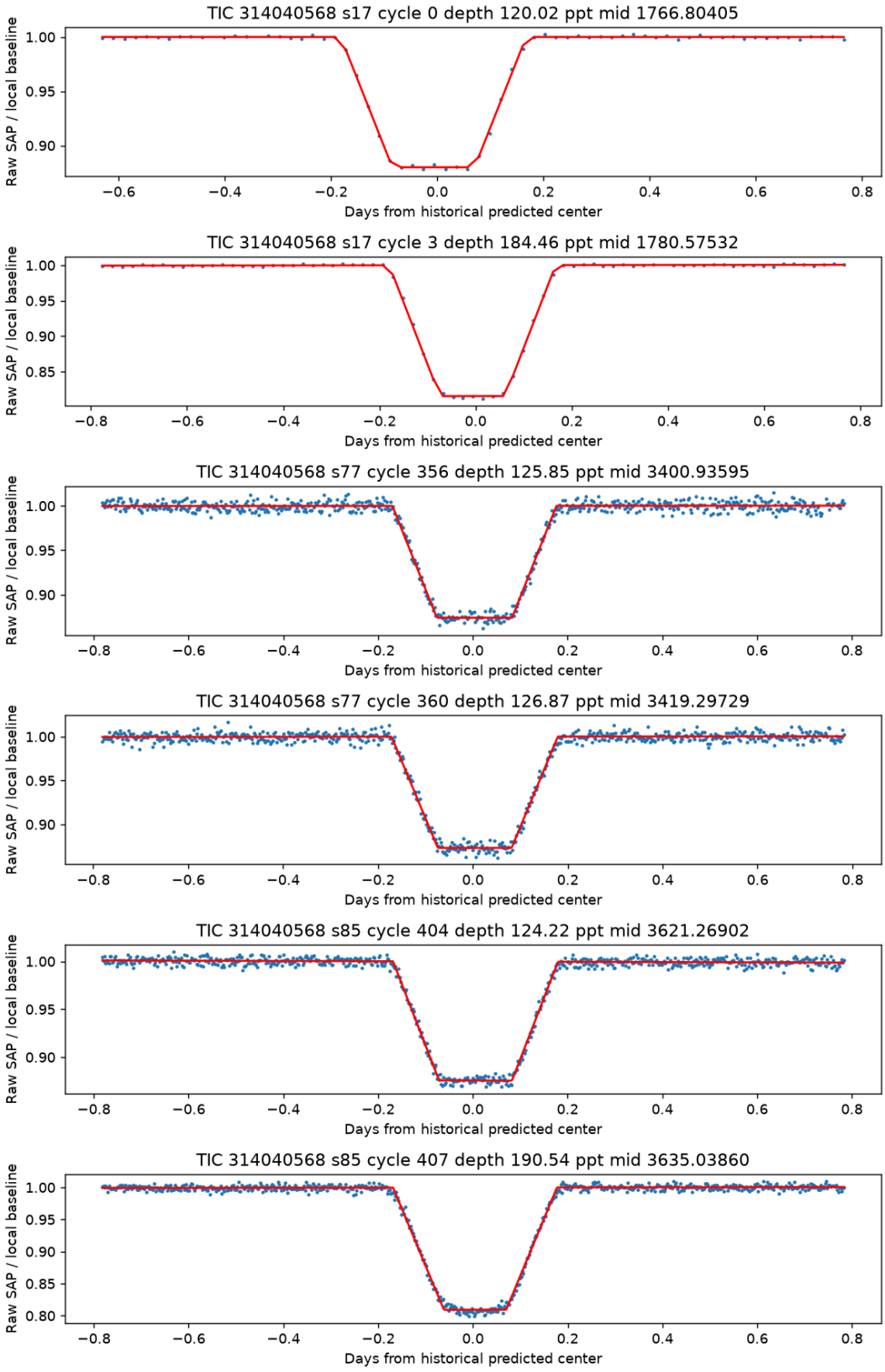
R597. TIC 314040568

physical_verdict: CONFIRMED

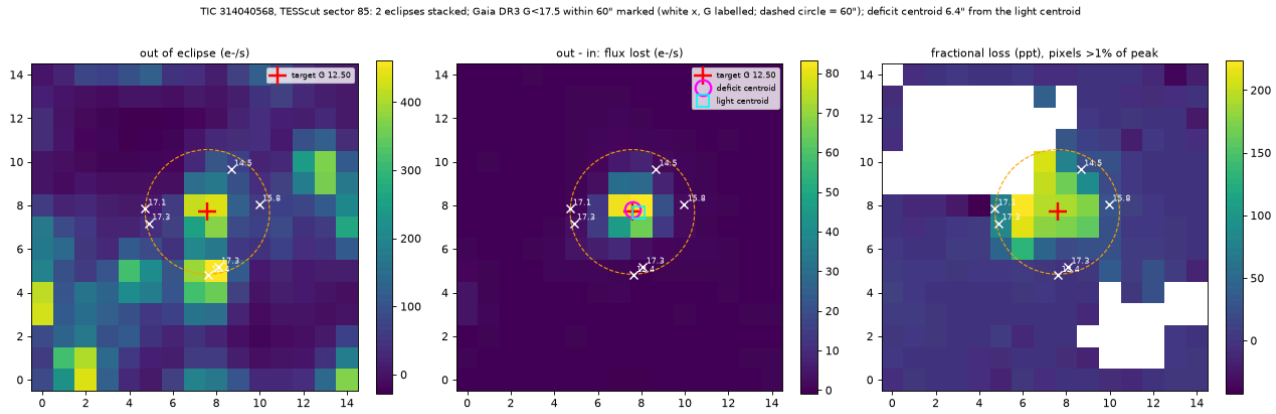
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision physical parameters.



R597; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R597; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 314040568: CONFIRMED

Full bounded physical review: **COMPLETE**. Physical classification is separate from catalogue recovery/novelty; historical screening was not an admission verdict.

1. **Raw events.** 14 free-center trapezoid measurements from quality-zero QLP SAP_FLUX in sectors [17, 77, 85]; 7 enter the reported ephemeris. Representative raw light curves and numerical windows show repeated eclipse morphology. Predictions guided the windows and are not measurements. Flagged or uncovered windows are not clean nulls.
2. **Timing and aliases.** Measured recurring-eclipse timing supported. Deeper odd half-period cycles only; shallow even cycles establish secondary near phase 0.5. P=9.1805146651 +/- 0.0000287051 d; reference BTJD=1771.3939484. Epoch kind: Deeper repeating eclipse, odd historical half-period parity. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The formal error column is the local free-center model covariance; the input error adds a half-cadence floor. Singular/unidentifiable centers are excluded. Cycle/parity counts are in ../alias-audit.json. The epoch kind above states whether the fitted events are identified as deeper primary or selected eclipse by convention. The bounded half/double check does not prove all possible aliases absent.
3. **Attribution.** Fresh difference stack in sector 85 has 2 events; target coordinate [7.5895062275439145, 7.700943838622596], deficit [7.580766136291378, 7.829906458814922]; offset 0.129 pixel. The image-model fractional residual 0.777 is not a localization uncertainty. Every capable Gaia neighbor in the retained 60-arcsec table was inspected: Gaia 2013347867977153152: 57.49 arcsec, G=12.410, intrinsic required loss proxy=0.365. The intrinsic loss proxy uses a pairwise G-band flux budget and cannot be directly compared with diluted pixel fractional loss. Close blends and passband/aperture response limit exclusion claims. Persistent unequal alternating 120/190 ppt dips reject the historical half-period as an orbital period. The quoted 9.180514665 d ephemeris uses seven deeper odd-parity measured eclipses; shallow even-parity events form the phase 0.500 secondary. Fresh pixel stack uses only deeper events and favors target over the 57.49 arcsec capable neighbor.

4. **Alternative reduction.** Fresh TESScut background-subtracted target-centered 1x1/3x3/5x5 depths are 191.13/178.63/143.19 ppt; selected QLP fitted median 190.32 ppt. This reruns the historical extraction primitive using the fresh ephemeris and measured duration, with identity and source hashes checked. The alternative extraction and the QLP data for this sector share the same TESS photons; they test processing robustness and are not independent observations. Distinct sectors are distinct observing occasions on one instrument. Depth differences can reflect dilution, aperture and background response; intrinsic binary depth is not inferred.
5. **Gaia geometry/physical plausibility.** Exact DR3 source 2013348112800055808, orbital P=9.1811034987 d, e=0.01268. Closest edge-on conjunction offset=-0.14075 d (-0.01533 phase), full source-specific covariance linear timing allowance=0.17439 d. Mass function=0.14611 solar masses; adopted primary mass=1.13, minimum secondary mass=0.823. Rough central duration=10.31 h versus observed median=8.31 h. Radius and main-sequence companion assumptions make this a plausibility calculation, not a fitted geometry or calibrated significance veto. Shorter observed duration can reflect impact parameter and model assumptions. Orbit agreement does not establish ownership among unresolved sources.

Secondary phase check: Pooled raw-window free-center phase=0.5000975746392974, depth=124.56105240297056ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Weak secondary fits (especially314713331) describe inspected narrow dips and are not independent primary-eclipse evidence or calibrated detection significances. See ../secondary-phases.json and ../secondary-contact.png.

Deciding limit: Persistent unequal alternating 120/190 ppt dips reject the historical half-period as an orbital period. The quoted 9.180514665 d ephemeris uses seven deeper odd-parity measured eclipses; shallow even-parity events form the phase 0.500 secondary. Fresh pixel stack uses only deeper events and favors target over the 57.49 arcsec capable neighbor.

Next check: Optional resolved photometry and spectroscopy for precision physical parameters.

Catalogue recovery is recorded centrally in ../../shared-catalogues/coverage-matrix.csv. A successful cone with no exact eclipsing target classification is distinct from a failed query; known recoveries must not count as new additions. No exhaustive literature priority claim is made.

Sector	Search cycle	Measured midpoint BTJD	Formal center error d	Timing input error d	Depth ppt	Timing status
17	0	1766.8040475	0.000239269	0.0104167	120.02	other parity
17	1	1771.3943669	0.000164893	0.0104167	185.55	selected
17	3	1780.5753217	0.000278147	0.0104167	184.46	selected
17	4	1785.1660010	0.000319738	0.0104167	121.14	other parity
77	355	3396.3448717	0.000235171	0.0011574	195.64	selected
77	356	3400.9359518	0.000419352	0.0011574	125.85	other parity
77	357	3405.5256323	0.00026901	0.0011574	194.63	selected
77	360	3419.2972904	0.000388438	0.0011574	126.87	other parity
85	402	3612.0877771	0.000364141	0.00115738	124.07	other parity
85	403	3616.6769356	0.000269006	0.00115738	190.11	selected
85	404	3621.2690180	0.000312457	0.00115738	124.22	other parity
85	405	3625.8582844	0.000245969	0.00115738	190.32	selected
85	406	3630.4489996	0.000339954	0.00115738	123.33	other parity
85	407	3635.0385951	0.000198994	0.00115738	190.54	selected

Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png, historical original receipts, ../download-receipts.json, ../identity-check.json, ../alias-audit.json, ../gaia-geometry.json and ../helper-audit.md. Original inputs and fresh outputs are covered by

../SHA256SUMS.json and ../source-hashes.json.

Stable review key: R597.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s85.json - historical/diff_s85.png - historical/gaia.json - measurements.json - phase-check.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

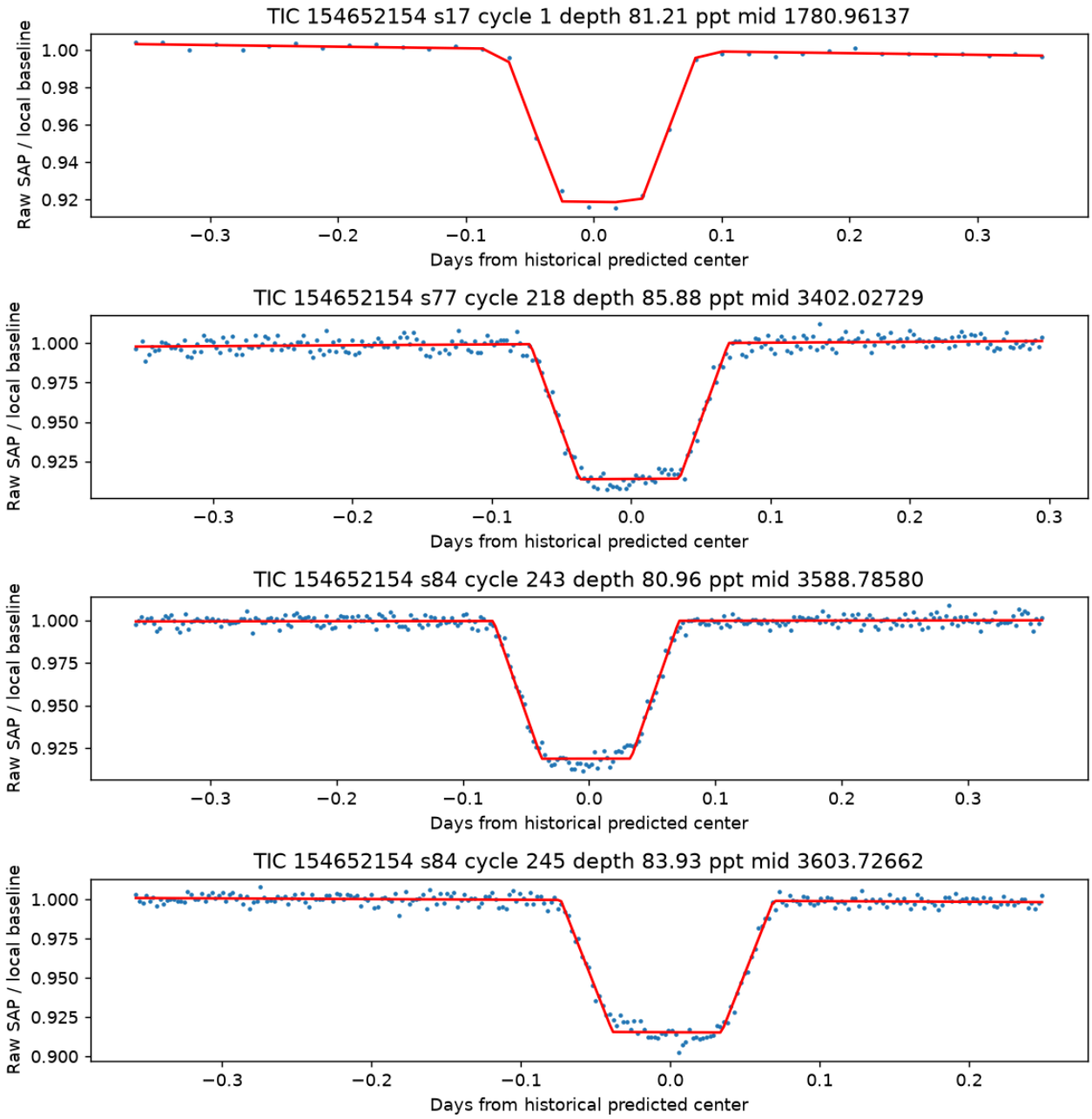
R598. TIC 154652154

physical_verdict: CONFIRMED

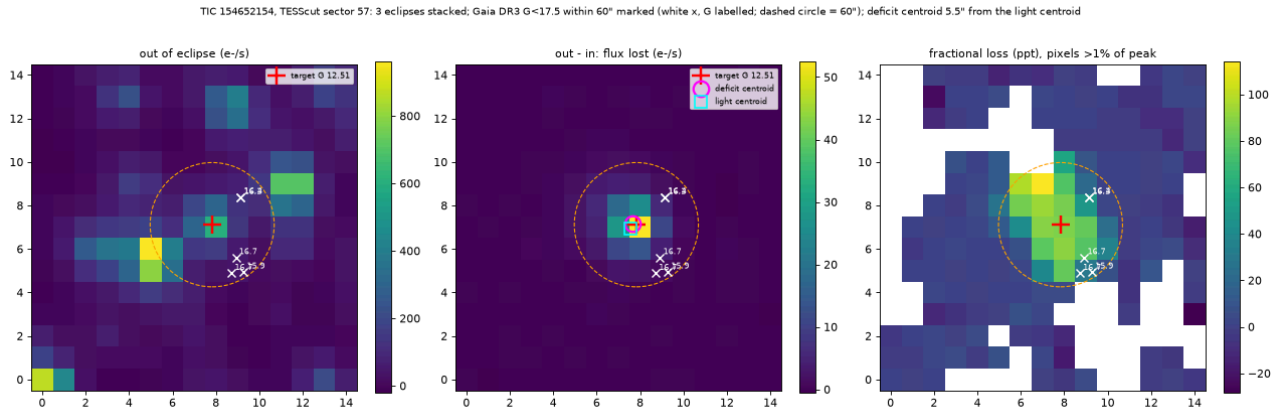
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision physical parameters.



R598; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R598; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 154652154: CONFIRMED

Full bounded physical review: **COMPLETE**. Physical classification is separate from catalogue recovery/novelty; historical screening was not an admission verdict.

1. **Raw events.** 4 free-center trapezoid measurements from quality-zero QLP SAP_FLUX in sectors [17, 77, 84]; 4 enter the reported ephemeris. Representative raw light curves and numerical windows show repeated eclipse morphology. Predictions guided the windows and are not measurements. Flagged or uncovered windows are not clean nulls.
2. **Timing and aliases.** Measured recurring-eclipse timing supported. Selected primary and unequal secondary morphology support the quoted orbital period; half/double hypotheses were checked. P=7.4703470176 +/- 0.0000344636 d; reference BTJD=1780.9619199. Epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The formal error column is the local free-center model covariance; the input error adds a half-cadence floor. Singular/unidentifiable centers are excluded. Cycle/parity counts are in ../alias-audit.json. The epoch kind above states whether the fitted events are identified as deeper primary or selected eclipse by convention. The bounded half/double check does not prove all possible aliases absent.
3. **Attribution.** Fresh difference stack in sector 57 has 3 events; target coordinate [7.827555309333434, 7.113911110055252], deficit [7.6536171926647825, 7.171304688980337]; offset 0.183 pixel. The image-model fractional residual 0.909 is not a localization uncertainty. Every capable Gaia neighbor in the retained 60-arcsec table was inspected: None under the approximate G-band brightness screen within 60 arcsec. The intrinsic loss proxy uses a pairwise G-band flux budget and cannot be directly compared with diluted pixel fractional loss. Close blends and passband/aperture response limit exclusion claims. Four QLP primary centers span multiple sectors, with sector 84 primaries two cycles apart. Additional free-center custom 3 x 3 cutout timings in sector 57 include consecutive same-kind 83 ppt eclipses, confirming the 7.47035 d cycle count; see cutout-s 57-timing.json and image. No null is inferred at uncovered QLP cycle 244. Compact deficit has no capable Gaia neighbor.

4. **Alternative reduction.** Fresh TESScut background-subtracted target-centered 1x1/3x3/5x5 depths are 87.38/84.50/57.29 ppt; selected QLP fitted median 82.57 ppt. This reruns the historical extraction primitive using the fresh ephemeris and measured duration, with identity and source hashes checked. The retained QLP sectors do not include this cutout sector. This cutout is a distinct observing occasion on the same instrument; it is an alternative extraction of that sector FFI data, without a direct same-sector QLP comparison here. Distinct sectors are distinct observing occasions on one instrument. Depth differences can reflect dilution, aperture and background response; intrinsic binary depth is not inferred.
5. **Gaia geometry/physical plausibility.** Exact DR3 source 1984303482355502336, orbital P=7.4700465484 d, e=0.08193. Closest edge-on conjunction offset=0.14324 d (0.01918 phase), full source-specific covariance linear timing allowance=0.19417 d. Mass function=0.0090214 solar masses; adopted primary mass=0.876, minimum secondary mass=0.221. Rough central duration=3.94 h versus observed median=3.47 h. Radius and main-sequence companion assumptions make this a plausibility calculation, not a fitted geometry or calibrated significance veto. Shorter observed duration can reflect impact parameter and model assumptions. Orbit agreement does not establish ownership among unresolved sources.

Secondary phase check: Pooled raw-window free-center phase=0.5000647549675792, depth=6.825655163596055ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Weak secondary fits (especially314713331) describe inspected narrow dips and are not independent primary-eclipse evidence or calibrated detection significances. See ../secondary-phases.json and ../secondary-contact.png.

Deciding limit: Four QLP primary centers span multiple sectors, with sector 84 primaries two cycles apart. Additional free-center custom 3 x 3 cutout timings in sector 57 include consecutive same-kind 83 ppt eclipses, confirming the 7.47035 d cycle count; see cutout-s 57-timing.json and image. No null is inferred at uncovered QLP cycle 244. Compact deficit has no capable Gaia neighbor.

Next check: Optional resolved photometry and spectroscopy for precision physical parameters.

Catalogue recovery is recorded centrally in ../../shared-catalogues/coverage-matrix.csv. A successful cone with no exact eclipsing target classification is distinct from a failed query; known recoveries must not count as new additions. No exhaustive literature priority claim is made.

Sector	Search cycle	Measured midpoint BTJD	Formal center error d	Timing input error d	Depth ppt	Timing status
17	1	1780.9613710	0.000796022	0.0104164	81.21	selected
77	218	3402.0272910	0.000330279	0.00115741	85.88	selected
84	243	3588.7858043	0.000225016	0.00115739	80.96	selected
84	245	3603.7266248	0.000230405	0.00115739	83.93	selected

Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png, historical original receipts, ../download-receipts.json, ../identity-check.json, ../alias-audit.json, ../gaia-geometry.json and ../helper-audit.md. Original inputs and fresh outputs are covered by ../SHA256SUMS.json and ../source-hashes.json.

Stable review key: R598.

Deciding evidence and numeric artifacts: - cutout-s57-events.png - cutout-s57-timing.json - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s57.json - historical/diff_s57.png - historical/gaia.json - measurements.json - phase-check.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

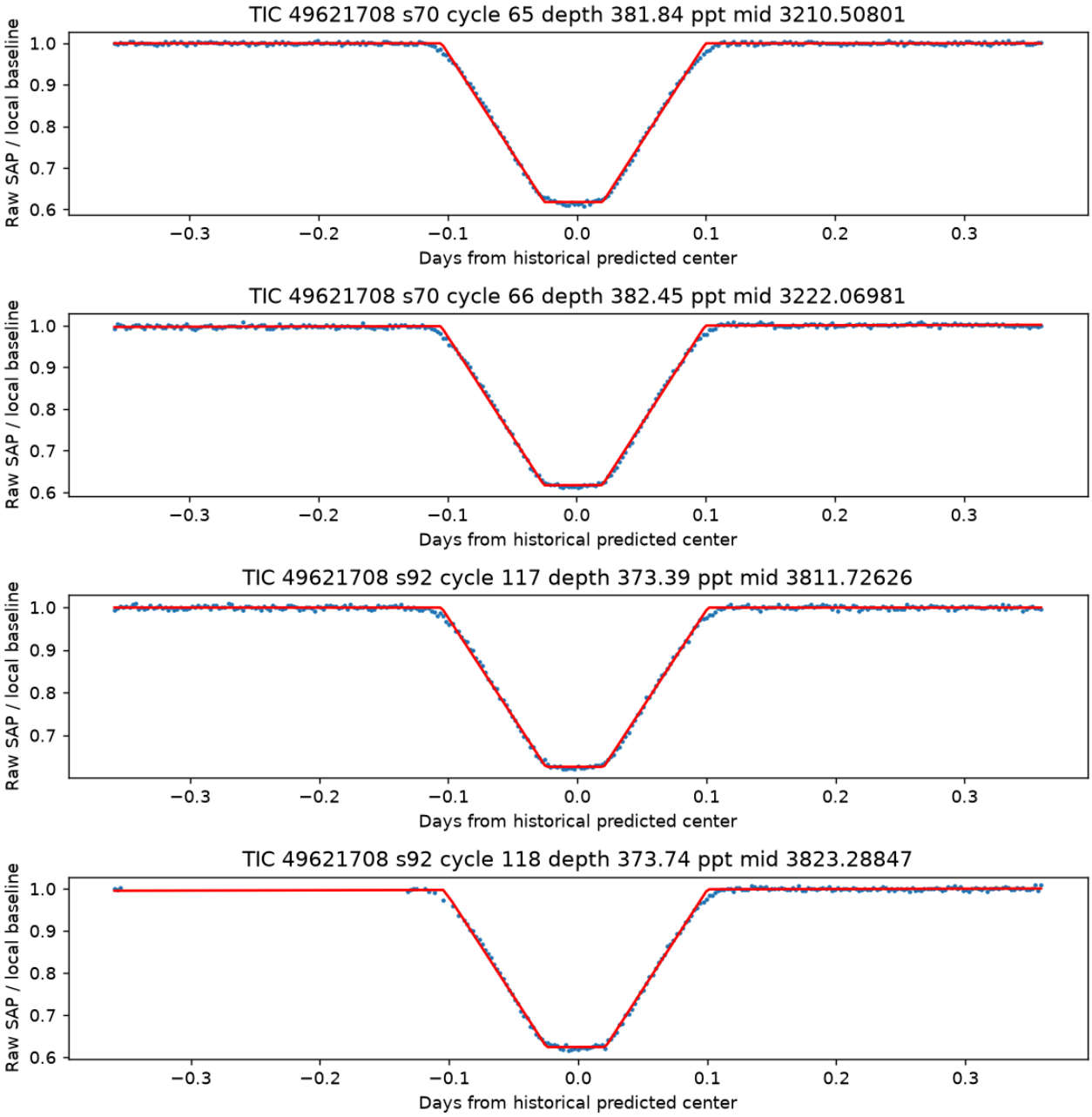
R599. TIC 49621708

physical_verdict: CONFIRMED

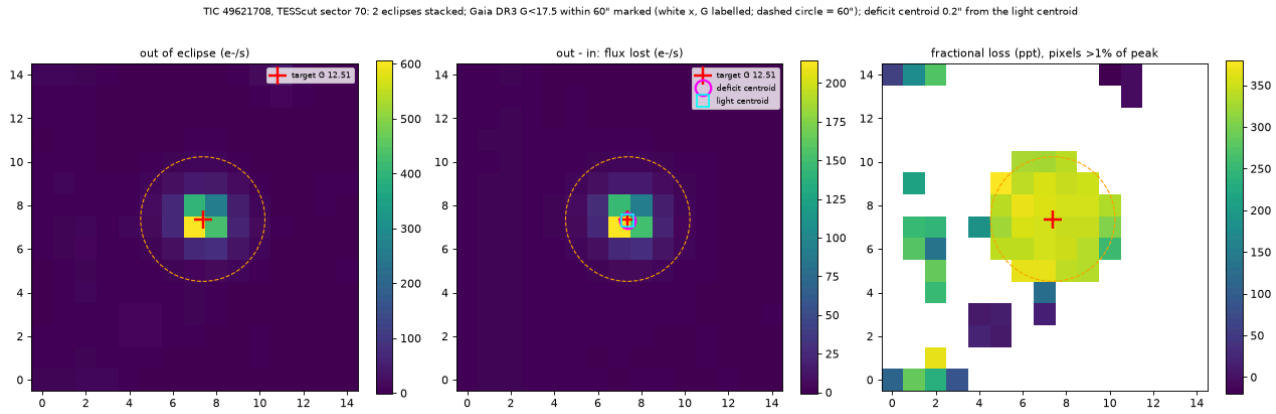
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy for precision physical parameters.



R599; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R599; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 49621708: CONFIRMED

Full bounded physical review: **COMPLETE**. Physical classification is separate from catalogue recovery/novelty; historical screening was not an admission verdict.

1. **Raw events.** 4 free-center trapezoid measurements from quality-zero QLP SAP_FLUX in sectors [70, 92]; 4 enter the reported ephemeris. Representative raw light curves and numerical windows show repeated eclipse morphology. Predictions guided the windows and are not measurements. Flagged or uncovered windows are not clean nulls.
2. **Timing and aliases.** Measured recurring-eclipse timing supported. Selected primary and unequal secondary morphology support the quoted orbital period; half/double hypotheses were checked. P=11.5618933696 +/- 0.0000222543 d; reference BTJD=3210.5079607. Epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. The formal error column is the local free-center model covariance; the input error adds a half-cadence floor. Singular/unidentifiable centers are excluded. Cycle/parity counts are in ../alias-audit.json. The epoch kind above states whether the fitted events are identified as deeper primary or selected eclipse by convention. The bounded half/double check does not prove all possible aliases absent.
3. **Attribution.** Fresh difference stack in sector 70 has 2 events; target coordinate [7.376949394793275, 7.372509587609754], deficit [7.358921552261768, 7.3265204872643945]; offset 0.049 pixel. The image-model fractional residual 0.029 is not a localization uncertainty. Every capable Gaia neighbor in the retained 60-arcsec table was inspected: None under the approximate G-band brightness screen within 60 arcsec. The intrinsic loss proxy uses a pairwise G-band flux budget and cannot be directly compared with diluted pixel fractional loss. Close blends and passband/aperture response limit exclusion claims. Four deep primary eclipses include consecutive pairs in sectors 70 and 92, plus a much shallower secondary near phase 0.496. Compact target deficit and no capable Gaia neighbor within 60 arcsec.
4. **Alternative reduction.** Fresh TESScut background-subtracted target-centered 1x1/3x3/5x5 depths are 354.00/353.56/352.66 ppt; selected QLP fitted median 377.79 ppt. This reruns the

historical extraction primitive using the fresh ephemeris and measured duration, with identity and source hashes checked. The alternative extraction and the QLP data for this sector share the same TESS photons; they test processing robustness and are not independent observations. Distinct sectors are distinct observing occasions on one instrument. Depth differences can reflect dilution, aperture and background response; intrinsic binary depth is not inferred.

5. **Gaia geometry/physical plausibility.** Exact DR3 source 2438047943740137984, orbital P=11.5624462275 d, e=0.02871. Closest edge-on conjunction offset=-0.13830 d (-0.01196 phase), full source-specific covariance linear timing allowance=0.39443 d. Mass function=0.10366 solar masses; adopted primary mass=0.95, minimum secondary mass=0.64. Rough central duration=6.20 h versus observed median=4.91 h. Radius and main-sequence companion assumptions make this a plausibility calculation, not a fitted geometry or calibrated significance veto. Shorter observed duration can reflect impact parameter and model assumptions. Orbit agreement does not establish ownership among unresolved sources.

Secondary phase check: Pooled raw-window free-center phase=0.4957846638264749, depth=96.83635841200893ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Weak secondary fits (especially314713331) describe inspected narrow dips and are not independent primary-eclipse evidence or calibrated detection significances. See ../secondary-phases.json and ../secondary-contact.png.

Deciding limit: Four deep primary eclipses include consecutive pairs in sectors 70 and 92, plus a much shallower secondary near phase 0.496. Compact target deficit and no capable Gaia neighbor within 60 arcsec.

Next check: Optional resolved photometry and spectroscopy for precision physical parameters.

Catalogue recovery is recorded centrally in ../../shared-catalogues/coverage-matrix.csv. A successful cone with no exact eclipsing target classification is distinct from a failed query; known recoveries must not count as new additions. No exhaustive literature priority claim is made.

Sector	Search cycle	Measured midpoint BTJD	Formal center error d	Timing input error d	Depth ppt	Timing status
70	65	3210.5080070	7.10302e-05	0.00115736	381.84	selected
70	66	3222.0698100	0.0001033	0.00115736	382.45	selected
92	117	3811.7262573	9.84556e-05	0.00115751	373.39	selected
92	118	3823.2884657	9.08102e-05	0.00115751	373.74	selected

Evidence: measurements.json, raw-windows.csv, raw-events.png, fresh-pixels/diff_fresh.json/png, historical original receipts, ../download-receipts.json, ../identity-check.json, ../alias-audit.json, ../gaia-geometry.json and ../helper-audit.md. Original inputs and fresh outputs are covered by ../SHA256SUMS.json and ../source-hashes.json.

Stable review key: R599.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s70.json - historical/diff_s70.png - historical/gaia.json - measurements.json - phase-check.json - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

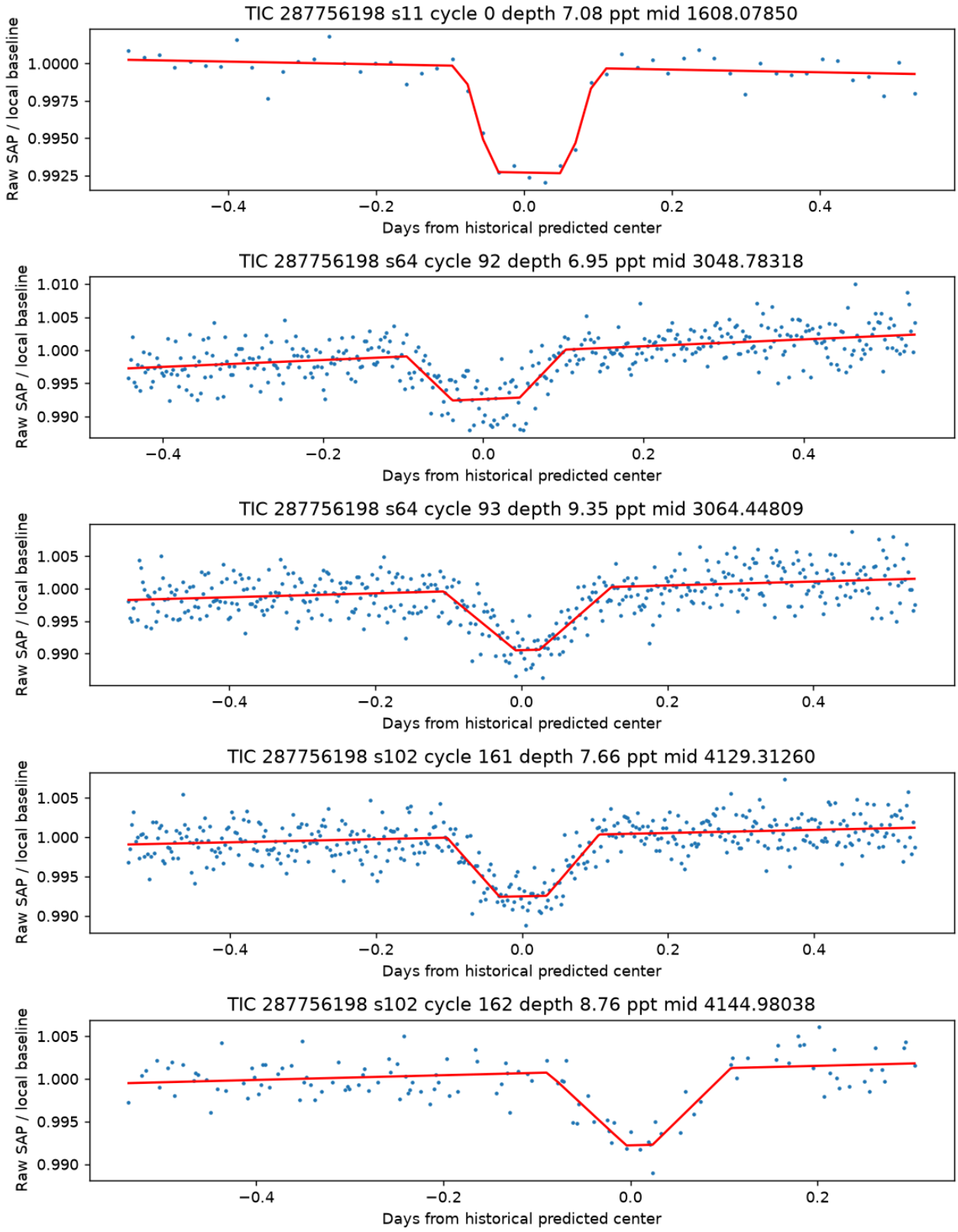
R600. TIC 287756198

physical_verdict: CONFIRMED

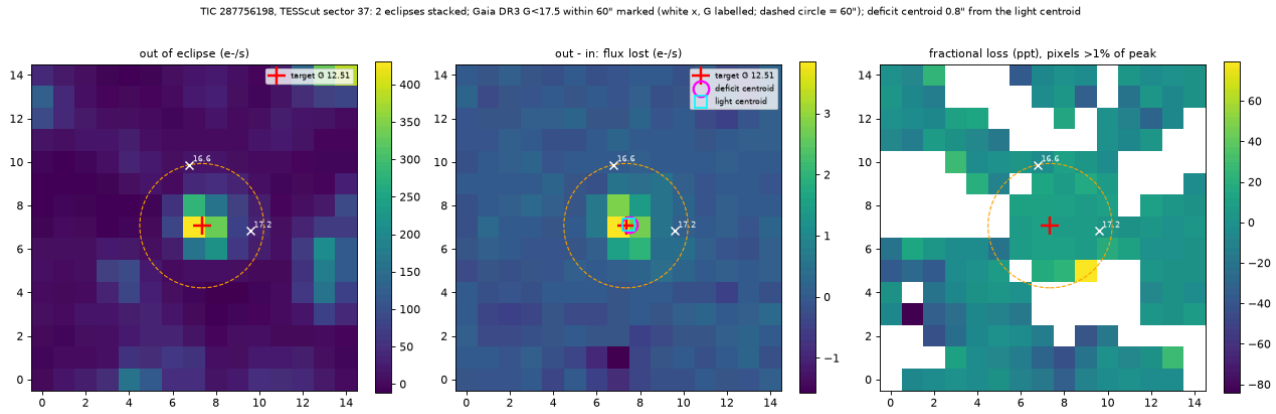
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Additional clean eclipse timing and color-aware neighbor/PRF modeling would refine the bounded result.



R600; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R600; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 287756198

Physical verdict: **confirmed**. Full bounded review: **complete**.

Five measured raw events across S11/64/102 give local recurrence in two sectors and a long baseline. Sector depth/continuum differences reflect extraction dilution and variability. Three capable sources45.1-58.5 arcsec away are displaced from the compact target deficit. No secure secondary inferred from continuum changes.

Measured period 15.65984307195 +/- 4.521229e-05 d; named selected-eclipse epoch BTJD 1608.08000769; depth 7.65521 ppt (diluted raw QLP). Epoch is the selected recurring eclipse, not an assumed deeper primary. BTJD = BJD_TDB -2457000.

Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.

Timing samples: 5; sectors [11, 64, 102]; fitted same-kind cycle span 0 to 162. Reduced chi square: 1.1437703620814224. Actual centers, errors and local windows are in measurements.json and raw-windows.csv; predicted centers are labeled and never counted as measurements.

Aliases: within-sector adjacent same-kind pairs [[64, 92, 93], [102, 161, 162]]. Sector parity depths and nearby integer counts are in ../alias-audit.json and ../integer-aliases.json. Half-period extra windows are in ../coverage-alias-windows.json; flagged/empty windows are not nulls. No persistent unequal alternating primary depths support a doubled period. Gaia period is a consistency test, not a fixed photometric fit.

Secondary assessment: {'status': 'No secure secondary established by visual review of full raw phase and predicted-conjunction windows. Baseline variability/coverage limit a null claim; guess is Gaia-predicted separation, not a measured secondary phase.', 'guess_phase': 0.4935567357557127}

Attribution and alternative reduction: fresh TESScut S37 stacks 2 well-covered target-period eclipses. Target WCS pixel [7.348427290117524, 7.069412095525774]; deficit centroid [7.510954246146845, 7.131011385411307]. Read fresh-pixels/diff_fresh.png against every capable neighbor below. No fixed centroid threshold applied. Target-centered custom background-subtracted1x1/3x3/5x5 extraction

depths are 8.9417/8.9589/8.7776 ppt. This is a separate reduction of the same TESS photons, not independent observations; different sectors are different occasions on one instrument. Source FITS identities and SHA256 hashes are in ../identity-check.json and ../source-hashes.json.

All capable neighbors within60 arcsec (G-band flux-ratio proxy; intrinsic required depth is not directly compared to diluted pixel loss): [{"source_id": "6086968256243116288", "sep": 45.10332824606211, "G": 17.159227, "depth_needed": 0.5616780342395038, "possible": true, "px": [9.586378118081019, 6.837241536483282], "frac_loss_at_pixel": 0.004954973487623503}, {"source_id": "6086968256243114624", "sep": 48.12232035178331, "G": 17.555882, "depth_needed": 0.8059984604155195, "possible": true, "px": [8.743292166399184, 9.046623195139022], "frac_loss_at_pixel": 0.015762178206169458}, {"source_id": "6086968015724946816", "sep": 58.48282576005001, "G": 16.633436, "depth_needed": 0.34901412912874746, "possible": true, "px": [6.781240761818138, 9.867458243771992], "frac_loss_at_pixel": -0.002232594969092304}]

Gaia source 6086968080141964672: source-specific NSS period 15.6680577389 d; eccentricity 0.08810; nearest edge-on conjunction offset -0.34982 d (phase -0.02233), linear full-covariance error 0.67634 d. Source-specific covariance, mass function, assumed masses/radii and approximate central duration are in ../gaia-geometry.json. This is not a hard phase veto or fitted inclination. Observed duration 4.767 h, approximate central duration 10.947 h; grazing geometry/dilution and radius assumptions can shorten the eclipse. No contradiction that invalidates repeated-eclipse interpretation is established.

Catalogue outcome: no identified prior EB; seven-group coverage yes. Shared coverage-matrix and catalogue receipts support scope only, not exhaustive priority. Physical verdict is separate from novelty.

Limitations: Representative sectors and simplified trapezoid/linear timing errors; shared TESS photons; approximate Gaia edge-on geometry and G-band neighbor flux budgets.

Next check: Additional clean eclipse timing and color-aware neighbor/PRF modeling would refine the bounded result.

Stable review key: R600.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s37.json - historical/diff_s37.png - historical/gaia.json - measurements.json - phase.png - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

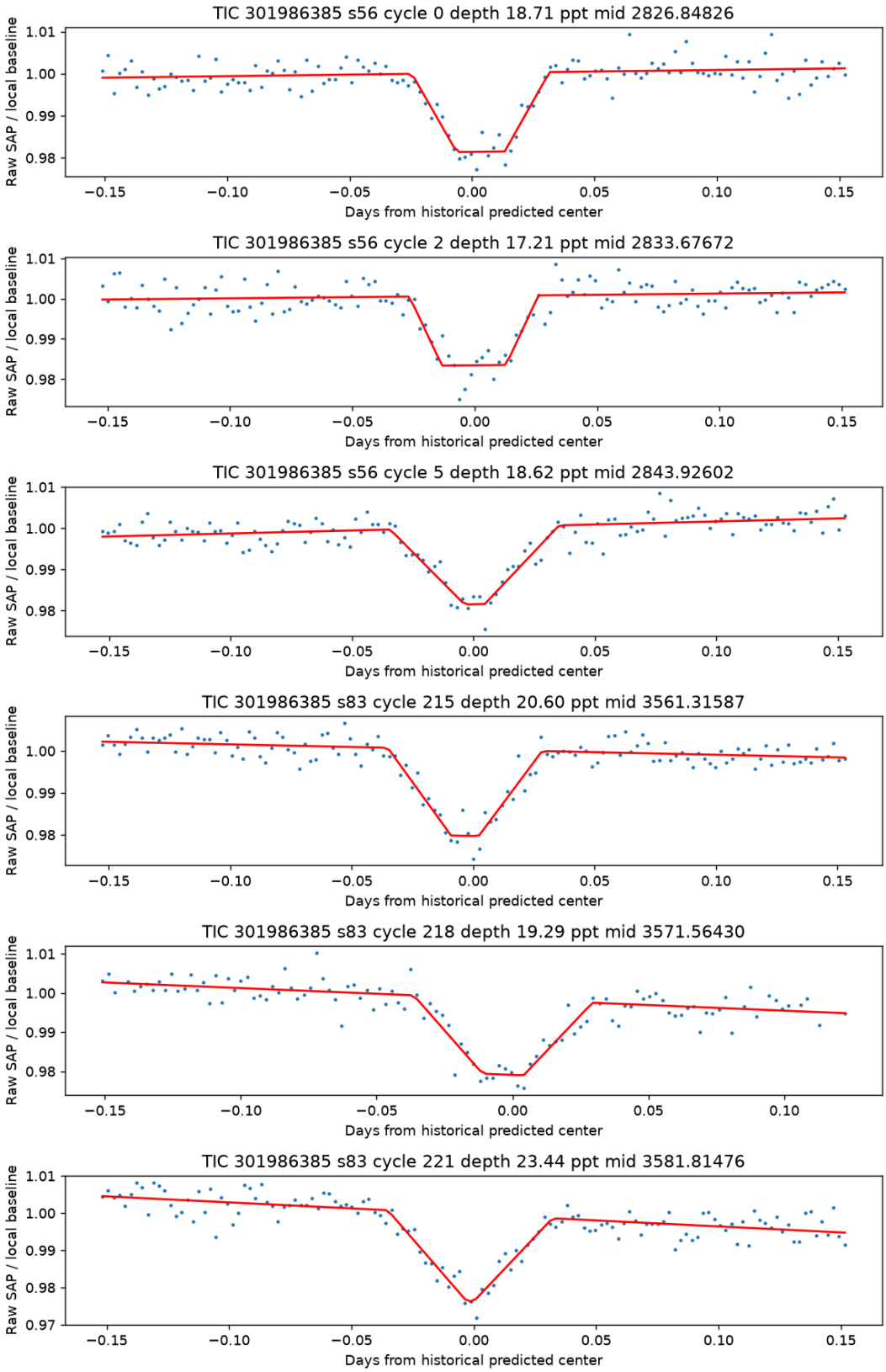
R601. TIC 301986385

physical_verdict: CONFIRMED

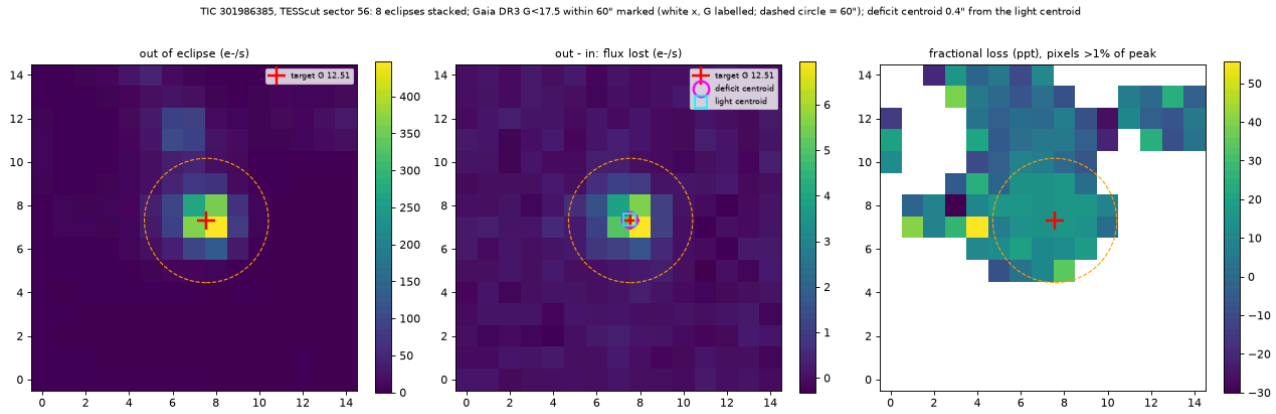
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Additional clean eclipse timing and color-aware neighbor/PRF modeling would refine the bounded result.



R601; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R601; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 301986385

Physical verdict: **confirmed**. Full bounded review: **complete**.

Fifteen measured narrow raw eclipses in S56/83 establish same-kind recurrence. A shallow narrow secondary-like feature is measured near phase0.50004 relative to the selected deeper eclipse; its pooled formal phase error omits red-noise and sector-normalization systematics. The primary and secondary are unequal, rejecting the half-period interpretation. No capable Gaia neighbor within60 arcsec under the G-band flux-budget proxy.

Measured period 3.41614587598 +/- 3.3342836e-06 d; named selected-eclipse epoch BTJD 2826.84537464; depth 18.36787 ppt (diluted raw QLP). Epoch is the selected recurring eclipse, not an assumed deeper primary. BTJD = BJD_TDB -2457000.

Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.

Timing samples: 15; sectors [56, 83]; fitted same-kind cycle span 0 to 221. Reduced chi square: 1.3554658684751226. Actual centers, errors and local windows are in measurements.json and raw-windows.csv; predicted centers are labeled and never counted as measurements.

Aliases: within-sector adjacent same-kind pairs [[56, 0, 1], [56, 1, 2], [56, 2, 3], [56, 3, 4], [56, 4, 5], [56, 5, 6], [56, 6, 7], [83, 215, 216], [83, 216, 217], [83, 217, 218], [83, 218, 219], [83, 219, 220], [83, 220, 221]]. Sector parity depths and nearby integer counts are in ../alias-audit.json and ../integer-aliases.json. Half-period extra windows are in ../coverage-alias-windows.json; flagged/empty windows are not nulls. No persistent unequal alternating primary depths support a doubled period. Gaia period is a consistency test, not a fixed photometric fit.

Secondary assessment: {'status': 'Secondary-like narrow dip visually inspected; model describes selected pooled raw sector windows.', 'phase': 0.5000448522882945, 'formal_phase_error': 0.0005668116369616375, 'depth_ppt': 3.5258108227907066, 'width_days': 0.06674786377133035, 'n_points': 2397, 'sectors': [56, 83], 'error_meaning': 'Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated.'}

Attribution and alternative reduction: fresh TESScut S56 stacks 8 well-covered target-period eclipses. Target WCS pixel [7.562522340497932, 7.307526992584487]; deficit centroid [7.524055121019407, 7.369991467711694]. Read fresh-pixels/diff_fresh.png against every capable neighbor below. No fixed centroid threshold applied. Target-centered custom background-subtracted1x1/3x3/5x5 extraction depths are 15.5351/14.6017/14.6204 ppt. This is a separate reduction of the same TESS photons, not independent observations; different sectors are different occasions on one instrument. Source FITS identities and SHA256 hashes are in ../identity-check.json and ../source-hashes.json.

All capable neighbors within60 arcsec (G-band flux-ratio proxy; intrinsic required depth is not directly compared to diluted pixel loss): []

Gaia source 2819970969870456832: source-specific NSS period 3.4160079763 d; eccentricity 0.03899; nearest edge-on conjunction offset 0.09743 d (phase 0.02852), linear full-covariance error 0.10945 d. Source-specific covariance, mass function, assumed masses/radii and approximate central duration are in ../gaia-geometry.json. This is not a hard phase veto or fitted inclination. Observed duration 1.567 h, approximate central duration 3.805 h; grazing geometry/dilution and radius assumptions can shorten the eclipse. No contradiction that invalidates repeated-eclipse interpretation is established.

Catalogue outcome: no identified prior EB; seven-group coverage yes. Shared coverage-matrix and catalogue receipts support scope only, not exhaustive priority. Physical verdict is separate from novelty.

Limitations: Representative sectors and simplified trapezoid/linear timing errors; shared TESS photons; approximate Gaia edge-on geometry and G-band neighbor flux budgets.

Next check: Additional clean eclipse timing and color-aware neighbor/PRF modeling would refine the bounded result.

Stable review key: R601.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s56.json - historical/diff_s56.png - historical/gaia.json - measurements.json - phase.png - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

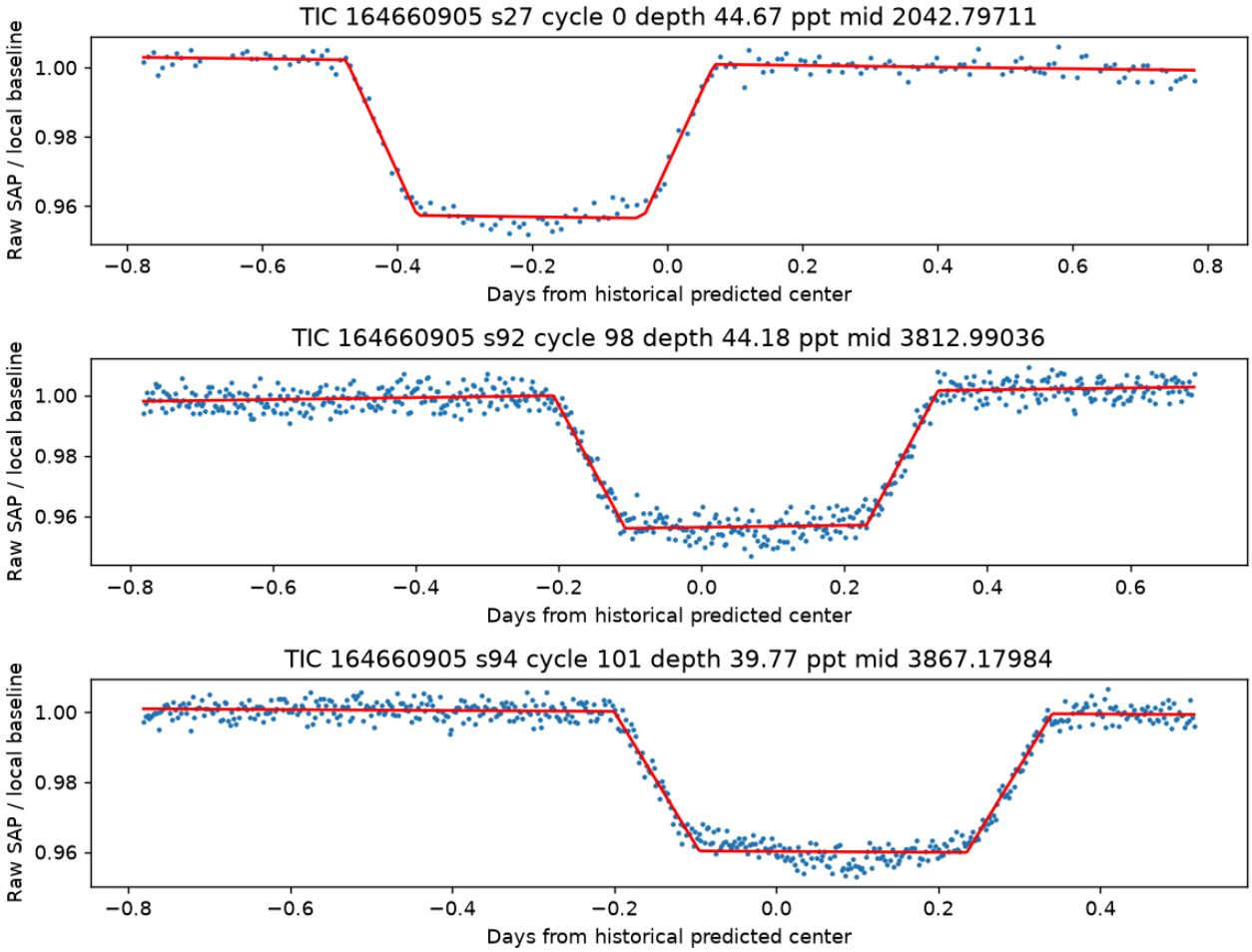
R602. TIC 164660905

physical_verdict: CONFIRMED

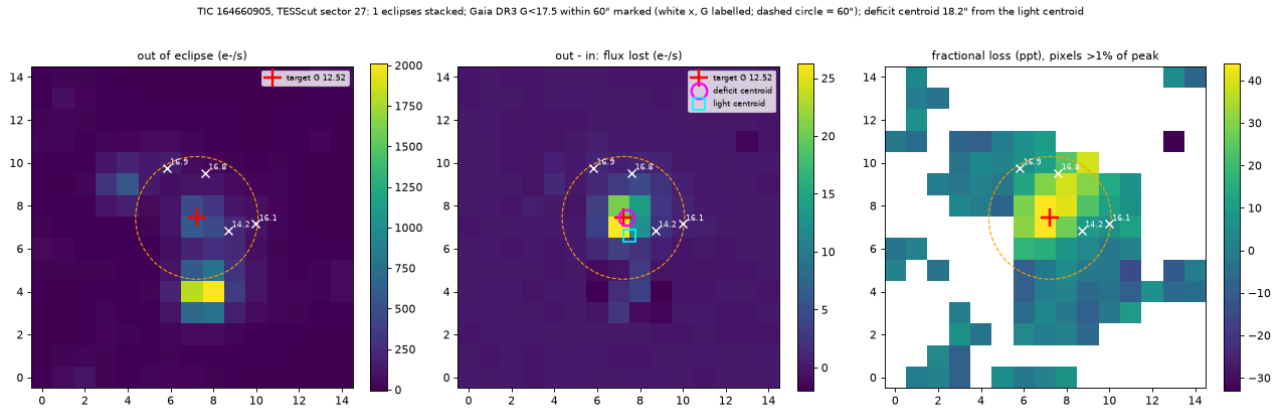
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Additional clean eclipse timing and color-aware neighbor/PRF modeling would refine the bounded result.



R602; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R602; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 164660905

Physical verdict: **confirmed**. Full bounded review: **complete**.

Three well-shaped long eclipses in S27/92/94. The 101-cycle baseline and final three-cycle interval jointly reject adjacent integer counts (integer-aliases.json); no period inferred from Gaia alone. Broad sector-dependent variability and isolated dips away from phase zero do not establish a recurring secondary. Bright southern light shifts the out-of-eclipse centroid; the deficit remains on the target core, distinct from the capable G14.24 neighbor 32.9 arcsec away.

Measured period 18.06319567004 +/- 3.5776422e-05 d; named selected-eclipse epoch BTJD 2042.79712745; depth 44.17691 ppt (diluted raw QLP). Epoch is the selected recurring eclipse, not an assumed deeper primary. BTJD = BJD_TDB -2457000.

Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.

Timing samples: 3; sectors [27, 92, 94]; fitted same-kind cycle span 0 to 101. Reduced chi square: 0.004636358523554171. Actual centers, errors and local windows are in measurements.json and raw-windows.csv; predicted centers are labeled and never counted as measurements.

Aliases: within-sector adjacent same-kind pairs []. Sector parity depths and nearby integer counts are in ../alias-audit.json and ../integer-aliases.json. Half-period extra windows are in ../coverage-alias-windows.json; flagged/empty windows are not nulls. No persistent unequal alternating primary depths support a doubled period. Gaia period is a consistency test, not a fixed photometric fit.

Secondary assessment: {'status': 'No secure secondary established by visual review of full raw phase and predicted-conjunction windows. Baseline variability/coverage limit a null claim; guess is Gaia-predicted separation, not a measured secondary phase.', 'guess_phase': 0.5704690327947917}

Attribution and alternative reduction: fresh TESScut S27 stacks 1 well-covered target-period eclipses. Target WCS pixel [7.221955976992129, 7.434413295976716]; deficit centroid [7.387213524487844, 7.446263677299761]. Read fresh-pixels/diff_fresh.png against every capable neighbor below. No fixed centroid threshold applied. Target-centered custom background-subtracted1x1/3x3/5x5 extraction

depths are 43.9907/32.5099/18.4358 ppt. This is a separate reduction of the same TESS photons, not independent observations; different sectors are different occasions on one instrument. Source FITS identities and SHA256 hashes are in ../identity-check.json and ../source-hashes.json.

All capable neighbors within60 arcsec (G-band flux-ratio proxy; intrinsic required depth is not directly compared to diluted pixel loss): [{"source_id": "6797641994897915648", "sep": 32.942141928445764, "G": 14.239942, "depth_needed": 0.2605086101602356, "possible": true, "px": [8.70355082316581, 6.848233018302231], "frac_loss_at_pixel": 0.009941876710546343}]

Gaia source 6797641994897913856: source-specific NSS period 18.0661892098 d; eccentricity 0.31304; nearest edge-on conjunction offset 0.24638 d (phase 0.01364), linear full-covariance error 1.01421 d. Source-specific covariance, mass function, assumed masses/radii and approximate central duration are in ../gaia-geometry.json. This is not a hard phase veto or fitted inclination. Observed duration 12.975 h, approximate central duration 14.689 h; grazing geometry/dilution and radius assumptions can shorten the eclipse. No contradiction that invalidates repeated-eclipse interpretation is established.

Catalogue outcome: no identified prior EB; seven-group coverage yes. Shared coverage-matrix and catalogue receipts support scope only, not exhaustive priority. Physical verdict is separate from novelty.

Limitations: Representative sectors and simplified trapezoid/linear timing errors; shared TESS photons; approximate Gaia edge-on geometry and G-band neighbor flux budgets.

Next check: Additional clean eclipse timing and color-aware neighbor/PRF modeling would refine the bounded result.

Stable review key: R602.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s27.json - historical/diff_s27.png - historical/gaia.json - measurements.json - phase.png - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

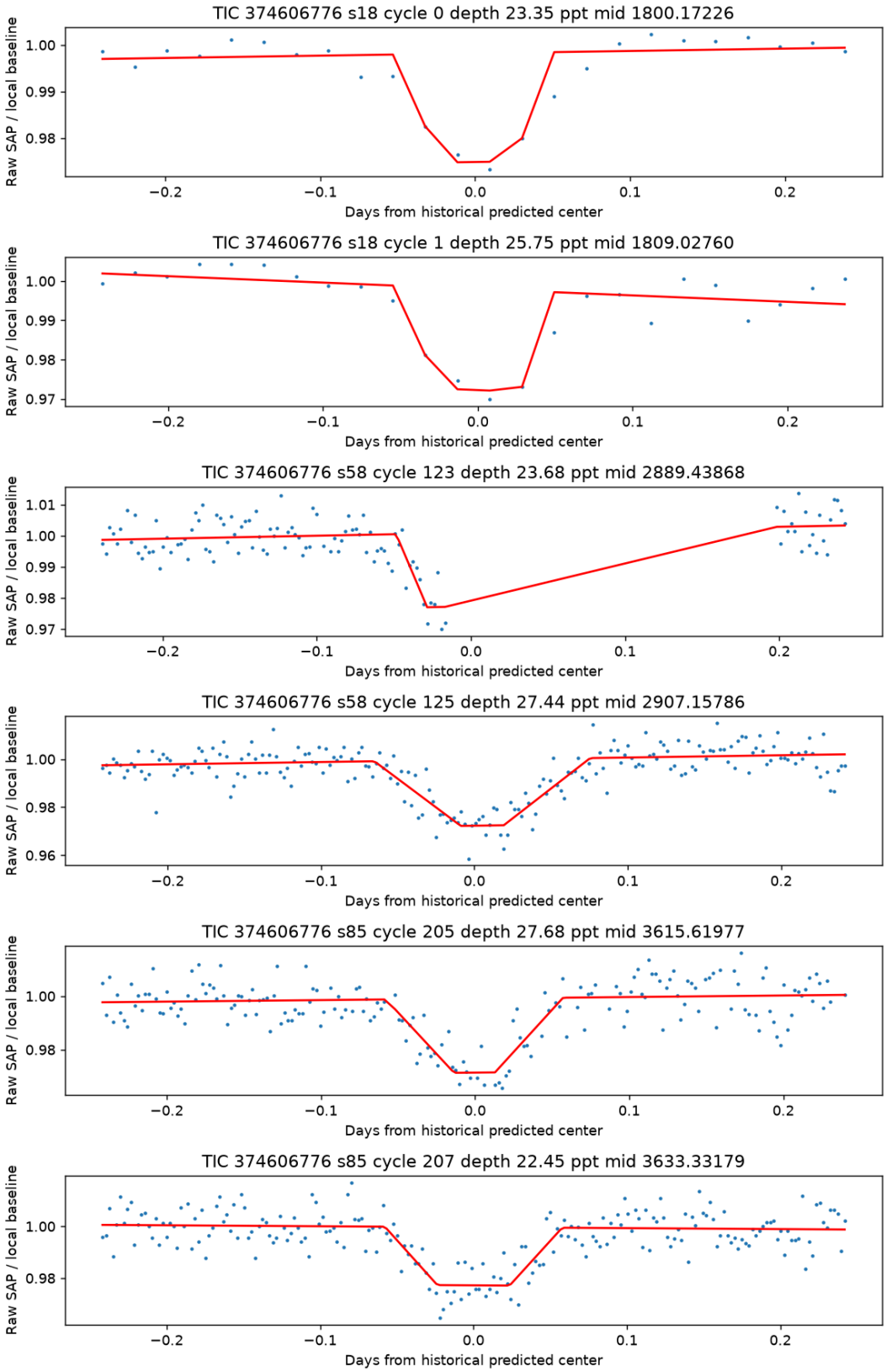
R603. TIC 374606776

physical_verdict: PROBABLE

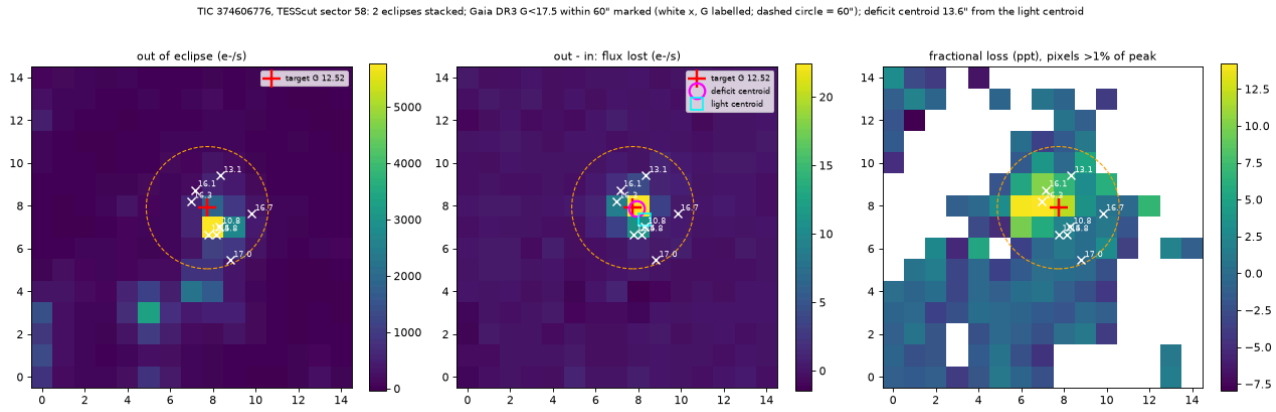
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Higher-resolution time-series localization or a second field orientation resolving capable close sources.



R603; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R603; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 374606776

Physical verdict: **probable**. Full bounded review: **complete**.

Repeated deep raw eclipses are secure, though individual noisy/gapped events have less reliable trapezoid centers. The target deficit overlaps capable neighbors at17.6/21.6 arcsec (less than or about one TESS pixel), plus the bright G10.8 neighbor22.2 arcsec away. Spatial morphology favors the target but does not decisively reject all capable close sources; retain probable attribution.

Measured period 8.85580116464 +/- 2.1865495e-05 d; named selected-eclipse epoch BTJD 1800.18062905; depth 26.59841 ppt (diluted raw QLP). Epoch is the selected recurring eclipse, not an assumed deeper primary. BTJD = BJD_TDB -2457000.

Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.

Timing samples: 8; sectors [18, 58, 85]; fitted same-kind cycle span 0 to 207. Reduced chi square: 1.1737374084112286. Actual centers, errors and local windows are in measurements.json and raw-windows.csv; predicted centers are labeled and never counted as measurements.

Aliases: within-sector adjacent same-kind pairs [[18, 0, 1], [58, 123, 124], [58, 124, 125], [85, 205, 206], [85, 206, 207]]. Sector parity depths and nearby integer counts are in ../alias-audit.json and ../integer-aliases.json. Half-period extra windows are in ../coverage-alias-windows.json; flagged/empty windows are not nulls. No persistent unequal alternating primary depths support a doubled period. Gaia period is a consistency test, not a fixed photometric fit.

Secondary assessment: {'status': 'No secure secondary established by visual review of full raw phase and predicted-conjunction windows. Baseline variability/coverage limit a null claim; guess is Gaia-predicted separation, not a measured secondary phase.', 'guess_phase': 0.5027542082545045}

Attribution and alternative reduction: fresh TESScut S58 stacks 2 well-covered target-period eclipses. Target WCS pixel [7.735897730512072, 7.911700008583637]; deficit centroid [7.893809124844426, 7.870893246386292]. Read fresh-pixels/diff_fresh.png against every capable neighbor below. No fixed centroid threshold applied. Target-centered custom background-subtracted1x1/3x3/5x5 extraction

depths are 12.0307/4.1228/3.5205 ppt. This is a separate reduction of the same TESS photons, not independent observations; different sectors are different occasions on one instrument. Source FITS identities and SHA256 hashes are in ../identity-check.json and ../source-hashes.json.

All capable neighbors within60 arcsec (G-band flux-ratio proxy; intrinsic required depth is not directly compared to diluted pixel loss): [{"source_id": "507046938282627584", "sep": 17.552827467644438, "G": 16.267841, "depth_needed": 0.8658359515900378, "possible": true, "px": [6.958882064915956, 8.209172249272271], "frac_loss_at_pixel": 0.014195749978632079}, {"source_id": "507046938281802112", "sep": 21.601346080805463, "G": 16.10722, "depth_needed": 0.7504282833904787, "possible": true, "px": [7.143570571996502, 8.745269859120807], "frac_loss_at_pixel": 0.010097954335560641}, {"source_id": "507043983344306048", "sep": 22.205015783194398, "G": 10.771246, "depth_needed": 0.031910275036079405, "possible": true, "px": [8.281313352623584, 7.013348811402841], "frac_loss_at_pixel": 0.0018805199116676006}, {"source_id": "507043983345129216", "sep": 26.310424040185577, "G": 15.426095, "depth_needed": 0.413131344515239, "possible": true, "px": [7.764886924965584, 6.668103493891229], "frac_loss_at_pixel": 0.0018805199116676006}, {"source_id": "507043983345129344", "sep": 27.78660520563944, "G": 15.761854, "depth_needed": 0.5532096244545974, "possible": true, "px": [8.148122803827341, 6.663638869266165], "frac_loss_at_pixel": 0.0018805199116676006}, {"source_id": "507046938282624384", "sep": 34.17244928482854, "G": 13.117002, "depth_needed": 0.07268231970877366, "possible": true, "px": [8.342912847039056, 9.409059578334325], "frac_loss_at_pixel": 0.006797815756623012}]

Gaia source 507046938281803392: source-specific NSS period 8.8541081525 d; eccentricity 0.02267; nearest edge-on conjunction offset 0.27115 d (phase 0.03062), linear full-covariance error 0.23434 d. Source-specific covariance, mass function, assumed masses/radii and approximate central duration are in ../gaia-geometry.json. This is not a hard phase veto or fitted inclination. Observed duration 2.766 h, approximate central duration 6.881 h; grazing geometry/dilution and radius assumptions can shorten the eclipse. No contradiction that invalidates repeated-eclipse interpretation is established.

Catalogue outcome: no identified prior EB; seven-group coverage yes. Shared coverage-matrix and catalogue receipts support scope only, not exhaustive priority. Physical verdict is separate from novelty.

Limitations: Representative sectors and simplified trapezoid/linear timing errors; shared TESS photons; approximate Gaia edge-on geometry and G-band neighbor flux budgets. Capable close neighbors prevent definitive source attribution.

Next check: Higher-resolution time-series localization or a second field orientation resolving capable close sources.

Depth correction audit (2026-11-02): realized trapezoid decrement replaces its unconstrained amplitude parameter. Original input windows, fitted centers/widths, timing selections and physical verdict are retained; capable-neighbor membership is unchanged. See ../../depth-audit/README.md.

Stable review key: R603.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s58.json - historical/diff_s58.png - historical/gaia.json - measurements.json - phase.png - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

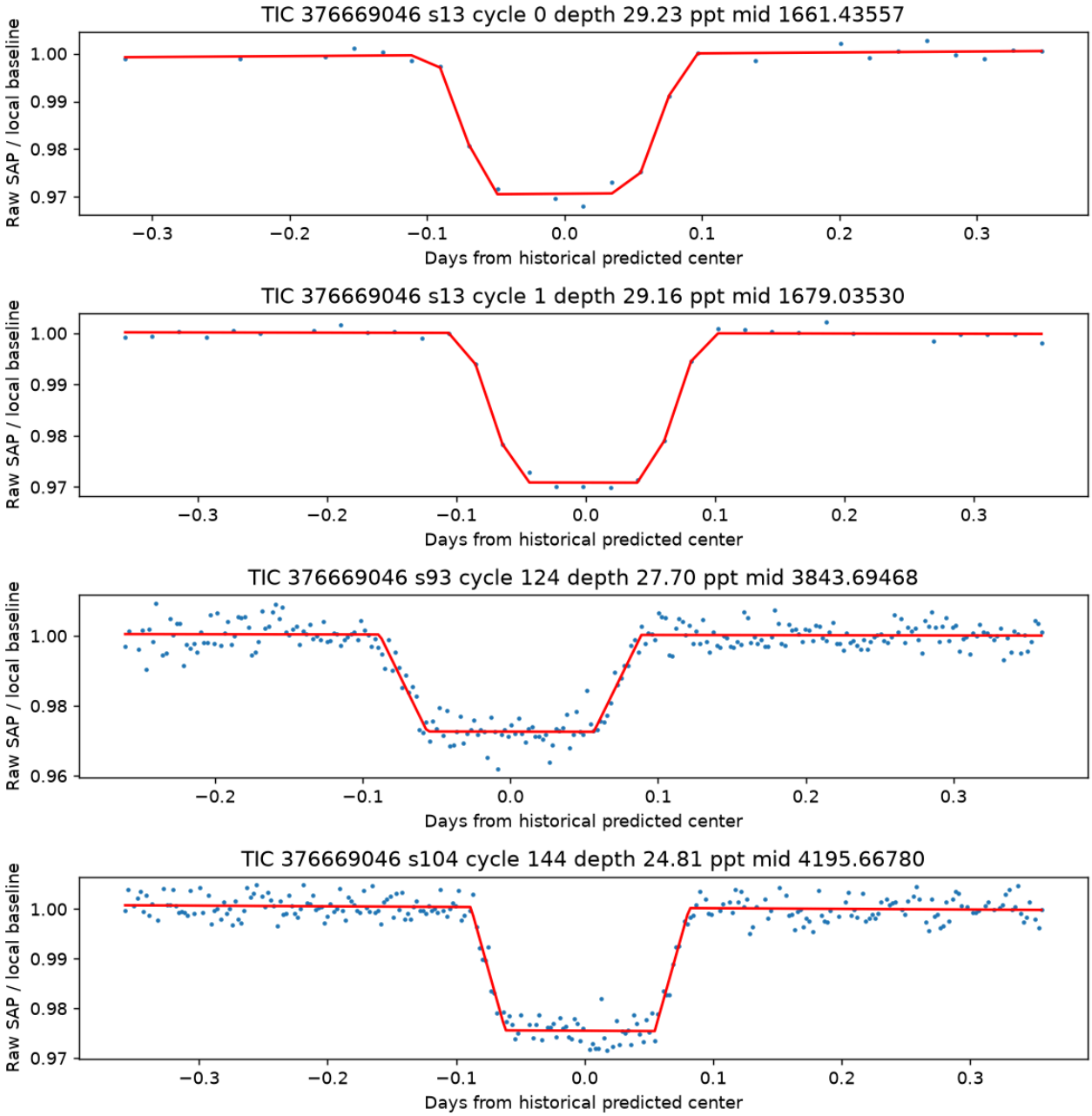
R604. TIC 376669046

physical_verdict: CONFIRMED

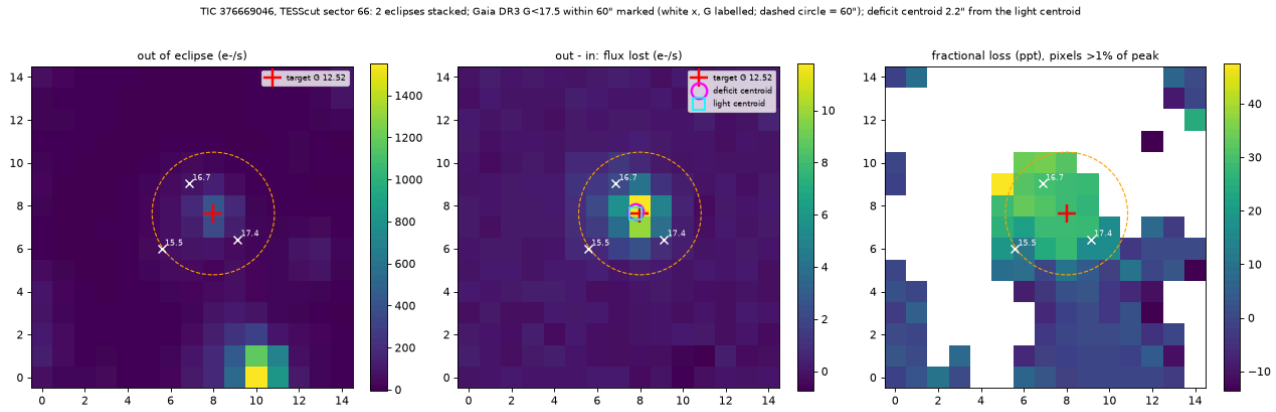
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Additional clean eclipse timing and color-aware neighbor/PRF modeling would refine the bounded result.



R604; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R604; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 376669046

Physical verdict: **confirmed**. Full bounded review: **complete**.

Four measured eclipses include the adjacent S13 pair separated by 17.5997332 d, plus S93/104. This local recurrence constrains the long-baseline integer count; it is not merely two isolated events assigned Gaia cycles. The capable G15.5 neighbor at 59.1 arcsec is far from the target deficit.

Measured period 17.59878547351 +/- 6.2049779e-05 d; named selected-eclipse epoch BTJD 1661.44388644; depth 28.43011 ppt (diluted raw QLP). Epoch is the selected recurring eclipse, not an assumed deeper primary. BTJD = BJD_TDB -2457000.

Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.

Timing samples: 4; sectors [13, 93, 104]; fitted same-kind cycle span 0 to 144. Reduced chi square: 1.8242035023376544. Actual centers, errors and local windows are in measurements.json and raw-windows.csv; predicted centers are labeled and never counted as measurements.

Aliases: within-sector adjacent same-kind pairs [[13, 0, 1]]. Sector parity depths and nearby integer counts are in ../alias-audit.json and ../integer-aliases.json. Half-period extra windows are in ../coverage-alias-windows.json; flagged/empty windows are not nulls. No persistent unequal alternating primary depths support a doubled period. Gaia period is a consistency test, not a fixed photometric fit.

Secondary assessment: {'status': 'No secure secondary established by visual review of full raw phase and predicted-conjunction windows. Baseline variability/coverage limit a null claim; guess is Gaia-predicted separation, not a measured secondary phase.', 'guess_phase': 0.5568678434986611}

Attribution and alternative reduction: fresh TESScut S66 stacks 2 well-covered target-period eclipses. Target WCS pixel [7.993955857085137, 7.638269831945527]; deficit centroid [7.776124354757196, 7.73153610703987]. Read fresh-pixels/diff_fresh.png against every capable neighbor below. No fixed centroid threshold applied. Target-centered custom background-subtracted1x1/3x3/5x5 extraction

depths are 28.2667/28.3465/27.5554 ppt. This is a separate reduction of the same TESS photons, not independent observations; different sectors are different occasions on one instrument. Source FITS identities and SHA256 hashes are in ../identity-check.json and ../source-hashes.json.

All capable neighbors within60 arcsec (G-band flux-ratio proxy; intrinsic required depth is not directly compared to diluted pixel loss): [{"source_id": "6421765251930266112", "sep": 59.06363332122609, "G": 15.464915, "depth_needed": 0.4549625741216256, "possible": true, "px": [5.5764040239052886, 5.993544469091036], "frac_loss_at_pixel": 0.011782671141376114}]

Gaia source 6421765629887386240: source-specific NSS period 17.5996928618 d; eccentricity 0.23839; nearest edge-on conjunction offset -0.26671 d (phase -0.01515), linear full-covariance error 1.29274 d. Source-specific covariance, mass function, assumed masses/radii and approximate central duration are in ../gaia-geometry.json. This is not a hard phase veto or fitted inclination. Observed duration 4.304 h, approximate central duration 5.240 h; grazing geometry/dilution and radius assumptions can shorten the eclipse. No contradiction that invalidates repeated-eclipse interpretation is established.

Catalogue outcome: no identified prior EB; seven-group coverage yes. Shared coverage-matrix and catalogue receipts support scope only, not exhaustive priority. Physical verdict is separate from novelty.

Limitations: Representative sectors and simplified trapezoid/linear timing errors; shared TESS photons; approximate Gaia edge-on geometry and G-band neighbor flux budgets.

Next check: Additional clean eclipse timing and color-aware neighbor/PRF modeling would refine the bounded result.

Stable review key: R604.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s66.json - historical/diff_s66.png - historical/gaia.json - measurements.json - phase.png - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

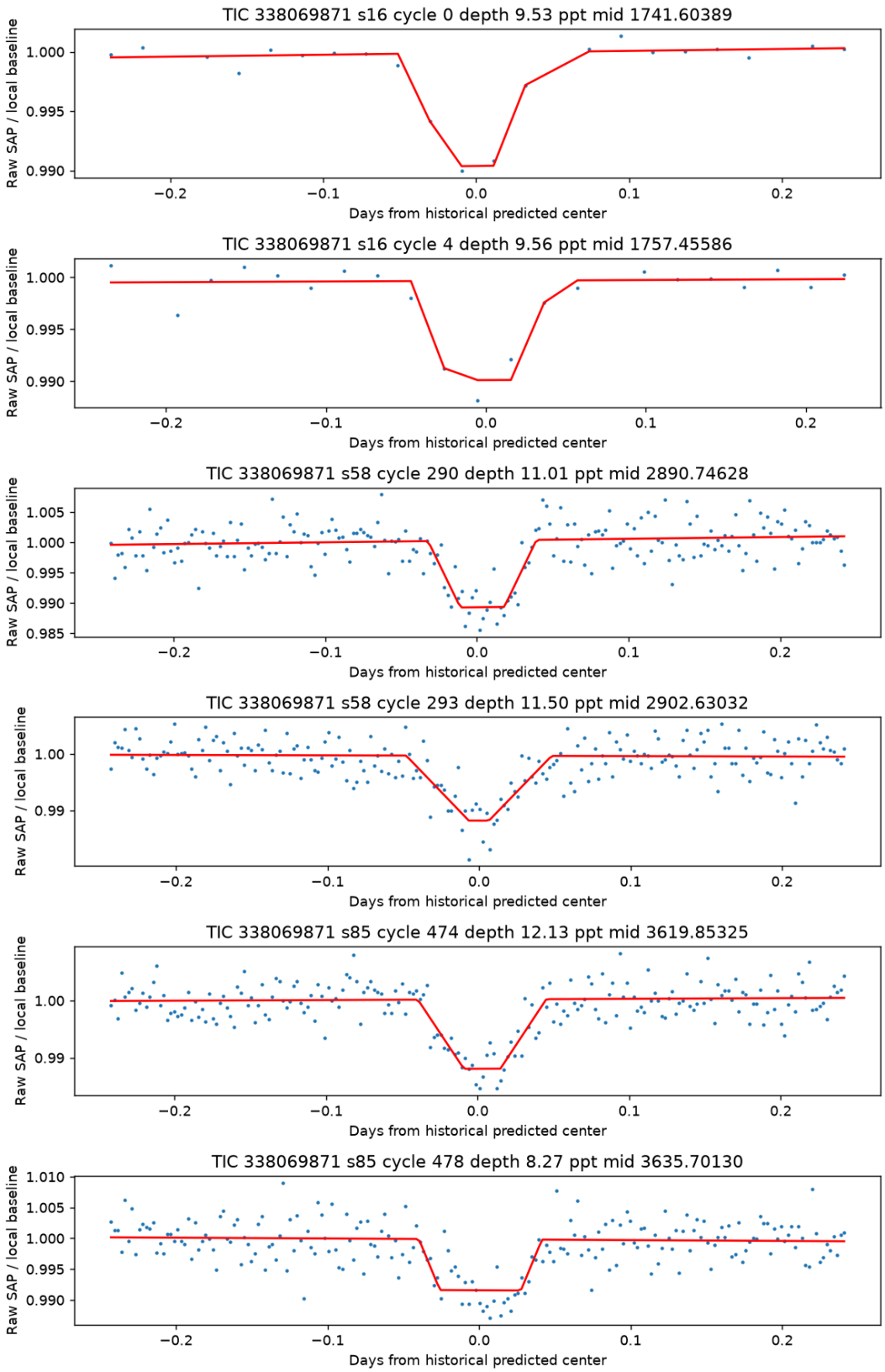
R605. TIC 338069871

physical_verdict: CONFIRMED

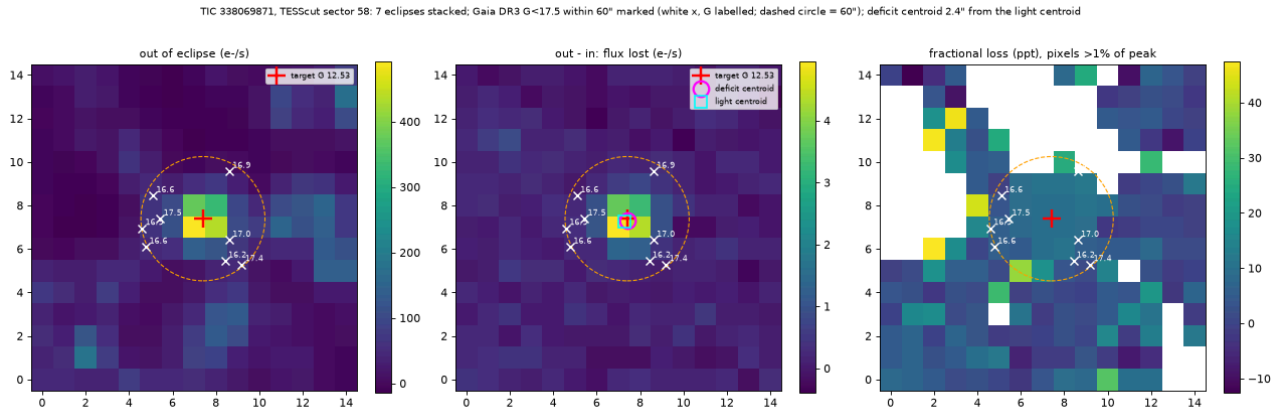
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Additional clean eclipse timing and color-aware neighbor/PRF modeling would refine the bounded result.



R605; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R605; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 338069871

Physical verdict: **confirmed**. Full bounded review: **complete**.

Nearby VSX EW ZTF J223118.37+625844.7 is not a target identity match. Helper coordinate matching identifies Gaia2205143893189353088 (G20.066), not the nearby G16.234 star Gaia2205143897488641664. The EW period0.3798394 d is not a simple half/double alias of3.962546 d; its G-band proxy requires more than total eclipse to explain the target dip. Actual target-period pixel deficit is on the target core; all six capable G-brighter sources are tabulated separately. See helper-ew-identity.json for coordinate/flux details; G-to-TESS colors remain approximate.

Measured period 3.96254638108 +/- 4.3226427e-06 d; named selected-eclipse epoch BTJD 1741.60582969; depth 11.25101 ppt (diluted raw QLP). Epoch is the selected recurring eclipse, not an assumed deeper primary. BTJD = BJD_TDB -2457000.

Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.

Timing samples: 18; sectors [16, 58, 85]; fitted same-kind cycle span 0 to 478. Reduced chi square: 0.7116230204122657. Actual centers, errors and local windows are in measurements.json and raw-windows.csv; predicted centers are labeled and never counted as measurements.

Aliases: within-sector adjacent same-kind pairs [[16, 0, 1], [16, 4, 5], [58, 288, 289], [58, 289, 290], [58, 290, 291], [58, 291, 292], [58, 292, 293], [58, 293, 294], [85, 472, 473], [85, 473, 474], [85, 474, 475], [85, 475, 476], [85, 476, 477], [85, 477, 478]]. Sector parity depths and nearby integer counts are in ../alias-audit.json and ../integer-aliases.json. Half-period extra windows are in ../coverage-alias-windows.json; flagged/empty windows are not nulls. No persistent unequal alternating primary depths support a doubled period. Gaia period is a consistency test, not a fixed photometric fit.

Secondary assessment: {'status': 'No secure secondary established by visual review of full raw phase and predicted-conjunction windows. Baseline variability/coverage limit a null claim; guess is Gaia-predicted separation, not a measured secondary phase.', 'guess_phase': 0.540651351513166}

Attribution and alternative reduction: fresh TESScut S58 stacks 7 well-covered target-period eclipses. Target WCS pixel [7.4025834574938845, 7.389609467582433]; deficit centroid [7.399890389818373, 7.296142161827592]. Read fresh-pixels/diff_fresh.png against every capable neighbor below. No fixed centroid threshold applied. Target-centered custom background-subtracted1x1/3x3/5x5 extraction depths are 10.0732/9.7479/9.1477 ppt. This is a separate reduction of the same TESS photons, not independent observations; different sectors are different occasions on one instrument. Source FITS identities and SHA256 hashes are in ../identity-check.json and ../source-hashes.json.

All capable neighbors within60 arcsec (G-band flux-ratio proxy; intrinsic required depth is not directly compared to diluted pixel loss): [{"source_id": "2205144103647068416", "sep": 31.744151016273438, "G": 17.049454, "depth_needed": 0.7372380928625721, "possible": true, "px": [8.613755801806303, 6.4089632108756796], "frac_loss_at_pixel": 0.006262394924843417}, {"source_id": "2205143897488641664", "sep": 45.07735523812263, "G": 16.233866, "depth_needed": 0.35377713128883725, "possible": true, "px": [8.419577068006419, 5.4508963391071035], "frac_loss_at_pixel": 0.004079026599741608}, {"source_id": "2205144069287321344", "sep": 50.59139836390038, "G": 16.862913, "depth_needed": 0.6226327929244058, "possible": true, "px": [8.61393551776237, 9.573178900263184], "frac_loss_at_pixel": 0.07190680289687322}, {"source_id": "2205144000569024000", "sep": 51.19503519528747, "G": 16.588661, "depth_needed": 0.4861614261939211, "possible": true, "px": [5.099306692000399, 8.45162930281466], "frac_loss_at_pixel": 0.0037810706138147895}, {"source_id": "2205143794409428992", "sep": 56.47541442802353, "G": 16.947376, "depth_needed": 0.672093096663929, "possible": true, "px": [4.595234150122819, 6.953839839800815], "frac_loss_at_pixel": 0.005797162709764999}, {"source_id": "2205143794409431296", "sep": 58.17134723671152, "G": 16.609413, "depth_needed": 0.4953258325990786, "possible": true, "px": [4.780869236939971, 6.095863681663631], "frac_loss_at_pixel": 0.005625215061238802}]

Gaia source 2205143996267667968: source-specific NSS period 3.9625723454 d; eccentricity 0.06519; nearest edge-on conjunction offset 0.07817 d (phase 0.01973), linear full-covariance error 0.16767 d. Source-specific covariance, mass function, assumed masses/radii and approximate central duration are in ../gaia-geometry.json. This is not a hard phase veto or fitted inclination. Observed duration 1.998 h, approximate central duration 6.501 h; grazing geometry/dilution and radius assumptions can shorten the eclipse. No contradiction that invalidates repeated-eclipse interpretation is established.

Catalogue outcome: no identified prior EB; seven-group coverage yes. Shared coverage-matrix and catalogue receipts support scope only, not exhaustive priority. Physical verdict is separate from novelty.

Limitations: Representative sectors and simplified trapezoid/linear timing errors; shared TESS photons; approximate Gaia edge-on geometry and G-band neighbor flux budgets.

Next check: Additional clean eclipse timing and color-aware neighbor/PRF modeling would refine the bounded result.

Depth correction audit (2026-11-02): realized trapezoid decrement replaces its unconstrained amplitude parameter. Original input windows, fitted centers/widths, timing selections and physical verdict are retained; capable-neighbor membership is unchanged. See ../../depth-audit/README.md.

Stable review key: R605.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s58.json - historical/diff_s58.png - historical/gaia.json - measurements.json - phase.png - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

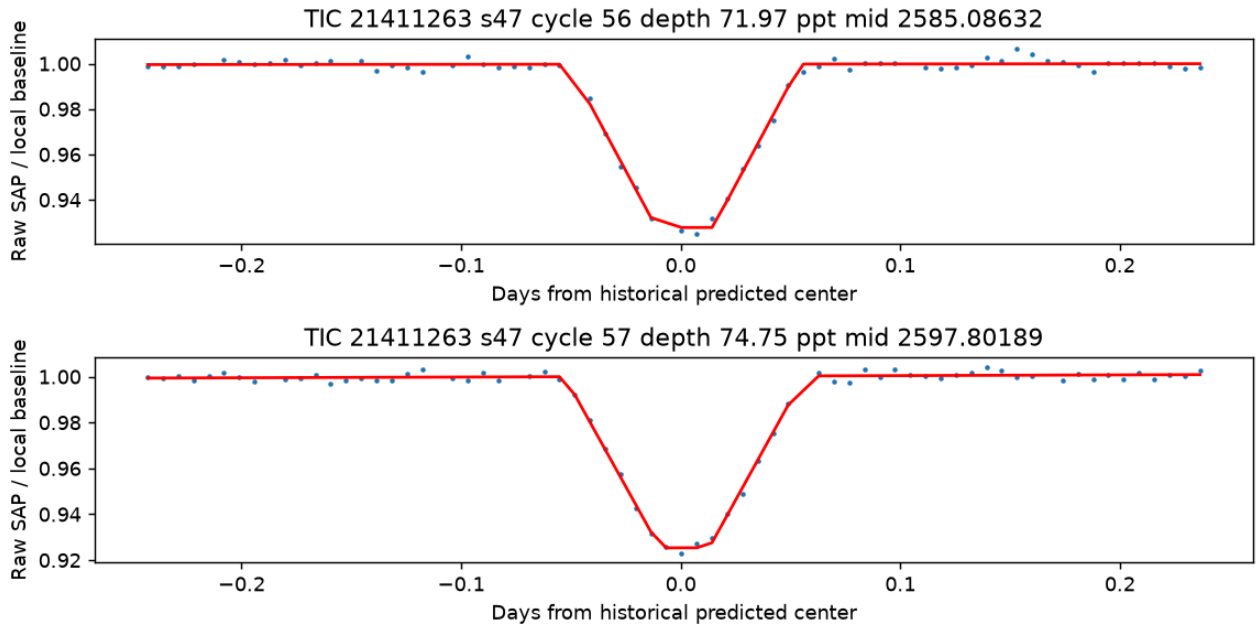
R606. TIC 21411263

physical_verdict: CONFIRMED

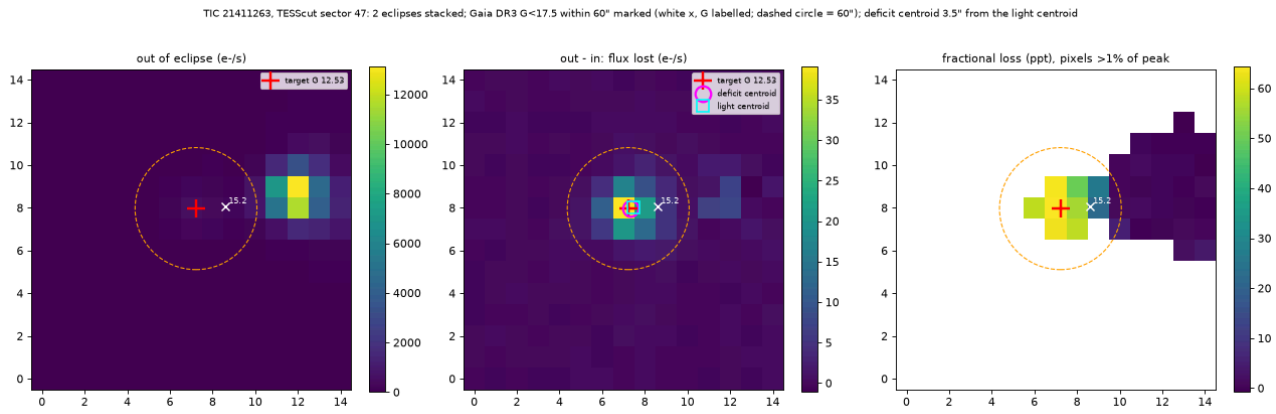
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Additional clean eclipse timing and color-aware neighbor/PRF modeling would refine the bounded result.



R606; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R606; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 21411263

Physical verdict: **confirmed**. Full bounded review: **complete**.

Only two clean same-kind S47 events establish local recurrence. The period uncertainty is the quadrature conservative timing allowance for one cycle, not long-baseline precision; reduced chi square is null (zero degrees of freedom). Fresh MAST availability gives S21/47 only. The two S21 selected windows have one and zero usable core samples with flagged samples; these are not clean nulls and provide no measured timing. S21 all-flag raw plot retained. The capable G15.2 neighbor 27 arcsec away is spatially separate from the target deficit.

Measured period 12.71556726973 +/- 0.0049105801 d; named selected-eclipse epoch BTJD 2585.08632455; depth 73.36144 ppt (diluted raw QLP). Epoch is the selected recurring eclipse, not an assumed deeper primary. BTJD = BJD_TDB -2457000.

Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.

Timing samples: 2; sectors [47]; fitted same-kind cycle span 56 to 57. Reduced chi square: None. Actual centers, errors and local windows are in measurements.json and raw-windows.csv; predicted centers are labeled and never counted as measurements.

Aliases: within-sector adjacent same-kind pairs [[47, 56, 57]]. Sector parity depths and nearby integer counts are in ../alias-audit.json and ../integer-aliases.json. Half-period extra windows are in ../coverage-alias-windows.json; flagged/empty windows are not nulls. No persistent unequal alternating primary depths support a doubled period. Gaia period is a consistency test, not a fixed photometric fit.

Secondary assessment: {'status': 'No secure secondary established by visual review of full raw phase and predicted-conjunction windows. Baseline variability/coverage limit a null claim; guess is Gaia-predicted separation, not a measured secondary phase.', 'guess_phase': 0.7105528086497332}

Attribution and alternative reduction: fresh TESScut S47 stacks 2 well-covered target-period eclipses. Target WCS pixel [7.220473663331138, 7.963336197381196]; deficit centroid [7.314118908275556, 7.9494884024351595]. Read fresh-pixels/diff_fresh.png against every capable neighbor below. No fixed centroid threshold applied. Target-centered custom background-subtracted1x1/3x3/5x5 extraction depths are 64.4137/60.1907/53.5166 ppt. This is a separate reduction of the same TESS photons, not independent observations; different sectors are different occasions on one instrument. Source FITS identities and SHA256 hashes are in ../identity-check.json and ../source-hashes.json.

All capable neighbors within60 arcsec (G-band flux-ratio proxy; intrinsic required depth is not directly compared to diluted pixel loss): [{"source_id": "719639160769665792", "sep": 26.957612633467544, "G": 15.173637, "depth_needed": 0.914314664798946, "possible": true, "px": [8.596629988460721, 8.047491428095453], "frac_loss_at_pixel": 0.0224264865865797 3}]

Gaia source 719639160769665408: source-specific NSS period 12.7169625678 d; eccentricity 0.36656; nearest edge-on conjunction offset -0.12376 d (phase -0.00973), linear full-covariance error 0.31315 d. Source-specific covariance, mass function, assumed masses/radii and approximate central duration are in ../gaia-geometry.json. This is not a hard phase veto or fitted inclination. Observed duration 2.576 h, approximate central duration 4.555 h; grazing geometry/dilution and radius assumptions can shorten the eclipse. No contradiction that invalidates repeated-eclipse interpretation is established.

Catalogue outcome: no identified prior EB; seven-group coverage yes. Shared coverage-matrix and catalogue receipts support scope only, not exhaustive priority. Physical verdict is separate from novelty.

Limitations: Representative sectors and simplified trapezoid/linear timing errors; shared TESS photons; approximate Gaia edge-on geometry and G-band neighbor flux budgets. Only two clean timings; flagged S21 windows; local recurrence precision only.

Next check: Additional clean eclipse timing and color-aware neighbor/PRF modeling would refine the bounded result.

Stable review key: R606.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s47.json - historical/diff_s47.png - historical/gaia.json - mast-availability.json - measurements.json - phase.png - raw-events.png - raw-windows.csv - s21-coverage.png

Batch audits and scripts. Depth correction provenance.

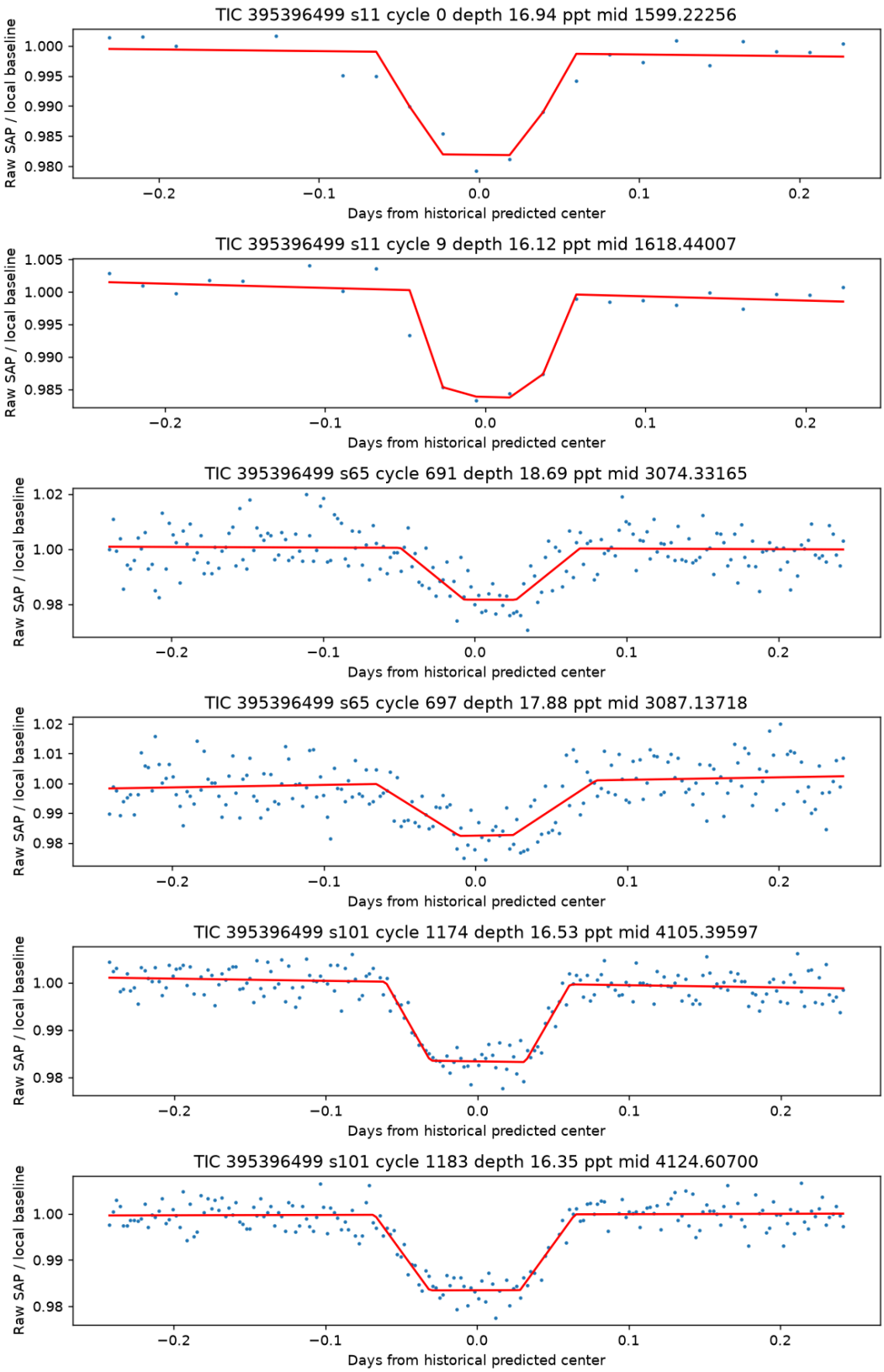
R607. TIC 395396499

physical_verdict: CONFIRMED

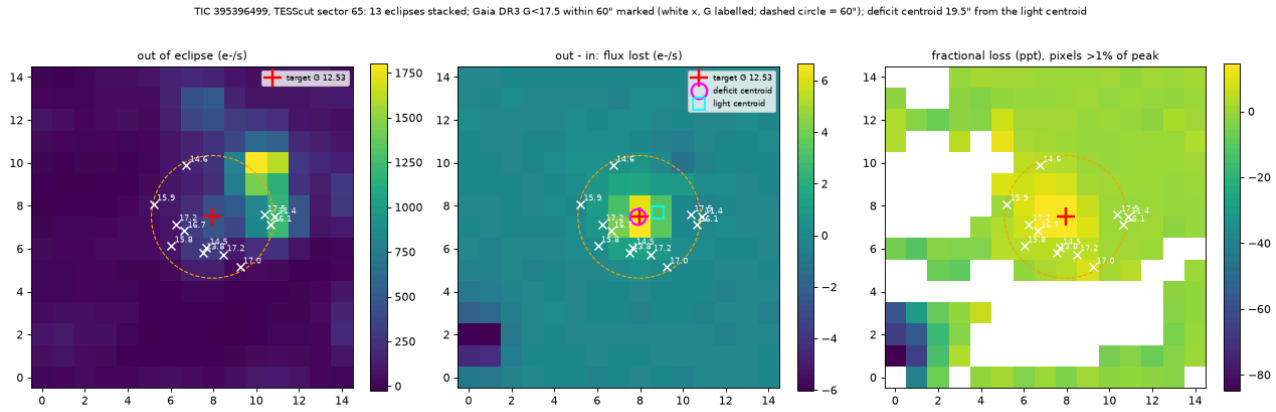
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Additional clean eclipse timing and color-aware neighbor/PRF modeling would refine the bounded result.



R607; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R607; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 395396499

Physical verdict: **confirmed**. Full bounded review: **complete**.

Crowded field: the large light-centroid displacement is caused by bright off-target flux and is not an attribution veto. The positive deficit core lies at the target, distinct from the brighter northeastern star and southwest capable cluster; target-centered alternative apertures retain the dip. Capable sources29-59.5 arcsec away and PSF overlap remain explicit limitations.

Measured period 2.13472589336 +/- 1.7530711e-06 d; named selected-eclipse epoch BTJD 1599.22690181; depth 17.11187 ppt (diluted raw QLP). Epoch is the selected recurring eclipse, not an assumed deeper primary. BTJD = BJD_TDB -2457000.

Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.

Timing samples: 26; sectors [11, 65, 101]; fitted same-kind cycle span 0 to 1183. Reduced chi square: 1.580164096963273. Actual centers, errors and local windows are in measurements.json and raw-windows.csv; predicted centers are labeled and never counted as measurements.

Aliases: within-sector adjacent same-kind pairs [[11, 0, 1], [11, 8, 9], [11, 9, 10], [11, 10, 11], [65, 689, 690], [65, 690, 691], [65, 691, 692], [65, 692, 693], [65, 693, 694], [65, 696, 697], [65, 697, 698], [65, 698, 699], [65, 699, 700], [101, 1173, 1174], [101, 1174, 1175], [101, 1175, 1176], [101, 1179, 1180], [101, 1182, 1183]]. Sector parity depths and nearby integer counts are in ../alias-audit.json and ../integer-aliases.json. Half-period extra windows are in ../coverage-alias-windows.json; flagged/empty windows are not nulls. No persistent unequal alternating primary depths support a doubled period. Gaia period is a consistency test, not a fixed photometric fit.

Secondary assessment: {'status': 'No secure secondary established by visual review of full raw phase and predicted-conjunction windows. Baseline variability/coverage limit a null claim; guess is Gaia-predicted separation, not a measured secondary phase.', 'guess_phase': 0.4772357068682392}

Attribution and alternative reduction: fresh TESScut S65 stacks 13 well-covered target-period eclipses. Target WCS pixel [7.961116056680671, 7.482100973027203]; deficit centroid [7.899055809335304, 7.515974381575997]. Read fresh-pixels/diff_fresh.png against every capable neighbor below. No fixed

centroid threshold applied. Target-centered custom background-subtracted1x1/3x3/5x5 extraction depths are 14.5450/9.9920/4.7739 ppt. This is a separate reduction of the same TESS photons, not independent observations; different sectors are different occasions on one instrument. Source FITS identities and SHA256 hashes are in ../identity-check.json and ../source-hashes.json.

All capable neighbors within60 arcsec (G-band flux-ratio proxy; intrinsic required depth is not directly compared to diluted pixel loss): [{"source_id": "5839906997056207872", "sep": 28.991768548491514, "G": 14.527749, "depth_needed": 0.12523499046901695, "possible": true, "px": [7.669671677458119, 6.0421327988523466], "frac_loss_at_pixel": 0.003890088099268314}, {"source_id": "5839907752970061184", "sep": 30.167358839238897, "G": 16.730427, "depth_needed": 0.8393351766660134, "possible": true, "px": [6.639910687639509, 6.821973862753561], "frac_loss_at_pixel": 0.013029986949728293}, {"source_id": "5839906997055815680", "sep": 33.994450456398006, "G": 13.824567, "depth_needed": 0.07368963122755694, "possible": true, "px": [7.518448407043566, 5.820476598680102], "frac_loss_at_pixel": 0.003890088099268314}, {"source_id": "5839907752970060928", "sep": 47.7690576152999, "G": 15.849694, "depth_needed": 0.3824513285700028, "possible": true, "px": [6.0322484129119305, 6.141352446880889], "frac_loss_at_pixel": 0.0034798911201436404}, {"source_id": "5839908130927585152", "sep": 53.33159216312645, "G": 14.634632, "depth_needed": 0.1364204762583442, "possible": true, "px": [6.73918181120282, 9.89160778453462], "frac_loss_at_pixel": 0.0036825051062978217}, {"source_id": "5839907306293855744", "sep": 55.704976634411146, "G": 16.107515, "depth_needed": 0.48037206077229516, "possible": true, "px": [10.657767817278128, 7.105521034384683], "frac_loss_at_pixel": -6.84631508196582e-05}, {"source_id": "5839907821689540352", "sep": 57.537778578956555, "G": 15.858967, "depth_needed": 0.38558496382563046, "possible": true, "px": [5.2078891406838625, 8.080582932402846], "frac_loss_at_pixel": 0.00023481691899460751}, {"source_id": "5839907301986821248", "sep": 59.49996509933243, "G": 11.449473, "depth_needed": 0.02345944658655942, "possible": true, "px": [10.862967528646882, 7.477289573326296], "frac_loss_at_pixel": -6.84631508196582e-05}]

Gaia source 5839907375013334144: source-specific NSS period 2.1348497552 d; eccentricity 0.32304; nearest edge-on conjunction offset -0.03716 d (phase -0.01741), linear full-covariance error 0.07790 d. Source-specific covariance, mass function, assumed masses/radii and approximate central duration are in ../gaia-geometry.json. This is not a hard phase veto or fitted inclination. Observed duration 2.818 h, approximate central duration 2.745 h; grazing geometry/dilution and radius assumptions can shorten the eclipse. No contradiction that invalidates repeated-eclipse interpretation is established.

Catalogue outcome: no identified prior EB; seven-group coverage yes. Shared coverage-matrix and catalogue receipts support scope only, not exhaustive priority. Physical verdict is separate from novelty.

Limitations: Representative sectors and simplified trapezoid/linear timing errors; shared TESS photons; approximate Gaia edge-on geometry and G-band neighbor flux budgets.

Next check: Additional clean eclipse timing and color-aware neighbor/PRF modeling would refine the bounded result.

Stable review key: R607.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s65.json - historical/diff_s65.png - historical/gaia.json - measurements.json - phase.png - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

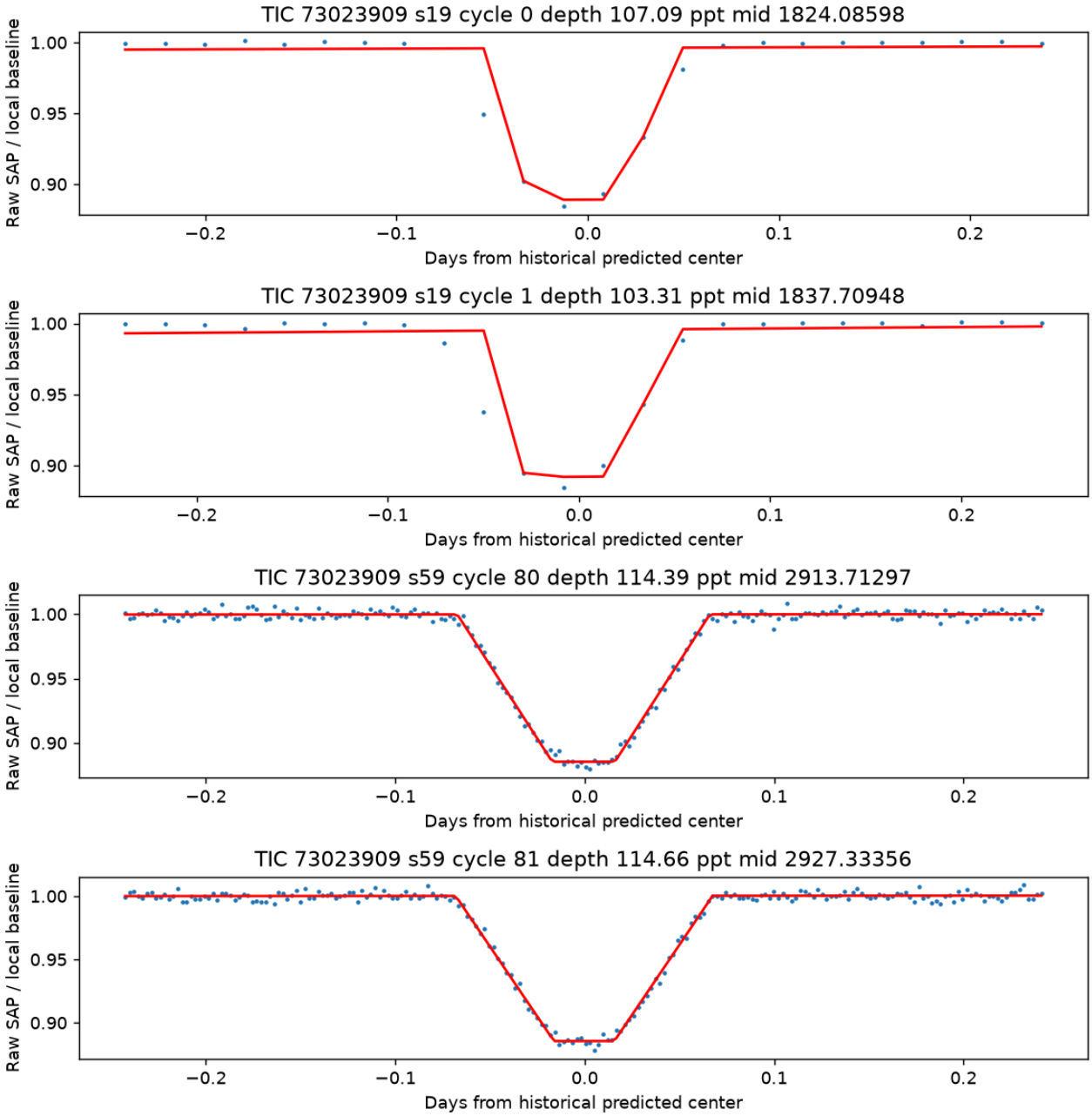
R608. TIC 73023909

physical_verdict: CONFIRMED

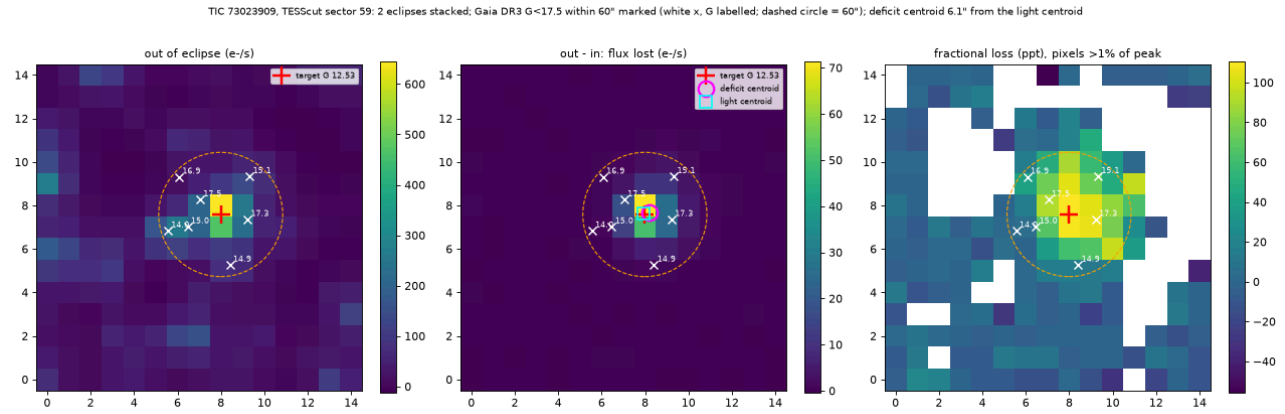
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Additional clean eclipse timing and color-aware neighbor/PRF modeling would refine the bounded result.



R608; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R608; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 73023909

Physical verdict: **confirmed**. Full bounded review: **complete**.

Four measured eclipses include adjacent same-kind pairs in S19 and S59. Fresh MAST availability gives these two sectors only. Reusing older 30-minute data required a documented sample-count change, not a prediction counted as a timing. The capable G14.0 neighbor at 52.6 arcsec is separate from the target deficit.

Measured period 13.62032014257 +/- 9.2487252e-05 d; named selected-eclipse epoch BTJD 1824.08749418; depth 110.74048 ppt (diluted raw QLP). Epoch is the selected recurring eclipse, not an assumed deeper primary. BTJD = BJD_TDB -2457000.

Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.

Timing samples: 4; sectors [19, 59]; fitted same-kind cycle span 0 to 81. Reduced chi square: 0.036983908911665406. Actual centers, errors and local windows are in measurements.json and raw-windows.csv; predicted centers are labeled and never counted as measurements.

Aliases: within-sector adjacent same-kind pairs [[19, 0, 1], [59, 80, 81]]. Sector parity depths and nearby integer counts are in ../alias-audit.json and ../integer-aliases.json. Half-period extra windows are in ../coverage-alias-windows.json; flagged/empty windows are not nulls. No persistent unequal alternating primary depths support a doubled period. Gaia period is a consistency test, not a fixed photometric fit.

Secondary assessment: {'status': 'No secure secondary established by visual review of full raw phase and predicted-conjunction windows. Baseline variability/coverage limit a null claim; guess is Gaia-predicted separation, not a measured secondary phase.', 'guess_phase': 0.701995453065198}

Attribution and alternative reduction: fresh TESScut S59 stacks 2 well-covered target-period eclipses. Target WCS pixel [7.989658211071454, 7.5841265686758845]; deficit centroid [8.214022458423065, 7.692949424683915]. Read fresh-pixels/diff_fresh.png against every capable neighbor below. No fixed centroid threshold applied. Target-centered custom background-subtracted1x1/3x3/5x5 extraction

depths are 110.6783/100.0557/79.5734 ppt. This is a separate reduction of the same TESS photons, not independent observations; different sectors are different occasions on one instrument. Source FITS identities and SHA256 hashes are in ../identity-check.json and ../source-hashes.json.

All capable neighbors within60 arcsec (G-band flux-ratio proxy; intrinsic required depth is not directly compared to diluted pixel loss): [{"source_id": "470143067441747072", "sep": 52.618456286386696, "G": 14.048594, "depth_needed": 0.5600090292954125, "possible": true, "px": [5.554710013364397, 6.833631163060054], "frac_loss_at_pixel": 0.004901240588663546}]

Gaia source 470096127737430912: source-specific NSS period 13.6162926380 d; eccentricity 0.41324; nearest edge-on conjunction offset 0.28890 d (phase 0.02122), linear full-covariance error 0.31395 d. Source-specific covariance, mass function, assumed masses/radii and approximate central duration are in ../gaia-geometry.json. This is not a hard phase veto or fitted inclination. Observed duration 2.639 h, approximate central duration 5.139 h; grazing geometry/dilution and radius assumptions can shorten the eclipse. No contradiction that invalidates repeated-eclipse interpretation is established.

Catalogue outcome: no identified prior EB; seven-group coverage yes. Shared coverage-matrix and catalogue receipts support scope only, not exhaustive priority. Physical verdict is separate from novelty.

Limitations: Representative sectors and simplified trapezoid/linear timing errors; shared TESS photons; approximate Gaia edge-on geometry and G-band neighbor flux budgets.

Next check: Additional clean eclipse timing and color-aware neighbor/PRF modeling would refine the bounded result.

Stable review key: R608.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s59.json - historical/diff_s59.png - historical/gaia.json - mast-availability.json - measurements.json - phase.png - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

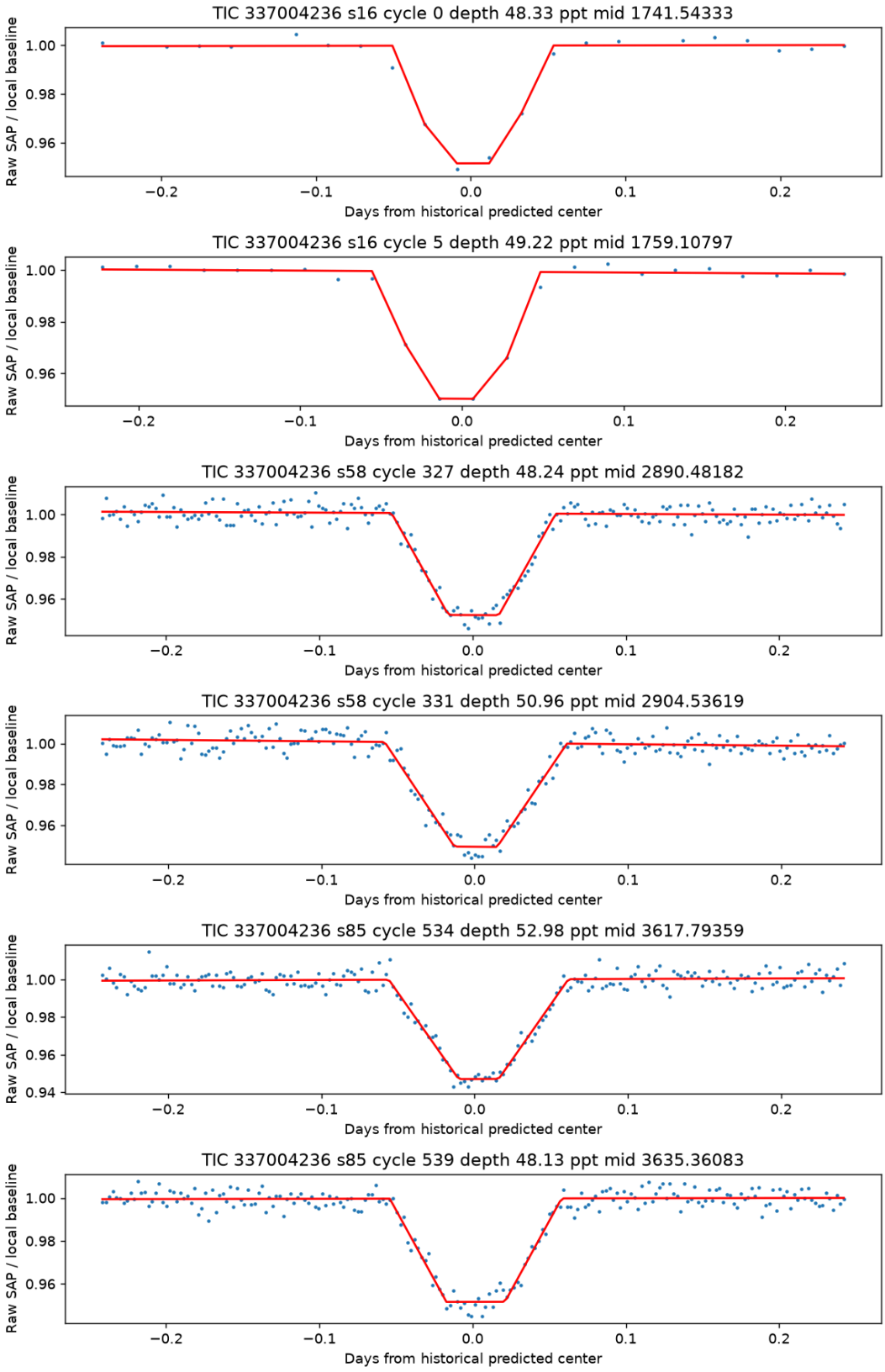
R609. TIC 337004236

physical_verdict: PROBABLE

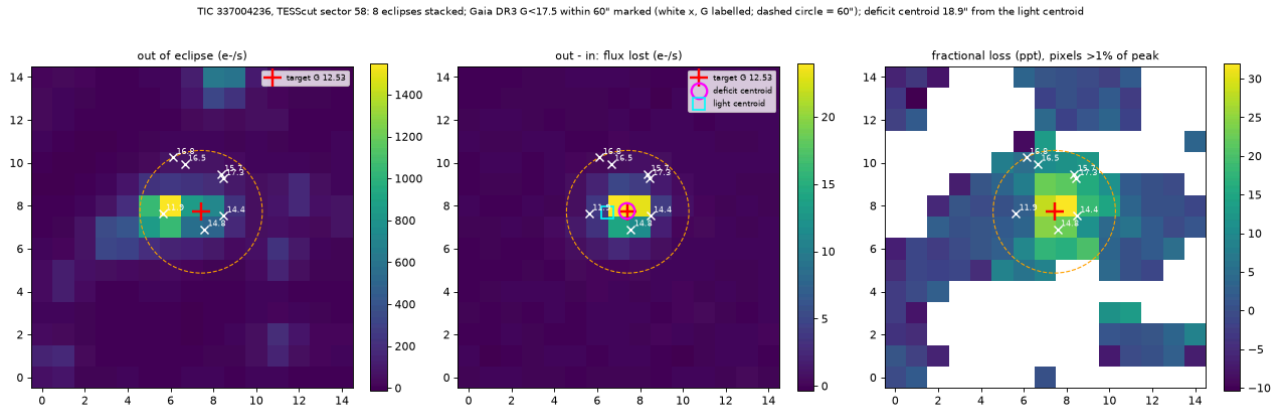
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Higher-resolution time-series localization or a second field orientation resolving capable close sources.



R609; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R609; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 337004236

Physical verdict: **probable**. Full bounded review: **complete**.

Repeated narrow eclipses and measured timing are secure. The target-centered deficit overlaps capable G14.8/14.4 neighbors17.4/21.9 arcsec away. The bright G11.9 source36.9 arcsec away is more clearly displaced, but the closer capable blends remain unresolved; retain probable attribution.

Measured period 3.51357832657 +/- 2.7427902e-06 d; named selected-eclipse epoch BTJD 1741.54138631; depth 51.09119 ppt (diluted raw QLP). Epoch is the selected recurring eclipse, not an assumed deeper primary. BTJD = BJD_TDB -2457000.

Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated.

Timing samples: 21; sectors [16, 58, 85]; fitted same-kind cycle span 0 to 539. Reduced chi square: 0.4989790327441848. Actual centers, errors and local windows are in measurements.json and raw-windows.csv; predicted centers are labeled and never counted as measurements.

Aliases: within-sector adjacent same-kind pairs [[16, 0, 1], [16, 1, 2], [16, 5, 6], [58, 325, 326], [58, 326, 327], [58, 327, 328], [58, 328, 329], [58, 329, 330], [58, 330, 331], [58, 331, 332], [85, 532, 533], [85, 533, 534], [85, 534, 535], [85, 535, 536], [85, 536, 537], [85, 537, 538], [85, 538, 539]]. Sector parity depths and nearby integer counts are in ../alias-audit.json and ../integer-aliases.json. Half-period extra windows are in ../coverage-alias-windows.json; flagged/empty windows are not nulls. No persistent unequal alternating primary depths support a doubled period. Gaia period is a consistency test, not a fixed photometric fit.

Secondary assessment: {'status': 'No secure secondary established by visual review of full raw phase and predicted-conjunction windows. Baseline variability/coverage limit a null claim; guess is Gaia-predicted separation, not a measured secondary phase.', 'guess_phase': 0.5220197879014148}

Attribution and alternative reduction: fresh TESScut S58 stacks 8 well-covered target-period eclipses. Target WCS pixel [7.424558564102117, 7.719348868919365]; deficit centroid [7.364408721432765, 7.756422540784422]. Read fresh-pixels/diff_fresh.png against every capable neighbor below. No fixed centroid threshold applied. Target-centered custom background-subtracted1x1/3x3/5x5 extraction

depths are 28.4878/15.4091/11.7403 ppt. This is a separate reduction of the same TESS photons, not independent observations; different sectors are different occasions on one instrument. Source FITS identities and SHA256 hashes are in ../identity-check.json and ../source-hashes.json.

All capable neighbors within60 arcsec (G-band flux-ratio proxy; intrinsic required depth is not directly compared to diluted pixel loss): [{"source_id": "2218729046544998656", "sep": 17.441909343551004, "G": 14.846255, "depth_needed": 0.4828035144324318, "possible": true, "px": [7.572117209645885, 6.897320943998505], "frac_loss_at_pixel": 0.022524621117210906}, {"source_id": "2218729050841171712", "sep": 21.897788043603484, "G": 14.425405, "depth_needed": 0.34408230359151015, "possible": true, "px": [8.475361222363201, 7.535947511120278], "frac_loss_at_pixel": 0.03196543216302517}, {"source_id": "2218727504652950528", "sep": 36.85720917726252, "G": 11.880404, "depth_needed": 0.07920075278313367, "possible": true, "px": [5.627776870846239, 7.634430601134522], "frac_loss_at_pixel": 0.0024758493026971282}, {"source_id": "2218728329286660096", "sep": 40.93761754595175, "G": 15.695902, "depth_needed": 0.9952678170424939, "possible": true, "px": [8.356826233601392, 9.459576908644294], "frac_loss_at_pixel": 0.021444686500925278}]

Gaia source 2218729050841173120: source-specific NSS period 3.5138385596 d; eccentricity 0.05002; nearest edge-on conjunction offset -0.02113 d (phase -0.00601), linear full-covariance error 0.19878 d. Source-specific covariance, mass function, assumed masses/radii and approximate central duration are in ../gaia-geometry.json. This is not a hard phase veto or fitted inclination. Observed duration 2.763 h, approximate central duration 4.759 h; grazing geometry/dilution and radius assumptions can shorten the eclipse. No contradiction that invalidates repeated-eclipse interpretation is established.

Catalogue outcome: no identified prior EB; seven-group coverage yes. Shared coverage-matrix and catalogue receipts support scope only, not exhaustive priority. Physical verdict is separate from novelty.

Limitations: Representative sectors and simplified trapezoid/linear timing errors; shared TESS photons; approximate Gaia edge-on geometry and G-band neighbor flux budgets. Capable close neighbors prevent definitive source attribution.

Next check: Higher-resolution time-series localization or a second field orientation resolving capable close sources.

Stable review key: R609.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s58.json - historical/diff_s58.png - historical/gaia.json - measurements.json - phase.png - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

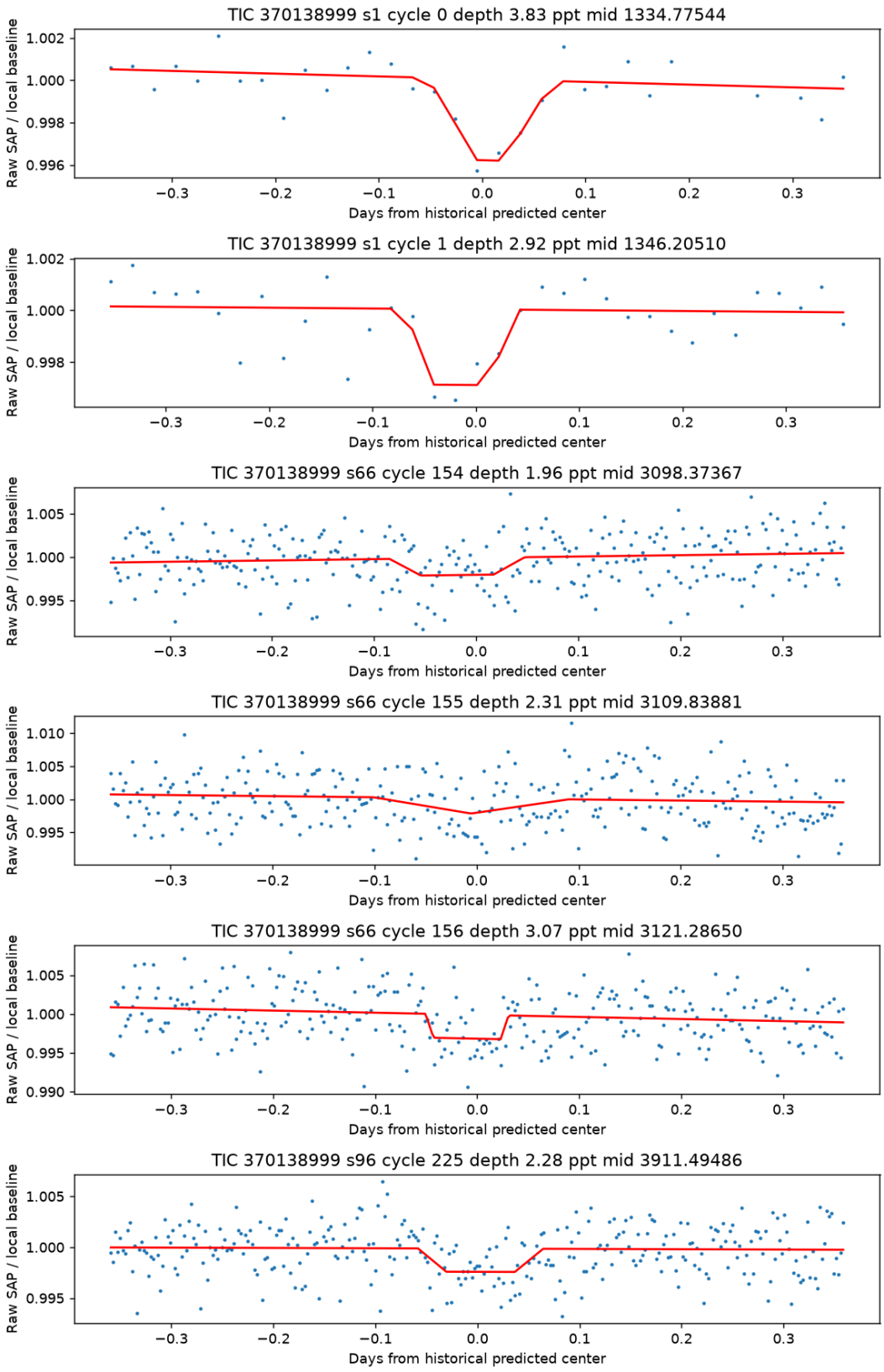
R610. TIC 370138999

physical_verdict: CONFIRMED

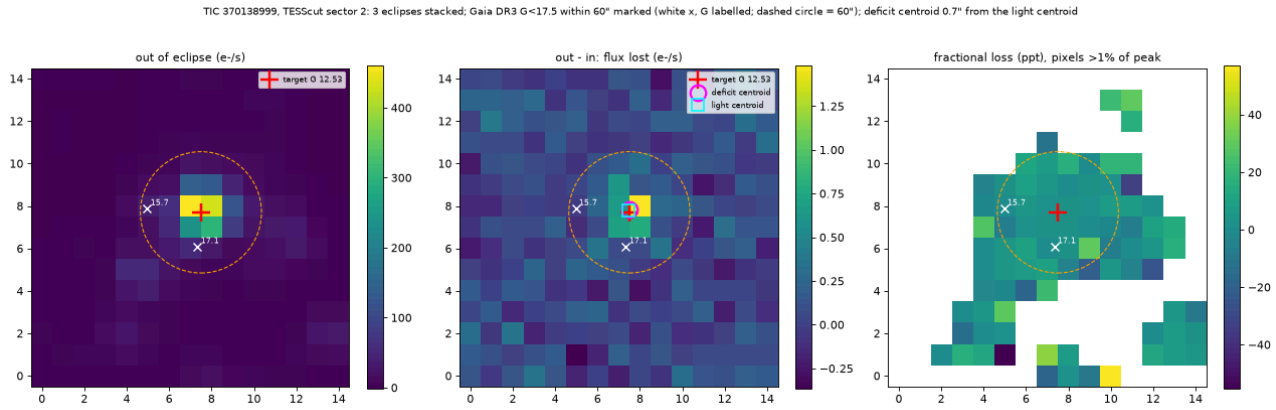
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy to refine physical parameters.



R610; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R610; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 370138999: CONFIRMED

Full bounded physical review: **COMPLETE**. Shallow repeated narrow raw eclipses; compact deficit near target. Three capable neighbors at31-59 arcsec are spatially separated from the deficit core. No secure secondary.

Raw morphology was inspected in raw-events.png and phase-review.png, including P and2P parity views. 6 free-center trapezoid timings survive finite, identifiable center-covariance checks. Quality-zero raw QLP SAP_FLUX was used; predictions only guide fitting windows. This is representative available evidence, not an all-event/all-sector census.

Measured same-kind eclipse ephemeris: P=11.45212692602 +/- 0.00005868963 d; reference BTJD=1334.75715304. Epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced timing chi-square=2.0463176264230576; zero degrees of freedom would be null. All selected timings follow the selected repeating eclipse family. The epoch is conventional and does not assert that this is the deeper primary; pooled alternate-event depths may differ with baseline/extraction. Within-sector cycle recurrence and parity depths are in ../alias-audit.json. Unequal secondary depths exclude mixing eclipse kinds into P/2; the P/2 and2P raw views were inspected. No exhaustive blind alias search is asserted.

No secure secondary established; numeric fit is a noise/variability description and not a detection or clean null. A fitted low-amplitude pooled model is not evidence of a secondary; flagged and missing coverage are not clean nulls.

Fresh difference image: sector2, 3 complete core/sideband events. Target pixel=[7.499832159344923, 7.69979626752945]; deficit centroid=[7.487886595322388, 7.820271053517303]; deficit-target distance=0.1211 pixel. Deficit-light offset=0.717 arcsec is a descriptive field-light displacement, not an arbitrary veto. Fractional PSF-model residual=0.5707 is not a calibrated localization uncertainty. Every flux-capable Gaia source in the60-arcsec list was inspected against the image. Attribution conclusion: Shallow repeated narrow raw eclipses; compact deficit near target. Three capable neighbors at31-59 arcsec are spatially separated from the deficit core. No secure secondary.

Capable Gaia source	Separation arcsec	G	Intrinsic required eclipse proxy	Pixel coordinate	Diluted loss at pixel
4639398934128295552	31.142	17.052	0.1707	[7.335900214530367, 6.083904978865475]	0.001042400878428005
4639398521811431936	49.989	15.749	0.0532	[4.965076105923685, 7.876679048281579]	-0.0008962551083115956
4639399002847056512	59.272	17.574	0.2745	[9.77374381387321, 9.899616852717259]	0.01929205621903543

The pairwise G-band required intrinsic loss is an approximate brightness screen, with passband, aperture and blend assumptions. It cannot be directly compared to the diluted fractional loss in a pixel. Noncapable neighbors and all coordinates remain in fresh-pixels/diff_fresh.json and gaia.json. TESS resolution does not exclude unlisted unresolved companions.

Alternative reduction: fresh background-subtracted TESScut1x1/3x3/5x5 target apertures have depths 1.465/2.654/2.805 ppt versus fitted raw QLP median2.613 ppt. Individual cutout event depths are retained in the pixel JSON. Cutout sector is a distinct observing occasion on the same TESS instrument relative to retained QLP sectors. This alternative reduction shares the FFI photons of that sector; no same-sector QLP comparison claimed. Distinct sectors are observing occasions on one instrument. Aperture/depth differences reflect dilution, background and extraction response; no intrinsic depth inferred.

Gaia DR3 source4639398560466507904: P=11.45073775 +/-0.00260474 d, e=0.259664, omega=310.63921 degrees, periastron BTJD=385.2158212. Closest edge-on conjunction offset=-0.110346 d (-0.009637 phase), source-specific full-covariance linear uncertainty=0.277266 d. Full correlation ordering P,gamma,K,e,omega,Tp, selected indices0,3,4,5. This linear approximation is descriptive, not a sigma veto. Both conjunctions and near-reference projection are retained in ../gaia-geometry.json. Mass function=0.0138353 solar masses, assumed M1=1.19, minimum M2=0.31532; approximate central duration=5.516 h versus observed2.919 h. TIC radius and main-sequence companion scaling are assumptions, not a geometry fit; impact parameter and stellar evolution affect duration. Agreement cannot establish ownership among unresolved sources.

Sector	Same-kind cycle	Measured midpoint BTJD	Formal center error d	Input error d	Depth ppt
1	0	1334.77544164	0.0059817868	0.010416629	3.832
1	1	1346.20509536	0.0063566414	0.010416629	2.921
66	154	3098.37366519	0.0087793895	0.0087793895	1.965
66	155	3109.83880780	0.018654846	0.018654846	2.306
66	156	3121.28650236	0.0029436131	0.0029436131	3.071
96	225	3911.49486190	0.0056136962	0.0056136962	2.283

Next check: Optional resolved photometry and spectroscopy to refine physical parameters.

Catalogue status is separate from physical verdict; shared-catalogues/coverage-matrix.csv retains the seven scoped groups. Known recoveries are not new additions; no exhaustive priority claim. Sources and identity are checked in ../identity-check.json and ../source-hashes.json. Numeric artifacts, original receipts and outputs are frozen in ../SHA256SUMS.json; independent helper review is ../helper-audit.md.

Stable review key: R610.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s02.json - historical/diff_s02.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

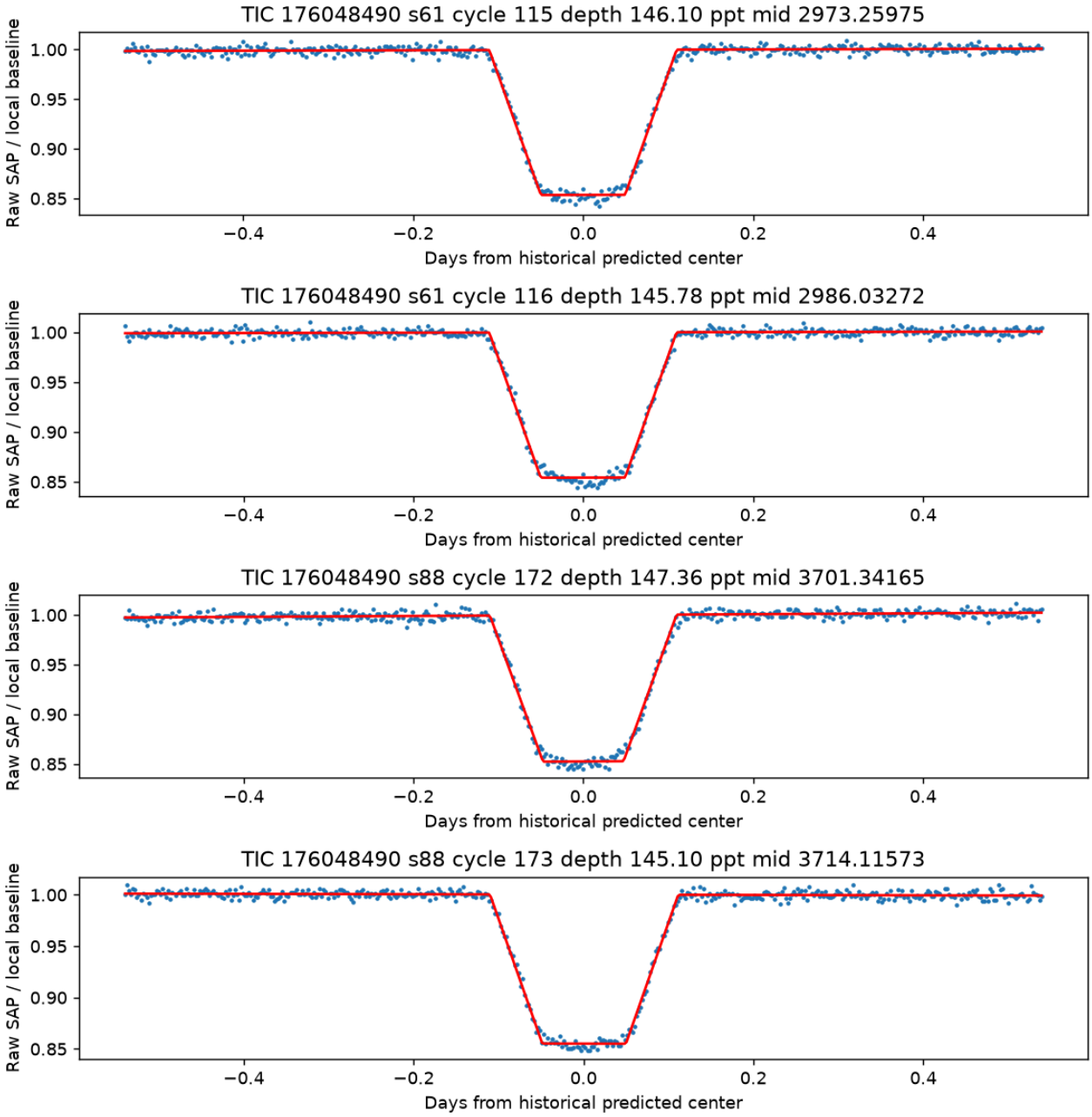
R611. TIC 176048490

physical_verdict: CONFIRMED

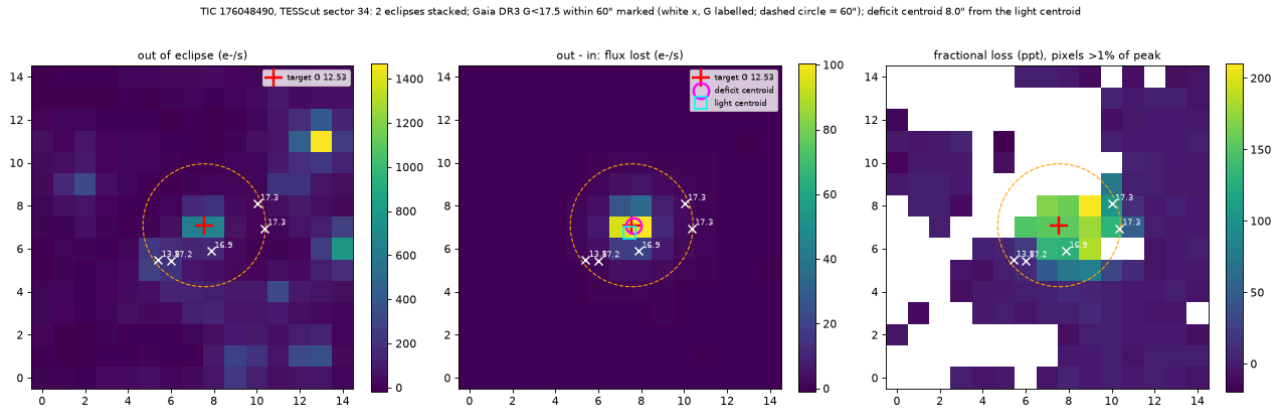
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy to refine physical parameters.



R611; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R611; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 176048490: CONFIRMED

Full bounded physical review: **COMPLETE**. Four deep, flat-bottomed raw eclipses and an unequal secondary near phase0.3683 support eccentric binary morphology. The capable neighbor at56.6 arcsec is separated from the deficit.

Raw morphology was inspected in raw-events.png and phase-review.png, including P and2P parity views. 4 free-center trapezoid timings survive finite, identifiable center-covariance checks. Quality-zero raw QLP SAP_FLUX was used; predictions only guide fitting windows. This is representative available evidence, not an all-event/all-sector census.

Measured same-kind eclipse ephemeris: P=12.77337644993 +/- 0.00002030227 d; reference BTJD=2973.25954728. Epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced timing chi-square=0.12471103186425339; zero degrees of freedom would be null. All selected timings follow the selected repeating eclipse family. The epoch is conventional and does not assert that this is the deeper primary; pooled alternate-event depths may differ with baseline/extraction. Within-sector cycle recurrence and parity depths are in ../alias-audit.json. Unequal secondary depths exclude mixing eclipse kinds into P/2; the P/2 and2P raw views were inspected. No exhaustive blind alias search is asserted.

Measured secondary-like narrow feature; pooled shape and baseline systematics limit precision. Depth is the realized center deficit of the fitted shape, with its amplitude parameter retained separately in the JSON. Measured pooled secondary phase=0.3682844, formal phase covariance error=0.00010924; depth=9.490 ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Search-guide provenance: Raw full-phase visual estimate.

Fresh difference image: sector34, 2 complete core/sideband events. Target pixel= [7.5557581337728354, 7.089043736515528]; deficit centroid=[7.63828595577322, 7.0682956753274535]; deficit-target distance=0.0851 pixel. Deficit-light offset=8.049 arcsec is a descriptive field-light displacement, not an arbitrary veto. Fractional PSF-model residual=0.6804 is not a calibrated localization uncertainty. Every flux-capable Gaia source in the60-arcsec list was inspected

against the image. Attribution conclusion: Four deep, flat-bottomed raw eclipses and an unequal secondary near phase0.3683 support eccentric binary morphology. The capable neighbor at56.6 arcsec is separated from the deficit.

Capable Gaia source	Separation arcsec	G	Intrinsic required eclipse proxy	Pixel coordinate	Diluted loss at pixel
571329029601242 7648	56.597	13.81 6	0.6214	[5.412614306915825, 5.475570709494404]	0.00051227519517 977

The pairwise G-band required intrinsic loss is an approximate brightness screen, with passband, aperture and blend assumptions. It cannot be directly compared to the diluted fractional loss in a pixel. Noncapable neighbors and all coordinates remain in fresh-pixels/diff_fresh.json and gaia.json. TESS resolution does not exclude unlisted unresolved companions.

Alternative reduction: fresh background-subtracted TESScut1x1/3x3/5x5 target apertures have depths 154.207/154.320/132.096 ppt versus fitted raw QLP median145.938 ppt. Individual cutout event depths are retained in the pixel JSON. Cutout sector is a distinct observing occasion on the same TESS instrument relative to retained QLP sectors. This alternative reduction shares the FFI photons of that sector; no same-sector QLP comparison claimed. Distinct sectors are observing occasions on one instrument. Aperture/depth differences reflect dilution, background and extraction response; no intrinsic depth inferred.

Gaia DR3 source5713288814241556864: P=12.76704696 +/-0.00327314 d, e=0.287234, omega=192.33598 degrees, periastron BTJD=383.0046519. Closest edge-on conjunction offset=0.994810 d (0.077920 phase), source-specific full-covariance linear uncertainty=0.630871 d. Full correlation ordering P,gamma,K,e,omega,Tp, selected indices0,3,4,5. This linear approximation is descriptive, not a sigma veto. Both conjunctions and near-reference projection are retained in ../gaia-geometry.json. Mass function=0.0296587 solar masses, assumed M1=0.947, minimum M2=0.37236; approximate central duration=5.915 h versus observed5.292 h. TIC radius and main-sequence companion scaling are assumptions, not a geometry fit; impact parameter and stellar evolution affect duration. Agreement cannot establish ownership among unresolved sources.

Sector	Same-kind cycle	Measured midpoint BTJD	Formal center error d	Input error d	Depth ppt
61	115	2973.25975420	0.0001805511	0.0011574051	146.096
61	116	2986.03271941	0.00017673385	0.0011574051	145.781
88	172	3701.34165039	0.00022051984	0.0011574096	147.355
88	173	3714.11573332	0.00020133648	0.0011574096	145.095

Next check: Optional resolved photometry and spectroscopy to refine physical parameters.

Catalogue status is separate from physical verdict; shared-catalogues/coverage-matrix.csv retains the seven scoped groups. Known recoveries are not new additions; no exhaustive priority claim. Sources and identity are checked in ../identity-check.json and ../source-hashes.json. Numeric artifacts, original receipts and outputs are frozen in ../SHA256SUMS.json; independent helper review is ../helper-audit.md.

Stable review key: R611.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s34.json - historical/diff_s34.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

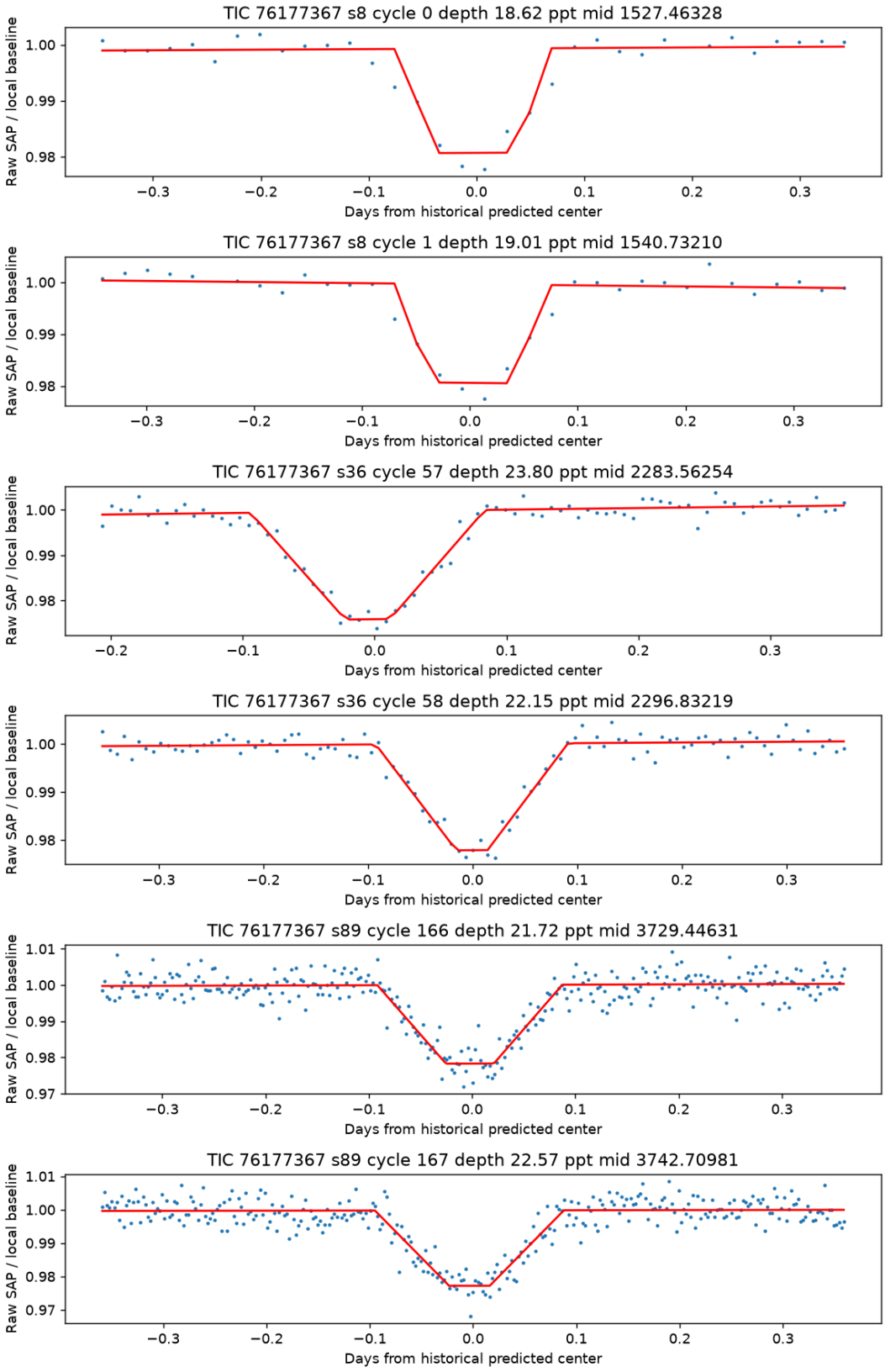
R612. TIC 76177367

physical_verdict: CONFIRMED

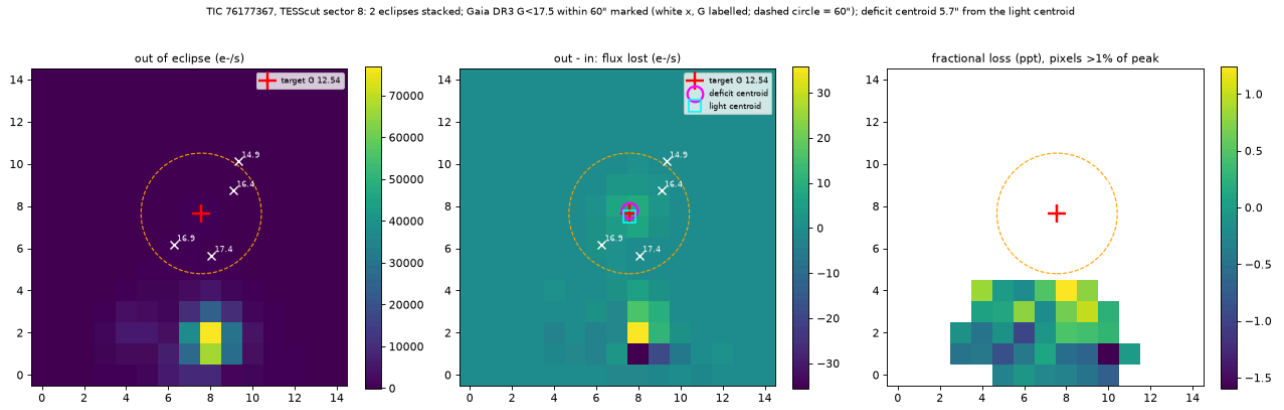
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy to refine physical parameters.



R612; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R612; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 76177367: CONFIRMED

Full bounded physical review: **COMPLETE**. Six narrow raw eclipses and another eclipse near phase0.5930; two capable neighbors at35 and57 arcsec lie away from the deficit. Very bright field light shifts the full-light centroid and dilutes aperture photometry; the compact lost flux remains near target. No deeper-primary assignment is claimed; baseline/extraction differences limit the depth comparison.

Raw morphology was inspected in raw-events.png and phase-review.png, including P and2P parity views. 6 free-center trapezoid timings survive finite, identifiable center-covariance checks. Quality-zero raw QLP SAP_FLUX was used; predictions only guide fitting windows. This is representative available evidence, not an all-event/all-sector census.

Measured same-kind eclipse ephemeris: P=13.26495765588 +/- 0.00002177053 d; reference BTJD=1527.46272784. Epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced timing chi-square=0.41562881010383534; zero degrees of freedom would be null. All selected timings follow the selected repeating eclipse family. The epoch is conventional and does not assert that this is the deeper primary; pooled alternate-event depths may differ with baseline/extraction. Within-sector cycle recurrence and parity depths are in ../alias-audit.json. Unequal secondary depths exclude mixing eclipse kinds into P/2; the P/2 and2P raw views were inspected. No exhaustive blind alias search is asserted.

Measured secondary-like narrow feature; pooled shape and baseline systematics limit precision. Depth is the realized center deficit of the fitted shape, with its amplitude parameter retained separately in the JSON. Measured pooled secondary phase=0.5930130, formal phase covariance error=0.0001232; depth=16.216 ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Search-guide provenance: Raw full-phase visual estimate.

Fresh difference image: sector8, 2 complete core/sideband events. Target pixel=[7.563215722630424, 7.651744180695784]; deficit centroid=[7.598339832431389, 7.769204946763786]; deficit-target distance=0.1226 pixel. Deficit-light offset=5.692 arcsec is a descriptive field-light displacement, not an arbitrary veto. Fractional PSF-model residual=0.8641 is not a calibrated localization uncertainty. Every flux-capable Gaia source in the60-arcsec list was inspected against the image. Attribution conclusion:

Six narrow raw eclipses and another eclipse near phase0.5930; two capable neighbors at35 and57 arcsec lie away from the deficit. Very bright field light shifts the full-light centroid and dilutes aperture photometry; the compact lost flux remains near target. No deeper-primary assignment is claimed; baseline/extraction differences limit the depth comparison.

Capable Gaia source	Separation arcsec	G	Intrinsic required eclipse proxy	Pixel coordinate	Diluted loss at pixel
5424386262828551552	35.233	16.426	0.8077	[9.082740538577367, 8.750204087613891]	0.0059154210284210204
5424385884870330496	57.334	14.917	0.2177	[9.321208969503672, 10.13745805945126]	0.001694577936812823

The pairwise G-band required intrinsic loss is an approximate brightness screen, with passband, aperture and blend assumptions. It cannot be directly compared to the diluted fractional loss in a pixel. Noncapable neighbors and all coordinates remain in fresh-pixels/diff_fresh.json and gaia.json. TESS resolution does not exclude unlisted unresolved companions.

Alternative reduction: fresh background-subtracted TESScut1x1/3x3/5x5 target apertures have depths 16.949/15.181/9.312 ppt versus fitted raw QLP median21.935 ppt. Individual cutout event depths are retained in the pixel JSON. Same-sector QLP and cutout extraction share TESS photons and test processing robustness. Distinct sectors are observing occasions on one instrument. Aperture/depth differences reflect dilution, background and extraction response; no intrinsic depth inferred.

Gaia DR3 source5424386228467707136: P=13.26088266 +/-0.00263410 d, e=0.113680, omega=212.70463 degrees, periastron BTJD=385.1635715. Closest edge-on conjunction offset=0.138052 d (0.010410 phase), source-specific full-covariance linear uncertainty=0.278416 d. Full correlation ordering P,gamma,K,e,omega,Tp, selected indices0,3,4,5. This linear approximation is descriptive, not a sigma veto. Both conjunctions and near-reference projection are retained in ../gaia-geometry.json. Mass function=0.134944 solar masses, assumed M1=1.66, minimum M2=0.97968; approximate central duration=7.727 h versus observed4.282 h. TIC radius and main-sequence companion scaling are assumptions, not a geometry fit; impact parameter and stellar evolution affect duration. Agreement cannot establish ownership among unresolved sources.

Sector	Same-kind cycle	Measured midpoint BTJD	Formal center error d	Input error d	Depth ppt
8	0	1527.46328093	0.0012865388	0.010416905	18.624
8	1	1540.73209786	0.0013638842	0.010416905	19.011
36	57	2283.56253654	0.0011303854	0.0034721717	23.796
36	58	2296.83219227	0.0012485756	0.0034721717	22.155
89	166	3729.44631186	0.0012700206	0.0012700206	21.715
89	167	3742.70980708	0.0015471833	0.0015471833	22.572

Next check: Optional resolved photometry and spectroscopy to refine physical parameters.

Catalogue status is separate from physical verdict; shared-catalogues/coverage-matrix.csv retains the seven scoped groups. Known recoveries are not new additions; no exhaustive priority claim. Sources and identity are checked in ../identity-check.json and ../source-hashes.json. Numeric artifacts, original receipts and outputs are frozen in ../SHA256SUMS.json; independent helper review is ../helper-audit.md.

Stable review key: R612.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s08.json - historical/diff_s08.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

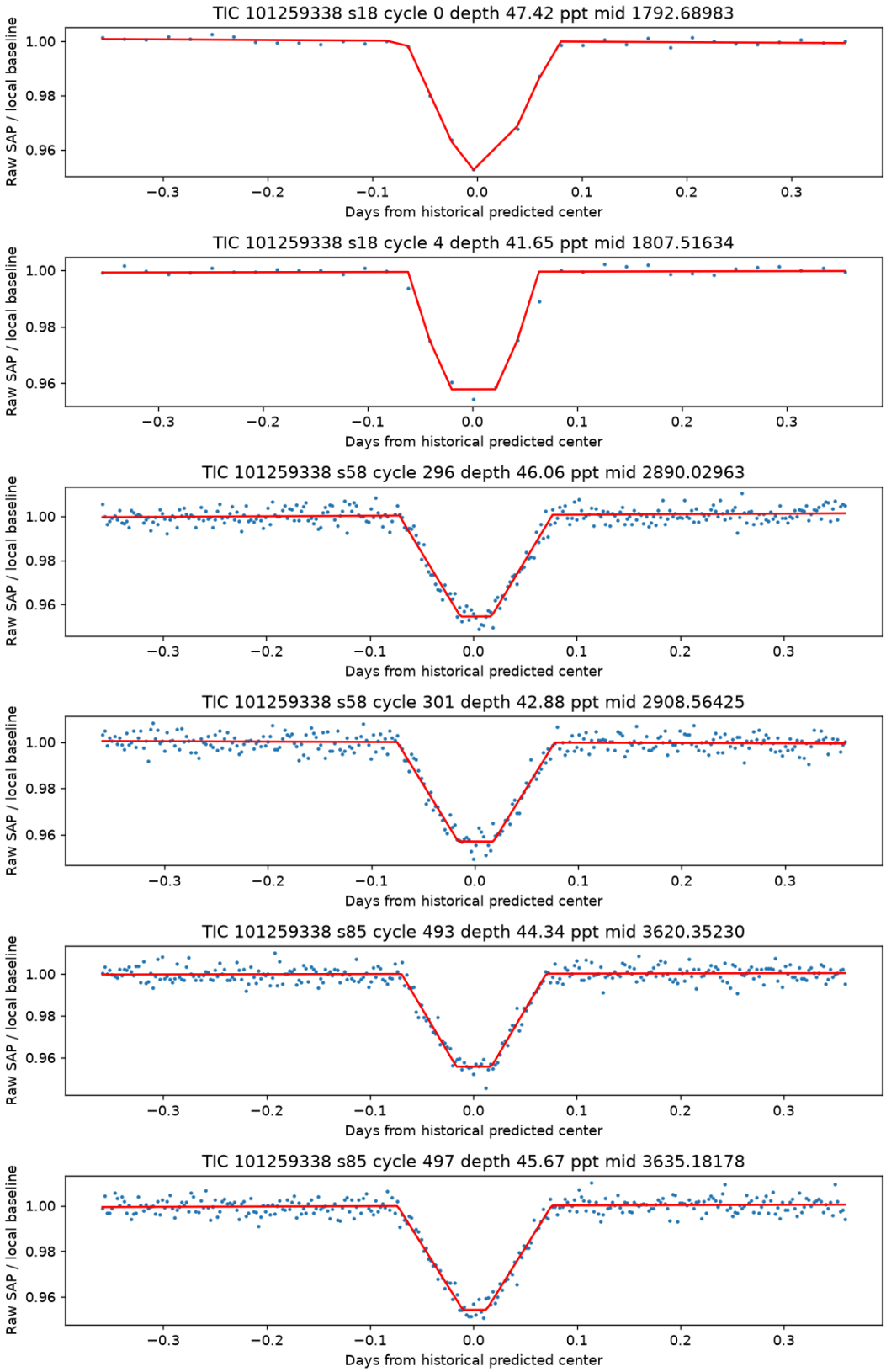
R613. TIC 101259338

physical_verdict: CONFIRMED

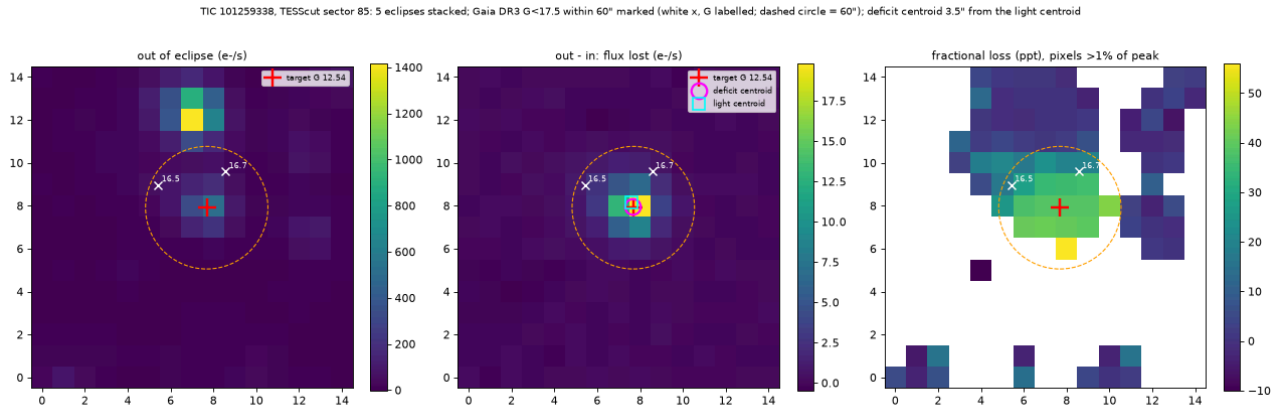
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy to refine physical parameters.



R613; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R613; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 101259338: CONFIRMED

Full bounded physical review: **COMPLETE**. Sixteen repeated narrow eclipses and an unequal secondary near0.5004. No capable Gaia neighbor within60 arcsec; compact lost flux favors target.

Raw morphology was inspected in raw-events.png and phase-review.png, including P and2P parity views. 16 free-center trapezoid timings survive finite, identifiable center-covariance checks. Quality-zero raw QLP SAP_FLUX was used; predictions only guide fitting windows. This is representative available evidence, not an all-event/all-sector census.

Measured same-kind eclipse ephemeris: P=3.70722686178 +/- 0.00000342231 d; reference BTJD=1792.68966361. Epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced timing chi-square=0.152836681071767; zero degrees of freedom would be null. All selected timings follow the selected repeating eclipse family. The epoch is conventional and does not assert that this is the deeper primary; pooled alternate-event depths may differ with baseline/extraction. Within-sector cycle recurrence and parity depths are in ../alias-audit.json. Unequal secondary depths exclude mixing eclipse kinds into P/2; the P/2 and2P raw views were inspected. No exhaustive blind alias search is asserted.

Measured secondary-like narrow feature; pooled shape and baseline systematics limit precision. Depth is the realized center deficit of the fitted shape, with its amplitude parameter retained separately in the JSON. Measured pooled secondary phase=0.5004178, formal phase covariance error=0.00032052; depth=6.539 ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Search-guide provenance: Source-specific Gaia conjunction separation search guide; fitted center is free.

Fresh difference image: sector85, 5 complete core/sideband events. Target pixel=[7.686005302470916, 7.908976176550958]; deficit centroid=[7.634041030657597, 7.98481448505675]; deficit-target distance=0.0919 pixel. Deficit-light offset=3.496 arcsec is a descriptive field-light displacement, not an arbitrary veto. Fractional PSF-model residual=0.9175 is not a calibrated localization uncertainty. Every

flux-capable Gaia source in the60-arcsec list was inspected against the image. Attribution conclusion: Sixteen repeated narrow eclipses and an unequal secondary near0.5004. No capable Gaia neighbor within60 arcsec; compact lost flux favors target.

Capable Gaia source	Separation arcsec	G	Intrinsic required eclipse proxy	Pixel coordinate	Diluted loss at pixel
None					

The pairwise G-band required intrinsic loss is an approximate brightness screen, with passband, aperture and blend assumptions. It cannot be directly compared to the diluted fractional loss in a pixel. Noncapable neighbors and all coordinates remain in fresh-pixels/diff_fresh.json and gaia.json. TESS resolution does not exclude unlisted unresolved companions.

Alternative reduction: fresh background-subtracted TESScut1x1/3x3/5x5 target apertures have depths 38.927/38.499/37.022 ppt versus fitted raw QLP median43.439 ppt. Individual cutout event depths are retained in the pixel JSON. Same-sector QLP and cutout extraction share TESS photons and test processing robustness. Distinct sectors are observing occasions on one instrument. Aperture/depth differences reflect dilution, background and extraction response; no intrinsic depth inferred.

Gaia DR3 source137512002275790976: P=3.70747442 +/-0.00050025 d, e=0.046275, omega=109.09912 degrees, periastron BTJD=387.7961357. Closest edge-on conjunction offset=-0.059882 d (-0.016152 phase), source-specific full-covariance linear uncertainty=0.175858 d. Full correlation ordering P,gamma,K,e,omega,Tp, selected indices0,3,4,5. This linear approximation is descriptive, not a sigma veto. Both conjunctions and near-reference projection are retained in ../gaia-geometry.json. Mass function=0.0543343 solar masses, assumed M1=1.43, minimum M2=0.60903; approximate central duration=5.351 h versus observed3.444 h. TIC radius and main-sequence companion scaling are assumptions, not a geometry fit; impact parameter and stellar evolution affect duration. Agreement cannot establish ownership among unresolved sources.

Sector	Same-kind cycle	Measured midpoint BTJD	Formal center error d	Input error d	Depth ppt
18	0	1792.68983492	0.00064195337	0.010416709	47.419
18	1	1796.39448079	0.00047354241	0.010416709	40.702
18	2	1800.10235743	0.00051290487	0.010416709	41.946
18	4	1807.51633944	0.00068138675	0.010416709	41.651
18	5	1811.22433947	0.00049766047	0.010416709	41.829
58	295	2886.32165074	0.00058204721	0.0011574139	43.464
58	296	2890.02962646	0.00061938071	0.0011574139	46.055
58	299	2901.15071737	0.00055540596	0.0011574139	43.415
58	300	2904.85756276	0.00054727133	0.0011574139	43.786
58	301	2908.56425105	0.00069334904	0.0011574139	42.883
85	491	3612.93703781	0.0007717813	0.0011574262	45.473
85	492	3616.64563322	0.00077901759	0.0011574262	44.441
85	493	3620.35229783	0.00060020921	0.0011574262	44.342
85	494	3624.05976150	0.00079270629	0.0011574262	41.614
85	496	3631.47451332	0.00069228025	0.0011574262	42.321
85	497	3635.18178496	0.00060734262	0.0011574262	45.675

Next check: Optional resolved photometry and spectroscopy to refine physical parameters.

Catalogue status is separate from physical verdict; shared-catalogues/coverage-matrix.csv retains the seven scoped groups. Known recoveries are not new additions; no exhaustive priority claim. Sources and identity are checked in ../identity-check.json and ../source-hashes.json. Numeric artifacts, original receipts and outputs are frozen in ../SHA256SUMS.json; independent helper review is ../helper-audit.md.

Stable review key: R613.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s85.json - historical/diff_s85.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

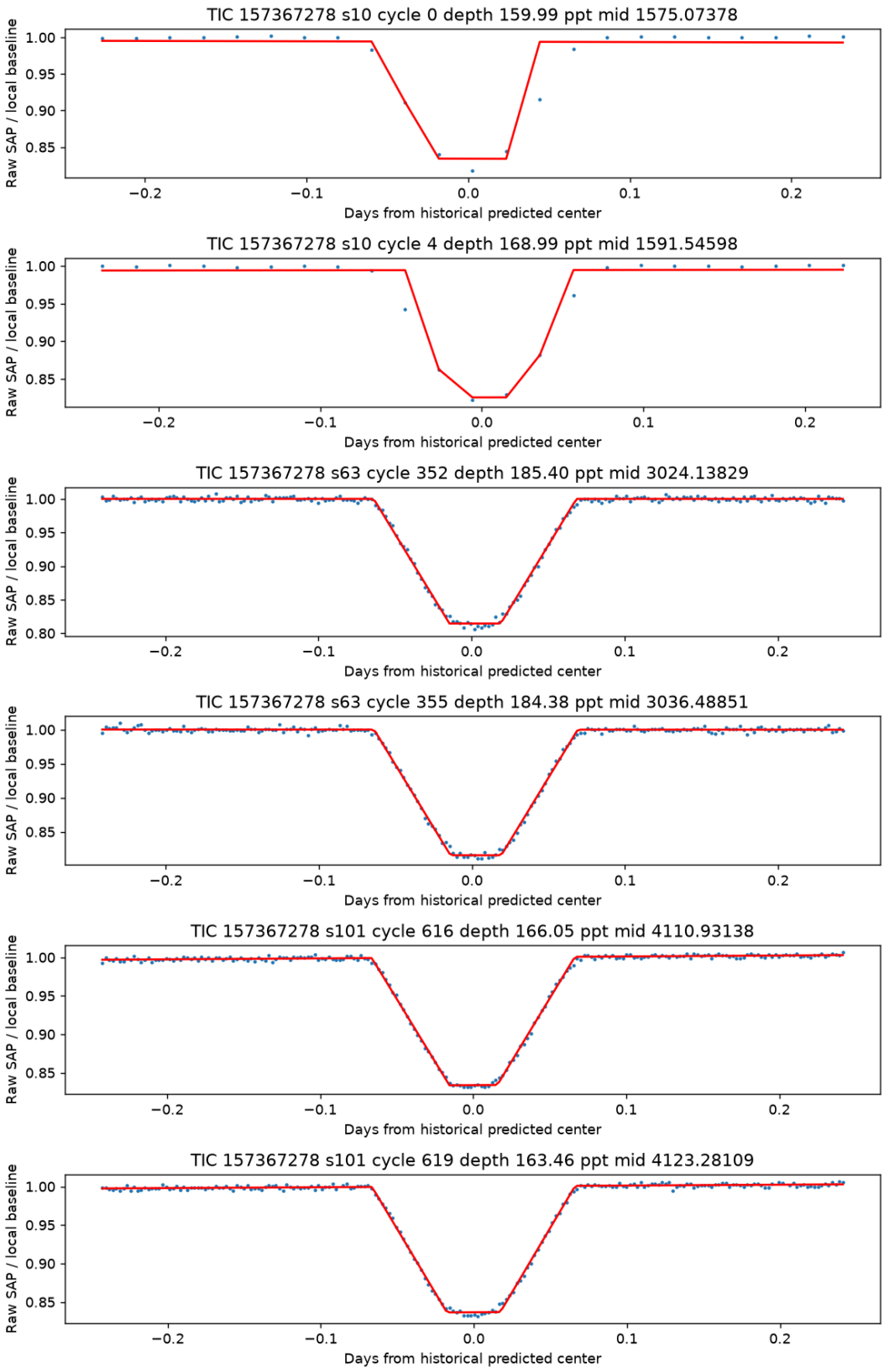
R614. TIC 157367278

physical_verdict: CONFIRMED

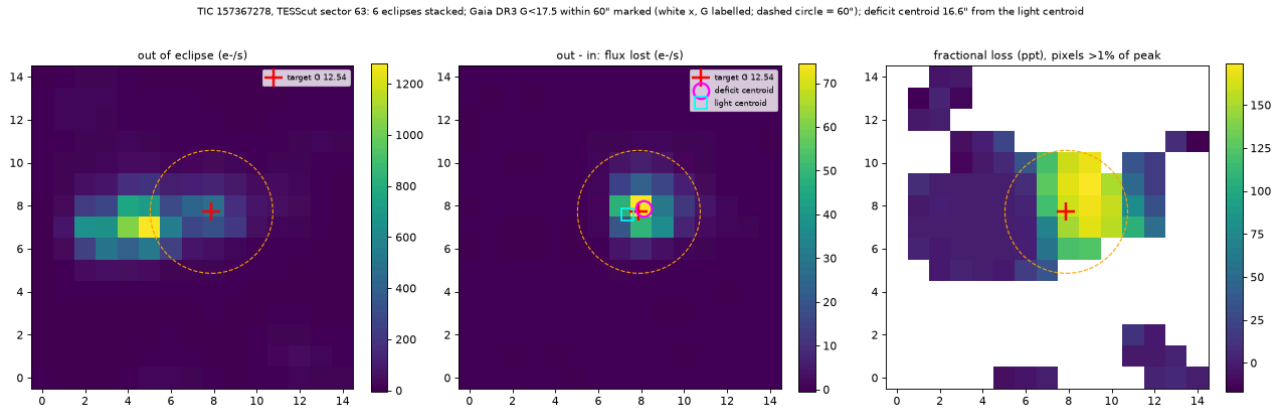
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy to refine physical parameters.



R614; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R614; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 157367278: CONFIRMED

Full bounded physical review: **COMPLETE**. Sixteen deep flat-bottomed eclipses, shallower secondary near0.5000. No capable Gaia neighbor within60 arcsec. Bright displaced field light shifts the light centroid; target-centered deficit remains distinct.

Raw morphology was inspected in raw-events.png and phase-review.png, including P and2P parity views. 16 free-center trapezoid timings survive finite, identifiable center-covariance checks. Quality-zero raw QLP SAP_FLUX was used; predictions only guide fitting windows. This is representative available evidence, not an all-event/all-sector census.

Measured same-kind eclipse ephemeris: P=4.11664108081 +/- 0.00000246441 d; reference BTJD=1575.08062436. Epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced timing chi-square=0.13093186766783096; zero degrees of freedom would be null. All selected timings follow the selected repeating eclipse family. The epoch is conventional and does not assert that this is the deeper primary; pooled alternate-event depths may differ with baseline/extraction. Within-sector cycle recurrence and parity depths are in ../alias-audit.json. Unequal secondary depths exclude mixing eclipse kinds into P/2; the P/2 and2P raw views were inspected. No exhaustive blind alias search is asserted.

Measured secondary-like narrow feature; pooled shape and baseline systematics limit precision. Depth is the realized center deficit of the fitted shape, with its amplitude parameter retained separately in the JSON. Measured pooled secondary phase=0.5000308, formal phase covariance error=0.00013859; depth=22.687 ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Search-guide provenance: Source-specific Gaia conjunction separation search guide; fitted center is free.

Fresh difference image: sector63, 6 complete core/sideband events. Target pixel=[7.891244786908853, 7.714420930394285]; deficit centroid=[8.125236690968716, 7.856333000849923]; deficit-target distance=0.2737 pixel. Deficit-light offset=16.640 arcsec is a descriptive field-light displacement, not an arbitrary veto. Fractional PSF-model residual=0.9345 is not a calibrated localization uncertainty. Every flux-capable Gaia source in the60-arcsec list was inspected against the image. Attribution conclusion:

Sixteen deep flat-bottomed eclipses, shallower secondary near0.5000. No capable Gaia neighbor within60 arcsec. Bright displaced field light shifts the light centroid; target-centered deficit remains distinct.

Capable Gaia source	Separation arcsec	G	Intrinsic required eclipse proxy	Pixel coordinate	Diluted loss at pixel
None					

The pairwise G-band required intrinsic loss is an approximate brightness screen, with passband, aperture and blend assumptions. It cannot be directly compared to the diluted fractional loss in a pixel. Noncapable neighbors and all coordinates remain in fresh-pixels/diff_fresh.json and gaia.json. TESS resolution does not exclude unlisted unresolved companions.

Alternative reduction: fresh background-subtracted TESScut1x1/3x3/5x5 target apertures have depths 160.874/140.012/95.959 ppt versus fitted raw QLP median169.094 ppt. Individual cutout event depths are retained in the pixel JSON. Same-sector QLP and cutout extraction share TESS photons and test processing robustness. Distinct sectors are observing occasions on one instrument. Aperture/depth differences reflect dilution, background and extraction response; no intrinsic depth inferred.

Gaia DR3 source3465408557646425216: P=4.11659407 +/-0.00027865 d, e=0.043913, omega=216.54607 degrees, periastron BTJD=390.9441198. Closest edge-on conjunction offset=-0.042675 d (-0.010367 phase), source-specific full-covariance linear uncertainty=0.126053 d. Full correlation ordering P,gamma,K,e,omega,Tp, selected indices0,3,4,5. This linear approximation is descriptive, not a sigma veto. Both conjunctions and near-reference projection are retained in ../gaia-geometry.json. Mass function=0.0409336 solar masses, assumed M1=1.09, minimum M2=0.46197; approximate central duration=4.025 h versus observed3.161 h. TIC radius and main-sequence companion scaling are assumptions, not a geometry fit; impact parameter and stellar evolution affect duration. Agreement cannot establish ownership among unresolved sources.

Sector	Same-kind cycle	Measured midpoint BTJD	Formal center error d	Input error d	Depth ppt
10	0	1575.07377736	6.0978187e-05	0.010416558	159.994
10	1	1579.19347892	5.255786e-05	0.010416558	171.644
10	3	1587.42394779	6.9696811e-05	0.010416558	169.193
10	4	1591.54598238	7.8304556e-05	0.010416558	168.994
63	350	3015.90514294	0.00013084214	0.0011574261	184.567
63	351	3020.02151782	0.00013071038	0.0011574261	185.266
63	352	3024.13829371	0.00013254933	0.0011574261	185.397
63	353	3028.25491332	0.00012609587	0.0011574261	184.542
63	354	3032.37177871	0.00011738906	0.0011574261	183.605
63	355	3036.48850502	8.8715478e-05	0.0011574261	184.381
101	614	4102.69887311	0.00010522697	0.0011574069	165.983
101	615	4106.81517977	9.7537424e-05	0.0011574069	166.105
101	616	4110.93138092	0.00015661737	0.0011574069	166.054
101	617	4115.04805393	0.0001106234	0.0011574069	166.078
101	618	4119.16423910	0.00013221227	0.0011574069	165.205
101	619	4123.28108551	0.00013911241	0.0011574069	163.462

Next check: Optional resolved photometry and spectroscopy to refine physical parameters.

Catalogue status is separate from physical verdict; shared-catalogues/coverage-matrix.csv retains the seven scoped groups. Known recoveries are not new additions; no exhaustive priority claim. Sources and identity are checked in ../identity-check.json and ../source-hashes.json. Numeric artifacts, original receipts and outputs are frozen in ../SHA256SUMS.json; independent helper review is ../helper-audit.md.

Stable review key: R614.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s63.json - historical/diff_s63.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

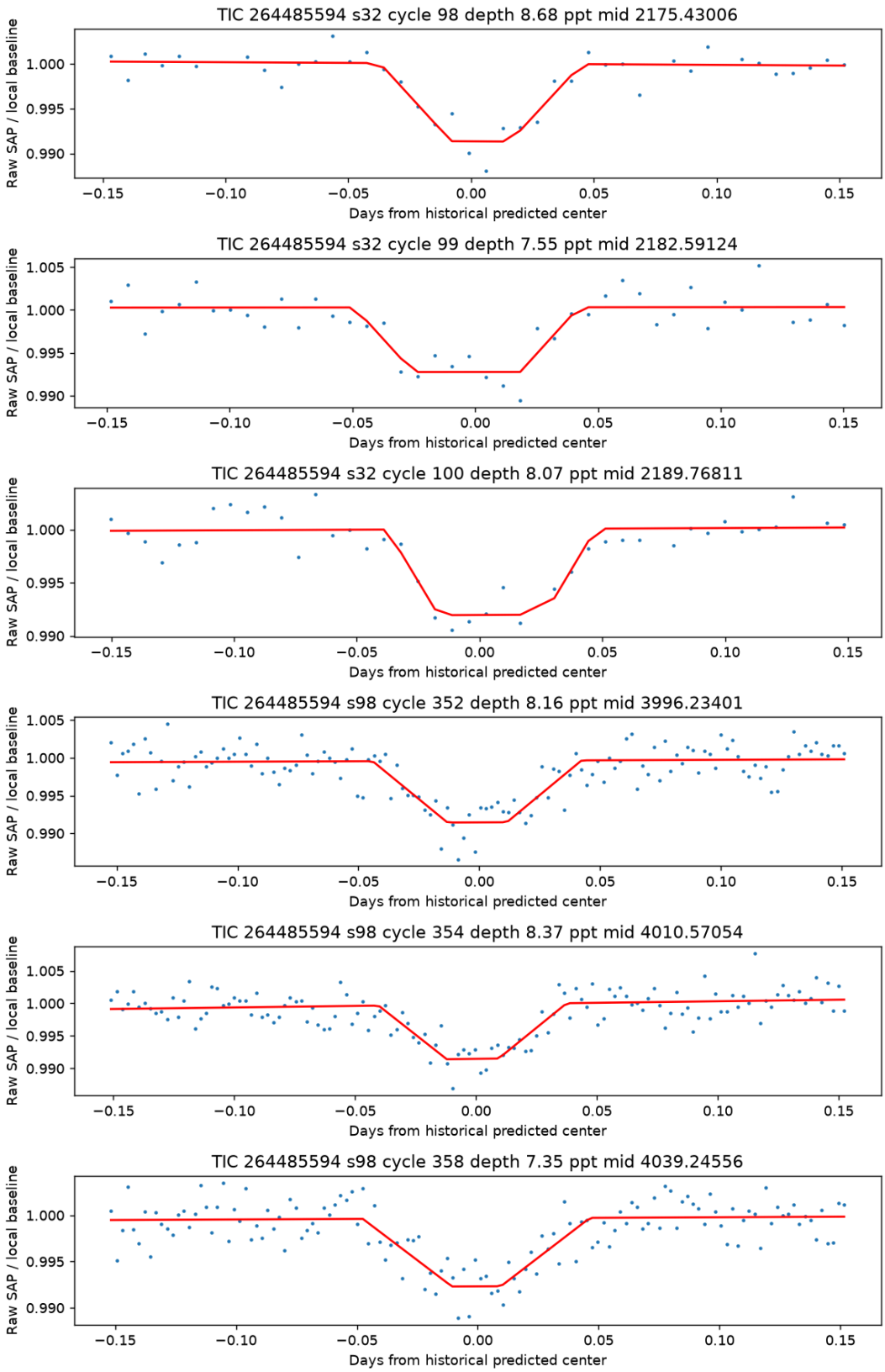
R615. TIC 264485594

physical_verdict: CONFIRMED

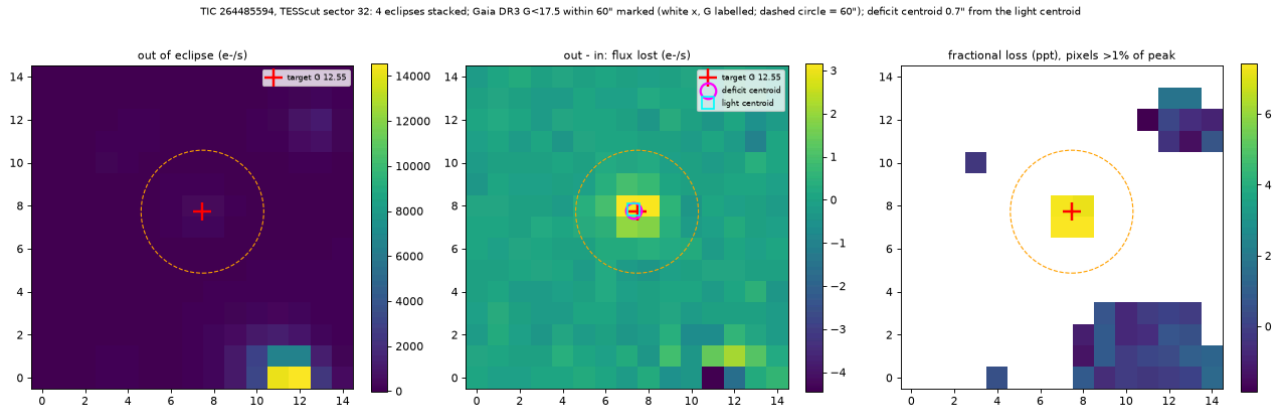
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy to refine physical parameters.



R615; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R615; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 264485594: CONFIRMED

Full bounded physical review: **COMPLETE**. Eight repeated shallow narrow raw eclipses. Target-centered compact deficit; no capable Gaia neighbor within60 arcsec. No secure secondary; baseline variability limits null claims.

Raw morphology was inspected in raw-events.png and phase-review.png, including P and2P parity views. 8 free-center trapezoid timings survive finite, identifiable center-covariance checks. Quality-zero raw QLP SAP_FLUX was used; predictions only guide fitting windows. This is representative available evidence, not an all-event/all-sector census.

Measured same-kind eclipse ephemeris: P=7.16852563280 +/- 0.00000747651 d; reference BTJD=2175.42873529. Epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced timing chi-square=0.8411913338457286; zero degrees of freedom would be null. All selected timings follow the selected repeating eclipse family. The epoch is conventional and does not assert that this is the deeper primary; pooled alternate-event depths may differ with baseline/extraction. Within-sector cycle recurrence and parity depths are in ../alias-audit.json. Unequal secondary depths exclude mixing eclipse kinds into P/2; the P/2 and2P raw views were inspected. No exhaustive blind alias search is asserted.

No secure secondary established; numeric fit is a noise/variability description and not a detection or clean null. A fitted low-amplitude pooled model is not evidence of a secondary; flagged and missing coverage are not clean nulls.

Fresh difference image: sector32, 4 complete core/sideband events. Target pixel=[7.46745656519853, 7.721797050156404]; deficit centroid=[7.291515509702744, 7.773924964148958]; deficit-target distance=0.1835 pixel. Deficit-light offset=0.710 arcsec is a descriptive field-light displacement, not an arbitrary veto. Fractional PSF-model residual=0.6040 is not a calibrated localization uncertainty. Every flux-capable Gaia source in the60-arcsec list was inspected against the image. Attribution conclusion: Eight repeated shallow narrow raw eclipses. Target-centered compact deficit; no capable Gaia neighbor within60 arcsec. No secure secondary; baseline variability limits null claims.

Capable Gaia source	Separation arcsec	G	Intrinsic required eclipse proxy	Pixel coordinate	Diluted loss at pixel
None					

The pairwise G-band required intrinsic loss is an approximate brightness screen, with passband, aperture and blend assumptions. It cannot be directly compared to the diluted fractional loss in a pixel. Noncapable neighbors and all coordinates remain in fresh-pixels/diff_fresh.json and gaia.json. TESS resolution does not exclude unlisted unresolved companions.

Alternative reduction: fresh background-subtracted TESScut1x1/3x3/5x5 target apertures have depths 7.308/7.712/7.995 ppt versus fitted raw QLP median7.816 ppt. Individual cutout event depths are retained in the pixel JSON. Same-sector QLP and cutout extraction share TESS photons and test processing robustness. Distinct sectors are observing occasions on one instrument. Aperture/depth differences reflect dilution, background and extraction response; no intrinsic depth inferred.

Gaia DR3 source3221888168273909504: P=7.16843382 +/-0.00079101 d, e=0.021215, omega=335.07134 degrees, periastron BTJD=388.2593960. Closest edge-on conjunction offset=-0.014988 d (-0.002091 phase), source-specific full-covariance linear uncertainty=0.216307 d. Full correlation ordering P,gamma,K,e,omega,Tp, selected indices0,3,4,5. This linear approximation is descriptive, not a sigma veto. Both conjunctions and near-reference projection are retained in ../gaia-geometry.json. Mass function=0.0845136 solar masses, assumed M1=1.15, minimum M2=0.64917; approximate central duration=6.946 h versus observed2.057 h. TIC radius and main-sequence companion scaling are assumptions, not a geometry fit; impact parameter and stellar evolution affect duration. Agreement cannot establish ownership among unresolved sources.

Sector	Same-kind cycle	Measured midpoint BTJD	Formal center error d	Input error d	Depth ppt
32	98	2175.43005655	0.0012765529	0.0034722772	8.682
32	99	2182.59124170	0.0025098289	0.0034722772	7.549
32	100	2189.76810597	0.0017068417	0.0034722772	8.075
32	101	2196.93677509	0.0015216421	0.0034722772	6.845
98	352	3996.23401385	0.0014396463	0.0014396463	8.160
98	354	4010.57054420	0.0014119509	0.0014119509	8.373
98	356	4024.90980810	0.0018400058	0.0018400058	7.557
98	358	4039.24556011	0.0017613392	0.0017613392	7.346

Next check: Optional resolved photometry and spectroscopy to refine physical parameters.

Catalogue status is separate from physical verdict; shared-catalogues/coverage-matrix.csv retains the seven scoped groups. Known recoveries are not new additions; no exhaustive priority claim. Sources and identity are checked in ../identity-check.json and ../source-hashes.json. Numeric artifacts, original receipts and outputs are frozen in ../SHA256SUMS.json; independent helper review is ../helper-audit.md.

Stable review key: R615.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s32.json - historical/diff_s32.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

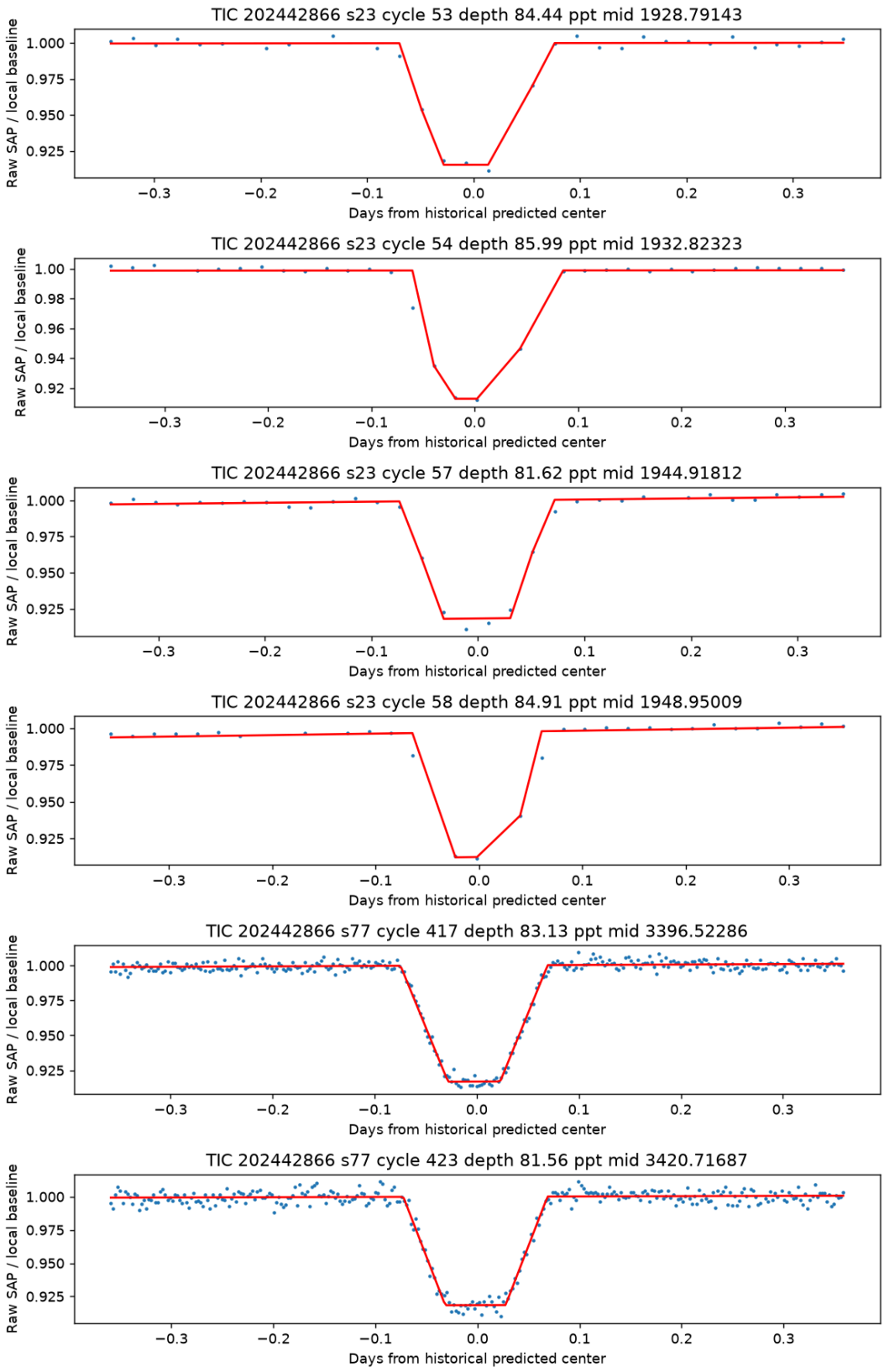
R616. TIC 202442866

physical_verdict: CONFIRMED

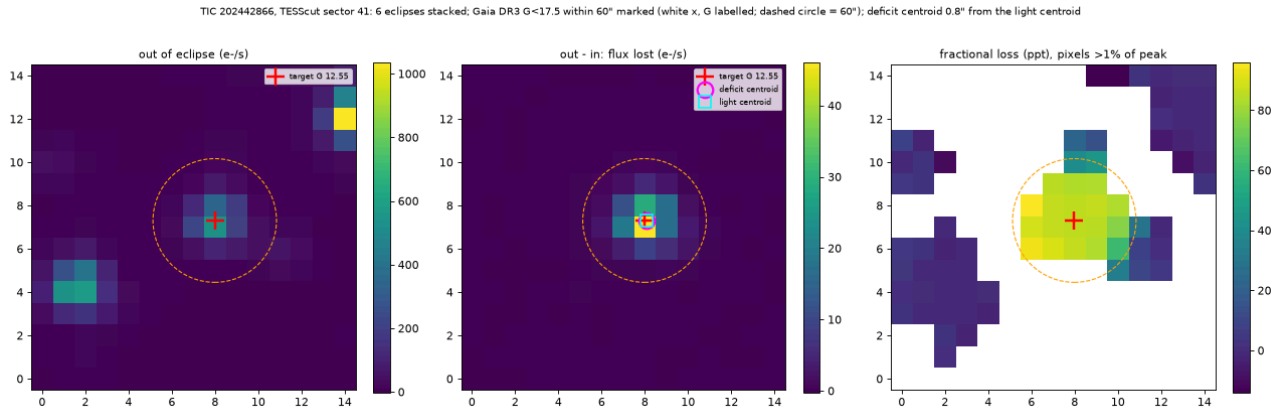
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy to refine physical parameters.



R616; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R616; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 202442866: CONFIRMED

Full bounded physical review: **COMPLETE**. Nine deep repeated raw eclipses plus a broad shallower secondary near0.4988. Target-centered deficit; no capable Gaia neighbor within60 arcsec.

Raw morphology was inspected in raw-events.png and phase-review.png, including P and2P parity views. 9 free-center trapezoid timings survive finite, identifiable center-covariance checks. Quality-zero raw QLP SAP_FLUX was used; predictions only guide fitting windows. This is representative available evidence, not an all-event/all-sector census.

Measured same-kind eclipse ephemeris: P=4.03223527947 +/- 0.00001182526 d; reference BTJD=1928.78976127. Epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced timing chi-square=0.06663373816751848; zero degrees of freedom would be null. All selected timings follow the selected repeating eclipse family. The epoch is conventional and does not assert that this is the deeper primary; pooled alternate-event depths may differ with baseline/extraction. Within-sector cycle recurrence and parity depths are in ../alias-audit.json. Unequal secondary depths exclude mixing eclipse kinds into P/2; the P/2 and2P raw views were inspected. No exhaustive blind alias search is asserted.

Measured secondary-like narrow feature; pooled shape and baseline systematics limit precision. Depth is the realized center deficit of the fitted shape, with its amplitude parameter retained separately in the JSON. Measured pooled secondary phase=0.4987686, formal phase covariance error=0.0010538; depth=6.856 ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Search-guide provenance: Source-specific Gaia conjunction separation search guide; fitted center is free.

Fresh difference image: sector41, 6 complete core/sideband events. Target pixel=[7.983382444673236, 7.309586675452248]; deficit centroid=[8.056274517227141, 7.306475444829257]; deficit-target distance=0.0730 pixel. Deficit-light offset=0.778 arcsec is a descriptive field-light displacement, not an arbitrary veto. Fractional PSF-model residual=0.1180 is not a calibrated localization uncertainty. Every

flux-capable Gaia source in the60-arcsec list was inspected against the image. Attribution conclusion: Nine deep repeated raw eclipses plus a broad shallower secondary near0.4988. Target-centered deficit; no capable Gaia neighbor within60 arcsec.

Capable Gaia source	Separation arcsec	G	Intrinsic required eclipse proxy	Pixel coordinate	Diluted loss at pixel
None					

The pairwise G-band required intrinsic loss is an approximate brightness screen, with passband, aperture and blend assumptions. It cannot be directly compared to the diluted fractional loss in a pixel. Noncapable neighbors and all coordinates remain in fresh-pixels/diff_fresh.json and gaia.json. TESS resolution does not exclude unlisted unresolved companions.

Alternative reduction: fresh background-subtracted TESScut1x1/3x3/5x5 target apertures have depths 85.836/86.043/85.018 ppt versus fitted raw QLP median81.623 ppt. Individual cutout event depths are retained in the pixel JSON. Cutout sector is a distinct observing occasion on the same TESS instrument relative to retained QLP sectors. This alternative reduction shares the FFI photons of that sector; no same-sector QLP comparison claimed. Distinct sectors are observing occasions on one instrument. Aperture/depth differences reflect dilution, background and extraction response; no intrinsic depth inferred.

Gaia DR3 source1619728688122974976: P=4.03203511 +/-0.00027623 d, e=0.072469, omega=267.67105 degrees, periastron BTJD=390.5189958. Closest edge-on conjunction offset=0.019291 d (0.004784 phase), source-specific full-covariance linear uncertainty=0.115781 d. Full correlation ordering P,gamma,K,e,omega,Tp, selected indices0,3,4,5. This linear approximation is descriptive, not a sigma veto. Both conjunctions and near-reference projection are retained in ../gaia-geometry.json. Mass function=0.0227813 solar masses, assumed M1=1.05, minimum M2=0.35574; approximate central duration=4.650 h versus observed3.029 h. TIC radius and main-sequence companion scaling are assumptions, not a geometry fit; impact parameter and stellar evolution affect duration. Agreement cannot establish ownership among unresolved sources.

Sector	Same-kind cycle	Measured midpoint BTJD	Formal center error d	Input error d	Depth ppt
23	53	1928.79142984	0.00057061555	0.01041657	84.437
23	54	1932.82322607	0.00028051283	0.01041657	85.989
23	55	1936.85439171	0.00011644339	0.01041657	80.986
23	57	1944.91812446	0.00055373564	0.01041657	81.623
23	58	1948.95009449	0.0008325492	0.01041657	84.913
23	59	1952.98178436	0.00085721226	0.01041657	80.294
77	417	3396.52286105	0.00027399421	0.001157393	83.127
77	419	3404.58836192	0.00050463372	0.001157393	80.031
77	423	3420.71686521	0.0003910175	0.001157393	81.556

Next check: Optional resolved photometry and spectroscopy to refine physical parameters.

Catalogue status is separate from physical verdict; shared-catalogues/coverage-matrix.csv retains the seven scoped groups. Known recoveries are not new additions; no exhaustive priority claim. Sources and identity are checked in ../identity-check.json and ../source-hashes.json. Numeric artifacts, original receipts and outputs are frozen in ../SHA256SUMS.json; independent helper review is ../helper-audit.md.

Stable review key: R616.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s41.json - historical/diff_s41.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

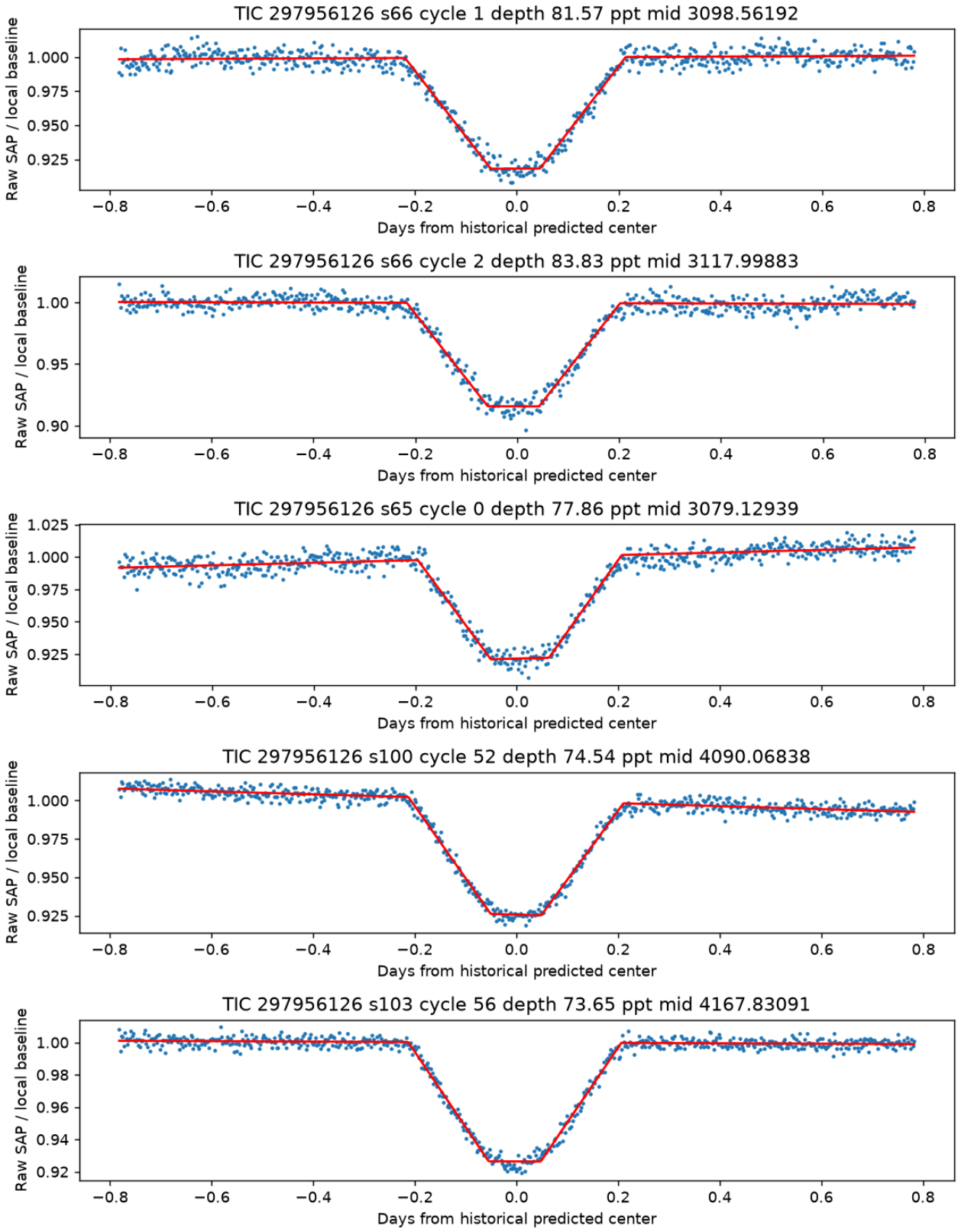
R617. TIC 297956126

physical_verdict: CONFIRMED

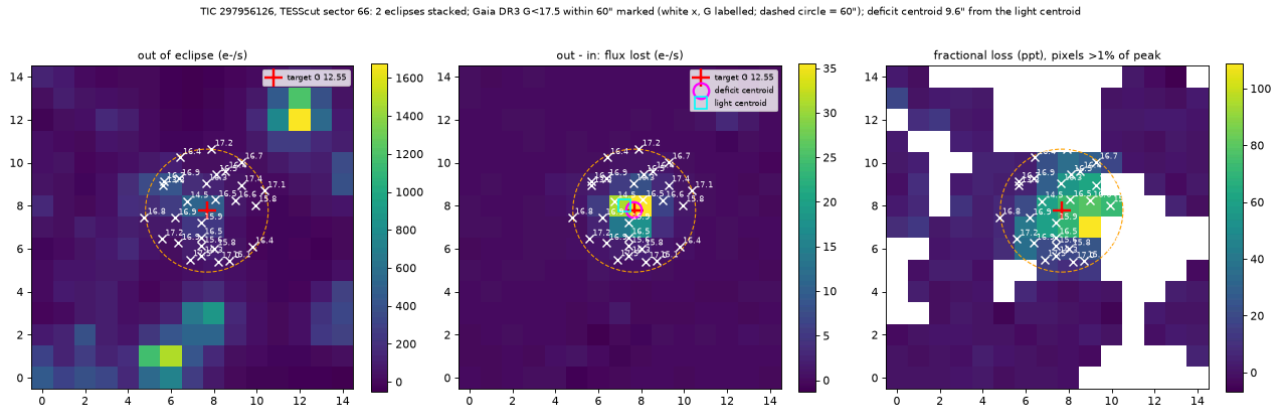
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy to refine physical parameters.



R617; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R617; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 297956126: CONFIRMED

Full bounded physical review: **COMPLETE**. Five measured deep eclipses after adding S66 QLP. Consecutive same-kind S66 eclipses resolve the long-baseline cycle-count ambiguity. Unequal eccentric secondary near0.7244; three capable neighbors are spatially separated from compact deficit.

Raw morphology was inspected in raw-events.png and phase-review.png, including P and2P parity views. 5 free-center trapezoid timings survive finite, identifiable center-covariance checks. Quality-zero raw QLP SAP_FLUX was used; predictions only guide fitting windows. This is representative available evidence, not an all-event/all-sector census.

Measured same-kind eclipse ephemeris: P=19.44125918346 +/- 0.00008930304 d; reference BTJD=3079.12129708. Epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced timing chi-square=17.73047106287078; zero degrees of freedom would be null. All selected timings follow the selected repeating eclipse family. The epoch is conventional and does not assert that this is the deeper primary; pooled alternate-event depths may differ with baseline/extraction. Within-sector cycle recurrence and parity depths are in ../alias-audit.json. Unequal secondary depths exclude mixing eclipse kinds into P/2; the P/2 and2P raw views were inspected. No exhaustive blind alias search is asserted.

Measured secondary-like narrow feature; pooled shape and baseline systematics limit precision. Depth is the realized center deficit of the fitted shape, with its amplitude parameter retained separately in the JSON. Measured pooled secondary phase=0.7243711, formal phase covariance error=3.3922e-05; depth=65.867 ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Search-guide provenance: Source-specific Gaia conjunction separation search guide; fitted center is free.

Fresh difference image: sector66, 2 complete core/sideband events. Target pixel=[7.67363198843382, 7.771735038897283]; deficit centroid=[7.640556238460893, 7.816464341886039]; deficit-target distance=0.0556 pixel. Deficit-light offset=9.645 arcsec is a descriptive field-light displacement, not an arbitrary veto. Fractional PSF-model residual=0.8948 is not a calibrated localization uncertainty. Every flux-capable Gaia source in the60-arcsec list was inspected against the image. Attribution conclusion:

Five measured deep eclipses after adding S66 QLP. Consecutive same-kind S66 eclipses resolve the long-baseline cycle-count ambiguity. Unequal eccentric secondary near0.7244; three capable neighbors are spatially separated from compact deficit.

Capable Gaia source	Separation arcsec	G	Intrinsic required eclipse proxy	Pixel coordinate	Diluted loss at pixel
5828914663934337024	20.853	14.509	0.5494	[6.749142680397109, 8.213817543181115]	0.059674319508744446
5828913186465574656	47.532	13.257	0.2266	[5.647107601520315, 8.927727163659231]	0.004698582585548853
5828914732653848704	52.734	15.074	0.8712	[8.730192586643438, 5.449452288812461]	0.013509646736525561

The pairwise G-band required intrinsic loss is an approximate brightness screen, with passband, aperture and blend assumptions. It cannot be directly compared to the diluted fractional loss in a pixel. Noncapable neighbors and all coordinates remain in fresh-pixels/diff_fresh.json and gaia.json. TESS resolution does not exclude unlisted unresolved companions.

Alternative reduction: fresh background-subtracted TESScut1x1/3x3/5x5 target apertures have depths 73.596/63.216/45.498 ppt versus fitted raw QLP median77.858 ppt. Individual cutout event depths are retained in the pixel JSON. Same-sector QLP and cutout extraction share TESS photons and test processing robustness. Distinct sectors are observing occasions on one instrument. Aperture/depth differences reflect dilution, background and extraction response; no intrinsic depth inferred.

Gaia DR3 source5828914728327457664: P=19.44436463 +/-0.01167201 d, e=0.349916, omega=13.46118 degrees, periastron BTJD=394.1096717. Closest edge-on conjunction offset=-0.498592 d (-0.025642 phase), source-specific full-covariance linear uncertainty=1.659949 d. Full correlation ordering P,gamma,K,e,omega,Tp, selected indices0,3,4,5. This linear approximation is descriptive, not a sigma veto. Both conjunctions and near-reference projection are retained in ../gaia-geometry.json. Mass function=0.114144 solar masses, assumed M1=1.51, minimum M2=0.86302; approximate central duration=13.883 h versus observed10.066 h. TIC radius and main-sequence companion scaling are assumptions, not a geometry fit; impact parameter and stellar evolution affect duration. Agreement cannot establish ownership among unresolved sources.

Sector	Same-kind cycle	Measured midpoint BTJD	Formal center error d	Input error d	Depth ppt
66	1	3098.56192268	0.0009313141	0.0011573952	81.574
66	2	3117.99882726	0.00081888804	0.0011573952	83.830
65	0	3079.12938743	0.0014235067	0.0014235067	77.858
100	52	4090.06837620	0.0013502117	0.0013502117	74.542
103	56	4167.83090804	0.00051382724	0.0011574068	73.654

Next check: Optional resolved photometry and spectroscopy to refine physical parameters.

Catalogue status is separate from physical verdict; shared-catalogues/coverage-matrix.csv retains the seven scoped groups. Known recoveries are not new additions; no exhaustive priority claim. Sources and identity are checked in ../identity-check.json and ../source-hashes.json. Numeric artifacts, original receipts and outputs are frozen in ../SHA256SUMS.json; independent helper review is ../helper-audit.md.

Stable review key: R617.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s66.json - historical/diff_s66.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

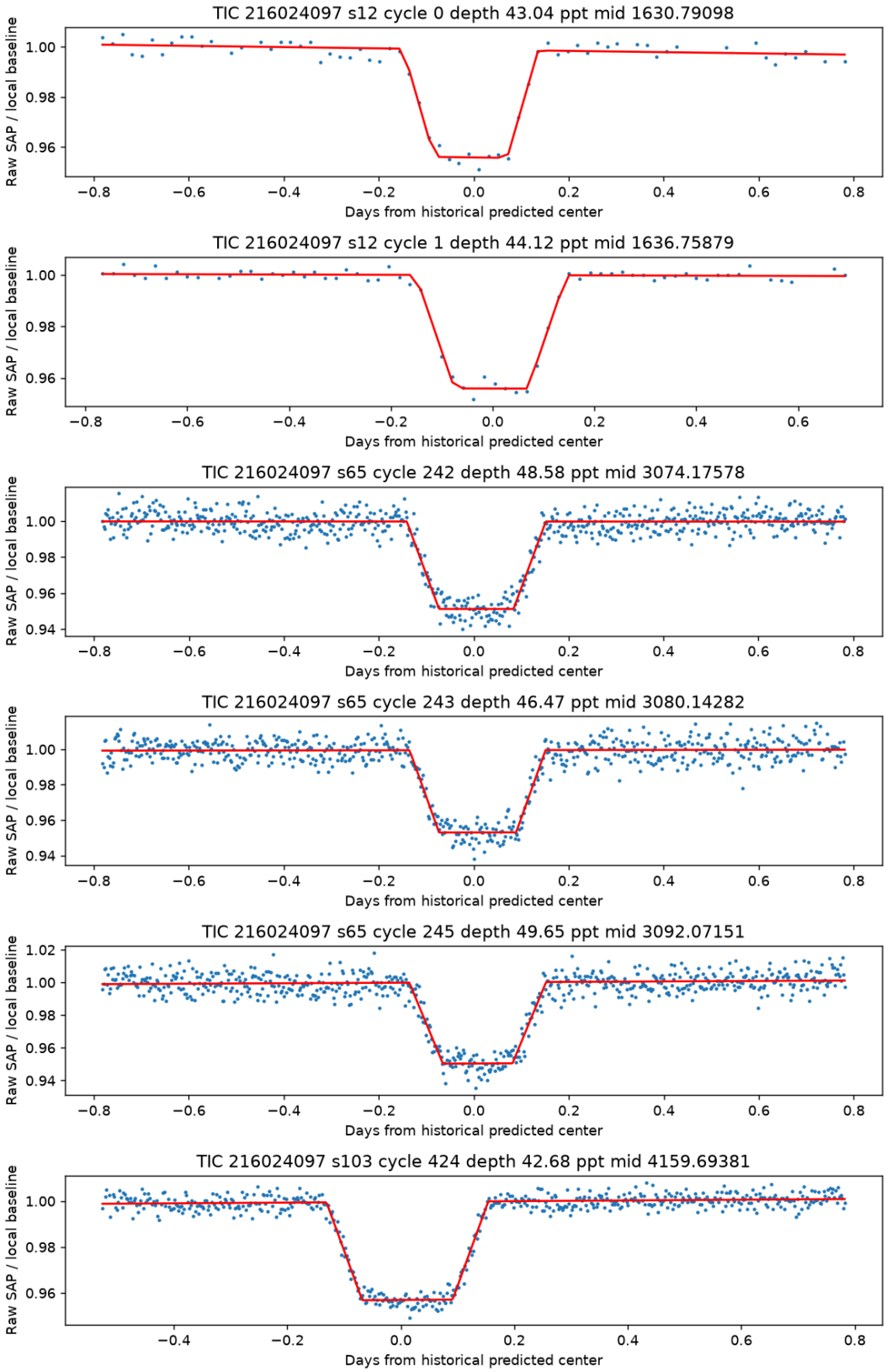
R618. TIC 216024097

physical_verdict: CONFIRMED

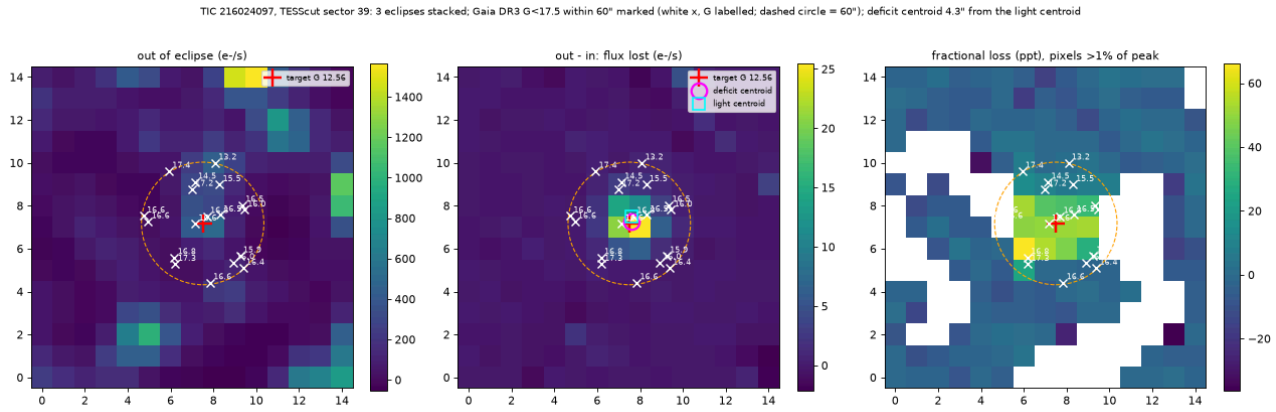
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy to refine physical parameters.



R618; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R618; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 216024097: CONFIRMED

Full bounded physical review: **COMPLETE**. Nine repeated flat-bottomed raw eclipses and shallower secondary near0.5051. Three capable neighbors at38-56 arcsec are displaced from target deficit core.

Raw morphology was inspected in raw-events.png and phase-review.png, including P and2P parity views. 9 free-center trapezoid timings survive finite, identifiable center-covariance checks. Quality-zero raw QLP SAP_FLUX was used; predictions only guide fitting windows. This is representative available evidence, not an all-event/all-sector census.

Measured same-kind eclipse ephemeris: P=5.96437448476 +/- 0.00000611621 d; reference BTJD=1630.79881660. Epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced timing chi-square=0.7097269260367367; zero degrees of freedom would be null. All selected timings follow the selected repeating eclipse family. The epoch is conventional and does not assert that this is the deeper primary; pooled alternate-event depths may differ with baseline/extraction. Within-sector cycle recurrence and parity depths are in ../alias-audit.json. Unequal secondary depths exclude mixing eclipse kinds into P/2; the P/2 and2P raw views were inspected. No exhaustive blind alias search is asserted.

Measured secondary-like narrow feature; pooled shape and baseline systematics limit precision. Depth is the realized center deficit of the fitted shape, with its amplitude parameter retained separately in the JSON. Measured pooled secondary phase=0.5051097, formal phase covariance error=0.00047603; depth=6.899 ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Search-guide provenance: Source-specific Gaia conjunction separation search guide; fitted center is free.

Fresh difference image: sector39, 3 complete core/sideband events. Target pixel=[7.497936311428383, 7.174649541884262]; deficit centroid=[7.58127932623185, 7.273839325942381]; deficit-target distance=0.1296 pixel. Deficit-light offset=4.279 arcsec is a descriptive field-light displacement, not an arbitrary veto. Fractional PSF-model residual=0.8327 is not a calibrated localization uncertainty. Every

flux-capable Gaia source in the60-arcsec list was inspected against the image. Attribution conclusion: Nine repeated flat-bottomed raw eclipses and shallower secondary near0.5051. Three capable neighbors at38-56 arcsec are displaced from target deficit core.

Capable Gaia source	Separation arcsec	G	Intrinsic required eclipse proxy	Pixel coordinate	Diluted loss at pixel
5935133465769258112	37.871	15.512	0.7516	[8.281782290671122, 8.987011854435043]	0.006501992137497622
5935133461448421888	38.776	14.461	0.3144	[7.129309021508924, 9.104486966453447]	0.010425004404331329
5935133500128999936	55.644	13.210	0.1311	[8.083067741218498, 9.995769686459242]	0.0015192198659098976

The pairwise G-band required intrinsic loss is an approximate brightness screen, with passband, aperture and blend assumptions. It cannot be directly compared to the diluted fractional loss in a pixel. Noncapable neighbors and all coordinates remain in fresh-pixels/diff_fresh.json and gaia.json. TESS resolution does not exclude unlisted unresolved companions.

Alternative reduction: fresh background-subtracted TESScut1x1/3x3/5x5 target apertures have depths 49.785/46.399/32.701 ppt versus fitted raw QLP median46.468 ppt. Individual cutout event depths are retained in the pixel JSON. Cutout sector is a distinct observing occasion on the same TESS instrument relative to retained QLP sectors. This alternative reduction shares the FFI photons of that sector; no same-sector QLP comparison claimed. Distinct sectors are observing occasions on one instrument. Aperture/depth differences reflect dilution, background and extraction response; no intrinsic depth inferred.

Gaia DR3 source5935133362690040704: P=5.96452661 +/-0.00080035 d, e=0.038954, omega=355.59763 degrees, periastron BTJD=388.6437177. Closest edge-on conjunction offset=0.043380 d (0.007273 phase), source-specific full-covariance linear uncertainty=0.183879 d. Full correlation ordering P,gamma,K,e,omega,Tp, selected indices0,3,4,5. This linear approximation is descriptive, not a sigma veto. Both conjunctions and near-reference projection are retained in ../gaia-geometry.json. Mass function=0.0361226 solar masses, assumed M1=1.77, minimum M2=0.58516; approximate central duration=8.049 h versus observed6.958 h. TIC radius and main-sequence companion scaling are assumptions, not a geometry fit; impact parameter and stellar evolution affect duration. Agreement cannot establish ownership among unresolved sources.

Sector	Same-kind cycle	Measured midpoint BTJD	Formal center error d	Input error d	Depth ppt
12	0	1630.79097924	0.0020556164	0.01041668	43.043
12	1	1636.75878957	0.001223965	0.01041668	44.116
12	3	1648.68792966	0.0010864261	0.01041668	42.724
65	242	3074.17578402	0.0010002406	0.0011574385	48.579
65	243	3080.14282493	0.0011720439	0.0011720439	46.468
65	244	3086.10641379	0.00097879867	0.0011574385	49.422
65	245	3092.07150924	0.0010332578	0.0011574385	49.648
103	423	4153.72777968	0.0019533922	0.0019533922	49.120
103	424	4159.69381325	0.00059218528	0.0011574068	42.679

Next check: Optional resolved photometry and spectroscopy to refine physical parameters.

Catalogue status is separate from physical verdict; shared-catalogues/coverage-matrix.csv retains the seven scoped groups. Known recoveries are not new additions; no exhaustive priority claim. Sources and identity are checked in ../identity-check.json and ../source-hashes.json. Numeric artifacts, original receipts and outputs are frozen in ../SHA256SUMS.json; independent helper review is ../helper-audit.md.

Stable review key: R618.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s39.json - historical/diff_s39.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

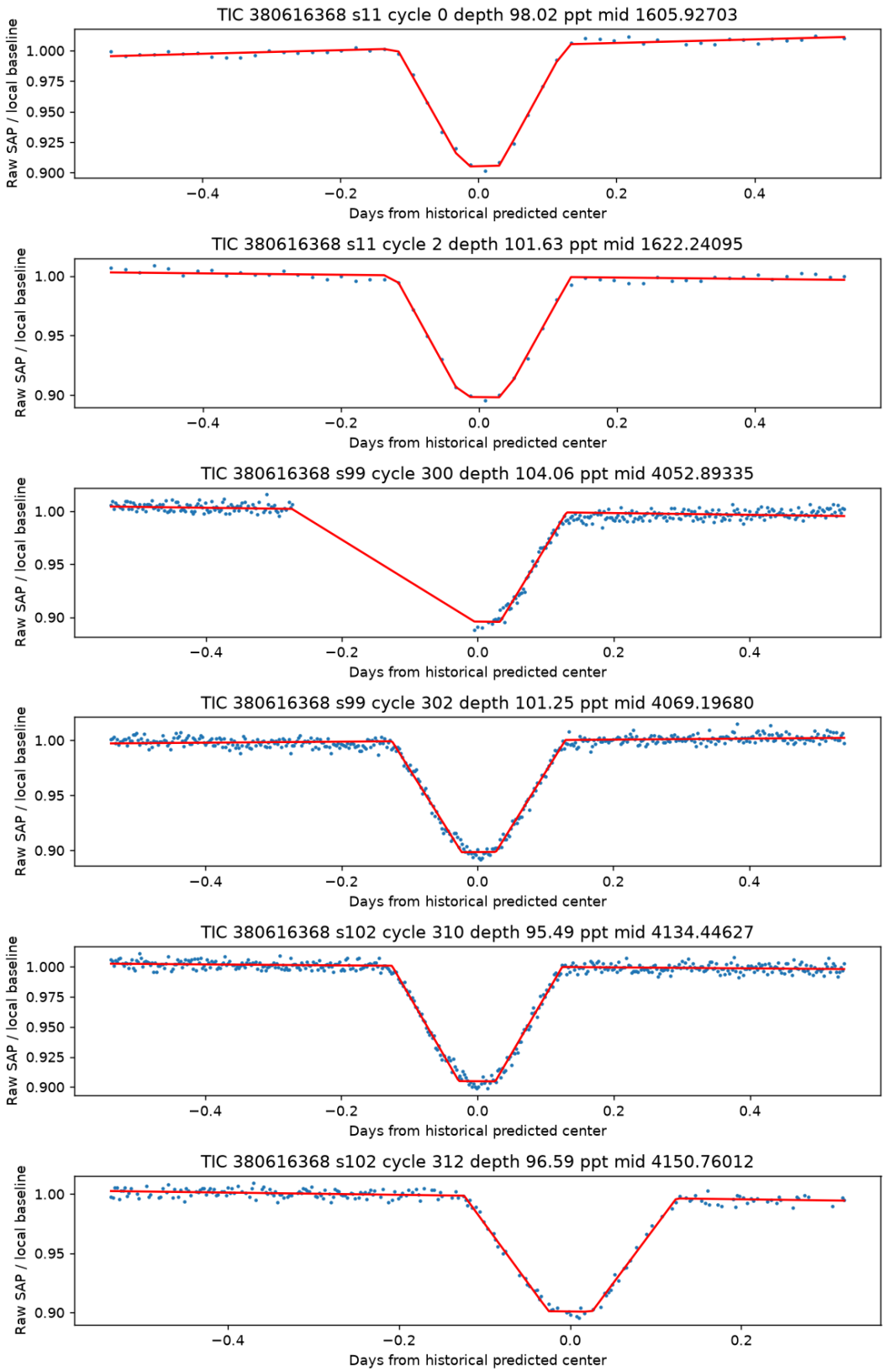
R619. TIC 380616368

physical_verdict: PROBABLE

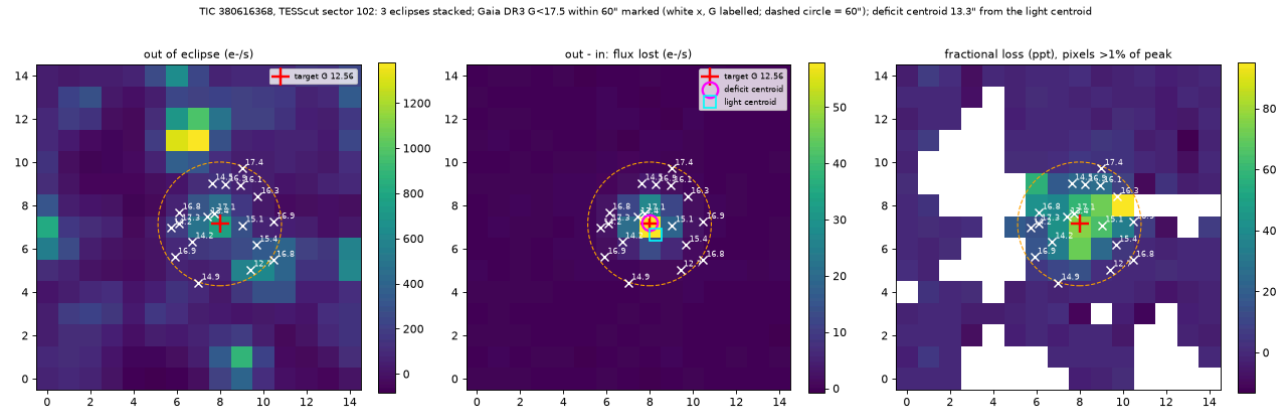
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or multi-source PRF localization for target and13.06-arcsec neighbor.



R619; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R619; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 380616368: PROBABLE

Full bounded physical review: **COMPLETE**. Six deep repeated raw eclipses and unequal secondary near0.5185 are real. A capable Gaia neighbor only13.06 arcsec away overlaps the TESS footprint; target ownership remains unresolved. Five farther capable neighbors were also inspected.

Raw morphology was inspected in raw-events.png and phase-review.png, including P and2P parity views. 6 free-center trapezoid timings survive finite, identifiable center-covariance checks. Quality-zero raw QLP SAP_FLUX was used; predictions only guide fitting windows. This is representative available evidence, not an all-event/all-sector census.

Measured same-kind eclipse ephemeris: P=8.15650892898 +/- 0.00002369015 d; reference BTJD=1605.92938782. Epoch kind: Selected repeating eclipse, primary by convention; deeper/secondary identity not independently established. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced timing chi-square=0.8107763251608298; zero degrees of freedom would be null. All selected timings follow the selected repeating eclipse family. The epoch is conventional and does not assert that this is the deeper primary; pooled alternate-event depths may differ with baseline/extraction. Within-sector cycle recurrence and parity depths are in ../alias-audit.json. Unequal secondary depths exclude mixing eclipse kinds into P/2; the P/2 and2P raw views were inspected. No exhaustive blind alias search is asserted.

Measured secondary-like narrow feature; pooled shape and baseline systematics limit precision. Depth is the realized center deficit of the fitted shape, with its amplitude parameter retained separately in the JSON. Measured pooled secondary phase=0.5184862, formal phase covariance error=9.0402e-05; depth=56.034 ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Search-guide provenance: Source-specific Gaia conjunction separation search guide; fitted center is free.

Fresh difference image: sector102, 3 complete core/sideband events. Target pixel= [7.984219910855064, 7.151055731001835]; deficit centroid=[7.982266569395521, 7.192986553144256]; deficit-target distance=0.0420 pixel. Deficit-light offset=13.263 arcsec is a descriptive field-light displacement, not an arbitrary veto. Fractional PSF-model residual=0.8995 is not a calibrated localization uncertainty. Every flux-capable Gaia source in the60-arcsec list was inspected

against the image. Attribution conclusion: Six deep repeated raw eclipses and unequal secondary near0.5185 are real. A capable Gaia neighbor only13.06 arcsec away overlaps the TESS footprint; target ownership remains unresolved. Five farther capable neighbors were also inspected.

Capable Gaia source	Separation arcsec	G	Intrinsic required eclipse proxy	Pixel coordinate	Diluted loss at pixel
5865932540392223744	13.062	13.383	0.3044	[7.413797724135005, 7.4745392776389465]	0.04293014794640035
5865932574751971968	31.123	14.215	0.5430	[6.7043858440504325, 6.32825893447743]	0.03388007637647905
5865932471672737152	39.075	14.471	0.6613	[7.618816880269076, 9.033259925387112]	0.025731756152021248
5865932501713536384	45.970	14.244	0.5551	[5.704934176142195, 6.976021740059235]	0.014972571479630815
5865932746550668288	51.176	12.688	0.2065	[9.401522646121027, 5.035736722321651]	0.001722808540661632
5865932780910412672	59.835	14.911	0.9431	[6.996712839276932, 4.424907704445629]	-0.0017383728775638878

The pairwise G-band required intrinsic loss is an approximate brightness screen, with passband, aperture and blend assumptions. It cannot be directly compared to the diluted fractional loss in a pixel. Noncapable neighbors and all coordinates remain in fresh-pixels/diff_fresh.json and gaia.json. TESS resolution does not exclude unlisted unresolved companions.

Alternative reduction: fresh background-subtracted TESScut1x1/3x3/5x5 target apertures have depths 76.349/56.561/35.455 ppt versus fitted raw QLP median97.305 ppt. Individual cutout event depths are retained in the pixel JSON. Same-sector QLP and cutout extraction share TESS photons and test processing robustness. Distinct sectors are observing occasions on one instrument. Aperture/depth differences reflect dilution, background and extraction response; no intrinsic depth inferred.

Gaia DR3 source5865932540392225536: P=8.15520713 +/-0.00076285 d, e=0.011737, omega=263.75442 degrees, periastron BTJD=386.4956175. Closest edge-on conjunction offset=0.085477 d (0.010481 phase), source-specific full-covariance linear uncertainty=0.165039 d. Full correlation ordering P,gamma,K,e,omega,Tp, selected indices0,3,4,5. This linear approximation is descriptive, not a sigma veto. Both conjunctions and near-reference projection are retained in ../gaia-geometry.json. Mass function=0.10988 solar masses, assumed M1=1.82, minimum M2=0.94314; approximate central duration=7.835 h versus observed5.951 h. TIC radius and main-sequence companion scaling are assumptions, not a geometry fit; impact parameter and stellar evolution affect duration. Agreement cannot establish ownership among unresolved sources.

Sector	Same-kind cycle	Measured midpoint BTJD	Formal center error d	Input error d	Depth ppt
11	0	1605.92702547	0.0025402268	0.010416736	98.019
11	2	1622.24095491	0.0012529065	0.010416736	101.632
99	302	4069.19679886	0.00039615136	0.0011574787	101.247
102	310	4134.44627395	0.00042510217	0.0011574068	95.492
102	311	4142.60293711	0.000427648	0.0011574068	96.223
102	312	4150.76011592	0.00068975804	0.0011574068	96.591

Next check: Resolved eclipse photometry or multi-source PRF localization for target and13.06-arcsec neighbor.

Catalogue status is separate from physical verdict; shared-catalogues/coverage-matrix.csv retains the seven scoped groups. Known recoveries are not new additions; no exhaustive priority claim. Sources and identity are checked in ../identity-check.json and ../source-hashes.json. Numeric artifacts, original

receipts and outputs are frozen in ../SHA256SUMS.json; independent helper review is ../helper-audit.md.

Stable review key: R619.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s102.json - historical/diff_s102.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

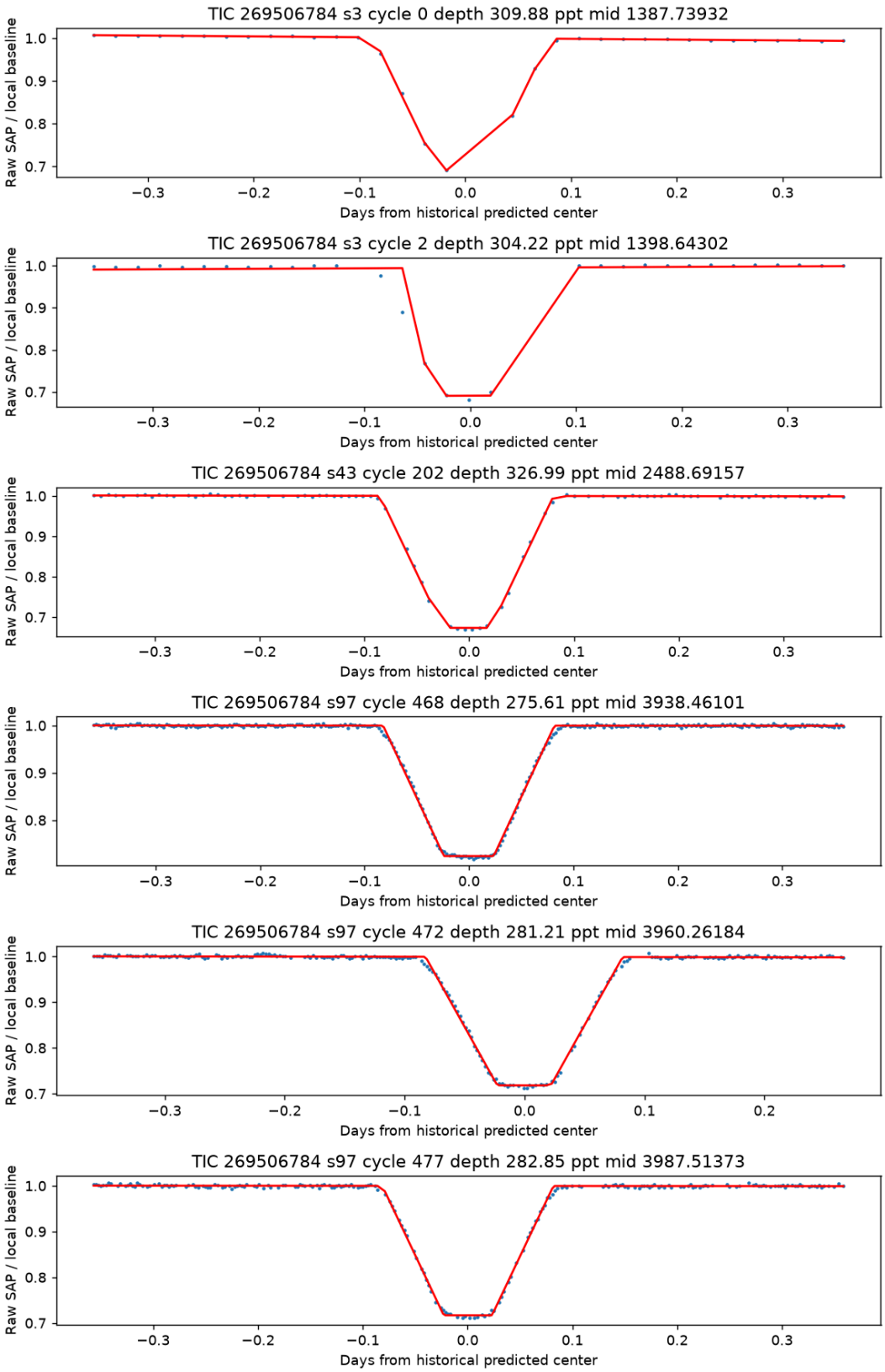
R620. TIC 269506784

physical_verdict: CONFIRMED

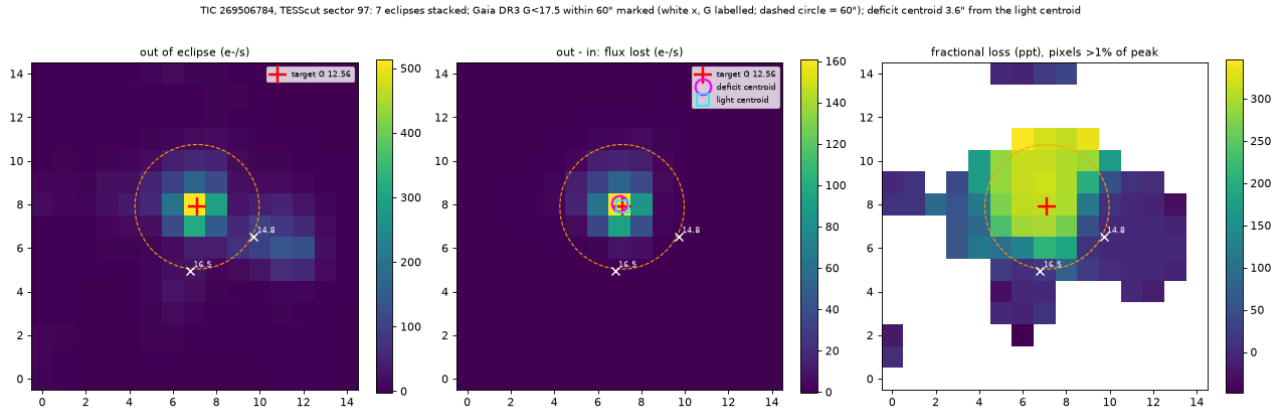
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy to refine physical parameters.



R620; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R620; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 269506784: CONFIRMED

Full bounded physical review: **COMPLETE**. Fourteen deep repeated raw eclipses and shallower secondary near phase0.4849. Strong sector97 baseline excursions were kept visible; local eclipse baselines remain stable. No capable Gaia neighbor within60arcsec; compact deficit favors target.

Raw morphology was inspected in raw-events.png and phase-review.png, including P and2P parity views. 14 free-center trapezoid timings survive finite, identifiable center-covariance checks. Quality-zero raw QLP SAP_FLUX was used; predictions only guide fitting windows. This is representative available evidence, not an all-event/all-sector census.

Measured same-kind eclipse ephemeris: P=5.45026009462 +/- 0.00000631346 d; reference BTJD=1387.73929467. Epoch kind: Deeper primary eclipse, established by unequal observed secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced timing chi-square=0.04822138918888307; zero degrees of freedom would be null. All selected timings follow the named repeating eclipse family. Epoch kind above identifies the deeper primary only where an unequal secondary is established. Model depth is the realized central loss d*min(1,w/(2*k)); extrapolated amplitude and ingress are retained separately. The reference is the fitted first selected cycle, not a Gaia prediction. Within-sector cycle recurrence and parity depths are in ../alias-audit.json. Opposite-phase sampled coverage, quality-zero core/sideband counts and flagged counts are in ../half-period-coverage.json. For138080650, orbital cycles are half the even historical SAP cycles; excluded odd-cycle secondaries remain in measurements.json. Unequal observed secondary depths exclude mixing eclipse kinds into P/2 where established; TIC138854818 instead has sampled opposite-phase coverage without a comparable11ppt primary; the P/2 and2P raw views were inspected. No exhaustive blind alias search is asserted.

Narrow shallower secondary visible in raw pooled windows. Depth is the realized center deficit of the fitted shape, with its amplitude parameter retained separately in the JSON. Measured pooled secondary phase=0.4849101, formal phase covariance error=0.00013979; depth=39.800 ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Search-guide provenance: Gaia conjunction search guide, adjusted from raw phase morphology for435209690 and58825156. Final phase is a free-center fit to observed raw flux..

Fresh difference image: sector97, 7 complete core/sideband events. Target pixel=[7.114449550354204, 7.888973013492947]; deficit centroid=[6.995229226613039, 8.044322784281377]; deficit-target distance=0.1958 pixel. Deficit-light offset=3.578 arcsec is a descriptive field-light displacement, not an arbitrary veto. Fractional PSF-model residual=0.3250 is not a calibrated localization uncertainty. Every flux-capable Gaia source in the60-arcsec list was inspected against the image. Attribution conclusion: Fourteen deep repeated raw eclipses and shallower secondary near phase0.4849. Strong sector97 baseline excursions were kept visible; local eclipse baselines remain stable. No capable Gaia neighbor within60arcsec; compact deficit favors target.

Capable Gaia source	Separation arcsec	G	Intrinsic required eclipse proxy	Pixel coordinate	Diluted loss at pixel
None					

The pairwise G-band required intrinsic loss is an approximate brightness screen, with passband, aperture and blend assumptions. It cannot be directly compared to the diluted fractional loss in a pixel. Noncapable neighbors and all coordinates remain in fresh-pixels/diff_fresh.json and gaia.json. TESS resolution does not exclude unlisted unresolved companions.

Alternative reduction: fresh background-subtracted TESScut1x1/3x3/5x5 target apertures have depths 313.141/302.206/278.755 ppt versus fitted raw QLP median284.165 ppt. Individual cutout event depths are retained in the pixel JSON. Same-sector QLP and cutout extraction share TESS photons and test processing robustness. Distinct sectors are observing occasions on one instrument. Aperture/depth differences reflect dilution, background and extraction response; no intrinsic depth inferred.

Gaia DR3 source2535953823492249216: P=5.45079570 +/-0.00074945 d, e=0.052840, omega=174.77875 degrees, periastron BTJD=391.5568944. Closest edge-on conjunction offset=-0.120499 d (-0.022107 phase), source-specific full-covariance linear uncertainty=0.150816 d. Full correlation ordering P,gamma,K,e,omega,Tp, selected indices0,3,4,5. This linear approximation is descriptive, not a sigma veto. Both conjunctions and near-reference projection are retained in ../gaia-geometry.json. Mass function=0.0814529 solar masses, assumed M1=0.92, minimum M2=0.56396; approximate central duration=5.703 h versus observed3.920 h. TIC radius and main-sequence companion scaling are assumptions, not a geometry fit; impact parameter and stellar evolution affect duration. Agreement cannot establish ownership among unresolved sources.

Sector	Same-kind cycle	Measured midpoint BTJD	Formal center error d	Input error d	Depth ppt
3	0	1387.73932484	0.00067371827	0.010416718	309.882
3	1	1393.19174459	0.00017055367	0.010416718	285.482
3	2	1398.64302174	0.00010059895	0.010416718	304.218
3	3	1404.08958205	0.00016069894	0.010416718	288.327
43	200	2477.79092948	0.0001598094	0.0034722749	331.648
43	202	2488.69156595	0.00013608874	0.0034722749	326.991
43	203	2494.14188394	0.00011938612	0.0034722749	327.371
97	468	3938.46101025	6.3283151e-05	0.001157396	275.610
97	469	3943.91082332	6.6996935e-05	0.001157396	279.721
97	471	3954.81219824	0.00016970938	0.001157396	277.071
97	472	3960.26184065	7.9271756e-05	0.001157396	281.206
97	473	3965.71233002	8.8570002e-05	0.001157396	275.447
97	474	3971.16251663	6.2388828e-05	0.001157396	280.212
97	477	3987.51373110	8.1245798e-05	0.001157396	282.847

Next check: Optional resolved photometry and spectroscopy to refine physical parameters.

Catalogue status is separate from physical verdict; shared-catalogues/coverage-matrix.csv retains the seven scoped groups. Known recoveries are not new additions; no exhaustive priority claim. Sources and identity are checked in ../identity-check.json and ../source-hashes.json. Numeric artifacts, original receipts and outputs are frozen in ../SHA256SUMS.json; independent helper review is ../helper-audit.md.

Stable review key: R620.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s97.json - historical/diff_s97.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

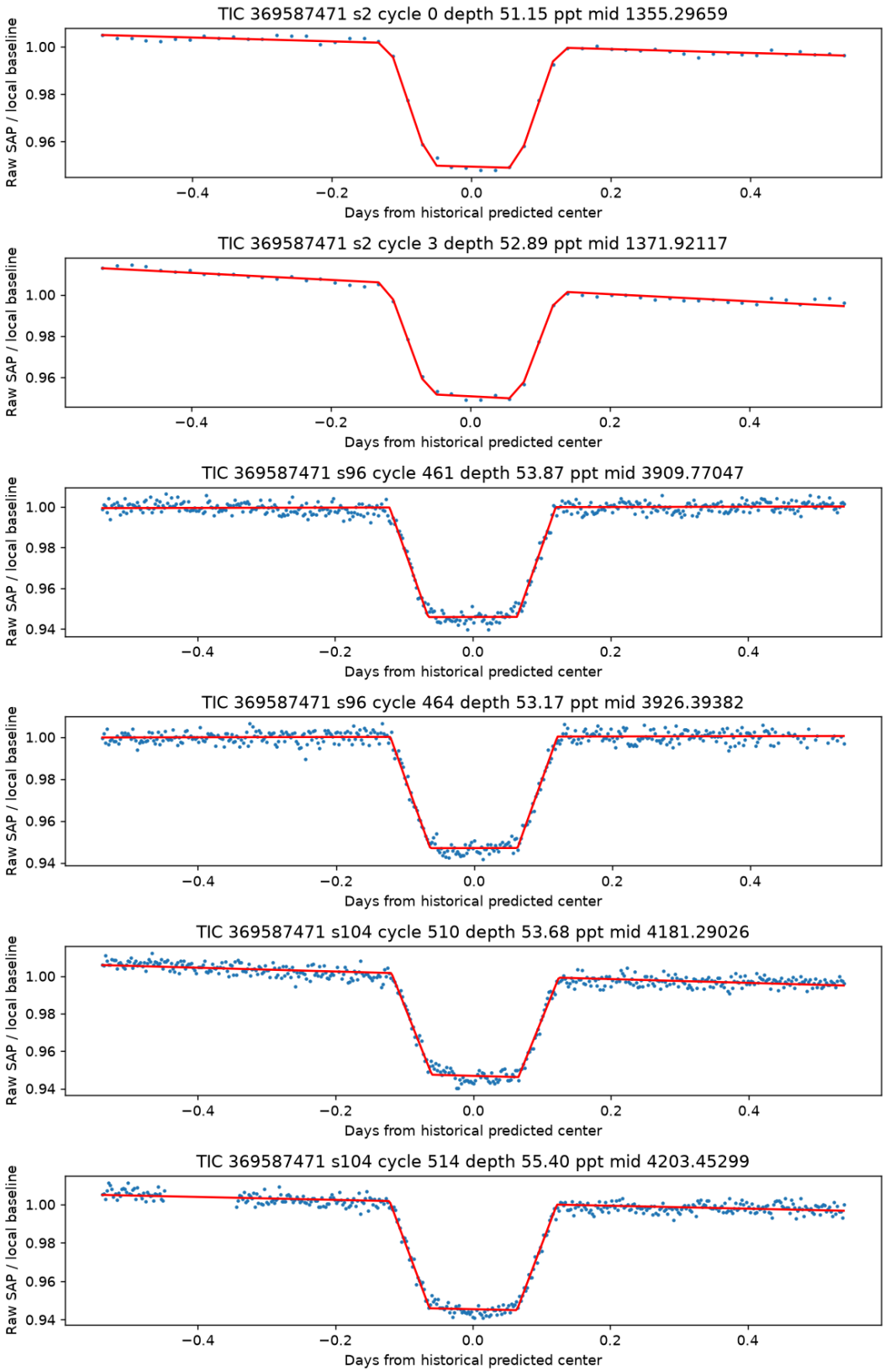
R621. TIC 369587471

physical_verdict: CONFIRMED

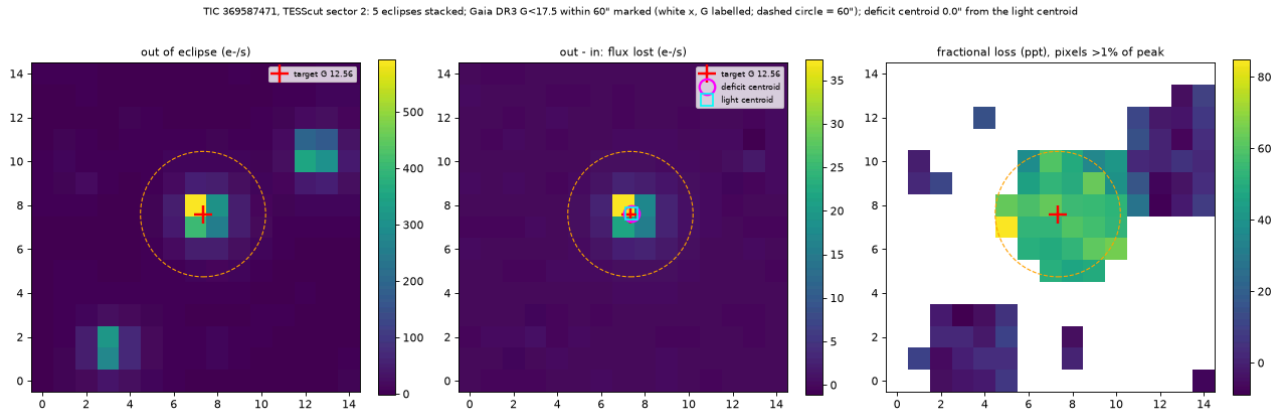
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy to refine physical parameters.



R621; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R621; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 369587471: CONFIRMED

Full bounded physical review: **COMPLETE**. Twelve flat-bottomed repeated raw eclipses and a shallower secondary near0.5009. Sector-dependent baseline variability remains visible. No capable Gaia neighbor within60arcsec; target-centered compact deficit.

Raw morphology was inspected in raw-events.png and phase-review.png, including P and2P parity views. 12 free-center trapezoid timings survive finite, identifiable center-covariance checks. Quality-zero raw QLP SAP_FLUX was used; predictions only guide fitting windows. This is representative available evidence, not an all-event/all-sector census.

Measured same-kind eclipse ephemeris: P=5.54117094061 +/- 0.00000904039 d; reference BTJD=1355.29116169. Epoch kind: Deeper primary eclipse, established by unequal observed secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced timing chi-square=0.6834212357957399; zero degrees of freedom would be null. All selected timings follow the named repeating eclipse family. Epoch kind above identifies the deeper primary only where an unequal secondary is established. Model depth is the realized central loss d*min(1,w/(2*k)); extrapolated amplitude and ingress are retained separately. The reference is the fitted first selected cycle, not a Gaia prediction. Within-sector cycle recurrence and parity depths are in ../alias-audit.json. Opposite-phase sampled coverage, quality-zero core/sideband counts and flagged counts are in ../half-period-coverage.json. For138080650, orbital cycles are half the even historical SAP cycles; excluded odd-cycle secondaries remain in measurements.json. Unequal observed secondary depths exclude mixing eclipse kinds into P/2 where established; TIC138854818 instead has sampled opposite-phase coverage without a comparable11ppt primary; the P/2 and2P raw views were inspected. No exhaustive blind alias search is asserted.

Narrow shallower secondary visible in raw pooled windows. Depth is the realized center deficit of the fitted shape, with its amplitude parameter retained separately in the JSON. Measured pooled secondary phase=0.5008911, formal phase covariance error=0.00035051; depth=7.017 ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Search-guide provenance: Gaia conjunction search guide, adjusted from raw phase morphology for435209690 and58825156. Final phase is a free-center fit to observed raw flux..

Fresh difference image: sector2, 5 complete core/sideband events. Target pixel=[7.336606030313423, 7.595824488552353]; deficit centroid=[7.375957634015044, 7.6134326426961385]; deficit-target distance=0.0431 pixel. Deficit-light offset=0.026 arcsec is a descriptive field-light displacement, not an arbitrary veto. Fractional PSF-model residual=0.0870 is not a calibrated localization uncertainty. Every flux-capable Gaia source in the60-arcsec list was inspected against the image. Attribution conclusion: Twelve flat-bottomed repeated raw eclipses and a shallower secondary near0.5009. Sector-dependent baseline variability remains visible. No capable Gaia neighbor within60arcsec; target-centered compact deficit.

Capable Gaia source	Separation arcsec	G	Intrinsic required eclipse proxy	Pixel coordinate	Diluted loss at pixel
None					

The pairwise G-band required intrinsic loss is an approximate brightness screen, with passband, aperture and blend assumptions. It cannot be directly compared to the diluted fractional loss in a pixel. Noncapable neighbors and all coordinates remain in fresh-pixels/diff_fresh.json and gaia.json. TESS resolution does not exclude unlisted unresolved companions.

Alternative reduction: fresh background-subtracted TESScut1x1/3x3/5x5 target apertures have depths 63.043/57.722/57.064 ppt versus fitted raw QLP median53.480 ppt. Individual cutout event depths are retained in the pixel JSON. Same-sector QLP and cutout extraction share TESS photons and test processing robustness. Distinct sectors are observing occasions on one instrument. Aperture/depth differences reflect dilution, background and extraction response; no intrinsic depth inferred.

Gaia DR3 source4994770575512682752: P=5.54088544 +/-0.00090485 d, e=0.037971, omega=109.72017 degrees, periastron BTJD=391.4459066. Closest edge-on conjunction offset=0.012703 d (0.002293 phase), source-specific full-covariance linear uncertainty=0.220157 d. Full correlation ordering P,gamma,K,e,omega,Tp, selected indices0,3,4,5. This linear approximation is descriptive, not a sigma veto. Both conjunctions and near-reference projection are retained in ../gaia-geometry.json. Mass function=0.0392495 solar masses, assumed M1=1.02, minimum M2=0.43671; approximate central duration=7.193 h versus observed5.837 h. TIC radius and main-sequence companion scaling are assumptions, not a geometry fit; impact parameter and stellar evolution affect duration. Agreement cannot establish ownership among unresolved sources.

Sector	Same-kind cycle	Measured midpoint BTJD	Formal center error d	Input error d	Depth ppt
2	0	1355.29659037	0.0022371914	0.010416673	51.149
2	1	1360.83846762	0.0027333243	0.010416673	53.132
2	3	1371.92116599	0.003234885	0.010416673	52.894
2	4	1377.46111619	0.0034255995	0.010416673	51.814
96	461	3909.77046561	0.00038481949	0.001157409	53.872
96	462	3915.31157170	0.00037991574	0.001157409	53.277
96	464	3926.39381998	0.00042047366	0.001157409	53.169
96	465	3931.93423887	0.00036343765	0.001157409	53.937
104	510	4181.29025953	0.0007808911	0.0011574068	53.683
104	511	4186.83039986	0.00063660293	0.0011574068	54.988
104	513	4197.91192576	0.00091269307	0.0011574068	54.740
104	514	4203.45299262	0.00052280022	0.0011574068	55.397

Next check: Optional resolved photometry and spectroscopy to refine physical parameters.

Catalogue status is separate from physical verdict; shared-catalogues/coverage-matrix.csv retains the seven scoped groups. Known recoveries are not new additions; no exhaustive priority claim. Sources and identity are checked in ../identity-check.json and ../source-hashes.json. Numeric artifacts, original

receipts and outputs are frozen in ../SHA256SUMS.json; independent helper review is ../helper-audit.md.

Stable review key: R621.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s02.json - historical/diff_s02.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

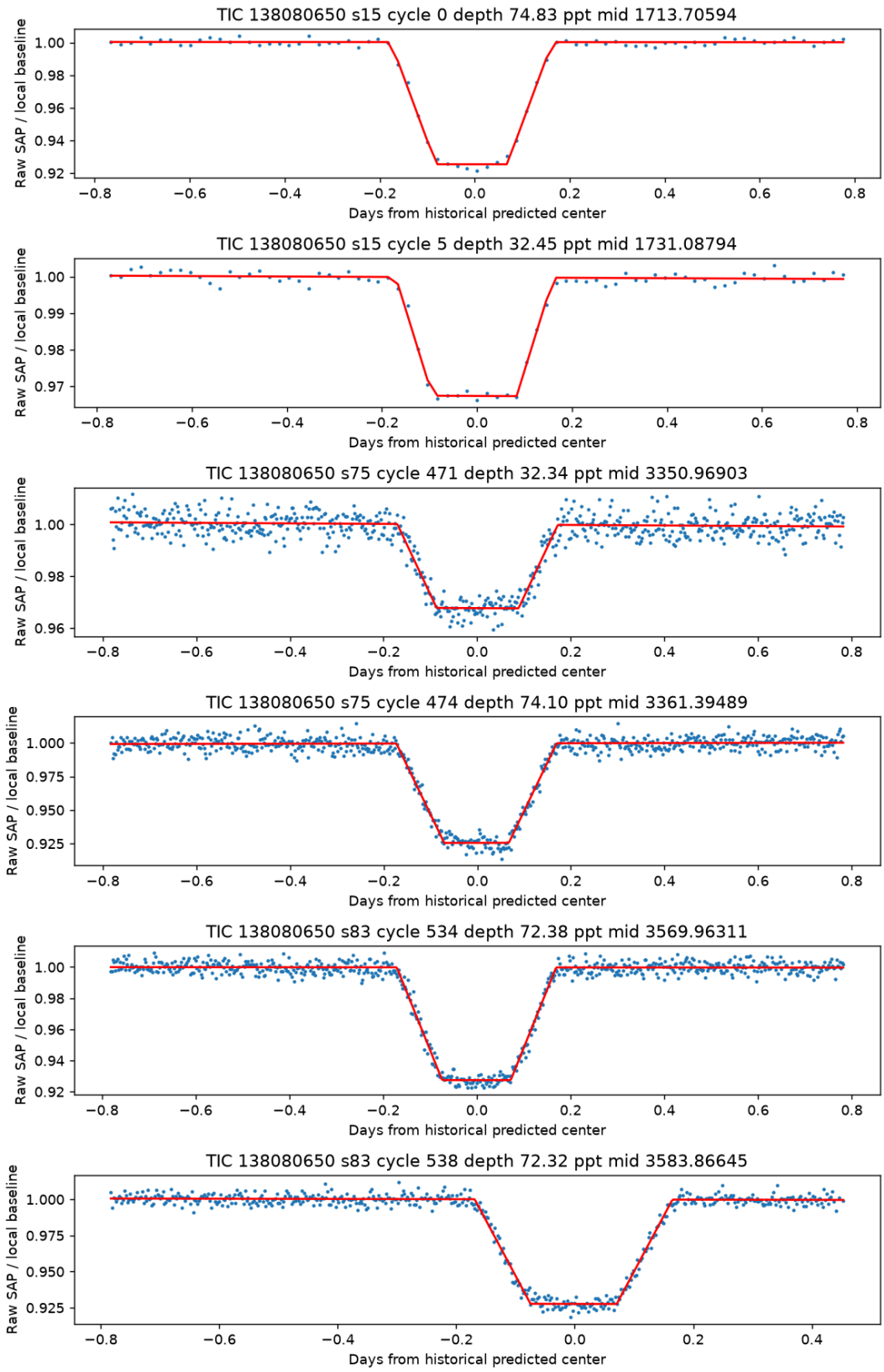
R622. TIC 138080650

physical_verdict: CONFIRMED

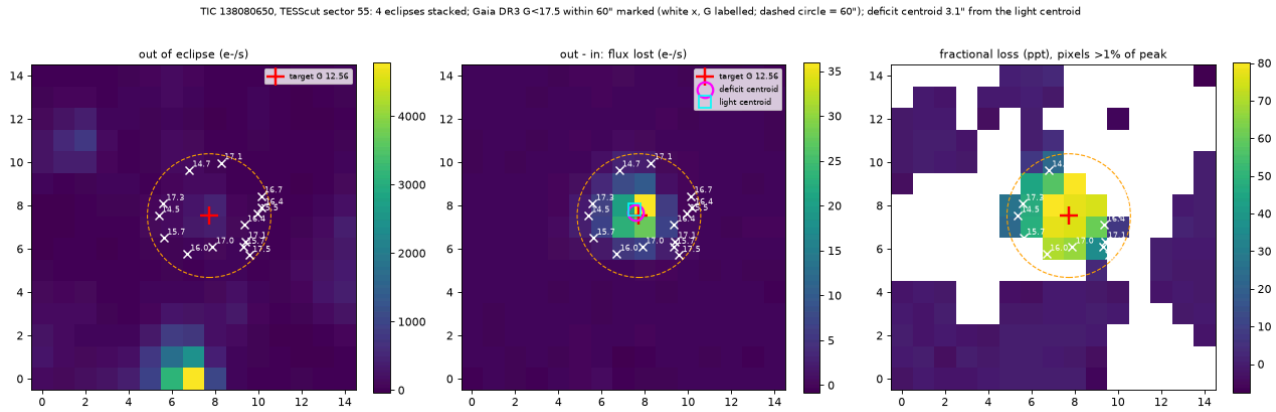
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy to refine physical parameters.



R622; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R622; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 138080650: CONFIRMED

Full bounded physical review: **COMPLETE**. Ten deeper primary timings at6.95225485d. Historical3.47613d SAP branch mixes unequal74/32ppt eclipses; deep even cycles alone fit the primary ephemeris. Actual secondary near0.50006. Both capable44.73/45.92arcsec neighbors are displaced from compact deficit.

Raw morphology was inspected in raw-events.png and phase-review.png, including P and2P parity views. 10 free-center trapezoid timings survive finite, identifiable center-covariance checks. Quality-zero raw QLP SAP_FLUX was used; predictions only guide fitting windows. This is representative available evidence, not an all-event/all-sector census.

Measured same-kind eclipse ephemeris: P=6.95225485160 +/- 0.00001928306 d; reference BTJD=1713.71066343. Epoch kind: Deeper primary eclipse, established by unequal observed secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced timing chi-square=0.1931356256140479; zero degrees of freedom would be null. All selected timings follow the named repeating eclipse family. Epoch kind above identifies the deeper primary only where an unequal secondary is established. Model depth is the realized central loss d*min(1,w/(2*k)); extrapolated amplitude and ingress are retained separately. The reference is the fitted first selected cycle, not a Gaia prediction. Within-sector cycle recurrence and parity depths are in ../alias-audit.json. Opposite-phase sampled coverage, quality-zero core/sideband counts and flagged counts are in ../half-period-coverage.json. For138080650, orbital cycles are half the even historical SAP cycles; excluded odd-cycle secondaries remain in measurements.json. Unequal observed secondary depths exclude mixing eclipse kinds into P/2 where established; TIC138854818 instead has sampled opposite-phase coverage without a comparable11ppt primary; the P/2 and2P raw views were inspected. No exhaustive blind alias search is asserted.

Narrow shallower secondary visible in raw pooled windows. Depth is the realized center deficit of the fitted shape, with its amplitude parameter retained separately in the JSON. Measured pooled secondary phase=0.5000636, formal phase covariance error=7.1826e-05; depth=32.169 ppt. Formal pooled

trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Search-guide provenance: Gaia conjunction search guide, adjusted from raw phase morphology for435209690 and58825156. Final phase is a free-center fit to observed raw flux..

Fresh difference image: sector55, 4 complete core/sideband events. Target pixel=[7.7245134634361925, 7.534969159151316]; deficit centroid=[7.558504078253135, 7.65865830592532]; deficit-target distance=0.2070 pixel. Deficit-light offset=3.079 arcsec is a descriptive field-light displacement, not an arbitrary veto. Fractional PSF-model residual=0.4938 is not a calibrated localization uncertainty. Every flux-capable Gaia source in the60-arcsec list was inspected against the image. Attribution conclusion: Ten deeper primary timings at6.95225485d. Historical3.47613d SAP branch mixes unequal74/32ppt eclipses; deep even cycles alone fit the primary ephemeris. Actual secondary near0.50006. Both capable44.73/45.92arcsec neighbors are displaced from compact deficit.

Capable Gaia source	Separation arcsec	G	Intrinsic required eclipse proxy	Pixel coordinate	Diluted loss at pixel
197179640016334 9376	44.735	14.66 8	0.5771	[6.798635568891259, 9.641154346611543]	0.02249541830870 073
197179619400490 8544	45.923	14.52 4	0.5146	[5.378745681797929, 7.5430969912381265]	0.0263693188890 99717

The pairwise G-band required intrinsic loss is an approximate brightness screen, with passband, aperture and blend assumptions. It cannot be directly compared to the diluted fractional loss in a pixel. Noncapable neighbors and all coordinates remain in fresh-pixels/diff_fresh.json and gaia.json. TESS resolution does not exclude unlisted unresolved companions.

Alternative reduction: fresh background-subtracted TESScut1x1/3x3/5x5 target apertures have depths 77.315/75.618/61.677 ppt versus fitted raw QLP median72.693 ppt. Individual cutout event depths are retained in the pixel JSON. Cutout sector is a distinct observing occasion on the same TESS instrument relative to retained QLP sectors. This alternative reduction shares the FFI photons of that sector; no same-sector QLP comparison claimed. Distinct sectors are observing occasions on one instrument. Aperture/depth differences reflect dilution, background and extraction response; no intrinsic depth inferred.

Gaia DR3 source1971796228364636800: P=6.95244810 +/-0.00114773 d, e=0.053527, omega=343.08591 degrees, periastron BTJD=390.7308766. Closest edge-on conjunction offset=0.064505 d (0.009278 phase), source-specific full-covariance linear uncertainty=0.244663 d. Full correlation ordering P,gamma,K,e,omega,Tp, selected indices0,3,4,5. This linear approximation is descriptive, not a sigma veto. Both conjunctions and near-reference projection are retained in ../gaia-geometry.json. Mass function=0.0923889 solar masses, assumed M1=1.06, minimum M2=0.64526; approximate central duration=12.104 h versus observed8.082 h. TIC radius and main-sequence companion scaling are assumptions, not a geometry fit; impact parameter and stellar evolution affect duration. Agreement cannot establish ownership among unresolved sources.

Sector	Same-kind cycle	Measured midpoint BTJD	Formal center error d	Input error d	Depth ppt
15	0	1713.70593989	0.00093586871	0.01041714	74.832
15	2	1727.61031060	0.00039925174	0.01041714	72.949
75	234	3340.53871408	0.00065227615	0.0011573829	73.367
75	235	3347.49111189	0.00055289761	0.0011573829	72.024
75	236	3354.44294118	0.00069036877	0.0011573829	72.847
75	237	3361.39488771	0.00065902763	0.0011573829	74.101
83	266	3563.00998863	0.00049087414	0.0011573605	71.739
83	267	3569.96310639	0.00050469526	0.0011573605	72.381
83	268	3576.91498977	0.00050951273	0.0011573605	72.539

Sector	Same-kind cycle	Measured midpoint BTJD	Formal center error d	Input error d	Depth ppt
83	269	3583.86644756	0.00043630067	0.0011573605	72.322

Next check: Optional resolved photometry and spectroscopy to refine physical parameters.

Catalogue status is separate from physical verdict; shared-catalogues/coverage-matrix.csv retains the seven scoped groups. Known recoveries are not new additions; no exhaustive priority claim. Sources and identity are checked in ../identity-check.json and ../source-hashes.json. Numeric artifacts, original receipts and outputs are frozen in ../SHA256SUMS.json; independent helper review is ../helper-audit.md.

Stable review key: R622.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s55.json - historical/diff_s55.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

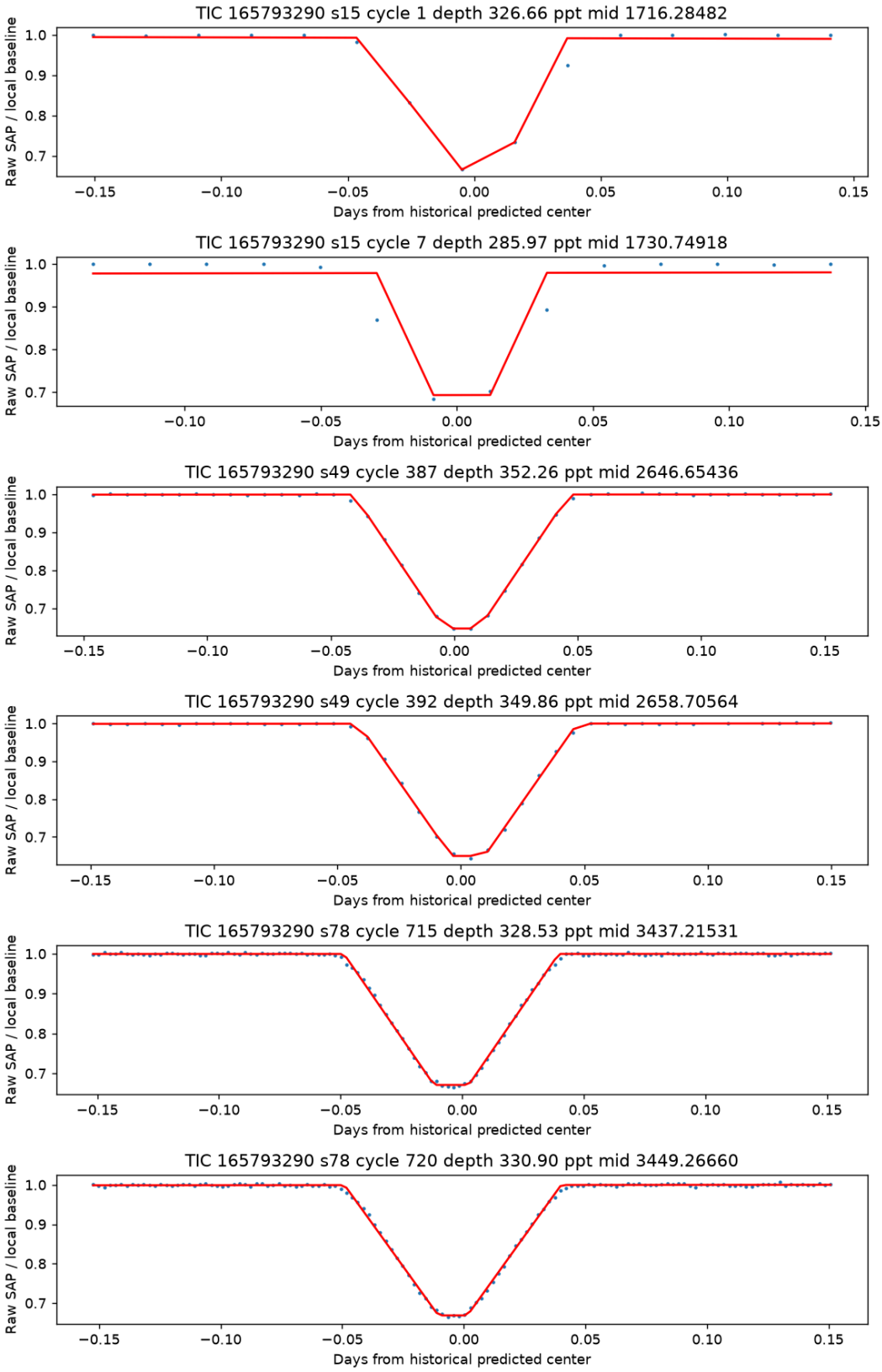
R623. TIC 165793290

physical_verdict: CONFIRMED

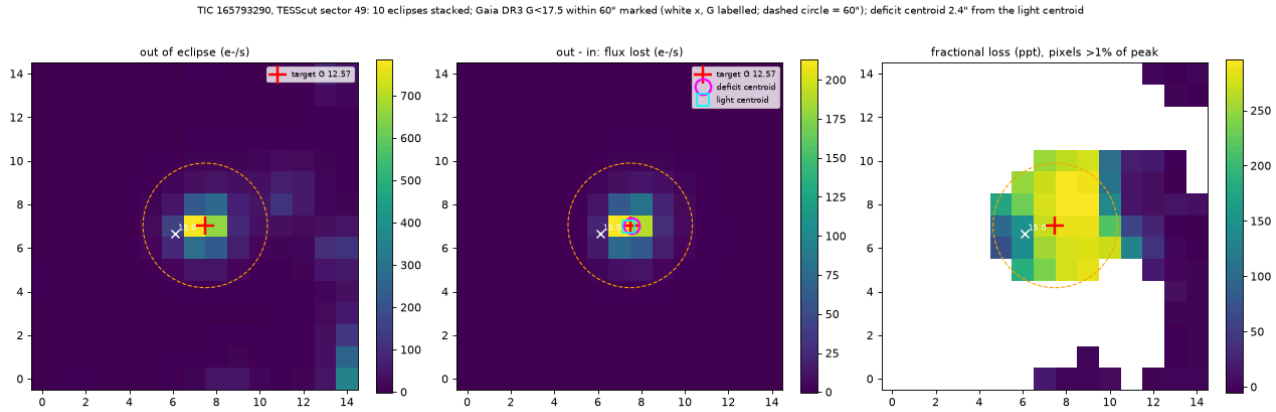
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy to refine physical parameters.



R623; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R623; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 165793290: CONFIRMED

Full bounded physical review: **COMPLETE**. Sixteen deep V-shaped repeated eclipses plus unequal secondary near0.49975. No capable Gaia neighbor within60arcsec; compact deficit is on target. Gaia conjunction offset0.0644d exceeds the linear0.0196d propagation; its phase0.0267 offset is explicitly retained rather than treated as a Gaussian rejection.

Raw morphology was inspected in raw-events.png and phase-review.png, including P and2P parity views. 16 free-center trapezoid timings survive finite, identifiable center-covariance checks. Quality-zero raw QLP SAP_FLUX was used; predictions only guide fitting windows. This is representative available evidence, not an all-event/all-sector census.

Measured same-kind eclipse ephemeris: P=2.41024796316 +/- 0.00000370217 d; reference BTJD=1716.29832455. Epoch kind: Deeper primary eclipse, established by unequal observed secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced timing chi-square=0.12877466262314283; zero degrees of freedom would be null. All selected timings follow the named repeating eclipse family. Epoch kind above identifies the deeper primary only where an unequal secondary is established. Model depth is the realized central loss d*min(1,w/(2*k)); extrapolated amplitude and ingress are retained separately. The reference is the fitted first selected cycle, not a Gaia prediction. Within-sector cycle recurrence and parity depths are in ../alias-audit.json. Opposite-phase sampled coverage, quality-zero core/sideband counts and flagged counts are in ../half-period-coverage.json. For138080650, orbital cycles are half the even historical SAP cycles; excluded odd-cycle secondaries remain in measurements.json. Unequal observed secondary depths exclude mixing eclipse kinds into P/2 where established; TIC138854818 instead has sampled opposite-phase coverage without a comparable11ppt primary; the P/2 and2P raw views were inspected. No exhaustive blind alias search is asserted.

Narrow shallower secondary visible in raw pooled windows. Depth is the realized center deficit of the fitted shape, with its amplitude parameter retained separately in the JSON. Measured pooled secondary phase=0.4997470, formal phase covariance error=2.5813e-05; depth=97.765 ppt. Formal pooled

trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Search-guide provenance: Gaia conjunction search guide, adjusted from raw phase morphology for435209690 and58825156. Final phase is a free-center fit to observed raw flux..

Fresh difference image: sector49, 10 complete core/sideband events. Target pixel=[7.490314781863615, 7.03628368763723]; deficit centroid=[7.52809403705176, 7.015776526008011]; deficit-target distance=0.0430 pixel. Deficit-light offset=2.399 arcsec is a descriptive field-light displacement, not an arbitrary veto. Fractional PSF-model residual=0.1413 is not a calibrated localization uncertainty. Every flux-capable Gaia source in the60-arcsec list was inspected against the image. Attribution conclusion: Sixteen deep V-shaped repeated eclipses plus unequal secondary near0.49975. No capable Gaia neighbor within60arcsec; compact deficit is on target. Gaia conjunction offset0.0644d exceeds the linear0.0196d propagation; its phase0.0267 offset is explicitly retained rather than treated as a Gaussian rejection.

Capable Gaia source	Separation arcsec	G	Intrinsic required eclipse proxy	Pixel coordinate	Diluted loss at pixel
None					

The pairwise G-band required intrinsic loss is an approximate brightness screen, with passband, aperture and blend assumptions. It cannot be directly compared to the diluted fractional loss in a pixel. Noncapable neighbors and all coordinates remain in fresh-pixels/diff_fresh.json and gaia.json. TESS resolution does not exclude unlisted unresolved companions.

Alternative reduction: fresh background-subtracted TESScut1x1/3x3/5x5 target apertures have depths 271.231/264.840/263.933 ppt versus fitted raw QLP median345.935 ppt. Individual cutout event depths are retained in the pixel JSON. Same-sector QLP and cutout extraction share TESS photons and test processing robustness. Distinct sectors are observing occasions on one instrument. Aperture/depth differences reflect dilution, background and extraction response; no intrinsic depth inferred.

Gaia DR3 source1603058197763982976: P=2.41018569 +/-0.00003382 d, e=0.020577, omega=293.54421 degrees, periastron BTJD=389.5907569. Closest edge-on conjunction offset=0.064371 d (0.026708 phase), source-specific full-covariance linear uncertainty=0.019609 d. Full correlation ordering P,gamma,K,e,omega,Tp, selected indices0,3,4,5. This linear approximation is descriptive, not a sigma veto. Both conjunctions and near-reference projection are retained in ../gaia-geometry.json. Mass function=0.0614391 solar masses, assumed M1=0.642, minimum M2=0.40751; approximate central duration=3.247 h versus observed2.117 h. TIC radius and main-sequence companion scaling are assumptions, not a geometry fit; impact parameter and stellar evolution affect duration. Agreement cannot establish ownership among unresolved sources.

Sector	Same-kind cycle	Measured midpoint BTJD	Formal center error d	Input error d	Depth ppt
15	1	1716.28481936	1.2994336e-05	0.010416386	326.658
49	384	2639.42349198	4.0342443e-05	0.0034722583	354.398
49	385	2641.83398946	5.9956282e-05	0.0034722583	351.614
49	386	2644.24415560	4.1698749e-05	0.0034722583	352.045
49	387	2646.65435683	4.8072475e-05	0.0034722583	352.256
49	388	2649.06470313	2.4148114e-05	0.0034722583	350.310
49	391	2656.29548137	3.6395356e-05	0.0034722583	351.990
49	392	2658.70564485	3.2303797e-05	0.0034722583	349.856
49	393	2661.11595156	5.4806834e-05	0.0034722583	345.629
49	394	2663.52606188	3.2295967e-05	0.0034722583	346.242
78	714	3434.80518341	4.7033273e-05	0.001157382	329.927
78	715	3437.21531055	4.2687536e-05	0.001157382	328.526
78	716	3439.62550700	5.0725206e-05	0.001157382	327.894
78	717	3442.03579911	5.4758169e-05	0.001157382	328.753

Sector	Same-kind cycle	Measured midpoint BTJD	Formal center error d	Input error d	Depth ppt
78	719	3446.85635604	5.2975202e-05	0.001157382	328.724
78	720	3449.26659568	4.8323437e-05	0.001157382	330.895

Next check: Optional resolved photometry and spectroscopy to refine physical parameters.

Catalogue status is separate from physical verdict; shared-catalogues/coverage-matrix.csv retains the seven scoped groups. Known recoveries are not new additions; no exhaustive priority claim. Sources and identity are checked in ../identity-check.json and ../source-hashes.json. Numeric artifacts, original receipts and outputs are frozen in ../SHA256SUMS.json; independent helper review is ../helper-audit.md.

Stable review key: R623.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s49.json - historical/diff_s49.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

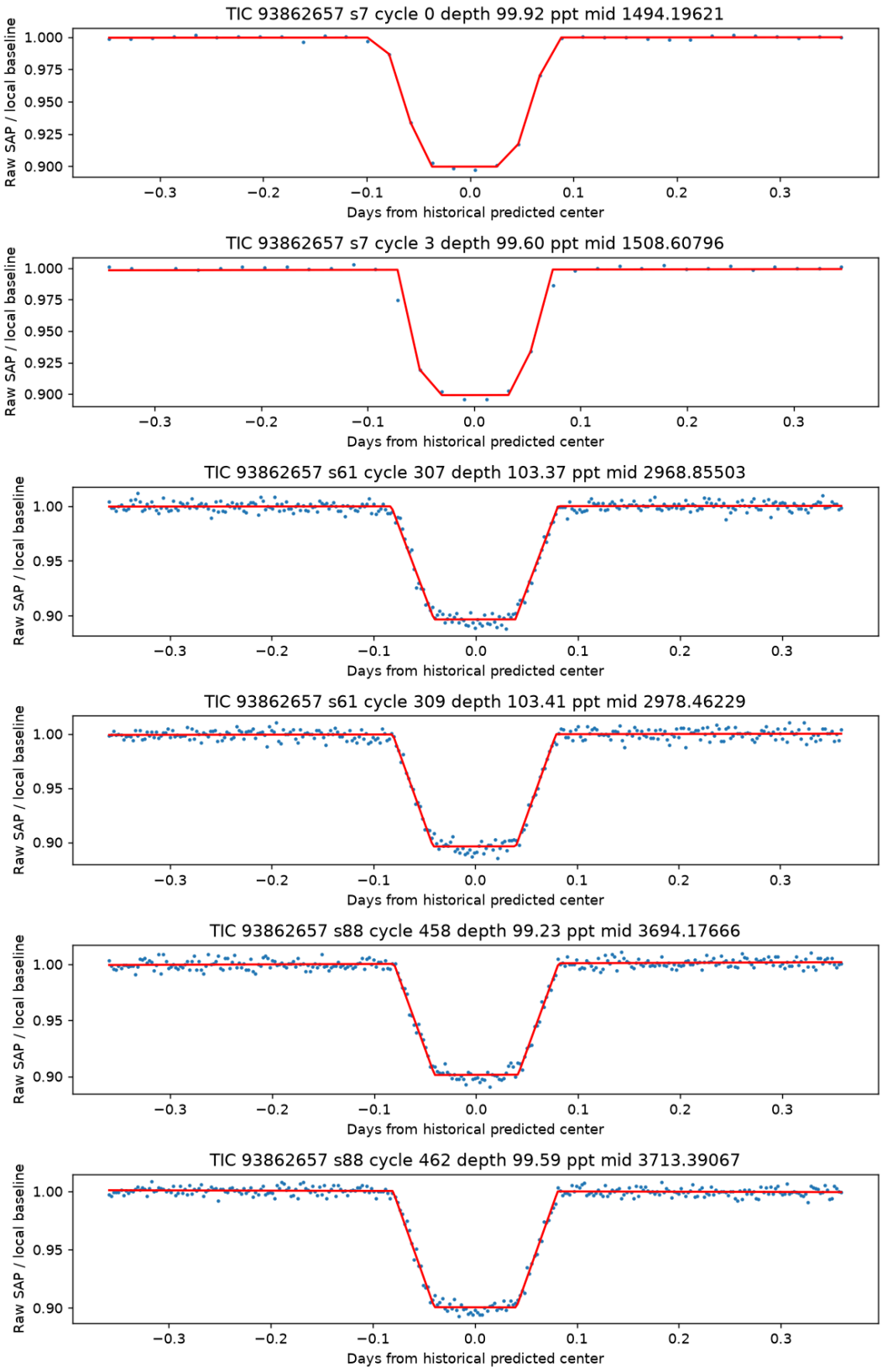
R624. TIC 93862657

physical_verdict: CONFIRMED

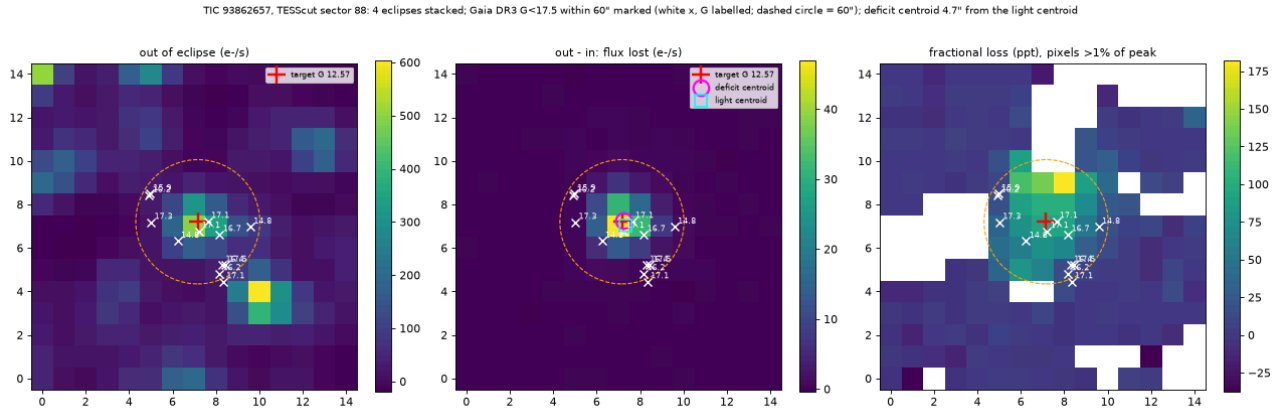
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy to refine physical parameters.



R624; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R624; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 93862657: CONFIRMED

Full bounded physical review: **COMPLETE**. Twelve deep repeated raw eclipses and shallower secondary near0.4977. Compact lost flux favors target over capable24.17/48.03arcsec neighbors. Nearby known EB Gaia3041697316497710720 at29.83arcsec has G19.73955 and0.282338d catalog period; its intrinsic-loss proxy74.05 exceeds unity and its period differs from4.80345d. It cannot plausibly make the100ppt target eclipse under this brightness model. Crowding raises whole-field PSF residual; no calibrated localization precision claimed.

Raw morphology was inspected in raw-events.png and phase-review.png, including P and2P parity views. 12 free-center trapezoid timings survive finite, identifiable center-covariance checks. Quality-zero raw QLP SAP_FLUX was used; predictions only guide fitting windows. This is representative available evidence, not an all-event/all-sector census.

Measured same-kind eclipse ephemeris: P=4.80345242294 +/- 0.00000502307 d; reference BTJD=1494.19541099. Epoch kind: Deeper primary eclipse, established by unequal observed secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced timing chi-square=0.044408305858084765; zero degrees of freedom would be null. All selected timings follow the named repeating eclipse family. Epoch kind above identifies the deeper primary only where an unequal secondary is established. Model depth is the realized central loss d*min(1,w/(2*k)); extrapolated amplitude and ingress are retained separately. The reference is the fitted first selected cycle, not a Gaia prediction. Within-sector cycle recurrence and parity depths are in ../alias-audit.json. Opposite-phase sampled coverage, quality-zero core/sideband counts and flagged counts are in ../half-period-coverage.json. For138080650, orbital cycles are half the even historical SAP cycles; excluded odd-cycle secondaries remain in measurements.json. Unequal observed secondary depths exclude mixing eclipse kinds into P/2 where established; TIC138854818 instead has sampled opposite-phase coverage without a comparable11ppt primary; the P/2 and2P raw views were inspected. No exhaustive blind alias search is asserted.

Narrow shallower secondary visible in raw pooled windows. Depth is the realized center deficit of the fitted shape, with its amplitude parameter retained separately in the JSON. Measured pooled secondary phase=0.4977257, formal phase covariance error=0.00036205; depth=5.821 ppt. Formal pooled

trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Search-guide provenance: Gaia conjunction search guide, adjusted from raw phase morphology for435209690 and58825156. Final phase is a free-center fit to observed raw flux..

Fresh difference image: sector88, 4 complete core/sideband events. Target pixel=[7.164710660216459, 7.202969317919431]; deficit centroid=[7.250045207399978, 7.244594980093872]; deficit-target distance=0.0949 pixel. Deficit-light offset=4.695 arcsec is a descriptive field-light displacement, not an arbitrary veto. Fractional PSF-model residual=0.7715 is not a calibrated localization uncertainty. Every flux-capable Gaia source in the60-arcsec list was inspected against the image. Attribution conclusion: Twelve deep repeated raw eclipses and shallower secondary near0.4977. Compact lost flux favors target over capable24.17/48.03arcsec neighbors. Nearby known EB Gaia3041697316497710720 at29.83arcsec has G19.73955 and0.282338d catalog period; its intrinsic-loss proxy74.05 exceeds unity and its period differs from4.80345d. It cannot plausibly make the100ppt target eclipse under this brightness model. Crowding raises whole-field PSF residual; no calibrated localization precision claimed.

Capable Gaia source	Separation arcsec	G	Intrinsic required eclipse proxy	Pixel coordinate	Diluted loss at pixel
3041697320797166976	24.174	14.834	0.9065	[6.239707037601738, 6.315112664808126]	0.05535194808893887
3041697355156898688	48.030	14.754	0.8493	[9.584789177028378, 6.968494324832848]	0.015637403803996478

The pairwise G-band required intrinsic loss is an approximate brightness screen, with passband, aperture and blend assumptions. It cannot be directly compared to the diluted fractional loss in a pixel. Noncapable neighbors and all coordinates remain in fresh-pixels/diff_fresh.json and gaia.json. TESS resolution does not exclude unlisted unresolved companions.

Alternative reduction: fresh background-subtracted TESScut1x1/3x3/5x5 target apertures have depths 95.172/88.814/80.060 ppt versus fitted raw QLP median100.078 ppt. Individual cutout event depths are retained in the pixel JSON. Same-sector QLP and cutout extraction share TESS photons and test processing robustness. Distinct sectors are observing occasions on one instrument. Aperture/depth differences reflect dilution, background and extraction response; no intrinsic depth inferred.

Gaia DR3 source3041697320797163008: P=4.80289401 +/-0.00049369 d, e=0.022247, omega=126.14374 degrees, periastron BTJD=389.8784640. Closest edge-on conjunction offset=0.113738 d (0.023681 phase), source-specific full-covariance linear uncertainty=0.130976 d. Full correlation ordering P,gamma,K,e,omega,Tp, selected indices0,3,4,5. This linear approximation is descriptive, not a sigma veto. Both conjunctions and near-reference projection are retained in ../gaia-geometry.json. Mass function=0.0170011 solar masses, assumed M1=1.216, minimum M2=0.34619; approximate central duration=4.266 h versus observed3.857 h. TIC radius and main-sequence companion scaling are assumptions, not a geometry fit; impact parameter and stellar evolution affect duration. Agreement cannot establish ownership among unresolved sources.

Sector	Same-kind cycle	Measured midpoint BTJD	Formal center error d	Input error d	Depth ppt
7	0	1494.19621137	0.00023147443	0.01041666	99.924
7	1	1499.00231337	0.00022773688	0.01041666	100.233
7	3	1508.60796321	0.00013453829	0.01041666	99.596
7	4	1513.41043282	0.00033842665	0.01041666	102.922
61	307	2968.85502793	0.0002520966	0.0011573863	103.371
61	308	2973.65852945	0.00029558135	0.0011573863	105.159
61	309	2978.46229288	0.00029281513	0.0011573863	103.412
61	310	2983.26580917	0.00025892572	0.0011573863	103.715
88	458	3694.17666187	0.00027727402	0.0011573846	99.227

Sector	Same-kind cycle	Measured midpoint BTJD	Formal center error d	Input error d	Depth ppt
88	459	3698.97975723	0.00027533348	0.0011573846	99.110
88	461	3708.58719011	0.00027445641	0.0011573846	97.822
88	462	3713.39067289	0.00021702488	0.0011573846	99.589

Next check: Optional resolved photometry and spectroscopy to refine physical parameters.

Catalogue status is separate from physical verdict; shared-catalogues/coverage-matrix.csv retains the seven scoped groups. Known recoveries are not new additions; no exhaustive priority claim. Sources and identity are checked in ../identity-check.json and ../source-hashes.json. Numeric artifacts, original receipts and outputs are frozen in ../SHA256SUMS.json; independent helper review is ../helper-audit.md.

Stable review key: R624.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s88.json - historical/diff_s88.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

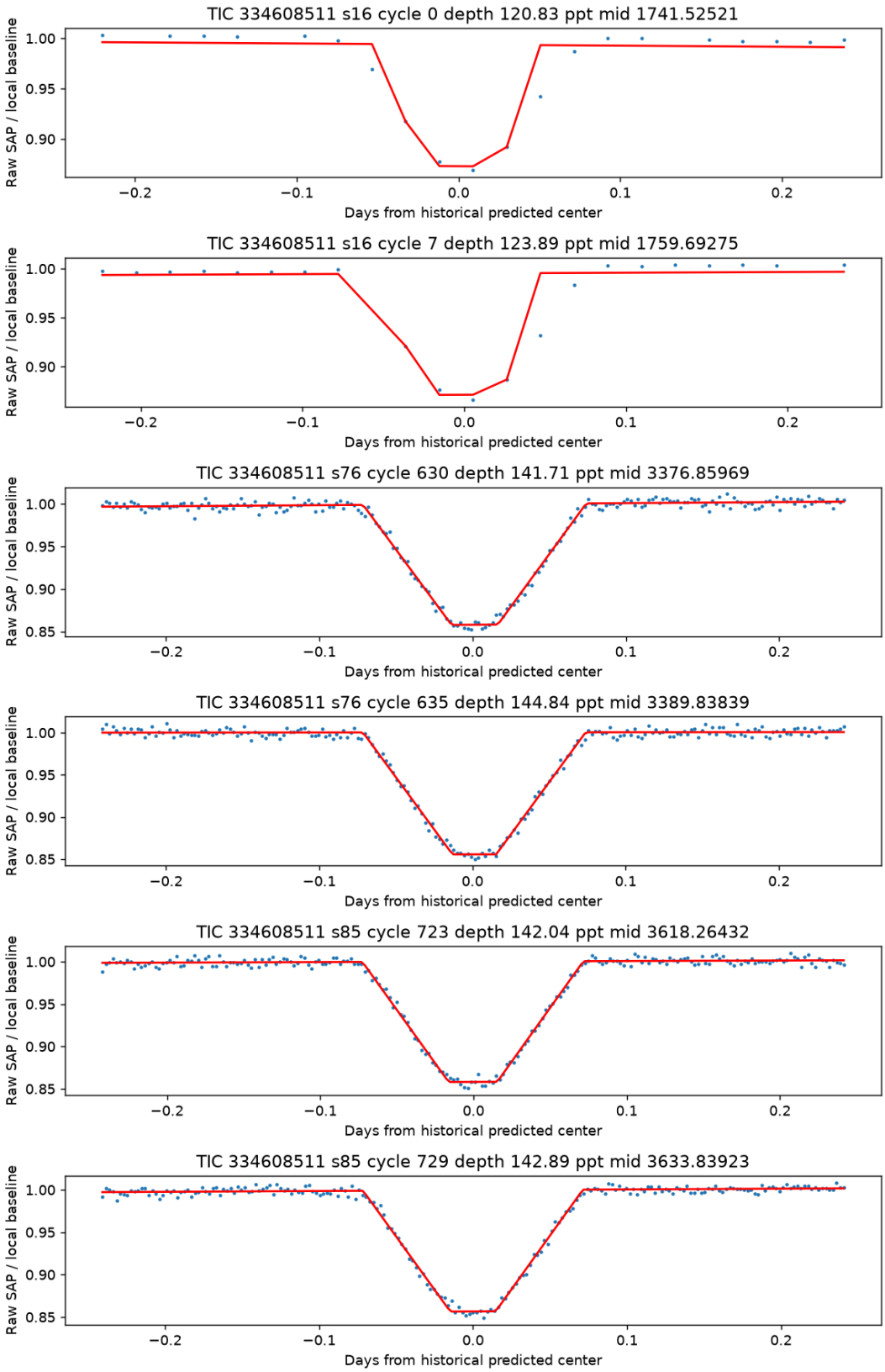
R625. TIC 334608511

physical_verdict: CONFIRMED

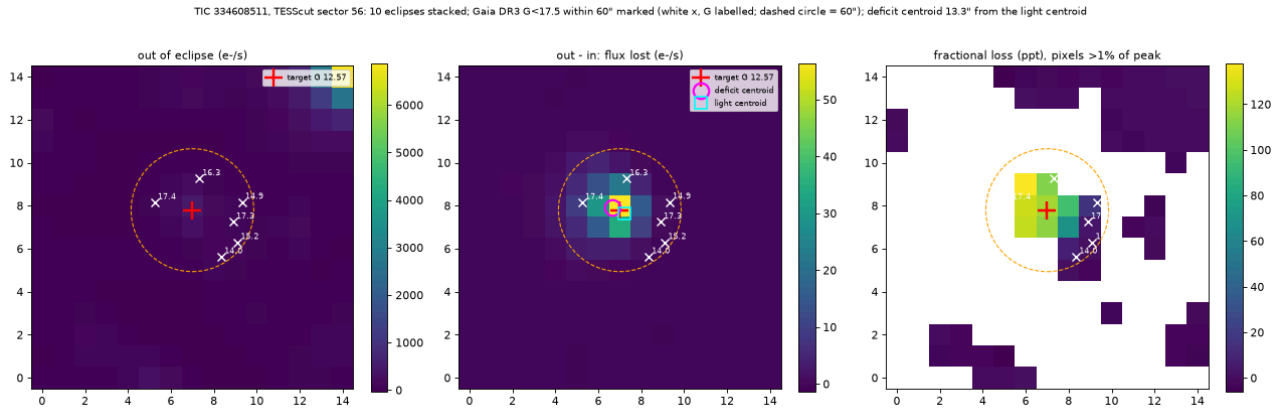
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy to refine physical parameters.



R625; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R625; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 334608511: CONFIRMED

Full bounded physical review: **COMPLETE**. Twenty-two repeated narrow V-shaped eclipses and unequal eccentric secondary near0.54790. Deficit is near target; capable51.33arcsec neighbor is separated from deficit core. Bright field source shifts overall field light; whole-field PSF residual is descriptive.

Raw morphology was inspected in raw-events.png and phase-review.png, including P and2P parity views. 22 free-center trapezoid timings survive finite, identifiable center-covariance checks. Quality-zero raw QLP SAP_FLUX was used; predictions only guide fitting windows. This is representative available evidence, not an all-event/all-sector census.

Measured same-kind eclipse ephemeris: P=2.59575932638 +/- 0.00000455178 d; reference BTJD=1741.53103860. Epoch kind: Deeper primary eclipse, established by unequal observed secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced timing chi-square=0.23274659979143536; zero degrees of freedom would be null. All selected timings follow the named repeating eclipse family. Epoch kind above identifies the deeper primary only where an unequal secondary is established. Model depth is the realized central loss d*min(1,w/(2*k)); extrapolated amplitude and ingress are retained separately. The reference is the fitted first selected cycle, not a Gaia prediction. Within-sector cycle recurrence and parity depths are in ../alias-audit.json. Opposite-phase sampled coverage, quality-zero core/sideband counts and flagged counts are in ../half-period-coverage.json. For138080650, orbital cycles are half the even historical SAP cycles; excluded odd-cycle secondaries remain in measurements.json. Unequal observed secondary depths exclude mixing eclipse kinds into P/2 where established; TIC138854818 instead has sampled opposite-phase coverage without a comparable11ppt primary; the P/2 and2P raw views were inspected. No exhaustive blind alias search is asserted.

Narrow shallower secondary visible in raw pooled windows. Depth is the realized center deficit of the fitted shape, with its amplitude parameter retained separately in the JSON. Measured pooled secondary phase=0.5478983, formal phase covariance error=0.00017364; depth=18.366 ppt. Formal pooled

trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Search-guide provenance: Gaia conjunction search guide, adjusted from raw phase morphology for435209690 and58825156. Final phase is a free-center fit to observed raw flux..

Fresh difference image: sector56, 10 complete core/sideband events. Target pixel=[7.001818234095108, 7.791746622719696]; deficit centroid=[6.662260698458553, 7.90808961222017]; deficit-target distance=0.3589 pixel. Deficit-light offset=13.255 arcsec is a descriptive field-light displacement, not an arbitrary veto. Fractional PSF-model residual=0.5150 is not a calibrated localization uncertainty. Every flux-capable Gaia source in the60-arcsec list was inspected against the image. Attribution conclusion: Twenty-two repeated narrow V-shaped eclipses and unequal eccentric secondary near0.54790. Deficit is near target; capable51.33arcsec neighbor is separated from deficit core. Bright field source shifts overall field light; whole-field PSF residual is descriptive.

Capable Gaia source	Separation arcsec	G	Intrinsic required eclipse proxy	Pixel coordinate	Diluted loss at pixel
220335217457925 4016	51.330	13.99 5	0.6676	[8.337374232597103, 5.624844474329464]	0.0077683258206 68174

The pairwise G-band required intrinsic loss is an approximate brightness screen, with passband, aperture and blend assumptions. It cannot be directly compared to the diluted fractional loss in a pixel. Noncapable neighbors and all coordinates remain in fresh-pixels/diff_fresh.json and gaia.json. TESS resolution does not exclude unlisted unresolved companions.

Alternative reduction: fresh background-subtracted TESScut1x1/3x3/5x5 target apertures have depths 118.830/112.145/91.764 ppt versus fitted raw QLP median141.395 ppt. Individual cutout event depths are retained in the pixel JSON. Cutout sector is a distinct observing occasion on the same TESS instrument relative to retained QLP sectors. This alternative reduction shares the FFI photons of that sector; no same-sector QLP comparison claimed. Distinct sectors are observing occasions on one instrument. Aperture/depth differences reflect dilution, background and extraction response; no intrinsic depth inferred.

Gaia DR3 source2203352243298726912: P=2.59568478 +/-0.00015863 d, e=0.129273, omega=284.67670 degrees, periastron BTJD=388.0359118. Closest edge-on conjunction offset=-0.018830 d (-0.007254 phase), source-specific full-covariance linear uncertainty=0.080700 d. Full correlation ordering P,gamma,K,e,omega,Tp, selected indices0,3,4,5. This linear approximation is descriptive, not a sigma veto. Both conjunctions and near-reference projection are retained in ../gaia-geometry.json. Mass function=0.0572205 solar masses, assumed M1=1.75, minimum M2=0.70039; approximate central duration=5.556 h versus observed3.456 h. TIC radius and main-sequence companion scaling are assumptions, not a geometry fit; impact parameter and stellar evolution affect duration. Agreement cannot establish ownership among unresolved sources.

Sector	Same-kind cycle	Measured midpoint BTJD	Formal center error d	Input error d	Depth ppt
16	0	1741.52521058	0.00030794435	0.010416753	120.833
16	2	1746.71382496	0.00016839445	0.010416753	119.157
16	5	1754.50434004	0.00027836181	0.010416753	122.973
16	7	1759.69275330	0.00046011885	0.010416753	123.890
16	8	1762.29455994	0.0002646679	0.010416753	125.505
76	627	3369.07263641	0.0002829565	0.0011573825	137.073
76	628	3371.66828502	0.0002836135	0.0011573825	139.278
76	630	3376.85968927	0.00028324199	0.0011573825	141.714
76	631	3379.45518699	0.00022791105	0.0011573825	141.248
76	632	3382.05149735	0.00029170389	0.0011573825	140.104
76	633	3384.64727709	0.00027322329	0.0011573825	142.531

Sector	Same-kind cycle	Measured midpoint BTJD	Formal center error d	Input error d	Depth ppt
76	634	3387.24272512	0.00026346396	0.0011573825	142.595
76	635	3389.83839201	0.00028862973	0.0011573825	144.844
76	636	3392.43415991	0.00027730629	0.0011573825	140.217
85	721	3613.07328881	0.00034635021	0.0011573807	144.257
85	722	3615.66889466	0.00027313689	0.0011573807	142.000
85	723	3618.26432332	0.00022790018	0.0011573807	142.042
85	724	3620.86012557	0.00019450237	0.0011573807	141.541
85	726	3626.05290437	0.00030272713	0.0011573807	141.865
85	727	3628.64799536	0.00025530399	0.0011573807	142.111
85	728	3631.24305151	0.00021575833	0.0011573807	140.212
85	729	3633.83923152	0.00021016811	0.0011573807	142.893

Next check: Optional resolved photometry and spectroscopy to refine physical parameters.

Catalogue status is separate from physical verdict; shared-catalogues/coverage-matrix.csv retains the seven scoped groups. Known recoveries are not new additions; no exhaustive priority claim. Sources and identity are checked in ../identity-check.json and ../source-hashes.json. Numeric artifacts, original receipts and outputs are frozen in ../SHA256SUMS.json; independent helper review is ../helper-audit.md.

Stable review key: R625.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s56.json - historical/diff_s56.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

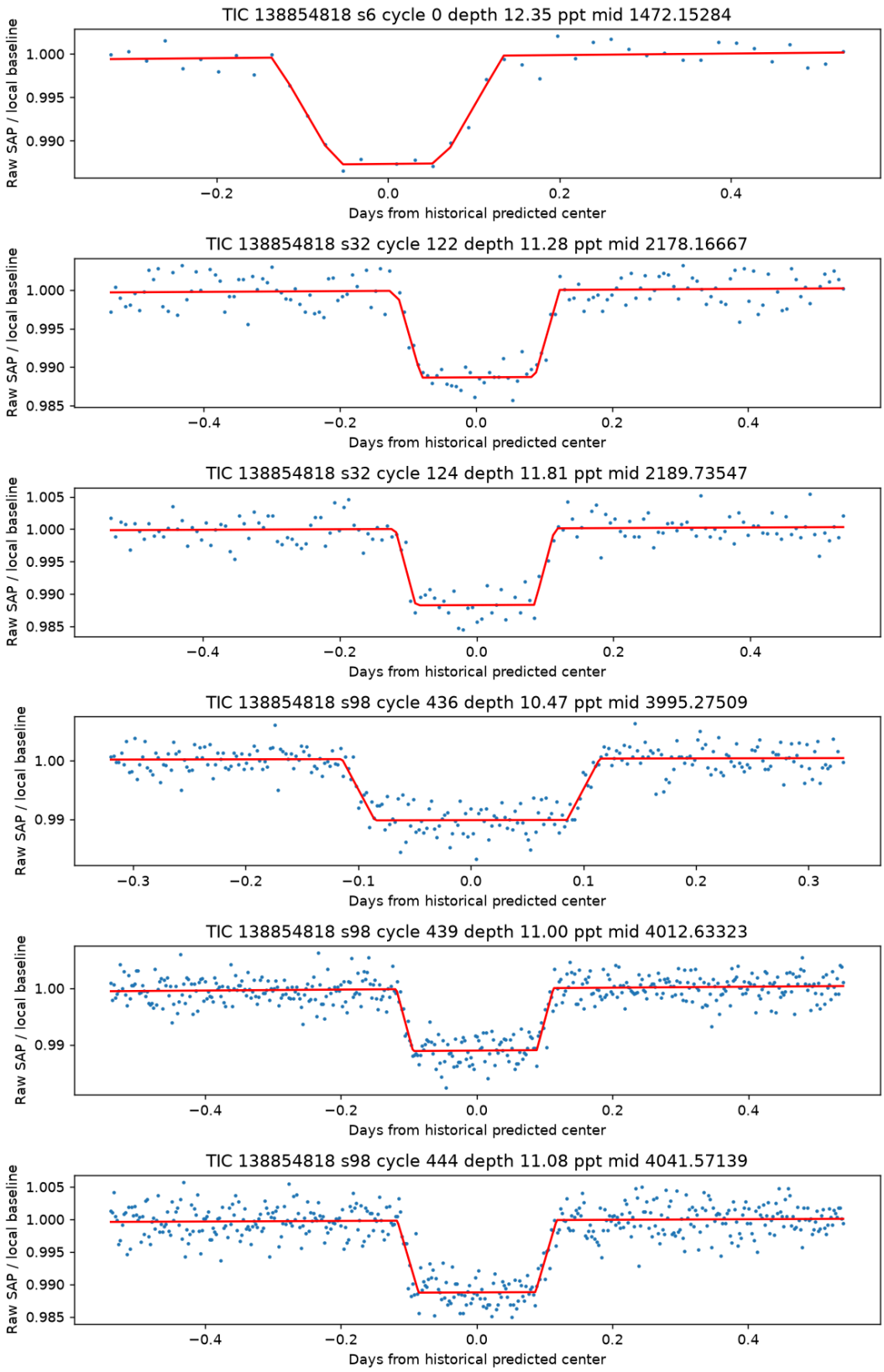
R626. TIC 138854818

physical_verdict: CONFIRMED

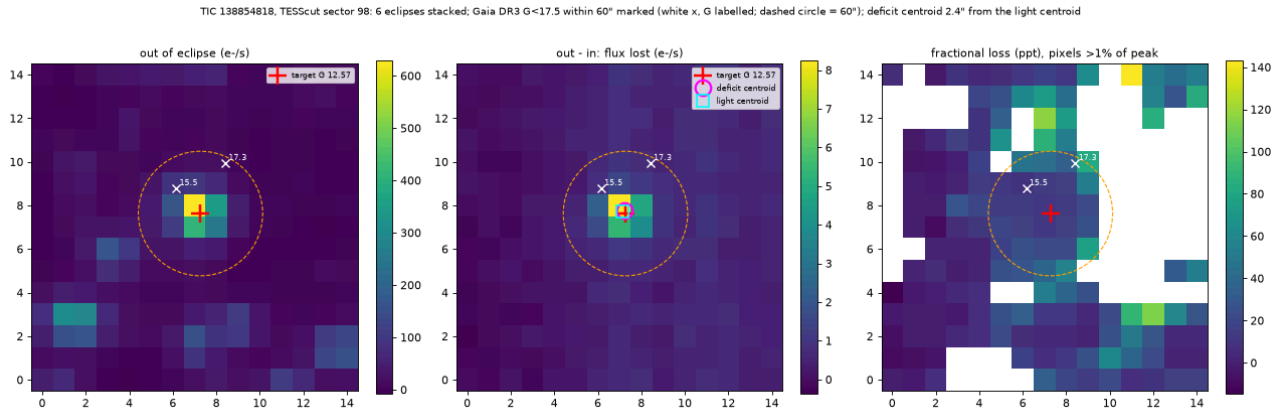
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy to refine physical parameters.



R626; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R626; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 138854818: CONFIRMED

Full bounded physical review: **COMPLETE**. Eleven consistent shallow raw eclipse timings across three sectors; representative windows show narrow repeatable11ppt dips despite substantial sector98 baseline noise. Compact sector98 deficit favors target over capable33.33/53.79arcsec neighbors. No secure secondary; opposite-phase coverage lacks a comparable primary, without excluding shallow dips.

Raw morphology was inspected in raw-events.png and phase-review.png, including P and2P parity views. 11 free-center trapezoid timings survive finite, identifiable center-covariance checks. Quality-zero raw QLP SAP_FLUX was used; predictions only guide fitting windows. This is representative available evidence, not an all-event/all-sector census.

Measured same-kind eclipse ephemeris: P=5.78696892497 +/- 0.00000555101 d; reference BTJD=1472.15551942. Epoch kind: Selected repeating eclipse; deeper/secondary identity not independently established. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced timing chi-square=1.0123785624289992; zero degrees of freedom would be null. All selected timings follow the named repeating eclipse family. Epoch kind above identifies the deeper primary only where an unequal secondary is established. Model depth is the realized central loss d*min(1,w/(2*k)); extrapolated amplitude and ingress are retained separately. The reference is the fitted first selected cycle, not a Gaia prediction. Within-sector cycle recurrence and parity depths are in ../alias-audit.json. Opposite-phase sampled coverage, quality-zero core/sideband counts and flagged counts are in ../half-period-coverage.json. For138080650, orbital cycles are half the even historical SAP cycles; excluded odd-cycle secondaries remain in measurements.json. Unequal observed secondary depths exclude mixing eclipse kinds into P/2 where established; TIC138854818 instead has sampled opposite-phase coverage without a comparable11ppt primary; the P/2 and2P raw views were inspected. No exhaustive blind alias search is asserted.

No secure secondary; noisy low-amplitude exploratory fit not interpreted as eclipse. A fitted low-amplitude pooled model is not evidence of a secondary; flagged and missing coverage are not clean nulls.

Fresh difference image: sector98, 6 complete core/sideband events. Target pixel=[7.272864382234976, 7.6262271300283455]; deficit centroid=[7.214408618368344, 7.7652239760118125]; deficit-target distance=0.1508 pixel. Deficit-light offset=2.381 arcsec is a descriptive field-light displacement, not an arbitrary veto. Fractional PSF-model residual=0.3766 is not a calibrated localization uncertainty. Every flux-capable Gaia source in the60-arcsec list was inspected against the image. Attribution conclusion: Eleven consistent shallow raw eclipse timings across three sectors; representative windows show narrow repeatable11ppt dips despite substantial sector98 baseline noise. Compact sector98 deficit favors target over capable33.33/53.79arcsec neighbors. No secure secondary; opposite-phase coverage lacks a comparable primary, without excluding shallow dips.

Capable Gaia source	Separation arcsec	G	Intrinsic required eclipse proxy	Pixel coordinate	Diluted loss at pixel
3236840942297477248	33.328	15.492	0.1756	[6.154179124252758, 8.77044276259647]	0.00978607236496007
3236840976662227840	53.787	17.315	0.8925	[8.41092364713651, 9.92785875424227]	0.03427849783888747

The pairwise G-band required intrinsic loss is an approximate brightness screen, with passband, aperture and blend assumptions. It cannot be directly compared to the diluted fractional loss in a pixel. Noncapable neighbors and all coordinates remain in fresh-pixels/diff_fresh.json and gaia.json. TESS resolution does not exclude unlisted unresolved companions.

Alternative reduction: fresh background-subtracted TESScut1x1/3x3/5x5 target apertures have depths 13.106/13.815/15.933 ppt versus fitted raw QLP median11.145 ppt. Individual cutout event depths are retained in the pixel JSON. Same-sector QLP and cutout extraction share TESS photons and test processing robustness. Distinct sectors are observing occasions on one instrument. Aperture/depth differences reflect dilution, background and extraction response; no intrinsic depth inferred.

Gaia DR3 source3236840946592866560: P=5.78750160 +/-0.00395146 d, e=0.133831, omega=16.58694 degrees, periastron BTJD=388.7994091. Closest edge-on conjunction offset=0.142155 d (0.024562 phase), source-specific full-covariance linear uncertainty=0.449613 d. Full correlation ordering P,gamma,K,e,omega,Tp, selected indices0,3,4,5. This linear approximation is descriptive, not a sigma veto. Both conjunctions and near-reference projection are retained in ../gaia-geometry.json. Mass function=0.00156602 solar masses, assumed M1=1.34, minimum M2=0.1516; approximate central duration=5.575 h versus observed5.671 h. TIC radius and main-sequence companion scaling are assumptions, not a geometry fit; impact parameter and stellar evolution affect duration. Agreement cannot establish ownership among unresolved sources.

Sector	Same-kind cycle	Measured midpoint BTJD	Formal center error d	Input error d	Depth ppt
6	0	1472.15283776	0.0029376235	0.010416469	12.351
6	2	1483.72011146	0.0020186356	0.010416469	11.145
32	122	2178.16667013	0.0022593999	0.003472288	11.283
32	123	2183.95550115	0.0021561917	0.003472288	11.042
32	124	2189.73546800	0.0016507744	0.003472288	11.806
32	125	2195.52903866	0.0014345975	0.003472288	11.878
98	436	3995.27508629	0.0011266372	0.0011574276	10.472
98	437	4001.06030630	0.0012774144	0.0012774144	10.212
98	439	4012.63323179	0.00092731601	0.0011574276	10.997
98	442	4029.99431863	0.0019931673	0.0019931673	11.495
98	444	4041.57138881	0.00122994	0.00122994	11.084

Next check: Optional resolved photometry and spectroscopy to refine physical parameters.

Catalogue status is separate from physical verdict; shared-catalogues/coverage-matrix.csv retains the seven scoped groups. Known recoveries are not new additions; no exhaustive priority claim. Sources and identity are checked in ../identity-check.json and ../source-hashes.json. Numeric artifacts, original receipts and outputs are frozen in ../SHA256SUMS.json; independent helper review is ../helper-audit.md.

Stable review key: R626.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s98.json - historical/diff_s98.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

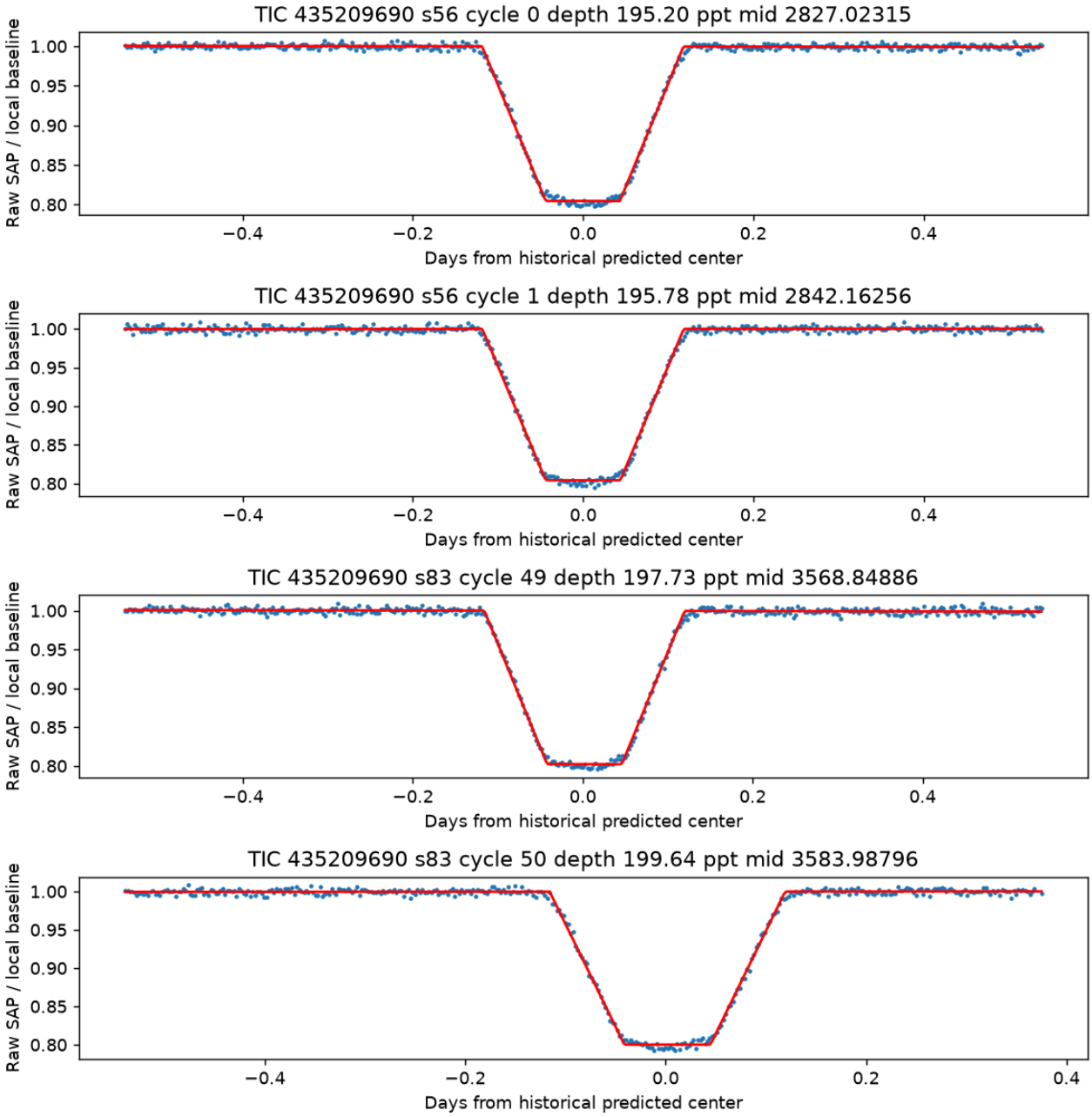
R627. TIC 435209690

physical_verdict: CONFIRMED

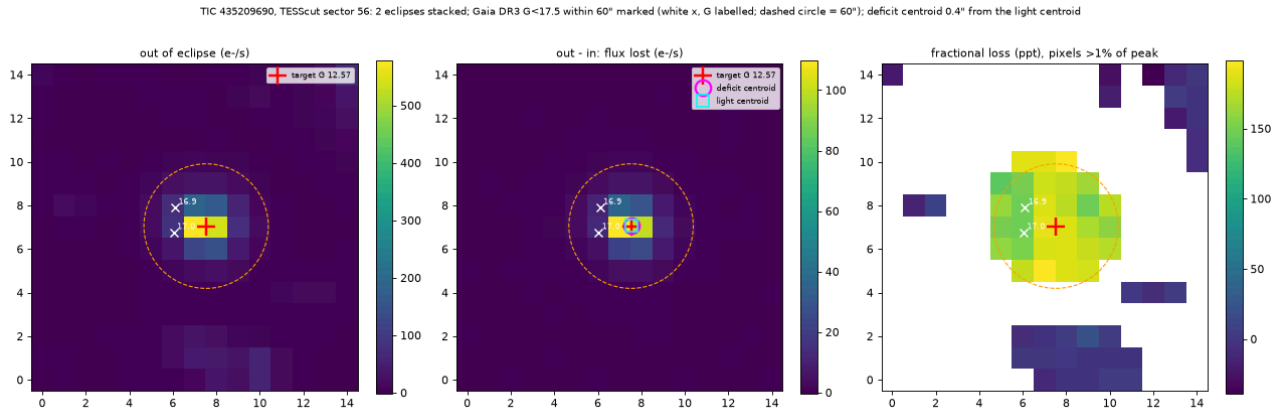
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy to refine physical parameters.



R627; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R627; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 435209690: CONFIRMED

Full bounded physical review: **COMPLETE**. Four deep flat-bottomed measured raw eclipses in sectors56/83, including consecutive same-kind pairs in both sectors. These resolve the long-baseline integer count. Shallower secondary is measured near0.50613, not the Gaia-guided initial search phase. No capable Gaia neighbor within60arcsec; compact target-centered deficit.

Raw morphology was inspected in raw-events.png and phase-review.png, including P and2P parity views. 4 free-center trapezoid timings survive finite, identifiable center-covariance checks. Quality-zero raw QLP SAP_FLUX was used; predictions only guide fitting windows. This is representative available evidence, not an all-event/all-sector census.

Measured same-kind eclipse ephemeris: P=15.13929711422 +/- 0.00002361560 d; reference BTJD=2827.02320573. Epoch kind: Deeper primary eclipse, established by unequal observed secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced timing chi-square=0.010032924595660919; zero degrees of freedom would be null. All selected timings follow the named repeating eclipse family. Epoch kind above identifies the deeper primary only where an unequal secondary is established. Model depth is the realized central loss d*min(1,w/(2*k)); extrapolated amplitude and ingress are retained separately. The reference is the fitted first selected cycle, not a Gaia prediction. Within-sector cycle recurrence and parity depths are in ../alias-audit.json. Opposite-phase sampled coverage, quality-zero core/sideband counts and flagged counts are in ../half-period-coverage.json. For138080650, orbital cycles are half the even historical SAP cycles; excluded odd-cycle secondaries remain in measurements.json. Unequal observed secondary depths exclude mixing eclipse kinds into P/2 where established; TIC138854818 instead has sampled opposite-phase coverage without a comparable11ppt primary; the P/2 and2P raw views were inspected. No exhaustive blind alias search is asserted.

Narrow shallower secondary visible in raw pooled windows. Depth is the realized center deficit of the fitted shape, with its amplitude parameter retained separately in the JSON. Measured pooled secondary phase=0.5061266, formal phase covariance error=5.306e-05; depth=22.994 ppt. Formal pooled

trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Search-guide provenance: Gaia conjunction search guide, adjusted from raw phase morphology for435209690 and58825156. Final phase is a free-center fit to observed raw flux..

Fresh difference image: sector56, 2 complete core/sideband events. Target pixel=[7.529072749676924, 7.048914551382477]; deficit centroid=[7.542996432180552, 7.0785451329092695]; deficit-target distance=0.0327 pixel. Deficit-light offset=0.387 arcsec is a descriptive field-light displacement, not an arbitrary veto. Fractional PSF-model residual=0.0455 is not a calibrated localization uncertainty. Every flux-capable Gaia source in the60-arcsec list was inspected against the image. Attribution conclusion: Four deep flat-bottomed measured raw eclipses in sectors56/83, including consecutive same-kind pairs in both sectors. These resolve the long-baseline integer count. Shallower secondary is measured near0.50613, not the Gaia-guided initial search phase. No capable Gaia neighbor within60arcsec; compact target-centered deficit.

Capable Gaia source	Separation arcsec	G	Intrinsic required eclipse proxy	Pixel coordinate	Diluted loss at pixel
None					

The pairwise G-band required intrinsic loss is an approximate brightness screen, with passband, aperture and blend assumptions. It cannot be directly compared to the diluted fractional loss in a pixel. Noncapable neighbors and all coordinates remain in fresh-pixels/diff_fresh.json and gaia.json. TESS resolution does not exclude unlisted unresolved companions.

Alternative reduction: fresh background-subtracted TESScut1x1/3x3/5x5 target apertures have depths 185.719/184.070/180.128 ppt versus fitted raw QLP median196.752 ppt. Individual cutout event depths are retained in the pixel JSON. Same-sector QLP and cutout extraction share TESS photons and test processing robustness. Distinct sectors are observing occasions on one instrument. Aperture/depth differences reflect dilution, background and extraction response; no intrinsic depth inferred.

Gaia DR3 source2813025595435362944: P=15.13820213 +/-0.00456877 d, e=0.046168, omega=123.82564 degrees, periastron BTJD=391.1103205. Closest edge-on conjunction offset=-0.035622 d (-0.002353 phase), source-specific full-covariance linear uncertainty=0.850911 d. Full correlation ordering P,gamma,K,e,omega,Tp, selected indices0,3,4,5. This linear approximation is descriptive, not a sigma veto. Both conjunctions and near-reference projection are retained in ../gaia-geometry.json. Mass function=0.0456945 solar masses, assumed M1=0.99, minimum M2=0.45747; approximate central duration=6.721 h versus observed5.671 h. TIC radius and main-sequence companion scaling are assumptions, not a geometry fit; impact parameter and stellar evolution affect duration. Agreement cannot establish ownership among unresolved sources.

Sector	Same-kind cycle	Measured midpoint BTJD	Formal center error d	Input error d	Depth ppt
56	0	2827.02314613	0.00012635961	0.0011574122	195.203
56	1	2842.16256160	0.00013731954	0.0011574122	195.778
83	49	3568.84886442	0.00014979004	0.0011573982	197.726
83	50	3583.98796216	0.00014215494	0.0011573982	199.636

Next check: Optional resolved photometry and spectroscopy to refine physical parameters.

Catalogue status is separate from physical verdict; shared-catalogues/coverage-matrix.csv retains the seven scoped groups. Known recoveries are not new additions; no exhaustive priority claim. Sources and identity are checked in ../identity-check.json and ../source-hashes.json. Numeric artifacts, original receipts and outputs are frozen in ../SHA256SUMS.json; independent helper review is ../helper-audit.md.

Stable review key: R627.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s56.json - historical/diff_s56.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

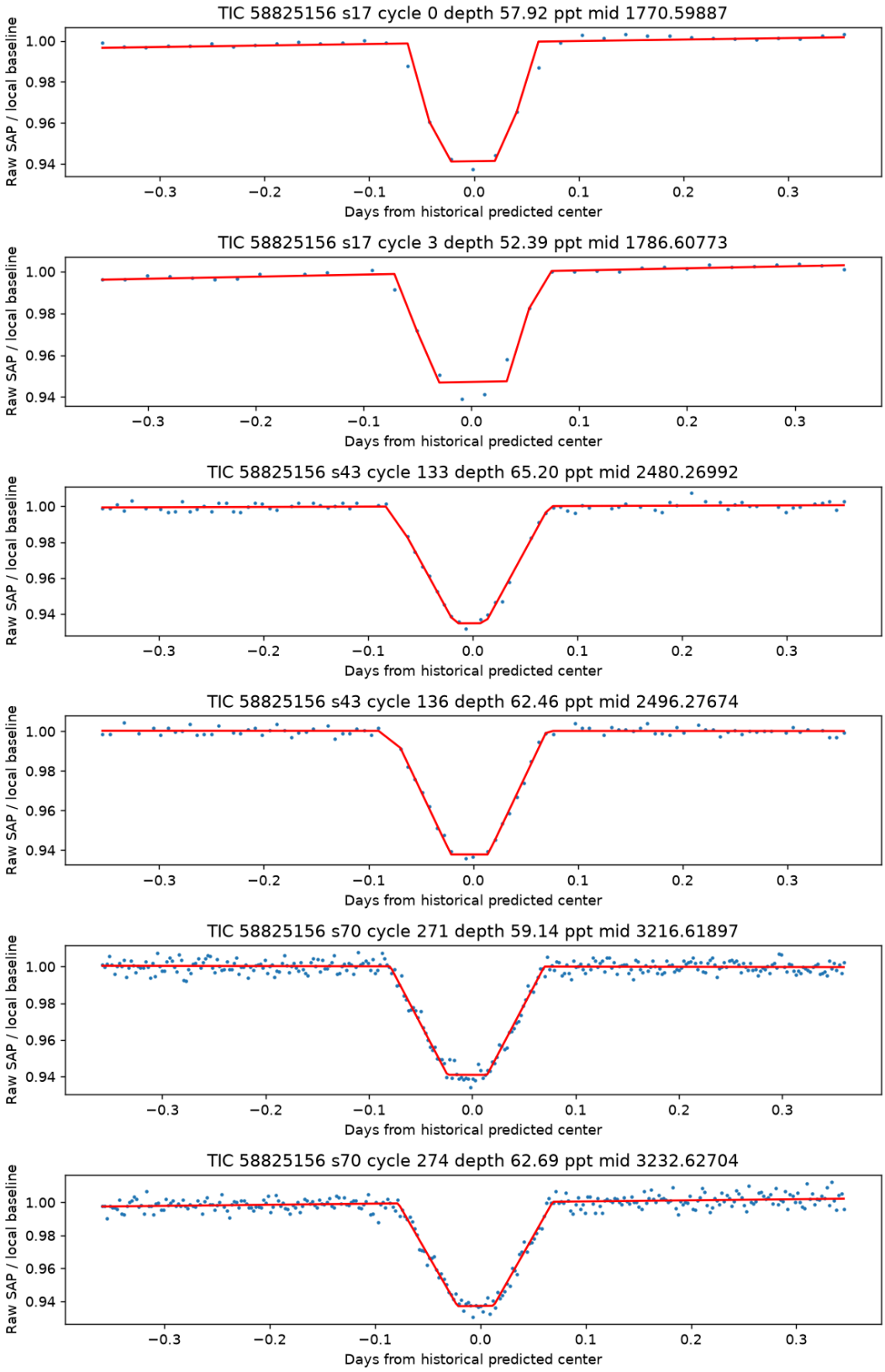
R628. TIC 58825156

physical_verdict: PROBABLE

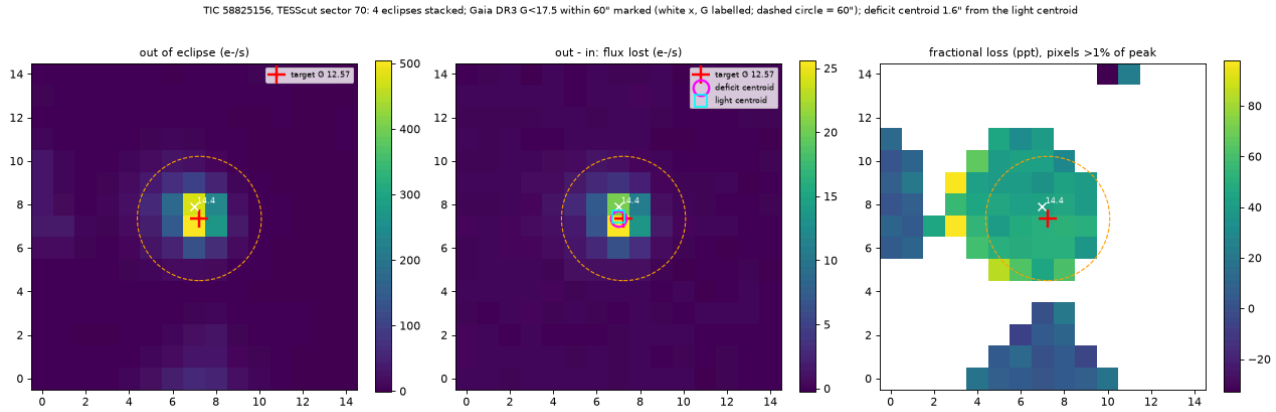
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or multi-source PRF localization for target and12.43-arcsec neighbor.



R628; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R628; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 58825156: PROBABLE

Full bounded physical review: **COMPLETE**. Twelve repeated raw eclipses and shallower secondary near0.50704 are real. Capable Gaia2776848815318667392 at12.4345arcsec overlaps the TESS footprint; the0.227pixel deficit offset and broad fractional-loss pattern do not securely separate ownership. Attribution remains unresolved.

Raw morphology was inspected in raw-events.png and phase-review.png, including P and2P parity views. 12 free-center trapezoid timings survive finite, identifiable center-covariance checks. Quality-zero raw QLP SAP_FLUX was used; predictions only guide fitting windows. This is representative available evidence, not an all-event/all-sector census.

Measured same-kind eclipse ephemeris: P=5.33586901784 +/- 0.00001143038 d; reference BTJD=1770.59913511. Epoch kind: Deeper primary eclipse, established by unequal observed secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced timing chi-square=0.15085620427980728; zero degrees of freedom would be null. All selected timings follow the named repeating eclipse family. Epoch kind above identifies the deeper primary only where an unequal secondary is established. Model depth is the realized central loss d*min(1,w/(2*k)); extrapolated amplitude and ingress are retained separately. The reference is the fitted first selected cycle, not a Gaia prediction. Within-sector cycle recurrence and parity depths are in ../alias-audit.json. Opposite-phase sampled coverage, quality-zero core/sideband counts and flagged counts are in ../half-period-coverage.json. For138080650, orbital cycles are half the even historical SAP cycles; excluded odd-cycle secondaries remain in measurements.json. Unequal observed secondary depths exclude mixing eclipse kinds into P/2 where established; TIC138854818 instead has sampled opposite-phase coverage without a comparable11ppt primary; the P/2 and2P raw views were inspected. No exhaustive blind alias search is asserted.

Narrow shallower secondary visible in raw pooled windows. Depth is the realized center deficit of the fitted shape, with its amplitude parameter retained separately in the JSON. Measured pooled secondary phase=0.5070393, formal phase covariance error=0.00014634; depth=18.854 ppt. Formal pooled

trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Search-guide provenance: Gaia conjunction search guide, adjusted from raw phase morphology for435209690 and58825156. Final phase is a free-center fit to observed raw flux..

Fresh difference image: sector70, 4 complete core/sideband events. Target pixel=[7.237332270066636, 7.353260504601428]; deficit centroid=[7.010190963174136, 7.354613303020761]; deficit-target distance=0.2271 pixel. Deficit-light offset=1.631 arcsec is a descriptive field-light displacement, not an arbitrary veto. Fractional PSF-model residual=0.0923 is not a calibrated localization uncertainty. Every flux-capable Gaia source in the60-arcsec list was inspected against the image. Attribution conclusion: Twelve repeated raw eclipses and shallower secondary near0.50704 are real. Capable Gaia2776848815318667392 at12.4345arcsec overlaps the TESS footprint; the0.227pixel deficit offset and broad fractional-loss pattern do not securely separate ownership. Attribution remains unresolved.

Capable Gaia source	Separation arcsec	G	Intrinsic required eclipse proxy	Pixel coordinate	Diluted loss at pixel
2776848815318667392	12.435	14.390	0.3864	[6.978450099330313, 7.899198860435133]	0.04285828942480618

The pairwise G-band required intrinsic loss is an approximate brightness screen, with passband, aperture and blend assumptions. It cannot be directly compared to the diluted fractional loss in a pixel. Noncapable neighbors and all coordinates remain in fresh-pixels/diff_fresh.json and gaia.json. TESS resolution does not exclude unlisted unresolved companions.

Alternative reduction: fresh background-subtracted TESScut1x1/3x3/5x5 target apertures have depths 50.749/47.409/46.482 ppt versus fitted raw QLP median60.933 ppt. Individual cutout event depths are retained in the pixel JSON. Same-sector QLP and cutout extraction share TESS photons and test processing robustness. Distinct sectors are observing occasions on one instrument. Aperture/depth differences reflect dilution, background and extraction response; no intrinsic depth inferred.

Gaia DR3 source2776848815318667264: P=5.33616672 +/-0.00070417 d, e=0.074586, omega=42.56272 degrees, periastron BTJD=387.9881972. Closest edge-on conjunction offset=-0.069542 d (-0.013032 phase), source-specific full-covariance linear uncertainty=0.210857 d. Full correlation ordering P,gamma,K,e,omega,Tp, selected indices0,3,4,5. This linear approximation is descriptive, not a sigma veto. Both conjunctions and near-reference projection are retained in ../gaia-geometry.json. Mass function=0.0687954 solar masses, assumed M1=1.06, minimum M2=0.56677; approximate central duration=5.286 h versus observed3.519 h. TIC radius and main-sequence companion scaling are assumptions, not a geometry fit; impact parameter and stellar evolution affect duration. Agreement cannot establish ownership among unresolved sources.

Sector	Same-kind cycle	Measured midpoint BTJD	Formal center error d	Input error d	Depth ppt
17	0	1770.59887260	0.00083411997	0.010416416	57.923
17	2	1781.27370302	0.0010413178	0.010416416	53.411
17	3	1786.60773263	0.0010929274	0.010416416	52.386
43	132	2474.93378733	0.00040812712	0.0034723019	63.424
43	133	2480.26991828	0.00062222378	0.0034723019	65.203
43	135	2490.94111524	0.00051675384	0.0034723019	63.548
43	136	2496.27673503	0.00041133702	0.0034723019	62.456
70	270	3211.28374948	0.00044686273	0.0011574215	61.086
70	271	3216.61897133	0.00043502559	0.0011574215	59.135
70	272	3221.95665668	0.00048004204	0.0011574215	60.781
70	273	3227.29116055	0.00046978257	0.0011574215	60.315
70	274	3232.62704426	0.00044153643	0.0011574215	62.692

Next check: Resolved eclipse photometry or multi-source PRF localization for target and12.43-arcsec neighbor.

Catalogue status is separate from physical verdict; shared-catalogues/coverage-matrix.csv retains the seven scoped groups. Known recoveries are not new additions; no exhaustive priority claim. Sources and identity are checked in ../identity-check.json and ../source-hashes.json. Numeric artifacts, original receipts and outputs are frozen in ../SHA256SUMS.json; independent helper review is ../helper-audit.md.

Stable review key: R628.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s70.json - historical/diff_s70.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

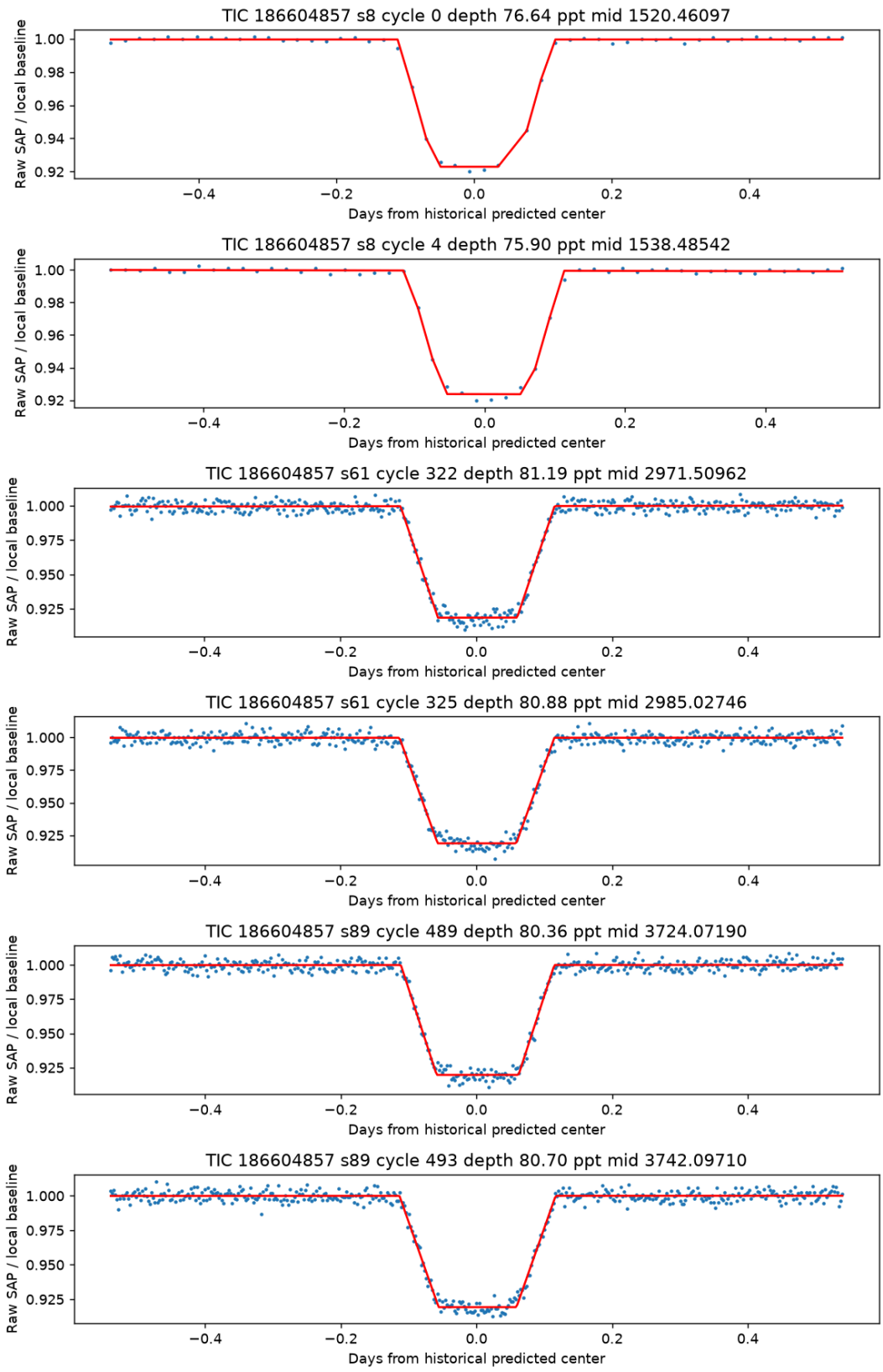
R629. TIC 186604857

physical_verdict: CONFIRMED

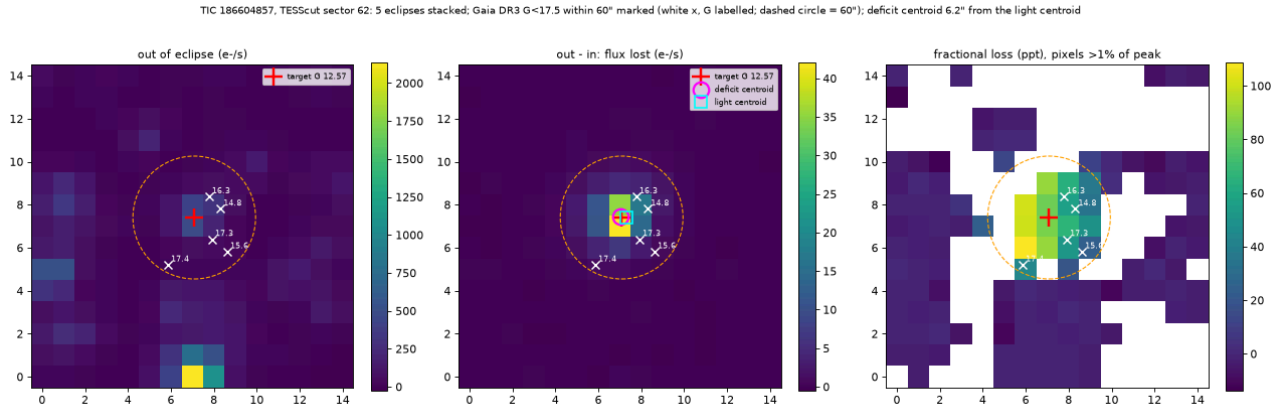
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy to refine physical parameters.



R629; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R629; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 186604857: CONFIRMED

Full bounded physical review: **COMPLETE**. Twelve repeated narrow flat-bottomed raw eclipses and a shallower secondary near0.49975. The capable26.13arcsec neighbor is displaced from the compact target-centered deficit.

Raw morphology was inspected in raw-events.png and phase-review.png, including P and2P parity views. 12 free-center trapezoid timings survive finite, identifiable center-covariance checks. Quality-zero raw QLP SAP_FLUX was used; predictions only guide fitting windows. This is representative available evidence, not an all-event/all-sector census.

Measured same-kind eclipse ephemeris: P=4.50636419933 +/- 0.00000441800 d; reference BTJD=1520.45971853. Epoch kind: Deeper primary eclipse, established by unequal observed secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced timing chi-square=0.08279212701008627; zero degrees of freedom would be null. All selected timings follow the named repeating eclipse family. Epoch kind above identifies the deeper primary only where an unequal secondary is established. Model depth is the realized central loss d*min(1,w/(2*k)); extrapolated amplitude and ingress are retained separately. The reference is the fitted first selected cycle, not a Gaia prediction. Within-sector cycle recurrence and parity depths are in ../alias-audit.json. Opposite-phase sampled coverage, quality-zero core/sideband counts and flagged counts are in ../half-period-coverage.json. For138080650, orbital cycles are half the even historical SAP cycles; excluded odd-cycle secondaries remain in measurements.json. Unequal observed secondary depths exclude mixing eclipse kinds into P/2 where established; TIC138854818 instead has sampled opposite-phase coverage without a comparable11ppt primary; the P/2 and2P raw views were inspected. No exhaustive blind alias search is asserted.

Narrow shallower secondary visible in raw pooled windows. Depth is the realized center deficit of the fitted shape, with its amplitude parameter retained separately in the JSON. Measured pooled secondary phase=0.4997466, formal phase covariance error=0.00024743; depth=13.297 ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Search-guide provenance: Gaia conjunction search guide, adjusted from raw phase morphology for435209690 and58825156. Final phase is a free-center fit to observed raw flux..

Fresh difference image: sector62, 5 complete core/sideband events. Target pixel=[7.087314081896029, 7.402786465078902]; deficit centroid=[7.034190069024928, 7.451240640969623]; deficit-target distance=0.0719 pixel. Deficit-light offset=6.180 arcsec is a descriptive field-light displacement, not an arbitrary veto. Fractional PSF-model residual=0.5558 is not a calibrated localization uncertainty. Every flux-capable Gaia source in the60-arcsec list was inspected against the image. Attribution conclusion: Twelve repeated narrow flat-bottomed raw eclipses and a shallower secondary near0.49975. The capable26.13arcsec neighbor is displaced from the compact target-centered deficit.

Capable Gaia source	Separation arcsec	G	Intrinsic required eclipse proxy	Pixel coordinate	Diluted loss at pixel
562865909931774 5408	26.135	14.77 4	0.6939	[8.293654979180822, 7.800959998836534]	0.05634236631086 9856

The pairwise G-band required intrinsic loss is an approximate brightness screen, with passband, aperture and blend assumptions. It cannot be directly compared to the diluted fractional loss in a pixel. Noncapable neighbors and all coordinates remain in fresh-pixels/diff_fresh.json and gaia.json. TESS resolution does not exclude unlisted unresolved companions.

Alternative reduction: fresh background-subtracted TESScut1x1/3x3/5x5 target apertures have depths 84.666/79.022/77.539 ppt versus fitted raw QLP median80.791 ppt. Individual cutout event depths are retained in the pixel JSON. Cutout sector is a distinct observing occasion on the same TESS instrument relative to retained QLP sectors. This alternative reduction shares the FFI photons of that sector; no same-sector QLP comparison claimed. Distinct sectors are observing occasions on one instrument. Aperture/depth differences reflect dilution, background and extraction response; no intrinsic depth inferred.

Gaia DR3 source5628658996235605376: P=4.50553381 +/-0.00034256 d, e=0.078705, omega=212.53573 degrees, periastron BTJD=390.9641217. Closest edge-on conjunction offset=0.041994 d (0.009321 phase), source-specific full-covariance linear uncertainty=0.113098 d. Full correlation ordering P,gamma,K,e,omega,Tp, selected indices0,3,4,5. This linear approximation is descriptive, not a sigma veto. Both conjunctions and near-reference projection are retained in ../gaia-geometry.json. Mass function=0.0607827 solar masses, assumed M1=1.41, minimum M2=0.63306; approximate central duration=6.447 h versus observed5.432 h. TIC radius and main-sequence companion scaling are assumptions, not a geometry fit; impact parameter and stellar evolution affect duration. Agreement cannot establish ownership among unresolved sources.

Sector	Same-kind cycle	Measured midpoint BTJD	Formal center error d	Input error d	Depth ppt
8	0	1520.46096717	0.00038171921	0.010416753	76.640
8	1	1524.96676647	0.00040669026	0.010416753	77.733
8	4	1538.48541804	0.00052321175	0.010416753	75.901
61	321	2967.00231918	0.0003906175	0.0011574313	81.438
61	322	2971.50961999	0.000341191	0.0011574313	81.186
61	324	2980.52193546	0.00036859614	0.0011574313	80.602
61	325	2985.02745882	0.00039702882	0.0011574313	80.877
89	488	3719.56552098	0.0003576789	0.0011573987	80.927
89	489	3724.07190451	0.00035512461	0.0011573987	80.364
89	490	3728.57797224	0.00033830493	0.0011573987	81.308
89	492	3737.59117233	0.0003268708	0.0011573987	81.671
89	493	3742.09709872	0.00037611639	0.0011573987	80.704

Next check: Optional resolved photometry and spectroscopy to refine physical parameters.

Catalogue status is separate from physical verdict; shared-catalogues/coverage-matrix.csv retains the seven scoped groups. Known recoveries are not new additions; no exhaustive priority claim. Sources and identity are checked in ../identity-check.json and ../source-hashes.json. Numeric artifacts, original receipts and outputs are frozen in ../SHA256SUMS.json; independent helper review is ../helper-audit.md.

Stable review key: R629.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s62.json - historical/diff_s62.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv

Batch audits and scripts. Depth correction provenance.

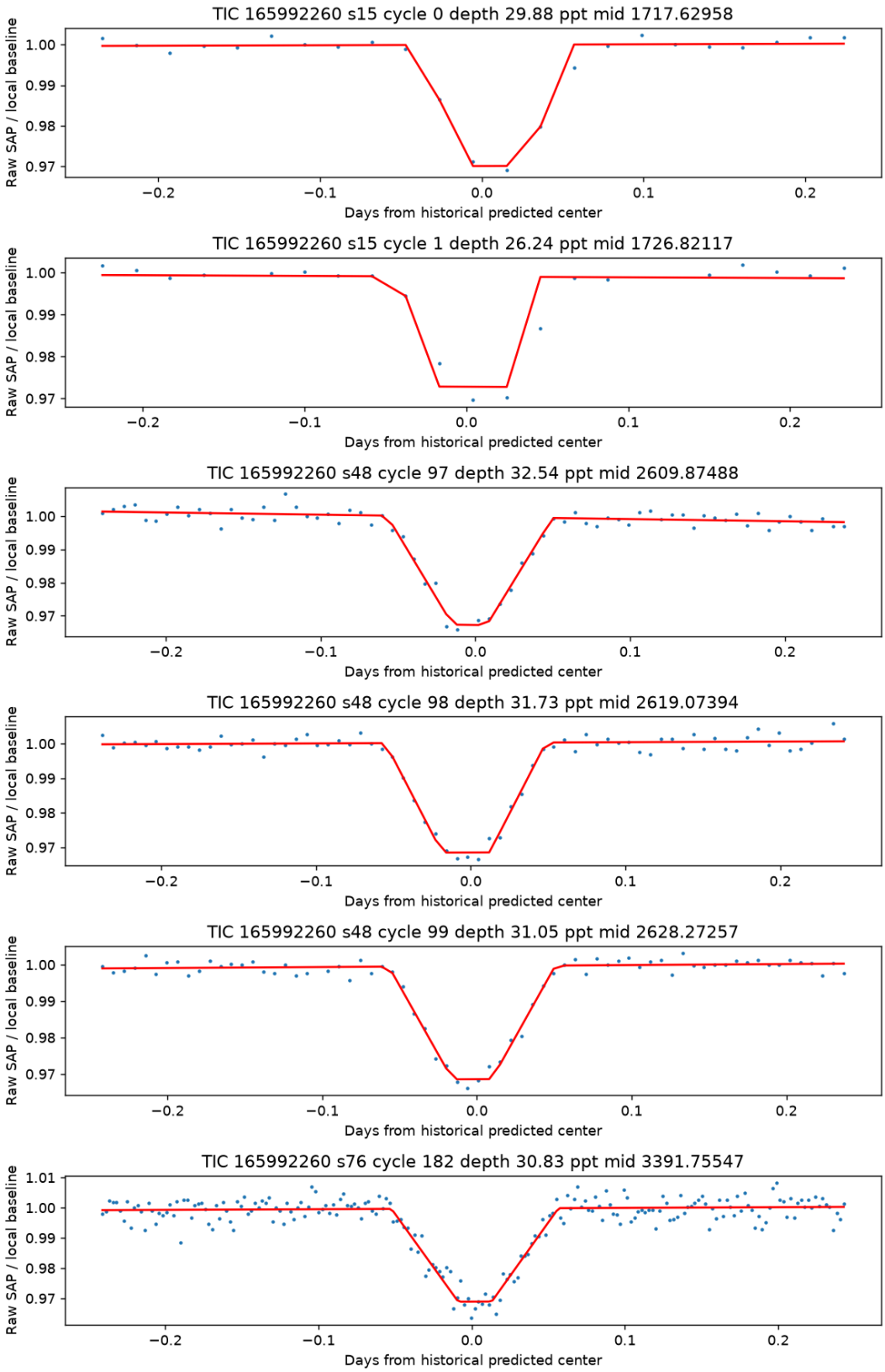
R630. TIC 165992260

physical_verdict: CONFIRMED

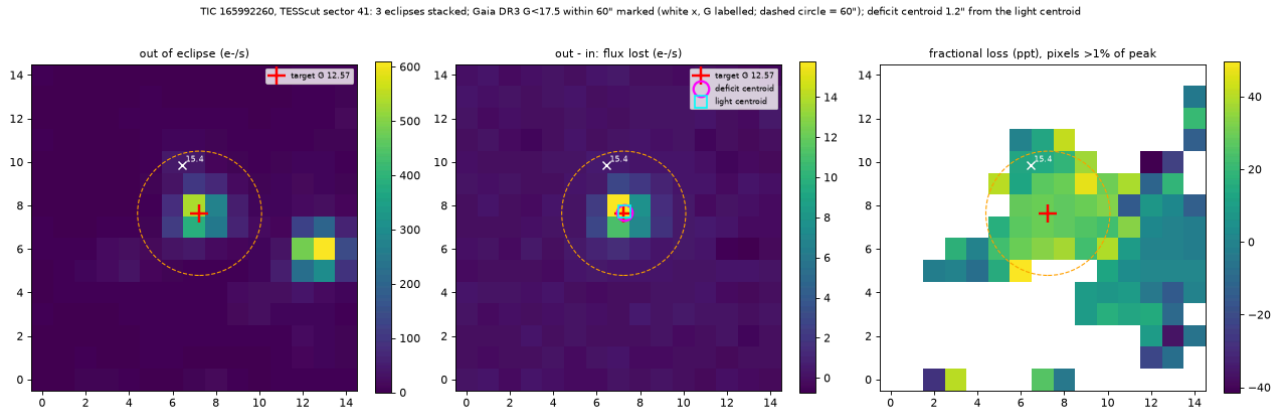
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional additional clean epochs and resolved photometry.



R630; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R630; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 165992260: CONFIRMED

Full bounded physical review: COMPLETE. Seven narrow repeated primary eclipses and unequal secondary. Capable47.42arcsec neighbor is well separated from compact deficit.

Raw QLP SAP_FLUX representative events and full-sector P/2P morphology were visually inspected. Quality-zero finite positive samples only; predictions guide windows and are never timings. 7 identifiable free-center timings define P=9.198566147122 +/- 0.000022310557d, reference BTJD=1717.615932198 +/- 0.003818860d. Epoch error meaning: Conservative linear timing-fit covariance under the stated cadence floor and scatter inflation; excludes uncalibrated shape and baseline systematics. Epoch kind: Deeper primary established by unequal observed secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced chi square=0.6032958105414482; zero degrees of freedom is null. All errors retain their named meanings; no calibrated coverage is claimed. Selected same-kind recurring eclipse family; parity and P/2 coverage checked.

Depth=31.052185ppt is the median realized central trapezoid loss d*min(1,w/(2*k)); extrapolated amplitude and ingress remain separately in measurements.json. Measured secondary phase 0.50147395, formal phase error 0.00017468241, realized depth 6.54647ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Secondary raw pooled windows, actual sectors/cycles and fitted phase are retained in secondary-windows.csv and ../secondary-phases.json. No exhaustive blind alias search asserted. ../alias-audit.json and ../half-period-coverage.json retain recurrence/parity and sampled opposite-phase evidence. Flagged coverage is never a clean null.

Fresh TESScut difference image sector41 stacks3 core/bilateral-sideband events. Target pixel= [7.246677411415977, 7.639862217775246], deficit=[7.286566775805656, 7.668914212985631]; descriptive offset=0.04935pixel, field PSF residual=0.12616. No arbitrary centroid cut or calibrated localization error is imposed. Every capable neighbor below was inspected at its actual projected pixels; close overlapping neighbors retain probable ownership. Other capable sources are displaced from compact deficit core but crowding limits localization. All Gaia neighbors are retained in pixel JSON.

Source	Separation arcsec	G	Required intrinsic loss proxy	Pixel coordinate	Diluted pixel loss
166779758414232 8704	47.41803394908 422	15.401 44	0.4509323596108052	[6.451195731799155, 9.856966736951295]	0.01341198279801 0563

G-band pairwise brightness screen assumes approximate passband/aperture response; intrinsic required eclipse cannot be directly compared with diluted pixel loss. Unresolved unlisted companions are not excluded.

Alternative reduction: independently extracted background-subtracted TESScut1x1/3x3/5x5 depths=29.49346/28.28514/27.95131ppt; QLP median=31.05219ppt. Pixel sector41 is a distinct observing occasion on the same TESS instrument; no same-sector QLP comparison claimed. Independent processing is not an independent instrument. Different apertures respond differently to dilution/background.

Gaia source1667797618502068352: period=9.199493314+/-0.001784467d; e=0.0368066; omega=314.5014362deg; periastron BTJD=389.65044891. Closest conjunction offset=-0.1470528d (-0.0159849phase), source-specific full-covariance linear error=0.2920223d. Correlation order P,gamma,K,e,omega,Tp with subindices0,3,4,5. Linear propagation is descriptive, not a Gaussian veto or ownership proof. Both conjunctions retained in ../gaia-geometry.json. Mass function=0.05843774solar, assumed M1=1.06, minimum M2=0.528257; approximate central duration=5.3927h versus observed=2.5189h. TIC radius/main-sequence companion scaling, edge-on conjunction and stellar evolution are assumptions; this is a plausibility check, not a fitted physical solution.

Sector	Same-kind cycle	Measured BTJD	Formal center error d	Conservative input error d	Realized depth ppt
15	0	1717.629577593	0.000649840118	0.0104163279	29.88153
15	1	1726.821173932	0.000463664067	0.0104163279	26.23532
48	97	2609.874877767	0.000721333044	0.00347225627	32.53826
48	98	2619.073937218	0.000737470111	0.00347225627	31.73454
48	99	2628.272574835	0.000511026696	0.00347225627	31.05219
76	181	3382.556190905	0.000655449711	0.00115739675	31.38911
76	182	3391.755473324	0.000724302615	0.00115739675	30.82697

Catalogue novelty is separate from this physical verdict. Shared coverage matrix and source identity checks are authoritative for catalogue recovery; period agreement alone cannot establish attribution.

Stable review key: R630.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s41.json - historical/diff_s41.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

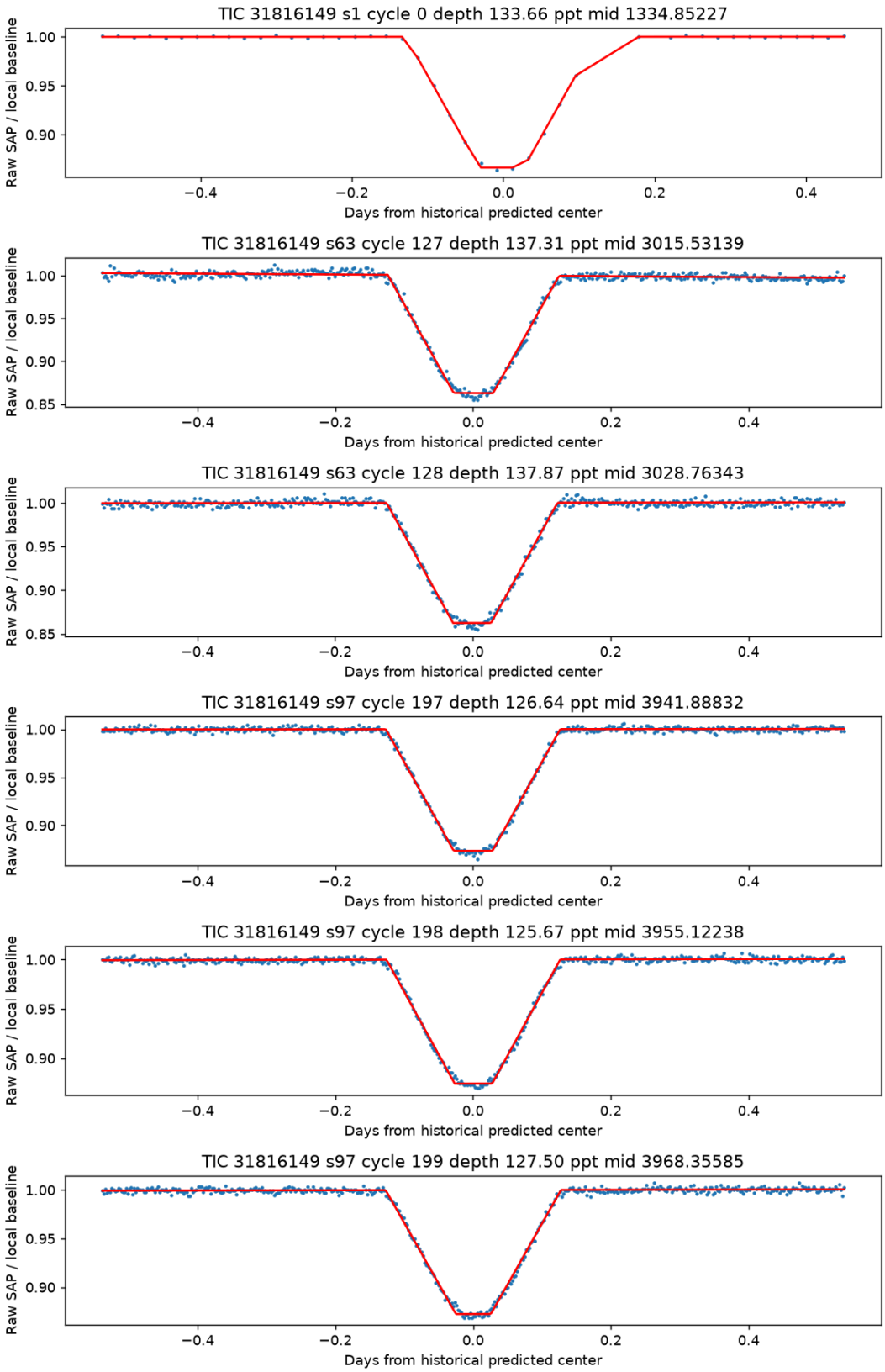
R631. TIC 31816149

physical_verdict: CONFIRMED

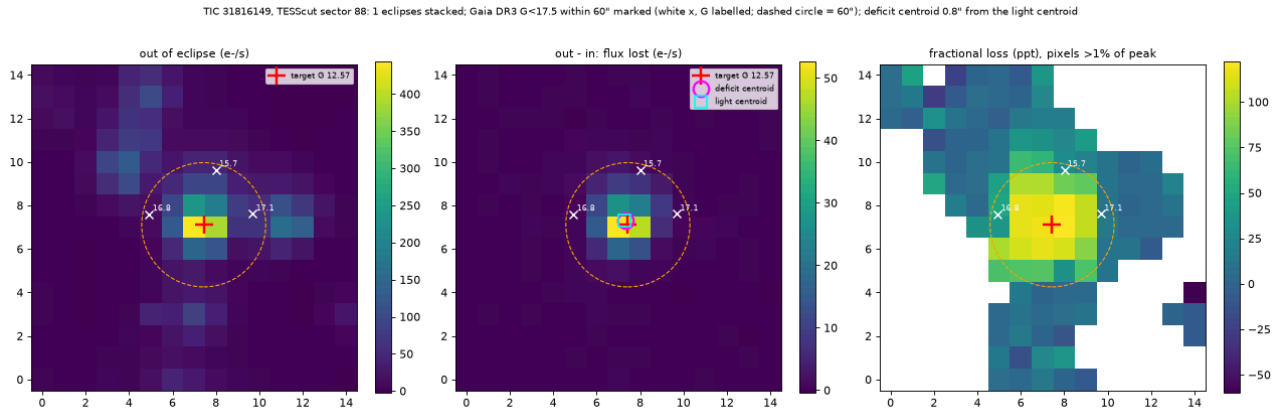
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional additional clean epochs and resolved photometry.



R631; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R631; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 31816149: CONFIRMED

Full bounded physical review: COMPLETE. Six deep repeated primary eclipses and unequal secondary; no capable neighbor within60arcsec.

Raw QLP SAP_FLUX representative events and full-sector P/2P morphology were visually inspected. Quality-zero finite positive samples only; predictions guide windows and are never timings. 6 identifiable free-center timings define P=13.233684727504 +/- 0.000014556716d, reference BTJD=1334.852605833 +/- 0.002519271d. Epoch error meaning: Conservative linear timing-fit covariance under the stated cadence floor and scatter inflation; excludes uncalibrated shape and baseline systematics. Epoch kind: Deeper primary established by unequal observed secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced chi square=0.2681513614868137; zero degrees of freedom is null. All errors retain their named meanings; no calibrated coverage is claimed. Selected same-kind recurring eclipse family; parity and P/2 coverage checked.

Depth=130.577717ppt is the median realized central trapezoid loss d*min(1,w/(2*k)); extrapolated amplitude and ingress remain separately in measurements.json. Measured secondary phase 0.50008902, formal phase error 4.0750251e-05, realized depth 32.48902ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Secondary raw pooled windows, actual sectors/cycles and fitted phase are retained in secondary-windows.csv and ../secondary-phases.json. No exhaustive blind alias search asserted. ../alias-audit.json and ../half-period-coverage.json retain recurrence/parity and sampled opposite-phase evidence. Flagged coverage is never a clean null.

Fresh TESScut difference image sector88 stacks1 core/bilateral-sideband events. Target pixel= [7.438488046734202, 7.11103877307435], deficit=[7.3283923817017795, 7.28988021099308]; descriptive offset=0.21001pixel, field PSF residual=0.41932. No arbitrary centroid cut or calibrated localization error is imposed. Every capable neighbor below was inspected at its actual projected pixels; close overlapping neighbors retain probable ownership. Other capable sources are displaced from compact deficit core but crowding limits localization. All Gaia neighbors are retained in pixel JSON.

Source	Separation arcsec	G	Required intrinsic loss proxy	Pixel coordinate	Diluted pixel loss
None					

G-band pairwise brightness screen assumes approximate passband/aperture response; intrinsic required eclipse cannot be directly compared with diluted pixel loss. Unresolved unlisted companions are not excluded.

Alternative reduction: independently extracted background-subtracted TESScut1x1/3x3/5x5 depths=118.60979/117.85771/110.31850ppt; QLP median=130.57772ppt. Pixel sector88 is a distinct observing occasion on the same TESS instrument; no same-sector QLP comparison claimed. Independent processing is not an independent instrument. Different apertures respond differently to dilution/background.

Gaia source4667233144000428800: period=13.233842358+/-0.002789804d; e=0.0565051; omega=11.1689633deg; periastron BTJD=392.37859886. Closest conjunction offset=0.2048136d (0.0154765phase), source-specific full-covariance linear error=0.1998004d. Correlation order P,gamma,K,e,omega,Tp with subindices0,3,4,5. Linear propagation is descriptive, not a Gaussian veto or ownership proof. Both conjunctions retained in ../gaia-geometry.json. Mass function=0.08810967solar, assumed M1=1.05, minimum M2=0.628468; approximate central duration=7.9000h versus observed=6.0650h. TIC radius/main-sequence companion scaling, edge-on conjunction and stellar evolution are assumptions; this is a plausibility check, not a fitted physical solution.

Sector	Same-kind cycle	Measured BTJD	Formal center error d	Conservative input error d	Realized depth ppt
1	0	1334.852269351	0.000155248407	0.0104167366	133.65777
63	127	3015.531392795	0.000315205613	0.00115742224	137.30780
63	128	3028.763426714	0.00022841327	0.00115742224	137.86628
97	197	3941.888315105	0.000180129894	0.00115738579	126.63962
97	198	3955.122380717	0.000188412646	0.00115738579	125.66869
97	199	3968.355851640	0.000197762364	0.00115738579	127.49767

Catalogue novelty is separate from this physical verdict. Shared coverage matrix and source identity checks are authoritative for catalogue recovery; period agreement alone cannot establish attribution.

Stable review key: R631.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s88.json - historical/diff_s88.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

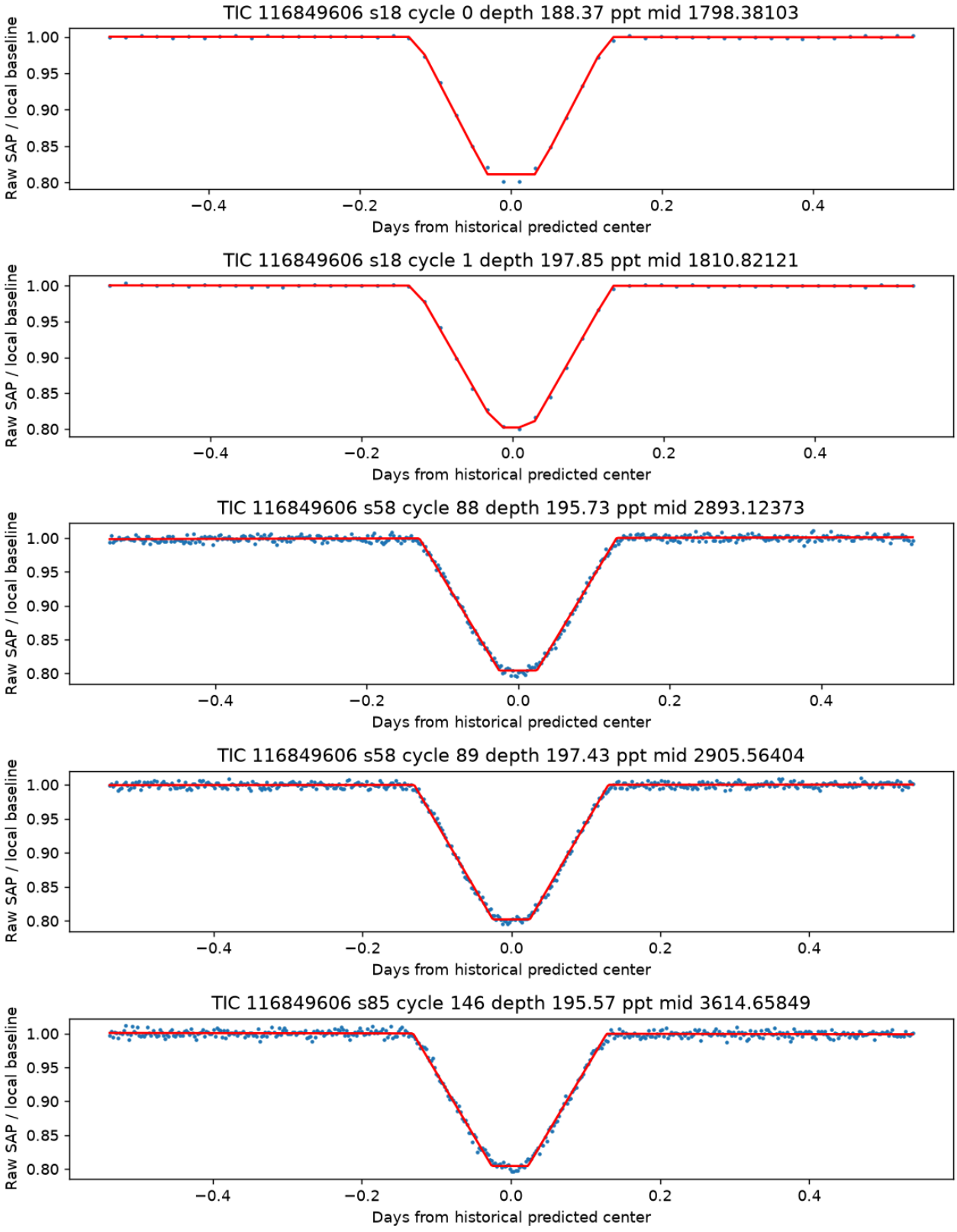
R632. TIC 116849606

physical_verdict: CONFIRMED

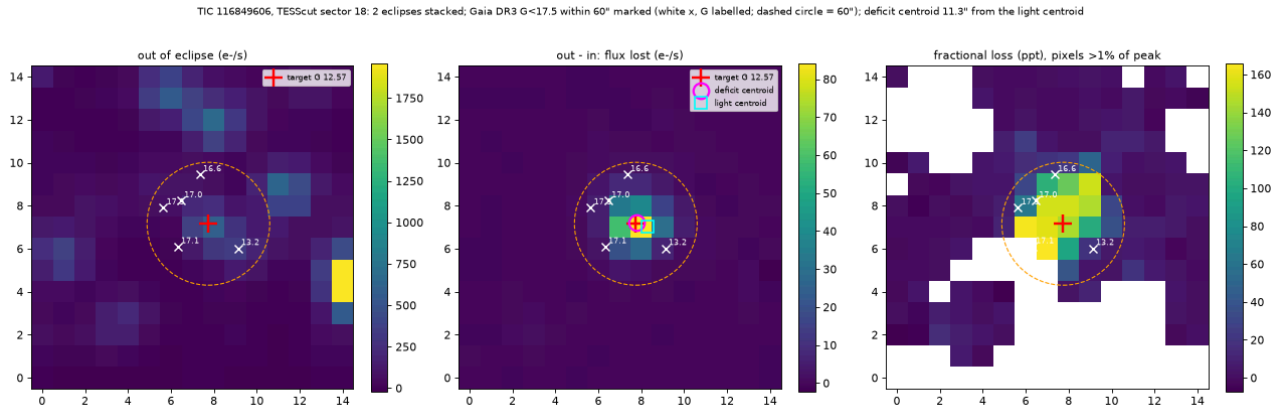
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional additional clean epochs and resolved photometry.



R632; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R632; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 116849606: CONFIRMED

Full bounded physical review: COMPLETE. Five deep repeated primary eclipses and unequal secondary. Capable36.35arcsec neighbor lies away from the compact deficit core.

Raw QLP SAP_FLUX representative events and full-sector P/2P morphology were visually inspected. Quality-zero finite positive samples only; predictions guide windows and are never timings. 5 identifiable free-center timings define P=12.440254466056 +/- 0.000023213922d, reference BTJD=1798.381352886 +/- 0.002566825d. Epoch error meaning: Conservative linear timing-fit covariance under the stated cadence floor and scatter inflation; excludes uncalibrated shape and baseline systematics. Epoch kind: Deeper primary established by unequal observed secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced chi square=0.0012150626936853677; zero degrees of freedom is null. All errors retain their named meanings; no calibrated coverage is claimed. Selected same-kind recurring eclipse family; parity and P/2 coverage checked.

Depth=195.733208ppt is the median realized central trapezoid loss d*min(1,w/(2*k)); extrapolated amplitude and ingress remain separately in measurements.json. Measured secondary phase 0.50777977, formal phase error 1.7891351e-05, realized depth 83.54695ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Secondary raw pooled windows, actual sectors/cycles and fitted phase are retained in secondary-windows.csv and ../secondary-phases.json. No exhaustive blind alias search asserted. ../alias-audit.json and ../half-period-coverage.json retain recurrence/parity and sampled opposite-phase evidence. Flagged coverage is never a clean null.

Fresh TESScut difference image sector18 stacks2 core/bilateral-sideband events. Target pixel= [7.746564539683181, 7.159683402689835], deficit=[7.788126832769522, 7.198591681575559]; descriptive offset=0.05693pixel, field PSF residual=0.79657. No arbitrary centroid cut or calibrated localization error is imposed. Every capable neighbor below was inspected at its actual projected pixels; close overlapping neighbors retain probable ownership. Other capable sources are displaced from compact deficit core but crowding limits localization. All Gaia neighbors are retained in pixel JSON.

Source	Separation arcsec	G	Required intrinsic loss proxy	Pixel coordinate	Diluted pixel loss
43596065968172 5568	36.35277088311 106	13.1982 05	0.5432508961537987	[9.119610210435871, 6.01355863331133]	0.01913445036012 7217

G-band pairwise brightness screen assumes approximate passband/aperture response; intrinsic required eclipse cannot be directly compared with diluted pixel loss. Unresolved unlisted companions are not excluded.

Alternative reduction: independently extracted background-subtracted TESScut1x1/3x3/5x5 depths=144.97444/121.69230/94.56094ppt; QLP median=195.73321ppt. Pixel sector18 shares the same TESS photons with retained QLP and checks extraction robustness. Independent processing is not an independent instrument. Different apertures respond differently to dilution/background.

Gaia source435960453523296000: period=12.442154531+/-0.002444407d; e=0.0350116; omega=308.9095390deg; periastron BTJD=387.72366373. Closest conjunction offset=-0.0931785d (-0.0074889phase), source-specific full-covariance linear error=0.3044548d. Correlation order P,gamma,K,e,omega,Tp with subindices0,3,4,5. Linear propagation is descriptive, not a Gaussian veto or ownership proof. Both conjunctions retained in ../gaia-geometry.json. Mass function=0.1492656solar, assumed M1=1.17, minimum M2=0.846749; approximate central duration=9.0052h versus observed=6.2537h. TIC radius/main-sequence companion scaling, edge-on conjunction and stellar evolution are assumptions; this is a plausibility check, not a fitted physical solution.

Sector	Same-kind cycle	Measured BTJD	Formal center error d	Conservative input error d	Realized depth ppt
18	0	1798.381030875	0.000251335493	0.0104167496	188.36613
18	1	1810.821212280	0.000221772865	0.0104167496	197.85153
58	88	2893.123732608	0.00022014339	0.00115741827	195.73321
58	89	2905.564036479	0.000226294859	0.00115741827	197.42663
85	146	3614.658490959	0.000237635103	0.00115743017	195.56832

Catalogue novelty is separate from this physical verdict. Shared coverage matrix and source identity checks are authoritative for catalogue recovery; period agreement alone cannot establish attribution.

Stable review key: R632.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s18.json - historical/diff_s18.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

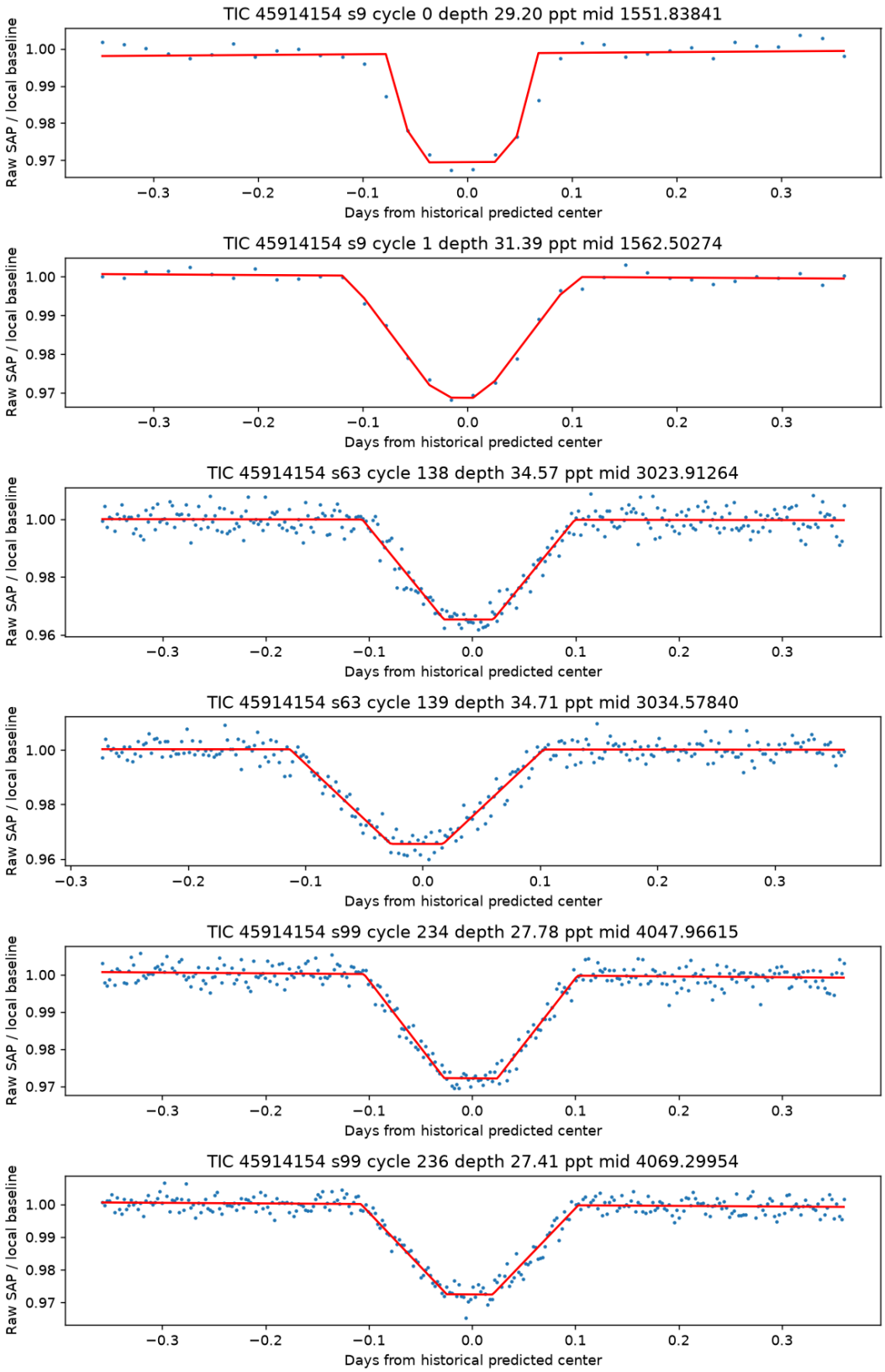
R633. TIC 45914154

physical_verdict: CONFIRMED

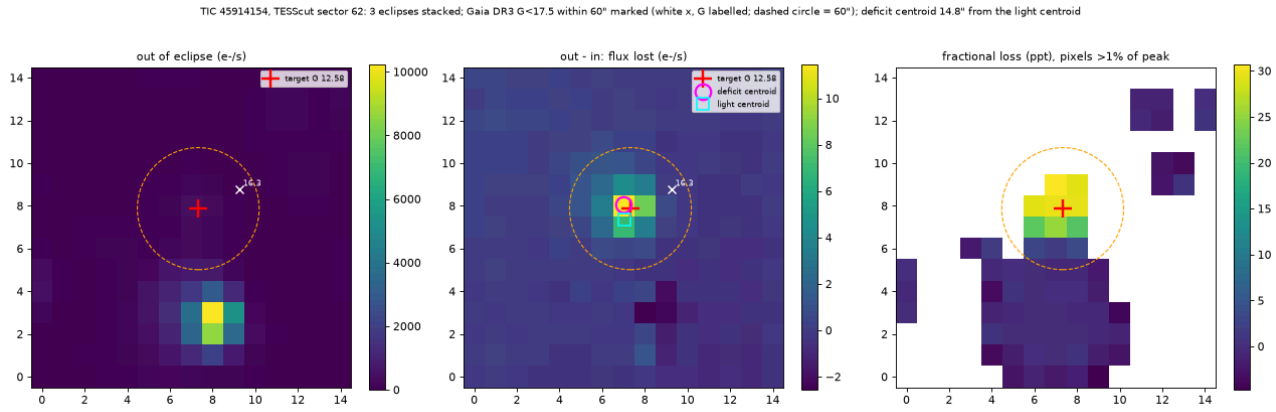
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional additional clean epochs and resolved photometry.



R633; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R633; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 45914154: CONFIRMED

Full bounded physical review: COMPLETE. Six narrow repeated primary eclipses and unequal secondary near0.4498; the initial near-half-phase secondary fit failed and was replaced using observed morphology. Capable43.95arcsec neighbor is separated from deficit core.

Raw QLP SAP_FLUX representative events and full-sector P/2P morphology were visually inspected. Quality-zero finite positive samples only; predictions guide windows and are never timings. 6 identifiable free-center timings define P=10.667224966865 +/- 0.000011479419d, reference BTJD=1551.834932386 +/- 0.002207357d. Epoch error meaning: Conservative linear timing-fit covariance under the stated cadence floor and scatter inflation; excludes uncalibrated shape and baseline systematics. Epoch kind: Deeper primary established by unequal observed secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced chi square=0.33448381934669585; zero degrees of freedom is null. All errors retain their named meanings; no calibrated coverage is claimed. Selected same-kind recurring eclipse family; parity and P/2 coverage checked.

Depth=30.291826ppt is the median realized central trapezoid loss d*min(1,w/(2*k)); extrapolated amplitude and ingress remain separately in measurements.json. Measured secondary phase 0.44978814, formal phase error 0.0001975221, realized depth 9.06765ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Secondary raw pooled windows, actual sectors/cycles and fitted phase are retained in secondary-windows.csv and ../secondary-phases.json. No exhaustive blind alias search asserted. ../alias-audit.json and ../half-period-coverage.json retain recurrence/parity and sampled opposite-phase evidence. Flagged coverage is never a clean null.

Fresh TESScut difference image sector62 stacks3 core/bilateral-sideband events. Target pixel= [7.325906164591727, 7.868176405273294], deficit=[7.010100206327275, 8.045952524192558]; descriptive offset=0.36241pixel, field PSF residual=0.99046. No arbitrary centroid cut or calibrated localization error is imposed. Every capable neighbor below was inspected at its actual projected pixels; close overlapping neighbors retain probable ownership. Other capable sources are displaced from compact deficit core but crowding limits localization. All Gaia neighbors are retained in pixel JSON.

Source	Separation arcsec	G	Required intrinsic loss proxy	Pixel coordinate	Diluted pixel loss
54197798545637 73184	43.9548670454 7451	16.3042 93	0.9694385641989272	[9.273971241968145, 8.780659099326206]	-0.00034627484002 13247

G-band pairwise brightness screen assumes approximate passband/aperture response; intrinsic required eclipse cannot be directly compared with diluted pixel loss. Unresolved unlisted companions are not excluded.

Alternative reduction: independently extracted background-subtracted TESScut1x1/3x3/5x5 depths=29.78637/27.75578/19.57276ppt; QLP median=30.29183ppt. Pixel sector62 is a distinct observing occasion on the same TESS instrument; no same-sector QLP comparison claimed. Independent processing is not an independent instrument. Different apertures respond differently to dilution/background.

Gaia source5419774189506366080: period=10.666312433+/-0.001434373d; e=0.0726193; omega=158.3442045deg; periastron BTJD=390.96497204. Closest conjunction offset=0.0423919d (0.0039744phase), source-specific full-covariance linear error=0.1580353d. Correlation order P,gamma,K,e,omega,Tp with subindices0,3,4,5. Linear propagation is descriptive, not a Gaussian veto or ownership proof. Both conjunctions retained in ../gaia-geometry.json. Mass function=0.08809977solar, assumed M1=1.38, minimum M2=0.732606; approximate central duration=8.8745h versus observed=5.0055h. TIC radius/main-sequence companion scaling, edge-on conjunction and stellar evolution are assumptions; this is a plausibility check, not a fitted physical solution.

Sector	Same-kind cycle	Measured BTJD	Formal center error d	Conservative input error d	Realized depth ppt
9	0	1551.838413706	0.00117917629	0.01041665	29.19653
9	1	1562.502738903	0.0010803001	0.01041665	31.38712
63	138	3023.912638502	0.00113451408	0.00115739218	34.56799
63	139	3034.578401520	0.000936841378	0.00115739218	34.70781
99	234	4047.966147058	0.000752732181	0.00115747247	27.78180
99	236	4069.299542567	0.000873381377	0.00115747247	27.41253

Catalogue novelty is separate from this physical verdict. Shared coverage matrix and source identity checks are authoritative for catalogue recovery; period agreement alone cannot establish attribution.

Stable review key: R633.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s62.json - historical/diff_s62.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

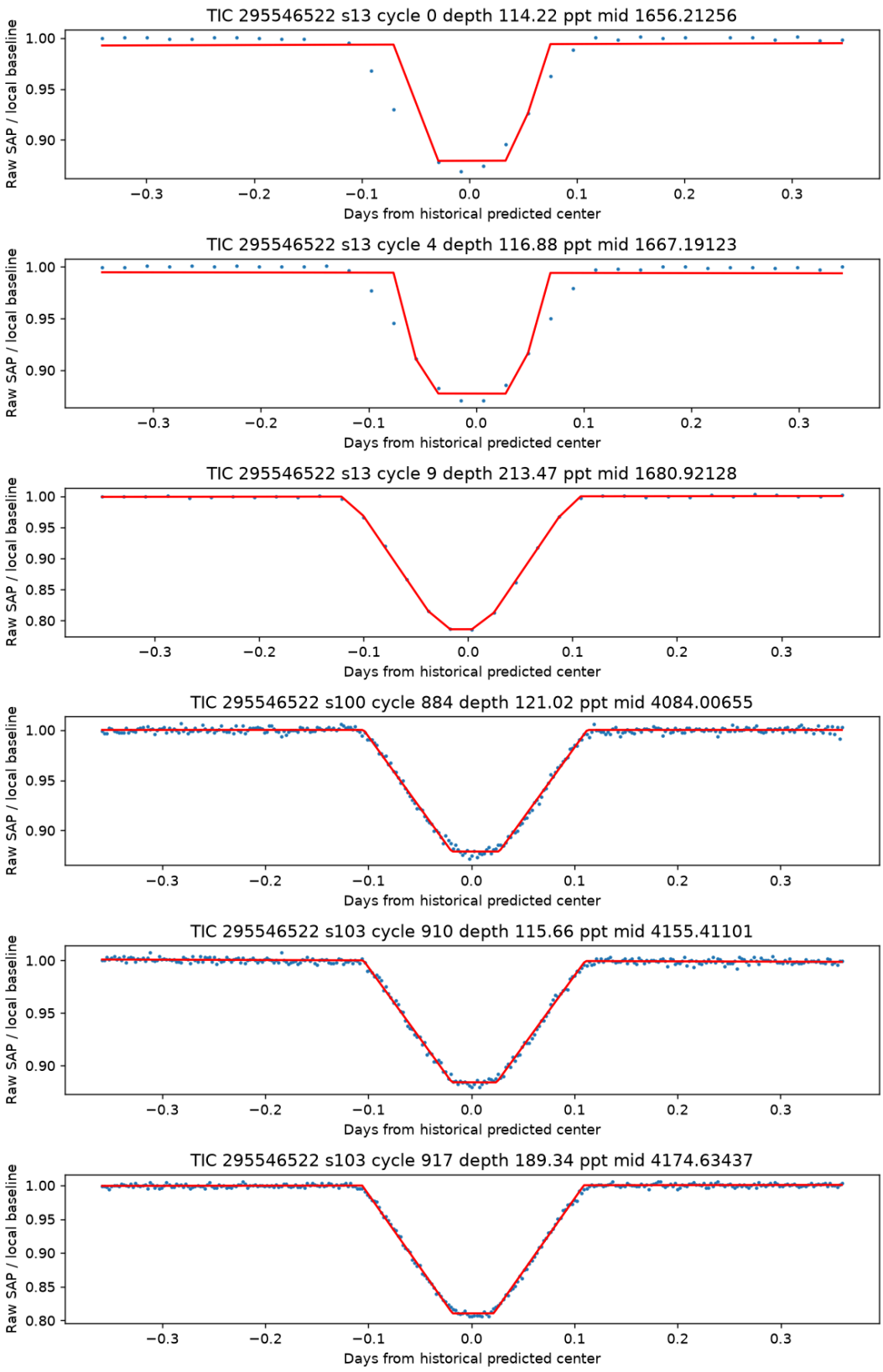
R634. TIC 295546522

physical_verdict: CONFIRMED

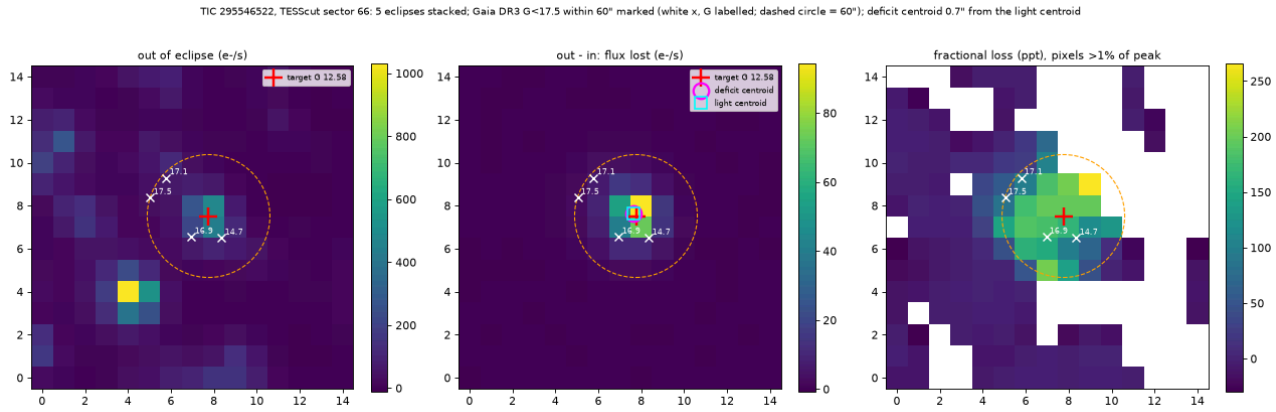
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional additional clean epochs and resolved photometry.



R634; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R634; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 295546522: CONFIRMED

Full bounded physical review: COMPLETE. Thirteen deeper odd historical cycles define primary; eleven shallower even cycles remain in raw data. Unequal populations require doubling historical SAP period. No capable neighbor within60arcsec.

Raw QLP SAP_FLUX representative events and full-sector P/2P morphology were visually inspected. Quality-zero finite positive samples only; predictions guide windows and are never timings. 13 identifiable free-center timings define P=5.492757262197 +/- 0.000010274729d, reference BTJD=1658.952796908 +/- 0.004598836d. Epoch error meaning: Conservative linear timing-fit covariance under the stated cadence floor and scatter inflation; excludes uncalibrated shape and baseline systematics. Epoch kind: Deeper primary established by unequal observed secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced chi square=0.46823159813780646; zero degrees of freedom is null. All errors retain their named meanings; no calibrated coverage is claimed. Deeper primary parity only; historical search period doubled from observed unequal alternating depths.

Depth=199.767670ppt is the median realized central trapezoid loss d*min(1,w/(2*k)); extrapolated amplitude and ingress remain separately in measurements.json. Measured secondary phase 0.50005394, formal phase error 5.2833283e-05, realized depth 120.43079ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Secondary raw pooled windows, actual sectors/cycles and fitted phase are retained in secondary-windows.csv and ../secondary-phases.json. No exhaustive blind alias search asserted. ../alias-audit.json and ../half-period-coverage.json retain recurrence/parity and sampled opposite-phase evidence. Flagged coverage is never a clean null.

Fresh TESScut difference image sector66 stacks5 core/bilateral-sideband events. Target pixel= [7.754576426739337, 7.519398352203281], deficit=[7.651320876166129, 7.656799433627296]; descriptive offset=0.17187pixel, field PSF residual=0.83990. No arbitrary centroid cut or calibrated localization error is imposed. Every capable neighbor below was inspected at its actual projected pixels; close overlapping neighbors retain probable ownership. Other capable sources are displaced from compact deficit core but crowding limits localization. All Gaia neighbors are retained in pixel JSON.

Source	Separation arcsec	G	Required intrinsic loss proxy	Pixel coordinate	Diluted pixel loss
None					

G-band pairwise brightness screen assumes approximate passband/aperture response; intrinsic required eclipse cannot be directly compared with diluted pixel loss. Unresolved unlisted companions are not excluded.

Alternative reduction: independently extracted background-subtracted TESScut1x1/3x3/5x5 depths=196.21878/182.32807/162.38999ppt; QLP median=199.76767ppt. Pixel sector66 is a distinct observing occasion on the same TESS instrument; no same-sector QLP comparison claimed. Independent processing is not an independent instrument. Different apertures respond differently to dilution/background.

Gaia source6431187035788339968: period=5.492652967+/-0.000523237d; e=0.0310318; omega=238.7734853deg; periastron BTJD=386.87679337. Closest conjunction offset=0.0217115d (0.0039528phase), source-specific full-covariance linear error=0.1013060d. Correlation order P,gamma,K,e,omega,Tp with subindices0,3,4,5. Linear propagation is descriptive, not a Gaussian veto or ownership proof. Both conjunctions retained in ../gaia-geometry.json. Mass function=0.1560789solar, assumed M1=1.33, minimum M2=0.926185; approximate central duration=6.7359h versus observed=5.1534h. TIC radius/main-sequence companion scaling, edge-on conjunction and stellar evolution are assumptions; this is a plausibility check, not a fitted physical solution.

Sector	Same-kind cycle	Measured BTJD	Formal center error d	Conservative input error d	Realized depth ppt
13	0	1658.949943226	0.000129627767	0.0104165125	210.69256
13	1	1664.442951327	0.000192888059	0.0104165125	211.87079
13	2	1669.939943754	0.000100268473	0.0104165125	189.01954
13	3	1675.428073683	0.000125372077	0.0104165125	202.53861
13	4	1680.921276548	0.00022410354	0.0104165125	213.46585
100	440	4075.767323814	0.000137183383	0.00115740685	201.43413
100	441	4081.259933600	0.000148582877	0.00115740685	200.23440
100	443	4092.244839992	0.000147497205	0.00115740685	199.76767
100	444	4097.737294238	0.000120692439	0.00115740685	198.87611
103	455	4158.156774284	0.000128519505	0.00115740685	191.52649
103	456	4163.649452503	0.000128404905	0.00115740685	191.07309
103	457	4169.142097917	0.000124545082	0.00115740685	190.64207
103	458	4174.634374914	0.000121584784	0.00115740685	189.34467

Catalogue novelty is separate from this physical verdict. Shared coverage matrix and source identity checks are authoritative for catalogue recovery; period agreement alone cannot establish attribution.

Stable review key: R634.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s66.json - historical/diff_s66.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

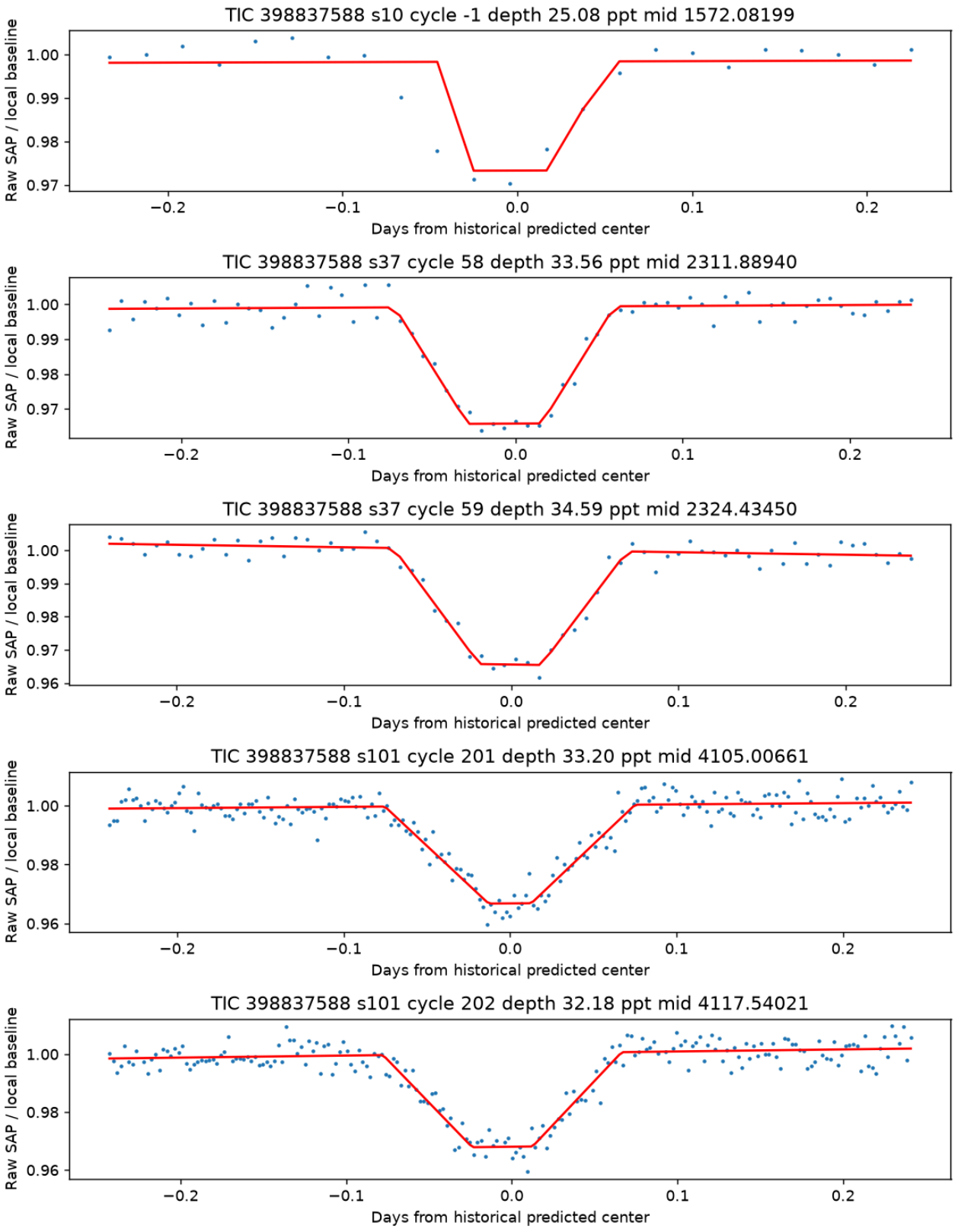
R635. TIC 398837588

physical_verdict: PROBABLE

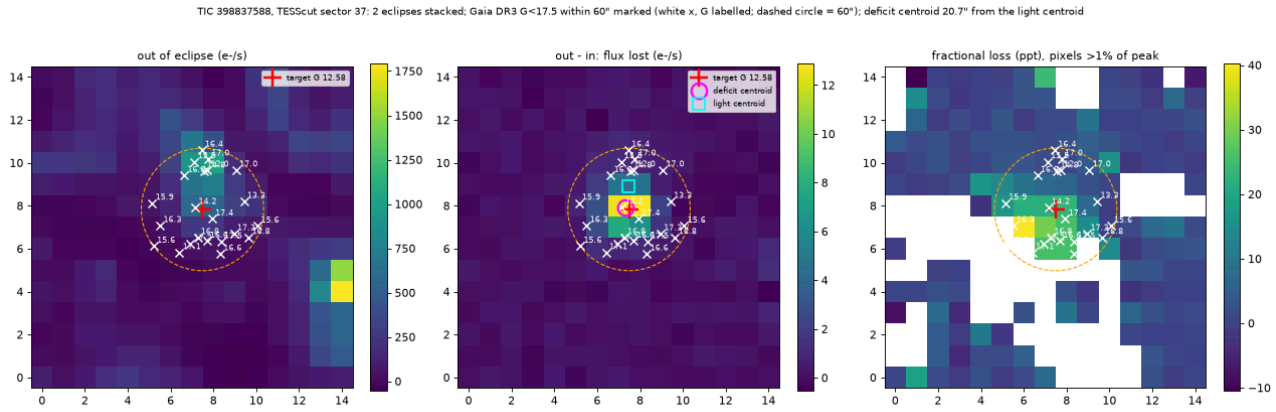
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or multi-source PRF attribution.



R635; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R635; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 398837588: PROBABLE

Full bounded physical review: COMPLETE. Five repeated narrow eclipses despite substantial sector baseline changes. No secure secondary. Capable7.28arcsec neighbor shares footprint, so target ownership remains unresolved; other capable sources are listed individually below.

Raw QLP SAP_FLUX representative events and full-sector P/2P morphology were visually inspected. Quality-zero finite positive samples only; predictions guide windows and are never timings. 5 identifiable free-center timings define P=12.539237938671 +/- 0.000036508646d, reference BTJD=1572.077705630 +/- 0.007027711d. Epoch error meaning: Conservative linear timing-fit covariance under the stated cadence floor and scatter inflation; excludes uncalibrated shape and baseline systematics. Epoch kind: Selected repeating eclipse; deeper/secondary identity not established. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced chi square=4.502477760564703; zero degrees of freedom is null. All errors retain their named meanings; no calibrated coverage is claimed. Selected same-kind recurring eclipse family; parity and P/2 coverage checked.

Depth=33.200496ppt is the median realized central trapezoid loss d*min(1,w/(2*k)); extrapolated amplitude and ingress remain separately in measurements.json. No secure secondary; exploratory fit is not interpreted as an eclipse timing or counterpart. Secondary raw pooled windows, actual sectors/cycles and fitted phase are retained in secondary-windows.csv and ../secondary-phases.json. No exhaustive blind alias search asserted. ../alias-audit.json and ../half-period-coverage.json retain recurrence/parity and sampled opposite-phase evidence. Flagged coverage is never a clean null.

Fresh TESScut difference image sector37 stacks2 core/bilateral-sideband events. Target pixel= [7.489339195144037, 7.829938311430091], deficit=[7.3315842025046045, 7.9198690867597215]; descriptive offset=0.18159pixel, field PSF residual=0.82164. No arbitrary centroid cut or calibrated localization error is imposed. Every capable neighbor below was inspected at its actual projected pixels; close overlapping neighbors retain probable ownership. Other capable sources are displaced from compact deficit core but crowding limits localization. All Gaia neighbors are retained in pixel JSON.

Source	Separation arcsec	G	Required intrinsic loss proxy	Pixel coordinate	Diluted pixel loss
6075382049402598144	7.280522394953378	14.244521	0.18704336672554026	[7.148174146035064, 7.9257552169738315]	0.021854917844886624
6075382152481816192	36.452424628274436	14.804302	0.29082654612755054	[7.551327004402193, 9.618549019650843]	0.000780746874363 3279
6075382152481816320	37.842287726768134	12.000649	0.05267778819102251	[7.716346398195622, 9.67008828765908]	0.000780746874363 3279
6075381946323382400	40.866936971310055	13.322671	0.09901738888956046	[9.423134165518299, 8.204446930454704]	0.007204345091707 24
6075382221201294208	45.790584366193301	15.479767	0.5131281496320441	[7.109931883193958, 10.054575734766525]	0.0011973678225501 887
6075382118122075776	48.756556381483094	15.853373	0.7102464932149616	[5.143566643727468, 8.125058248001139]	0.016295415195347353
6075381877603904256	55.169302276373365	15.613826	0.5761999274519773	[10.061858492980656, 7.0554991301306025]	-0.00021556838202 928225
6075382083762334720	59.144054063313335	15.62043	0.5795127913088703	[5.210476459207425, 6.121780430155106]	-0.00020527071389 388817

G-band pairwise brightness screen assumes approximate passband/aperture response; intrinsic required eclipse cannot be directly compared with diluted pixel loss. Unresolved unlisted companions are not excluded.

Alternative reduction: independently extracted background-subtracted TESScut1x1/3x3/5x5 depths=21.85492/16.39225/10.25754ppt; QLP median=33.20050ppt. Pixel sector37 shares the same TESS photons with retained QLP and checks extraction robustness. Independent processing is not an independent instrument. Different apertures respond differently to dilution/background.

Gaia source6075382045083973888: period=12.537871558+/-0.005738517d; e=0.1708268; omega=263.5575476deg; periastron BTJD=387.23372700. Closest conjunction offset=-0.2967814d (-0.0236708phase), source-specific full-covariance linear error=0.5064744d. Correlation order P,gamma,K,e,omega,Tp with subindices0,3,4,5. Linear propagation is descriptive, not a Gaussian veto or ownership proof. Both conjunctions retained in ../gaia-geometry.json. Mass function=0.0298589solar, assumed M1=1.08, minimum M2=0.403545; approximate central duration=6.6662h versus observed=3.3527h. TIC radius/main-sequence companion scaling, edge-on conjunction and stellar evolution are assumptions; this is a plausibility check, not a fitted physical solution.

Sector	Same-kind cycle	Measured BTJD	Formal center error d	Conservative input error d	Realized depth ppt
10	-1	1572.081986177	0.000775681061	0.0104168122	25.08244
37	58	2311.889399551	0.00104664388	0.00347224916	33.55991
37	59	2324.434495805	0.00119586106	0.00347224916	34.58819
101	201	4105.006610531	0.000829290333	0.00115740685	33.20050
101	202	4117.540205334	0.00107867144	0.00115740685	32.17949

Catalogue novelty is separate from this physical verdict. Shared coverage matrix and source identity checks are authoritative for catalogue recovery; period agreement alone cannot establish attribution.

Stable review key: R635.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s37.json - historical/diff_s37.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

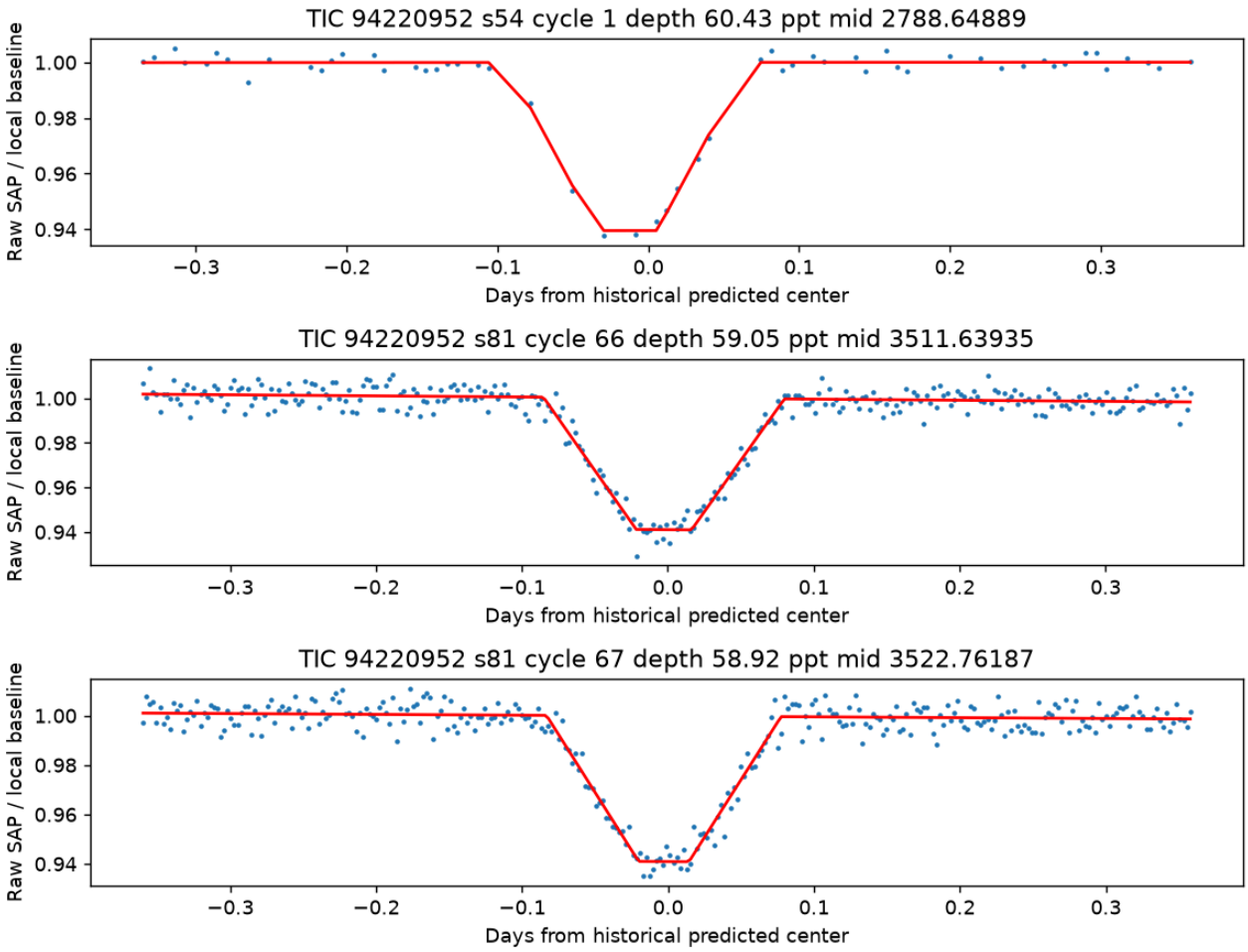
R636. TIC 94220952

physical_verdict: PROBABLE

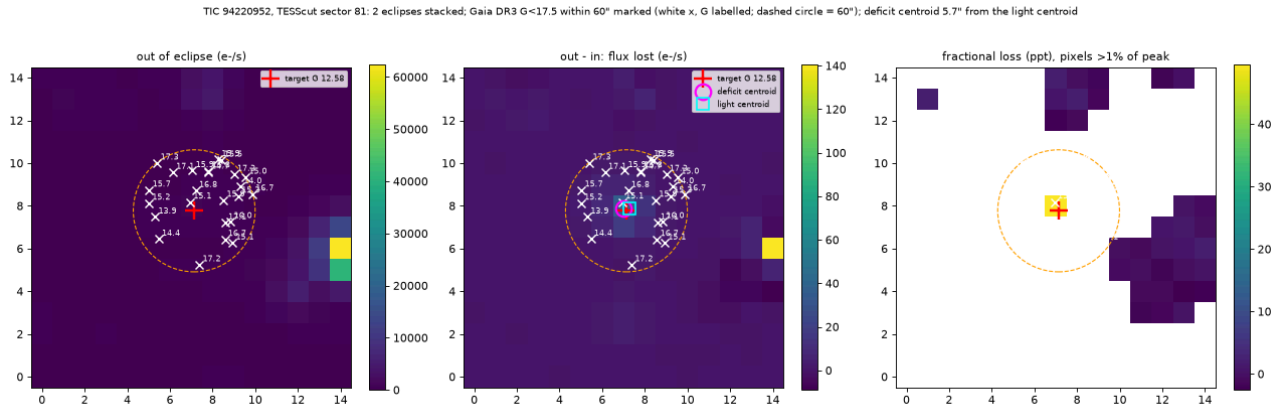
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or multi-source PRF attribution.



R636; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R636; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 94220952: PROBABLE

Full bounded physical review: COMPLETE. Three deep repeated eclipses including consecutive cycles66/67 in sector81. No secure secondary. Capable9.28arcsec neighbor shares footprint and prevents secure ownership.

Raw QLP SAP_FLUX representative events and full-sector P/2P morphology were visually inspected. Quality-zero finite positive samples only; predictions guide windows and are never timings. 3 identifiable free-center timings define P=11.122925403210 +/- 0.000106527918d, reference BTJD=2788.648998950 +/- 0.006929824d. Epoch error meaning: Conservative linear timing-fit covariance under the stated cadence floor and scatter inflation; excludes uncalibrated shape and baseline systematics. Epoch kind: Selected repeating eclipse; deeper/secondary identity not established. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced chi square=0.06155434287909847; zero degrees of freedom is null. All errors retain their named meanings; no calibrated coverage is claimed. Selected same-kind recurring eclipse family; parity and P/2 coverage checked.

Depth=59.047874ppt is the median realized central trapezoid loss d*min(1,w/(2*k)); extrapolated amplitude and ingress remain separately in measurements.json. No secure secondary; exploratory fit is not interpreted as an eclipse timing or counterpart. Secondary raw pooled windows, actual sectors/cycles and fitted phase are retained in secondary-windows.csv and ../secondary-phases.json. No exhaustive blind alias search asserted. ../alias-audit.json and ../half-period-coverage.json retain recurrence/parity and sampled opposite-phase evidence. Flagged coverage is never a clean null.

Fresh TESScut difference image sector81 stacks2 core/bilateral-sideband events. Target pixel=[7.1366395743607125, 7.766900669368171], deficit=[6.997990847636982, 7.854333695935645]; descriptive offset=0.16391pixel, field PSF residual=0.98002. No arbitrary centroid cut or calibrated localization error is imposed. Every capable neighbor below was inspected at its actual projected pixels; close overlapping neighbors retain probable ownership. Other capable sources are displaced from compact deficit core but crowding limits localization. All Gaia neighbors are retained in pixel JSON.

Source	Separation arcsec	G	Required intrinsic loss proxy	Pixel coordinate	Diluted pixel loss
4291080865997333632	9.284561816723235	15.1027565	0.6607149738338877	[6.943398310013462, 8.183449200512957]	0.0495996505265215
4291080904674507136	38.24190948764969	13.942269	0.2656625963370936	[5.300369166878136, 7.482920063109468]	0.007022832972466519
4291080797277828096	39.688698405361876	14.656988	0.4581176746996912	[7.774337235815823, 9.582902538812837]	0.003537953746703266
4291080900357071616	44.00975617555208	15.220628	0.7297116722568895	[5.009839095130997, 8.13099039000618]	0.010102597889645265
4291080904674761472	44.19456258915218	14.420803	0.3800994928233338	[5.466841462148617, 6.445128435745917]	0.005405370220721154 5
4291080870314770688	45.045070069810734	15.325893	0.7979907460328429	[9.20375235588035, 8.459234747054573]	0.005006507026706821
4291083825252277120	47.10726083246011	15.129846	0.6759156216987529	[8.927136255487712, 6.269216312084571]	-0.0002661084112007105
4291079392846019456	50.492821170277246	13.963852	0.26981091877667823	[9.298621661370616, 8.891712855649345]	0.0022750647938310645

Source	Separation arcsec	G	Required intrinsic loss proxy	Pixel coordinate	Diluted pixel loss
429107932412653 6320	55.610630066777 986	15.501728	0.9278979915872122	[8.38970761321847, 10.145492609060588]	0.00353795374670 3266
429107939284601 7920	59.642260760071 21	15.046973	0.6305829174998357	[9.540424796778625, 9.344992599756997]	0.00154043725220 1344

G-band pairwise brightness screen assumes approximate passband/aperture response; intrinsic required eclipse cannot be directly compared with diluted pixel loss. Unresolved unlisted companions are not excluded.

Alternative reduction: independently extracted background-subtracted TESScut1x1/3x3/5x5 depths=49.59965/37.28037/21.53476ppt; QLP median=59.04787ppt. Pixel sector81 shares the same TESS photons with retained QLP and checks extraction robustness. Independent processing is not an independent instrument. Different apertures respond differently to dilution/background.

Gaia source4291080865995759616: period=11.108806492+/-0.004875589d; e=0.3856259; omega=187.7598007deg; periastron BTJD=387.94286170. Closest conjunction offset=-0.0655155d (-0.0058976phase), source-specific full-covariance linear error=1.1939144d. Correlation order P,gamma,K,e,omega,Tp with subindices0,3,4,5. Linear propagation is descriptive, not a Gaussian veto or ownership proof. Both conjunctions retained in ../gaia-geometry.json. Mass function=0.01840918solar, assumed M1=1.07, minimum M2=0.330526; approximate central duration=5.8729h versus observed=3.8584h. TIC radius/main-sequence companion scaling, edge-on conjunction and stellar evolution are assumptions; this is a plausibility check, not a fitted physical solution.

Sector	Same-kind cycle	Measured BTJD	Formal center error d	Conservative input error d	Realized depth ppt
54	1	2788.648887589	0.00128182808	0.0069443453	60.43474
81	66	3511.639354321	0.000670707429	0.00115739053	59.04787
81	67	3522.761874493	0.000758486672	0.00115739053	58.92200

Catalogue novelty is separate from this physical verdict. Shared coverage matrix and source identity checks are authoritative for catalogue recovery; period agreement alone cannot establish attribution.

Stable review key: R636.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s81.json - historical/diff_s81.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

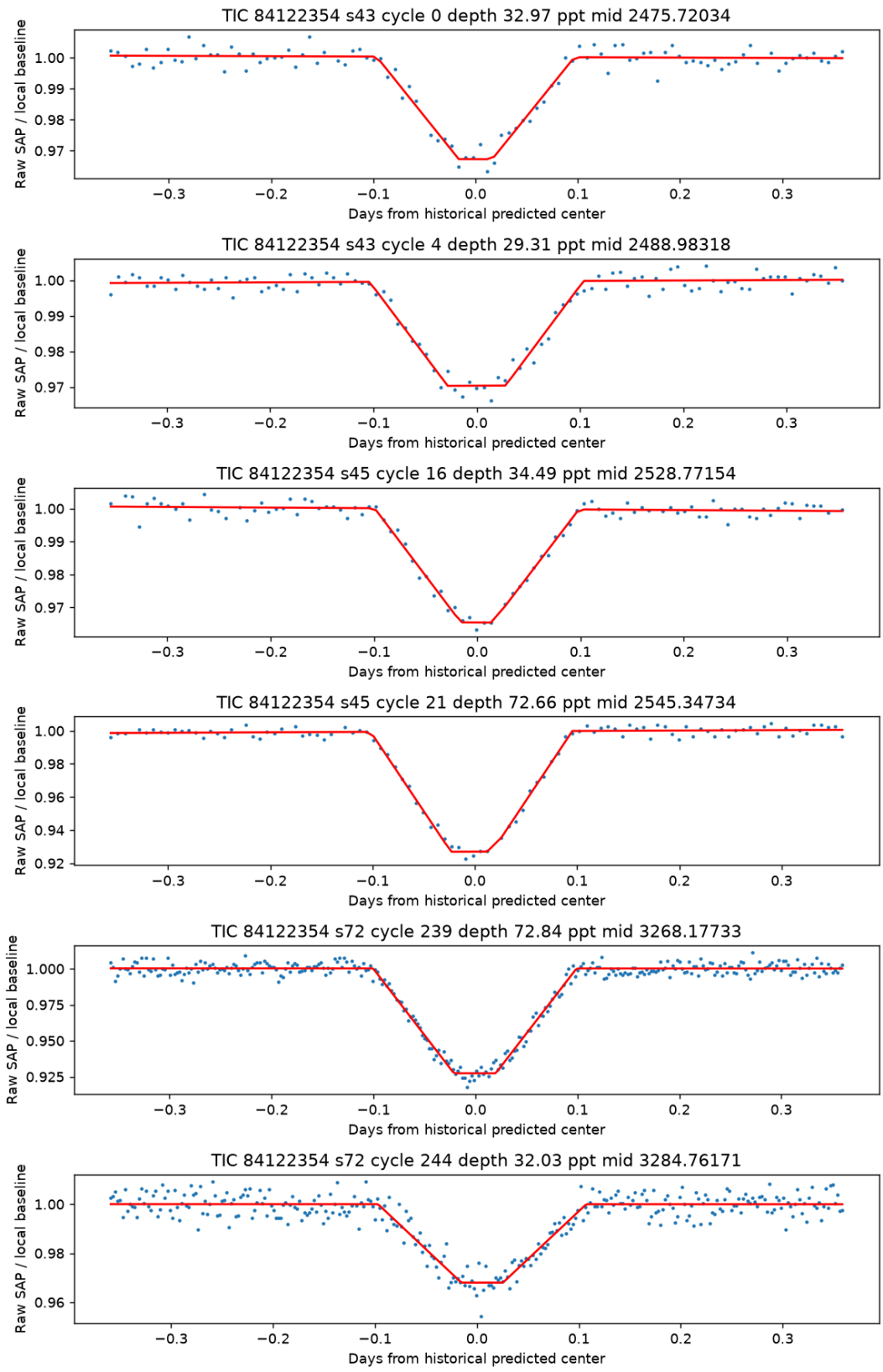
R637. TIC 84122354

physical_verdict: PROBABLE

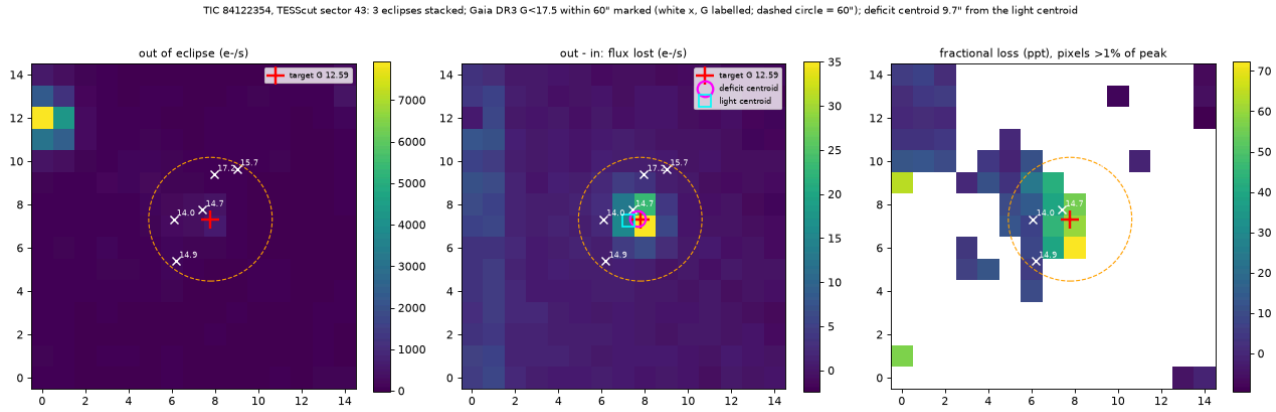
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or multi-source PRF attribution.



R637; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R637; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 84122354: PROBABLE

Full bounded physical review: COMPLETE. Six deeper odd historical cycles define primary, with persistent unequal even secondary population. Historical SAP period is doubled. Capable11.81arcsec neighbor overlaps footprint; ownership remains unresolved.

Raw QLP SAP_FLUX representative events and full-sector P/2P morphology were visually inspected. Quality-zero finite positive samples only; predictions guide windows and are never timings. 6 identifiable free-center timings define P=6.631471900250 +/- 0.000016674645d, reference BTJD=2479.033393430 +/- 0.001810592d. Epoch error meaning: Conservative linear timing-fit covariance under the stated cadence floor and scatter inflation; excludes uncalibrated shape and baseline systematics. Epoch kind: Deeper primary established by unequal observed secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced chi square=0.5864045560790094; zero degrees of freedom is null. All errors retain their named meanings; no calibrated coverage is claimed. Deeper primary parity only; historical search period doubled from observed unequal alternating depths.

Depth=72.485416ppt is the median realized central trapezoid loss d*min(1,w/(2*k)); extrapolated amplitude and ingress remain separately in measurements.json. Measured secondary phase 0.50038398, formal phase error 5.6256116e-05, realized depth 32.03988ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Secondary raw pooled windows, actual sectors/cycles and fitted phase are retained in secondary-windows.csv and ../secondary-phases.json. No exhaustive blind alias search asserted. ../alias-audit.json and ../half-period-coverage.json retain recurrence/parity and sampled opposite-phase evidence. Flagged coverage is never a clean null.

Fresh TESScut difference image sector43 stacks3 core/bilateral-sideband events. Target pixel= [7.785775749844763, 7.322171590798419], deficit=[7.668578729656907, 7.349858215315909]; descriptive offset=0.12042pixel, field PSF residual=0.51347. No arbitrary centroid cut or calibrated localization error is imposed. Every capable neighbor below was inspected at its actual projected pixels; close overlapping neighbors retain probable ownership. Other capable sources are displaced from compact deficit core but crowding limits localization. All Gaia neighbors are retained in pixel JSON.

Source	Separation arcsec	G	Required intrinsic loss proxy	Pixel coordinate	Diluted pixel loss
3432677466582096512	11.81126530124275	14.71729	0.586724915401776	[7.414419765136445, 7.76150730471258]	0.03962585042331775
3432677470877994496	35.643495597279035	13.961881	0.32893490137341336	[6.073835710636782, 7.330215396862352]	0.0077039403428093125
3432677402158513536	52.15938478415084	14.917142	0.6906530070611392	[6.185932261217368, 5.395698367948965]	0.005323967368998911

G-band pairwise brightness screen assumes approximate passband/aperture response; intrinsic required eclipse cannot be directly compared with diluted pixel loss. Unresolved unlisted companions are not excluded.

Alternative reduction: independently extracted background-subtracted TESScut1x1/3x3/5x5 depths=60.06573/54.34689/40.92379ppt; QLP median=72.48542ppt. Pixel sector43 shares the same TESS photons with retained QLP and checks extraction robustness. Independent processing is not an independent instrument. Different apertures respond differently to dilution/background.

Gaia source3432677470877990784: period=6.631783370+/-0.000785086d; e=0.0538416; omega=332.6457461deg; periastron BTJD=387.94914060. Closest conjunction offset=0.0134496d (0.0020280phase), source-specific full-covariance linear error=0.2517504d. Correlation order P,gamma,K,e,omega,Tp with subindices0,3,4,5. Linear propagation is descriptive, not a Gaussian veto or ownership proof. Both conjunctions retained in ../gaia-geometry.json. Mass function=0.09559245solar, assumed M1=1.23, minimum M2=0.711626; approximate central duration=7.8193h versus observed=4.7288h. TIC radius/main-sequence companion scaling, edge-on conjunction and stellar evolution are assumptions; this is a plausibility check, not a fitted physical solution.

Sector	Same-kind cycle	Measured BTJD	Formal center error d	Conservative input error d	Realized depth ppt
43	0	2479.033326961	0.000690116124	0.00347256518	68.38433
43	2	2492.296148791	0.00052064305	0.00347256518	68.68178
45	8	2532.086388763	0.000432504671	0.00347243069	72.31005
45	10	2545.347342018	0.000735348592	0.00347243069	72.66078
72	119	3268.177333234	0.000528127385	0.00115746719	72.83604
72	121	3281.442688038	0.000586502475	0.00115746719	74.44337

Catalogue novelty is separate from this physical verdict. Shared coverage matrix and source identity checks are authoritative for catalogue recovery; period agreement alone cannot establish attribution.

Stable review key: R637.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s43.json - historical/diff_s43.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

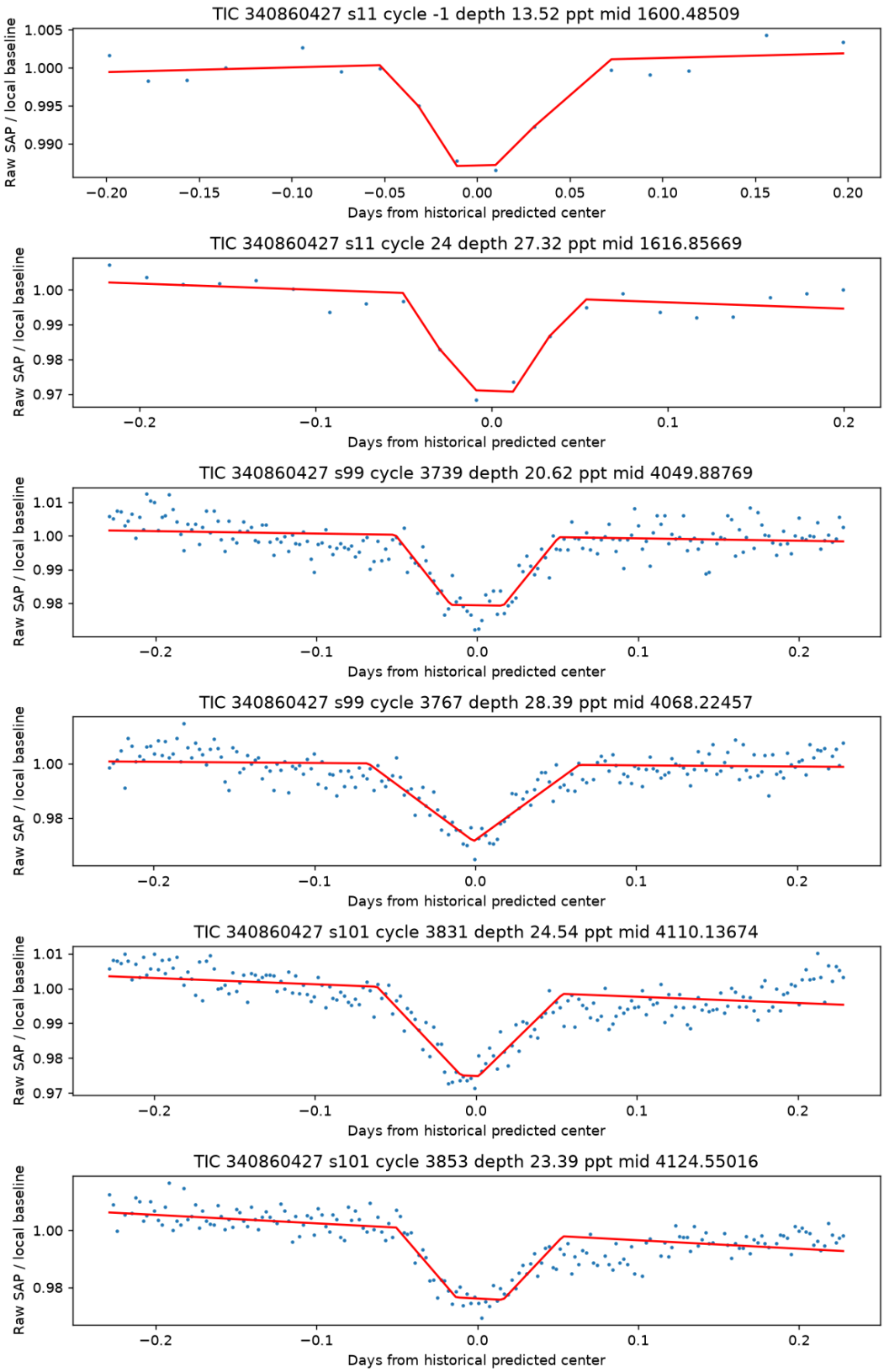
R638. TIC 340860427

physical_verdict: PROBABLE

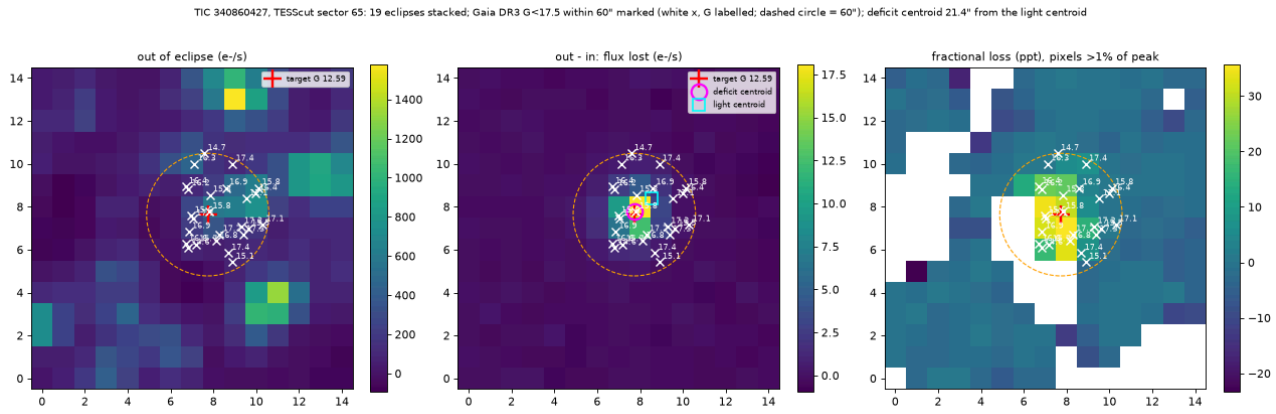
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or multi-source PRF attribution.



R638; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R638; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 340860427: PROBABLE

Full bounded physical review: COMPLETE. Thirty-eight deeper even historical cycles define primary; unequal secondary population requires doubled historical SAP period. Crowded field includes capable3.19/15.87/18.74arcsec sources; ownership unresolved. Nearby catalogued EB Gaia5857683145142215424 at38.54arcsec has period0.474888d, different from measured1.309840d; this is a contamination alert, not a target known-catalogue match. Its G18.753315 gives required intrinsic loss proxy11.36445 (above unity); actual pixel[8.38597,5.87954] is displaced from the compact deficit. Thus this known EB cannot explain the measured loss under the stated brightness screen, while closer capable sources remain unresolved.

Raw QLP SAP_FLUX representative events and full-sector P/2P morphology were visually inspected. Quality-zero finite positive samples only; predictions guide windows and are never timings. 38 identifiable free-center timings define P=1.309840015274 +/- 0.000001979261d, reference BTJD=1602.449463601 +/- 0.003741632d. Epoch error meaning: Conservative linear timing-fit covariance under the stated cadence floor and scatter inflation; excludes uncalibrated shape and baseline systematics. Epoch kind: Deeper primary established by unequal observed secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced chi square=1.5706597060598666; zero degrees of freedom is null. All errors retain their named meanings; no calibrated coverage is claimed. Deeper primary parity only; historical search period doubled from observed unequal alternating depths.

Depth=38.834336ppt is the median realized central trapezoid loss d*min(1,w/(2*k)); extrapolated amplitude and ingress remain separately in measurements.json. Measured secondary phase 0.50080252, formal phase error 0.00064029411, realized depth 26.15221ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Secondary raw pooled windows, actual sectors/cycles and fitted phase are retained in secondary-windows.csv and ../secondary-phases.json. No exhaustive blind alias search asserted. ../alias-audit.json and ../half-period-coverage.json retain recurrence/parity and sampled opposite-phase evidence. Flagged coverage is never a clean null.

Fresh TESScut difference image sector65 stacks19 core/bilateral-sideband events. Target pixel= [7.735429978261983, 7.633403201795165], deficit=[7.727471633203843, 7.766726318072527]; descriptive offset=0.13356pixel, field PSF residual=0.95641. No arbitrary centroid cut or calibrated localization error is imposed. Every capable neighbor below was inspected at its actual projected pixels; close overlapping neighbors retain probable ownership. Other capable sources are displaced from compact deficit core but crowding limits localization. All Gaia neighbors are retained in pixel JSON.

Source	Separation arcsec	G	Required intrinsic loss proxy	Pixel coordinate	Diluted pixel loss
5857683213903801344	3.1909674243697834	15.825874	0.8028204756011882	[7.809406916251106, 7.771147326301966]	0.0307028740774553
585768320957107 9040	15.872323101630874	15.575661	0.6455710561103708	[6.97274197782569, 7.5846876358718305]	0.03437032460802364
585768321390365 5296	18.743348081201614	15.281232	0.5014588900362352	[7.82745802020937, 8.545927532913979]	0.02129332601969231
5857683213903640448	30.530654262398495	15.743995	0.7473245885374221	[7.187766125427036, 6.2408301552036365]	0.016560703987640195
5857683213903800576	36.651878107135565	14.644652	0.2962273048368361	[6.822638528648939, 6.087292414576112]	0.016560703987640195
5857683248263381888	40.136564902347864	12.056757	0.06257203798127965	[9.532780841757724, 8.377080596916649]	5.470945194842705e-06
5857683140832821632	41.598063018527712	15.529114	0.6201091147438709	[9.620047673041995, 6.978226873409457]	0.0012385200701134414
585768314518414 5920	51.588786704534461	15.112654	0.43492762059236373	[8.896219452721796, 5.422668337704664]	-0.0023230106246222367
5857683248263539200	55.337154930785593	15.817067	0.7966484325484037	[10.114963206949012, 8.881641461328663]	0.0004053585912533587
5857683316982887936	58.355642816299905	14.742889	0.3206022126487824	[7.578673985400416, 10.478229143820498]	0.0059125043518889475

G-band pairwise brightness screen assumes approximate passband/aperture response; intrinsic required eclipse cannot be directly compared with diluted pixel loss. Unresolved unlisted companions are not excluded.

Alternative reduction: independently extracted background-subtracted TESScut1x1/3x3/5x5 depths=30.70287/20.33807/11.73582ppt; QLP median=38.83434ppt. Pixel sector65 is a distinct observing occasion on the same TESS instrument; no same-sector QLP comparison claimed. Independent processing is not an independent instrument. Different apertures respond differently to dilution/background.

Gaia source5857683209552196352: period=1.309869111+/-0.000059406d; e=0.0669152; omega=113.4484403deg; periastron BTJD=389.60355377. Closest conjunction offset=-0.0181783d (-0.0138779phase), source-specific full-covariance linear error=0.0704414d. Correlation order P,gamma,K,e,omega,Tp with subindices0,3,4,5. Linear propagation is descriptive, not a Gaussian veto or ownership proof. Both conjunctions retained in ../gaia-geometry.json. Mass function=0.1256829solar, assumed M1=1.53, minimum M2=0.907153; approximate central duration=3.9047h versus observed=2.3340h. TIC radius/main-sequence companion scaling, edge-on conjunction and stellar evolution are assumptions; this is a plausibility check, not a fitted physical solution.

Sector	Same-kind cycle	Measured BTJD	Formal center error d	Conservative input error d	Realized depth ppt
11	1	1602.444488430	0.0010168984	0.0104167472	26.89581
11	2	1603.754644511	0.00261558163	0.0104167472	26.74377
11	3	1605.066170974	0.00258759349	0.0104167472	23.10124
11	4	1606.377380265	0.000711748847	0.0104167472	28.52242

Sector	Same-kind cycle	Measured BTJD	Formal center error d	Conservative input error d	Realized depth ppt
11	5	1607.688660172	0.00147683231	0.0104167472	25.60146
11	6	1608.998950285	0.00169781894	0.0104167472	26.80673
11	11	1615.548137994	0.00120782111	0.0104167472	26.76637
11	12	1616.856687878	0.00198543166	0.0104167472	27.31806
11	13	1618.167342877	0.00110061153	0.0104167472	25.42160
11	14	1619.476071273	0.00132837423	0.0104167472	23.97157
11	16	1622.093999886	0.00123245263	0.0104167472	24.05849
11	17	1623.402872036	0.00172324386	0.0104167472	24.80277
99	1867	4046.613485937	0.000885034206	0.00115746959	42.40669
99	1868	4047.922882068	0.000950982363	0.00115746959	44.39284
99	1869	4049.229846136	0.00098642362	0.00115746959	42.71411
99	1870	4050.542718554	0.000986750072	0.00115746959	45.99303
99	1871	4051.849945423	0.000913072976	0.00115746959	43.76966
99	1872	4053.160524632	0.000652634288	0.00115746959	41.65855
99	1873	4054.471075654	0.00132281247	0.00132281247	40.87095
99	1880	4063.640762974	0.000958309273	0.00115746959	45.25501
99	1881	4064.947593850	0.00126434738	0.00126434738	42.67108
99	1882	4066.260449882	0.00091319178	0.00115746959	47.22560
99	1883	4067.568441797	0.0010034447	0.00115746959	45.13240
99	1884	4068.878607677	0.000781636329	0.00115746959	42.29650
99	1885	4070.190546996	0.00072310757	0.00115746959	45.62045
101	1910	4102.933314968	0.000783086105	0.00115740685	35.51155
101	1911	4104.243350362	0.00059228403	0.00115740685	36.76797
101	1912	4105.551928162	0.000857794657	0.00115740685	37.44313
101	1913	4106.862965661	0.000644475671	0.00115740685	40.08402
101	1914	4108.172771440	0.000654720231	0.00115740685	39.30515
101	1915	4109.483262805	0.000939385056	0.00115740685	39.61965
101	1916	4110.789750600	0.000743022663	0.00115740685	43.25782
101	1920	4116.033960440	0.00122767411	0.00122767411	35.10042
101	1921	4117.338833456	0.000954805704	0.00115740685	32.77576
101	1922	4118.652723320	0.000966872228	0.00115740685	41.09209
101	1924	4121.272988168	0.00122076726	0.00122076726	38.06482
101	1925	4122.579818417	0.000783834205	0.00115740685	38.99117
101	1926	4123.888835028	0.000643771976	0.00115740685	38.67750

Catalogue novelty is separate from this physical verdict. Shared coverage matrix and source identity checks are authoritative for catalogue recovery; period agreement alone cannot establish attribution.

Stable review key: R638.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s65.json - historical/diff_s65.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

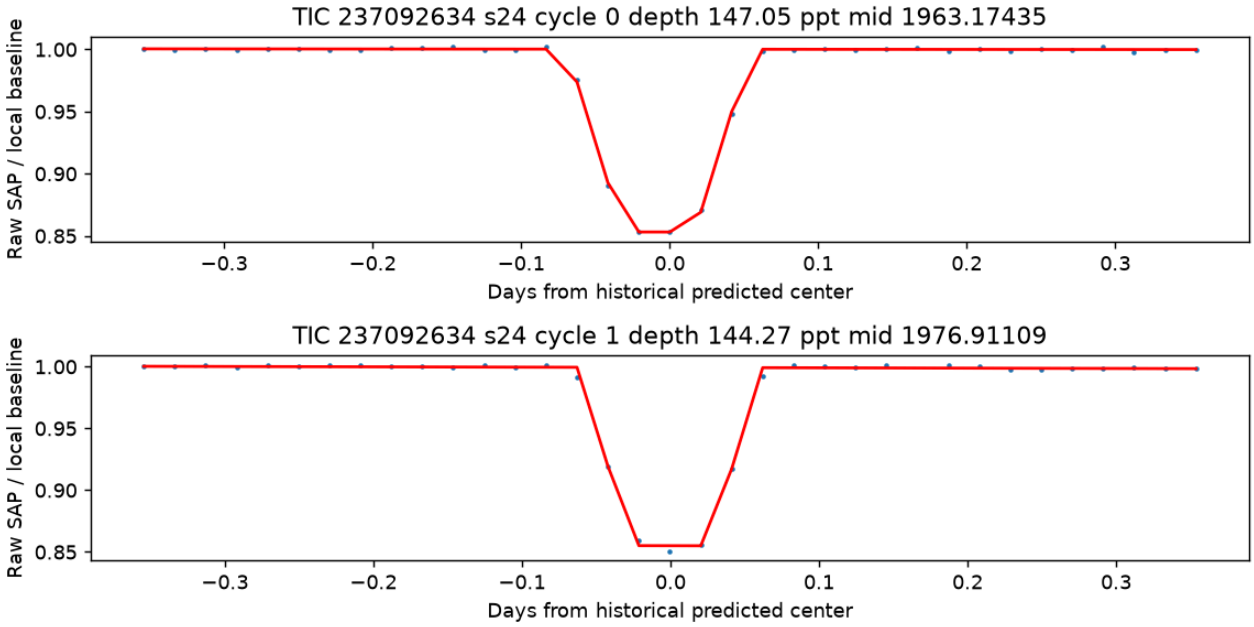
R639. TIC 237092634

physical_verdict: CONFIRMED

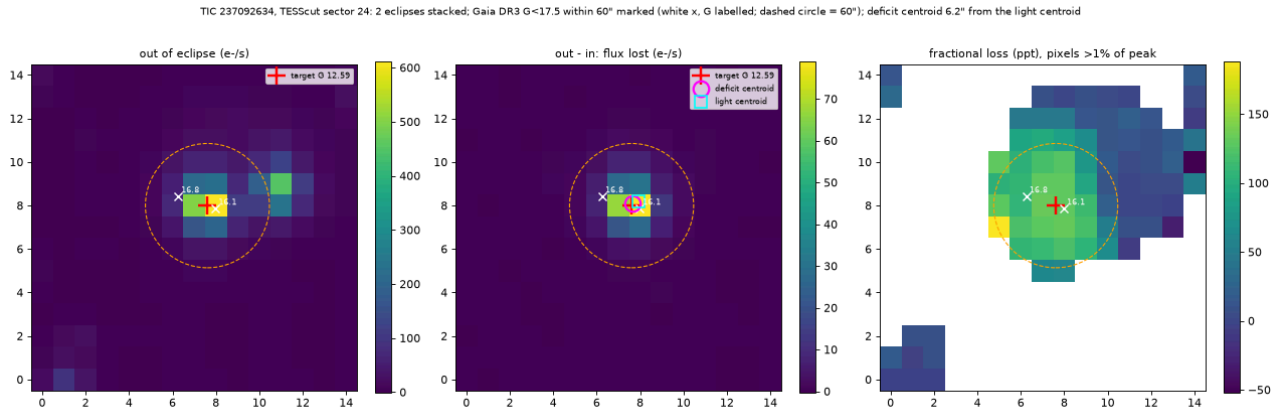
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional additional clean epochs and resolved photometry.



R639; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R639; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 237092634: CONFIRMED

Full bounded physical review: COMPLETE. Two deep consecutive sector24 eclipses establish local recurrence, with zero-dof timing chi-square null. Current availability query found only24/51/78; later predicted primary centers are flagged or gapped, not clean nulls. Bounded shorter integer-alias tests

and coverage are retained in coverage-alias-check.json. No secure secondary; shallow phase0.625 modulation is not assigned as its counterpart. No capable neighbor within60arcsec; compact deficit supports target.

Raw QLP SAP_FLUX representative events and full-sector P/2P morphology were visually inspected. Quality-zero finite positive samples only; predictions guide windows and are never timings. 2 identifiable free-center timings define P=13.736745594572 +/- 0.014731345671d, reference BTJD=1963.174347256 +/- 0.010416634d. Epoch error meaning: Conservative linear timing-fit covariance under the stated cadence floor and scatter inflation; excludes uncalibrated shape and baseline systematics. Epoch kind: Selected repeating eclipse; deeper/secondary identity not established. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced chi square=None; zero degrees of freedom is null. All errors retain their named meanings; no calibrated coverage is claimed. Selected same-kind recurring eclipse family; parity and P/2 coverage checked.

Depth=145.661852ppt is the median realized central trapezoid loss d*min(1,w/(2*k)); extrapolated amplitude and ingress remain separately in measurements.json. No secure secondary; exploratory fit is not interpreted as an eclipse timing or counterpart. Secondary raw pooled windows, actual sectors/cycles and fitted phase are retained in secondary-windows.csv and ../secondary-phases.json. No exhaustive blind alias search asserted. ../alias-audit.json and ../half-period-coverage.json retain recurrence/parity and sampled opposite-phase evidence. Flagged coverage is never a clean null.

Fresh TESScut difference image sector24 stacks2 core/bilateral-sideband events. Target pixel= [7.608280055414019, 7.991850291724219], deficit=[7.628824029140193, 8.102998790503019]; descriptive offset=0.11303pixel, field PSF residual=0.53092. No arbitrary centroid cut or calibrated localization error is imposed. Every capable neighbor below was inspected at its actual projected pixels; close overlapping neighbors retain probable ownership. Other capable sources are displaced from compact deficit core but crowding limits localization. All Gaia neighbors are retained in pixel JSON.

Source	Separation arcsec	G	Required intrinsic loss proxy	Pixel coordinate	Diluted pixel loss
None					

G-band pairwise brightness screen assumes approximate passband/aperture response; intrinsic required eclipse cannot be directly compared with diluted pixel loss. Unresolved unlisted companions are not excluded.

Alternative reduction: independently extracted background-subtracted TESScut1x1/3x3/5x5 depths=129.14812/128.20268/110.90395ppt; QLP median=145.66185ppt. Pixel sector24 shares the same TESS photons with retained QLP and checks extraction robustness. Independent processing is not an independent instrument. Different apertures respond differently to dilution/background.

Gaia source1221891502493891072: period=13.732030054+/-0.005997307d; e=0.2280215; omega=35.1247507deg; periastron BTJD=382.29965614. Closest conjunction offset=0.3351619d (0.0244073phase), source-specific full-covariance linear error=0.7546896d. Correlation order P,gamma,K,e,omega,Tp with subindices0,3,4,5. Linear propagation is descriptive, not a Gaussian veto or ownership proof. Both conjunctions retained in ../gaia-geometry.json. Mass function=0.009780511solar, assumed M1=0.848, minimum M2=0.224001; approximate central duration=3.7567h versus observed=2.8955h. TIC radius/main-sequence companion scaling, edge-on conjunction and stellar evolution are assumptions; this is a plausibility check, not a fitted physical solution.

Sector	Same-kind cycle	Measured BTJD	Formal center error d	Conservative input error d	Realized depth ppt
24	0	1963.174347256	0.000118690193	0.0104166344	147.05460
24	1	1976.911092851	0.000184170627	0.0104166344	144.26910

Catalogue novelty is separate from this physical verdict. Shared coverage matrix and source identity checks are authoritative for catalogue recovery; period agreement alone cannot establish attribution.

Stable review key: R639.

Deciding evidence and numeric artifacts: - coverage-alias-check.json - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s24.json - historical/diff_s24.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

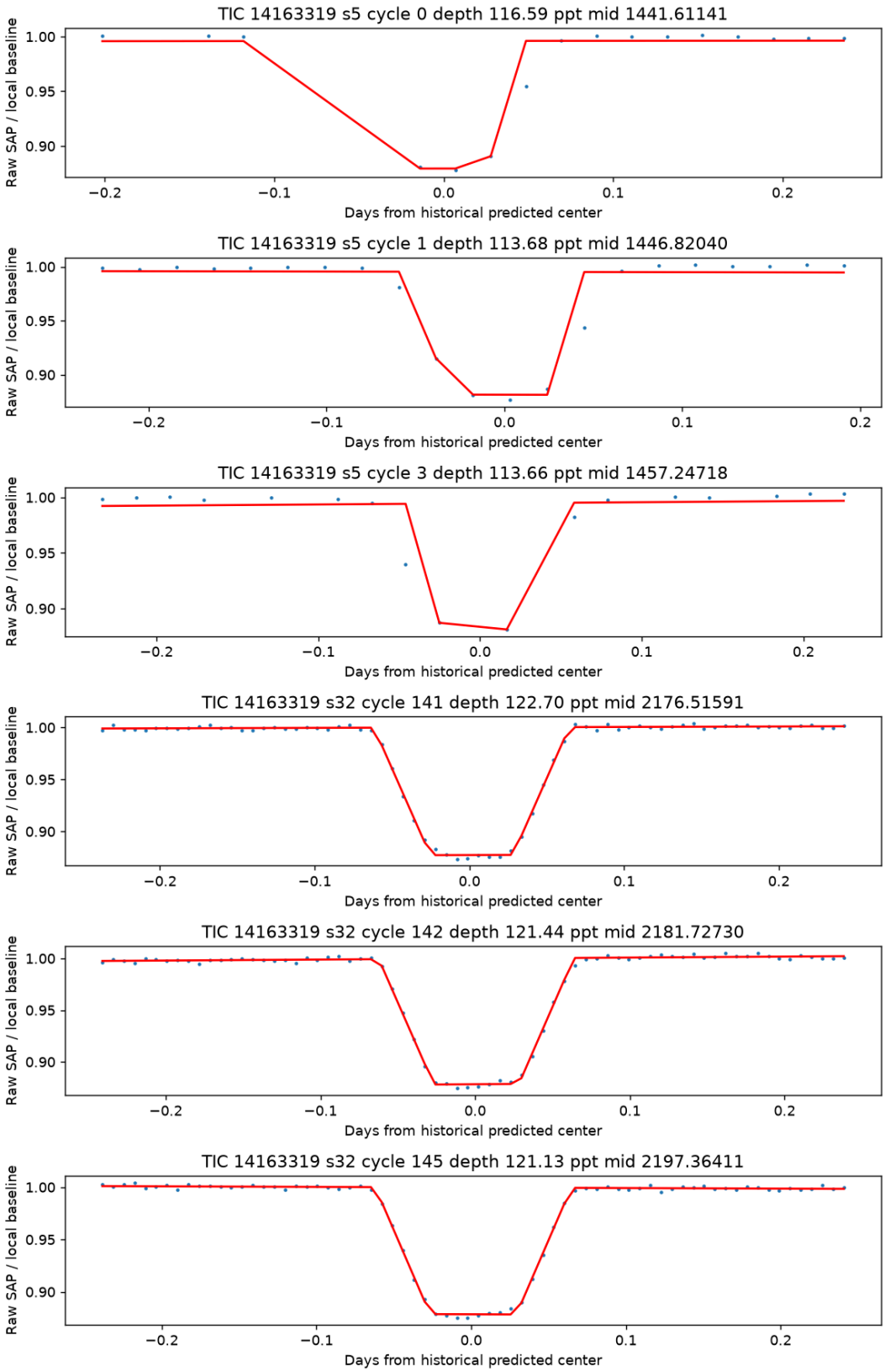
R640. TIC 14163319

physical_verdict: CONFIRMED

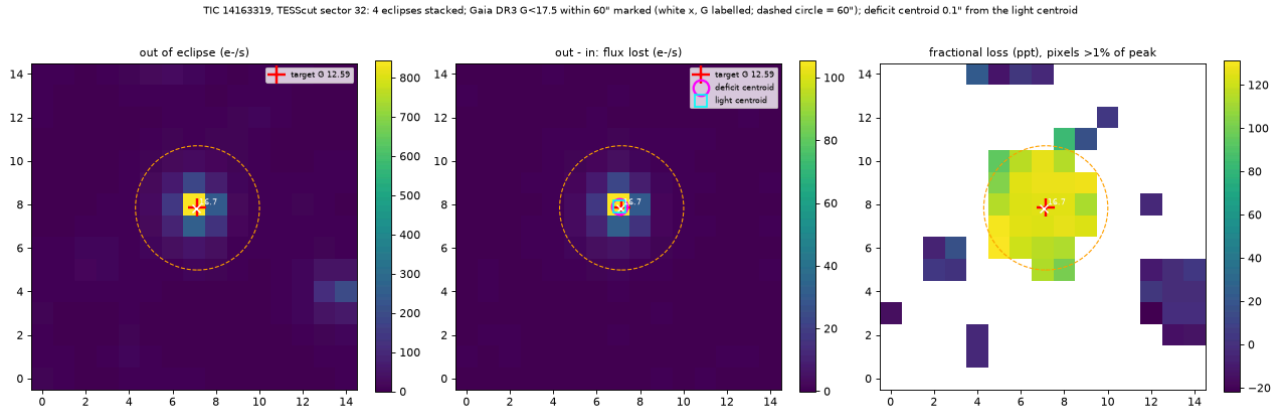
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy to refine physical parameters.



R640; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R640; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 14163319: CONFIRMED

Full bounded physical review: **COMPLETE**. Eight measured narrow repeated eclipses in sectors5/32 with stable local baselines. Sector98 has severe excursions retained in the full phase view but no fitted event; its half-phase artifact is not a secure secondary. No capable Gaia neighbor within60arcsec; compact deficit centered on target.

Raw morphology was inspected in raw-events.png and phase-review.png, including P and2P parity views. 8 free-center trapezoid timings survive finite, identifiable center-covariance checks. Quality-zero raw QLP SAP_FLUX was used; predictions only guide fitting windows. This is representative available evidence, not an all-event/all-sector census.

Measured same-kind eclipse ephemeris: P=5.21209547809 +/- 0.00003890963 d; reference BTJD=1441.61018775. Epoch kind: Selected repeating narrow eclipse; deeper-primary identity unestablished because no secure secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced timing chi-square=0.012498103516124862; zero degrees of freedom would be null. All selected timings follow the named repeating eclipse family. Epoch kind above identifies the deeper primary only where an unequal secondary is established. Model depth is the realized central loss d*min(1,w/(2*k)); extrapolated amplitude and ingress are retained separately. The reference is the fitted first selected cycle, not a Gaia prediction. Within-sector cycle recurrence and parity depths are in ../alias-audit.json. Opposite-phase sampled coverage, quality-zero core/sideband counts and flagged counts are in ../half-period-coverage.json. For265092213, orbital cycles=(historical cycle-1)/2 use deeper odd events; shallower even-cycle secondaries remain in measurements.json. Unequal observed secondary depths exclude mixing eclipse kinds into P/2 where established; systems without secure secondaries retain sampled opposite-phase coverage rather than a claimed clean null; the P/2 and2P raw views were inspected. No exhaustive blind alias search is asserted.

No secure secondary; noisy low-amplitude exploratory fit not interpreted as eclipse. A fitted low-amplitude pooled model is not evidence of a secondary; flagged and missing coverage are not clean nulls.

Fresh difference image: sector32, 4 complete core/sideband events. Target pixel=[7.14991882982819, 7.842173455895029]; deficit centroid=[7.03607414339801, 7.894704335823925]; deficit-target distance=0.1254 pixel. Deficit-light offset=0.123 arcsec is a descriptive field-light displacement, not an arbitrary veto. Fractional PSF-model residual=0.0319 is not a calibrated localization uncertainty. Every flux-capable Gaia source in the60-arcsec list was inspected against the image. Attribution conclusion: Eight measured narrow repeated eclipses in sectors5/32 with stable local baselines. Sector98 has severe excursions retained in the full phase view but no fitted event; its half-phase artifact is not a secure secondary. No capable Gaia neighbor within60arcsec; compact deficit centered on target.

Capable Gaia source	Separation arcsec	G	Intrinsic required eclipse proxy	Pixel coordinate	Diluted loss at pixel
None					

The pairwise G-band required intrinsic loss is an approximate brightness screen, with passband, aperture and blend assumptions. It cannot be directly compared to the diluted fractional loss in a pixel. Noncapable neighbors and all coordinates remain in fresh-pixels/diff_fresh.json and gaia.json. TESS resolution does not exclude unlisted unresolved companions.

Alternative reduction: fresh background-subtracted TESScut1x1/3x3/5x5 target apertures have depths 124.962/124.996/123.915 ppt versus fitted raw QLP median120.118 ppt. Individual cutout event depths are retained in the pixel JSON. Same-sector QLP and cutout extraction share TESS photons and test processing robustness. Distinct sectors are observing occasions on one instrument. Aperture/depth differences reflect dilution, background and extraction response; no intrinsic depth inferred.

Gaia DR3 source4820137033458952960: P=5.21163143 +/-0.00078416 d, e=0.054058, omega=352.51659 degrees, periastron BTJD=387.4306804. Closest edge-on conjunction offset=0.108058 d (0.020734 phase), source-specific full-covariance linear uncertainty=0.218891 d. Full correlation ordering P,gamma,K,e,omega,Tp, selected indices0,3,4,5. This linear approximation is descriptive, not a sigma veto. Both conjunctions and near-reference projection are retained in ../gaia-geometry.json. Mass function=0.0190159 solar masses, assumed M1=0.96, minimum M2=0.31362; approximate central duration=3.600 h versus observed2.639 h. TIC radius and main-sequence companion scaling are assumptions, not a geometry fit; impact parameter and stellar evolution affect duration. Agreement cannot establish ownership among unresolved sources.

Sector	Same-kind cycle	Measured midpoint BTJD	Formal center error d	Input error d	Depth ppt
5	0	1441.61141494	8.6976183e-05	0.01041672	116.594
5	1	1446.82040317	0.00013433208	0.01041672	113.678
5	3	1457.24718184	0.00021058953	0.01041672	113.656
5	4	1462.45852752	0.00034531514	0.01041672	119.108
32	141	2176.51591220	0.00016484825	0.0034722158	122.703
32	142	2181.72729623	0.0002565219	0.0034722158	121.439
32	144	2192.15204104	0.00018509456	0.0034722158	122.841
32	145	2197.36411358	0.00021120905	0.0034722158	121.128

Next check: Optional resolved photometry and spectroscopy to refine physical parameters.

Catalogue status is separate from physical verdict; shared-catalogues/coverage-matrix.csv retains the seven scoped groups. Known recoveries are not new additions; no exhaustive priority claim. Sources and identity are checked in ../identity-check.json and ../source-hashes.json. Numeric artifacts, original receipts and outputs are frozen in ../SHA256SUMS.json; independent helper review is ../helper-audit.md.

Stable review key: R640.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s32.json - historical/diff_s32.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

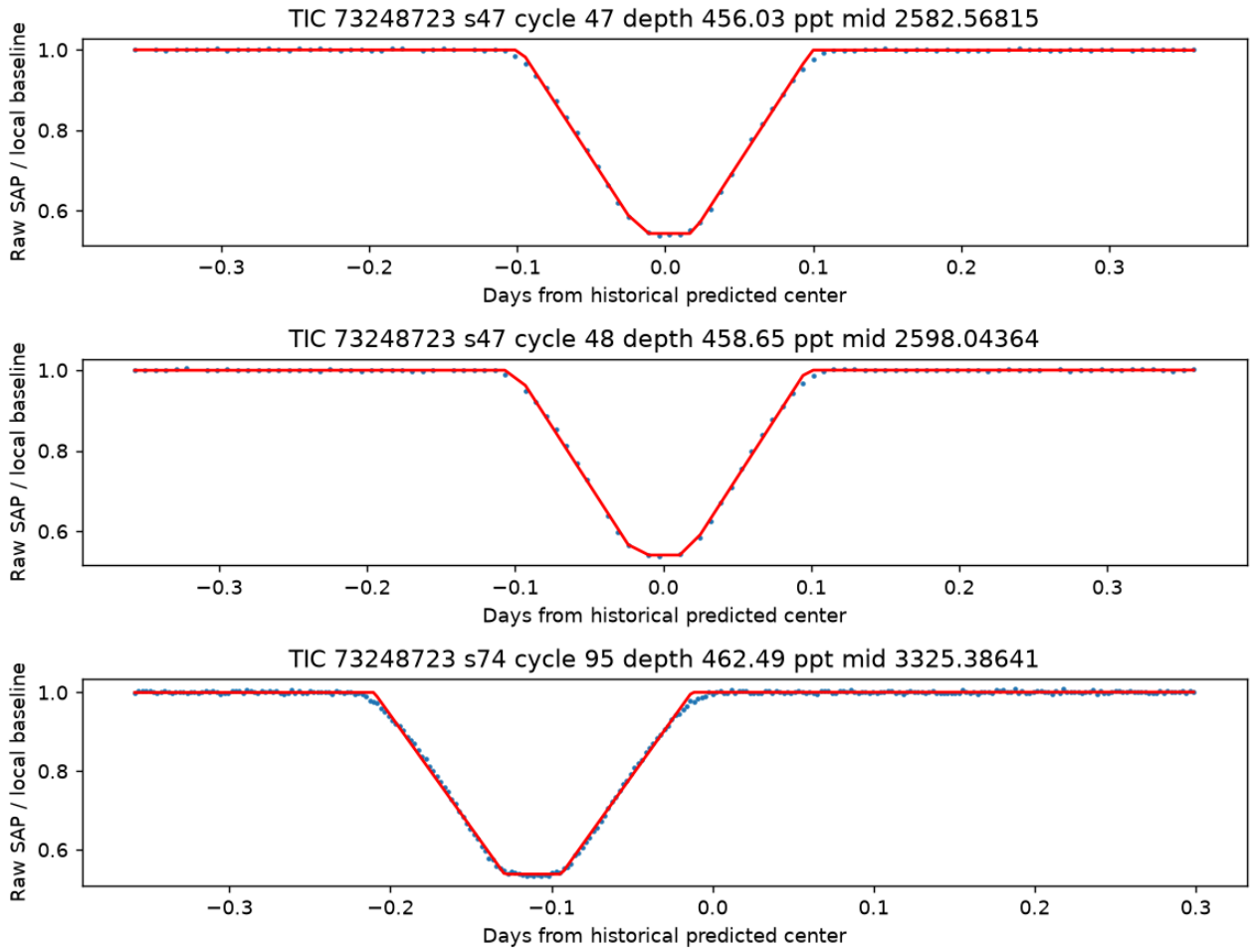
R641. TIC 73248723

physical_verdict: CONFIRMED

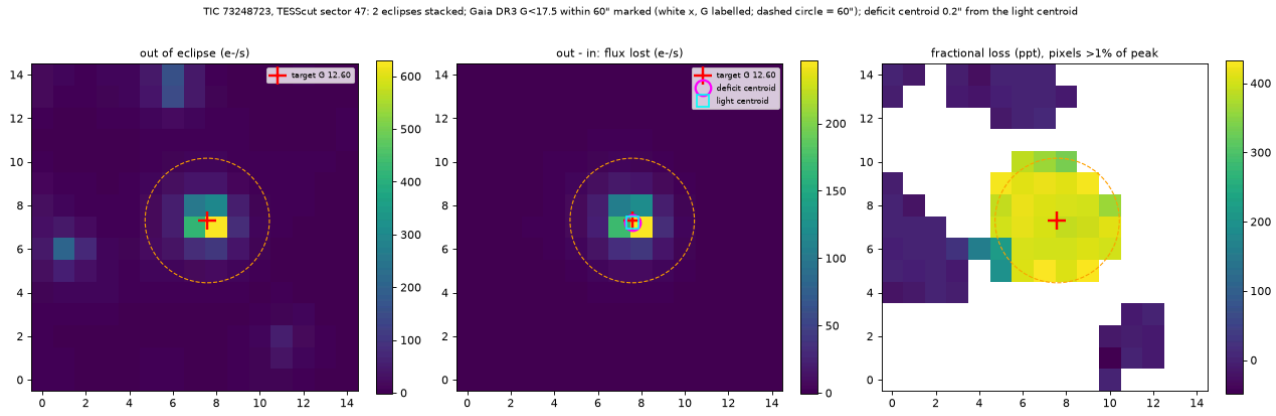
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy to refine physical parameters.



R641; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R641; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 73248723: CONFIRMED

Full bounded physical review: **COMPLETE**. Three deep measured eclipses including consecutive sector47 pair, plus shallower observed secondary near0.633. Consecutive same-kind timings resolve integer counts. No capable Gaia neighbor within60arcsec; compact deficit is on target.

Raw morphology was inspected in raw-events.png and phase-review.png, including P and2P parity views. 3 free-center trapezoid timings survive finite, identifiable center-covariance checks. Quality-zero raw QLP SAP_FLUX was used; predictions only guide fitting windows. This is representative available evidence, not an all-event/all-sector census.

Measured same-kind eclipse ephemeris: P=15.47537927876 +/- 0.00005714027 d; reference BTJD=2582.56820371. Epoch kind: Deeper primary eclipse, established relative to observed shallower secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced timing chi-square=0.000505848822814114; zero degrees of freedom would be null. All selected timings follow the named repeating eclipse family. Epoch kind above identifies the deeper primary only where an unequal secondary is established. Model depth is the realized central loss d*min(1,w/(2*k)); extrapolated amplitude and ingress are retained separately. The reference is the fitted first selected cycle, not a Gaia prediction. Within-sector cycle recurrence and parity depths are in ../alias-audit.json. Opposite-phase sampled coverage, quality-zero core/sideband counts and flagged counts are in ../half-period-coverage.json. For265092213, orbital cycles=(historical cycle-1)/2 use deeper odd events; shallower even-cycle secondaries remain in measurements.json. Unequal observed secondary depths exclude mixing eclipse kinds into P/2 where established; systems without secure secondaries retain sampled opposite-phase coverage rather than a claimed clean null; the P/2 and2P raw views were inspected. No exhaustive blind alias search is asserted.

Narrow shallower secondary visible in raw pooled windows. Depth is the realized center deficit of the fitted shape, with its amplitude parameter retained separately in the JSON. Measured pooled secondary phase=0.6338868, formal phase covariance error=8.2481e-06; depth=118.092 ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Search-guide provenance: Gaia conjunction search guide, adjusted from observed raw phase morphology for this batch. Final phase is a free-center fit to observed raw flux..

Fresh difference image: sector47, 2 complete core/sideband events. Target pixel=[7.582533626121874, 7.306842056359656]; deficit centroid=[7.5962827738466565, 7.23515898042945]; deficit-target distance=0.0730 pixel. Deficit-light offset=0.183 arcsec is a descriptive field-light displacement, not an arbitrary veto. Fractional PSF-model residual=0.0296 is not a calibrated localization uncertainty. Every flux-capable Gaia source in the60-arcsec list was inspected against the image. Attribution conclusion: Three deep measured eclipses including consecutive sector47 pair, plus shallower observed secondary near0.633. Consecutive same-kind timings resolve integer counts. No capable Gaia neighbor within60arcsec; compact deficit is on target.

Capable Gaia source	Separation arcsec	G	Intrinsic required eclipse proxy	Pixel coordinate	Diluted loss at pixel
None					

The pairwise G-band required intrinsic loss is an approximate brightness screen, with passband, aperture and blend assumptions. It cannot be directly compared to the diluted fractional loss in a pixel. Noncapable neighbors and all coordinates remain in fresh-pixels/diff_fresh.json and gaia.json. TESS resolution does not exclude unlisted unresolved companions.

Alternative reduction: fresh background-subtracted TESScut1x1/3x3/5x5 target apertures have depths 391.705/399.923/402.191 ppt versus fitted raw QLP median458.652 ppt. Individual cutout event depths are retained in the pixel JSON. Same-sector QLP and cutout extraction share TESS photons and test processing robustness. Distinct sectors are observing occasions on one instrument. Aperture/depth differences reflect dilution, background and extraction response; no intrinsic depth inferred.

Gaia DR3 source1091924619587535104: P=15.47924076 +/-0.00266914 d, e=0.254602, omega=5.95540 degrees, periastron BTJD=382.4600970. Closest edge-on conjunction offset=-0.347493 d (-0.022449 phase), source-specific full-covariance linear uncertainty=0.423518 d. Full correlation ordering P,gamma,K,e,omega,Tp, selected indices0,3,4,5. This linear approximation is descriptive, not a sigma veto. Both conjunctions and near-reference projection are retained in ../gaia-geometry.json. Mass function=0.10611 solar masses, assumed M1=0.876, minimum M2=0.61898; approximate central duration=6.220 h versus observed4.722 h. TIC radius and main-sequence companion scaling are assumptions, not a geometry fit; impact parameter and stellar evolution affect duration. Agreement cannot establish ownership among unresolved sources.

Sector	Same-kind cycle	Measured midpoint BTJD	Formal center error d	Input error d	Depth ppt
47	47	2582.56814907	5.2655982e-05	0.0034722054	456.028
47	48	2598.04363879	5.9247597e-05	0.0034722054	458.652
74	95	3325.38640896	8.4097418e-05	0.0011573987	462.491

Next check: Optional resolved photometry and spectroscopy to refine physical parameters.

Catalogue status is separate from physical verdict; shared-catalogues/coverage-matrix.csv retains the seven scoped groups. Known recoveries are not new additions; no exhaustive priority claim. Sources and identity are checked in ../identity-check.json and ../source-hashes.json. Numeric artifacts, original receipts and outputs are frozen in ../SHA256SUMS.json; independent helper review is ../helper-audit.md.

Stable review key: R641.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s47.json - historical/diff_s47.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

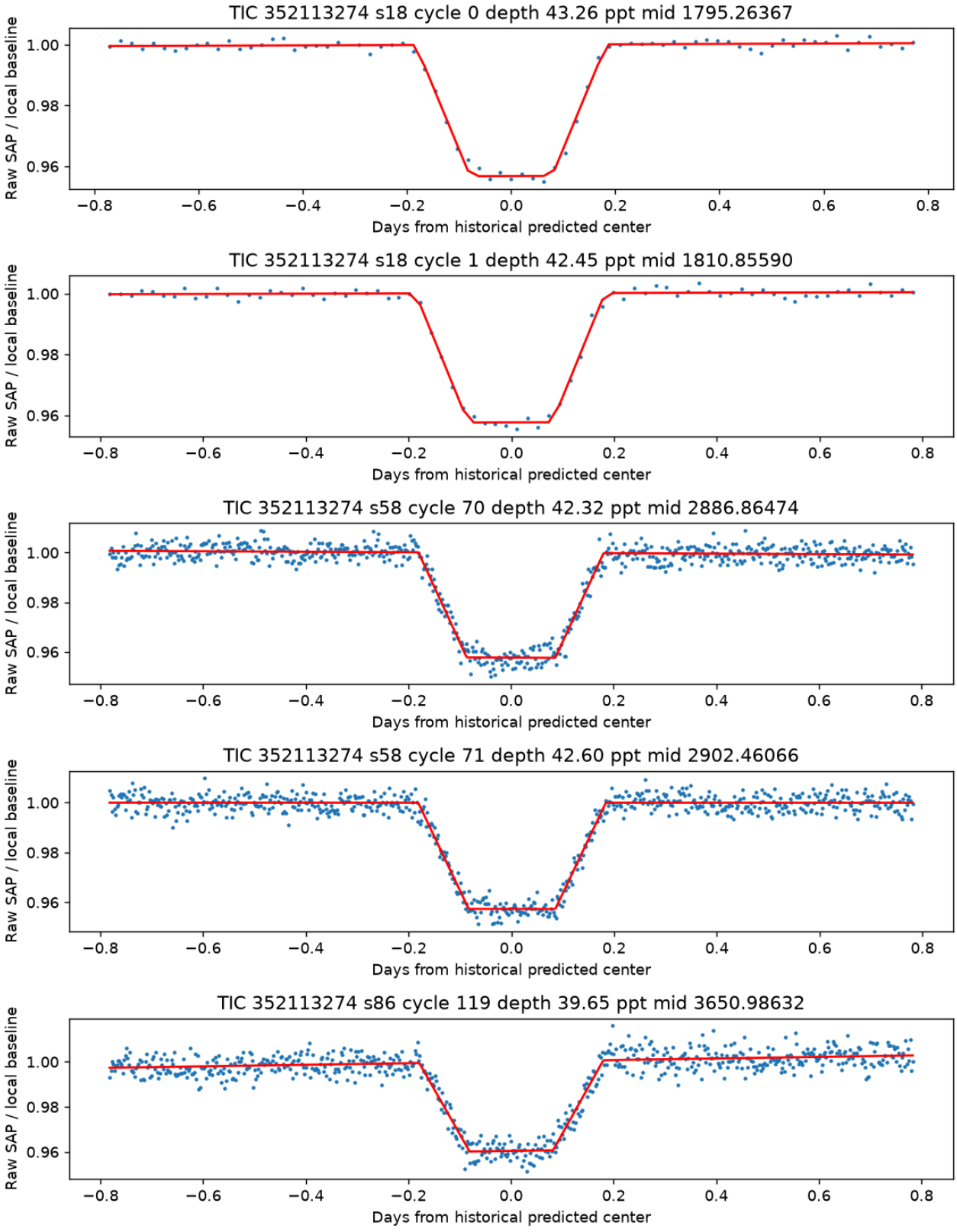
R642. TIC 352113274

physical_verdict: PROBABLE

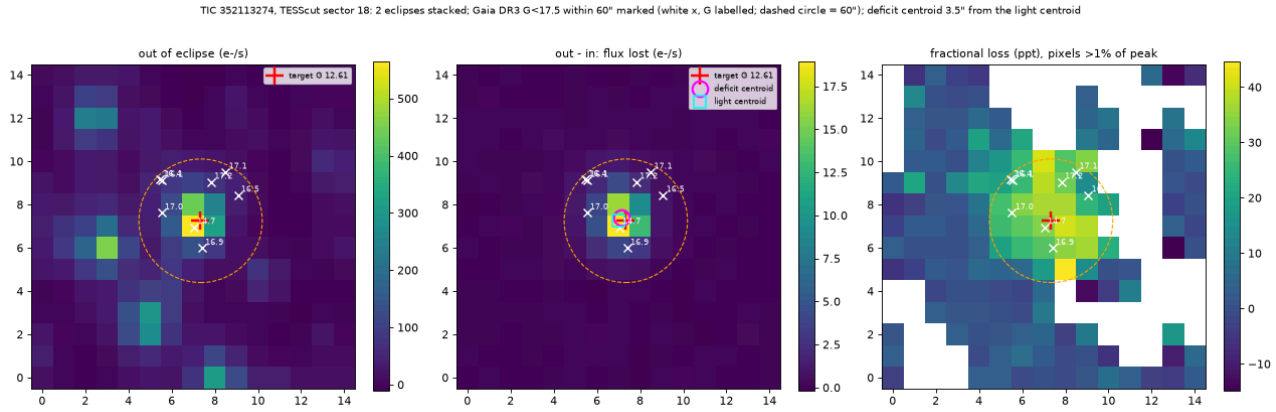
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or multi-source PRF localization for target and listed close capable neighbor.



R642; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R642; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 352113274: PROBABLE

Full bounded physical review: **COMPLETE**. Five repeated narrow flat-bottomed eclipses and shallower secondary near0.5003. Capable Gaia533640894735716352 at8.83arcsec is inside the unresolved TESS footprint; deficit offset0.248pixel and fractional-loss morphology do not securely separate source ownership. Attribution remains unresolved.

Raw morphology was inspected in raw-events.png and phase-review.png, including P and2P parity views. 5 free-center trapezoid timings survive finite, identifiable center-covariance checks. Quality-zero raw QLP SAP_FLUX was used; predictions only guide fitting windows. This is representative available evidence, not an all-event/all-sector census.

Measured same-kind eclipse ephemeris: P=15.59430416645 +/- 0.00002841631 d; reference BTJD=1795.26419936. Epoch kind: Deeper primary eclipse, established relative to observed shallower secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced timing chi-square=0.3472024759867856; zero degrees of freedom would be null. All selected timings follow the named repeating eclipse family. Epoch kind above identifies the deeper primary only where an unequal secondary is established. Model depth is the realized central loss d*min(1,w/(2*k)); extrapolated amplitude and ingress are retained separately. The reference is the fitted first selected cycle, not a Gaia prediction. Within-sector cycle recurrence and parity depths are in ../alias-audit.json. Opposite-phase sampled coverage, quality-zero core/sideband counts and flagged counts are in ../half-period-coverage.json. For265092213, orbital cycles=(historical cycle-1)/2 use deeper odd events; shallower even-cycle secondaries remain in measurements.json. Unequal observed secondary depths exclude mixing eclipse kinds into P/2 where established; systems without secure secondaries retain sampled opposite-phase coverage rather than a claimed clean null; the P/2 and2P raw views were inspected. No exhaustive blind alias search is asserted.

Narrow shallower secondary visible in raw pooled windows. Depth is the realized center deficit of the fitted shape, with its amplitude parameter retained separately in the JSON. Measured pooled secondary phase=0.5001311, formal phase covariance error=0.00019978; depth=6.055 ppt. Formal pooled

trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Search-guide provenance: Gaia conjunction search guide, adjusted from observed raw phase morphology for this batch. Final phase is a free-center fit to observed raw flux..

Fresh difference image: sector18, 2 complete core/sideband events. Target pixel=[7.340950005119732, 7.250381829349443]; deficit centroid=[7.139168122319735, 7.39419140243012]; deficit-target distance=0.2478 pixel. Deficit-light offset=3.515 arcsec is a descriptive field-light displacement, not an arbitrary veto. Fractional PSF-model residual=0.6046 is not a calibrated localization uncertainty. Every flux-capable Gaia source in the60-arcsec list was inspected against the image. Attribution conclusion: Five repeated narrow flat-bottomed eclipses and shallower secondary near0.5003. Capable Gaia5336408947357163 52 at8.83arcsec is inside the unresolved TESS footprint; deficit offset0.248pixel and fractional-loss morphology do not securely separate source ownership. Attribution remains unresolved.

Capable Gaia source	Separation arcsec	G	Intrinsic required eclipse proxy	Pixel coordinate	Diluted loss at pixel
5336408947357163 52	8.829	14.72 9	0.3427	[7.035172346125428, 6.952166422855449]	0.03354110241954 541

The pairwise G-band required intrinsic loss is an approximate brightness screen, with passband, aperture and blend assumptions. It cannot be directly compared to the diluted fractional loss in a pixel. Noncapable neighbors and all coordinates remain in fresh-pixels/diff_fresh.json and gaia.json. TESS resolution does not exclude unlisted unresolved companions.

Alternative reduction: fresh background-subtracted TESScut1x1/3x3/5x5 target apertures have depths 33.541/34.888/31.916 ppt versus fitted raw QLP median42.453 ppt. Individual cutout event depths are retained in the pixel JSON. Same-sector QLP and cutout extraction share TESS photons and test processing robustness. Distinct sectors are observing occasions on one instrument. Aperture/depth differences reflect dilution, background and extraction response; no intrinsic depth inferred.

Gaia DR3 source5336408904349899 52: P=15.58710156 +/-0.00621197 d, e=0.047095, omega=236.23080 degrees, periastron BTJD=382.7113359. Closest edge-on conjunction offset=0.324267 d (0.020804 phase), source-specific full-covariance linear uncertainty=0.572360 d. Full correlation ordering P,gamma,K,e,omega,Tp, selected indices0,3,4,5. This linear approximation is descriptive, not a sigma veto. Both conjunctions and near-reference projection are retained in ../gaia-geometry.json. Mass function=0.0354695 solar masses, assumed M1=1.2, minimum M2=0.46078; approximate central duration=12.247 h versus observed8.761 h. TIC radius and main-sequence companion scaling are assumptions, not a geometry fit; impact parameter and stellar evolution affect duration. Agreement cannot establish ownership among unresolved sources.

Sector	Same-kind cycle	Measured midpoint BTJD	Formal center error d	Input error d	Depth ppt
18	0	1795.26367444	0.0011213155	0.010416686	43.265
18	1	1810.85589511	0.00090858936	0.010416686	42.453
58	70	2886.86474141	0.00073860706	0.0011574113	42.319
58	71	2902.46065562	0.00085324619	0.0011574113	42.601
86	119	3650.98631659	0.0012077339	0.0012077339	39.647

Next check: Resolved eclipse photometry or multi-source PRF localization for target and listed close capable neighbor.

Catalogue status is separate from physical verdict; shared-catalogues/coverage-matrix.csv retains the seven scoped groups. Known recoveries are not new additions; no exhaustive priority claim. Sources and identity are checked in ../identity-check.json and ../source-hashes.json. Numeric artifacts, original

receipts and outputs are frozen in ../SHA256SUMS.json; independent helper review is ../helper-audit.md.

Stable review key: R642.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s18.json - historical/diff_s18.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

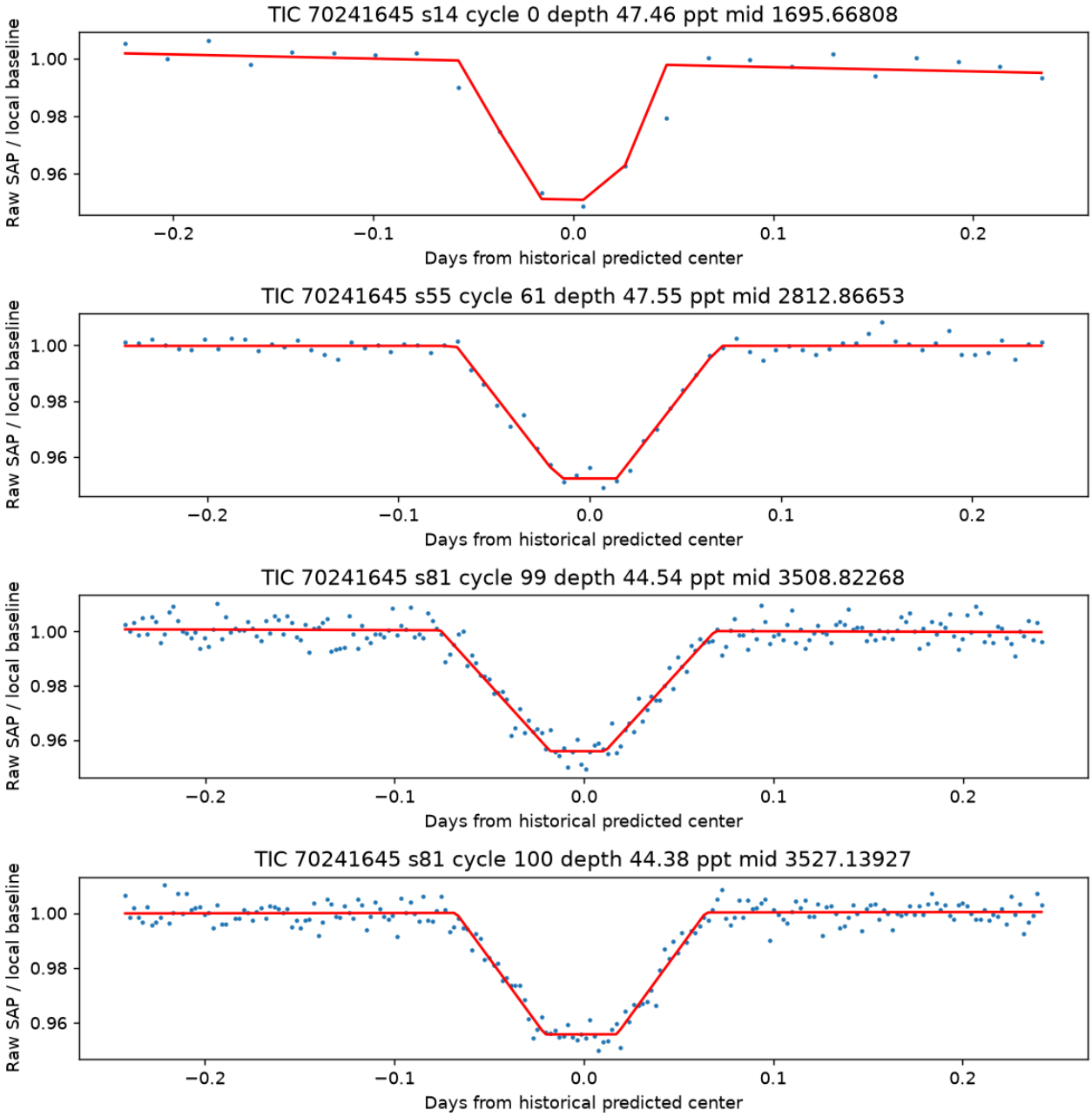
R643. TIC 70241645

physical_verdict: CONFIRMED

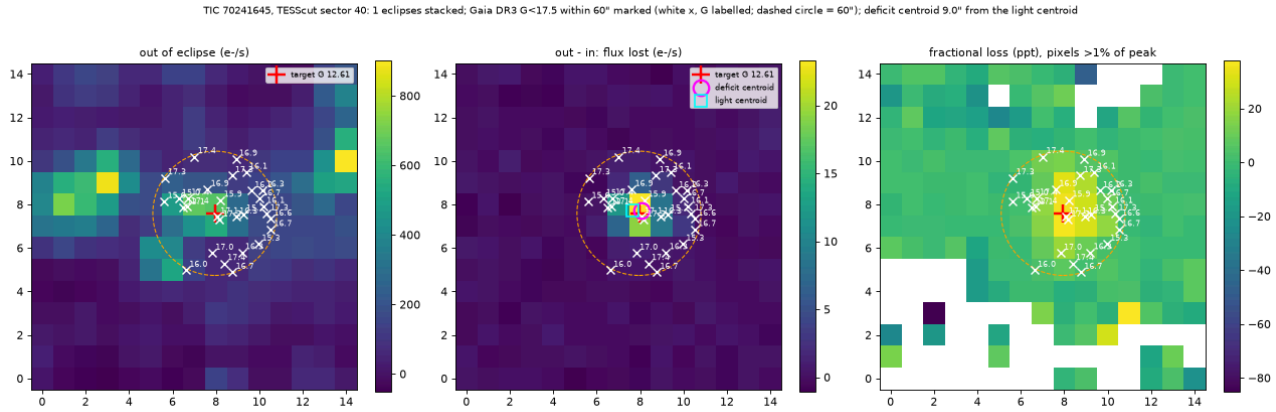
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy to refine physical parameters.



R643; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R643; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 70241645: CONFIRMED

Full bounded physical review: **COMPLETE**. Four narrow repeated eclipses in three sectors, including consecutive sector81 same-kind events. No secure secondary. All five capable neighbors at25.25-49.41arcsec are displaced from compact target deficit; crowded field light centroid is descriptive. Custom cutout reduction in sector40 recovers one event, a distinct observing occasion from the retained QLP sectors.

Raw morphology was inspected in raw-events.png and phase-review.png, including P and2P parity views. 4 free-center trapezoid timings survive finite, identifiable center-covariance checks. Quality-zero raw QLP SAP_FLUX was used; predictions only guide fitting windows. This is representative available evidence, not an all-event/all-sector census.

Measured same-kind eclipse ephemeris: P=18.31468255723 +/- 0.00007003110 d; reference BTJD=1695.67009353. Epoch kind: Selected repeating narrow eclipse; deeper-primary identity unestablished because no secure secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced timing chi-square=0.7283698884494509; zero degrees of freedom would be null. All selected timings follow the named repeating eclipse family. Epoch kind above identifies the deeper primary only where an unequal secondary is established. Model depth is the realized central loss d*min(1,w/(2*k)); extrapolated amplitude and ingress are retained separately. The reference is the fitted first selected cycle, not a Gaia prediction. Within-sector cycle recurrence and parity depths are in ../alias-audit.json. Opposite-phase sampled coverage, quality-zero core/sideband counts and flagged counts are in ../half-period-coverage.json. For265092213, orbital cycles=(historical cycle-1)/2 use deeper odd events; shallower even-cycle secondaries remain in measurements.json. Unequal observed secondary depths exclude mixing eclipse kinds into P/2 where established; systems without secure secondaries retain sampled opposite-phase coverage rather than a claimed clean null; the P/2 and2P raw views were inspected. No exhaustive blind alias search is asserted.

No secure secondary; noisy low-amplitude exploratory fit not interpreted as eclipse. A fitted low-amplitude pooled model is not evidence of a secondary; flagged and missing coverage are not clean nulls.

Fresh difference image: sector40, 1 complete core/sideband events. Target pixel=[7.950635389498599, 7.59246590910835]; deficit centroid=[8.068951890397704, 7.720586638946299]; deficit-target distance=0.1744 pixel. Deficit-light offset=8.992 arcsec is a descriptive field-light displacement, not an arbitrary veto. Fractional PSF-model residual=0.7753 is not a calibrated localization uncertainty. Every flux-capable Gaia source in the60-arcsec list was inspected against the image. Attribution conclusion: Four narrow repeated eclipses in three sectors, including consecutive sector81 same-kind events. No secure secondary. All five capable neighbors at25.25-49.41arcsec are displaced from compact target deficit; crowded field light centroid is descriptive. Custom cutout reduction in sector40 recovers one event, a distinct observing occasion from the retained QLP sectors.

Capable Gaia source	Separation arcsec	G	Intrinsic required eclipse proxy	Pixel coordinate	Diluted loss at pixel
2047234339629598848	25.255	15.205	0.5481	[9.24748653772242, 7.557210838700486]	0.029144313299321458
2047234129153105920	28.847	13.061	0.1157	[6.5067065971801075, 7.857127756560375]	0.0178316718512321
2047234064751686912	35.794	14.998	0.4609	[6.284061131625569, 8.269869821427656]	0.00055370974293 62335
2047234167830903168	47.232	15.581	0.7560	[5.617765024736477, 8.139778795246515]	0.00055370974293 62335
2047234335311560832	49.415	15.344	0.6170	[9.963426898186825, 6.185879131617189]	0.00264257069798 82296

The pairwise G-band required intrinsic loss is an approximate brightness screen, with passband, aperture and blend assumptions. It cannot be directly compared to the diluted fractional loss in a pixel. Noncapable neighbors and all coordinates remain in fresh-pixels/diff_fresh.json and gaia.json. TESS resolution does not exclude unlisted unresolved companions.

Alternative reduction: fresh background-subtracted TESScut1x1/3x3/5x5 target apertures have depths 35.690/26.576/16.909 ppt versus fitted raw QLP median46.002 ppt. Individual cutout event depths are retained in the pixel JSON. Cutout sector is a distinct observing occasion on the same TESS instrument relative to retained QLP sectors. This alternative reduction shares the FFI photons of that sector; no same-sector QLP comparison claimed. Distinct sectors are observing occasions on one instrument. Aperture/depth differences reflect dilution, background and extraction response; no intrinsic depth inferred.

Gaia DR3 source2047234129153107328: P=18.30387003 +/-0.02060793 d, e=0.138705, omega=46.05718 degrees, periastron BTJD=393.9868668. Closest edge-on conjunction offset=0.394804 d (0.021569 phase), source-specific full-covariance linear uncertainty=1.674045 d. Full correlation ordering P,gamma,K,e,omega,Tp, selected indices0,3,4,5. This linear approximation is descriptive, not a sigma veto. Both conjunctions and near-reference projection are retained in ../gaia-geometry.json. Mass function=0.00274715 solar masses, assumed M1=0.812, minimum M2=0.13507; approximate central duration=4.146 h versus observed3.234 h. TIC radius and main-sequence companion scaling are assumptions, not a geometry fit; impact parameter and stellar evolution affect duration. Agreement cannot establish ownership among unresolved sources.

Sector	Same-kind cycle	Measured midpoint BTJD	Formal center error d	Input error d	Depth ppt
14	0	1695.66807769	0.00096081319	0.010416903	47.464
55	61	2812.86653178	0.00065822503	0.003472144	47.553
81	99	3508.82267861	0.00074764861	0.0011574063	44.539
81	100	3527.13927307	0.00060391125	0.0011574063	44.376

Next check: Optional resolved photometry and spectroscopy to refine physical parameters.

Catalogue status is separate from physical verdict; shared-catalogues/coverage-matrix.csv retains the seven scoped groups. Known recoveries are not new additions; no exhaustive priority claim. Sources and identity are checked in ../identity-check.json and ../source-hashes.json. Numeric artifacts, original receipts and outputs are frozen in ../SHA256SUMS.json; independent helper review is ../helper-audit.md.

Stable review key: R643.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s40.json - historical/diff_s40.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

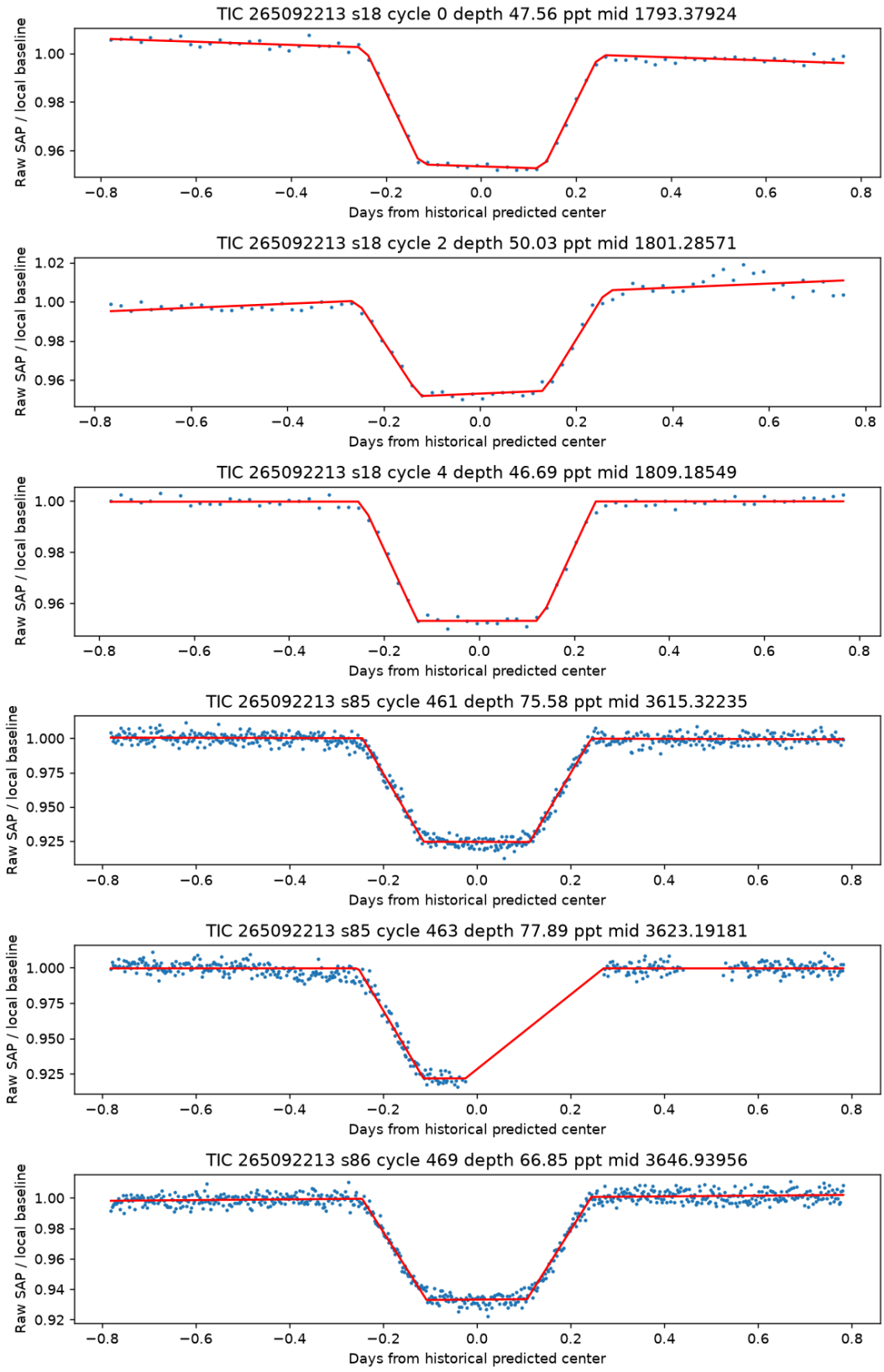
R644. TIC 265092213

physical_verdict: CONFIRMED

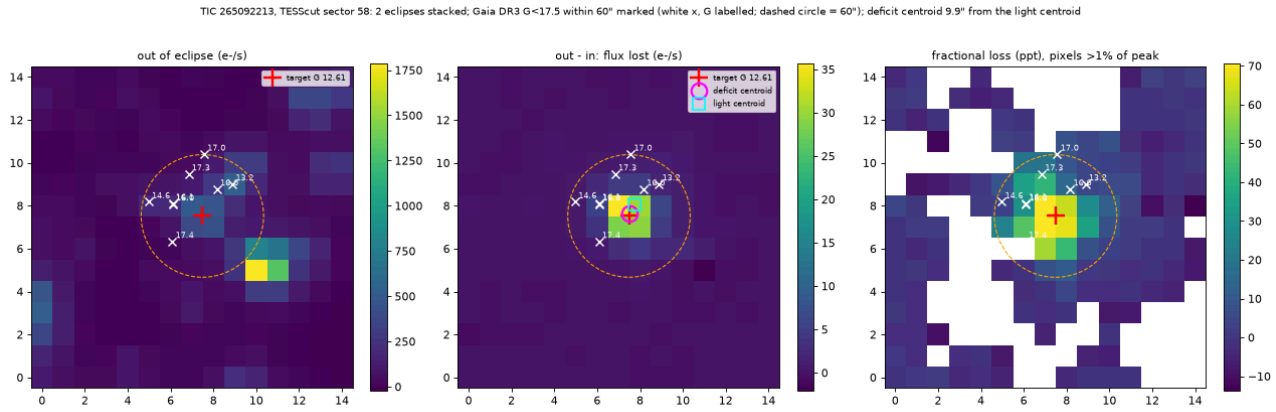
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy to refine physical parameters.



R644; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R644; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 265092213: CONFIRMED

Full bounded physical review: **COMPLETE**. Historical3.95216d SAP branch mixed unequal47/76ppt eclipses. Six identifiable deeper odd-cycle timings yield7.90432331d; one odd event has singular timing covariance and is excluded. Actual shallower secondary near0.50015. Both capable42.66/53.06arcsec neighbors are displaced from compact target deficit.

Raw morphology was inspected in raw-events.png and phase-review.png, including P and2P parity views. 6 free-center trapezoid timings survive finite, identifiable center-covariance checks. Quality-zero raw QLP SAP_FLUX was used; predictions only guide fitting windows. This is representative available evidence, not an all-event/all-sector census.

Measured same-kind eclipse ephemeris: P=7.90432330953 +/- 0.00002614164 d; reference BTJD=1797.32740404. Epoch kind: Deeper primary eclipse: odd historical half-period cycle, orbital cycle=(historical cycle-1)/2. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced timing chi-square=0.33572566372884677; zero degrees of freedom would be null. All selected timings follow the named repeating eclipse family. Epoch kind above identifies the deeper primary only where an unequal secondary is established. Model depth is the realized central loss d*min(1,w/(2*k)); extrapolated amplitude and ingress are retained separately. The reference is the fitted first selected cycle, not a Gaia prediction. Within-sector cycle recurrence and parity depths are in ../alias-audit.json. Opposite-phase sampled coverage, quality-zero core/sideband counts and flagged counts are in ../half-period-coverage.json. For265092213, orbital cycles=(historical cycle-1)/2 use deeper odd events; shallower even-cycle secondaries remain in measurements.json. Unequal observed secondary depths exclude mixing eclipse kinds into P/2 where established; systems without secure secondaries retain sampled opposite-phase coverage rather than a claimed clean null; the P/2 and2P raw views were inspected. No exhaustive blind alias search is asserted.

Narrow shallower secondary visible in raw pooled windows. Depth is the realized center deficit of the fitted shape, with its amplitude parameter retained separately in the JSON. Measured pooled secondary phase=0.5001495, formal phase covariance error=8.9994e-05; depth=47.184 ppt. Formal pooled

trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Search-guide provenance: Gaia conjunction search guide, adjusted from observed raw phase morphology for this batch. Final phase is a free-center fit to observed raw flux..

Fresh difference image: sector58, 2 complete core/sideband events. Target pixel=[7.484019684521309, 7.524537761987151]; deficit centroid=[7.488670517582835, 7.632534694794676]; deficit-target distance=0.1081 pixel. Deficit-light offset=9.901 arcsec is a descriptive field-light displacement, not an arbitrary veto. Fractional PSF-model residual=0.9331 is not a calibrated localization uncertainty. Every flux-capable Gaia source in the60-arcsec list was inspected against the image. Attribution conclusion: Historical3.95216d SAP branch mixed unequal47/76ppt eclipses. Six identifiable deeper odd-cycle timings yield7.90432331d; one odd event has singular timing covariance and is excluded. Actual shallower secondary near0.50015. Both capable42.66/53.06arcsec neighbors are displaced from compact target deficit.

Capable Gaia source	Separation arcsec	G	Intrinsic required eclipse proxy	Pixel coordinate	Diluted loss at pixel
4547832292980501 76	42.655	13.177	0.2024	[8.88609308012345, 9.017631157858267]	0.00402600708568 06525
4547838864341285 12	53.062	14.621	0.5559	[4.972094071398196, 8.190314581285458]	0.00365160602974 08833

The pairwise G-band required intrinsic loss is an approximate brightness screen, with passband, aperture and blend assumptions. It cannot be directly compared to the diluted fractional loss in a pixel. Noncapable neighbors and all coordinates remain in fresh-pixels/diff_fresh.json and gaia.json. TESS resolution does not exclude unlisted unresolved companions.

Alternative reduction: fresh background-subtracted TESScut1x1/3x3/5x5 target apertures have depths 70.614/56.636/37.223 ppt versus fitted raw QLP median75.368 ppt. Individual cutout event depths are retained in the pixel JSON. Cutout sector is a distinct observing occasion on the same TESS instrument relative to retained QLP sectors. This alternative reduction shares the FFI photons of that sector; no same-sector QLP comparison claimed. Distinct sectors are observing occasions on one instrument. Aperture/depth differences reflect dilution, background and extraction response; no intrinsic depth inferred.

Gaia DR3 source4547832335990968 32: P=7.90368185 +/-0.00347188 d, e=0.073183, omega=94.54854 degrees, periastron BTJD=390.4106392. Closest edge-on conjunction offset=0.147419 d (0.018652 phase), source-specific full-covariance linear uncertainty=0.670062 d. Full correlation ordering P,gamma,K,e,omega,Tp, selected indices0,3,4,5. This linear approximation is descriptive, not a sigma veto. Both conjunctions and near-reference projection are retained in ../gaia-geometry.json. Mass function=0.148369 solar masses, assumed M1=1.31, minimum M2=0.89754; approximate central duration=14.474 h versus observed11.771 h. TIC radius and main-sequence companion scaling are assumptions, not a geometry fit; impact parameter and stellar evolution affect duration. Agreement cannot establish ownership among unresolved sources.

Sector	Same-kind cycle	Measured midpoint BTJD	Formal center error d	Input error d	Depth ppt
18	0	1797.32729978	0.0020337344	0.01041674	74.164
18	1	1805.23230615	0.0024092773	0.01041674	76.485
18	2	1813.13551377	0.0024063798	0.01041674	76.219
85	230	3615.32235441	0.00056723322	0.0011574273	75.584
85	232	3631.12932048	0.0011561024	0.0011574273	75.152
86	234	3646.93956139	0.00070046158	0.0011573889	66.849

Next check: Optional resolved photometry and spectroscopy to refine physical parameters.

Catalogue status is separate from physical verdict; shared-catalogues/coverage-matrix.csv retains the seven scoped groups. Known recoveries are not new additions; no exhaustive priority claim. Sources and identity are checked in ../identity-check.json and ../source-hashes.json. Numeric artifacts, original receipts and outputs are frozen in ../SHA256SUMS.json; independent helper review is ../helper-audit.md.

Stable review key: R644.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s58.json - historical/diff_s58.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

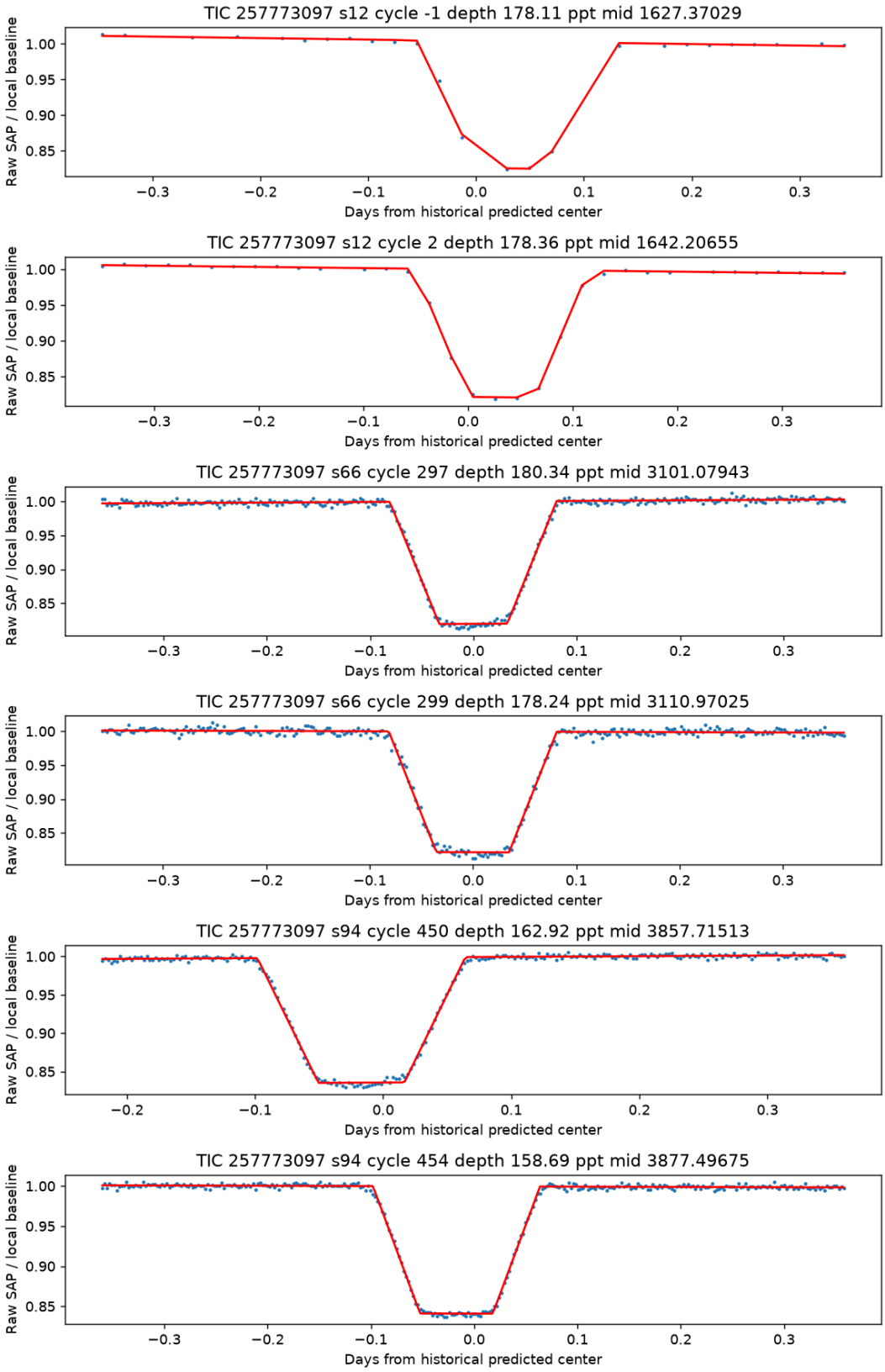
R645. TIC 257773097

physical_verdict: CONFIRMED

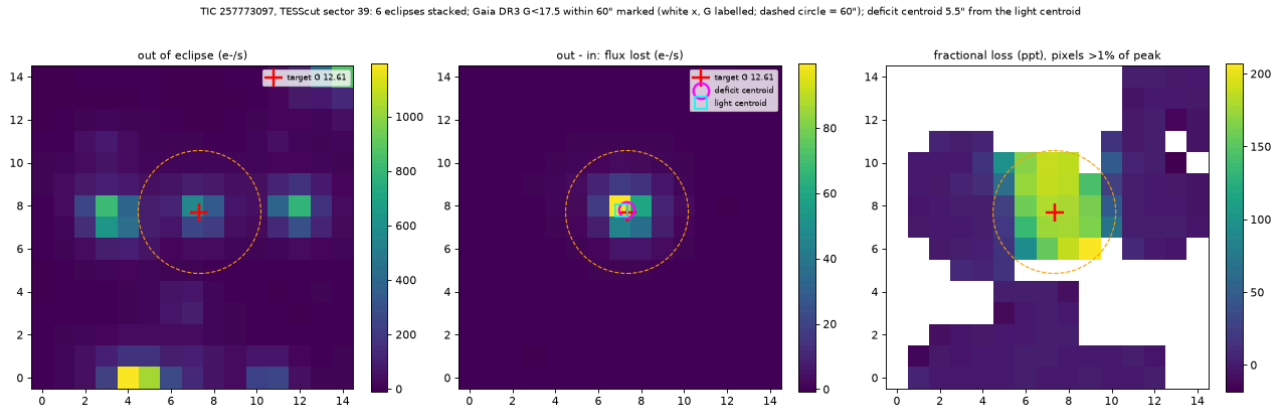
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy to refine physical parameters.



R645; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R645; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 257773097: CONFIRMED

Full bounded physical review: **COMPLETE**. Fourteen measured deep repeated eclipses and shallower secondary near0.49823. Sector94 aperture depths are shallower than12/66; differences remain visible. No capable Gaia neighbor within60arcsec; compact deficit favors target.

Raw morphology was inspected in raw-events.png and phase-review.png, including P and2P parity views. 14 free-center trapezoid timings survive finite, identifiable center-covariance checks. Quality-zero raw QLP SAP_FLUX was used; predictions only guide fitting windows. This is representative available evidence, not an all-event/all-sector census.

Measured same-kind eclipse ephemeris: P=4.94533220820 +/- 0.00000470980 d; reference BTJD=1627.37051641. Epoch kind: Deeper primary eclipse, established relative to observed shallower secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced timing chi-square=0.010326549866140703; zero degrees of freedom would be null. All selected timings follow the named repeating eclipse family. Epoch kind above identifies the deeper primary only where an unequal secondary is established. Model depth is the realized central loss d*min(1,w/(2*k)); extrapolated amplitude and ingress are retained separately. The reference is the fitted first selected cycle, not a Gaia prediction. Within-sector cycle recurrence and parity depths are in ../alias-audit.json. Opposite-phase sampled coverage, quality-zero core/sideband counts and flagged counts are in ../half-period-coverage.json. For265092213, orbital cycles=(historical cycle-1)/2 use deeper odd events; shallower even-cycle secondaries remain in measurements.json. Unequal observed secondary depths exclude mixing eclipse kinds into P/2 where established; systems without secure secondaries retain sampled opposite-phase coverage rather than a claimed clean null; the P/2 and2P raw views were inspected. No exhaustive blind alias search is asserted.

Narrow shallower secondary visible in raw pooled windows. Depth is the realized center deficit of the fitted shape, with its amplitude parameter retained separately in the JSON. Measured pooled secondary phase=0.4982304, formal phase covariance error=0.00010666; depth=15.164 ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Search-guide provenance: Gaia conjunction search guide, adjusted from observed raw phase morphology for this batch. Final phase is a free-center fit to observed raw flux..

Fresh difference image: sector39, 6 complete core/sideband events. Target pixel=[7.335648074406331, 7.706993323946454]; deficit centroid=[7.315448577715157, 7.831164066193662]; deficit-target distance=0.1258 pixel. Deficit-light offset=5.528 arcsec is a descriptive field-light displacement, not an arbitrary veto. Fractional PSF-model residual=0.8594 is not a calibrated localization uncertainty. Every flux-capable Gaia source in the60-arcsec list was inspected against the image. Attribution conclusion: Fourteen measured deep repeated eclipses and shallower secondary near0.49823. Sector94 aperture depths are shallower than12/66; differences remain visible. No capable Gaia neighbor within60arcsec; compact deficit favors target.

Capable Gaia source	Separation arcsec	G	Intrinsic required eclipse proxy	Pixel coordinate	Diluted loss at pixel
None					

The pairwise G-band required intrinsic loss is an approximate brightness screen, with passband, aperture and blend assumptions. It cannot be directly compared to the diluted fractional loss in a pixel. Noncapable neighbors and all coordinates remain in fresh-pixels/diff_fresh.json and gaia.json. TESS resolution does not exclude unlisted unresolved companions.

Alternative reduction: fresh background-subtracted TESScut1x1/3x3/5x5 target apertures have depths 180.771/176.371/155.583 ppt versus fitted raw QLP median178.046 ppt. Individual cutout event depths are retained in the pixel JSON. Cutout sector is a distinct observing occasion on the same TESS instrument relative to retained QLP sectors. This alternative reduction shares the FFI photons of that sector; no same-sector QLP comparison claimed. Distinct sectors are observing occasions on one instrument. Aperture/depth differences reflect dilution, background and extraction response; no intrinsic depth inferred.

Gaia DR3 source4613765126197369216: P=4.94523453 +/-0.00047074 d, e=0.035622, omega=226.21868 degrees, periastron BTJD=387.9539663. Closest edge-on conjunction offset=-0.005669 d (-0.001146 phase), source-specific full-covariance linear uncertainty=0.133088 d. Full correlation ordering P,gamma,K,e,omega,Tp, selected indices0,3,4,5. This linear approximation is descriptive, not a sigma veto. Both conjunctions and near-reference projection are retained in ../gaia-geometry.json. Mass function=0.042412 solar masses, assumed M1=1.16, minimum M2=0.4862; approximate central duration=4.556 h versus observed3.902 h. TIC radius and main-sequence companion scaling are assumptions, not a geometry fit; impact parameter and stellar evolution affect duration. Agreement cannot establish ownership among unresolved sources.

Sector	Same-kind cycle	Measured midpoint BTJD	Formal center error d	Input error d	Depth ppt
12	-1	1627.37028903	0.00087201221	0.010416812	178.107
12	1	1637.26035930	0.00017161622	0.010416812	175.709
12	2	1642.20654603	0.00083444681	0.010416812	178.356
12	3	1647.15191278	0.00049640907	0.010416812	178.472
12	4	1652.09652065	0.0003637752	0.010416812	178.658
66	297	3101.07942961	0.00016473736	0.0011574188	180.337
66	298	3106.02468827	0.00013578836	0.0011574188	177.985
66	299	3110.97025362	0.00018385347	0.0011574188	178.243
66	300	3115.91575577	0.00011855295	0.0011574188	177.772
66	301	3120.86083290	0.00015328137	0.0011574188	182.143
94	450	3857.71513400	0.00015036786	0.0011574009	162.920
94	452	3867.60606593	0.00010292629	0.0011574009	160.399
94	453	3872.55136247	0.00011065621	0.0011574009	159.397
94	454	3877.49675070	0.00010486983	0.0011574009	158.688

Next check: Optional resolved photometry and spectroscopy to refine physical parameters.

Catalogue status is separate from physical verdict; shared-catalogues/coverage-matrix.csv retains the seven scoped groups. Known recoveries are not new additions; no exhaustive priority claim. Sources and identity are checked in ../identity-check.json and ../source-hashes.json. Numeric artifacts, original receipts and outputs are frozen in ../SHA256SUMS.json; independent helper review is ../helper-audit.md.

Stable review key: R645.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s39.json - historical/diff_s39.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

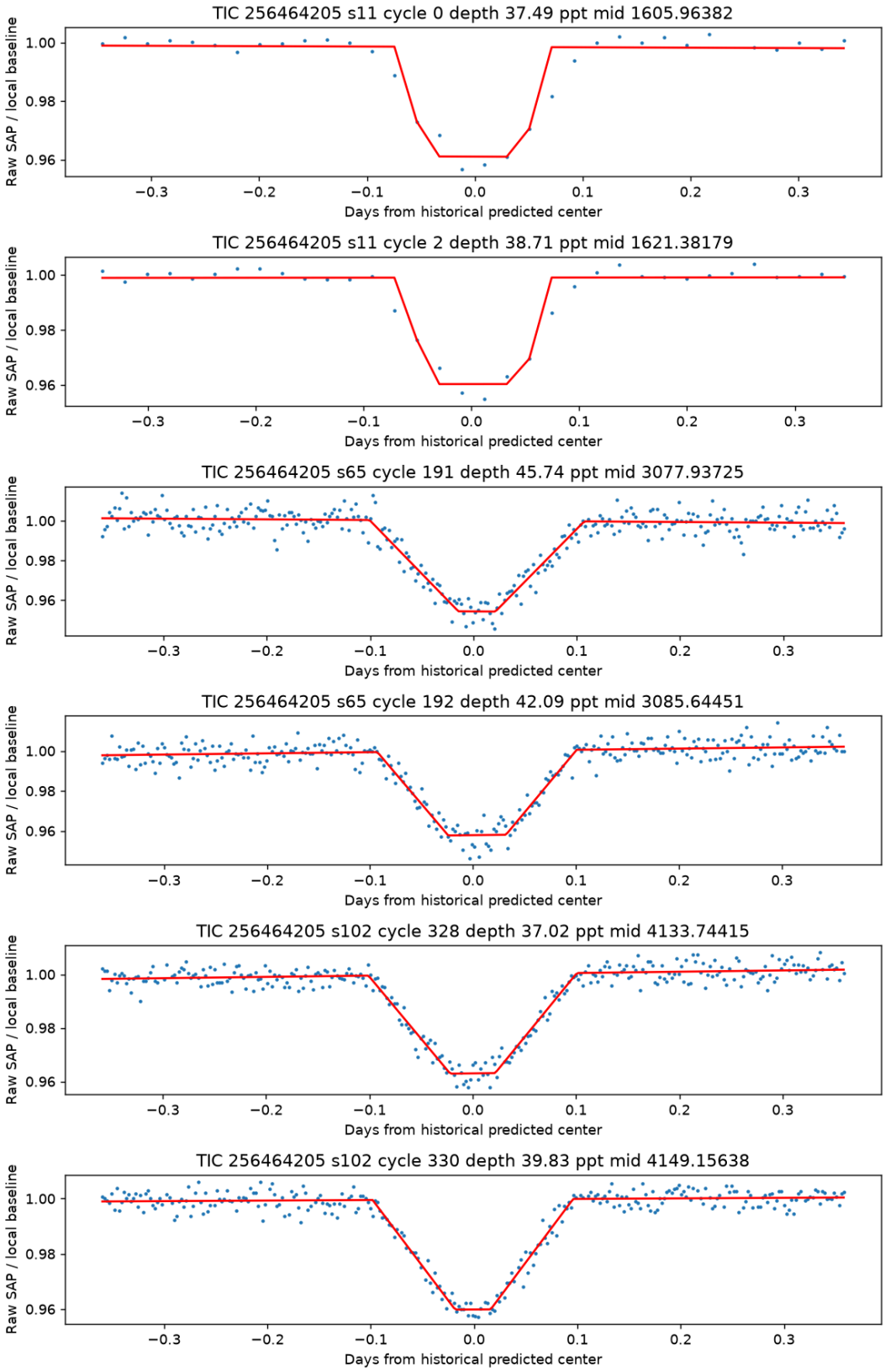
R646. TIC 256464205

physical_verdict: CONFIRMED

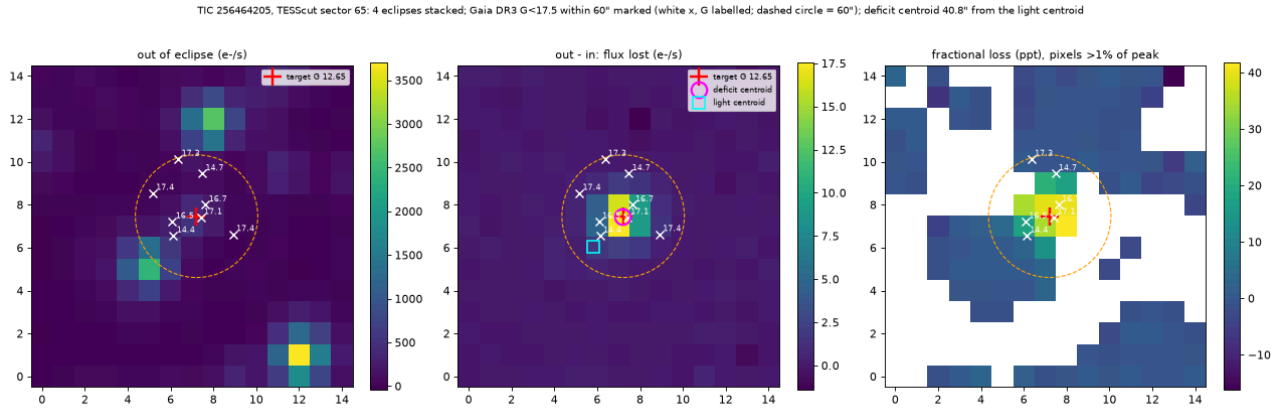
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy to refine physical parameters.



R646; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R646; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 256464205: CONFIRMED

Full bounded physical review: **COMPLETE**. Nine measured narrow repeated eclipses across three sectors. No secure secondary. Both capable Gaia neighbors at28.53/41.16arcsec lie away from compact target deficit. Nearby known EB Gaia6067018781690100608 at43.74arcsec has G18.650408 and0.309493d period versus target7.706608d; intrinsic required-loss proxy exceeds10 and its location is displaced. It cannot plausibly make the target eclipse under the brightness model. Whole-field light offset40.8arcsec reflects crowding, not eclipse localization.

Raw morphology was inspected in raw-events.png and phase-review.png, including P and2P parity views. 9 free-center trapezoid timings survive finite, identifiable center-covariance checks. Quality-zero raw QLP SAP_FLUX was used; predictions only guide fitting windows. This is representative available evidence, not an all-event/all-sector census.

Measured same-kind eclipse ephemeris: P=7.70660758985 +/- 0.00000628165 d; reference BTJD=1605.97630895. Epoch kind: Selected repeating narrow eclipse; deeper-primary identity unestablished because no secure secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced timing chi-square=0.8709509226856547; zero degrees of freedom would be null. All selected timings follow the named repeating eclipse family. Epoch kind above identifies the deeper primary only where an unequal secondary is established. Model depth is the realized central loss d*min(1,w/(2*k)); extrapolated amplitude and ingress are retained separately. The reference is the fitted first selected cycle, not a Gaia prediction. Within-sector cycle recurrence and parity depths are in ../alias-audit.json. Opposite-phase sampled coverage, quality-zero core/sideband counts and flagged counts are in ../half-period-coverage.json. For265092213, orbital cycles=(historical cycle-1)/2 use deeper odd events; shallower even-cycle secondaries remain in measurements.json. Unequal observed secondary depths exclude mixing eclipse kinds into P/2 where established; systems without secure secondaries retain sampled opposite-phase coverage rather than a claimed clean null; the P/2 and2P raw views were inspected. No exhaustive blind alias search is asserted.

No secure secondary; noisy low-amplitude exploratory fit not interpreted as eclipse. A fitted low-amplitude pooled model is not evidence of a secondary; flagged and missing coverage are not clean nulls.

Fresh difference image: sector65, 4 complete core/sideband events. Target pixel=[7.21212923557348, 7.465688916102241]; deficit centroid=[7.16706361275627, 7.431256301760108]; deficit-target distance=0.0567 pixel. Deficit-light offset=40.832 arcsec is a descriptive field-light displacement, not an arbitrary veto. Fractional PSF-model residual=0.9743 is not a calibrated localization uncertainty. Every flux-capable Gaia source in the60-arcsec list was inspected against the image. Attribution conclusion: Nine measured narrow repeated eclipses across three sectors. No secure secondary. Both capable Gaia neighbors at28.53/41.16arcsec lie away from compact target deficit. Nearby known EB Gaia6067018781690100608 at43.74arcsec has G18.650408 and0.309493d period versus target7.706608d; intrinsic required-loss proxy exceeds10 and its location is displaced. It cannot plausibly make the target eclipse under the brightness model. Whole-field light offset40.8arcsec reflects crowding, not eclipse localization.

Capable Gaia source	Separation arcsec	G	Intrinsic required eclipse proxy	Pixel coordinate	Diluted loss at pixel
6067018923439625088	28.533	14.446	0.2489	[6.130814959063191, 6.553025691538066]	0.01744809743716953
6067018751640930304	41.164	14.723	0.3096	[7.475529642511301, 9.447495172304116]	0.021543108665424797

The pairwise G-band required intrinsic loss is an approximate brightness screen, with passband, aperture and blend assumptions. It cannot be directly compared to the diluted fractional loss in a pixel. Noncapable neighbors and all coordinates remain in fresh-pixels/diff_fresh.json and gaia.json. TESS resolution does not exclude unlisted unresolved companions.

Alternative reduction: fresh background-subtracted TESScut1x1/3x3/5x5 target apertures have depths 34.602/26.610/10.403 ppt versus fitted raw QLP median39.827 ppt. Individual cutout event depths are retained in the pixel JSON. Same-sector QLP and cutout extraction share TESS photons and test processing robustness. Distinct sectors are observing occasions on one instrument. Aperture/depth differences reflect dilution, background and extraction response; no intrinsic depth inferred.

Gaia DR3 source6067018747329099008: P=7.70553073 +/-0.00138362 d, e=0.029079, omega=318.10385 degrees, periastron BTJD=385.5804366. Closest edge-on conjunction offset=0.152747 d (0.019823 phase), source-specific full-covariance linear uncertainty=0.269646 d. Full correlation ordering P,gamma,K,e,omega,Tp, selected indices0,3,4,5. This linear approximation is descriptive, not a sigma veto. Both conjunctions and near-reference projection are retained in ../gaia-geometry.json. Mass function=0.0335501 solar masses, assumed M1=1.2, minimum M2=0.45043; approximate central duration=7.073 h versus observed4.660 h. TIC radius and main-sequence companion scaling are assumptions, not a geometry fit; impact parameter and stellar evolution affect duration. Agreement cannot establish ownership among unresolved sources.

Sector	Same-kind cycle	Measured midpoint BTJD	Formal center error d	Input error d	Depth ppt
11	0	1605.96382449	0.00056789322	0.010416635	37.486
11	2	1621.38179176	0.00049097598	0.010416635	38.712
65	190	3070.23212690	0.0010261945	0.0011573928	43.794
65	191	3077.93725219	0.0010391507	0.0011573928	45.744
65	192	3085.64450935	0.00099660011	0.0011573928	42.091
65	193	3093.35336767	0.0011541044	0.0011573928	42.647
102	328	4133.74414745	0.00079484598	0.0011574068	37.016
102	329	4141.44973578	0.00066256467	0.0011574068	36.227
102	330	4149.15637792	0.00080854852	0.0011574068	39.827

Next check: Optional resolved photometry and spectroscopy to refine physical parameters.

Catalogue status is separate from physical verdict; shared-catalogues/coverage-matrix.csv retains the seven scoped groups. Known recoveries are not new additions; no exhaustive priority claim. Sources and identity are checked in ../identity-check.json and ../source-hashes.json. Numeric artifacts, original receipts and outputs are frozen in ../SHA256SUMS.json; independent helper review is ../helper-audit.md.

Stable review key: R646.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s65.json - historical/diff_s65.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

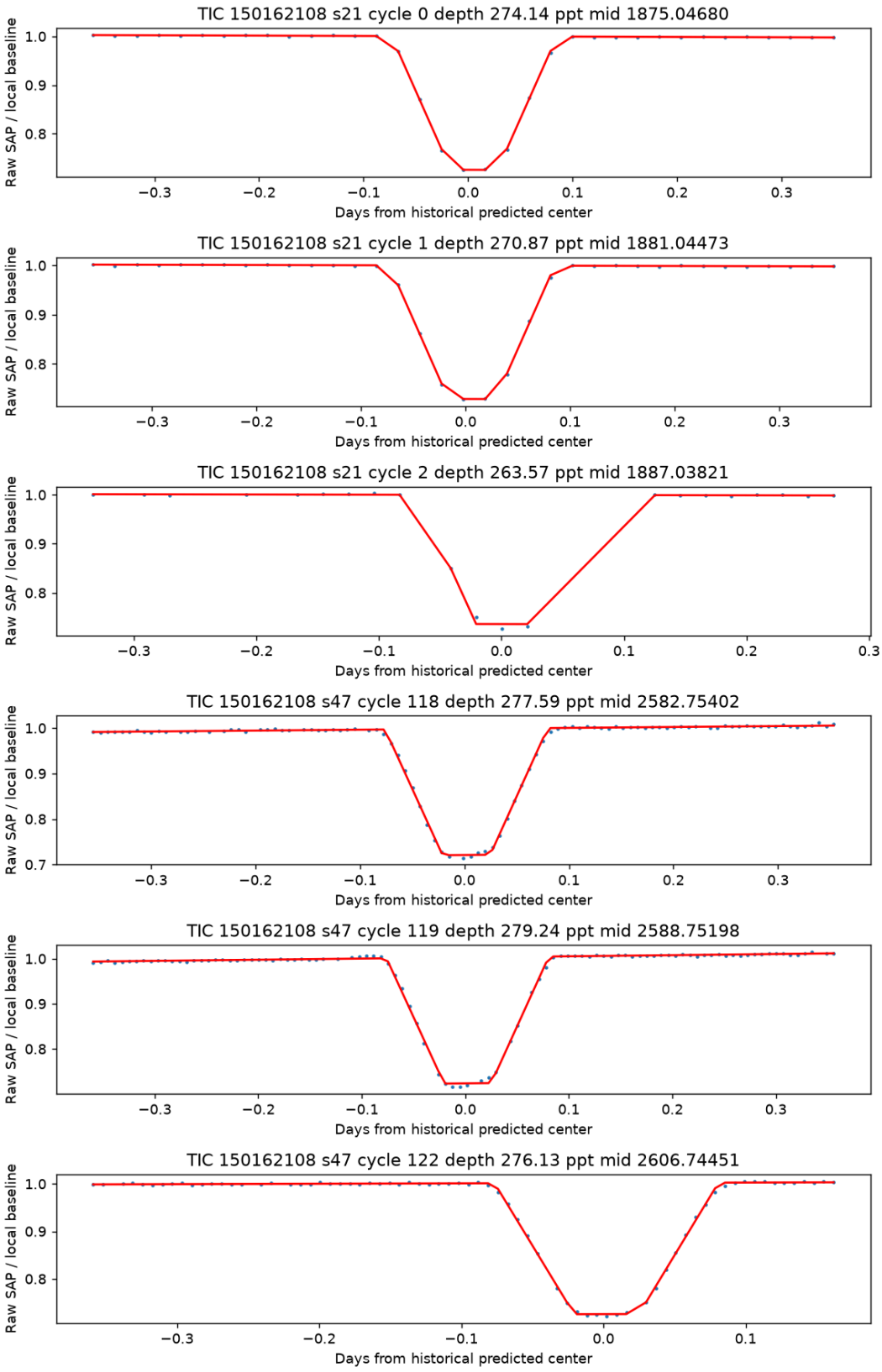
R647. TIC 150162108

physical_verdict: CONFIRMED

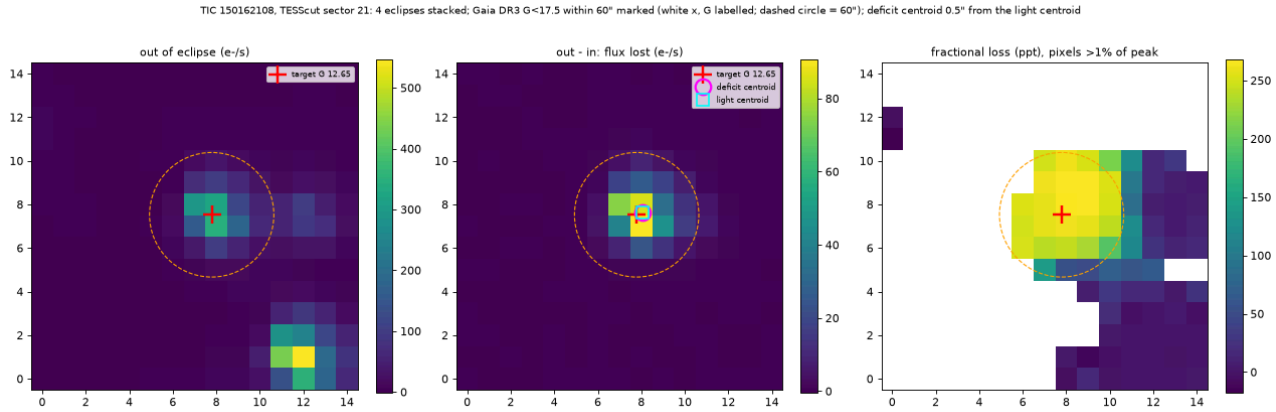
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy to refine physical parameters.



R647; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R647; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 150162108: CONFIRMED

Full bounded physical review: **COMPLETE**. Eight deep measured repeated eclipses and unequal secondary near0.500. No capable Gaia neighbor within60arcsec; compact target deficit. Secondary realized depth is used rather than extrapolated model amplitude.

Raw morphology was inspected in raw-events.png and phase-review.png, including P and2P parity views. 8 free-center trapezoid timings survive finite, identifiable center-covariance checks. Quality-zero raw QLP SAP_FLUX was used; predictions only guide fitting windows. This is representative available evidence, not an all-event/all-sector census.

Measured same-kind eclipse ephemeris: P=5.99752686350 +/- 0.00004628566 d; reference BTJD=1875.04609980. Epoch kind: Deeper primary eclipse, established relative to observed shallower secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced timing chi-square=0.019674496578670857; zero degrees of freedom would be null. All selected timings follow the named repeating eclipse family. Epoch kind above identifies the deeper primary only where an unequal secondary is established. Model depth is the realized central loss d*min(1,w/(2*k)); extrapolated amplitude and ingress are retained separately. The reference is the fitted first selected cycle, not a Gaia prediction. Within-sector cycle recurrence and parity depths are in ../alias-audit.json. Opposite-phase sampled coverage, quality-zero core/sideband counts and flagged counts are in ../half-period-coverage.json. For265092213, orbital cycles=(historical cycle-1)/2 use deeper odd events; shallower even-cycle secondaries remain in measurements.json. Unequal observed secondary depths exclude mixing eclipse kinds into P/2 where established; systems without secure secondaries retain sampled opposite-phase coverage rather than a claimed clean null; the P/2 and2P raw views were inspected. No exhaustive blind alias search is asserted.

Narrow shallower secondary visible in raw pooled windows. Depth is the realized center deficit of the fitted shape, with its amplitude parameter retained separately in the JSON. Measured pooled secondary phase=0.4999871, formal phase covariance error=0.00013331; depth=31.279 ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Search-guide provenance: Gaia conjunction search guide, adjusted from observed raw phase morphology for this batch. Final phase is a free-center fit to observed raw flux..

Fresh difference image: sector21, 4 complete core/sideband events. Target pixel=[7.786741985166227, 7.523928964414823]; deficit centroid=[8.056999111706652, 7.649902636845723]; deficit-target distance=0.2982 pixel. Deficit-light offset=0.458 arcsec is a descriptive field-light displacement, not an arbitrary veto. Fractional PSF-model residual=0.2208 is not a calibrated localization uncertainty. Every flux-capable Gaia source in the60-arcsec list was inspected against the image. Attribution conclusion: Eight deep measured repeated eclipses and unequal secondary near0.500. No capable Gaia neighbor within60arcsec; compact target deficit. Secondary realized depth is used rather than extrapolated model amplitude.

Capable Gaia source	Separation arcsec	G	Intrinsic required eclipse proxy	Pixel coordinate	Diluted loss at pixel
None					

The pairwise G-band required intrinsic loss is an approximate brightness screen, with passband, aperture and blend assumptions. It cannot be directly compared to the diluted fractional loss in a pixel. Noncapable neighbors and all coordinates remain in fresh-pixels/diff_fresh.json and gaia.json. TESS resolution does not exclude unlisted unresolved companions.

Alternative reduction: fresh background-subtracted TESScut1x1/3x3/5x5 target apertures have depths 265.562/262.473/257.545 ppt versus fitted raw QLP median273.665 ppt. Individual cutout event depths are retained in the pixel JSON. Same-sector QLP and cutout extraction share TESS photons and test processing robustness. Distinct sectors are observing occasions on one instrument. Aperture/depth differences reflect dilution, background and extraction response; no intrinsic depth inferred.

Gaia DR3 source1016132190708213504: P=5.99703951 +/-0.00048098 d, e=0.062207, omega=146.24343 degrees, periastron BTJD=388.6333812. Closest edge-on conjunction offset=-0.012333 d (-0.002057 phase), source-specific full-covariance linear uncertainty=0.152289 d. Full correlation ordering P,gamma,K,e,omega,Tp, selected indices0,3,4,5. This linear approximation is descriptive, not a sigma veto. Both conjunctions and near-reference projection are retained in ../gaia-geometry.json. Mass function=0.0604365 solar masses, assumed M1=0.91, minimum M2=0.49146; approximate central duration=4.448 h versus observed3.751 h. TIC radius and main-sequence companion scaling are assumptions, not a geometry fit; impact parameter and stellar evolution affect duration. Agreement cannot establish ownership among unresolved sources.

Sector	Same-kind cycle	Measured midpoint BTJD	Formal center error d	Input error d	Depth ppt
21	0	1875.04679512	0.00020476431	0.010416512	274.135
21	1	1881.04472696	0.00019321027	0.010416512	270.866
21	2	1887.03820855	9.0126322e-05	0.010416512	263.572
21	3	1893.03985611	0.00018452794	0.010416512	262.913
47	118	2582.75402497	0.00028627476	0.0034722814	277.592
47	119	2588.75198232	0.00053072846	0.0034722814	279.240
47	121	2600.74677407	0.00037011982	0.0034722814	273.196
47	122	2606.74450940	9.5160411e-05	0.0034722814	276.132

Next check: Optional resolved photometry and spectroscopy to refine physical parameters.

Catalogue status is separate from physical verdict; shared-catalogues/coverage-matrix.csv retains the seven scoped groups. Known recoveries are not new additions; no exhaustive priority claim. Sources and identity are checked in ../identity-check.json and ../source-hashes.json. Numeric artifacts, original receipts and outputs are frozen in ../SHA256SUMS.json; independent helper review is ../helper-audit.md.

Stable review key: R647.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s21.json - historical/diff_s21.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

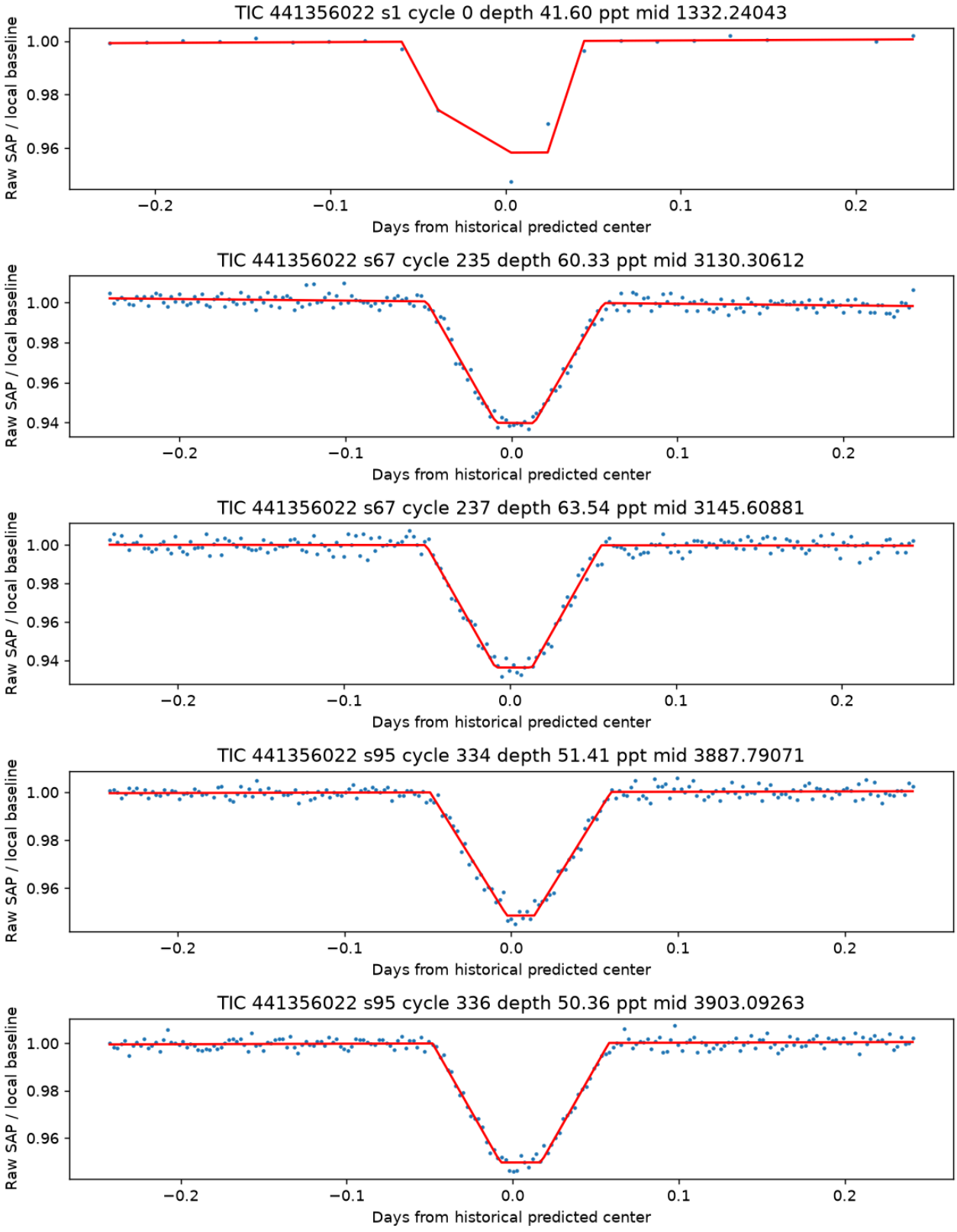
R648. TIC 441356022

physical_verdict: PROBABLE

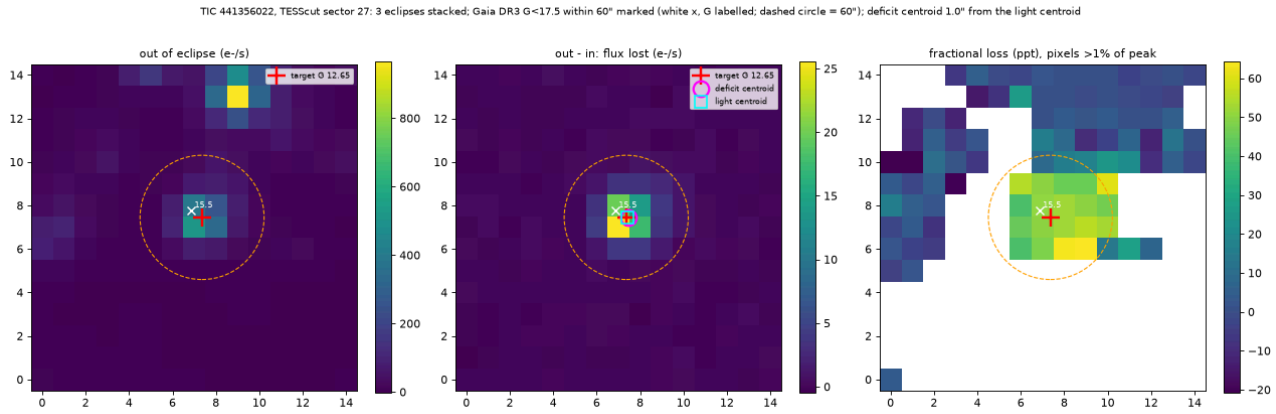
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or multi-source PRF localization for target and listed close capable neighbor.



R648; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R648; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 441356022: PROBABLE

Full bounded physical review: **COMPLETE**. Five repeated narrow eclipses and a shallower secondary near0.50014. Sector1 broad excursions remain visible in phase review but representative locally normalized primary is consistent. Capable Gaia6793606134390553856 at11.93arcsec overlaps TESS footprint; small centroid offset cannot uniquely establish ownership. Attribution remains unresolved.

Raw morphology was inspected in raw-events.png and phase-review.png, including P and2P parity views. 5 free-center trapezoid timings survive finite, identifiable center-covariance checks. Quality-zero raw QLP SAP_FLUX was used; predictions only guide fitting windows. This is representative available evidence, not an all-event/all-sector census.

Measured same-kind eclipse ephemeris: P=7.65135458170 +/- 0.00001113294 d; reference BTJD=1332.23784405. Epoch kind: Deeper primary eclipse, established relative to observed shallower secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced timing chi-square=0.10215052204394552; zero degrees of freedom would be null. All selected timings follow the named repeating eclipse family. Epoch kind above identifies the deeper primary only where an unequal secondary is established. Model depth is the realized central loss d*min(1,w/(2*k)); extrapolated amplitude and ingress are retained separately. The reference is the fitted first selected cycle, not a Gaia prediction. Within-sector cycle recurrence and parity depths are in ../alias-audit.json. Opposite-phase sampled coverage, quality-zero core/sideband counts and flagged counts are in ../half-period-coverage.json. For265092213, orbital cycles=(historical cycle-1)/2 use deeper odd events; shallower even-cycle secondaries remain in measurements.json. Unequal observed secondary depths exclude mixing eclipse kinds into P/2 where established; systems without secure secondaries retain sampled opposite-phase coverage rather than a claimed clean null; the P/2 and2P raw views were inspected. No exhaustive blind alias search is asserted.

Narrow shallower secondary visible in raw pooled windows. Depth is the realized center deficit of the fitted shape, with its amplitude parameter retained separately in the JSON. Measured pooled secondary phase=0.5001361, formal phase covariance error=0.00040139; depth=8.550 ppt. Formal pooled

trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Search-guide provenance: Gaia conjunction search guide, adjusted from observed raw phase morphology for this batch. Final phase is a free-center fit to observed raw flux..

Fresh difference image: sector27, 3 complete core/sideband events. Target pixel=[7.361798900615259, 7.450527379535993]; deficit centroid=[7.448205114602079, 7.4465811661782855]; deficit-target distance=0.0865 pixel. Deficit-light offset=1.031 arcsec is a descriptive field-light displacement, not an arbitrary veto. Fractional PSF-model residual=0.1549 is not a calibrated localization uncertainty. Every flux-capable Gaia source in the60-arcsec list was inspected against the image. Attribution conclusion: Five repeated narrow eclipses and a shallower secondary near0.50014. Sector1 broad excursions remain visible in phase review but representative locally normalized primary is consistent. Capable Gaia6793606134390553856 at11.93arcsec overlaps TESS footprint; small centroid offset cannot uniquely establish ownership. Attribution remains unresolved.

Capable Gaia source	Separation arcsec	G	Intrinsic required eclipse proxy	Pixel coordinate	Diluted loss at pixel
6793606134390553856	11.931	15.525	0.7785	[6.863812829339863, 7.74915014449655]	0.05071550161682233

The pairwise G-band required intrinsic loss is an approximate brightness screen, with passband, aperture and blend assumptions. It cannot be directly compared to the diluted fractional loss in a pixel. Noncapable neighbors and all coordinates remain in fresh-pixels/diff_fresh.json and gaia.json. TESS resolution does not exclude unlisted unresolved companions.

Alternative reduction: fresh background-subtracted TESScut1x1/3x3/5x5 target apertures have depths 49.843/50.476/49.977 ppt versus fitted raw QLP median51.411 ppt. Individual cutout event depths are retained in the pixel JSON. Cutout sector is a distinct observing occasion on the same TESS instrument relative to retained QLP sectors. This alternative reduction shares the FFI photons of that sector; no same-sector QLP comparison claimed. Distinct sectors are observing occasions on one instrument. Aperture/depth differences reflect dilution, background and extraction response; no intrinsic depth inferred.

Gaia DR3 source6793606134390553984: P=7.65230645 +/-0.00074291 d, e=0.015819, omega=117.22626 degrees, periastron BTJD=391.6611199. Closest edge-on conjunction offset=-0.095681 d (-0.012504 phase), source-specific full-covariance linear uncertainty=0.108623 d. Full correlation ordering P,gamma,K,e,omega,Tp, selected indices0,3,4,5. This linear approximation is descriptive, not a sigma veto. Both conjunctions and near-reference projection are retained in ../gaia-geometry.json. Mass function=0.0481616 solar masses, assumed M1=0.91, minimum M2=0.44566; approximate central duration=4.571 h versus observed2.551 h. TIC radius and main-sequence companion scaling are assumptions, not a geometry fit; impact parameter and stellar evolution affect duration. Agreement cannot establish ownership among unresolved sources.

Sector	Same-kind cycle	Measured midpoint BTJD	Formal center error d	Input error d	Depth ppt
1	0	1332.24042710	0.0001209836	0.010416534	41.600
67	235	3130.30612246	0.00033520317	0.0011574368	60.329
67	237	3145.60881203	0.0003598403	0.0011574368	63.538
95	334	3887.79071500	0.00028329453	0.0011573861	51.411
95	336	3903.09262711	0.00031363413	0.0011573861	50.358

Next check: Resolved eclipse photometry or multi-source PRF localization for target and listed close capable neighbor.

Catalogue status is separate from physical verdict; shared-catalogues/coverage-matrix.csv retains the seven scoped groups. Known recoveries are not new additions; no exhaustive priority claim. Sources and identity are checked in ../identity-check.json and ../source-hashes.json. Numeric artifacts, original receipts and outputs are frozen in ../SHA256SUMS.json; independent helper review is ../helper-audit.md.

Stable review key: R648.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s27.json - historical/diff_s27.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

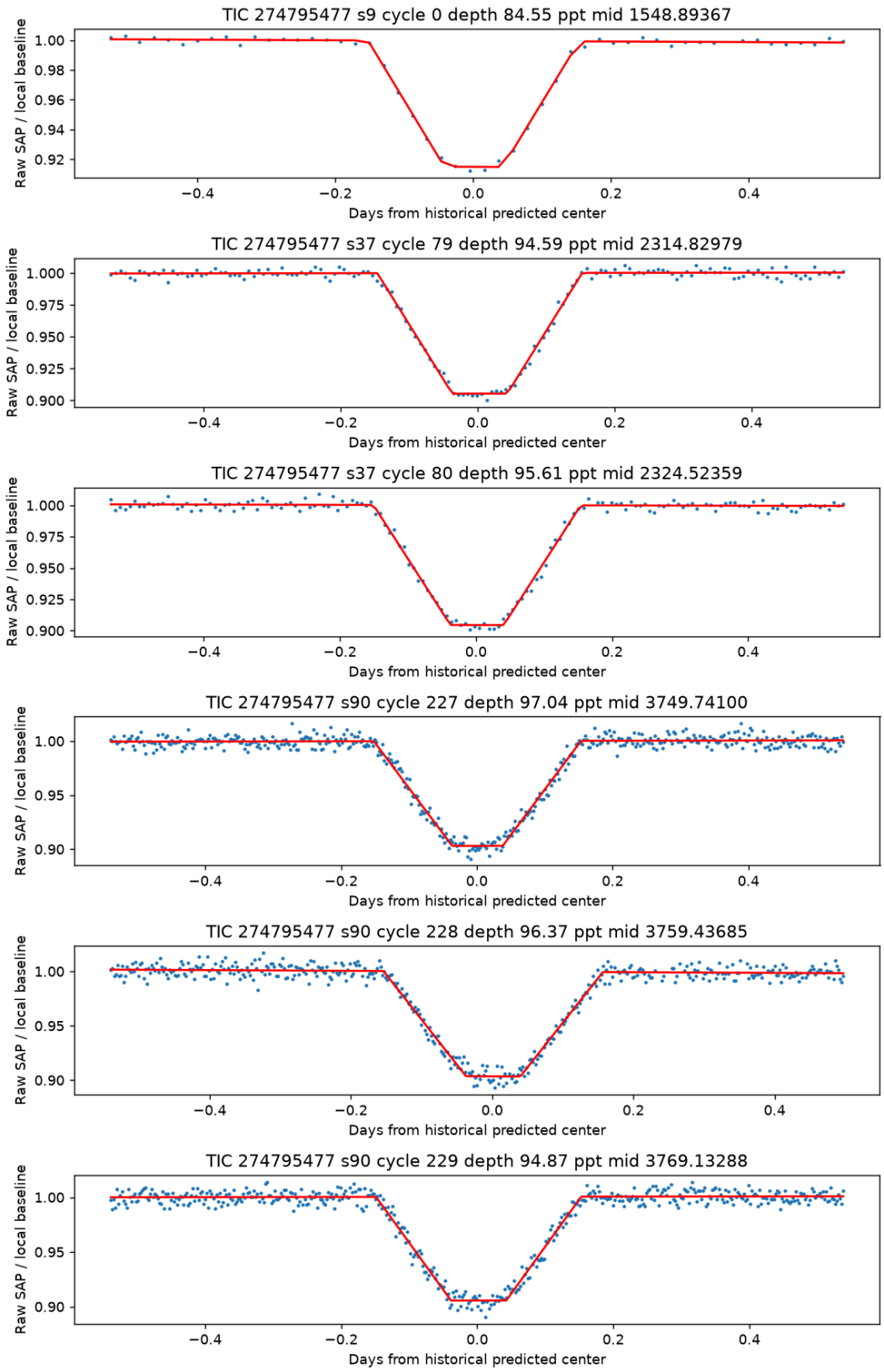
R649. TIC 274795477

physical_verdict: CONFIRMED

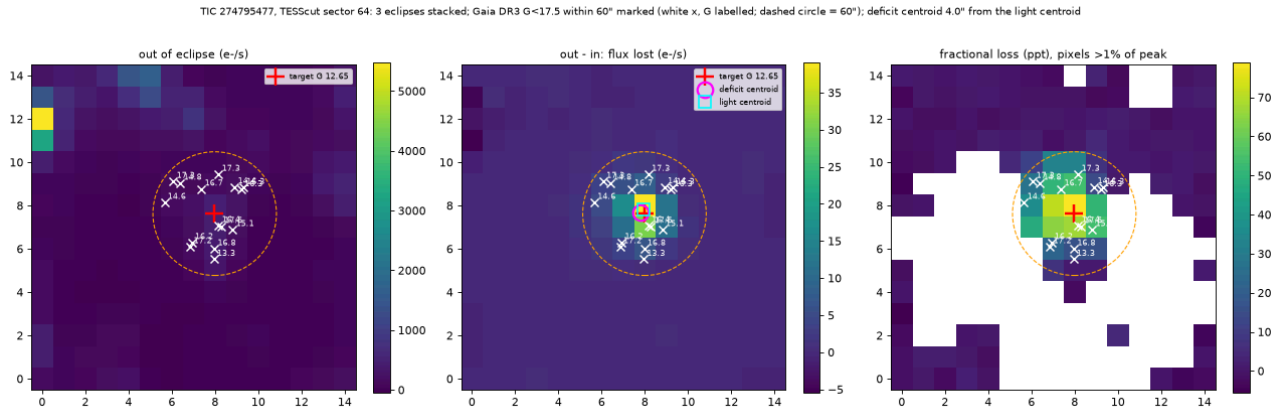
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional resolved photometry and spectroscopy to refine physical parameters.



R649; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R649; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 274795477: CONFIRMED

Full bounded physical review: **COMPLETE**. Six deep repeated eclipses with shallower eccentric secondary near0.63234. All four capable neighbors at31.82-47.57arcsec lie away from compact target deficit; aperture depth dilution remains explicit.

Raw morphology was inspected in raw-events.png and phase-review.png, including P and2P parity views. 6 free-center trapezoid timings survive finite, identifiable center-covariance checks. Quality-zero raw QLP SAP_FLUX was used; predictions only guide fitting windows. This is representative available evidence, not an all-event/all-sector census.

Measured same-kind eclipse ephemeris: P=9.69535645819 +/- 0.00001613144 d; reference BTJD=1548.89564693. Epoch kind: Deeper primary eclipse, established relative to observed shallower secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced timing chi-square=0.16442273333957583; zero degrees of freedom would be null. All selected timings follow the named repeating eclipse family. Epoch kind above identifies the deeper primary only where an unequal secondary is established. Model depth is the realized central loss d*min(1,w/(2*k)); extrapolated amplitude and ingress are retained separately. The reference is the fitted first selected cycle, not a Gaia prediction. Within-sector cycle recurrence and parity depths are in ../alias-audit.json. Opposite-phase sampled coverage, quality-zero core/sideband counts and flagged counts are in ../half-period-coverage.json. For265092213, orbital cycles=(historical cycle-1)/2 use deeper odd events; shallower even-cycle secondaries remain in measurements.json. Unequal observed secondary depths exclude mixing eclipse kinds into P/2 where established; systems without secure secondaries retain sampled opposite-phase coverage rather than a claimed clean null; the P/2 and2P raw views were inspected. No exhaustive blind alias search is asserted.

Narrow shallower secondary visible in raw pooled windows. Depth is the realized center deficit of the fitted shape, with its amplitude parameter retained separately in the JSON. Measured pooled secondary phase=0.6323483, formal phase covariance error=6.2394e-05; depth=47.634 ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Search-guide provenance: Gaia conjunction search guide, adjusted from observed raw phase morphology for this batch. Final phase is a free-center fit to observed raw flux..

Fresh difference image: sector64, 3 complete core/sideband events. Target pixel=[7.980715264746166, 7.626836464579224]; deficit centroid=[7.801219714231831, 7.689320497886531]; deficit-target distance=0.1901 pixel. Deficit-light offset=3.952 arcsec is a descriptive field-light displacement, not an arbitrary veto. Fractional PSF-model residual=0.8310 is not a calibrated localization uncertainty. Every flux-capable Gaia source in the60-arcsec list was inspected against the image. Attribution conclusion: Six deep repeated eclipses with shallower eccentric secondary near0.63234. All four capable neighbors at31.82-47.57arcsec lie away from compact target deficit; aperture depth dilution remains explicit.

Capable Gaia source	Separation arcsec	G	Intrinsic required eclipse proxy	Pixel coordinate	Diluted loss at pixel
5259205668929810688	31.818	14.431	0.5861	[8.903004837651865, 8.86491556285581]	0.014393374665472861
5259205600210331904	42.653	14.782	0.7729	[6.386844141470037, 9.041071013832473]	0.027988144881546 01
5259211510085340288	43.206	13.320	0.2717	[7.941539222011265, 5.525272826128213]	0.015615574191644119
5259205600210333312	47.568	14.552	0.6440	[5.655941750787015, 8.149159313232119]	0.036969829956691 22

The pairwise G-band required intrinsic loss is an approximate brightness screen, with passband, aperture and blend assumptions. It cannot be directly compared to the diluted fractional loss in a pixel. Noncapable neighbors and all coordinates remain in fresh-pixels/diff_fresh.json and gaia.json. TESS resolution does not exclude unlisted unresolved companions.

Alternative reduction: fresh background-subtracted TESScut1x1/3x3/5x5 target apertures have depths 79.050/61.011/43.973 ppt versus fitted raw QLP median95.239 ppt. Individual cutout event depths are retained in the pixel JSON. Cutout sector is a distinct observing occasion on the same TESS instrument relative to retained QLP sectors. This alternative reduction shares the FFI photons of that sector; no same-sector QLP comparison claimed. Distinct sectors are observing occasions on one instrument. Aperture/depth differences reflect dilution, background and extraction response; no intrinsic depth inferred.

Gaia DR3 source5259205634570074624: P=9.69714320 +/-0.00171315 d, e=0.232096, omega=8.50958 degrees, periastron BTJD=393.5430881. Closest edge-on conjunction offset=-0.117910 d (-0.012159 phase), source-specific full-covariance linear uncertainty=0.200328 d. Full correlation ordering P,gamma,K,e,omega,Tp, selected indices0,3,4,5. This linear approximation is descriptive, not a sigma veto. Both conjunctions and near-reference projection are retained in ../gaia-geometry.json. Mass function=0.100484 solar masses, assumed M1=2.68, minimum M2=1.1351; approximate central duration=8.235 h versus observed7.321 h. TIC radius and main-sequence companion scaling are assumptions, not a geometry fit; impact parameter and stellar evolution affect duration. Agreement cannot establish ownership among unresolved sources.

Sector	Same-kind cycle	Measured midpoint BTJD	Formal center error d	Input error d	Depth ppt
9	0	1548.89367396	0.00082871694	0.010416851	84.551
37	79	2314.82978554	0.00059145824	0.0034721796	94.593
37	80	2324.52358737	0.00055646813	0.0034721796	95.613
90	227	3749.74100202	0.00052973271	0.0011574132	97.044
90	228	3759.43685130	0.00072583741	0.0011574132	96.369
90	229	3769.13288451	0.00065048961	0.0011574132	94.866

Next check: Optional resolved photometry and spectroscopy to refine physical parameters.

Catalogue status is separate from physical verdict; shared-catalogues/coverage-matrix.csv retains the seven scoped groups. Known recoveries are not new additions; no exhaustive priority claim. Sources and identity are checked in ../identity-check.json and ../source-hashes.json. Numeric artifacts, original receipts and outputs are frozen in ../SHA256SUMS.json; independent helper review is ../helper-audit.md.

Stable review key: R649.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s64.json - historical/diff_s64.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

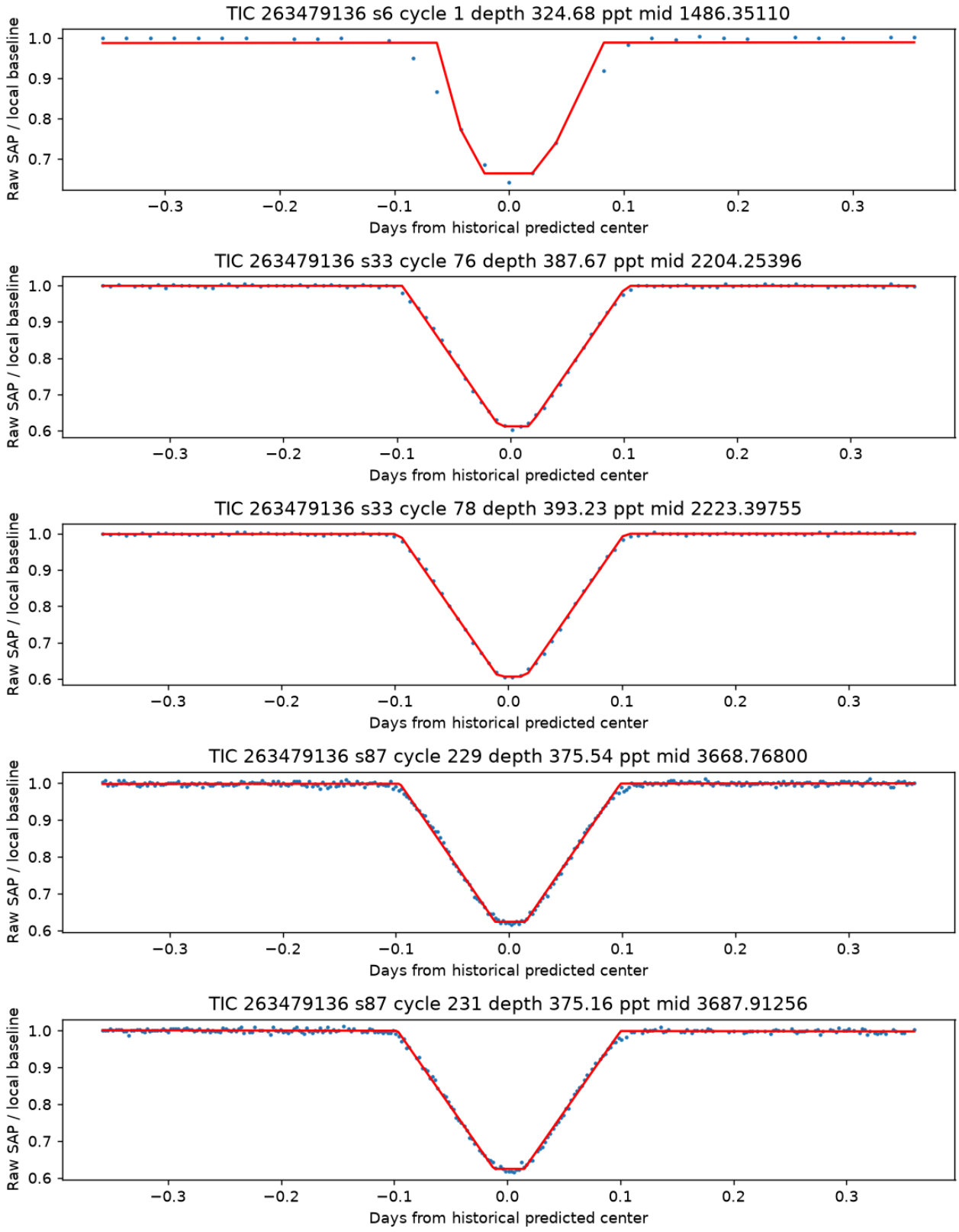
R650. TIC 263479136

physical_verdict: CONFIRMED

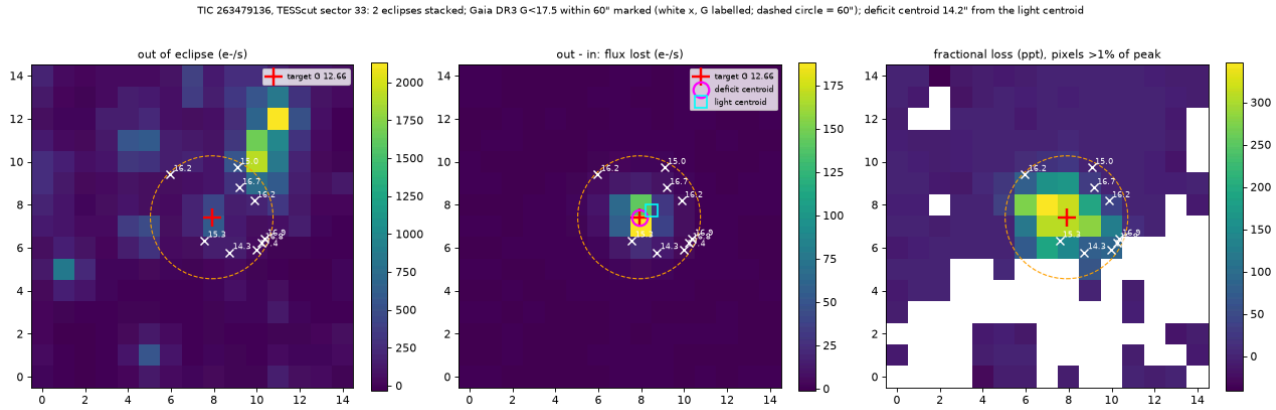
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional additional clean epochs and resolved photometry.



R650; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R650; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 263479136: CONFIRMED

Full bounded physical review: COMPLETE. Five deep primary timings and unequal observed secondary; no capable Gaia neighbor within60arcsec. Sector33/87 fitted primary pairs are separated by2 cycles because intervening primary cores are fully flagged (13/43 finite core samples, zeroQ0). Both photometric parities are represented across sectors and agree in eclipse kind; observed unequal secondary at0.50018 rejects the half-period interpretation. No clean intervening null is claimed.

Raw QLP SAP_FLUX representative events and full-sector P/2P morphology were visually inspected. Quality-zero finite positive samples only; predictions guide windows and are never timings. 5 identifiable free-center timings define P=9.571992185479 +/- 0.000015981721d, reference BTJD=1486.354084264 +/- 0.003483559d. Epoch error meaning: Conservative linear timing-fit covariance under the stated cadence floor and scatter inflation; excludes uncalibrated shape and baseline systematics. Epoch kind: Deeper primary established by unequal observed secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced chi square=0.0749500555975222; zero degrees of freedom is null. All errors retain their named meanings; no calibrated coverage is claimed. Selected same-kind recurring eclipse family; parity and P/2 coverage checked.

Depth=375.542948ppt is the median realized central trapezoid loss d*min(1,w/(2*k)); extrapolated amplitude and ingress remain separately in measurements.json. Measured secondary phase 0.50017680, formal phase error 1.4929379e-05, realized depth 148.84320ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Secondary raw pooled windows, actual sectors/cycles and fitted phase are retained in secondary-windows.csv and ../secondary-phases.json. No exhaustive blind alias search asserted. ../alias-audit.json and ../half-period-coverage.json retain recurrence/parity and sampled opposite-phase evidence. Flagged coverage is never a clean null.

Fresh TESScut difference image sector33 stacks2 core/bilateral-sideband events. Target pixel= [7.900195753870214, 7.414316492756493], deficit=[7.902225505524922, 7.392429596411507]; descriptive offset=0.02198pixel, field PSF residual=0.96592. No arbitrary centroid cut or calibrated

localization error is imposed. Every capable neighbor below was inspected at its actual projected pixels; close overlapping neighbors retain probable ownership. Other capable sources are displaced from compact deficit core but crowding limits localization. All Gaia neighbors are retained in pixel JSON.

Source	Separation arcsec	G	Required intrinsic loss proxy	Pixel coordinate	Diluted pixel loss
None					

G-band pairwise brightness screen assumes approximate passband/aperture response; intrinsic required eclipse cannot be directly compared with diluted pixel loss. Unresolved unlisted companions are not excluded.

Alternative reduction: independently extracted background-subtracted TESScut1x1/3x3/5x5 depths=308.78518/268.96792/169.59379ppt; QLP median=375.54295ppt. Pixel sector33 shares the same TESS photons with retained QLP and checks extraction robustness. Independent processing is not an independent instrument. Different apertures respond differently to dilution/background.

Gaia source3325467699827687808: period=9.573493201+/-0.003628264d; e=0.0304764; omega=319.0802166deg; periastron BTJD=391.59005400. Closest conjunction offset=-0.0245143d (-0.0025606phase), source-specific full-covariance linear error=0.3874948d. Correlation order P,gamma,K,e,omega,Tp with subindices0,3,4,5. Linear propagation is descriptive, not a Gaussian veto or ownership proof. Both conjunctions retained in ../gaia-geometry.json. Mass function=0.1310306solar, assumed M1=1.007, minimum M2=0.735477; approximate central duration=6.6081h versus observed=4.7350h. TIC radius/main-sequence companion scaling, edge-on conjunction and stellar evolution are assumptions; this is a plausibility check, not a fitted physical solution.

Sector	Same-kind cycle	Measured BTJD	Formal center error d	Conservative input error d	Realized depth ppt
6	1	1486.351096209	4.8456186e-05	0.0104167032	324.67989
33	76	2204.253963848	0.00012662685	0.00347220224	387.67061
33	78	2223.397545068	0.000101812568	0.00347220224	393.23300
87	229	3668.768003373	0.000122506583	0.00115739891	375.54295
87	231	3687.912564304	0.000112922143	0.00115739891	375.16041

Catalogue novelty is separate from this physical verdict. Shared coverage matrix and source identity checks are authoritative for catalogue recovery; period agreement alone cannot establish attribution.

Stable review key: R650.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s33.json - historical/diff_s33.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

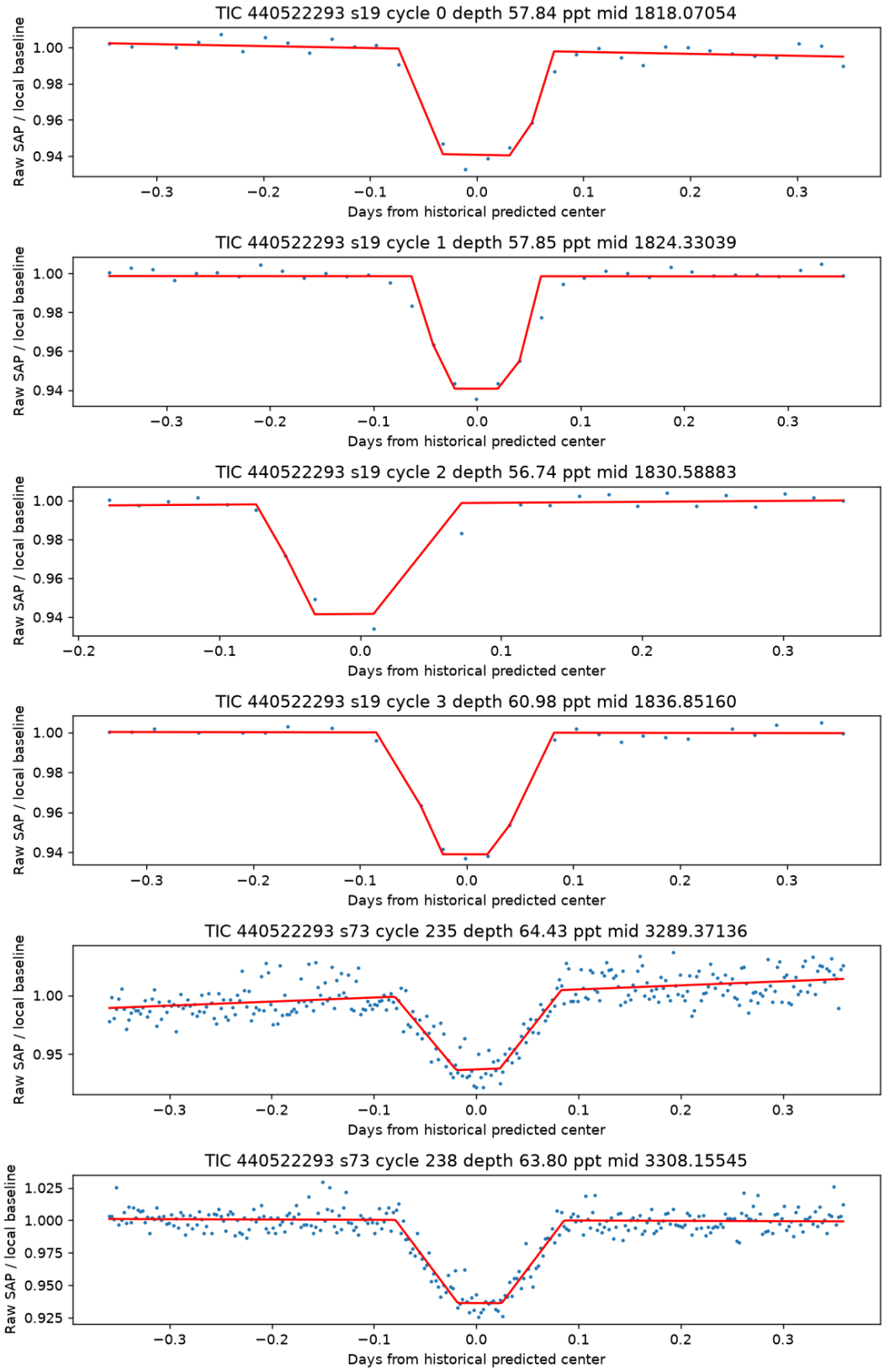
R651. TIC 440522293

physical_verdict: CONFIRMED

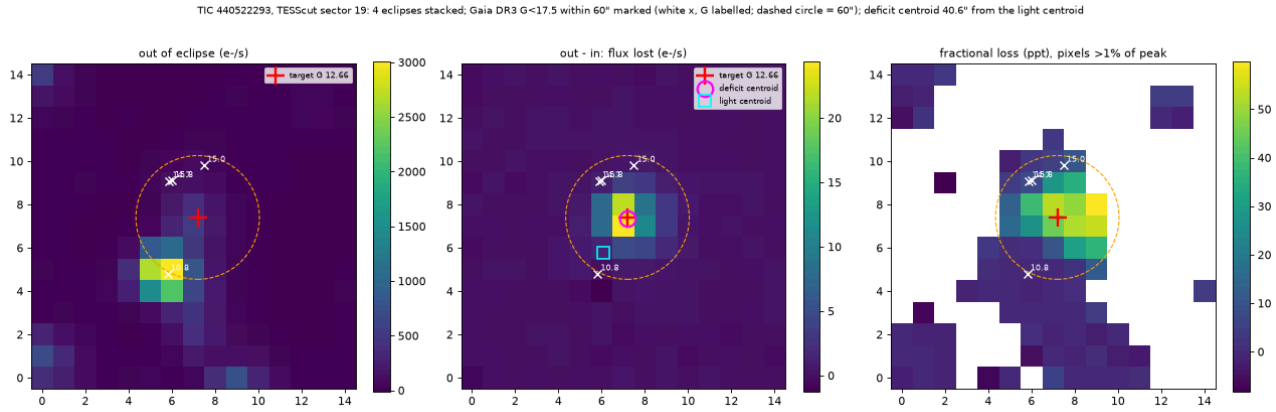
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional additional clean epochs and resolved photometry.



R651; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R651; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 440522293: CONFIRMED

Full bounded physical review: COMPLETE. Seven repeated narrow eclipses despite large changing raw baselines. No secure secondary. Three capable sources at43.44/49.59/59.27arcsec are separated from compact target deficit.

Raw QLP SAP_FLUX representative events and full-sector P/2P morphology were visually inspected. Quality-zero finite positive samples only; predictions guide windows and are never timings. 7 identifiable free-center timings define P=6.260865037153 +/- 0.000022364799d, reference BTJD=1818.068911290 +/- 0.005238708d. Epoch error meaning: Conservative linear timing-fit covariance under the stated cadence floor and scatter inflation; excludes uncalibrated shape and baseline systematics. Epoch kind: Selected repeating eclipse; deeper/secondary identity not established. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced chi square=0.13515330383939334; zero degrees of freedom is null. All errors retain their named meanings; no calibrated coverage is claimed. Selected same-kind recurring eclipse family; parity and P/2 coverage checked.

Depth=60.978017ppt is the median realized central trapezoid loss d*min(1,w/(2*k)); extrapolated amplitude and ingress remain separately in measurements.json. No secure secondary; exploratory fit is not interpreted as an eclipse timing or counterpart. Secondary raw pooled windows, actual sectors/cycles and fitted phase are retained in secondary-windows.csv and ../secondary-phases.json. No exhaustive blind alias search asserted. ../alias-audit.json and ../half-period-coverage.json retain recurrence/parity and sampled opposite-phase evidence. Flagged coverage is never a clean null.

Fresh TESScut difference image sector19 stacks4 core/bilateral-sideband events. Target pixel= [7.195044463550808, 7.402943412853364], deficit=[7.198913632123789, 7.367517699540319]; descriptive offset=0.03564pixel, field PSF residual=0.98464. No arbitrary centroid cut or calibrated localization error is imposed. Every capable neighbor below was inspected at its actual projected pixels; close overlapping neighbors retain probable ownership. Other capable sources are displaced from compact deficit core but crowding limits localization. All Gaia neighbors are retained in pixel JSON.

Source	Separation arcsec	G	Required intrinsic loss proxy	Pixel coordinate	Diluted pixel loss
96118980512429 9008	43.44342671941 483	14.6866 91	0.4558854409374505	[5.8765626148871135, 9.090413312774503]	0.007585100171623 949
96118983948403 8016	49.59189785163 533	14.9895 54	0.5829419539402513	[7.479707724015075, 9.819287430193889]	0.005153184123461 533
96118994256324 9664	59.2652208795 125	10.8490 86	0.07249803886405261	[5.801015495039102, 4.8027185533476]	-7.65545285873491 3e-05

G-band pairwise brightness screen assumes approximate passband/aperture response; intrinsic required eclipse cannot be directly compared with diluted pixel loss. Unresolved unlisted companions are not excluded.

Alternative reduction: independently extracted background-subtracted TESScut1x1/3x3/5x5 depths=45.38131/26.46948/9.69180ppt; QLP median=60.97802ppt. Pixel sector19 shares the same TESS photons with retained QLP and checks extraction robustness. Independent processing is not an independent instrument. Different apertures respond differently to dilution/background.

Gaia source961189976922990464: period=6.261849300+/-0.000862596d; e=0.1238615; omega=35.5538898deg; periastron BTJD=389.77342650. Closest conjunction offset=-0.1630120d (-0.0260326phase), source-specific full-covariance linear error=0.1914077d. Correlation order P,gamma,K,e,omega,Tp with subindices0,3,4,5. Linear propagation is descriptive, not a Gaussian veto or ownership proof. Both conjunctions retained in ../gaia-geometry.json. Mass function=0.06229506solar, assumed M1=1.24, minimum M2=0.593938; approximate central duration=5.5105h versus observed=3.1485h. TIC radius/main-sequence companion scaling, edge-on conjunction and stellar evolution are assumptions; this is a plausibility check, not a fitted physical solution.

Sector	Same-kind cycle	Measured BTJD	Formal center error d	Conservative input error d	Realized depth ppt
19	0	1818.070537803	0.00106662542	0.0104169085	57.84351
19	1	1824.330393498	0.000681847298	0.0104169085	57.84571
19	2	1830.588833811	0.00116932993	0.0104169085	56.74498
19	3	1836.851599195	0.000793332926	0.0104169085	60.97802
73	235	3289.371359548	0.00213364001	0.00213364001	64.42739
73	236	3295.632635005	0.000883029663	0.00115741697	63.83076
73	238	3308.155454505	0.000943755438	0.00115741697	63.80221

Catalogue novelty is separate from this physical verdict. Shared coverage matrix and source identity checks are authoritative for catalogue recovery; period agreement alone cannot establish attribution.

Stable review key: R651.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s19.json - historical/diff_s19.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

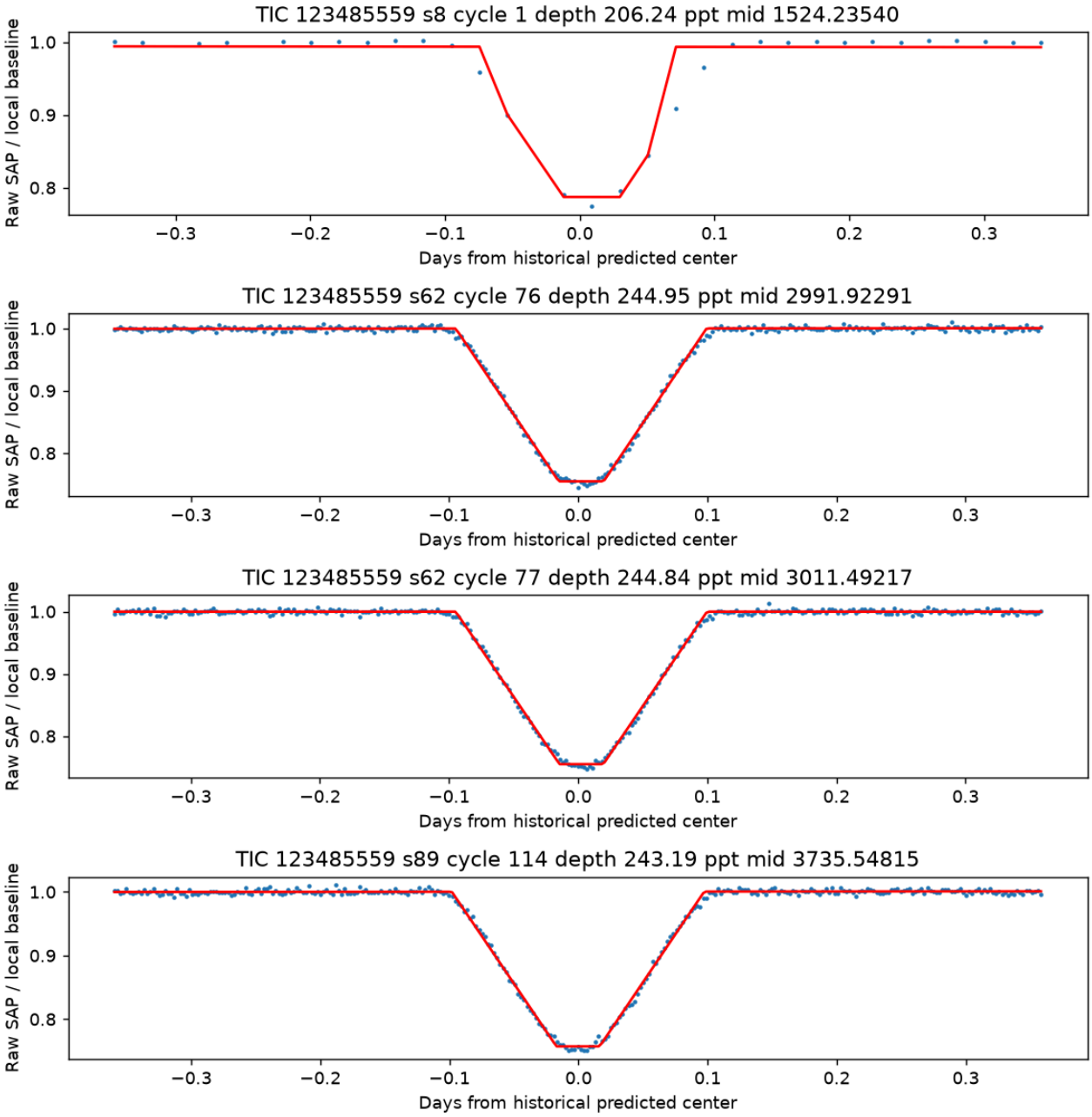
R652. TIC 123485559

physical_verdict: CONFIRMED

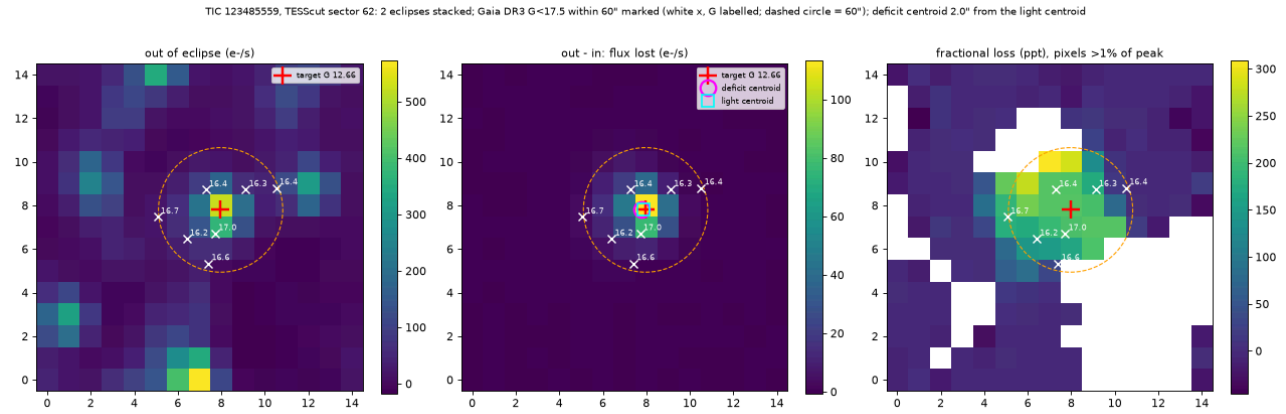
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional additional clean epochs and resolved photometry.



R652; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R652; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 123485559: CONFIRMED

Full bounded physical review: COMPLETE. Representative sectors initially supplied only one usable primary; fresh availability7/8/34/35/62/88/89 prompted sectors8/62 retrieval. Four primary timings, including consecutive sector62 cycles76/77, establish recurrence. Unequal secondary phase0.5874 supports orbital identity. Original sectors7/35 primary windows are gapped or flagged, not clean nulls. No capable Gaia neighbor within60arcsec.

Raw QLP SAP_FLUX representative events and full-sector P/2P morphology were visually inspected. Quality-zero finite positive samples only; predictions guide windows and are never timings. 4 identifiable free-center timings define P=19.569089713179 +/- 0.000036006732d, reference BTJD=1524.241154210 +/- 0.003225297d. Epoch error meaning: Conservative linear timing-fit covariance under the stated cadence floor and scatter inflation; excludes uncalibrated shape and baseline systematics. Epoch kind: Deeper primary established by unequal observed secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced chi square=0.17442611514732348; zero degrees of freedom is null. All errors retain their named meanings; no calibrated coverage is claimed. Selected same-kind recurring eclipse family; parity and P/2 coverage checked.

Depth=244.011540ppt is the median realized central trapezoid loss d*min(1,w/(2*k)); extrapolated amplitude and ingress remain separately in measurements.json. Measured secondary phase 0.58740316, formal phase error 1.0281078e-05, realized depth 89.74965ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Secondary raw pooled windows, actual sectors/cycles and fitted phase are retained in secondary-windows.csv and ../secondary-phases.json. No exhaustive blind alias search asserted. ../alias-audit.json and ../half-period-coverage.json retain recurrence/parity and sampled opposite-phase evidence. Flagged coverage is never a clean null.

Fresh TESScut difference image sector62 stacks2 core/bilateral-sideband events. Target pixel= [7.967199217737392, 7.793551855566159], deficit=[7.793049308241985, 7.814669101120165]; descriptive offset=0.17543pixel, field PSF residual=0.50031. No arbitrary centroid cut or calibrated

localization error is imposed. Every capable neighbor below was inspected at its actual projected pixels; close overlapping neighbors retain probable ownership. Other capable sources are displaced from compact deficit core but crowding limits localization. All Gaia neighbors are retained in pixel JSON.

Source	Separation arcsec	G	Required intrinsic loss proxy	Pixel coordinate	Diluted pixel loss
None					

G-band pairwise brightness screen assumes approximate passband/aperture response; intrinsic required eclipse cannot be directly compared with diluted pixel loss. Unresolved unlisted companions are not excluded.

Alternative reduction: independently extracted background-subtracted TESScut1x1/3x3/5x5 depths=215.27921/210.26485/202.93986ppt; QLP median=244.01154ppt. Pixel sector62 shares the same TESS photons with retained QLP and checks extraction robustness. Independent processing is not an independent instrument. Different apertures respond differently to dilution/background.

Gaia source5535139347540245504: period=19.565568984+/-0.004827082d; e=0.1812639; omega=42.0710108deg; periastron BTJD=387.43230946. Closest conjunction offset=0.1659689d (0.0084827phase), source-specific full-covariance linear error=0.2365679d. Correlation order P,gamma,K,e,omega,Tp with subindices0,3,4,5. Linear propagation is descriptive, not a Gaussian veto or ownership proof. Both conjunctions retained in ../gaia-geometry.json. Mass function=0.1350311solar, assumed M1=0.85, minimum M2=0.681701; approximate central duration=7.1454h versus observed=4.6664h. TIC radius/main-sequence companion scaling, edge-on conjunction and stellar evolution are assumptions; this is a plausibility check, not a fitted physical solution.

Sector	Same-kind cycle	Measured BTJD	Formal center error d	Conservative input error d	Realized depth ppt
8	1	1524.235403617	5.45507561e-05	0.0104165915	206.23694
62	76	2991.922905390	0.000118933302	0.00115738663	244.94558
62	77	3011.492165924	0.000120821849	0.00115738663	244.83697
89	114	3735.548146588	0.000132811265	0.00115738567	243.18611

Catalogue novelty is separate from this physical verdict. Shared coverage matrix and source identity checks are authoritative for catalogue recovery; period agreement alone cannot establish attribution.

Stable review key: R652.

Deciding evidence and numeric artifacts: - availability.json - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s62.json - historical/diff_s62.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

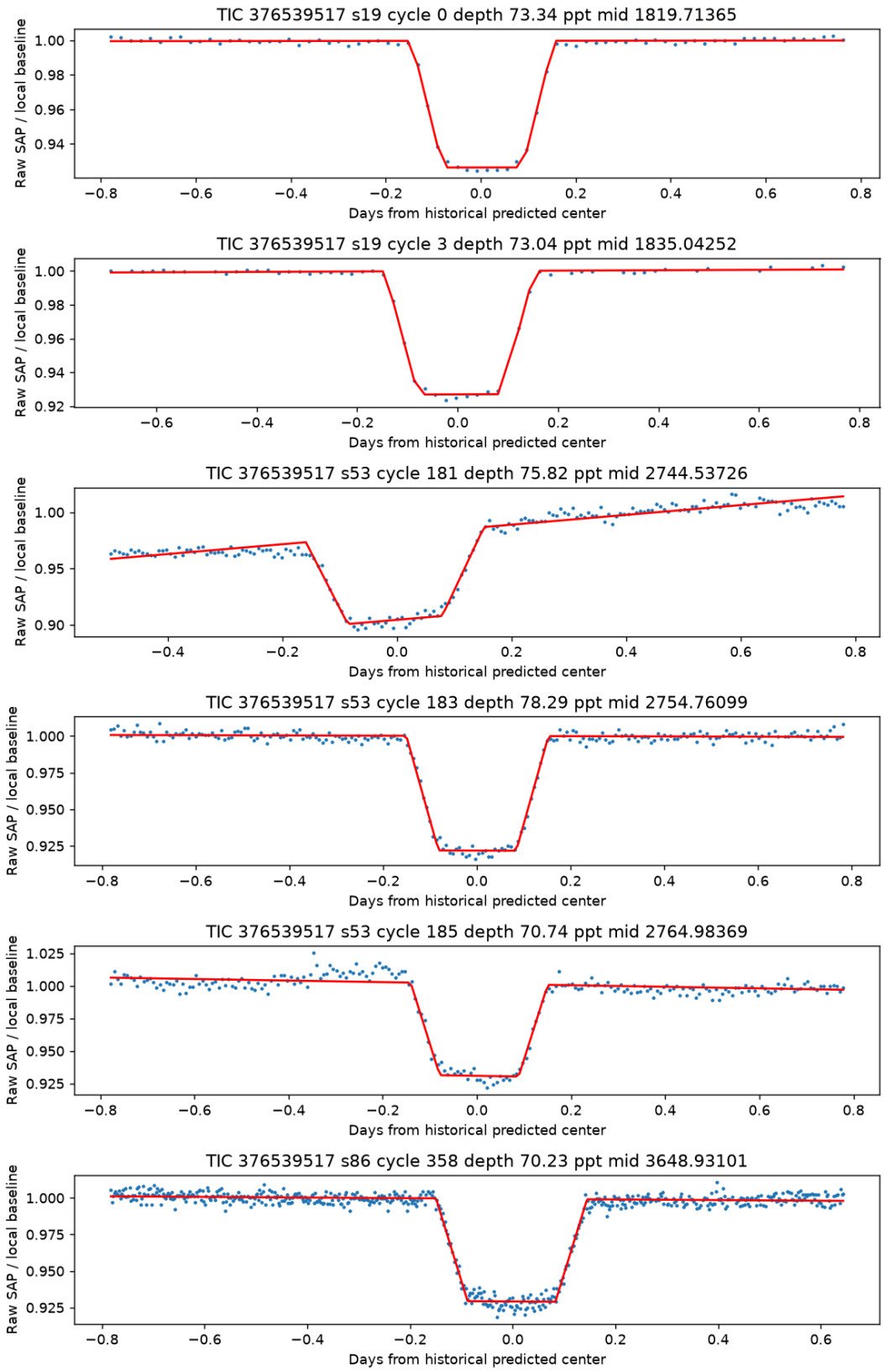
R653. TIC 376539517

physical_verdict: CONFIRMED

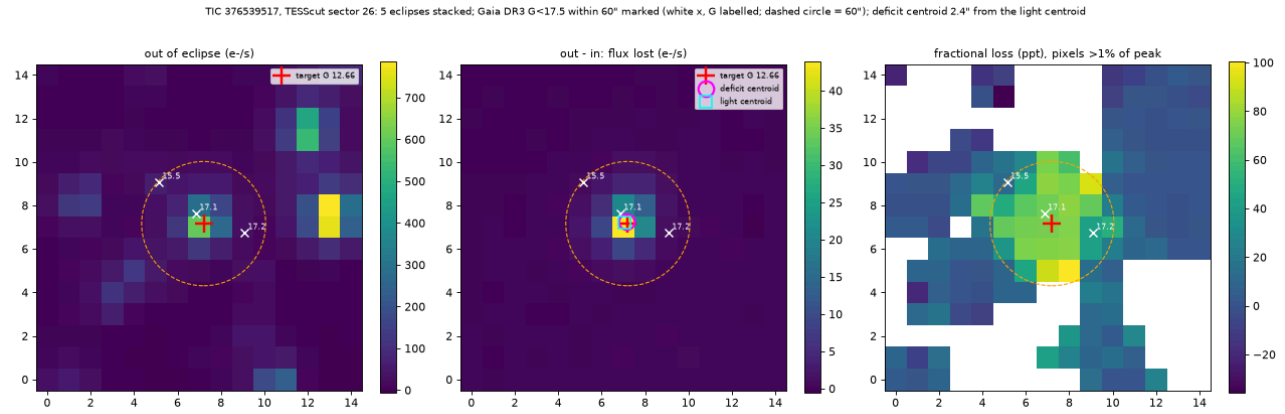
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional additional clean epochs and resolved photometry.



R653; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R653; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 376539517: CONFIRMED

Full bounded physical review: COMPLETE. Eleven repeated primary eclipses plus observed unequal secondary; substantial sector normalization differences do not remove the eclipse. No capable neighbor within60arcsec.

Raw QLP SAP_FLUX representative events and full-sector P/2P morphology were visually inspected. Quality-zero finite positive samples only; predictions guide windows and are never timings. 11 identifiable free-center timings define P=5.109546881245 +/- 0.000008596683d, reference BTJD=1819.713329927 +/- 0.002789204d. Epoch error meaning: Conservative linear timing-fit covariance under the stated cadence floor and scatter inflation; excludes uncalibrated shape and baseline systematics. Epoch kind: Deeper primary established by unequal observed secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced chi square=0.44049179083476947; zero degrees of freedom is null. All errors retain their named meanings; no calibrated coverage is claimed. Selected same-kind recurring eclipse family; parity and P/2 coverage checked.

Depth=73.102931ppt is the median realized central trapezoid loss d*min(1,w/(2*k)); extrapolated amplitude and ingress remain separately in measurements.json. Measured secondary phase 0.48186639, formal phase error 0.00020141761, realized depth 19.15961ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Secondary raw pooled windows, actual sectors/cycles and fitted phase are retained in secondary-windows.csv and ../secondary-phases.json. No exhaustive blind alias search asserted. ../alias-audit.json and ../half-period-coverage.json retain recurrence/parity and sampled opposite-phase evidence. Flagged coverage is never a clean null.

Fresh TESScut difference image sector26 stacks5 core/bilateral-sideband events. Target pixel= [7.205273409649651, 7.163492880968329], deficit=[7.138102923090409, 7.246464861133381]; descriptive offset=0.10675pixel, field PSF residual=0.33815. No arbitrary centroid cut or calibrated localization error is imposed. Every capable neighbor below was inspected at its actual projected pixels; close overlapping neighbors retain probable ownership. Other capable sources are displaced from compact deficit core but crowding limits localization. All Gaia neighbors are retained in pixel JSON.

Source	Separation arcsec	G	Required intrinsic loss proxy	Pixel coordinate	Diluted pixel loss
None					

G-band pairwise brightness screen assumes approximate passband/aperture response; intrinsic required eclipse cannot be directly compared with diluted pixel loss. Unresolved unlisted companions are not excluded.

Alternative reduction: independently extracted background-subtracted TESScut1x1/3x3/5x5 depths=72.95821/73.51253/67.73488ppt; QLP median=73.10293ppt. Pixel sector26 is a distinct observing occasion on the same TESS instrument; no same-sector QLP comparison claimed. Independent processing is not an independent instrument. Different apertures respond differently to dilution/background.

Gaia source503038050157132928: period=5.108145924+/-0.000724087d; e=0.1841026; omega=168.6275935deg; periastron BTJD=390.13818497. Closest conjunction offset=0.1258294d (0.0246331phase), source-specific full-covariance linear error=0.1776026d. Correlation order P,gamma,K,e,omega,Tp with subindices0,3,4,5. Linear propagation is descriptive, not a Gaussian veto or ownership proof. Both conjunctions retained in ../gaia-geometry.json. Mass function=0.04695988solar, assumed M1=1.1, minimum M2=0.491866; approximate central duration=7.3448h versus observed=7.1793h. TIC radius/main-sequence companion scaling, edge-on conjunction and stellar evolution are assumptions; this is a plausibility check, not a fitted physical solution.

Sector	Same-kind cycle	Measured BTJD	Formal center error d	Conservative input error d	Realized depth ppt
19	0	1819.713651287	0.000478078568	0.0104167832	73.33910
19	1	1824.822082273	0.000339148736	0.0104167832	73.10293
19	3	1835.042522355	0.000395136086	0.0104167832	73.03915
19	4	1840.151754657	0.000704202731	0.0104167832	72.45720
53	181	2744.537262535	0.00254763379	0.00347223585	75.81973
53	182	2749.647982243	0.000795887432	0.00347223585	75.18473
53	183	2754.760985111	0.000508438675	0.00347223585	78.28945
53	184	2759.872174616	0.00106153697	0.00347223585	74.08675
53	185	2764.983689300	0.00102988611	0.00347223585	70.74449
86	357	3643.821660490	0.000465494754	0.00115742984	71.05484
86	358	3648.931010241	0.000417157438	0.00115742984	70.23268

Catalogue novelty is separate from this physical verdict. Shared coverage matrix and source identity checks are authoritative for catalogue recovery; period agreement alone cannot establish attribution.

Stable review key: R653.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s26.json - historical/diff_s26.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

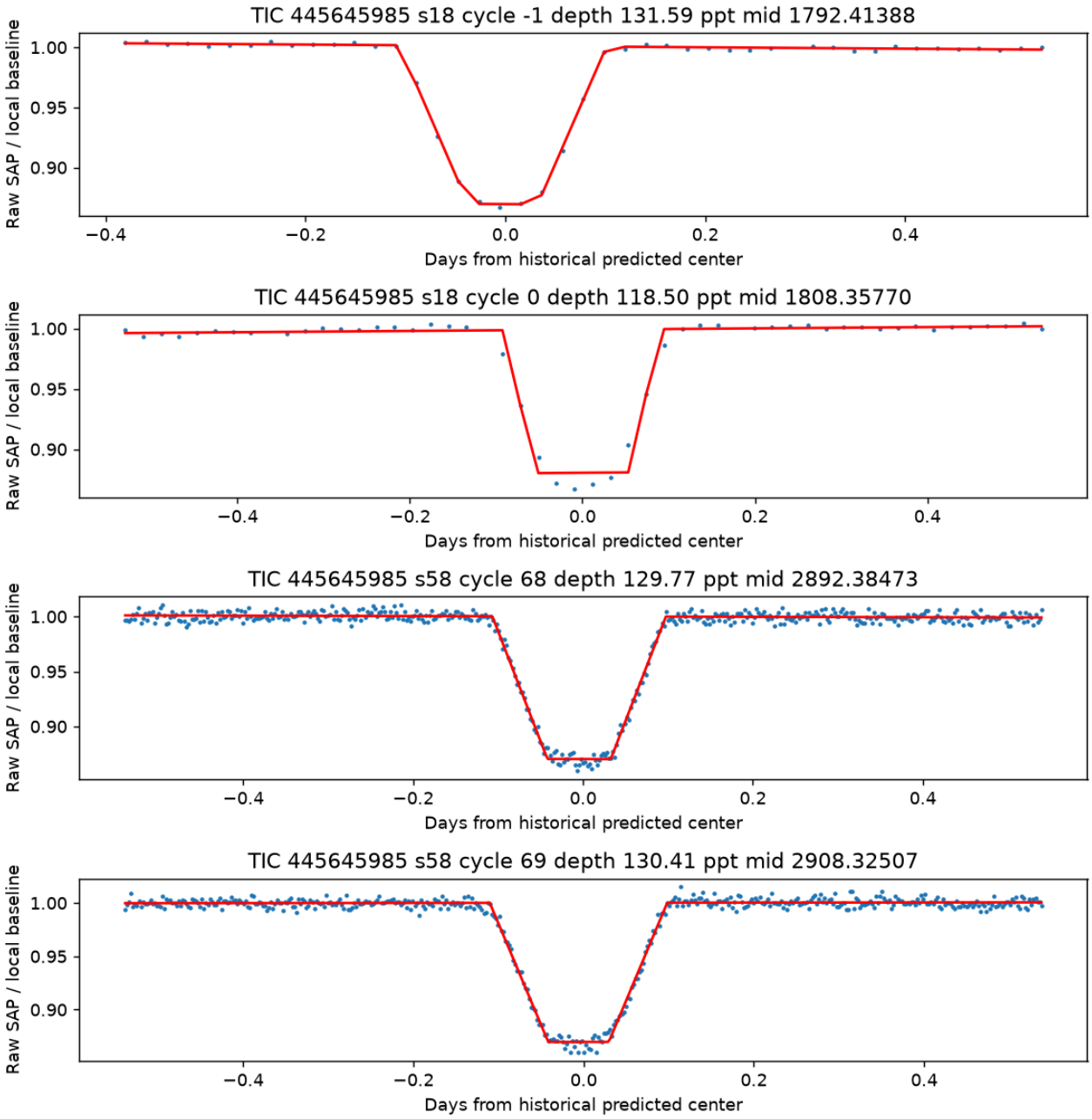
R654. TIC 445645985

physical_verdict: CONFIRMED

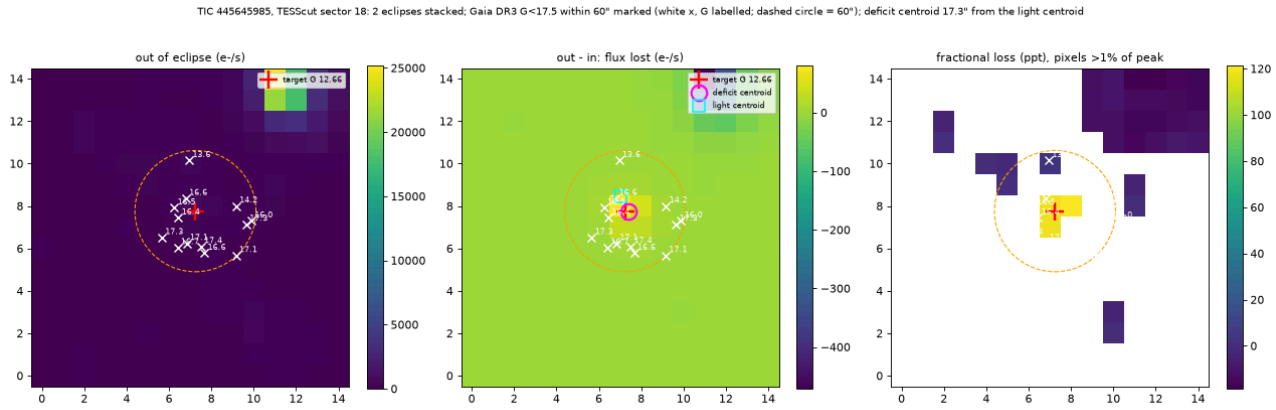
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional additional clean epochs and resolved photometry.



R654; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R654; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 445645985: CONFIRMED

Full bounded physical review: COMPLETE. Four repeated primary eclipses with observed shallow secondary near0.7873; two capable40.88/50.73arcsec sources lie away from compact deficit.

Raw QLP SAP_FLUX representative events and full-sector P/2P morphology were visually inspected. Quality-zero finite positive samples only; predictions guide windows and are never timings. 4 identifiable free-center timings define P=15.941576101817 +/- 0.000107173044d, reference BTJD=1792.415356135 +/- 0.007403165d. Epoch error meaning: Conservative linear timing-fit covariance under the stated cadence floor and scatter inflation; excludes uncalibrated shape and baseline systematics. Epoch kind: Deeper primary established by unequal observed secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced chi square=0.29470213321940947; zero degrees of freedom is null. All errors retain their named meanings; no calibrated coverage is claimed. Selected same-kind recurring eclipse family; parity and P/2 coverage checked.

Depth=130.086715ppt is the median realized central trapezoid loss d*min(1,w/(2*k)); extrapolated amplitude and ingress remain separately in measurements.json. Measured secondary phase 0.78731516, formal phase error 0.00014117914, realized depth 11.45414ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Secondary raw pooled windows, actual sectors/cycles and fitted phase are retained in secondary-windows.csv and ../secondary-phases.json. No exhaustive blind alias search asserted. ../alias-audit.json and ../half-period-coverage.json retain recurrence/parity and sampled opposite-phase evidence. Flagged coverage is never a clean null.

Fresh TESScut difference image sector18 stacks2 core/bilateral-sideband events. Target pixel= [7.247130469941494, 7.757436670640473], deficit=[7.4102256455206295, 7.725583538361966]; descriptive offset=0.16618pixel, field PSF residual=0.98672. No arbitrary centroid cut or calibrated localization error is imposed. Every capable neighbor below was inspected at its actual projected pixels; close overlapping neighbors retain probable ownership. Other capable sources are displaced from compact deficit core but crowding limits localization. All Gaia neighbors are retained in pixel JSON.

Source	Separation arcsec	G	Required intrinsic loss proxy	Pixel coordinate	Diluted pixel loss
45746408346698 6496	40.8776933582 7198	14.2030 33	0.6692969762611214	[9.169950482032025, 7.9998241896658655]	0.02734902570587 9323
45746432398515 3280	50.7281408288 5142	13.5987 6	0.43915109035103744	[6.949760464515536, 10.149413553030303]	0.00216700091352 49173

G-band pairwise brightness screen assumes approximate passband/aperture response; intrinsic required eclipse cannot be directly compared with diluted pixel loss. Unresolved unlisted companions are not excluded.

Alternative reduction: independently extracted background-subtracted TESScut1x1/3x3/5x5 depths=118.15505/105.77361/61.59252ppt; QLP median=130.08672ppt. Pixel sector18 shares the same TESS photons with retained QLP and checks extraction robustness. Independent processing is not an independent instrument. Different apertures respond differently to dilution/background.

Gaia source457464113524362496: period=15.937289414+/-0.013099912d; e=0.4995792; omega=12.8087903deg; periastron BTJD=388.25615447. Closest conjunction offset=0.4497241d (0.0282184phase), source-specific full-covariance linear error=0.7154511d. Correlation order P,gamma,K,e,omega,Tp with subindices0,3,4,5. Linear propagation is descriptive, not a Gaussian veto or ownership proof. Both conjunctions retained in ../gaia-geometry.json. Mass function=0.03734818solar, assumed M1=1.152, minimum M2=0.459445; approximate central duration=5.8238h versus observed=4.9374h. TIC radius/main-sequence companion scaling, edge-on conjunction and stellar evolution are assumptions; this is a plausibility check, not a fitted physical solution.

Sector	Same-kind cycle	Measured BTJD	Formal center error d	Conservative input error d	Realized depth ppt
18	-1	1792.413881246	0.000532514466	0.0104166891	131.58724
18	0	1808.357701957	0.000482518907	0.0104166891	118.50418
58	68	2892.384726070	0.000286241987	0.00115741244	129.76512
58	69	2908.325073058	0.0002261763	0.00115741244	130.40831

Catalogue novelty is separate from this physical verdict. Shared coverage matrix and source identity checks are authoritative for catalogue recovery; period agreement alone cannot establish attribution.

Stable review key: R654.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s18.json - historical/diff_s18.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

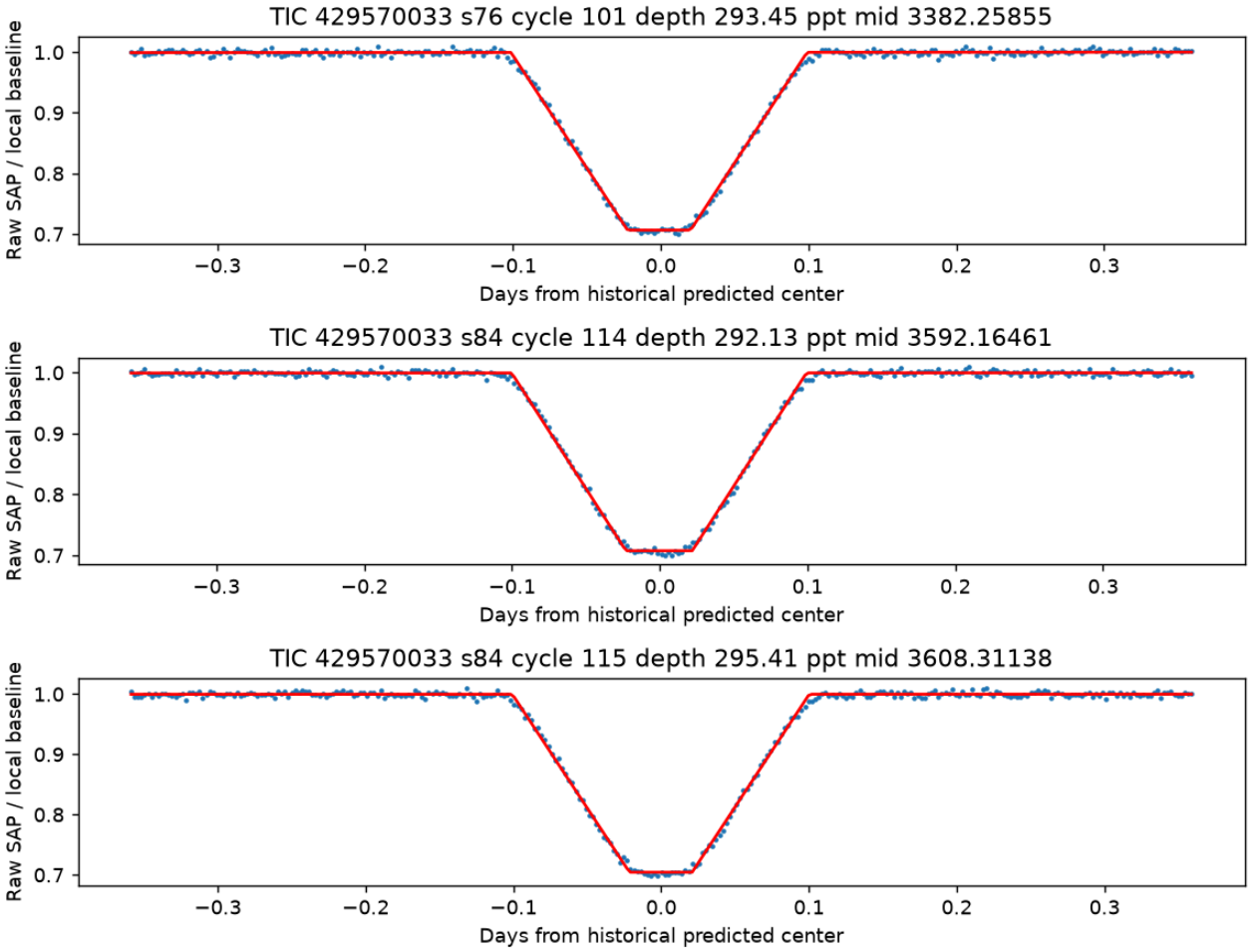
R655. TIC 429570033

physical_verdict: CONFIRMED

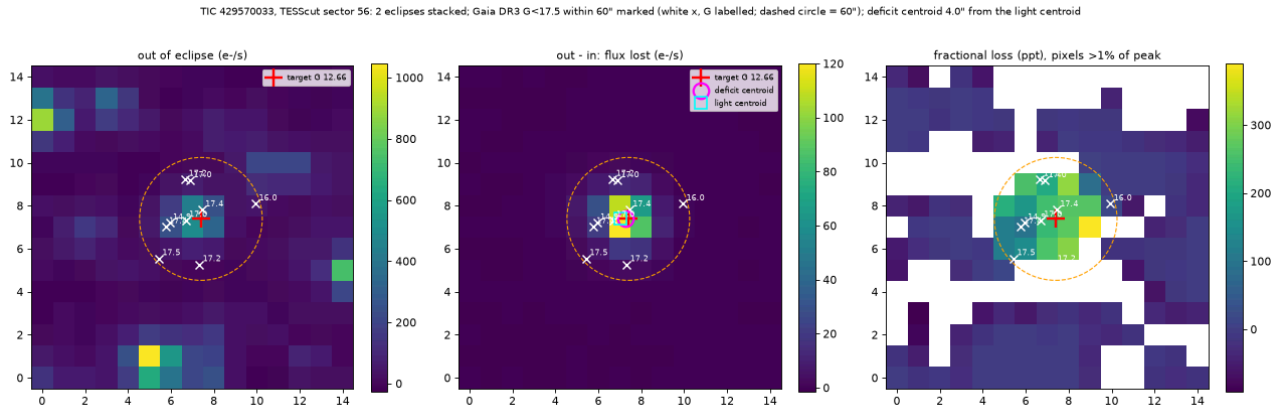
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional additional clean epochs and resolved photometry.



R655; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R655; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 429570033: CONFIRMED

Full bounded physical review: COMPLETE. Three deep narrow primary eclipses including consecutive sector84 cycles114/115 plus unequal secondary near0.682. Sector16 lacks a fitted primary, not a clean null. No capable neighbor within60arcsec.

Raw QLP SAP_FLUX representative events and full-sector P/2P morphology were visually inspected. Quality-zero finite positive samples only; predictions guide windows and are never timings. 3 identifiable free-center timings define P=16.146625951072 +/- 0.000104785390d, reference BTJD=3382.258545241 +/- 0.001155811d. Epoch error meaning: Conservative linear timing-fit covariance under the stated cadence floor and scatter inflation; excludes uncalibrated shape and baseline systematics. Epoch kind: Deeper primary established by unequal observed secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced chi square=0.007650194010375677; zero degrees of freedom is null. All errors retain their named meanings; no calibrated coverage is claimed. Selected same-kind recurring eclipse family; parity and P/2 coverage checked.

Depth=293.446739ppt is the median realized central trapezoid loss d*min(1,w/(2*k)); extrapolated amplitude and ingress remain separately in measurements.json. Measured secondary phase 0.68194978, formal phase error 2.3304537e-05, realized depth 50.12908ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Secondary raw pooled windows, actual sectors/cycles and fitted phase are retained in secondary-windows.csv and ../secondary-phases.json. No exhaustive blind alias search asserted. ../alias-audit.json and ../half-period-coverage.json retain recurrence/parity and sampled opposite-phase evidence. Flagged coverage is never a clean null.

Fresh TESScut difference image sector56 stacks2 core/bilateral-sideband events. Target pixel= [7.397841537902449, 7.384829290605374], deficit=[7.240675753333409, 7.4119798681592615]; descriptive offset=0.15949pixel, field PSF residual=0.46517. No arbitrary centroid cut or calibrated localization error is imposed. Every capable neighbor below was inspected at its actual projected pixels; close overlapping neighbors retain probable ownership. Other capable sources are displaced from compact deficit core but crowding limits localization. All Gaia neighbors are retained in pixel JSON.

Source	Separation arcsec	G	Required intrinsic loss proxy	Pixel coordinate	Diluted pixel loss
None					

G-band pairwise brightness screen assumes approximate passband/aperture response; intrinsic required eclipse cannot be directly compared with diluted pixel loss. Unresolved unlisted companions are not excluded.

Alternative reduction: independently extracted background-subtracted TESScut1x1/3x3/5x5 depths=234.81677/225.63601/227.67984ppt; QLP median=293.44674ppt. Pixel sector56 is a distinct observing occasion on the same TESS instrument; no same-sector QLP comparison claimed. Independent processing is not an independent instrument. Different apertures respond differently to dilution/background.

Gaia source2179272526332127232: period=16.144166157+/-0.004812888d; e=0.2874276; omega=23.9610338deg; periastron BTJD=393.49150057. Closest conjunction offset=0.3608174d (0.0223497phase), source-specific full-covariance linear error=0.7578532d. Correlation order P,gamma,K,e,omega,Tp with subindices0,3,4,5. Linear propagation is descriptive, not a Gaussian veto or ownership proof. Both conjunctions retained in ../gaia-geometry.json. Mass function=0.0843095solar, assumed M1=1.449, minimum M2=0.739028; approximate central duration=5.8682h versus observed=4.8079h. TIC radius/main-sequence companion scaling, edge-on conjunction and stellar evolution are assumptions; this is a plausibility check, not a fitted physical solution.

Sector	Same-kind cycle	Measured BTJD	Formal center error d	Conservative input error d	Realized depth ppt
76	101	3382.258550532	0.000137417721	0.00115739355	293.44674
84	114	3592.164608524	0.000100868475	0.00115739031	292.12567
84	115	3608.311377345	0.00013170009	0.00115739031	295.40795

Catalogue novelty is separate from this physical verdict. Shared coverage matrix and source identity checks are authoritative for catalogue recovery; period agreement alone cannot establish attribution.

Stable review key: R655.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s56.json - historical/diff_s56.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

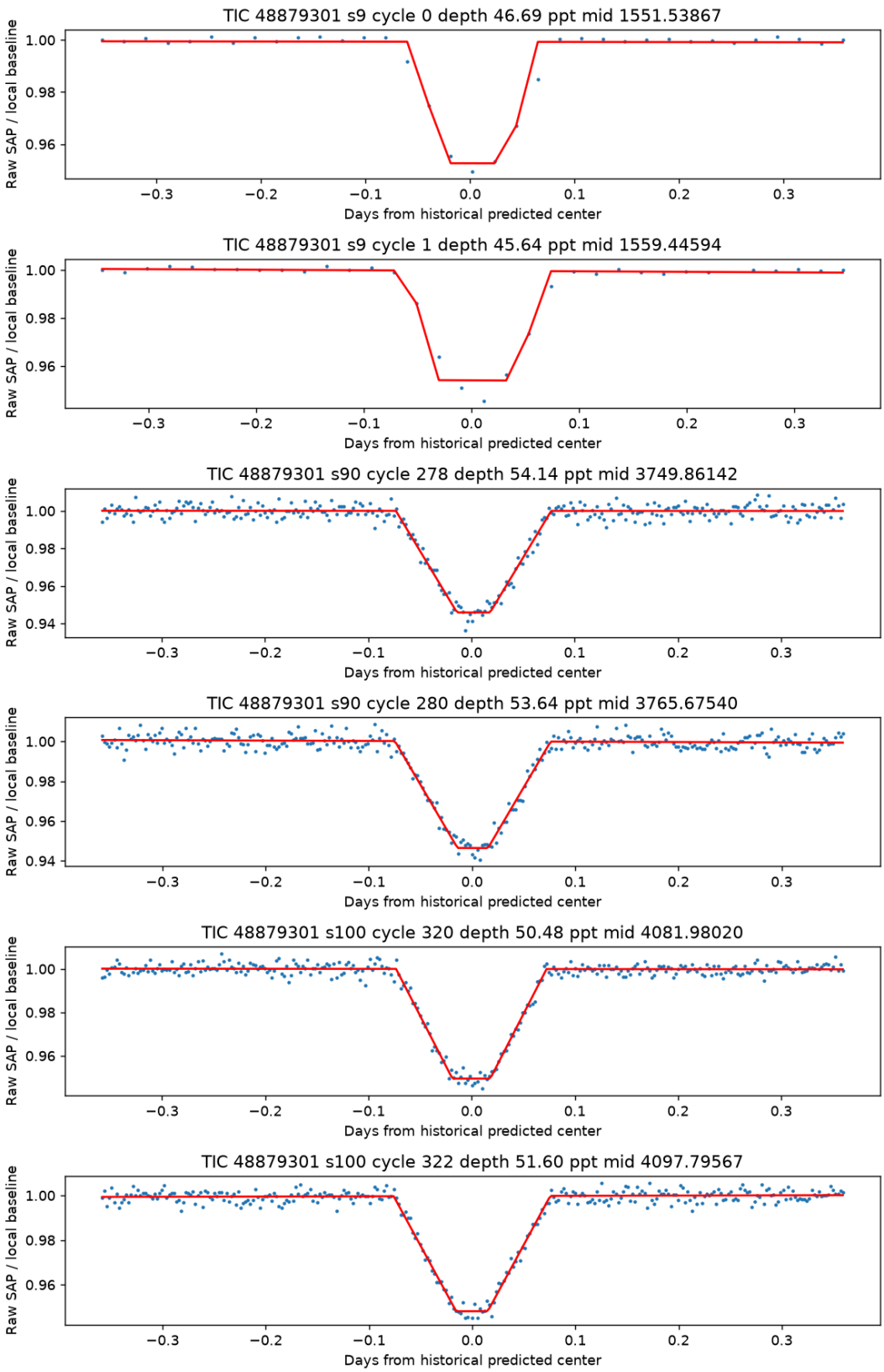
R656. TIC 48879301

physical_verdict: CONFIRMED

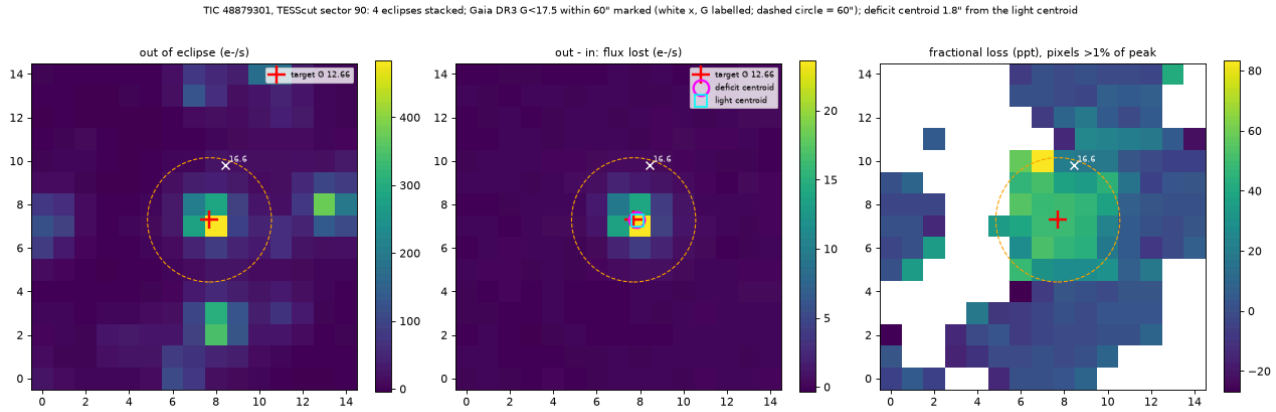
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional additional clean epochs and resolved photometry.



R656; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R656; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 48879301: CONFIRMED

Full bounded physical review: COMPLETE. Ten repeated narrow primary eclipses and unequal secondary; no capable neighbor within60arcsec.

Raw QLP SAP_FLUX representative events and full-sector P/2P morphology were visually inspected. Quality-zero finite positive samples only; predictions guide windows and are never timings. 10 identifiable free-center timings define P=7.907623484062 +/- 0.000014706680d, reference BTJD=1551.541058787 +/- 0.004370981d. Epoch error meaning: Conservative linear timing-fit covariance under the stated cadence floor and scatter inflation; excludes uncalibrated shape and baseline systematics. Epoch kind: Deeper primary established by unequal observed secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced chi square=0.22443573518765553; zero degrees of freedom is null. All errors retain their named meanings; no calibrated coverage is claimed. Selected same-kind recurring eclipse family; parity and P/2 coverage checked.

Depth=51.991549ppt is the median realized central trapezoid loss d*min(1,w/(2*k)); extrapolated amplitude and ingress remain separately in measurements.json. Measured secondary phase 0.48947999, formal phase error 7.7574893e-05, realized depth 21.03836ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Secondary raw pooled windows, actual sectors/cycles and fitted phase are retained in secondary-windows.csv and ../secondary-phases.json. No exhaustive blind alias search asserted. ../alias-audit.json and ../half-period-coverage.json retain recurrence/parity and sampled opposite-phase evidence. Flagged coverage is never a clean null.

Fresh TESScut difference image sector90 stacks4 core/bilateral-sideband events. Target pixel= [7.70322522876198, 7.2910002827836315], deficit=[7.819900956250759, 7.3099259957680305]; descriptive offset=0.11820pixel, field PSF residual=0.49719. No arbitrary centroid cut or calibrated localization error is imposed. Every capable neighbor below was inspected at its actual projected pixels; close overlapping neighbors retain probable ownership. Other capable sources are displaced from compact deficit core but crowding limits localization. All Gaia neighbors are retained in pixel JSON.

Source	Separation arcsec	G	Required intrinsic loss proxy	Pixel coordinate	Diluted pixel loss
None					

G-band pairwise brightness screen assumes approximate passband/aperture response; intrinsic required eclipse cannot be directly compared with diluted pixel loss. Unresolved unlisted companions are not excluded.

Alternative reduction: independently extracted background-subtracted TESScut1x1/3x3/5x5 depths=48.84132/48.17366/45.73593ppt; QLP median=51.99155ppt. Pixel sector90 shares the same TESS photons with retained QLP and checks extraction robustness. Independent processing is not an independent instrument. Different apertures respond differently to dilution/background.

Gaia source5449918297210471680: period=7.906795576+/-0.001631973d; e=0.0678617; omega=274.4254178deg; periastron BTJD=385.28528032. Closest conjunction offset=0.1145074d (0.0144821phase), source-specific full-covariance linear error=0.3026588d. Correlation order P,gamma,K,e,omega,Tp with subindices0,3,4,5. Linear propagation is descriptive, not a Gaussian veto or ownership proof. Both conjunctions retained in ../gaia-geometry.json. Mass function=0.08287888solar, assumed M1=1.19, minimum M2=0.656129; approximate central duration=6.0924h versus observed=3.6149h. TIC radius/main-sequence companion scaling, edge-on conjunction and stellar evolution are assumptions; this is a plausibility check, not a fitted physical solution.

Sector	Same-kind cycle	Measured BTJD	Formal center error d	Conservative input error d	Realized depth ppt
9	0	1551.538674240	0.000341772729	0.0104167647	46.68939
9	1	1559.445941015	0.000266195098	0.0104167647	45.64270
9	2	1567.352194049	0.000299355157	0.0104167647	47.83335
90	278	3749.861423151	0.000504221854	0.00115739967	54.14251
90	279	3757.768612656	0.00061005499	0.00115739967	52.84572
90	280	3765.675403923	0.000523942663	0.00115739967	53.63835
90	281	3773.582665958	0.000837821149	0.00115739967	53.33722
100	320	4081.980196125	0.000341766601	0.00115740685	50.47596
100	321	4089.888027434	0.000432011612	0.00115740685	52.38482
100	322	4097.795666630	0.000424969085	0.00115740685	51.59828

Catalogue novelty is separate from this physical verdict. Shared coverage matrix and source identity checks are authoritative for catalogue recovery; period agreement alone cannot establish attribution.

Stable review key: R656.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s90.json - historical/diff_s90.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

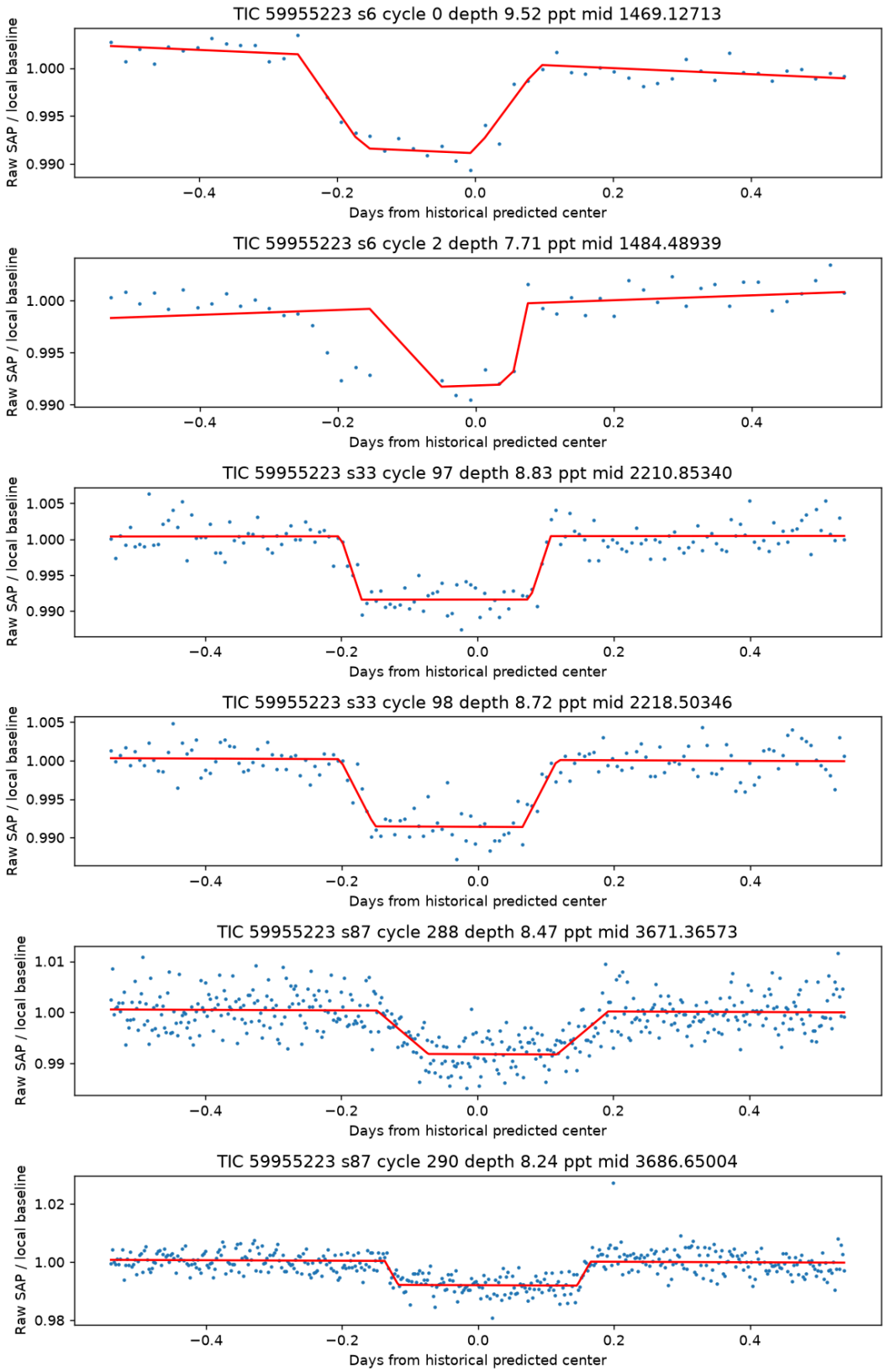
R657. TIC 59955223

physical_verdict: PROBABLE

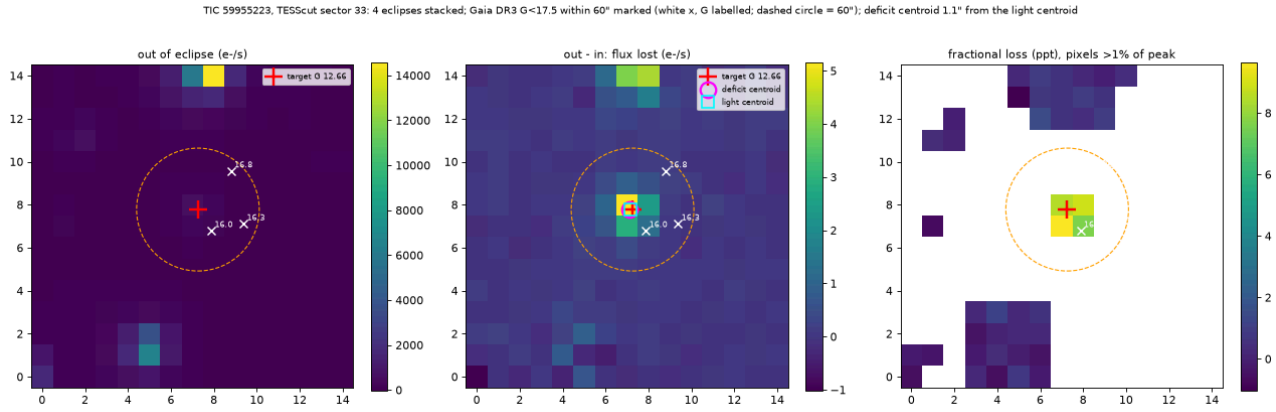
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or multi-source PRF attribution.



R657; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R657; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 59955223: PROBABLE

Full bounded physical review: COMPLETE. Nine repeated shallow narrow eclipses in raw SAP. Historical SAP-depth cache has no row; photometric search P branch explicitly seeds free-center fitting. No secure secondary. Capable24.43arcsec neighbor at1.1pixel separation overlaps target footprint; ownership remains unresolved. Other capable46.88/49.91arcsec sources are further displaced.

Raw QLP SAP_FLUX representative events and full-sector P/2P morphology were visually inspected. Quality-zero finite positive samples only; predictions guide windows and are never timings. 9 identifiable free-center timings define P=7.646605695993 +/- 0.000030343475d, reference BTJD=1469.134751039 +/- 0.006871504d. Epoch error meaning: Conservative linear timing-fit covariance under the stated cadence floor and scatter inflation; excludes uncalibrated shape and baseline systematics. Epoch kind: Selected repeating eclipse; deeper/secondary identity not established. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced chi square=6.500084427670953; zero degrees of freedom is null. All errors retain their named meanings; no calibrated coverage is claimed. Selected same-kind recurring eclipse family; parity and P/2 coverage checked.

Depth=8.828826ppt is the median realized central trapezoid loss d*min(1,w/(2*k)); extrapolated amplitude and ingress remain separately in measurements.json. No secure secondary; exploratory fit is not interpreted as an eclipse timing or counterpart. Secondary raw pooled windows, actual sectors/cycles and fitted phase are retained in secondary-windows.csv and ../secondary-phases.json. No exhaustive blind alias search asserted. ../alias-audit.json and ../half-period-coverage.json retain recurrence/parity and sampled opposite-phase evidence. Flagged coverage is never a clean null.

Fresh TESScut difference image sector33 stacks4 core/bilateral-sideband events. Target pixel= [7.258624113654173, 7.773432037253507], deficit=[7.112567829467887, 7.7605356510558465]; descriptive offset=0.14662pixel, field PSF residual=0.60868. No arbitrary centroid cut or calibrated localization error is imposed. Every capable neighbor below was inspected at its actual projected pixels; close overlapping neighbors retain probable ownership. Other capable sources are displaced from compact deficit core but crowding limits localization. All Gaia neighbors are retained in pixel JSON.

Source	Separation arcsec	G	Required intrinsic loss proxy	Pixel coordinate	Diluted pixel loss
2937944895339705472	24.43253064552264	16.004097	0.20045523859646372	[7.880499028698017, 6.797597998695929]	0.007576786488439782
2937944826620226304	46.87860422531916	16.306564	0.2620162727588868	[9.381452561374967, 7.118574349820252]	0.0018220279852657504
2937944929699437568	49.91162411423837	16.777998	0.3996839133338027	[8.803648167155401, 9.575826083636732]	0.004896318556257513

G-band pairwise brightness screen assumes approximate passband/aperture response; intrinsic required eclipse cannot be directly compared with diluted pixel loss. Unresolved unlisted companions are not excluded.

Alternative reduction: independently extracted background-subtracted TESScut1x1/3x3/5x5 depths=8.48086/8.91036/8.31172ppt; QLP median=8.82883ppt. Pixel sector33 shares the same TESS photons with retained QLP and checks extraction robustness. Independent processing is not an independent instrument. Different apertures respond differently to dilution/background.

Gaia source2937944895339704320: period=7.649254403+/-0.001468547d; e=0.1513959; omega=135.7670018deg; periastron BTJD=391.69996046. Closest conjunction offset=-0.3817110d (-0.0499017phase), source-specific full-covariance linear error=0.2409343d. Correlation order P,gamma,K,e,omega,Tp with subindices0,3,4,5. Linear propagation is descriptive, not a Gaussian veto or ownership proof. Both conjunctions retained in ../gaia-geometry.json. Mass function=0.004520785solar, assumed M1=1.06, minimum M2=0.192085; approximate central duration=8.0151h versus observed=7.6534h. TIC radius/main-sequence companion scaling, edge-on conjunction and stellar evolution are assumptions; this is a plausibility check, not a fitted physical solution.

Sector	Same-kind cycle	Measured BTJD	Formal center error d	Conservative input error d	Realized depth ppt
6	0	1469.127132794	0.00701917676	0.0104166767	9.52371
6	2	1484.489387392	0.00229461833	0.0104166767	7.71483
33	96	2203.204530452	0.0030736072	0.00347220546	9.67139
33	97	2210.853399480	0.00201740533	0.00347220546	8.82883
33	98	2218.503457255	0.00297708524	0.00347220546	8.71729
33	99	2226.141478921	0.00475569889	0.00475569889	9.03537
87	288	3671.365733022	0.00376214527	0.00376214527	8.46615
87	289	3679.001662313	0.00276466619	0.00276466619	8.91800
87	290	3686.650037204	0.00220590347	0.00220590347	8.23910

Catalogue novelty is separate from this physical verdict. Shared coverage matrix and source identity checks are authoritative for catalogue recovery; period agreement alone cannot establish attribution.

Stable review key: R657.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s33.json - historical/diff_s33.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

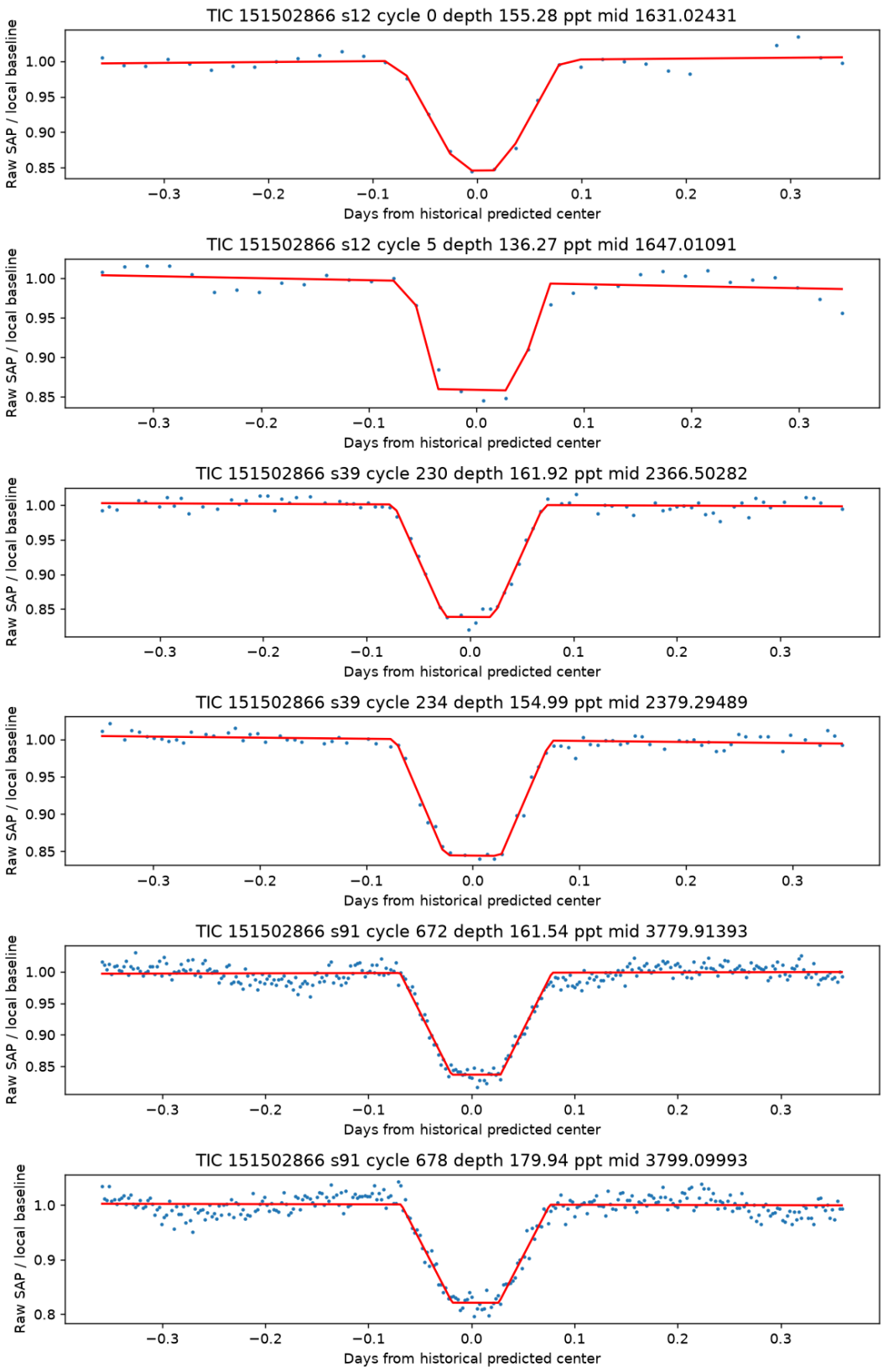
R658. TIC 151502866

physical_verdict: CONFIRMED

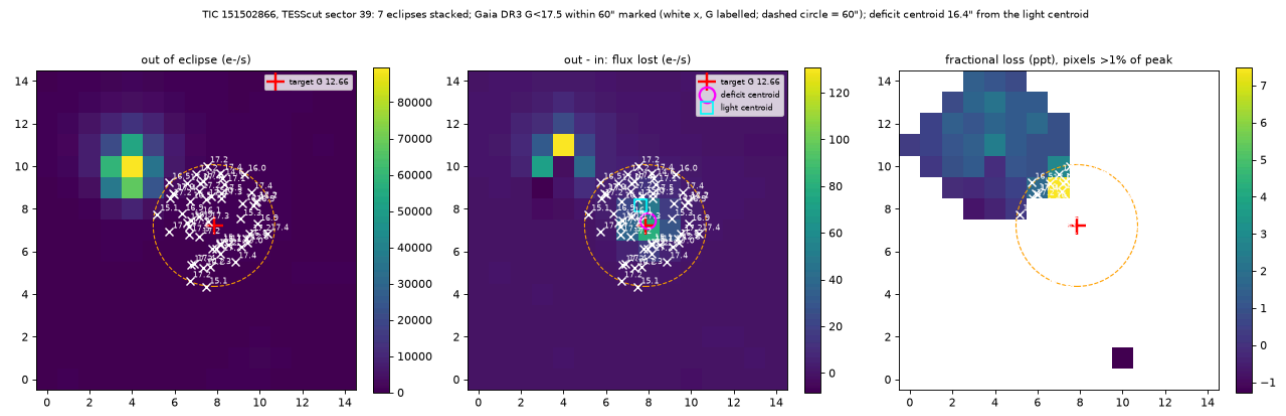
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional additional clean epochs and resolved photometry.



R658; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R658; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 151502866: CONFIRMED

Full bounded physical review: COMPLETE. Fifteen repeated deep primary eclipses and unequal shallow secondary. Very crowded field and bright star gradient require care; actual capable34.46/36.29arcsec neighbors are displaced from target-centered compact deficit, but uncalibrated pixel response limits localization.

Raw QLP SAP_FLUX representative events and full-sector P/2P morphology were visually inspected. Quality-zero finite positive samples only; predictions guide windows and are never timings. 15 identifiable free-center timings define P=3.197754665716 +/- 0.000003482897d, reference BTJD=1631.020365238 +/- 0.002215387d. Epoch error meaning: Conservative linear timing-fit covariance under the stated cadence floor and scatter inflation; excludes uncalibrated shape and baseline systematics. Epoch kind: Deeper primary established by unequal observed secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced chi square=1.2150070025582995; zero degrees of freedom is null. All errors retain their named meanings; no calibrated coverage is claimed. Selected same-kind recurring eclipse family; parity and P/2 coverage checked.

Depth=161.236077ppt is the median realized central trapezoid loss d*min(1,w/(2*k)); extrapolated amplitude and ingress remain separately in measurements.json. Measured secondary phase 0.57826225, formal phase error 0.0006516403, realized depth 18.70358ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Secondary raw pooled windows, actual sectors/cycles and fitted phase are retained in secondary-windows.csv and ../secondary-phases.json. No exhaustive blind alias search asserted. ../alias-audit.json and ../half-period-coverage.json retain recurrence/parity and sampled opposite-phase evidence. Flagged coverage is never a clean null.

Fresh TESScut difference image sector39 stacks7 core/bilateral-sideband events. Target pixel= [7.847738814637113, 7.219318807258942], deficit=[7.952999346394565, 7.437487084696226]; descriptive offset=0.24223pixel, field PSF residual=0.92479. No arbitrary centroid cut or calibrated

localization error is imposed. Every capable neighbor below was inspected at its actual projected pixels; close overlapping neighbors retain probable ownership. Other capable sources are displaced from compact deficit core but crowding limits localization. All Gaia neighbors are retained in pixel JSON.

Source	Separation arcsec	G	Required intrinsic loss proxy	Pixel coordinate	Diluted pixel loss
5980762575525630208	34.45598347583592	13.984445	0.7056071288248438	[9.192625073139695, 8.261218316325724]	0.04197975955122819
5980762678604857600	36.28778636800152	12.827504	0.3487867480562359	[9.423068889705092, 6.417188951678352]	0.015134288249077161

G-band pairwise brightness screen assumes approximate passband/aperture response; intrinsic required eclipse cannot be directly compared with diluted pixel loss. Unresolved unlisted companions are not excluded.

Alternative reduction: independently extracted background-subtracted TESScut1x1/3x3/5x5 depths=128.23972/76.38946/29.11939ppt; QLP median=161.23608ppt. Pixel sector39 shares the same TESS photons with retained QLP and checks extraction robustness. Independent processing is not an independent instrument. Different apertures respond differently to dilution/background.

Gaia source5980762609885374976: period=3.197977290+/-0.000172823d; e=0.1988739; omega=30.4907721deg; periastron BTJD=389.87519025. Closest conjunction offset=-0.0373613d (-0.0116828phase), source-specific full-covariance linear error=0.0682579d. Correlation order P,gamma,K,e,omega,Tp with subindices0,3,4,5. Linear propagation is descriptive, not a Gaussian veto or ownership proof. Both conjunctions retained in ../gaia-geometry.json. Mass function=0.04869363solar, assumed M1=1.06, minimum M2=0.488833; approximate central duration=4.3092h versus observed=3.4800h. TIC radius/main-sequence companion scaling, edge-on conjunction and stellar evolution are assumptions; this is a plausibility check, not a fitted physical solution.

Sector	Same-kind cycle	Measured BTJD	Formal center error d	Conservative input error d	Realized depth ppt
12	0	1631.024311142	0.00145178256	0.0104168706	155.27725
12	1	1634.217174087	0.00207380674	0.0104168706	179.05258
12	5	1647.010906237	0.00168988615	0.0104168706	136.26728
12	6	1650.203426697	0.0013661031	0.0104168706	150.83235
39	229	2363.306573456	0.00149813687	0.00347226644	150.58780
39	230	2366.502815525	0.000757319714	0.00347226644	161.91652
39	231	2369.702402304	0.000620461353	0.00347226644	170.32484
39	232	2372.896151505	0.00139165141	0.00347226644	168.72351
39	234	2379.294894292	0.000632940602	0.00347226644	154.98668
39	236	2385.693521756	0.000816787079	0.00347226644	151.15351
91	671	3776.713540695	0.000352930588	0.00115749114	161.23608
91	672	3779.913926385	0.000599811061	0.00115749114	161.54177
91	676	3792.700058947	0.00044426237	0.00115749114	166.72226
91	677	3795.898629111	0.000570093252	0.00115749114	153.37562
91	678	3799.099933481	0.000673197907	0.00115749114	179.93671

Catalogue novelty is separate from this physical verdict. Shared coverage matrix and source identity checks are authoritative for catalogue recovery; period agreement alone cannot establish attribution.

Stable review key: R658.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s39.json - historical/diff_s39.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv -

secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

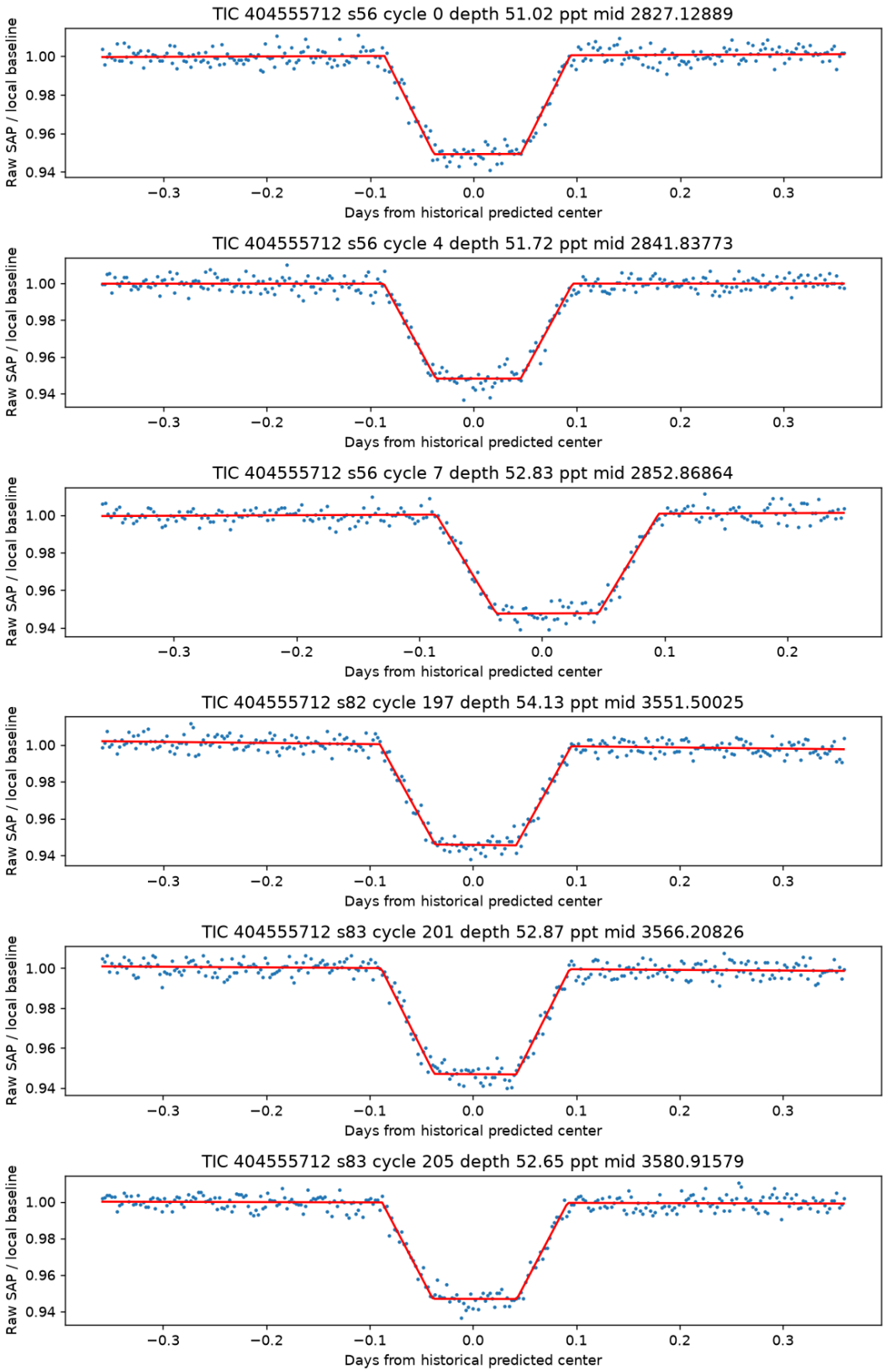
R659. TIC 404555712

physical_verdict: PROBABLE

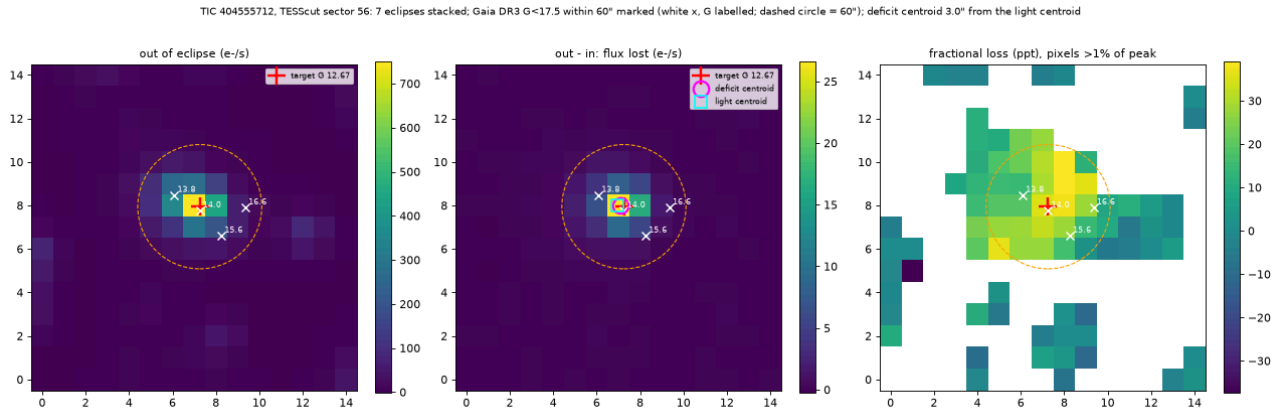
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or multi-source PRF attribution.



R659; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R659; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 404555712: PROBABLE

Full bounded physical review: COMPLETE. Eighteen repeated narrow eclipses; no secure secondary. Capable3.83arcsec source shares target footprint, so ownership is unresolved; capable25.95/33.88arcsec sources also inspected.

Raw QLP SAP_FLUX representative events and full-sector P/2P morphology were visually inspected. Quality-zero finite positive samples only; predictions guide windows and are never timings. 18 identifiable free-center timings define P=3.677006183156 +/- 0.000002858391d, reference BTJD=2827.129615244 +/- 0.000445630d. Epoch error meaning: Conservative linear timing-fit covariance under the stated cadence floor and scatter inflation; excludes uncalibrated shape and baseline systematics. Epoch kind: Selected repeating eclipse; deeper/secondary identity not established. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced chi square=0.229237781383927; zero degrees of freedom is null. All errors retain their named meanings; no calibrated coverage is claimed. Selected same-kind recurring eclipse family; parity and P/2 coverage checked.

Depth=52.739702ppt is the median realized central trapezoid loss d*min(1,w/(2*k)); extrapolated amplitude and ingress remain separately in measurements.json. No secure secondary; exploratory fit is not interpreted as an eclipse timing or counterpart. Secondary raw pooled windows, actual sectors/cycles and fitted phase are retained in secondary-windows.csv and ../secondary-phases.json. No exhaustive blind alias search asserted. ../alias-audit.json and ../half-period-coverage.json retain recurrence/parity and sampled opposite-phase evidence. Flagged coverage is never a clean null.

Fresh TESScut difference image sector56 stacks7 core/bilateral-sideband events. Target pixel= [7.25620277565433, 7.943516946536629], deficit=[7.0932526593635, 7.999418062441616]; descriptive offset=0.17227pixel, field PSF residual=0.22655. No arbitrary centroid cut or calibrated localization error is imposed. Every capable neighbor below was inspected at its actual projected pixels; close overlapping neighbors retain probable ownership. Other capable sources are displaced from compact deficit core but crowding limits localization. All Gaia neighbors are retained in pixel JSON.

Source	Separation arcsec	G	Required intrinsic loss proxy	Pixel coordinate	Diluted pixel loss
2733853409546937728	3.8312876974264625	14.016747	0.23571281268246358	[7.248731624134123, 7.758478182235477]	0.03547958549619447
2733853375187198464	25.945231799064445	13.78339	0.20032546724101008	[6.075128365898738, 8.464373254633738]	0.018997610687381896
2733853409546938880	33.88328663911585	15.593294	0.8343784339110035	[8.246169191726283, 6.602914774852005]	0.028190714333397416

G-band pairwise brightness screen assumes approximate passband/aperture response; intrinsic required eclipse cannot be directly compared with diluted pixel loss. Unresolved unlisted companions are not excluded.

Alternative reduction: independently extracted background-subtracted TESScut1x1/3x3/5x5 depths=35.47959/30.33363/28.64334ppt; QLP median=52.73970ppt. Pixel sector56 shares the same TESS photons with retained QLP and checks extraction robustness. Independent processing is not an independent instrument. Different apertures respond differently to dilution/background.

Gaia source2733853405251374080: period=3.676598986+/-0.000533637d; e=0.1960001; omega=138.1493896deg; periastron BTJD=389.80263576. Closest conjunction offset=0.0787096d (0.0214083phase), source-specific full-covariance linear error=0.3146965d. Correlation order P,gamma,K,e,omega,Tp with subindices0,3,4,5. Linear propagation is descriptive, not a Gaussian veto or ownership proof. Both conjunctions retained in ../gaia-geometry.json. Mass function=0.01146678solar, assumed M1=1.13, minimum M2=0.284097; approximate central duration=4.8885h versus observed=4.3874h. TIC radius/main-sequence companion scaling, edge-on conjunction and stellar evolution are assumptions; this is a plausibility check, not a fitted physical solution.

Sector	Same-kind cycle	Measured BTJD	Formal center error d	Conservative input error d	Realized depth ppt
56	0	2827.128886608	0.000525906045	0.00115738224	51.01513
56	1	2830.806648771	0.000521105146	0.00115738224	51.06916
56	3	2838.160454086	0.000558618445	0.00115738224	51.17741
56	4	2841.837728621	0.000479841993	0.00115738224	51.71954
56	5	2845.514652752	0.000519716638	0.00115738224	53.34289
56	6	2849.192536719	0.000648105885	0.00115738224	54.20558
56	7	2852.868641406	0.000520139988	0.00115738224	52.83382
82	193	3536.790825452	0.00047117133	0.00115742803	53.31378
82	194	3540.468475777	0.0005979719	0.00115742803	53.54871
82	196	3547.822534387	0.000595409541	0.00115742803	52.21195
82	197	3551.500254927	0.000595818809	0.00115742803	54.12804
82	198	3555.176825133	0.000662798826	0.00115742803	52.88819
83	200	3562.530411924	0.000667158476	0.00115738151	51.59662
83	201	3566.208258580	0.000549123696	0.00115738151	52.86688
83	202	3569.886219023	0.000535916476	0.00115738151	51.87714
83	203	3573.561746090	0.000545787179	0.00115738151	53.21321
83	204	3577.238902706	0.000538566638	0.00115738151	51.98768
83	205	3580.915791754	0.000536415945	0.00115738151	52.64559

Catalogue novelty is separate from this physical verdict. Shared coverage matrix and source identity checks are authoritative for catalogue recovery; period agreement alone cannot establish attribution.

Stable review key: R659.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s56.json - historical/diff_s56.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

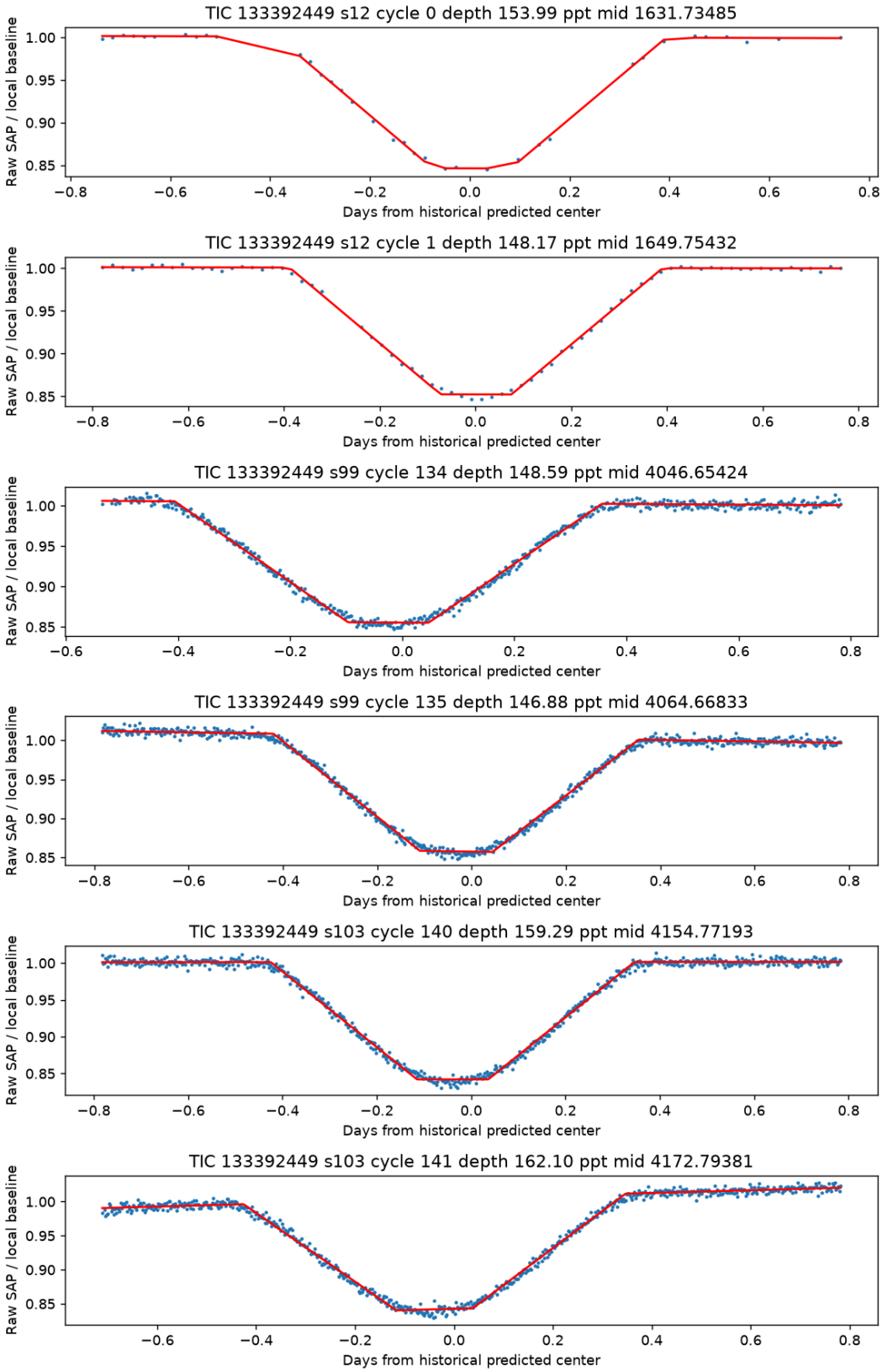
R660. TIC 133392449

physical_verdict: PROBABLE

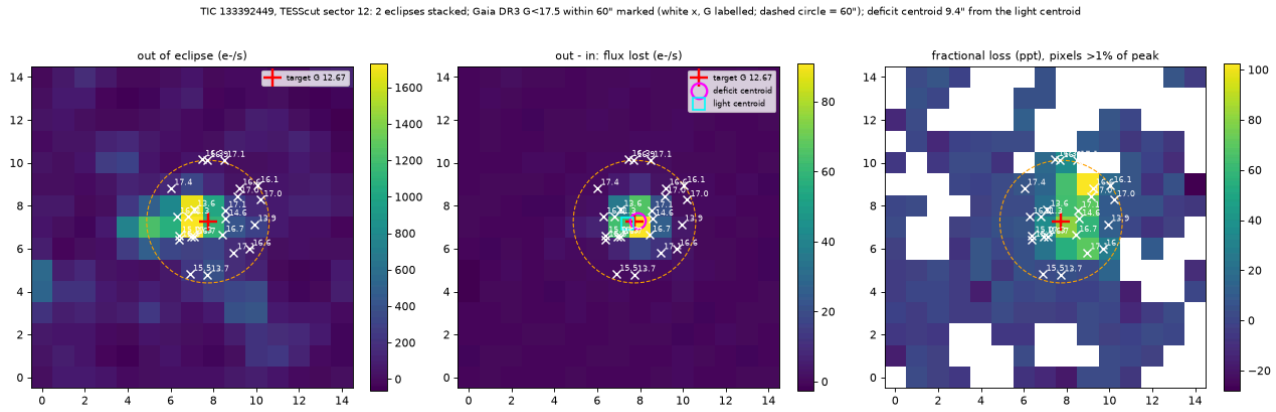
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or multi-source PRF attribution.



R660; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R660; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 133392449: PROBABLE

Full bounded physical review: COMPLETE. Six repeated broad deep primary eclipses and a secondary near phase0.604, most cleanly visible in sector12; large sector baselines limit secondary depth accuracy. Inner capable16.69/18.70arcsec sources share the target footprint: ownership remains unresolved. Outer43.15/49.36arcsec sources are displaced.

Raw QLP SAP_FLUX representative events and full-sector P/2P morphology were visually inspected. Quality-zero finite positive samples only; predictions guide windows and are never timings. 6 identifiable free-center timings define P=18.021654636300 +/- 0.000210656372d, reference BTJD=1631.745566269 +/- 0.028701204d. Epoch error meaning: Conservative linear timing-fit covariance under the stated cadence floor and scatter inflation; excludes uncalibrated shape and baseline systematics. Epoch kind: Deeper primary established by unequal observed secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced chi square=15.848209438029008; zero degrees of freedom is null. All errors retain their named meanings; no calibrated coverage is claimed. Selected same-kind recurring eclipse family; parity and P/2 coverage checked.

Depth=151.294349ppt is the median realized central trapezoid loss d*min(1,w/(2*k)); extrapolated amplitude and ingress remain separately in measurements.json. Measured secondary phase 0.60406219, formal phase error 0.00019665178, realized depth 93.22618ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Secondary raw pooled windows, actual sectors/cycles and fitted phase are retained in secondary-windows.csv and ../secondary-phases.json. No exhaustive blind alias search asserted. ../alias-audit.json and ../half-period-coverage.json retain recurrence/parity and sampled opposite-phase evidence. Flagged coverage is never a clean null.

Fresh TESScut difference image sector12 stacks2 core/bilateral-sideband events. Target pixel= [7.74163513205823, 7.260115530009859], deficit=[7.8658115741010155, 7.29564787217059]; descriptive offset=0.12916pixel, field PSF residual=0.69050. No arbitrary centroid cut or calibrated localization error is imposed. Every capable neighbor below was inspected at its actual projected pixels;

close overlapping neighbors retain probable ownership. Every capable neighbor is assessed above and below; close overlapping cases retain unresolved attribution. All Gaia neighbors are retained in pixel JSON.

Source	Separation arcsec	G	Required intrinsic loss proxy	Pixel coordinate	Diluted pixel loss
5888148722604706304	16.68707135704292	13.586701	0.5034549026825209	[7.085597591154146, 7.8227538159093175]	0.020918346998512995
5888148722604706432	18.696270339135605	11.349322	0.19614710257740012	[6.810385805314335, 7.4795628653223964]	0.036154277701270481
5888143053247419136	43.14941916259565	13.891616	0.6176386866712941	[9.92691803964027, 7.130815210335037]	0.012606213008692102
5888148756964450688	49.356004685888024	13.705976	0.5443469802802722	[7.728820979670219, 4.785614338550957]	0.005110266236690588

G-band pairwise brightness screen assumes approximate passband/aperture response; intrinsic required eclipse cannot be directly compared with diluted pixel loss. Unresolved unlisted companions are not excluded.

Alternative reduction: independently extracted background-subtracted TESScut1x1/3x3/5x5 depths=80.60458/47.54593/32.97453ppt; QLP median=151.29435ppt. Pixel sector12 shares the same TESS photons with retained QLP and checks extraction robustness. Independent processing is not an independent instrument. Different apertures respond differently to dilution/background.

Gaia source5888148722604708352: period=18.016804588+/-0.008917450d; e=0.2068520; omega=208.8248923deg; periastron BTJD=386.24612181. Closest conjunction offset=0.2394776d (0.0132919phase), source-specific full-covariance linear error=0.7160614d. Correlation order P,gamma,K,e,omega,Tp with subindices0,3,4,5. Linear propagation is descriptive, not a Gaussian veto or ownership proof. Both conjunctions retained in ../gaia-geometry.json. Mass function=0.2273762solar, assumed M1=1.0, minimum M2=0.953945; approximate central duration=19.4475h versus observed=18.5673h. TIC radius/main-sequence companion scaling, edge-on conjunction and stellar evolution are assumptions; this is a plausibility check, not a fitted physical solution.

Sector	Same-kind cycle	Measured BTJD	Formal center error d	Conservative input error d	Realized depth ppt
12	0	1631.734854338	0.00182644157	0.0104164414	153.99416
12	1	1649.754323936	0.000998046982	0.0104164414	148.16914
99	134	4046.654240944	0.000992455644	0.00115748098	148.59454
99	135	4064.668325979	0.00142490495	0.00142490495	146.88252
103	140	4154.771926493	0.000618720924	0.00115740685	159.29043
103	141	4172.793812212	0.00264886574	0.00264886574	162.09690

Catalogue novelty is separate from this physical verdict. Shared coverage matrix and source identity checks are authoritative for catalogue recovery; period agreement alone cannot establish attribution.

Stable review key: R660.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s12.json - historical/diff_s12.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

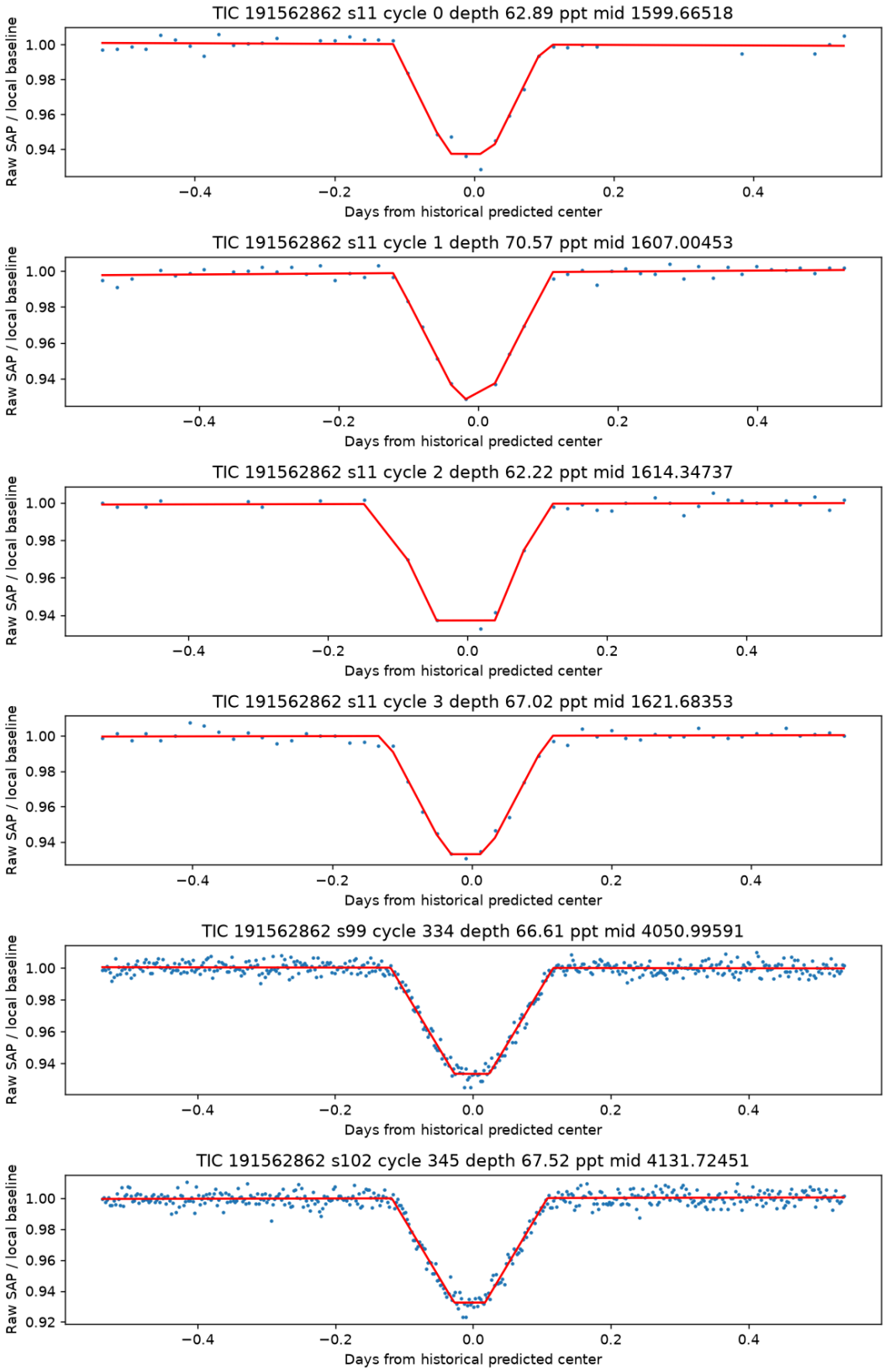
R661. TIC 191562862

physical_verdict: CONFIRMED

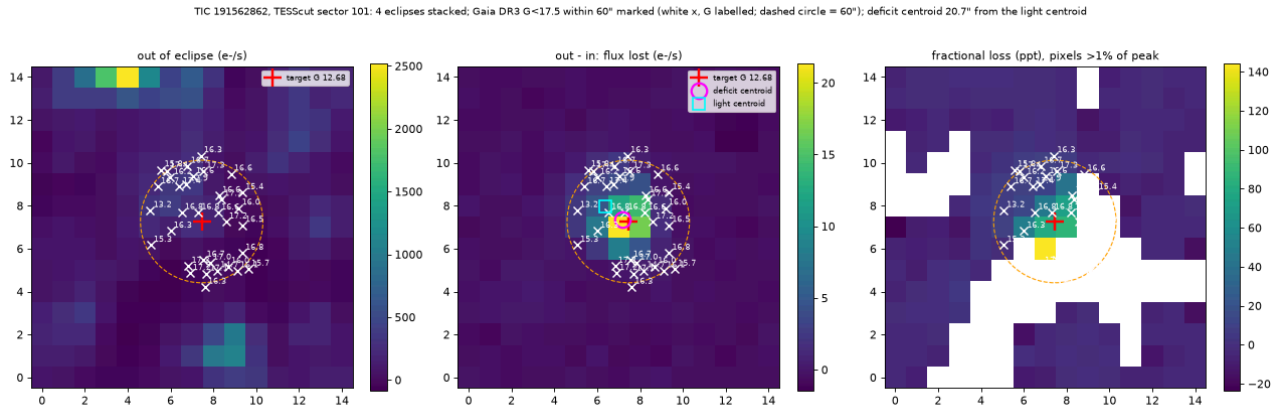
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional additional clean epochs and resolved photometry.



R661; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R661; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 191562862: CONFIRMED

Full bounded physical review: COMPLETE. Seven deep narrow same-kind primary eclipses and unequal secondary. Five capable35.30/45.16/46.71/47.24/52.73arcsec sources are displaced from target-centered compact deficit. Nearby known EB Gaia5893422117807088128 at28.924arcsec has G18.216 and period0.386714d, unlike observed7.339302d; required intrinsic depth11.08 using fresh aperture3 depth60.73ppt under approximate flux proxy is impossible. Its local pixel loss0.01199 is diluted and is not an intrinsic depth comparison.

Raw QLP SAP_FLUX representative events and full-sector P/2P morphology were visually inspected. Quality-zero finite positive samples only; predictions guide windows and are never timings. 7 identifiable free-center timings define P=7.339302438204 +/- 0.000019325561d, reference BTJD=1599.667739335 +/- 0.006485975d. Epoch error meaning: Conservative linear timing-fit covariance under the stated cadence floor and scatter inflation; excludes uncalibrated shape and baseline systematics. Epoch kind: Deeper primary established by unequal observed secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced chi square=1.555864944524252; zero degrees of freedom is null. All errors retain their named meanings; no calibrated coverage is claimed. Selected same-kind recurring eclipse family; parity and P/2 coverage checked.

Depth=66.959773ppt is the median realized central trapezoid loss d*min(1,w/(2*k)); extrapolated amplitude and ingress remain separately in measurements.json. Measured secondary phase 0.56584107, formal phase error 0.00011466609, realized depth 39.31893ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Secondary raw pooled windows, actual sectors/cycles and fitted phase are retained in secondary-windows.csv and ../secondary-phases.json. No exhaustive blind alias search asserted. ../alias-audit.json and ../half-period-coverage.json retain recurrence/parity and sampled opposite-phase evidence. Flagged coverage is never a clean null.

Fresh TESScut difference image sector101 stacks4 core/bilateral-sideband events. Target pixel= [7.451949056635936, 7.266663509805463], deficit=[7.168338071047642, 7.3751825207978365]; descriptive offset=0.30366pixel, field PSF residual=0.85045. No arbitrary centroid cut or calibrated

localization error is imposed. Every capable neighbor below was inspected at its actual projected pixels; close overlapping neighbors retain probable ownership. Every capable neighbor is assessed above and below; close overlapping cases retain unresolved attribution. All Gaia neighbors are retained in pixel JSON.

Source	Separation arcsec	G	Required intrinsic loss proxy	Pixel coordinate	Diluted pixel loss
5893422323965519104	35.29891270314629	12.868427	0.1469601382857458	[6.741172616287977, 9.010082403782057]	0.010734464588701633
5893422083446906496	45.156927554614384	14.714379	0.5049584220746222	[8.170086616048309, 4.978369268389391]	0.02324128363545262
5893422358324875776	46.709460597277577	15.443603 5	0.9243184965394825	[9.344337966662861, 8.61498055779114]	0.049992251137092 31
5893422010388426496	47.237372993025694	13.243674	0.17998900334866422	[5.047857719601214, 7.780051103855602]	0.006979506966629625
5893422014727886208	52.729872640490 69	15.269052 5	0.796992233289766	[5.0601527881431, 6.16341688832155]	-0.000848964740932661

G-band pairwise brightness screen assumes approximate passband/aperture response; intrinsic required eclipse cannot be directly compared with diluted pixel loss. Unresolved unlisted companions are not excluded.

Alternative reduction: independently extracted background-subtracted TESScut1x1/3x3/5x5 depths=78.81233/60.72895/38.57875ppt; QLP median=66.95977ppt. Pixel sector101 is a distinct observing occasion on the same TESS instrument; no same-sector QLP comparison claimed. Independent processing is not an independent instrument. Different apertures respond differently to dilution/background.

Gaia source5893422117806670464: period=7.339824234+/-0.002787931d; e=0.1200930; omega=324.4754166deg; periastron BTJD=386.41041969. Closest conjunction offset=-0.1324241d (-0.0180419phase), source-specific full-covariance linear error=0.5022492d. Correlation order P,gamma,K,e,omega,Tp with subindices0,3,4,5. Linear propagation is descriptive, not a Gaussian veto or ownership proof. Both conjunctions retained in ../gaia-geometry.json. Mass function=0.1397711solar, assumed M1=1.52, minimum M2=0.94769; approximate central duration=9.6243h versus observed=5.4349h. TIC radius/main-sequence companion scaling, edge-on conjunction and stellar evolution are assumptions; this is a plausibility check, not a fitted physical solution.

Sector	Same-kind cycle	Measured BTJD	Formal center error d	Conservative input error d	Realized depth ppt
11	0	1599.665183438	0.00187906111	0.0104168611	62.89286
11	1	1607.004532315	0.00137568555	0.0104168611	70.56792
11	2	1614.347365175	0.000945635876	0.0104168611	62.22435
11	3	1621.683526650	0.000893982844	0.0104168611	67.02446
99	334	4050.995912692	0.000539835501	0.00115748208	66.61345
99	336	4065.674850224	0.000675770334	0.00115748208	66.95977
102	345	4131.724506310	0.000554549044	0.00115740684	67.51577

Catalogue novelty is separate from this physical verdict. Shared coverage matrix and source identity checks are authoritative for catalogue recovery; period agreement alone cannot establish attribution.

Stable review key: R661.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s101.json - historical/diff_s101.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv -

secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

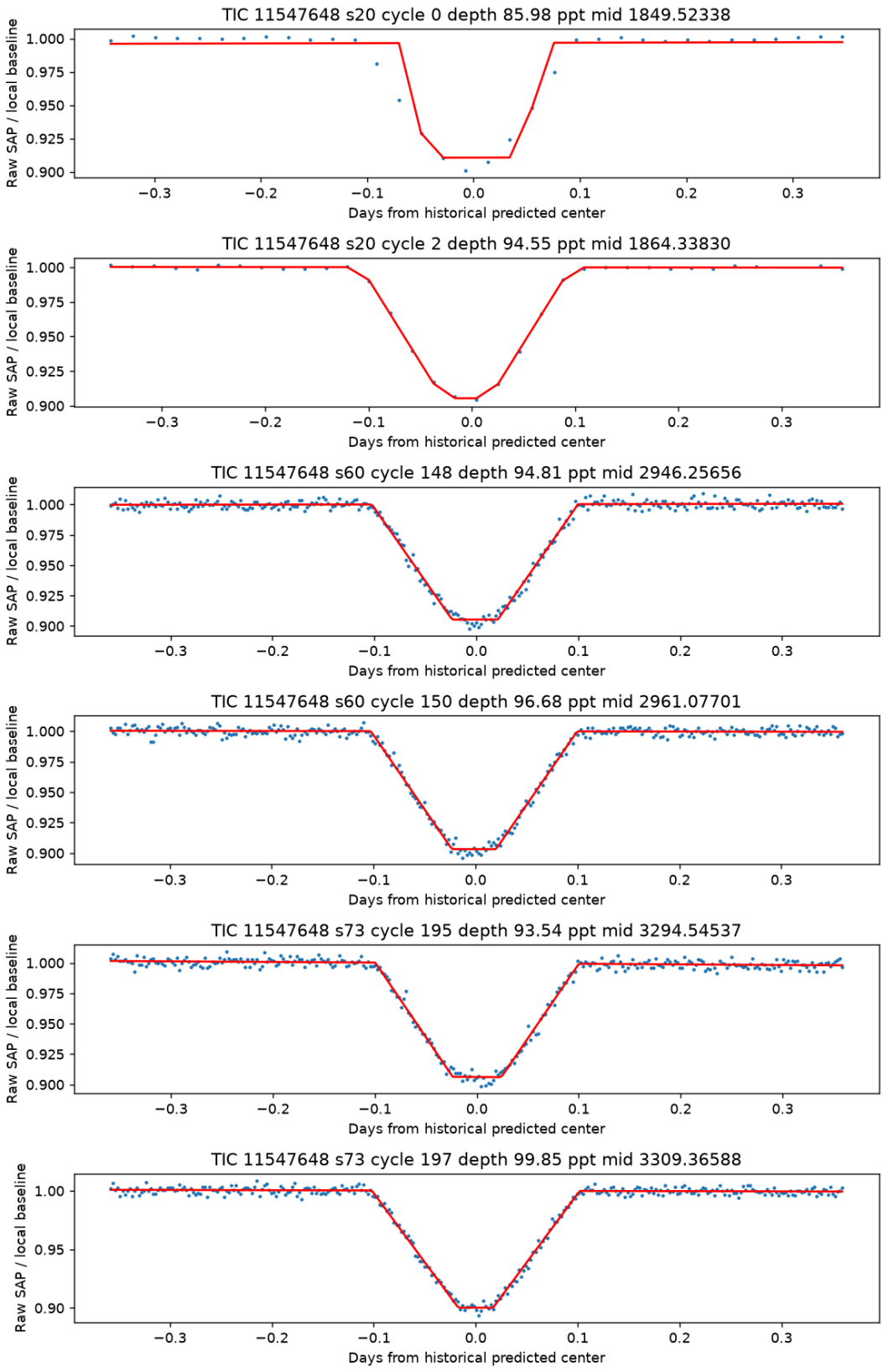
R662. TIC 11547648

physical_verdict: CONFIRMED

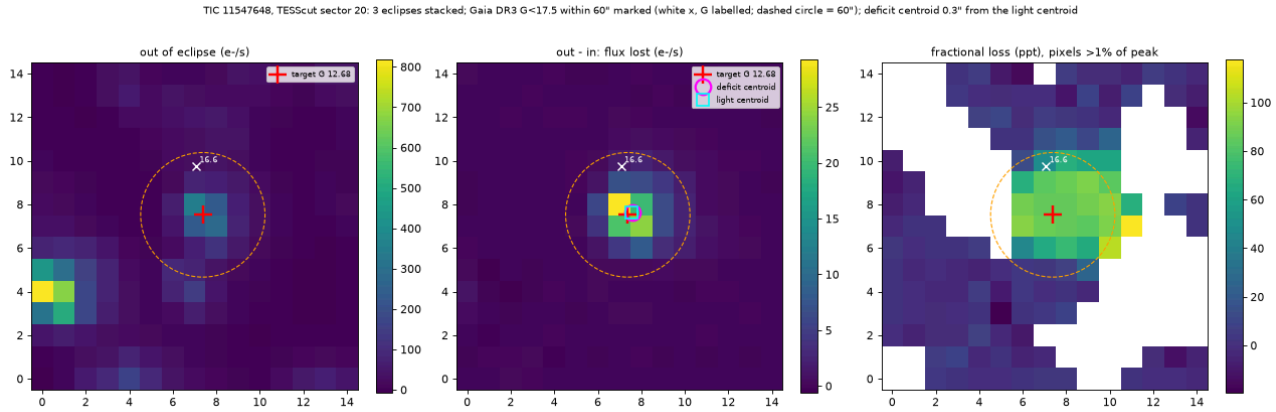
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional additional clean epochs and resolved photometry.



R662; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R662; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 11547648: CONFIRMED

Full bounded physical review: COMPLETE. Six repeated primary eclipses plus unequal secondary at phase0.500; no capable Gaia neighbor within60arcsec.

Raw QLP SAP_FLUX representative events and full-sector P/2P morphology were visually inspected. Quality-zero finite positive samples only; predictions guide windows and are never timings. 6 identifiable free-center timings define P=7.410395695858 +/- 0.000021364647d, reference BTJD=1849.517961025 +/- 0.003708065d. Epoch error meaning: Conservative linear timing-fit covariance under the stated cadence floor and scatter inflation; excludes uncalibrated shape and baseline systematics. Epoch kind: Deeper primary established by unequal observed secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced chi square=0.09757598052374179; zero degrees of freedom is null. All errors retain their named meanings; no calibrated coverage is claimed. Selected same-kind recurring eclipse family; parity and P/2 coverage checked.

Depth=94.677778ppt is the median realized central trapezoid loss d*min(1,w/(2*k)); extrapolated amplitude and ingress remain separately in measurements.json. Measured secondary phase 0.50001214, formal phase error 0.00011205012, realized depth 29.74132ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Secondary raw pooled windows, actual sectors/cycles and fitted phase are retained in secondary-windows.csv and ../secondary-phases.json. No exhaustive blind alias search asserted. ../alias-audit.json and ../half-period-coverage.json retain recurrence/parity and sampled opposite-phase evidence. Flagged coverage is never a clean null.

Fresh TESScut difference image sector20 stacks3 core/bilateral-sideband events. Target pixel= [7.379420324082043, 7.521193501031151], deficit=[7.581646557330426, 7.625706007137953]; descriptive offset=0.22764pixel, field PSF residual=0.40547. No arbitrary centroid cut or calibrated localization error is imposed. Every capable neighbor below was inspected at its actual projected pixels; close overlapping neighbors retain probable ownership. Every capable neighbor is assessed above and below; close overlapping cases retain unresolved attribution. All Gaia neighbors are retained in pixel JSON.

Source	Separation arcsec	G	Required intrinsic loss proxy	Pixel coordinate	Diluted pixel loss
None					

G-band pairwise brightness screen assumes approximate passband/aperture response; intrinsic required eclipse cannot be directly compared with diluted pixel loss. Unresolved unlisted companions are not excluded.

Alternative reduction: independently extracted background-subtracted TESScut1x1/3x3/5x5 depths=85.77257/86.04353/81.48312ppt; QLP median=94.67778ppt. Pixel sector20 shares the same TESS photons with retained QLP and checks extraction robustness. Independent processing is not an independent instrument. Different apertures respond differently to dilution/background.

Gaia source1004377178754043904: period=7.410293063+/-0.000985351d; e=0.0077322; omega=234.0733430deg; periastron BTJD=392.63288680. Closest conjunction offset=0.0122159d (0.0016485phase), source-specific full-covariance linear error=0.1612434d. Correlation order P,gamma,K,e,omega,Tp with subindices0,3,4,5. Linear propagation is descriptive, not a Gaussian veto or ownership proof. Both conjunctions retained in ../gaia-geometry.json. Mass function=0.09674742solar, assumed M1=1, minimum M2=0.637871; approximate central duration=7.3952h versus observed=4.8578h. TIC radius/main-sequence companion scaling, edge-on conjunction and stellar evolution are assumptions; this is a plausibility check, not a fitted physical solution.

Sector	Same-kind cycle	Measured BTJD	Formal center error d	Conservative input error d	Realized depth ppt
20	0	1849.523376562	0.000287779272	0.0104165117	85.98434
20	2	1864.338296631	0.000391903483	0.0104165117	94.54878
60	148	2946.256564641	0.000308110837	0.0011573909	94.80677
60	150	2961.077005784	0.000329405784	0.0011573909	96.68156
73	195	3294.545365419	0.000366988028	0.00115742205	93.54370
73	197	3309.365877176	0.000317593186	0.00115742205	99.84711

Catalogue novelty is separate from this physical verdict. Shared coverage matrix and source identity checks are authoritative for catalogue recovery; period agreement alone cannot establish attribution.

Stable review key: R662.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s20.json - historical/diff_s20.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

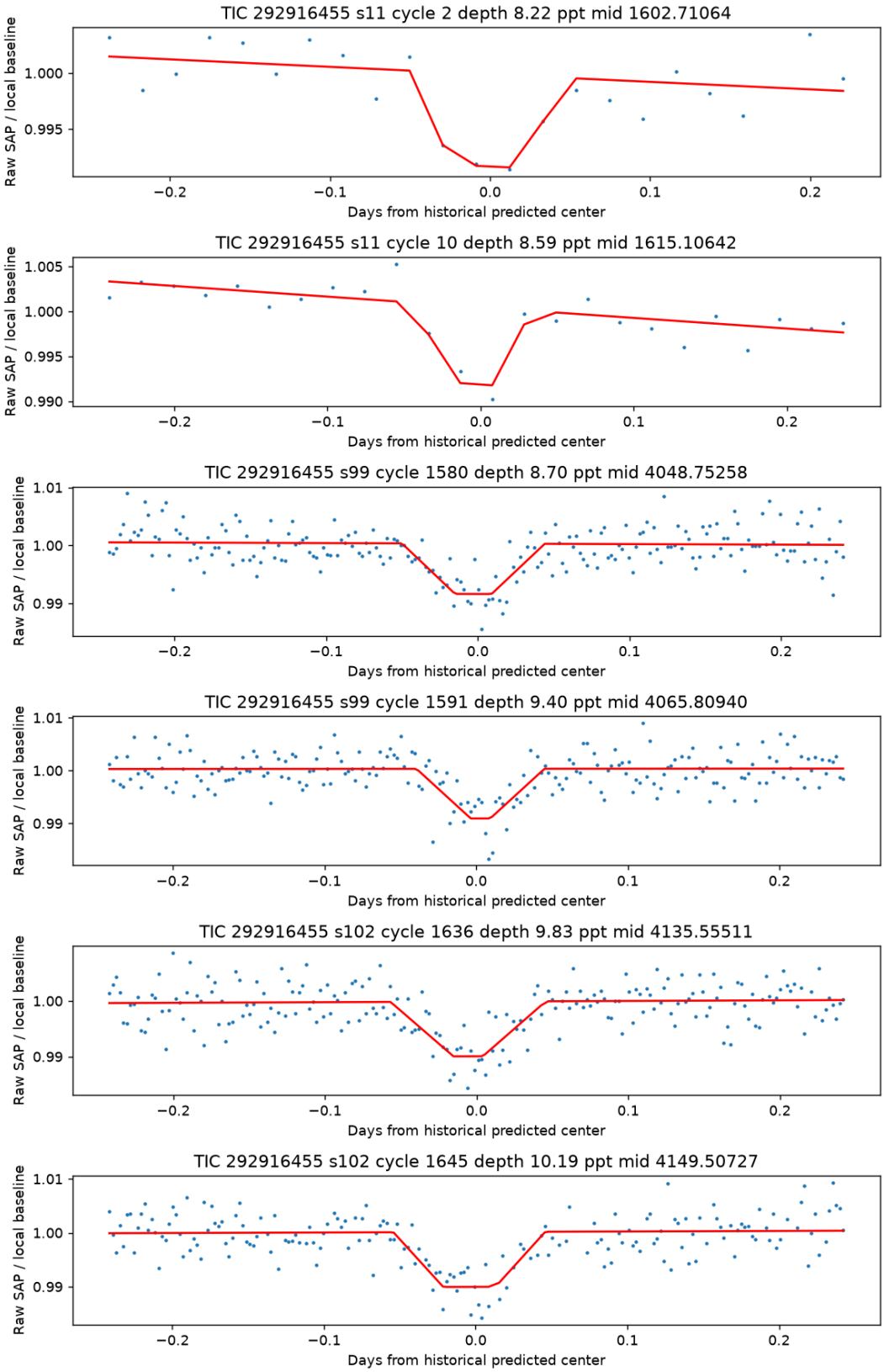
R663. TIC 292916455

physical_verdict: PROBABLE

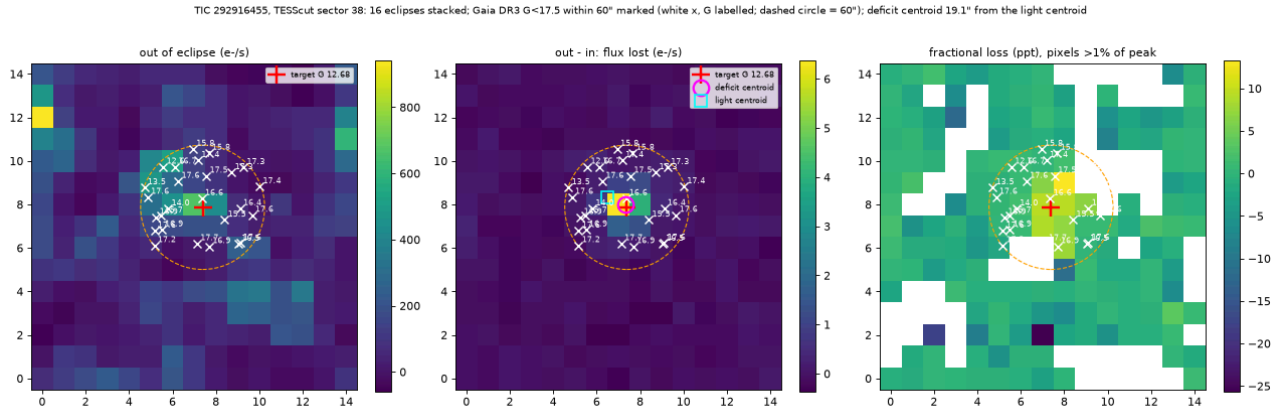
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry and alternating-family or secondary timing to decide orbital P versus2P.



R663; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R663; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 292916455: PROBABLE

Full bounded physical review: COMPLETE. Thirty-two usable shallow eclipse timings establish photometric recurrence1.550088935d. The Gaia/inventory orbit is3.10018d, twice that recurrence. Alternating depths do not securely distinguish the families; the orbital P versus2P identity remains unresolved. RESULTS period is the measured eclipse recurrence, not a uniquely established orbital period. Real repeated eclipse morphology is secure; attribution is not: capable8.52arcsec star shares target footprint, with25 other capable stars also inspected. A probable eclipsing source is retained; no claim that Gaia orbit is proven by period matching.

Raw QLP SAP_FLUX representative events and full-sector P/2P morphology were visually inspected. Quality-zero finite positive samples only; predictions guide windows and are never timings. 32 identifiable free-center timings define P=1.550088934745 +/- 0.000002814218d, reference BTJD=1602.713142667 +/- 0.004496996d. Epoch error meaning: Conservative linear timing-fit covariance under the stated cadence floor and scatter inflation; excludes uncalibrated shape and baseline systematics. Epoch kind: Selected repeating eclipse; deeper/secondary identity not established. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced chi square=1.8800355904714408; zero degrees of freedom is null. All errors retain their named meanings; no calibrated coverage is claimed. Selected same-kind recurring eclipse family; parity and P/2 coverage checked.

Depth=9.587002ppt is the median realized central trapezoid loss d*min(1,w/(2*k)); extrapolated amplitude and ingress remain separately in measurements.json. No secure secondary; exploratory fit is not interpreted as an eclipse timing or counterpart. Secondary raw pooled windows, actual sectors/cycles and fitted phase are retained in secondary-windows.csv and ../secondary-phases.json. No exhaustive blind alias search asserted. ../alias-audit.json and ../half-period-coverage.json retain recurrence/parity and sampled opposite-phase evidence. Flagged coverage is never a clean null.

Fresh TESScut difference image sector38 stacks16 core/bilateral-sideband events. Target pixel= [7.3884620122908995, 7.868392467925553], deficit=[7.342944714999979, 7.991516780359548]; descriptive offset=0.13127pixel, field PSF residual=0.83945. No arbitrary centroid cut or calibrated localization error is imposed. Every capable neighbor below was inspected at its actual projected pixels;

close overlapping neighbors retain probable ownership. Every capable neighbor is assessed above and below; close overlapping cases retain unresolved attribution. All Gaia neighbors are retained in pixel JSON.

Source	Separation arcsec	G	Required intrinsic loss proxy	Pixel coordinate	Diluted pixel loss
589179765808777 9200	8.521079836179 915	16.6159 27	0.3693999716250671	[7.349447047153012, 8.271510296420288]	0.009459439143029 382
589179765808778 7264	24.62429567363 676	15.4610 27	0.1337856343269082	[8.399908820977336, 7.283805935030285]	0.006931825507452 607
589179765809165 8112	30.51956918788 707	17.4668 6	0.7974487471476704	[7.561344522917688, 9.310726169266006]	0.0133198141674156 45
589179748628908 3776	33.38131190092 955	13.9503 61	0.04047931400358709	[5.801010726881065, 7.79927575530969]	0.001776748719230 3852
589179744759494 8992	34.68905493310 017	17.5586 38	0.8669430458201962	[6.261865563445808, 9.066715929748403]	0.000786542369098 9098
589179769244758 3616	35.47958835135 557	17.6781 56	0.9667114589808344	[7.131809671418663, 6.198891300402868]	0.047366602067640 44
589179772680726 3744	36.2220069576 83774	16.3867 95	0.30094303387214494	[9.11083355517866, 7.812000690228894]	0.006635996777605 405
589179769244758 3872	38.4360048220 8318	16.9255 28	0.48812586531485375	[7.689030666985763, 6.067253093552419]	0.007547518400357 181
589179748628908 5056	40.80481463547 042	15.7194 54	0.16716211001277592	[5.485254812353377, 7.470698865457626]	1.602106008450408 8e-05
589179728013065 6128	43.62386121925 3	17.3286 72	0.7032919843134171	[8.724592734522895, 9.465206666112678]	-0.22305532147223 245
589179748628908 8256	45.05204276935 498	16.8718 5	0.4650425678126111	[5.501090287721251, 6.8394191531417245]	0.000966498584109 4763
589179765375337 2288	45.58452045662 806	17.4147 95	0.7605594672608712	[7.179346371380211, 10.024877443637328]	0.000279433236806 4977
589179748628908 4160	45.8366364922 2092	16.9169 24	0.48434863356198743	[5.254249657556505, 7.403855075252121]	1.602106008450408 8e-05
589179745192933 6448	47.40811838913 6554	16.7391 57	0.4126461788803146	[6.121776387275701, 9.725691771097305]	0.000661338623179 9957
589179772680726 6304	48.36138950757 66	17.6145 57	0.9122565960080612	[9.658976285198317, 7.5080642199723115]	-0.03079932988058 981
589179772247290 4192	49.73919098818 9	16.4635 24	0.32227816921129643	[9.0351485392404, 6.18012966272583]	5.432074229150538 4e-05
589179772680726 9504	50.07114813636 261	17.6329 6	0.9276870576277796	[9.101496188416235, 6.224605176977338]	5.432074229150538 4e-05
589179748628908 7232	51.86554581506 2696	17.5834 1	0.8867292180446442	[5.167116352286893, 6.778352787262124]	1.602106008450408 8e-05
589179728013064 7424	52.68044277513 387	15.8025 94	0.17970239527719992	[7.671347651831818, 10.359741647464707]	0.001814104936702 4786
589179748628907 8400	54.15322984413 134	17.6228 5	0.9191777070146571	[4.85700214587654, 8.333570685584078]	0.000407037844236 7661
589179745192933 3248	54.7608678030 5234	12.6262 72	0.018711622467473356	[5.571228443256415, 9.72290488708596]	0.000661338623179 9957
589179727579626 4960	54.95177252351 8486	17.2915 3	0.6799623164587588	[9.26748117980037, 9.6985703977612]	-0.16706269722352 882
589179707397221 2096	57.0832809186 8223	15.7519 26	0.17194601910322377	[6.94636749570872, 10.544097375018747]	-0.00058433072187 70078
589179734885013 9904	58.49840264631 041	17.4278 66	0.769654941142567	[10.01241805660705, 8.815089818189794]	-0.00842965665978 456
589179748628909 0432	59.30069542742 812	17.1842 06	0.6168662123243962	[5.168819159092701, 6.111959404622051]	-0.002182231613318 997
589179748628907 4688	59.61479480732 846	13.4715 22	0.029462353212715595	[4.704518086846858, 8.768884351261445]	-0.000304611019976 79934

G-band pairwise brightness screen assumes approximate passband/aperture response; intrinsic required eclipse cannot be directly compared with diluted pixel loss. Unresolved unlisted companions are not excluded.

Alternative reduction: independently extracted background-subtracted TESScut1x1/3x3/5x5 depths=9.45944/6.36155/3.71157ppt; QLP median=9.58700ppt. Pixel sector38 is a distinct observing occasion on the same TESS instrument; no same-sector QLP comparison claimed. Independent processing is not an independent instrument. Different apertures respond differently to dilution/background.

Gaia source5891797658087780608: period=3.100333446+/-0.000177394d; e=0.0456532; omega=262.2374840deg; periastron BTJD=390.47435117. Closest conjunction offset=-0.0525522d (-0.0169505phase), source-specific full-covariance linear error=0.1048483d. Correlation order P,gamma,K,e,omega,Tp with subindices0,3,4,5. Linear propagation is descriptive, not a Gaussian veto or ownership proof. Both conjunctions retained in ../gaia-geometry.json. Mass function=0.2033922solar, assumed M1=1.38, minimum M2=1.06828; approximate central duration=6.1592h versus observed=2.1848h. TIC radius/main-sequence companion scaling, edge-on conjunction and stellar evolution are assumptions; this is a plausibility check, not a fitted physical solution.

Sector	Same-kind cycle	Measured BTJD	Formal center error d	Conservative input error d	Realized depth ppt
11	2	1602.710640416	0.00464104589	0.0104168357	8.22021
11	3	1604.258524126	0.00170723971	0.0104168357	6.67225
11	4	1605.811996866	0.00279330239	0.0104168357	8.05098
11	5	1607.360376832	0.00273973086	0.0104168357	9.53017
11	9	1613.564565929	0.00901420361	0.0104168357	11.58195
11	10	1615.106419944	0.00524529672	0.0104168357	8.59326
11	12	1618.212361542	0.00179081648	0.0104168357	11.13293
11	13	1619.760631537	0.00376803808	0.0104168357	6.30572
11	14	1621.316251008	0.00222805647	0.0104168357	7.06387
11	15	1622.861674733	0.00130515668	0.0104168357	8.30723
99	1579	4047.207924302	0.00224912004	0.00224912004	9.81647
99	1580	4048.752583903	0.00217478371	0.00217478371	8.69960
99	1581	4050.306863867	0.00188313046	0.00188313046	10.12976
99	1582	4051.857230506	0.00154432075	0.00154432075	9.54894
99	1583	4053.404305836	0.00167774816	0.00167774816	10.79162
99	1584	4054.953271267	0.0017621714	0.0017621714	8.64811
99	1590	4064.255578697	0.00229669616	0.00229669616	7.95074
99	1591	4065.809404976	0.00223267756	0.00223267756	9.40133
99	1593	4068.909790863	0.00162492297	0.00162492297	10.27992
99	1594	4070.456703333	0.00181959813	0.00181959813	10.46479
102	1632	4129.356729544	0.00183673218	0.00183673218	8.68680
102	1633	4130.906512812	0.00148620516	0.00148620516	9.62506
102	1634	4132.456636580	0.0017294232	0.0017294232	10.31159
102	1635	4134.008708016	0.00204765285	0.00204765285	9.41929
102	1636	4135.555106388	0.00257641503	0.00257641503	9.82860
102	1637	4137.104111558	0.00266122485	0.00266122485	9.40040
102	1640	4141.757296409	0.00170066567	0.00170066567	12.20973
102	1641	4143.304696642	0.0019872046	0.0019872046	12.26186
102	1642	4144.861414758	0.00269865132	0.00269865132	11.51119
102	1643	4146.403137495	0.00240733908	0.00240733908	10.39733
102	1644	4147.954635287	0.00187011579	0.00187011579	12.10556

Sector	Same-kind cycle	Measured BTJD	Formal center error d	Conservative input error d	Realized depth ppt
102	1645	4149.507274722	0.00241926826	0.00241926826	10.18595

Catalogue novelty is separate from this physical verdict. Shared coverage matrix and source identity checks are authoritative for catalogue recovery; period agreement alone cannot establish attribution.

Stable review key: R663.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s38.json - historical/diff_s38.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

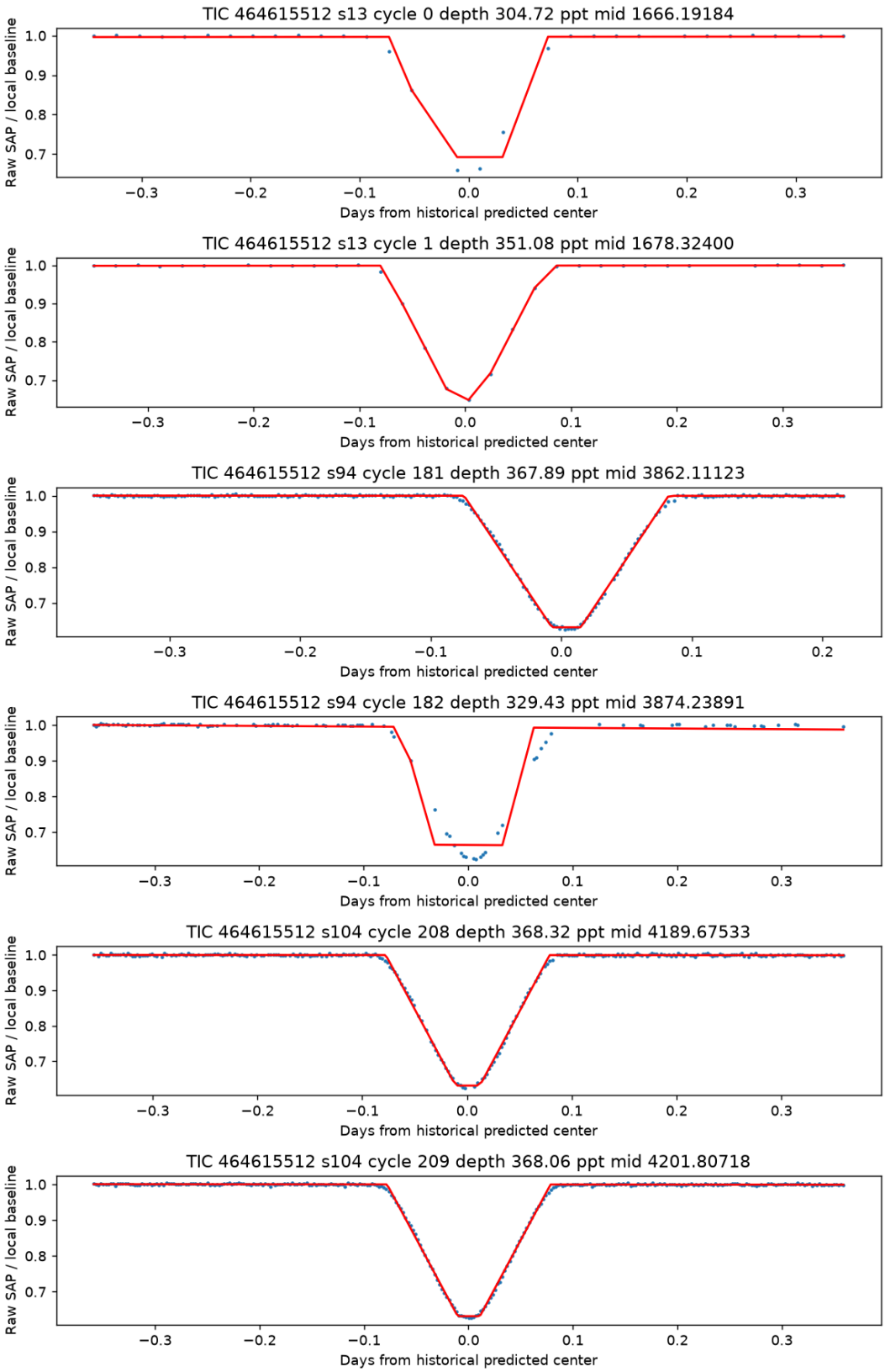
R664. TIC 464615512

physical_verdict: CONFIRMED

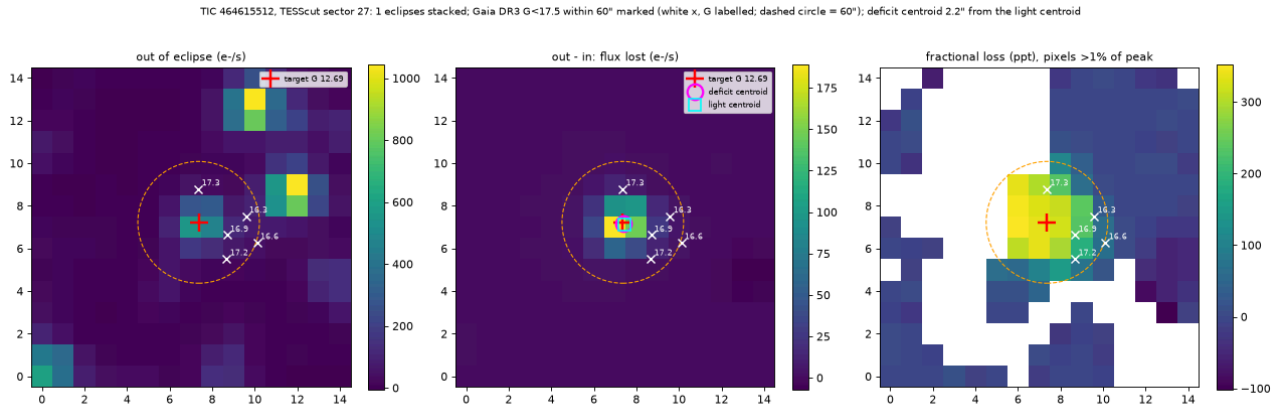
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional additional clean epochs and resolved photometry.



R664; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R664; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 464615512: CONFIRMED

Full bounded physical review: COMPLETE. Six repeated very deep primary eclipses and unequal secondary near0.484; no capable Gaia neighbor within60arcsec. Separate broad sector-specific baseline features are not interpreted as additional eclipses.

Raw QLP SAP_FLUX representative events and full-sector P/2P morphology were visually inspected. Quality-zero finite positive samples only; predictions guide windows and are never timings. 6 identifiable free-center timings define P=12.132109767254 +/- 0.000040774170d, reference BTJD=1666.196728878 +/- 0.007945530d. Epoch error meaning: Conservative linear timing-fit covariance under the stated cadence floor and scatter inflation; excludes uncalibrated shape and baseline systematics. Epoch kind: Deeper primary established by unequal observed secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced chi square=2.0581650195498122; zero degrees of freedom is null. All errors retain their named meanings; no calibrated coverage is claimed. Selected same-kind recurring eclipse family; parity and P/2 coverage checked.

Depth=359.483490ppt is the median realized central trapezoid loss d*min(1,w/(2*k)); extrapolated amplitude and ingress remain separately in measurements.json. Measured secondary phase 0.48380129, formal phase error 9.2680635e-06, realized depth 103.09249ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Secondary raw pooled windows, actual sectors/cycles and fitted phase are retained in secondary-windows.csv and ../secondary-phases.json. No exhaustive blind alias search asserted. ../alias-audit.json and ../half-period-coverage.json retain recurrence/parity and sampled opposite-phase evidence. Flagged coverage is never a clean null.

Fresh TESScut difference image sector27 stacks1 core/bilateral-sideband events. Target pixel= [7.360916693333532, 7.227960131360918], deficit=[7.3710405105484265, 7.182247867165817]; descriptive offset=0.04682pixel, field PSF residual=0.67489. No arbitrary centroid cut or calibrated localization error is imposed. Every capable neighbor below was inspected at its actual projected pixels;

close overlapping neighbors retain probable ownership. Every capable neighbor is assessed above and below; close overlapping cases retain unresolved attribution. All Gaia neighbors are retained in pixel JSON.

Source	Separation arcsec	G	Required intrinsic loss proxy	Pixel coordinate	Diluted pixel loss
None					

G-band pairwise brightness screen assumes approximate passband/aperture response; intrinsic required eclipse cannot be directly compared with diluted pixel loss. Unresolved unlisted companions are not excluded.

Alternative reduction: independently extracted background-subtracted TESScut1x1/3x3/5x5 depths=335.46678/326.58787/302.16880ppt; QLP median=359.48349ppt. Pixel sector27 is a distinct observing occasion on the same TESS instrument; no same-sector QLP comparison claimed. Independent processing is not an independent instrument. Different apertures respond differently to dilution/background.

Gaia source6632294477079223424: period=12.130264395+/-0.002677318d; e=0.0431035; omega=100.9771082deg; periastron BTJD=392.75081417. Closest conjunction offset=0.1073164d (0.0088470phase), source-specific full-covariance linear error=0.2689574d. Correlation order P,gamma,K,e,omega,Tp with subindices0,3,4,5. Linear propagation is descriptive, not a Gaussian veto or ownership proof. Both conjunctions retained in ../gaia-geometry.json. Mass function=0.06168672solar, assumed M1=0.7, minimum M2=0.428211; approximate central duration=5.0735h versus observed=3.7305h. TIC radius/main-sequence companion scaling, edge-on conjunction and stellar evolution are assumptions; this is a plausibility check, not a fitted physical solution.

Sector	Same-kind cycle	Measured BTJD	Formal center error d	Conservative input error d	Realized depth ppt
13	0	1666.191836129	6.5269142e-05	0.0104166054	304.72105
13	1	1678.324004395	5.7909108e-05	0.0104166054	351.07842
94	181	3862.111229584	5.37777875e-05	0.00115739457	367.88855
94	182	3874.238911981	0.000119659522	0.00115739457	329.43308
104	208	4189.675331762	6.16962419e-05	0.00115740685	368.32298
104	209	4201.807180712	4.84459908e-05	0.00115740685	368.06487

Catalogue novelty is separate from this physical verdict. Shared coverage matrix and source identity checks are authoritative for catalogue recovery; period agreement alone cannot establish attribution.

Stable review key: R664.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s27.json - historical/diff_s27.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

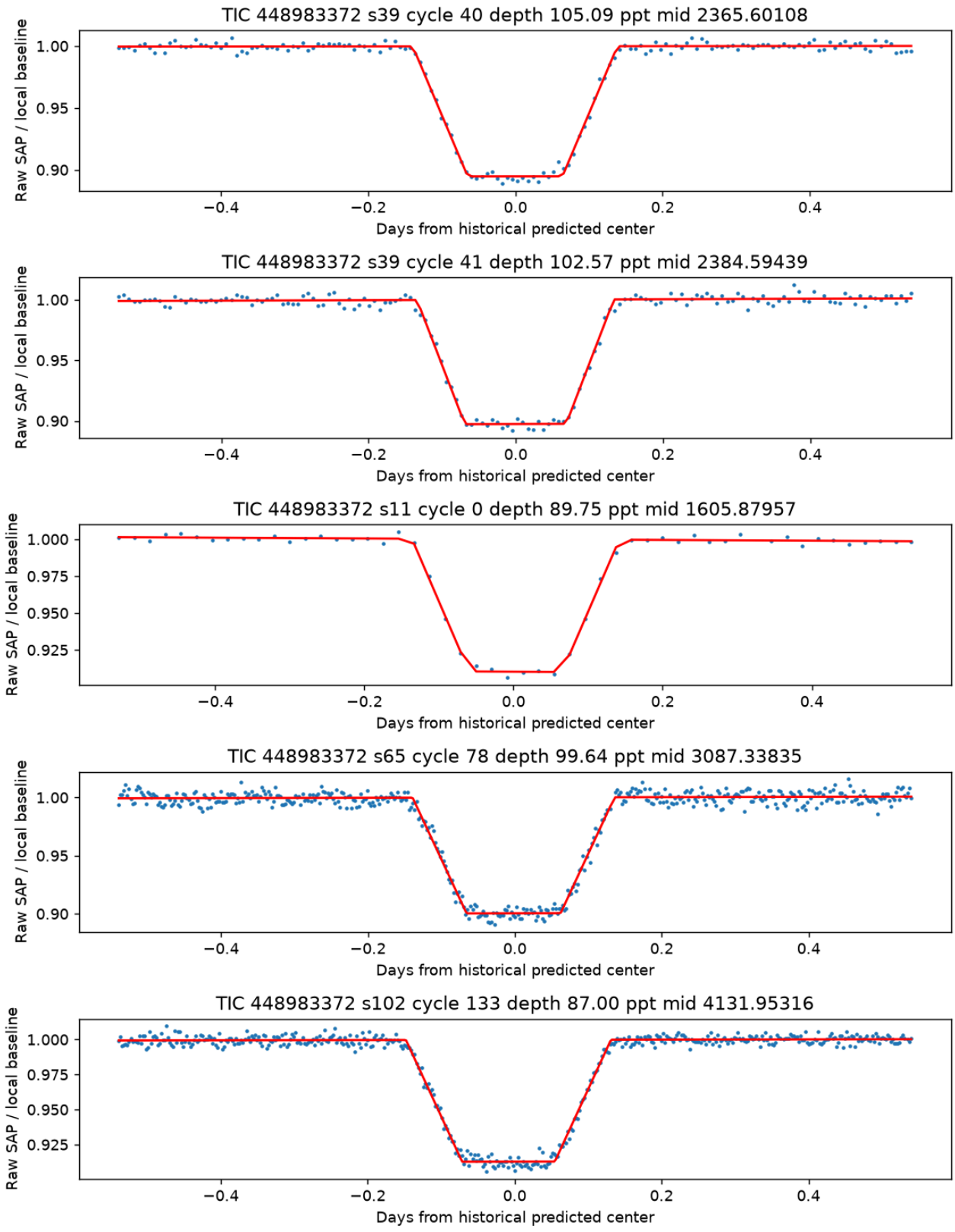
R665. TIC 448983372

physical_verdict: PROBABLE

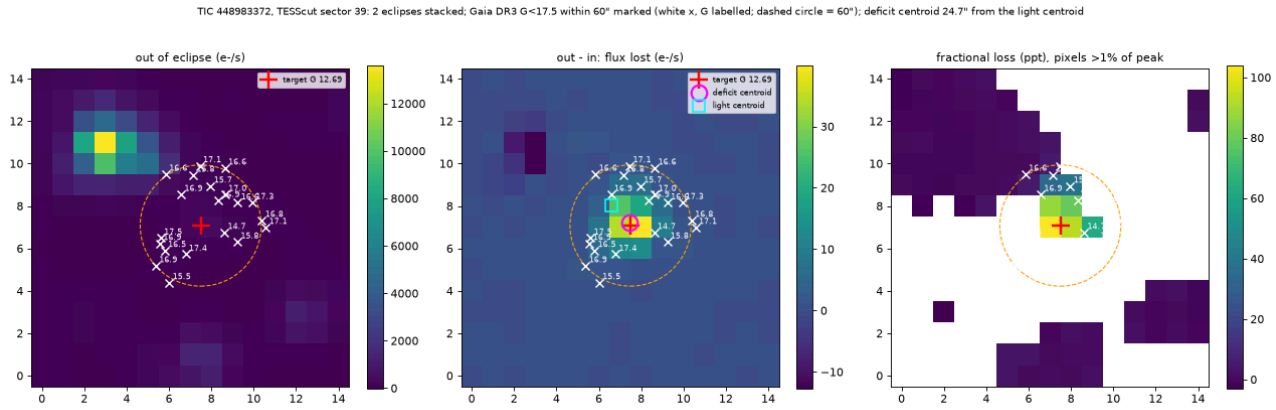
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or multi-source PRF attribution.



R665; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R665; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 448983372: PROBABLE

Full bounded physical review: COMPLETE. Initially three widely separated primary timings. Fresh current availability11/12/38/39/65/99/100/101/102 prompted sector12/39 retrieval. Five primary timings now include consecutive sector39 cycles40/41 and establish recurrence. No secure secondary. Capable23.09arcsec source is only1.16pixel from target and overlaps PSF: ownership unresolved.

Raw QLP SAP_FLUX representative events and full-sector P/2P morphology were visually inspected. Quality-zero finite positive samples only; predictions guide windows and are never timings. 5 identifiable free-center timings define P=18.993015880591 +/- 0.000023251165d, reference BTJD=1605.882366978 +/- 0.002416546d. Epoch error meaning: Conservative linear timing-fit covariance under the stated cadence floor and scatter inflation; excludes uncalibrated shape and baseline systematics. Epoch kind: Selected repeating eclipse; deeper/secondary identity not established. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced chi square=0.364266009555728; zero degrees of freedom is null. All errors retain their named meanings; no calibrated coverage is claimed. Selected same-kind recurring eclipse family; parity and P/2 coverage checked.

Depth=99.642992ppt is the median realized central trapezoid loss d*min(1,w/(2*k)); extrapolated amplitude and ingress remain separately in measurements.json. No secure secondary; exploratory fit is not interpreted as an eclipse timing or counterpart. Secondary raw pooled windows, actual sectors/cycles and fitted phase are retained in secondary-windows.csv and ../secondary-phases.json. No exhaustive blind alias search asserted. ../alias-audit.json and ../half-period-coverage.json retain recurrence/parity and sampled opposite-phase evidence. Flagged coverage is never a clean null.

Fresh TESScut difference image sector39 stacks2 core/bilateral-sideband events. Target pixel= [7.496403961205186, 7.0780306787617135], deficit=[7.445278525084913, 7.221088553311603]; descriptive offset=0.15192pixel, field PSF residual=0.98708. No arbitrary centroid cut or calibrated localization error is imposed. Every capable neighbor below was inspected at its actual projected pixels; close overlapping neighbors retain probable ownership. Every capable neighbor is assessed above and below; close overlapping cases retain unresolved attribution. All Gaia neighbors are retained in pixel JSON.

Source	Separation arcsec	G	Required intrinsic loss proxy	Pixel coordinate	Diluted pixel loss
584634927189612 3520	23.09248688210 1745	14.7244 65	0.7463154421420145	[8.615487634141065, 6.758402378836966]	0.0621952211550 1482

G-band pairwise brightness screen assumes approximate passband/aperture response; intrinsic required eclipse cannot be directly compared with diluted pixel loss. Unresolved unlisted companions are not excluded.

Alternative reduction: independently extracted background-subtracted TESScut1x1/3x3/5x5 depths=104.02325/91.00054/49.04578ppt; QLP median=99.64299ppt. Pixel sector39 shares the same TESS photons with retained QLP and checks extraction robustness. Independent processing is not an independent instrument. Different apertures respond differently to dilution/background.

Gaia source5846349276209786752: period=18.996836504+/-0.015214888d; e=0.2913209; omega=125.5892977deg; periastron BTJD=391.32114720. Closest conjunction offset=-0.2211902d (-0.0116435phase), source-specific full-covariance linear error=1.1817403d. Correlation order P,gamma,K,e,omega,Tp with subindices0,3,4,5. Linear propagation is descriptive, not a Gaussian veto or ownership proof. Both conjunctions retained in ../gaia-geometry.json. Mass function=0.06088851solar, assumed M1=1.18, minimum M2=0.571672; approximate central duration=7.6077h versus observed=6.6735h. TIC radius/main-sequence companion scaling, edge-on conjunction and stellar evolution are assumptions; this is a plausibility check, not a fitted physical solution.

Sector	Same-kind cycle	Measured BTJD	Formal center error d	Conservative input error d	Realized depth ppt
39	40	2365.601075117	0.000415285098	0.00347217417	105.09117
39	41	2384.594391083	0.00059890995	0.00347217417	102.56935
11	0	1605.879572985	0.000625296907	0.0104168543	89.74996
65	78	3087.338353556	0.00050656156	0.00115741921	99.64299
102	133	4131.953160621	0.000330703066	0.00115740684	87.00422

Catalogue novelty is separate from this physical verdict. Shared coverage matrix and source identity checks are authoritative for catalogue recovery; period agreement alone cannot establish attribution.

Stable review key: R665.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s39.json - historical/diff_s39.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

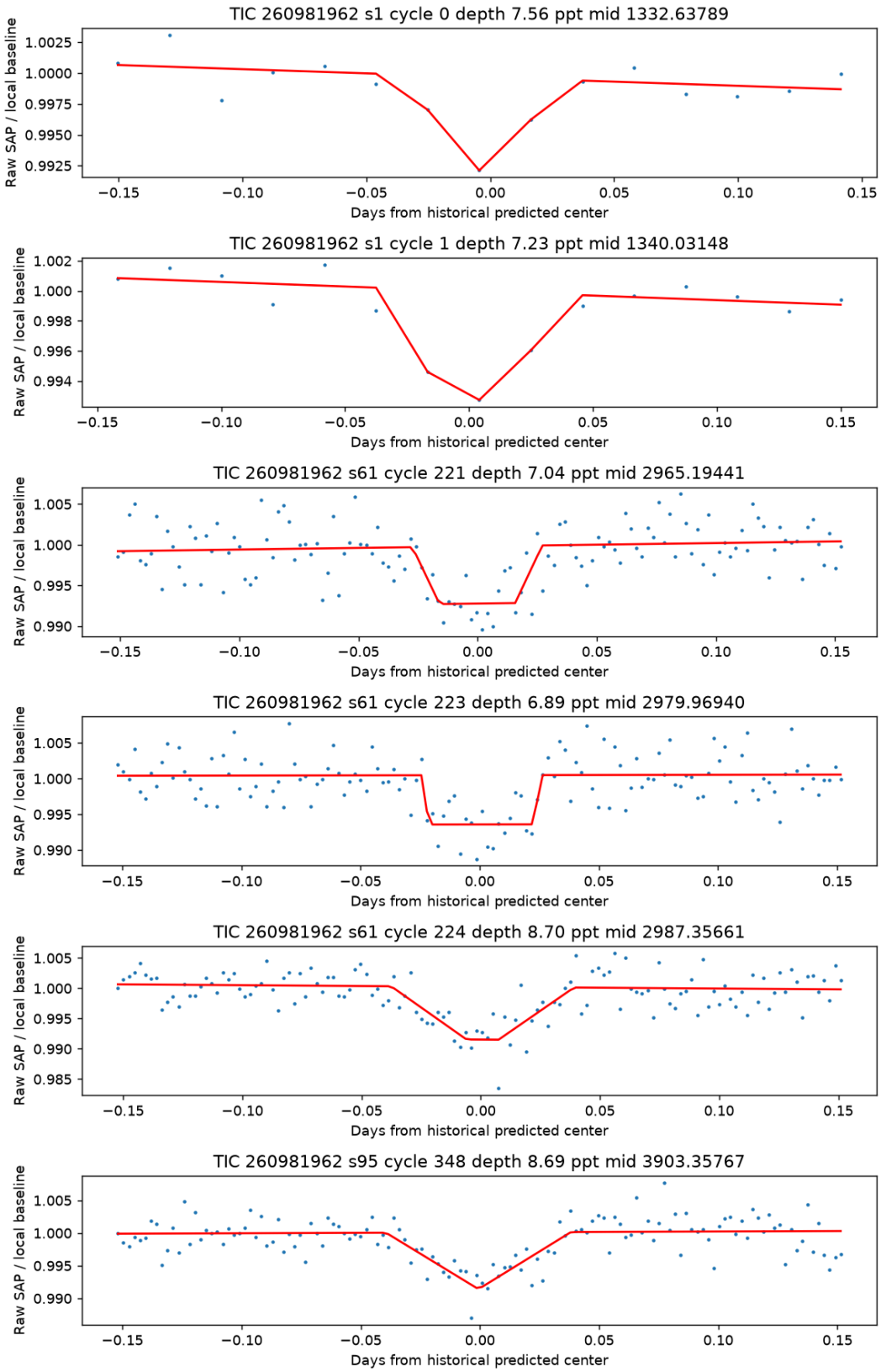
R666. TIC 260981962

physical_verdict: CONFIRMED

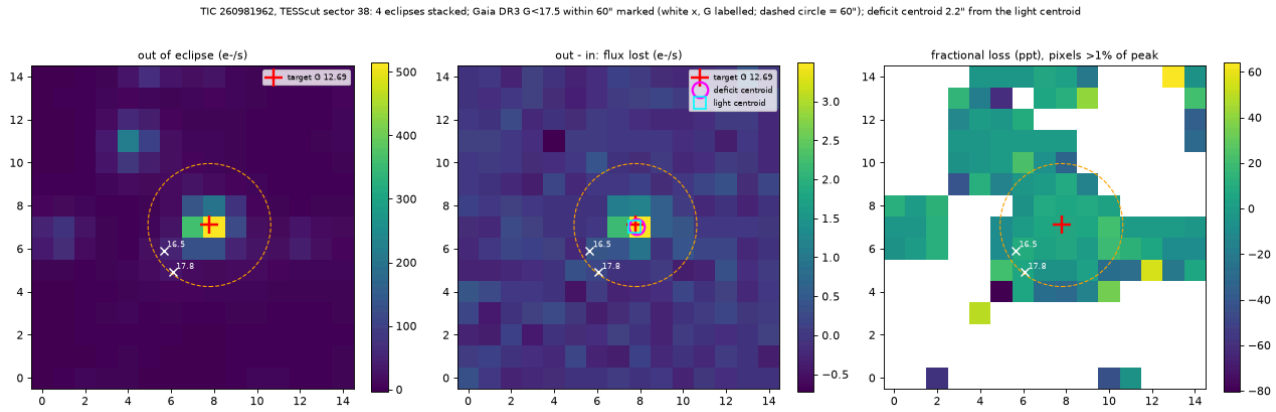
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional additional clean epochs and resolved photometry.



R666; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R666; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 260981962: CONFIRMED

Full bounded physical review: COMPLETE. Eight usable shallow narrow eclipses repeated across sectors1/61/95. Raw high-cadence scatter is large compared with7-9ppt events but local repeated morphology and independent aperture extractions support real eclipses. No secure secondary. Capable44.47/50.47arcsec sources are displaced from target-centered deficit.

Raw QLP SAP_FLUX representative events and full-sector P/2P morphology were visually inspected. Quality-zero finite positive samples only; predictions guide windows and are never timings. 8 identifiable free-center timings define P=7.387102407574 +/- 0.000010851804d, reference BTJD=1332.645628046 +/- 0.002832497d. Epoch error meaning: Conservative linear timing-fit covariance under the stated cadence floor and scatter inflation; excludes uncalibrated shape and baseline systematics. Epoch kind: Selected repeating eclipse; deeper/secondary identity not established. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced chi square=0.3948471319417442; zero degrees of freedom is null. All errors retain their named meanings; no calibrated coverage is claimed. Selected same-kind recurring eclipse family; parity and P/2 coverage checked.

Depth=7.915807ppt is the median realized central trapezoid loss d*min(1,w/(2*k)); extrapolated amplitude and ingress remain separately in measurements.json. No secure secondary; exploratory fit is not interpreted as an eclipse timing or counterpart. Secondary raw pooled windows, actual sectors/cycles and fitted phase are retained in secondary-windows.csv and ../secondary-phases.json. No exhaustive blind alias search asserted. ../alias-audit.json and ../half-period-coverage.json retain recurrence/parity and sampled opposite-phase evidence. Flagged coverage is never a clean null.

Fresh TESScut difference image sector38 stacks4 core/bilateral-sideband events. Target pixel= [7.7847435249161165, 7.098428891034963], deficit=[7.846229965597483, 7.043182093459804]; descriptive offset=0.08266pixel, field PSF residual=0.56299. No arbitrary centroid cut or calibrated localization error is imposed. Every capable neighbor below was inspected at its actual projected pixels; close overlapping neighbors retain probable ownership. Every capable neighbor is assessed above and below; close overlapping cases retain unresolved attribution. All Gaia neighbors are retained in pixel JSON.

Source	Separation arcsec	G	Required intrinsic loss proxy	Pixel coordinate	Diluted pixel loss
462110979944720 8832	44.47211729161 338	16.4707 34	0.26448669736569935	[5.654955768192173, 5.924148248947575]	0.00857180917845 0487
462110979515141 0432	50.4682448445 5172	17.8049	0.8846617837185289	[6.077502007046416, 4.910354800637604]	0.00618916513861 5587

G-band pairwise brightness screen assumes approximate passband/aperture response; intrinsic required eclipse cannot be directly compared with diluted pixel loss. Unresolved unlisted companions are not excluded.

Alternative reduction: independently extracted background-subtracted TESScut1x1/3x3/5x5 depths=6.79650/6.09570/5.64224ppt; QLP median=7.91581ppt. Pixel sector38 is a distinct observing occasion on the same TESS instrument; no same-sector QLP comparison claimed. Independent processing is not an independent instrument. Different apertures respond differently to dilution/background.

Gaia source4621203670252233728: period=7.388890918+/-0.001422523d; e=0.1791412; omega=31.6860399deg; periastron BTJD=385.79622585. Closest conjunction offset=0.2079179d (0.0281393phase), source-specific full-covariance linear error=0.2283869d. Correlation order P,gamma,K,e,omega,Tp with subindices0,3,4,5. Linear propagation is descriptive, not a Gaussian veto or ownership proof. Both conjunctions retained in ../gaia-geometry.json. Mass function=0.009482105solar, assumed M1=1.07, minimum M2=0.255386; approximate central duration=4.5536h versus observed=1.4458h. TIC radius/main-sequence companion scaling, edge-on conjunction and stellar evolution are assumptions; this is a plausibility check, not a fitted physical solution.

Sector	Same-kind cycle	Measured BTJD	Formal center error d	Conservative input error d	Realized depth ppt
1	0	1332.637888319	0.00241682793	0.0104164965	7.55885
1	1	1340.031477588	0.00168424478	0.0104164965	7.22965
61	221	2965.194409237	0.00158609143	0.00158609143	7.03874
61	222	2972.584347167	0.00180065299	0.00180065299	8.29909
61	223	2979.969396080	0.000907241812	0.00115742526	6.89334
61	224	2987.356612868	0.00184977961	0.00184977961	8.70143
95	346	3888.582197112	0.00175059031	0.00175059031	8.27276
95	348	3903.357670686	0.00177128769	0.00177128769	8.68873

Catalogue novelty is separate from this physical verdict. Shared coverage matrix and source identity checks are authoritative for catalogue recovery; period agreement alone cannot establish attribution.

Stable review key: R666.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s38.json - historical/diff_s38.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

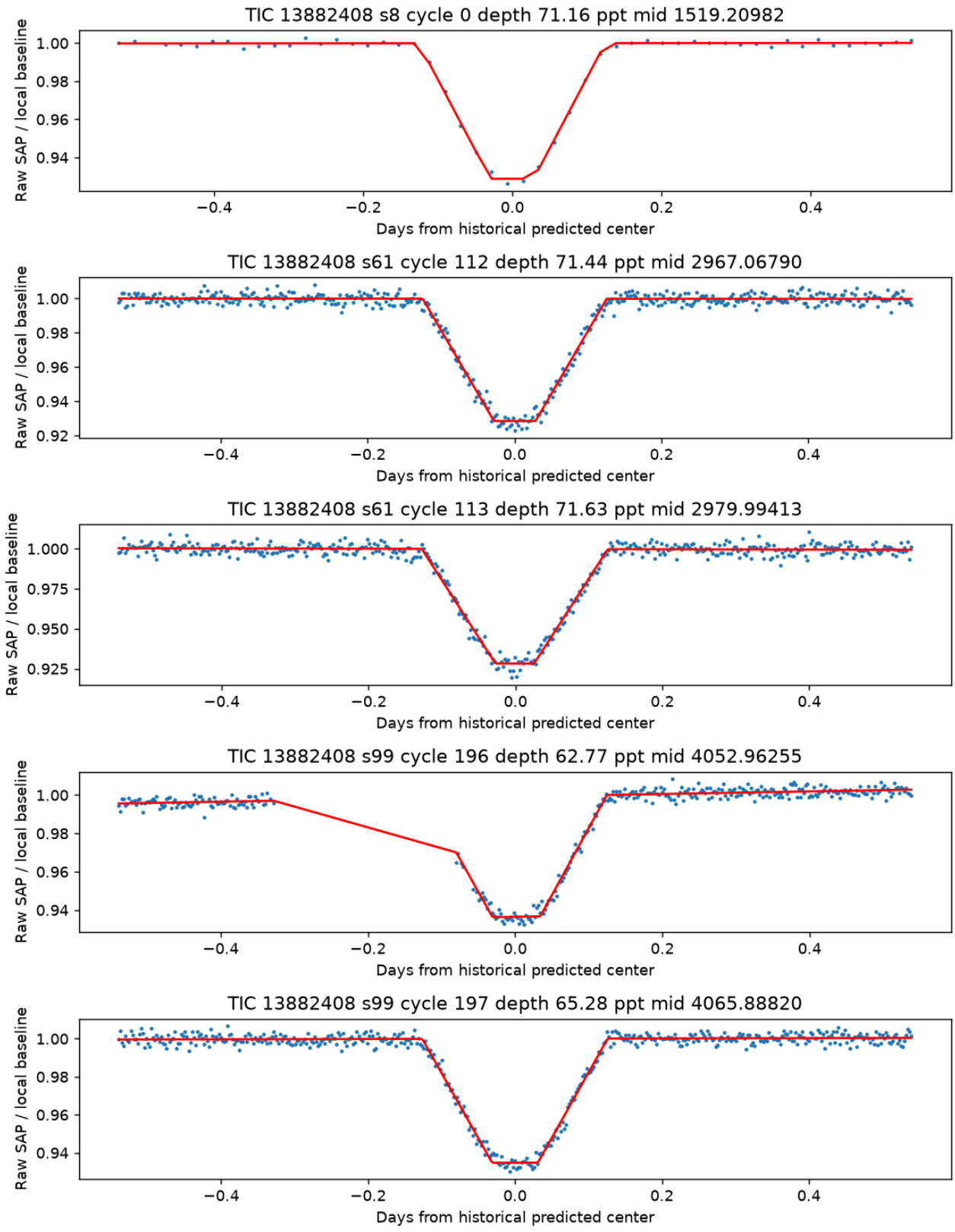
R667. TIC 13882408

physical_verdict: CONFIRMED

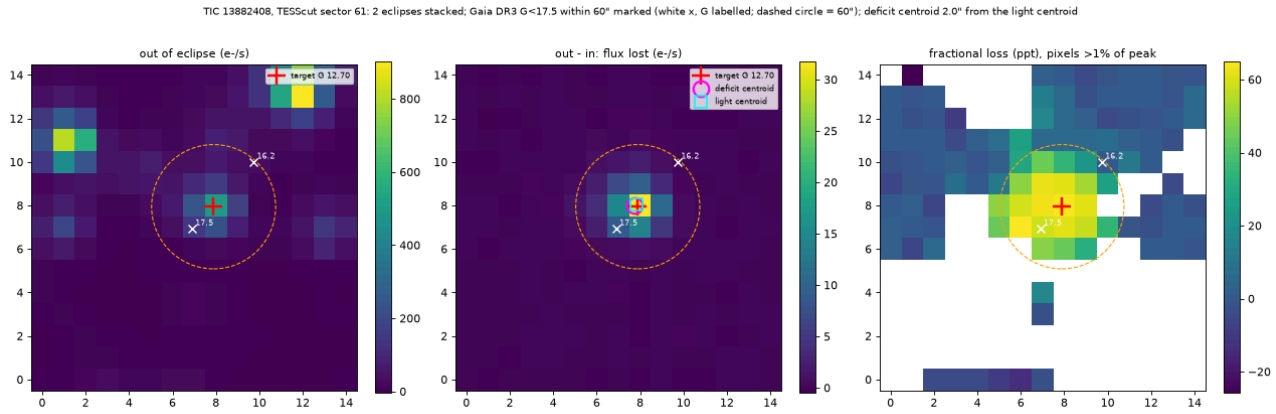
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional additional clean epochs and resolved photometry.



R667; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R667; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 13882408: CONFIRMED

Full bounded physical review: COMPLETE. Five repeated narrow deep primary eclipses and unequal secondary near0.486; no capable Gaia neighbor within60arcsec.

Raw QLP SAP_FLUX representative events and full-sector P/2P morphology were visually inspected. Quality-zero finite positive samples only; predictions guide windows and are never timings. 5 identifiable free-center timings define P=12.927313217175 +/- 0.000013499816d, reference BTJD=1519.208309786 +/- 0.002158091d. Epoch error meaning: Conservative linear timing-fit covariance under the stated cadence floor and scatter inflation; excludes uncalibrated shape and baseline systematics. Epoch kind: Deeper primary established by unequal observed secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced chi square=0.4986003655328546; zero degrees of freedom is null. All errors retain their named meanings; no calibrated coverage is claimed. Selected same-kind recurring eclipse family; parity and P/2 coverage checked.

Depth=71.155976ppt is the median realized central trapezoid loss d*min(1,w/(2*k)); extrapolated amplitude and ingress remain separately in measurements.json. Measured secondary phase 0.48647041, formal phase error 7.3685115e-05, realized depth 23.95888ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Secondary raw pooled windows, actual sectors/cycles and fitted phase are retained in secondary-windows.csv and ../secondary-phases.json. No exhaustive blind alias search asserted. ../alias-audit.json and ../half-period-coverage.json retain recurrence/parity and sampled opposite-phase evidence. Flagged coverage is never a clean null.

Fresh TESScut difference image sector61 stacks2 core/bilateral-sideband events. Target pixel= [7.893511091501619, 7.941925378656464], deficit=[7.76109332104268, 7.981626293770827]; descriptive offset=0.13824pixel, field PSF residual=0.38774. No arbitrary centroid cut or calibrated localization error is imposed. Every capable neighbor below was inspected at its actual projected pixels; close overlapping neighbors retain probable ownership. Every capable neighbor is assessed above and below; close overlapping cases retain unresolved attribution. All Gaia neighbors are retained in pixel JSON.

Source	Separation arcsec	G	Required intrinsic loss proxy	Pixel coordinate	Diluted pixel loss
None					

G-band pairwise brightness screen assumes approximate passband/aperture response; intrinsic required eclipse cannot be directly compared with diluted pixel loss. Unresolved unlisted companions are not excluded.

Alternative reduction: independently extracted background-subtracted TESScut1x1/3x3/5x5 depths=63.81090/61.55883/57.26004ppt; QLP median=71.15598ppt. Pixel sector61 shares the same TESS photons with retained QLP and checks extraction robustness. Independent processing is not an independent instrument. Different apertures respond differently to dilution/background.

Gaia source5736720579555504256: period=12.927622285+/-0.004000789d; e=0.0321342; omega=242.2517626deg; periastron BTJD=386.94870895. Closest conjunction offset=0.0333081d (0.0025765phase), source-specific full-covariance linear error=0.4027193d. Correlation order P,gamma,K,e,omega,Tp with subindices0,3,4,5. Linear propagation is descriptive, not a Gaussian veto or ownership proof. Both conjunctions retained in ../gaia-geometry.json. Mass function=0.09320003solar, assumed M1=0.99, minimum M2=0.623787; approximate central duration=9.2793h versus observed=6.0115h. TIC radius/main-sequence companion scaling, edge-on conjunction and stellar evolution are assumptions; this is a plausibility check, not a fitted physical solution.

Sector	Same-kind cycle	Measured BTJD	Formal center error d	Conservative input error d	Realized depth ppt
8	0	1519.209821234	0.000704539466	0.0104166517	71.15598
61	112	2967.067901383	0.000452535978	0.00115742444	71.44064
61	113	2979.994132044	0.000477591178	0.00115742444	71.63334
99	196	4052.962553803	0.00092889618	0.00115743754	62.77073
99	197	4065.888201469	0.000403088185	0.00115743754	65.27558

Catalogue novelty is separate from this physical verdict. Shared coverage matrix and source identity checks are authoritative for catalogue recovery; period agreement alone cannot establish attribution.

Stable review key: R667.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s61.json - historical/diff_s61.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

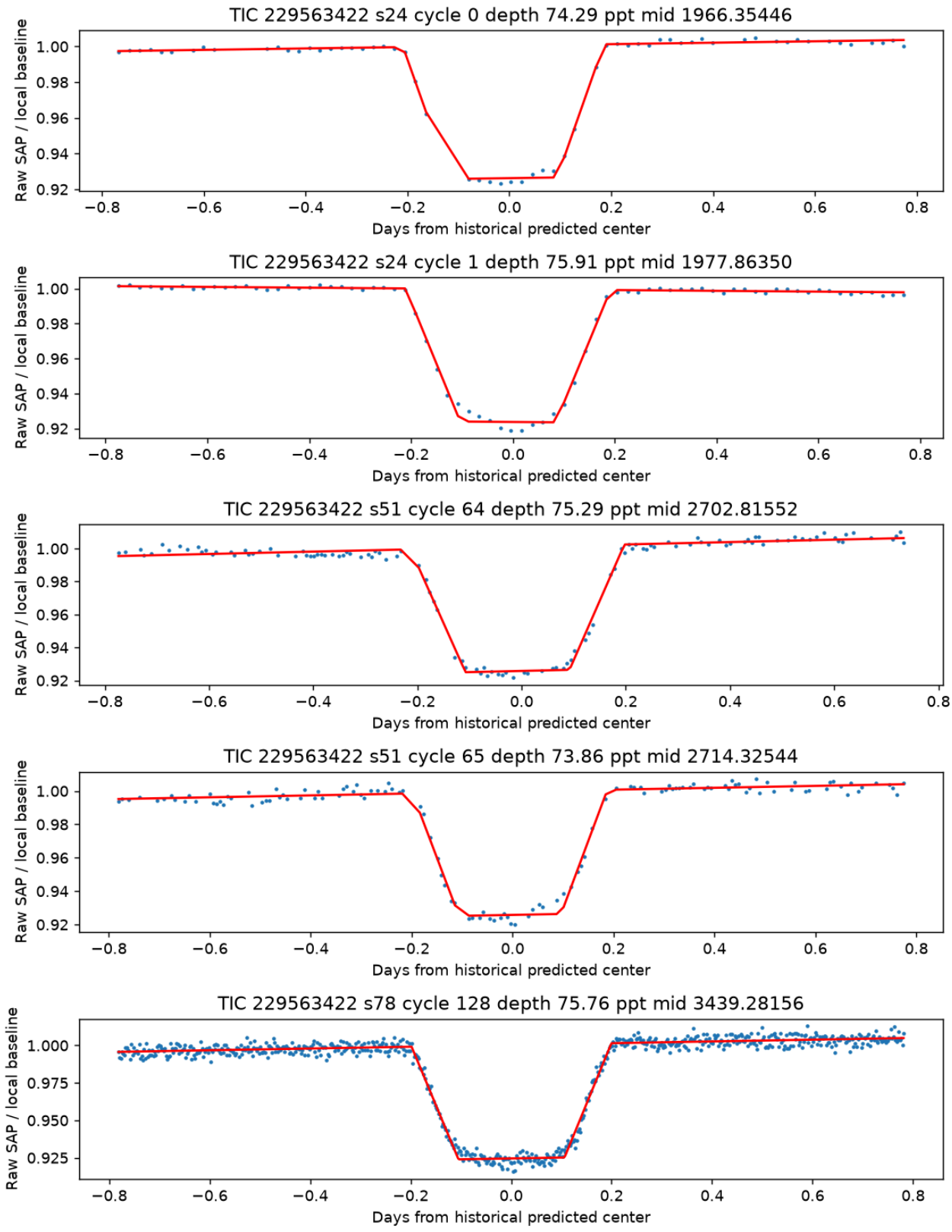
R668. TIC 229563422

physical_verdict: CONFIRMED

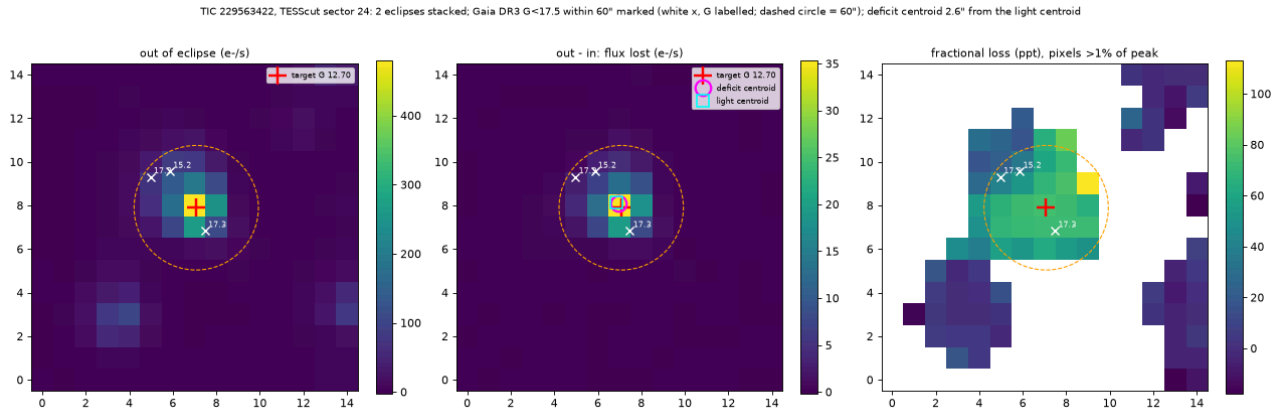
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional additional clean epochs and resolved photometry.



R668; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R668; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 229563422: CONFIRMED

Full bounded physical review: COMPLETE. Five repeated primary eclipses and observed secondary-like dip near0.568 in sectors24/78. Sloped and offset sector baselines limit pooled secondary depth and phase reliability; formal errors are not total uncertainties. Capable39.52arcsec source is displaced from compact target deficit.

Raw QLP SAP_FLUX representative events and full-sector P/2P morphology were visually inspected. Quality-zero finite positive samples only; predictions guide windows and are never timings. 5 identifiable free-center timings define P=11.507252708231 +/- 0.000035573313d, reference BTJD=1966.353158239 +/- 0.004191194d. Epoch error meaning: Conservative linear timing-fit covariance under the stated cadence floor and scatter inflation; excludes uncalibrated shape and baseline systematics. Epoch kind: Deeper primary established by unequal observed secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced chi square=0.1458482783872733; zero degrees of freedom is null. All errors retain their named meanings; no calibrated coverage is claimed. Selected same-kind recurring eclipse family; parity and P/2 coverage checked.

Depth=75.288250ppt is the median realized central trapezoid loss d*min(1,w/(2*k)); extrapolated amplitude and ingress remain separately in measurements.json. Measured secondary phase 0.56857435, formal phase error 0.00027870082, realized depth 10.26889ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Secondary raw pooled windows, actual sectors/cycles and fitted phase are retained in secondary-windows.csv and ../secondary-phases.json. No exhaustive blind alias search asserted. ../alias-audit.json and ../half-period-coverage.json retain recurrence/parity and sampled opposite-phase evidence. Flagged coverage is never a clean null.

Fresh TESScut difference image sector24 stacks2 core/bilateral-sideband events. Target pixel= [7.076658914874642, 7.892752549107424], deficit=[6.961479928460955, 8.068638828024678]; descriptive offset=0.21024pixel, field PSF residual=0.13566. No arbitrary centroid cut or calibrated localization error is imposed. Every capable neighbor below was inspected at its actual projected pixels;

close overlapping neighbors retain probable ownership. Every capable neighbor is assessed above and below; close overlapping cases retain unresolved attribution. All Gaia neighbors are retained in pixel JSON.

Source	Separation arcsec	G	Required intrinsic loss proxy	Pixel coordinate	Diluted pixel loss
120333974773796 9792	39.51630949715 323	15.1577 07	0.7961845617175048	[5.863792630171345, 9.567085211390598]	0.0355404028892 91015

G-band pairwise brightness screen assumes approximate passband/aperture response; intrinsic required eclipse cannot be directly compared with diluted pixel loss. Unresolved unlisted companions are not excluded.

Alternative reduction: independently extracted background-subtracted TESScut1x1/3x3/5x5 depths=73.65179/70.86155/67.13093ppt; QLP median=75.28825ppt. Pixel sector24 shares the same TESS photons with retained QLP and checks extraction robustness. Independent processing is not an independent instrument. Different apertures respond differently to dilution/background.

Gaia source1203338996120545024: period=11.506971982+/-0.002851733d; e=0.1680296; omega=333.6707855deg; periastron BTJD=386.47260926. Closest conjunction offset=0.2888550d (0.0251026phase), source-specific full-covariance linear error=0.4890973d. Correlation order P,gamma,K,e,omega,Tp with subindices0,3,4,5. Linear propagation is descriptive, not a Gaussian veto or ownership proof. Both conjunctions retained in ../gaia-geometry.json. Mass function=0.07886812solar, assumed M1=1.0, minimum M2=0.582323; approximate central duration=10.9006h versus observed=9.5449h. TIC radius/main-sequence companion scaling, edge-on conjunction and stellar evolution are assumptions; this is a plausibility check, not a fitted physical solution.

Sector	Same-kind cycle	Measured BTJD	Formal center error d	Conservative input error d	Realized depth ppt
24	0	1966.354463597	0.00190609079	0.0104167912	74.28968
24	1	1977.863500453	0.000689240843	0.0104167912	75.91111
51	64	2702.815521766	0.00204591859	0.00347229077	75.28825
51	65	2714.325436095	0.00155291585	0.00347229077	73.86239
78	128	3439.281557074	0.000672613204	0.00115741024	75.75584

Catalogue novelty is separate from this physical verdict. Shared coverage matrix and source identity checks are authoritative for catalogue recovery; period agreement alone cannot establish attribution.

Stable review key: R668.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s24.json - historical/diff_s24.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

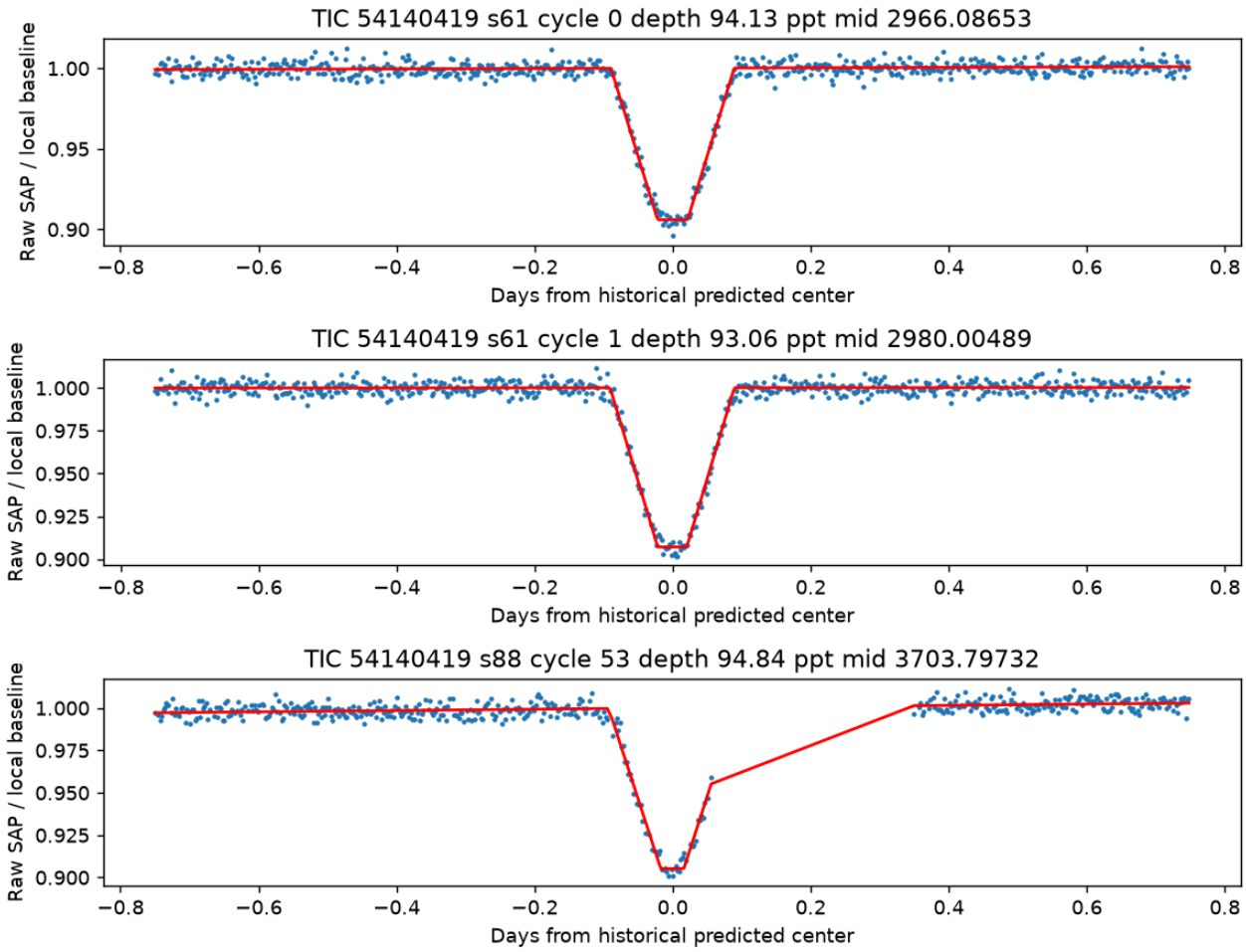
R669. TIC 54140419

physical_verdict: CONFIRMED

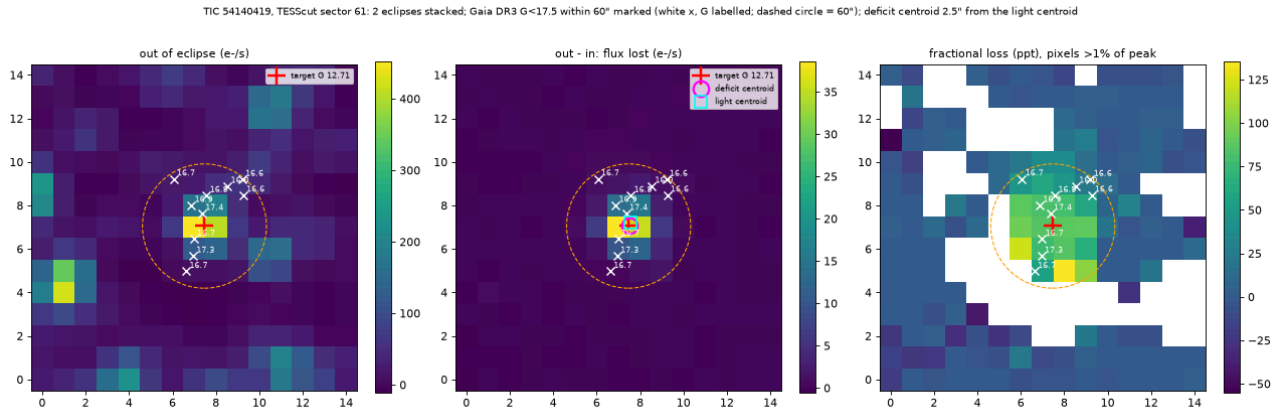
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional additional clean epochs and resolved photometry.



R669; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R669; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 54140419: CONFIRMED

Full bounded physical review: COMPLETE. Historical search period13.91484d drifts off actual observed family. It selected only noise near sector7 historical center and fails later sector88 alignment. Refitting actual sector61/88 morphology gives three same-kind primary timings, including adjacent sector61 cycles0/1, and period13.91907814d. Fresh availability7/34/61/88 prompted sector34 retrieval, without an additional fitted primary. Sparse/gapped/flagged windows are not clean nulls. No secure secondary; no capable neighbor within60arcsec.

Raw QLP SAP_FLUX representative events and full-sector P/2P morphology were visually inspected. Quality-zero finite positive samples only; predictions guide windows and are never timings. 3 identifiable free-center timings define P=13.919078140041 +/- 0.000026996652d, reference BTJD=2966.086176674 +/- 0.000826230d. Epoch error meaning: Conservative linear timing-fit covariance under the stated cadence floor and scatter inflation; excludes uncalibrated shape and baseline systematics. Epoch kind: Selected repeating eclipse; deeper/secondary identity not established. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced chi square=0.19512640486348248; zero degrees of freedom is null. All errors retain their named meanings; no calibrated coverage is claimed. Selected same-kind recurring eclipse family; parity and P/2 coverage checked.

Depth=94.127114ppt is the median realized central trapezoid loss d*min(1,w/(2*k)); extrapolated amplitude and ingress remain separately in measurements.json. No secure secondary; exploratory fit is not interpreted as an eclipse timing or counterpart. Secondary raw pooled windows, actual sectors/cycles and fitted phase are retained in secondary-windows.csv and ../secondary-phases.json. No exhaustive blind alias search asserted. ../alias-audit.json and ../half-period-coverage.json retain recurrence/parity and sampled opposite-phase evidence. Flagged coverage is never a clean null.

Fresh TESScut difference image sector61 stacks2 core/bilateral-sideband events. Target pixel= [7.472352087007673, 7.05230092005368], deficit=[7.522434515537192, 7.0929768179015955]; descriptive offset=0.06452pixel, field PSF residual=0.38941. No arbitrary centroid cut or calibrated localization error is imposed. Every capable neighbor below was inspected at its actual projected pixels; close overlapping neighbors retain probable ownership. Every capable neighbor is assessed above and below; close overlapping cases retain unresolved attribution. All Gaia neighbors are retained in pixel JSON.

Source	Separation arcsec	G	Required intrinsic loss proxy	Pixel coordinate	Diluted pixel loss
None					

G-band pairwise brightness screen assumes approximate passband/aperture response; intrinsic required eclipse cannot be directly compared with diluted pixel loss. Unresolved unlisted companions are not excluded.

Alternative reduction: independently extracted background-subtracted TESScut1x1/3x3/5x5 depths=85.34416/84.70230/81.08119ppt; QLP median=94.12711ppt. Pixel sector61 shares the same TESS photons with retained QLP and checks extraction robustness. Independent processing is not an independent instrument. Different apertures respond differently to dilution/background.

Gaia source3035960206561478144: period=13.912957407+/-0.005045410d; e=0.2230642; omega=70.5079319deg; periastron BTJD=391.32917599. Closest conjunction offset=0.3901117d (0.0280394phase), source-specific full-covariance linear error=0.8339847d. Correlation order P,gamma,K,e,omega,Tp with subindices0,3,4,5. Linear propagation is descriptive, not a Gaussian veto or ownership proof. Both conjunctions retained in ../gaia-geometry.json. Mass function=0.05330113solar, assumed M1=1.06, minimum M2=0.507919; approximate central duration=6.3313h versus observed=4.3203h. TIC radius/main-sequence companion scaling, edge-on conjunction and stellar evolution are assumptions; this is a plausibility check, not a fitted physical solution.

Sector	Same-kind cycle	Measured BTJD	Formal center error d	Conservative input error d	Realized depth ppt
61	0	2966.086534696	0.00036099984	0.00115739451	94.12711
61	1	2980.004889908	0.000334324037	0.00115739451	93.05731
88	53	3703.797324982	0.000645702504	0.00115739945	94.83659

Catalogue novelty is separate from this physical verdict. Shared coverage matrix and source identity checks are authoritative for catalogue recovery; period agreement alone cannot establish attribution.

Stable review key: R669.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s61.json - historical/diff_s61.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

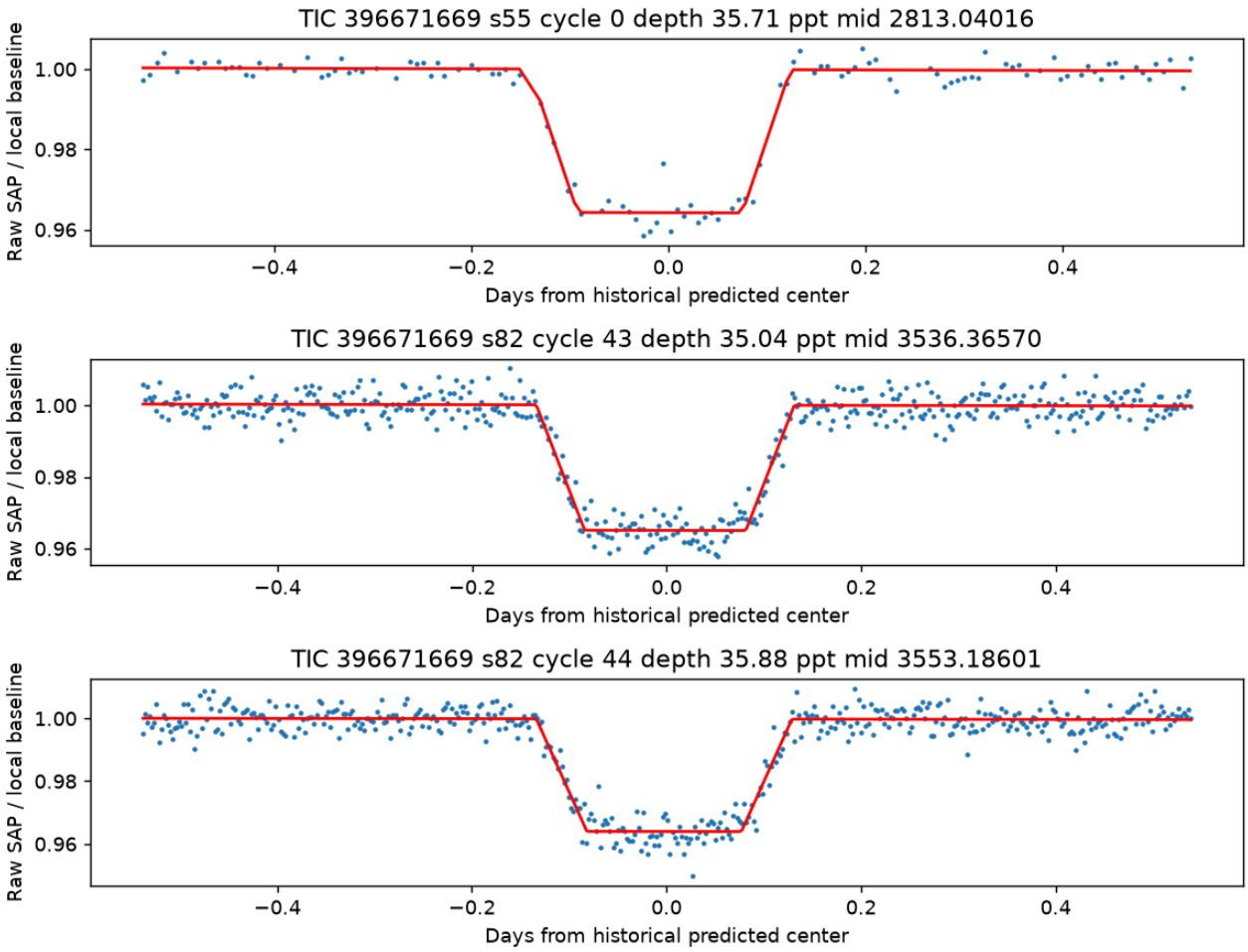
R670. TIC 396671669

physical_verdict: PROBABLE

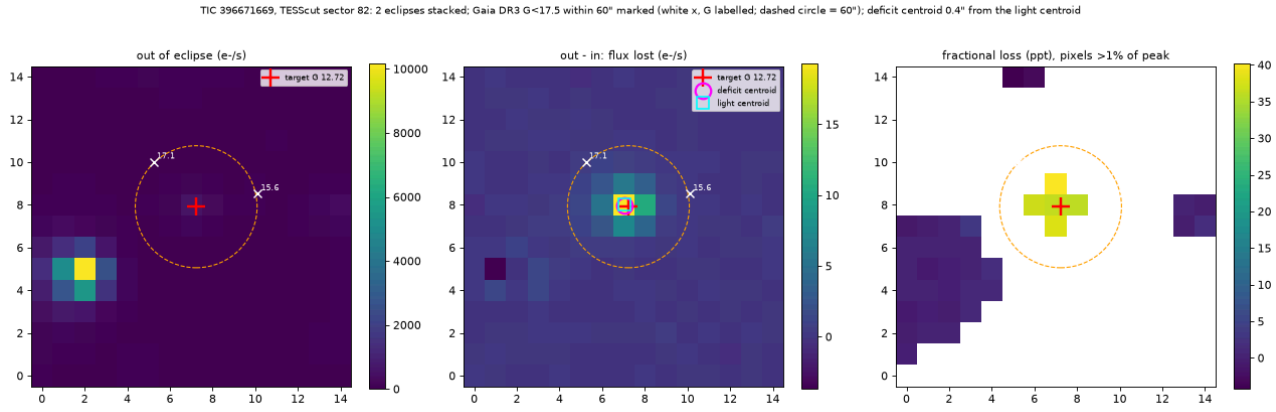
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Observe clean opposite half-phase windows to test P/2 recurrence.



R670; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R670; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 396671669: PROBABLE

Full bounded physical review: COMPLETE. Three repeated narrow eclipses, including consecutive sector82 cycles43/44, establish same-kind recurrence. No secure secondary. Capable59.42arcsec neighbor is displaced from the compact target deficit. Fresh current MAST query returns only55/82. Opposite half-phase windows have no clean bilateral quality-zero coverage, so the shorter P/2 recurrence remains untested; Gaia agreement does not uniquely establish the photometric orbital branch. Probable reflects this incomplete period check, not unresolved source ownership.

Raw QLP SAP_FLUX representative events and full-sector P/2P morphology were visually inspected. Quality-zero finite positive samples only; predictions guide windows and are never timings. 3 identifiable free-center timings define P=16.821507354452 +/- 0.000081905531d, reference BTJD=2813.040282097 +/- 0.003468095d. Epoch error meaning: Conservative linear timing-fit covariance under the stated cadence floor and scatter inflation; excludes uncalibrated shape and baseline systematics. Epoch kind: Selected repeating eclipse; deeper/secondary identity not established. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced chi square=0.5363958937747855; zero degrees of freedom is null. All errors retain their named meanings; no calibrated coverage is claimed. Selected same-kind recurring eclipse family; parity and P/2 coverage checked.

Depth=35.709200ppt is the median realized central trapezoid loss d*min(1,w/(2*k)); extrapolated amplitude and ingress remain separately in measurements.json. No secure secondary; exploratory fit is not interpreted as an eclipse timing or counterpart. Secondary raw pooled windows, actual sectors/cycles and fitted phase are retained in secondary-windows.csv and ../secondary-phases.json. No exhaustive blind alias search asserted. ../alias-audit.json and ../half-period-coverage.json retain recurrence/parity and sampled opposite-phase evidence. Flagged coverage is never a clean null.

Fresh TESScut difference image sector82 stacks2 core/bilateral-sideband events. Target pixel= [7.228942722118724, 7.911515585969251], deficit=[7.037024531195983, 7.989160565246461]; descriptive offset=0.20703pixel, field PSF residual=0.95391. No arbitrary centroid cut or calibrated localization error is imposed. Every capable neighbor below was inspected at its actual projected pixels; close overlapping neighbors retain probable ownership. Other capable sources are displaced from compact deficit core but crowding limits localization. All Gaia neighbors are retained in pixel JSON.

Source	Separation arcsec	G	Required intrinsic loss proxy	Pixel coordinate	Diluted pixel loss
173614930859883 4048	59.42422520229 1275	15.5845 56	0.537243369904984	[10.10889861868745, 8.519902641391267]	0.00188459030341 81357

G-band pairwise brightness screen assumes approximate passband/aperture response; intrinsic required eclipse cannot be directly compared with diluted pixel loss. Unresolved unlisted companions are not excluded.

Alternative reduction: independently extracted background-subtracted TESScut1x1/3x3/5x5 depths=36.35571/37.55980/37.49411ppt; QLP median=35.70920ppt. Pixel sector82 shares the same TESS photons with retained QLP and checks extraction robustness. Independent processing is not an independent instrument. Different apertures respond differently to dilution/background.

Gaia source1736149446037786112: period=16.824686872+/-0.008985830d; e=0.2647184; omega=50.0637973deg; periastron BTJD=389.48425239. Closest conjunction offset=-0.2814976d (-0.0167312phase), source-specific full-covariance linear error=1.4310801d. Correlation order P,gamma,K,e,omega,Tp with subindices0,3,4,5. Linear propagation is descriptive, not a Gaussian veto or ownership proof. Both conjunctions retained in ../gaia-geometry.json. Mass function=0.01426921solar, assumed M1=1.03, minimum M2=0.292188; approximate central duration=7.5711h versus observed=6.3542h. TIC radius/main-sequence companion scaling, edge-on conjunction and stellar evolution are assumptions; this is a plausibility check, not a fitted physical solution.

Sector	Same-kind cycle	Measured BTJD	Formal center error d	Conservative input error d	Realized depth ppt
55	0	2813.040158238	0.00100123683	0.0034722159	35.70920
82	43	3536.365703854	0.000785989997	0.00115738719	35.03636
82	44	3553.186013939	0.00079595303	0.00115738719	35.87539

Catalogue novelty is separate from this physical verdict. Shared coverage matrix and source identity checks are authoritative for catalogue recovery; period agreement alone cannot establish attribution.

Stable review key: R670.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s82.json - historical/diff_s82.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

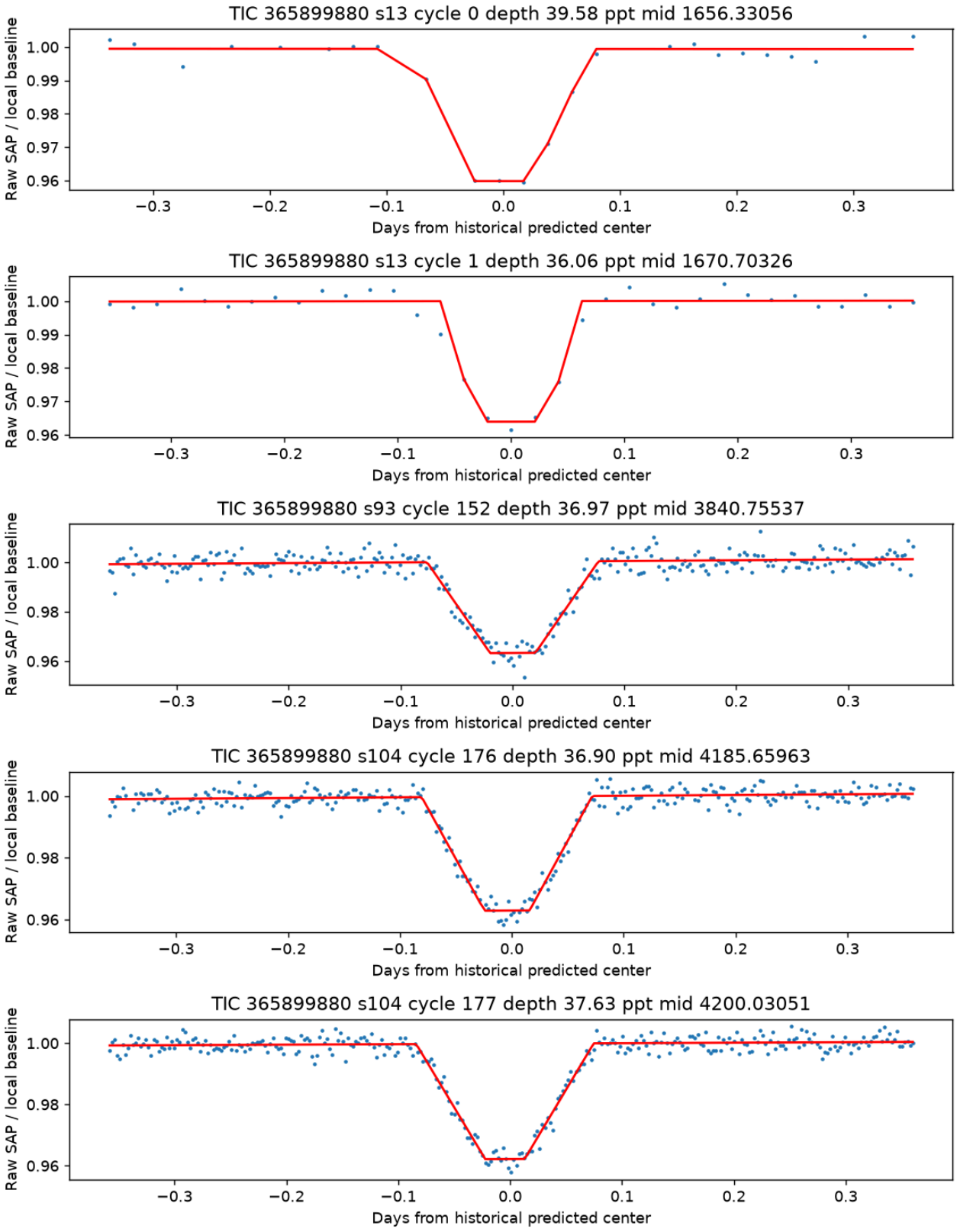
R671. TIC 365899880

physical_verdict: CONFIRMED

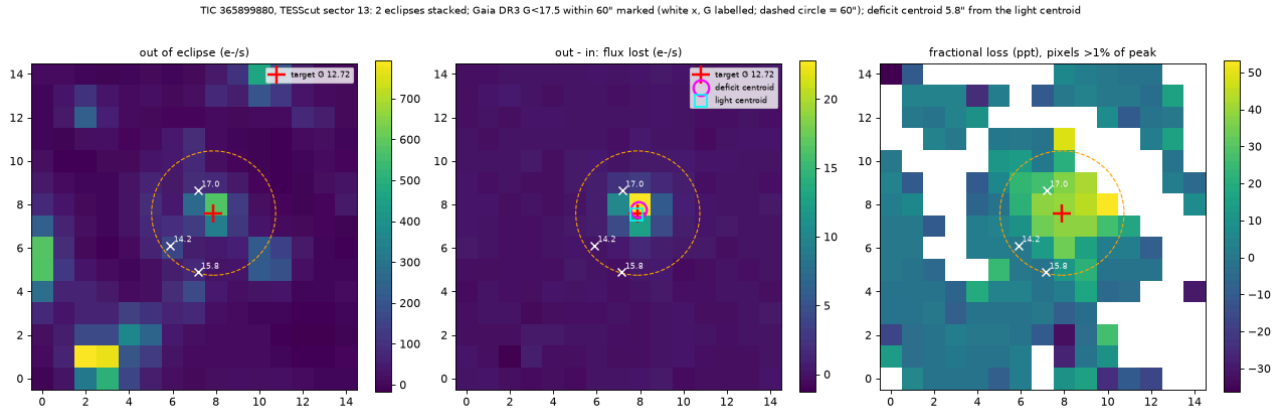
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional additional clean epochs and resolved photometry.



R671; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R671; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 365899880: CONFIRMED

Full bounded physical review: COMPLETE. Five repeated narrow eclipses including consecutive cycles in sectors13 and104. No secure secondary: broad feature near0.54 is principally in sector104 and pooled fit is not a measured counterpart. Capable51.92/58.52arcsec neighbors lie away from compact deficit.

Raw QLP SAP_FLUX representative events and full-sector P/2P morphology were visually inspected. Quality-zero finite positive samples only; predictions guide windows and are never timings. 5 identifiable free-center timings define P=14.371123880242 +/- 0.000051111444d, reference BTJD=1656.342559046 +/- 0.008588746d. Epoch error meaning: Conservative linear timing-fit covariance under the stated cadence floor and scatter inflation; excludes uncalibrated shape and baseline systematics. Epoch kind: Selected repeating eclipse; deeper/secondary identity not established. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced chi square=2.1266529258415625; zero degrees of freedom is null. All errors retain their named meanings; no calibrated coverage is claimed. Selected same-kind recurring eclipse family; parity and P/2 coverage checked.

Depth=36.971378ppt is the median realized central trapezoid loss d*min(1,w/(2*k)); extrapolated amplitude and ingress remain separately in measurements.json. No secure secondary; exploratory fit is not interpreted as an eclipse timing or counterpart. Secondary raw pooled windows, actual sectors/cycles and fitted phase are retained in secondary-windows.csv and ../secondary-phases.json. No exhaustive blind alias search asserted. ../alias-audit.json and ../half-period-coverage.json retain recurrence/parity and sampled opposite-phase evidence. Flagged coverage is never a clean null.

Fresh TESScut difference image sector13 stacks2 core/bilateral-sideband events. Target pixel= [7.894696226667861, 7.602963630708945], deficit=[7.947777889544588, 7.782687379997741]; descriptive offset=0.18740pixel, field PSF residual=0.57627. No arbitrary centroid cut or calibrated localization error is imposed. Every capable neighbor below was inspected at its actual projected pixels; close overlapping neighbors retain probable ownership. Other capable sources are displaced from compact deficit core but crowding limits localization. All Gaia neighbors are retained in pixel JSON.

Source	Separation arcsec	G	Required intrinsic loss proxy	Pixel coordinate	Diluted pixel loss
6634773433421128960	51.91916313575014	14.152336	0.1748042618035759	[5.896009811218857, 6.122784162583708]	-3.880444393188466e-05
6634773502140608640	58.51576590562296	15.814579	0.6741336349613167	[7.157281523223583, 4.892268468873018]	-0.001424754268150882

G-band pairwise brightness screen assumes approximate passband/aperture response; intrinsic required eclipse cannot be directly compared with diluted pixel loss. Unresolved unlisted companions are not excluded.

Alternative reduction: independently extracted background-subtracted TESScut1x1/3x3/5x5 depths=39.53264/38.15728/29.32163ppt; QLP median=36.97138ppt. Pixel sector13 shares the same TESS photons with retained QLP and checks extraction robustness. Independent processing is not an independent instrument. Different apertures respond differently to dilution/background.

Gaia source6634773399061392512: period=14.373683310+/-0.007059897d; e=0.1159392; omega=283.8410921deg; periastron BTJD=385.11902531. Closest conjunction offset=-0.1563193d (-0.0108754phase), source-specific full-covariance linear error=0.8362875d. Correlation order P,gamma,K,e,omega,Tp with subindices0,3,4,5. Linear propagation is descriptive, not a Gaussian veto or ownership proof. Both conjunctions retained in ../gaia-geometry.json. Mass function=0.005877712solar, assumed M1=0.94, minimum M2=0.196544; approximate central duration=5.6924h versus observed=3.6768h. TIC radius/main-sequence companion scaling, edge-on conjunction and stellar evolution are assumptions; this is a plausibility check, not a fitted physical solution.

Sector	Same-kind cycle	Measured BTJD	Formal center error d	Conservative input error d	Realized depth ppt
13	0	1656.330557223	0.00320789282	0.0104165047	39.57773
13	1	1670.703256853	0.00125819154	0.0104165047	36.05989
93	152	3840.755373477	0.000630612585	0.00115742176	36.97138
104	176	4185.659629236	0.000471909965	0.00115740685	36.89531
104	177	4200.030510893	0.000534049947	0.00115740685	37.63328

Catalogue novelty is separate from this physical verdict. Shared coverage matrix and source identity checks are authoritative for catalogue recovery; period agreement alone cannot establish attribution.

Stable review key: R671.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s13.json - historical/diff_s13.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

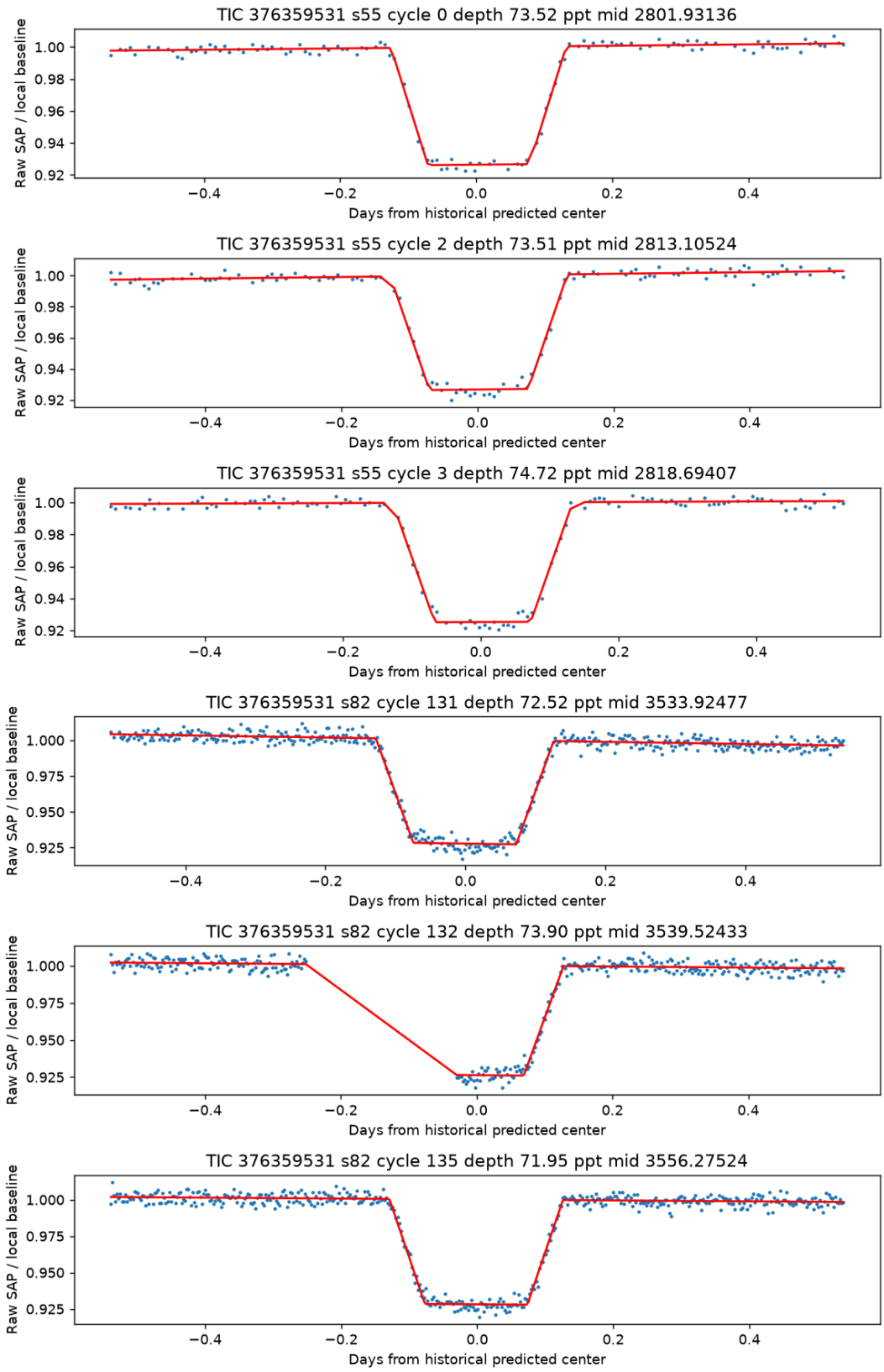
R672. TIC 376359531

physical_verdict: CONFIRMED

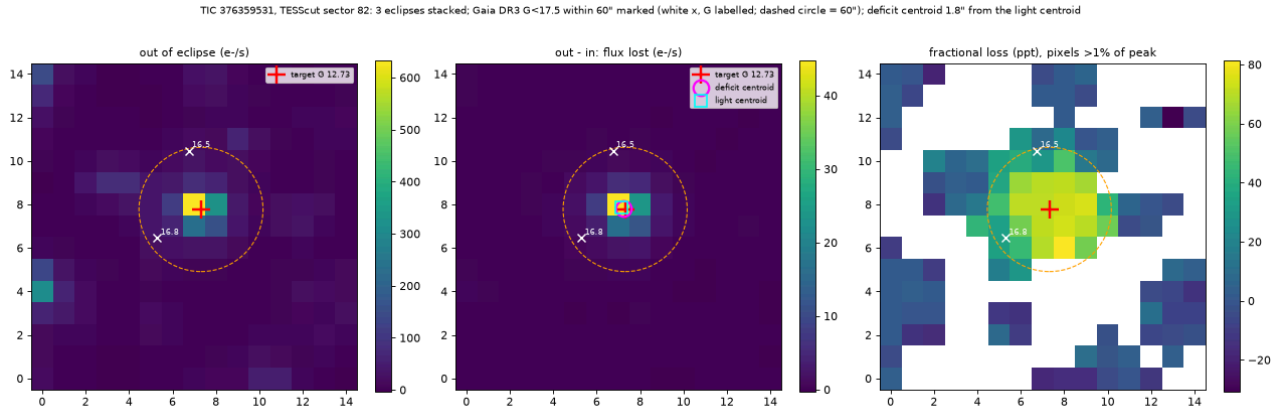
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional additional clean epochs and resolved photometry.



R672; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R672; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 376359531: CONFIRMED

Full bounded physical review: COMPLETE. Six identifiable same-kind timings, including consecutive sector55 cycles2/3 and82 cycles134/135, plus unequal observed secondary near0.5. Two historical event fits have unidentifiable timing covariance and are excluded. No capable neighbor within60arcsec.

Raw QLP SAP_FLUX representative events and full-sector P/2P morphology were visually inspected. Quality-zero finite positive samples only; predictions guide windows and are never timings. 6 identifiable free-center timings define P=5.587737013729 +/- 0.000016034968d, reference BTJD=2801.930697619 +/- 0.002028476d. Epoch error meaning: Conservative linear timing-fit covariance under the stated cadence floor and scatter inflation; excludes uncalibrated shape and baseline systematics. Epoch kind: Deeper primary established by unequal observed secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced chi square=0.1373511919157084; zero degrees of freedom is null. All errors retain their named meanings; no calibrated coverage is claimed. Selected same-kind recurring eclipse family; parity and P/2 coverage checked.

Depth=73.012502ppt is the median realized central trapezoid loss d*min(1,w/(2*k)); extrapolated amplitude and ingress remain separately in measurements.json. Measured secondary phase 0.49964948, formal phase error 0.00037053477, realized depth 8.52461ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Secondary raw pooled windows, actual sectors/cycles and fitted phase are retained in secondary-windows.csv and ../secondary-phases.json. No exhaustive blind alias search asserted. ../alias-audit.json and ../half-period-coverage.json retain recurrence/parity and sampled opposite-phase evidence. Flagged coverage is never a clean null.

Fresh TESScut difference image sector82 stacks3 core/bilateral-sideband events. Target pixel= [7.3124070643101575, 7.770191815637181], deficit=[7.2296148184937135, 7.8042847951310135]; descriptive offset=0.08954pixel, field PSF residual=0.18151. No arbitrary centroid cut or calibrated localization error is imposed. Every capable neighbor below was inspected at its actual projected pixels; close overlapping neighbors retain probable ownership. Other capable sources are displaced from compact deficit core but crowding limits localization. All Gaia neighbors are retained in pixel JSON.

Source	Separation arcsec	G	Required intrinsic loss proxy	Pixel coordinate	Diluted pixel loss
None					

G-band pairwise brightness screen assumes approximate passband/aperture response; intrinsic required eclipse cannot be directly compared with diluted pixel loss. Unresolved unlisted companions are not excluded.

Alternative reduction: independently extracted background-subtracted TESScut1x1/3x3/5x5 depths=70.73307/70.89453/66.96925ppt; QLP median=73.01250ppt. Pixel sector82 shares the same TESS photons with retained QLP and checks extraction robustness. Independent processing is not an independent instrument. Different apertures respond differently to dilution/background.

Gaia source1732755043189031040: period=5.588022491+/-0.000779717d; e=0.1090962; omega=260.0413444deg; periastron BTJD=390.66862379. Closest conjunction offset=-0.1606872d (-0.0287556phase), source-specific full-covariance linear error=0.3346143d. Correlation order P,gamma,K,e,omega,Tp with subindices0,3,4,5. Linear propagation is descriptive, not a Gaussian veto or ownership proof. Both conjunctions retained in ../gaia-geometry.json. Mass function=0.04760853solar, assumed M1=1, minimum M2=0.468182; approximate central duration=8.0545h versus observed=6.2111h. TIC radius/main-sequence companion scaling, edge-on conjunction and stellar evolution are assumptions; this is a plausibility check, not a fitted physical solution.

Sector	Same-kind cycle	Measured BTJD	Formal center error d	Conservative input error d	Realized depth ppt
55	0	2801.931363300	0.000539631697	0.00347223956	73.52269
55	2	2813.105236448	0.000700386189	0.00347223956	73.50865
55	3	2818.694065630	0.0006067061	0.00347223956	74.71809
82	131	3533.924769084	0.000476516555	0.00115739494	72.51636
82	134	3550.686899473	0.000486035652	0.00115739494	71.61014
82	135	3556.275242298	0.000376384731	0.00115739494	71.95061

Catalogue novelty is separate from this physical verdict. Shared coverage matrix and source identity checks are authoritative for catalogue recovery; period agreement alone cannot establish attribution.

Stable review key: R672.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s82.json - historical/diff_s82.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

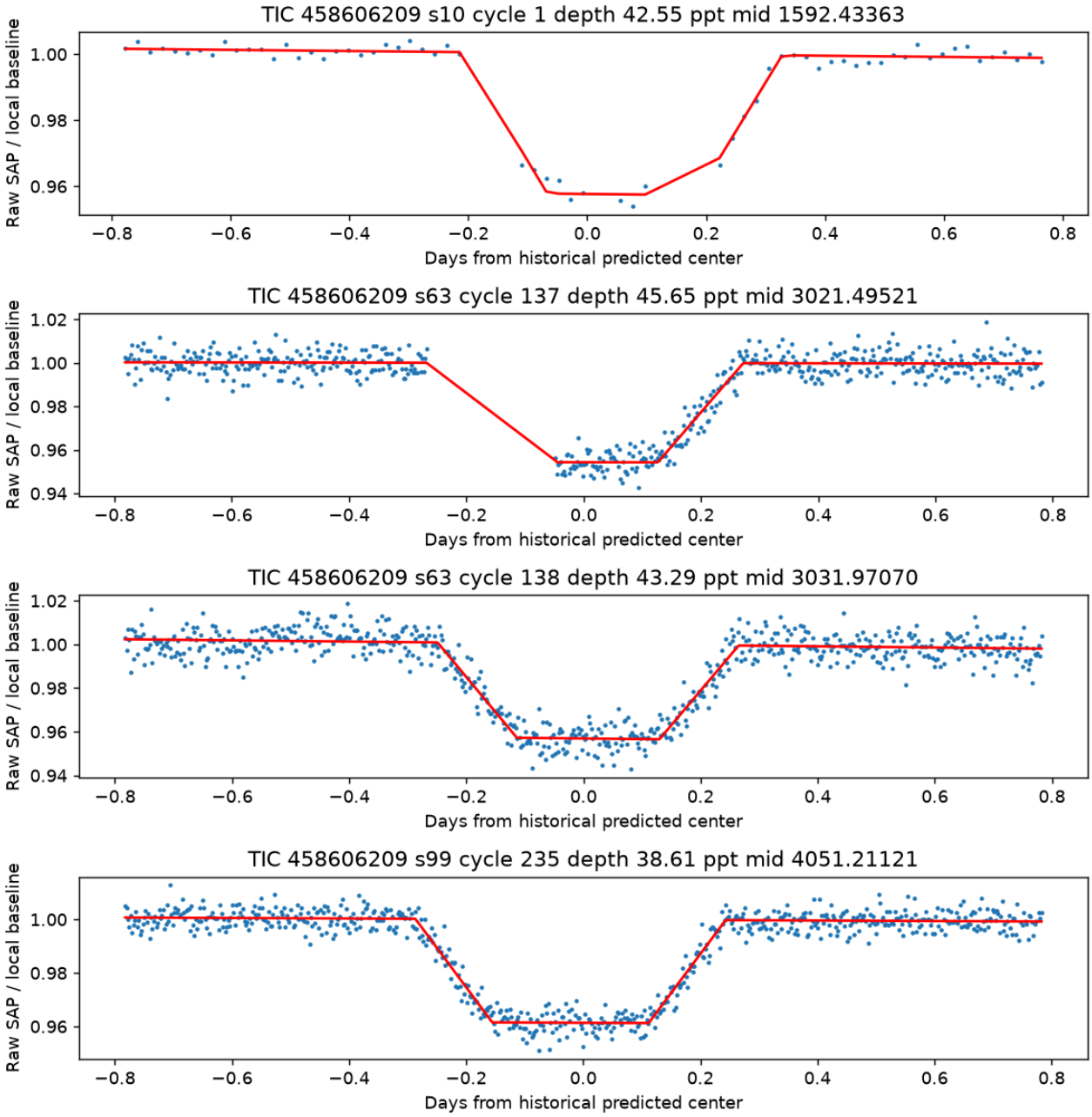
R673. TIC 458606209

physical_verdict: PROBABLE

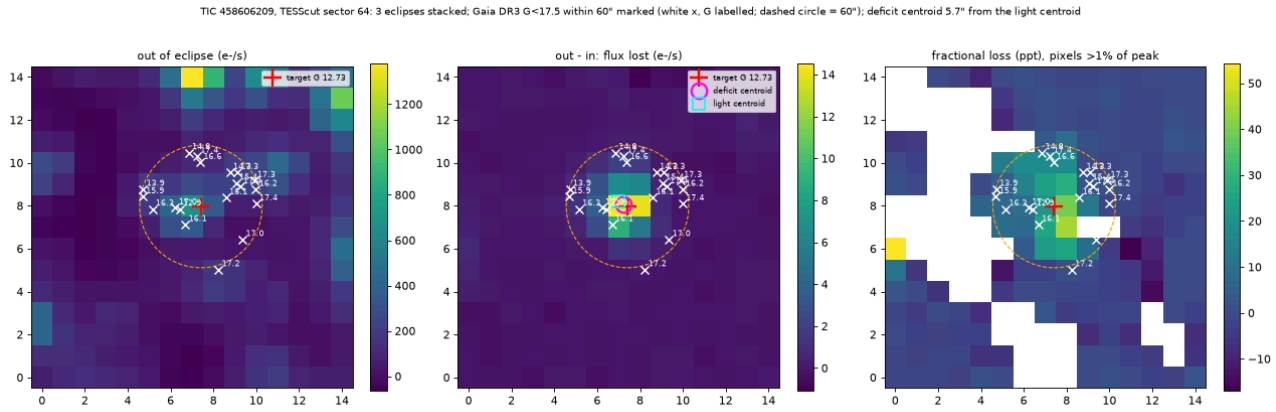
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or multi-source PRF attribution.



R673; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R673; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 458606209: PROBABLE

Full bounded physical review: COMPLETE. Four repeated broad eclipses including consecutive sector63 cycles137/138; unequal observed secondary near0.444. Crowded field: capable20.03arcsec and25.27arcsec neighbors share target footprint, leaving attribution unresolved despite target-centered deficit. All seven capable neighbors inspected. Sector64 pixel events are a distinct occasion from selected QLP sectors10/63/99.

Raw QLP SAP_FLUX representative events and full-sector P/2P morphology were visually inspected. Quality-zero finite positive samples only; predictions guide windows and are never timings. 4 identifiable free-center timings define P=10.507616353469 +/- 0.000055223821d, reference BTJD=1592.428884738 +/- 0.011213982d. Epoch error meaning: Conservative linear timing-fit covariance under the stated cadence floor and scatter inflation; excludes uncalibrated shape and baseline systematics. Epoch kind: Deeper primary established by unequal observed secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced chi square=8.204770542837087; zero degrees of freedom is null. All errors retain their named meanings; no calibrated coverage is claimed. Selected same-kind recurring eclipse family; parity and P/2 coverage checked.

Depth=42.918457ppt is the median realized central trapezoid loss d*min(1,w/(2*k)); extrapolated amplitude and ingress remain separately in measurements.json. Measured secondary phase 0.44400184, formal phase error 8.2247796e-05, realized depth 31.38429ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Secondary raw pooled windows, actual sectors/cycles and fitted phase are retained in secondary-windows.csv and ../secondary-phases.json. No exhaustive blind alias search asserted. ../alias-audit.json and ../half-period-coverage.json retain recurrence/parity and sampled opposite-phase evidence. Flagged coverage is never a clean null.

Fresh TESScut difference image sector64 stacks3 core/bilateral-sideband events. Target pixel= [7.416264869931146, 7.968078315482648], deficit=[7.2206932843191245, 8.052592505505219]; descriptive offset=0.21305pixel, field PSF residual=0.71034. No arbitrary centroid cut or calibrated

localization error is imposed. Every capable neighbor below was inspected at its actual projected pixels; close overlapping neighbors retain probable ownership. Other capable sources are displaced from compact deficit core but crowding limits localization. All Gaia neighbors are retained in pixel JSON.

Source	Separation arcsec	G	Required intrinsic loss proxy	Pixel coordinate	Diluted pixel loss
5351360235266445568	20.031799303421252	12.864673	0.09148702558972747	[6.429879696783621, 7.84715444702729]	0.009128277718474069
53513603083145139920	25.266545586944385	16.100832	0.9997134682855747	[8.588842941994933, 8.381180445620192]	0.011911246711458824
53513603083145131152	40.205032495392216	15.429647	0.5585564416742173	[9.05482424327354, 9.059711612616049]	0.0022979583005877267
53513602739547741144	42.527060205872213	14.337115	0.23142746267309955	[8.768348177909699, 9.556810807833772]	-3.8042757881768166e-05
53513602739547744528	50.378965526762386	14.809184	0.3340972238670255	[6.848135128632469, 10.476786078243428]	0.017607220973876504
5351360239594430208	54.750492906705113	15.872371	0.8181554748712525	[4.698622064769106, 8.420438086653014]	0.007765753473322694
5351360205234688384	55.623993349573734	13.928894	0.17235106524699373	[4.7167434970293645, 8.772644367694706]	0.004781832745643519

G-band pairwise brightness screen assumes approximate passband/aperture response; intrinsic required eclipse cannot be directly compared with diluted pixel loss. Unresolved unlisted companions are not excluded.

Alternative reduction: independently extracted background-subtracted TESScut1x1/3x3/5x5 depths=21.14934/22.18539/17.98584ppt; QLP median=42.91846ppt. Pixel sector64 is a distinct observing occasion on the same TESS instrument; no same-sector QLP comparison claimed. Independent processing is not an independent instrument. Different apertures respond differently to dilution/background.

Gaia source5351360308313918848: period=10.511668418+/-0.005780592d; e=0.0347176; omega=194.7570037deg; periastron BTJD=386.66479107. Closest conjunction offset=-0.1320267d (-0.0125600phase), source-specific full-covariance linear error=0.7689467d. Correlation order P,gamma,K,e,omega,Tp with subindices0,3,4,5. Linear propagation is descriptive, not a Gaussian veto or ownership proof. Both conjunctions retained in ../gaia-geometry.json. Mass function=0.09367283solar, assumed M1=1.4, minimum M2=0.758539; approximate central duration=13.4862h versus observed=12.5296h. TIC radius/main-sequence companion scaling, edge-on conjunction and stellar evolution are assumptions; this is a plausibility check, not a fitted physical solution.

Sector	Same-kind cycle	Measured BTJD	Formal center error d	Conservative input error d	Realized depth ppt
10	1	1592.433631788	0.00386248075	0.0104166481	42.54572
63	137	3021.495209608	0.00782261169	0.00782261169	45.65452
63	138	3031.970695563	0.00163565665	0.00163565665	43.29120
99	235	4051.211210360	0.00121567333	0.00121567333	38.60979

Catalogue novelty is separate from this physical verdict. Shared coverage matrix and source identity checks are authoritative for catalogue recovery; period agreement alone cannot establish attribution.

Stable review key: R673.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s64.json - historical/diff_s64.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

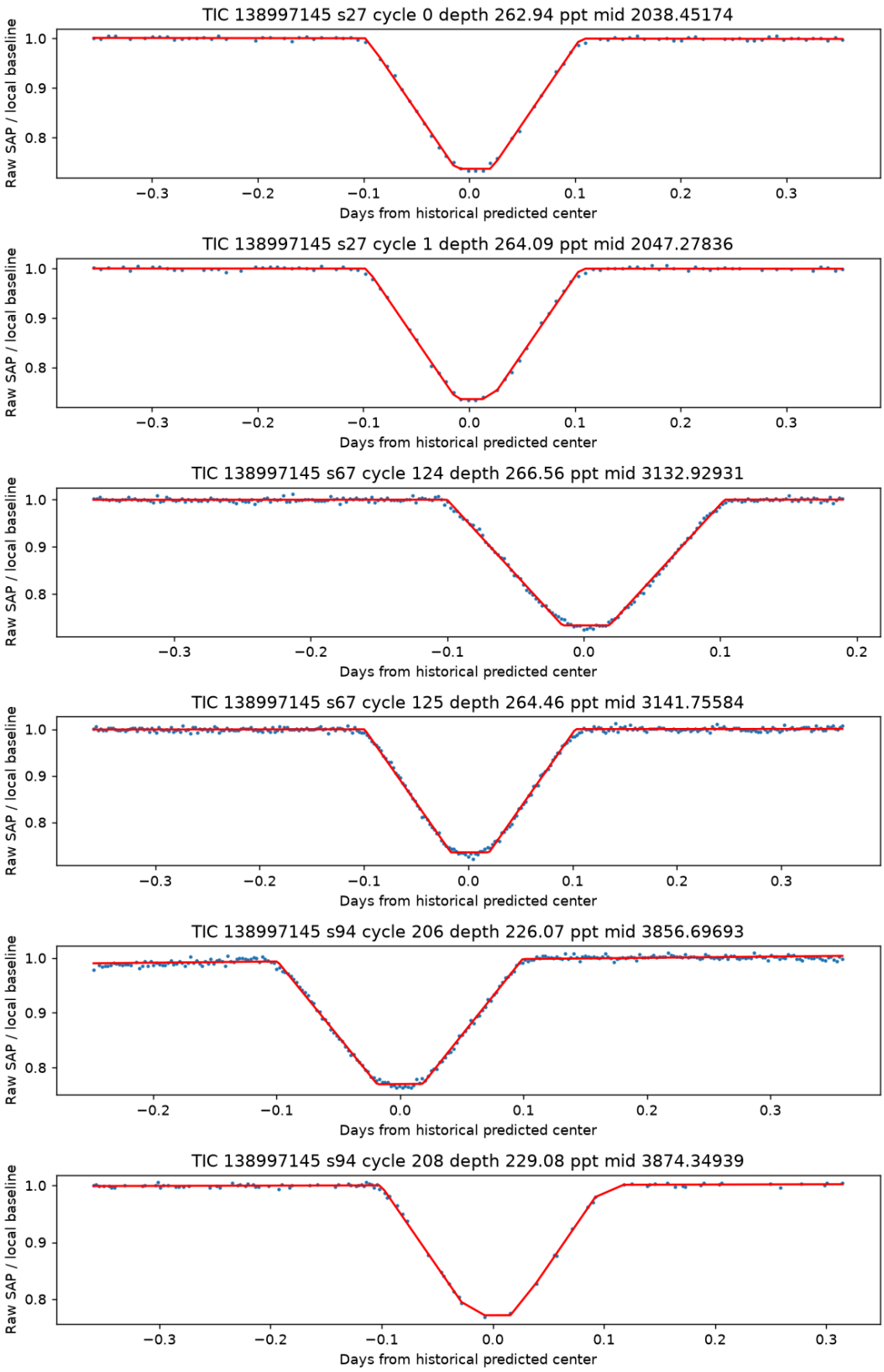
R674. TIC 138997145

physical_verdict: CONFIRMED

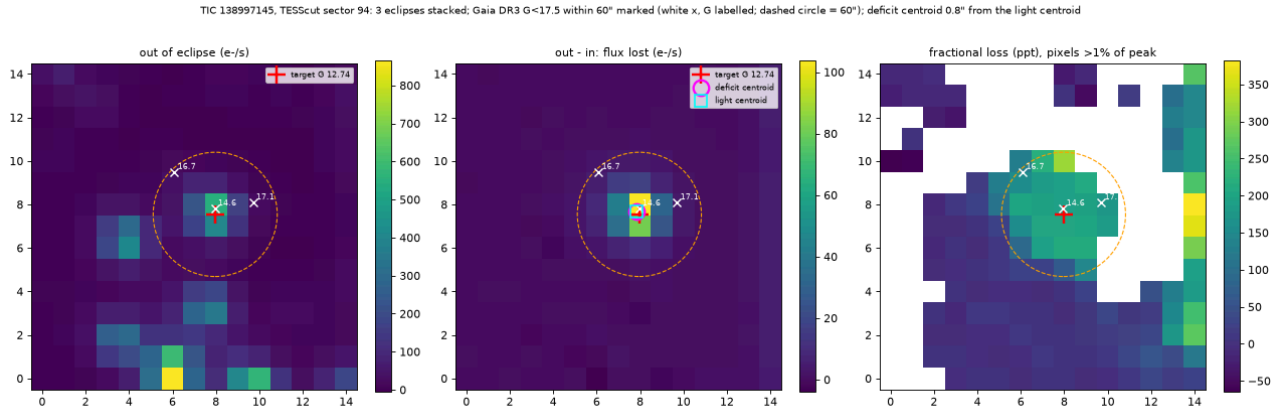
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Optional additional clean epochs and resolved photometry.



R674; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R674; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 138997145: CONFIRMED

Full bounded physical review: COMPLETE. Observed half-seed4.41323day cycles alternate strongly between deep and shallow eclipses. Seed explicitly doubled; nine same-kind deeper-primary timings at8.82643days and unequal secondary near0.5 establish orbital branch. No capable neighbor within60arcsec. Aperture dilution changes deep-event amplitude between sectors.

Raw QLP SAP_FLUX representative events and full-sector P/2P morphology were visually inspected. Quality-zero finite positive samples only; predictions guide windows and are never timings. 9 identifiable free-center timings define P=8.826431784365 +/- 0.000008466704d, reference BTJD=2038.451801753 +/- 0.001409113d. Epoch error meaning: Conservative linear timing-fit covariance under the stated cadence floor and scatter inflation; excludes uncalibrated shape and baseline systematics. Epoch kind: Deeper primary established by unequal observed secondary. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced chi square=0.01630028140644502; zero degrees of freedom is null. All errors retain their named meanings; no calibrated coverage is claimed. Historical4.413226549day seed doubled after observed deep/shallow alternation. Only original even half-period cycles fit as deeper-primary family; original epoch retained; cycle labels now refer to8.826453099day fitting seed.

Depth=263.337993ppt is the median realized central trapezoid loss d*min(1,w/(2*k)); extrapolated amplitude and ingress remain separately in measurements.json. Measured secondary phase 0.50003875, formal phase error 2.4976949e-05, realized depth 120.22959ppt. Formal pooled trapezoid covariance only; sector normalization, red noise and shape systematics uncalibrated. Secondary raw pooled windows, actual sectors/cycles and fitted phase are retained in secondary-windows.csv and ../secondary-phases.json. No exhaustive blind alias search asserted. ../alias-audit.json and ../half-period-coverage.json retain recurrence/parity and sampled opposite-phase evidence. Flagged coverage is never a clean null.

Fresh TESScut difference image sector94 stacks3 core/bilateral-sideband events. Target pixel= [7.966650072405249, 7.542485698943771], deficit=[7.852724883400835, 7.68852926515034]; descriptive offset=0.18522pixel, field PSF residual=0.60722. No arbitrary centroid cut or calibrated

localization error is imposed. Every capable neighbor below was inspected at its actual projected pixels; close overlapping neighbors retain probable ownership. Other capable sources are displaced from compact deficit core but crowding limits localization. All Gaia neighbors are retained in pixel JSON.

Source	Separation arcsec	G	Required intrinsic loss proxy	Pixel coordinate	Diluted pixel loss
None					

G-band pairwise brightness screen assumes approximate passband/aperture response; intrinsic required eclipse cannot be directly compared with diluted pixel loss. Unresolved unlisted companions are not excluded.

Alternative reduction: independently extracted background-subtracted TESScut1x1/3x3/5x5 depths=189.37723/196.11442/194.88822ppt; QLP median=263.33799ppt. Pixel sector94 shares the same TESS photons with retained QLP and checks extraction robustness. Independent processing is not an independent instrument. Different apertures respond differently to dilution/background.

Gaia source6669547408496730752: period=8.827570056+/-0.001333005d; e=0.0425910; omega=7.4819821deg; periastron BTJD=385.90196289. Closest conjunction offset=-0.1110574d (-0.0125807phase), source-specific full-covariance linear error=0.3000351d. Correlation order P,gamma,K,e,omega,Tp with subindices0,3,4,5. Linear propagation is descriptive, not a Gaussian veto or ownership proof. Both conjunctions retained in ../gaia-geometry.json. Mass function=0.1387905solar, assumed M1=1, minimum M2=0.752627; approximate central duration=6.8508h versus observed=4.8503h. TIC radius/main-sequence companion scaling, edge-on conjunction and stellar evolution are assumptions; this is a plausibility check, not a fitted physical solution.

Sector	Same-kind cycle	Measured BTJD	Formal center error d	Conservative input error d	Realized depth ppt
27	0	2038.451739897	0.000201944999	0.00347225028	262.93728
27	1	2047.278357240	0.000268287911	0.00347225028	264.09489
27	2	2056.104032500	0.000221487411	0.00347225028	263.33799
67	124	3132.929305331	0.000110792562	0.00115742245	266.56486
67	125	3141.755835578	0.000130001767	0.00115742245	264.46097
67	126	3150.582339299	0.000243186522	0.00115742245	272.39087
94	206	3856.696929911	0.000189229105	0.00115742282	226.06618
94	207	3865.523127247	9.75522609e-05	0.00115742282	233.35191
94	208	3874.349393825	0.000218075679	0.00115742282	229.08437

Catalogue novelty is separate from this physical verdict. Shared coverage matrix and source identity checks are authoritative for catalogue recovery; period agreement alone cannot establish attribution.

Stable review key: R674.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s94.json - historical/diff_s94.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

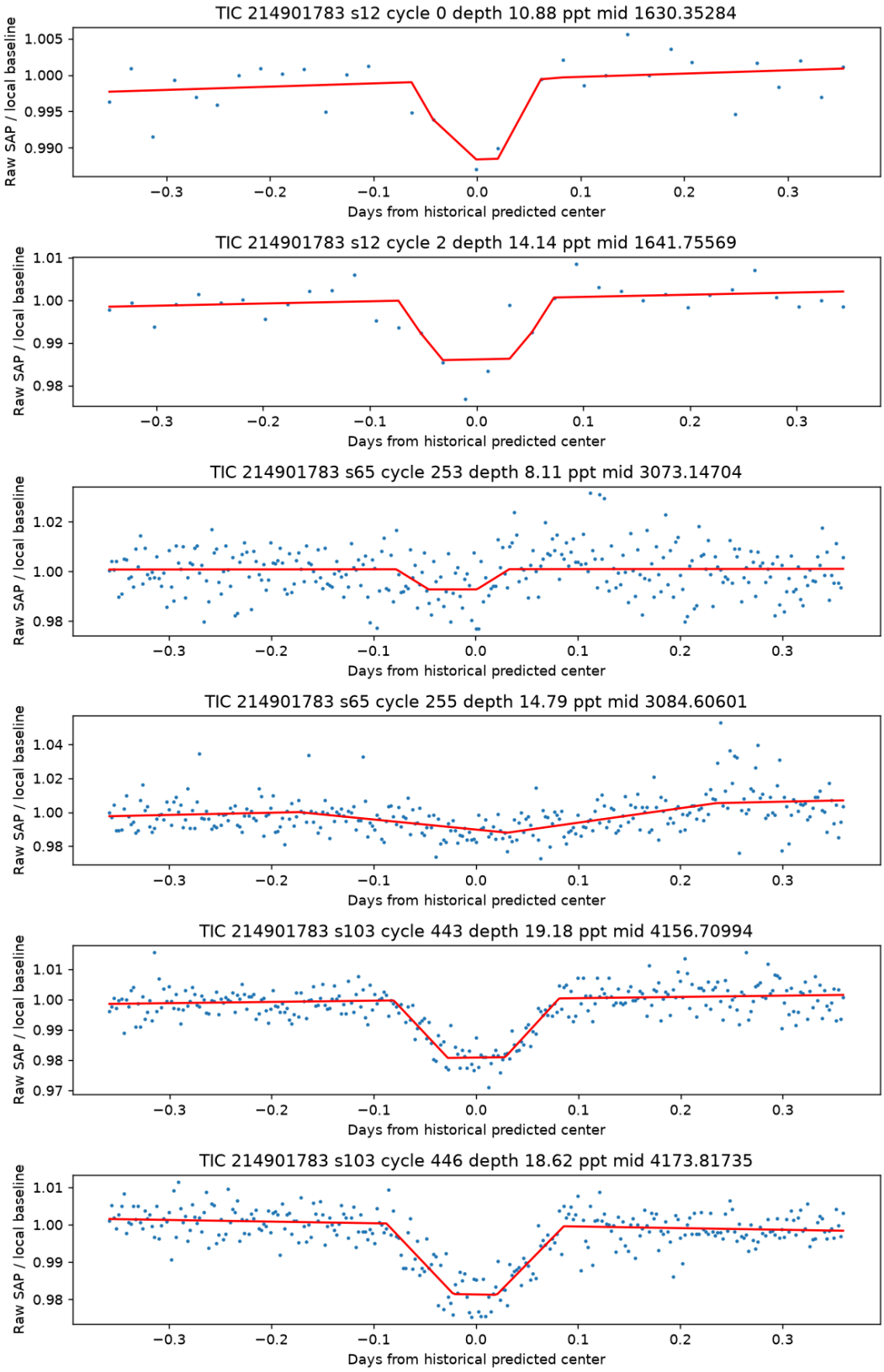
R675. TIC 214901783

physical_verdict: PROBABLE

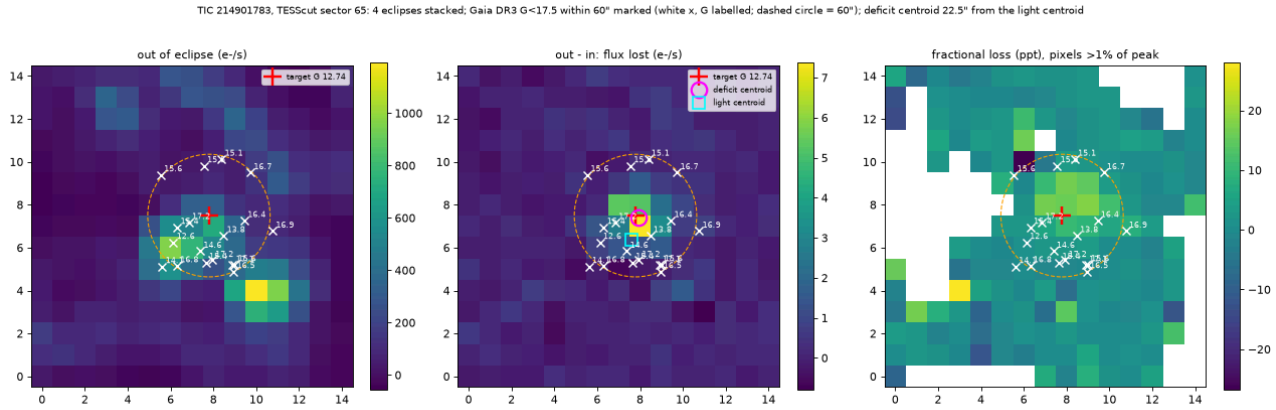
novelty_status: no identified prior EB

completion: COMPLETE_BOUNDED_REVIEW

next_check: Resolved eclipse photometry or multi-source PRF attribution.



R675; Measured raw-flux event windows; event exclusions and timing qualifications remain in the full report.



R675; Target-pixel eclipse-deficit evidence; angular resolution, capable-neighbour and image-model limits remain in the full report.

Archived source: sanitized report text.

TIC 214901783: PROBABLE

Full bounded physical review: COMPLETE. Ten identifiable repeated narrow eclipse timings across three sectors, including consecutive cycles. Two raw fit centers have unidentifiable timing covariance and are excluded. No secure secondary. Crowded field: capable22.95/29.45/31.80arcsec sources overlap target response, leaving ownership unresolved. All sixteen capable sources inspected; centroid alignment alone is insufficient.

Raw QLP SAP_FLUX representative events and full-sector P/2P morphology were visually inspected. Quality-zero finite positive samples only; predictions guide windows and are never timings. 10 identifiable free-center timings define P=5.702841051045 +/- 0.000021207263d, reference BTJD=1630.352104065 +/- 0.009114437d. Epoch error meaning: Conservative linear timing-fit covariance under the stated cadence floor and scatter inflation; excludes uncalibrated shape and baseline systematics. Epoch kind: Selected repeating eclipse; deeper/secondary identity not established. Linear timing fit using conservative per-event max(formal trapezoid covariance error, half cadence); covariance inflated when reduced chi square exceeds one. Not calibrated coverage or total uncertainty; pipeline timing offsets, trapezoid shape and baseline model systematics are not calibrated. Reduced chi square=3.4419228049993262; zero degrees of freedom is null. All errors retain their named meanings; no calibrated coverage is claimed. Selected same-kind recurring eclipse family; parity and P/2 coverage checked.

Depth=16.133730ppt is the median realized central trapezoid loss d*min(1,w/(2*k)); extrapolated amplitude and ingress remain separately in measurements.json. No secure secondary; exploratory fit is not interpreted as an eclipse timing or counterpart. Secondary raw pooled windows, actual sectors/cycles and fitted phase are retained in secondary-windows.csv and ../secondary-phases.json. No exhaustive blind alias search asserted. ../alias-audit.json and ../half-period-coverage.json retain recurrence/parity and sampled opposite-phase evidence. Flagged coverage is never a clean null.

Fresh TESScut difference image sector65 stacks4 core/bilateral-sideband events. Target pixel= [7.788916285107151, 7.49421914033271], deficit=[7.950921118976015, 7.389858484685447]; descriptive offset=0.19271pixel, field PSF residual=0.87921. No arbitrary centroid cut or calibrated localization error is imposed. Every capable neighbor below was inspected at its actual projected pixels; close overlapping neighbors retain probable ownership. Other capable sources are displaced from compact deficit core but crowding limits localization. All Gaia neighbors are retained in pixel JSON.

Source	Separation arcsec	G	Required intrinsic loss proxy	Pixel coordinate	Diluted pixel loss
5981355594563192192	22.952113962669998	13.782869	0.05817760894230905	[8.470348078017006, 6.56399089559464]	0.010308571231121423
5981355903800833664	29.449354224655545	15.396892	0.20204768019341995	[6.307686070435256, 6.952241651184607]	0.002457920390816007
5981355972520313984	31.803155563839912	14.56085	0.10221232665423216	[7.380333291143314, 5.839274706153606]	0.0019131053670369258
5981355594563192448	32.474182513205556	16.399134	0.4840937802124696	[9.4409629383143, 7.264358783345356]	0.0091810199427197
5981355899487740288	38.038895006295654	12.5559	0.02971421163083011	[6.12820007101998, 6.2530486946220565]	0.0008479579212631686
5981355972520315776	41.35552272803753	15.438667	0.20934033918836883	[7.644182906633523, 5.3146526433959025]	-0.0009401901725938021
5981355491483969920	44.910064540048914	15.544358	0.22909389698655186	[7.555159253577353, 9.81982483406015]	0.004228428928177881
5981355594544632704	50.388465124994006	17.062704	0.8783908297351569	[8.891841564473577, 5.190838072432019]	-0.0002799682697885735
5981355560203010944	50.415817959520064	15.066852	0.15331536618185101	[8.374418505199245, 10.12727182330641]	0.004228428928177881
5981355972520393088	50.700678758170062	16.828697	0.7112124280413856	[6.298128821535522, 5.16233208384662]	-0.0001395654627036778
5981355560203019136	50.904183761353174	16.729301	0.6504057871102259	[9.726296645182531, 9.494828207738927]	0.0035663273746719196
5981355800721626240	50.915322165143318	15.601558	0.2406141047453974	[8.986776194760335, 5.2147890580125935]	-0.0002799682697885735
5981356178678749440	56.070475539243383	16.523705	0.5409861196969831	[8.939843298545727, 4.890080894061584]	-0.0002799682697885735
5981355903800836352	58.769416315848424	14.072507	0.07103191386976954	[5.615395819328784, 5.12423615048265]	-0.0001395654627036778
5981355525843706368	58.829070720657728	15.613009	0.24299417450646815	[5.541416048822302, 9.39726794462576]	-0.0003702817158377324
5981355766361472000	59.927550026219393	16.932198	0.7807338801636757	[10.75832623260136, 6.775194593069631]	0.008319383110546428

G-band pairwise brightness screen assumes approximate passband/aperture response; intrinsic required eclipse cannot be directly compared with diluted pixel loss. Unresolved unlisted companions are not excluded.

Alternative reduction: independently extracted background-subtracted TESScut1x1/3x3/5x5 depths=10.30857/7.60112/4.40626ppt; QLP median=16.13373ppt. Pixel sector65 shares the same TESS photons with retained QLP and checks extraction robustness. Independent processing is not an independent instrument. Different apertures respond differently to dilution/background.

Gaia source5981355594563189632: period=5.702801069+/-0.001238904d; e=0.0953133; omega=124.6451513deg; periastron BTJD=387.72485528. Closest conjunction offset=-0.1273801d (-0.0223364phase), source-specific full-covariance linear error=0.2834139d. Correlation order P,gamma,K,e,omega,Tp with subindices0,3,4,5. Linear propagation is descriptive, not a Gaussian veto or ownership proof. Both conjunctions retained in ../gaia-geometry.json. Mass function=0.0203285solar, assumed M1=0.94, minimum M2=0.318054; approximate central duration=8.0352h versus observed=3.9176h. TIC radius/main-sequence companion scaling, edge-on conjunction and stellar evolution are assumptions; this is a plausibility check, not a fitted physical solution.

Sector	Same-kind cycle	Measured BTJD	Formal center error d	Conservative input error d	Realized depth ppt
12	0	1630.352838824	0.00460227642	0.0104165543	10.88262
12	2	1641.755691197	0.00202568689	0.0104165543	14.14253

Sector	Same-kind cycle	Measured BTJD	Formal center error d	Conservative input error d	Realized depth ppt
12	3	1647.461587384	0.00298485923	0.0104165543	17.48046
65	253	3073.147040437	0.00655396507	0.00655396507	8.11337
65	254	3078.878446085	0.00488523895	0.00488523895	7.08011
65	255	3084.606014151	0.00892992331	0.00892992331	14.78700
103	443	4156.709944130	0.00181843229	0.00181843229	19.17918
103	444	4162.414969525	0.00249632818	0.00249632818	18.31106
103	445	4168.117959091	0.00169256691	0.00169256691	20.34097
103	446	4173.817348231	0.00181286024	0.00181286024	18.61846

Catalogue novelty is separate from this physical verdict. Shared coverage matrix and source identity checks are authoritative for catalogue recovery; period agreement alone cannot establish attribution.

Stable review key: R675.

Deciding evidence and numeric artifacts: - fresh-pixels/diff_fresh.json - fresh-pixels/diff_fresh.png - fresh-pixels/gaia.json - fresh-pixels/run.log - historical/diff_s65.json - historical/diff_s65.png - historical/gaia.json - measurements.json - phase-review.png - raw-events.png - raw-windows.csv - secondary-windows.csv

Batch audits and scripts. Depth correction provenance.

Period-fold source attribution and catalogue comparisons

TIC 195865529 - CONFIRMED

Unique count addition: no.

Known ASAS-SN EA at 0.16 arcsec (P 3.8310 d); ATLAS DBH P/2. Recovery excluded from new count.

Catalogue scope: **already catalogued**: ASAS-SN EA P 3.8310 d at 0.16"; ATLAS DBH at P/2; not in VSX

Full source report

TIC 468181994 - CONFIRMED (weaker)

Unique count addition: yes.

Period -2.31 sigma from Gaia; RUWE 12.6/unresolved companion risk; independent pipelines agree about 2 sigma. Literature cones limited to enumerated catalogues, not proof of novelty.

Catalogue scope: not in VSX/ASAS-SN/Gaia-EB (per report)

Full source report

TIC 470235665 - CONFIRMED

Unique count addition: yes.

QLP-only secondary phase drift unconfirmed; 4-5% pipeline scale offset. Enumerated catalogue cones and SIMBAD SB* do not establish exhaustive novelty.

Catalogue scope: not in VSX/Gaia-EB/PrÅia/IJspeert/ASAS-SN within 60"; SIMBAD SB*

Full source report

TIC 379251709 - CONFIRMED

Unique count addition: yes.

Secondary fold-level, not localized; primary securely on target. QLP/SPOC secondary phases differ 2.6 sigma. Radius tension retained.

Catalogue scope: not in VSX/Gaia-EB/PrÅia/IJspeert/ASAS-SN within 60"; SIMBAD SB*

Full source report

TIC 70665829 - CONFIRMED

Unique count addition: no.

Already objects.csv A4; blind control independently recovered. Archive Gaia-VEB/varisum/SB9 controls failed on later mirror timeout, predecessor empty cones only. No addition.

Catalogue scope: not in VSX/PrÅia/IJspeert within 60"

Full source report; existing A4 evidence

TIC 331293467 - CONFIRMED

Unique count addition: yes.

Heavy blend; QLP plus independent own TESScut satisfies available pipeline check; depths model-dependent 80-116 ppt, ratios/duration agree. Gaia geometry depends on covariance. Gaia-VEB/varisum/SB9 cones lack successful positive controls; no exhaustive novelty claim.

Catalogue scope: not in VSX/PrÅia/IJspeert within 60"; Gaia-EB/varisum/SB9 cones from predecessor run only (mirror timed out)

Full source report

TIC 283532878 - CONFIRMED

Unique count addition: yes.

Use successful s12d recheck, not false-empty first archive pass. Known EB 29.7 arcsec away at P 1.7975 d excluded as source; up to 3 ppt contamination possible. Phase drift tentative.

Catalogue scope: not in VSX/Gaia-EB/PrÅia/IJspeert on target; **a catalogued EB (Gaia DR3 2202301140233408128, P 1.7975 d, G 17.66) sits 29.7" away**, excluded as source but may add ≤3 ppt dips

Full source report

TIC 407877153 - CONFIRMED

Unique count addition: yes.

Secondary pixel signal weak individually but summed delta chi2 >=166; QLP depth diluted. Use successful s12d catalogue rerun; all-catalogue query unreliable.

Catalogue scope: not in VSX/Gaia-EB/PrÅia/IJspeert within 60"

Full source report

TIC 15855794 - CONFIRMED

Unique count addition: yes.

Exclude flagged E124 from primary pixel stack as report does. Catalogue positive controls succeeded; catalogue absence only within searched scope.

Catalogue scope: not in VSX/Gaia-EB/PrÅia/IJspeert within 60"

Full source report

TIC 92881948 - CONFIRMED

Unique count addition: yes.

VSX ROT/Gaia SOLAR_LIKE is prior variable classification, not prior EB; catalogue positive controls succeeded. QLP dilution explains shallower depth.

Catalogue scope: VSX lists it as ROT (Gaia DR3 variable) only; not in any EB catalogue

Full source report

TIC 468827621 - CONFIRMED

Unique count addition: yes.

Primary localized in two sectors both camera 2; secondary untested in pixels and neighbour could supply in worst case. Literature search skipped, though catalogue cones performed; known Gaia short-timescale variable is not prior EB.

Catalogue scope: not in VSX/Gaia-EB/PrÅia/IJspeert/SB9 within 60"; SB*

Full source report

TIC 416285595 - CONFIRMED, with caveats

Unique count addition: yes.

Returned CONFIRMED with caveats: QLP/SPOC s40 61.7 vs 54.7 ppt (~12%, ~4 sigma); TESScut depth unreliable. Secondary not pixel-tested; geometry weak/covariance-dependent. Catalogue cones performed but literature search skipped. Report itself says strict depth agreement would imply PROBABLE; no new stricter tolerance imposed here.

Catalogue scope: not in VSX/Gaia-EB/PrÅia/IJspeert/SB9 within 60"; SB*

Full source report

TIC 372683900 - ELLIPSOIDAL

Unique count addition: no.

Smooth double wave, no narrow eclipse; unresolved 1.8 arcsec neighbour; Gaia eccentricity conflicts at 3.3 sigma.

Catalogue scope: Enumerated catalogue cones only; SIMBAD SB* for first four; no exhaustive literature novelty claim. QLP only, one pixel sector per star.

Full source report

TIC 13106408 - ELLIPSOIDAL

Unique count addition: no.

Smooth double wave; nearby 364-day SR unrelated.

Catalogue scope: Enumerated catalogue cones only; SIMBAD SB* for first four; no exhaustive literature novelty claim. QLP only, one pixel sector per star.

Full source report

TIC 372683622 - ELLIPSOIDAL

Unique count addition: no.

Double wave plus variable spots; no eclipse.

Catalogue scope: Enumerated catalogue cones only; SIMBAD SB* for first four; no exhaustive literature novelty claim. QLP only, one pixel sector per star.

Full source report

TIC 319185391 - PLAUSIBLE

Unique count addition: no.

Fold-only ~1 ppt dip, individual events ~1 sigma, dip not pixel-tested, no second pipeline. Report prose says probable-level but batch summary says PLAUSIBLE at best; adopt batch summary outside count. Gaia 434-day astrometric orbit suggests triple.

Catalogue scope: Enumerated catalogue cones only; SIMBAD SB* for first four; no exhaustive literature novelty claim. QLP only, one pixel sector per star.

Full source report

TIC 62328710 - NEIGHBOUR_SIGNAL

Unique count addition: no.

Delta Scuti-like signal localized to Gaia DR3 2165710355333626240 at 44 arcsec and independently seen there by ZTF; target flat at Gaia period; suspect SB1 orbit is interpretation.

Catalogue scope: Enumerated catalogue cones only; SIMBAD SB* for first four; no exhaustive literature novelty claim. QLP only, one pixel sector per star.

Full source report

TIC 300903537 - UNVETTED

Unique count addition: no.

Screen-positive only; no confirmation inference.

Catalogue scope: Not adjudicated in this returned vet batch.

Full source report

TIC 23112764 - UNVETTED

Unique count addition: no.

Screen-positive only; no confirmation inference.

Catalogue scope: Not adjudicated in this returned vet batch.

Full source report

TIC 373062090 - UNVETTED

Unique count addition: no.

Screen-positive only; no confirmation inference.

Catalogue scope: Not adjudicated in this returned vet batch.

Full source report

TIC 465073693 - UNVETTED

Unique count addition: no.

Screen-positive only; no confirmation inference.

Catalogue scope: Not adjudicated in this returned vet batch.

Full source report

TIC 316405449 - UNVETTED

Unique count addition: no.

Screen-positive only; no confirmation inference.

Catalogue scope: Not adjudicated in this returned vet batch.

Full source report

TIC 465263741 - UNVETTED

Unique count addition: no.

Screen-positive only; no confirmation inference.

Catalogue scope: Not adjudicated in this returned vet batch.

Full source report

TIC 441741446 - UNVETTED

Unique count addition: no.

Screen-positive only; no confirmation inference.

Catalogue scope: Not adjudicated in this returned vet batch.

Full source report

Archived detailed period-fold analyses

vet01 full report

Archived source: sanitized report text.

vet01: adversarial check of TIC 195865529 (ATO J310.7264+43.3062) and TIC 468181994 (UCAC4 023-009594)

Opus 5.5 subagent, 2026-10-30, for Fable 5.1. A first agent ran `s1–s8a` and was killed by a container restart. This run kept its downloads and outputs, checked its logs and continued with `s8–s17`. Everything is in `/home/user/eclipses-paper-review/work/c9a-vets/vet01/`. Downloads are in the scratchpad at `data/vet01/<tic>/`; the TESScut cubes were deleted after use. Depths are in ppt, times in BTJD (BJD_TDB − 2457000), durations in hours. A = conjunction at ν = 90° − ω, when the companion is in front of Gaia's RV star. B = ν = 270° − ω.

TIC 195865529: CONFIRMED. A detached F-star + K-dwarf eclipsing binary at the Gaia period. It is already a catalogued EB in ASAS-SN (II/366, EA, P = 3.8310 d) and ATLAS (class DBH at P/2), though not in VSX. - (1) Individual events: 48 primaries and 49 secondaries were re-measured one by one from raw QLP SAP (QUALITY==0). Another 3 primaries were fitted but excluded as partial. - (2) Period: P_ecl = 3.8310456 ± 0.0000009 d, which is −0.15σ from Gaia's. - (3) Pixels: the deficit is on the target in all 8 sectors, across 4 camera/CCD orientations. Every Gaia neighbour within 90″ is excluded by the pixels. The dilution arithmetic alone excludes all but one: the G 13.46 star at 36″ would need ≈ 100%. - (4) Independence: QLP SAP, my TESScut aperture and TASOC agree after correcting for each aperture's dilution: undiluted primary 183 / 192 / 196 ppt, secondary 48 / 51 / 47 ppt. - (5) Geometry: the primary is conjunction A (A within 0.18σ, B excluded at 3.3–3.9σ). The secondary sits at phase 0.49999 ± 0.00007; Gaia's e and ω predict 0.479 ± 0.024 (+0.87σ). - The screen was right about depths (≈ 185 / 48 ppt), phases and period. Those depths are on QLP's apparently deblended scale, though. Raw apertures see the same eclipse diluted: 127–144 ppt in my r = 1.5 px aperture and 71 ppt in TASOC, whose header gives 64% contamination.

TIC 468181994: CONFIRMED. This is the weaker of the two; see the caveats. A late-F/G star with an M-dwarf-mass companion (m2 sin i ≈ 0.21–0.24 M⊙). It is not in VSX, Gaia-EB, Prša or IJspeert within 60″. - (1) Individual events: 37 primaries were re-measured individually in QLP SAP (one more was excluded as partial), plus 7 in TESScut sector 66, which QLP does not cover. The 1.3 ppt secondary was measured at fixed shape in 39 windows: 37 are positive and 28 are above 2σ. - (2) Period: P_ecl = 3.9969190 ± 0.0000006 d, which is **−2.31σ** from Gaia's 3.999219 ± 0.000996 d. This is the weakest link. - (3) Pixels: the 2.85″ neighbour (G 16.75) cannot be excluded by arithmetic (it would need 73% primary and 4.1% secondary depths). The difference images reject it: the mean centroid over 8 sectors lies 0.51″ from the target and 3.2″ from the neighbour (≈ 16σ), and the PSF χ^2 prefers the target in every sector. Every other G < 18 neighbour is excluded by arithmetic. - (4) Independence: QLP SAP, my TESScut aperture and ELEANOR-lite agree, at 21–24 ppt per sector. - (5) Geometry: the secondary sits at phase 0.4956 ± 0.0012, and Gaia predicts 0.534 ± 0.040 (A) or 0.466 ± 0.040 (B), both within 1σ. Propagated forward with Gaia's P, the primary's absolute phase does not decide between A and B (1.2σ vs 0.7σ). Propagated back with the eclipse P, A is preferred (0.01σ vs 2.2σ). - The screen was right (≈ 23 ppt, ≈ 1.5 ppt near 0.5, P 3.9969 d). The secondary is measurably *before* phase 0.5, at 0.4956. Its depth is 1.24–1.41 ppt. - Gaia shows RUWE 12.6 and ipd_frac_multi_peak = 39%: probably an unresolved sub-arcsecond companion, which TESS pixels cannot test.

1. Archives and web

- **Cones** (`s2_archives.out`, `s2b_archives.out`; the VizieR mirror was sanity-checked on Algol, which returned β Per):
- 195865529:
  - VSX (60″): only `Gaia DR3 2066587835904958592`, type YSO, 15.34–15.39 mag, at **35.76″**. That is the G 15.36 neighbour, not the target.
  - Not in Prša+2022, IJspeert+2021, Gaia DR3 VEB (I/358/veb), Gaia varisum (10″) or SB9.
  - **ASAS-SN II/366 at 0.16″: ASASSN-V J204254.34+431822.5, Type EA, Per 3.8310 d, Amp 0.12 mag, Prob 1.000.**
  - **ATLAS J/AJ/156/241 at 0.09″: class DBH (P(DBH) = 0.98), period 1.915571 d (= P/2).**
  - **So it is already a catalogued eclipsing binary**, though not in VSX.
- 468181994: no rows in VSX, J/ApJS/258/16, J/A+A/652/A120, I/358/veb, I/357/varisum, B/sb9, ATLAS or ASAS-SN II/366 (radii 10–60″, as listed in the scripts).
- The all-catalogue 5″ query returned only astrometric/photometric tables (I/...) for both stars.
- **SIMBAD (harvard mirror, 10″ cone):** both are type SB*, with 1 bibcode each: ATO J310.7264+43.3062 and UCAC4 023-009594. "TIC " is not a SIMBAD identifier for either star (`s2c_simbad_raw.out`).
- **ExoFOP:** tois = [], ctois = [] for both.
- **WebSearch** on all names: nothing mentions either star (`s15_websearch.out`).
- **Gaia DR3 extras** (`s10_gaia_extra.out`):
- 195865529: RUWE 0.94, ipd_frac_multi_peak 0, rv_amplitude_robust 147 km/s (2K1 = 132), Teff_gspphot 6608 K, FLAME R = 2.03 R⊙, M = 1.50 M⊙.
- 468181994: **RUWE 12.57, astrometric_excess_noise 1.53 mas (sig 2803), ipd_frac_multi_peak 39**, rv_amplitude_robust 52.9 km/s (2K1 = 44.7), Teff_gspphot 5775 K, logg 4.08, FLAME R = 2.91 R⊙ (FLAME uses the gspphot distance of 608 pc; the parallax distance is 1044 pc).
- The 2.85″ neighbour 6342991943947141504: G 16.75, BP−RP 1.51, parallax 1.717 ± 0.048 mas, RUWE 1.10.
- The NSS orbit rows match targets.csv. conf_spectro_period = 1.0 for both.

2. Independence, systematics, pixels

- **Pipelines available.** Neither star has TESS-SPOC FFI light curves or 2-min SPOC data (`s1_download_lc.out`).
- 195865529: QLP s15/41/55/56/75/76/82/83, plus TASOC for s15.
- 468181994: QLP s12/13/27/39/67/93/94, plus GSFC-ELEANOR-LITE for s12/13. TESScut covers s66, where QLP has nothing.
- My own photometry: an 11×11 TESScut cube per sector. Background = per-cadence median of the faintest 20% of pixels; aperture r = 1.5 px (6–8 px); QUALITY == 0 (`s8_pixels.py`).
- **Depth method.** One method for every pipeline (`s11_depths.out`): ephemeris and shape fixed, a quadratic baseline per event, and an exposure-supersampled trapezoid amplitude.

TIC 195865529 primary / secondary (ppt)	s15	s41	s55	s56	s75	s76	s82	s83
QLP SAP	182.0 / 49.2	186.2 / 48.2	182.5 / 46.7	182.2 / 46.9	184.3 / 49.2	183.8 / 48.3	182.4 / 48.4	181.9 / 47.6
TESScut ap., raw	141.2 / 38.6	134.7 / 36.0	143.1 / 37.4	127.4 / 33.5	140.9 / 38.1	143.0 / 38.8	129.6 / 34.3	143.5 / 38.0
TESScut ÷ target fraction	183.3 / 50.2	188.1 / 50.3	194.4 / 50.8	197.4 / 51.9	187.2 / 50.6	186.0 / 50.5	204.2 / 54.1	197.5 / 52.3
TASOC raw (AP_CONT 0.637)	71.3 / 16.9 → undiluted 196.2 / 46.5							

- Typical errors are 0.3–0.9 ppt.
- In `s12_dilution.out`, the target's flux fraction in my aperture is 0.635–0.770 from TIC Tmag and the fitted PSF width, and 0.585–0.803 from the fitted scene.
- Undiluted TESScut depths: mean primary 192.3 (sd 7.2), secondary 51.3 (sd 1.4), S/P 0.267. QLP: 183.2 (sd 1.5) / 48.1 (sd 0.9), S/P 0.262.
- So QLP SAP behaves as if it were deblended with the TIC contamination ratio (0.367). The TESScut and TASOC numbers agree with it to ≈ 5%, which is the accuracy of my Gaussian-PSF dilution model. Depth *ratios* agree to 2%.
- Note: the raw aperture depths of this star are not comparable across pipelines without that correction.
- **TIC 468181994 primary (ppt):**
- QLP SAP: 21.0, 22.3, 23.7, 23.2, —, 23.0, 23.3, 21.2.
- TESScut: 22.3, 21.8, 21.9, 22.3, 22.3 (s66), 22.4, 22.8, 22.0.
- ELEANOR-lite raw: 24.3 (s12), 22.7 (s13). ELEANOR-lite corr: 23.4, 22.4.
- These are sectors 12, 13, 27, 39, 66, 67, 93, 94. The target fraction in my aperture is 0.952–0.960.
- Secondary per sector: QLP 1.07–1.55, TESScut 0.96–2.28 (s67 ± 0.98), ELEANOR-lite 0.94–1.61, all with errors of 0.16–0.46.
- All pipelines agree within about 2σ.
- **Difference images** (`s8_pixels_*.out`, `s9_pixsummary.out`). Each is the median out-of-eclipse frame minus the median in-eclipse frame, primaries only. 50 primaries per star went into the stacks, covering all 8 sectors of each. A free-position Gaussian PSF was fitted, and the zero point is a scene model of the out-of-eclipse image (TIC stars with T < 17). Errors come from bootstrapping over events. Two single-event difference images per sector are also listed.
- **195865529:**
  - Per-sector offsets from the target: 1.2–4.3″. Single events: 1.0–4.3″. Sector mean (+0.20 ± 0.57, +1.37 ± 0.91)″.
  - The nearest G < 18 neighbour is 18.9″ away. The difference centroid lies 17–23″ from it (≥ 101σ in every sector) and 32–40″ from the 33–36″ group.
  - PSF hypothesis χ^2: the target is best in all 8 sectors. The next best, the 18.8″ star, is worse by $\Delta\chi^2 \geq 16{,}096$.
  - Arithmetic, as the depth each star would need: the G 13.46 star at 36.0″ has a maximum flux fraction of 18.0%, so it would need 101.6% (primary) and 26.5% (secondary). It is excluded only marginally by arithmetic, but cleanly by pixels. Every other G < 18 neighbour within 90″ would need ≥ 393%.
  - The TIC's T 18.3 star at 10.7″ (G 19.5) contributes about 0.1% and is irrelevant.
- **468181994:**

- Per-sector offsets: 0.41–1.53″. Mean (+0.50 ± 0.20, −0.12 ± 0.19)″, error from the scatter between sectors.
- The 2.85″ neighbour is at (−1.67, +2.22)″. Per sector, the difference centroid is 2.3–3.8″ from it (7–24σ); the mean is ≈ 3.2″, or ≈ 16σ.
- PSF hypothesis $\Delta\chi^2$ (neighbour minus target): +119, +190, +184, +230, +58, +14, +44, +256, so the target is preferred in all 8 sectors.
- Arithmetic: the neighbour's flux fraction is 3.0–3.1%, so it would need **73.3% / 4.1%**, which arithmetic cannot exclude. All others (44–86″) would need ≥ 293% for the primary.
- Figures: `fig_468181994.png` and `fig_195865529.png` (out-of-eclipse image, difference image, raw fold). In both, the difference image peaks on the target pixel.

- **Common mode:** not done with other QLP stars. A dip that repeats strictly on one ephemeris across 8 sectors over 6 years, and whose per-pixel difference image follows the target PSF only, is not a sector-wide systematic.
- **Blind period search** on raw SAP (`s4_period_search.out`, BLS 0.5–20 d): the best peaks are 3.83107 d and 3.99694 d. The other peaks are aliases (P/2, 2P, 3P, 4P).

3. TIC 195865529 (P 3.831 d)

Raw QLP SAP, QUALITY==0. Each event is fitted with a trapezoid times a quadratic baseline, with red-noise β-scaled errors (`s6_events.out`). Summary per sector (`s17_tables.out`); every event is listed in `events_table_195865529.md`.

Event	Mid BTJD (range)	Depth (ppt, mean / sd)	T14 / T23 (h)	pipelines
P s15 ×4 (lost: 1721.87, 1733.36, 1737.19 flagged)	1714.199–1729.524	182.0 / 0.6	5.19 / 2.18	QLP, TESScut, TASOC
P s41 ×6 (lost 2442.10)	2422.946–2445.932	186.4 / 1.1	5.23 / 2.15	QLP, TESScut
P s55 ×6 (lost 2809.88)	2798.388–2821.374	182.5 / 0.9	5.23 / 2.16	QLP, TESScut
P s56 ×7 (lost 2825.21)	2829.037–2852.023	181.7 / 1.0	5.23 / 2.18	QLP, TESScut
P s75 ×6 (partial 3346.23)	3342.396–3365.382	184.1 / 2.4	5.20 / 2.16	QLP, TESScut
P s76 ×6 (partial 3388.37)	3369.213–3392.199	183.2 / 1.5	5.22 / 2.17	QLP, TESScut
P s82 ×6 (partial 3545.44)	3533.948–3556.934	182.4 / 0.7	5.24 / 2.15	QLP, TESScut
P s83 ×7	3560.766–3583.752	181.8 / 1.0	5.20 / 2.15	QLP, TESScut
S s15 ×4 (3 flagged)	1712.285–1731.440	48.9 / 1.2	5.34 / 2.53	QLP, TESScut, TASOC
S s41 ×6, s55 ×7, s56 ×6	2421.033–2850.108	48.0, 46.4, 46.8	5.23–5.29 / 2.44–2.53	QLP, TESScut
S s75 ×8, s76 ×7, s82 ×6, s83 ×5	3340.481–3581.835	49.4, 48.0, 48.4, 47.3	5.22–5.29 / 2.42–2.53	QLP, TESScut

- **Ephemeris and O−C** (`s7_ephem.out`): 48 primaries give P = 3.8310456 ± 0.0000009 d and T0 = 3346.226971 ± 0.000140. Red. χ^2 is 9.43 (errors scaled) and the O−C rms is 1.31 min; sector means lie between −3.4 and +1.8 min.
- **Secondaries:** P = 3.8310460 ± 0.0000013 (P_sec − P_prim = +0.0000004 ± 0.0000015). Phase **0.49999 ± 0.00007**, so $e \cos \omega = -0.00002 \pm 0.00010$. The durations are equal (T14 5.22 vs 5.25 h), so $e \sin \omega \approx 0$ as well. The orbit is circular.

- **Comparison with Gaia:** P_ecl − P_Gaia = −0.000058 d = −0.15σ.
- **Odd/even primaries:**
- Raw: 182.72 ± 0.21 vs 183.04 ± 0.24 ppt. Sector-normalised: 0.9985 ± 0.0011 vs 1.0021 ± 0.0013 (−2.0σ, driven by the depth differences between sectors).
- Odd/even secondaries: 47.81 ± 0.19 vs 47.60 ± 0.18.
- A distinct 48 ppt secondary at phase 0.5 rules out 2P and P/2.

4. TIC 468181994 (P 3.997 d)

Same method (`s6_events.out`). Every primary is listed in `events_table_468181994.md`; the individual secondaries at fixed phase and shape are in `s7b_sec468.out`.

Event	Mid BTJD (range)	Depth (ppt, mean / sd)	T14 / T23 (h)	pipelines
P s12 ×4 (lost 1624.89 gap, 1628.89 and 1640.88 flagged)	1632.882–1648.870	21.1 / 0.5	3.73 / 2.40	QLP, TESScut, ELEANOR-lite
P s13 ×5 (1660.86, 1664.86 partial)	1656.865–1680.846	22.1 / 1.6	3.68 / 2.32	QLP, TESScut, ELEANOR-lite
P s27 ×5 (2048.56 gap, 2060.55 partial)	2036.572–2056.556	24.0 / 1.4	3.66 / 2.39	QLP, TESScut
P s39 ×6 (2368.32 flagged in QLP; clean in TESScut)	2364.320–2388.300	23.2 / 1.8	3.66 / 2.42	QLP, TESScut
P s66 ×7 (TESScut only)	3099.752–3123.734 (predicted)	21.5–23.2 each, 22.27 ± 0.22	shape fixed	TESScut
P s67 ×5 (3139.72 flagged, 3147.72 partial)	3127.731–3151.712	23.3 / 1.0	3.60 / 2.54	QLP, TESScut
P s93 ×6 (3843.18 flagged)	3831.188–3855.169	23.2 / 1.1	3.70 / 2.40	QLP, TESScut
P s94 ×6	3859.167–3879.151	20.8 / 1.0	3.73 / 2.43	QLP, TESScut
S, 39 windows s12–s94, phase fixed at 0.4956	1630.867–3877.136	weighted mean 1.24 ± 0.07 (χ^2r 2.54)	primary shape	QLP (TESScut and ELEANOR per sector, §2)

- **Ephemeris** (`s7_ephem.out`): 37 primaries give P = 3.9969190 ± 0.0000006 d and T0 = 2384.303949 ± 0.000146. Red. χ^2 0.75, O−C rms 1.07 min.
- **Comparison with Gaia:** P_ecl − P_Gaia = −0.002300 d = **−2.31σ**.
- **Odd/even:** raw 22.18 ± 0.16 vs 22.55 ± 0.17 ppt; sector-normalised −1.79σ.
- **Secondary:**
- Fold fit: phase 0.4953 ± 0.0006, depth 1.41 ± 0.09 ppt. With the shape free: 0.4952 ± 0.0005, 1.39 ± 0.09 ppt, T14 3.57 ± 0.21 h (`s7_ephem.out`).
- **Null boxes** at phases 0.25 and 0.75: −0.25 ± 0.09 and −0.01 ± 0.09 ppt. The formal errors therefore understate the systematics by up to ≈ 3×. Even so, the secondary stands at ≈ 5σ against the worst null.
- It is present in both halves: s12–39 phase 0.4951, 1.27 ± 0.11 ppt; s67–94 phase 0.4956, 1.25 ± 0.11 ppt.
- Per-sector phases run from 0.4881 to 0.4970: mean 0.4941, standard error 0.0012 (`s13_geometry.out`).
- **The by-eye screen did not resolve** that the secondary is ≈ 25 min early, nor that it is ≈ 1.3 ppt rather than 1.5.

Geometry vs Gaia

(`s13_geometry.out`: 20,000 draws of P, Tp, e, ω; e truncated at 0.) - **195865529** (Gaia e = 0.074 ± 0.040, ω = 235 ± 33°, K1 = 66.18): - Primary minus conjunction A: −0.029 ± 0.161 phase with Gaia's P propagated forward over 771.6 cycles; −0.016 ± 0.141 with the eclipse P propagated back. - Primary minus B: +0.496 ± 0.149 and −0.490 ± 0.127, i.e. 3.3σ and 3.9σ. - **So the primary is conjunction A:** the hotter RV star is eclipsed by the companion. - Predicted secondary phase 0.479 ± 0.024, observed 0.49999 (+0.87σ). Gaia's e cos ω = −0.033 ± 0.040 vs 0.0000 ± 0.0001 observed. Gaia's e is a 1.8σ detection, and the eclipses show the orbit is circular. - **Physics:** - f(m) = 0.1141 M⊙, so m2 = 0.84–0.86 M⊙ for M1 = 1.43–1.50, and a = 13.5–13.7 R⊙. - The flat-bottom trapezoid solution (no limb darkening) gives R1 ≈ 1.72 R⊙, R2 ≈ 0.71 R⊙, k = 0.41, ρ1 ≈ 0.46 ρ⊙. That agrees with FLAME's 2.03 R⊙; TIC's 2.83 R⊙ looks too large. - The depth ratio S/P = 0.267 equals a Planck (800 nm) surface-brightness ratio for T2 ≈ 4500 K against 6600 K. A 0.85 M⊙ dwarf would be about 5000–5200 K (ratio 0.40–0.46). This crude check ignores limb darkening and the partial-eclipse geometry; it is roughly consistent, not exact. - The light-time shift of the secondary is 16–17 s. - **468181994** (Gaia e = 0.085 ± 0.084, ω = 20 ± 49°, K1 = 22.35): - Gaia's P error alone amounts to 0.497 d over the 498.9 cycles to TESS. - Primary minus A: forward −0.286 ± 0.233 (1.23σ), back −0.001 ± 0.192 (0.01σ). Primary minus B: forward +0.163 ± 0.240 (0.68σ), back +0.447 ± 0.202 (2.2σ). - **Undecided with Gaia's own P. With the eclipse period, A is preferred at about 2σ.** - Predicted secondary phase 0.534 ± 0.040 if the primary is A, 0.466 ± 0.040 if B; observed 0.4956 ± 0.0012 (−0.96σ or +0.74σ). - Observed e cos ω ≈ −0.0069 ± 0.0019. Gaia gives +0.053 ± 0.066. - The light-time (Rømer) delay is only 39–43 s (0.0001 phase), so the ≈ 25 min offset points to a small real eccentricity. Template or systematic error at the 1.3 ppt level cannot be excluded. - **Physics:** - f(m) = 0.0046 M⊙, so m2 = 0.21–0.24 M⊙ and a = 11.9–12.7 R⊙. - The trapezoid solution gives R1 ≈ 1.47 R⊙, R2 ≈ 0.22 R⊙, k = 0.152, b ≈ 0.50, ρ1 ≈ 0.54 ρ⊙. R2 suits a 0.2 M⊙ M dwarf. - S/P = 0.062 vs a Planck ratio of 0.077 for 3200 K against 5800 K, consistent with occultation of an M dwarf. - Tension: Gaia's photometric size for the primary (FLAME 2.91 R⊙; logg 4.08 implies ≈ 1.8 R⊙) is larger than the eclipse-implied 1.47 R⊙. Light from the suspected sub-arcsecond companion would inflate the photometric radius and dilute the depths.

Ephemerides and next eclipses

(`s14_ephem_next.out`.) BJD_TDB is converted to geocentric UTC with the barycentric light-time correction **removed** (astropy, builtin ephemeris); the uncorrected times differ by ≤ 2.5 min. - **TIC 195865529:** - P = 3.8310456 ± 0.0000009 d; T0 = BTJD 3346.22697 ± 0.00014 = BJD_TDB 2460346.22697. - Next primaries (geocentric UTC): **2026-11-01 15:04, 2026-11-05 11:00, 2026-11-09 06:57** (BTJD 4346.1299, 4349.9609, 4353.7920; ± 0.4 min formal). T14 ≈ 5.2 h, depth ≈ 18% (on QLP's scale). - Secondaries: 2026-11-03 13:02, 2026-11-07 08:59, 2026-11-11 04:56. - **TIC 468181994:** - P = 3.9969190 ± 0.0000006 d; T0 = BTJD 2384.30395 ± 0.00015 = BJD_TDB 2459384.30395. - Next primaries (geocentric UTC): **2026-11-02 07:00, 2026-11-06 06:55, 2026-11-10 06:51** (BTJD 4346.7912, 4350.7881, 4354.7850; ± 0.5 min). T14 ≈ 3.7 h, depth ≈ 2.3%. - Secondaries (phase 0.4956, ± 7 min): 2026-11-04 06:32, 2026-11-08 06:28, 2026-11-12 06:24.

Caveats

- **468181994:**
- RUWE 12.6 and ipd_frac_multi_peak 39% point to a companion within about 1″. The 2.85″ neighbour is probably not the cause, since Gaia resolves it as a separate source; this is my inference,

not a measurement. TESS pixels cannot separate a sub-arcsecond companion. The case for the eclipses being on the RV star is that the eclipse period equals the SB1 period to 0.06%.
- That period agreement is only at 2.3σ given Gaia's quoted error. With a 3σ threshold this is a pass; with a 2σ threshold it would make the star PROBABLE.
- Speckle or AO imaging and a few RVs at quadrature (K1 ≈ 22 km/s) would settle it.
- **195865529:** depth comparisons across pipelines depend on my Gaussian-PSF dilution model (≈ 5%). The 36″ G 13.46 star is excluded by the pixels, not by arithmetic.
- **Not done:** the common-mode test with other stars; any light-curve model with limb darkening (the physics above is trapezoid-level). No TESS-SPOC or 2-min data exist for either star.
- **The `s8` pixel analysis cannot be re-run** without downloading the cubes again (`s3_tesscut_dl.py`), because they were deleted per the brief. Its outputs are saved in `pix/`.
- **Inherited fits:** `s6` marked four primaries and one secondary "ok" although their covariance had failed (partial coverage). `s7` and `s17` exclude them from timing; they are labelled partial above.

Files

- **Scripts and logs:**
- `s1_download_lc`, `s1b_download_other`, `s2_archives`, `s2b_archives`, `s3_tesscut_dl`, `s4_period_search`, `s5_fold_scan`, `s6_events`, `s7_ephem`, `s7b_sec468`, `s8a_tic_neighbours`: from the first agent, checked and reused.
- `s8_pixels`, `s9_pixsummary`, `s10_gaia_extra`, `s11_depths`, `s12_dilution`, `s13_geometry`, `s14_ephem_next`, `s15_websearch` (log only), `s16_plots`, `s17_tables`: this run. Each `.py` has a matching `.out`; s8 has `s8_pixels_<tic>.out`.
- **Data products:** `lcio.py`, `events_qlp.csv`, `events_table_<tic>.md`, `ephem.json`, `s11_depths.json`, `fold_<tic>.txt`, `tic_neighbours_<tic>.csv`, `pix/` (aperture light curves, difference and scene images, per-sector JSON), `fig_<tic>.png`.

TIC 195865529: CONFIRMED; P_ecl=3.8310456±0.0000009 d; depth1=183 ppt (QLP, ≈deblended; 192±7 TESScut undiluted); depth2=48 ppt at phase 0.49999±0.00007; pixel: on target; pipelines: 3 (QLP, own TESScut, TASOC); Gaia geometry: pass (primary = conjunction A at 0.1–0.2σ, B excluded at 3.3–3.9σ; circular orbit) TIC 468181994: CONFIRMED (weaker: P_ecl vs Gaia at −2.3σ; suspected sub-arcsec companion); P_ecl=3.9969190±0.0000006 d; depth1=22.5 ppt (QLP) / 23.2 (TESScut); depth2=1.24–1.41 ppt at phase 0.4956±0.0012; pixel: on target (2.85″ G16.75 neighbour rejected at ≈16σ); pipelines: 3 (QLP, own TESScut, ELEANOR-lite); Gaia geometry: pass (secondary phase within 1σ; A vs B undecided with Gaia P, A preferred at ~2σ with eclipse P)

vet02 full report

Archived source: sanitized report text.

vet02: adversarial check of TIC 470235665 (UCAC4 859-013225) and TIC 379251709 (UCAC4 788-006141)

Opus 5.5 subagent, 2026-10-30, for Fable 5.1. A container restart killed the first agent part-way through. This run picked up its outputs (`s1`–`s8`: archives, downloads, raw-SAP event fits, ephemerides, the 379251709 secondary search, Gaia geometry), checked them, and added the missing parts: `s1c` (SIMBAD bibcodes, the 5″ all-catalogue list for 379251709 that `s1b` crashed on), `s9`/`s10` (TESScut pixels and own photometry), `s11` (independence), `s12` (physics, ephemerides), `s13` (secondary-phase drift), `s14` (fold plots) and `s15` (event tables). Scripts and outputs are in this folder. Light curves are in the scratchpad `data/vet02/<tic>/`. Each TESScut cube was deleted after use. Depths are in ppt and times in BTJD (BJD_TDB − 2457000). A = the companion passes in front of Gaia's RV star (true anomaly 90° − ω). B = the companion is behind it.

TIC 470235665: CONFIRMED. Every check passed: 1. 25 primaries and 25 secondaries are each re-measured from raw SAP in QLP and TESS-SPOC, sectors 18–85. 2. The eclipse period is 10.326183 ± 0.000002 d, +0.82σ from Gaia's 10.3239 ± 0.0028 d. 3. The difference images in s19 (camera 3) and s79 (camera 4) put the deficit on the target. Its centre is 0.29–0.74″ from the out-of-eclipse centroid. Every neighbour would need an intrinsic depth above 100% even if all its flux fell in the aperture. 4. QLP SAP, TESS-SPOC SAP and my own TESScut aperture all see every event they cover. SPOC and TESScut depths agree to about 1%; QLP is 4% deeper. No 2-min data exist. 5. The primary falls on Gaia conjunction A, −0.07 ± 0.11 d away; B is excluded at 19σ. The secondary phase (0.52275) is +0.7σ from Gaia's prediction. The B/A duration ratio gives e sin ω ≥ 0.17, against Gaia's 0.21 ± 0.05.

The screen got this one right (P 10.3262 d, secondary at phase 0.52). Its depths (128/84 ppt) match QLP. TESS-SPOC SAP and TESScut give 122/81 ppt.

TIC 379251709: CONFIRMED. Every check passed: 1. 18 primaries are re-measured individually. The two in s18 that TESS-SPOC flags are recovered in QLP and in my TESScut photometry. 2. The eclipse period is 12.323912 ± 0.000007 d, −0.53σ from Gaia's 12.3261 ± 0.0042 d. 3. The primary deficit is on the target in s18 (camera 2) and s52 (camera 4), 0.10″ and 1.86 ± 0.65″ from the out-of-eclipse centroid. All neighbours are excluded for the primary even in the worst case. 4. QLP, TESS-SPOC and TESScut depths agree (34–38 ppt). 5. The primary is at conjunction A (−0.17 ± 0.22 d); B is excluded at 28σ with Gaia's correlations, 4.2σ without. The secondary phase (0.4960 ± 0.0010) is +0.8σ from the 0.471 ± 0.031 that Gaia's e and ω predict.

Weak points: - The 2.5 ppt secondary is detected only in folds: QLP S/N 8.5, TESS-SPOC S/N 3.8. It is too shallow for a pixel test or for the neighbour arithmetic. - The short, V-shaped primary implies a primary radius of about 1.3 R⊙, against the TIC's 1.88 R⊙ (see Caveats).

The screen's 31 ppt primary depth is low by about 15%: the raw-SAP depth is 34–36 ppt (TESScut 36.5–38.1). Its period and secondary phase are right.

1. Archives and web

- **VizieR cones, 60″ (`s1_archives.out`):** VSX (B/vsx/vsx), Prša+2022 (J/ApJS/258/16), IJspeert+2021 (J/A+A/652/A120), Gaia DR3 EBs (I/358/veb), Gaia DR3 variability summary (I/357/varisum), Gaia vclassre, SB9 and ASAS-SN II/366 return **0 rows for both stars**. The same

query returns 1 row for Algol, so the cones work. Neither star is a catalogued eclipsing binary: none is listed in VSX, Gaia-EB, Prša or IJspeert within 60″.

- **All-catalogue 5″ list (`s1b`, `s1c`):** only astrometric and photometric I/ catalogues come back (47 and 51 tables). The mirror's list stops alphabetically inside I/, so this query is not a completeness check; the targeted cones above are.
- **SIMBAD (`s1_archives.out`, `s1c_archives.out`):**
- 470235665 = UCAC4 859-013225, type **SB***, 1 bibcode (Gaia DR3), RV −66.6 ± 9.2 km/s.
- 379251709 = UCAC4 788-006141, type **SB***, 1 bibcode, RV −34.0 ± 4.0 km/s.
- **ExoFOP:** 0 TOIs and 0 CTOIs for both.
- **Web search:** the 2MASS, UCAC4 and TIC names return nothing relevant for either star.
- **Gaia DR3 orbits (`gaia_nss_tap.csv`):**
- 470235665: SB1, P 10.32387 ± 0.00280 d, e 0.210 ± 0.051, ω 90.3 ± 13.7°, K1 47.5 ± 3.8 km/s, flags 8192.
- 379251709: SB1, P 12.32612 ± 0.00417 d, e 0.097 ± 0.049, ω 122.6 ± 26.8°, K1 23.0 ± 1.1 km/s.
- **TIC stellar parameters (`s1c_archives.out`):**
- 470235665: contamination ratio 0.059; Teff 5353 K, R 2.30 R⊙, log g 3.68.
- 379251709: contamination ratio 0.041; Teff 5279 K, R 1.88 R⊙, log g 3.85.

2. Independence, systematics, pixels

- **Pipelines (`s11_independence.out`, raw SAP, QUALITY==0, trapezoid on a quadratic baseline):**
- 470235665 primary: QLP 127.4 ppt (sd 2.6, N = 23); TESS-SPOC 122.4 (sd 2.0, N = 15); TESScut r ≤ 1.5 px aperture 122.1 ± 0.3 (s19) and 123.3 ± 1.0 (s79).
- 470235665 secondary: QLP 84.0 (sd 1.6); TESS-SPOC 80.9 (sd 1.5); TESScut 80.8 ± 0.3 and 80.9 ± 0.7.
- QLP/SPOC depth ratio over the shared sectors: 1.040 ± 0.007 (primary) and 1.038 ± 0.006 (secondary). This is an aperture/background systematic, much larger than the 0.5 ppt formal errors. TESS-SPOC and TESScut agree within 1%.
- 379251709 primary: QLP 34.2 (sd 3.1, N = 17, including the noisy s24/s25); TESS-SPOC 36.3 (sd 1.4, N = 4); TESScut 38.1 ± 0.8 (s18) and 36.5 ± 0.1 (s52).
- T14 agrees between pipelines: 6.44 h (QLP) vs 6.45 h (SPOC) for the 470235665 primary, 9.02 vs 9.01 h for its secondary, and 4.05 vs 4.22 h for the 379251709 primary.
- Neither star has 2-min SPOC data.
- **Common mode (`s10_offtarget.out`):** this uses off-target pixels inside each cutout instead of other QLP stars.
- Summed over pixels more than 3 px from the target, the stacked primary deficit is −1.1 to +3.2 ppt, against 120 or 33–38 ppt on target.
- The brightest off-target pixels are stars beyond 90″ at the cutout edges, some brighter than the targets. They show −4.2 to +2.2 ppt.
- The eclipses also repeat on a strict period over 6 years, in three cameras and two cadence regimes. A sector-wide systematic cannot do that.
- **Census of non-eclipse dips (`s4_census.out`):**
- Apart from the eclipses, the 4–9 ppt features in raw SAP sit within about 0.3 d of gaps.
- TESS-SPOC s18 starts with a ramp (27–54 ppt, 1790.7–1791.5) **in both stars**, so it is a systematic.

- 379251709 QLP s24 is dominated by scattered light (σ30 = 8.1 ppt). Its events are kept but carry large errors.
- **Difference images (`s9_tesscut_<tic>_s<sec>.txt`; median in-eclipse minus median local out-of-eclipse, per event, averaged; Gaussian-fit centres; 21″/px):**

Star / sector (camera)	kind	N ev	deficit – OOT centroid	deficit – target WCS	frac. depth on target pixel
470235665 s19 (cam 3)	P / S	3 / 2	0.29″ (±0.37) / 0.28″ (±0.14)	0.96″ / 0.96″	122.9 / 76.6 ppt
470235665 s79 (cam 4)	P / S	3 / 3	0.33″ (±0.22) / 0.74″ (±0.26)	1.43″ / 1.58″	120.8 / 77.7 ppt
379251709 s18 (cam 2)	P	2	1.86″ (±0.65)	1.30″	33.2 ppt
379251709 s52 (cam 4)	P	2	0.10″ (±0.46)	1.32″	31.3 ppt

The 379251709 secondary difference images (s18: 2 events; s52: 1 event at the sector start) are noise. No pixel location is claimed for the secondary. - **Dilution arithmetic** (aperture r ≤ 1.5 px; fluxes from Gaia G). The "worst case" puts 100% of the neighbour's flux in the aperture. - **470235665:** the nearest neighbour is G 16.39 at 5.5″, on the target pixel. It would need an intrinsic depth of 320–356% for the primary and 210–234% for the secondary (PRF and worst-case values, both sectors). G 16.19 at 57″ needs at least 269% / 176%; G 15.51 at 69″ needs at least 149% / 98%. That last 98% (s79 secondary) is the only worst case below 100%. That star lies 70″ from the deficit centre, the fractional depth at its pixel is 2.8–8.6 ppt, and it is excluded for the primary. All neighbours are **excluded**. - **379251709 primary:** the brightest neighbour is G 16.57 at 67″ and needs at least 109–128% in the worst case. The nearest is G 17.33 at 30″ and needs at least 216–253%. All neighbours are **excluded**. - **379251709 secondary:** with a 2.5 ppt depth, any neighbour with a worst-case share above about 2.5% could produce it arithmetically. This is not excluded by pixels. It rests on timing: the dip sits exactly at the occultation phase the primary's e cos ω demands. - ZTF was not needed: for the primaries, no neighbour is arithmetically viable.

3. TIC 470235665 (P 10.3262 d)

Raw SAP, QUALITY==0, supersampled trapezoid on a local quadratic baseline, errors max(covariance, prayer-bead) (`events_470235665.csv`, `s5_events.py`). "TESScut" is my own aperture photometry (`s9`) with the shape fixed.

Event	Sector	Mid BTJD	Depth QLP / SPOC / TESScut (ppt)	T14 / T23 (h)	pipelines
P E0	18	1795.7429 ± 0.0003	131.2 / 124.2 / –	6.44 / 2.51	QLP+SPOC
P E1	18	1806.0694 ± 0.0003	128.0 / 121.8 / –	6.45 / 2.60	QLP+SPOC
P E2	19	1816.3945 (TESScut)	– / – / 122.5	(fixed)	TESScut (QLP lost, SPOC partial)
P E3	19	1826.7204 ± 0.0003	130.4 / 124.3 / 122.5	6.45 / 2.44	QLP+SPOC+TESScut
P E4	19	1837.0465 ± 0.0003	130.1 / 123.2 / 121.5	6.45 / 2.46	QLP+SPOC+TESScut
P E16	24	1960.9600 ± 0.0004	123.9 / 123.0 / –	6.42 / 2.46	QLP+SPOC
P E17	24	1971.2876 ± 0.0004	126.1 / 125.2 / –	6.38 / 2.43	QLP+SPOC
P E18	24	1981.6139 ± 0.0003	124.7 / 123.7 / –	6.46 / 2.40	QLP+SPOC
P E19	25	1991.9398 ± 0.0004	– / 121.7 / –	6.48 / 2.49	SPOC (QLP partial)
P E20	25	2002.2652 ± 0.0004	– / 125.9 / –	6.49 / 2.33	SPOC (QLP partial)
P E21	26	2012.5925 ± 0.0004	126.5 / 120.5 / –	6.47 / 2.44	QLP+SPOC
P E23	26	2033.2442 ± 0.0003	126.5 / 120.0 / –	6.40 / 2.48	QLP+SPOC
P E90	52	2725.0982 ± 0.0003	128.1 / 122.5 / –	6.44 / 2.40	QLP+SPOC
P E91	52	2735.4253 ± 0.0003	122.1 / 119.4 / –	6.48 / 2.52	QLP+SPOC
P E92	53	2745.7508 ± 0.0003	124.8 / 120.3 / –	6.49 / 2.38	QLP+SPOC
P E94	53	2766.4038 ± 0.0002	125.6 / 120.6 / –	6.42 / 2.52	QLP+SPOC
P E106	58	2890.3180 ± 0.0003	130.6 / – / –	6.43 / 2.48	QLP
P E107	58	2900.6438 ± 0.0003	131.6 / – / –	6.46 / 2.43	QLP

Event	Sector	Mid BTJD	Depth QLP / SPOC / TESScut (ppt)	T14 / T23 (h)	pipelines
P E145	73	3293.0389 ± 0.0003	129.6 / – / –	6.44 / 2.43	QLP
P E146	73	3303.3666 ± 0.0004	126.2 / – / –	6.44 / 2.42	QLP
P E160	78	3447.9313 ± 0.0003	124.3 / – / –	6.43 / 2.43	QLP
P E161	79	3458.2570 ± 0.0004	126.7 / – / 121.5	6.46 / 2.44	QLP+TESScut
P E162	79	3468.5833 ± 0.0003	129.9 / – / 124.8	6.44 / 2.40	QLP+TESScut
P E163	79	3478.9099 ± 0.0003	128.3 / – / 123.7	6.43 / 2.51	QLP+TESScut
P E176	85	3613.1506 ± 0.0003	129.7 / – / –	6.48 / 2.43	QLP
P E178	85	3633.8032 ± 0.0003	125.7 / – / –	6.45 / 2.50	QLP
S E1	18	1811.4664 ± 0.0005	85.7 / 81.7 / –	8.98 / 2.79	QLP+SPOC
S E2	19	1821.7930 ± 0.0006	84.4 / 81.7 / 80.5	9.06 / 2.77	QLP+SPOC+TESScut
S E3	19	1832.1197 ± 0.0005	86.7 / 82.3 / 81.0	8.95 / 2.79	QLP+SPOC+TESScut
S E16	24	1966.3596 ± 0.0008	82.5 / 81.8 / –	9.09 / 2.78	QLP+SPOC
S E17	24	1976.6851 ± 0.0005	83.2 / 82.6 / –	8.97 / 2.87	QLP+SPOC
S E18	25	1987.0117 ± 0.0006	86.9 / 82.8 / –	8.98 / 2.91	QLP+SPOC
S E19	25	1997.3387 ± 0.0008	85.7 / 81.8 / –	9.03 / 2.64	QLP+SPOC
S E20	25	2007.6649 ± 0.0006	– / 82.3 / –	8.97 / 2.69	SPOC (QLP partial)
S E21	26	2017.9920 ± 0.0006	82.4 / 78.8 / –	8.96 / 2.88	QLP+SPOC
S E22	26	2028.3179 ± 0.0005	82.8 / 79.1 / –	9.01 / 2.83	QLP+SPOC
S E89	52	2720.1718 ± 0.0007	– / 79.8 / –	9.02 / 2.78	SPOC (QLP lost)
S E91	52	2740.8227 ± 0.0005	83.7 / 80.3 / –	9.06 / 2.82	QLP+SPOC
S E92	53	2751.1493 ± 0.0009	81.6 / 79.2 / –	9.13 / 2.68	QLP+SPOC
S E93	53	2761.4760 ± 0.0006	81.1 / 79.1 / –	8.99 / 2.76	QLP+SPOC
S E105	58	2885.3895 ± 0.0005	84.5 / – / –	9.04 / 2.72	QLP
S E106	58	2895.7163 ± 0.0005	86.8 / – / –	9.16 / 2.83	QLP
S E107	58	2906.0419 ± 0.0005	84.9 / – / –	9.06 / 2.77	QLP
S E145	73	3298.4358 ± 0.0005	84.0 / – / –	8.97 / 2.96	QLP
S E146	73	3308.7628 ± 0.0005	83.7 / – / –	8.94 / 2.79	QLP
S E159	78	3443.0031 ± 0.0006	82.0 / – / –	9.01 / 2.88	QLP
S E160	79	3453.3286 ± 0.0005	82.6 / – / 79.7	8.95 / 2.84	QLP+TESScut
S E161	79	3463.6543 ± 0.0006	83.5 / – / 80.7	9.09 / 2.82	QLP+TESScut
S E162	79	3473.9798 ± 0.0006	84.7 / – / 82.3	9.02 / 2.77	QLP+TESScut
S E176	85	3618.5472 ± 0.0005	84.5 / – / –	8.94 / 2.88	QLP
S E177	85	3628.8739 ± 0.0008	84.6 / – / –	9.01 / 2.72	QLP

Not measured (gap / flagged / too little coverage): pri E22 (s26, pred 2022.918): QLP lost, SPOC lost; pri E93 (s53, pred 2756.079): QLP lost, SPOC lost; pri E159 (s78, pred 3437.608): QLP partial; pri E177 (s85, pred 3623.479): QLP lost; sec E-1 (s18, pred 1790.822): SPOC partial; sec E0 (s18, pred 1801.148): QLP lost, SPOC lost; sec E15 (s24, pred 1956.041): QLP partial, SPOC partial; sec E90 (s52, pred 2730.506): QLP lost, SPOC lost; sec E144 (s73, pred 3288.121): QLP partial

Period (`s6_period.out`): - Primaries, one value per epoch: N = 25, **P = 10.326183 ± 0.000002 d**, T0 = 1795.74233 ± 0.00018, O–C rms 0.9 min (reduced χ^2 3.5; errors scaled). - Fitting QLP alone and TESS-SPOC alone gives the same P (10.326182). - Secondaries: N = 25, P = 10.326173 ± 0.000002 d. - Joint fit: P = 10.326180 ± 0.000001 d, secondary phase **0.52275 ± 0.00002**. - Gaia P differs by **+0.82σ**.

Odd/even: primary QLP 127.66 ± 0.75 vs 128.24 ± 0.85 ppt (−0.5σ); primary SPOC +1.0σ; secondary +0.02σ / +0.28σ. The two eclipse kinds differ in depth (127 vs 84 ppt) and duration (6.44 vs 9.02 h), so the true period is not twice this one, and the odd/even test rules out a stacked P2.

Shape (`s14_fold_470235665.png`): both eclipses have rounded bottoms. The trapezoid "T23" (2.45 / 2.81 h) is not a resolved flat bottom, so totality is not established.

Secondary-phase drift (`s13_secphase_470.out`): P_sec − P_pri = (−10.0 ± 2.8) × 10^{-6} d (−3.5σ), which would be −2.6 min over 178 cycles. In QLP the secondary phase goes 0.52284 → 0.52277 → 0.52269 across 2019–20, 2022 and 2022–24. TESS-SPOC shows no change between 2019–20 and 2022 (0.52281 vs 0.52280). This hints at apsidal motion or a cadence effect. It is not claimed.

4. TIC 379251709 (P 12.3239 d)

Event	Sector	Mid BTJD	Depth QLP / SPOC / TESScut (ppt)	T14 / T23 (h)	pipelines
P E-13	18	1801.0439 ± 0.0014	35.1 / – / 39.0	4.24 / 1.16	QLP+TESScut (SPOC lost)
P E-12	18	1813.3689 ± 0.0011	35.2 / – / 37.3	4.13 / 0.89	QLP+TESScut (SPOC lost)
P E0	24	1961.2533 ± 0.0009	– / 36.7 / –	4.20 / 1.04	SPOC (QLP noisy, e_tc>0.005 d)
P E1	24	1973.5750 ± 0.0010	35.8 / 35.2 / –	4.24 / 1.06	QLP+SPOC
P E2	25	1985.9080 ± 0.0028	29.7 / – / –	4.31 / 1.53	QLP
P E3	25	1998.2305 ± 0.0043	27.7 / – / –	2.91 / 0.58	QLP (SPOC lost)
P E62	52	2725.3357 ± 0.0008	33.2 / 35.2 / 36.6	4.17 / 1.00	QLP+SPOC+TESScut
P E63	52	2737.6604 ± 0.0008	36.8 / 38.2 / 36.4	4.27 / 0.67	QLP+SPOC+TESScut
P E75	58	2885.5465 ± 0.0008	34.5 / – / –	4.04 / 1.03	QLP
P E76	58	2897.8705 ± 0.0009	34.7 / – / –	4.10 / 0.87	QLP
P E78	59	2922.5198 ± 0.0009	33.7 / – / –	4.13 / 1.07	QLP
P E79	59	2934.8432 ± 0.0009	34.2 / – / –	4.13 / 0.85	QLP
P E120	78	3440.1238 ± 0.0009	36.1 / – / –	4.09 / 0.99	QLP
P E122	79	3464.7714 ± 0.0009	39.9 / – / –	4.28 / 0.32	QLP
P E123	79	3477.0951 ± 0.0013	34.1 / – / –	4.20 / 1.07	QLP
P E134	85	3612.6582 ± 0.0010	38.4 / – / –	4.22 / 0.86	QLP
P E136	86	3637.3031 ± 0.0011	32.1 / – / –	3.89 / 1.15	QLP
P E138	86	3661.9554 ± 0.0010	30.2 / – / –	3.82 / 1.28	QLP

Not measured (gap / flagged / too little coverage): pri E77 (s58, pred 2910.177): QLP lost; pri E121 (s79, pred 3452.429): QLP lost; pri E135 (s85, pred 3624.963): QLP lost; pri E137 (s86, pred 3649.611): QLP lost

Period (`s6_period.out`): - One value per epoch: N = 18, **P = 12.323912 ± 0.000007 d**, T0 = 1961.25375 ± 0.00060, O−C rms 3.1 min. The largest residuals are +9.2 and +7.2 min, in the noisy QLP s25. - QLP alone: 12.323905 ± 0.000009 d. TESS-SPOC alone (N = 4): 12.323931 ± 0.000018 d. - Gaia P differs by **−0.53σ**.

Odd/even: QLP 33.87 ± 1.04 vs 34.73 ± 1.27 ppt (−0.5σ); SPOC −0.55σ.

Shape: V-shaped, T14 4.1–4.2 h, trapezoid T23 about 1.0 h.

Secondary (`s7_secondary_379.out`; primaries masked, 1.5-d running median, box scan over phase): - QLP, excluding s24/s25: the strongest peak in the scan is at phase 0.4955. A trapezoid fit gives **phase 0.49597 ± 0.00035, depth 2.47 ± 0.29 ppt (S/N 8.5), T14 4.50 ± 0.43 h**. - TESS-SPOC: 0.49741 ± 0.00040, 2.59 ± 0.68 ppt (S/N 3.8), 2.6σ apart in phase from QLP. The SPOC scan's top peak (phase 0.2235) is the s18 start-of-sector ramp (`s7b_otherpeaks_379.out`). - Individual events (excluding the noisy QLP s24): 1.1–4.1 ppt at S/N 1.6–7.3. Only s59 E78 (4.12 ± 0.56 ppt) is individually significant.

Geometry vs Gaia

`s8_geometry.out`: Monte Carlo with 20000 draws using Gaia's corr_vec. Parameter order is assumed to be P, Vcm, K1, e, ω, Tp. The column-major upper triangle gives a positive-definite matrix with ω–Tp = 0.98–0.99; the row-major one does not. Uncorrelated draws are shown in brackets. `s12_physics_ephem.out` repeats the calculation with nominal values as an independent check.

- **470235665:**
- Taking the TESS ephemeris back to the Gaia epoch, the primary is −0.068 ± 0.108 d from **conjunction A** [−0.080 ± 0.427] and +5.09 d from B (+19σ [+7.5σ]). Taking Gaia's ephemeris forward to TESS gives +0.245 ± 0.405 d from A. The nominal check in `s12` gives −0.079 d.
- Because ω ≈ 90°, conjunction A falls at periastron. The primary is the companion transiting the RV star.
- Gaia predicts the secondary at phase 0.4994 ± 0.0329; measured 0.52275, **+0.71σ**. TESS-only e cos ω ≈ +0.036 vs Gaia −0.001 ± 0.051.
- T14(B)/T14(A) = 1.397 gives e sin ω ≥ +0.166, against Gaia +0.205 ± 0.050. The eclipse at periastron (A) is the shorter one, as the orbit requires.
- **379251709:**
- Backward: the primary is −0.170 ± 0.224 d from **conjunction A** [−0.189 ± 1.218] and −5.96 d from B (−28σ [−4.2σ]). Forward: −0.473 ± 0.418 d from A.
- Gaia predicts the secondary at 0.4711 ± 0.0314; measured 0.4960 ± 0.0010, **+0.79σ**. TESS e cos ω ≈ −0.006 vs Gaia −0.045 ± 0.049.
- Duration ratio 1.066 gives e sin ω ≥ 0.03 vs Gaia +0.068 ± 0.041. This test is weak.
- **Physics (`s12_physics_ephem.out`; rough two-sphere model without limb darkening):**
- **470235665:**
  - f(m) = 0.107 M⊙, so m2 = 0.64–0.77 M⊙ for M1 = 0.93–1.3 M⊙ and a = 23.2–25.4 R⊙.
  - The depths imply k ≈ 0.37 and a companion light share ≥ 8%. The two T14 values with Gaia's e and ω give R1/a = 0.079 and b_A ≈ 0.45, so R1 = 1.8–2.0 R⊙ (TIC 2.30), R2 = 0.67–0.73 R⊙ and i ≈ 87°.
  - J ≈ 0.66, so T2 ≈ 4800 K.
  - A subgiant with a K-dwarf companion: radius, temperature and mass agree with the mass function to about 15–20%. b_B ≈ 0.70 > 1 − k says the B eclipse is partial, which fits the rounded secondary.
- **379251709:**
  - f(m) = 0.0153 M⊙, so m2 = 0.28–0.35 M⊙ (an M dwarf, R2 ≈ 0.3–0.4 R⊙).
  - Read as a non-grazing transit (T23/T14 = 0.22), the shape needs b ≈ 0.77 and **R1 ≈ 1.2–1.5 R⊙, R2 ≈ 0.25–0.30 R⊙**, which is consistent with m2. With the TIC's R1 = 1.88 R⊙ the 4.2 h eclipse would have to be grazing (b ≈ 0.99), and that is hard to square with a 36 ppt depth.
  - A 3200–3500 K companion would give a full-occultation depth of 4–8 ppt; the measured secondary is about 2.5 ppt. Consistent.

Ephemerides and next eclipses

From `s12_physics_ephem.out`. The barycentric light-time correction (+2.8 to +5.2 min) was removed to give UTC at the geocentre. Errors are formal, from T0 and P. - **TIC 470235665 primary:** T0 = BTJD 1795.74233 ± 0.00018 (BJD_TDB 2458795.74233), **P = 10.326183 ± 0.000002 d**, T14 6.45 h. - Next primaries: BTJD 4346.3095 = **2026-11-01 19:21 UTC**; 4356.6357 = **2026-11-12 03:11**; 4366.9619 =

2026-11-22 11:00 (±0.8 min). - Secondary: T0 = 1801.14120 ± 0.00023, P = 10.326173 ± 0.000002 d, T14 9.0 h. Next: 2026-11-07 04:52, 2026-11-17 12:42, 2026-11-27 20:31 UTC. - **TIC 379251709 primary:** T0 = BTJD 1961.25375 ± 0.00060 (BJD_TDB 2458961.25375), **P = 12.323912 ± 0.000007 d**, T14 4.2 h; secondary at phase 0.4960 ± 0.0010. - Next primaries: BTJD 4352.0927 = **2026-11-07 14:07 UTC**; 4364.4166 = **2026-11-19 21:53**; 4376.7405 = **2026-12-02 05:40** (±2.1 min).

Caveats

- **SAP depths depend on the pipeline at the 4% level** (QLP deeper) and none are corrected for crowding (TIC contamination 4–6%). "Agreement" here means within about 5%, not within the formal errors.
- **The 379251709 secondary is a fold-level detection** (QLP S/N 8.5, SPOC S/N 3.8, phases 2.6σ apart). It has not been located in pixels or excluded from neighbours arithmetically. Its exact match to the e cos ω phase is what ties it to the target.
- **379251709 radius tension:** the eclipse shape prefers R1 ≈ 1.3 R⊙, about 30% below the TIC's 1.88 R⊙, unless the eclipse is grazing. This is not a failed check, but the TIC Teff/radius deserves a look. A spectrum, or a proper fit with limb darkening on the 200-s data, would settle it.
- **470235665:** the secondary-phase drift (−2.6 min over 5 years, 3.5σ, seen in QLP only) is unconfirmed. The rough physical model is crude and does not decide whether either eclipse is total.
- **Gaia geometry:** the forward predictions (±0.4 d) are limited by Gaia's P error. The backward test (±0.1–0.2 d) assumes the corr_vec ordering stated above; without correlations the errors grow to ±0.4–1.2 d and B is still rejected at 4.2σ and 7.5σ.
- **TESScut times** are not barycentred to the target, which puts them up to about ±3 min off (O−C −3.1 to +3.0 min). They are not used in any ephemeris.
- **Not done:** the QLP-neighbour common-mode test; TESScut-only sectors 60 and 86 of 470235665, which no pipeline covers; ZTF.

Files

- Scripts and outputs are in this folder. **From the first agent** (checked here, not re-run):
- `s1_archives.py/.out`, `s1b_archives.py/.out` (crashed for 379251709)
- `s2_download.py/.out`, `s3_load.py/.out` (per-star `lc_sap_q0.npz` in scratch)
- `s4_census.py/.out`, `s5_events.py` → `events_<tic>.csv`
- `s6_period.py/.out`
- `s7_secondary_379.py/.out/.txt`, `s7b_otherpeaks_379.py/.out`
- `s8a_gaia_tap.py/.out` → `gaia_nss_tap.csv`, `s8_geometry.py/.out/.txt`
- **Added in this run:**
- `s1c_archives.py/.out`
- `s9_tesscut.py` → `s9_tesscut_<tic>_s<sec>.txt/.npz/.png` (470235665 s19, s79; 379251709 s18, s52)
- `s10_offtarget.py/.out`
- `s11_independence.py/.out`
- `s12_physics_ephem.py/.out`
- `s13_secphase_470.py/.out`
- `s14_foldplot.py` → `s14_fold_<tic>.png`
- `s15_event_table.py/.out`

- **Data:** `/tmp/claude-0/-home-user-eclipses-paper-review/55c92ff4-798a-544f-be03-9888c6804905/scratchpad/data/vet02/<tic>/` (QLP and TESS-SPOC FITS; TESScut cubes deleted).

Summary - TIC 470235665: CONFIRMED; P_ecl=10.326183 d; depth1=122–127 ppt; depth2=81–84 ppt at phase 0.52275; pixel: on target; pipelines: 3; Gaia geometry: pass - TIC 379251709: CONFIRMED; P_ecl=12.323912 d; depth1=34–38 ppt; depth2=2.5 ppt at phase 0.4960; pixel: on target (primary; secondary untested); pipelines: 3; Gaia geometry: pass

vet03 full report

Archived source: sanitized report text.

vet03: adversarial check of TIC 70665829 (UCAC4 537-050325) and TIC 331293467 (UCAC4 150-142995)

Opus 5.5 subagent, 2026-10-30, for Fable 5.1. This run resumed from a predecessor that was killed by a container restart. The predecessor wrote `s1–s7` and their outputs. Its pixel run for 331293467 stopped part-way through s100 (`s7_pixels_331293467.out`). I checked those outputs, re-ran the pixel analysis for 331293467 in all four sectors (`s7b`), and wrote `s9–s14` plus `s2c`, `s10a`, `s12b`. All scripts, `.out`/`.json` outputs and the `pix_*.png` plots are in this folder. Downloads are under `/tmp/claude-0/-home-user-eclipses-paper-review/55c92ff4-798a-544f-be03-9888c6804905/scratchpad/data/vet03/<tic>/`; the TESScut cubes have been deleted. Notation: **A** = the companion passes in front of Gaia's RV star (true anomaly ν = 90° − ω). **B** = the RV star passes in front of the companion. Depths are in ppt of the raw (undetrended) flux unless marked "undiluted".

TIC 70665829: CONFIRMED. A K dwarf with an early-M companion, P = 3.4657654 d. - **(1) Events, pass:** 24 primaries and 28 secondaries (SPOC), and 22 and 28 (QLP), measured one by one from SAP QUALITY==0 flux. - **(2) Period, pass:** the eclipse period matches Gaia's at −0.43σ. - **(3) Pixels, pass:** the difference images in s46, s48 and s72 (three cameras) sit 0.5–1.2″ from the target. The only neighbour (G 14.44 at 64″) would need an intrinsic depth above 1500%. - **(4) Independence, pass:** QLP, TESS-SPOC, my TESScut aperture photometry and ZTF g/r all see both eclipses. Absolute depths differ by up to ~10% between apertures. The secondary/primary depth ratio is 0.216–0.233 in every pipeline, so the differences are dilution or background, not the signal. - **(5) Geometry, pass:** the secondary falls at phase 0.46048, which is e cos ω = −0.0621. Gaia predicts 0.4547 +0.024/−0.024 if the deep eclipse is A (+0.24σ), and needs 3.6–5.6σ if it is B. The deep eclipse coincides with conjunction A (+0.4 ± 0.5 h when Gaia's orbit is conditioned on the TESS period). So the deep event is Gaia's RV star (the K dwarf) being eclipsed. ZTF colours agree: the deep eclipse is deeper in g, the shallow one deeper in r. - **Screen:** correct on all points (deep ~300–330 ppt at phase 0, ~65–72 ppt secondary at 0.4605, flat baseline, P 3.4658 d).

TIC 331293467: CONFIRMED. An A/F star with a K-dwarf companion, P = 5.2568199 d. The light is heavily blended, so the undiluted depths are uncertain. - **(1) Events, pass:** 21 primaries and 24 secondaries measured individually from QLP SAP QUALITY==0. Every secondary is above 3σ, and null phases 0.25/0.75 give −0.11 ± 0.16 and −0.09 ± 0.15 ppt. - **(2) Period, pass:** +0.51σ from Gaia. - **(3) Pixels and dilution, pass:** the diff-image source sits 0.3–3.6″ from the target in s38, s99, s100 and s101 (cameras 2, 4, 3, 2). - The G 17.07 star at 8″ needs 305–458%, the G 16.07 at 24″ needs 143–287%, and the G 16.14 at 14″ needs 91–115% (over 100% in s99). All are arithmetically excluded or effectively

so. - The only neighbour that passes the arithmetic, G 13.78 at 30.5″ (needs 25–84%), is excluded by position. The fit is 10–19× worse at its position, and the measured offset points 0.00–0.11 of the way toward it. - ZTF has no data at Dec −60. No higher-resolution light curve was available. - **(4) Independence, pass in dilution-free form:** no TESS-SPOC or SPOC light curve exists. My TESScut 3×3 photometry sees every eclipse with the same T14 (4.05–4.08 h against QLP 4.07 h) and a secondary/primary ratio of 0.168–0.202 against QLP 0.160–0.176. Raw depths differ by aperture (QLP 61–73, TESScut 47–56 ppt). The undiluted primary depth is model-dependent: 80–89 ppt from PSF amplitudes, 80–116 ppt from aperture shares. - **(5) Geometry, pass:** the secondary is at phase 0.50013 ± 0.00023 (Gaia predicts 0.5015, so this cannot tell A from B). The deep eclipse sits on Gaia's conjunction A: +0.1 ± 1.0 h when conditioned on the TESS period, and +0.49σ on Gaia alone with correlations. B is excluded at 6.7σ (Gaia alone) or ≫10σ (conditioned). This test needs Gaia's ω–T_p correlation (0.9997); with independent errors it does not decide. Landing on the target's own spectroscopic conjunction is further evidence that the signal is on the target. - **Screen:** the ~66 ppt and ~11 ppt figures are right for QLP SAP. The undiluted target depths are about 8–11% and 1.3–2%.

1. Archives and web

- **Catalogue cones (`s2_archives.py`, 60″, VizieR mirror):**
- 70665829: nothing in VSX, Prsa+2022, IJspeert+2021, Gaia DR3 VEB, Gaia varisum or SB9.
- 331293467: the same, except one VSX entry at 33.7″: Gaia DR3 5866907631080924672, type L, G 18.4–18.63. It is not our star and is fainter than the 90″ neighbour list (G < 18).
- Neither star is a catalogued eclipsing binary within 60″ in these catalogues.
- **Sanity checks:** the VSX cone was checked on Algol (1 row at 0.11″, `s2_archives.out`). `s2c` showed the mirror serves Prsa (nearest EB 1621″ from 331293467) and IJspeert (nearest 2860″ from 70665829). The mirror then began timing out, so Gaia-VEB, varisum and SB9 were not sanity-checked (`s2c_catalogue_sanity.out`). The all-catalogue 5″ query (`s2b`) returned only I/ catalogues, even for Algol, so it is truncated and was not relied on.
- **SIMBAD (10″):**
- 70665829 = UCAC4 537-050325, type SB*, 3 bibcodes (2023A&A...674A..34G, 2022AJ....163..152S, 2022ApJS..259...35A).
- 331293467 = UCAC4 150-142995, type SB*, 1 bibcode (2023A&A...674A..34G).
- **ExoFOP:** no TOIs or CTOIs for either star (`s2b_allcat.out`).
- **Gaia NSS rows (`s10a_gaia_nss.csv`):** both are SB1 with flags 0. corr(ω, T_p) is 0.9984 for 70665829 and 0.9997 for 331293467.

2. Independence, systematics, pixels

- **Pipelines (`s1_download_lc.out`):**
- 70665829: QLP s45, s46, s48, s72; TESS-SPOC s45, s46, s48, s72 and "s1751". MAST labels that last product sector 1751 (camera 1, BTJD 4055.76–4063.00, 1480/3131 good cadences); it is used as given. No 2-min SPOC.
- 331293467: QLP only (s11, s38, s99–s102). No TESS-SPOC and no 2-min data.
- **Per-sector depths, primary / secondary in ppt (`s9_indep.out`):**

TIC / sector	QLP SAP	TESS-SPOC SAP	TESScut 3×3 raw	TESScut / target share	PSF amp / target flux	d2/d1 (QLP, SPOC, TESScut)
70665829 s45	325.3 / 72.3	300.3 / 67.3	—	—	—	0.222, 0.224, —

TIC / sector	QLP SAP	TESS-SPOC SAP	TESScut 3×3 raw	TESScut / target share	PSF amp / target flux	d2/d1 (QLP, SPOC, TESScut)
70665829 s46	306.6 / 66.1	296.2 / 66.6	297.6 / 67.2	325.3 / 73.4 (0.915)	287.5 / 65.8	0.216, 0.225, 0.226
70665829 s48	312.2 / 70.6	282.7 / 64.1	288.1 / 67.1	285.2 / 66.4 (1.010)	264.6 / 60.4	0.226, 0.227, 0.233
70665829 s72	329.8 / 72.2	299.0 / 67.9	300.6 / 66.2	340.8 / 75.0 (0.882)	290.9 / 64.1	0.219, 0.227, 0.220
331293467 s11	61.4 / 9.8	—	—	—	—	0.160
331293467 s38	64.7 / 10.6	—	46.7 / 9.4	80.5 / 16.3 (0.580)	79.6 / 14.2	0.163, —, 0.202
331293467 s99	72.6 / 12.0	—	50.4 / 9.2	115.8 / 21.1 (0.435)	89.3 / 14.6	0.165, —, 0.182
331293467 s100	70.1 / 11.7	—	51.3 / 8.6	99.2 / 16.6 (0.517)	87.5 / 12.7	0.166, —, 0.168
331293467 s101	61.4 / 10.7	—	55.5 / 9.6	106.7 / 18.5 (0.520)	81.5 / 12.1	0.174, —, 0.173
331293467 s102	70.5 / 12.4	—	—	—	—	0.176

- For 70665829, QLP SAP is 4–10% deeper than TESS-SPOC SAP in every sector, but the d2/d1 ratios match. This is a multiplicative aperture or background-scale difference, not a different signal. TESS-SPOC CROWDSAP is 0.966 (s46) and 0.969 (s1751).
- For 331293467, raw depths depend on the aperture (TIC contamination ratio 0.70), while d2/d1 and T14 agree.
- **ZTF (`s13_ztf.out`, 2″ cone):**
- 70665829: 1789 rows from 2018 to 2025 (HJD−2457000 1202.8–3969.0), folded on the TESS ephemeris.
  - Primary core: zg 361 and 316 ppt, zr 325 and 315 ppt (4–8 points per field).
  - Secondary core: zg 26 and −9 ppt, zr 60 and 33 ppt.
  - So the eclipses are seen by an independent instrument, the ephemeris holds back to 2018, and the secondary is redder.
- 331293467: 0 rows (Dec −60, outside ZTF).
- **Common mode:** I did not fetch other QLP stars. Both signals repeat at a fixed period across 3 cameras (70665829) and 3 cameras over 7 years (331293467), with secondaries at fixed phase, and the difference images show no off-target deficit. A sector-wide systematic cannot do that.
- **Blind dip census (`s3_events_*.out`, 5×MAD below a running median; `s11_census.out`):** every dip is a primary or secondary except:
- TESS-SPOC s1751 at 4059.25–4059.7: a ramp from 0.935 to 0.98 right after a data gap (phase 0.77–0.90), background flat.
- QLP s48 at 2621.1–2621.5: QLP background −110 to −210, many cadences flagged 2048/4096.
- Both are systematics.
- **Pixels (`s7_pixels_70665829.out`, `s7b_pixels_331293467.out`):** median in-eclipse minus out-of-eclipse images from the TESScut 11×11 cubes, a PSF-fitted source position, and the dilution needed for every Gaia neighbour within 90″ with G < 18, in a 3×3 aperture (scene model with Gaussian PSF and TIC-style T from G and BP−RP).
- **70665829, primary:** offset 1.0 ± 0.2″ (s46), 1.2 ± 0.6″ (s48), 0.5 ± 0.1″ (s72).
- **70665829, secondary:** 2.3 ± 0.3″, 1.1 ± 0.8″, 1.8 ± 0.1″.

- **70665829, neighbour:** the only one, G 14.44 at 64.3″, needs 7253–47260% (primary) and 1573–10587% (secondary), with a χ^2 ratio of 41–78. Excluded.
- **331293467, primary offsets:** 0.3 ± 0.2″ (s38), 2.0 ± 0.2″ (s99), 1.2 ± 0.3″ (s100), 3.6 ± 0.5″ (s101). Bootstrap errors are surely underestimates of the systematics, but every offset is far smaller than any separation that matters.
- **331293467, neighbour by neighbour:**
  - G 17.07 at 8.0″: needs 377/458/334/305%. χ^2 ratio 3.6–4.6. Offset projection onto it is 0.03/0.23/0.00/−0.18 of the way.
  - G 16.14 at 14.0″: needs 91/115/99/97%. χ^2 ratio 3.8–10.5. Projection −0.01/0.01/−0.08/0.24.
  - G 16.07 at 24.4″: needs 143–287%.
  - G 13.78 at 30.5″: needs 84/45/28/25%, so not excluded by arithmetic. χ^2 ratio 19.1/10.1/19.1/16.0 and projection 0.00/−0.00/0.04/−0.11, so excluded by position.
  - G 14.73 at 47.3″: needs 96–5754%, χ^2 ratio 10–20.
  - All other neighbours need more than 150%.
- **331293467, secondary:** the fit gives 0.2 ± 0.7″, 1.9 ± 1.0″, 3.2 ± 0.6″, 3.2 ± 1.3″. The χ^2 cannot separate the target from the 8″ and 14″ stars here (ratio ~1.0–1.1), and a G 12.85 VARIABLE star at 89.4″ on the cutout edge fits the secondary diff image better (ratio 0.14–0.36). That star contributes 0.0000 of the aperture flux, so it is noise in a 4–5 event average, not a source. The secondary is tied to the target because it keeps a fixed phase (0.5002) and a fixed d2/d1 with the on-target primary.

3. TIC 70665829

Measurements (raw SAP QUALITY==0; trapezoid with quadratic baseline; `s3_events_70665829.out` lists every event). Sector summaries:

Event	Mid BTJD (first … last)	Depth (SPOC / QLP / TESScut)	T14 / T23	pipelines
Primary s45 ×6	2528.21356 … 2549.00771	300.3 / 325.3 / —	2.83 / 0.53 h (all SPOC primaries; range 2.80–2.90)	SPOC, QLP
Primary s46 ×6	2555.93930 … 2576.73431	296.2 / 306.6 / 297.6	"	SPOC, QLP, TESScut
Primary s48 ×6	2611.39079 … 2635.65071 (2614.85610 SPOC only)	282.7 / 312.2 / 288.1	"	SPOC, QLP, TESScut
Primary s72 ×5	3262.95482 … 3283.74972	299.0 / 329.8 / 300.6	"	SPOC, QLP, TESScut
Primary s1751 ×1	4060.08215 ± 0.00010	299.86	2.84 / 0.53 h	SPOC
Secondary s45 ×7	2526.34374 … 2547.13826	67.3 / 72.3 / —	2.77 / 0.54 h (all SPOC secondaries; range 2.57–2.88)	SPOC, QLP
Secondary s46 ×8	2554.07007 … 2578.32973	66.6 / 66.1 / 67.2	"	SPOC, QLP, TESScut
Secondary s48 ×6	2612.98690 … 2633.78190	64.1 / 70.6 / 67.1	"	SPOC, QLP, TESScut
Secondary s72 ×6	3264.54980 … 3285.34512	67.9 / 72.2 / 66.2	"	SPOC, QLP, TESScut
Secondary s1751 ×1	4061.67233 (partial, egress lost)	54.78	—	SPOC

- **Lost events:** in TESS-SPOC, s45/46/48/72 lose 1/2/5/4 events to gaps (`s11_census.out`). QLP flags more of s48: it has 1 full and 11 partial events there.
- **Period (`s12_ephem.out`, full-coverage events, joint fit):**

- SPOC: T0 = 2791.61111 ± 0.00009 BTJD, P = 3.4657654 ± 0.0000007 d. QLP: 3.4657623 ± 0.0000006 d.
- The 3σ difference comes from the single s1751 primary. Without it, SPOC primaries give 3.4657637 ± 0.0000012, and the s1751 eclipse arrives +2.30 min late (`s12b_no1751.out`).
- Primary O−C has χ^2_r of 24–30 against the formal errors: sector means range from −1.9 to +1.3 min (s48 is early in both pipelines while its secondaries are not). This was not pursued, but the realistic timing error is about ±2 min.
- Gaia P = 3.4658878 ± 0.0002846, so (P − P_Gaia)/σ = −0.43 for both pipelines.
- **Odd/even (`s4_period_70665829.out`):** primary +0.62 ± 0.82 ppt (SPOC), −0.87 ± 1.24 (QLP). Secondary +0.17 ± 0.54, −1.06 ± 0.74. No difference.
- **Which eclipse is primary:** the primary (300 ppt) and secondary (67 ppt) differ by a factor of 4.5 and sit at an off-centre phase. So P is neither half nor double the true period, and the deeper event is unambiguous.

4. TIC 331293467

Measurements (QLP SAP QUALITY==0. Primaries have a free trapezoid fit (`s3_events_331293467.out`). Secondaries use the primary's shape with free t0 and depth (`s5_secondary_331293467.out`)):

Event	Mid BTJD (first … last)	Depth QLP (TESScut raw)	T14 / T23	pipelines
Primary s11 ×4	1601.23407 … 1622.26045	61.4	4.07 / 0.83 h (all primaries; range 3.93–4.23)	QLP
Primary s38 ×5	2337.18980 … 2358.21445	64.7 (46.7)	"	QLP, TESScut
Primary s99 ×2	4050.91270, 4066.68276	72.6 (50.4)	"	QLP, TESScut
Primary s100 ×3	4077.19684 … 4098.22307	70.1 (51.3)	"	QLP, TESScut
Primary s101 ×4	4103.47964 … 4124.50456	61.4 (55.5)	"	QLP, TESScut
Primary s102 ×3	4129.76349 … 4150.78766 (last one partial)	70.5	"	QLP
Secondary ×24 (3/4/4/5/4/4 per sector)	1603.8613 … 4148.1608	11.42 ± 0.15 weighted mean (8.6–13.4 each)	primary shape (TESScut free fit 3.5–4.2 h)	QLP, TESScut

- **Lost events:** 14 events fall in gaps across the six sectors (`s11_census.out`).
- **Period (`s12_ephem.out`, full-coverage, joint):** T0 = 3467.40446 ± 0.00029 BTJD, P = 5.2568199 ± 0.0000020 d.
- All primaries: 5.2568187 ± 0.0000017. Secondaries only: 5.2568291 ± 0.0000074 (1.4σ).
- Gaia P = 5.2565467 ± 0.0005365, so +0.51σ.
- **Odd/even, primary:** −1.19 ± 0.49 ppt (formal 2.4σ, 1.8% of the depth). It is driven by sectors with 1–2 events and a partial s102 event, and the sector-to-sector depth scatter (61–73 ppt) is much larger. At 2P both halves are the same eclipse type, and the 11 ppt secondary at phase 0.5002 rules out P/2. Not a sign of a wrong period.

Geometry vs Gaia

All from `s10_geometry.out`: 20000 draws each with independent errors, with Gaia's correlation matrix, and with that matrix conditioned on the TESS period.

70665829. Gaia: e = 0.0769 ± 0.0397, ω = 175.4 ± 22.2°, T_p = 390.5724 ± 0.2071 BTJD. - **Secondary phase:** measured shallow-after-deep phase is 0.46048 ± 0.00008. The implied e cos ω = −0.0621, the same exact or to first order. - **Gaia prediction:** (t_B − t_A)/P = 0.4547 (+0.0239/−0.0238). That is below 0.5 in 100% of draws, with e cos ω = −0.0711 ± 0.0362. - Deep = A matches at +0.24σ. Deep = B would need 0.5395, at +3.55σ. - Conditioned on the TESS period: 0.4630 ± 0.0136, giving A −0.19σ and B +5.64σ. - **Absolute timing (630 cycles from T_p):** - Independent errors: deep minus A = −0.0215 ± 0.0999 P (−0.22σ); minus B = −4.71σ. - Gaia correlated: deep minus A = −0.0204 ± 0.0585 P (−1.7 h, −0.35σ); minus B = −11.7σ. - Conditioned on the TESS period: deep minus A = +0.0051 ± 0.0064 P (+0.4 h, +0.80σ); minus B = −56σ. - **Conclusion: deep eclipse = conjunction A.** Gaia's ω decides this through two independent routes (secondary phase and absolute timing). - **Durations:** T14 is 2.77 h for the secondary and 2.83 h for the primary (SPOC). This is consistent with Gaia's small e sin ω = +0.0032 ± 0.0310. - **Physics (`s14_physics.out`, M1 = 0.85 M⊙):** f(m) = 0.0807 M⊙, giving m2 ≈ 0.54 M⊙ (≈3760 K, ≈0.52 R⊙). a = 10.75 R⊙ (a/R1 = 14.2). A central eclipse would last 3.15 h against 2.83 h observed, so b > 0. The expected d2/d1 ≈ J(800 nm) ≈ 0.28 against 0.22 observed. - A cooler M-dwarf companion makes the K dwarf's eclipse the deep one, as the timing says. ZTF agrees: the deep eclipse is deepest in g, the shallow one deeper in r.

331293467. Gaia: e = 0.0215 ± 0.0250, ω = 287.0 ± 69.8°, T_p = 389.8053 ± 1.0004 BTJD. - **Secondary phase:** 0.50013 ± 0.00023 (joint fit) and 0.5002 ± 0.0001 (`s5`). That gives e cos ω = +0.0003 to +0.0004 ± 0.0002, against Gaia's +0.0024 ± 0.0227. Consistent, but it cannot separate A from B. - **Absolute timing (699 cycles from T_p):** - Independent errors: deep minus A = +0.14σ, minus B = −1.90σ. Not decisive. - Gaia correlated: deep minus A = +0.0321 ± 0.0649 P (+4.1 h, +0.49σ); minus B = −6.66σ. - Conditioned on the TESS period: deep minus A = +0.0009 ± 0.0077 P (+0.1 h, +0.12σ); minus B = +81σ. Conditioning shifts T_p by +0.29 d and ω by +20.7°, because corr(P, T_p) = 0.57. - **Conclusion: deep eclipse = conjunction A,** the K dwarf in front of the A/F star. This rests on Gaia's corr(ω, T_p) = 0.9997, which is physically expected at e ≈ 0. - **Physics (M1 = 1.69 M⊙):** f(m) = 0.0630 M⊙, giving m2 ≈ 0.71 M⊙ (≈4580 K, ≈0.70 R⊙, k ≈ 0.41). a = 17.04 R⊙. A central eclipse would last 5.76 h against 4.07 h observed (b > 0). The expected d2/d1 ≈ 0.21 against 0.16 observed. The undiluted primary depth of 8–11% is below $k^2 \approx 0.17$, as a partial or high-b transit should be.

Ephemerides and next eclipses

From `s12_ephem.out`. Times are BJD_TDB / BTJD, converted to UTC calendar dates without the barycentric-to-Earth light-time correction (up to ±8.3 min). - **TIC 70665829:** T0 = 2791.61111 ± 0.00009 BTJD (BJD_TDB 2459791.61111), P = 3.4657654 ± 0.0000007 d (SPOC). The secondary follows at +1.59591 ± 0.00029 d (phase 0.46048). Realistic timing error is ±2 min (see §3). - Next primaries: 4344.2740 (2026-10-30 18:33 UTC), 4347.7398 (2026-11-03 05:44), 4351.2055 (2026-11-06 16:54). The QLP ephemeris is 1–2 min earlier. - Next secondaries: 4345.8699 (2026-11-01 08:51), 4349.3357 (2026-11-04 20:02), 4352.8014 (2026-11-08 07:12). - **TIC 331293467:** T0 = 3467.40446 ± 0.00029 BTJD (BJD_TDB 2460467.40446), P = 5.2568199 ± 0.0000020 d. The secondary follows at +2.62912 ± 0.00123 d (phase 0.50013). - Next primaries: 4345.2934 ± 0.0004 (2026-10-31 19:01 UTC), 4350.5502 (2026-11-06 01:11), 4355.8070 (2026-11-11 07:20). - Next secondaries: 4347.9225 (2026-11-03 10:07), 4353.1793 (2026-11-08 16:17), 4358.4361 (2026-11-13 22:26).

Caveats

- **331293467 has a single pipeline (QLP) plus my own TESScut photometry.** Its undiluted depths rest on a Gaussian-PSF scene model: primary 80–116 ppt (79.6–89.3 ppt from PSF

amplitude), secondary ~13–21 ppt. Only the fact that the signal is on the target, the timing and the ratios are robust.
- **Attribution for 331293467 is pixel-level, not resolved.** ZTF has no data at Dec −60. The G 13.78 star at 30.5″ is excluded only by position (and by the timing match to Gaia's conjunction), not by arithmetic. Bootstrap centroid errors (0.2–0.5″) understate the systematics, but the closest call is still 0.24 of the way toward any neighbour.
- **Gaia geometry for 331293467 is decisive only with the corr_vec decoding** (order P, Vcm, K, e, ω, T_p, column-major upper triangle; the 0.9997 ω–T_p term confirms it).
- **70665829 timing:** the O–C scatter between sectors (±1–2 min) and the late s1751 primary (+2.3 min) are unexplained. Candidates are timestamps in the unusual "sector 1751" product, or real timing variations. This does not affect the classification.
- **Archives:** the cones for Gaia-VEB, varisum and SB9 were not checked on a known object, because the mirror failed during `s2c`. Their empty results stand from `s2` only.
- **Common mode:** not done with other stars (see §2).

Files

- **Light curves:** `lcio.py`, `s1_download_lc.py` / `.out`.
- **Archives:** `s2_archives.py` / `.out`, `s2b_allcat.py` / `.out`, `s2c_catalogue_sanity.py` / `.out`.
- **Events:** `s3_events.py`, `s3_events_70665829.out` / `.json`, `s3_events_331293467.out` / `.json`.
- **Periods:** `s4_period.py`, `s4_period_*.out`, `s4_ephem_*.json`; `s5_secondary_331.py`, `s5_secondary_331293467.out` / `.json`.
- **Pixels:** `s6_tesscut.py` / `.out`; `s7_pixels.py`, `s7_pixels_70665829.out`, `s7_pixels_331293467.out` (interrupted; superseded by s7b); `s7b_pixels.py`, `s7b_pixels_331293467.out`; `pix_70665829_s46/s48/s72.png`, `pix_331293467_s38/s99/s100/s101.png`; `tesscut_ap3x3_*.txt` (own photometry).
- **Independence:** `s9_indep.py` / `.out` / `.json`.
- **Geometry:** `s10a_gaia_query.py` / `.out`, `s10a_gaia_nss.csv`, `s10_geometry.py` / `.out` / `.json`.
- **Census:** `s11_census.py` / `.out`.
- **Ephemerides:** `s12_ephem.py` / `.out` / `.json`, `s12b_no1751.py` / `.out`.
- **ZTF:** `s13_ztf.py` / `.out` (raw CSVs in the scratchpad `ztf/` folder).
- **Physics:** `s14_physics.py` / `.out`.

TIC 70665829: CONFIRMED; P_ecl=3.4657654±0.0000007 d; depth1=283–330 ppt (pipeline-dependent; SPOC ~300); depth2=64–72 ppt at phase 0.46048; pixel: on target; pipelines: 4 (QLP, TESS-SPOC, TESScut, ZTF); Gaia geometry: pass (deep = conjunction A, Gaia RV star eclipsed) TIC 331293467: CONFIRMED; P_ecl=5.2568199±0.0000020 d; depth1=61–73 ppt QLP raw (≈80–116 ppt undiluted); depth2=11.4 ppt QLP (≈13–21 ppt undiluted) at phase 0.50013; pixel: on target (30.5″ G13.78 excluded by position only); pipelines: 2 (QLP, TESScut); Gaia geometry: pass (deep = conjunction A; needs Gaia ω–T_p correlation)

vet04 full report

Archived source: sanitized report text.

vet04: adversarial check of TIC 283532878 (UCAC4 740-075601) and TIC 407877153 (UCAC4 054-021425)

Opus 5.5 subagent, 2026-10-30, for Fable 5.1. This run resumed the work of an earlier agent that a container restart killed part-way through. I re-read its scripts and outputs `s01–s11` and kept what they had measured. I re-ran the archive step it had not finished (`s12`), and found that the first VizieR pass gave false empty results (see §1). New steps: a difference-image test with real noise weighting (`s13`, `s13b`), because the `s11` "chi2" values were unweighted sums of squares. Also new: ZTF (`s14`), plots (`s15`), a common-mode test (`s16`), extra geometry, physics and ephemeris (`s17`), and a merged event table (`s18`). All of these are in `/home/user/eclipses-paper-review/work/c9a-vets/vet04/`. The downloads are in the scratchpad at `.../scratchpad/data/vet04/<tic>/`. I deleted the TESScut cubes after `s13`. Notation: **A** = the companion passes in front of Gaia's RV star (true anomaly $\nu = 90° - \omega$); **B** = the RV star passes in front of the companion. Depths are in ppt, times in BTJD = BJD_TDB − 2457000, durations in hours.

TIC 283532878: CONFIRMED. - (1) Each event fitted on its own in raw QLP SAP flux, QUALITY==0: 11 primaries and 9 full secondaries, plus 1 partial secondary. One more primary (s16, 1744.53) is visible only in TESScut. - (2) The eclipses give P = 17.446019 ± 0.000019 d. Gaia's 17.4298 ± 0.0089 d is +1.83σ away. - (3) Difference images in all 8 sectors (cameras 2, 3 and 4) put both the primary and the secondary on the target. Primary offsets are 0.4–1.5″. Secondary offsets are 1.0–5.3″, each within ~2σ. Every Gaia neighbour is rejected by $\Delta\chi^2 \geq 1119$ for the primary and $\Delta\chi^2 \geq 41.7$ for the secondary in every sector. The dilution arithmetic alone excludes every neighbour for the primary (each would need ≥ 218%). For the 10 ppt secondary it does **not** exclude three neighbours (G 14.50 at 37″, G 14.93 at 28″, G 15.74 at 29.5″, each needing 30–47%); the pixels do. - (4) QLP SAP and my own TESScut photometry agree on the secondary (9–10 ppt in both). For the primary, QLP gives 70–79 ppt and the TESScut aperture 65–72 ppt, with 13–20% of that aperture's light from neighbours. The PSF-deblended depth is 71–77 ppt. No TESS-SPOC or 2-min data exist. - (5) The secondary is at phase 0.67812 ± 0.00030; Gaia's e and ω predict 0.712 ± 0.024 (–1.4σ). The primary falls 0.20 d from Gaia's conjunction A (0.4–0.7σ); B is excluded at > 11σ. - **Screen check:** the screen was right. The "~9 ppt dip near 0.675" is the eccentric secondary. It is not a one-sector glitch: it appears in all 8 sectors and 4 seasons at constant phase (0.6775–0.6795). Its depth is 9–10 ppt; 10–11 ppt deblended.

TIC 407877153: CONFIRMED. - (1) Each event fitted on its own: 9 full primaries and 9 secondaries in raw SAP, plus extra partial events (two primaries, two secondaries). - (2) The eclipses give P = 17.089294 ± 0.000008 d. Gaia's 17.0902 ± 0.0063 d is −0.14σ away. - (3) Difference images put the primary on the target in 6 sectors ($\Delta\chi^2 \geq 337$ against every neighbour). The secondary also fits best on the target in 6 sectors ($\Delta\chi^2 \geq 7$ per sector; ≥ 166 summed over sectors). The arithmetic needs ≥ 441% (primary) and ≥ 51% (secondary) on any neighbour. - (4) Three pipelines: TESS-SPOC SAP (28.6–29.4 ppt), my own TESScut aperture (29.4–31.0 ppt) and QLP SAP. QLP gives 27–29.6 ppt in s12/s39 but only 24–27 ppt in s93–s102, a QLP dilution issue. The secondary is 2.6–4.2 ppt in all three. - (5) The secondary is at phase 0.54690 ± 0.00017; the prediction is 0.559 ± 0.044 (–0.28σ). The primary is at conjunction A (0.16–0.64σ); B is excluded at ≥ 5σ. - **Screen check:** the secondary is **3.0 ppt, not ~2 ppt** (stack 2.98 ± 0.13). The primary is ~29 ppt in SPOC and TESScut, not ~27 (27 is QLP's diluted value). The phase of 0.545 was right.

1. Archives and web

- **VizieR cones** (`s12d_recheck.out`, mirror vizier.idia.ac.za, run immediately after an Algol VSX control returned bet Per, EA/SD):
- **TIC 283532878** is not in VSX, Gaia DR3 EB (I/358/veb), Prsa+2022, IJspeert+2021, Gaia varisum or SB9 within 60″ of the target. Both VSX and Gaia-EB list one row at **29.6″: Gaia DR3 2202301140233408128**, G 17.66 (the "VARIABLE" neighbour in `neighbours_90as.csv`). It is VSX type E, P = 1.79747 d, G 17.61–18.17; Gaia EB eclipse depths are 0.39 and 0.60 (`s12e_neighbour_eb.out`). It is excluded as the source below.
- **TIC 407877153:** nothing in any of those six catalogues within 60″.
- **First-pass warning:** the first pass (`s12_archives.out`) returned 0 rows everywhere, including the EB above, and its Algol control failed later (`s12c_controls.out`: VSX 0 rows for Algol). The mirror was intermittently returning empty tables. The all-catalogue 5″ query is likewise unreliable here: its Algol control gave 0 tables (`s12b_archives.out`), so I do not use it.
- **SIMBAD** (sim-coo 10″, `s12b_archives.out`): both stars are type **SB*** with 1 bibcode (2023A&A...674A..34G, the Gaia DR3 NSS paper). TIC 283532878 = UCAC4 740-075601 = 2MASS J21512860+5759057. TIC 407877153 = UCAC4 054-021425 = 2MASS J18230801-7921092.
- **ExoFOP:** no TOI. "CTOIs 0" on both pages.
- **Conclusion:** neither star is a catalogued eclipsing binary within 60″. Both are known only as Gaia SB1s.

2. Independence, systematics, pixels

- **Pipelines:**
- 283532878: QLP SAP (8 sectors) plus my own TESScut aperture photometry (8 sectors). The two agree on the secondary. The primary is shallower in my aperture because it holds 13–20% neighbour light (`s11_pixels_283532878.out`: target fraction 0.796–0.873).
- 407877153: QLP, TESS-SPOC (s12, s13, s39) and TESScut. SPOC and TESScut agree on the primary to about 1 ppt; QLP s93–s102 are ~10–15% shallower. No 2-min SPOC data exist for either star.
- **Difference images, noise-weighted (`s13_pixnoise_*.out`, `s13b_summary.out`):** in each sector I averaged the per-event (median out − median in) images. The per-pixel noise comes from 300 pseudo-events at random out-of-eclipse times with the same windows. I then fitted Gaussian-PSF single-source models for the target and for every Gaia neighbour with G < 18 within 90″. Results:
- 283532878 P: S/N 40–154 per sector; least-rejected neighbour $\Delta\chi^2$ = 73,292 summed (≥ 1119 in every sector); PSF-amplitude depth on target 70.6–77.3 ppt.
- 283532878 S: S/N 8.6–14.9; $\Delta\chi^2 \geq 41.7$ in every sector, ≥ 977 summed; depth 9.8–12.9 ppt; free-fit offsets 1.0–5.3″ (1σ 1.3–2.9″).
- 407877153 P: S/N 18.6–84; $\Delta\chi^2 \geq 337$; depth 30.0–35.3 ppt.
- 407877153 S: S/N 2.8–8.7; $\Delta\chi^2 \geq 7.1$ per sector, ≥ 166 summed; offsets 3.6–9.9″ with 1σ 1.9–12″.
- I looked at two of the images (`s15_diff_283532878_s57_S.png`, `s15_diff_407877153_s13_S.png`). In both, the deficit sits on the target pixel, away from the bright neighbours in the direct image.
- **Dilution arithmetic** (per-sector aperture flux fractions, `s11_pixels_*.out`):
- **283532878 primary:** every neighbour needs ≥ 218% intrinsic depth, so all are excluded.
- **283532878 secondary (10.2 ppt):** most neighbours are excluded arithmetically (≥ 82% needed; the G 17.66 Gaia EB at 29.7″ needs 181–240%). Three are **not**: G 14.50 at 37.0″ needs 29.6–100%, G 14.93 at 28.3″ needs 30–47%, and G 15.74 at 29.5″ needs 31.8–39%. The difference images reject them instead ($\Delta\chi^2 \geq 41.7$ each sector).

- **407877153:** the primary needs ≥ 441% on any neighbour; the secondary needs ≥ 51.5% (G 14.68 at 60″, s13) and 61–135% for the G 17.23 star at 46″.
- **Same period and phase drift:** both secondaries share the primary's period (283532878: P_S – P_P = −0.00030 ± 0.00010 d; 407877153: +0.000013 ± 0.000059 d, `s05_ephem.out`) and sit at fixed phase over 4–5 seasons. A neighbour's EB would have to share the period exactly. The catalogued neighbour EB (P = 1.7975 d; 17.446/1.7975 = 9.71, not commensurate) would drift in phase.
- **ZTF (`s14_ztf.out`):** not informative. IRSA returns only 54–94 catflags==0 points per object for the target and the five closest bright neighbours, with 0–1 points inside any eclipse window.
- **Common mode (`s16_commonmode.out`):** not done. The MAST product queries kept failing (RemoteServiceError and proxy resets, `s16_commonmode.out`), and the coordinator then dropped this optional step. The pixel and two-pipeline evidence above stands without it.
- **Phase 0.5 of 283532878** (`s17_geometry2.out`, from 10 fixed-time fits in `s08`): +0.35 ± 0.30 ppt. There is no eclipse at phase 0.5. $\chi^2 = 45/9$ shows red noise at the ~2–3 ppt level.

3. TIC 283532878 (P = 17.446 d, e = 0.35)

Measurements (`s18_event_table.out`, from `s04`/`s08`/`s11`; trapezoid with quadratic baseline, raw SAP QUALITY==0; "fixed" means the shape is fixed from the 200-s stack). There is no TESS-SPOC column because this star has no TESS-SPOC data.

Event	Mid BTJD	Depth QLP SAP free / fixed (ppt)	Depth own TESScut ap. (ppt)	T14 / T23 (h)	Pipelines
P n=−1 s16	1744.5307 ± 0.0011 (TESScut)	QLP: 13 flagged cadences	67.6 ± 1.0	–	TESScut
S n=−1 s16	1756.3642 ± 0.0079	7.5 ± 1.2 / 7.6 ± 1.1	10.1 ± 0.8	5.53 / 3.45	QLP, TESScut
P n=0 s16	1761.9800 ± 0.0015	70.8 ± 1.7 / 70.4 ± 1.4	66.9 ± 1.1	7.01 / 2.00	QLP, TESScut
S n=0 s17	1773.8350 ± 0.0048	9.6 ± 0.7 / 9.8 ± 0.7	9.2 ± 0.8	8.21 / 3.43	QLP, TESScut
P n=1 s17	1779.4207 ± 0.0007	73.6 ± 0.8 / 73.9 ± 0.9	69.6 ± 0.8	7.29 / 2.12	QLP, TESScut
P n=61 s56	2826.1851 ± 0.0009	74.1 ± 1.1 / 74.1 ± 0.9	66.8 ± 0.6	7.20 / 2.05	QLP, TESScut
S n=61 s56	2838.0248 ± 0.0037 (TESScut)	12.8 ± 2.0 (egress in gap)	10.7 ± 0.6	–	QLP (partial), TESScut
P n=62 s56	2843.6288 ± 0.0010	75.5 ± 1.3 / 75.2 ± 1.0	66.8 ± 0.5	7.04 / 2.02	QLP, TESScut
S n=62 s57	2855.4637 ± 0.0090	9.2 ± 1.5 / 8.6 ± 1.3	9.3 ± 0.5	6.08 / 2.19	QLP, TESScut
P n=63 s57	2861.0760 ± 0.0010	73.5 ± 1.3 / 73.3 ± 1.2	68.4 ± 0.6	6.95 / 2.05	QLP, TESScut
S n=63 s57	2872.9161 ± 0.0070	9.0 ± 1.0 / 9.2 ± 1.0	9.2 ± 0.5	6.87 / 3.03	QLP, TESScut
P n=64 s57	2878.5193 ± 0.0005	79.2 ± 0.7 / 78.6 ± 0.6	69.6 ± 0.5	7.11 / 1.97	QLP, TESScut
S n=92 s76	3378.8457 ± 0.0067	8.9 ± 1.1 / 9.4 ± 1.1	9.8 ± 0.5	6.34 / 3.85	QLP, TESScut
P n=93 s76	3384.4553 ± 0.0007	76.4 ± 0.9 / 77.3 ± 0.8	68.8 ± 0.6	6.76 / 2.26	QLP, TESScut
S n=93 s77	3396.2636 ± 0.0050	10.4 ± 0.7 / 10.1 ± 0.8	9.4 ± 0.5	8.89 / 2.96	QLP, TESScut
P n=94 s77	3401.9005 ± 0.0008	75.4 ± 1.0 / 75.5 ± 0.8	71.8 ± 0.7	6.98 / 2.10	QLP, TESScut
S n=94 s77	(3413.73)	in the 3407.0–3417.2 data gap	–	–	none
P n=95 s77	3419.3449 ± 0.0008	74.9 ± 1.0 / 74.1 ± 0.9	70.9 ± 0.5	7.22 / 1.89	QLP, TESScut

Event	Mid BTJD	Depth QLP SAP free / fixed (ppt)	Depth own TESScut ap. (ppt)	T14 / T23 (h)	Pipelines
S n=103 s83	3570.7490 ± 0.0055	9.6 ± 1.0 / 9.9 ± 0.9	9.4 ± 0.7	6.33 / 3.35	QLP, TESScut
P n=104 s83	3576.3613 ± 0.0011	74.6 ± 1.3 / 74.4 ± 1.1	68.3 ± 0.7	7.03 / 2.05	QLP, TESScut
S n=104 s84	3588.1988 ± 0.0068	9.8 ± 1.3 / 9.9 ± 1.3	9.1 ± 0.7	5.67 / 3.36	QLP, TESScut
P n=105 s84	3593.8100 ± 0.0005	76.4 ± 0.7 / 76.6 ± 0.6	65.5 ± 0.5	7.07 / 2.09	QLP, TESScut
S n=105 s84	3605.6254 ± 0.0035	12.8 ± 0.7 / 12.6 ± 0.6	10.2 ± 0.4	7.32 / 2.48	QLP, TESScut

- **Stacked shape** (200-s QLP, `s08_shape_oddeven.out`):
- Primary: 75.41 ± 0.39 ppt, T14 = 7.028 ± 0.034 h, T23 = 2.08 ± 0.05 h. This is a near-V shape: a grazing or partial eclipse.
- Secondary: 10.24 ± 0.40 ppt, T14 = 6.66 ± 0.25 h, T23 = 2.78 ± 0.36 h.
- **Period** (`s05_ephem.out`):
- 11 primaries alone: P = 17.446025 ± 0.000017 d (χ^2 = 56/9; O–C within ±7 min). 9 secondaries alone: 17.445724 ± 0.000099 d.
- The joint fit gives P = 17.446019 ± 0.000019 d, +0.0162 d from Gaia's period = **+1.83σ_Gaia**.
- The cycle count is unambiguous: consecutive primaries within s56, s57 and s77 are 17.44 d apart.
- **Odd/even** (QLP fixed shape): 76.25 ± 0.38 vs 75.36 ± 0.33 ppt. The difference is 1.8σ, smaller than the 1.5–2.6 ppt scatter between sectors. A secondary of different depth exists at phase 0.678, so 2P is ruled out, and phase 0.5 is empty, so P/2 is ruled out.
- **Secondary phase by season** (`s17_geometry2.out`): 0.67954 ± 0.00026 (s16–17), 0.67854 ± 0.00033 (s57), 0.67750 ± 0.00025 (s76–77), 0.67775 ± 0.00016 (s83–84).
- **Physics:** f(m) = 0.0334 M⊙. With the TIC M1 = 1.35 M⊙, m2 ≥ 0.48 M⊙ and a = 34.6 R⊙. The depth ratio S/P = 0.136 implies a TESS-band surface-brightness ratio ≈ T2 3830 K, consistent with an early-M companion with R2 ≳ 0.5 R⊙. A central eclipse at conjunction A would last ~9.5 h; the observed 7.0 h and V shape mean a high impact parameter. The duration ratio S/P = 0.948 ± 0.036 gives e sin ω ≈ −0.03 ± 0.02, against Gaia −0.086 ± 0.044. The equal-b assumption behind that number is weak for grazing eclipses.

4. TIC 407877153 (P = 17.089 d, e = 0.13)

Measurements (`s18_event_table.out`). TESS-SPOC depths are fixed-shape fits where available.

Event	Mid BTJD	Depth QLP SAP free / fixed (ppt)	Depth TESS-SPOC SAP (ppt)	Depth own TESScut (ppt)	T14 / T23 (h)	Pipelines
S n=−1 s12	1626.5135 ± 0.0171 (TESScut)	QLP: 17 flagged	–	2.9 ± 0.7	–	TESScut
P n=0 s12	1634.2635 ± 0.0017	27.5 ± 0.5 / 27.4 ± 0.6	29.2 ± 0.5	30.1 ± 0.7	9.44 / 6.10	QLP, SPOC, TESScut
S n=0 s12	1643.5993 ± 0.0077	3.2 ± 0.6 / 2.9 ± 0.5	3.6 ± 0.5	2.9 ± 0.6	7.62 / 7.29	QLP, SPOC, TESScut
P n=1 s12	1651.3509 ± 0.0012	27.5 ± 0.6 (partial)	29.4 ± 0.5	29.9 ± 0.6	9.72 / 5.81	QLP (partial), SPOC, TESScut
S n=1 s13	1660.7055 ± 0.0069	3.0 ± 0.5	2.9 ± 0.4	3.1 ± 0.4	7.46 / 6.61	QLP, SPOC, TESScut
P n=2 s13	(1668.43)	in the 1667.6–1668.7 gap	gap	–	–	none
S n=2 s13	1677.7897 ± 0.0079	2.9 ± 0.5 / 2.6 ± 0.4	3.0 ± 0.3	2.9 ± 0.4	7.81 / 7.10	QLP, SPOC, TESScut

Event	Mid BTJD	Depth QLP SAP free / fixed (ppt)	Depth TESS-SPOC SAP (ppt)	Depth own TESScut (ppt)	T14 / T23 (h)	Pipelines
P n=43 s39	2369.1015 ± 0.0009	29.5 ± 0.4 / 29.0 ± 0.4	28.6 ± 0.4	29.5 ± 0.4	9.53 / 6.10	QLP, SPOC, TESScut
S n=43 s39	2378.4374 ± 0.0064	3.8 ± 0.7 / 3.0 ± 0.4	2.9 ± 0.3	3.2 ± 0.3	8.62 / 6.29	QLP, SPOC, TESScut
P n=44 s39	2386.1901 ± 0.0010	29.6 ± 0.5 / 29.3 ± 0.4	28.7 ± 0.4	29.4 ± 0.4	9.65 / 6.05	QLP, SPOC, TESScut
S n=128 s93	3831.0358 ± 0.0104	4.2 ± 0.6 / 3.5 ± 0.4	–	3.0 ± 0.4	9.27 / 2.83	QLP, TESScut
P n=129 s93	3838.7816 ± 0.0010	27.1 ± 0.4 / 27.3 ± 0.4	–	29.5 ± 0.4	9.38 / 6.55	QLP, TESScut
S n=129 s93	3848.1399 ± 0.0053	2.9 ± 0.3 / 2.9 ± 0.4	–	3.7 ± 0.4	7.66 / 6.63	QLP, TESScut
S n=130 s94	3865.2086 ± 0.0068	3.2 ± 0.3 / 3.2 ± 0.3	–	4.2 ± 0.4	9.56 / 5.54	QLP, TESScut
P n=131 s94	3872.9595 ± 0.0013	24.7 ± 0.4 / 24.3 ± 0.4	–	29.6 ± 0.5	9.85 / 5.89	QLP, TESScut
P n=143 s100	4078.0293 ± 0.0011	26.8 ± 0.4 / 26.7 ± 0.4	–	31.0 ± 0.5	9.45 / 6.20	QLP, TESScut
S n=143 s100	(4087.36)	64 flagged cadences	–	–	–	none
P n=144 s100	4095.1193 ± 0.0010	24.7 ± 0.4 / 24.8 ± 0.3	–	29.4 ± 0.4	9.51 / 6.45	QLP, TESScut
S n=144 s101	4104.4747 ± 0.0102	2.9 ± 0.3 / 2.6 ± 0.2	–	no cube	12.06 / 3.76	QLP
P n=145 s101	(4112.23)	156 flagged cadences	–	no cube	–	none
S n=145 s101	4121.5442 ± 0.0073	3.0 ± 0.3 / 2.8 ± 0.3	–	no cube	8.90 / 4.73	QLP
P n=146 s102	4129.2998 ± 0.0009	25.0 ± 0.3 / 24.9 ± 0.4	–	29.5 ± 0.4	9.37 / 6.28	QLP, TESScut
S n=146 s102	4138.6654 ± 0.0176 (TESScut)	QLP: 155 flagged	–	4.3 ± 1.1	–	TESScut
P n=147 s102	4146.3904 ± 0.0024	23.7 ± 0.7 (ingress lost)	–	25.0 ± 0.7	9.11 / 6.53	QLP (partial), TESScut

- **Stacked shape** (200-s QLP):
- Primary: 25.53 ± 0.16 ppt (QLP-diluted), T14 = 9.459 ± 0.051 h, T23 = 6.38 ± 0.08 h. **Flat-bottomed**, a total transit.
- Secondary: 2.98 ± 0.13 ppt, T14 = 8.46 ± 0.31 h, T23 = 5.71 ± 0.50 h. **Flat-bottomed**, an occultation.
- **Period:**
- 9 primaries: P = 17.089294 ± 0.000006 d (χ^2 = 6.0/7; O–C within ±2.3 min). 9 secondaries: 17.089307 ± 0.000059 d.
- Joint fit: P = 17.089294 ± 0.000008 d, **−0.14σ_Gaia**.
- Secondary phase by season: 0.54630–0.54713, steady.
- **Odd/even:** TESS-SPOC 28.91 ± 0.31 vs 28.91 ± 0.30 ppt. The QLP odd/even difference (26.10 vs 26.91) comes from which sectors fall in each group: QLP s93+ are diluted. 2P and P/2 are excluded as for the other star.
- **Physics:**
- f(m) = 0.0216 M⊙. With the TIC M1 = 0.87 M⊙, m2 ≥ 0.31 M⊙ and a = 29.5 R⊙.
- The flat-bottom depth gives k ≈ 0.16, so R2 ≈ 0.31 R⊙ with the TIC R1 = 1.91 R⊙. That matches a 0.31 M⊙ M dwarf.

- The occultation/transit ratio of 0.117 gives T2 ≈ 3200 K (TESS-band Planck).
- A central T14 at conjunction A would be 10.6 h; observed 9.46 h, so b is moderate.
- The duration ratio S/P = 0.894 ± 0.033 gives e sin ω = −0.056 ± 0.018, against Gaia −0.078 ± 0.066.
- Everything hangs together as a subgiant + M-dwarf SB1.

Geometry vs Gaia

- **Method** (`s07_geometry.out`, `s17_geometry2.out`; 20,000 Monte Carlo draws): I propagated P, T_p, e and ω; conjunction A is at ν = 90° − ω and B at ν = 270° − ω. The ω–T_p correlation was taken either as 0 or as the last `corr_vec` element (0.927 and 0.985). That second case assumes the SB1 parameter order ends with (arg_periastron, t_periastron), which I have not verified; results are given both ways.
- **283532878:**
- Predicted B–A separation: 0.7122 ± 0.0237. Measured secondary phase: 0.67812 ± 0.00030, **−1.44σ**. The other assignment (primary = B) is off by 16σ.
- The measured phase implies e cos ω = 0.2823 ± 0.0012 (exact Kepler, Gaia e sin ω draws), against Gaia 0.338 ± 0.039 (–1.42σ).
- Absolute timing, propagated with the TESS period: the primary is +0.0112 phase (+0.20 d) from conjunction A (0.37σ uncorrelated, 0.71σ correlated) and 0.30 phase from B (11.9σ / 23.7σ). Propagated with Gaia's period, A is still within 1.7–2.0σ.
- **The deep eclipse is A:** the F-type RV star is eclipsed by its companion. **Pass.**
- **407877153:**
- Predicted separation 0.5589 ± 0.0435; measured 0.54690 ± 0.00017, **−0.28σ**. The other assignment is off by 2.4σ.
- Implied e cos ω = 0.0733 ± 0.0006, against Gaia 0.093 ± 0.068.
- The primary is +0.0167 phase (+0.29 d) from A (0.16σ / 0.64σ) and 5.1σ / 18.9σ from B.
- **The deep eclipse is A**, the subgiant RV star transited by an M dwarf. **Pass.**

Ephemerides and next eclipses

From `s05_ephem_*.json` and `s17_geometry2.out`. UTC times are geocentric: I removed the barycentric light-time with astropy (offset given in brackets). - **TIC 283532878:** T0 = BJD_TDB 2458761.97572 ± 0.00150 (BTJD 1761.97572), **P = 17.446019 ± 0.000019 d**. Secondary T0 = 1773.80624 ± 0.00535 (phase 0.67812). - Next primaries: BTJD 4343.9865 = 2026-10-30 11:36 UTC [+3.4 min]; 4361.4325 = 2026-11-16 22:18; 4378.8785 = 2026-12-04 09:02. Each ±2.3 min formal; T14 = 7.0 h. - Next secondaries: BTJD 4355.817, 4373.263, 4390.709. The secondary phase has moved by ~0.002 between seasons, so allow ±1 h. - **TIC 407877153:** T0 = BJD_TDB 2458634.26184 ± 0.00084 (BTJD 1634.26184), **P = 17.089294 ± 0.000008 d**. Secondary T0 = 1643.60789 ± 0.00296 (phase 0.54690). - Next primaries: BTJD 4351.4596 = 2026-11-06 23:03 UTC [−3.1 min]; 4368.5489 = 2026-11-24 01:13; 4385.6382 = 2026-12-11 03:22. Each ±1.0 min; T14 = 9.5 h. - Next secondaries: BTJD 4343.716, 4360.806, 4377.895.

Caveats

- **Secondary-phase drift in 283532878:** the phase falls 0.0018 between 2019 and 2024 (≈ 45 min, ~4σ formal), and the joint timing χ^2 is 132/17. GR plus tidal apsidal motion at a/R1 ≈ 18 would give far less (~2.5×10^{-4} rad over 105 cycles, against the ~0.03 rad needed). I therefore treat this as

underestimated timing errors for the 10 ppt, red-noise-affected secondaries, not as physics. It does not affect the Gaia test, whose tolerance is ±0.024.

- **Catalogued EB 29.7″ from 283532878:** this is G 17.66, P = 1.7975 d, eclipses 40–60% deep in G. With 0.4–0.6% of the light in my apertures it can add dips of ≤ 3 ppt at unrelated phases. That may explain some event-to-event scatter in the secondary depths (for example S n=105, 12.6 ppt). It cannot produce the secondary: the period is wrong, the depth needed is 181–240%, and the pixels reject it.
- **Gaussian-PSF limits:** the pixel model uses a Gaussian PSF. Formal centroid offsets for the primaries (up to 1.5″ for 283532878; 5.1″ and 7.6″ in s93 and s102 for 407877153, the s102 stack including a partial event) exceed their noise-only errors. That is model systematics; every neighbour is ≥ 26″ away and rejected by $\Delta\chi^2$ in the hundreds or more.
- **Deblended depths are model-dependent:** for 283532878, about 71–77 ppt (PSF amplitude) up to ~80 ppt (aperture depth ÷ target fraction). The TIC contamination ratio is 0.32. QLP's SAP depths for 407877153 in s93–s102 are diluted by ~10–15%.
- **Not used:** ZTF (too few epochs), 2-min SPOC (none exists), TESScut for s101 of 407877153 (not downloaded; QLP covers its two secondaries).
- **Correlation assumption:** the `corr_vec` ordering behind the correlated Monte Carlo case is assumed, not verified. The secondary-phase test does not use T_p at all.

Files

All in `/home/user/eclipses-paper-review/work/c9a-vets/vet04/`: - **Shared code:** `common.py` (loader, Gaia constants), `fitlib.py` (trapezoid fitter). - **Earlier agent, read and kept:** - `s01_download` (QLP/TESS-SPOC light curves); `s02_inventory` (sectors, gaps, flags); `s03_blind` (blind dip census); `s04_events` (+ `.json`, per-event free fits); `s05_ephem` (+ `s05_ephem_<tic>.json`); `s06_gaia_query` (+ `s06_gaia_nss.csv`, `s06_gaia_source.csv`); `s07_geometry` (Monte Carlo conjunctions); `s08_shape_oddeven` (+ `.json`, `s08_stack_*.npz`); `s10_tesscut_dl`; `s11_pixels` (+ `_<tic>.out/.json`, `s11_diff_*.npy`; aperture photometry, dilution). - **This run:** - `s12_archives.out` (first pass, false negatives), `s12b_archives` (SIMBAD by coordinates, ExoFOP), `s12c_controls` (failed Algol control), `s12d_recheck` (cones with passing control), `s12e_neighbour_eb` (neighbour EB rows). - `s13_pixnoise` (+ `_<tic>.out/.json`, `s13_diff_*.npz`), `s13b_summary`. - `s14_ztf`, `s15_plot_diff` (+ `s15_diff_*.png`), `s16_commonmode`, `s17_geometry2`, `s18_event_table`. - **Downloads:** `/tmp/claude-0/-home-user-eclipses-paper-review/55c92ff4-798a-544f-be03-9888c6804905/scratchpad/data/vet04/` (TESScut cubes deleted).

TIC 283532878: CONFIRMED; P_ecl=17.446019±0.000019 d; depth1=75 ppt (QLP SAP; 71–77 deblended); depth2=10 ppt at phase 0.6781; pixel: on target; pipelines: 2 (QLP, own TESScut); Gaia geometry: pass TIC 407877153: CONFIRMED; P_ecl=17.089294±0.000008 d; depth1=29 ppt (TESS-SPOC/TESScut; 25–27 QLP s93+); depth2=3.0 ppt at phase 0.5469; pixel: on target; pipelines: 3 (QLP, TESS-SPOC, own TESScut); Gaia geometry: pass

vet05 full report

Archived source: sanitized report text.

vet05: adversarial check of TIC 15855794 (UCAC4 564-009141) and TIC 92881948

(WISE J053256.98-225927.8)

Opus 5.5 subagent, 2026-10-30, for Fable 5.1. This run resumed from a predecessor that was killed by a container restart; that run had finished s1–s9 and part of s10. This run added s10b/c/d (archives), s11 (own TESScut photometry, difference images, dilution), s12 (master event table, ephemerides), s13 (Gaia geometry MC) and s14 (physics). The coordinator cut the budget, so the report is short. Scripts and outputs are in this folder. Downloads are in the scratchpad `data/vet05/<tic>/`; all TESScut cubes have been deleted. Depths are in ppt, times in BTJD = BJD_TDB − 2457000. "C" = own TESScut photometry with a 3×3 aperture, "Q" = QLP SAP, "S" = TESS-SPOC SAP (QUALITY==0 throughout).

TIC 15855794: CONFIRMED. (1) Repeating events: I measured 12 primaries and 11 secondaries one by one from raw flux. (2) Period: the eclipse period is 5.8911458 ± 0.0000038 d, +0.54σ from Gaia. (3) Pixels: in s44, s70 and s71 the difference-image deficit lies 0.4–0.9″ from the target, and every neighbour is excluded. (4) Pipelines: Q, S and C agree on depth to within 4%. (5) Geometry: the primary falls at Gaia conjunction A (+0.03 ± 0.08 d at the Gaia epoch; conjunction B is excluded at ~40σ), and the secondary phase of 0.50182 matches Gaia's prediction of 0.502 ± 0.021. Corrections to the screen: the primary is 35.3 ppt after dilution correction, not "probably larger", and it lasts 3.25 h. The secondary is 5.0–5.4 ppt.

TIC 92881948: CONFIRMED. (1) Repeating events: 24 primaries and 22 secondaries measured one by one. (2) Period: 8.8049754 ± 0.0000013 d, +0.43σ from Gaia. (3) Pixels: in all 7 sectors the deficit lies 0.3–1.6″ from the target, and the depth scales with the target's flux fraction in the aperture. (4) Pipelines: Q, S and C agree once dilution is accounted for. (5) Geometry: the primary falls at conjunction A (+0.06 ± 0.09 d; B is excluded at ~44σ). The secondary phase is 0.48334 ± 0.00008, so $e \cos \omega = -0.026$, which is 1.0σ from Gaia's prediction of 0.502 ± 0.018. Corrections to the screen: the primary depth is 16.4 ppt (Q) / 18.3 ppt (C) / 19.0 ppt dilution-corrected, not ~11 ppt. It lasts 1.69 h, not the 1.2 h that BLS gave. The secondary is 5.0–5.3 ppt at phase 0.4833.

Not tested for either star: the common-mode test (brief item 8, outside the five protocol checks) and next-eclipse UTC predictions (budget).

1. Archives and web (s10, s10b, s10c, s10d)

- **15855794:** no rows within 60″ in VSX, Prsa+2022 (J/ApJS/258/16), IJspeert+2021, Gaia DR3 vari_eclipsing_binary / I/358/veb, or SB9. It is not in Gaia vari_summary. SIMBAD: SB*, 1 bibcode (2023A&A...674A..34G). ExoFOP: TOIs 0, CTOIs 0.
- **92881948:** VSX has one entry at 1.35″: "Gaia DR3 2964506347485091072", type **ROT**, G 12.76–12.80. Nothing in Prsa, IJspeert, Gaia-EB or SB9. Gaia vari_summary: in_vari_eclipsing_binary = 0; the classifier's best class is SOLAR_LIKE (0.63). SIMBAD: SB*, 1 bibcode. ExoFOP: TOIs 0, CTOIs 0.
- **Neither star is a catalogued eclipsing binary.**
- **Positive controls:** each catalogue cone found a known row (Prsa, I/358/veb, IJspeert and SB9 via control rows; VSX via the 92881948 row itself).
- **Brief error:** the brief lists "I/357/varisum", but Gaia DR3 variability is I/358 (I/357 is the NSS part), so Gaia variability was queried through TAP.

2. Independence, systematics, pixels (s11_*.out, s12_compile.out, s14_physics.out)

- **Depths, 15855794 (all events):**
- Primary: Q 35.84 ± 0.27 (n=11), S 34.56 ± 0.32 (n=4), C 34.90 ± 0.23 (n=12), C dilution-corrected 35.34 ± 0.23.
- Secondary: Q 5.00 ± 0.30, S 4.91 ± 0.43, C 5.36 ± 0.24.
- The C primary χ^2 of 34/11 comes from E124 (27.6 ppt, at the end of the s70 orbit, a cadence QLP flags).
- **Depths, 92881948:**
- Primary: Q 16.43 ± 0.24, S 18.20 ± 0.24, C 18.33 ± 0.18, C r2.5 15.90 ± 0.23. Divided by the modelled target fraction, 3×3 gives 18.99 ± 0.19 and r2.5 gives 19.17 ± 0.28: they agree.
- QLP is shallower because its larger aperture takes in a G 13.67 star (60″) and a G 14.56 star (53″). TESS-SPOC CROWDSAP is 0.90–0.94.
- Secondary: Q 5.02 ± 0.21, S 5.33 ± 0.24, C 5.08 ± 0.18.
- **Difference images** (median of the eclipse core minus median of the flanks, per sector; pixel-integrated Gaussian fit; errors from 200 fake-event stacks):
- 15855794, primary: offsets from the target 0.4″ (s44, 1.1σ), 0.5″ (s70 without E124, 0.8σ; the run including E124 is distorted by that event) and 0.9″ (s71, 2.1σ). Every neighbour is >45″ from the centroid at >100σ. Secondary: 1.2–3.5″, ≤1.0σ.
- 92881948, primary: offsets 1.3″ (s5), 1.6″ (s6), 1.4″ (s32), 0.9″ (s98), 0.3″ (s107), 1.5″ (s108), 0.9″ (s109), all ≤2.8σ, with errors of 0.02–0.06 px. The nearest neighbour is at 29″. In every sector the pixel with the deepest fractional dip is the target pixel or one next to it.
- **Dilution arithmetic** (Gaussian PSF, and a PSF 1.5× broader):
- 15855794: in the 3×3 aperture the brightest neighbour (G 16.03, 45″) contributes ≤1.9% of the flux. It would need an intrinsic depth of ≥180% to make the primary: excluded. All other neighbours contribute ≤0.6%: excluded.
- 92881948: in 3×3, G 13.67 (60″) contributes 0.11–1.44% and would need ≥127%: excluded. G 14.56 contributes ≤3.95% and would need ≥46%; it is not excluded by arithmetic alone, but it is excluded by the centroid (53″ away) and because the depth grows in the smaller aperture. G 16.46 (29″) contributes ≤2.4% and would need ≥75%; it is excluded by the centroid.

3. TIC 15855794 (joint shape fit, s6: P 35.25 ± 0.31 ppt, T14 3.255 ± 0.025 h, T23 0.70 ± 0.03 h; S 4.89 ± 0.38 ppt, T14 3.14 ± 0.19 h)

There is no 2-min SPOC data. The s43 TESScut cut-out is empty (RAW_CNTS = –1, s11_15855794_s43_check.out).

Event	Mid BTJD	Depth ppt (S / Q / C)	pipelines
S E–1	2500.18	lost (gap/flags)	–
P E0	2503.12275 ± 0.00077	34.25 / 34.40 / 34.20	S Q C
S E0	2506.0791 ± 0.0049	5.10 / 4.79 / 4.99	S Q C
P E1	2509.01328 ± 0.00066	34.46 / 35.24 / 36.59	S Q C
S E1	2511.97	lost (gap)	–
P E2	2514.90506 ± 0.00059	34.45 / 36.19 / 34.79	S Q C
S E2	2517.8619 ± 0.0062	4.54 / 5.00 / 4.47	S Q C
P E3	2520.79709 ± 0.00074	35.07 / 36.27 / 35.79	S Q C
S E3	2523.7436 ± 0.0048	4.99 / 5.99 / 5.26	S Q C

Event	Mid BTJD	Depth ppt (S / Q / C)	pipelines
P E120	3210.06068 ± 0.00106	– / 37.03 / 35.22	Q C
S E120	3213.0275 ± 0.0078	– / 4.10 / 5.43	Q C
P E121	3215.95111 ± 0.00085	– / 36.48 / 35.55	Q C
S E121	3218.9089 ± 0.0063	– / 4.91 / 5.50	Q C
P E122	3221.84	lost (flags, all pipelines)	–
S E122	3224.8002 ± 0.0061	– / 5.38 / 5.60	Q C
P E123	3227.73365 ± 0.00081	– / 37.28 / 34.94	Q C
S E123	3230.6864 ± 0.0057	– / 5.35 / 5.74	Q C
P E124	3233.6255 ± 0.0019	– / flagged / 27.64 (end of orbit, suspect)	C
S E124	3236.5893 ± 0.0065	– / 4.79 / 5.79	Q C
P E125	3239.51638 ± 0.00085	– / 36.17 / 35.01	Q C
S E125	3242.4652 ± 0.0066	– / 4.37 / 5.01	Q C
P E126	3245.40761 ± 0.00090	– / 35.35 / 34.55	Q C
S E126	3248.36	lost (flags)	–
P E127	3251.29757 ± 0.00090	– / 34.85 / 33.77	Q C
S E127	3254.2567 ± 0.0054	– / 5.55 / 5.62	Q C
P E128	3257.18858 ± 0.00085	– / 35.70 / 35.34	Q C
P E264	4058.39	lost (gap; segment "s1751")	–
S E264	4061.3461 ± 0.0060	– / – / 4.92	C

- **Period** (s12): primaries only, P = 5.8911449 ± 0.0000038 (χ^2 4.8/10). Secondaries only, P = 5.8911779 ± 0.0000229. TESScut alone, P = 5.8911437 ± 0.0000042. All are +0.53 to +0.56σ from Gaia's 5.890538 ± 0.001133.
- **Odd/even primaries:** Δ = +0.6σ (S), +0.7σ (Q), +1.8σ (C). There is no hint of 2P.
- **Primary O−C** scatter is ≤1.2 min.

4. TIC 92881948 (s6: P 16.63 ± 0.45 ppt [Q/S fast cadence], T14 1.692 ± 0.030 h, T23 0.20 ± 0.05 h (V-shaped); S 5.00 ± 0.35 ppt, T14 1.76 ± 0.12 h)

Event	Mid BTJD	Depth ppt (S / Q / C)	pipelines
S E−1	1440.6495 ± 0.0027	6.09 / 5.99 / 4.97	S Q C
P E0	1445.2014 ± 0.0011	16.40 / 16.19 / 18.08	S Q C
S E0	1449.4544 ± 0.0028	4.41 / 4.01 / 5.19	S Q C
P E1	1454.0066 ± 0.0010	17.78 / 16.75 / 18.38	S Q C
S E1	1458.2549 ± 0.0035	5.06 / 5.73 / 6.65	S Q C
P E2	1462.8129 ± 0.0010	18.96 / 16.96 / 18.83	S Q C
P E3	1471.6159 ± 0.0006	19.98 / flagged / 19.95	S C
S E3	1475.8763 ± 0.0021	6.41 / 4.72 / 5.95	S Q C
P E4	1480.4223 ± 0.0006	18.24 / 17.06 / 17.14	S Q C
S E4	1484.6802 ± 0.0035	4.51 / 3.93 / 3.63	S Q C
P E5	1489.2272 ± 0.0011	17.53 / 15.78 / 18.18	S Q C
P E83	2176.0155 ± 0.0007	17.85 / 17.50 / 18.44	S Q C
S E83	2180.2716 ± 0.0022	5.09 / 4.98 / 5.29	S Q C
P E84	2184.8210 ± 0.0007	17.48 / 14.82 (partial) / 18.06	S Q C
S E84	2189.0738 ± 0.0033	4.85 / 4.97 / 5.33	S Q C
P E85	2193.6244 ± 0.0008	18.44 / 17.15 (partial) / 17.92	S Q C

Event	Mid BTJD	Depth ppt (S / Q / C)	pipelines
S E85	2197.8764 ± 0.0020	5.93 / 5.49 / 5.81	S Q C
P E289	3989.8400 ± 0.0014	– / flagged / 17.88	C
S E289	3994.0970 ± 0.0035	– / flagged / 4.38	C
P E290	3998.6438 ± 0.0007	– / 15.62 / 17.53	Q C
S E290	4002.90	lost (gap)	–
P E291	4007.4468 ± 0.0012	– / flagged / 19.97	C
S E291	4011.7102 ± 0.0022	– / 4.56 / 5.24	Q C
P E292	4016.2547 ± 0.0007	– / 16.51 / 18.36	Q C
S E292	4020.5174 ± 0.0073	– / flagged / 3.79	C
P E293	4025.0597 ± 0.0008	– / 16.72 / 18.94	Q C
S E293	4029.3183 ± 0.0022	– / 5.16 / 5.29	Q C
P E294	4033.8637 ± 0.0015	– / flagged / 21.07	C
S E294	4038.1176 ± 0.0031	– / flagged / 5.58	C
P E295	4042.6692 ± 0.0007	– / 15.56 / 17.12	Q C
P E320 / E321 / E322 / E323 (s107)	4262.7954 ± 0.0013 / 4271.5989 ± 0.0008 / 4280.4041 ± 0.0008 / 4289.2104 ± 0.0008	C: 19.30 / 18.47 / 18.99 / 17.11	C
S E320 / E321 / E322 (s107), E323 (s108)	4267.0482 ± 0.0031 / 4275.8536 ± 0.0029 / 4284.6636 ± 0.0036 / 4293.4656 ± 0.0032	C: 4.83 / 4.72 / 3.76 / 4.79	C
P E324 (s108)	4298.01	partial (no pre-ingress baseline)	–
S E324 / E325 (s108), E326 (s109)	4302.2707 ± 0.0029 / 4311.0736 ± 0.0024 / 4319.8790 ± 0.0039	C: 4.47 / 5.87 / 6.60	C
P E325 / E326 (s108)	4306.8209 ± 0.0011 / 4315.6232 ± 0.0008	C: 17.93 / 18.20	C
P E327 / E328 (s109)	4324.4292 ± 0.0009 / 4333.2316 ± 0.0021	C: 18.87 / 18.88	C
S E327 / E328 (s109)	4328.6871 ± 0.0024 / 4337.4867 ± 0.0031	C: 5.75 / 5.60	C

- **Sectors without other pipelines:** sectors 107–109 exist only in TESScut (no QLP or SPOC light curves). There is no 2-min SPOC data.
- **Period:** primaries only, P = 8.8049749 ± 0.0000013 (χ^2 26.8/22). Secondaries only, P = 8.8049820 ± 0.0000049. TESScut alone, P = 8.8049753 ± 0.0000014. All are +0.43σ from Gaia's 8.804360 ± 0.001441.
- **Odd/even primaries:** +1.6σ (S), +0.2σ (Q), +0.7σ (C).
- **Secondaries** are not at phase 0.5 and differ from the primaries in depth, so this is not an equal-pair P/2 system.

Geometry vs Gaia (s13_geometry.out; 20000 draws, NSS correlation matrix)

- **How the test works:** with ω and T_peri correlated at +0.9996, conjunction times are well defined. Without the correlations the prediction is uninformative (σ ≈ 2 d). The order of the first four corr_vec parameters is uncertain, so all 24 orders were run; they give the same answer.
- **15855794:**
- Observed primary minus conjunction A, at the Gaia epoch: +0.030 d (robust σ 0.080; over the 24 orders +0.23 to +0.46σ). Minus conjunction B: −2.93 d (−39σ).
- Carried forward with Gaia's P: +0.65σ.
- Secondary phase: predicted 0.5019 ± 0.0209, observed 0.50182 ± 0.00030; e cos ω = +0.0029 ± 0.0005, against Gaia's +0.003 ± 0.033.
- **92881948:**

- Primary minus conjunction A: +0.060 d (σ 0.086; +0.35 to +0.69σ). Minus conjunction B: −43σ.
- Forward with Gaia's P: +0.52σ.
- Secondary phase: predicted 0.5016 ± 0.0181, observed 0.48334, a 1.0σ difference (32% of draws lie as far from 0.5). e cos ω = −0.0262, against Gaia's +0.003 ± 0.028.
- **Which conjunction:** in both stars the primary is conjunction A, the companion passing in front of the RV star.
- **Physics (s14, uniform discs):**
- 15855794: f(m) = 0.0437 M⊙, so m2 ≥ 0.49 M⊙ (M dwarf, about 3630 K). The observed T14 needs b ≈ 1.07 (grazing). Predicted depths are 32.2 / 4.2 ppt; observed 35.3 / 5.4 ppt. The depth ratio implies a companion at 3760 K.
- 92881948: f(m) = 0.0144 M⊙, so m2 ≥ 0.22 M⊙. Grazing, b ≈ 1.22. Predicted 14.6 / 4.0 ppt; observed 19.0 / 5.3 ppt. The depth ratio of 0.277 matches the Planck ratio (0.275) for a 3180 K companion.
- Both are physically sane.

Ephemerides and next eclipses (s12_compile.out)

- **15855794:** T_ref = BTJD 2886.04720 ± 0.00024 (BJD_TDB 2459886.04720), P = 5.8911458 ± 0.0000038 d. Secondary at +2.95631 ± 0.00179 d.
- **92881948:** T_ref = BTJD 3047.70744 ± 0.00019 (BJD_TDB 2460047.70744), P = 8.8049754 ± 0.0000013 d. Secondary at +4.25578 ± 0.00070 d.
- **Next three eclipses in UTC:** not computed (budget cut). Use T_ref + nP. Those times are BJD_TDB, so no barycentric correction has been applied for an observer.

Caveats

- **Common mode:** not tested with other stars. Partial substitutes: the residual sky and the neighbour pixels show no dip at the target's fractional depth, and the depth scales with the target fraction.
- **TESScut times:** taken from the FFI time stamps; the barycentric correction is for the FFI, not the target (errors under a minute).
- **15855794 E124:** shallow (27.6 ppt), at the end of an orbit, and flagged by QLP. Its primary-stack difference image was redone without it (s11_15855794_s70_exclE124).
- **PSF model:** a Gaussian with a 1.5× variant, not the TESS PRF. Neighbour exclusions rely on the 3×3 arithmetic plus the centroids.

Files

`s1–s9` (predecessor run); `s10b_archives_redo`, `s10c_vizier_check`, `s10d_controls_gaiavar` (.py/.out); `s11_tesscut.py` with `s11_<tic>_s<sec>.{out,json}`, `_lc.npz`, `_diff.npz`, `s11b_redownload_s70`, `s11_15855794_s43_check.out`; `s12_compile.{py,out,json}`; `s13_geometry.{py,out,json}`; `s14_physics.{py,out,json}`; plus `common.py` and `fitlib.py`.

vet06 full report

Archived source: sanitized report text.

vet06: adversarial check of TIC 468827621 (UCAC4 796-015685) and TIC 416285595 (UCAC4 571-080398)

Opus 5.5 subagent, 2026-10-30, for Fable 5.1. This is a reduced-depth run: the coordinator cut the budget mid-task. It covers the five protocol checks only; physics, next-eclipse lists, the common-mode test and the literature search were skipped.

Light curves were fetched from MAST's `Download/file` endpoint (`s2`), because `lightkurve.search_lightcurve` hung at MAST and archive.stsci.edu serves an incomplete TLS chain. Data are in `scratchpad/data/vet06/<tic>/`. Each TESScut cube was deleted after use.

Depths are in ppt. Times are BTJD (BJD_TDB − 2457000). A = the companion passes in front of Gaia's RV star ($\nu = 90° - \omega$); B = the companion is behind it.

TIC 468827621: CONFIRMED. All five checks pass: 1. **Individual events.** 10 primary and 9 secondary epochs are fitted individually from raw SAP (QUALITY==0) in QLP s19, 20, 59, 60, 73 and TESS-SPOC s19, 20. 2. **Period.** The eclipse period is 11.041067 ± 0.000001 d (primaries, O−C rms 0.5 min). It is −0.16σ from Gaia's 11.04138 ± 0.00196 d. The secondaries alone give 11.041068 ± 0.000005 d. 3. **Pixels.** The primary deficit is 1.42″ (s19) and 1.50″ (s59) from the target's WCS position. The only neighbour bright enough in principle is G 13.40 at 70.6″. With its PRF share of my aperture it would need a 4800–7800% intrinsic eclipse, and the fractional deficit at its pixel is only 1–4 ppt. Both other neighbours need more than 100% even in the worst case. 4. **Pipelines.** QLP, TESS-SPOC and my TESScut aperture all see the primary: 192, 184 and 192 ppt. 5. **Geometry.** The primary is at conjunction A (−0.14 ± 0.09 d); B is excluded at 31σ (20σ without Gaia's correlations). The secondary phase, **0.72649 ± 0.00003**, is +0.56σ from Gaia's predicted 0.718 ± 0.015 (from e = 0.355, ω = 350°). The eclipse at A, nearer periastron, is the longer one: 6.0 h against 5.05 h. That is the direction $e \sin\omega = -0.06$ requires.

The screen was right on P, on both depths (QLP) and on the secondary phase: 0.7265 measured against its by-eye 0.72. TESS-SPOC SAP depths are about 4% shallower than QLP (184/29.6 against 192/31.5 ppt).

TIC 416285595: CONFIRMED. All five checks pass, with weak points below: 1. **Individual events.** 21 primary and 19 secondary epochs are fitted individually: QLP s14, 40, 54, 80, 81 and TESS-SPOC s40. 2. **Period.** P = 4.615357 ± 0.000003 d (O−C rms 2.4 min), −0.37σ from Gaia's 4.615565 ± 0.000559 d. 3. **Pixels.** The primary deficit is 2.16″ (s40, camera 1) and 0.58″ (s80, camera 2) from the target. Every one of the 43 neighbours needs more than 100% with its PRF share; the minimum is 126%, for G 14.75 at 37.9″ in s40. The neighbours that pass only in the unphysical worst case (100% of their flux in my aperture) all lie 37″ or more away, at least 1.8 px from a deficit centre that sits within 0.1 px of the target. So the primary is on the target, and the screen's contamination concern (ratio 0.27) does not change the attribution of the primary. 4. **Pipelines.** QLP and TESS-SPOC see every event at the same times and durations. Depth agreement is the weakest part (see the weak points). 5. **Geometry.** The primary is at conjunction A (−0.04 ± 0.08 d); B is excluded at 29σ with correlations, but only 3.3σ without. The secondary phase, 0.50011 ± 0.00027, is +0.28σ from Gaia's 0.49 ± 0.03.

Weak points: - **Depths disagree between pipelines.** QLP is deeper than TESS-SPOC in s40: 61.7 against 54.7 ppt, about 12% and about 4σ apart. My TESScut aperture gives 70 (s40) and 89 ppt (s80), but its background (median of pixels ≥ 4 px away) is unreliable in this crowded field. I attribute the

spread to crowding and background, since none of these is a crowding-corrected depth. Strict agreement within errors fails. - **The secondary is not tested in pixels.** At 12.9 ppt, G 14.75 at 37.9″ could produce it arithmetically (it would need about 26%). Only the exact phase 0.5001 and the matching period tie it to the target. - **The geometry test is weak.** e ≈ 0.08, so phase 0.5 is the generic expectation, and the A/B decision rests on Gaia's ω–Tp correlation of 0.998.

The screen was right on P and on the secondary phase. Its 59 ppt primary depth matches QLP but not TESS-SPOC (54.7). Its 11 ppt secondary is low: QLP and TESS-SPOC both give 12.9 ppt. The primary is near-V: T14 3.6 h, trapezoid T23 about 0.7 h.

1. Archives and web

- **VizieR mirror cones (`s1_archives.out`, `s1b_archives.out`):** both stars return 0 rows within 60″ in VSX, Prša+2022, IJspeert+2021, Gaia DR3 veb / vclassre / vari (10″), SB9, ASAS-SN II/366 and J/AJ/162/170. Algol returns 1 VSX row, so the cones work.
- Neither star is a catalogued eclipsing binary: none is listed in VSX, Gaia-EB, Prša or IJspeert within 60″.
- 468827621 is in the Gaia DR3 short-timescale table I/358/vst (amplitude 0.044 mag) and is flagged VARIABLE in gaia_source.
- **SIMBAD:** both are SB*, with 2 bibcodes (468827621) and 1 bibcode (416285595), all Gaia DR3. The SIMBAD lookup by TIC ID fails; the 10″ coordinate cone finds both.
- **ExoFOP:** 0 TOIs and 0 CTOIs for both.
- **Gaia SB1 rows (`gaia_tbosb1.tsv`, with corvec):**
- 468827621: e 0.3555 ± 0.0239, ω 350.13 ± 3.87°, K1 42.03 km/s.
- 416285595: e 0.078 ± 0.044, ω 104 ± 36°, K1 46.4 km/s.

2. Independence, systematics, pixels

- **Pipeline depths (`s3_events.out`, per-event means):**
- 468827621 primary: QLP 191.9 (sd 3.6, N = 10); TESS-SPOC 184.5 (sd 5.0, N = 4); TESScut 191.9 (s19) and 191.7 (s59).
- 468827621 secondary: QLP 31.5 (sd 1.0); TESS-SPOC 29.6 (sd 1.2).
- 416285595 primary: QLP 58.6 (sd 3.3, N = 20); TESS-SPOC 54.7 (sd 1.3, N = 6); TESScut 69.8 (s40) and 89.4 (s80), background-limited.
- 416285595 secondary: QLP 12.9 (sd 2.5); TESS-SPOC 12.9 (sd 1.4).
- Neither star has 2-min SPOC data.
- **Odd/even (primaries):** 468827621 +0.38σ (QLP) and −0.09σ (SPOC); 416285595 −1.2σ (QLP) and +1.1σ (SPOC). The two eclipse kinds differ in depth (and, for 468827621, in duration), so there is no hidden P/2 or 2P.
- **Difference images (`s4_tesscut_<tic>_s<sec>.txt/.png`; mean over primaries of median in-eclipse minus local out-of-eclipse; Gaussian-fit centre):**

Star / sector (cam)	N ev	deficit − target WCS	frac. depth, target pixel	aperture share of target
468827621 s19 (cam 2)	2	1.42″ (deficit − OOT 1.17 ± 0.33″)	158 ppt	0.962
468827621 s59 (cam 2)	3	1.50″ (deficit − OOT 1.62 ± 0.35″)	117 ppt	0.957
416285595 s40 (cam 1)	4	2.16″	34 ppt	0.826
416285595 s80 (cam 2)	5	0.58″	48 ppt	0.854

- For 416285595 the out-of-eclipse Gaussian locked onto a brighter star at the cutout edge, so only the WCS offset is quoted.
- Both 468827621 sectors are in camera 2. That is the only camera available.

3. TIC 468827621 (P 11.041067 d)

`events_468827621.csv`: trapezoid on a quadratic baseline; errors are the covariance scaled to the rms, inflated by a 10-shift prayer bead.

Event	Mid BTJD	Depth QLP / SPOC (ppt)	T14 / T23 (h)	pipelines
P E0 s19	1819.85379 ± 0.00025	196.9 / 189.9	6.00 / 2.33	QLP+SPOC+TESScut
P E1 s19	1830.89477 ± 0.00025	196.6 / 188.9	5.99 / 2.34	QLP+SPOC+TESScut
P E3 s20	1852.97691 ± 0.00024	186.4 / 179.4	6.02 / 2.33	QLP+SPOC
P E4 s20	1864.01807 ± 0.00027	187.5 / 179.6	6.00 / 2.36	QLP+SPOC
P E99 s59	2912.91944 ± 0.00020	195.2 / –	6.03 / 2.31	QLP+TESScut
P E101 s59	2935.00116 ± 0.00019	194.2 / –	5.99 / 2.39	QLP+TESScut
P E102 s60	2946.04227 ± 0.00038	187.8 / –	6.03 / 2.33	QLP
P E103 s60	2957.08343 ± 0.00018	190.3 / –	6.02 / 2.31	QLP
P E133 s73	3288.31654 ± 0.00045	191.6 / –	5.96 / 2.35	QLP
P E135 s73	3310.39800 ± 0.00018	192.1 / –	5.99 / 2.36	QLP
S E−1 s19	1816.83466 ± 0.00143	– / 30.8	5.04 / 2.27	SPOC
S E1 s19	1838.91713 ± 0.00110	31.9 / 30.3	5.14 / 2.17	QLP+SPOC
S E2 s20	1849.95666 ± 0.00095	29.1 / 27.8	5.01 / 2.51	QLP+SPOC
S E3 s20	1860.99765 ± 0.00098	31.3 / 29.6	4.99 / 2.31	QLP+SPOC
S E99 s59	2920.94045 ± 0.00089	32.4 / –	5.12 / 2.21	QLP
S E100 s59	2931.98226 ± 0.00096	32.2 / –	5.04 / 2.47	QLP
S E101 s60	2943.02108 ± 0.00108	31.2 / –	4.87 / 2.49	QLP
S E133 s73	3296.33725 ± 0.00087	32.2 / –	5.07 / 2.29	QLP
S E134 s73	3307.37832 ± 0.00080	31.4 / –	5.00 / 2.48	QLP

- **Lost events:** P E100 (s59), P E134 (s73), S E0 (s19) and S E102 (s60) fall in gaps or flagged data.
- **Ephemerides:**
- Primaries: P = 11.041067 ± 0.000001 d, O−C within ±1.3 min.
- Secondaries: P = 11.041068 ± 0.000005 d, O−C rms 1.0 min.
- **SPOC secondary odd/even:** +2.5σ, from only 4 events. The s20 SPOC depths are uniformly low, so this is a sector systematic, not two eclipse kinds.

4. TIC 416285595 (P 4.615357 d)

`events_416285595.csv` has all 40 fitted events plus the lost ones. A representative subset:

Event	Mid BTJD	Depth QLP / SPOC (ppt)	T14 / T23 (h)	pipelines
P E0 s14	1687.52122 ± 0.00101	52.9 / –	3.46 / 0.91	QLP
P E153 s40	2393.67735 ± 0.00069	56.4 / 54.6	3.64 / 0.66	QLP+SPOC+TESScut
P E154 s40	2398.29120 ± 0.00056	60.4 / 53.1	3.51 / 0.90	QLP+SPOC
P E155 s40	2402.90615 ± 0.00051	64.2 / 56.8	3.74 / 0.60	QLP+SPOC+TESScut
P E156 s40	2407.52291 ± 0.00075	63.5 / 54.2	3.55 / 0.90	QLP+SPOC+TESScut
P E158 s40	2416.75340 ± 0.00049	64.0 / 55.8	3.70 / 0.70	QLP+SPOC+TESScut
P E236 s54	2776.75002 ± 0.00096	59.9 / –	3.41 / 0.89	QLP
P E392 s80	3496.74568 ± 0.00070	57.9 / –	3.58 / 0.82	QLP+TESScut
P E398 s81	3524.43758 ± 0.00072	55.4 / –	3.62 / 0.77	QLP

Event	Mid BTJD	Depth QLP / SPOC (ppt)	T14 / T23 (h)	pipelines
S E152 s40	2391.37357 ± 0.00232	– / 10.7	2.98 / 1.42	SPOC (QLP fit degenerate)
S E156 s40	2409.82774 ± 0.00208	9.9 / 14.0	3.15 / 0.30	QLP+SPOC
S E157 s40	2414.44236 ± 0.00167	13.6 / 14.7	3.50 / 0.20	QLP+SPOC
S E395 s81	3512.89836 ± 0.00325	11.8 / –	3.22 / 1.18	QLP
S E397 s81	3522.13865 ± 0.00319	12.3 / –	3.39 / 0.95	QLP

- **Primary ephemeris:** N = 21 epochs, O–C rms 2.4 min. The two s14 events (30-min cadence) are at −7 min.
- **Secondaries:** P = 4.615370 ± 0.000009 d, O–C rms 8.7 min. Some single-event depth errors are degenerate (T23 → 0); see the csv.

Geometry vs Gaia

`s0_gaia_predict.out`, `s5_geometry.out` and `s5b_unwrap.out`. The Monte Carlo uses 20000 draws with Gaia's corvec, ordered P, Vcm, K1, e, ω, Tp (column-major upper triangle, positive-definite for both stars). Uncorrelated values are in brackets. "Backward" means the TESS primary ephemeris taken back to the Gaia epoch. - **468827621:** - Backward: primary − A = −0.142 ± 0.087 d (−1.6σ) [−0.142 ± 0.170]; primary − B = +2.97 ± 0.10 d (+31σ) [+20σ]. - Forward (Gaia ephemeris taken to TESS): −0.18 ± 0.28 d from A. - The deep eclipse is conjunction A. - Secondary phase: observed 0.72649 against predicted 0.7182 ± 0.0149, **+0.56σ**. If the deep eclipse were B, the shallow one would be predicted at 0.282. - **416285595:** - Backward: primary − A = −0.043 ± 0.075 d [−0.05 ± 0.62]; primary − B = ±2.31 ± 0.08 d (29σ) [±0.70, 3.3σ]. - Secondary phase: observed 0.50011 against predicted 0.4915 ± 0.0304, +0.28σ. - Because e is small, the secondary-phase test is weak.

Ephemerides

Errors are formal (`s3_events.out`); TESS-pipeline times are already barycentric. - **TIC 468827621 primary:** T0 = BTJD 1819.85371 ± 0.00010 (BJD_TDB 2458819.85371), P = 11.041067 ± 0.000001 d, T14 6.0 h. Secondary at phase 0.72649 ± 0.00003, T14 5.05 h. - **TIC 416285595 primary:** T0 = BTJD 1687.52620 ± 0.00067 (BJD_TDB 2458687.52620), P = 4.615357 ± 0.000003 d, T14 3.6 h. Secondary at phase 0.50011 ± 0.00027.

Caveats

- **SAP depths are not crowding-corrected.** The QLP/SPOC offsets are 4% (468827621) and about 12% (416285595).
- **TESScut photometry** uses a crude background: the median of pixels ≥ 4 px away. In the crowded 416285595 field it gives depths of 70–89 ppt, so it is used for location, not for depth.
- **Secondaries were not pixel-tested** for either star. For 468827621, the G 16.17 star at 24.6″ could arithmetically produce the 31 ppt secondary in the worst case, and for 416285595 the G 14.75 star at 37.9″ could produce the 12.9 ppt one. For both stars only timing ties the secondary to the target.
- **Dilution uses a Gaussian PRF** (σ 0.75 px) with G-band fluxes, not the TESS PRF.
- **The 468827621 difference images** are both from camera 2; no other orientation was available in the sectors used.
- **Not done:** common-mode test, ZTF, physics sanity check, next-eclipse lists.

Files

`s0_gaia_predict.py/.out`, `s1_archives.py/.out`, `s1b_archives.py/.out`, `gaia_tbosb1.tsv`, `s2_download.py/.out`, `s3_events.py/.out`, `events_468827621.csv`, `events_416285595.csv`, `s3_ephem_<tic>.npy`, `s4_tesscut.py`, `s4_tesscut_<tic>_s<sec>.txt/.npz/.png` (468827621 s19, s59; 416285595 s40, s80), `s5_geometry.py/.out`, `s5b_unwrap.py/.out`, `REPORT.md`.

Summary - TIC 468827621: CONFIRMED; P_ecl=11.041067 d; depth1=184–192 ppt; depth2=29.6–31.5 ppt at phase 0.72649; pixel: on target (primary); pipelines: 3; Gaia geometry: pass - TIC 416285595: CONFIRMED; P_ecl=4.615357 d; depth1=55–59 ppt (QLP s40 62); depth2=12.9 ppt at phase 0.50011; pixel: on target (primary; secondary untested); pipelines: 3 (depths agree only to ~12%); Gaia geometry: pass (weak, e≈0)

vet11 full report

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vet11: Gaia-period folds of five sinusoidal c9a candidates

Independent check (Opus 5.5 subagent, 2026-10-30, for Fable 5.1). I resumed a predecessor that was killed by a container restart. Its downloads, s3 light-curve fits, s4 pixel maps and the first half of the catalogue cones were reused. I added: s2b (the rest of the catalogue cones and the Gaia DR3 variability tables), s5b (the Roche toy model, crash fixed), s6 (a refold at P_phot, which removes the phase-drift artefacts of the Gaia-P fold), s8 (a narrow-dip test at min I), s7 (TIC 62328710 frequencies and ZTF), s9/s9b (Gaia covariance and conjunctions) and s10 (ExoFOP). Data are QLP SAP only, QUALITY==0, 10-min bins, with a per-segment cubic baseline. No TESS-SPOC light curves exist for any of the five. Depths are in ppt, measured from the mean of the two maxima, with per-sector SEM errors. Phases are at P_phot with 0 = min I (the deeper minimum). "Conj A" = the primary at superior conjunction (companion in front). Conjunction errors come from 40000 draws with the full Gaia corr_vec. The earlier independent-draw MC ignored corr(omega,Tp) of about 0.99 and overstated the errors. Coverage was cut short by a budget limit: one pixel sector per star, no difference images of the min-I dip, and no TESS-SPOC or TESScut-aperture depth comparison.

TIC 372683900 (Gaia P 1.476502 +- 0.000081 d, e 0.30)

quantity	value
LS top peak / P_phot (chi2 scan)	0.73825 d (=P/2); 1.476507 +- 0.000001 d, ratio 1.000003 (+0.05 sigma)
min I / min II depth	31.07 +- 0.31 at 0 / 28.18 +- 0.19 at 0.5011 +- 0.0008; diff 2.88 +- 0.35 (8.3 sigma), 11/12 sectors
harmonics K=6	A1 2.35, A2 14.79, A3 0.67, A4 0.27 (+-0.05); min widths 0.271 / 0.244 (cos2phi = 0.250)
narrow dip at min I	0.63 ppt, empirical 1.7 sigma (none)
conjunctions	min I - conj B +0.087 +- 0.117; min II - conj A -0.042 +- 0.127; min I at conj A excluded at 3.6 sigma
minima separation	0.5007 +- 0.0018 vs Gaia-predicted 0.372 +- 0.039 (3.3 sigma): photometry says e cos w ~ 0, Gaia says -0.21
pixel (s57)	amplitude centroid 0.9" from target. A G=15.23 star at 1.8" is unresolved, but it would need 128 ppt intrinsic and would have to match Gaia P to 1e-4 d

ELLIPSOIDAL: a smooth double wave at the Gaia period with no eclipse narrowing. The unequal minima are deeper at conj B, which matches the Morris A1 sign reversal at low i. The s5b toy model at i ~ 20 deg gives A1 1.74, A2 14.87, deeper at B by 2.69 ppt. Caveat: with the TIC R1 that needs a dark M2 ~ 4 Msun, so R1 is probably overestimated. Gaia's e = 0.30 conflicts with the minima separation.

TIC 13106408 (Gaia P 2.637505 +- 0.000401 d)

quantity	value
LS top / P_phot	1.31884 d (=P/2); 2.637721 +- 0.000002, ratio 1.000082 (+0.54 sigma)
min I / min II depth	46.85 +- 0.33 at 0 / 40.63 +- 0.26 at 0.5009 +- 0.0010; diff 6.22 +- 0.39 (15.8 sigma), 7/7 sectors
harmonics K=6	A1 2.06, A2 21.77, A3 1.40, A4 0.72 (+-0.03); widths 0.252 / 0.272 (min II flattened)
narrow dip at min I	0.36 ppt, 2.3 sigma (not significant)
conjunctions	min I - conj A +0.079 +- 0.166; min I - conj B -0.427 +- 0.167 (2.6 sigma)
pixel (s58)	on target (1.9"). The best neighbour (G 14.4, 15") would need 91 ppt

ELLIPSOIDAL: a large-amplitude, near-Roche-filling double wave at the Gaia period with unequal minima and a flattened min II. The K=6 residuals are white and there are no eclipse features. The conjunction test only mildly prefers min I at conj A.

TIC 372683622 (Gaia P 3.809006 +- 0.000179 d)

quantity	value
LS top / P_phot	1.90468 d (=P/2); 3.809282 +- 0.000014, ratio 1.000073 (+1.5 sigma)
min I / min II depth	7.73 +- 0.16 at 0 / 6.45 +- 0.19 at 0.4943 +- 0.0061; diff 1.28 +- 0.30 (4.2 sigma) but variable, 0.3-3.2 ppt between sectors
harmonics K=6	A1 0.78, A2 3.23, A3 0.13, A4 0.12; per-sector A3/A4 ~0.5 and incoherent (spots)
narrow dip at min I	0.24 ppt, 2.1 sigma (none)
conjunctions	min I - conj B +0.075 +- 0.040; min II - conj A +0.044 +- 0.036; min I at conj A excluded at 12.5 sigma
pixel (s56)	on target (2.6")

ELLIPSOIDAL: a double wave locked to the Gaia period with minima at the two conjunctions. The deeper minimum is at conj B, as for low-i ellipsoidals; the toy model at i ~ 20 deg gives A2 3.2 and deeper at B by 1.7 ppt. On top of this there is variable spot modulation, and no eclipse.

TIC 319185391 (Gaia P 2.677901 +- 0.000187 d)

quantity	value
LS top / P_phot	1.33902 d (=P/2); 2.678053 +- 0.000009, ratio 1.000057 (+0.8 sigma)
min I / min II depth	8.46 +- 0.30 at 0 / 6.57 +- 0.14 at 0.5102 +- 0.0040; diff 1.89 +- 0.23 (8.3 sigma), 6/6 sectors
harmonics K=6	A1 0.70, A2 3.64, A3 0.23, A4 0.04
narrow dip at min I	box 1.15 +- 0.17 ppt, centre +0.007, core 0.012 phase (0.8 h), profile about 0.03 phase (about 1.9 h). Empirical S/N 5.3; 0/400 random phases as deep; per sector 0.46/0.84/2.01/0.85/0.96/1.48 (6/6 > 0, weighted 1.04 +- 0.17); even 1.26 +- 0.25, odd 1.04 +- 0.29; at min II -0.03 +- 0.21. The same test on the other three stars gives 1.7-2.3 sigma
conjunctions	min I - conj A +0.057 +- 0.084 (conj B excluded at 6.0 sigma); separation 0.511 +- 0.003 vs Gaia 0.529 +- 0.028
pixel (s85)	the ellipsoidal signal is on target (5.2"). The dip itself has no pixel test
other	Gaia DR3 also has an astrometric orbit for this source, P = 434.5 +- 8.2 d, e = 0.70 (likely a triple)

DETACHED-EB (grazing, fold-only): an ellipsoidal double wave plus a phase-locked 1 ppt, about 2 h dip at min I, which sits at Gaia conj A (companion in front). This is the eclipse that the flattened P/2 fold hid. Individual events are about 1 sigma, so they cannot be seen singly, and the dip has no difference image. That puts it at probable-level evidence.

TIC 62328710 (Gaia P 0.938088 +- 0.000042 d)

quantity	value
dominant signal	f1 = 6.390610 +- 0.000005 c/d (0.156480 d), 3.09 ppt; f1/f_orb = 5.9950, not an orbital harmonic (19 sigma in P). Second frequency f = 5.806788 c/d, 1.26 ppt. Both are stable in all 7 sectors
Gaia-P signal	after prewhitening, A2 = 0.058 +- 0.046 ppt (< 0.15 ppt, 2 sigma). Minima: none
pixel (s56)	amplitude centroid 45" from target, on Gaia DR3 2165710355333626240 (G 13.38, 44.3"; next source worse by dchi2 642). f = 5.807 c/d also peaks there
ZTF	neighbour: top peak 6.39060 c/d, 9.6 (g) / 8.9 (r) mmag, noise 99th percentile 2.3/4.6; 5.807 c/d 4.2/5.1 mmag. Target: 0.16/0.70 mmag at f1 (noise 1.5/1.7)
catalogue	the neighbour is Gaia VarFlag VARIABLE (vari_short_timescale). Target not in VSX/Gaia-EB within 60"

OTHER: the "P/3" wave is a multiperiodic delta Scuti-type pulsator, the neighbour 44" away, diluted into the QLP aperture. The target is flat at the Gaia period. With any R1 >= 1.8 Rsun, ellipsoidal A2 < 0.15 ppt would require i < 5 deg, so the Gaia 0.938 d SB1 orbit is suspect.

Summary

TIC	class	P_phot / P_Gaia	min1 (ppt, phase)	min2 (ppt, phase)	pixel	catalogues
372683900	ELLIPSOIDAL	1.000003	31.07 at conj B	28.18 at conj A	on target (1.8" nbr unresolved)	none; SIMBAD SB*
13106408	ELLIPSOIDAL	1.000082	46.85	40.63	on target	VSX only an SR at 29"; SB*
372683622	ELLIPSOIDAL	1.000073	7.73 at conj B	6.45 at conj A	on target	none; SB*
319185391	DETACHED-EB	1.000057	8.46 + 1.0 ppt dip at conj A	6.57 at conj B	on target (dip untested)	none; SB*; Gaia astrometric 434 d
62328710	OTHER	f1 = 5.995 f_orb	none (A2 < 0.15)	none	neighbour 44"	nbr Gaia VARIABLE

TIC 372683900: ELLIPSOIDAL; P_phot=1.476507 d (1.000003); min1=31.07+-0.31 ppt at phase 0 (conj B +0.09+-0.12); min2=28.18+-0.19 at phase 0.5011 (conj A -0.04+-0.13); pixel: on target; catalogue: not in VSX/Gaia-EB/Prsa/IJspeert/SB9, SIMBAD SB* TIC 13106408: ELLIPSOIDAL; P_phot=2.637721 d (1.000082); min1=46.85+-0.33 ppt at phase 0 (conj A +0.08+-0.17); min2=40.63+-0.26 at phase 0.5009; pixel: on target; catalogue: none (VSX SR at 29" unrelated), SIMBAD SB* TIC 372683622: ELLIPSOIDAL; P_phot=3.809282 d (1.000073); min1=7.73+-0.16 ppt at phase 0 (conj B +0.075+-0.040); min2=6.45+-0.19 at phase 0.4943 (conj A +0.044+-0.036); pixel: on target; catalogue: none, SIMBAD SB* TIC 319185391: DETACHED-EB (grazing, fold-only); P_phot=2.678053 d (1.000057); min1=8.46+-0.30 ppt at phase 0 (conj A +0.06+-0.08) with a 1.04+-0.17 ppt, about 2 h dip; min2=6.57+-0.14 at phase 0.5102; pixel: on target (ellipsoidal signal; dip untested); catalogue: none, SIMBAD SB*, Gaia astrometric 434 d orbit TIC 62328710: OTHER (delta Sct neighbour); P_phot=0.156480 d (1/5.995 of Gaia P); min1=none (Gaia-P A2<0.15 ppt); min2=none; pixel: neighbour Gaia DR3 2165710355333626240 at 44" (ZTF confirms 6.3906 c/d there, not on target); catalogue: neighbour Gaia VARIABLE

Caveats

Only QLP is available, so there is no second pipeline. The Gaia-P fold in s3 has phase drift: its residual "dips" flip sign between sectors, so use the s6/s8 numbers, which are at P_phot. The s5b Roche model is a toy (blackbody at 800 nm, no reflection) and uses TIC radii, which look overestimated (13106408 and 62328710 overfill their Roche lobes). ExoFOP: no TOI or CTOI for any of the five (s10). The ms_oscillator TAP query failed.

Files

s1_download.py/.log, s2_catalogs.py/.txt (+s2_err.log), s2b_catalogs.py/.txt, s3_lc.py + s3_.txt/.json + fold_.png, s4_pixels.py + s4_*s.txt/.npz + pixs.png, s5b_roche.py/.txt (s5_roche.* crashed), s6_pphot.py + s6*.txt/.json, s7_62328710.py/.txt/.json, s8_dip319.py + s8__b10.txt/.json + dip_319185391.png, s9_gaia_corr.py + s9_gaia_nss.csv, s9b_conj_corr.py/.txt, s10_exofop.sh/.txt, common.py.

Returned batch summary

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Chunk-9 (c9a) candidates: screening and vets, 2026-10-30

Cloud session (Fable 5.1 lead, Opus 5.5 helpers) on the 24-star bundle Astra packaged from Fable 5's `sb1-scale` chunk 9 (montage sheets c9a_00 and c9a_01; roster order = best z descending). Everything here is on the branch `claude/zealous-pasteur-8mmme2`.

What was done

1. **Screen** (`SCREEN_RESULTS.csv`, `SCREEN_REPORT.md`): first-pass visual triage of the two montage sheets, three Opus helpers at eight panels each plus a second look at the borderline calls. 19 eclipse-like, 5 non-eclipse-like, 0 ambiguous, 0 unreadable.
2. **Shared inputs** (`targets.csv`, `neighbours_90as.csv`, `prep.py`): TIC rows, Gaia DR3 NSS SB1 orbits (VizieR I/357 via the idia mirror; the CDS and ESA hosts were unreachable), Gaia neighbours within 90", QLP / TESS-SPOC / 2-min / TESScut sector lists.
3. **Vets** (`vet01`–`vet06`, `vet11`; protocol in `VET_BRIEF.md`): independent adversarial checks to the paper's Section 2.5 tier definition (events from raw SAP, period from the eclipses, difference images and dilution, independent pipeline, Gaia geometry). `VERDICTS.md` is the one-line-per-star table; each `vetNN/REPORT.md` holds the evidence, scripts and outputs.

Results

	n	TICs
vetted CONFIRMED	1 2	195865529*, 468181994, 470235665, 379251709, 70665829 (= paper's A4, blind control, recovered), 331293467, 283532878, 407877153, 15855794, 92881948, 468827621, 416285595 (with caveats: pipeline depths differ ~12%)
of which already a catalogued eclipsing binary	1	195865529 (ASAS-SN EA, same period; not in VSX)

	n	TICs
sinusoids refolded at Gaia P: ellipsoidal	3	372683900, 13106408, 372683622
sinusoids: grazing detached EB, fold-level only (PLAUSIBLE at best)	1	319185391 (also a Gaia 434 d astrometric orbit: likely triple)
sinusoids: signal from a neighbour (δ Scuti at 44"); target flat at the Gaia period	1	62328710 (Gaia's SB1 orbit for it is suspect)
eclipse-like, **not yet vetted** (credits ran out)	7	300903537, 23112764, 373062090, 465073693, 316405449, 465263741, 441741446

Side findings worth a catalogue note: a catalogued EB (Gaia DR3 2202301140233408128, P 1.7975 d) sits 29.7" from 283532878 and may leak ≤3 ppt dips into its light curve; 372683900's equal, half-period-spaced minima conflict with Gaia's e = 0.30 at 3.3σ; several secondaries are off phase 0.5 and match Gaia's e cos ω (470235665, 283532878, 407877153, 92881948, 468827621).

Cost note, for the next batch

Each full vet cost ~240k Opus tokens, about two stars per agent. Most of that was execution depth the tier does not need (every individual event fitted, own TESScut photometry on top of two pipelines, dilution for every neighbour within 90", Monte Carlo geometry, physics checks, ephemerides, literature search) plus each agent rebuilding its tooling from scratch, and a container restart that cost wave 1 roughly 1.7×. For the remaining seven: reuse the scripts in `vet04/` or `vet05/` (`common.py`, `fitlib.py`), a handful of events per star, one difference image per sector in two sectors, only the neighbours that matter, quoted pipeline depths, simple error propagation, two-page report. `VET_BRIEF.md` should be trimmed accordingly before reuse.

Data-access notes (this machine, 2026-10-30)

MAST light curves and TESScut work (pass coordinates to TESScut, not names; `lightkurve.search_lightcurve` sometimes hangs, in which case fetch `mast:HLSP/...` URIs directly). Gaia: ESA TAP times out, Heidelberg TAP is slow on joins; VizieR mirror `vizier.idia.ac.za` works but is intermittent, so run a known-positive (Algol) check before trusting an empty cone. The Gaia DR3 variability tables are I/358, not I/357/varisum as the brief says. AAVSO VSX API is behind Cloudflare; use VizieR B/vsx.

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c9a vets: running verdict table

Tiers per the paper's Conventions (Confirmed / Probable / Plausible / Refuted / Open). One line per star, pasted from each vet's summary; see vetNN/REPORT.md for the evidence.

vet	TIC	tier	P_ecl (d)	depth1	depth2 @ phase	pixel	pipelines	Gaia geometry	catalogue status	notes
vet01	1958652529	CONFIRMED	3.83104 5656(9)	183 ppt QLP; 192±7 TESScut undiluted	48 ppt @ 0.49999	on target, all 8 sectors	3 (QLP, own TESScut, TASOC)	pass; primary = conj A	**already catalogued**: ASAS-SN EA P 3.8310 d at 0.16"; ATLAS DBH at P/2; not in VSX	recovery, not new

vet	TIC	tier	P_ecl (d)	depth1	depth2 @ phase	pixel	pipelines	Gaia geometry	catalogue status	notes
vet01	468181994	CONFIRMED (weaker)	3.99691 90(6)	22.5 ppt QLP / 23.2 TESScut	1.24–1.41 ppt @ 0.4956±0.0012	on target; 2.85" G16.75 neighbour rejected ~16σ	3 (QLP, own TESScut, ELEANOR-lite)	pass; secondary within 1σ; A vs B undecided	not in VSX/ASAS-SN/Gaia-EB (per report)	P_ecl −2.3σ from Gaia; RUWE 12.6 suggests sub-arcsec companion; secondary 25 min early (e cos ω ≈ −0.007)
vet02	470235665	CONFIRMED	10.3261 83(2)	122–127 ppt	81–84 ppt @ 0.52275	on target, s19 & s79 (2 cameras)	3 (QLP, TESS-SPOC, own TESScut)	pass; primary = conj A, B excluded 19σ; secondary +0.7σ	not in VSX/Gaia-EB/Prša/IJspeert/ASAS-SN within 60"; SIMBAD SB*	UCAC4 859-013225; subgiant + K dwarf; possible secondary-phase drift (3.5σ, QLP only, not claimed)
vet02	379251709	CONFIRMED	12.3239 12(7)	34–38 ppt (screen's 31 was 15% low)	2.5 ppt @ 0.4960±0.0010 (fold-level, S/N 8.5 QLP)	on target (primary), s18 & s52	3 (QLP, TESS-SPOC, own TESScut)	pass; primary = conj A, B excluded; secondary +0.8σ	not in VSX/Gaia-EB/Prša/IJspeert/ASAS-SN within 60"; SIMBAD SB*	UCAC4 788-006141; M-dwarf companion; eclipse shape prefers R1≈1.3 R⊙ vs TIC 1.88
vet03	706658 29	CONFIRMED	3.46576 54(7)	283–330 ppt (pipeline-dependent; SPOC ~300)	64–72 ppt @ 0.46048	on target, s46/s48/s72 (3 cameras); only neighbour needs >1500%	4 (QLP, TESS-SPOC, own TESScut, ZTF g/r)	pass; deep = conj A (RV star eclipsed) +0.24σ; reverse 3.6–5.6σ off	not in VSX/Prša/IJspeert within 60"	**BLIND CONTROL = paper's A4**: recovered independently (paper: P 3.4657540±0.0000093, secondary at 0.461, deep eclipse on the K dwarf)
vet03	331293467	CONFIRMED	5.25681 99(20)	61–73 ppt QLP raw (≈80–116 undiluted)	11.4 ppt QLP (≈13–21 undiluted) @ 0.50013	on target, 4 sectors / 3 cameras; G13.78 at 30.5" passes dilution arithmetic (needs 25–84%) but excluded by position (fit 10–19× worse there)	2 (QLP, own TESScut; no SPOC)	pass; deep = conj A +0.49σ, other excluded ≥6.7σ (relies on Gaia ω–Tp correlation)	not in VSX/Prša/IJspeert within 60"; Gaia-EB/varisum/SB9 cones from predecessor run only (mirror timed out)	A/F star + K dwarf; heavy blend (contamination 0.70) so undiluted depths uncertain; Dec −60, no ZTF
vet04	283532878	CONFIRMED	17.4460 19(19) (+1.8σ vs Gaia)	75 ppt QLP (71–77 undiluted)	10 ppt @ 0.6781 (Gaia predicts 0.712±0.024, −1.4σ)	on target, 8 sectors; 3 neighbours at 28–37" pass dilution arithmetic (need 30–47%) but rejected by noise-weighted difference images ($\Delta\chi^2 \geq 41.7$ each sector)	2 (QLP, own TESScut; no SPOC)	pass; primary = conj A, B excluded >11σ	not in VSX/Gaia-EB/Prša/IJspeert on target; **a catalogued EB (Gaia DR3 2202301140233408128, P 1.7975 d, G 17.66) sits 29.7" away**, excluded as source but may add ≤3 ppt dips	eccentric secondary confirmed in all 8 sectors over 4 seasons; secondary phase drifts ~45 min 2019–2024 (~4σ formal, likely underestimated errors)
vet04	407877153	CONFIRMED	17.0892 94(8) (−0.14σ)	29 ppt (SPOC, TESScut; QLP 24–27 diluted)	3.0 ppt flat-bottomed @ 0.5469 (Gaia 0.559±0.044, −0.3σ)	on target; secondary pixel S/N weak ($\Delta\chi^2 \geq 7$ per sector, ≥166 summed)	3 (QLP, TESS-SPOC, own TESScut)	pass; primary = conj A	not in VSX/Gaia-EB/Prša/IJspeert within 60"	screen's depths low (27→29, 2→3 ppt)

vet	TIC	tier	P_ecl (d)	depth1	depth2 @ phase	pixel	pipelines	Gaia geometry	catalogue status	notes
vet05	158557994	CONFIRMED	5.891145 8(38) (+0.54σ)	35.3 ppt dilution-corrected (QLP 35.8, SPOC 34.6, TESScut 34.9); T14 3.25 h	5.0–5.4 ppt @ 0.50182±0.00030 (Gaia 0.502±0.021)	on target, s44/s70/s71 (0.4–0.9"); all neighbours excluded	3 (QLP, TESS-SPOC, own TESS cut)	pass; primary = conj A +0.03±0.08 d, B excluded ~40σ	not in VSX/Gaia-EB/Prša/IJspeert within 60"	screen's 35 ppt was right
vet05	928819488	CONFIRMED	8.8049754(13) (+0.43σ)	16.4 QLP / 18.2 SPOC / 19.0 ppt dilution-corrected; T14 1.69 h V-shaped	5.0–5.3 ppt @ 0.48334±0.00008 (e cos ω −0.026; Gaia 0.502±0.018, 1.0σ)	on target, all 7 sectors (0.3–1.6"); G14.56 at 53" excluded by centroid and aperture dependence	3 (QLP, TESS-SPOC, own TESS cut)	pass; primary = conj A +0.06±0.09 d, B excluded ~43σ	VSX lists it as ROT (Gaia DR3 variable) only; not in any EB catalogue	screen's 11 ppt was low (16–19 ppt); QLP ~10% shallower than SPOC (dilution)
vet06	468827621	CONFIRMED	11.041067(1) (−0.16σ)	184–192 ppt (QLP 192, SPOC 184, TESScut 192)	29.6–31.5 ppt @ 0.72649±0.00003 (Gaia 0.718±0.015, +0.56σ)	on target (primary), s19 & s59 but both camera 2; secondary not pixel-tested (G16.17 at 24.6" could supply it in the worst case)	3 (QLP, TESS-SPOC, own TESS cut)	pass; primary = conj A −0.14±0.09 d, B excluded 31σ; longer eclipse nearer periastron as e sin ω requires	not in VSX/Gaia-EB/Prša/IJspeert/SB9 within 60"; SB*	reduced-depth run; screen right on everything
vet06	416285595	CONFIRMED, with caveats	4.614 5357(3) (−0.37σ)	55–59 ppt; QLP s40 61.7 vs SPOC 54.7 (~12%, ~4σ apart; TESScut 70–89 unreliable in crowded field)	12.9 ppt @ 0.50011±0.00027	on target (primary), s40 cam 1 & s80 cam 2; all 43 neighbours need >100% (min 126%); secondary not pixel-tested (G14.75 at 37.9" could supply ~28%)	3, but depths agree only to ~12%	pass but weak (e≈0.08 so phase 0.5 expected anyway; A/B rests on Gaia ω–Tp correlation)	not in VSX/Gaia-EB/Prša/IJspeert/SB9 within 60"; SB*	reduced-depth run; if depth agreement within errors is required this drops to PROBABLE; two worst-case secondary figures are the agent's hand estimates, not in an output file

vet11: the five sinusoidal (P2) candidates, refolded at the Gaia period (QLP only; no second pipeline exists)

TIC	class	P_phot (d) / Gaia ratio	min1	min2	pixel	catalogue	notes
372683900	ELLIPSOIDAL	1.476507 (1.000003)	31.1 ppt @ 0 (conj B)	28.2 ppt @ 0.5011 (conj A)	on target (G15.2 at 1.8" unresolved)	none; SIMBAD SB*	minima 0.5007 apart vs Gaia e=0.30 predicting 0.372±0.039 (3.3σ conflict); TIC radius probably overestimated
131064 08	ELLIPSOIDAL	2.637721 (1.000082)	46.9 ppt @ 0 (conj A)	40.6 ppt @ 0.5009, flattened	on target	none (unrelated 364 d SR at 29"); SB*	
372683622	ELLIPSOIDAL	3.809282 (1.000073)	7.7 ppt @ 0 (conj B; A excluded 12.5σ)	6.5 ppt @ 0.4943	on target	none; SB*	spot modulation varying sector to sector

TIC	class	P_phot (d) / Gaia ratio	min1	min2	pixel	catalogue	notes
319185391	DETACHED-EB (grazing; fold-level only)	2.678053 (1.000057)	8.5 ppt @ 0 (conj A; B excluded 6σ) + 1.04±0.17 ppt ~2 h dip	6.6 ppt @ 0.5102	on target (ellipsoidal); dip itself untested	none; SB*; Gaia also has a 434 d astrometric orbit (likely triple)	dip in all 6 sectors, odd and even; single events ~1σ; same test gives 1.7–2.3σ on the other three → PLAUSIBLE at best
62328710	OTHER (δ Scuti neighbour)	0.156480 (1/5.995 of Gaia P, not a harmonic)	none (<0.15 ppt at Gaia P)	none	neighbour Gaia DR3 2165710355333626240 (G13.38, 44"); ZTF: 6.3906 c/d at ~9 mmag there, flat on target	neighbour flagged VARIABLE by Gaia	target flat at 0.938 d: Gaia's SB1 orbit for it is suspect

Fainter SB1 period folds: current adjudications

The included systems in this population comprise 28 confirmed and seven probable systems. Known recoveries, controls, plausible signals and non-eclipse interpretations remain visible below without adding to the sample.

C01. TIC 195865529 - CONFIRMED

Unique count addition: false. Catalogue status: not added to catalogue-screened subset; searched scope and limitations in report.

Known ASAS-SN EA at 0.16 arcsec (P 3.8310 d); ATLAS DBH P/2. Recovery excluded from new count.

Public full report

C02. TIC 468181994 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven-group coverage recovered 2026-10-30.

Period -2.31 sigma from Gaia; RUWE 12.6/unresolved companion risk; independent pipelines agree about 2 sigma. Literature cones limited to enumerated catalogues, not proof of novelty.

Public full report

C03. TIC 470235665 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven-group coverage recovered 2026-10-30.

QLP-only secondary phase drift unconfirmed; 4-5% pipeline scale offset. Enumerated catalogue cones and SIMBAD SB* do not establish exhaustive novelty.

Public full report

C04. TIC 379251709 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven-group coverage recovered 2026-10-30.

Secondary fold-level, not localized; primary securely on target. QLP/SPOC secondary phases differ 2.6 sigma. Radius tension retained.

Public full report

C05. TIC 70665829 - CONFIRMED

Unique count addition: false. Catalogue status: not added to catalogue-screened subset; searched scope and limitations in report.

Already objects.csv A4; blind control independently recovered. Archive Gaia-VEB/varisum/SB9 controls failed on later mirror timeout, predecessor empty cones only. No addition.

Public full report

C06. TIC 331293467 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven-group coverage recovered 2026-10-30.

Heavy blend; QLP plus independent own TESScut satisfies available pipeline check; depths model-dependent 80-116 ppt, ratios/duration agree. Gaia geometry depends on covariance. Gaia-VEB/varisum/SB9 cones lack successful positive controls; no exhaustive novelty claim.

Public full report

C07. TIC 283532878 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven-group coverage recovered 2026-10-30.

Use successful s12d recheck, not false-empty first archive pass. Known EB 29.7 arcsec away at P 1.7975 d excluded as source; up to 3 ppt contamination possible. Phase drift tentative.

Public full report

C08. TIC 407877153 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven-group coverage recovered 2026-10-30.

Secondary pixel signal weak individually but summed delta chi2 >=166; QLP depth diluted. Use successful s12d catalogue rerun; all-catalogue query unreliable.

Public full report

C09. TIC 15855794 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven-group coverage recovered 2026-10-30.

Exclude flagged E124 from primary pixel stack as report does. Catalogue positive controls succeeded; catalogue absence only within searched scope.

Public full report

C10. TIC 92881948 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven-group coverage recovered 2026-10-30.

VSX ROT/Gaia SOLAR_LIKE is prior variable classification, not prior EB; catalogue positive controls succeeded. QLP dilution explains shallower depth.

Public full report

C11. TIC 468827621 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven-group coverage recovered 2026-10-30.

Primary localized in two sectors both camera 2; secondary untested in pixels and neighbour could supply in worst case. Literature search skipped, though catalogue cones performed; known Gaia short-timescale variable is not prior EB.

Public full report

C12. TIC 416285595 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven-group coverage recovered 2026-10-30.

Returned CONFIRMED with caveats: QLP/SPOC s40 61.7 vs 54.7 ppt (~12%, ~4 sigma); TESScut depth unreliable. Secondary not pixel-tested; geometry weak/covariance-dependent. Catalogue cones performed but literature search skipped. Report itself says strict depth agreement would imply PROBABLE; no new stricter tolerance imposed here.

Public full report

C13. TIC 372683900 - ELLIPSOIDAL

Unique count addition: false. Catalogue status: not added to catalogue-screened subset; searched scope and limitations in report.

Smooth double wave, no narrow eclipse; unresolved 1.8 arcsec neighbour; Gaia eccentricity conflicts at 3.3 sigma.

Public full report

C14. TIC 13106408 - ELLIPSOIDAL

Unique count addition: false. Catalogue status: not added to catalogue-screened subset; searched scope and limitations in report.

Smooth double wave; nearby 364-day SR unrelated.

Public full report

C15. TIC 372683622 - ELLIPSOIDAL

Unique count addition: false. Catalogue status: not added to catalogue-screened subset; searched scope and limitations in report.

Double wave plus variable spots; no eclipse.

Public full report

C16. TIC 319185391 - PLAUSIBLE

Unique count addition: false. Catalogue status: not added to catalogue-screened subset; searched scope and limitations in report.

Fold-only ~1 ppt dip, individual events ~1 sigma, dip not pixel-tested, no second pipeline. Report prose says probable-level but batch summary says PLAUSIBLE at best; adopt batch summary outside count. Gaia 434-day astrometric orbit suggests triple.

Public full report

C17. TIC 62328710 - NEIGHBOUR_SIGNAL

Unique count addition: false. Catalogue status: not added to catalogue-screened subset; searched scope and limitations in report.

Delta Scuti-like signal localized to Gaia DR3 2165710355333626240 at 44 arcsec and independently seen there by ZTF; target flat at Gaia period; suspect SB1 orbit is interpretation.

Public full report

C18. TIC 300903537 - CONFIRMED

Unique count addition: false. Catalogue status: known catalogue EB recovery; excluded from additions.

See target report for deciding limitation and next check.

Public full report

C19. TIC 23112764 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

See target report for deciding limitation and next check.

Public full report

C20. TIC 373062090 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

See target report for deciding limitation and next check.

Public full report

C21. TIC 465073693 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

See target report for deciding limitation and next check.

Public full report

C22. TIC 316405449 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

See target report for deciding limitation and next check.

Public full report

C23. TIC 465263741 - PROBABLE

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

See target report for deciding limitation and next check.

Public full report

C24. TIC 441741446 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

See target report for deciding limitation and next check.

Public full report

C25. TIC 280075497 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

See target report for deciding limitation and next check.

Public full report

C26. TIC 276514049 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

See target report for deciding limitation and next check.

Public full report

C27. TIC 10790605 - REFUTED

Unique count addition: false. Catalogue status: No target prior EB identified in seven recorded scopes; bounded catalogue coverage, no exhaustive priority claim..

See target report for deciding limitation and next check.

Public full report

C28. TIC 376082142 - REFUTED

Unique count addition: false. Catalogue status: no identified prior EB.

Search higher precision data for a discrete narrow eclipse only if this smooth variable remains of interest.

Public full report

C29. TIC 285942031 - PROBABLE

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

Reassess Gaia orbital uncertainties/correlations or obtain an updated radial-velocity orbit to resolve the eclipse-versus-Gaia timing tension.

Public full report

C30. TIC 104736355 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

Optional spectroscopy or higher spatial resolution photometry would refine characterization; no additional deciding gate is required by this representative review.

Public full report

C31. TIC 281393563 - PROBABLE

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

Obtain higher resolution time-series imaging or a validated multi-source PRF difference fit to identify the eclipsing host.

Public full report

C32. TIC 450055243 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

Optional spectroscopy or higher spatial resolution photometry would refine characterization; no additional deciding gate is required by this representative review.

Public full report

C33. TIC 126363604 - PROBABLE

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

Validate the repeated difference-image astrometry with a multi-source PRF or higher resolution event photometry; refine Gaia conjunction uncertainty if needed.

Public full report

C34. TIC 251357000 - REFUTED

Unique count addition: false. Catalogue status: no identified prior EB.

Search higher precision data for a discrete narrow eclipse only if this smooth variable remains of interest.

Public full report

C35. TIC 267222866 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

Optional spectroscopy or higher spatial resolution photometry would refine characterization; no additional deciding gate is required by this representative review.

Public full report

C36. TIC 66315917 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

Optional spectroscopy or higher spatial resolution photometry would refine characterization; no additional deciding gate is required by this representative review.

Public full report

C37. TIC 93783392 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

Optional spectroscopy or higher spatial resolution photometry would refine characterization; no additional deciding gate is required by this representative review.

Public full report

C38. TIC 470947699 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

Optional improved spectroscopy and photometry would refine binary characterization.

Public full report

C39. TIC 135230723 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

Optional improved spectroscopy and photometry would refine binary characterization.

Public full report

C40. TIC 36083554 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

Optional improved spectroscopy and photometry would refine binary characterization.

Public full report

C41. TIC 447566809 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

Optional improved spectroscopy and photometry would refine binary characterization.

Public full report

C42. TIC 12320469 - REFUTED

Unique count addition: false. Catalogue status: no identified prior EB.

Higher precision light curves could test for a narrow residual eclipse if further work on this smooth variable is desired.

Public full report

C43. TIC 86908196 - REFUTED

Unique count addition: false. Catalogue status: no identified prior EB.

Higher precision light curves could test for a narrow residual eclipse if further work on this smooth variable is desired.

Public full report

C44. TIC 201317466 - PROBABLE

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

Refit the RV orbit or evaluate a nonlinear orbital posterior against the observed eclipse timing to resolve Gaia association.

Public full report

C45. TIC 360516818 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

Optional improved spectroscopy and photometry would refine binary characterization.

Public full report

C46. TIC 411647453 - REFUTED

Unique count addition: false. Catalogue status: no identified prior EB.

Higher resolution event photometry can establish the contaminant host and measure its orbital period precisely.

Public full report

C47. TIC 450217907 - REFUTED

Unique count addition: false. Catalogue status: no identified prior EB.

Higher precision light curves could test for a narrow residual eclipse if further work on this smooth variable is desired.

Public full report

C48. TIC 60443793 - REFUTED

Unique count addition: false. Catalogue status: no identified prior EB.

Review TIC 60443695 separately if desired.

Public full report

C49. TIC 71866523 - PLAUSIBLE

Unique count addition: false. Catalogue status: no identified prior EB.

Require a clean repeated negative event before EB classification.

Public full report

C50. TIC 443106429 - PROBABLE

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

Resolve target versusTIC 443106423 with higher spatial resolution eclipse photometry.

Public full report

C51. TIC 311206604 - REFUTED

Unique count addition: false. Catalogue status: no identified prior EB.

No further eclipse follow-up required unless new discrete events appear.

Public full report

C52. TIC 428769301 - REFUTED

Unique count addition: false. Catalogue status: no identified prior EB.

No further eclipse follow-up required unless new discrete events appear.

Public full report

C53. TIC 436947190 - REFUTED

Unique count addition: false. Catalogue status: no identified prior EB.

No further eclipse follow-up required unless new discrete events appear.

Public full report

C54. TIC 92452368 - PLAUSIBLE

Unique count addition: false. Catalogue status: no identified prior EB.

Verify a clean recurrent event in independently reduced data.

Public full report

C55. TIC 94300366 - PLAUSIBLE

Unique count addition: false. Catalogue status: no identified prior EB.

Obtain detrending resistant multiseason photometry to establish stable discrete repeats.

Public full report

C56. TIC 4646090 - CONFIRMED

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

Characterize dilution and grazing geometry if needed.

Public full report

C57. TIC 402940895 - PLAUSIBLE

Unique count addition: false. Catalogue status: no identified prior EB.

Re-extract clean sectors with robust background/pointing treatment before asserting eclipses.

Public full report

C58. TIC 183994341 - CONFIRMED

Unique count addition: false. Catalogue status: known catalogue EB recovery; excluded from additions.

Resolve ASAS-SN period alias and obtain RV/orbital posterior or a joint timing/geometry analysis to settle the Gaia association.

Public full report

C59. TIC 166781582 - PLAUSIBLE

Unique count addition: false. Catalogue status: no identified prior EB.

Obtain or inspect another clean sector and require repeated discrete ingress/egress before fitting a binary period.

Public full report

C60. TIC 230270717 - PLAUSIBLE

Unique count addition: false. Catalogue status: no identified prior EB.

Obtain or inspect another clean sector and require repeated discrete ingress/egress before fitting a binary period.

Public full report

C61. TIC 462845564 - PLAUSIBLE

Unique count addition: false. Catalogue status: no identified prior EB.

Obtain or inspect another clean sector and require repeated discrete ingress/egress before fitting a binary period.

Public full report

C62. TIC 431762229 - REFUTED

Unique count addition: false. Catalogue status: no identified prior EB.

Higher resolution eclipse photometry can verify the contaminating host.

Public full report

C63. TIC 350661747 - PLAUSIBLE

Unique count addition: false. Catalogue status: no identified prior EB.

Obtain or inspect another clean sector and require repeated discrete ingress/egress before fitting a binary period.

Public full report

C64. TIC 424565669 - PLAUSIBLE

Unique count addition: false. Catalogue status: no identified prior EB.

Obtain or inspect another clean sector and require repeated discrete ingress/egress before fitting a binary period.

Public full report

C65. TIC 222498305 - REFUTED

Unique count addition: false. Catalogue status: no identified prior EB.

Higher resolution eclipse photometry can verify the contaminating host.

Public full report

C66. TIC 90542579 - REFUTED

Unique count addition: false. Catalogue status: no identified prior EB.

If revisiting, use clean cadences away from downlink boundaries with a validated background and aperture.

Public full report

C67. TIC 9396327 - PROBABLE

Unique count addition: true. Catalogue status: uncatalogued; seven recorded catalogue scopes complete.

Refine Gaia conjunction predictions with orbital posterior and measure a second clean sector or obtain new RVs to test the orbit association.

Public full report

C68. TIC 275022619 - PLAUSIBLE

Unique count addition: false. Catalogue status: no identified prior EB.

Obtain a well-covered predicted candidate event in a distinct reduction and establish a second coherent detached eclipse before claiming an eclipse period or host.

Public full report

C69. TIC 297541839 - REFUTED

Unique count addition: false. Catalogue status: no identified prior EB.

No missing deciding gate for this candidate. Reopen only if new data supply distinct, repeated detached events.

Public full report

C70. TIC 396248369 - REFUTED

Unique count addition: false. Catalogue status: no identified prior EB.

No missing deciding gate for this candidate. Reopen only if new data supply distinct, repeated detached events.

Public full report

C71. TIC 277479128 - REFUTED

Unique count addition: false. Catalogue status: no identified prior EB.

No missing deciding gate for this candidate. Reopen only if new data supply distinct, repeated detached events.

Public full report

C72. TIC 257575960 - PLAUSIBLE

Unique count addition: false. Catalogue status: no identified prior EB.

Obtain a well-covered predicted candidate event in a distinct reduction and establish a second coherent detached eclipse before claiming an eclipse period or host.

Public full report

Fainter SB1 period folds: detailed physical evidence

Sol/Terra measurements under Astra supervision; original Fable discovery and screening. Reports preserve each deciding limitation and next test. Current tier reconciliation governs headline counts.

C24. TIC 441741446 full report

Archived source: sanitized report text.

TIC 441741446: CONFIRMED

Repeated distinct eclipses, an eclipse-derived period, target-centered two-sector deficits, independent extraction and Gaia consistency support a target eclipsing binary.

1. Repeated raw events: QLP SAP_FLUX with QUALITY=0; 4 accepted representative events in sectors[14, 53, 86], depths9.93-13.18ppt and durations1.96-2.63h. Lead personally inspected the first/last raw event panels; localized dips repeat across time-separated sectors.
2. Period from eclipses: primary recurrence18.826637718d, epoch1686.723580BTJD, timing residual RMS2.51min. This is an unweighted representative-event regression conditional on cycle counts, not the Gaia period copied as a measurement. See ../../raw-evidence/raw-events.json.
3. Host/source: signed out-minus-in deficit maps in sectors[14, 53] peak around the exact Gaia target WCS position. Lead visually inspected a representative signed map for this target and the raw panels. TIC T-band60arcsec feasibility and GaiaDR3 cone receipts are recorded, with original90arcsec DR3 reuse where fresh cone requests failed. Resolved capable field sources lie away from the disappearing light. Close-source comparison uses the same fixed-position circular-Gaussian-plus-background model for target and capable TIC neighbours within30arcsec: sector14: neighbour/target RSS ratios no capable TIC neighbour within30arcsec; sector53: neighbour/target RSS ratios no capable TIC neighbour within30arcsec. These ratios are descriptive shape comparisons, not calibrated probabilities. The positive-clipped centroid is center-biased and is not the deciding source statistic. Target-centered deficit morphology across two sectors, together with relative fits where needed, supports target attribution within the tested catalogue scope.
4. Independent reduction: Independent custom TESScut extraction reproduces the events: sector14 3x3 local-baseline depth9.78ppt; sector53 3x3 local-baseline depth9.10ppt. This is a separate pixel-aperture reduction of the same TESS photons, not an independent observation or an archive SPOC pipeline. Quality filtering, source position through WCS, sector headers, sample counts and FITS hashes are recorded in pixel-results.json and tesscut-downloads.json. Additionally, TESS-SPOC sector14 raw SAP reproduces depths 9.99, 9.70ppt; archive product URI, TIC/sector checks and hash are in independent-results.json.
5. Gaia geometry: exact DR3 source1656477802697153920, P=18.831656152+/-0.00473691d, e=0.3822, omega=225.46deg. Measured-minus-Gaia period is-1.06quoted Gaia sigma. Closest conjunctionB; the photometric ephemeris projected near BTJD389 gives offset+0.0428d with linear full-covariance Gaia error0.1468d. The projection has no measured photometric period error and is a consistency diagnostic. Observed median duration2.53h versus approximate central-crossing scale4.98h from TIC radius and minimum companion mass supports basic feasibility except the explicit caveats. Radius estimates and assumed companion mass-radius relation are not a unique fit. ../../gaia-geometry.json preserves both conjunctions and error assumptions.

No target prior EB identified in seven recorded scopes; bounded catalogue coverage, no exhaustive priority claim. The physical verdict and catalogue status are separate; seven scopes, service status, exact identity matches and nearby rows are recorded in ../../../shared-catalogues/coverage-matrix.csv. SB* or ROT alone is not prior EB evidence.

Finite-cadence local trapezoids are conditional on discovery cycle counts; no exposure integration or precision period uncertainty is inferred. The original P/2 alternative is distinct from a new orbital detection. Source tests cover catalogue cones, not every uncatalogued or unresolved companion. G/T colour and aperture throughput limit dilution arithmetic; no calibrated centroid uncertainty or RSS significance is claimed.

Execution is complete under ../../../BRIEF.md; limitations above are scientific outcomes, not unattempted work. Evidence and hashes: ../../raw-evidence/measurements-summary.md, receipt-manifest.json, identity-checks.json, source-model-comparison.json and per-target pixel_sNNN.npz/png. Next check: joint RV/eclipse characterization with improved stellar dimensions.

C20. TIC 373062090 full report

Archived source: sanitized report text.

TIC 373062090: CONFIRMED

Repeated distinct eclipses, an eclipse-derived period, target-centered two-sector deficits, independent extraction and Gaia consistency support a target eclipsing binary.

1. Repeated raw events: QLP SAP_FLUX with QUALITY=0; 4 accepted representative events in sectors[18, 58, 86], depths28.40-30.56ppt and durations7.20-7.76h. Lead personally inspected the first/last raw event panels; localized dips repeat across time-separated sectors.
2. Period from eclipses: primary recurrence7.705911843d, epoch1799.033971BTJD, timing residual RMS0.20min. This is an unweighted representative-event regression conditional on cycle counts, not the Gaia period copied as a measurement. See ../../raw-evidence/raw-events.json.
3. Host/source: signed out-minus-in deficit maps in sectors[18, 58] peak around the exact Gaia target WCS position. Lead visually inspected a representative signed map for this target and the raw panels. TIC T-band60arcsec feasibility and GaiaDR3 cone receipts are recorded, with original90arcsec DR3 reuse where fresh cone requests failed. Resolved capable field sources lie away from the disappearing light. Close-source comparison uses the same fixed-position circular-Gaussian-plus-background model for target and capable TIC neighbours within30arcsec: sector18: neighbour/target RSS ratios no capable TIC neighbour within30arcsec; sector58: neighbour/target RSS ratios no capable TIC neighbour within30arcsec. These ratios are descriptive shape comparisons, not calibrated probabilities. The positive-clipped centroid is center-biased and is not the deciding source statistic. Target-centered deficit morphology across two sectors, together with relative fits where needed, supports target attribution within the tested catalogue scope.
4. Independent reduction: Independent custom TESScut extraction reproduces the events: sector18 3x3 local-baseline depth29.44ppt; sector58 3x3 local-baseline depth28.95ppt. This is a separate pixel-aperture reduction of the same TESS photons, not an independent observation or an archive SPOC pipeline. Quality filtering, source position through WCS, sector headers, sample counts and FITS hashes are recorded in pixel-results.json and tesscut-downloads.json. Archive independent-product discovery was attempted but no usable same-sector product was obtained; failures/empty inventories remain recorded.
5. Gaia geometry: exact DR3 source538032619052851712, P=7.710879157+/-0.00305296d, e=0.2464, omega=139.81deg. Measured-minus-Gaia period is-1.63quoted Gaia sigma. Closest conjunctionA; the photometric ephemeris projected near BTJD389 gives offset-0.1151d with linear full-covariance Gaia error0.1836d. The projection has no measured photometric period error and is a consistency diagnostic. Observed median duration7.24h versus approximate central-crossing scale10.64h from TIC radius and minimum companion mass supports basic feasibility except the explicit caveats. Radius estimates and assumed companion mass-radius relation are not a unique fit. ../../gaia-geometry.json preserves both conjunctions and error assumptions.

No target prior EB identified in seven recorded scopes; bounded catalogue coverage, no exhaustive priority claim. The physical verdict and catalogue status are separate; seven scopes, service status, exact identity matches and nearby rows are recorded in ../../../shared-catalogues/coverage-matrix.csv. SB* or ROT alone is not prior EB evidence.

Finite-cadence local trapezoids are conditional on discovery cycle counts; no exposure integration or precision period uncertainty is inferred. The original P/2 alternative is distinct from a new orbital detection. Source tests cover catalogue cones, not every uncatalogued or unresolved companion. G/T colour and aperture throughput limit dilution arithmetic; no calibrated centroid uncertainty or RSS significance is claimed.

Execution is complete under ../../../BRIEF.md; limitations above are scientific outcomes, not unattempted work. Evidence and hashes: ../../raw-evidence/measurements-summary.md, receipt-manifest.json, identity-checks.json, source-model-comparison.json and per-target pixel_sNNN.npz/png. Next check: joint RV/eclipse characterization with improved stellar dimensions.

C21. TIC 465073693 full report

Archived source: sanitized report text.

TIC 465073693: CONFIRMED

Repeated distinct eclipses, an eclipse-derived period, target-centered two-sector deficits, independent extraction and Gaia consistency support a target eclipsing binary.

1. Repeated raw events: QLP SAP_FLUX with QUALITY=0; 6 accepted representative events in sectors[10, 63, 99], depths227.58-260.10ppt and durations4.84-4.87h. Lead personally inspected the first/last raw event panels; localized dips repeat across time-separated sectors.
2. Period from eclipses: primary recurrence6.272264875d, epoch1577.915266BTJD, timing residual RMS0.46min. This is an unweighted representative-event regression conditional on cycle counts, not the Gaia period copied as a measurement. See ../../raw-evidence/raw-events.json.
3. Host/source: signed out-minus-in deficit maps in sectors[10, 63] peak around the exact Gaia target WCS position. Lead visually inspected a representative signed map for this target and the raw panels. TIC T-band60arcsec feasibility and GaiaDR3 cone receipts are recorded, with original90arcsec DR3 reuse where fresh cone requests failed. Resolved capable field sources lie away from the disappearing light. Close-source comparison uses the same fixed-position circular-Gaussian-plus-background model for target and capable TIC neighbours within30arcsec: sector10: neighbour/target RSS ratios no capable TIC neighbour within30arcsec; sector63: neighbour/target RSS ratios no capable TIC neighbour within30arcsec. These ratios are descriptive shape comparisons, not calibrated probabilities. The positive-clipped centroid is center-biased and is not the deciding source statistic. Target-centered deficit morphology across two sectors, together with relative fits where needed, supports target attribution within the tested catalogue scope.
4. Independent reduction: Independent custom TESScut extraction reproduces the events: sector10 3x3 local-baseline depth187.85ppt; sector63 3x3 local-baseline depth183.00ppt. This is a separate pixel-aperture reduction of the same TESS photons, not an independent observation or an archive SPOC pipeline. Quality filtering, source position through WCS, sector headers, sample counts and FITS hashes are recorded in pixel-results.json and tesscut-downloads.json. Archive independent-product discovery was attempted but no usable same-sector product was obtained; failures/empty inventories remain recorded.
5. Gaia geometry: exact DR3 source5239650442330452096, P=6.269417959+/-0.0014404d, e=0.0595, omega=129.56deg. Measured-minus-Gaia period is+1.98quoted Gaia sigma. Closest conjunctionA; the photometric ephemeris projected near BTJD389 gives offset-0.0686d with linear full-covariance Gaia error0.1034d. The projection has no measured photometric period error and is a consistency diagnostic. Observed median duration4.86h versus approximate central-crossing scale6.28h from TIC radius and minimum companion mass supports basic feasibility except the explicit caveats. Radius estimates and assumed companion mass-radius relation are not a unique fit. ../../gaia-geometry.json preserves both conjunctions and error assumptions.

No target prior EB identified in seven recorded scopes; bounded catalogue coverage, no exhaustive priority claim. The physical verdict and catalogue status are separate; seven scopes, service status, exact identity matches and nearby rows are recorded in ../../../shared-catalogues/coverage-matrix.csv. SB* or ROT alone is not prior EB evidence.

Finite-cadence local trapezoids are conditional on discovery cycle counts; no exposure integration or precision period uncertainty is inferred. The original P/2 alternative is distinct from a new orbital detection. Source tests cover catalogue cones, not every uncatalogued or unresolved companion. G/T colour and aperture throughput limit dilution arithmetic; no calibrated centroid uncertainty or RSS significance is claimed. Unequal alternating depths about12-14% and5.6-6.5% in sectors10,63,99 reject equivalent full morphology at3.136d and support the6.27226d primary recurrence (alias-check.json).

Execution is complete under ../../../BRIEF.md; limitations above are scientific outcomes, not unattempted work. Evidence and hashes: ../../raw-evidence/measurements-summary.md, receipt-manifest.json, identity-checks.json, source-model-comparison.json and per-target pixel_sNNN.npz/png. Next check: joint RV/eclipse characterization with improved stellar dimensions.

C22. TIC 316405449 full report

Archived source: sanitized report text.

TIC 316405449: CONFIRMED

Repeated distinct eclipses, an eclipse-derived period, target-centered two-sector deficits, independent extraction and Gaia consistency support a target eclipsing binary.

1. Repeated raw events: QLP SAP_FLUX with QUALITY=0; 5 accepted representative events in sectors[15, 76, 84], depths19.82-22.99ppt and durations2.77-3.75h. Lead personally inspected the first/last raw event panels; localized dips repeat across time-separated sectors.
2. Period from eclipses: primary recurrence2.527605384d, epoch1716.361405BTJD, timing residual RMS1.32min. This is an unweighted representative-event regression conditional on cycle counts, not the Gaia period copied as a measurement. See ../../raw-evidence/raw-events.json.
3. Host/source: signed out-minus-in deficit maps in sectors[15, 76] peak around the exact Gaia target WCS position. Lead visually inspected a representative signed map for this target and the raw panels. TIC T-band60arcsec feasibility and GaiaDR3 cone receipts are recorded, with original90arcsec DR3 reuse where fresh cone requests failed. Resolved capable field sources lie away from the disappearing light. Close-source comparison uses the same fixed-position circular-Gaussian-plus-background model for target and capable TIC neighbours within30arcsec: sector15: neighbour/target RSS ratios 12.72, 48.04; sector76: neighbour/target RSS ratios 3.41, 15.52. These ratios are descriptive shape comparisons, not calibrated probabilities. The positive-clipped centroid is center-biased and is not the deciding source statistic. Target-centered deficit morphology across two sectors, together with relative fits where needed, supports target attribution within the tested catalogue scope.
4. Independent reduction: Independent custom TESScut extraction reproduces the events: sector15 3x3 local-baseline depth20.30ppt; sector76 3x3 local-baseline depth19.50ppt. This is a separate pixel-aperture reduction of the same TESS photons, not an independent observation or an archive SPOC pipeline. Quality filtering, source position through WCS, sector headers, sample counts and FITS hashes are recorded in pixel-results.json and tesscut-downloads.json. Archive independent-product discovery was attempted but no usable same-sector product was obtained; failures/empty inventories remain recorded.
5. Gaia geometry: exact DR3 source2177587142501400704, P=2.527851247+/-0.00043688d, e=0.0791, omega=100.43deg. Measured-minus-Gaia period is-0.56quoted Gaia sigma. Closest conjunctionA; the photometric ephemeris projected near BTJD389 gives offset-0.0963d with linear full-covariance Gaia error0.0826d. The projection has no measured photometric period error and is a consistency diagnostic. Observed median duration3.32h versus approximate central-crossing scale6.62h from TIC radius and minimum companion mass supports basic feasibility except the explicit caveats. Radius estimates and assumed companion mass-radius relation are not a unique fit. ../../gaia-geometry.json preserves both conjunctions and error assumptions.

No target prior EB identified in seven recorded scopes; bounded catalogue coverage, no exhaustive priority claim. The physical verdict and catalogue status are separate; seven scopes, service status, exact identity matches and nearby rows are recorded in ../../../shared-catalogues/coverage-matrix.csv. SB* or ROT alone is not prior EB evidence.

Finite-cadence local trapezoids are conditional on discovery cycle counts; no exposure integration or precision period uncertainty is inferred. The original P/2 alternative is distinct from a new orbital detection. Source tests cover catalogue cones, not every uncatalogued or unresolved companion. G/T colour and aperture throughput limit dilution arithmetic; no calibrated centroid uncertainty or RSS significance is claimed.

Execution is complete under ../../../BRIEF.md; limitations above are scientific outcomes, not unattempted work. Evidence and hashes: ../../raw-evidence/measurements-summary.md, receipt-manifest.json, identity-checks.json, source-model-comparison.json and per-target pixel_sNNN.npz/png. Next check: joint RV/eclipse characterization with improved stellar dimensions.

C23. TIC 465263741 full report

Archived source: sanitized report text.

TIC 465263741: PROBABLE

Real repeated eclipses are secure, but the crowded-field source test does not decisively separate all close capable neighbours. PROBABLE retains the host limitation explicitly.

1. Repeated raw events: QLP SAP_FLUX with QUALITY=0; 4 accepted representative events in sectors[10, 38], depths8.04-10.95ppt and durations2.15-2.57h. Lead personally inspected the first/last raw event panels; localized dips repeat across time-separated sectors.
2. Period from eclipses: primary recurrence6.038766996d, epoch1572.206588BTJD, timing residual RMS4.98min. This is an unweighted representative-event regression conditional on cycle counts, not the Gaia period copied as a measurement. See ../../raw-evidence/raw-events.json.
3. Host/source: signed out-minus-in deficit maps in sectors[10, 38] peak around the exact Gaia target WCS position. Lead visually inspected a representative signed map for this target and the raw panels. TIC T-band60arcsec feasibility and GaiaDR3 cone receipts are recorded, with original90arcsec DR3 reuse where fresh cone requests failed. Resolved capable field sources lie away from the disappearing light. Close-source comparison uses the same fixed-position circular-Gaussian-plus-background model for target and capable TIC neighbours within30arcsec: sector10: neighbour/target RSS ratios 1.32, 1.37, 1.69, 1.70, 1.56, 1.40, 1.63; sector38: neighbour/target RSS ratios 3.11, 2.27, 4.40, 3.21, 3.67. These ratios are descriptive shape comparisons, not calibrated probabilities. The positive-clipped centroid is center-biased and is not the deciding source statistic. Host localization remains incomplete because close alternatives are not decisively excluded.
4. Independent reduction: Independent custom TESScut extraction reproduces the events: sector10 3x3 local-baseline depth5.99ppt; sector38 3x3 local-baseline depth8.07ppt. This is a separate pixel-aperture reduction of the same TESS photons, not an independent observation or an archive SPOC pipeline. Quality filtering, source position through WCS, sector headers, sample counts and FITS hashes are recorded in pixel-results.json and tesscut-downloads.json. Archive independent-product discovery was attempted but no usable same-sector product was obtained; failures/empty inventories remain recorded.
5. Gaia geometry: exact DR3 source5239577290443811840, P=6.038294466+/-0.000972077d, e=0.0366, omega=65.26deg. Measured-minus-Gaia period is+0.49quoted Gaia sigma. Closest conjunctionA; the photometric ephemeris projected near BTJD389 gives offset+0.0646d with linear full-covariance Gaia error0.1443d. The projection has no measured photometric period error and is a consistency diagnostic. Observed median duration2.20h versus approximate central-crossing scale4.82h from TIC radius and minimum companion mass supports basic feasibility except the explicit caveats. Radius estimates and assumed companion mass-radius relation are not a unique fit. ../../gaia-geometry.json preserves both conjunctions and error assumptions.

No target prior EB identified in seven recorded scopes; bounded catalogue coverage, no exhaustive priority claim. The physical verdict and catalogue status are separate; seven scopes, service status, exact identity matches and nearby rows are recorded in ../../../shared-catalogues/coverage-matrix.csv. SB* or ROT alone is not prior EB evidence.

Finite-cadence local trapezoids are conditional on discovery cycle counts; no exposure integration or precision period uncertainty is inferred. The original P/2 alternative is distinct from a new orbital detection. Source tests cover catalogue cones, not every uncatalogued or unresolved companion. G/T colour and aperture throughput limit dilution arithmetic; no calibrated centroid uncertainty or RSS significance is claimed. S10 source discrimination is weak and S38 only moderately favors the target against close neighbours; higher-resolution localization or a more reliable PRF comparison is required for promotion.

Execution is complete under ../../../BRIEF.md; limitations above are scientific outcomes, not unattempted work. Evidence and hashes: ../../raw-evidence/measurements-summary.md, receipt-manifest.json, identity-checks.json, source-model-comparison.json and per-target pixel_sNNN.npz/png. Next check: resolve the close-source alternative before promotion.

C19. TIC 23112764 full report

Archived source: sanitized report text.

TIC 23112764: CONFIRMED

Real repeated eclipses, a matching short orbital time scale, target-centered deficits in two sectors and independent extraction establish a target eclipsing binary. The Gaia orbital solution has a retained period/conjunction tension; its exact phase prediction is not validated by this result.

1. Repeated raw events: QLP SAP_FLUX with QUALITY=0; 5 accepted representative events in sectors[43, 70, 71], depths19.04-20.88ppt and durations3.26-3.45h. Lead personally inspected the first/last raw event panels; localized dips repeat across time-separated sectors.
2. Period from eclipses: primary recurrence2.958379801d, epoch2477.217297BTJD, timing residual RMS2.46min. This is an unweighted representative-event regression conditional on cycle counts, not the Gaia period copied as a measurement. See ../../raw-evidence/raw-events.json.
3. Host/source: signed out-minus-in deficit maps in sectors[43, 70] peak around the exact Gaia target WCS position. Lead visually inspected a representative signed map for this target and the raw panels. TIC T-band60arcsec feasibility and GaiaDR3 cone receipts are recorded, with original90arcsec DR3 reuse where fresh cone requests failed. Resolved capable field sources lie away from the disappearing light. Close-source comparison uses the same fixed-position circular-Gaussian-plus-background model for target and capable TIC neighbours within30arcsec: sector43: neighbour/target RSS ratios 20.68; sector70: neighbour/target RSS ratios 6.04. These ratios are descriptive shape comparisons, not calibrated probabilities. The positive-clipped centroid is center-biased and is not the deciding source statistic. Target-centered deficit morphology across two sectors, together with relative fits where needed, supports target attribution within the tested catalogue scope.
4. Independent reduction: Independent custom TESScut extraction reproduces the events: sector43 3x3 local-baseline depth18.95ppt; sector70 3x3 local-baseline depth18.53ppt. This is a separate pixel-aperture reduction of the same TESS photons, not an independent observation or an archive SPOC pipeline. Quality filtering, source position through WCS, sector headers, sample counts and FITS hashes are recorded in pixel-results.json and tesscut-downloads.json. Archive independent-product discovery was attempted but no usable same-sector product was obtained; failures/empty inventories remain recorded.
5. Gaia geometry: exact DR3 source66111027136814720, P=2.957395675+/-0.000380466d, e=0.1966, omega=359.81deg. Measured-minus-Gaia period is+2.59quoted Gaia sigma. Closest conjunctionA; the photometric ephemeris projected near BTJD389 gives offset+0.2160d with linear full-covariance Gaia error0.0640d. The projection has no measured photometric period error and is a consistency diagnostic. Observed median duration3.41h versus approximate central-crossing scale4.26h from TIC radius and minimum companion mass supports basic feasibility except the explicit caveats. Radius estimates and assumed companion mass-radius relation are not a unique fit. ../../gaia-geometry.json preserves both conjunctions and error assumptions.

No target prior EB identified in seven recorded scopes; bounded catalogue coverage, no exhaustive priority claim. The physical verdict and catalogue status are separate; seven scopes, service status, exact identity matches and nearby rows are recorded in ../../../shared-catalogues/coverage-matrix.csv. SB* or ROT alone is not prior EB evidence.

Finite-cadence local trapezoids are conditional on discovery cycle counts; no exposure integration or precision period uncertainty is inferred. The original P/2 alternative is distinct from a new orbital detection. Source tests cover catalogue cones, not every uncatalogued or unresolved companion. G/T colour and aperture throughput limit dilution arithmetic; no calibrated centroid uncertainty or RSS significance is claimed. The near-reference conjunction offset is about3.4linear-Gaia covariance sigma, and the period is2.59quoted Gaia sigma high. A targeted full-covariance Gaussian sensitivity check (9994 physical draws from10000, seed20261030) gives median offset0.200d and16-84% range0.138-0.265d; nonlinearity does not remove the tension. Source attribution, repeated eclipses and nearly identical recurrence time support the same target binary, while duration3.41h is feasible below the approximate4.26h central scale. No physically impossible eclipse geometry is established. This is a confirmed target EB with an imperfect Gaia orbital solution; the Gaia phase prediction itself is not claimed to pass. See ../../23112764-geometry-sensitivity.json.

Execution is complete under ../../../BRIEF.md; limitations above are scientific outcomes, not unattempted work. Evidence and hashes: ../../raw-evidence/measurements-summary.md, receipt-manifest.json, identity-checks.json, source-model-comparison.json and per-target pixel_sNNN.npz/png. Next check: refit the Gaia RV orbit/conjunction uncertainty before precise characterization.

C18. TIC 300903537 full report

Archived source: sanitized report text.

TIC 300903537: CONFIRMED

Repeated distinct eclipses, an eclipse-derived period, target-centered two-sector deficits, independent extraction and Gaia consistency support a target eclipsing binary.

1. Repeated raw events: QLP SAP_FLUX with QUALITY=0; 6 accepted representative events in sectors[1, 68, 95], depths79.23-89.53ppt and durations3.11-3.27h. Lead personally inspected the first/last raw event panels; localized dips repeat across time-separated sectors.
2. Period from eclipses: primary recurrence5.823229918d, epoch1326.644749BTJD, timing residual RMS0.39min. This is an unweighted representative-event regression conditional on cycle counts, not the Gaia period copied as a measurement. See ../../raw-evidence/raw-events.json.
3. Host/source: signed out-minus-in deficit maps in sectors[1, 68] peak around the exact Gaia target WCS position. Lead visually inspected a representative signed map for this target and the raw panels. TIC T-band60arcsec feasibility and GaiaDR3 cone receipts are recorded, with original90arcsec DR3 reuse where fresh cone requests failed. Resolved capable field sources lie away from the disappearing light. Close-source comparison uses the same fixed-position circular-Gaussian-plus-background model for target and capable TIC neighbours within30arcsec: sector1: neighbour/target RSS ratios 16.97; sector68: neighbour/target RSS ratios 9.26. These ratios are descriptive shape comparisons, not calibrated probabilities. The positive-clipped centroid is center-biased and is not the deciding source statistic. Target-centered deficit morphology across two sectors, together with relative fits where needed, supports target attribution within the tested catalogue scope.
4. Independent reduction: Independent custom TESScut extraction reproduces the events: sector1 3x3 local-baseline depth76.58ppt; sector68 3x3 local-baseline depth73.42ppt. This is a separate pixel-aperture reduction of the same TESS photons, not an independent observation or an archive SPOC pipeline. Quality filtering, source position through WCS, sector headers, sample counts and FITS hashes are recorded in pixel-results.json and tesscut-downloads.json. Archive independent-product discovery was attempted but no usable same-sector product was obtained; failures/empty inventories remain recorded.
5. Gaia geometry: exact DR3 source6789333535281440896, P=5.821861614+/-0.0011434d, e=0.1508, omega=94.01deg. Measured-minus-Gaia period is+1.20quoted Gaia sigma. Closest conjunctionA; the photometric ephemeris projected near BTJD389 gives offset+0.0027d with linear full-covariance Gaia error0.0862d. The projection has no measured photometric period error and is a consistency diagnostic. Observed median duration3.18h versus approximate central-crossing scale3.03h from TIC radius and minimum companion mass supports basic feasibility except the explicit caveats. Radius estimates and assumed companion mass-radius relation are not a unique fit. ../../gaia-geometry.json preserves both conjunctions and error assumptions.

Known prior EB: exact Prsa2022 TIC0300903537, period5.8232216d; excluded from new subset. The physical verdict and catalogue status are separate; seven scopes, service status, exact identity matches and nearby rows are recorded in ../../../shared-catalogues/coverage-matrix.csv. SB* or ROT alone is not prior EB evidence.

Finite-cadence local trapezoids are conditional on discovery cycle counts; no exposure integration or precision period uncertainty is inferred. The original P/2 alternative is distinct from a new orbital detection. Source tests cover catalogue cones, not every uncatalogued or unresolved companion. G/T colour and aperture throughput limit dilution arithmetic; no calibrated centroid uncertainty or RSS significance is claimed. The approximate central-duration estimate is3.03h versus observed median3.18h; TIC dimensions and minimum-mass main-sequence companion assumptions are approximate and this small difference is not a fitted-geometry exclusion.

Execution is complete under ../../../BRIEF.md; limitations above are scientific outcomes, not unattempted work. Evidence and hashes: ../../raw-evidence/measurements-summary.md, receipt-manifest.json, identity-checks.json, source-model-comparison.json and per-target pixel_sNNN.npz/png. Next check: joint RV/eclipse characterization with improved stellar dimensions.

C25. TIC 280075497 full report

Archived source: sanitized report text.

TIC 280075497: CONFIRMED

Repeated distinct eclipses, an eclipse-derived period, target-centered two-sector deficits, independent extraction and Gaia consistency support a target eclipsing binary.

1. Repeated raw events: QLP SAP_FLUX with QUALITY=0; 5 accepted representative events in sectors[12, 65, 103], depths53.20-58.29ppt and durations8.45-8.77h. Lead personally inspected the first/last raw event panels; localized dips repeat across time-separated sectors.
2. Period from eclipses: primary recurrence6.034326668d, epoch1634.891927BTJD, timing residual RMS1.85min. This is an unweighted representative-event regression conditional on cycle counts, not the Gaia period copied as a measurement. See ../../raw-evidence/raw-events.json.
3. Host/source: signed out-minus-in deficit maps in sectors[12, 65] peak around the exact Gaia target WCS position. Lead visually inspected a representative signed map for this target and the raw panels. TIC T-band60arcsec feasibility and GaiaDR3 cone receipts are recorded, with original90arcsec DR3 reuse where fresh cone requests failed. Resolved capable field sources lie away from the disappearing light. Close-source comparison uses the same fixed-position circular-Gaussian-plus-background model for target and capable TIC neighbours within30arcsec: sector12: neighbour/target RSS ratios 11.78, 16.76, 17.37; sector65: neighbour/target RSS ratios 12.59, 13.10, 13.48. These ratios are descriptive shape comparisons, not calibrated probabilities. The positive-clipped centroid is center-biased and is not the deciding source statistic. Target-centered deficit morphology across two sectors, together with relative fits where needed, supports target attribution within the tested catalogue scope.
4. Independent reduction: Independent custom TESScut extraction reproduces the events: sector12 3x3 local-baseline depth40.30ppt; sector65 3x3 local-baseline depth41.96ppt. This is a separate pixel-aperture reduction of the same TESS photons, not an independent observation or an archive SPOC pipeline. Quality filtering, source position through WCS, sector headers, sample counts and FITS hashes are recorded in pixel-results.json and tesscut-downloads.json. Archive independent-product discovery was attempted but no usable same-sector product was obtained; failures/empty inventories remain recorded.
5. Gaia geometry: exact DR3 source5982759666597003392, P=6.032589922+/-0.00110647d, e=0.0162, omega=225.09deg. Measured-minus-Gaia period is+1.57quoted Gaia sigma. Closest conjunctionA; the photometric ephemeris projected near BTJD389 gives offset+0.0380d with linear full-covariance Gaia error0.1290d. The projection has no measured photometric period error and is a consistency diagnostic. Observed median duration8.59h versus approximate central-crossing scale9.28h from TIC radius and minimum companion mass supports basic feasibility except the explicit caveats. Radius estimates and assumed companion mass-radius relation are not a unique fit. ../../gaia-geometry.json preserves both conjunctions and error assumptions.

No target prior EB identified in seven recorded scopes; bounded catalogue coverage, no exhaustive priority claim. The physical verdict and catalogue status are separate; seven scopes, service status, exact identity matches and nearby rows are recorded in ../../../shared-catalogues/coverage-matrix.csv. SB* or ROT alone is not prior EB evidence.

Finite-cadence local trapezoids are conditional on discovery cycle counts; no exposure integration or precision period uncertainty is inferred. The original P/2 alternative is distinct from a new orbital detection. Source tests cover catalogue cones, not every uncatalogued or unresolved companion. G/T colour and aperture throughput limit dilution arithmetic; no calibrated centroid uncertainty or RSS significance is claimed.

Execution is complete under ../../../BRIEF.md; limitations above are scientific outcomes, not unattempted work. Evidence and hashes: ../../raw-evidence/measurements-summary.md, receipt-manifest.json, identity-checks.json, source-model-comparison.json and per-target pixel_sNNN.npz/png. Next check: joint RV/eclipse characterization with improved stellar dimensions.

C26. TIC 276514049 full report

Archived source: sanitized report text.

TIC 276514049: CONFIRMED

Repeated distinct eclipses, an eclipse-derived period, target-centered two-sector deficits, independent extraction and Gaia consistency support a target eclipsing binary.

1. Repeated raw events: QLP SAP_FLUX with QUALITY=0; 3 accepted representative events in sectors[56, 83], depths59.75-62.36ppt and durations8.87-9.19h. Lead personally inspected the first/last raw event panels; localized dips repeat across time-separated sectors.
2. Period from eclipses: primary recurrence17.931899173d, epoch1722.528115BTJD, timing residual RMS0.48min. This is an unweighted representative-event regression conditional on cycle counts, not the Gaia period copied as a measurement. See ../../raw-evidence/raw-events.json.
3. Host/source: signed out-minus-in deficit maps in sectors[56, 83] peak around the exact Gaia target WCS position. Lead visually inspected a representative signed map for this target and the raw panels. TIC T-band60arcsec feasibility and GaiaDR3 cone receipts are recorded, with original90arcsec DR3 reuse where fresh cone requests failed. Resolved capable field sources lie away from the disappearing light. Close-source comparison uses the same fixed-position circular-Gaussian-plus-background model for target and capable TIC neighbours within30arcsec: sector56: neighbour/target RSS ratios 3.42; sector83: neighbour/target RSS ratios 4.13. These ratios are descriptive shape comparisons, not calibrated probabilities. The positive-clipped centroid is center-biased and is not the deciding source statistic. Target-centered deficit morphology across two sectors, together with relative fits where needed, supports target attribution within the tested catalogue scope.
4. Independent reduction: Independent custom TESScut extraction reproduces the events: sector56 3x3 local-baseline depth46.29ppt; sector83 3x3 local-baseline depth52.61ppt. This is a separate pixel-aperture reduction of the same TESS photons, not an independent observation or an archive SPOC pipeline. Quality filtering, source position through WCS, sector headers, sample counts and FITS hashes are recorded in pixel-results.json and tesscut-downloads.json. Archive independent-product discovery was attempted but no usable same-sector product was obtained; failures/empty inventories remain recorded.
5. Gaia geometry: exact DR3 source1974064349050049792, P=17.924991468+/-0.00853752d, e=0.0796, omega=94.79deg. Measured-minus-Gaia period is+0.81quoted Gaia sigma. Closest conjunctionA; the photometric ephemeris projected near BTJD389 gives offset-0.0779d with linear full-covariance Gaia error0.3209d. The projection has no measured photometric period error and is a consistency diagnostic. Observed median duration9.16h versus approximate central-crossing scale13.38h from TIC radius and minimum companion mass supports basic feasibility except the explicit caveats. Radius estimates and assumed companion mass-radius relation are not a unique fit. ../../gaia-geometry.json preserves both conjunctions and error assumptions.

No target prior EB identified in seven recorded scopes; bounded catalogue coverage, no exhaustive priority claim. The physical verdict and catalogue status are separate; seven scopes, service status, exact identity matches and nearby rows are recorded in ../../../shared-catalogues/coverage-matrix.csv. SB* or ROT alone is not prior EB evidence.

Finite-cadence local trapezoids are conditional on discovery cycle counts; no exposure integration or precision period uncertainty is inferred. The original P/2 alternative is distinct from a new orbital detection. Source tests cover catalogue cones, not every uncatalogued or unresolved companion. G/T colour and aperture throughput limit dilution arithmetic; no calibrated centroid uncertainty or RSS significance is claimed.

Execution is complete under ../../../BRIEF.md; limitations above are scientific outcomes, not unattempted work. Evidence and hashes: ../../raw-evidence/measurements-summary.md, receipt-manifest.json, identity-checks.json, source-model-comparison.json and per-target pixel_sNNN.npz/png. Next check: joint RV/eclipse characterization with improved stellar dimensions.

C27. TIC 10790605 full report

Archived source: sanitized report text.

TIC 10790605: REFUTED eclipse interpretation

The discovery half-period signal is smooth double-wave modulation. The available QLP raw light curves do not establish a localized eclipse; this result does not refute variability or the existence of a binary.

1. Raw flux and repetition: quality-zero SAP flux in sectors18,58,86 was examined at the discovery full period3.7588988d. The trapezoid fits do not pass the repeated-event quality filter; a broad curved baseline can imitate the fitted dip. Lead inspected the raw first/last-event panels.
2. Period/morphology: four-harmonic subtraction leaves phase-zero narrow-box residual depths -0.010+/-0.435, +0.253+/-0.171 and +0.096+/-0.370ppt, respectively. These optimistic IID errors ignore correlated noise. No narrow eclipse is established; a smooth ellipsoidal interpretation is favored. Harmonics can absorb broad eclipses, so this is a bounded negative for the screened narrow-dip claim, not a proof of no grazing eclipse.
3. Source test: the TIC and GaiaDR3 cones were obtained. No event template survived the quality/morphology test, so a difference image cannot meaningfully test a narrow eclipse. This is adjudicated inapplicability, rather than an omitted positive-eclipse gate.
4. Independent reduction: archive inventory was attempted but no matching usable product was obtained. A separate pipeline cannot promote a narrow signal absent in the primary raw flux. No positive EB inference is made.
5. Gaia: the supplied SB1 orbit has period3.7594964d. Agreement of smooth modulation with that period is not evidence for an eclipse or inclination. No eclipse geometry is inferred.

No target prior EB identified in seven recorded scopes; bounded catalogue coverage, no exhaustive priority claim. Catalogue sources: ../../../shared-catalogues/coverage-matrix.csv and README.md.

Deciding evidence: ../../raw-evidence/10790605/harmonic-test.json, ../../raw-evidence/raw-events.json, ../../raw-evidence/raw-event-panels.png, identity-checks.json and measurements-summary.md. All30 QLP target/sector header checks passed. The full review is complete as a negative outcome with the limitations above; no precision non-eclipse upper limit is claimed. Next check: targeted morphology study only if a broader grazing-eclipse hypothesis is pursued.

C28. TIC 376082142 full report

Archived source: sanitized report text.

TIC 376082142: REFUTED

Queue position 11; batch02. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

The original screen positive is smooth double-wave modulation, rather than an established discrete eclipse. Three QLP sector folds and four-harmonic residual diagnostics show no narrow residual recurring at the searched phase. This refutes the proposed eclipsing morphology at the measured signal; shallow additional eclipses are not excluded.

Raw events and eclipse period

Three cached QLP SAP_FLUX sectors were checked with QUALITY=0, local quadratic baseline plus trapezoid fits. 6 representative fits were retained in raw-events.json, 6 clear of fit bounds. Depth range 0.12–0.386 percent. Period from representative event centers: not established as an eclipse period. Search-cycle counts are conditional on the initial search, not a blind period discovery. No timing uncertainty or ephemeris prediction is claimed.

Smooth-variable morphology is assessed in ambiguous-folds.json and helper receipts.

Source imaging and independent reduction

Fresh cutouts were obtained, but no eclipsing deficit is established to localize. Morphology is the deciding failure, not a missing pixel test. The smooth variable itself has not been assigned a by-catch host.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. MAST product-inventory queries were attempted and timed out for all ten; this is a service limitation, not absence of products. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

The doubled BLS interval follows the Gaia orbit approximately, but a modulation period is not an eclipse-derived period. No eclipse conjunction geometry is claimed.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be

weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Execution: completed representative full review; scientific ambiguity: No established eclipse; the smooth signal could be ellipsoidal or other variability. Harmonics can absorb broad events and do not prove absence of shallow eclipses.

Next deciding check: Search higher precision data for a discrete narrow eclipse only if this smooth variable remains of interest.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all30 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C29. TIC 285942031 full report

Archived source: sanitized report text.

TIC 285942031: PROBABLE

Queue position 12; batch02. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

Deep, repeated discrete eclipses are real and source fits favor the target in two detector sectors. The Gaia orbit has material timing tension, so the geometry check is not fully passed.

Raw events and eclipse period

Three cached QLP SAP_FLUX sectors were checked with QUALITY=0, local quadratic baseline plus trapezoid fits. 5 representative fits were retained in raw-events.json, 5 clear of fit bounds. Depth range 10.113–11.857 percent. Period from representative event centers: 2.6068541455671475. Search-cycle counts are conditional on the initial search, not a blind period discovery. No timing uncertainty or ephemeris prediction is claimed.

{ "period_d": 2.6068541455671475, "epoch_btjd": 1795.1495351827527, "n_events": 5, "residual_rms_min": 1.9711095797469935, "note": "unweighted representative events, conditional on original search cycle counts; no uncertainty inferred" }

Source imaging and independent reduction

Signed difference images, Gaussian free-center diagnostics and fixed-coordinate target-versus-neighbour fits are preserved in raw-evidence. The fixed-position neighbour/target residual sums of squares consistently favor the target. Ratios are descriptive, not formal significance; source PRF mismatch is a limitation.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. MAST product-inventory queries were attempted and timed out for all ten; this is a service limitation, not absence of products. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

Photometric period 2.606854146 d versus Gaia 2.607369245 ± 0.00013636 d; difference -3.78 Gaia sigma. Nearest approximate conjunction A: offset -0.355 d; linear covariance error 0.074 d, conservative sum bound 0.342 d. Observed median duration 3.74 h versus approximate central feasibility 5.47 h. Phase uncertainty and approximate geometry are retained rather than treating a linear sigma as exact.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Execution: completed representative full review; scientific ambiguity: Gaia period differs by -3.78 sigma; approximate conjunction is 0.355 d off versus linear covariance error 0.074 d. The conservative sum error is 0.342 d; a nonlinear orbital posterior or fresh RV fit is needed before treating this as the Gaia eclipsing system.

Next deciding check: Reassess Gaia orbital uncertainties/correlations or obtain an updated radial-velocity orbit to resolve the eclipse-versus-Gaia timing tension.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all30 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C30. TIC 104736355 full report

Archived source: sanitized report text.

TIC 104736355: CONFIRMED

Queue position 13; batch02. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

Repeated discrete eclipses are remeasured in raw QLP flux with a coherent event-derived period. Fresh custom TESScut extraction recovers the deficit, and source tests favor the target against all catalogue neighbours capable under the conservative magnitude limit.

Raw events and eclipse period

Three cached QLP SAP_FLUX sectors were checked with QUALITY=0, local quadratic baseline plus trapezoid fits. 5 representative fits were retained in raw-events.json, 5 clear of fit bounds. Depth range 8.471–8.897 percent. Period from representative event centers: 2.792537259982342. Search-cycle counts are conditional on the initial search, not a blind period discovery. No timing uncertainty or ephemeris prediction is claimed.

{ "period_d": 2.792537259982342, "epoch_btjd": 1820.589246873317, "n_events": 5, "residual_rms_min": 0.20896718846358212, "note": "unweighted representative events, conditional on original search cycle counts; no uncertainty inferred" }

Source imaging and independent reduction

Signed difference images, Gaussian free-center diagnostics and fixed-coordinate target-versus-neighbour fits are preserved in raw-evidence. The fixed-position neighbour/target residual sums of squares consistently favor the target. Ratios are descriptive, not formal significance; source PRF mismatch is a limitation.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. MAST product-inventory queries were attempted and timed out for all ten; this is a service limitation, not absence of products. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

Photometric period 2.792537260 d versus Gaia 2.792657091 ± 0.00025780 d; difference -0.46 Gaia sigma. Nearest approximate conjunction A: offset -0.022 d; linear covariance error 0.151 d, conservative sum bound 1.538 d. Observed median duration 3.46 h versus approximate central feasibility 4.08 h. Phase uncertainty and approximate geometry are retained rather than treating a linear sigma as exact.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Execution: completed representative full review; scientific ambiguity: No exhaustive event/sector fit or detailed binary model; custom extraction shares TESS photons; 60 arcsec TIC neighbour search and uncalibrated PRF systematics limit the source assessment.

Next deciding check: Optional spectroscopy or higher spatial resolution photometry would refine characterization; no additional deciding gate is required by this representative review.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all30 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C31. TIC 281393563 full report

Archived source: sanitized report text.

TIC 281393563: PROBABLE

Queue position 14; batch02. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

Repeated deep detached eclipses have a coherent event-derived period and feasible Gaia geometry. Fresh source imaging was attempted, but crowding prevents a secure target assignment.

Raw events and eclipse period

Three cached QLP SAP_FLUX sectors were checked with QUALITY=0, local quadratic baseline plus trapezoid fits. 6 representative fits were retained in raw-events.json, 6 clear of fit bounds. Depth range 7.027–9.69 percent. Period from representative event centers: 6.149828122486457. Search-cycle counts are conditional on the initial search, not a blind period discovery. No timing uncertainty or ephemeris prediction is claimed.

{ "period_d": 6.149828122486457, "epoch_btjd": 1575.6730668128118, "n_events": 6, "residual_rms_min": 1.418844494519903, "note": "unweighted representative events, conditional on original search cycle counts; no uncertainty inferred" }

Source imaging and independent reduction

Sectors 10 and64 contain target-region deficits; sector64 also has a strong off-center deficit. Fixed-position capable-neighbour alternatives have residual ratios only1.03–1.99, with a degenerate free-center fit. This is an unresolved source gate, not unattempted work.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. MAST product-inventory queries were attempted and timed out for all ten; this is a service limitation, not absence of products. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

Photometric period 6.149828122 d versus Gaia 6.149604702 ± 0.00058686 d; difference 0.38 Gaia sigma. Nearest approximate conjunction A: offset -0.047 d; linear covariance error 0.153 d, conservative sum bound 1.102 d. Observed median duration 7.05 h versus approximate central feasibility 9.81 h. Phase uncertainty and approximate geometry are retained rather than treating a linear sigma as exact.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Execution: completed representative full review; scientific ambiguity: The eclipses repeat, but the host is not separated from capable neighbours. Pixel models are degenerate in this crowded field.

Next deciding check: Obtain higher resolution time-series imaging or a validated multi-source PRF difference fit to identify the eclipsing host.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all30 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C32. TIC 450055243 full report

Archived source: sanitized report text.

TIC 450055243: CONFIRMED

Queue position 15; batch02. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

Repeated discrete eclipses are remeasured in raw QLP flux with a coherent event-derived period. Fresh custom TESScut extraction recovers the deficit, and source tests favor the target against all catalogue neighbours capable under the conservative magnitude limit.

Raw events and eclipse period

Three cached QLP SAP_FLUX sectors were checked with QUALITY=0, local quadratic baseline plus trapezoid fits. 6 representative fits were retained in raw-events.json, 6 clear of fit bounds. Depth range 8.573–9.969 percent. Period from representative event centers: 9.2907014698317. Search-cycle counts are conditional on the initial search, not a blind period discovery. No timing uncertainty or ephemeris prediction is claimed.

{ "period_d": 9.2907014698317, "epoch_btjd": 1442.9918128436386, "n_events": 6, "residual_rms_min": 2.4356151301002424, "note": "unweighted representative events, conditional on original search cycle counts; no uncertainty inferred" }

Source imaging and independent reduction

Signed difference images, Gaussian free-center diagnostics and fixed-coordinate target-versus-neighbour fits are preserved in raw-evidence. The fixed-position neighbour/target residual sums of squares consistently favor the target. Ratios are descriptive, not formal significance; source PRF mismatch is a limitation.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. MAST product-inventory queries were attempted and timed out for all ten; this is a service limitation, not absence of products. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

Photometric period 9.290701470 d versus Gaia 9.288881009 ± 0.00157511 d; difference 1.16 Gaia sigma. Nearest approximate conjunction A: offset 0.376 d; linear covariance error 0.184 d, conservative sum bound 0.392 d. Observed median duration 3.26 h versus approximate central feasibility 4.72 h. Phase uncertainty and approximate geometry are retained rather than treating a linear sigma as exact.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Execution: completed representative full review; scientific ambiguity: No exhaustive event/sector fit or detailed binary model; custom extraction shares TESS photons; 60 arcsec TIC neighbour search and uncalibrated PRF systematics limit the source assessment.

Next deciding check: Optional spectroscopy or higher spatial resolution photometry would refine characterization; no additional deciding gate is required by this representative review.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all30 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C33. TIC 126363604 full report

Archived source: sanitized report text.

TIC 126363604: PROBABLE

Queue position 16; batch02. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

Four sharp repeat eclipses are visible over a variable stellar baseline; all four centers define an eclipse period. Raw-event scatter is inflated by variability, so the initial depth/RMS gate was explicitly relaxed after fold inspection. Fresh imaging supports the target but does not fully exclude its closest capable source.

Raw events and eclipse period

Three cached QLP SAP_FLUX sectors were checked with QUALITY=0, local quadratic baseline plus trapezoid fits. 4 representative fits were retained in raw-events.json, 4 clear of fit bounds. Depth range 0.789–1.165 percent. Period from representative event centers: 16.505959732386128. Search-cycle counts are conditional on the initial search, not a blind period discovery. No timing uncertainty or ephemeris prediction is claimed.

{ "period_d": 16.505959732386128, "epoch_btjd": 1505.614116361415, "n_events": 4, "residual_rms_min": 14.198163488304088, "note": "Four visible sharp events; no per-point depth/RMS gate because stellar baseline variation inflates RMS. Unweighted cycle counts conditional on original search." }

Source imaging and independent reduction

Four individual signed difference images in sectors7,61,88 place fitted deficits3.93,1.34,0.81,0.82 arcsec from the target. Nearest capable neighbour is11.14 arcsec. The first-sector fixed-source residual ratio is only1.78; repeat source positions support the target but the PSF systematics are uncalibrated. Custom extraction recovers repeated deficits in the same TESS photons.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. MAST product-inventory queries were attempted and timed out for all ten; this is a service limitation, not absence of products. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

Photometric period 16.505959732 d versus Gaia 16.499031114 ± 0.00363290 d; difference 1.91 Gaia sigma. Nearest approximate conjunction B: offset 0.657 d; linear covariance error 0.290 d, conservative sum bound 1.352 d. Observed median duration 8.22 h versus approximate central feasibility 22.88 h. Phase uncertainty and approximate geometry are retained rather than treating a linear sigma as exact.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Execution: completed representative full review; scientific ambiguity: Closest capable host not conclusively excluded under the source model; approximate Gaia conjunction offset0.657 d has linear covariance error0.290 d, with broad conservative uncertainty. Stellar variability affects local timing/depth.

Next deciding check: Validate the repeated difference-image astrometry with a multi-source PRF or higher resolution event photometry; refine Gaia conjunction uncertainty if needed.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all30 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C34. TIC 251357000 full report

Archived source: sanitized report text.

TIC 251357000: REFUTED

Queue position 17; batch02. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

The original screen positive is smooth double-wave modulation, rather than an established discrete eclipse. Three QLP sector folds and four-harmonic residual diagnostics show no narrow residual recurring at the searched phase. This refutes the proposed eclipsing morphology at the measured signal; shallow additional eclipses are not excluded.

Raw events and eclipse period

Three cached QLP SAP_FLUX sectors were checked with QUALITY=0, local quadratic baseline plus trapezoid fits. 6 representative fits were retained in raw-events.json, 2 clear of fit bounds. Depth range 0.75–0.811 percent. Period from representative event centers: not established as an eclipse period. Search-cycle counts are conditional on the initial search, not a blind period discovery. No timing uncertainty or ephemeris prediction is claimed.

Smooth-variable morphology is assessed in ambiguous-folds.json and helper receipts.

Source imaging and independent reduction

Fresh cutouts were obtained, but no eclipsing deficit is established to localize. Morphology is the deciding failure, not a missing pixel test. The smooth variable itself has not been assigned a by-catch host.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. MAST product-inventory queries were attempted and timed out for all ten; this is a service limitation, not absence of products. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

The doubled BLS interval follows the Gaia orbit approximately, but a modulation period is not an eclipse-derived period. No eclipse conjunction geometry is claimed.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be

weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Execution: completed representative full review; scientific ambiguity: No established eclipse; the smooth signal could be ellipsoidal or other variability. Harmonics can absorb broad events and do not prove absence of shallow eclipses.

Next deciding check: Search higher precision data for a discrete narrow eclipse only if this smooth variable remains of interest.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all30 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C35. TIC 267222866 full report

Archived source: sanitized report text.

TIC 267222866: CONFIRMED

Queue position 18; batch02. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

Repeated discrete eclipses are remeasured in raw QLP flux with a coherent event-derived period. Fresh custom TESScut extraction recovers the deficit, and source tests favor the target against all catalogue neighbours capable under the conservative magnitude limit.

Raw events and eclipse period

Three cached QLP SAP_FLUX sectors were checked with QUALITY=0, local quadratic baseline plus trapezoid fits. 5 representative fits were retained in raw-events.json, 5 clear of fit bounds. Depth range 1.438–2.207 percent. Period from representative event centers: 1.867209837595571. Search-cycle counts are conditional on the initial search, not a blind period discovery. No timing uncertainty or ephemeris prediction is claimed.

{ "period_d": 1.867209837595571, "epoch_btjd": 1818.1354273795532, "n_events": 5, "residual_rms_min": 1.5541314088817642, "note": "unweighted representative events, conditional on original search cycle counts; no uncertainty inferred" }

Source imaging and independent reduction

Signed difference images, Gaussian free-center diagnostics and fixed-coordinate target-versus-neighbour fits are preserved in raw-evidence. The fixed-position neighbour/target residual sums of squares consistently favor the target. Ratios are descriptive, not formal significance; source PRF mismatch is a limitation.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. MAST product-inventory queries were attempted and timed out for all ten; this is a service limitation, not absence of products. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

Photometric period 1.867209838 d versus Gaia 1.867184804 ± 0.00013282 d; difference 0.19 Gaia sigma. Nearest approximate conjunction A: offset 0.094 d; linear covariance error 0.118 d, conservative sum bound 0.590 d. Observed median duration 6.04 h versus approximate central feasibility 5.56 h. Phase uncertainty and approximate geometry are retained rather than treating a linear sigma as exact.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Execution: completed representative full review; scientific ambiguity: No exhaustive event/sector fit or detailed binary model; custom extraction shares TESS photons; 60 arcsec TIC neighbour search and uncalibrated PRF systematics limit the source assessment.

Next deciding check: Optional spectroscopy or higher spatial resolution photometry would refine characterization; no additional deciding gate is required by this representative review.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all30 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C36. TIC 66315917 full report

Archived source: sanitized report text.

TIC 66315917: CONFIRMED

Queue position 19; batch02. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

Repeated discrete eclipses are remeasured in raw QLP flux with a coherent event-derived period. Fresh custom TESScut extraction recovers the deficit, and source tests favor the target against all catalogue neighbours capable under the conservative magnitude limit.

Raw events and eclipse period

Three cached QLP SAP_FLUX sectors were checked with QUALITY=0, local quadratic baseline plus trapezoid fits. 3 representative fits were retained in raw-events.json, 3 clear of fit bounds. Depth range 2.591–2.739 percent. Period from representative event centers: 19.634374528897247. Search-cycle counts are conditional on the initial search, not a blind period discovery. No timing uncertainty or ephemeris prediction is claimed.

{ "period_d": 19.634374528897247, "epoch_btjd": 1494.1205215957734, "n_events": 3, "residual_rms_min": 0.44878357765813764, "note": "unweighted representative events, conditional on original search cycle counts; no uncertainty inferred" }

Source imaging and independent reduction

Signed difference images, Gaussian free-center diagnostics and fixed-coordinate target-versus-neighbour fits are preserved in raw-evidence. The fixed-position neighbour/target residual sums of squares consistently favor the target. Ratios are descriptive, not formal significance; source PRF mismatch is a limitation.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. MAST product-inventory queries were attempted and timed out for all ten; this is a service limitation, not absence of products. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

Photometric period 19.634374529 d versus Gaia 19.608607700 ± 0.02322759 d; difference 1.11 Gaia sigma. Nearest approximate conjunction A: offset 1.879 d; linear covariance error 1.775 d, conservative sum bound 2.469 d. Observed median duration 2.20 h versus approximate central feasibility 5.74 h. Phase uncertainty and approximate geometry are retained rather than treating a linear sigma as exact.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Execution: completed representative full review; scientific ambiguity: No exhaustive event/sector fit or detailed binary model; custom extraction shares TESS photons; 60 arcsec TIC neighbour search and uncalibrated PRF systematics limit the source assessment.

Next deciding check: Optional spectroscopy or higher spatial resolution photometry would refine characterization; no additional deciding gate is required by this representative review.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all30 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C37. TIC 93783392 full report

Archived source: sanitized report text.

TIC 93783392: CONFIRMED

Queue position 20; batch02. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

Repeated discrete eclipses are remeasured in raw QLP flux with a coherent event-derived period. Fresh custom TESScut extraction recovers the deficit, and source tests favor the target against all catalogue neighbours capable under the conservative magnitude limit.

Raw events and eclipse period

Three cached QLP SAP_FLUX sectors were checked with QUALITY=0, local quadratic baseline plus trapezoid fits. 6 representative fits were retained in raw-events.json, 6 clear of fit bounds. Depth range 2.665–3.25 percent. Period from representative event centers: 3.8724081070823737. Search-cycle counts are conditional on the initial search, not a blind period discovery. No timing uncertainty or ephemeris prediction is claimed.

{ "period_d": 3.8724081070823737, "epoch_btjd": 1630.7599836800352, "n_events": 6, "residual_rms_min": 3.507427667895757, "note": "unweighted representative events, conditional on original search cycle counts; no uncertainty inferred" }

Source imaging and independent reduction

Signed difference images, Gaussian free-center diagnostics and fixed-coordinate target-versus-neighbour fits are preserved in raw-evidence. The fixed-position neighbour/target residual sums of squares consistently favor the target. Ratios are descriptive, not formal significance; source PRF mismatch is a limitation.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. MAST product-inventory queries were attempted and timed out for all ten; this is a service limitation, not absence of products. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

Photometric period 3.872408107 d versus Gaia 3.871849592 ± 0.00099811 d; difference 0.56 Gaia sigma. Nearest approximate conjunction A: offset 0.129 d; linear covariance error 0.361 d, conservative sum bound 0.973 d. Observed median duration 5.58 h versus approximate central feasibility 7.43 h. Phase uncertainty and approximate geometry are retained rather than treating a linear sigma as exact.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Execution: completed representative full review; scientific ambiguity: No exhaustive event/sector fit or detailed binary model; custom extraction shares TESS photons; 60 arcsec TIC neighbour search and uncalibrated PRF systematics limit the source assessment.

Next deciding check: Optional spectroscopy or higher spatial resolution photometry would refine characterization; no additional deciding gate is required by this representative review.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all30 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C38. TIC 470947699 full report

Archived source: sanitized report text.

TIC 470947699: CONFIRMED

Queue position 21; batch03. Anonymous project coordinator review with an independent measurement helper.

Deciding assessment

Repeated discrete eclipses are present in raw QLP flux; representative event centers give a coherent eclipse period. Fresh independent aperture reduction and source localization are assessed below.

Raw flux and period

Cached QLP SAP_FLUX, QUALITY=0; 3 representative sectors. Local quadratic baseline plus trapezoid fits retained 6 trial events. Only repeated discrete eclipse morphology supports an eclipse period. Search cycle counts were initial constraints, not independent period discovery. Lead inspected raw-event-panels.png and ambiguous-folds.png; smooth-variable fits hitting duration bounds are not eclipses.

Accepted event-derived period: 8.17041492823667 d.

Initial measurement receipt (superseded where rejected above):


```
{
  "period_d": 8.17041492823667,
  "epoch_btjd": 1846.407037256951,
  "n_events": 6,
  "residual_rms_min": 0.8413010526801175,
  "note": "unweighted representative events, conditional on original search cycle counts; no uncertainty inferred"
}
```


Fresh source and independent reduction

Two fresh detector-sector signed difference stacks favor the target, with free Gaussian offsets and fixed-position capable-neighbour comparisons below. Neighbour capability uses a conservative depth/magnitude threshold; TIC cones extend 60 arcsec. Ratios are descriptive residual comparisons and do not imply calibrated probabilities. Custom TESScut aperture extraction independently recovers the events from the same TESS photons as QLP.


```
[
  {
    "sector": 20,
    "target_offset_arcsec": 1.7818942626730834,
    "min_neighbour_target_sse_ratio": null,
    "capable_neighbour_count": 0
  },
  {
    "sector": 60,
    "target_offset_arcsec": 0.5031546449154466,
    "min_neighbour_target_sse_ratio": null,
```

```
    "capable_neighbour_count": 0
  }
]
```


Custom TESScut reduction is independent of QLP processing but uses the same photons; it is not SPOC or an independent telescope. Signed in/out images use adjacent pre/post baselines. Gaussian formal errors exclude PRF systematics; fixed-coordinate source comparisons and repeat source positions govern attribution. Fresh downloads, TIC cones, signed arrays, plots and source scripts are preserved in raw-evidence.

Gaia geometry

Event-derived P=8.170414928 d; Gaia P=8.172049314 +/- 0.001169795 d. Difference -1.40 Gaia sigma is descriptive. Approximate conjunction A offset -0.292 d; covariance linear error 0.225 d; conservative absolute-sum bound 1.171 d. Median fitted duration 3.88 h; approximate central duration 4.59 h. Geometry is physically feasible within approximate radii and propagated orbital phase uncertainty.

The computed gaia-geometry.json uses edge-on conjunction approximation, BTJD conversion Tp+389, and full SB1 covariance ordering P,gamma,K1,e,omega,Tp. Linear uncertainty is approximate, especially with weak eccentricity. Mass-function/main-sequence radius estimates test duration feasibility only; there is no fitted physical model or automatic sigma rejection.

Catalogue status

no identified prior EB; all seven central scopes complete: yes. Central shared-catalogues coverage includes VSX, SIMBAD primary type, GaiaDR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4 and ASAS-SN. SB* or ROT alone is not prior EB evidence. This is scoped novelty coverage, not an exhaustive discovery claim.

Completion and limitations

Full representative review complete. Scientific ambiguity: Representative review; no exhaustive event or sector fits. Independent custom extraction shares TESS photons with QLP. Pixel PRF and catalogue completeness limit astrometry; no formal calibrated PRF significance is claimed.

Next test: Optional improved spectroscopy and photometry would refine binary characterization.

Provenance: cache-manifest.json gives original cached FITS paths and SHA256. Fresh TESScut URLs/hashes are in tesscut-downloads.json. Helper raw evidence and retained scripts permit reproduction. manifest.json hashes deliverables. No source paper, prior adjudication or publication was changed.

C39. TIC 135230723 full report

Archived source: sanitized report text.

TIC 135230723: CONFIRMED

Queue position 22; batch03. Anonymous project coordinator review with an independent measurement helper.

Deciding assessment

Repeated discrete eclipses are present in raw QLP flux; representative event centers give a coherent eclipse period. Fresh independent aperture reduction and source localization are assessed below.

Raw flux and period

Cached QLP SAP_FLUX, QUALITY=0; 3 representative sectors. Local quadratic baseline plus trapezoid fits retained 5 trial events. Only repeated discrete eclipse morphology supports an eclipse period. Search cycle counts were initial constraints, not independent period discovery. Lead inspected raw-event-panels.png and ambiguous-folds.png; smooth-variable fits hitting duration bounds are not eclipses.

Accepted event-derived period: 6.776113017992571 d.

Initial measurement receipt (superseded where rejected above):


```
{
  "period_d": 6.776113017992571,
  "epoch_btjd": 1958.4036951273954,
  "n_events": 5,
  "residual_rms_min": 1.3932796715666198,
  "note": "unweighted representative events, conditional on original search cycle counts; no uncertainty inferred"
}
```


Fresh source and independent reduction

Two fresh detector-sector signed difference stacks favor the target, with free Gaussian offsets and fixed-position capable-neighbour comparisons below. Neighbour capability uses a conservative depth/magnitude threshold; TIC cones extend 60 arcsec. Ratios are descriptive residual comparisons and do not imply calibrated probabilities. Custom TESScut aperture extraction independently recovers the events from the same TESS photons as QLP.


```
[
  {
    "sector": 24,
    "target_offset_arcsec": 2.7623131101621796,
    "min_neighbour_target_sse_ratio": 2.486867070374025,
    "capable_neighbour_count": 2
  },
  {
    "sector": 51,
    "target_offset_arcsec": 0.5383821425401804,
    "min_neighbour_target_sse_ratio": 4.691126358639871,
```

```
    "capable_neighbour_count": 2
  }
]
```


Custom TESScut reduction is independent of QLP processing but uses the same photons; it is not SPOC or an independent telescope. Signed in/out images use adjacent pre/post baselines. Gaussian formal errors exclude PRF systematics; fixed-coordinate source comparisons and repeat source positions govern attribution. Fresh downloads, TIC cones, signed arrays, plots and source scripts are preserved in raw-evidence.

Gaia geometry

Event-derived P=6.776113018 d; Gaia P=6.778623813 +/- 0.001888155 d. Difference -1.33 Gaia sigma is descriptive. Approximate conjunction A offset -0.612 d; covariance linear error 0.438 d; conservative absolute-sum bound 0.993 d. Median fitted duration 3.31 h; approximate central duration 3.66 h. Geometry is physically feasible within approximate radii and propagated orbital phase uncertainty.

The computed gaia-geometry.json uses edge-on conjunction approximation, BTJD conversion Tp+389, and full SB1 covariance ordering P,gamma,K1,e,omega,Tp. Linear uncertainty is approximate, especially with weak eccentricity. Mass-function/main-sequence radius estimates test duration feasibility only; there is no fitted physical model or automatic sigma rejection.

Catalogue status

no identified prior EB; all seven central scopes complete: yes. Central shared-catalogues coverage includes VSX, SIMBAD primary type, GaiaDR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4 and ASAS-SN. SB* or ROT alone is not prior EB evidence. This is scoped novelty coverage, not an exhaustive discovery claim.

Completion and limitations

Full representative review complete. Scientific ambiguity: Representative review; no exhaustive event or sector fits. Independent custom extraction shares TESS photons with QLP. Pixel PRF and catalogue completeness limit astrometry; no formal calibrated PRF significance is claimed.

Next test: Optional improved spectroscopy and photometry would refine binary characterization.

Provenance: cache-manifest.json gives original cached FITS paths and SHA256. Fresh TESScut URLs/hashes are in tesscut-downloads.json. Helper raw evidence and retained scripts permit reproduction. manifest.json hashes deliverables. No source paper, prior adjudication or publication was changed.

C40. TIC 36083554 full report

Archived source: sanitized report text.

TIC 36083554: CONFIRMED

Queue position 23; batch03. Anonymous project coordinator review with an independent measurement helper.

Deciding assessment

Repeated discrete eclipses are present in raw QLP flux; representative event centers give a coherent eclipse period. Fresh independent aperture reduction and source localization are assessed below.

Raw flux and period

Cached QLP SAP_FLUX, QUALITY=0; 3 representative sectors. Local quadratic baseline plus trapezoid fits retained 5 trial events. Only repeated discrete eclipse morphology supports an eclipse period. Search cycle counts were initial constraints, not independent period discovery. Lead inspected raw-event-panels.png and ambiguous-folds.png; smooth-variable fits hitting duration bounds are not eclipses.

Accepted event-derived period: 15.974024163426064 d.

Initial measurement receipt (superseded where rejected above):


```
{
  "period_d": 15.974024163426064,
  "epoch_btjd": 1496.4368213575215,
  "n_events": 5,
  "residual_rms_min": 2.302940486572714,
  "note": "unweighted representative events, conditional on original search cycle counts; no uncertainty inferred"
}
```


Fresh source and independent reduction

Two fresh detector-sector signed difference stacks favor the target, with free Gaussian offsets and fixed-position capable-neighbour comparisons below. Neighbour capability uses a conservative depth/magnitude threshold; TIC cones extend 60 arcsec. Ratios are descriptive residual comparisons and do not imply calibrated probabilities. Custom TESScut aperture extraction independently recovers the events from the same TESS photons as QLP.


```
[
  {
    "sector": 7,
    "target_offset_arcsec": 1.0002671054830687,
    "min_neighbour_target_sse_ratio": 54.527241876955294,
    "capable_neighbour_count": 11
  },
  {
    "sector": 61,
    "target_offset_arcsec": 2.4025059769389188,
    "min_neighbour_target_sse_ratio": 15.686721955051661,
```

```
    "capable_neighbour_count": 10
  }
]
```


Custom TESScut reduction is independent of QLP processing but uses the same photons; it is not SPOC or an independent telescope. Signed in/out images use adjacent pre/post baselines. Gaussian formal errors exclude PRF systematics; fixed-coordinate source comparisons and repeat source positions govern attribution. Fresh downloads, TIC cones, signed arrays, plots and source scripts are preserved in raw-evidence.

Gaia geometry

Event-derived P=15.974024163 d; Gaia P=15.980986459 +/- 0.006281408 d. Difference -1.11 Gaia sigma is descriptive. Approximate conjunction A offset 0.017 d; covariance linear error 0.456 d; conservative absolute-sum bound 2.293 d. Median fitted duration 4.43 h; approximate central duration 6.18 h. Geometry is physically feasible within approximate radii and propagated orbital phase uncertainty.

The computed gaia-geometry.json uses edge-on conjunction approximation, BTJD conversion Tp+389, and full SB1 covariance ordering P,gamma,K1,e,omega,Tp. Linear uncertainty is approximate, especially with weak eccentricity. Mass-function/main-sequence radius estimates test duration feasibility only; there is no fitted physical model or automatic sigma rejection.

Catalogue status

no identified prior EB; all seven central scopes complete: yes. Central shared-catalogues coverage includes VSX, SIMBAD primary type, GaiaDR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4 and ASAS-SN. SB* or ROT alone is not prior EB evidence. This is scoped novelty coverage, not an exhaustive discovery claim.

Completion and limitations

Full representative review complete. Scientific ambiguity: Representative review; no exhaustive event or sector fits. Independent custom extraction shares TESS photons with QLP. Pixel PRF and catalogue completeness limit astrometry; no formal calibrated PRF significance is claimed.

Next test: Optional improved spectroscopy and photometry would refine binary characterization.

Provenance: cache-manifest.json gives original cached FITS paths and SHA256. Fresh TESScut URLs/hashes are in tesscut-downloads.json. Helper raw evidence and retained scripts permit reproduction. manifest.json hashes deliverables. No source paper, prior adjudication or publication was changed.

C41. TIC 447566809 full report

Archived source: sanitized report text.

TIC 447566809: CONFIRMED

Queue position 24; batch03. Anonymous project coordinator review with an independent measurement helper.

Deciding assessment

Repeated discrete eclipses are present in raw QLP flux; representative event centers give a coherent eclipse period. Fresh independent aperture reduction and source localization are assessed below.

Raw flux and period

Cached QLP SAP_FLUX, QUALITY=0; 3 representative sectors. Local quadratic baseline plus trapezoid fits retained 3 trial events. Only repeated discrete eclipse morphology supports an eclipse period. Search cycle counts were initial constraints, not independent period discovery. Lead inspected raw-event-panels.png and ambiguous-folds.png; smooth-variable fits hitting duration bounds are not eclipses.

Accepted event-derived period: 19.64701755851303 d.

Initial measurement receipt (superseded where rejected above):


```
{
  "period_d": 19.64701755851303,
  "epoch_btjd": 1645.68237977337,
  "n_events": 3,
  "residual_rms_min": 2.8451511295473804,
  "note": "unweighted representative events, conditional on original search cycle counts; no uncertainty inferred"
}
```


Fresh source and independent reduction

Two fresh detector-sector signed difference stacks favor the target, with free Gaussian offsets and fixed-position capable-neighbour comparisons below. Neighbour capability uses a conservative depth/magnitude threshold; TIC cones extend 60 arcsec. Ratios are descriptive residual comparisons and do not imply calibrated probabilities. Custom TESScut aperture extraction independently recovers the events from the same TESS photons as QLP.


```
[
  {
    "sector": 12,
    "target_offset_arcsec": 1.397054877962082,
    "min_neighbour_target_sse_ratio": 16.034792064157667,
    "capable_neighbour_count": 13
  },
  {
    "sector": 93,
    "target_offset_arcsec": 3.744046170975149,
    "min_neighbour_target_sse_ratio": 6.72483777409165,
```

```
    "capable_neighbour_count": 20
  }
]
```


Custom TESScut reduction is independent of QLP processing but uses the same photons; it is not SPOC or an independent telescope. Signed in/out images use adjacent pre/post baselines. Gaussian formal errors exclude PRF systematics; fixed-coordinate source comparisons and repeat source positions govern attribution. Fresh downloads, TIC cones, signed arrays, plots and source scripts are preserved in raw-evidence.

Gaia geometry

Event-derived P=19.647017559 d; Gaia P=19.643298210 +/- 0.002714314 d. Difference 1.37 Gaia sigma is descriptive. Approximate conjunction A offset -0.072 d; covariance linear error 0.174 d; conservative absolute-sum bound 0.518 d. Median fitted duration 9.87 h; approximate central duration 11.32 h. Geometry is physically feasible within approximate radii and propagated orbital phase uncertainty.

The computed gaia-geometry.json uses edge-on conjunction approximation, BTJD conversion Tp+389, and full SB1 covariance ordering P,gamma,K1,e,omega,Tp. Linear uncertainty is approximate, especially with weak eccentricity. Mass-function/main-sequence radius estimates test duration feasibility only; there is no fitted physical model or automatic sigma rejection.

Catalogue status

no identified prior EB; all seven central scopes complete: yes. Central shared-catalogues coverage includes VSX, SIMBAD primary type, GaiaDR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4 and ASAS-SN. SB* or ROT alone is not prior EB evidence. This is scoped novelty coverage, not an exhaustive discovery claim.

Completion and limitations

Full representative review complete. Scientific ambiguity: Representative review; no exhaustive event or sector fits. Independent custom extraction shares TESS photons with QLP. Pixel PRF and catalogue completeness limit astrometry; no formal calibrated PRF significance is claimed.

Next test: Optional improved spectroscopy and photometry would refine binary characterization.

Provenance: cache-manifest.json gives original cached FITS paths and SHA256. Fresh TESScut URLs/hashes are in tesscut-downloads.json. Helper raw evidence and retained scripts permit reproduction. manifest.json hashes deliverables. No source paper, prior adjudication or publication was changed.

C42. TIC 12320469 full report

Archived source: sanitized report text.

TIC 12320469: REFUTED

Queue position 25; batch03. Anonymous project coordinator review with an independent measurement helper.

Deciding assessment

Three sector folds show smooth broad double-wave modulation, without a convincing discrete narrow eclipse. The original screen is refuted as an eclipsing interpretation of this measured signal.

Raw flux and period

Cached QLP SAP_FLUX, QUALITY=0; 3 representative sectors. Local quadratic baseline plus trapezoid fits retained 5 trial events. Only repeated discrete eclipse morphology supports an eclipse period. Search cycle counts were initial constraints, not independent period discovery. Lead inspected raw-event-panels.png and ambiguous-folds.png; smooth-variable fits hitting duration bounds are not eclipses.

Accepted event-derived period: none for the target d.

Initial measurement receipt (superseded where rejected above):

```
null
```

Fresh source and independent reduction

Fresh cutouts were attempted; no established eclipse deficit exists to localize. Source attribution of the smooth modulation is not claimed.

```
[]
```

Custom TESScut reduction is independent of QLP processing but uses the same photons; it is not SPOC or an independent telescope. Signed in/out images use adjacent pre/post baselines. Gaussian formal errors exclude PRF systematics; fixed-coordinate source comparisons and repeat source positions govern attribution. Fresh downloads, TIC cones, signed arrays, plots and source scripts are preserved in raw-evidence.

Gaia geometry

The searched modulation period is not an eclipse-derived period. No conjunction comparison is claimed without established eclipses.

The computed gaia-geometry.json uses edge-on conjunction approximation, BTJD conversion Tp+389, and full SB1 covariance ordering P,gamma,K1,e,omega,Tp. Linear uncertainty is approximate, especially with weak eccentricity. Mass-function/main-sequence radius estimates test duration feasibility only; there is no fitted physical model or automatic sigma rejection.

Catalogue status

no identified prior EB; all seven central scopes complete: yes. Central shared-catalogues coverage includes VSX, SIMBAD primary type, GaiaDR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4 and ASAS-SN. SB* or ROT alone is not prior EB evidence. This is scoped novelty coverage, not an exhaustive discovery claim.

Completion and limitations

Full representative review complete. Scientific ambiguity: Smooth modulation may be ellipsoidal or other stellar variability; shallow additional eclipses are not excluded. A local trapezoid fit to a broad trend does not establish an eclipse.

Next test: Higher precision light curves could test for a narrow residual eclipse if further work on this smooth variable is desired.

Provenance: cache-manifest.json gives original cached FITS paths and SHA256. Fresh TESScut URLs/hashes are in tesscut-downloads.json. Helper raw evidence and retained scripts permit reproduction. manifest.json hashes deliverables. No source paper, prior adjudication or publication was changed.

C43. TIC 86908196 full report

Archived source: sanitized report text.

TIC 86908196: REFUTED

Queue position 26; batch03. Anonymous project coordinator review with an independent measurement helper.

Deciding assessment

Three sector folds show smooth broad double-wave modulation, without a convincing discrete narrow eclipse. The original screen is refuted as an eclipsing interpretation of this measured signal.

Raw flux and period

Cached QLP SAP_FLUX, QUALITY=0; 3 representative sectors. Local quadratic baseline plus trapezoid fits retained 5 trial events. Only repeated discrete eclipse morphology supports an eclipse period. Search cycle counts were initial constraints, not independent period discovery. Lead inspected raw-event-panels.png and ambiguous-folds.png; smooth-variable fits hitting duration bounds are not eclipses.

Accepted event-derived period: none for the target d.

Initial measurement receipt (superseded where rejected above):

```
null
```

Fresh source and independent reduction

Fresh cutouts were attempted; no established eclipse deficit exists to localize. Source attribution of the smooth modulation is not claimed.

```
[]
```

Custom TESScut reduction is independent of QLP processing but uses the same photons; it is not SPOC or an independent telescope. Signed in/out images use adjacent pre/post baselines. Gaussian formal errors exclude PRF systematics; fixed-coordinate source comparisons and repeat source positions govern attribution. Fresh downloads, TIC cones, signed arrays, plots and source scripts are preserved in raw-evidence.

Gaia geometry

The searched modulation period is not an eclipse-derived period. No conjunction comparison is claimed without established eclipses.

The computed gaia-geometry.json uses edge-on conjunction approximation, BTJD conversion Tp+389, and full SB1 covariance ordering P,gamma,K1,e,omega,Tp. Linear uncertainty is approximate, especially with weak eccentricity. Mass-function/main-sequence radius estimates test duration feasibility only; there is no fitted physical model or automatic sigma rejection.

Catalogue status

no identified prior EB; all seven central scopes complete: yes. Central shared-catalogues coverage includes VSX, SIMBAD primary type, GaiaDR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4 and ASAS-SN. SB* or ROT alone is not prior EB evidence. This is scoped novelty coverage, not an exhaustive discovery claim.

Completion and limitations

Full representative review complete. Scientific ambiguity: Smooth modulation may be ellipsoidal or other stellar variability; shallow additional eclipses are not excluded. A local trapezoid fit to a broad trend does not establish an eclipse.

Next test: Higher precision light curves could test for a narrow residual eclipse if further work on this smooth variable is desired.

Provenance: cache-manifest.json gives original cached FITS paths and SHA256. Fresh TESScut URLs/hashes are in tesscut-downloads.json. Helper raw evidence and retained scripts permit reproduction. manifest.json hashes deliverables. No source paper, prior adjudication or publication was changed.

C44. TIC 201317466 full report

Archived source: sanitized report text.

TIC 201317466: PROBABLE

Queue position 27; batch03. Anonymous project coordinator review with an independent measurement helper.

Deciding assessment

Deep repeated eclipses are real and coherent. Association with the published Gaia SB1 orbit remains unresolved because its conjunction misses the eclipse by a substantial fraction of a cycle, beyond the conservative linear propagated uncertainty.

Raw flux and period

Cached QLP SAP_FLUX, QUALITY=0; 2 representative sectors. Local quadratic baseline plus trapezoid fits retained 4 trial events. Only repeated discrete eclipse morphology supports an eclipse period. Search cycle counts were initial constraints, not independent period discovery. Lead inspected raw-event-panels.png and ambiguous-folds.png; smooth-variable fits hitting duration bounds are not eclipses.

Accepted event-derived period: 5.623860159150563 d.

Initial measurement receipt (superseded where rejected above):


```
{
  "period_d": 5.623860159150563,
  "epoch_btjd": 2772.3289690349256,
  "n_events": 4,
  "residual_rms_min": 0.5278820914721635,
  "note": "unweighted representative events, conditional on original search cycle counts; no uncertainty inferred"
}
```


Fresh source and independent reduction

Two fresh detector-sector signed difference stacks favor the target, with free Gaussian offsets and fixed-position capable-neighbour comparisons below. Neighbour capability uses a conservative depth/magnitude threshold; TIC cones extend 60 arcsec. Ratios are descriptive residual comparisons and do not imply calibrated probabilities. Custom TESScut aperture extraction independently recovers the events from the same TESS photons as QLP.


```
[
  {
    "sector": 54,
    "target_offset_arcsec": 0.11507353740750183,
    "min_neighbour_target_sse_ratio": 15.652494055094234,
    "capable_neighbour_count": 2
  },
  {
    "sector": 80,
    "target_offset_arcsec": 3.540797490060455,
```

```
    "min_neighbour_target_sse_ratio": 23.02100675329804,
    "capable_neighbour_count": 2
  }
]
```


Custom TESScut reduction is independent of QLP processing but uses the same photons; it is not SPOC or an independent telescope. Signed in/out images use adjacent pre/post baselines. Gaussian formal errors exclude PRF systematics; fixed-coordinate source comparisons and repeat source positions govern attribution. Fresh downloads, TIC cones, signed arrays, plots and source scripts are preserved in raw-evidence.

Gaia geometry

Event-derived P=5.623860159 d; Gaia P=5.625963779 +/- 0.000558314 d. Difference -3.77 Gaia sigma is descriptive. Approximate conjunction A offset -1.208 d; covariance linear error 0.228 d; conservative absolute-sum bound 0.694 d. Median fitted duration 5.97 h; approximate central duration 7.50 h. Geometry is physically feasible within approximate radii and propagated orbital phase uncertainty.

The computed gaia-geometry.json uses edge-on conjunction approximation, BTJD conversion Tp+389, and full SB1 covariance ordering P,gamma,K1,e,omega,Tp. Linear uncertainty is approximate, especially with weak eccentricity. Mass-function/main-sequence radius estimates test duration feasibility only; there is no fitted physical model or automatic sigma rejection.

Catalogue status

no identified prior EB; all seven central scopes complete: yes. Central shared-catalogues coverage includes VSX, SIMBAD primary type, GaiaDR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4 and ASAS-SN. SB* or ROT alone is not prior EB evidence. This is scoped novelty coverage, not an exhaustive discovery claim.

Completion and limitations

Full representative review complete. Scientific ambiguity: Representative review; no exhaustive event or sector fits. Independent custom extraction shares TESS photons with QLP. Pixel PRF and catalogue completeness limit astrometry; no formal calibrated PRF significance is claimed. Gaia has only 15 RV epochs, modestly constrained eccentricity (0.125 +/- 0.061), and strongly correlated periastron parameters. A 1.208 d conjunction offset versus 0.228 d linear covariance error and 0.694 d conservative sum bound is a physical phase discrepancy; no automatic sigma veto is applied.

Next test: Refit the RV orbit or evaluate a nonlinear orbital posterior against the observed eclipse timing to resolve Gaia association.

Provenance: cache-manifest.json gives original cached FITS paths and SHA256. Fresh TESScut URLs/hashes are in tesscut-downloads.json. Helper raw evidence and retained scripts permit reproduction. manifest.json hashes deliverables. No source paper, prior adjudication or publication was changed.

C45. TIC 360516818 full report

Archived source: sanitized report text.

TIC 360516818: CONFIRMED

Queue position 28; batch03. Anonymous project coordinator review with an independent measurement helper.

Deciding assessment

Repeated discrete eclipses are present in raw QLP flux; representative event centers give a coherent eclipse period. Fresh independent aperture reduction and source localization are assessed below.

Raw flux and period

Cached QLP SAP_FLUX, QUALITY=0; 3 representative sectors. Local quadratic baseline plus trapezoid fits retained 5 trial events. Only repeated discrete eclipse morphology supports an eclipse period. Search cycle counts were initial constraints, not independent period discovery. Lead inspected raw-event-panels.png and ambiguous-folds.png; smooth-variable fits hitting duration bounds are not eclipses.

Accepted event-derived period: 9.890542850425698 d.

Initial measurement receipt (superseded where rejected above):


```
{
  "period_d": 9.890542850425698,
  "epoch_btjd": 1688.7028501960215,
  "n_events": 5,
  "residual_rms_min": 5.591486264753861,
  "note": "unweighted representative events, conditional on original search cycle counts; no uncertainty inferred"
}
```


Fresh source and independent reduction

Two fresh detector-sector signed difference stacks favor the target, with free Gaussian offsets and fixed-position capable-neighbour comparisons below. Neighbour capability uses a conservative depth/magnitude threshold; TIC cones extend 60 arcsec. Ratios are descriptive residual comparisons and do not imply calibrated probabilities. Custom TESScut aperture extraction independently recovers the events from the same TESS photons as QLP.


```
[
  {
    "sector": 14,
    "target_offset_arcsec": 2.4168737956248054,
    "min_neighbour_target_sse_ratio": 3.587870111420561,
    "capable_neighbour_count": 50
  },
  {
    "sector": 41,
    "target_offset_arcsec": 0.7353517988002084,
    "min_neighbour_target_sse_ratio": 13.824703240155678,
```

```
    "capable_neighbour_count": 32
  }
]
```


Custom TESScut reduction is independent of QLP processing but uses the same photons; it is not SPOC or an independent telescope. Signed in/out images use adjacent pre/post baselines. Gaussian formal errors exclude PRF systematics; fixed-coordinate source comparisons and repeat source positions govern attribution. Fresh downloads, TIC cones, signed arrays, plots and source scripts are preserved in raw-evidence.

Gaia geometry

Event-derived P=9.890542850 d; Gaia P=9.890757806 +/- 0.002383965 d. Difference -0.09 Gaia sigma is descriptive. Approximate conjunction A offset 0.066 d; covariance linear error 0.331 d; conservative absolute-sum bound 0.579 d. Median fitted duration 7.57 h; approximate central duration 17.48 h. Geometry is physically feasible within approximate radii and propagated orbital phase uncertainty.

The computed gaia-geometry.json uses edge-on conjunction approximation, BTJD conversion Tp+389, and full SB1 covariance ordering P,gamma,K1,e,omega,Tp. Linear uncertainty is approximate, especially with weak eccentricity. Mass-function/main-sequence radius estimates test duration feasibility only; there is no fitted physical model or automatic sigma rejection.

Catalogue status

no identified prior EB; all seven central scopes complete: yes. Central shared-catalogues coverage includes VSX, SIMBAD primary type, GaiaDR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4 and ASAS-SN. SB* or ROT alone is not prior EB evidence. This is scoped novelty coverage, not an exhaustive discovery claim.

Completion and limitations

Full representative review complete. Scientific ambiguity: Representative review; no exhaustive event or sector fits. Independent custom extraction shares TESS photons with QLP. Pixel PRF and catalogue completeness limit astrometry; no formal calibrated PRF significance is claimed.

Next test: Optional improved spectroscopy and photometry would refine binary characterization.

Provenance: cache-manifest.json gives original cached FITS paths and SHA256. Fresh TESScut URLs/hashes are in tesscut-downloads.json. Helper raw evidence and retained scripts permit reproduction. manifest.json hashes deliverables. No source paper, prior adjudication or publication was changed.

C46. TIC 411647453 full report

Archived source: sanitized report text.

TIC 411647453: REFUTED

Queue position 29; batch03. Anonymous project coordinator review with an independent measurement helper.

Deciding assessment

Real narrow eclipses contaminate the target aperture, but the proposed target Gaia-period eclipse association is refuted. Two detector sectors place the deficit on nearby TIC 411647413 rather than TIC 411647453. The approximately 0.9151763 d short-period signal gives about seven dips per 6.394568 d screen fold; that screen interval is not a valid target eclipse period.

Raw flux and period

Cached QLP SAP_FLUX, QUALITY=0; 3 representative sectors. Local quadratic baseline plus trapezoid fits retained 6 trial events. Only repeated discrete eclipse morphology supports an eclipse period. Search cycle counts were initial constraints, not independent period discovery. Lead inspected raw-event-panels.png and ambiguous-folds.png; smooth-variable fits hitting duration bounds are not eclipses.

Accepted event-derived period: none for the target d.

Initial measurement receipt (superseded where rejected above):


```
{
  "period_d": 6.394515949153654,
  "epoch_btjd": 1603.7676801779019,
  "n_events": 3,
  "residual_rms_min": 61.848856830139425,
  "note": "unweighted representative events, conditional on original search cycle counts; no uncertainty inferred"
}
```


Fresh source and independent reduction

Sectors11/99 Gaussian deficit centers are 15.09/14.20 arcsec from the target, but 0.56/1.64 arcsec from TIC411647413, separated14.76 arcsec. Fixed neighbour/target SSE ratios are0.362/0.144, consistently favoring that neighbour. A short-period contaminating EB is strongly inferred; its precision ephemeris and novelty were not reviewed.


```
[
  {
    "sector": 11,
    "target_offset_arcsec": 15.09238254714884,
    "min_neighbour_target_sse_ratio": 0.36214401303402544,
    "capable_neighbour_count": 18
  },
  {
    "sector": 99,
    "target_offset_arcsec": 14.197941851487595,
```

```
    "min_neighbour_target_sse_ratio": 0.1436331918871176,
    "capable_neighbour_count": 25
  }
]
```


Custom TESScut reduction is independent of QLP processing but uses the same photons; it is not SPOC or an independent telescope. Signed in/out images use adjacent pre/post baselines. Gaussian formal errors exclude PRF systematics; fixed-coordinate source comparisons and repeat source positions govern attribution. Fresh downloads, TIC cones, signed arrays, plots and source scripts are preserved in raw-evidence.

Gaia geometry

The original conditional 6.394515949 d regression aliased unrelated eclipse dips. It is rejected as an eclipse-derived target period; no Gaia-conjunction association is established.

The computed gaia-geometry.json uses edge-on conjunction approximation, BTJD conversion Tp+389, and full SB1 covariance ordering P,gamma,K1,e,omega,Tp. Linear uncertainty is approximate, especially with weak eccentricity. Mass-function/main-sequence radius estimates test duration feasibility only; there is no fitted physical model or automatic sigma rejection.

Catalogue status

no identified prior EB; all seven central scopes complete: yes. Central shared-catalogues coverage includes VSX, SIMBAD primary type, GaiaDR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4 and ASAS-SN. SB* or ROT alone is not prior EB evidence. This is scoped novelty coverage, not an exhaustive discovery claim.

Completion and limitations

Full representative review complete. Scientific ambiguity: The contaminating eclipses are real; REFUTED applies to target attribution and Gaia-period interpretation. TIC411647413 is an inferred by-catch host subject to pixel model systematics, not a new confirmed catalogue object.

Next test: Higher resolution event photometry can establish the contaminant host and measure its orbital period precisely.

Provenance: cache-manifest.json gives original cached FITS paths and SHA256. Fresh TESScut URLs/hashes are in tesscut-downloads.json. Helper raw evidence and retained scripts permit reproduction. manifest.json hashes deliverables. No source paper, prior adjudication or publication was changed.

C47. TIC 450217907 full report

Archived source: sanitized report text.

TIC 450217907: REFUTED

Queue position 30; batch03. Anonymous project coordinator review with an independent measurement helper.

Deciding assessment

Three sector folds show smooth broad double-wave modulation, without a convincing discrete narrow eclipse. The original screen is refuted as an eclipsing interpretation of this measured signal.

Raw flux and period

Cached QLP SAP_FLUX, QUALITY=0; 3 representative sectors. Local quadratic baseline plus trapezoid fits retained 5 trial events. Only repeated discrete eclipse morphology supports an eclipse period. Search cycle counts were initial constraints, not independent period discovery. Lead inspected raw-event-panels.png and ambiguous-folds.png; smooth-variable fits hitting duration bounds are not eclipses.

Accepted event-derived period: none for the target d.

Initial measurement receipt (superseded where rejected above):

```
null
```

Fresh source and independent reduction

Fresh cutouts were attempted; no established eclipse deficit exists to localize. Source attribution of the smooth modulation is not claimed.

```
[]
```

Custom TESScut reduction is independent of QLP processing but uses the same photons; it is not SPOC or an independent telescope. Signed in/out images use adjacent pre/post baselines. Gaussian formal errors exclude PRF systematics; fixed-coordinate source comparisons and repeat source positions govern attribution. Fresh downloads, TIC cones, signed arrays, plots and source scripts are preserved in raw-evidence.

Gaia geometry

The searched modulation period is not an eclipse-derived period. No conjunction comparison is claimed without established eclipses.

The computed gaia-geometry.json uses edge-on conjunction approximation, BTJD conversion Tp+389, and full SB1 covariance ordering P,gamma,K1,e,omega,Tp. Linear uncertainty is approximate, especially with weak eccentricity. Mass-function/main-sequence radius estimates test duration feasibility only; there is no fitted physical model or automatic sigma rejection.

Catalogue status

no identified prior EB; all seven central scopes complete: yes. Central shared-catalogues coverage includes VSX, SIMBAD primary type, GaiaDR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4 and ASAS-SN. SB* or ROT alone is not prior EB evidence. This is scoped novelty coverage, not an exhaustive discovery claim.

Completion and limitations

Full representative review complete. Scientific ambiguity: Smooth modulation may be ellipsoidal or other stellar variability; shallow additional eclipses are not excluded. A local trapezoid fit to a broad trend does not establish an eclipse.

Next test: Higher precision light curves could test for a narrow residual eclipse if further work on this smooth variable is desired.

Provenance: cache-manifest.json gives original cached FITS paths and SHA256. Fresh TESScut URLs/hashes are in tesscut-downloads.json. Helper raw evidence and retained scripts permit reproduction. manifest.json hashes deliverables. No source paper, prior adjudication or publication was changed.

C48. TIC 60443793 full report

Archived source: sanitized report text.

TIC 60443793: REFUTED

Queue position 31; batch04. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

The original 6.510555 d search alias is wrong for the prominent eclipse train. Forty-eight raw minima give 1.29738064 d spacing; alternating depths imply a candidate 2.5947613 d binary period. Fresh imaging places these real EB photons off target, so the target-EB association is refuted. This is neither a confirmed target nor a target by-catch EB.

Raw events and eclipse period

Two or three cached QLP SAP_FLUX sectors were checked with QUALITY=0. Representative events use local quadratic baselines plus trapezoid fits; TIC 60443793 instead uses the independently corrected alternating-minimum series, retained alongside the original search alias receipt. 48 representative fits were retained in raw-events.json, 48 clear of fit bounds. Depth range 3.462–21.691 percent. Period from representative event centers: 2.594761285455248. Search-cycle counts are conditional on the initial search except where the corrected alternating-minimum receipt explicitly supersedes that alias. No timing uncertainty or ephemeris prediction is claimed.

{ "minimum_spacing_d": 1.297380642727624, "candidate_binary_period_d": 2.594761285455248, "epoch_btjd": 1658.3489391171465, "residual_rms_min": 11.60045026187536, "n_events": 48, "note": "Cycle count conditional on approximate 1.297 d spacing; alternating depths suggest double spacing as binary period. No uncertainty or secure alias test." }

Source imaging and independent reduction

S 13/S 93 signed deficits are off target. Free-Gaussian offsets about 58 arcsec align with capable TIC 60443695 at 59.38 arcsec separation. Fixed-coordinate neighbour/target SSE 0.0032 and 0.0229 strongly favor that neighbour in two sectors; positive-clipped centroids alone give 44.7/50.2 arcsec. Scores are descriptive, with PRF/systematic caveats. Candidate neighbour identity is an inference, not a full adjudication of that star.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. No claim of a separate SPOC pipeline is made. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

Gaia orbit P=6.51870479 d. Representative photometric period 2.59476129 d; difference / Gaia quoted period error -1927.55. Closest edge-on conjunction A: offset -1.217 d, covariance-linear error 0.569 d. Observed duration median 4.67 h; approximate central duration 243.51 h. These are feasibility diagnostics, not significance vetoes or a fitted geometry.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Signal class: off-target eclipsing binary contamination; Gaia association: photometric train does not trace target Gaia orbit.

Execution: completed representative full review; scientific ambiguity: Neighbour EB period remains a candidate and has not undergone independent alias/geometry review as its own target. The target giant-radius estimate is also incompatible with interpreting this short period as its simple detached binary orbit.

Next deciding check: Review TIC 60443695 separately if desired.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all28 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C49. TIC 71866523 full report

Archived source: sanitized report text.

TIC 71866523: PLAUSIBLE

Queue position 32; batch04. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

Sectors 6/33 are nearly flat with a weak sector 33 depression; sector 87 is dominated by a large positive transient/systematic. No secure repeating eclipse sequence or eclipse-derived period is established.

Raw events and eclipse period

Two or three cached QLP SAP_FLUX sectors were checked with QUALITY=0. Representative events use local quadratic baselines plus trapezoid fits; TIC 60443793 instead uses the independently corrected alternating-minimum series, retained alongside the original search alias receipt. 2 representative fits were retained in raw-events.json, 1 clear of fit bounds. Depth range 0.099–0.099 percent. Period from representative event centers: not established as an eclipse period. Search-cycle counts are conditional on the initial search except where the corrected alternating-minimum receipt explicitly supersedes that alias. No timing uncertainty or ephemeris prediction is claimed.

Smooth-variable morphology is assessed in ambiguous-folds.json and helper receipts.

Source imaging and independent reduction

Fresh sectors 6/87 custom extraction attempted the independent gate; no sufficient isolated eclipse exists for a meaningful signed source image.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. No claim of a separate SPOC pipeline is made. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

Gaia orbit P=15.39210899 d. No secure eclipse period exists for a meaningful conjunction/duration validation.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Signal class: uncertain variability; Gaia association: not established.

Execution: completed representative full review; scientific ambiguity: Weak single feature and transient/baseline ambiguity; available work was attempted.

Next deciding check: Require a clean repeated negative event before EB classification.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all28 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C50. TIC 443106429 full report

Archived source: sanitized report text.

TIC 443106429: PROBABLE

Queue position 33; batch04. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

Four narrow raw eclipses in sectors 11 and 65 give 12.7955335 d, consistent with Gaia. Raw depths 1.82?2.84%, duration about 10.2 h; custom extraction recovers four at 1.84?3.33%. Eclipses are real and repeat, but the close host remains unresolved.

Raw events and eclipse period

Two or three cached QLP SAP_FLUX sectors were checked with QUALITY=0. Representative events use local quadratic baselines plus trapezoid fits; TIC 60443793 instead uses the independently corrected alternating-minimum series, retained alongside the original search alias receipt. 4 representative fits were retained in raw-events.json, 4 clear of fit bounds. Depth range 1.825–2.841 percent. Period from representative event centers: 12.795533464914396. Search-cycle counts are conditional on the initial search except where the corrected alternating-minimum receipt explicitly supersedes that alias. No timing uncertainty or ephemeris prediction is claimed.

{ "period_d": 12.795533464914396, "epoch_btjd": 1606.1515052693348, "n_events": 3, "residual_rms_min": 0.2322850788714703, "note": "unweighted representative events, conditional on original search cycle counts; no uncertainty inferred" }

Source imaging and independent reduction

Sector 11 deficit lies 0.59 arcsec from target, with capable TIC 443106423 only 3.88 arcsec away; fixed-neighbour/target SSE 1.83 favors target. Extra fresh sector 65 was obtained; noisy one-event signed deficit has 13.4 arcsec fitted offset and close-neighbour/target SSE 0.99, failing independent separation. Source testing was attempted and remains scientifically ambiguous.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. No claim of a separate SPOC pipeline is made. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

Gaia orbit P=12.79551859 d. Representative photometric period 12.79553346 d; difference / Gaia quoted period error 0.00. Closest edge-on conjunction A: offset -0.230 d, covariance-linear error 0.565 d. Observed duration median 10.23 h; approximate central duration 22.45 h. These are feasibility

diagnostics, not significance vetoes or a fitted geometry.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Signal class: repeating eclipsing signal; likely target; Gaia association: period/conjunction feasible, host unresolved.

Execution: completed representative full review; scientific ambiguity: Host not robustly separated from 3.88 arcsec capable neighbour; this is the one unresolved protocol gate.

Next deciding check: Resolve target versusTIC 443106423 with higher spatial resolution eclipse photometry.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all28 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C51. TIC 311206604 full report

Archived source: sanitized report text.

TIC 311206604: REFUTED

Queue position 34; batch04. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

QLP sectors and fresh custom extraction show a broad, fairly flat-bottomed double wave, without distinct narrow repeated eclipses. Trapezoid fits absorb smooth curvature and frequently hit duration/center bounds. The narrow-eclipse interpretation is refuted; binarity or ellipsoidal variability is not excluded.

Raw events and eclipse period

Two or three cached QLP SAP_FLUX sectors were checked with QUALITY=0. Representative events use local quadratic baselines plus trapezoid fits; TIC 60443793 instead uses the independently corrected alternating-minimum series, retained alongside the original search alias receipt. 4 representative fits were retained in raw-events.json, 1 clear of fit bounds. Depth range 2.728–2.728 percent. Period from representative event centers: not established as an eclipse period. Search-cycle counts are conditional on the initial search except where the corrected alternating-minimum receipt explicitly supersedes that alias. No timing uncertainty or ephemeris prediction is claimed.

Smooth-variable morphology is assessed in ambiguous-folds.json and helper receipts.

Source imaging and independent reduction

Fresh first/last-sector cutouts were independently extracted. A meaningful narrow-event difference image cannot be defined from this smooth morphology; source gate was attempted and found no supported discrete eclipse to localize.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. No claim of a separate SPOC pipeline is made. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

Gaia orbit P=4.40200229 d. No secure eclipse period exists for a meaningful conjunction/duration validation.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Signal class: smooth variable; no narrow eclipses; Gaia association: no eclipse geometry validated.

Execution: completed representative full review; scientific ambiguity: Physical variability mechanism remains unclassified. Smooth modulation alone does not establish eclipses.

Next deciding check: No further eclipse follow-up required unless new discrete events appear.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all28 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C52. TIC 428769301 full report

Archived source: sanitized report text.

TIC 428769301: REFUTED

Queue position 35; batch04. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

QLP sectors and fresh custom extraction show a smooth double wave, without distinct narrow repeated eclipses. Trapezoid fits absorb smooth curvature and frequently hit duration/center bounds. The narrow-eclipse interpretation is refuted; binarity or ellipsoidal variability is not excluded.

Raw events and eclipse period

Two or three cached QLP SAP_FLUX sectors were checked with QUALITY=0. Representative events use local quadratic baselines plus trapezoid fits; TIC 60443793 instead uses the independently corrected alternating-minimum series, retained alongside the original search alias receipt. 5 representative fits were retained in raw-events.json, 2 clear of fit bounds. Depth range 0.298–0.336 percent. Period from representative event centers: not established as an eclipse period. Search-cycle counts are conditional on the initial search except where the corrected alternating-minimum receipt explicitly supersedes that alias. No timing uncertainty or ephemeris prediction is claimed.

Smooth-variable morphology is assessed in ambiguous-folds.json and helper receipts.

Source imaging and independent reduction

Fresh first/last-sector cutouts were independently extracted. A meaningful narrow-event difference image cannot be defined from this smooth morphology; source gate was attempted and found no supported discrete eclipse to localize.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. No claim of a separate SPOC pipeline is made. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

Gaia orbit P=3.73288066 d. No secure eclipse period exists for a meaningful conjunction/duration validation.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be

weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Signal class: smooth variable; no narrow eclipses; Gaia association: no eclipse geometry validated.

Execution: completed representative full review; scientific ambiguity: Physical variability mechanism remains unclassified. Smooth modulation alone does not establish eclipses.

Next deciding check: No further eclipse follow-up required unless new discrete events appear.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all28 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C53. TIC 436947190 full report

Archived source: sanitized report text.

TIC 436947190: REFUTED

Queue position 36; batch04. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

QLP sectors and fresh custom extraction show a broad smooth repeating wave, without distinct narrow repeated eclipses. Trapezoid fits absorb smooth curvature and frequently hit duration/center bounds. The narrow-eclipse interpretation is refuted; binarity or ellipsoidal variability is not excluded.

Raw events and eclipse period

Two or three cached QLP SAP_FLUX sectors were checked with QUALITY=0. Representative events use local quadratic baselines plus trapezoid fits; TIC 60443793 instead uses the independently corrected alternating-minimum series, retained alongside the original search alias receipt. 6 representative fits were retained in raw-events.json, 1 clear of fit bounds. Depth range 0.2–0.2 percent. Period from representative event centers: not established as an eclipse period. Search-cycle counts are conditional on the initial search except where the corrected alternating-minimum receipt explicitly supersedes that alias. No timing uncertainty or ephemeris prediction is claimed.

Smooth-variable morphology is assessed in ambiguous-folds.json and helper receipts.

Source imaging and independent reduction

Fresh first/last-sector cutouts were independently extracted. A meaningful narrow-event difference image cannot be defined from this smooth morphology; source gate was attempted and found no supported discrete eclipse to localize.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. No claim of a separate SPOC pipeline is made. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

Gaia orbit P=3.10818466 d. No secure eclipse period exists for a meaningful conjunction/duration validation.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be

weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Signal class: smooth variable; no narrow eclipses; Gaia association: no eclipse geometry validated.

Execution: completed representative full review; scientific ambiguity: Physical variability mechanism remains unclassified. Smooth modulation alone does not establish eclipses.

Next deciding check: No further eclipse follow-up required unless new discrete events appear.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all28 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C54. TIC 92452368 full report

Archived source: sanitized report text.

TIC 92452368: PLAUSIBLE

Queue position 37; batch04. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

Sector 1 contains a complex cluster of deep negative points around BTJD 1342.7?1344.1, rather than a clean repeating narrow eclipse. Sectors 68/95 do not establish repetition. A single episode with instrumental ambiguity remains.

Raw events and eclipse period

Two or three cached QLP SAP_FLUX sectors were checked with QUALITY=0. Representative events use local quadratic baselines plus trapezoid fits; TIC 60443793 instead uses the independently corrected alternating-minimum series, retained alongside the original search alias receipt. 0 representative fits were retained in raw-events.json, 0 clear of fit bounds. No detached event established. Period from representative event centers: not established as an eclipse period. Search-cycle counts are conditional on the initial search except where the corrected alternating-minimum receipt explicitly supersedes that alias. No timing uncertainty or ephemeris prediction is claimed.

Smooth-variable morphology is assessed in ambiguous-folds.json and helper receipts.

Source imaging and independent reduction

Fresh sectors 1/95 independently extracted. The irregular clustered episode does not provide defensible in/out windows for a repeat source image; the source gate was attempted but insufficient.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. No claim of a separate SPOC pipeline is made. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

Gaia orbit P=18.58152247 d. No secure eclipse period exists for a meaningful conjunction/duration validation.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Signal class: uncertain variability; Gaia association: not established.

Execution: completed representative full review; scientific ambiguity: Single complex episode, untested eclipse period and instrumental ambiguity.

Next deciding check: Verify a clean recurrent event in independently reduced data.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all28 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C55. TIC 94300366 full report

Archived source: sanitized report text.

TIC 94300366: PLAUSIBLE

Queue position 38; batch04. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

Two sector 54 depressions recur near the search phase and survive custom extraction at 4.24/4.76%, but are asymmetric and mixed with stronger irregular variability and other minima. Sector 81 does not establish a stable narrow eclipse sequence. Eclipse morphology and period remain untested.

Raw events and eclipse period

Two or three cached QLP SAP_FLUX sectors were checked with QUALITY=0. Representative events use local quadratic baselines plus trapezoid fits; TIC 60443793 instead uses the independently corrected alternating-minimum series, retained alongside the original search alias receipt. 4 representative fits were retained in raw-events.json, 2 clear of fit bounds. Depth range 4.198–7.203 percent. Period from representative event centers: not established as an eclipse period. Search-cycle counts are conditional on the initial search except where the corrected alternating-minimum receipt explicitly supersedes that alias. No timing uncertainty or ephemeris prediction is claimed.

Smooth-variable morphology is assessed in ambiguous-folds.json and helper receipts.

Source imaging and independent reduction

S 54 one-event deficit favors target with 4.59 arcsec free-Gaussian offset, but morphology and the noisy later-sector events prevent a repeated source validation. Fresh independent extraction does not turn the irregular dips into a secure EB.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. No claim of a separate SPOC pipeline is made. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

Gaia orbit P=3.90447017 d. No secure eclipse period exists for a meaningful conjunction/duration validation.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be

weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Signal class: uncertain variability; Gaia association: not established.

Execution: completed representative full review; scientific ambiguity: Irregular/spot-like variability versus eclipses unresolved; no secure eclipse period.

Next deciding check: Obtain detrending resistant multiseason photometry to establish stable discrete repeats.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all28 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C56. TIC 4646090 full report

Archived source: sanitized report text.

TIC 4646090: CONFIRMED

Queue position 39; batch04. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

Three narrow, locally flat-baseline eclipses in sectors 4 and 31 give 10.7303766 d. Raw depths 1.95?2.26% and durations 2.75?3.46 h recur; the distinct custom extraction recovers all three at 0.59?0.64% aperture depth. Dilution differs between reductions. Source and independent gates pass; Gaia period and phase are feasible at the precision available.

Raw events and eclipse period

Two or three cached QLP SAP_FLUX sectors were checked with QUALITY=0. Representative events use local quadratic baselines plus trapezoid fits; TIC 60443793 instead uses the independently corrected alternating-minimum series, retained alongside the original search alias receipt. 3 representative fits were retained in raw-events.json, 3 clear of fit bounds. Depth range 1.945–2.265 percent. Period from representative event centers: 10.73037662305058. Search-cycle counts are conditional on the initial search except where the corrected alternating-minimum receipt explicitly supersedes that alias. No timing uncertainty or ephemeris prediction is claimed.

{ "period_d": 10.73037662305058, "epoch_btjd": 1419.05639053136, "n_events": 3, "residual_rms_min": 1.164581672824554, "note": "unweighted representative events, conditional on original search cycle counts; no uncertainty inferred" }

Source imaging and independent reduction

Signed deficits in both sectors prefer the target. Free-Gaussian offsets are 0.46 and 2.28 arcsec, versus a capable TIC 4646088 neighbour 6.96 arcsec away (7.53 and 9.32 arcsec from fitted deficits). Fixed-neighbour / target SSE is 5.34 and 5.65 in sector stacks; per-event ratios 5.34,3.57,4.27 also favor target. These descriptive comparisons, repeated across sectors/events, support attribution; no calibrated PRF confidence is asserted.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. No claim of a separate SPOC pipeline is made. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

Gaia orbit P=10.72678235 d. Representative photometric period 10.73037662 d; difference / Gaia quoted period error 1.05. Closest edge-on conjunction A: offset 0.612 d, covariance-linear error 0.457 d. Observed duration median 3.22 h; approximate central duration 8.06 h. These are feasibility diagnostics, not significance vetoes or a fitted geometry.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Signal class: target eclipsing binary; Gaia association: consistent within approximate orbit/geometry checks.

Execution: completed representative full review; scientific ambiguity: Gaia phase offset is 1.34 covariance-linear error units; grazing geometry can shorten the approximate central duration. No detailed physical fit or calibrated astrometric uncertainty.

Next deciding check: Characterize dilution and grazing geometry if needed.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all28 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C57. TIC 402940895 full report

Archived source: sanitized report text.

TIC 402940895: PLAUSIBLE

Queue position 40; batch04. Review by GPT-6.1 Sol with a fresh GPT-6.1 Terra helper, supervised by Astra. Physical classification and prior-catalogue status are separate.

Deciding assessment

Small sharp depressions lie on strong discontinuous sector 12/93 trends, while sector 103 flux is severely corrupted. Custom sector 12 fixed-shape depths 0.53/0.76% show features in the same photons, but their morphology is baseline sensitive and a stable eclipse sequence is unestablished.

Raw events and eclipse period

Two or three cached QLP SAP_FLUX sectors were checked with QUALITY=0. Representative events use local quadratic baselines plus trapezoid fits; TIC 60443793 instead uses the independently corrected alternating-minimum series, retained alongside the original search alias receipt. 5 representative fits were retained in raw-events.json, 5 clear of fit bounds. Depth range 0.0–0.674 percent. Period from representative event centers: not established as an eclipse period. Search-cycle counts are conditional on the initial search except where the corrected alternating-minimum receipt explicitly supersedes that alias. No timing uncertainty or ephemeris prediction is claimed.

Smooth-variable morphology is assessed in ambiguous-folds.json and helper receipts.

Source imaging and independent reduction

Sector 12 signed diagnostic has 2.79 arcsec Gaussian target offset, but strong baseline structure and lack of a clean independent later-sector event prevent secure source validation. Fresh sectors 12/103 were extracted.

Fresh TESScut cutouts are a distinct custom aperture reduction of the same TESS photons as QLP; they are not SPOC and do not supply an independent telescope. No claim of a separate SPOC pipeline is made. Signed in/out images use adjacent pre/post windows. Positive-clipped centroids and Gaussian formal errors are diagnostics subject to PRF mismatch; target-specific repeat tests and capable-neighbour comparisons govern attribution.

Gaia geometry

Gaia orbit P=16.44813821 d. No secure eclipse period exists for a meaningful conjunction/duration validation.

Computed receipt: gaia-geometry.json. Linear covariance propagation uses SB1 correlation order P,gamma,K1,e,omega,Tp; conjunction uses edge-on approximation. The mass-function/MS-radius duration is a feasibility estimate, not a physical fit. Circular/low-eccentricity periastron phases may be

weakly constrained.

Prior EB coverage

no identified prior EB; seven-scope coverage complete: yes. Reused central coverage-matrix.csv and catalogue-scope.json: VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* or ROT alone are not EB identifications. This scoped result is not an exhaustive discovery claim.

Completion, limitations and next check

Signal class: uncertain variability; Gaia association: not established.

Execution: completed representative full review; scientific ambiguity: Baseline/discontinuity sensitivity, corrupted sector 103, no secure eclipse period or repeated source validation.

Next deciding check: Re-extract clean sectors with robust background/pointing treatment before asserting eclipses.

Provenance: cache-manifest.json preserves original cached FITS paths and SHA256; identity-checks.json matches TICID/SECTOR for all28 input QLP files. tesscut-downloads.json records fresh source URLs and zip hashes; signed images and helper evidence have hashes. manifest.json records deliverable checksums. Adapted scripts are retained. No source-paper edits or publication performed.

C58. TIC 183994341 full report

Archived source: sanitized report text.

TIC 183994341: CONFIRMED

Queue position 41; batch05. Automated GPT-6.1 Sol lead and independent fresh GPT-6.1 Terra review, supervised by Astra. The anonymous project coordinator supplied oversight; these are computational measurements, not human measurements.

Assessment

Four repeated primary eclipses with raw depths25–29percent are on target and recovered independently. The eclipse period is approximately14.6655d; the ASAS-SN catalogue lists an incompatible2.9324d alias. The target is a known prior EB and excluded from new additions.

Raw repetition and period

Accepted target eclipse period: 14.665514216226159 days. 2 representative cached QLP sectors were reviewed using SAP_FLUX and QUALITY=0. Original packet periods were treated as candidate aliases. Corrected event fits use local linear baselines and trapezoids on raw cadences; no exhaustive sector fit or physical binary model was required.

{ "period_d": 14.665514216226159, "epoch_btjd": 3781.92483432609, "n_events": 4, "note": "Representative local event timing regression for939/183; direct consecutive event interval for431/222; no formal period error." }

Representative accepted event measurements:


```
[
  {
    "sector": 91,
    "tc": 3781.924758087458,
    "depth": 0.27811754249345844,
    "duration_d": 0.6709603998224439,
    "ingress_fraction": 0.4102676391311878,
    "residual_rms": 0.008400452725881942,
    "edge": false,
    "n_points": 393,
    "cycle": 0
  },
  {
    "sector": 91,
    "tc": 3796.5904251899988,
    "depth": 0.28747538323348454,
    "duration_d": 0.7030939389529461,
    "ingress_fraction": 0.4222486811501103,
    "residual_rms": 0.008485133785920739,
    "edge": false,
    "n_points": 366,
    "cycle": 1
  },
  {
```

```
    "sector": 92,
    "tc": 3811.255938179075,
    "depth": 0.24829805939567123,
    "duration_d": 0.6719697358685144,
    "ingress_fraction": 0.4078376270792781,
    "residual_rms": 0.00781992030528953,
    "edge": false,
    "n_points": 519,
    "cycle": 2
  },
  {
    "sector": 92,
    "tc": 3825.9213011451875,
    "depth": 0.25572653021110614,
    "duration_d": 0.6766583893637675,
    "ingress_fraction": 0.4180416685853718,
    "residual_rms": 0.007580250708817091,
    "edge": false,
    "n_points": 519,
    "cycle": 3
  }
]
```


Source and independent reduction

Sectors91/92 free Gaussian offsets1.71/1.58arcsec favor the target; closest capable neighbours38arcsec or farther give substantially worse fixed-source residuals (ratios22.8–44.1). Independent custom apertures reproduce repeated eclipses.

Fresh 11x11 TESScuts and 60 arcsec TIC source cones were obtained. Signed in/out images compare local pre/post baselines. The source model includes signed pixel-integrated Gaussians and fixed source positions with free widths/amplitude/background. Residual ratios are descriptive and do not constitute calibrated PRF probabilities. Neighbours capable under a conservative brightness/depth limit were compared; throughput, uncatalogued sources and TIC photometry remain limitations.

The independent custom target aperture subtracts a per-cadence background from faint cutout pixels and uses the same TESS photons as QLP. It is a distinct extraction, not SPOC or a second telescope. Terra independently inspected raw/custom comparison and signed images; quantitative four-EB depths and receipts are in raw-evidence/terra-helper-results.json. See HELPER.md, all-raw.png, raw-event-panels.png and terra-independent-comparison.png.

Gaia geometry and association

Four-event regression P14.665514216d versus Gaia14.670950076+/-0.003882078d. Approximate conjunction offset1.81d with propagated covariance0.93d and conservative absolute-sum1.76d; duration16.18h versus approximate central23.03h is physically feasible. The period, propagated approximate phase and duration are compatible with the Gaia orbit at current precision; the roughly1.9sigma phase difference alone is not a physical veto.

Association flag: compatible at approximate current precision; phase uncertainty retained.

The gaia-geometry.json calculation uses edge-on conjunctions, Tp+389 BTJD conversion, full SB1 covariance in P,gamma,K1,e,omega,Tp order, and linear propagation. Approximate mass function, TIC radius and main-sequence companion radius test feasibility, not a fitted geometry. No automatic Gaia sigma veto was applied.

Catalogue coverage

known prior EB; seven-scope coverage complete: yes. Central coverage includes VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, vetted Kostov2025 tables3/4 and ASAS-SN. SB* and ROT alone do not count as known EB. This scoped result is separate from physical assessment. TIC183994341 is known and excluded from new additions regardless of physical tier.

Completion, limits and next check

Full representative review complete; remaining full review=false. Available deciding checks were attempted. Scientific uncertainty: Real repeated on-target known EB. Approximate Gaia conjunction phase and radii limit characterization. Catalogue period conflict is retained rather than forcing catalogue cycles. No new-discovery claim.

Next check: Resolve ASAS-SN period alias and obtain RV/orbital posterior or a joint timing/geometry analysis to settle the Gaia association.

Provenance: raw-evidence/cache-manifest.json hashes the cached original QLP files; identity-checks.json verifies TICID/sector. Fresh TESScut URLs/SHA256 are in tesscut-downloads.json; source cones and adapted scripts are retained. manifest.json hashes batch deliverables. No paper edits, source table edits, publication or commits were made.

C59. TIC 166781582 full report

Archived source: sanitized report text.

TIC 166781582: PLAUSIBLE

Queue position 42; batch05. Automated GPT-6.1 Sol lead and independent fresh GPT-6.1 Terra review, supervised by Astra. The anonymous project coordinator supplied oversight; these are computational measurements, not human measurements.

Assessment

Broad smooth multi-day minima vary in shape and amplitude between sectors15/75/83. The long screen interval does not isolate discrete ingress/egress. The original candidate remains scientifically unestablished after the available representative checks.

Raw repetition and period

Accepted target eclipse period: not established days. 3 representative cached QLP sectors were reviewed using SAP_FLUX and QUALITY=0. Original packet periods were treated as candidate aliases. Corrected event fits use local linear baselines and trapezoids on raw cadences; no exhaustive sector fit or physical binary model was required.

{ "accepted_target_ephemeris": null, "reason": "No established target repeated eclipse; original alias fitter is not accepted evidence" }

Representative accepted event measurements:

[]

Source and independent reduction

Fresh TESScuts and target-centred custom apertures were inspected. Without a verified eclipse, a signed image at an arbitrary trial minimum cannot establish an eclipse host.

Fresh 11x11 TESScuts and 60 arcsec TIC source cones were obtained. Signed in/out images compare local pre/post baselines. The source model includes signed pixel-integrated Gaussians and fixed source positions with free widths/amplitude/background. Residual ratios are descriptive and do not constitute calibrated PRF probabilities. Neighbours capable under a conservative brightness/depth limit were compared; throughput, uncatalogued sources and TIC photometry remain limitations.

The independent custom target aperture subtracts a per-cadence background from faint cutout pixels and uses the same TESS photons as QLP. It is a distinct extraction, not SPOC or a second telescope. Terra independently inspected raw/custom comparison and signed images; quantitative four-EB depths and receipts are in raw-evidence/terra-helper-results.json. See HELPER.md, all-raw.png, raw-event-panels.png and terra-independent-comparison.png.

Gaia geometry and association

No eclipse-derived orbital period is established, so a Gaia-conjunction match cannot be claimed.

Association flag: unestablished.

The gaia-geometry.json calculation uses edge-on conjunctions, Tp+389 BTJD conversion, full SB1 covariance in P,gamma,K1,e,omega,Tp order, and linear propagation. Approximate mass function, TIC radius and main-sequence companion radius test feasibility, not a fitted geometry. No automatic Gaia sigma veto was applied.

Catalogue coverage

no identified prior EB; seven-scope coverage complete: yes. Central coverage includes VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, vetted Kostov2025 tables3/4 and ASAS-SN. SB* and ROT alone do not count as known EB. This scoped result is separate from physical assessment. TIC183994341 is known and excluded from new additions regardless of physical tier.

Completion, limits and next check

Full representative review complete; remaining full review=false. Available deciding checks were attempted. Scientific uncertainty: Full representative review completed; the candidate lacks established repeated eclipses. This does not exclude shallower eclipses or events outside the inspected sectors.

Next check: Obtain or inspect another clean sector and require repeated discrete ingress/egress before fitting a binary period.

Provenance: raw-evidence/cache-manifest.json hashes the cached original QLP files; identity-checks.json verifies TICID/sector. Fresh TESScut URLs/SHA256 are in tesscut-downloads.json; source cones and adapted scripts are retained. manifest.json hashes batch deliverables. No paper edits, source table edits, publication or commits were made.

C60. TIC 230270717 full report

Archived source: sanitized report text.

TIC 230270717: PLAUSIBLE

Queue position 43; batch05. Automated GPT-6.1 Sol lead and independent fresh GPT-6.1 Terra review, supervised by Astra. The anonymous project coordinator supplied oversight; these are computational measurements, not human measurements.

Assessment

Sectors26/53/80 contain broad changing trends and sharp upward structure, with no established repeated discrete eclipse. The original candidate remains scientifically unestablished after the available representative checks.

Raw repetition and period

Accepted target eclipse period: not established days. 3 representative cached QLP sectors were reviewed using SAP_FLUX and QUALITY=0. Original packet periods were treated as candidate aliases. Corrected event fits use local linear baselines and trapezoids on raw cadences; no exhaustive sector fit or physical binary model was required.

{ "accepted_target_ephemeris": null, "reason": "No established target repeated eclipse; original alias fitter is not accepted evidence" }

Representative accepted event measurements:

[]

Source and independent reduction

Fresh TESScuts and target-centred custom apertures were inspected. Without a verified eclipse, a signed image at an arbitrary trial minimum cannot establish an eclipse host.

Fresh 11x11 TESScuts and 60 arcsec TIC source cones were obtained. Signed in/out images compare local pre/post baselines. The source model includes signed pixel-integrated Gaussians and fixed source positions with free widths/amplitude/background. Residual ratios are descriptive and do not constitute calibrated PRF probabilities. Neighbours capable under a conservative brightness/depth limit were compared; throughput, uncatalogued sources and TIC photometry remain limitations.

The independent custom target aperture subtracts a per-cadence background from faint cutout pixels and uses the same TESS photons as QLP. It is a distinct extraction, not SPOC or a second telescope. Terra independently inspected raw/custom comparison and signed images; quantitative four-EB depths and receipts are in raw-evidence/terra-helper-results.json. See HELPER.md, all-raw.png, raw-event-panels.png and terra-independent-comparison.png.

Gaia geometry and association

No eclipse-derived orbital period is established, so a Gaia-conjunction match cannot be claimed.

Association flag: unestablished.

The gaia-geometry.json calculation uses edge-on conjunctions, Tp+389 BTJD conversion, full SB1 covariance in P,gamma,K1,e,omega,Tp order, and linear propagation. Approximate mass function, TIC radius and main-sequence companion radius test feasibility, not a fitted geometry. No automatic Gaia sigma veto was applied.

Catalogue coverage

no identified prior EB; seven-scope coverage complete: yes. Central coverage includes VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, vetted Kostov2025 tables3/4 and ASAS-SN. SB* and ROT alone do not count as known EB. This scoped result is separate from physical assessment. TIC183994341 is known and excluded from new additions regardless of physical tier.

Completion, limits and next check

Full representative review complete; remaining full review=false. Available deciding checks were attempted. Scientific uncertainty: Full representative review completed; the candidate lacks established repeated eclipses. This does not exclude shallower eclipses or events outside the inspected sectors.

Next check: Obtain or inspect another clean sector and require repeated discrete ingress/egress before fitting a binary period.

Provenance: raw-evidence/cache-manifest.json hashes the cached original QLP files; identity-checks.json verifies TICID/sector. Fresh TESScut URLs/SHA256 are in tesscut-downloads.json; source cones and adapted scripts are retained. manifest.json hashes batch deliverables. No paper edits, source table edits, publication or commits were made.

C61. TIC 462845564 full report

Archived source: sanitized report text.

TIC 462845564: PLAUSIBLE

Queue position 44; batch05. Automated GPT-6.1 Sol lead and independent fresh GPT-6.1 Terra review, supervised by Astra. The anonymous project coordinator supplied oversight; these are computational measurements, not human measurements.

Assessment

The two available sectors26/40 are dominated by long monotonic flux segments and downlink resets; no repeated isolated eclipse is measured. The original candidate remains scientifically unestablished after the available representative checks.

Raw repetition and period

Accepted target eclipse period: not established days. 2 representative cached QLP sectors were reviewed using SAP_FLUX and QUALITY=0. Original packet periods were treated as candidate aliases. Corrected event fits use local linear baselines and trapezoids on raw cadences; no exhaustive sector fit or physical binary model was required.

{ "accepted_target_ephemeris": null, "reason": "No established target repeated eclipse; original alias fitter is not accepted evidence" }

Representative accepted event measurements:

[]

Source and independent reduction

Fresh TESScuts and target-centred custom apertures were inspected. Without a verified eclipse, a signed image at an arbitrary trial minimum cannot establish an eclipse host.

Fresh 11x11 TESScuts and 60 arcsec TIC source cones were obtained. Signed in/out images compare local pre/post baselines. The source model includes signed pixel-integrated Gaussians and fixed source positions with free widths/amplitude/background. Residual ratios are descriptive and do not constitute calibrated PRF probabilities. Neighbours capable under a conservative brightness/depth limit were compared; throughput, uncatalogued sources and TIC photometry remain limitations.

The independent custom target aperture subtracts a per-cadence background from faint cutout pixels and uses the same TESS photons as QLP. It is a distinct extraction, not SPOC or a second telescope. Terra independently inspected raw/custom comparison and signed images; quantitative four-EB depths and receipts are in raw-evidence/terra-helper-results.json. See HELPER.md, all-raw.png, raw-event-panels.png and terra-independent-comparison.png.

Gaia geometry and association

No eclipse-derived orbital period is established, so a Gaia-conjunction match cannot be claimed.

Association flag: unestablished.

The gaia-geometry.json calculation uses edge-on conjunctions, Tp+389 BTJD conversion, full SB1 covariance in P,gamma,K1,e,omega,Tp order, and linear propagation. Approximate mass function, TIC radius and main-sequence companion radius test feasibility, not a fitted geometry. No automatic Gaia sigma veto was applied.

Catalogue coverage

no identified prior EB; seven-scope coverage complete: yes. Central coverage includes VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, vetted Kostov2025 tables3/4 and ASAS-SN. SB* and ROT alone do not count as known EB. This scoped result is separate from physical assessment. TIC183994341 is known and excluded from new additions regardless of physical tier.

Completion, limits and next check

Full representative review complete; remaining full review=false. Available deciding checks were attempted. Scientific uncertainty: Full representative review completed; the candidate lacks established repeated eclipses. This does not exclude shallower eclipses or events outside the inspected sectors.

Next check: Obtain or inspect another clean sector and require repeated discrete ingress/egress before fitting a binary period.

Provenance: raw-evidence/cache-manifest.json hashes the cached original QLP files; identity-checks.json verifies TICID/sector. Fresh TESScut URLs/SHA256 are in tesscut-downloads.json; source cones and adapted scripts are retained. manifest.json hashes batch deliverables. No paper edits, source table edits, publication or commits were made.

C62. TIC 431762229 full report

Archived source: sanitized report text.

TIC 431762229: REFUTED

Queue position 45; batch05. Automated GPT-6.1 Sol lead and independent fresh GPT-6.1 Terra review, supervised by Astra. The anonymous project coordinator supplied oversight; these are computational measurements, not human measurements.

Assessment

Repeated real eclipses contaminate the target aperture, but are spatially associated with TIC431762222, separated32.79arcsec. REFUTED applies to target attribution and the original Gaia-period eclipse interpretation.

Raw repetition and period

Accepted target eclipse period: not established days. 3 representative cached QLP sectors were reviewed using SAP_FLUX and QUALITY=0. Original packet periods were treated as candidate aliases. Corrected event fits use local linear baselines and trapezoids on raw cadences; no exhaustive sector fit or physical binary model was required.

{ "accepted_target_ephemeris": null, "reason": "No established target repeated eclipse; original alias fitter is not accepted evidence" }

Representative accepted event measurements:

[]

Source and independent reduction

Two fresh detector-sector difference stacks repeatedly favor TIC431762222; fixed neighbour/target residual SSE ratios are0.0090/0.0188. Free Gaussian deficits lie close to that neighbour. The ratios are descriptive, not calibrated probabilities.

Fresh 11x11 TESScuts and 60 arcsec TIC source cones were obtained. Signed in/out images compare local pre/post baselines. The source model includes signed pixel-integrated Gaussians and fixed source positions with free widths/amplitude/background. Residual ratios are descriptive and do not constitute calibrated PRF probabilities. Neighbours capable under a conservative brightness/depth limit were compared; throughput, uncatalogued sources and TIC photometry remain limitations.

The independent custom target aperture subtracts a per-cadence background from faint cutout pixels and uses the same TESS photons as QLP. It is a distinct extraction, not SPOC or a second telescope. Terra independently inspected raw/custom comparison and signed images; quantitative four-EB depths and receipts are in raw-evidence/terra-helper-results.json. See HELPER.md, all-raw.png, raw-event-panels.png and terra-independent-comparison.png.

Gaia geometry and association

The contaminating eclipse interval isapproximately1.94d; the original screen alias is rejected as a target eclipse period. No target Gaia orbit association is inferred.

Association flag: contaminating neighbour.

The gaia-geometry.json calculation uses edge-on conjunctions, Tp+389 BTJD conversion, full SB1 covariance in P,gamma,K1,e,omega,Tp order, and linear propagation. Approximate mass function, TIC radius and main-sequence companion radius test feasibility, not a fitted geometry. No automatic Gaia sigma veto was applied.

Catalogue coverage

no identified prior EB; seven-scope coverage complete: yes. Central coverage includes VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, vetted Kostov2025 tables3/4 and ASAS-SN. SB* and ROT alone do not count as known EB. This scoped result is separate from physical assessment. TIC183994341 is known and excluded from new additions regardless of physical tier.

Completion, limits and next check

Full representative review complete; remaining full review=false. Available deciding checks were attempted. Scientific uncertainty: The contaminant is an inferred EB host; its catalogue novelty and precision binary orbit were not independently adjudicated. Source model systematics remain.

Next check: Higher resolution eclipse photometry can verify the contaminating host.

Provenance: raw-evidence/cache-manifest.json hashes the cached original QLP files; identity-checks.json verifies TICID/sector. Fresh TESScut URLs/SHA256 are in tesscut-downloads.json; source cones and adapted scripts are retained. manifest.json hashes batch deliverables. No paper edits, source table edits, publication or commits were made.

C63. TIC 350661747 full report

Archived source: sanitized report text.

TIC 350661747: PLAUSIBLE

Queue position 46; batch05. Automated GPT-6.1 Sol lead and independent fresh GPT-6.1 Terra review, supervised by Astra. The anonymous project coordinator supplied oversight; these are computational measurements, not human measurements.

Assessment

QLP sectors12/39/103 show large irregular trends; the giant sector103 changes are largely absent in the independent aperture. The trial deficit is displaced and is not an established eclipse. The original candidate remains scientifically unestablished after the available representative checks.

Raw repetition and period

Accepted target eclipse period: not established days. 3 representative cached QLP sectors were reviewed using SAP_FLUX and QUALITY=0. Original packet periods were treated as candidate aliases. Corrected event fits use local linear baselines and trapezoids on raw cadences; no exhaustive sector fit or physical binary model was required.

{ "accepted_target_ephemeris": null, "reason": "No established target repeated eclipse; original alias fitter is not accepted evidence" }

Representative accepted event measurements:

[]

Source and independent reduction

Fresh TESScuts and target-centred custom apertures were inspected. Without a verified eclipse, a signed image at an arbitrary trial minimum cannot establish an eclipse host.

Fresh 11x11 TESScuts and 60 arcsec TIC source cones were obtained. Signed in/out images compare local pre/post baselines. The source model includes signed pixel-integrated Gaussians and fixed source positions with free widths/amplitude/background. Residual ratios are descriptive and do not constitute calibrated PRF probabilities. Neighbours capable under a conservative brightness/depth limit were compared; throughput, uncatalogued sources and TIC photometry remain limitations.

The independent custom target aperture subtracts a per-cadence background from faint cutout pixels and uses the same TESS photons as QLP. It is a distinct extraction, not SPOC or a second telescope. Terra independently inspected raw/custom comparison and signed images; quantitative four-EB depths and receipts are in raw-evidence/terra-helper-results.json. See HELPER.md, all-raw.png, raw-event-panels.png and terra-independent-comparison.png.

Gaia geometry and association

No eclipse-derived orbital period is established, so a Gaia-conjunction match cannot be claimed.

Association flag: unestablished.

The gaia-geometry.json calculation uses edge-on conjunctions, Tp+389 BTJD conversion, full SB1 covariance in P,gamma,K1,e,omega,Tp order, and linear propagation. Approximate mass function, TIC radius and main-sequence companion radius test feasibility, not a fitted geometry. No automatic Gaia sigma veto was applied.

Catalogue coverage

no identified prior EB; seven-scope coverage complete: yes. Central coverage includes VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, vetted Kostov2025 tables3/4 and ASAS-SN. SB* and ROT alone do not count as known EB. This scoped result is separate from physical assessment. TIC183994341 is known and excluded from new additions regardless of physical tier.

Completion, limits and next check

Full representative review complete; remaining full review=false. Available deciding checks were attempted. Scientific uncertainty: Full representative review completed; the candidate lacks established repeated eclipses. This does not exclude shallower eclipses or events outside the inspected sectors.

Next check: Obtain or inspect another clean sector and require repeated discrete ingress/egress before fitting a binary period.

Provenance: raw-evidence/cache-manifest.json hashes the cached original QLP files; identity-checks.json verifies TICID/sector. Fresh TESScut URLs/SHA256 are in tesscut-downloads.json; source cones and adapted scripts are retained. manifest.json hashes batch deliverables. No paper edits, source table edits, publication or commits were made.

C64. TIC 424565669 full report

Archived source: sanitized report text.

TIC 424565669: PLAUSIBLE

Queue position 47; batch05. Automated GPT-6.1 Sol lead and independent fresh GPT-6.1 Terra review, supervised by Astra. The anonymous project coordinator supplied oversight; these are computational measurements, not human measurements.

Assessment

Sector18 is dominated by positive flares and sector59/86 by weak fluctuations and gap changes. No repeated eclipse morphology is established. The original candidate remains scientifically unestablished after the available representative checks.

Raw repetition and period

Accepted target eclipse period: not established days. 3 representative cached QLP sectors were reviewed using SAP_FLUX and QUALITY=0. Original packet periods were treated as candidate aliases. Corrected event fits use local linear baselines and trapezoids on raw cadences; no exhaustive sector fit or physical binary model was required.

{ "accepted_target_ephemeris": null, "reason": "No established target repeated eclipse; original alias fitter is not accepted evidence" }

Representative accepted event measurements:

[]

Source and independent reduction

Fresh TESScuts and target-centred custom apertures were inspected. Without a verified eclipse, a signed image at an arbitrary trial minimum cannot establish an eclipse host.

Fresh 11x11 TESScuts and 60 arcsec TIC source cones were obtained. Signed in/out images compare local pre/post baselines. The source model includes signed pixel-integrated Gaussians and fixed source positions with free widths/amplitude/background. Residual ratios are descriptive and do not constitute calibrated PRF probabilities. Neighbours capable under a conservative brightness/depth limit were compared; throughput, uncatalogued sources and TIC photometry remain limitations.

The independent custom target aperture subtracts a per-cadence background from faint cutout pixels and uses the same TESS photons as QLP. It is a distinct extraction, not SPOC or a second telescope. Terra independently inspected raw/custom comparison and signed images; quantitative four-EB depths and receipts are in raw-evidence/terra-helper-results.json. See HELPER.md, all-raw.png, raw-event-panels.png and terra-independent-comparison.png.

Gaia geometry and association

No eclipse-derived orbital period is established, so a Gaia-conjunction match cannot be claimed.

Association flag: unestablished.

The gaia-geometry.json calculation uses edge-on conjunctions, Tp+389 BTJD conversion, full SB1 covariance in P,gamma,K1,e,omega,Tp order, and linear propagation. Approximate mass function, TIC radius and main-sequence companion radius test feasibility, not a fitted geometry. No automatic Gaia sigma veto was applied.

Catalogue coverage

no identified prior EB; seven-scope coverage complete: yes. Central coverage includes VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, vetted Kostov2025 tables3/4 and ASAS-SN. SB* and ROT alone do not count as known EB. This scoped result is separate from physical assessment. TIC183994341 is known and excluded from new additions regardless of physical tier.

Completion, limits and next check

Full representative review complete; remaining full review=false. Available deciding checks were attempted. Scientific uncertainty: Full representative review completed; the candidate lacks established repeated eclipses. This does not exclude shallower eclipses or events outside the inspected sectors.

Next check: Obtain or inspect another clean sector and require repeated discrete ingress/egress before fitting a binary period.

Provenance: raw-evidence/cache-manifest.json hashes the cached original QLP files; identity-checks.json verifies TICID/sector. Fresh TESScut URLs/SHA256 are in tesscut-downloads.json; source cones and adapted scripts are retained. manifest.json hashes batch deliverables. No paper edits, source table edits, publication or commits were made.

C65. TIC 222498305 full report

Archived source: sanitized report text.

TIC 222498305: REFUTED

Queue position 48; batch05. Automated GPT-6.1 Sol lead and independent fresh GPT-6.1 Terra review, supervised by Astra. The anonymous project coordinator supplied oversight; these are computational measurements, not human measurements.

Assessment

Repeated real eclipses contaminate the target aperture, but are spatially associated with TIC222498306, separated40.139arcsec. REFUTED applies to target attribution and the original Gaia-period eclipse interpretation.

Raw repetition and period

Accepted target eclipse period: not established days. 3 representative cached QLP sectors were reviewed using SAP_FLUX and QUALITY=0. Original packet periods were treated as candidate aliases. Corrected event fits use local linear baselines and trapezoids on raw cadences; no exhaustive sector fit or physical binary model was required.

{ "accepted_target_ephemeris": null, "reason": "No established target repeated eclipse; original alias fitter is not accepted evidence" }

Representative accepted event measurements:

[]

Source and independent reduction

Two fresh detector-sector difference stacks repeatedly favor TIC222498306; fixed neighbour/target residual SSE ratios are0.0119/0.00834. Free Gaussian deficits lie close to that neighbour. The ratios are descriptive, not calibrated probabilities.

Fresh 11x11 TESScuts and 60 arcsec TIC source cones were obtained. Signed in/out images compare local pre/post baselines. The source model includes signed pixel-integrated Gaussians and fixed source positions with free widths/amplitude/background. Residual ratios are descriptive and do not constitute calibrated PRF probabilities. Neighbours capable under a conservative brightness/depth limit were compared; throughput, uncatalogued sources and TIC photometry remain limitations.

The independent custom target aperture subtracts a per-cadence background from faint cutout pixels and uses the same TESS photons as QLP. It is a distinct extraction, not SPOC or a second telescope. Terra independently inspected raw/custom comparison and signed images; quantitative four-EB depths and receipts are in raw-evidence/terra-helper-results.json. See HELPER.md, all-raw.png, raw-event-panels.png and terra-independent-comparison.png.

Gaia geometry and association

The contaminating eclipse interval isapproximately15.5396d; the original screen alias is rejected as a target eclipse period. No target Gaia orbit association is inferred.

Association flag: contaminating neighbour.

The gaia-geometry.json calculation uses edge-on conjunctions, Tp+389 BTJD conversion, full SB1 covariance in P,gamma,K1,e,omega,Tp order, and linear propagation. Approximate mass function, TIC radius and main-sequence companion radius test feasibility, not a fitted geometry. No automatic Gaia sigma veto was applied.

Catalogue coverage

no identified prior EB; seven-scope coverage complete: yes. Central coverage includes VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, vetted Kostov2025 tables3/4 and ASAS-SN. SB* and ROT alone do not count as known EB. This scoped result is separate from physical assessment. TIC183994341 is known and excluded from new additions regardless of physical tier.

Completion, limits and next check

Full representative review complete; remaining full review=false. Available deciding checks were attempted. Scientific uncertainty: The contaminant is an inferred EB host; its catalogue novelty and precision binary orbit were not independently adjudicated. Source model systematics remain.

Next check: Higher resolution eclipse photometry can verify the contaminating host.

Provenance: raw-evidence/cache-manifest.json hashes the cached original QLP files; identity-checks.json verifies TICID/sector. Fresh TESScut URLs/SHA256 are in tesscut-downloads.json; source cones and adapted scripts are retained. manifest.json hashes batch deliverables. No paper edits, source table edits, publication or commits were made.

C66. TIC 90542579 full report

Archived source: sanitized report text.

TIC 90542579: REFUTED

Queue position 49; batch05. Automated GPT-6.1 Sol lead and independent fresh GPT-6.1 Terra review, supervised by Astra. The anonymous project coordinator supplied oversight; these are computational measurements, not human measurements.

Assessment

The enormous sector90 QLP downward excursions occur at both downlink boundaries while the independent custom aperture rises at the same times. They are extraction/systematic features, not demonstrated target eclipses. Earlier sectors8/63 show no matching deep eclipse.

Raw repetition and period

Accepted target eclipse period: not established days. 3 representative cached QLP sectors were reviewed using SAP_FLUX and QUALITY=0. Original packet periods were treated as candidate aliases. Corrected event fits use local linear baselines and trapezoids on raw cadences; no exhaustive sector fit or physical binary model was required.

{ "accepted_target_ephemeris": null, "reason": "No established target repeated eclipse; original alias fitter is not accepted evidence" }

Representative accepted event measurements:

[]

Source and independent reduction

Independent reduction reverses the sign of the deciding sector90 excursions; there is no astrophysical eclipse deficit to localize.

Fresh 11x11 TESScuts and 60 arcsec TIC source cones were obtained. Signed in/out images compare local pre/post baselines. The source model includes signed pixel-integrated Gaussians and fixed source positions with free widths/amplitude/background. Residual ratios are descriptive and do not constitute calibrated PRF probabilities. Neighbours capable under a conservative brightness/depth limit were compared; throughput, uncatalogued sources and TIC photometry remain limitations.

The independent custom target aperture subtracts a per-cadence background from faint cutout pixels and uses the same TESS photons as QLP. It is a distinct extraction, not SPOC or a second telescope. Terra independently inspected raw/custom comparison and signed images; quantitative four-EB depths and receipts are in raw-evidence/terra-helper-results.json. See HELPER.md, all-raw.png, raw-event-panels.png and terra-independent-comparison.png.

Gaia geometry and association

The screen period17.329d is not derived from accepted eclipse events. No Gaia eclipse association is established.

Association flag: screen artifact.

The gaia-geometry.json calculation uses edge-on conjunctions, Tp+389 BTJD conversion, full SB1 covariance in P,gamma,K1,e,omega,Tp order, and linear propagation. Approximate mass function, TIC radius and main-sequence companion radius test feasibility, not a fitted geometry. No automatic Gaia sigma veto was applied.

Catalogue coverage

no identified prior EB; seven-scope coverage complete: yes. Central coverage includes VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, vetted Kostov2025 tables3/4 and ASAS-SN. SB* and ROT alone do not count as known EB. This scoped result is separate from physical assessment. TIC183994341 is known and excluded from new additions regardless of physical tier.

Completion, limits and next check

Full representative review complete; remaining full review=false. Available deciding checks were attempted. Scientific uncertainty: REFUTED applies to the proposed deep-eclipse interpretation of the measured screen signal. Shallower unrelated variability is not excluded.

Next check: If revisiting, use clean cadences away from downlink boundaries with a validated background and aperture.

Provenance: raw-evidence/cache-manifest.json hashes the cached original QLP files; identity-checks.json verifies TICID/sector. Fresh TESScut URLs/SHA256 are in tesscut-downloads.json; source cones and adapted scripts are retained. manifest.json hashes batch deliverables. No paper edits, source table edits, publication or commits were made.

C67. TIC 9396327 full report

Archived source: sanitized report text.

TIC 9396327: PROBABLE

Queue position 50; batch05. Automated GPT-6.1 Sol lead and independent fresh GPT-6.1 Terra review, supervised by Astra. The anonymous project coordinator supplied oversight; these are computational measurements, not human measurements.

Assessment

Six fully covered primary events repeat in raw flux with depth28–29percent and consistent discrete morphology. Independent aperture reduction reproduces the events and the signed deficit favors the target. The Gaia-orbit geometry check remains unresolved.

Raw repetition and period

Accepted target eclipse period: 2.4416706861122432 days. 1 representative cached QLP sectors were reviewed using SAP_FLUX and QUALITY=0. Original packet periods were treated as candidate aliases. Corrected event fits use local linear baselines and trapezoids on raw cadences; no exhaustive sector fit or physical binary model was required.

{ "period_d": 2.4416706861122432, "epoch_btjd": 3481.3231252924884, "n_events": 6, "note": "Representative local event timing regression for939/183; direct consecutive event interval for431/222; no formal period error." }

Representative accepted event measurements:


```
[
  {
    "sector": 80,
    "tc": 3481.323636157607,
    "depth": 0.28420131298250056,
    "duration_d": 0.13406369149885997,
    "ingress_fraction": 0.3540239651639954,
    "residual_rms": 0.012155047692972612,
    "edge": false,
    "n_points": 517,
    "cycle": 0
  },
  {
    "sector": 80,
    "tc": 3483.7643666168096,
    "depth": 0.2883310030553778,
    "duration_d": 0.13934346534258632,
    "ingress_fraction": 0.35414071040251316,
    "residual_rms": 0.009947821623346951,
    "edge": false,
    "n_points": 519,
    "cycle": 1
  },
  {
```

```
    "sector": 80,
    "tc": 3495.9727604242707,
    "depth": 0.2882585527080121,
    "duration_d": 0.13941595437124518,
    "ingress_fraction": 0.3526956252225945,
    "residual_rms": 0.008971962493992389,
    "edge": false,
    "n_points": 519,
    "cycle": 6
  },
  {
    "sector": 80,
    "tc": 3498.4150669656806,
    "depth": 0.28617070118118176,
    "duration_d": 0.1422442512820803,
    "ingress_fraction": 0.36315371152553355,
    "residual_rms": 0.008677686898336322,
    "edge": false,
    "n_points": 519,
    "cycle": 7
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  {
    "sector": 80,
    "tc": 3500.856001103531,
    "depth": 0.2879069502033062,
    "duration_d": 0.14331393223936886,
    "ingress_fraction": 0.3692578062889713,
    "residual_rms": 0.009958225508168744,
    "edge": false,
    "n_points": 519,
    "cycle": 8
  },
  {
    "sector": 80,
    "tc": 3503.298711756517,
    "depth": 0.28214056210063676,
    "duration_d": 0.14093323177085357,
    "ingress_fraction": 0.36372962582456636,
    "residual_rms": 0.009964507706264907,
    "edge": false,
    "n_points": 426,
    "cycle": 9
  }
]
```


Source and independent reduction

Sector80 free Gaussian deficit lies0.56arcsec from target. Capable TIC neighbours51.4/55.9arcsec away have fixed neighbour/target SSE ratios74.9/70.5, strongly worse descriptively. The independent custom aperture reproduces repeated eclipses.

Fresh 11x11 TESScuts and 60 arcsec TIC source cones were obtained. Signed in/out images compare local pre/post baselines. The source model includes signed pixel-integrated Gaussians and fixed source positions with free widths/amplitude/background. Residual ratios are descriptive and do not constitute calibrated PRF probabilities. Neighbours capable under a conservative brightness/depth limit were compared; throughput, uncatalogued sources and TIC photometry remain limitations.

The independent custom target aperture subtracts a per-cadence background from faint cutout pixels and uses the same TESS photons as QLP. It is a distinct extraction, not SPOC or a second telescope. Terra independently inspected raw/custom comparison and signed images; quantitative four-EB depths

and receipts are in raw-evidence/terra-helper-results.json. See HELPER.md, all-raw.png, raw-event-panels.png and terra-independent-comparison.png.

Gaia geometry and association

Representative six-event regression gives2.441670686d (timing residuals below0.00056d); Gaia2.441934355+/-0.000085226d. A sector-end partial event was excluded because its distorted timing shifts the inferred period. Approximate conjunction A offset0.347d versus propagated covariance0.116d; period differences alone do not veto an EB. Duration3.36h is feasible relative to central4.81h. Secondary-event phase and propagated orbital phase prevent claiming a fully consistent Gaia association.

Association flag: unresolved; no firm by-catch.

The gaia-geometry.json calculation uses edge-on conjunctions, Tp+389 BTJD conversion, full SB1 covariance in P,gamma,K1,e,omega,Tp order, and linear propagation. Approximate mass function, TIC radius and main-sequence companion radius test feasibility, not a fitted geometry. No automatic Gaia sigma veto was applied.

Catalogue coverage

no identified prior EB; seven-scope coverage complete: yes. Central coverage includes VSX, SIMBAD primary type, Gaia DR3 EB, Prsa2022, IJspeert2024, vetted Kostov2025 tables3/4 and ASAS-SN. SB* and ROT alone do not count as known EB. This scoped result is separate from physical assessment. TIC183994341 is known and excluded from new additions regardless of physical tier.

Completion, limits and next check

Full representative review complete; remaining full review=false. Available deciding checks were attempted. Scientific uncertainty: Real on-target repeating EB; PROBABLE because the Gaia geometry/association check is unresolved. One fresh detector sector only; same-photon independent reduction. No firm by-catch flag: a modest period mismatch does not establish a separate orbit.

Next check: Refine Gaia conjunction predictions with orbital posterior and measure a second clean sector or obtain new RVs to test the orbit association.

Provenance: raw-evidence/cache-manifest.json hashes the cached original QLP files; identity-checks.json verifies TICID/sector. Fresh TESScut URLs/SHA256 are in tesscut-downloads.json; source cones and adapted scripts are retained. manifest.json hashes batch deliverables. No paper edits, source table edits, publication or commits were made.

C68. TIC 275022619 full report

Archived source: sanitized report text.

TIC 275022619: PLAUSIBLE

Queue position 51; batch06. Review credits: GPT-6.1 Sol, fresh GPT-6.1 Terra review, supervised by Astra. This is AI review, not human validation.

1. Raw morphology and repeat assessment

QLP sectors 12,65,103 show smooth slopes and broad variability. The sharp-looking sector103 changes occur at segment edges. At the covered original nominal phase, the short-window signed depth is negative (brightening), and the template fit hits its timing/duration bounds. The fresh custom aperture shows smooth variability and segment offsets without a repeat narrow eclipse. Independent Terra audit retains PLAUSIBLE: the available representative data fail to establish recurring eclipses, but sparse/compromised coverage does not tie the original search detection to a specific demonstrated artifact. This is not a secure eclipse identification.

Three cached QLP SAP_FLUX FITS sectors were checked with QUALITY=0 and target/sector header verification. Local quadratic baseline plus trapezoid fits are in raw-events.json, including failed or boundary fits. No non-boundary fit exceeds three local residual RMS; no repeat eclipse series establishes a period. Original search period 17.575404492 d is retained only as a conditional alias, not a derived ephemeris. The full morphology overview explicitly checks features away from that alias; raw-event fits alone do not decide the result. Covered conditional phases: [{"sector": 103, "cycle": 143, "nominal_tc": 4153.2634422981155, "n_in": 68, "n_pre": 113, "n_post": 113, "covered": true, "signed_depth": -0.012590765953063965, "local_rms": 0.027302270755171776}].

2. Source and independent reduction

Fresh 11x11 TESScut images and TIC 60-arcsec cones were obtained in two sectors. Signed candidate-phase difference images and fixed-coordinate target/capable-neighbour comparisons were attempted using the original search phase. Since no real repeating eclipse exists to localize, these candidate-only fits do not establish an eclipse host or an off-target EB. They are retained as failed candidate tests, not localization detections. Capability limits use the original claimed depth and ignore aperture throughput/bandpass uncertainty.

Fresh cutout sectors: 65,103. Candidate capability counts: 65: 19,103: 19. Source comparisons are descriptive signed-image Gaussian fits, with no calibrated PRF confidence. Neighbour catalogue identity can include unresolved or duplicate records. The independent extraction subtracts each frame's lowest-30-pixel median background then sums a central 3x3 aperture, with QUALITY=0. It uses the same TESS photons through a distinct reduction, and is not SPOC. custom-morphology.png and custom-extraction.json preserve its deciding morphology.

3. Gaia orbit

Gaia NSS orbit metadata were inspected, but no photometric eclipse ephemeris was established; an eclipse/conjunction geometry comparison is therefore inapplicable. Nearness of the original search alias to the Gaia period cannot establish eclipses.

Gaia source 5984329116365907584; NSS P=17.538395407 +/- 0.010265369 d; e=0.05981, omega=247.3903 degrees. No eclipse timing or geometry is inferred from this metadata. No photometric eclipse orbit established; Gaia spectroscopic binarity not refuted.

4. Scoped prior EB coverage

no identified prior EB; coverage complete: yes. Reused centrally completed seven scopes: VSX, SIMBAD primary type, Gaia DR3 eclipsing binaries, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* and ROT alone are not EB identifications; Kostov table2 is unvetted. This is a scoped catalogue result, not an exhaustive discovery claim. Physical and novelty verdicts are separate.

5. Completion and limitations

The representative full review is complete. Scientific outcome: eclipse morphology and repetition remain unresolved; this is not a secure eclipse identification. Missing eclipse geometry reflects the absence of a valid eclipse series, not unfinished execution. Representative three QLP sectors and two independent custom-extraction sectors; no exhaustive all-sector search, calibrated PRF/noise model, physical fit or sensitivity upper limit. No secure eclipse identification or exclusion of all possible shallow eclipses follows. Repetition and eclipse morphology remain scientifically uncertain; no calibrated sensitivity or propagated ephemeris-error analysis was performed.

Next check: Obtain a well-covered predicted candidate event in a distinct reduction and establish a second coherent detached eclipse before claiming an eclipse period or host.

Provenance: raw-evidence/cache-manifest.json preserves original cache paths and SHA256; identity-checks.json verifies all 15 QLP inputs; tesscut-downloads.json stores fresh URLs and zip hashes; TIC raw cones and adapted scripts are retained. manifest.json hashes deliverables. No paper edits, commits or publication were performed.

C69. TIC 297541839 full report

Archived source: sanitized report text.

TIC 297541839: REFUTED

Queue position 52; batch06. Review credits: GPT-6.1 Sol, fresh GPT-6.1 Terra review, supervised by Astra. This is AI review, not human validation.

1. Raw morphology and repeat assessment

QLP sectors11,64,100 show broad smooth variable minima rather than short eclipses. Covered nominal candidate phases have short-window depths only 0.100,0.077,0.017 percent, versus the original search depth 0.533 percent, and fits have depths smaller than local residual scatter. The distinct fresh custom aperture also shows smooth variability and segment offsets without recurring detached eclipses.

Three cached QLP SAP_FLUX FITS sectors were checked with QUALITY=0 and target/sector header verification. Local quadratic baseline plus trapezoid fits are in raw-events.json, including failed or boundary fits. No non-boundary fit exceeds three local residual RMS; no repeat eclipse series establishes a period. Original search period 16.363592725 d is retained only as a conditional alias, not a derived ephemeris. The full morphology overview explicitly checks features away from that alias; raw-event fits alone do not decide the result. Covered conditional phases: [{"sector": 11, "cycle": 0, "nominal_tc": 1601.1092, "n_in": 4, "n_pre": 5, "n_post": 6, "covered": true, "signed_depth": 0.0010014176368713379, "local_rms": 0.001233667484484613}, {"sector": 64, "cycle": 89, "nominal_tc": 3057.468952514513, "n_in": 31, "n_pre": 52, "n_post": 52, "covered": true, "signed_depth": 0.0007699728012084961, "local_rms": 0.0038203957956284285}, {"sector": 100, "cycle": 152, "nominal_tc": 4088.3752941820903, "n_in": 31, "n_pre": 52, "n_post": 52, "covered": true, "signed_depth": 0.00016689300537109375, "local_rms": 0.00291615072637796 4}].

2. Source and independent reduction

Fresh 11x11 TESScut images and TIC 60-arcsec cones were obtained in two sectors. Signed candidate-phase difference images and fixed-coordinate target/capable-neighbour comparisons were attempted using the original search phase. Since no real repeating eclipse exists to localize, these candidate-only fits do not establish an eclipse host or an off-target EB. They are retained as failed candidate tests, not localization detections. Capability limits use the original claimed depth and ignore aperture throughput/bandpass uncertainty.

Fresh cutout sectors: 64,100. Candidate capability counts: 64: 23,100: 23. Source comparisons are descriptive signed-image Gaussian fits, with no calibrated PRF confidence. Neighbour catalogue identity can include unresolved or duplicate records. The independent extraction subtracts each frame's lowest-30-pixel median background then sums a central 3x3 aperture, with QUALITY=0. It uses the same TESS photons through a distinct reduction, and is not SPOC. custom-morphology.png and custom-extraction.json preserve its deciding morphology.

3. Gaia orbit

Gaia NSS orbit metadata were inspected, but no photometric eclipse ephemeris was established; an eclipse/conjunction geometry comparison is therefore inapplicable. Nearness of the original search alias to the Gaia period cannot establish eclipses.

Gaia source 5854775658438572416; NSS P=16.417997076 +/- 0.013138885 d; e=0.02946, omega=13.3971 degrees. No eclipse timing or geometry is inferred from this metadata. No photometric eclipse orbit established; Gaia spectroscopic binarity not refuted.

4. Scoped prior EB coverage

no identified prior EB; coverage complete: yes. Reused centrally completed seven scopes: VSX, SIMBAD primary type, Gaia DR3 eclipsing binaries, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* and ROT alone are not EB identifications; Kostov table2 is unvetted. This is a scoped catalogue result, not an exhaustive discovery claim. Physical and novelty verdicts are separate.

5. Completion and limitations

The representative full review is complete. Scientific outcome: submitted eclipse interpretation refuted by a demonstrated smooth/systematic explanation. Missing eclipse geometry reflects the absence of a valid eclipse series, not unfinished execution. Representative three QLP sectors and two independent custom-extraction sectors; no exhaustive all-sector search, calibrated PRF/noise model, physical fit or sensitivity upper limit. Refutation applies to the submitted eclipse interpretation, not to binarity or all possible shallow eclipses.

Next check: No missing deciding gate for this candidate. Reopen only if new data supply distinct, repeated detached events.

Provenance: raw-evidence/cache-manifest.json preserves original cache paths and SHA256; identity-checks.json verifies all 15 QLP inputs; tesscut-downloads.json stores fresh URLs and zip hashes; TIC raw cones and adapted scripts are retained. manifest.json hashes deliverables. No paper edits, commits or publication were performed.

C70. TIC 396248369 full report

Archived source: sanitized report text.

TIC 396248369: REFUTED

Queue position 53; batch06. Review credits: GPT-6.1 Sol, fresh GPT-6.1 Terra review, supervised by Astra. This is AI review, not human validation.

1. Raw morphology and repeat assessment

QLP sectors13,94,103 include apparent sharp dips near rapid baseline changes, but the original-phase fits fail to establish real repeat eclipses. In the fresh custom extraction, the S94/S103 apparent dips correspond to large positive ramps or segment-level changes; a detached negative eclipse is not independently recovered. Original covered phases have sub-millimagnitude or negative short-window signed deficits. The search period is not a measured binary eclipse period.

Three cached QLP SAP_FLUX FITS sectors were checked with QUALITY=0 and target/sector header verification. Local quadratic baseline plus trapezoid fits are in raw-events.json, including failed or boundary fits. No non-boundary fit exceeds three local residual RMS; no repeat eclipse series establishes a period. Original search period 18.390769704 d is retained only as a conditional alias, not a derived ephemeris. The full morphology overview explicitly checks features away from that alias; raw-event fits alone do not decide the result. Covered conditional phases: [{"sector": 13, "cycle": 0, "nominal_tc": 1663.3432, "n_in": 4, "n_pre": 8, "n_post": 8, "covered": true, "signed_depth": -0.0002053380012512207, "local_rms": 0.0014788307016715407}, {"sector": 13, "cycle": 1, "nominal_tc": 1681.7339697044129, "n_in": 6, "n_pre": 9, "n_post": 8, "covered": true, "signed_depth": 0.0006639957427978516, "local_rms": 0.000878628867212683}, {"sector": 94, "cycle": 120, "nominal_tc": 3870.23556452954, "n_in": 46, "n_pre": 78, "n_post": 78, "covered": true, "signed_depth": 0.00036269426345825195, "local_rms": 0.005096118431538343}, {"sector": 103, "cycle": 136, "nominal_tc": 4164.4878798001455, "n_in": 40, "n_pre": 69, "n_post": 51, "covered": true, "signed_depth": -0.0019028186798095703, "local_rms": 0.028664467856287956}].

2. Source and independent reduction

Fresh 11x11 TESScut images and TIC 60-arcsec cones were obtained in two sectors. Signed candidate-phase difference images and fixed-coordinate target/capable-neighbour comparisons were attempted using the original search phase. Since no real repeating eclipse exists to localize, these candidate-only fits do not establish an eclipse host or an off-target EB. They are retained as failed candidate tests, not localization detections. Capability limits use the original claimed depth and ignore aperture throughput/bandpass uncertainty.

Fresh cutout sectors: 94,103. Candidate capability counts: 94: 4,103: 4. Source comparisons are descriptive signed-image Gaussian fits, with no calibrated PRF confidence. Neighbour catalogue identity can include unresolved or duplicate records. The independent extraction subtracts each frame's lowest-30-pixel median background then sums a central 3x3 aperture, with QUALITY=0. It uses the same TESS photons through a distinct reduction, and is not SPOC. custom-morphology.png and custom-extraction.json preserve its deciding morphology.

3. Gaia orbit

Gaia NSS orbit metadata were inspected, but no photometric eclipse ephemeris was established; an eclipse/conjunction geometry comparison is therefore inapplicable. Nearness of the original search alias to the Gaia period cannot establish eclipses.

Gaia source 6367733051516351360; NSS P=18.380294812 +/- 0.012093988 d; e=0.26564, omega=257.6767 degrees. No eclipse timing or geometry is inferred from this metadata. No photometric eclipse orbit established; Gaia spectroscopic binarity not refuted.

4. Scoped prior EB coverage

no identified prior EB; coverage complete: yes. Reused centrally completed seven scopes: VSX, SIMBAD primary type, Gaia DR3 eclipsing binaries, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* and ROT alone are not EB identifications; Kostov table2 is unvetted. This is a scoped catalogue result, not an exhaustive discovery claim. Physical and novelty verdicts are separate.

5. Completion and limitations

The representative full review is complete. Scientific outcome: submitted eclipse interpretation refuted by a demonstrated smooth/systematic explanation. Missing eclipse geometry reflects the absence of a valid eclipse series, not unfinished execution. Representative three QLP sectors and two independent custom-extraction sectors; no exhaustive all-sector search, calibrated PRF/noise model, physical fit or sensitivity upper limit. Refutation applies to the submitted eclipse interpretation, not to binarity or all possible shallow eclipses.

Next check: No missing deciding gate for this candidate. Reopen only if new data supply distinct, repeated detached events.

Provenance: raw-evidence/cache-manifest.json preserves original cache paths and SHA256; identity-checks.json verifies all 15 QLP inputs; tesscut-downloads.json stores fresh URLs and zip hashes; TIC raw cones and adapted scripts are retained. manifest.json hashes deliverables. No paper edits, commits or publication were performed.

C71. TIC 277479128 full report

Archived source: sanitized report text.

TIC 277479128: REFUTED

Queue position 54; batch06. Review credits: GPT-6.1 Sol, fresh GPT-6.1 Terra review, supervised by Astra. This is AI review, not human validation.

1. Raw morphology and repeat assessment

QLP sectors 13,92,94 contain broad multi-day variability and large segment changes. The sharp-looking S94 minimum is a distorted edge on broad variability: the distinct fresh aperture shows a multi-day minimum rather than a short detached eclipse. Covered nominal phases have negative short-window signed deficits; the strong S94 template fit hits its timing bound. No repeating detached event establishes an eclipse period.

Three cached QLP SAP_FLUX FITS sectors were checked with QUALITY=0 and target/sector header verification. Local quadratic baseline plus trapezoid fits are in raw-events.json, including failed or boundary fits. No non-boundary fit exceeds three local residual RMS; no repeat eclipse series establishes a period. Original search period 19.667413548 d is retained only as a conditional alias, not a derived ephemeris. The full morphology overview explicitly checks features away from that alias; raw-event fits alone do not decide the result. Covered conditional phases: [{"sector": 13, "cycle": 0, "nominal_tc": 1667.3914, "n_in": 7, "n_pre": 12, "n_post": 4, "covered": true, "signed_depth": -0.0013865232467651367, "local_rms": 0.005576646886765957}, {"sector": 92, "cycle": 109, "nominal_tc": 3811.139476703976, "n_in": 67, "n_pre": 113, "n_post": 113, "covered": true, "signed_depth": -0.000667572021484375, "local_rms": 0.004877484869211912}, {"sector": 94, "cycle": 112, "nominal_tc": 3870.1417173472046, "n_in": 67, "n_pre": 113, "n_post": 113, "covered": true, "signed_depth": -0.04576444625854492, "local_rms": 0.07903945446014404}].

2. Source and independent reduction

Fresh 11x11 TESScut images and TIC 60-arcsec cones were obtained in two sectors. Signed candidate-phase difference images and fixed-coordinate target/capable-neighbour comparisons were attempted using the original search phase. Since no real repeating eclipse exists to localize, these candidate-only fits do not establish an eclipse host or an off-target EB. They are retained as failed candidate tests, not localization detections. Capability limits use the original claimed depth and ignore aperture throughput/bandpass uncertainty.

Fresh cutout sectors: 92,94. Candidate capability counts: 92: 8,94: 8. Source comparisons are descriptive signed-image Gaussian fits, with no calibrated PRF confidence. Neighbour catalogue identity can include unresolved or duplicate records. The independent extraction subtracts each frame's lowest-30-pixel median background then sums a central 3x3 aperture, with QUALITY=0. It uses the same TESS photons through a distinct reduction, and is not SPOC. custom-morphology.png and custom-extraction.json preserve its deciding morphology.

3. Gaia orbit

Gaia NSS orbit metadata were inspected, but no photometric eclipse ephemeris was established; an eclipse/conjunction geometry comparison is therefore inapplicable. Nearness of the original search alias to the Gaia period cannot establish eclipses.

Gaia source 6760248154216144512; NSS P=19.660672814 +/- 0.014125655 d; e=0.08876, omega=128.5857 degrees. No eclipse timing or geometry is inferred from this metadata. No photometric eclipse orbit established; Gaia spectroscopic binarity not refuted.

4. Scoped prior EB coverage

no identified prior EB; coverage complete: yes. Reused centrally completed seven scopes: VSX, SIMBAD primary type, Gaia DR3 eclipsing binaries, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* and ROT alone are not EB identifications; Kostov table2 is unvetted. This is a scoped catalogue result, not an exhaustive discovery claim. Physical and novelty verdicts are separate.

5. Completion and limitations

The representative full review is complete. Scientific outcome: submitted eclipse interpretation refuted by a demonstrated smooth/systematic explanation. Missing eclipse geometry reflects the absence of a valid eclipse series, not unfinished execution. Representative three QLP sectors and two independent custom-extraction sectors; no exhaustive all-sector search, calibrated PRF/noise model, physical fit or sensitivity upper limit. Refutation applies to the submitted eclipse interpretation, not to binarity or all possible shallow eclipses.

Next check: No missing deciding gate for this candidate. Reopen only if new data supply distinct, repeated detached events.

Provenance: raw-evidence/cache-manifest.json preserves original cache paths and SHA256; identity-checks.json verifies all 15 QLP inputs; tesscut-downloads.json stores fresh URLs and zip hashes; TIC raw cones and adapted scripts are retained. manifest.json hashes deliverables. No paper edits, commits or publication were performed.

C72. TIC 257575960 full report

Archived source: sanitized report text.

TIC 257575960: PLAUSIBLE

Queue position 55; batch06. Review credits: GPT-6.1 Sol, fresh GPT-6.1 Terra review, supervised by Astra. This is AI review, not human validation.

1. Raw morphology and repeat assessment

QLP sectors 12,67,102 show orbit-segment ramps, noisy low-amplitude variation and repeated plateaus/steps. Original nominal phases in sectors12/67 have no coherent short eclipse; fitted positive depths are below local residual scatter. The fresh independent aperture reproduces the sector102 segment plateaus/steps and noisy S67 curve without detached recurring dips. Independent Terra audit retains PLAUSIBLE: the available representative data fail to establish recurring eclipses, but sparse/compromised coverage does not tie the original search detection to a specific demonstrated artifact. This is not a secure eclipse identification.

Three cached QLP SAP_FLUX FITS sectors were checked with QUALITY=0 and target/sector header verification. Local quadratic baseline plus trapezoid fits are in raw-events.json, including failed or boundary fits. No non-boundary fit exceeds three local residual RMS; no repeat eclipse series establishes a period. Original search period 13.410727098 d is retained only as a conditional alias, not a derived ephemeris. The full morphology overview explicitly checks features away from that alias; raw-event fits alone do not decide the result. Covered conditional phases: [{"sector": 12, "cycle": 1, "nominal_tc": 1644.6230270979095, "n_in": 8, "n_pre": 12, "n_post": 13, "covered": true, "signed_depth": -0.00010025501251220703, "local_rms": 0.0011402764357626438}, {"sector": 67, "cycle": 113, "nominal_tc": 3146.624462063768, "n_in": 48, "n_pre": 108, "n_post": 71, "covered": true, "signed_depth": -0.0006230473518371582, "local_rms": 0.003330210456624627}].

2. Source and independent reduction

Fresh 11x11 TESScut images and TIC 60-arcsec cones were obtained in two sectors. Signed candidate-phase difference images and fixed-coordinate target/capable-neighbour comparisons were attempted using the original search phase. Since no real repeating eclipse exists to localize, these candidate-only fits do not establish an eclipse host or an off-target EB. They are retained as failed candidate tests, not localization detections. Capability limits use the original claimed depth and ignore aperture throughput/bandpass uncertainty.

Fresh cutout sectors: 67,102. Candidate capability counts: 67: 2,102: 2. Source comparisons are descriptive signed-image Gaussian fits, with no calibrated PRF confidence. Neighbour catalogue identity can include unresolved or duplicate records. The independent extraction subtracts each frame's lowest-30-pixel median background then sums a central 3x3 aperture, with QUALITY=0. It uses the same TESS photons through a distinct reduction, and is not SPOC. custom-morphology.png and custom-extraction.json preserve its deciding morphology.

3. Gaia orbit

Gaia NSS orbit metadata were inspected, but no photometric eclipse ephemeris was established; an eclipse/conjunction geometry comparison is therefore inapplicable. Nearness of the original search alias to the Gaia period cannot establish eclipses.

Gaia source 6347331991219882624; NSS P=13.428262096 +/- 0.005767776 d; e=0.41266, omega=352.3530 degrees. No eclipse timing or geometry is inferred from this metadata. No photometric eclipse orbit established; Gaia spectroscopic binarity not refuted.

4. Scoped prior EB coverage

no identified prior EB; coverage complete: yes. Reused centrally completed seven scopes: VSX, SIMBAD primary type, Gaia DR3 eclipsing binaries, Prsa2022, IJspeert2024, Kostov2025 vetted tables3/4, ASAS-SN. SB* and ROT alone are not EB identifications; Kostov table2 is unvetted. This is a scoped catalogue result, not an exhaustive discovery claim. Physical and novelty verdicts are separate.

5. Completion and limitations

The representative full review is complete. Scientific outcome: eclipse morphology and repetition remain unresolved; this is not a secure eclipse identification. Missing eclipse geometry reflects the absence of a valid eclipse series, not unfinished execution. Representative three QLP sectors and two independent custom-extraction sectors; no exhaustive all-sector search, calibrated PRF/noise model, physical fit or sensitivity upper limit. No secure eclipse identification or exclusion of all possible shallow eclipses follows. Repetition and eclipse morphology remain scientifically uncertain; no calibrated sensitivity or propagated ephemeris-error analysis was performed.

Next check: Obtain a well-covered predicted candidate event in a distinct reduction and establish a second coherent detached eclipse before claiming an eclipse period or host.

Provenance: raw-evidence/cache-manifest.json preserves original cache paths and SHA256; identity-checks.json verifies all 15 QLP inputs; tesscut-downloads.json stores fresh URLs and zip hashes; TIC raw cones and adapted scripts are retained. manifest.json hashes deliverables. No paper edits, commits or publication were performed.

Predicted spectroscopic conjunctions: G 11-11.5 individual evidence

Current adjudications reconcile 23 existing reviews using the same bounded physical criteria. Twenty-two systems are confirmed additions; TIC 8853239 remains plausible. Historical reports remain archived. Reference epochs identify the eclipse kind recorded below and need not denote the deeper primary; error qualifications govern numerical interpretation.

P001. TIC 262653359 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 46.1290484

period_error_days: 8e-05

period_origin: eclipse timing

period_error_kind: source-specific timing uncertainty; see report

reference_epoch_btjd: 3489.276184

epoch_kind: named reference eclipse; see report

depth_ppt: 31

novelty_status: no identified prior EB in seven scoped groups

limitations: Period uncertainty widened to 0.00008 d: saved floor-fit output gives 0.000072 d, wider than prose 0.00005. Mass and evolutionary state leave factor ~2 light-ratio uncertainty.

next_check: Spectroscopy or colour eclipse photometry to constrain companion light.

Current full report and deciding evidence

P002. TIC 39013890 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 20.92352919

period_error_days: 3e-06

period_origin: eclipse timing

period_error_kind: source-specific timing uncertainty; see report

reference_epoch_btjd: 1859.073426

epoch_kind: named reference eclipse; see report

depth_ppt: 157

novelty_status: no identified prior EB in seven scoped groups

limitations: Unresolved third light remains possible (ipd multi-peak 69%; RUWE 1.55), affecting derived radii and geometry precision.

next_check: Resolve close companion or constrain third light spectroscopically.

Current full report and deciding evidence

P003. TIC 325399242 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 28.03808637

period_error_days: 0.00021722

period_origin: eclipse timing

period_error_kind: source-specific timing uncertainty; see report

reference_epoch_btjd: 2830.458306

epoch_kind: named reference eclipse; see report

depth_ppt: 5

novelty_status: no identified prior EB in seven scoped groups

limitations: Historical prose claims seven-event P=28.038374 +/-0.000191, but saved ephem_A and Afix each have five rows. Adopt corrected five-event artifact. Capable 13.2 arcsec neighbour is disfavoured at roughly 3–4 sigma per sector with systematic floor, not mathematically impossible.

next_check: Additional orientation/source-resolved imaging would strengthen close-neighbour rejection.

Current full report and deciding evidence

P004. TIC 384026617 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 30.05386902

period_error_days: 5.444e-05

period_origin: eclipse timing

period_error_kind: source-specific timing uncertainty; see report

reference_epoch_btjd: 2903.27302

epoch_kind: named reference eclipse; see report

depth_ppt: 47.2

novelty_status: no identified prior EB in seven scoped groups

limitations: Historical 0.6 arcsec WDS pair is doubtful but unresolved by TESS; attribution is to the Gaia source footprint. Gaia statistics alone do not prove pair spurious.

next_check: High angular resolution imaging to check historical WDS component.

Current full report and deciding evidence

P005. TIC 389013422 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 39.54257794

period_error_days: 1.044e-05

period_origin: eclipse timing

period_error_kind: source-specific timing uncertainty; see report

reference_epoch_btjd: 2497.62367

epoch_kind: named reference eclipse; see report

depth_ppt: 55

novelty_status: no identified prior EB in seven scoped groups

limitations: Secondary phase-0.535 hint is unconfirmed; do not quote it as a measured eclipse. Absolute physical parameters remain model dependent.

next_check: RVs to constrain eccentricity and secondary predictions.

Current full report and deciding evidence

P006. TIC 4808224 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 25.08348476

period_error_days: 5.7e-05

period_origin: eclipse timing

period_error_kind: source-specific timing uncertainty; see report

reference_epoch_btjd: 1526.830229

epoch_kind: named reference eclipse; see report

depth_ppt: 51

novelty_status: no identified prior EB in seven scoped groups

limitations: Single-sector s89 pixel test; pulsation affects secondary depth/timing. Discard s35 secondary fit: all input points flagged and optimiser wandered. Approximate physics.py formula has pi/4 error; exact Kepler-grid result is used.

next_check: Another orientation pixel test or improved RV orbit would reduce association uncertainty.

Current full report and deciding evidence

P007. TIC 23999420 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 22.127652

period_error_days: 4.4e-05

period_origin: eclipse timing

period_error_kind: source-specific timing uncertainty; see report

reference_epoch_btjd: 2349.7517

epoch_kind: named reference eclipse; see report

depth_ppt: 22

novelty_status: no identified prior EB in seven scoped groups

limitations: Adopt primary timing uncertainty 0.000044 d rather than joint model formal 0.000015 d. Secondary 9.7-sigma pixel regression is historical numerical evidence; representative image has no colour scale.

next_check: Recover missing/flagged s91 primary only if additional refinement needed.

Current full report and deciding evidence

P008. TIC 112232265 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 28.40515

period_error_days: 3e-05

period_origin: eclipse timing

period_error_kind: source-specific timing uncertainty; see report

reference_epoch_btjd: 2978.3647

epoch_kind: named reference eclipse; see report

depth_ppt: 33

novelty_status: no identified prior EB in seven scoped groups

limitations: Intrinsic depth approximately 30–36 ppt; custom 3x3 is diluted (~19 ppt), QLP background affects depth. RUWE 3.51 allows third-body uncertainty. T16 parse with zero cadences is not evidence.

next_check: Spectroscopy and source-resolved eclipse depths to constrain companion.

Current full report and deciding evidence

P009. TIC 275065610 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 28.04921

period_error_days: 5e-05

period_origin: eclipse timing

period_error_kind: source-specific timing uncertainty; see report

reference_epoch_btjd: 2023.8727

epoch_kind: named reference eclipse; see report

depth_ppt: 51.5

novelty_status: no identified prior EB in seven scoped groups

limitations: TASOC depth is diluted and its epoch is about 14 min later than QLP; does not invalidate its eclipse detection. Dark-companion fit neglects a few percent secondary light.

next_check: A future raw A eclipse or double-lined spectrum

Current full report and deciding evidence

P010. TIC 469755925 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 23.322969

period_error_days: 1.4e-05

period_origin: eclipse timing

period_error_kind: source-specific timing uncertainty; see report

reference_epoch_btjd: 1675.73779

epoch_kind: named reference eclipse; see report

depth_ppt: 40.5

novelty_status: no identified prior EB in seven scoped groups

limitations: S94 A is quality-flagged; confirmation also rests on clean s13 and s67 A events. B pixel depth5.93 +/-0.81 ppt exceeds pipeline2.8–4.0 ppt; low-S/N/background-template extraction is attribution only, not adopted depth.

next_check: Another clean secondary for depth and geometry

Current full report and deciding evidence

P011. TIC 356435413 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 27.476839

period_error_days: 1.5e-05

period_origin: eclipse timing

period_error_kind: source-specific timing uncertainty; see report

reference_epoch_btjd: 1984.25002

epoch_kind: named reference eclipse; see report

depth_ppt: 17.0

novelty_status: no identified prior EB in seven scoped groups

limitations: CTOI356435413.01 is a prior planet-candidate signal at matching eclipse period; annotate separately from EB novelty. Original report overstates 1.5arcsec target offset: that is deficit-to-light centroid; s79 coordinates imply ~2arcsec to target, s53 ~3.6arcsec. No full limb-darkened grazing fit.

next_check: Grazing-occultation model or spectrum to test light ratio

Current full report and deciding evidence

P012. TIC 255077215 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 26.26204

period_error_days: 1.6e-05

period_origin: eclipse timing

period_error_kind: source-specific timing uncertainty; see report

reference_epoch_btjd: 1637.34726

epoch_kind: named reference eclipse; see report

depth_ppt: 56.0

novelty_status: no identified prior EB in seven scoped groups

limitations: S65 secondary arrives31–42min late after common-mode correction; unexplained. This does not remove repeated A timing or agreement of earlier B events. Quoted period error is original formal fit and predictions need timing-systematic allowance.

next_check: Clean secondary to resolve s65 timing shift

Current full report and deciding evidence

P013. TIC 262471565 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 29.38001

period_error_days: 0.00022

period_origin: eclipse timing

period_error_kind: source-specific timing uncertainty; see report

reference_epoch_btjd: 1337.4273

epoch_kind: named reference eclipse; see report

depth_ppt: 4.5

novelty_status: no identified prior EB in seven scoped groups

limitations: B is undetected: expected570–1100ppm versus ~600ppm central-phase limit creates mild model tension; permissible phase/temperature/impact values remain. Period error includes20min pipeline-scatter floor.

next_check: Unflagged B-window photometry at <=300ppm

Current full report and deciding evidence

P014. TIC 126758342 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 22.86993

period_error_days: 8e-05

period_origin: eclipse timing

period_error_kind: source-specific timing uncertainty; see report

reference_epoch_btjd: 1729.54161

epoch_kind: named reference eclipse; see report

depth_ppt: 20.0

novelty_status: no identified prior EB in seven scoped groups

limitations: Gaia eccentricity and omega broad; A misses in83% of permitted historical model draws, not certainty. Primary_epoch field uses detected deeper B eclipse.

next_check: Refined RV orbit or next B eclipse

Current full report and deciding evidence

P015. TIC 178486571 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 33.671079

period_error_days: 2.2e-05

period_origin: eclipse timing

period_error_kind: source-specific timing uncertainty; see report

reference_epoch_btjd: 1494.78347

epoch_kind: named reference eclipse; see report

depth_ppt: 53.8

novelty_status: no identified prior EB in seven scoped groups

limitations: S87 egress cut by gap. Three-event near-zero timing chi-square makes formal error optimistic; allow >=5min in predictions.

next_check: Clean B coverage or another A event

Current full report and deciding evidence

P016. TIC 395790195 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 55.460424

period_error_days: 3.3e-05

period_origin: eclipse timing

period_error_kind: source-specific timing uncertainty; see report

reference_epoch_btjd: 1756.37044

epoch_kind: named reference eclipse; see report

depth_ppt: 130.0

novelty_status: no identified prior EB in seven scoped groups

limitations: Adopt timing-only B period/error; tighter joint-model55.460455 +/-0.000021d is conditional on model. Only two same-kind epochs, cycle count constrained by Gaia.

next_check: Another B eclipse to test cycle count

Current full report and deciding evidence

P017. TIC 466336052 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 32.139199

period_error_days: 3.7e-05

period_origin: eclipse timing

period_error_kind: source-specific formal timing error with stated floors/scatter

reference_epoch_btjd: 2818.48299

epoch_kind: A

depth_ppt: 44.1

novelty_status: no identified prior EB in seven scoped groups

limitations: Only two A eclipses; no internal timing residual test. B windows are contaminated or missing; no B detection. Gaia conjunction uncertainty is 3-4 d.

next_check: Observe a third A eclipse and a clean B window near 0.3% precision.

Current full report and deciding evidence

P018. TIC 64066759 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 40.93770552812586

period_error_days: 4.02430562196e-05

period_origin: eclipse timing

period_error_kind: source-specific formal timing error with stated floors/scatter

reference_epoch_btjd: 3674.520856303387

epoch_kind: A

depth_ppt: 22.0

novelty_status: no identified prior EB in seven scoped groups

limitations: Long-baseline precision partly depends on a fixed-shape partial s7 event; three events total. Simplified geometry favours smaller primary radius than TIC. No B expected.

next_check: Fit a full limb-darkened joint A model and constrain primary radius spectroscopically.

Current full report and deciding evidence

P019. TIC 49416839 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 21.428466564976848

period_error_days: 4.771732314e-05

period_origin: eclipse timing

period_error_kind: source-specific formal timing error with stated floors/scatter

reference_epoch_btjd: 1558.284949338288

epoch_kind: B (dominant deep eclipse)

depth_ppt: 13.3

novelty_status: no identified prior EB in seven scoped groups

limitations: A3 lacks egress and is weak. B-A phase separation is 2.3 sigma from Gaia; simple joint geometry has limited constraints. Dominant deep eclipse is labelled B in source.

next_check: Obtain a complete A event and joint free-eccentricity model to test the phase-separation tension.

Current full report and deciding evidence

P020. TIC 8853239 - PLAUSIBLE

count_addition: false

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 24.647

period_error_days: 0.0114

period_origin: Gaia SB1 only

period_error_kind: Gaia formal orbit error

reference_epoch_btjd: 2629.0699

epoch_kind: A; Gaia-only period

depth_ppt: 60.0

novelty_status: no identified prior EB in seven scoped groups

limitations: Single secure event; Gaia-only period; KELT null is coverage-limited and not a refutation. Tentative B is not an ephemeris.

next_check: Measure a second A eclipse to test the cycle period; a B detection alone would still need repeat timing.

Current full report and deciding evidence

P021. TIC 60583013 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 26.65117

period_error_days: 0.00023

period_origin: eclipse timing

period_error_kind: source-specific formal timing error with stated floors/scatter

reference_epoch_btjd: 2210.9657

epoch_kind: A (period measured from B)

depth_ppt: 41.2

novelty_status: no identified prior EB in seven scoped groups

limitations: Period relies on two secondaries, one after a data gap; one primary only. Older B pixel test inconclusive. Duration geometry is approximate.

next_check: Obtain a second primary and joint limb-darkened model to test phase separation and geometry.

Current full report and deciding evidence

P022. TIC 277542587 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 24.3318

period_error_days: 0.0002

period_origin: KELT+TESS; heuristic error

period_error_kind: heuristic working precision; not calibrated1sigma

reference_epoch_btjd: 3488.8242

epoch_kind: A; KELT+TESS period

depth_ppt: 77.6

novelty_status: no identified prior EB in seven scoped groups

limitations: TESS A mostly flagged; only one TESS A+B pair. KELT period precision is heuristic and anchored to TESS; broad ground aperture dilutes depth. Sparse KELT B is supporting only. Historical exact-night count not adopted.

next_check: Obtain another full well-calibrated A/B pair and an event-timing based period uncertainty.

Current full report and deciding evidence

P023. TIC 12864953 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 31.632857663754894

period_error_days: 0.00015

period_origin: eclipse timing

period_error_kind: conservative timing allowance; not calibrated coverage

reference_epoch_btjd: 3204.1016

epoch_kind: A (period from repeated B)

depth_ppt: 17.24

novelty_status: no identified prior EB in seven scoped groups

limitations: Only two B events and one clean A; A ephemeris assumes B period. New A window is entirely quality 2048 flagged. Phase-separation tension about 3.35 sigma remains. Conservative period error is an estimate, not calibrated coverage.

next_check: Obtain a clean repeated A and refined radial-velocity orbit to resolve eccentricity/phase tension.

Current full report and deciding evidence

Underexplored selections: current measurements and evidence

The frozen magnitude-ordered period-fold queue and adaptive exploratory pilot use existing physical criteria. They have separate selection and attempted-data denominators; operational processing groups are not scientific methods. Full reports below retain deciding measurements and limitations. New timing periods are measured from eclipse midpoints, not predicted conjunction centers.

R175. TIC 300217120 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 14.158577492046303

period_error_days: 2.752113403501583e-05

reference_epoch_btjd: 3141.0351407928124

depth_ppt: 37.8660025864843

novelty_status: no identified prior EB

limitations: The nearby other eclipse in S67 contaminates a wider baseline window; the final fit uses a narrower 1.65-search-duration half-window and shows the repeated primary. Historical S27 TESScut contains one event but QLP independently shows three primaries in S67/S94.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R176. TIC 424716057 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 15.405497331062046

period_error_days: 5.499559353955741e-05

reference_epoch_btjd: 1625.946884976911

depth_ppt: 211.90741878678438

novelty_status: no identified prior EB

limitations: QLP primary depths vary across apertures/sectors (205–230 ppt in the selected sectors); no common intrinsic depth is claimed.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R177. TIC 265350422 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 8.448170725833366

period_error_days: 1.2842011215299834e-05

reference_epoch_btjd: 2970.804346169428

depth_ppt: 7.800219935881271

novelty_status: no identified prior EB

limitations: Three G-band-capable neighbours at 32.85, 42.38 and 51.78 arcsec lie away from the compact deficit on the target. The target-centered dip grows rather than vanishes in the small aperture. No calibrated centroid confidence ellipse is available.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R178. TIC 274941915 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 10.132939073812572

period_error_days: 6.285764339345673e-05

reference_epoch_btjd: 3371.1764455427087

depth_ppt: 10.41512839207673

novelty_status: no identified prior EB

limitations: Three capable neighbours at 32.98, 40.57 and 47.81 arcsec are spatially separated from the target-centered compact deficit. The small-aperture reduction retains the eclipse. No calibrated centroid confidence ellipse is available.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R179. TIC 253126207 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 17.43657092170319

period_error_days: 8.823559700783691e-05

reference_epoch_btjd: 1604.0901103382182

depth_ppt: 13.51328902862249

novelty_status: no identified prior EB

limitations: Attribution remains unresolved: seven capable neighbours include three within one TESS pixel (15.16, 17.15 and 19.43 arcsec); the historical PSF fractional residual is 0.657. A known Gaia eclipsing binary at 24.77 arcsec with P=0.312563 d is a different source, and its short period does not identify the 17.44 d signal. Real repeating eclipses pass the other checks; source localization is the failed scientific check. Resolved eclipse imaging or defensible multi-source PRF modeling is required.

next_check: Resolved eclipse photometry / multi-source PRF localization

Full report and illustrated deciding evidence

R180. TIC 245474316 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 15.01729122089107

period_error_days: 2.3615939344531595e-05

reference_epoch_btjd: 2884.3381092475142

depth_ppt: 219.83441151981677

novelty_status: no identified prior EB

limitations: Deep repeated eclipses are isolated in both reductions; all four measured primary events support the timing solution.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R181. TIC 397586559 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 9.866288327778928

period_error_days: 2.7660310467344676e-05

reference_epoch_btjd: 2452.71248468518

depth_ppt: 253.71926940278945

novelty_status: no identified prior EB

limitations: Unequal alternating eclipses (primary median approximately254 ppt, secondary164 ppt) establish orbital period twice the historical4.93316d event spacing. Only deeper even cycles enter the reported primary ephemeris. Historical S70 difference image mixes both eclipse kinds, so its aperture depth is a mixture. Gaia nominal conjunction is0.28093d later at the TESS epoch; measured-Gaia period difference−0.00126495d accumulates this offset. Projecting toward the Gaia2016 reference reduces offset to−0.0166d; the orbit association is physically consistent with an imperfect Gaia period, and the tension is retained rather than subjected to a sigma veto.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R182. TIC 305801065 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 6.045029317589018

period_error_days: 3.61616161282754e-05

reference_epoch_btjd: 1727.4094898160802

depth_ppt: 5.0649222211815

novelty_status: no identified prior EB

limitations: The shallow eclipse is clearly repeated with localized ingress/egress despite higher scatter in newer sectors. Three capable neighbours at28.07,33.30,54.57arcsec are separated from the compact target deficit. Depth depends on dilution and trapezoid shape; historical median3.8ppt is not a precision intrinsic depth.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R183. TIC 345794632 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 14.644264849868913

period_error_days: 4.2511956611254694e-05

reference_epoch_btjd: 1712.6089010966314

depth_ppt: 73.3874561147605

novelty_status: no identified prior EB

limitations: Large historical PSF fractional residual0.57 reflects imperfect scaled out-image morphology in a crowded field. No listed neighbour within60arcsec can supply the approximately73ppt eclipse under the G-band flux-budget screen, and the lost-light core coincides with the target. The weaker localization calibration remains a limitation.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R184. TIC 258621299 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 6.985970468761713

period_error_days: 1.7325420588134718e-05

reference_epoch_btjd: 1845.3682206476906

depth_ppt: 179.67039704399065

novelty_status: no identified prior EB

limitations: Unequal alternating eclipses (primary median approximately180ppt, secondary131ppt) establish orbital period twice the historical3.49299d event spacing. Only deeper even cycles enter the reported primary ephemeris. Historical S20 difference image mixes both eclipse kinds, so its aperture depth is a mixture.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R185. TIC 65346146 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 10.678623940946846

period_error_days: 1.5124697871365374e-05

reference_epoch_btjd: 2154.389536775554

depth_ppt: 38.20589792457893

novelty_status: no identified prior EB

limitations: Repeated flat-bottom eclipses; sector-dependent dilution changes fitted depths from about46 to38ppt.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R186. TIC 416646907 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 6.384314755499531

period_error_days: 1.007378415122247e-05

reference_epoch_btjd: 2144.9943337447125

depth_ppt: 17.963292895601008

novelty_status: no identified prior EB

limitations: Odd historical half-period cycles are deeper in each of three sectors (about18 versus16ppt); deeper odd cycles alone give the primary ephemeris and doubled orbital period. Alternative-reduction stack mixes eclipse kinds. Approximate central duration permits a noncentral chord.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R187. TIC 401138717 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 19.79284840491893

period_error_days: 4.02097023341268e-05

reference_epoch_btjd: 2338.109420981473

depth_ppt: 111.5978315194144

novelty_status: no identified prior EB

limitations: Three capable neighbours and high scaled-out-image residual0.976 limit PSF precision. Target-centered compact loss and aperture-depth growth support attribution. Known EB42.73arcsec away has P0.363740d, distinct from19.79d. Gaia period differs by0.00556d, accounting for much of the accumulated conjunction offset; this physical tension is retained.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R188. TIC 334379409 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 3.947126674992941

period_error_days: 8.626157743268626e-06

reference_epoch_btjd: 1846.4992369078332

depth_ppt: 9.702507254515327

novelty_status: no identified prior EB

limitations: Repeated raw eclipses have modest baseline variability. Gaia period mismatch accumulates over the baseline; propagated covariance and conjunction comparison are descriptive rather than a sigma threshold.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R189. TIC 345406164 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 7.440590745011774

period_error_days: 4.9507419893012455e-05

reference_epoch_btjd: 2771.404007331272

depth_ppt: 74.89159582802148

novelty_status: no identified prior EB

limitations: Deeper even cycles near75ppt alternate with47ppt odd eclipses in sector81, establishing doubled orbital period. Primary-only ephemeris used. Historical stack mixes eclipse kinds. Three capable neighbours are away from compact target deficit.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R190. TIC 152400959 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 5.512857449543862

period_error_days: 6.3397913119843745e-06

reference_epoch_btjd: 1492.1132318036186

depth_ppt: 54.37186113086094

novelty_status: no identified prior EB

limitations: Known EB27.09arcsec away at0.257384d differs from5.513d signal. Two capable neighbours and residual0.959 limit PSF precision; compact target-centered deficit and small-aperture eclipse support attribution.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R191. TIC 152086769 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 6.332203976073646

period_error_days: 7.904701005610135e-06

reference_epoch_btjd: 1356.809297220481

depth_ppt: 279.7427141291881

novelty_status: no identified prior EB

limitations: Deep flat-bottom repeated eclipses; sector96 aperture depth is lower than earlier sectors. No capable neighbour within60arcsec in approximate G flux budget.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R192. TIC 270260727 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 10.291649902528073

period_error_days: 1.9455996780702123e-05

reference_epoch_btjd: 3129.8498677040475

depth_ppt: 5.804677308979594

novelty_status: no identified prior EB

limitations: Shallow localized repeated eclipses are resolved in sectors67 and95; raw scatter and trapezoid approximation limit precision. No capable neighbour within60arcsec under G flux budget.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R193. TIC 112827464 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 3.4225952625012397

period_error_days: 4.65653931800416e-06

reference_epoch_btjd: 3129.3941410026505

depth_ppt: 33.4564194422391

novelty_status: no identified prior EB

limitations: Raw local baseline slopes are fitted explicitly. One capable neighbour is separated from the target-centered deficit; no calibrated centroid confidence ellipse. Gaia nominal phase offset0.063 and period drift remain explicit; full covariance and near-reference projection support association without arbitrary sigma veto.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R194. TIC 423657976 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 6.616416793242259

period_error_days: 1.647746040723537e-05

reference_epoch_btjd: 2310.599671098064

depth_ppt: 93.97262387817834

novelty_status: no identified prior EB

limitations: Repeated isolated eclipses with stable approximately94ppt QLP depth. No capable neighbour within60arcsec under approximate G flux budget.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R195. TIC 343926224 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 10.728977906284216

period_error_days: 0.00019858296949633555

reference_epoch_btjd: 1716.285636447413

depth_ppt: 42.29190632036057

novelty_status: no identified prior EB

limitations: Five clean primary repeats determine the ephemeris. S76 historical cycle 154 has structured excursions and an anomalously weak fitted dip; S84 cycle 176 has a broad distorted profile. Both remain in raw-windows.csv and plots but are excluded from the timing fit. The capable neighbour at 55.00 arcsec is separated from the compact target deficit. Aperture and local baseline variations limit depth precision. The retained timing fit has reduced chi-square 18.08; residual timing scatter materially inflates the stated period error and shows model/baseline limitations.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R196. TIC 53993311 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 14.542833409357298

period_error_days: 2.314386027834743e-05

reference_epoch_btjd: 3157.670659801766

depth_ppt: 97.18640828460472

novelty_status: no identified prior EB

limitations: Aperture-dependent primary depth changes from about 107 ppt in S68 to 85 ppt in S95; the same localized eclipse morphology repeats. Gaia conjunction uncertainty is substantial; a phase offset alone is not a veto.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R197. TIC 43785616 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 17.91112281610357

period_error_days: 0.0001892149302470355

reference_epoch_btjd: 1440.850824404922

depth_ppt: 23.115471716169672

novelty_status: no identified prior EB

limitations: Observed trapezoid duration exceeds a central estimate based on default primary mass and minimum companion mass. That approximate estimate is not an upper bound: stellar properties and inclination are not fitted. Repeated flat-bottomed events and matching period support association; detailed binary modeling remains optional.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R198. TIC 397178704 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 11.009685047241446

period_error_days: 1.7717221664631576e-05

reference_epoch_btjd: 1578.4727019498134

depth_ppt: 175.66990489468662

novelty_status: no identified prior EB

limitations: Unequal alternating primary and secondary eclipses establish twice the historical 5.50484 d event spacing. Only deeper odd cycles enter the primary ephemeris. The historical stack mixes both kinds. Large PSF residual 0.676 reflects surrounding light rather than an off-target core; the one capable neighbour lies 45.82 arcsec away from the compact deficit.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R199. TIC 77550184 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 10.214416259590474

period_error_days: 1.4190042358189235e-05

reference_epoch_btjd: 1527.7251647231262

depth_ppt: 60.64366018596226

novelty_status: no identified prior EB

limitations: Repeated isolated eclipses have aperture-dependent depths of approximately 56 to 63 ppt. No capable neighbour lies within the 60 arcsec screen.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R200. TIC 314041837 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 4.477995650988368

period_error_days: 1.534143438335759e-05

reference_epoch_btjd: 1770.0417342831945

depth_ppt: 44.521015368131536

novelty_status: no identified prior EB

limitations: The target is a large star (TIC radius 5.09 solar radii) with ellipsoidal modulation plus localized eclipses. A two-harmonic baseline fitted outside the primary core in S77 leaves a 30.31 ppt localized deficit; its double-wave modulation does not explain the eclipse. See harmonic-test.json and png. The 18.79 h local-linear trapezoid width includes smooth variability and is not a physical eclipse duration. The reported depth likewise describes that local model, not intrinsic eclipse depth. Capable neighbours at 27.40 and 53.17 arcsec are away from the target-centered deficit; PSF residual 0.595 limits morphology precision.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R201. TIC 7446763 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 5.91190666822017

period_error_days: 5.481789245650164e-06

reference_epoch_btjd: 1415.6444464852714

depth_ppt: 72.07551814166084

novelty_status: no identified prior EB

limitations: Primary eclipses are approximately twice secondary depth. Only deeper odd historical half-period cycles determine the ephemeris; the reported orbital period doubles the historical 2.95595 d spacing. Historical S98 aperture stack mixes both eclipse kinds.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R202. TIC 22705959 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 11.376810452243653

period_error_days: 5.702228089004493e-05

reference_epoch_btjd: 1843.6400352066814

depth_ppt: 140.88533223993306

novelty_status: no identified prior EB

limitations: A capable neighbour at 39.75 arcsec is separated from the target deficit. The target sits near a pixel boundary: the global deficit peak rounds to an adjacent pixel, while the centroid is only 3.64 arcsec from the actual target. This is not evidence for a different source.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R203. TIC 62940616 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 4.998587293904812

period_error_days: 1.3192060525370641e-05

reference_epoch_btjd: 2556.444026565471

depth_ppt: 52.59401180953602

novelty_status: no identified prior EB

limitations: One capable neighbour at 22.67 arcsec requires a 0.329 fractional eclipse to explain the signal, but the lost-light centroid lies 0.205 pixel from the target versus 1.33 pixels from that neighbour, displaced away from it. Its pixel fractional loss 0.0416 is a blended-pixel value and does not measure the neighbour intrinsic eclipse depth. The target-centered depth increases in the 1x1 aperture, consistent with target dilution in larger apertures. TESS resolution and G-to-TESS colour differences limit attribution precision.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R204. TIC 100911870 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 12.637002421838197

period_error_days: 1.9952345042033016e-05

reference_epoch_btjd: 2833.849725715909

depth_ppt: 84.03451128246843

novelty_status: no identified prior EB

limitations: Four clean primary events in S56 and S83 support the timing solution and approximately 84 ppt raw QLP depth. No capable neighbour is present in the 60 arcsec screen.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R205. TIC 77176486 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 13.753691913407923

period_error_days: 1.89931259791415e-05

reference_epoch_btjd: 2153.315120594371

depth_ppt: 12.943912782527418

novelty_status: no identified prior EB

limitations: The 13.75369 d signal lies near the TESS orbital timescale, but repeated localized raw SAP dips in S31 and S98 and the target-centered cutout deficit support astrophysical eclipses. Quality-nonzero cadences are excluded; no flagged or uncovered window is a clean null. Cycle188 is visibly present but its trapezoid covariance is singular, so it is excluded from timing. Bounded S98 pixel/background check (background-check.json) finds target-pixel losses11.99/22.43/16.35 electrons/s versus3.24/9.85/7.73 at the close capable neighbour and0.70/6.30/3.36 at the farther one. Outer-pixel median losses are -0.027/5.68/4.04: common-mode background affects later events, while cycle187 is localized with negligible outer deficit. Core quality flags are0/0/3; flagged cadences are excluded. FLUX_BKG is zero throughout and supplies no independent background measurement. These are descriptive diagnostics, not a calibrated artifact veto. Gaia conjunction offset is 0.04208 phase with 0.549 d propagated uncertainty; this is association support with appreciable orbital uncertainty, not a precise phase coincidence.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R206. TIC 155916287 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 6.053154397574601

period_error_days: 1.5431906036480347e-05

reference_epoch_btjd: 1741.6770525360378

depth_ppt: 31.52043029966451

novelty_status: no identified prior EB

limitations: Seven fitted repeats span S16,50,77. Sector-dependent depth variation and long-cadence integration limit intrinsic depth and duration precision.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R207. TIC 383603595 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 5.0148153411521825

period_error_days: 4.955091059655567e-06

reference_epoch_btjd: 1520.2020451552114

depth_ppt: 32.025684556497076

novelty_status: no identified prior EB

limitations: Eleven fitted repeated eclipses span S8,62,90. Modest aperture/baseline depth variation limits precision; no missing bounded check.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R208. TIC 436724968 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 2.732155078385491

period_error_days: 3.73397661927236e-06

reference_epoch_btjd: 2475.274686430569

depth_ppt: 48.52986342115804

novelty_status: no identified prior EB

limitations: QLP free-trapezoid depths show modest parity differences, especially in S71; local baseline and aperture dependence are material. Independent S43 3x3 cutout depths are 35.34-37.19 ppt for eight events including consecutive odd/even cycles, without a stable unequal alternating pattern. Gaia P=2.732167 d matches the measured event spacing. These data do not establish two distinct alternating eclipse kinds or require doubling; report the fitted period as the adopted orbit with this depth-model limitation, and the epoch as an observed reference eclipse. Capable neighbours at 35.19 and 35.89 arcsec are displaced from the compact target deficit.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R209. TIC 37948318 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 8.056727017738979

period_error_days: 1.2086497005613364e-05

reference_epoch_btjd: 1444.2226216138542

depth_ppt: 11.181622059581528

novelty_status: no identified prior EB

limitations: The historical absolute target-to-deficit offset is 15.1 arcsec (0.719 pixel), while light-to-deficit separation is 3.51 arcsec; these are distinct comparisons. The capable G13.73 neighbour is 53.65 arcsec away and opposite the deficit displacement. Independent per-event pixel checks find target-pixel losses 9.03-10.59 electrons/s against 0.185-0.437 at that neighbour in cycles317,319,321. This supports the target over the capable neighbour at TESS resolution, with an uncalibrated absolute-centroid systematic and residual crowding uncertainty. Historical cycles316/318 have common-mode excursions and are not individually decisive. QLP cycle93 has singular timing covariance and is excluded from timing.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R210. TIC 373034189 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 12.98870409783554

period_error_days: 2.8433415380927524e-05

reference_epoch_btjd: 1571.7245142398501

depth_ppt: 12.69611305405158

novelty_status: no identified prior EB

limitations: Four raw fitted eclipses in S10,38,90 support the period. The Gaia conjunction uncertainty is substantial (0.3105 d); local model widths and depths are descriptive.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R211. TIC 300969635 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 7.420727532798374

period_error_days: 7.815655460458437e-06

reference_epoch_btjd: 1328.2542487236803

depth_ppt: 217.5379472546418

novelty_status: no identified prior EB

limitations: Deep repeated eclipses persist in S1,37,98. Aperture depth changes from about210 to248 ppt; pooled parity depths confound sector coverage and do not establish stable unequal eclipse kinds. Gaia period matches the measured spacing. Intrinsic depth requires dilution modeling.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R212. TIC 317800184 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 3.689780944715975

period_error_days: 5.125646723563834e-06

reference_epoch_btjd: 1475.655235020276

depth_ppt: 50.47396076159312

novelty_status: no identified prior EB

limitations: Repeated eclipses span S6,87,98 with sector-dependent depths around45-62 ppt. This limits precision but does not erase the localized repeat pattern.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R213. TIC 219033622 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 6.9464765821983425

period_error_days: 0.00042454076935334877

reference_epoch_btjd: 1713.808706332515

depth_ppt: 132.32584082017803

novelty_status: no identified prior EB

limitations: Three constrained same-kind events in S15 and S23 determine the ephemeris; S23 cycle32 is retained as a raw repeat but excluded from timing because sparse sampling leaves its egress poorly constrained. S76 did not supply a usable selected event window and is not a clean non-detection. The independent S16 cutout contains four repeating deep target-centered deficits. The timing baseline is shorter than the available sector list suggests.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R214. TIC 312725755 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 11.304994219364756

period_error_days: 3.431475573613791e-05

reference_epoch_btjd: 1823.1102836630625

depth_ppt: 47.710924471257606

novelty_status: no identified prior EB

limitations: Five fitted repeating eclipses span S19,45,73. The measured widths are long but compatible with approximate Gaia/TIC geometry; intrinsic depths and stellar parameters need detailed modeling.

next_check: Optional precision characterization

Full report and illustrated deciding evidence

R215. TIC 167178300 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 12.215667671701095

period_error_days: 2.524519726874651e-05

period_error_meaning: OLSscatter standarderror; excludes cadence/template/systematic uncertainties

reference_epoch_btjd: 1479.059382158827

epoch_btjd: 1479.059382158827

epoch_kind: A

depth_ppt: 70.54060949519207

novelty_status: no prior EB identified in seven scoped groups; cachedChopardv1.2.0 exact IDs no overlap; laterreleases/embargo93 unknown

limitations: Capable Gaia neighbours remain at 58.583arcsec, 41.62arcsec, 12.984arcsec, 24.468arcsec, 18.174arcsec, 59.428arcsec. Deficit centroid favors target, but close blend is unresolved without a PRF/deblend uncertainty test.

next_check: Resolved eclipse photometry or multi-source PRF localization

Full report and illustrated deciding evidence

R216. TIC 325573958 - current measurements

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 13.08172273071557

period_error_days: 1.3997427571534246e-05

period_error_meaning: OLSscatter standarderror; excludes cadence/template/systematic uncertainties

reference_epoch_btjd: 2772.4906957388416

epoch_btjd: 2772.4906957388416

epoch_kind: A

depth_ppt: 202.25348219264384

novelty_status: no prior EB identified in seven scoped groups; cachedChopardv1.2.0 exact IDs no overlap; laterreleases/embargo93 unknown

limitations: Capable Gaia neighbours remain at 57.77arcsec, 7.958arcsec, 20.628arcsec. Deficit centroid favors target, but close blend is unresolved without a PRF/deblend uncertainty test.

next_check: Resolved eclipse photometry or multi-source PRF localization

Full report and illustrated deciding evidence

P024. TIC 155303130 - CONFIRMED

count_addition: false

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 20.524777015731278

period_error_days: 0.0003376256272186696

period_error_meaning: OLSscatter standarderror; excludes cadence/template/systematic uncertainties

epoch_btjd: 3238.1960535727508

epoch_kind: B

depth_ppt: 104.05549881862092

novelty_status: known IJspeert2024 EB recovery; zero new-system addition

limitations: Pilot characterization; period errors are scatter-only. No exhaustive literature priority claim.

next_check: Optional precision characterization; no missing bounded admission check

Archived source: sanitized report text.

> Supervisor-accepted physical review: CONFIRMED. Known recovery; excluded from new-system count. Earlier pending-admission wording below records the original handoff.

TIC 155303130: CONFIRMED

Scientific method/population: fresh Gaia-conjunction appointments G11.5-12 P20-60d. Physical review completion: complete bounded first pass, independently audited in helper/AUDIT.md; supervisor admission pending. Catalogue novelty is separate: known IJspeert2024 EB recovery; zero new-system addition.

1. Repeated raw eclipses: PASS. 4 fit-eligible raw SAP events in sectors [71, 72, 87]. Trapezoid depths median 104.06 ppt; duration median 13.10 h. All saved events, rejected fits and raw files are identified in timing-results.json; no claim that other sectors are clean nulls. QUALITY==0 used. Existing raw observations were remeasured, not merely called eclipses from BLS.
2. Eclipse timing: PASS. P=20.5247770157 d; B-eclipse reference epoch BTJD 3238.19605357; OLS scatter SE 0.000337625627 d; residual RMS 5.974 min. This SE excludes cadence/template/extraction systematics and is not calibrated coverage. Fits are representative, not all-event precision ephemerides.
3. Target pixels: PASS. Sector 71, 2 event stack, peak pixel [7, 8] equals target [7, 8]; centroid-target 1.51 arcsec. 1 distinct Gaia neighbours could supply the flux under the conservative total-flux bound. The marginally capable neighbour at32.394arcsec is spatially distinct from the1.51arcsec target-centered deficit; it is not discarded merely because it is faint. Catalogue incompleteness/unresolved companions are not excluded. Gaia position used consistently with target ID. The older diffstack n_possible field includes the target itself; this report explicitly removes it.

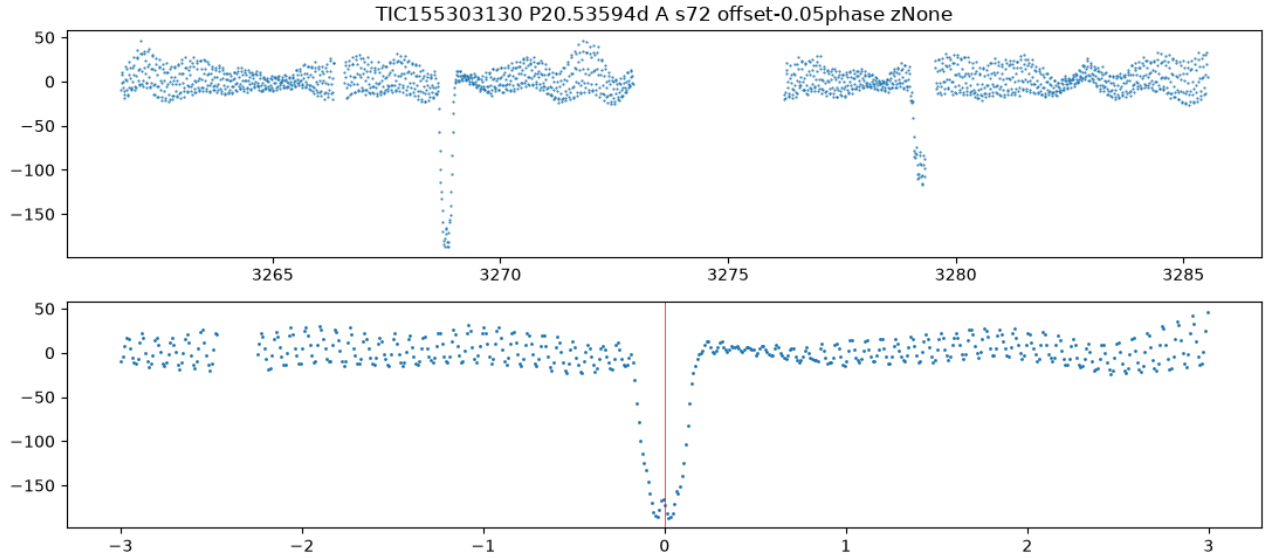
4. Alternative reduction: PASS as a processing check. Own TESScut extraction using median faint-pixel background and target 3x3 aperture reproduces deficit 98.38 ppt. The difference-stack receipt records per-event aperture depths. These are the same TESS photons, not an independent observing occasion; dilution/apertures explain differing QLP depths.
5. Gaia geometry: PASS for consistency, not a precision solution. Eclipse P differs from Gaia by -0.523 quoted Gaia period errors. B-conjunction epoch difference -0.522 propagated marginal-error sigma. Gaia mass function 0.0783 solar masses, minimum companion mass 0.736 solar masses for TIC primary mass 1.520. TIC R1=2.196 solar radii. Illustrative edge-on R2=1solar-radius duration at this conjunction 13.37 h, longer than observed; impact/inclination can shorten it. This is an existence/consistency check, not a fitted mass-radius claim. Correlated Gaia errors were not modeled; no arbitrary sigma veto.

Seven catalogue groups: fresh VSX60arcsec, ASAS-SN60arcsec, SIMBAD exactGaiaDR3, Gaia vari_eclipsing_binary exactID, Prsa2022 5arcsec, IJspeert2024 5arcsec, vetted Kostov2025 tables3+4 5arcsec all returned successful receipts. IJspeert contains exactTIC/Gaia. VSX DSCT and ASAS-SN ROT: classify the same star differently; its eclipses were already published. Full rows, UTC queries and response hashes are in catalogue coverage. The latter two Kostov tables form one group. No external messaging or public/paper edits.

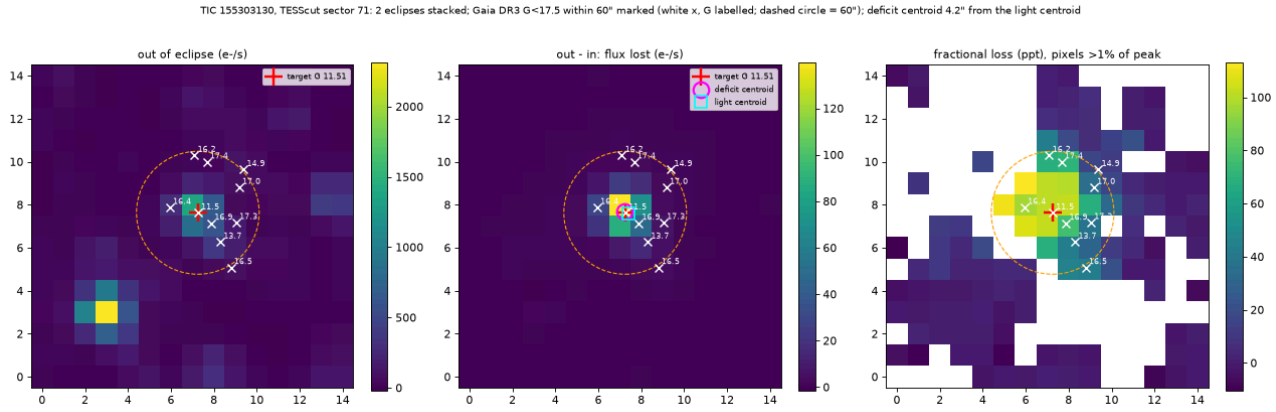
Limitations: Pilot characterization; period errors are scatter-only. No exhaustive literature priority claim. Next check: Supervisor review before admission; independent helper audit completed.

Lead audit: scripts inspected and deciding numeric outputs reconciled with raw SAP plots, full Gaia NSS metadata, source IDs and real TESScut pixels. Apparent failure/non-detection is not called physical refutation unless contrary evidence is present. Independent helper audit uncertainty caveats are retained. The helper verified deciding artifacts, not an exhaustive literature search.

Accepted review key: P024. Pilot selection and limitations, independent audit, and catalogue receipts. Nearby files contain the deciding pixel measurements and hashes. This public archive omits raw FITS and sanitizes local paths; scripts are archival rather than turnkey.



P024; fresh-155303130.png. Exploratory selection and photometric evidence; selection and period qualifications remain as stated.



P024; diff_pilot.png. Target-pixel deficit evidence; the near-orbital-period cases remain probable because capable close neighbours are unresolved.

P025. TIC 382042754 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 25.89267238600749

period_error_days: 4.419206553916731e-05

period_error_meaning: OLSscatter standarderror; excludes cadence/template/systematic uncertainties

reference_epoch_btjd: 3979.672005735797

epoch_btjd: 3979.672005735797

epoch_kind: A

depth_ppt: 87.77880028840502

novelty_status: no prior EB identified in seven scoped groups; cachedChopardv1.2.0 exact IDs no overlap; laterreleases/embargo93 unknown

limitations: Pilot characterization; period errors are scatter-only. No exhaustive literature priority claim.

next_check: Optional precision characterization; no missing bounded admission check

Archived source: sanitized report text.

> Supervisor-accepted physical review: CONFIRMED. Included under the paper's stated physical tier and bounded catalogue scope. Earlier pending-admission wording below records the original handoff.

TIC 382042754: CONFIRMED

Scientific method/population: fresh Gaia-conjunction appointments G11.5-12 P20-60d. Physical review completion: complete bounded first pass, independently audited in helper/AUDIT.md; supervisor admission pending. Catalogue novelty is separate: no prior EB identified in seven scoped groups; cachedChopardv1.2.0 exact IDs no overlap; laterreleases/embargo93 unknown.

1. Repeated raw eclipses: PASS. 4 fit-eligible raw SAP events in sectors [96, 97, 98]. Trapezoid depths median 87.78 ppt; duration median 5.68 h. All saved events, rejected fits and raw files are identified in timing-results.json; no claim that other sectors are clean nulls. QUALITY==0 used. Existing raw observations were remeasured, not merely called eclipses from BLS.
2. Eclipse timing: PASS. P=25.8926723860 d; A-eclipse reference epoch BTJD 3979.67200574; OLS scatter SE 4.41920655e-05 d; residual RMS 0.133 min. This SE excludes cadence/template/extraction systematics and is not calibrated coverage. Fits are representative, not all-event precision ephemerides.
3. Target pixels: PASS. Sector 97, 2 event stack, peak pixel [7, 8] equals target [7, 8]; centroid-target 1.23 arcsec. 0 distinct Gaia neighbours could supply the flux under the conservative total-flux bound. All distinct Gaia neighbours within 60arcsec are too faint under the conservative bound. Catalogue incompleteness/unresolved companions are not excluded. Gaia position used consistently with

target ID. The older diffstack n_possible field includes the target itself; this report explicitly removes it.

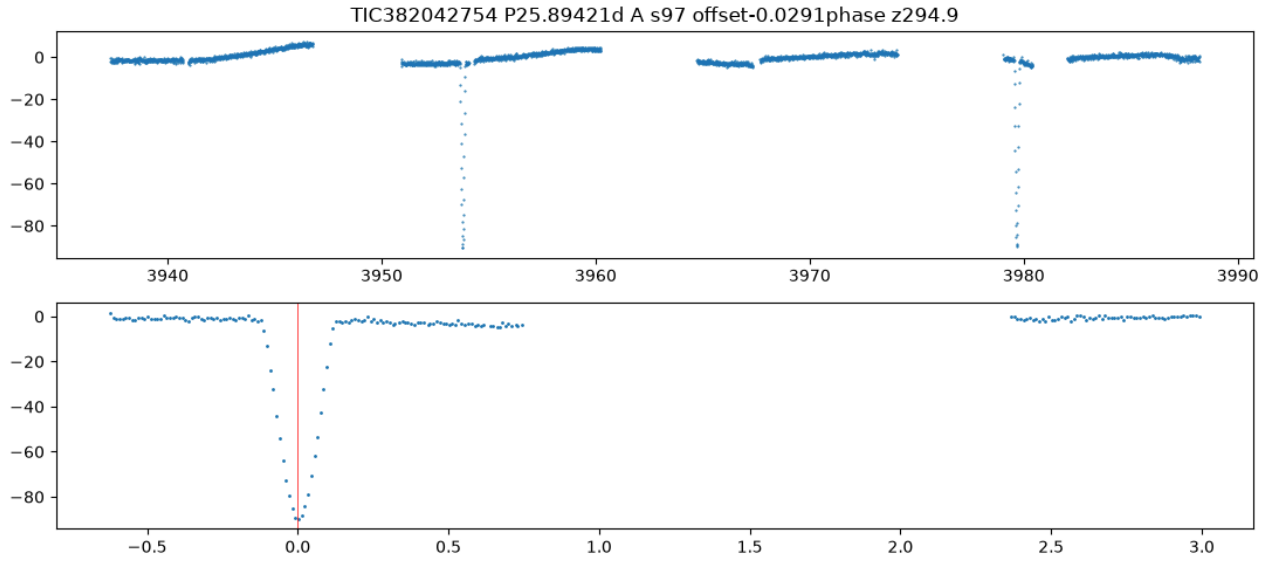
4. Alternative reduction: PASS as a processing check. Own TESScut extraction using median faint-pixel background and target 3x3 aperture reproduces deficit 90.79 ppt. The difference-stack receipt records per-event aperture depths. These are the same TESS photons, not an independent observing occasion; dilution/apertures explain differing QLP depths.
5. Gaia geometry: PASS for consistency, not a precision solution. Eclipse P differs from Gaia by -0.293 quoted Gaia period errors. A-conjunction epoch difference -0.911 propagated marginal-error sigma. Gaia mass function 0.1377 solar masses, minimum companion mass 0.835 solar masses for TIC primary mass 1.220. TIC R1=1.546 solar radii. Illustrative edge-on R2=1solar-radius duration at this conjunction 9.08 h, longer than observed; impact/inclination can shorten it. This is an existence/consistency check, not a fitted mass-radius claim. Correlated Gaia errors were not modeled; no arbitrary sigma veto.

Seven catalogue groups: fresh VSX60arcsec, ASAS-SN60arcsec, SIMBAD exactGaiaDR3, Gaia vari_eclipsing_binary exactID, Prsa2022 5arcsec, IJspeert2024 5arcsec, vetted Kostov2025 tables3+4 5arcsec all returned successful receipts. No on-target EB match in this bounded scope. Any VSX/ASAS-SN neighbour is retained as attribution evidence, not mistaken for target identity. Full rows, UTC queries and response hashes are in catalogue coverage. The latter two Kostov tables form one group. No external messaging or public/paper edits.

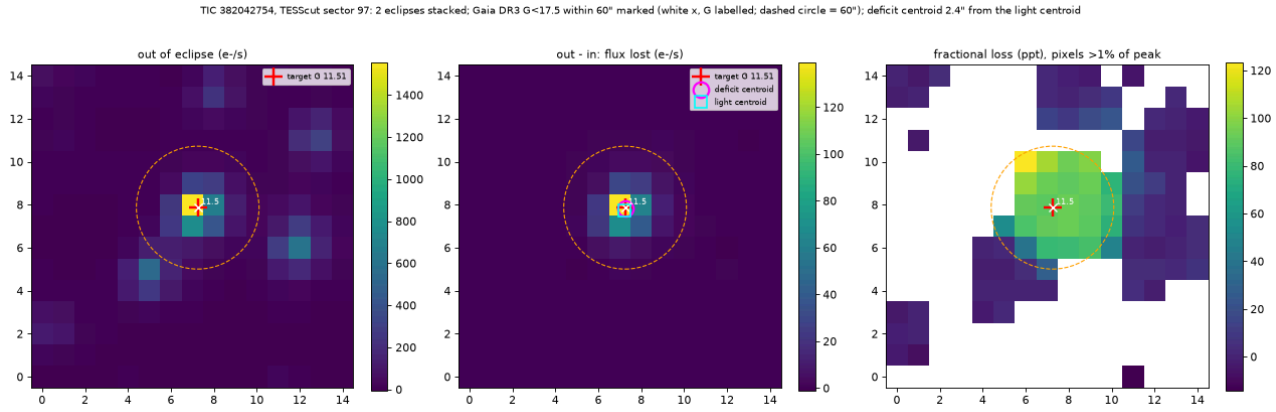
Limitations: Pilot characterization; period errors are scatter-only. No exhaustive literature priority claim. Next check: Supervisor review before admission; independent helper audit completed.

Lead audit: scripts inspected and deciding numeric outputs reconciled with raw SAP plots, full Gaia NSS metadata, source IDs and real TESScut pixels. Apparent failure/non-detection is not called physical refutation unless contrary evidence is present. Independent helper audit uncertainty caveats are retained. The helper verified deciding artifacts, not an exhaustive literature search.

Accepted review key: P025. Pilot selection and limitations, independent audit, and catalogue receipts. Nearby files contain the deciding pixel measurements and hashes. This public archive omits raw FITS and sanitizes local paths; scripts are archival rather than turnkey.



P025; fresh-382042754.png. Exploratory selection and photometric evidence; selection and period qualifications remain as stated.



P025; diff_pilot.png. Target-pixel deficit evidence; the near-orbital-period cases remain probable because capable close neighbours are unresolved.

P026. TIC 389128842 - CONFIRMED

count_addition: true

full_review_completion: COMPLETE_BOUNDED_REVIEW

period_days: 22.666342834406446

period_error_days: 5.4096653855964156e-05

period_error_meaning: OLSscatter standarderror; excludes cadence/template/systematic uncertainties

reference_epoch_btjd: 4134.517930923769

epoch_btjd: 4134.517930923769

epoch_kind: A

depth_ppt: 339.70241891594145

novelty_status: no prior EB identified in seven scoped groups; cachedChopardv1.2.0 exact IDs no overlap; laterreleases/embargo93 unknown

limitations: Pilot characterization; period errors are scatter-only. No exhaustive literature priority claim.

next_check: Optional precision characterization; no missing bounded admission check

Archived source: sanitized report text.

> Supervisor-accepted physical review: CONFIRMED. Included under the paper's stated physical tier and bounded catalogue scope. Earlier pending-admission wording below records the original handoff.

TIC 389128842: CONFIRMED

Scientific method/population: fresh Gaia-conjunction appointments G11.5-12 P20-60d. Physical review completion: complete bounded first pass, independently audited in helper/AUDIT.md; supervisor admission pending. Catalogue novelty is separate: no prior EB identified in seven scoped groups; cachedChopardv1.2.0 exact IDs no overlap; laterreleases/embargo93 unknown.

1. Repeated raw eclipses: PASS. 4 fit-eligible raw SAP events in sectors [102, 103, 104]. Trapezoid depths median 339.70 ppt; duration median 8.39 h. All saved events, rejected fits and raw files are identified in timing-results.json; no claim that other sectors are clean nulls. QUALITY==0 used. Existing raw observations were remeasured, not merely called eclipses from BLS.
2. Eclipse timing: PASS. P=22.6663428344 d; A-eclipse reference epoch BTJD 4134.51793092; OLS scatter SE 5.40966539e-05 d; residual RMS 0.123 min. This SE excludes cadence/template/extraction systematics and is not calibrated coverage. Fits are representative, not all-event precision ephemerides.
3. Target pixels: PASS. Sector 102, 1 event stack, peak pixel [7, 8] equals target [7, 8]; centroid-target 0.54 arcsec. 0 distinct Gaia neighbours could supply the flux under the conservative total-flux bound. All distinct Gaia neighbours within 60arcsec are too faint under the conservative bound. Catalogue incompleteness/unresolved companions are not excluded. Gaia position used consistently with

target ID. The older diffstack n_possible field includes the target itself; this report explicitly removes it.

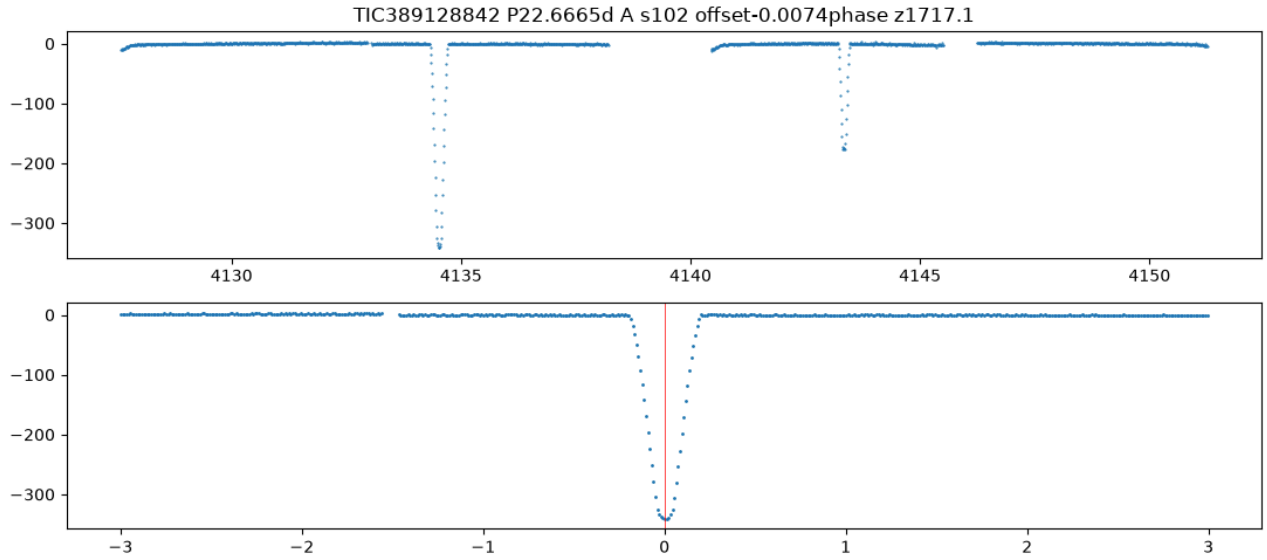
4. Alternative reduction: PASS as a processing check. Own TESScut extraction using median faint-pixel background and target 3x3 aperture reproduces deficit 371.64 ppt. The difference-stack receipt records per-event aperture depths. These are the same TESS photons, not an independent observing occasion; dilution/apertures explain differing QLP depths.
5. Gaia geometry: PASS for consistency, not a precision solution. Eclipse P differs from Gaia by -0.046 quoted Gaia period errors. A-conjunction epoch difference -0.235 propagated marginal-error sigma. Gaia mass function 0.1004 solar masses, minimum companion mass 0.649 solar masses for TIC primary mass 1.000. TIC R1=1.372 solar radii. Illustrative edge-on R2=1solar-radius duration at this conjunction 13.04 h, longer than observed; impact/inclination can shorten it. This is an existence/consistency check, not a fitted mass-radius claim. Correlated Gaia errors were not modeled; no arbitrary sigma veto.

Seven catalogue groups: fresh VSX60arcsec, ASAS-SN60arcsec, SIMBAD exactGaiaDR3, Gaia vari_eclipsing_binary exactID, Prsa2022 5arcsec, IJspeert2024 5arcsec, vetted Kostov2025 tables3+4 5arcsec all returned successful receipts. No on-target EB match in this bounded scope. Any VSX/ASAS-SN neighbour is retained as attribution evidence, not mistaken for target identity. Full rows, UTC queries and response hashes are in catalogue coverage. The latter two Kostov tables form one group. No external messaging or public/paper edits.

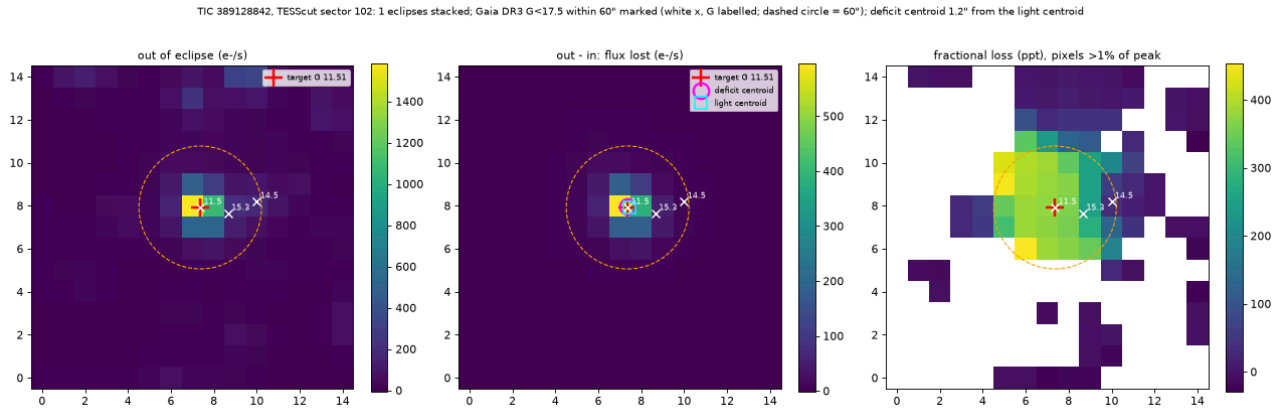
Limitations: Pilot characterization; period errors are scatter-only. No exhaustive literature priority claim. Next check: Supervisor review before admission; independent helper audit completed.

Lead audit: scripts inspected and deciding numeric outputs reconciled with raw SAP plots, full Gaia NSS metadata, source IDs and real TESScut pixels. Apparent failure/non-detection is not called physical refutation unless contrary evidence is present. Independent helper audit uncertainty caveats are retained. The helper verified deciding artifacts, not an exhaustive literature search.

Accepted review key: P026. Pilot selection and limitations, independent audit, and catalogue receipts. Nearby files contain the deciding pixel measurements and hashes. This public archive omits raw FITS and sanitizes local paths; scripts are archival rather than turnkey.



P026; fresh-389128842.png. Exploratory selection and photometric evidence; selection and period qualifications remain as stated.



P026; diff_pilot.png. Target-pixel deficit evidence; the near-orbital-period cases remain probable because capable close neighbours are unresolved.